# Introduction to Plane Geometry

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# Contents

# Notation and Conventions

Before beginning the main text, we explain some notation and conventions that may be unfamiliar to the reader.

**Intersections.** Since a line can be regarded as a set of points, we use $\cap$ for the intersection of two lines. For example, $a \cap b = P$ means that the lines $a, b$ meet at the point $P$. Strictly speaking, the intersection of two sets is itself a set, so the right-hand side would be $\{P\}$; we omit the braces for simplicity. Likewise, if a circle $\omega$ and a line $l$ meet at two points $A, B$, we may write $\omega \cap l = A, B$, again omitting the braces. This convention extends to higher dimensions: $\pi \cap \sigma = l$ means that the planes $\pi, \sigma$ intersect in the line $l$.

**Circles.** We write $\odot O$ for a circle with center $O$ whose radius is not specified, and $\odot(O, r)$ for the circle with center $O$ and radius $r$. In particular, $\odot(O, OA)$ denotes the circle centered at $O$ and passing through $A$. We write $\odot(AB)$ for the circle with diameter $AB$, and $\odot(ABC)$ for the circle through the three points $A, B, C$. The radius of $\odot O$ is written $r_{\odot O}$; for a named circle $\omega$, it is written $r_\omega$.

**Distances.** The symbol dist denotes distance. Thus $\operatorname{dist}(A, B)$ is the distance between the points $A, B$, and $\operatorname{dist}(P, l)$ is the distance from the point $P$ to the line $l$.

**Areas.** The area of a figure is denoted by $S$ or area. For example, $S_{\triangle ABC}$ or $\operatorname{area}(\triangle ABC)$ denotes the area of the triangle $\triangle ABC$. These are unsigned areas; directed areas are introduced separately in the main text.

**Other notation.** Notation not normally encountered in secondary school is explained when it first appears. An index of notation is provided at the end of the book. Pay attention to expressions such as "we shall henceforth write …": they introduce notation for convenience in the subsequent text of this book, without claiming that it is a universal convention.

**Terminology.** New terms are emphasized in italics when they are first introduced.

**Optional material.** A star $*$ marks sections or exercises that are secondary to the main development or more difficult, so that readers may choose whether to study them. This does not mean that every unstarred section or exercise is compulsory; readers may select material according to their interests and return to earlier results when needed. Passages in smaller type serve the same purpose as the star.

Part I

# Foundations of Plane Geometry

*Modern Euclidean Geometry and the Foundations of Conics*

In this part, building on the plane geometry studied in secondary school, we use synthetic methods to introduce further fundamental topics in plane geometry and to prepare for the developments that follow. The part covers the foundations of modern Euclidean geometry (named after Euclid, Εὐκλείδης), the concepts and basic properties of conics, elementary applications of projective geometry to plane geometry, and several familiar and celebrated centers of a triangle.

**Chapter 0**

# Preliminaries: Elementary Plane Geometry

In this chapter we first review the basic facts of elementary plane geometry. Material already presented explicitly in secondary-school textbooks will not be repeated. Instead, the chapter introduces several important elementary facts that are usually absent from those textbooks and prepares the ground for the developments that follow.

We also collect here a number of results needed later but only loosely connected with the subsequent material. The reader may skip these results at first and return to them when they are used, or study them as applications of elementary plane geometry.

## 0.1 Directed Language in Plane Geometry

This section introduces directed segments, directed angles, and directed areas. These notions help remove ambiguities that otherwise arise in geometric arguments.

### 0.1.1 Directed Angles

Among these directed notions, the directed angle is the most important. We therefore begin with it.

**Definition 0.1.1** (Directed angle)**.**

(1) For two lines $l_1, l_2$, the *directed angle* from $l_1$ to $l_2$ is defined as follows.

  a. If $l_1$ and $l_2$ are neither parallel nor coincident, it is the angle through which $l_1$ must be rotated counterclockwise about $l_1 \cap l_2$ until it coincides with $l_2$.[1]

  b. If the lines are parallel or coincident, it is $n \cdot 180^\circ$, where $n$ is any integer.

  We denote this angle by $\measuredangle(l_1, l_2)$. For points $A, O, B$, the directed angle $\measuredangle AOB$ is defined as $\measuredangle(OA, OB)$.

(2) A line admits two choices of positive direction. Once one of them has been chosen, $l$ becomes an *oriented line*, denoted in boldface or with an arrow, say $\boldsymbol{l}$ or $\vec{l}$. The symbol $-\boldsymbol{l}$ denotes the same line with the opposite positive direction. When a distinction is needed, an ordinary line may be called an *unoriented line*; unless the word "oriented" is present, a line is understood to be unoriented.

(3) The directed angle from an oriented line $\boldsymbol{l}_1$ to an oriented line $\boldsymbol{l}_2$ is defined as follows.

  a. If the underlying lines are neither parallel nor coincident, it is the angle through which $\boldsymbol{l}_1$ must be rotated about $l_1 \cap l_2$ until it coincides with $\boldsymbol{l}_2$ with the same positive direction.

  b. If they are parallel or coincident with the same positive direction, it is $n \cdot 360^\circ$, where $n$ is any integer.

  c. If they are parallel or coincident with opposite positive directions, it is $180^\circ + n \cdot 360^\circ$, where $n$ is any integer.

  We denote this angle by $\measuredangle(\boldsymbol{l}_1, \boldsymbol{l}_2)$.

(4) Let $\boldsymbol{a}, \boldsymbol{b}$ be vectors having the same directions as the oriented lines $\boldsymbol{l}_1, \boldsymbol{l}_2$, respectively. The directed angle $\measuredangle(\boldsymbol{a}, \boldsymbol{b})$ is defined to be $\measuredangle(\boldsymbol{l}_1, \boldsymbol{l}_2)$. The directed angle from the ray $AB$ to the ray $CD$ is $\measuredangle(\overrightarrow{AB}, \overrightarrow{CD})$. For points $A, O, B$, define $\overrightarrow{\measuredangle} AOB = \measuredangle(\overrightarrow{OA}, \overrightarrow{OB})$.

[1] A counterclockwise rotation through $-\alpha$, where $\alpha > 0$, means a clockwise rotation through $\alpha$; the same convention is used below.

**Remark.**

(1) We use $\measuredangle$, or $\overrightarrow{\measuredangle}$ when the sides themselves are oriented, to distinguish a directed angle from an ordinary angle $\angle$.

(2) In part (1), if a rotation through $\alpha$ brings the two lines into coincidence, so does a rotation through $\alpha + n \cdot 180^\circ$, $n \in \mathbb{Z}$. This explains why the angle between parallel lines was not defined simply as $0^\circ$. The parallel case is then a limiting case of the nonparallel one, and the definition is consistent. Thus the angle between parallel lines has the form $n \cdot 180^\circ$, $n \in \mathbb{Z}$, and the angles in part (1) are taken modulo $180^\circ$: two values are regarded as equal when they have the same remainder modulo $180^\circ$.[2]

Likewise, the directed angles in parts (3) and (4), whose sides carry directions, are taken modulo $360^\circ$: values with the same remainder on division by $360^\circ$ are regarded as equal.

(3) A directed angle is consequently multivalued. For example, $\measuredangle(l_1,l_2) = 60^\circ$, $\measuredangle(l_1,l_2) = -120^\circ$, and $\measuredangle(l_1,l_2) = 240^\circ$ represent the same directed angle. If $\overrightarrow{\measuredangle}AOB = 70^\circ$ and $\overrightarrow{\measuredangle}A'OB' = -10^\circ$, then one may write $5\overrightarrow{\measuredangle}AOB = \overrightarrow{\measuredangle}A'OB'$.[3] Although $a$ and $b$ are multivalued, elementary number theory shows that $a+b$, $a-b$, and $ka$, $k \in \mathbb{Z}$, are independent of the chosen representatives modulo $180^\circ$ (or modulo $360^\circ$).[4] Thus this multivaluedness causes no difficulty when adding or subtracting directed angles or multiplying them by integers.

(4) The directed angles defined in parts (3) and (4) differ from those in part (1) because their sides carry directions. To emphasize this distinction, the directed angles of part (1) may be called *directed angles of the first kind*, and those of parts (3) and (4) *directed angles of the second kind*. The notation $\overrightarrow{\measuredangle}$ may be used for angles between oriented lines or vectors. Most directed angles used later are of the first kind.

**Proposition 0.1.2.**

*(1) For any lines $l_1, l_2, l_3$, $\measuredangle(l_1,l_2) + \measuredangle(l_2,l_3) = \measuredangle(l_1,l_3)$.*

*(2) For any oriented lines $\boldsymbol{l}_1, \boldsymbol{l}_2, \boldsymbol{l}_3$, $\measuredangle(\boldsymbol{l}_1,\boldsymbol{l}_2) + \measuredangle(\boldsymbol{l}_2,\boldsymbol{l}_3) = \measuredangle(\boldsymbol{l}_1,\boldsymbol{l}_3)$.*

**Proof.** This is immediate. □

Directed angles allow the following definition of a rotation.

**Definition 0.1.3** (Rotation)**.** A *rotation* of the plane with center $O$ and arbitrary real angle $\alpha$ sends each point $A$ to the point $A'$ obtained by rotating $A$ about $O$ through $\alpha$ in the chosen positive direction, counterclockwise by default; equivalently, $\overrightarrow{\measuredangle}AOA' = \alpha$.[5] In what follows, write $\mathfrak{r}_{O,\alpha}$ for this rotation.

For a figure $\mathscr{A}$, the set of images of all its points under the rotation is the *image* $\mathscr{A}'$ of $\mathscr{A}$.

The following properties are immediate.

**Proposition 0.1.4.** *A figure $\mathscr{A}$ and its image $\mathscr{A}'$ under a rotation are congruent.*

**Proposition 0.1.5.** *If $\mathfrak{r}_{O,\alpha}$ sends a line $l$ to a line $l'$, then $\measuredangle(l,l') = \alpha$.*

We can similarly define similarity transformations.

**Definition 0.1.6** (Similarity and similarity transformation)**.**

(1) A *direct similarity transformation* of the plane sends points to points and satisfies both of the following conditions:

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[2]Congruence classes modulo $a$ are also discussed in section A.1.

[3]Notice that $\overrightarrow{\measuredangle}AOB$ is taken modulo $360^\circ$, not modulo $180^\circ$; thus $\overrightarrow{\measuredangle}AOB - 11\overrightarrow{\measuredangle}A'OB' = 0^\circ$ would be false in this example.

[4]The reader can prove this directly without any knowledge of elementary number theory.

[5]If $\alpha < 0$, the rotation is through $-\alpha$ in the direction opposite to the chosen positive direction.

   (i) For any three corresponding pairs $(A,A')$, $(B,B')$, $(C,C')$,[6] one has $\measuredangle ABC = \measuredangle A'B'C'$.
   (ii) There is a fixed positive number $k$ such that $\frac{A'B'}{AB} = k$ for every two corresponding pairs $(A,A')$, $(B,B')$.

   The real number $k$ in condition (ii) is this direct similarity transformation's *similarity ratio*.
(2) An *opposite similarity transformation* of the plane sends points to points and satisfies both of the following conditions:
   (i) For any three corresponding pairs $(A,A')$, $(B,B')$, $(C,C')$, one has $\measuredangle ABC = -\measuredangle A'B'C'$.
   (ii) There is a fixed positive number $k$ such that $\frac{A'B'}{AB} = k$ for every two corresponding pairs $(A,A')$, $(B,B')$.

   The real number $k$ in condition (ii) is the transformation's *similarity ratio*.
(3) Direct and opposite similarity transformations are collectively called *similarity transformations*. For a figure $\mathscr{A}$, construct the images of all its points under a direct or opposite similarity transformation. They form a figure $\mathscr{A}'$, called the *image* of $\mathscr{A}$ under that transformation.
(4) Figures $\mathscr{A}$ and $\mathscr{A}'$ are *similar*, or form a pair of *similar figures*, written $\mathscr{A} \sim \mathscr{A}'$, if a similarity transformation sends $\mathscr{A}$ to $\mathscr{A}'$. The ratio of that transformation is the similarity ratio of $\mathscr{A}'$ to $\mathscr{A}$. If the transformation is direct, the figures are *directly similar*, written $\mathscr{A} \overset{+}{\sim} \mathscr{A}'$; if it is opposite, they are *oppositely similar*, written $\mathscr{A} \overset{-}{\sim} \mathscr{A}'$.
(5) Given $\triangle ABC$, a line $l$ is *antiparallel* to $BC$ with respect to $\triangle ABC$ if the triangle bounded by $l, AB, AC$ is oppositely similar to $\triangle ABC$.

**Remark.** Every similarity transformation $f$ has an inverse $f^{-1}$, with $f^{-1}(f(A)) = A$ for every point $A$. The definition shows that the inverse is again a similarity, with ratio $1/k$. Thus if a similarity sends $\mathscr{A}'$ to $\mathscr{A}$, then $\mathscr{A}'$ is also similar to $\mathscr{A}$. Although the statements "$\mathscr{A}'$ is similar to $\mathscr{A}$" and "$\mathscr{A}$ is similar to $\mathscr{A}'$" are equivalent, the ratio is ordered: if the ratio of $\mathscr{A}'$ to $\mathscr{A}$ is $k$, then that of $\mathscr{A}$ to $\mathscr{A}'$ is $1/k$.

Secondary-school texts usually call two polygons similar when corresponding angles are equal and corresponding side lengths are proportional. This agrees with the transformation-based definition above. Similar triangles are the case used most often.

For polygons the distinction may be stated more simply: equal directed corresponding angles together with proportional corresponding sides gives direct similarity; opposite directed corresponding angles together with proportional corresponding sides gives opposite similarity.

For similar triangles, the following conclusion is immediate.

**Proposition 0.1.7.** *Triangles $ABC$ and $A'B'C'$ are directly similar if and only if*

$$\measuredangle CAB = \measuredangle C'A'B', \qquad \measuredangle ABC = \measuredangle A'B'C', \qquad \measuredangle BCA = \measuredangle B'C'A'.$$

*They are oppositely similar if and only if*

$$\measuredangle CAB = -\measuredangle C'A'B', \qquad \measuredangle ABC = -\measuredangle A'B'C', \qquad \measuredangle BCA = -\measuredangle B'C'A'.$$

*Here the corresponding vertices are understood in their stated order.*[7]

Directed angles reduce dependence on a particular drawing. For example, the familiar cyclicity criterion in theorem 0.1.8 does not require us to distinguish whether $C$ and $D$ lie on the same side of the chord $AB$.

**Theorem 0.1.8.** *Four points $A,B,C,D$ are concyclic if and only if $\measuredangle ACB = \measuredangle ADB$.*

[6] Primed points are the images under the transformation; the same convention is used below.

[7] When we say that $\triangle ABC$ and $\triangle A'B'C'$ are similar, $A,A'$, $B,B'$, and $C,C'$ are understood to be corresponding pairs. The same convention applies to polygons.

**Proof.** If $A, B, C, D$ lie on one circle, elementary geometry gives $\measuredangle ACB = \measuredangle ADB$, whether $D$ lies on the arc $\overset{\frown}{ACB}$ or on the other arc.

Conversely, suppose that $\measuredangle ACB = \measuredangle ADB$. Construct $\odot(ABC)$ to meet $AD$ at $A, D'$. Then $\measuredangle ACB = \measuredangle AD'B$, and hence $\measuredangle ADB = \measuredangle AD'B$. On the fixed line $AD$ and for the fixed point $B$, however, there is only one point $D'$ on $AD$ for which $\measuredangle AD'B$ has a given value (why?). Therefore $D = D'$, as required. □

**Remark.** In the second half of this proof, the hypothesis was awkward to use directly. We constructed another point $D'$ and then proved that it was the original point $D$, that is, $D = D'$. This method is called the *method of coincidence*; it will be used repeatedly below.

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## Exercises

**Exercise 0.1.** Prove that $\measuredangle(l_1, l_2) + \measuredangle(l_3, l_4) = \measuredangle(l_1, l_4) + \measuredangle(l_3, l_2)$.

**Exercise 0.2.** Prove that the area of the polygon $A_0A_1A_2 \cdots A_n$ is

$$S = \left| \frac{1}{2} \sum_{i=1}^{n-1} A_0A_i \cdot A_0A_{i+1} \sin \overrightarrow{\measuredangle} A_iA_0A_{i+1} \right|.$$

Does the result remain valid if the directed angles of the second kind are replaced by directed angles of the first kind?

**Exercise 0.3.** Prove propositions 0.1.4, 0.1.5 and 0.1.7.

**Exercise 0.4.** Prove that applying a direct similarity and then reflecting in a line is equivalent to applying an opposite similarity.

**Exercise 0.5.** Prove that a pair of similar triangles determines a unique similarity transformation: given $\triangle ABC$ and a similar triangle $\triangle A'B'C'$, there is a unique similarity sending the former to the latter.

**Exercise 0.6.** Prove that infinitely many similarity transformations exist between any two circles.

**Exercise 0.7*.** Let $\mathscr{X}$ be the set of all lines through $P$. Fix $l_0 \in \mathscr{X}$ and define an addition on $\mathscr{X}$ by

$$l_3 = l_1 + l_2 \quad \Longleftrightarrow \quad \measuredangle(l_0, l_1) + \measuredangle(l_0, l_2) = \measuredangle(l_0, l_3).$$

This definition applies to every $l_1, l_2, l_3 \in \mathscr{X}$. Prove that $\mathscr{X}$ is an abelian group (named after Abel) under this operation.

### 0.1.2 Directed Segments and Directed Areas

We begin with directed segments.

**Definition 0.1.9** (Directed segment)**.** A segment with a specified direction is called a *directed line segment*. The directed segment with initial point $A$ and terminal point $B$ is denoted by $\overline{AB}$. Once the positive direction has been fixed, its length may be positive or negative. If the direction from $A$ to $B$ is positive, then $\overline{AB} > 0$ and $\overline{BA} < 0$.

**Remark.** This notion differs essentially from a vector: the sign belongs only to the length and does not encode an additional spatial direction. The overline distinguishes a directed segment from an ordinary one. We use the following conventions.

(1) Collinear or parallel directed segments use the same positive direction. Unless needed, that direction is not stated explicitly, since directed lengths usually occur in products or ratios whose values are independent of the choice. For example, if $A, B, C$ occur in this order on a line, then $\overline{AB} \cdot \overline{BC}$ and $\frac{\overline{AB}}{\overline{BC}}$ are positive, whereas $\overline{AB} \cdot \overline{CB}$ and $\frac{\overline{AB}}{\overline{CB}}$ are negative, whichever of the two directions, from $A$ to $B$ or from $B$ to $A$, is chosen.

(2) Relative to a fixed reference triangle, suppose that $Q$ is the projection of $P$ on one of its sides. If $P$ lies on the exterior side of that sideline, set $\overline{PQ} < 0$ (and $\overline{QP} > 0$); if $P$ lies on the interior side, set $\overline{PQ} > 0$ (and $\overline{QP} < 0$). Here the interior side is the half-plane containing the interior of the triangle. The same rule applies to a reference polygon.

(3) Since all perpendiculars to a fixed line are parallel, a directed distance from a point to a line can be defined. If $Q$ is the projection of $P$ on $l$, then $\overline{PQ}$ is the *directed distance* from $P$ to $l$. In what follows, write $\mathrm{dist}_{\pm}(P, l)$ for this directed distance. Thus distances on one side of $l$ are positive and those on the other side are negative.

**Proposition 0.1.10.** *If $A, B, C$ are collinear, then $\overline{AB} + \overline{BC} = \overline{AC}$.*

**Proof.** This is immediate. □

We next introduce directed areas.

**Definition 0.1.11** (Directed area)**.** An area supplied with a sign convention is called a *directed area*. We write it in square brackets; for instance, the directed area of $\triangle ABC$ is $[\triangle ABC]$. If $A, B$ are fixed and $C$ crosses the line $AB$, then $[\triangle ABC]$ changes sign.

**Remark.** Directed areas are used primarily for triangles. We adopt the following conventions.

(1) As with directed segments, for triangles with a common base, the positive sign is left unstated unless needed. With $A, B$ fixed, moving $C$ across $AB$ reverses the sign of $[\triangle ABC]$.

(2) For triangles with a common altitude, we likewise leave the positive direction unstated unless needed. The sign agrees with the positive direction of the directed bases corresponding to that altitude. Thus, if $A, B, C, D$ lie on $l$ and $P \notin l$, then $[\triangle ABP] : [\triangle CDP] = \overline{AB} : \overline{CD}$.

(3) As with directed segments, relative to a reference triangle, the directed area of the triangle formed by $P$ and one side is negative when $P$ lies outside that side and positive when it lies inside. The same convention applies to a reference polygon.

Directed segments, angles, and areas reduce dependence on the drawing. Besides the example in the preceding subsection, they yield the following elegant identity. With directed areas, we need not distinguish whether $P$ lies inside or outside $\triangle ABC$. The reader is invited to prove it.

**Theorem 0.1.12.** *For a triangle $ABC$ and a point $P$,*

$$[\triangle BPC]\overrightarrow{PA} + [\triangle CPA]\overrightarrow{PB} + [\triangle APB]\overrightarrow{PC} = \mathbf{0}.$$

Here is a short proof using cross products. A reader unfamiliar with cross products may first consult section A.2.1, or prove the identity by elementary methods.

**Proof.** By the geometric meaning of the cross product, $2[\triangle BPC]\hat{z} = \overrightarrow{PB} \times \overrightarrow{PC}$, where $\hat{z}$ is the unit vector perpendicular to the plane. Hence

$$2[\triangle BPC]\overrightarrow{PA} = [(\overrightarrow{PB} \times \overrightarrow{PC}) \times \overrightarrow{PA}] \times \hat{z}.$$

The magnitude of the expression in square brackets is already correct; its direction is perpendicular to $\overrightarrow{PA}$ in the plane, and the final cross product with $\hat{z}$ returns it to the direction of $\overrightarrow{PA}$. Treating the other two terms similarly gives

$$\text{LHS} = \left[ (\overrightarrow{PB} \times \overrightarrow{PC}) \times \overrightarrow{PA} + (\overrightarrow{PC} \times \overrightarrow{PA}) \times \overrightarrow{PB} + (\overrightarrow{PA} \times \overrightarrow{PB}) \times \overrightarrow{PC} \right] \times \frac{\hat{z}}{2}.$$

The expression in square brackets vanishes by the Jacobi identity. □

Directed quantities also give convenient formulations of Ceva's theorem and the Menelaus theorem.

**Lemma 0.1.13.** *Let $D$ lie on the line $BC$ of $\triangle ABC$. For a point $P$, one has $P \in AD$ if and only if*

$$\frac{\overline{BD}}{\overline{CD}} = \frac{[\triangle BPA]}{[\triangle CPA]}.$$

**Proof.** Clearly,

$$\frac{\overline{BD}}{\overline{CD}} = \frac{[\triangle BPD]}{[\triangle CPD]} = \frac{[\triangle BAD]}{[\triangle CAD]}.$$

If $P \in AD$, the elementary properties of proportions[8] give

$$\frac{[\triangle BPA]}{[\triangle CPA]} = \frac{[\triangle BAD] - [\triangle BPD]}{[\triangle CAD] - [\triangle CPD]} = \frac{\overline{BD}}{\overline{CD}}.$$

Conversely, suppose $\frac{\overline{BD}}{\overline{CD}} = \frac{[\triangle BPA]}{[\triangle CPA]}$, and put $D' = AP \cap BC$. The preceding argument gives $\frac{\overline{BD'}}{\overline{CD'}} = \frac{[\triangle BPA]}{[\triangle CPA]}$. But $\frac{\overline{BD}}{\overline{CD}} \neq \frac{\overline{BD'}}{\overline{CD'}}$ unless $D = D'$. Thus $D = D'$, and $A, P, D$ are collinear. □

**Theorem 0.1.14** (Ceva's theorem)**.** *Given $\triangle ABC$, let $A_1, B_1, C_1$ lie on the lines $BC, CA, AB$, respectively. Then $AA_1, BB_1, CC_1$ meet at a point $P$[9] if and only if*

$$\frac{\overline{BA_1}}{\overline{CA_1}} \cdot \frac{\overline{CB_1}}{\overline{AB_1}} \cdot \frac{\overline{AC_1}}{\overline{BC_1}} = -1.$$

**Proof.** We prove only the forward implication. By lemma 0.1.13,

$$\frac{\overline{BA_1}}{\overline{CA_1}} \cdot \frac{\overline{CB_1}}{\overline{AB_1}} \cdot \frac{\overline{AC_1}}{\overline{BC_1}} = \frac{[\triangle APB]}{[\triangle APC]} \cdot \frac{[\triangle BPC]}{[\triangle BPA]} \cdot \frac{[\triangle CPA]}{[\triangle CPB]} = -1.$$ □

**Theorem 0.1.15** (Menelaus's theorem (Μενέλαος))**.** *Given $\triangle ABC$, let $D, E, F$ lie on the lines $BC, CA, AB$, respectively. Then $D, E, F$ are collinear if and only if*

$$\frac{\overline{BD}}{\overline{CD}} \cdot \frac{\overline{CE}}{\overline{AE}} \cdot \frac{\overline{AF}}{\overline{BF}} = 1.$$

**Proof.** Again we prove only the forward implication. Lemma 0.1.13 gives

$$\frac{\overline{BD}}{\overline{CD}} \cdot \frac{\overline{CE}}{\overline{AE}} \cdot \frac{\overline{AF}}{\overline{BF}} = \frac{[\triangle BED]}{[\triangle CED]} \cdot \frac{[\triangle CED]}{[\triangle AED]} \cdot \frac{[\triangle AED]}{[\triangle BED]} = 1.$$ □

Directed language is occasionally less intuitive. We shall therefore sometimes use undirected language and accompany it with a figure. A reader who prefers directed language may reformulate the argument accordingly. When a proof is illustrated and written in undirected language, it is understood to follow the configuration shown; the other configurations are analogous and will not be discussed separately.

[8] If $\frac{a}{b} = \frac{c}{d}$, then, whenever the displayed fractions are defined, (1) $\frac{a}{b} = \frac{a+c}{b+d}$ and (2) $\frac{a}{b} = \frac{a-c}{b-d}$.

[9] Strictly speaking, the three lines may instead be parallel; that case can be regarded as the limiting case in which their common point tends to infinity.

## Exercises

**Exercise 0.8.** Four collinear points $A,B,C,D$ satisfy $\overline{AB}\cdot\overline{CD}=\overline{BC}\cdot\overline{AD}$. If $O$ is the midpoint of $AC$, prove that $\overline{OA}^2=\overline{OB}\cdot\overline{OD}$.

**Exercise 0.9.** Given points $A,B,C,D$, let $E$ be the midpoint of $AD$. Prove that $[\triangle ABC]+[\triangle DBC]=2[\triangle EBC]$.

**Exercise 0.10.** Prove the theorems.

(1) Prove the reverse implications in Ceva's theorem and the Menelaus theorem.
(2) Prove theorem 0.1.12 without using cross products.

**Exercise 0.11.** Suppose $\triangle ABD \stackrel{\sim}{-} \triangle ACE$, with $D,E$ outside $\triangle ABC$. Let $U,V,X,Y,Q$ be the midpoints of $AD,AE,BD,CE,BC$, respectively, and put $P=UY\cap VX$. Prove that $A,P,Q$ are collinear.

**Exercise 0.12.** Let $A,B,P,Q,R$ be five distinct points with $R\in PQ$. Prove that

$$[RAB]=\frac{\overline{PR}}{\overline{PQ}}[QAB]+\frac{\overline{QR}}{\overline{QP}}[PAB].$$

**Exercise 0.13.** Let $A,B,C$ lie on $\odot O$. Prove that

$$\sin 2\angle CAB\cdot\overrightarrow{OA}+\sin 2\angle ABC\cdot\overrightarrow{OB}+\sin 2\angle BCA\cdot\overrightarrow{OC}=\mathbf{0}.$$

## 0.2 Triangles and Quadrilaterals

We first review several theorems concerning triangles and quadrilaterals.

**Theorem 0.2.1** (Right-triangle altitude theorem; Euclidean theorem). *Let $ABC$ be a right triangle, and let $CD$ be the altitude to the hypotenuse $AB$, where $D\in AB$. Then*

$$CD^2=AD\cdot BD,\qquad CA^2=AD\cdot AB,\qquad CB^2=BD\cdot AB.$$

**Proof.** Observe that $\triangle ACB\sim\triangle CDB\sim\triangle ADC$. The result follows immediately. □

**Theorem 0.2.2** (Generalized angle-bisector theorem). *Let $D$ lie on the line $BC$ of $\triangle ABC$. Then*

$$\frac{\overline{BD}}{\overline{CD}}=\frac{AB}{AC}\cdot\frac{\sin\measuredangle BAD}{\sin\measuredangle CAD}.$$

**Proof.**

$$\frac{\overline{BD}}{\overline{CD}}=\frac{[\triangle BAD]}{[\triangle CAD]}=\frac{AB\cdot AD\sin\measuredangle BAD}{AC\cdot AD\sin\measuredangle CAD}=\frac{AB\sin\measuredangle BAD}{AC\sin\measuredangle CAD}.$$ □

When $AD$ is an angle bisector, this gives two important special cases: the angle-bisector theorem and the external angle-bisector theorem.

**Theorem 0.2.3** (Angle-bisector theorem). *If the bisector of $\angle BAC$ in $\triangle ABC$ meets $BC$ at $D$, then* $\frac{AB}{AC}=\frac{BD}{CD}$.

**Theorem 0.2.4** (External angle-bisector theorem)**.** *If the external bisector of $\angle BAC$ in $\triangle ABC$ meets the line $BC$ at $D$, then $\frac{AB}{AC} = \frac{BD}{CD}$.*

We next recall the sine rule, the cosine rule, and some related consequences.

**Theorem 0.2.5** (Sine rule)**.** *In $\triangle ABC$,*

$$\frac{BC}{\sin A} = \frac{CA}{\sin B} = \frac{AB}{\sin C} = 2R,$$

*where $R$ is its circumradius.*

**Proof.** This is familiar. □

**Theorem 0.2.6** (Cosine rule)**.** *In $\triangle ABC$, $BC^2 = AB^2 + AC^2 - 2AB \cdot AC \cos A$.*

**Proof.** This is familiar. □

**Corollary 0.2.7.** *In $\triangle ABC$,*

$$\cot A = \frac{AB^2 + AC^2 - BC^2}{4S_{\triangle ABC}}.$$

**Proof.**

$$\cot A = \frac{\cos A}{\sin A} = \frac{AB^2 + AC^2 - BC^2}{2AB \cdot AC \sin A} = \frac{\cos A}{\sin A} = \frac{AB^2 + AC^2 - BC^2}{4S_{\triangle ABC}}.$$ □

**Theorem 0.2.8** (Stewart's theorem)**.** *Let $D$ lie on the line $BC$ of $\triangle ABC$. Then*

$$AD^2 = AB^2 \cdot \frac{\overline{DC}}{\overline{BC}} + AC^2 \cdot \frac{\overline{BD}}{\overline{BC}} - \overline{BD} \cdot \overline{DC}.$$

**Proof.** Put $\angle ADB = \theta$. By the cosine rule,

$$\cos\theta = \frac{AD^2 + BD^2 - AB^2}{2AD \cdot \overline{BD}}, \qquad -\cos\theta = \frac{AD^2 + CD^2 - AC^2}{2AD \cdot \overline{DC}}.$$

Adding and simplifying gives the result. □

Stewart's theorem is important in triangle geometry, but its formula is rather heavy and lacks geometric elegance. We shall not use it later, except once in the special case in which $D$ is a midpoint.

**Theorem 0.2.9** (Median-length formula)**.** *If $D$ is the midpoint of $BC$ in $\triangle ABC$, then*

$$AD^2 = \frac{AB^2}{2} + \frac{AC^2}{2} - \frac{BC^2}{4}.$$

There is also a special formula for the length of an angle bisector; see the exercises below.

The following useful criterion is frequently used to prove perpendicularity.

**Theorem 0.2.10** (Perpendicularity lemma)**.** *The lines $AB$ and $CD$ are perpendicular if and only if*

$$AC^2 - AD^2 = BC^2 - BD^2.$$

**Proof.** This follows immediately from the Pythagorean theorem (Pythagoras, Πυθαγόρας). □

We conclude this part with Echols's first theorem and a generalization.

**Theorem 0.2.11** (Echols's first theorem)**.** *As in Figure 0.1, let $\triangle ABC$ and $\triangle DEF$ be similarly oriented equilateral triangles.*[10] *If $X,Y,Z$ are the respective midpoints of the nonintersecting segments $AD,BE,CF$, then $\triangle XYZ$ is equilateral.*

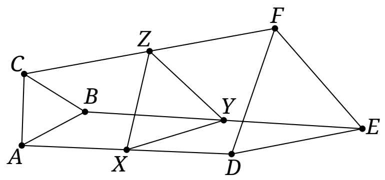


FIGURE 0.1

Rather than prove this result separately, we prove the following generalization.

**Theorem 0.2.12.** *As in Figure 0.2, suppose $\triangle ABC \overset{\pm}{\sim} \triangle A'B'C'$. Let $X,Y,Z$ lie on the lines $AA',BB',CC'$, respectively, with $\frac{\overline{AX}}{\overline{A'X}} = \frac{\overline{BY}}{\overline{B'Y}} = \frac{\overline{CZ}}{\overline{C'Z}}$. Then $\triangle ABC \overset{\pm}{\sim} \triangle XYZ$.*

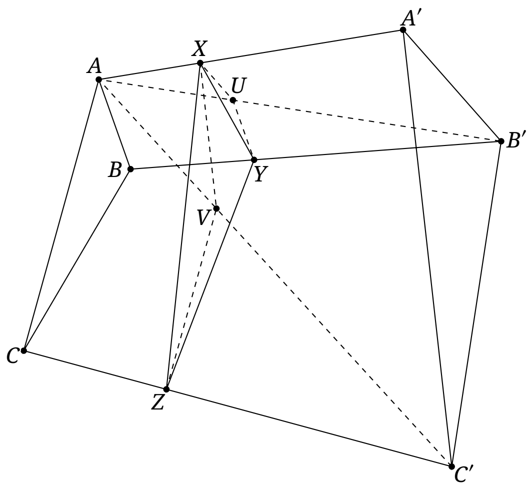


FIGURE 0.2

**Proof.** Choose $U \in AB'$ and $V \in AC'$ so that $\dfrac{\overline{AU}}{\overline{B'U}} = \dfrac{\overline{AV}}{\overline{C'V}} = \dfrac{\overline{AX}}{\overline{A'X}} = \dfrac{\overline{BY}}{\overline{B'Y}} = \dfrac{\overline{CZ}}{\overline{C'Z}}$. Then $XU \parallel A'B'$, $UY \parallel AB$, $XV \parallel A'C'$, and $ZV \parallel AC$. Hence

$$\frac{XV}{A'C'} = \frac{XU}{A'B'}, \qquad \frac{ZV}{AC} = \frac{YU}{AB}.$$

Since the original triangles are directly similar, $\frac{A'C'}{A'B'} = \frac{AC}{AB}$, and therefore $\frac{XV}{XU} = \frac{VZ}{UY}$.

On the other hand,

$$\begin{aligned}
\measuredangle XVZ &= \measuredangle XVA + \measuredangle AVZ = -(\measuredangle AXV + \measuredangle VAX) - \measuredangle CAV \\
&= -\measuredangle CAX - \measuredangle AXV = -\measuredangle CAX - \measuredangle AA'C', \\
\measuredangle XUY &= \measuredangle XUA + \measuredangle AUY = -(\measuredangle XAU + \measuredangle AXU) - \measuredangle BAU \\
&= -\measuredangle BAX - \measuredangle AXU = -\measuredangle BAX - \measuredangle AA'B'.
\end{aligned}$$

Thus

$$\measuredangle XVZ - \measuredangle XUY = (\measuredangle BAX - \measuredangle CAX) + (\measuredangle AA'B' - \measuredangle AA'C') = -\measuredangle CAB + \measuredangle C'A'B' = 0.$$

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[10] That is, $\triangle ABC \overset{\pm}{\sim} \triangle DEF$.

Thus $\measuredangle XVZ = \measuredangle XUY$. Hence $\triangle XVZ \overset{\pm}{\sim} \triangle XUY$, so $\frac{XY}{XZ} = \frac{YU}{ZV} = \frac{AB}{AC}$. Similarly, $\frac{AB}{CB} = \frac{XY}{ZY}$. Therefore $AB : BC : CA = XY : YZ : ZX$, proving the proposition. □

**Corollary 0.2.13.** *Let the n-gons satisfy $A_1A_2\cdots A_n \overset{\pm}{\sim} A_1'A_2'\cdots A_n'$. Choose $X_1, X_2, \ldots, X_n$ on $A_1A_1'$, $A_2A_2'$, ..., $A_nA_n'$, respectively, so that*

$$\frac{\overline{X_1A_1}}{\overline{X_1A_1'}} = \frac{\overline{X_2A_2}}{\overline{X_2A_2'}} = \cdots = \frac{\overline{X_nA_n}}{\overline{X_nA_n'}}.$$

*Then $A_1A_2\cdots A_n \overset{\pm}{\sim} X_1X_2\cdots X_n$.*

**Proof.** Triangulate the polygons and apply theorem 0.2.12. □

**Corollary 0.2.14** (Finsler–Hadwiger theorem)**.** *As in Figure 0.3, let the squares ABCD and EBGI have the common vertex B. Let $L, K$ be the midpoints of the nonintersecting segments $AE, CG$, and let $H, J$ be the centers of the two squares. Then $LHKJ$ is a square.*

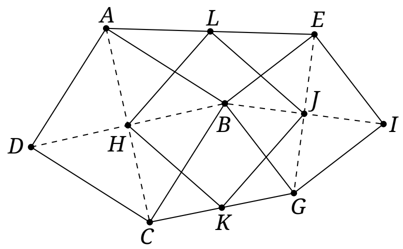


FIGURE 0.3

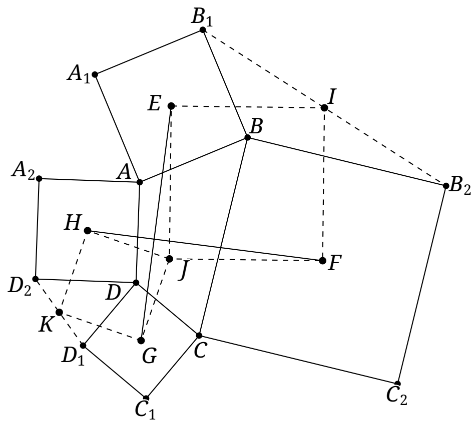


FIGURE 0.4

**Proof.** This is the special case of corollary 0.2.13 in which the polygons are squares and all the selected points $X_i$ are midpoints. □

The Finsler–Hadwiger theorem has the following application.

**Theorem 0.2.15** (van Aubel's theorem)**.** *Construct outward squares on the four sides of a quadrilateral. The segments joining the centers of opposite squares are equal and perpendicular.*

**Proof.** In Figure 0.4, the quadrilaterals $AA_1B_1B$, $BB_2C_2C$, $CC_1D_1D$, and $DD_2A_2A$ are squares, with centers $E, F, G, H$, respectively. We must prove that $EG$ and $HF$ are equal and perpendicular.

Let $I, J, K$ be the respective midpoints of $B_1B_2, AC, D_1D_2$. The Finsler–Hadwiger theorem shows that $HJKG$ and $EIFJ$ are squares. Under the rotation $\mathfrak{r}_{J,90^\circ}$, the points $F, H$ go to $E, G$, respectively. Hence the segments joining corresponding points, $HF$ and $EG$, are equal and form the rotation angle; the assertion follows. □

**Corollary 0.2.16** (Thébault's first theorem)**.** *Construct outward squares on the sides of a parallelogram. Their centers form a square.*

**Proof.** In the proof of theorem 0.2.15, additionally suppose that $ABCD$ is a parallelogram. Then $J$ is its center and the entire configuration is centrally symmetric about $J$. Thus $E,G$ and $H,F$ are opposite pairs under this symmetry. Since $EG$ and $HF$ are equal and perpendicular, $EFGH$ is a square. □

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## Exercises

**Exercise 0.14** (Thébault's second theorem)**.** Let $ABCD$ be a square. On its sides $AB,BC$, construct outward equilateral triangles $\triangle ABP, \triangle BCQ$, respectively, and inward equilateral triangles $\triangle ABP', \triangle BCQ'$, respectively. Prove that $D,P,Q'$ are collinear, that $D,P',Q$ are collinear, and that both $\triangle DP'Q'$ and $\triangle DPQ$ are equilateral.

**Exercise 0.15** (Trigonometric Ceva)**.**
Let $A_1,B_1,C_1$ lie on the lines containing the sides $BC,CA,AB$ of $\triangle ABC$, respectively. Prove that each of the following is necessary and sufficient for $AA_1,BB_1,CC_1$ to be concurrent (or mutually parallel):

(1) First trigonometric form:
$$\frac{\sin\measuredangle CAA_1}{\sin\measuredangle BAA_1}\frac{\sin\measuredangle ABB_1}{\sin\measuredangle CBB_1}\cdot\frac{\sin\measuredangle BCC_1}{\sin\measuredangle ACC_1}=-1;$$
(2) Second trigonometric form:
$$\frac{\sin\measuredangle BXA_1}{\sin\measuredangle CXA_1}\cdot\frac{\sin\measuredangle CXB_1}{\sin\measuredangle AXB_1}\cdot\frac{\sin\measuredangle AXC_1}{\sin\measuredangle BXC_1}=-1,$$
where $X$ lies on none of the lines $AB,BC,CA$.

**Exercise 0.16** (Trigonometric Menelaus's theorem)**.** Let $A_1,B_1,C_1$ lie on the lines containing the sides $BC,CA,AB$ of $\triangle ABC$, respectively. Prove that each of the following is necessary and sufficient for $A_1,B_1,C_1$ to be collinear:

(1) First trigonometric form:
$$\frac{\sin\measuredangle CAA_1}{\sin\measuredangle BAA_1}\frac{\sin\measuredangle ABB_1}{\sin\measuredangle CBB_1}\cdot\frac{\sin\measuredangle BCC_1}{\sin\measuredangle ACC_1}=1;$$
(2) Second trigonometric form:
$$\frac{\sin\measuredangle BXA_1}{\sin\measuredangle CXA_1}\cdot\frac{\sin\measuredangle CXB_1}{\sin\measuredangle AXB_1}\cdot\frac{\sin\measuredangle AXC_1}{\sin\measuredangle BXC_1}=1,$$
where $X$ lies on none of the lines $AB,BC,CA$.

**Exercise 0.17** (Length of an internal angle bisector)**.** In $\triangle ABC$, let the bisector of $\angle A$ meet $BC$ at $D$. If the side lengths are $a,b,c$ and $p=\frac{a+b+c}{2}$, prove that
$$AD=\sqrt{AB\cdot AC-BD\cdot CD}=\frac{2}{b+c}\sqrt{bcp(p-a)}.$$

**Exercise 0.18** (Length of an external angle bisector)**.** In $\triangle ABC$, let the external bisector of $\angle A$ meet $BC$ at $D$. Write $a,b,c$ for the side lengths and put $p=\frac{a+b+c}{2}$. Prove that
$$AD=\sqrt{BD\cdot CD-AB\cdot AC}=\frac{2}{|b-c|}\sqrt{bc(p-b)(p-c)}.$$

**Exercise 0.19** (Steiner–Lehmus theorem)**.** In $\triangle ABC$, let the bisectors of $\angle B$ and $\angle C$ meet the opposite sides at $D$ and $E$. If $BD=CE$, prove that $AB=AC$.

**Exercise 0.20** (Heron's formula (῞Ηρων))**.** If the side lengths of $\triangle ABC$ are $a,b,c$ and $p=\frac{a+b+c}{2}$, prove that

$$S_{\triangle ABC}=\sqrt{p(p-a)(p-b)(p-c)}.$$

**Exercise 0.21** (Brahmagupta's formula (ब्रह्मगुप्त))**.** Let the side lengths of $ABCD$ be $a,b,c,d$, let half the sum of a pair of opposite angles be $\theta$, and put $p=\frac{a+b+c+d}{2}$. Prove that

$$S_{ABCD}=\sqrt{(p-a)(p-b)(p-c)(p-d)-abcd\cos^2\theta}.$$

In particular, if $ABCD$ is cyclic, then

$$S_{ABCD}=\sqrt{(p-a)(p-b)(p-c)(p-d)}.$$

**Exercise 0.22** (Abī Khuzām's theorem (أبي خزام))**.** Prove that, for $\triangle ABC$,

$$\sin A\sin B\sin C\le\left(\frac{3\sqrt{3}}{2\pi}\right)^3 ABC.$$

**Exercise 0.23.** For a point $P$ inside a rectangle $ABCD$, prove that $PA^2+PC^2=PB^2+PD^2$.

**Exercise 0.24.** In a convex quadrilateral $ABCD$, suppose $AB\perp BC$, $AD\perp DC$, and $AB=AD$. Choose points $E,F$ on the sides $CD,CB$, respectively, so that $DF\perp AE$. Prove that $AF\perp BE$.

**Exercise 0.25.** Let $\triangle ABC$, $\triangle CDE$, $\triangle BFG$, and $\triangle AHI$ be four similarly oriented equilateral triangles. If $X,Y,Z$ are the midpoints of $DG,EH,FI$, respectively, prove that $\triangle XYZ$ is equilateral.

## 0.3 Properties of Circles

### 0.3.1 The Relative Position of a Line and a Circle

We begin with several of the best-known theorems concerning lines and circles.

**Theorem 0.3.1** (Tangent–chord theorem)**.** An angle whose vertex lies on a circle and one of whose sides is tangent to the circle is called a *tangent–chord angle. Its measure equals that of an inscribed angle subtending the intercepted arc. More explicitly, in Figure 0.5, AP is a chord of a circle* $\omega$*, B lies on the tangent to* $\omega$ *at P, and* $\angle PQA$ *is the inscribed angle subtending the arc* $\overset{\frown}{PA}$ *intercepted by* $\angle BPA$*; then* $\angle PQA=\angle BPA$.

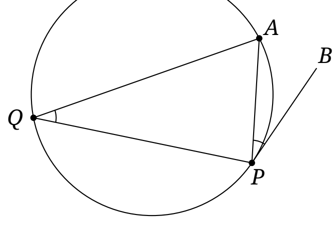


FIGURE 0.5

**Theorem 0.3.2** (Equal tangents theorem)**.** *Let $P$ be outside $\odot O$, and draw its two tangents from $P$, touching $\odot O$ at $A$ and $B$. Then $PA = PB$.*[11]

**Theorem 0.3.3** (Intersecting chords theorem)**.** *Let $P$ be inside $\odot O$. Two chords through $P$ meet $\odot O$ at $A,B$ and $C,D$, respectively. Then $PA \cdot PB = PC \cdot PD$.*

**Theorem 0.3.4** (Secant–secant theorem)**.** *Let $P$ be outside $\odot O$. Two secants through $P$ meet $\odot O$ at $A,B$ and $C,D$, respectively. Then $PA \cdot PB = PC \cdot PD$.*

**Theorem 0.3.5** (Tangent–secant theorem)**.** *Let $P$ be outside $\odot O$. A secant through $P$ meets $\odot O$ at $A,B$, and a tangent from $P$ touches $\odot O$ at $C$. Then $PA \cdot PB = PC^2$.*

The proofs of these five theorems are straightforward. The equal tangents theorem is immediate, and the tangent–chord theorem is left to the reader. For the remaining three, it suffices to observe that $\triangle PAC \overset{-}{\sim} \triangle PDB$.

The intersecting chords, secant–secant, tangent–secant, and equal tangents theorems are sometimes collectively called the *power of a point theorem*. The intersecting-chords and secant–secant statements have the following converse, whose proof is left to the reader.

**Theorem 0.3.6.** *Let the lines $AB$ and $CD$ meet at $P$. If $\overline{PA} \cdot \overline{PB} = \overline{PC} \cdot \overline{PD}$, then $A,B,C,D$ are concyclic.*

We now give several other important results about circles.

**Theorem 0.3.7** (Apollonius's circle theorem (Ἀπολλώνιος))**.** *Given points $A,B$ and a positive real number $k$,*[12] *the locus of points $P$ satisfying $\frac{PA}{PB} = k$ is a circle whose center lies on $AB$.* This is the *Apollonius circle* associated with $A,B$.

**Proof.** Let the internal and external bisectors of $\angle APB$ meet $AB$ at $X$ and $Y$, respectively. By the internal and external angle-bisector theorems, $X,Y$ are fixed. Since $\angle XPY = 90^\circ$, the point $P$ lies on the circle with diameter $XY$. □

**Proposition 0.3.8.** *A quadrilateral is cyclic if and only if the bisectors of the angles formed by its two pairs of opposite sides are perpendicular.*

**Proof.** We prove the forward implication; the converse follows by the method of coincidence. In Figure 0.6, the points $A,B,C,D$ are concyclic, $HE$ bisects $\angle AED$ and meets $\odot O$ at $P$, while $HF$ bisects $\angle BFA$ and meets $\odot O$ at $Q$. Then

$$\begin{aligned}\angle EHF = \angle ECF - \angle CEP - \angle CFH = \angle BCD - \frac{\angle BEC}{2} - \frac{\angle DFC}{2}\\ \overset{\mathrm{m}}{=} \frac{\overset{\frown}{BAD}}{2} - \frac{1}{2}\cdot\frac{\overset{\frown}{DA} - \overset{\frown}{BC}}{2} - \frac{1}{2}\cdot\frac{\overset{\frown}{BA} - \overset{\frown}{CD}}{2} = \frac{\overset{\frown}{BAD}}{2} - \frac{\overset{\frown}{BAD}}{4} + \frac{\overset{\frown}{BD}}{4} = \frac{\overset{\frown}{BAD}}{4} + \frac{\overset{\frown}{BD}}{4} \overset{\mathrm{m}}{=} 90^\circ.\end{aligned}$$

Here $\overset{\mathrm{m}}{=}$ indicates equality between the measure of an angle and the central angle corresponding to the arc. □

The following example will also be used later.

[11] The length of $PA$ (or $PB$) is called the *tangent length*.

[12] If $k = 1$, the locus below becomes the perpendicular bisector of $AB$. It may be viewed as a circle of infinite radius. In this sense a line can sometimes be treated as a special circle, and the two have analogous properties; we call lines and circles collectively *generalized circles*.

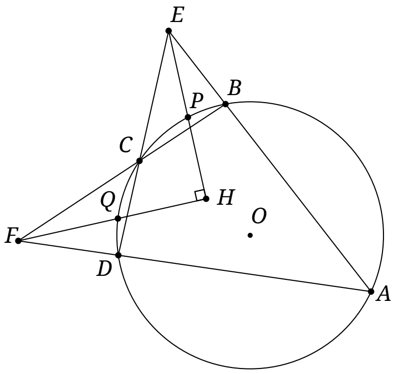


FIGURE 0.6

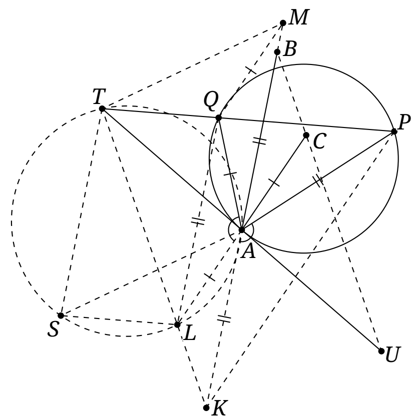


FIGURE 0.7

**Example 0.3.9.** Given $\triangle ABC$, let $P$ be the reflection of $B$ in $AC$ and $Q$ the reflection of $C$ in $AB$. The tangent at $A$ to $\odot(APQ)$ meets $PQ$ at $T$. Then the reflection $U$ of $T$ in $A$ lies on $BC$.

**Solution.** As in Figure 0.7, let $K,L$ be the reflections of $B,C$ in $A$. Then $CLKP$ is plainly an isosceles trapezoid. It is enough to prove that $T,L,K$ are collinear.

Since $AL = AC = AQ$, $AK = AB = AP$, and

$$\angle LAQ = 180^\circ - \angle QAC = 180^\circ - 2\angle BAC = 180^\circ - \angle BAP = \angle PAK,$$

the isosceles triangles $QAL$ and $PAK$ are similar.

Because $AT$ is tangent to $\odot(QAP)$, the tangent–chord theorem gives $\angle TAQ = \angle TPA$. Thus $\triangle TQA \sim \triangle TAP$, and a point $M$ can be chosen so that the quadrilaterals $TMAQ$ and $TKPA$ are similar.

Then $\triangle MQA \sim \triangle KAP \sim \triangle LAQ$. The triangles $MQA$ and $LAQ$ have a common pair of corresponding sides, so $\triangle MQA \cong \triangle LAQ$ and $MQLA$ is a parallelogram. Consequently,

$$\angle QAM = \angle AQL = \angle AKP = \angle BAC = \angle BAQ,$$

and $A,B,M$ are collinear.

Choose $S$ so that $TSLQ$ is a parallelogram. Then $TQ \parallel SL$ and $TQ = SL$; likewise, $QM \parallel LA$ and $QM = LA$. Hence $\triangle TQM \cong \triangle SLA$, and therefore

$$\angle SLA = \angle TQM = \angle TAK = 180^\circ - \angle TAM = 180^\circ - \angle STA.$$

The equality $\angle TQM = \angle TAK$ follows from $TQMA \sim TAKP$. Thus $L,A,T,S$ are concyclic. Therefore

$$\angle ATL = \angle ASL = \angle MTQ = \angle ATK.$$

The last equality follows from $TQMA \sim TAKP$. Hence $T \in KL$. □

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## Exercises

**Exercise 0.26.** Prove theorems 0.3.1 and 0.3.6.

**Exercise 0.27.** The diagonals of a cyclic quadrilateral are perpendicular. Prove that the sum of the squares of its four side lengths is eight times the square of the circumradius.

**Exercise 0.28.** The lines containing two chords $AB,CD$ of a circle meet at $E$. Prove that

$$\frac{AE}{BE} = \frac{AC}{BC} \cdot \frac{AD}{BD}.$$

**Exercise 0.29.** Let the convex quadrilateral $ABCD$ satisfy $\angle B + \angle D = 90^\circ$. Prove that

$$AC^2 \cdot BD^2 = AD^2 \cdot BC^2 + AB^2 \cdot CD^2.$$

**Exercise 0.30.** Let $ABCD$ be inscribed in $\odot O$, and put $E = AC \cap BD$. Through $E$ draw the line parallel to $BC$, meeting $AD$ at $F$. A tangent from $F$ to $\odot O$ touches it at $G$. Prove that $EF = GF$.

**Exercise 0.31.** Let $T$ be an interior point of the convex pentagon $ABCDE$, with $TB = TD$, $TE = TC$, and $\angle ABT = \angle AET$. Put $P = AB \cap CD$, $R = AE \cap CD$, $S = DT \cap EA$, and $Q = CT \cap AB$. Prove that $P, R, Q, S$ are concyclic.

### 0.3.2 The Relative Position of Two Circles

We conclude the section with some facts about the relative position of two circles. Some readers may not have encountered all the terminology in school, so we briefly introduce it.

Let the radii be $R, r$, where $R \geq r$, and let the distance between the centers be $d$. The possibilities are as follows.

(1) *One circle lies inside the other*: the circles are disjoint and one is contained in the other; this requires $R \neq r$ (otherwise the circles would coincide) and is characterized by $d < R - r$ (Figure 0.8a).
(2) *Externally disjoint*: the circles are disjoint and neither lies inside the other; $d > R + r$ (Figure 0.8b).
(3) *Internally tangent*: the circles are tangent and one lies inside the other; this requires $R \neq r$ (otherwise the circles would coincide) and is characterized by $d = R - r$ (Figure 0.8c).
(4) *Externally tangent*: the circles are tangent and neither lies inside the other; $d = R + r$ (Figure 0.8d).
(5) *Intersecting*: the circles meet at two distinct points; $R - r < d < R + r$ (Figure 0.8e).

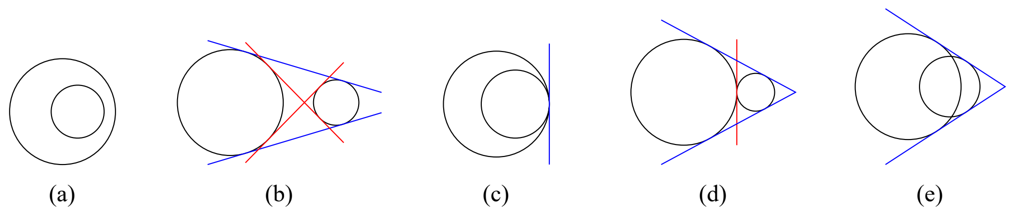


FIGURE 0.8

The first two cases are collectively called disjoint, and the two tangency cases collectively tangent. Semantically, tangency is a special form of intersection for general curves. Consequently, in a few later passages the phrase "two circles intersect" may include tangency; the intended meaning will always be clear from context.

A line tangent to two circles is a *common tangent*. If both circles lie on the same side of it, it is a *direct common tangent*; if they lie on opposite sides, it is a *transverse common tangent*.

The common tangents in the five cases are shown in Figure 0.8: direct common tangents are blue and transverse common tangents red. Nested disjoint circles have no common tangent. Externally disjoint circles have two of each kind. Internally tangent circles have one direct and no transverse common tangent. Externally tangent circles have two direct and one transverse common tangent. Intersecting circles have two direct and no transverse common tangent.

Applying the equal tangents theorem to common tangents gives the following.

**Theorem 0.3.10.** *For two circles:*

*(1) if their radii are unequal and they have two direct common tangents, those tangents are symmetric about the line of centers, meet at a point on that line, and the two direct common-tangent segments have equal lengths.*

*(2) if they have two transverse common tangents, those tangents are symmetric about the line of centers, meet at a point on that line, and the two transverse common-tangent segments have equal lengths.*

*Here a common-tangent segment means the portion of the tangent between the two circles.*

**Proof.** For part (1), use Figure 0.9. Let the direct common-tangent segments of $\odot O_1, \odot O_2$ be $IJ$ and $KL$, and let their lines meet at $Q$. By equality of tangent lengths,

$$IJ = QJ - QI = QL - QK = KL.$$

Since $QI, QK$ are tangents to $\odot O_1$, $O_1Q$ bisects $\angle IQK$. Similarly, $O_2Q$ bisects $\angle JQL$. These are the same angle, so $Q, O_1, O_2$ are collinear, which also proves the asserted symmetry.

For part (2), again use Figure 0.9. Let the transverse common-tangent segments of $\odot O_1, \odot O_2$ be $AB$ and $CD$, and let their lines meet at $P$. Equality of tangent lengths gives

$$AB = AP + PB = CP + DP = CD.$$

Since $PA, PC$ are tangents to $\odot O_1$, $O_1P$ bisects $\angle APC$. Similarly, $O_2P$ bisects $\angle BPD$. These two angles are vertically opposite, so $O_1, P, O_2$ are collinear, which proves the symmetry. □

**Proposition 0.3.11.** *Suppose two circles have two direct and two transverse common tangents. The portion of a direct common tangent between the transverse common tangents has the same length as a transverse common-tangent segment. Conversely, the portion of a transverse common tangent between the direct common tangents has the same length as a direct common-tangent segment.*

**Proof.** In Figure 0.9, the two direct common tangents touch the circles at $I, J$ and $K, L$, and the two transverse common tangents touch them at $A, B$ and $C, D$. The four tangent lines meet pairwise at $E, F, G, H, P, Q$. We prove $AB = GF$ and $EF = KL$.

The equal tangents theorem yields $BF + AB = AF = KF = KG + GF$, and

$$BF + GF = LF + GF = GL = GD = GC + CD = KG + CD = KG + AB,$$

where the last equality uses theorem 0.3.10. Subtracting gives $AB = GF$.

By symmetry, $IE = KG$. Together with $AB = GF$, equality of tangent lengths now gives

$$KL = KG + GF + FL = IE + AB + FL = EA + AB + BF = EF.$$ □

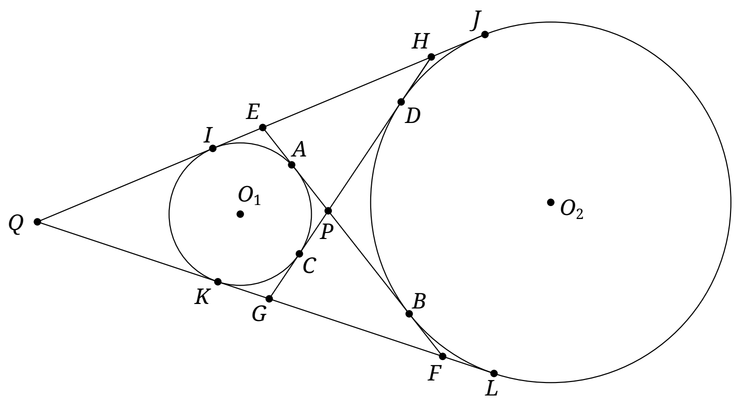


FIGURE 0.9

We finish with one more result for two circles.

**Theorem 0.3.12** (Reim's theorem)**.** *Let $\odot O_1, \odot O_2$ meet at $P, Q$. A line through P meets the circles again at $A_1, A_2$, respectively, and a line through Q meets them again at $B_1, B_2$. Then $A_1B_1 \parallel A_2B_2$.*

**Proof.** Consider the configuration in Figure 0.10. We have

$$\angle A_1B_1Q + \angle A_2B_2Q = \angle A_1B_1Q + (180^\circ - \angle QPA_2) = \angle A_1B_1Q + \angle A_1PQ = 180^\circ,$$

which proves the parallelism. □

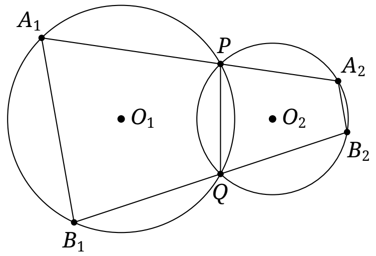


FIGURE 0.10

The converse form is also called Reim's theorem. Its proof is straightforward and is left to the reader.

**Theorem 0.3.13** (Reim's theorem, converse)**.** *Suppose $P, Q, A_1, B_1$ are concyclic, $A_2 \in PA_1$, $B_2 \in QB_1$, and $A_1B_1 \parallel A_2B_2$. Then $P, Q, A_2, B_2$ are also concyclic.*

---

### Exercises

**Exercise 0.32.** In Figure 0.9, prove that

$$IE = EA = HJ = DH = KG = GC = BF = FL.$$

**Exercise 0.33.** Two circles $\omega_1, \omega_2$ meet at $A, B$, and $P \in AB$. Tangents from $P$ to the two circles have lengths $l_1, l_2$. Prove that $l_1 = l_2$.

**Exercise 0.34.** Three distinct circles $\omega_1, \omega_2, \omega_3$ pass through the same two points $A, B$. Let $P$ vary on $\omega_3$, with $P \neq A, B$. Tangents from $P$ to $\omega_1, \omega_2$ have lengths $l_1, l_2$. Prove that $\frac{l_1}{l_2}$ is independent of $P$.

**Exercise 0.35.** Two circles $\omega_1, \omega_2$ meet at $A, B$, and $P$ varies on $\omega_1$. A tangent from $P$ to $\omega_2$ touches it at $T$. Prove that $\frac{PT^2}{PA \cdot PB}$ is constant.

**Exercise 0.36.** Two circles are internally tangent at $T$. A chord $AB$ of the larger circle meets the smaller at $C, D$. Prove that $\angle ATC = \angle BTD$.

## 0.4 Homothety

We have already introduced similarity and its elementary properties. We now turn to the related notion of homothety.

**Definition 0.4.1** (Homothety)**.**

(1) Given a point $O$ and a nonzero real number $k$, send each point $A$ of the plane to the point $A'$ satisfying
$$\overrightarrow{OA'} = k\overrightarrow{OA},$$
while $O$ remains fixed. This transformation is a *homothety*; $O$ is its *center* and $k$ its *ratio*. In what follows, write $\mathfrak{h}_{O,k}$ for this transformation.

(2) For a figure $\mathscr{A}$, the points obtained from all points of $\mathscr{A}$ under a homothety form its *image* $\mathscr{A}'$.

(3) Figures $\mathscr{A}$ and $\mathscr{A}'$ are *homothetic* if there is a homothety $\mathfrak{h}_{O,k}$ with $\mathfrak{h}_{O,k}(\mathscr{A}) = \mathscr{A}'$. The point $O$ and the number $k$ are the center and ratio of the homothety of $\mathscr{A}'$ to $\mathscr{A}$.

**Remark.** A homothety $f$ has an inverse $f^{-1}$ satisfying $f^{-1}(f(A)) = A$ for every $A$. The definition shows that $f^{-1}$ is also a homothety, with $f^{-1} = \mathfrak{h}_{O,1/k}$. Thus, in part (3) of the definition, if there is a homothety $\mathfrak{h}_{O,k'}$ such that $\mathfrak{h}_{O,k'}(\mathscr{A}') = \mathscr{A}$, then $\mathscr{A}'$ is also homothetic to $\mathscr{A}$. The statements "$\mathscr{A}'$ is homothetic to $\mathscr{A}$" and "$\mathscr{A}$ is homothetic to $\mathscr{A}'$" are equivalent, but the ratio is ordered: if the ratio of $\mathscr{A}'$ to $\mathscr{A}$ is $k$, then that of $\mathscr{A}$ to $\mathscr{A}'$ is $k' = 1/k$. The center is $O$ in both descriptions.

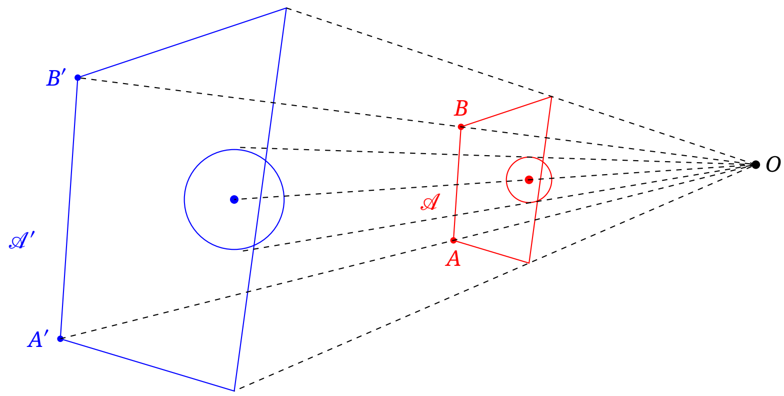


Figure 0.11

A homothety may be viewed as a dilation of a figure. In Figure 0.11, the blue figure $\mathscr{A}'$ is the image of the red figure $\mathscr{A}$ under a homothety with center $O$ and positive ratio $k$.

**Proposition 0.4.2.** *Suppose $\mathscr{A}'$ is homothetic to $\mathscr{A}$ with center $O$ and ratio $k$, and that $A'$,$A$ and $B'$,$B$ are corresponding pairs; that is, $\mathfrak{h}_{O,k}$ sends $A, B, \mathscr{A}$ to $A', B', \mathscr{A}'$, respectively. Then $AB \parallel A'B'$*[13] *and*
$$\frac{\overline{A'O}}{\overline{AO}} = \frac{\overline{B'O}}{\overline{BO}} = \frac{\overline{A'B'}}{\overline{AB}} = k.$$

**Proof.** The equality $\frac{\overline{A'O}}{\overline{AO}} = \frac{\overline{B'O}}{\overline{BO}} = k$ is the definition of homothety. Hence $\triangle A'OB' \stackrel{+}{\sim} \triangle AOB$, which also gives $AB \parallel A'B'$ and $\frac{\overline{A'B'}}{\overline{AB}} = k$. □

The following results describe the relation between homothety and similarity.

**Theorem 0.4.3.**

*(1) Every homothety is a special similarity transformation.*

*(2) Two homothetic figures are directly similar. Conversely, suppose $\mathscr{A}$ and $\mathscr{A}'$ are directly similar and the lines joining* all *corresponding points meet at one point $O$. Then $\mathscr{A}'$ and $\mathscr{A}$ are homothetic about $O$, and the absolute value of the homothety ratio equals the similarity ratio of $\mathscr{A}'$ to $\mathscr{A}$.*

---

[13] The lines may be coincident. Unlike the strict secondary-school convention, we shall often regard coincidence as a special case of parallelism. This is extremely convenient, and the context will make clear whether coincidence is included.

**Proof.** Part (1) is immediate. For part (2), suppose $\mathscr{A}'$ is homothetic to $\mathscr{A}$ with center $O$ and ratio $k$. For any three corresponding pairs $(A',A)$, $(B',B)$, $(C',C)$, the definition gives

$$\measuredangle ABC = \measuredangle AOB + \measuredangle OBC + \measuredangle BAO = \measuredangle A'OB' + \measuredangle OB'C' + \measuredangle B'A'O = \measuredangle A'B'C'.$$

Proposition 0.4.2 also gives $\frac{A'B'}{AB} = |k|$, so the defining conditions for a similarity hold. Conversely, the same argument shows that directly similar figures for which all corresponding-point lines pass through one point are homothetic. □

The proofs of the next two propositions are left to the reader.

**Proposition 0.4.4.** *Triangles whose corresponding sides are parallel or coincident are homothetic, and corresponding sides of homothetic triangles are parallel or coincident.* [14]

**Proposition 0.4.5.** *If corresponding sides of two similar figures are parallel or coincident, then the figures are homothetic.*[15]

Homothety between two circles has several special features.

**Definition 0.4.6.** Two circles can be regarded as homothetic in two ways. Their two possible centers are called the *internal center of homothety*, or *insimilicenter*, and the *external center of homothety*, or *outsimilicenter*. The homothety about the internal center has negative ratio, whereas that about the external center has positive ratio.

**Remark.** If the circles are concentric, their only center of homothety is their common center; the internal and external centers may then be regarded as coincident. If their radii are equal, the external center does not exist as a finite point and may be regarded as lying at infinity.

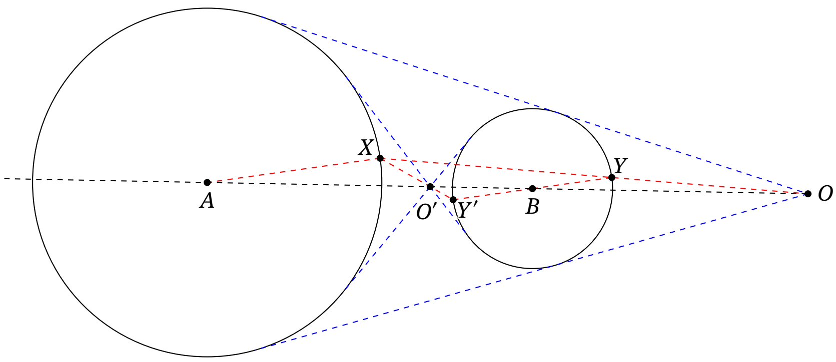


FIGURE 0.12

**Proposition 0.4.7.** *Join the endpoints of a pair of codirectional parallel radii of two circles. The joining line always passes through a fixed point on the line of centers, namely the external center of homothety. If the radii are oppositely directed and parallel, the corresponding joining line likewise passes through a fixed point on the line of centers, namely the internal center of homothety.*

**Proof.** In Figure 0.12, the radii $AX$ and $BY$ of $\odot A, \odot B$ are codirectional and parallel, and $O = XY \cap AB$. Clearly $\triangle AXO \sim \triangle BYO$, so

$$\frac{BO}{AO} = \frac{BY}{AX} = \text{const.}$$

[14]If two congruent triangles differ by a translation, their corresponding-point lines are parallel and, strictly speaking, the triangles are not homothetic. In a limiting sense, parallel lines meet at a point at infinity, so the center of homothety may be regarded as lying at infinity. See chapter 3 for a detailed introduction to points at infinity.

[15]As in proposition 0.4.4, if two congruent figures differ by a translation, their corresponding-point lines are parallel.

Thus $O$ is fixed. Moreover, $X,Y$ are corresponding points under one of the similarities between the circles and lie on the same side of $O$, so $O$ is the external center of homothety. The analogous argument for the oppositely directed radii $AX,BY'$ shows that $O' = XY' \cap AB$ is the internal center. □

**Proposition 0.4.8.** *If direct common tangents can be drawn to two circles, the intersection of the two direct common tangents is the external center of homothety. If transverse common tangents can be drawn, the intersection of the two transverse common tangents is the internal center of homothety; see Figure 0.12.*

**Proposition 0.4.9.** *For two internally tangent circles, the point of tangency is their external center of homothety. For two externally tangent circles, it is their internal center of homothety.*

**Proof.** Let the two circles in proposition 0.4.8 tend to tangency. □

**Proposition 0.4.10.** *Let three circles have distinct centers and pairwise different radii.*
 *(1) Their three pairwise external centers of homothety are collinear.*[16]
 *(2) The internal centers belonging to any two pairs and the external center belonging to the remaining pair are collinear.*

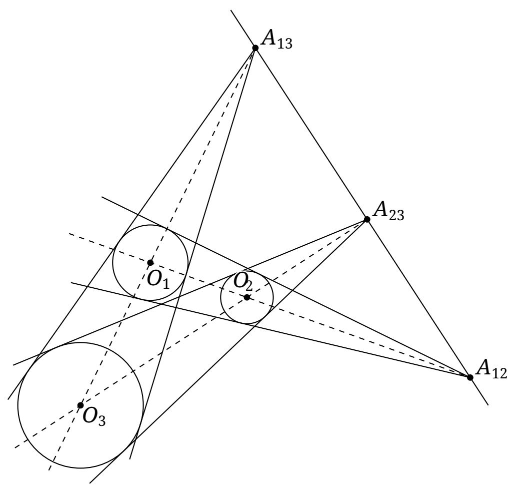


FIGURE 0.13

**Proof.** For part (1), let the radii of the circles with centers $O_1,O_2,O_3$ be $r_1,r_2,r_3$, and let the intersections of the pairwise direct common tangents be $A_{23},A_{13},A_{12}$. Similarity gives

$$\frac{A_{ij}O_i}{A_{ij}O_j} = \frac{r_i}{r_j}.$$

Apply the Menelaus theorem to $\triangle O_1O_2O_3$ and the points $A_{12},A_{13},A_{23}$. Since

$$\frac{A_{13}O_1}{A_{13}O_3} \cdot \frac{A_{23}O_3}{A_{23}O_2} \cdot \frac{A_{12}O_2}{A_{12}O_1} = \frac{r_1}{r_3} \cdot \frac{r_3}{r_2} \cdot \frac{r_2}{r_1},$$

the three points are collinear. The proof of part (2) is analogous. □

We end with a related definition.

[16]Equivalently, if three circles have distinct centers and the direct common tangents of each pair meet, then the three points of intersection are collinear. In this form the statement is called *Monge's theorem*.

**Definition 0.4.11** (Spiral similarity)**.**

(1) Let $\mathfrak{r}$ be a rotation about $O$ and $\mathfrak{h}$ a homothety centered at $O$. Their composition $\mathfrak{r} \circ \mathfrak{h}$ is a *spiral similarity transformation* with center $O$: first apply $\mathfrak{h}$, then $\mathfrak{r}$.
(2) The images of all points of a figure $\mathscr{A}$ under such a transformation form its image $\mathscr{A}'$.
(3) If a spiral similarity sends $\mathscr{A}$ to $\mathscr{A}'$, the two figures are said to be *spirally similar*.

---

## Exercises

**Exercise 0.37.** Prove propositions 0.4.4 and 0.4.5.

**Exercise 0.38.** If two triangles are similar and the lines joining corresponding vertices are concurrent, must the triangles be homothetic?

**Exercise 0.39.** Given homotheties $\mathfrak{h}_{O_1,k_1}$ and $\mathfrak{h}_{O_2,k_2}$, prove that:

(1) if $k_1 k_2 = 1$, then $\mathfrak{h}_{O_2,k_2} \circ \mathfrak{h}_{O_1,k_1}$ is a translation;
(2) if $k_1 k_2 \neq 1$, there is an $O \in O_1 O_2$ such that $\mathfrak{h}_{O_2,k_2} \circ \mathfrak{h}_{O_1,k_1} = \mathfrak{h}_{O,k_1 k_2}$.

**Exercise 0.40.** Bob claims that any two directly similar figures must be spirally similar. Is this correct? Prove your assertion.

**Exercise 0.41.** The circle $\omega_1$ is internally tangent to $\omega_2$ at $A$ and lies inside it. A chord $BC$ of $\omega_2$ is tangent to $\omega_1$ at $D$, and $AD$ meets $\omega_2$ again at $E$. Prove that the tangent to $\omega_2$ at $E$ is parallel to $BC$.[17]

**Exercise 0.42.** The circles $\omega_1, \omega_2, \omega_3$ are externally tangent to a circle $\omega$ at $A', B', C'$, respectively. Construct $\triangle ABC$ so that $\omega_1$ is tangent to both $AB, AC$, $\omega_2$ to both $BC, BA$, and $\omega_3$ to both $CA, CB$, with all three circles inside $\triangle ABC$. Prove that $AA', BB', CC'$ are concurrent.

**Exercise 0.43** (Pascal's theorem)**.** Let $A,B,C,D,E,F$ be concyclic, and put $R = AB \cap DE$, $S = BC \cap EF$, $T = CD \cap FA$. Prove that $R,S,T$ are collinear.

**Exercise 0.44.** Suppose $\odot A, \odot B$ are externally tangent at $S$, $\odot B, \odot C$ at $T$, $\odot C, \odot D$ at $U$, and $\odot D, \odot A$ at $V$, and that $ABCD$ is a convex quadrilateral. Choose $P$ on $\odot A$ and $Q$ on $\odot C$ so that $PS$ and $QT$ are both perpendicular to $ST$. Put $R = VP \cap UQ$. Prove that $B,D,R$ are collinear.

*Hint.* If a quadrilateral $ABCD$ is circumscribed about $\odot O$, then the intersection of the diagonals of the quadrilateral formed by its four points of tangency coincides with the intersection of the diagonals of $ABCD$.

## 0.5 Centroid, Circumcenter, and Orthocenter

We briefly review several elementary properties of the centroid, circumcenter, and orthocenter that will be used in later chapters. Some proofs are immediate and are left as exercises.

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[17] This immediately implies the result of Exercise 0.36, but in that earlier exercise we want the reader to use a method not based on homothety.

### 0.5.1 The Centroid and Circumcenter of a Triangle

#### The centroid

We begin by recalling the centroid.

**Definition 0.5.1.** The three *medians* of a triangle meet at a point called its *centroid*.

The centroid has the following familiar properties.

**Proposition 0.5.2.** *If $G$ is the centroid of $\triangle ABC$, then*

$$S_{\triangle AGB} = S_{\triangle BGC} = S_{\triangle CGA}.$$

**Proposition 0.5.3.** *In a triangle, the distances from the centroid to the three sides are proportional to the reciprocals of the corresponding side lengths.*

**Proposition 0.5.4.** *Among all points in the plane of $\triangle ABC$, the centroid minimizes the sum of the squared distances to the three vertices.*

**Proof.** Let $G$ be the centroid of $\triangle ABC$ and $P$ any point. Then

$$\begin{aligned} PA^2 + PB^2 + PC^2 &= (\overrightarrow{PG} + \overrightarrow{GA})^2 + (\overrightarrow{PG} + \overrightarrow{GB})^2 + (\overrightarrow{PG} + \overrightarrow{GC})^2 \\ &= 3PG^2 + GA^2 + GB^2 + GC^2 + 2\overrightarrow{PG} \cdot (\overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC}) \\ &= 3PG^2 + GA^2 + GB^2 + GC^2 \geq GA^2 + GB^2 + GC^2. \end{aligned}$$ □

The centroid also leads to the following definitions.

**Definition 0.5.5.** Given $\triangle ABC$, let $D, E, F$ be the midpoints of its sides $BC, CA, AB$, respectively. Then $\triangle DEF$ is the *medial triangle* of $\triangle ABC$, and $\triangle ABC$ is the *anticomplementary triangle* of $\triangle DEF$.

**Definition 0.5.6.** Given $\triangle ABC$ and a point $P$, let $P'$ satisfy $\overrightarrow{PG} = 2\overrightarrow{GP'}$, where $G$ is the centroid. Then $P'$ is the *complement* of $P$, and $P$ is the *anticomplement* of $P'$, with respect to $\triangle ABC$. The complement and anticomplement of $P$ with respect to $\triangle ABC$ are denoted by $\mathbf{c}_{\triangle ABC}P$ and $\mathbf{a}_{\triangle ABC}P$, respectively, abbreviated to $\mathbf{c}P$ and $\mathbf{a}P$ when no ambiguity can arise.

The following consequences are immediate.

**Proposition 0.5.7.** *The centroid of a triangle divides each median in the ratio $2:1$. Equivalently, the complement of a vertex is the midpoint of the opposite side.*

**Proposition 0.5.8.** *Let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, where $M_a, M_b, M_c$ are the midpoints of $BC, CA, AB$, respectively. Then the two triangles are homothetic about their common centroid $G$. Under the homothety from $\triangle ABC$ to $\triangle M_aM_bM_c$, the image $P'$ of a point $P$ is its complement.*

**Remark.** Throughout the chapters on triangle geometry, subscripts correspond cyclically to the vertices. For example, $M_a, M_b, M_c$ denote the midpoints of $BC, CA, AB$. Unless stated otherwise, analogous notation is used without further explanation. We use five triples to display cyclic symmetry: $(A, B, C)$, $(D, E, F)$, $(L, M, N)$, $(X, Y, Z)$, and $(1, 2, 3)$. Thus if the orthic triangle of $\triangle ABC$ (to be defined shortly) is denoted by $\triangle H_1H_2H_3$, then $H_1, H_2, H_3$ are the projections of the orthocenter on $BC, CA, AB$, respectively. Likewise, if a line $l$ cuts the three sides of $\triangle ABC$ at $X, Y, Z$, this means that $l$ intersects the lines $BC, CA, AB$ at $X, Y, Z$, respectively.

Finally, the centroid extends naturally to polygons.

**Definition 0.5.9.** The *centroid* of an $n$-gon $A_1A_2\cdots A_n$ is the point $G$ satisfying

$$\overrightarrow{GA_1}+\overrightarrow{GA_2}+\cdots+\overrightarrow{GA_n}=\mathbf{0}.$$

This agrees with the preceding definition for a triangle: the intersection of the three medians of $\triangle ABC$, denoted by $G$, satisfies $\overrightarrow{GA}+\overrightarrow{GB}+\overrightarrow{GC}=\mathbf{0}$. In Cartesian coordinates (named after Descartes), if the vertices of an $n$-gon are $(x_1,y_1),\ldots,(x_n,y_n)$, its centroid is

$$\Big(\frac{x_1+\cdots+x_n}{n},\frac{y_1+\cdots+y_n}{n}\Big).$$

### The circumcenter

**Definition 0.5.10.** The center of the circumcircle of a triangle is its *circumcenter*.

**Proposition 0.5.11.** *The circumcenter of a triangle is the common point of the perpendicular bisectors of the three sides.*

We record several further definitions concerning a triangle's circumcenter or circumcircle that will be used later.

**Definition 0.5.12.**

(1) Given $\triangle ABC$ and a point $P$, let $P_1,P_2,P_3$ be the circumcenters of $\triangle BCP$, $\triangle CAP$, $\triangle ABP$, respectively. Then $\triangle P_1P_2P_3$ is the *Carnot triangle* of $P$ with respect to $\triangle ABC$.

(2) The triangle formed by the tangents to the circumcircle at the three vertices is the *tangential triangle* of the original triangle.

(3) Given $\triangle ABC$, the midpoints of the arcs $\overset{\frown}{BC}$, $\overset{\frown}{CA}$, $\overset{\frown}{AB}$ on its circumcircle that do not contain $A,B,C$, respectively, form its *circumcircle mid-arc triangle*. The midpoints of the complementary arcs $\overset{\frown}{BAC}$, $\overset{\frown}{CBA}$, $\overset{\frown}{ABC}$ form its *complementary arc-midpoint triangle*. The points antipodal to $A,B,C$ on the circumcircle form the *antipodal triangle*.[18]

---

## Exercises

**Exercise 0.45.** Prove propositions 0.5.2, 0.5.3, 0.5.7 and 0.5.8.

**Exercise 0.46.** Given a polygon $A_1\cdots A_n$, a variable point $P$ in its plane satisfies $\sum_{i=1}^{n}PA_i^2=\text{const}$. Prove that $P$ lies on a fixed circle centered at the centroid of the polygon.

**Exercise 0.47.** Let $O$ be the circumcenter of $\triangle ABC$. Prove that $\measuredangle BAC+\measuredangle CBO=90°$.

**Exercise 0.48.** If $\triangle ABC$ has side lengths $a,b,c$ and circumradius $R$, prove that $4R\cdot S_{\triangle ABC}=abc$.

**Exercise 0.49.** Given $\triangle ABC$, choose $D,E,F$ on the segments $BC,CA,AB$, respectively, so that $\frac{DB}{DC}=\frac{EC}{EA}=\frac{FA}{FB}$. Prove that $\triangle ABC$ and $\triangle DEF$ have the same centroid.

**Exercise 0.50** (Echols's second theorem)**.** Let $\triangle ABC$, $\triangle DEF$, and $\triangle IJK$ be similarly oriented equilateral triangles.[19] Let $X,Y,Z$ be the centroids of $\triangle ADI$, $\triangle BEJ$, $\triangle CFK$, respectively. Prove that $\triangle XYZ$ is equilateral.

---

[18]A point $P'$ is the *antipodal point* of a point $P$ on a circle if $P'$ is on the same circle and $PP'$ is a diameter.

[19]That is, $\triangle ABC \overset{+}{\sim} \triangle DEF \overset{+}{\sim} \triangle IJK$.

### 0.5.2 The Orthocenter of a Triangle

We now study the elementary properties of the orthocenter.

**Definition 0.5.13.**

(1) The three *altitudes* of a triangle meet at a point called its *orthocenter*; see Figure 0.14.

(2) By part (1), if $H$ is the orthocenter of $\triangle ABC$, then $A,B,C$ are the orthocenters of $\triangle BHC$, $\triangle CHA$, $\triangle AHB$, respectively. The four points $A,B,C,H$ form an *orthocentric system*.

**Definition 0.5.14.** The triangle formed by the projections of the three vertices on their opposite sides is called the *orthic triangle*, or *orthotriangle*.

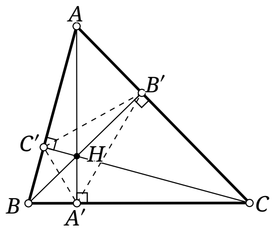


FIGURE 0.14

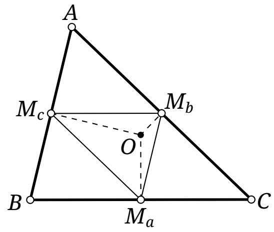


FIGURE 0.15

Most readers will know the proof that the orthocenter exists, but we include a short argument. In Figure 0.14, suppose $CC' \perp AB$, $BB' \perp AC$, $AA' \perp BC$, and $CC' \cap BB' = H$. We prove that $A,H,A'$ are collinear. The assumptions show that $A,B',H,C'$ are concyclic and that $B,B',C',C$ are concyclic. Hence

$$\measuredangle(AH,BC) = \measuredangle HAB' + \measuredangle B'CB = \measuredangle HC'B' + \measuredangle B'C'B = \measuredangle HC'B = 90^\circ.$$

Thus $AH \perp BC$. Since $AA' \perp BC$ by assumption, $A,H,A'$ are collinear, proving the existence of the orthocenter. $\square$

We shall use the following properties.

**Proposition 0.5.15.** *The circumcenter of a triangle is the orthocenter of its medial triangle; see Figure 0.15.*

**Proof.** Let $O$ be the circumcenter of $\triangle ABC$, and let $\triangle M_aM_bM_c$ be its medial triangle. Clearly $OM_a \perp BC$, while $M_bM_c \parallel BC$; hence $OM_a \perp M_bM_c$. The other two perpendicular relations are analogous, so $O$ is the orthocenter. $\square$

Proposition 0.5.15 also proves the existence of the orthocenter: take the circumcenter of the anticomplementary triangle. It satisfies the defining conditions for the orthocenter of the original triangle. The existence of the circumcenter therefore implies the existence of the orthocenter.

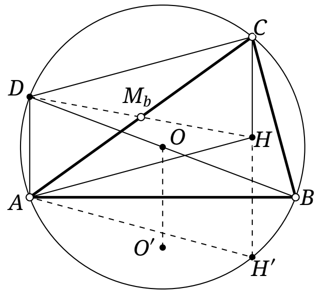


FIGURE 0.16

**Lemma 0.5.16.** *Let $H$ be the orthocenter of $\triangle ABC$, and let $D$ be the point antipodal to $B$ on $\mathscr{C}$. Then $ADCH$ is a parallelogram.*

**Proof.** In Figure 0.16, write $\odot O := \odot(ABC)$, and let $BO$ meet $\odot O$ again at $D$. Since $CH \perp AB$ and $DA \perp AB$, we have $AD \parallel CH$. Similarly $CD \parallel AH$, so $ADCH$ is a parallelogram. ☐

**Claim 0.5.17.** *If $H$ is the orthocenter of $\triangle ABC$, then*

$$CH = 2R|\cos\angle BCA|,$$

*where $R$ is the circumradius.*

**Proof.** In the proof of lemma 0.5.16, the quadrilateral $ADCH$ is a parallelogram. Hence

$$CH = AD = BD|\cos\angle BDA| = 2R|\cos\angle ACB|.$$ ☐

**Lemma 0.5.18.** *Let $H, O$ be the orthocenter and circumcenter of $\triangle ABC$, and let $O'$ be the reflection of $O$ in $AB$. Then $CHO'O$ is a parallelogram.*

**Proof.** The proof of claim 0.5.17 gives $CH \parallel AD$ and $CH = AD$, while plainly $OO' \parallel AD$ and $OO' = AD$. Thus $CH \parallel OO'$ and $CH = OO'$, and the assertion follows. ☐

**Alternative proof.** Let $\triangle M_aM_bM_c$ be the medial triangle. Under the homothety between the original and medial triangles, $H, O$ correspond. Hence $\overline{CH} = 2\overline{OM_c}$. Therefore $CH \parallel OO'$ and $CH = OO'$, as required. ☐

**Lemma 0.5.19.** *The reflections of the orthocenter $H$ of $\triangle ABC$ in the midpoints of its three sides lie on $\odot(ABC)$.*

**Proof.** In the proof of claim 0.5.17, the diagonals of the parallelogram $CDAH$ bisect one another; see Figure 0.16. Thus $DH$ passes through the midpoint $M_b$ of $AC$, and $D, H$ are symmetric about $M_b$. The other sides are analogous. ☐

**Lemma 0.5.20.** *The antipodal and medial triangles of a given triangle are homothetic about the orthocenter with ratio $2:1$.*

**Proof.** In the proof of lemma 0.5.19, $D$ is antipodal to $B$ and $M_b = \mathfrak{h}_{H,1/2}(D)$. The antipodes of $A$ and $C$ are treated similarly. ☐

**Lemma 0.5.21.** *The reflections of the orthocenter of a triangle in the three sidelines lie on the circumcircle.*

**Proof.** In Figure 0.16, extend $CH$ to meet the circumcircle at $H'$. Since $\angle BCH = \angle BAH = \angle BAH'$, the point $H'$ is the reflection of $H$ in $AB$. Thus the reflection of $H$ in $AB$ lies on the circumcircle. The other sides are analogous. ☐

**Lemma 0.5.22.** *Let $\triangle ABC$ have orthocenter $H$ and circumradius $R$, and let its three altitudes be $AA', BB', CC'$, where $A', B', C'$ lie on their corresponding sides. Then*

$$AH \cdot A'H = BH \cdot B'H = CH \cdot C'H = 4R^2|\cos A \cos B \cos C|.$$

**Proof.** In Figure 0.14, the perpendicularities imply that $A, A', B, B'$ are concyclic. The power of $H$ therefore gives $AH \cdot A'H = BH \cdot B'H$, and similarly $AH \cdot A'H = CH \cdot C'H$. Using claim 0.5.17,

$$AH \cdot A'H = AH \cdot BH|\cos\angle BHA'| = |(2R\cos A)| \cdot |(2R\cos B)||\cos C| = 4R^2|\cos A\cos B\cos C|. \qquad \square$$

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## Exercises

**Exercise 0.51.** Prove that the distance from the orthocenter of a triangle to a vertex is twice the distance from the circumcenter to the opposite side.

**Exercise 0.52.** If $O, H$ are the circumcenter and orthocenter of $\triangle ABC$, respectively, prove that $\overrightarrow{OH} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$.

**Exercise 0.53.** If $O, H$ are the circumcenter and orthocenter of $\triangle ABC$, respectively, prove that the lines $BO$ and $BH$ are symmetric about the bisector of $\angle ABC$.

**Exercise 0.54.** Given an acute triangle $ABC$, choose $D, E, F$ on the segments $BC, CA, AB$, respectively. Prove that the perimeter of $\triangle DEF$ is minimized when it is the orthic triangle of $\triangle ABC$.

**Exercise 0.55** (Hagge's theorem)**.** Let $H$ be the orthocenter of $\triangle ABC$. Choose $X, Y, Z$ on the three sides so that $AX, BY, CZ$ are concurrent at $P$. Prove the following.

(1) Let $\odot(AX), \odot(BY), \odot(CZ)$ meet $\odot(BC), \odot(CA), \odot(AB)$ at $A_1, A_2$; $B_1, B_2$; $C_1, C_2$, respectively. These six points are concyclic.
(2) Draw through $H$ perpendiculars to $AX, BY, CZ$. Their intersections with $\odot(BC), \odot(CA), \odot(AB)$ are $X_1, X_2$; $Y_1, Y_2$; $Z_1, Z_2$, respectively. These six points are concyclic.
(3) Draw through $H$ perpendiculars to $AX, BY, CZ$, meeting $\odot(AX), \odot(BY), \odot(CZ)$ at $X_1', X_2'$; $Y_1', Y_2'$; $Z_1', Z_2'$, respectively. These six points are concyclic.

### 0.5.3 Euler Line and Nine-point Circle

We conclude by studying the Euler line and the nine-point circle, which reveal the relation among the centroid, circumcenter, and orthocenter.

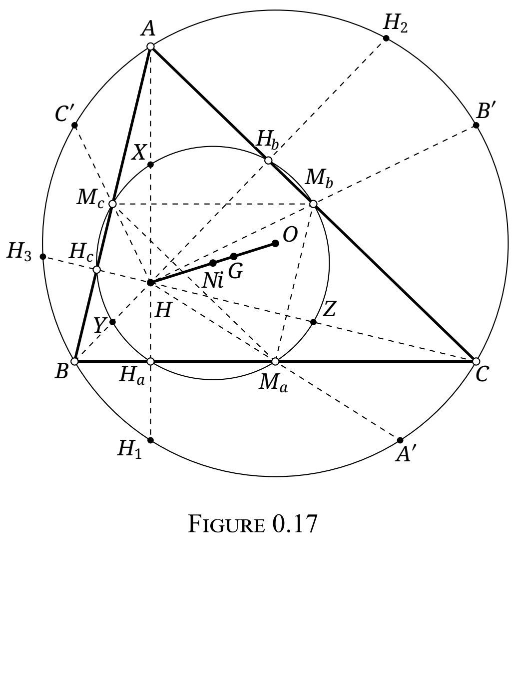


FIGURE 0.17

**Theorem 0.5.23** (Euler line theorem). *For every triangle, its circumcenter, centroid, and orthocenter lie on one line in that order, and the circumcenter is the complement of the orthocenter.* Their common line is the *Euler line*. In what follows, write $\mathscr{E}_{\triangle ABC}$ for the Euler line of $\triangle ABC$, abbreviated to $\mathscr{E}$ when no ambiguity can arise.

**Proof.** In Figure 0.17, consider the homothety between $\triangle ABC$ and its medial triangle $\triangle M_aM_bM_c$. By proposition 0.5.15, the orthocenters of the two triangles correspond: the orthocenter $H$ of the original triangle corresponds to its circumcenter $O$. The assertion now follows from proposition 0.5.8. $\square$

**Theorem 0.5.24** (Nine-point circle theorem; Euler circle theorem). *Let $H$ be the orthocenter of $\triangle ABC$. The three side midpoints, the three feet of the altitudes, and the three midpoints of $AH, BH, CH$ lie on one circle.* This circle is the *nine-point circle*, also called the *Euler circle*. In what follows, write $\mathscr{N}_{\triangle ABC}$ and $\mathscr{C}_{\triangle ABC}$ for the nine-point circle and circumcircle of $\triangle ABC$, abbreviated to $\mathscr{N}$ and $\mathscr{C}$ when no ambiguity can arise.

**Proof.** As in Figure 0.17, by lemmas 0.5.19 and 0.5.21, the reflections $H_1, H_2, H_3$ of $H$ in the three sides and the reflections $A', B', C'$ of $H$ in the three side midpoints all lie on the circumcircle. Apply the homothety with center $H$ and ratio $1/2$. $\square$

**Theorem 0.5.25.** *For any triangle, the center of the nine-point circle lies on the Euler line. It is the midpoint of the circumcenter and orthocenter, and is also the complement of the circumcenter; see Figure 0.17.*

**Proof.** Theorem 0.5.24 gives $\mathscr{N} = \mathfrak{h}_{H,1/2}(\mathscr{C})$. Hence the nine-point center *Ni* is the midpoint of the circumcenter $O$ and the orthocenter $H$.[20] Together with theorem 0.5.23, this shows that *Ni* is the complement of $O$. $\square$

**Corollary 0.5.26.** *The centroid of a triangle is the internal center of homothety of the nine-point circle and the circumcircle.*

**Proof.** The Euler-line and nine-point-circle results show that the centroid divides the segment joining the nine-point center and the circumcenter in the ratio $1:2$, which is precisely the ratio of the radii. $\square$

We use the nine-point circle to prove one further proposition. First we need a definition.

**Definition 0.5.27.** Given a point $P$ and $\triangle ABC$, the triangle formed by reflecting $P$ in the three side-lines is the *reflection triangle of $P$* with respect to $\triangle ABC$. For $\triangle ABC$, reflecting $A$ in $BC$, $B$ in $CA$, and $C$ in $AB$ gives the *reflection triangle of* $\triangle ABC$. Thus the object following "of"—a point in the first construction and a triangle in the second—distinguishes the two uses.

**Proposition 0.5.28.** *The reflection triangle of a triangle's circumcenter has its circumcenter at the original triangle's orthocenter.*[21]

[20] Later, after isogonal conjugacy has been developed, this relation also follows from corollary 4.4.5, since the circumcenter and orthocenter are isogonal conjugates.

[21] When the meaning is clear, the reference triangle is omitted. Here, for example, the reflection triangle is taken with respect to the original triangle. In general the omitted reference triangle is the first triangle specified, usually denoted $\triangle ABC$.

**Proof.** The projections of the circumcenter on the three sides form the medial triangle, whose circumcircle is the nine-point circle. The medial triangle and the reflection triangle of the circumcenter are homothetic about the circumcenter with ratio $1:2$. Under the homothety with center at the circumcenter and ratio 2, the circumcenter of the medial triangle, namely the nine-point center, goes to the circumcenter of that reflection triangle. Theorem 0.5.25 completes the proof. □

Further properties of the nine-point circle and its center will be studied in chapter 5.

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### Exercises

**Exercise 0.56.** Prove that the orthocenter of a triangle's anticomplementary triangle lies on the original triangle's Euler line and is the reflection of the original triangle's orthocenter in that triangle's circumcenter.

**Exercise 0.57.** Prove that the sum of the squares of the distances from the nine-point center to the three vertices and to the orthocenter equals three times the square of the circumradius.

**Exercise 0.58.** Let $G,H$ be the centroid and orthocenter of an obtuse triangle, respectively. Prove that the circle $\odot(GH)$ and the circumcircle of the triangle's tangential triangle both pass through the two intersection points of $\mathscr{N}$ and $\mathscr{C}$.

**Exercise 0.59.** For a given triangle, prove that its reflection triangle and the reflection triangle of the nine-point center are homothetic with ratio $2:1$.

**Exercise 0.60.** Given a point $H$, a circle $\omega$, and a point $A \in \omega$, construct a triangle $ABC$ such that $\omega = \mathscr{C}$ and $H$ is its orthocenter. Discuss all cases and determine the necessary and sufficient condition for a solution to exist.

## 0.6 Incenter and Excenters

We briefly introduce the incenter and excenters of a triangle.

### 0.6.1 Elementary Properties

**Definition 0.6.1** (Incenter and excenters)**.**

(1) For $\triangle ABC$, there is a circle tangent to all three segments $AB,BC,CA$; it is the *incircle* of $\triangle ABC$. There is also a circle tangent to the segment $BC$ and to the extensions of $AB,AC$; it is the $A$-*excircle* of $\triangle ABC$. The $B$- and $C$-excircles are defined analogously; these three circles are collectively the *excircles* of $\triangle ABC$.

In what follows, write $\mathscr{I}_{\triangle ABC}$, $\mathscr{I}_{a,\triangle ABC}$, $\mathscr{I}_{b,\triangle ABC}$, and $\mathscr{I}_{c,\triangle ABC}$ for its incircle and its $A$-, $B$-, and $C$-excircles, respectively. When no ambiguity can arise, omit only the reference-triangle designator and write $\mathscr{I}$, $\mathscr{I}_a$, $\mathscr{I}_b$, and $\mathscr{I}_c$.

(2) The center of the incircle is the *incenter*; the centers of the three excircles are the *excenters*. We shall often refer collectively to the incenter and excenters as the triangle's in- and excenters.

The incenter is the intersection of the three internal angle bisectors. An excenter is the intersection of two external angle bisectors and the remaining internal angle bisector. For $\triangle ABC$, the excenter on the internal bisector of $\angle A$ is the $A$-excenter. Its $B$- and $C$-excenters are defined similarly.

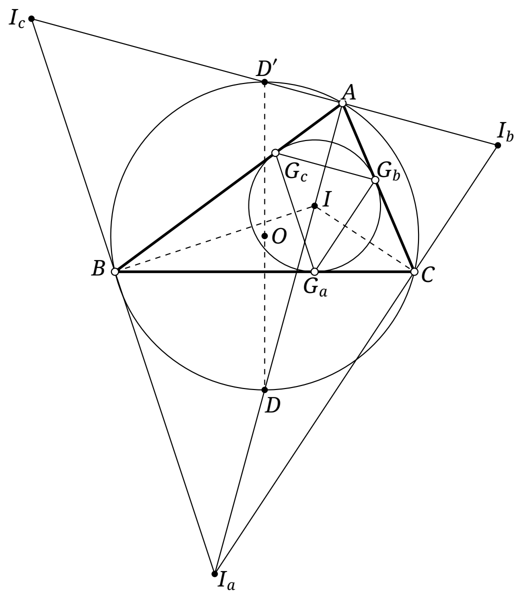


FIGURE 0.18

**Definition 0.6.2.** A triangle's *excentral triangle* is formed by its three excenters. Its *intouch triangle*, or *contact triangle*, is formed by the points where its incircle touches its three sides. Its *extouch triangle* is formed by the points where its three excircles touch their corresponding sides, here meaning the side segments.

We briefly recall several elementary properties of the incenter and excenters of a triangle that will be used in later chapters.

**Theorem 0.6.3.** *As in Figure 0.18, let $I$ be the incenter of $\triangle ABC$, and let $AI$ be extended to meet $\odot(ABC)$ at $D$. Then $BD = ID = CD$; equivalently, $D$ is the circumcenter of $\triangle BIC$.*

**Proof.** Since

$$\angle DBI = \angle DBC + \angle CBI = \angle DAC + \angle CBI = \angle BAI + \angle ABI = \angle BID,$$

we have $BD = ID$. Similarly, $CD = ID$. □

**Proposition 0.6.4.** *As in Figure 0.18, let $\triangle I_aI_bI_c$ be the excentral triangle of $\triangle ABC$, and let $I$ be the incenter. Each of the following triples is collinear: $I_a, I_b, C$; $I_a, I_c, B$; $I_b, I_c, A$; $I, A, I_a$; $I, B, I_b$; and $I, C, I_c$. Moreover, $I$ is the orthocenter of $\triangle I_aI_bI_c$. Equivalently, $\triangle ABC$ is the orthic triangle of $\triangle I_aI_bI_c$.*

**Proof.** This follows immediately from the definitions of the incenter and excenters. □

**Corollary 0.6.5.** *The orthocenter of a triangle is one of the in- or excenters of its orthic triangle. In particular, for an acute triangle it is the incenter of the orthic triangle. If one angle is obtuse, it is the excenter lying in the corresponding angle of the orthic triangle.*[22]

**Remark.** The proof is immediate; indeed, it is already suggested by our proof of the existence of the orthocenter. One should note, however, that although a triangle is the orthic triangle of its excentral triangle, it need not be the excentral triangle of its orthic triangle. The latter holds only when the original triangle is acute. In the orthocentric system consisting of a triangle and its orthocenter, suppose

[22] For example, if $\angle A$ of $\triangle ABC$ is obtuse, then its orthocenter is the $H_a$-excenter of the orthic triangle $\triangle H_aH_bH_c$.

that $A,B,C$ are the three points forming an acute triangle and that $D$ is the remaining point. Then $D$ is the orthocenter of $\triangle ABC$, and $A,B,C$ are the three excenters of the orthic triangle of $\triangle ABC$.

**Corollary 0.6.6.** *The circumcircle of a triangle is the nine-point circle of its excentral triangle, and its circumcenter is the nine-point center of that triangle.*

**Proof.** This is immediate from proposition 0.6.4. □

**Proposition 0.6.7.** *The excentral and intouch triangles of a given triangle are homothetic; see Figure 0.18.*

**Proof.** Let $\triangle G_aG_bG_c$ be the intouch triangle and $\triangle I_aI_bI_c$ the excentral triangle. Since $AG_b = AG_c$, the line $G_bG_c$ is perpendicular to the bisector $AI_a$ of $\angle BAC$. Also $I_bI_c \perp AI_a$, so $G_bG_c \parallel I_bI_c$. Similarly, $G_aG_c \parallel I_aI_c$ and $G_aG_b \parallel I_aI_b$. Hence the triangles are homothetic. □

**Proposition 0.6.8.** *Given $\triangle ABC$, let $\triangle I_aI_bI_c$ be its excentral triangle and $I$ its incenter, as in Figure 0.18. Let $D'$ be the second intersection of $I_bI_c$ with $\mathscr{C}$, and $D$ the second intersection of $AI_a$ with $\mathscr{C}$. Then:*

*(1) $D'$ and $D$ are the midpoints of the arcs $\overset{\frown}{BAC}$ and $\overset{\frown}{BC}$, respectively, and they are antipodal on $\mathscr{C}$;*
*(2) $D'$ is the midpoint of $I_bI_c$.*

**Proof.** For part (1), since $A,B,C,D'$ are concyclic, $\angle D'BC = \angle CAI_b = \angle BAD' = \angle BCD'$. Thus $D'$ bisects the arc $\overset{\frown}{BAC}$. Plainly $D$ bisects the arc $\overset{\frown}{BC}$, so $D,D'$ are antipodal.

For part (2), proposition 0.6.4 says that $\triangle ABC$ is the orthic triangle of $\triangle I_aI_bI_c$. Thus $\mathscr{C}_{\triangle ABC} = \mathscr{N}_{\triangle I_aI_bI_c}$. Apart from the foot of the altitude, the other intersection of a nine-point circle with a side is the midpoint of that side. Hence $D'$ is the midpoint of $I_bI_c$. □

**Claim 0.6.9.** *The complementary arc-midpoint triangle is the medial triangle of the excentral triangle.*

**Claim 0.6.10.** *The lines joining corresponding vertices of a triangle and its circumcircle mid-arc triangle concur at the original triangle's incenter.*[23]

**Proposition 0.6.11.** *For $\triangle ABC$, its intouch triangle $\triangle G_aG_bG_c$ and circumcircle mid-arc triangle $\triangle D_aD_bD_c$ are homothetic. Their ratio of homothety is the ratio of the inradius to the circumradius.*

**Proof.** Let $I$ be the incenter. Clearly $IA \perp G_bG_c$. From the arc-midpoint definition, $AD_a \perp D_bD_c$, and theorem 0.6.3 gives $I \in AD_a$, so $IA \perp D_bD_c$. Hence $G_bG_c \parallel D_bD_c$. The other pairs of sides are analogous, so the triangles are homothetic. Their circumcircles are the incircle and circumcircle of $\triangle ABC$, respectively; hence the ratio is the ratio of those radii. □

**Proposition 0.6.12.** *The incenter $I$ of $\triangle ABC$ is the orthocenter of its circumcircle mid-arc triangle $\triangle D_aD_bD_c$.*

**Proof.** The proof of proposition 0.6.11 gives $IA \perp D_bD_c$. The other two perpendicularities are analogous. □

[23] In the language of the perspective center introduced in section 3.4, the incenter is the perspective center of the original triangle and its circumcircle mid-arc triangle.

## Exercises

**Exercise 0.61.** Prove proposition 0.6.4.

**Exercise 0.62.** The $A$-excircle $\mathcal{I}_a$ of $\triangle ABC$ touches $BC$ at $D$. Prove that $AB+BD=AC+CD$.

**Exercise 0.63.** Prove that the circumradius of the excentral triangle is twice the circumradius of the original triangle.

**Exercise 0.64.** Let $\triangle H_aH_bH_c$ and $\triangle I_aI_bI_c$ be the orthic and excentral triangles of $\triangle ABC$, respectively. Let $X_a,X_b,X_c$ be the incenters of $\triangle AH_bH_c$, $\triangle BH_cH_a$, $\triangle CH_aH_b$, respectively. Prove that $X_bX_c \parallel I_bI_c$.

**Exercise 0.65.** Suppose $AB+BC=3AC$ in $\triangle ABC$, whose intouch triangle is $\triangle DEF$. Let $P,Q$ be the points on $\mathcal{I}$ antipodal to $D,F$, respectively. Prove that $A,C,P,Q$ are concyclic.

**Exercise 0.66.** Let $ABCD$ be cyclic. Prove that all the incenters and excenters of $\triangle ABC$, $\triangle BCD$, $\triangle CDA$, and $\triangle DAB$ (sixteen points in total) form a $4\times 4$ rectangular array.

**Exercise 0.67.** Suppose $AB<AC<BC$ in $\triangle ABC$. Let $I$ be its incenter and $\omega$ its incircle. Choose $X\neq C$ on the line $BC$ so that the line through $X$ parallel to $AC$ is tangent to $\omega$, and choose $Y\neq B$ on the line $BC$ so that the line through $Y$ parallel to $AB$ is tangent to $\omega$. Let $P$ be the second intersection of $AI$ with $\mathscr{C}$, and let $K,L$ be the midpoints of $AC,AB$. Prove that $\angle KIL+\angle YPX=180^\circ$.

**Exercise 0.68.** Let $O,I$ be the circumcenter and incenter of $\triangle ABC$, respectively, and let $\triangle DEF$ be the intouch triangle. If $X$ is the reflection of $D$ in $EF$, prove that $OI$, $AX$, and $BC$ are concurrent.

### 0.6.2 Metric Properties of the Incenter and Excenters

We now discuss several metric properties of the incenter and excenters.

**Proposition 0.6.13.** *In a triangle ABC, let the lengths of $BC,CA,AB$ be $a,b,c$, and let the semiperimeter be $p=\frac{a+b+c}{2}$.*

*(1) If $\mathcal{I}$ touches $BC,CA,AB$ at $G_a,G_b,G_c$, respectively, then*

$$AG_b=AG_c=p-a,\qquad BG_a=BG_c=p-b,\qquad CG_a=CG_b=p-c.$$

*(2) If $\mathcal{I}_a,\mathcal{I}_b,\mathcal{I}_c$ touch $BC,CA,AB$ at $N_a,N_b,N_c$, respectively, then*

$$AN_b=BN_a=p-c,\qquad CN_a=AN_c=p-b,\qquad CN_b=BN_c=p-a.$$

**Proof.** The proof is direct and is left to the reader. □

**Proposition 0.6.14.** *Let $\triangle ABC$ have side lengths $a=BC$, $b=CA$, $c=AB$, semiperimeter $p=\frac{a+b+c}{2}$, and area $S$.*

*(1) The radius of $\mathcal{I}$ is $r=\frac{S}{p}$.*

*(2) The radii of $\mathcal{I}_a,\mathcal{I}_b,\mathcal{I}_c$ are, respectively,*

$$r_a=\frac{S}{p-a},\qquad r_b=\frac{S}{p-b},\qquad r_c=\frac{S}{p-c}.$$

**Proof.** Clearly,

$$S = S_{\triangle AIB} + S_{\triangle BIC} + S_{\triangle CIA} = \frac{1}{2}(AB \cdot r + BC \cdot r + CA \cdot r) = pr,$$

which proves part (1). Part (2) is analogous and is left to the reader. $\square$

**Proposition 0.6.15.** *Let $\triangle G_aG_bG_c$ be the intouch triangle of $\triangle ABC$, and let $P$ be the projection of $G_c$ on $G_aG_b$. Then $\triangle AG_bP \sim \triangle BG_aP$.*

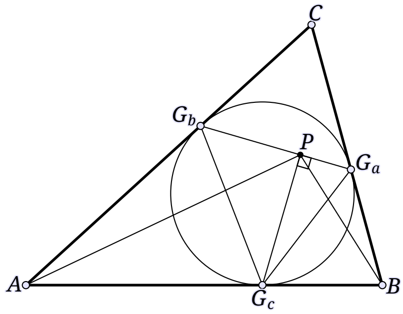


FIGURE 0.19

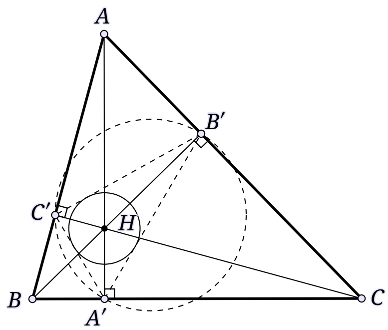


FIGURE 0.20

**Proof.** Since $AB$ is tangent to the incircle, $\angle PG_bG_c = \angle G_aG_cB$ and $\angle PG_aG_c = \angle G_bG_cA$. Thus,

$$\frac{PG_a}{PG_b} = \frac{G_aG_c \cos\angle PG_aG_c}{G_bG_c \cos\angle PG_bG_c} = \frac{(2BG_a \cos\angle BG_cG_a)\cos\angle AG_cG_b}{(2AG_b \cos\angle AG_cG_b)\cos\angle BG_cG_a} = \frac{BG_a}{AG_b}.$$

Together with $\angle BG_aP = \angle AG_bP$, this proves the similarity. $\square$

**Proposition 0.6.16.** *If $\triangle ABC$ has circumradius $R$, its inradius is $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$, and its $A$-exradius is $r_a = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$.*

**Proof.** We prove only the incircle case; the excircle case is analogous. From the geometric relations in Figure 0.18, with the penultimate equality using the sine rule,

$$r = IC\sin\angle BCI = II_a \sin\angle AI_aC \sin\angle BCI = 2BD\sin\angle IBC\sin\angle BCI$$
$$= 4R\sin\angle IAB\sin\angle IBC\sin\angle BCI = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

Here the third equality uses the cyclicity of $B, I, C, I_a$ and $BD = ID = I_aD$. $\square$

**Proposition 0.6.17.** *Let $R$ be the circumradius of $\triangle ABC$. The signed radius $\rho$ of the in- or excircle of the orthic triangle whose center is the orthocenter of $\triangle ABC$ is*

$$\rho = 2R\cos A\cos B\cos C.$$

*Here $\rho > 0$ when $\triangle ABC$ is acute and its orthocenter is the incenter of the orthic triangle; $\rho < 0$ when it is obtuse and the orthocenter is an excenter of the orthic triangle. In what follows, write $\rho_{\triangle ABC}$ for this signed radius.*

**Proof.** We prove only the case in which $\triangle ABC$ is acute; the other cases are analogous. Let $\triangle H_aH_bH_c$ be its orthic triangle. Its circumcircle is the nine-point circle and has radius $R/2$. Applying proposition 0.6.16, and noting that $A, H_b, H_c, H$ are concyclic, that $B, H_c, H_a, H$ are concyclic, and that

$C, H_a, H_b, H$ are concyclic, gives

$$\rho = 2R\prod_{\text{cyc}} \sin\frac{\angle H_bH_aH_c}{2} = 2R\sin\angle H_cH_aH \sin\angle H_aH_bH \sin\angle H_bH_cH$$
$$= 2R\sin\angle ABH \sin\angle BCH \sin\angle CAH = 2R\cos A\cos B\cos C. \qquad \square$$

We next investigate the distances from the orthocenter and circumcenter to the incenter and excenters.

**Proposition 0.6.18** (Distance from the orthocenter to an in- or excenter)**.** *Let $d$ be the distance from the orthocenter of $\triangle ABC$ to one of its in- or excenters, and let $r$ be the radius of the corresponding incircle or excircle. If $R$ is the circumradius, then*

$$d^2 = 2r^2 - 4R^2\cos A\cos B\cos C.$$

**Proof.** We treat the incenter $I$ and orthocenter $H$; the excenter cases are analogous. Observe that $\angle IAH = \frac{A}{2} - (\frac{\pi}{2} - C) = \frac{C-B}{2}$, and proposition 0.6.16 gives $AI = \frac{r}{\sin\frac{A}{2}} = 4R\sin\frac{B}{2}\sin\frac{C}{2}$. Applying the cosine rule in $\triangle AHI$,

$$\begin{aligned}
HI^2 &= AI^2 + AH^2 - 2AI\cdot AH\cos\angle IAH\\
&= 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2} + 4R^2\cos^2 A - 16R^2\sin\frac{B}{2}\sin\frac{C}{2}\cos A\cos\frac{B-C}{2}\\
&= 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2} + 4R^2\cos^2 A - 16R^2\sin\frac{B}{2}\sin\frac{C}{2}\cos A\left(\cos\frac{B}{2}\cos\frac{C}{2} + \sin\frac{B}{2}\sin\frac{C}{2}\right)\\
&= 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2}(1-\cos A) + 4R^2\cos^2 A - 16R^2\sin\frac{B}{2}\sin\frac{C}{2}\cos A\cos\frac{B}{2}\cos\frac{C}{2}\\
&= 32R^2\prod_{\text{cyc}}\sin^2\frac{A}{2} - 4R^2\cos A\cos(B+C) - 4R^2\cos A\sin B\sin C\\
&= 32R^2\prod_{\text{cyc}}\sin^2\frac{A}{2} - 4R^2\cos A(\cos B\cos C - \sin B\sin C) - 4R^2\cos A\sin B\sin C = 2r^2 - 4R^2\prod_{\text{cyc}}\cos A. \square
\end{aligned}$$

The preceding proof uses trigonometric computation, which some readers may find unfamiliar. A more geometric proof based on Feuerbach's theorem will be given later.

**Corollary 0.6.19** (Distance from the orthocenter to an in- or excenter)**.** *Let $d$ be the distance from the orthocenter of $\triangle ABC$ to one of its in- or excenters, and let $r$ be the radius of the corresponding incircle or excircle. If $R$ and $\rho$ are, respectively, the circumradius of $\triangle ABC$ and $\rho_{\triangle ABC}$, then*

$$d^2 = 2r^2 - 2R\rho.$$

**Proof.** This follows from propositions 0.6.17 and 0.6.18. $\square$

**Theorem 0.6.20** (Euler–Chapple formula; distance from the circumcenter to an in- or excenter)**.** *Let $R$ be the circumradius of $\triangle ABC$, let $r$ be the radius of its incircle or of one of its excircles, and let $d$ be the distance from the corresponding in- or excenter to the circumcenter. Then*

$$d^2 = R^2 \pm 2Rr,$$

*with the minus sign for the incircle and the plus sign for an excircle.*

**Proof.** We prove the incircle case; the excircle cases are analogous.

In Figure 0.21, let $O$ be the circumcenter and let $AI$ meet $\mathscr{C}$ again at $D$. By the sine rule, $DC = 2R\sin\angle DAC$. Theorem 0.6.3 gives $ID = DC = 2R\sin\angle DAC$, while $AI = \frac{IE}{\sin\angle DAC} = \frac{r}{\sin\angle DAC}$. Hence $AI\cdot DI = 2Rr$.

Put $OI \cap \mathscr{C} = P, Q$. Then

$$R^2 - d^2 = (R+d)(R-d) = PI\cdot QI = AI\cdot DI = 2Rr. \qquad \square$$

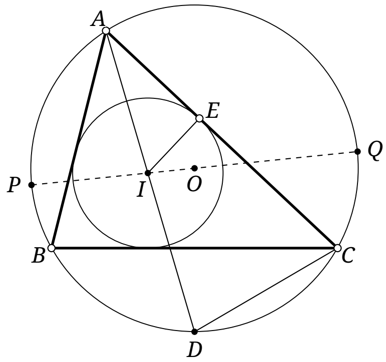


FIGURE 0.21

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## Exercises

**Exercise 0.69.** Prove proposition 0.6.13 and part (2) of proposition 0.6.14.

**Exercise 0.70.** Let $I_b$ be the $B$-excenter of $\triangle ABC$, and let $BI_b$ meet $\mathscr{C}$ again at $E$. Prove that the length of the projection of $EI_b$ on $AB$ equals $\frac{AC}{2}$.

**Exercise 0.71.** Prove that the inradius of the excentral triangle of $\triangle ABC$ is

$$8R\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\sin\frac{\pi-C}{4},$$

where $R$ is the circumradius of the original triangle.

**Exercise 0.72.** Prove that if a triangle exists which is inscribed in a circle $\omega_1$ and circumscribed about a circle $\omega_2$, then infinitely many such triangles exist.

**Exercise 0.73** (Distance between the orthocenter and circumcenter)**.** Let $R$ be the circumradius of $\triangle ABC$, let $\rho = \rho_{\triangle ABC}$, and let $d$ be the distance between the orthocenter and circumcenter. Prove that $d^2 = R^2 - 4R\rho$.

**Exercise 0.74.** For $\triangle ABC$, let the radii of $\mathscr{C}, \mathscr{I}, \mathscr{I}_a, \mathscr{I}_b, \mathscr{I}_c$ be $R, r, r_a, r_b, r_c$, respectively. Prove that $r_a + r_b + r_c = 4R + r$.

**Exercise 0.75.** Use the formulas for the orthocenter–circumcenter distance and the Euler–Chapple formula to prove that the nine-point circle is tangent to the incircle and all three excircles of the triangle.

**Exercise 0.76.** For $\triangle ABC$, suppose $r_{\mathscr{C}} = R$, $r_{\mathscr{I}} = r$, and $\rho_{\triangle ABC} = \rho$. Find the distances from its incenter to the nine-point center and to the centroid.

**Exercise 0.77.** Prove that the reflection triangle of $\triangle ABC$ is degenerate[24] if and only if $4\rho_{\triangle ABC} = -3r_{\mathscr{C}}$.

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[24]That is, its three vertices are collinear.

Chapter 1

# Basic Properties of Conics

With this chapter we begin the first part of the book proper. Unlike the usual order of presentation in plane geometry, we start with conics. We first study their definitions and basic properties, and throughout the chapters that follow we shall bring the viewpoint of conics to bear on other topics.

## 1.1 Definitions of Conics

**Definition 1.1.1** (Ellipse)**.**

(1) As shown in Figure 1.1, an *ellipse* is the locus of a point $P$ for which the sum of the distances from two distinct fixed points $F_1, F_2$ is constant. The points $F_1, F_2$ are the two *foci* (singular: focus) of the ellipse, and the distance between them is its *focal distance*.

(2) An ellipse has two axes of symmetry, denoted by $a$ and $b$. The line $a\colon F_1F_2$ is its *major axis*,[1] and the perpendicular bisector $b$ of $F_1F_2$ is its *minor axis*. Together, the major and minor axes are called the *axes* of the ellipse. The two intersections of the major axis with the ellipse are its *major-axis vertices*, and the two intersections of the minor axis with the ellipse are its *minor-axis vertices*; collectively they are the *vertices* of the ellipse. The intersection of $a$ and $b$ is its *center*.

(3) The circle whose diameter is the major axis of the ellipse is its *major auxiliary circle*, often simply its *auxiliary circle*. The circle whose diameter is the minor axis is its *minor auxiliary circle*.

(4) The ratio of the focal distance to the length of the major axis is the *eccentricity* of the ellipse.

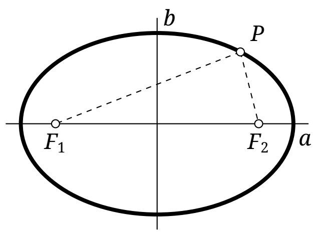


FIGURE 1.1

**Remark.** In the preceding definition, as $F_1$ and $F_2$ tend to the same point $O$, the eccentricity tends to 0 and the ellipse tends to a circle centered at $O$. A circle is therefore sometimes regarded as a special ellipse. In this book the word "ellipse" excludes circles in many contexts, although the corresponding properties of circles can then be obtained by passage to the limit; in other contexts it includes circles. The intended convention will always be clear from context. At least until chapter 3, an ellipse will never include a circle.

[1] The term "major axis" is sometimes used for the segment cut from this line by the ellipse. The same convention applies below to the minor axis of an ellipse and the transverse axis of a hyperbola. Context always makes clear whether a line or a segment is meant; for example, in part (4) below, "the length of the major axis" necessarily refers to the segment.

**Proposition 1.1.2.** *Let $F_1, F_2$ be the foci of an ellipse. If $X$ lies inside the ellipse, then $XF_1 + XF_2$ is less than the length of the major axis; if $X$ lies outside, then $XF_1 + XF_2$ is greater than that length.*

**Proof.** The case $X = F_1$ is immediate. Otherwise, let the ray $F_1X$ meet the ellipse at $Y$. If $X$ lies inside, then

$$XF_1 + XF_2 < XF_1 + XY + YF_2 = YF_1 + YF_2,$$

which is the length of the major axis. If $X$ lies outside, then

$$XF_1 + XF_2 = F_1Y + YX + XF_2 > YF_1 + YF_2,$$

where $YF_1 + YF_2$ again equals the length of the major axis. □

**Definition 1.1.3** (Hyperbola)**.**

(1) As shown in Figure 1.2, a *hyperbola* is the locus of a point $P$ for which the absolute value of the difference of the distances from two distinct fixed points $F_1, F_2$ is constant. The points $F_1, F_2$ are the two foci, and the distance between them is the focal distance.

(2) A hyperbola also has two axes of symmetry, denoted by $a$ and $b$. The line $a\colon F_1F_2$ is its *transverse axis*, and the perpendicular bisector $b$ of $F_1F_2$ is its *conjugate axis*. In some synthetic-geometry literature, the synonymous terms *real axis* and *imaginary axis* are used, respectively. Together, the transverse and conjugate axes are called its axes. The two intersections of the transverse axis with the hyperbola are its *transverse-axis vertices*. The conjugate axis separates the hyperbola into two disconnected parts, each called a *branch*. The intersection of $a$ and $b$ is the *center* of the hyperbola. We use *transverse axis* and *conjugate axis* below.

(3) A hyperbola has two *asymptotes* $l_1, l_2$.[2] If the asymptotes are perpendicular, the hyperbola is called *equilateral*.

(4) The circle whose diameter is the transverse axis of the hyperbola is its major auxiliary circle, often simply its auxiliary circle.

(5) The ratio of the focal distance to the length of the transverse axis is the eccentricity of the hyperbola.

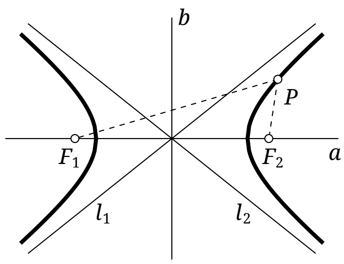


FIGURE 1.2

**Proposition 1.1.4.** *Let $F_1, F_2$ be the foci of a hyperbola. If $X$ lies outside the hyperbola,[3] then $|XF_1 - XF_2|$ is less than the length of the transverse axis; if $X$ lies inside, then $|XF_1 - XF_2|$ is greater than the length of the transverse axis.*

[2] That is, as a point $P$ recedes indefinitely from the center along the hyperbola, its distance from one of $l_1, l_2$ tends to zero.

[3] The terms interior and exterior of a hyperbola are used differently in different sources. In this book, the region between its two branches is called its exterior, and the two remaining regions form its interior. The conventions for all conics are stated together at the end of this section.

**Proof.** The proof is left to the reader. □

**Definition 1.1.5** (Parabola)**.**

(1) As shown in Figure 1.3, a *parabola* is the locus of a point $P$ whose distance from a fixed point $F$ equals its distance from a fixed line $l$. The point $F$ is the *focus* and the line $l$ the *directrix* of the parabola.

(2) A parabola has one axis of symmetry, denoted by $l'$; we call it simply the *axis*. Its intersection with the parabola is its *vertex*.

(3) Circles, ellipses, hyperbolas, and parabolas are collectively called *conics* or *conic sections*. Eccentricity is a fundamental geometric characteristic of a conic: a parabola has eccentricity 1, and a circle has eccentricity 0. Hyperbolas, ellipses, and circles are collectively called *central conics*. As uniform shorthand in this book, we call the transverse axis of a hyperbola, the major axis of an ellipse, and the axis of a parabola its *principal axis*; similarly, we call the conjugate axis of a hyperbola and the minor axis of an ellipse its *secondary axis*.

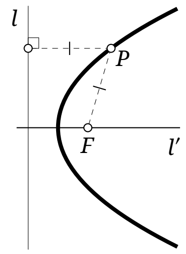


FIGURE 1.3

**Proposition 1.1.6.** *If $X$ lies inside a parabola—that is, on the same side of the parabola as its focus $F$—then $XF$ is less than the distance from $X$ to the directrix; if $X$ lies outside, then $XF$ is greater than that distance.*

**Proof.** The proof is left to the reader. □

The preceding approach is called the *first definition of a conic*. In this chapter we first present the most important properties of conics, which are closely related to the material developed later. Further elementary properties will be discussed in chapter 2.

The symmetry assertions in the three definitions above are immediate. Some of the other assertions, such as the existence of the asymptotes of a hyperbola, depend on a more rigorous analysis. These details will also be supplied in chapter 2; readers not interested in them may omit that discussion.

Finally, we make the following conventions. The interior of a conic is the union of the regions of the plane cut out by the conic that contain its focus or foci; thus the interior of a hyperbola has two components. Its exterior is the region containing no focus. Moreover, when we say in what follows that a polygon is circumscribed about a conic, or equivalently that the conic is inscribed in the polygon, we generally require only that the line containing each side be tangent to the conic; a point of tangency need not lie on the corresponding side segment. Similarly, when we say that a polygon is inscribed in a conic, or equivalently that the conic is circumscribed about the polygon, we generally require only that all its vertices lie on the conic; the entire figure need not lie in the interior of the conic. Of course, when we speak specifically of a triangle circumscribed about a circle or of the incircle of a triangle, the meanings are those given in chapter 0.

## Exercises

**Exercise 1.1.** Prove propositions 1.1.4 and 1.1.6.

**Exercise 1.2.**

(1) Let $F_1, F_2$ be the foci of an ellipse, and let $P$ be a point of the ellipse not on the major axis. Prove that the $F_1$-excircle of $\triangle PF_1F_2$ is tangent to the major axis at the major-axis vertex farther from $F_1$.

(2) Let $F_1, F_2$ be the foci of a hyperbola, and let $P$ be a point of the hyperbola not on the transverse axis. Prove that the incircle of $\triangle F_1PF_2$ is tangent to the transverse axis at the vertex on the same branch as $P$.

**Exercise 1.3.** Let $A, B$ lie on a parabola, and suppose that $AB$ passes through its focus $F$. Prove that the circle $\odot(AB)$ with diameter $AB$ is tangent to the directrix.

**Exercise 1.4.**

(1) Prove that all parabolas are similar.

(2) Prove that two parabolas with the same vertex and axis are homothetic.

## 1.2 Standard Equations and Classification of Quadratic Curves

In this section we discuss conics analytically. Although the material is not purely geometric, it clarifies two points in particular: conics are curves of degree two,[4] and the introduction of imaginary elements will simplify later arguments. Both points can also be treated synthetically, but the necessary tools have not yet been developed; we shall return to them in due course.

Analytically, we make the following definition.

**Definition 1.2.1.** A *conic*, or *quadratic curve*, is the locus of points whose coordinates in some rectangular coordinate system $xOy$ satisfy

$$a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + 2b_1x + 2b_2y + c = 0, \tag{1.1}$$

where $a_{11}, a_{12}, a_{22}, b_1, b_2, c$ are real and $a_{11}, a_{12}, a_{22}$ are not all zero. A quadratic curve is *degenerate* if the left-hand side of (1.1) factors over the complex numbers into a product of two linear expressions.

By rotating and translating the coordinate system, the equation of a quadratic curve can be reduced to one of the following standard forms.

a.) The equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a > b > 0) \tag{1.2}$$

represents a *real ellipse* whose major and minor axes are the $x$- and $y$-axes. The numbers $a, b$ are respectively its *semimajor axis* and *semiminor axis*, namely half the lengths of the corresponding axes, and its foci are $(\pm c, 0)$, where $c := \sqrt{a^2 - b^2}$.

b.) The equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \qquad (a, b > 0) \tag{1.3}$$

[4] The fundamental theorem of algebra may therefore be used to discuss the number of intersections with other curves. For example, a conic and a line have at most two intersections, and two conics have at most four.

represents a hyperbola whose transverse and conjugate axes are the $x$- and $y$-axes. The numbers $a,b$ are respectively its *semi-transverse axis* (half the length of the transverse axis) and *semi-conjugate axis*, and its foci are $(\pm c,0)$, where $c := \sqrt{a^2+b^2}$. The circle centered at the center of the hyperbola with radius $b$ is its minor auxiliary circle.

The asymptotes of this hyperbola are $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$. When $a = b$, the hyperbola is equilateral.

c.) The equation

$$y^2 = 2px \qquad (p > 0) \tag{1.4}$$

represents a parabola with vertex at the origin and axis the $x$-axis, opening to the right. Its focus is $(\frac{p}{2},0)$ and its directrix is $x = -\frac{p}{2}$.

d.) The equation

$$x^2 + y^2 = r^2 \qquad (r > 0) \tag{1.5}$$

represents a *real circle* centered at the origin with radius $r$.

e.) The equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \qquad (a,b > 0) \tag{1.6}$$

represents the two intersecting lines $\frac{x}{a} + \frac{y}{b} = 0$ and $\frac{x}{a} - \frac{y}{b} = 0$, because its left-hand side factors as $(\frac{x}{a} + \frac{y}{b})(\frac{x}{a} - \frac{y}{b})$. It is degenerate.

f.) The equation

$$y^2 = a^2 \qquad (a > 0) \tag{1.7}$$

represents the two parallel lines $y = a$ and $y = -a$, since it can be written $(y+a)(y-a) = 0$. It is degenerate.

g.) The equation

$$y^2 = 0 \tag{1.8}$$

represents the line $y = 0$ counted twice. Since $y = 0$ is a double root, this is described as a pair of coincident lines. It is degenerate.

h.) The equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = -1 \qquad (a > b > 0) \tag{1.9}$$

represents an *imaginary ellipse*.

i.) The equation

$$x^2 + y^2 = -r^2 \qquad (r > 0) \tag{1.10}$$

represents an *imaginary circle*.[5]

j.) The equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0 \qquad (a,b > 0) \tag{1.11}$$

represents a point, or equivalently two conjugate imaginary lines meeting at a real point. Indeed, it factors as

$$\left(\frac{x}{a} + \mathrm{i}\frac{y}{b}\right)\left(\frac{x}{a} - \mathrm{i}\frac{y}{b}\right) = 0,$$

so it is degenerate. When $a = b$, the equation becomes $x^2 + y^2 = 0$; it may be regarded as a circle of radius zero and is sometimes called a *point circle*.

k.) The equation

$$y^2 = -a^2 \qquad (a > 0) \tag{1.12}$$

represents two parallel conjugate imaginary lines. It can be written $(y + a\mathrm{i})(y - a\mathrm{i}) = 0$ and is degenerate.

[5]It is also sometimes regarded as a special imaginary ellipse, obtained by allowing $a,b$ in (1.9) to tend to the same positive number. The meaning of "imaginary" is explained below.

Every quadratic curve can be reduced by a translation and rotation of coordinates to one of these eleven standard forms. Forms (a), (b), (d), (e), (h), (i), and (j) are central, whereas (c), (f), (g), and (k) are noncentral. Forms (a)–(g) are real, whereas (h)–(k) are imaginary. Forms (a)–(d), (h), and (i) are nondegenerate, whereas (e)–(g), (j), and (k) are degenerate. Real and imaginary ellipses are collectively called ellipses; real circles, point circles, and imaginary circles are collectively called circles.

Hereafter, "quadratic curve" will usually mean a nondegenerate real quadratic curve. In some contexts it may include degenerate curves, or all eleven kinds above; the intended meaning will always be clear. Similarly, "ellipse" and "circle" will normally mean a real ellipse and a real circle; here too the intended meaning will always be clear from context.

We shall not make a strict distinction between the terms "quadratic curve" and "conic": they will be used synonymously, although "conic" is preferred in synthetic settings. Thus "conic" will no longer be restricted to the four curves in (a)–(d), even though those are what it denotes in most cases; again, context will make the intended class clear.

We conclude with a clarification concerning imaginary quadratic curves. In contrast to the ordinary Euclidean plane, these curves extend the ranges of $x,y$ to the complex numbers. The plane consisting of all points $(x,y)$ with $x,y\in\mathbb{C}$ is called the *complex Euclidean plane* $\mathbb{C}^2$. For $A(x_1,y_1)$ and $B(x_2,y_2)$, their distance is defined by the same formal expression as in the real Euclidean plane $\mathbb{R}^2$.[6]

$$\operatorname{dist}(A,B)=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}.$$

The resulting metric structure is called the *Euclidean metric*. Moreover, the equation $Ax+By+C=0$, where $A,B$ are complex and not both zero, still represents a line. Its slope is $k=-\frac{A}{B}$ when $B\neq 0$; when $B=0$, its slope is said to be undefined or infinite. Unless stated otherwise, "Euclidean plane" will continue to mean the ordinary real Euclidean plane.

Thus the imaginary ellipse (1.9) is the locus of points whose sum of distances from $(-\mathrm{i}\sqrt{a^2-b^2},0)$ and $(\mathrm{i}\sqrt{a^2-b^2},0)$ equals $2a\mathrm{i}$. Equation (1.10) represents a circle centered at the origin with radius $r\mathrm{i}$. Equation (1.11) represents the imaginary lines $\frac{x}{a}+\mathrm{i}\frac{y}{b}=0$ and $\frac{x}{a}-\mathrm{i}\frac{y}{b}=0$, which meet at the real point $(0,0)$.[7] Equation (1.12) represents the lines $y=a\mathrm{i}$ and $y=-a\mathrm{i}$. This explains the terminology "imaginary quadratic curve."

Although we have introduced the complex Euclidean plane, it is only a convenience. We shall still consider only those quadratic curves for which all coefficients in (1.1) are real. The most general conic in the complex Euclidean plane may have complex coefficients and admits a further classification, but this lies outside the scope of the book; interested readers may consult the literature.

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## Exercises

**Exercise 1.5.** Prove the following.

(1) Equation (1.2) represents an ellipse of focal distance $2\sqrt{a^2-b^2}$.
(2) Equation (1.3) represents a hyperbola of focal distance $2\sqrt{a^2+b^2}$.
(3) Equation (1.4) represents a parabola whose focus is at distance $p$ from its directrix.

---

[6] The usual metric on the complex Euclidean plane is $\|A-B\|=\sqrt{|x_1-x_2|^2+|y_1-y_2|^2}$, not the expression used here. Here "distance" merely extends the real Euclidean expression to complex values so that several statements can be written uniformly. Over the complex numbers the square-root function is multivalued, and one ordinarily selects a principal value, for example by taking arguments in $[0,2\pi)$. This issue will scarcely arise in this book. It is enough to know that for $a>0$ we take $\sqrt{-a}=\mathrm{i}\sqrt{a}$, not $-\mathrm{i}\sqrt{a}$.

[7] A point $(x,y)$ of the complex Euclidean plane is called *real* precisely when both $x$ and $y$ are real; otherwise it is *imaginary*. Lines are classified similarly. A real line contains infinitely many real points, and all other lines are imaginary. One may prove the following characterization: the line $Ax+By+C=0$, where $A,B$ are not both zero, is real if and only if there is a nonzero constant $\lambda$ for which $\lambda A,\lambda B,\lambda C$ are all real. An imaginary line contains at most one real point.

**Exercise 1.6.** For each of the following equations, prove that it represents a real central quadratic curve, determine its type, and find its eccentricity and the coordinates of its foci:

(1) $y = \frac{1}{x}$;
(2) $x^2 + xy + y^2 = 1$.

**Exercise 1.7**.** Prove that through any five points of the plane, no four of which are collinear, there passes exactly one quadratic curve, possibly degenerate; and that if no three of the five points are collinear, there passes exactly one nondegenerate conic.

## 1.3 Optical Properties of Conics

We now study the most important geometric properties of conics: their optical properties. Throughout this section, unless otherwise stated, $F_1, F_2$ denote the foci of an ellipse or hyperbola, and $F$ denotes the focus of a parabola.

### 1.3.1 The Optical Property

**Theorem 1.3.1** (Optical property of conics)**.**

*(1) Let the line $l$ pass through a point $P$ of an ellipse $\alpha$. Then $l$ is tangent to $\alpha$ if and only if it bisects an exterior angle of $\angle F_1PF_2$.*
*(2) Let the line $l$ pass through a point $P$ of a hyperbola $\alpha$. Then $l$ is tangent to $\alpha$ if and only if it bisects $\angle F_1PF_2$.*
*(3) Let the line $l$ pass through a point $P$ of a parabola $\alpha$, and let $P'$ be the projection of $P$ on the directrix. Then $l$ is tangent to $\alpha$ if and only if it bisects $\angle FPP'$.*

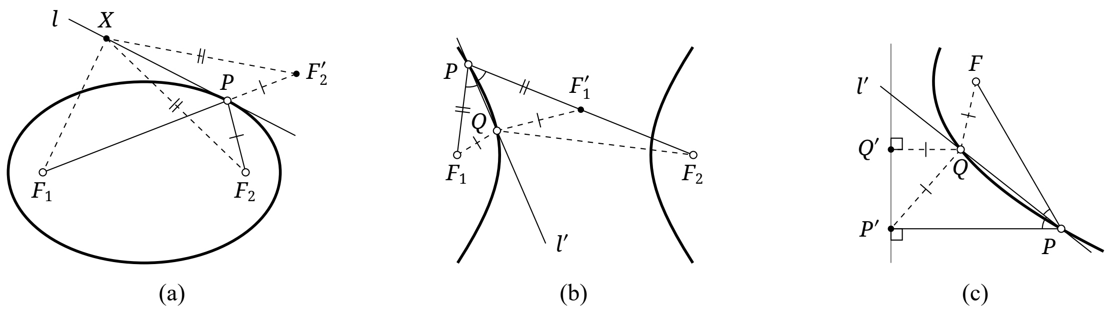


FIGURE 1.4

**Proof.** For part (1) we prove only that tangency implies the angle-bisector condition; for parts (2) and (3) we prove only the converse implications. The remaining arguments are analogous.

For (1), take any $X \in l$ as in Figure 1.4a. Since $l$ is tangent, $X$ lies outside $\alpha$, and proposition 1.1.2 gives

$$F_1X + F_2X \geq F_1P + F_2P$$

for every $X$, with equality if and only if $X = P$. Thus $P$ is the unique position of $X$ for which $F_1X + F_2X$ is least. This shows that $l$ bisects an exterior angle of $\angle F_1PF_2$. Otherwise, if $F_2'$ is the reflection of $F_2$ in $l$ and $X = F_1F_2' \cap l$, then $F_1, P, F_2'$ are not collinear and

$$F_1X + F_2X = F_1X + F_2'X = F_1F_2' < F_1P + F_2'P = F_1P + F_2P,$$

a contradiction.

For (2), argue by contradiction. If the bisector $l'$ of $\angle F_1PF_2$ met $\alpha$ again at $Q$, as in Figure 1.4b, let $F_1'$ be the reflection of $F_1$ in $l'$. Then $\triangle F_1QP \cong \triangle F_1'QP$, and $P, F_1', F_2$ are collinear. Consequently,

$$F_2Q - QF_1' = F_2Q - QF_1 = F_2P - PF_1 = F_2P - PF_1' = F_2F_1',$$

contrary to $F_2F_1' + QF_1' > QF_2$.

For (3), again argue by contradiction. Suppose the bisector $l'$ of $\angle FPP'$ met $\alpha$ again at $Q$, as in Figure 1.4c, and let $Q'$ be the projection of $Q$ on the directrix. The congruence $\triangle FPQ \cong \triangle P'PQ$ gives $QP' = QF$, whereas the definition of the parabola gives $QF = QQ'$, a contradiction. □

**Remark.** The proof has tacitly used the assertion that a line is tangent to a conic if and only if the line and conic have exactly one point in common. In the real Euclidean plane this is *almost* correct. For a hyperbola or parabola, however, a line with only one finite intersection need not be tangent: the exceptions are a line parallel to an asymptote of the hyperbola and a line parallel to the axis of the parabola. In these exceptional cases, one may start with a secant and let one of its intersections tend to infinity along the conic.

**Claim 1.3.2.** *An ellipse and a hyperbola having the same two foci meet orthogonally.*[8]

The following results are consequences of the optical property.

**Theorem 1.3.3.** *The tangents to an ellipse $\alpha$ at the endpoints of a chord $PQ$ meet at $R$. If $PQ$ passes through the focus $F_1$, then $R$ is the $F_2$-excenter of $\triangle PQF_2$, and the corresponding excircle touches $PQ$ at $F_1$; in other words, $RF_1 \perp PQ$.*

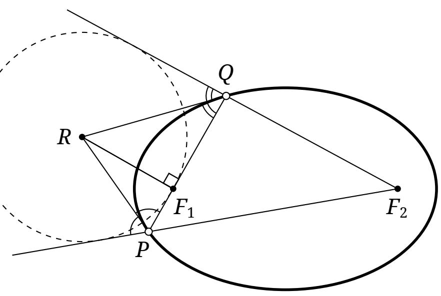


FIGURE 1.5

**Proof.** By theorem 1.3.1, $R$ is an excenter of the triangle. Let the corresponding excircle touch $PQ$ at $X$. The tangent-length relations give $XP + PF_2 = XQ + QF_2$, and $F_1$ is the only point $X$ satisfying this equality. □

There is an analogous result for a hyperbola, with an analogous proof.

**Theorem 1.3.4.** *A line through a focus $F_1$ of a hyperbola meets it at $P, Q$, and the tangents at $P, Q$ meet at $R$. Then $R$ is an excenter or the incenter of $\triangle PQF_2$, and $RF_1 \perp PQ$. More precisely, $R$ is the incenter when $P, Q$ lie on the same branch and an excenter when they lie on different branches.*

The auxiliary circle of a central conic has the following important property.

**Theorem 1.3.5.** *Let $F_1$ be either focus of an ellipse or hyperbola $\alpha$. The locus of the projection of $F_1$ on a variable tangent to $\alpha$ is the auxiliary circle of $\alpha$.*[9]

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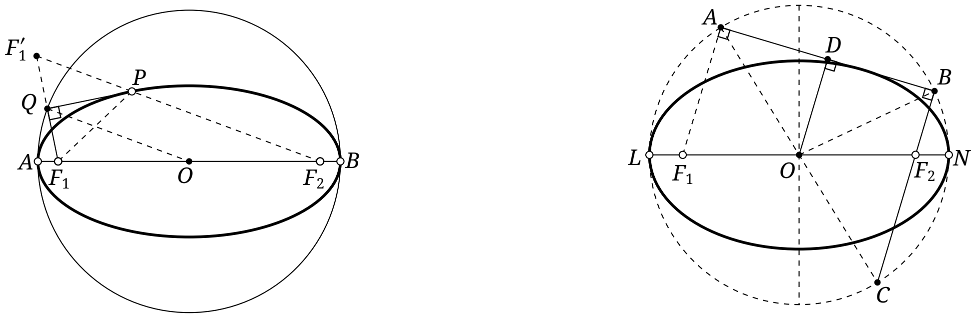
[8]The angle between two curves at an intersection is the angle between their tangents there. To meet orthogonally means that this angle is $90^\circ$.

[9]Here the asymptotes of a hyperbola are regarded as limiting tangents, so the corresponding limiting cases are included. The same convention applies below without further mention.




(a) (b)

FIGURE 1.6

**Proof.** It is enough to treat an ellipse; the hyperbolic case is analogous. In Figure 1.6a, let $F_2$ be the other focus, $O$ the center, $A,B$ the major-axis vertices, $P$ the point of contact of a tangent, and $Q$ the projection of $F_1$ on that tangent. Reflect $F_1$ in $PQ$ to $F_1'$. By the optical property, $F_1',P,F_2$ are collinear. The segment $QO$ is a midline of $\triangle F_1F_1'F_2$, and hence

$$OQ=\frac{F_1'F_2}{2}=\frac{PF_1+PF_2}{2}=\mathrm{const}=\frac{AF_1+AF_2}{2}=\frac{AF_1+BF_1}{2},$$

which is the semimajor axis. This proves the result. $\square$

The converse may be stated as follows.

**Theorem 1.3.6.** *Let $\omega=\odot O$ be a circle, and let $F$ be a point neither on $\omega$ nor equal to $O$. For a variable point $P\in\omega$, let $l$ be the line through $P$ perpendicular to $PF$. Then $l$ is always tangent to a fixed conic $\alpha$. The foci of $\alpha$ are $F,\mathfrak{f}_O(F)$, and its auxiliary circle is $\omega$. If $F$ lies inside $\omega$, then $\alpha$ is an ellipse; if $F$ lies outside, it is a hyperbola.*

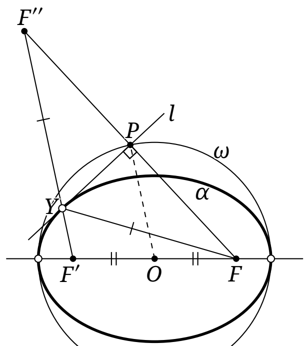



FIGURE 1.7

**Proof.** Consider first the case in which $F$ lies inside $\omega$, as in Figure 1.7. Write $\mathfrak{f}_O(F)=F'$ and $\mathfrak{f}_l(F)=F''$.[10] Put $Y=F'F''\cap l$. The rays $YF''$ and $YF$ are reflections of one another in $l$, and

$$FY+F'Y=F''Y+F'Y=F'F''=2OP=2r_\omega=\mathrm{const},$$

where $F'F''=2OP$ follows from the midline theorem. Thus $Y$ lies on a fixed ellipse $\alpha'$. By symmetry, $FY$ and $F'Y$ make equal angles with $l$, so the optical property shows that $l$ is tangent to $\alpha'$ at $Y$. Hence $\alpha'=\alpha$; it has foci $F,F'$, and $FY+F'Y=2r_\omega$ shows that its auxiliary circle is $\omega$. The case in which $F$ lies outside $\omega$ is analogous. $\square$

[10] Here and below, write $\mathfrak{f}_\gamma$ for reflection in an element $\gamma$: reflection in a point means central reflection, and reflection in a line means axial reflection.

We record one more property of the auxiliary circle.

**Proposition 1.3.7.** *Let $F_1, F_2$ be the foci and $O$ the center of a central conic $\alpha$. If $A, B$ are the respective projections of $F_1, F_2$ on a tangent to $\alpha$, then*

$$\overline{F_1A} \cdot \overline{F_2B} = a^2 - c^2,$$

*where $a$ is the semi-principal axis and $c$ is the semifocal distance.*

**Proof.** We prove only the elliptic case; the hyperbolic case is analogous. In Figure 1.6b, let $D$ be the projection of $O$ on $AB$, and let $L, N$ be the major-axis vertices. By theorem 1.3.5, $OA = OB = a$. By symmetry, the intersection $C = OA \cap BF_2$ also satisfies $OC = a$, so $A, B, C, L, N$ all lie on the auxiliary circle. Consequently,

$$AF_1 \cdot BF_2 = CF_2 \cdot BF_2 = LF_2 \cdot NF_2 = (a-c)(a+c) = a^2 - c^2. \qquad \square$$

---

## Exercises

**Exercise 1.8.**

(1) In theorem 1.3.1, prove the implication from tangency to the angle-bisector condition.
(2) Prove claim 1.3.2.
(3) Prove theorems 1.3.4 and 1.3.5 and proposition 1.3.7 in the hyperbolic case.

**Exercise 1.9.** Two ellipses have a common focus. Prove that the ellipses are tangent if and only if their major auxiliary circles are tangent.

**Exercise 1.10.** Let $A, B$ be the major-axis vertices of an ellipse $\alpha$. The tangent to $\alpha$ at a variable point $P$ meets the major auxiliary circle at $M, N$, and $AM$ and $BN$ meet outside the ellipse. Prove that $\tan \measuredangle BAM \cdot \tan \measuredangle ABN = \text{const}$.

**Exercise 1.11.** Let $A, B$ be points of an ellipse $\alpha$, and suppose that the tangent at $A$ is parallel to $OB$, where $O$ is the center. The circle centered at $A$ with radius equal to the semimajor axis meets $OB$ at $T, T'$. Prove that $AT$ and $AT'$ each pass through one of the two foci.

**Exercise 1.12.** Fix an angle $\theta$ and a central conic $\Gamma$. Let $F$ be one of its foci and $P$ a variable point on $\Gamma$. If $l$ is the tangent at $P$, choose $Q \in l$ so that $\measuredangle(FQ, l) = \theta$. Prove that the locus of $Q$ is a circle.

**Exercise 1.13*.** In the Rutherford scattering experiment, consider the classical, nonrelativistic case in which all particles arrive parallel to one another with the same initial speed. Prove that all their scattering trajectories are tangent to a single parabola.

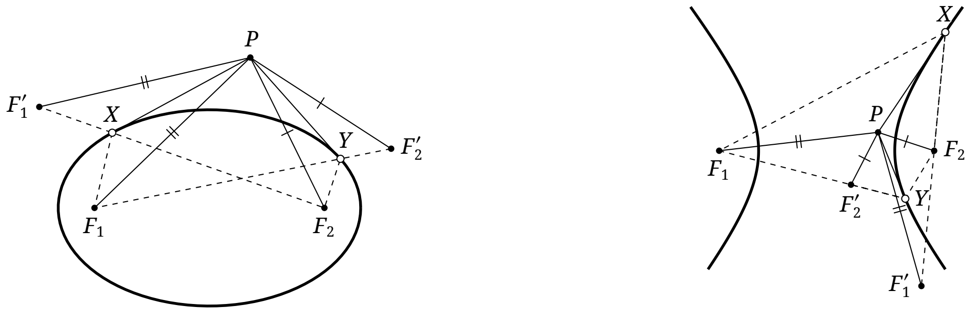
### 1.3.2 The Isogonal Property of Conics

In this subsection we give several further important theorems for central conics that follow from the optical property. Parabolas have analogous but somewhat special properties; these are treated together in section 1.5.

**Theorem 1.3.8** (Second little Poncelet theorem)**.** *From a point $P$ outside an ellipse or hyperbola $\alpha$, draw tangents $PX, PY$ with points of contact $X, Y$. Then $\measuredangle F_1PX + \measuredangle F_2PY = 0$. More explicitly:*

*(1) if $\alpha$ is an ellipse, then $\angle F_1PX = \angle F_2PY$, as in Figure 1.8a;*

(2) *if $\alpha$ is a hyperbola and $X,Y$ lie on the same branch, then $\angle F_1PX + \angle F_2PY = 180^\circ$, as in Figure 1.8b; if $X,Y$ lie on different branches, then $\angle F_1PX = \angle F_2PY$.*




(a) (b)

FIGURE 1.8

**Proof.** For an ellipse, as in Figure 1.8a, reflect $F_1$ in $PX$ to $F_1'$ and $F_2$ in $PY$ to $F_2'$. By the optical property, $F_1',X,F_2$ are collinear and $F_2',Y,F_1$ are collinear. It follows readily that $\triangle F_1'PF_2 \cong \triangle F_1PF_2'$. Hence $\angle F_1'PF_2 = \angle F_1PF_2'$, and therefore $\angle F_1'PF_1 = \angle F_2PF_2'$. Thus $\angle F_1PX = \angle F_2PY$.

The proof for a hyperbola is analogous. The auxiliary lines for one case are shown in Figure 1.8b; the details are left as an exercise. □

The same pair of congruent triangles in the preceding proof also gives the following two theorems.

**Theorem 1.3.9.** *From a point $P$ outside an ellipse or hyperbola $\alpha$, draw tangents $PX,PY$ with points of contact $X,Y$. Then $\measuredangle PF_1X + \measuredangle PF_1Y = 0$. More explicitly:*

(1) *if $\alpha$ is an ellipse, then $\angle PF_1X = \angle PF_1Y$;*
(2) *if $\alpha$ is a hyperbola, then $\angle PF_1X = \angle PF_1Y$ when $X,Y$ lie on the same branch, whereas $\angle PF_1X + \angle PF_1Y = 180^\circ$ when they lie on different branches.*

The next two theorems are applications of theorem 1.3.8.

**Theorem 1.3.10** (Monge circle theorem)**.** *Let $a$ denote the semimajor axis of an ellipse, or the semitransverse axis of a hyperbola. Let $O$ be the center of the conic and $c$ its semifocal distance.*

(1) *From a point $P$ outside an ellipse $\alpha$, draw the two tangents $PX,PY$, with points of contact $X,Y$. The locus of points $P$ for which $XP \perp YP$ is a circle centered at the center $O$, with squared radius $r^2 = 2a^2 - c^2$.*
(2) *From a point $P$ outside a hyperbola $\alpha$, draw the two tangents $PX,PY$, with points of contact $X,Y$. If the eccentricity of the hyperbola is less than $\sqrt{2}$, the locus of points $P$ for which $XP \perp YP$ is a circle centered at $O$, again with $r^2 = 2a^2 - c^2$. If the eccentricity is at least $\sqrt{2}$, no such real point $P$ exists.*[11]

The circle in either case is called the *Monge circle* or *director circle* of the ellipse or hyperbola.

**Proof.** For an ellipse, reflect $F_1$ in $PX$ to $F_1'$, as in Figure 1.9. Then

$$\angle F_1'PF_2 = \angle F_1'PX + \angle XPF_2 = \angle XPF_1 + \angle XPF_2 = \angle XPY = 90^\circ,$$

where the penultimate equality follows from theorem 1.3.8. Hence

$$F_1P^2 + F_2P^2 = PF_1'^2 + PF_2^2 = F_1'F_2^2 = (2a)^2 = \text{const}.$$

[11]Only the ordinary real Euclidean plane is being considered here. If the complex Euclidean plane is allowed, such points always exist.

The median-length formula now gives

$$PO^2 = \frac{PF_1^2 + PF_2^2}{2} - \frac{F_1F_2^2}{4} = \text{const} = 2a^2 - c^2.$$

This proves the assertion for an ellipse. The hyperbolic case is analogous. □

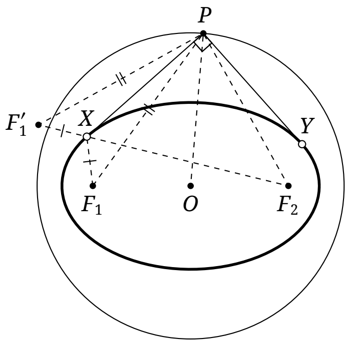



FIGURE 1.9

**Remark.** The proof inferred a circular locus from $PF_1^2 + PF_2^2 = \text{const}$. More generally, one may prove the following.[12]

Given points $P_1, \ldots, P_n$ and constants $k_1, \ldots, k_n$ with $k_1 + \cdots + k_n \neq 0$, the locus of points $X$ satisfying

$$k_1XP_1^2 + \cdots + k_nXP_n^2 = \text{const}$$

is a circle—real, imaginary, or a point circle. It is called a *Fermat–Apollonius circle*.

The following result is due to Graves.

**Theorem 1.3.11** (Graves's theorem). *As shown in Figure 1.10a, draw the two tangents from a point $X$ outside an ellipse $\alpha$, with points of contact $M,N$. If $\widehat{MN} + MX + XN = \text{const}$, where $\widehat{MN}$ denotes the length of the longer arc of $\alpha$ between $M$ and $N$, then the locus of $X$ is an ellipse having the same two foci as $\alpha$.*[13]

**Proof.** The curve $\beta$ is smooth. It is enough to prove that at every point $X \in \beta$, the tangent is an exterior angle bisector of $\angle F_1XF_2$. By theorem 1.3.8, $\angle F_1XM = \angle F_2XN$; hence the exterior angle bisectors of $\angle MXN$ and $\angle F_1XF_2$ coincide. Denote this common line by $l$. Take any $Y \in l$, and draw the tangents from $Y$ to $\alpha$, with points of contact $L,R$. We may suppose that $Y$ lies to the left of $X$, and put $P = YL \cap XM$.

We have $YN < \widehat{RN} + YR$ and $\widehat{LM} < LP + PM$.[14] Since $l$ is an exterior angle bisector of $\angle NXP$, we also have $PX + XN < PY + YN$. Therefore

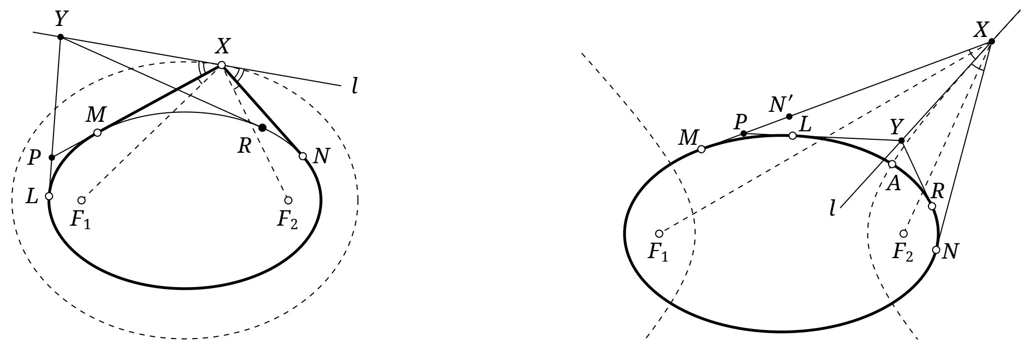
$$\begin{aligned} MX + XN + \widehat{NLM} &< MX + XN + \widehat{NL} + LP + PM = PX + XN + \widehat{NL} + LP \\ &< PY + YN + \widehat{NL} + LP = LY + YN + \widehat{NL} \\ &< LY + YR + \widehat{RN} + \widehat{NL} = LY + YR + \widehat{RNL}. \end{aligned}$$

Thus $Y$ lies outside $\beta$. The case in which $Y$ lies to the right of $X$ is analogous. It follows that $l$ has only one point in common with $\beta$, and hence is tangent to it. This proves the theorem. □

[12]Why is this true?

[13]In less formal language, place an inextensible loop of string around the ellipse $\alpha$, insert a pencil into the loop, and pull the string taut. As the pencil makes one circuit of $\alpha$ under this constraint, it traces an ellipse $\beta$ having the same two foci as $\alpha$.

[14]The second inequality is intuitively clear. A fully rigorous proof lies outside the scope of this course; for readers familiar with the Cauchy–Crofton theorem, it is an immediate consequence.




(a) (b)

FIGURE 1.10

There is an analogous theorem for a hyperbola, although the relative positions of the points change substantially. We include its proof.

**Theorem 1.3.12** (MacCullagh's theorem)**.** *Let an ellipse $\alpha$ and a hyperbola $\beta$ have the same two foci, and choose an intersection point $A$ of the two conics. Let $X$ vary along the part of $\beta$ that starts at $A$ and lies outside $\alpha$.*[15] *Draw the two tangents from $X$ to $\alpha$, with points of contact $M,N$. Then $XM-\overset{\frown}{AM}=XN-\overset{\frown}{AN}$.*

**Proof.** Conversely, omit the condition $X\in\beta$. Draw the tangents $XM,XN$ as above, and let $\beta'$ be the locus, on the side of $A$ exterior to $\alpha$, defined by $XM-\overset{\frown}{AM}=XN-\overset{\frown}{AN}$. As before, $\beta'$ is smooth. It is therefore enough to show that at each $X\in\beta'$, the tangent to $\beta'$ is an angle bisector of $\angle F_1XF_2$. By the second little Poncelet theorem, $\angle F_1XM=\angle F_2XN$; hence $\angle MXN$ and $\angle F_1XF_2$ have the same exterior angle bisector. Denote it by $l$. Take any $Y\in l$, draw the tangents from $Y$ to $\alpha$ with points of contact $L,R$, put $\delta(Y):=YL-YR-\overset{\frown}{LA}+\overset{\frown}{RA}$, and let $P=YL\cap XM$.

Suppose first that $Y$ lies inside $\angle MXN$. The points $M,L,A,R,N$ then occur in this order on $\alpha$, and $YN<\overset{\frown}{RN}+YR$, $\overset{\frown}{LM}<LP+PM$. Let $N'$ be the reflection of $N$ in $l$. Then $N'\in PX$, and hence $PX-NX=PN'>PY-YN'=PY-YN$. Consequently,

$$\begin{aligned}\delta(X)&=MX-XN-\overset{\frown}{MA}+\overset{\frown}{NA}=PX-XN+PM-\overset{\frown}{ML}-\overset{\frown}{LA}+\overset{\frown}{NR}+\overset{\frown}{RA}\\&>PX-XN-PL-\overset{\frown}{LA}+\overset{\frown}{NR}+\overset{\frown}{RA}>PY-YN-PL-\overset{\frown}{LA}+\overset{\frown}{NR}+\overset{\frown}{RA}\\&=YL-YN+\overset{\frown}{NR}-\overset{\frown}{LA}+\overset{\frown}{RA}>YL-YR-\overset{\frown}{LA}+\overset{\frown}{RA}=\delta(Y).\end{aligned}$$

The same inequality $\delta(Y)<\delta(X)$ follows when $Y$ lies on the other side of $X$ along $l$.

The locus $\beta'$ is not closed, so the fact that $l$ has only one point in common with it does not by itself prove tangency. If $l$ were not tangent, however, the order of $\delta(Y)$ and $\delta(X)$ would have to reverse as $Y$ crossed from one side of $\beta'$ to the other.[16] Since $\delta(Y)<\delta(X)$ for every $Y\neq X$ on $l$, all points of $l$ lie on the same side of $\beta'$. Thus $l$ is tangent to $\beta'$, and the theorem follows. □

Analogous results can be stated when tangents are drawn to a hyperbola; their proofs are left as an exercise.

**Theorem 1.3.13.** *Let two hyperbolas $\alpha,\beta$ have the same two foci, with $\alpha$ lying inside $\beta$ (so that tangents to $\alpha$ can be drawn from points of $\beta$). For a variable point $X\in\beta$, draw the two tangents from $X$ to $\alpha$, with points of contact $M,N$. Then:*

*(1) if $M,N$ lie on the same branch of $\alpha$, then $XM+XN-\overset{\frown}{MN}$ is constant;*

[15] More precisely, $X$ and $A$ lie on the same branch of $\beta$, and the hyperbolic arc joining $A$ to $X$ does not meet $\alpha$.

[16] This is the intuition of synthetic geometry; in the synthetic part of the book we do not give a modern rigorous treatment.

(2) *if $M,N$ lie on different branches of $\alpha$, assume that $M$ is closer to $X$, and let $N'$ be the reflection of $N$ in the conjugate axis. Then $N',M$ lie on the same branch of $\alpha$, and $XN - XM - \widehat{N'M}$ is constant.*

**Theorem 1.3.14.** *Let a hyperbola $\alpha$ and an ellipse $\beta$ have the same two foci. For a variable point $X \in \beta$ lying outside $\alpha$, draw the two tangents from $X$ to $\alpha$, with points of contact $M,N$. Then:*

(1) *if $M,N$ lie on the same branch of $\alpha$, let $A$ be the intersection of $\alpha$ and $\beta$ on the arc $\widehat{MN}$. Then $XM - \widehat{AM} = XN - \widehat{AN}$;*
(2) *if $M,N$ lie on different branches of $\alpha$, let $N'$ be the reflection of $N$ in their common center. Then $M,N'$ lie on the same branch, and $XM + XN - \widehat{MN'}$ is constant.*

Conics can also solve problems that at first seem unrelated to them. The next result is an example.

**Theorem 1.3.15** (Urquhart's theorem)**.** *Let $ABCD$ be a convex quadrilateral, with $E = AB \cap CD$ and $F = AD \cap CB$. If $AB + BC = AD + DC$, then $AE + EC = AF + FC$.*

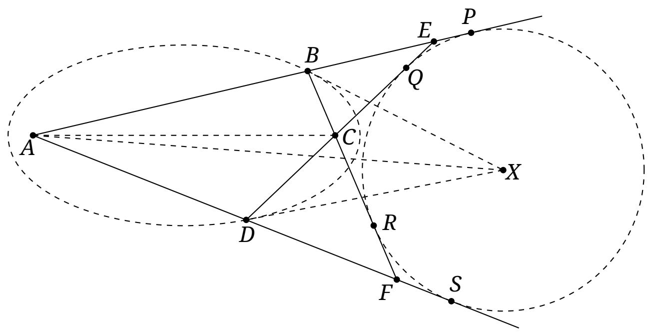



FIGURE 1.11

**Proof.** The hypothesis ensures that there is an ellipse $\alpha$ with foci $A,C$ passing through both $B,D$, as in Figure 1.11. Let the exterior angle bisectors of $\angle ABC$ and $\angle ADC$ meet at $X$. The optical property shows that $BX$ and $DX$ are tangents to $\alpha$, and theorem 1.3.9 then shows that $AX$ bisects $\angle EAF$.

Since $BX, DX, AX$ bisect $\angle CBE, \angle CDF, \angle BAD$, respectively, $X$ is equidistant from the four lines $AB, BC, CD, DA$. Thus $ABCD$ has an excircle centered at $X$. If its points of contact are $P,Q,R,S$, the tangent-length theorem gives

$$AE + EC = AE + EP - EQ + EC = AP + QC = AS + RC = AS - FS + FR + CR = AF + CF. \qquad \square$$

The interested reader may also try to find a conventional plane-geometric proof.

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## Exercises

**Exercise 1.14.**

(1) Prove theorem 1.3.9.
(2) Prove the hyperbolic cases of theorems 1.3.8 and 1.3.10.
(3) Prove theorems 1.3.13 and 1.3.14.

**Exercise 1.15.** A $2n$-gon is circumscribed about a conic with focus $F$. Color its sides alternately black and white. Prove that the sum of the angles subtended at $F$ by all the black sides is $180°$.

**Exercise 1.16.** An ellipse is inscribed in a convex quadrilateral, and its two foci lie on the two diagonals of the quadrilateral, respectively. Prove that the products of the lengths of opposite sides are equal.

**Exercise 1.17.** Inside a smooth elliptical mirror is a point source that sends a narrow beam to the mirror along a line through neither focus. The beam then undergoes infinitely many reflections. Prove that all its subsequent reflected rays are tangent to one fixed central conic.

**Exercise 1.18.** Two triangles are both circumscribed about an ellipse $\alpha_1$ and inscribed in an ellipse $\alpha_2$. If $\alpha_1$ and $\alpha_2$ have the same two foci, prove that the two triangles have equal perimeters.

**Exercise 1.19.** In $\triangle ABC$, let $D$ be the midpoint of the arc $\widehat{BC}$ of the circumcircle that does not contain $A$. Choose $E$ on the segment $BD$ and $F$ on the segment $CD$ so that $2\angle EAF = \angle BAC$, and let $AG \perp EF$ at $G$. Prove that $G$ lies on a fixed circle of radius $(AB+AC)/2$.

## 1.4 A Three-Dimensional View of Conics

In this section we use solid geometry to investigate conics. We learned in section 1.1 that circles, ellipses, hyperbolas, and parabolas are all called conics. What, then, do these four kinds of curve have to do with a cone? The answer will also explain the origin of the name "conic section."

### Central Projection

Consider the following problem.

*Given a circle $\omega$, draw through its center the line perpendicular to its plane and choose a point $S$ on that line. All lines joining $S$ to points of $\omega$ form a* cone*; each such line is called a* generator*, or ruling, of the cone. If a plane $\pi$ cuts the cone, what is the shape of the section $\alpha$?*[17]

a.) If $\pi$ is parallel to the plane of $\omega$, then $\alpha$ is plainly a circle.

b.) Suppose $\pi$ meets every generator of the cone. As shown in Figure 1.12a, inscribe two spheres in the cone, tangent to the cutting plane $\pi$ at $F_1, F_2$. For any $X \in \alpha$, let $Y_1, Y_2$ be the points where $SX$ touches the two spheres. Then $Y_1X = F_1X$ and $Y_2X = F_2X$, whence
$$F_1X + F_2X = Y_1X + Y_2X = Y_1Y_2 = \text{const}.$$
Thus $\alpha$ is an ellipse.

c.) Suppose exactly two generators are parallel to $\pi$. The auxiliary lines are drawn in Figure 1.12b; the same argument shows that $\alpha$ is a hyperbola. The details are left to the reader.

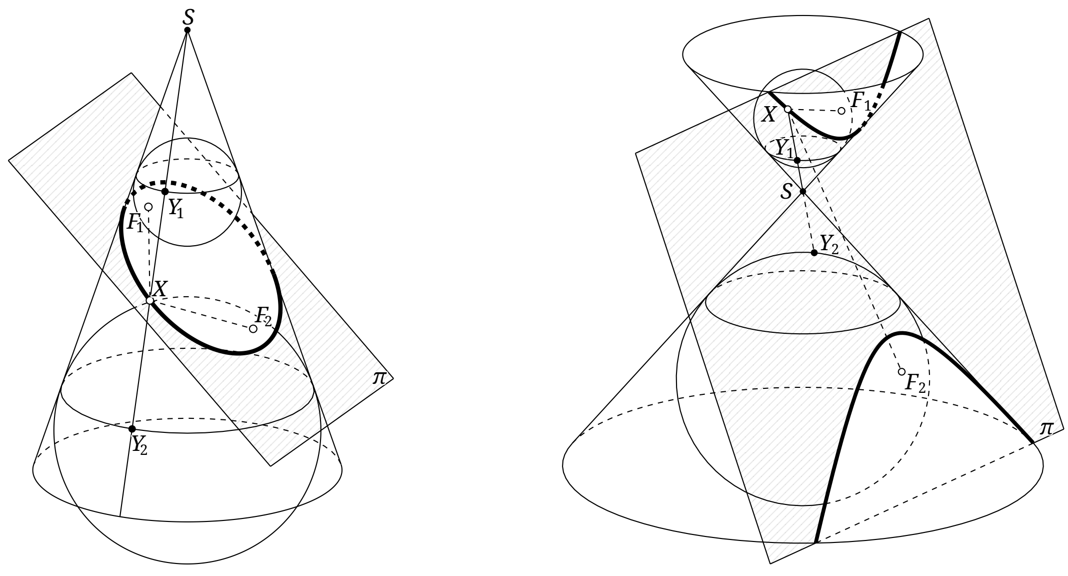
In parts (b) and (c), the ellipse and hyperbola were obtained by means of two inscribed spheres. This construction was introduced by Dandelin (Germinal Pierre Dandelin), and the two spheres are called the *Dandelin spheres*.

d.) Suppose exactly one generator is parallel to $\pi$. We show that the section is a parabola.

As in Figure 1.13, inscribe a sphere in the cone tangent to $\pi$ at $F$. The sphere touches the cone along a circle; denote the plane of this circle by $\sigma$, and put $l = \sigma \cap \pi$. For any $X \in \alpha$, let $Y$ be the point where $SX$ touches the sphere. Then $XY = XF$. Let $Z$ be the projection of $X$ on $l$. Since $\pi$ is parallel to a generator of the cone, the angle between $XZ$ and $\sigma$ equals the angle between a generator and $\sigma$. In particular, $XZ$ and $XY$ make equal angles with $\sigma$, so $XZ = XY$. Hence $XF = XZ$, proving that $\alpha$ is a parabola.

This discussion explains the term "conic section." It also shows that if a point source is placed at the vertex $S$, it projects the base circle of the cone onto $\pi$ as a general conic. We therefore say that a conic is a *central projection* of a circle.

[17]The section is the set of all common points of $\pi$ and the cone $S-\omega$.




(a) (b)

FIGURE 1.12

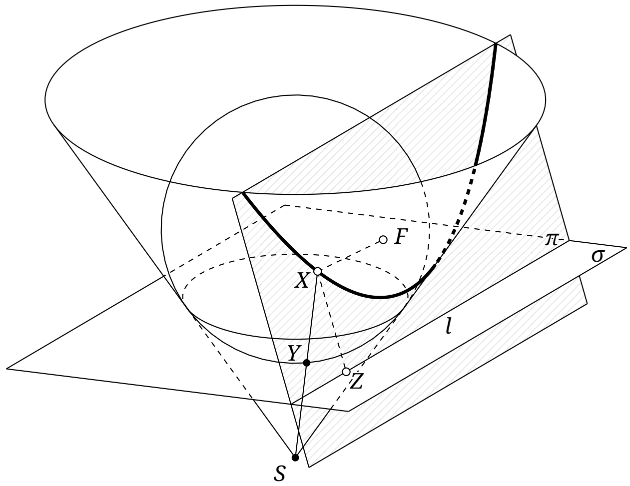



FIGURE 1.13

**Parallel Projection**

Replace the cone by a *cylinder*, as in Figure 1.14a. Two spheres can again be inscribed, proving in the same way that an oblique plane section of the cylinder is an ellipse. A source situated at infinity directly above the cylinder projects its base circle onto $\pi$ as that ellipse. Thus an ellipse may also be regarded as a *parallel projection* of a circle.

Parallel projection gives several properties of an ellipse.

**Proposition 1.4.1.** *The locus of the midpoints of a family of parallel chords of an ellipse is a line through its center.* This line is called a *diameter* of the ellipse.[18]

[18] As for a circle, a segment whose endpoints both lie on a conic is called a *chord* of the conic. Thus when $AB$ is called a chord of a conic $\alpha$, it is understood that $A, B \in \alpha$.

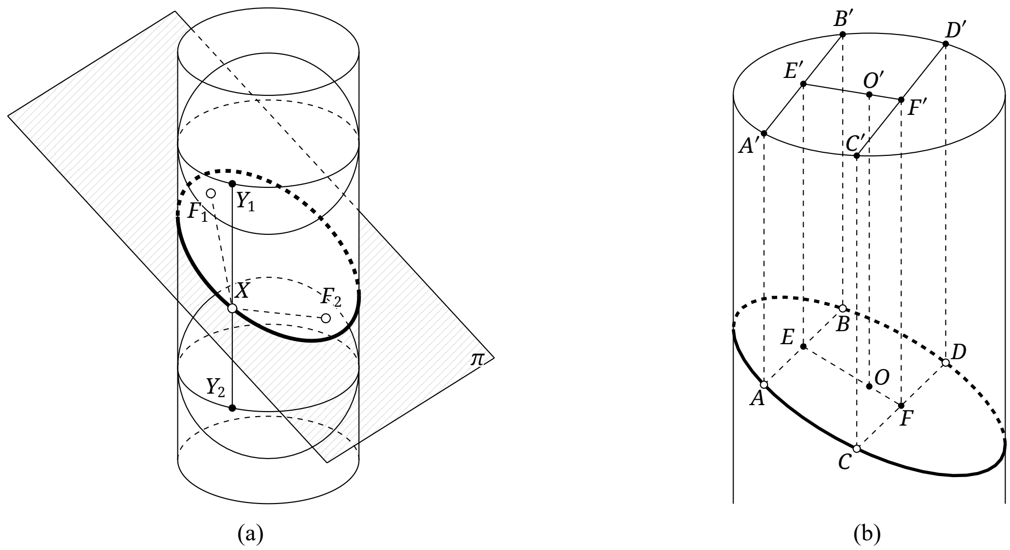



FIGURE 1.14

**Proof.** Let $AB$ and $CD$ be parallel chords of an ellipse $\alpha$, let $O$ be the center of $\alpha$, and let $E,F$ be the midpoints of $AB,CD$. It is enough to prove that $E,F,O$ are collinear.

Regard $\alpha$ as the parallel projection of $\odot O'$, as shown in Figure 1.14b: take a cylinder whose base is $\odot O'$ and whose intersection with a suitable plane is $\alpha$. Through $A$, $B$, $C$, $D$, $E$, $F$, and $O$ draw lines parallel to the generators of the cylinder,[19] meeting the plane of $\odot O'$ at $A'$, $B'$, $C'$, $D'$, $E'$, $F'$, and $O'$, respectively. These primed points project to their corresponding unprimed points.

The planes $A'ABB'$ and $C'CDD'$ are parallel and are both perpendicular to the plane of $\odot O'$. Their traces $A'B'$ and $C'D'$ on that plane are therefore parallel. Call this observation (A).

Since $AA' \parallel EE' \parallel BB'$, $\frac{A'E'}{B'E'} = \frac{AE}{BE} = 1$. Thus $E'$ is the midpoint of $A'B'$; similarly, $F'$ is the midpoint of $C'D'$. Call this observation (B).

Because $A'B'$ and $C'D'$ are chords of $\odot O'$, the perpendicular bisector theorem gives $O'E' \perp A'B'$ and $O'F' \perp C'D'$. Observation (A) now implies that $E',O',F'$ are collinear. After parallel projection to the ellipse, $E,O,F$ are therefore collinear. □

The steps in the preceding proof yield the following general facts.

**Proposition 1.4.2.** *Parallel projection has the following properties.*
*(1) Collinear points remain collinear, and concurrent lines remain concurrent.*
*(2) The ratio of the lengths of collinear segments is preserved.*
*(3) Parallel lines remain parallel.*

**Proof.** Part (1) is immediate. Part (2) is the general form of observation (B) in the proof of proposition 1.4.1, and part (3) is the general form of observation (A) in the proof of proposition 1.4.1. □

### The Second Definition

The construction with Dandelin spheres gives further useful conclusions. Suppose that a plane $\pi$ cuts a cone with vertex $S$ in a conic, as in Figure 1.15.[20] We continue to investigate this section.

An inscribed sphere is tangent to $\pi$ at $F$. As in the construction for the parabola above, let $\sigma$ be the plane of the circle along which the sphere touches the cone, put $l = \sigma \cap \pi$, let $Y$ be the point where $SX$ touches the sphere, and let $Z$ be the projection of $X$ on $l$.

[19] A generator of a cylinder is a line on its surface parallel to its axis.

[20] The figure shows an ellipse, but the following argument applies to every conic.

Draw $XT \perp \sigma$ at $T$. Then $XF = XY$. Moreover, $\alpha := \angle YXT$ and $\beta := \angle ZXT$ are constant, and hence

$$\frac{XY}{XZ} = \frac{XY}{XT} \cdot \frac{XT}{XZ} = \frac{\cos\beta}{\cos\alpha} = \text{const.}$$

Thus the ratio of the distance from $X$ to $F$ to the distance from $X$ to $l$ is constant.

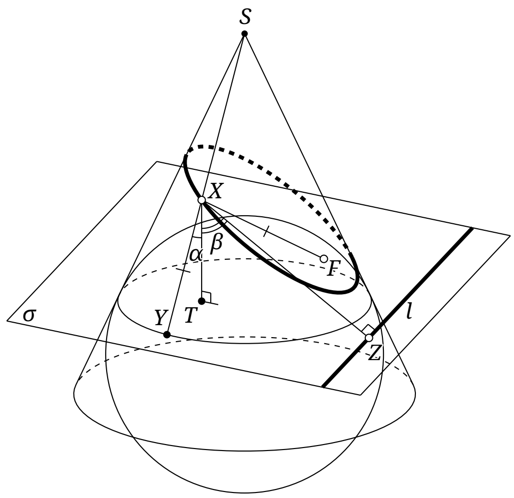



FIGURE 1.15

This leads to another definition of a conic, called the *second definition*, or the *unified definition*. The definition in section 1.1 is then called the first definition.

**Definition 1.4.3** (Second, or unified, definition of a conic)**.**

(1) Given a line $l$ and a point $F$, the locus of points $P$ for which the ratio of the distance from $P$ to $F$ to the distance from $P$ to $l$ is a constant $e$ is a conic. The point $F$ is a *focus*, the line $l$ a *directrix*, and the constant $e$ the *eccentricity* of the conic.
(2) A conic with $0 < e < 1$ is an ellipse; one with $e = 1$ is a parabola; and one with $e > 1$ is a hyperbola.
(3) The distance from a focus of a conic to its corresponding directrix is the *focal parameter* of the conic.

**Remark.** The eccentricity of a circle is defined to be $e = 0$. A circle may be regarded as the limiting case in which the two foci of an ellipse coalesce at its center and the directrices recede to infinity. The distance from a point of the curve to a directrix then tends to infinity, which explains the definition.

For a parabola, the construction in Figure 1.15 shows that the second definition is identical with the first. For an ellipse or hyperbola, $F$ in the second definition is one of the two foci in the first, and either focus may be used.[21] Thus an ellipse or hyperbola has two focus–directrix pairs.

We finish by interpreting the asymptotes of a hyperbola in three dimensions. Suppose a plane $\pi$ cuts a cone with vertex $S$ in a hyperbola $\alpha$. We shall prove that the two generators parallel to $\pi$ are parallel to the two asymptotes of $\alpha$.

As in Figure 1.16, let $A, B$ be the vertices of $\alpha$, $O$ its center, and $F$ the focus corresponding to $A$. Let the corresponding directrix $l$ meet $AB$ at $X$. For $P \in \alpha$, let $PM$ be perpendicular to the plane $SAB$ at $M$. Let $SC, SD$ be the two generators of the cone in the plane $SAB$, and suppose $C, M, D, P$ are coplanar in a plane perpendicular to $SAB$.

As before, inscribe a sphere in the cone tangent to the plane of $\alpha$ at $F$. Let the circle of contact of the sphere and cone meet $CD$ at $K$. The preceding construction shows that $KX$ is parallel to the plane $PCMD$, and hence $KX \parallel CD$.

[21] In the preceding construction the sphere was placed below $\pi$. Placing the other Dandelin sphere above $\pi$ gives the other focus and its corresponding directrix.

Draw $SN \parallel OM$ to meet $CD$ at $N$, and draw $NQ \parallel CD$ to meet the cone at $Q$. Since $\triangle SND \sim \triangle AXK$,

$$\csc \angle NSQ = \frac{SQ}{SN} = \frac{SD}{SN} = \frac{KA}{AX} = \frac{FA}{AX} = e.$$

This is precisely the cosecant of the angle between an asymptote of the hyperbola and its transverse axis.[22] Since the plane of $\alpha$ is parallel to the plane $SNQ$ and $SN \parallel OM$, one asymptote is parallel to $SQ$. The other relation follows in the same way.

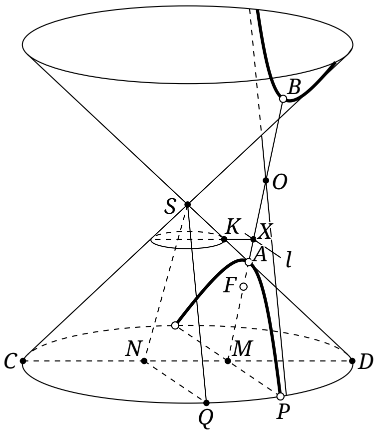



FIGURE 1.16

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## Exercises

**Exercise 1.20.**
(1) Prove that the section $\alpha$ in case (c) of the central-projection discussion is a hyperbola.
(2) Prove that a parallel projection maps a circle to an ellipse.

**Exercise 1.21.** Prove the following properties of parallel projection.
(1) Ratios of lengths of parallel segments are preserved.
(2) Ratios of areas of corresponding triangles are preserved.

**Exercise 1.22.** Draw the two tangents from a point $P$ outside an ellipse $\alpha$, with points of contact $X, Y$. If $O$ is the center of $\alpha$, prove that $OP$ bisects the segment $XY$.

**Exercise 1.23*.** In central and parallel projection, a plane cuts a cone or cylinder in a conic. What property do the two solids being cut have in common? More generally, under what conditions on a solid with axial symmetry is every plane section a conic?

**Exercise 1.24.** Given an ellipse $\alpha$, construct its two foci with straightedge and compass.

**Exercise 1.25.** Let $AB$ be a chord through a focus $F$ of a conic $\alpha$. Prove that

$$\left| \frac{1}{\overline{FA}} - \frac{1}{\overline{FB}} \right| = \frac{2}{ep},$$

where $e$ and $p$ are respectively the eccentricity and focal parameter of $\alpha$.

---

[22]This is familiar from the standard equation (1.3): its asymptotes are $y = \pm \frac{b}{a}x$, and its semifocal distance is $c = \sqrt{a^2 + b^2}$. A purely geometric proof is also possible, but the required details will be established rigorously in chapter 2.

**Exercise 1.26.** Let $A,B$ be fixed points. A variable point $P$ satisfies $\measuredangle PBA + 2\measuredangle PAB = 0$. Prove that $P$ moves on a hyperbola of eccentricity 2.

**Exercise 1.27.** Consider all equilateral hyperbolas having a fixed point $F$ as one focus and passing through a fixed point $P$. Prove that their asymptotes are tangent to two fixed circles. A hyperbola has two asymptotes, so this produces two families of asymptotes, one corresponding to each circle.

**Exercise 1.28.** Given a circle and a line, prove that the locus of points for which the ratio of the tangent length to the circle to the distance from the line is $e$ is a conic, and determine its eccentricity.

**Exercise 1.29*.** In $\triangle ABC$, let $I$ be the incenter and $J$ the $A$-excenter. The $A$-excircle $\mathcal{I}_a$ touches $BC$ and $AB$ at $D,E$, respectively, and $M$ is the midpoint of $BC$. Choose $G$ on the ray $DJ$ so that $AG = AE$, and put $H = MJ \cap AG$. Prove that $HD = HG$. *Hint: construct an ellipse and use exercise 1.22.*

## 1.5 Basic Properties of the Parabola

We conclude the chapter by studying the parabola, the most special of the conics. Throughout this section, $F$ denotes the focus and $l$ the directrix.

**Lemma 1.5.1.** *The reflection of the focus $F$ of a parabola in any tangent $l'$ lies on the directrix. It is precisely the projection on the directrix of the point at which that tangent touches the parabola.*

**Proof.** Let $P'$ be the projection on the directrix of a point $P$ on the parabola, as in Figure 1.17a. The optical property says that the tangent $l'$ at $P$ bisects $\angle FPP'$, while the definition of the parabola gives $PF = PP'$. Hence $F$ and $P'$ are reflections of one another in $l'$. □


(a) (b)

FIGURE 1.17

**Claim 1.5.2.** *The projections of the focus of a parabola on its tangents all lie on a fixed line. This line is tangent to the parabola at its vertex, as shown in Figure 1.17b.*

**Proof.** In the proof of lemma 1.5.1, $F$ and $P'$ are reflections in $l'$, so the projection $T$ of the focus on $l'$ is the midpoint of $FP'$. As $P$ varies on the parabola, the locus of $T$ is therefore $\mathfrak{h}_{F,1/2}(l)$. □

The property in claim 1.5.2 is entirely analogous to the auxiliary-circle property of an ellipse or hyperbola. To state properties of conics uniformly, we therefore also call the tangent to a parabola at its vertex its "*auxiliary circle*," although it is a line—and hence only a generalized circle. Indeed, this assertion is the limit of the corresponding one for an ellipse as its center tends to infinity: the center of the auxiliary circle also tends to infinity, and the circle becomes a line.

**Lemma 1.5.3.** *The tangent at a point $P$ of a parabola meets its axis at $T$. Then $TF = FP$, as shown in Figure 1.18a.*

**Proof.** Let $P'$ be the projection of $P$ on $l$. Since $FT \parallel PP'$, $\angle FTP = \angle TPP' = \angle FPT$, and hence $TF = FP$. □

**Lemma 1.5.4.** *Let $X', Y'$ be the respective projections on the directrix of points $X, Y$ of a parabola. If the tangents at $X, Y$ meet at $P$, then $P$ is the circumcenter of $\triangle X'Y'F$, as shown in Figure 1.18b.*

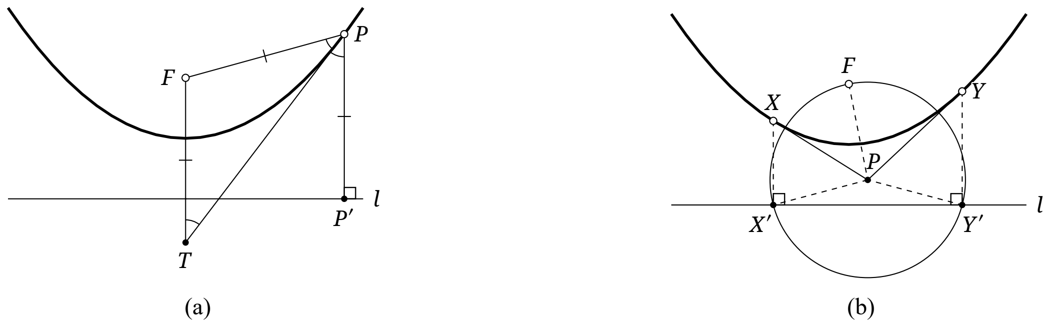


FIGURE 1.18

**Proof.** By lemma 1.5.1, $PY$ and $XP$ are the perpendicular bisectors of $FY'$ and $FX'$, respectively. Thus $P$ is the circumcenter of $\triangle X'FY'$. □

**Claim 1.5.5.** *From a point $P$, draw two tangents to a parabola, with points of contact $X, Y$. Let $X', Y'$ be the respective projections of $X, Y$ on the directrix, and let $P'$ be the projection of $P$ on the directrix. Then $P'$ bisects $X'Y'$, as shown in Figure 1.19a.*

**Proof.** By lemma 1.5.4, $P$ lies on the perpendicular bisector of $X'Y'$, which proves the claim. □

**Claim 1.5.6.** *From a point $P$, draw two tangents to a parabola, with points of contact $X, Y$. If $V$ is the midpoint of $XY$, then $PV$ is parallel to the axis of the parabola.*


FIGURE 1.19

**Theorem 1.5.7.** *From a point $P$ outside a parabola, draw two tangents with points of contact $X, Y$. The locus of points $P$ for which $XP \perp YP$ is the directrix of the parabola. Under this condition, $F \in XY$ and $FP \perp XY$, as shown in Figure 1.19b.*

**Proof.** Let $X', Y'$ be the projections of $X, Y$ on the directrix. Since $F$ and $X'$ are reflections in $XP$, we have $\angle X'PX = \angle FPX$. Similarly, $\angle Y'PY = \angle FPY$, so $\angle X'PY' = 2\angle XPF + 2\angle YPF = 180°$. Thus $P$ lies on the directrix $X'Y'$. Moreover, the reflection of $X'$ in $XP$ is $F$, and hence $\angle XFP = \angle XX'P = 90°$. Similarly, $\angle YFP = \angle YY'P = 90°$. Therefore $\angle XFY = \angle XFP + \angle YFP = 180°$, so the chord $XY$ passes through $F$ and $PF \perp XY$. ☐

**Theorem 1.5.8.** *Fix an obtuse angle $\phi$. From a point $P$ outside a parabola $\alpha$, draw the two tangents $PX, PY$, with points of contact $X, Y$. The locus of points $P$ for which $\angle XPY = \phi$ or $180° - \phi$ is a hyperbola having one focus–directrix pair in common with $\alpha$, as shown in Figure 1.20a.*

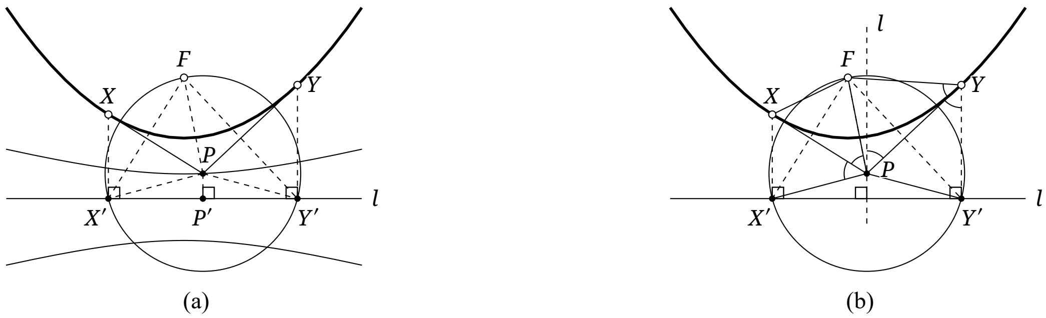


FIGURE 1.20

**Proof.** First suppose $\angle XPY = \phi$. Let $X', Y', P'$ be the projections of $X, Y, P$ on the directrix. Since $FX' \perp XP$ and $FY' \perp YP$, $\angle X'FY' = 180° - \phi$. By lemma 1.5.4, $P$ is the circumcenter of $\triangle X'Y'F$, and hence $\angle X'PP' = \angle X'PY'/2 = \angle X'FY' = \text{const}$. It follows that

$$\frac{PF}{PP'} = \frac{PX'}{PP'} = \sec\angle X'PP' = -\sec\phi = \text{const}.$$

Thus $P$ lies on a hyperbola.

The same argument applies when $\angle XPY = 180° - \phi$. The point $P$ then lies on the other branch of the same hyperbola, the branch farther from the parabola. ☐

The final theorem is the parabolic counterpart of theorems 1.3.8 and 1.3.9.

**Theorem 1.5.9.** *As in Figure 1.20b, draw two tangents from a point $P$ outside a parabola $\alpha$, with points of contact $X, Y$. Let $l$ be the line through $P$ parallel to the axis of the parabola. Then:*

*(1) the angle between $PY$ and $l$ equals $\angle XPF$;*
*(2) $\angle XFP = \angle PFY$;*
*(3) $\triangle XFP \sim \triangle PFY$.*

**Proof.** We have

$$\angle XPF = \angle X'Y'F = \angle(l, PY) = \angle PYY' = \angle PYF.$$

Here the first and last equalities follow from lemmas 1.5.1 and 1.5.4, respectively. The equality $\angle X'Y'F = \angle(l, PY)$ follows from $l \perp X'Y'$ and $FY' \perp PY$. This also proves part (1). Similarly, $\angle FXP = \angle FPY$, so the triangles in part (3) are similar, and part (2) follows. ☐

## Exercises

**Exercise 1.30.** At a point $X$ of a parabola $\alpha$, the normal[23] meets the axis of $\alpha$ at $Y$. Let $Z$ be the projection of $X$ on the axis. Prove that the length $YZ$ is independent of $X$.

**Exercise 1.31.** Prove that the line joining the intersection of the tangents at the endpoints of a chord of a parabola to the midpoint of that chord is parallel to the axis of the parabola.

**Exercise 1.32.** Two moving points travel with constant speeds on two intersecting lines and do not pass through the intersection simultaneously. Prove that the line joining them is always tangent to a fixed parabola.

**Exercise 1.33.** Let $F$ be the focus of a parabola $\alpha$. The two tangents from $P$ touch $\alpha$ at $X, Y$, and the bisector of $\angle XPY$ meets the axis of $\alpha$ at $S$. Prove that $PF = FS$.

**Exercise 1.34.** Let $\alpha$ be a fixed hyperbola of eccentricity 2, and let an equilateral triangle $ABC$ be inscribed in $\alpha$. Prove that $AB$ is always tangent to one of two fixed parabolas.

[23] The normal to a curve at a point is the line through that point perpendicular to the tangent there.

## Chapter 2

# Elementary Properties of Conics

In the preceding chapter we surveyed the principal geometric properties of conics, necessarily leaving aside some details. The present chapter derives these properties rigorously by elementary plane geometry, beginning in every case from the unified definition of a conic.

## 2.1 Elementary Properties of the Hyperbola

We begin with the hyperbola.

### 2.1.1 Basic Properties of the Hyperbola

**Definition 2.1.1.** A *hyperbola* is the locus $\alpha$ of points for which the ratio of the distance from a fixed point $F$ to the distance from a fixed line $l$ is a constant $e > 1$. The point $F$ is called a *focus* of $\alpha$, and the line $l$ its *directrix*.

Draw through the focus $F$ of the hyperbola $\alpha$ the perpendicular to $l$, with foot $X$. The definition shows at once that the hyperbola is symmetric about $FX$. Moreover, two points $A, A'$ of $FX$ lie on the hyperbola. They are its *transverse-axis vertices*, often simply its *vertices*; their midpoint $O$ is the *center* of the hyperbola, and $AA'$ is its *transverse axis*. The points $A, A'$ lie on opposite sides of the directrix $l$, as shown in Figure 2.1. The circle with diameter $AA'$ is the *major auxiliary circle* of the hyperbola, often simply its *auxiliary circle*.

For the remainder of this discussion we retain this notation and take $A'$ to be the vertex on the left, called the *left vertex*, and $A$ the *right vertex*.[1]

**Proposition 2.1.2.** *Let $P, P'$ lie on the hyperbola $\alpha$, and let $PP'$ meet the directrix $l$ corresponding to the focus $F$ at $K$. Then $FK$ bisects either $\angle PFP'$ or its exterior angle. More precisely, it bisects the exterior angle when the segment $PP'$ does not meet $l$, and it bisects $\angle PFP'$ when the segment does meet $l$.*

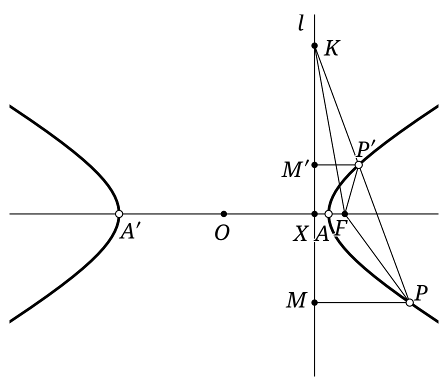


FIGURE 2.1

[1] These names depend only on orientation and are appropriate when the transverse axis is horizontal. When it is vertical, the corresponding names are upper and lower vertex.

**Proof.** It suffices to consider the case in which the segment $PP'$ does not meet $l$; the other case is analogous. As in Figure 2.1, let $M,M'$ be the respective projections of $P,P'$ on $l$. By the definition, $\frac{FP}{FP'} = \frac{PM}{P'M'} = \frac{PK}{P'K}$. The exterior-angle-bisector theorem now shows that $FK$ bisects the exterior angle of $\angle PFP'$. □

**Proposition 2.1.3.** *Let the perpendicular from a point $P \in \alpha$ to the transverse axis have foot $N$. Then* $\frac{PN^2}{AN \cdot A'N} = \text{const.}$

**Proof.** As in Figure 2.2, suppose without loss of generality that $P$ lies on the right branch. Let $PA$ and $PA'$ meet the right directrix $l$ at $K$ and $K'$, respectively, and let $l$ meet the transverse axis at $X$. By proposition 2.1.2, $FK$ bisects the exterior angle of $\angle AFP$, whereas $FK'$ bisects $\angle AFP$; hence $\angle KFK' = 90^\circ$. The right-triangle altitude theorem gives $KX \cdot K'X = FX^2$. Since $PN \parallel l$,

$$\frac{PN^2}{AN \cdot A'N} = \frac{PN}{AN} \cdot \frac{PN}{A'N} = \frac{K'X}{A'X} \cdot \frac{KX}{AX} = \frac{FX^2}{AX \cdot A'X} = \text{const.}$$ □

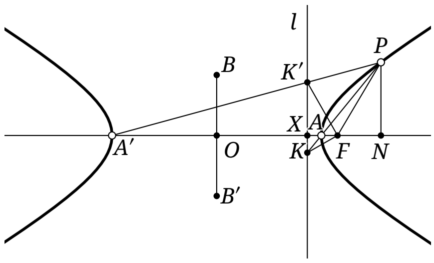


FIGURE 2.2

By proposition 2.1.3, there are two points $B,B'$ on the perpendicular bisector of $AA'$, symmetric about $O$, such that

$$\frac{OB^2}{OA^2} = \frac{OB'^2}{OA^2} = \frac{FX^2}{AX \cdot A'X},$$

the constant appearing in proposition 2.1.3. The points $B,B'$ are the two *conjugate-axis vertices* of the hyperbola. In what follows, take $B$ to be the upper one and call it the *upper vertex*; correspondingly, $B'$ is the *lower vertex*.[2] The line $BB'$ is called the *conjugate axis*.[3] See Figure 2.2. The circle with diameter $BB'$ is the *minor auxiliary circle*.

**Proposition 2.1.4.** *Through $B,B'$ draw the lines perpendicular to the conjugate axis, and through $A,A'$ the lines perpendicular to the transverse axis. These four lines form a rectangle, and its diagonals are the asymptotes of the hyperbola.*

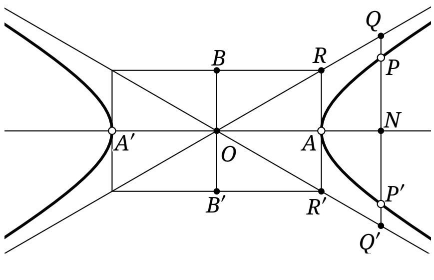


FIGURE 2.3

[2]Again, these names apply when the transverse axis is horizontal. When it is vertical, the corresponding names are left and right vertex.

[3]As noted in section 1.1, the expression "conjugate axis" sometimes also means the segment $BB'$; the context makes the meaning clear.

**Proof.** As in Figure 2.3, draw through $P$ the perpendicular to the transverse axis. Let it meet that axis at $N$, the hyperbola again at $P'$, and the two diagonals in question at $Q, Q'$. Let $R$ be one vertex of the rectangle, with $RB \parallel OA$ and $RA \parallel OB$. By symmetry, $N$ is the midpoint of both $PP'$ and $QQ'$.

By proposition 2.1.3,

$$\frac{PN^2}{ON^2 - OA^2} = \frac{PN^2}{(ON + OA)(ON - OA)} = \frac{PN^2}{AN \cdot A'N} = \frac{OB^2}{OA^2},$$

whereas $\frac{QN^2}{ON^2} = \frac{AR^2}{OA^2} = \frac{OB^2}{OA^2}$. Taking differences in these proportions gives

$$\frac{QN^2 - PN^2}{OA^2} = \frac{QN^2 - PN^2}{ON^2 - (ON^2 - OA^2)} = \frac{OB^2}{OA^2}.$$

Consequently,

$$OB^2 = QN^2 - PN^2 = (QN - PN) \cdot (QN + PN) = PQ \cdot Q'P.$$

Thus $PQ \cdot PQ'$ is constant. As $P$ recedes without bound,[4] $PQ'$ can be made arbitrarily large, and hence $PQ$ arbitrarily small but positive. The distance from $P$ to $OR$ can therefore be made arbitrarily small but positive, so $OR$ is an asymptote. The other diagonal $OR'$ is likewise an asymptote. □

**Proposition 2.1.5.** *The hyperbola $\alpha$ is symmetric about its conjugate axis.*

**Proof.** Let $N$ be the projection on the transverse axis of a point $P \in \alpha$, and let $N'$ be the reflection of $N$ in $O$. Since $ON > OA$, we have $ON' > OA'$, so the perpendicular to the transverse axis through $N'$ meets $\alpha$ in two points $P', P''$. By proposition 2.1.3, $P'N' = P''N' = PN$, which proves the required symmetry. □

**Corollary 2.1.6.** *The hyperbola has a second corresponding focus and directrix. Every chord of $\alpha$ through $O$ is bisected by $O$.*

**Proof.** Both assertions follow immediately from the symmetry. □

Below, write $F'$ for the focus symmetric to $F$, and write $X'$ for the intersection with the transverse axis of the directrix $l'$ symmetric to $l$, as in Figure 2.4.

**Proposition 2.1.7.** *For the hyperbola $\alpha$,*[5]

*(1)* $FA = e \cdot AX$;
*(2)* $OA = e \cdot OX$;
*(3)* $OF = e \cdot OA$;
*(4)* $OA^2 = OF \cdot OX$;
*(5)* $OF^2 = OA^2 + OB^2$;
*(6)* $FX = \frac{OB^2}{OF}$.

**Proof.** For (1), apply the second definition at $A$: $FA = e \cdot \operatorname{dist}(A, l) = e \cdot AX$.

For (2), $e \cdot AX' = AF'$ and $e \cdot AX = AF$, whence $e \cdot 2OX = e(AX' - AX) = AF' - AF = AA' = 2OA$.

For (3), $OF = OA + AF = e(OX + AX) = e \cdot OA$.

Part (4) follows by dividing (2) by (3).

For (5), the definition of $B$ gives

$$\begin{aligned} OB^2 &= \frac{FX^2}{AX \cdot A'X} \cdot OA^2 = \frac{(OF - OX)^2 \cdot OA^2}{(OA - OX)(OA + OX)} = \frac{(e \cdot OA - e^{-1} \cdot OA)^2 \cdot OA^2}{OA^2 - OX^2} \\ &= \frac{(e - e^{-1})^2 \cdot OA^4}{OA^2 - e^{-2} \cdot OA^2} = (e^2 - 1) OA^2 = OF^2 - OA^2. \end{aligned}$$

[4] The reader may wish to justify explicitly why $P$ can tend to infinity.

[5] Thus, in the customary notation, if $a, b, c$ are respectively the semi-transverse axis, the semi-conjugate axis, and the semifocal distance, then $c^2 = a^2 + b^2$, the eccentricity is $e = \frac{c}{a}$, and the focal parameter is $p = \frac{b^2}{c}$.

For (6),

$$FX = OF - OX = e \cdot OA - e^{-1} \cdot OA = \left(\frac{OF}{OA} - \frac{OA}{OF}\right) \cdot OA = \frac{OF^2 - OA^2}{OA \cdot OF} \cdot OA = \frac{OB^2}{OF}. \qquad \square$$

**Proposition 2.1.8.** *For every point $P$ on the hyperbola, $|PF' - PF| = AA'$.*

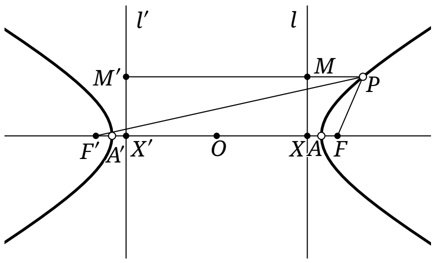


FIGURE 2.4

**Proof.** As in Figure 2.4, let $M'$ and $M$ be the projections of $P$ on the left and right directrices, respectively. By corollary 2.1.6,

$$|PF' - PF| = |e \cdot PM - e \cdot PM'| = e \cdot MM' = AA',$$

where the last equality uses proposition 2.1.7. $\square$

We have now established the basic structure of the hyperbola. We next give elementary proofs of several further important properties.

**Proposition 2.1.9.** *Let the tangent to $\alpha$ at $P$ meet $l$ at $K$. Then $KF \perp PF$.*

**Proof.** As in Figure 2.5, choose a point $P'$ of $\alpha$ near $P$, and put $PP' \cap l = K'$. By proposition 2.1.2, $FK'$ bisects the exterior angle of $\angle PFP'$. As $P' \longrightarrow P$, the line $PP'$ tends to the tangent and $K' \longrightarrow K$, while $\angle K'FP \longrightarrow 90°$. $\square$

**Corollary 2.1.10.** *The tangents to $\alpha$ at the endpoints of a chord $PP'$ through the focus $F$ meet on the directrix $l$.*

**Proof.** Let the two tangents meet the directrix at $K, K'$. Both $KF$ and $K'F$ are perpendicular to $PP'$, and therefore $K = K'$. $\square$

**Proposition 2.1.11** (Adams property)**.** *Let $C$ lie on the tangent to $\alpha$ at $P$. From $C$ draw $CI \perp l$ and $CU \perp FP$, with feet $I$ and $U$, respectively. Then $FU = e \cdot CI$.*

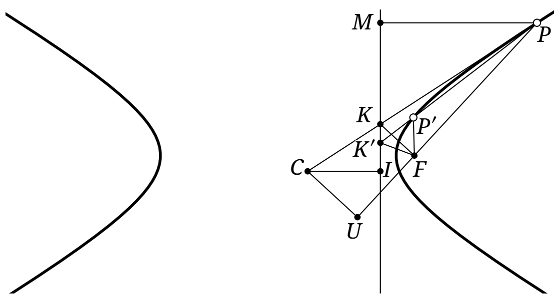


FIGURE 2.5

**Proof.** As in Figure 2.5, put $PC \cap l = K$. By proposition 2.1.9, $FK \perp FP$, and hence $FK \parallel CU$. If $M$ is the projection of $P$ on $l$, then $\frac{FU}{FP} = \frac{KC}{KP} = \frac{CI}{MP}$. It follows that $\frac{FU}{CI} = \frac{FP}{MP} = e$. □

There is, however, an important detail that has so far been left implicit. Every point of a hyperbola other than its vertices lies outside the vertices $A, A'$.[6] This is what distinguishes the behavior of the hyperbola from that of the ellipse treated below: the ellipse is obtained by taking $e < 1$, yet all of its points lie between its two major-axis vertices. We therefore supply a rigorous argument for the hyperbola; the analogous argument for the ellipse will not be repeated.

Suppose $A', X, A, F$ have been fixed in this order on a line. For a point $P$ of the curve other than $A, A'$, let $N$ be its projection on $AA'$; then $N \neq A, A'$. If $N = F$, then $PF \perp AA'$ and we require $PF = e \cdot XF$; such points $P$ exist and lie outside the vertices $A, A'$. If $N \neq F$, choose $K$ on the ray $FN$ so that $FK = e \cdot XN$, where $e$ is the eccentricity. One possible configuration is shown in Figure 2.6.

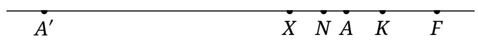


FIGURE 2.6

Since $XN$ is the distance from $P$ to the directrix, the point $P$ can exist only if the circle $\odot(F, FK)$ meets the line $PN$, that is, only if $FK \geq FN$.

(1) If $N$ lies between $A$ and $X$, then $FK = e \cdot NX < e \cdot AX = FA < FN$, which is impossible.
(2) If $N = X$, the defining condition gives $PF = e \cdot XN = 0$, hence $P = F$. But the projection of $P$ on $AA'$ cannot then be $X$, so this is impossible.
(3) If $N$ lies between $X$ and $A'$, then
$$KA' = FA' - FK = e \cdot XA' - e \cdot XN = e \cdot NA' > NA'.$$
Thus $N$ is closer to $A'$ and correspondingly farther from $F$, so $FK < FN$; again this is impossible.
(4) If $N$ lies on the extension of $FA'$ beyond the segment $FA'$, then
$$A'K = FK - FA' = e \cdot XN - e \cdot XA' = e \cdot A'N > A'N.$$
Now $K$ is farther from $A'$ and hence also farther from $F$, so $FK > FN$; this case is possible.
(5) If $N$ lies between $A$ and $F$, then $FK = e \cdot NX > e \cdot AX = FA > FN$, so this case is possible.
(6) If $N$ lies on the extension of $AF$ beyond the segment $AF$, then $FK = e \cdot XN > XN > FN$, so this case too is possible. □

---

## Exercises

**Exercise 2.1.** Let a hyperbola $\alpha$ have center $O$, a focus $F$, and corresponding directrix $l$. Suppose $l$ meets the two asymptotes at $U, V$, and that $OA, OB$ are respectively the semi-transverse and semi-conjugate axes. Prove that $OU = OV = OA$ and $FU = FV = OB$.

**Exercise 2.2.** Let $\alpha$ be a hyperbola of eccentricity $e$, with center $O$, and let $P$ vary on $\alpha$. Let $H$ be the projection of $P$ on the transverse axis. From $H$ draw tangents to the auxiliary circle, and let $Q$ be one of the points of contact. Prove that $\frac{PH}{QH} = \sqrt{e^2 - 1}$.

**Exercise 2.3.** The two asymptotes of a hyperbola and one point on it are given. Construct, with straightedge and compass, its foci and its transverse-axis vertices.

**Exercise 2.4.** Let $a, b$ be respectively the semi-transverse and semi-conjugate axes of a hyperbola $\alpha$, and let $F$ be one of its foci. Prove that among all chords of $\alpha$ through $F$, the shortest is perpendicular to the transverse axis and has length $\frac{2b^2}{a}$. This chord is called a *latus rectum*.

[6] Here "outside" means that the point's projection on the line $AA'$ lies outside the segment $AA'$, with "inside" understood analogously. These terms do not refer to the previously agreed interior and exterior of a conic.

**Exercise 2.5.** Let $O$ be the center of a hyperbola $\alpha$, and let $A,B \in \alpha$ satisfy $OA \perp OB$.[7] Prove that:

(1) $\frac{1}{OA^2} + \frac{1}{OB^2} = \text{const}$;
(2) the line $AB$ envelops a fixed circle.

### 2.1.2 Conjugate Hyperbolas and Diameters

We now introduce several further concepts and properties of the hyperbola.

**Definition 2.1.12.** Let $\alpha$ be a hyperbola. If the transverse axis of a hyperbola $\beta$ is precisely the conjugate axis of $\alpha$, and the conjugate axis of $\beta$ is precisely the transverse axis of $\alpha$—here the words "axis" refer to the corresponding segments—then $\beta$ is called the *conjugate hyperbola* of $\alpha$. Evidently $\alpha$ is also the conjugate hyperbola of $\beta$, and the two hyperbolas are said to be *conjugate*.

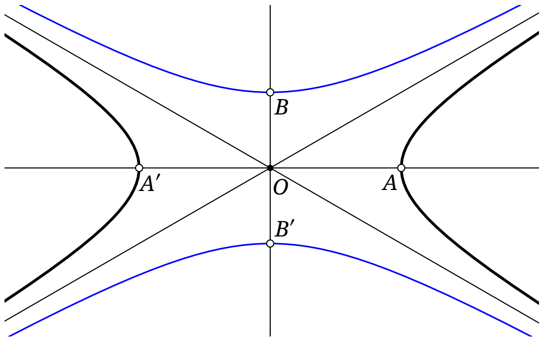

FIGURE 2.7

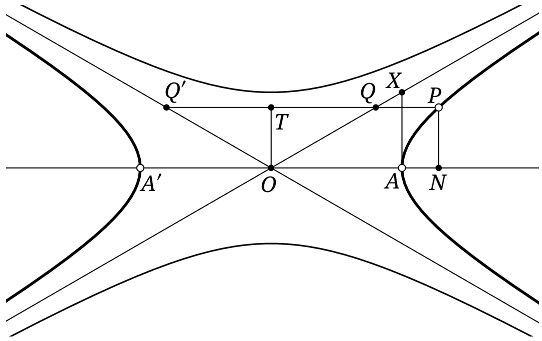

FIGURE 2.8

For example, in Figure 2.7 the blue hyperbola is the conjugate of the black hyperbola $\alpha$.

**Proposition 2.1.13.** *Two conjugate hyperbolas have the same asymptotes, as in Figure 2.7.*

**Proof.** This follows immediately from proposition 2.1.4. □

We next develop some further properties of conjugate hyperbolas.

**Proposition 2.1.14.** *Through a point $P$ on $\alpha$ or its conjugate hyperbola $\beta$, draw the line parallel to the transverse axis of $\alpha$, meeting the two asymptotes at $Q,Q'$. Then $PQ \cdot PQ' = OA^2$. If instead the line through $P$ is parallel to the conjugate axis of $\alpha$, meeting the two asymptotes at $Q,Q'$, then $PQ \cdot PQ' = OB^2$.*

**Proof.** The proof of proposition 2.1.4 already established the second assertion when $P \in \alpha$. By the definition of $\beta$, the first assertion for $P \in \beta$ is the same result after a rotation through 90°. We prove the first assertion for $P \in \alpha$; the second assertion for $P \in \beta$ then follows in the same way.

As in Figure 2.8, let $T$ be the midpoint of $QQ'$, so that $OT \perp QQ'$. Let $N$ be the projection of $P$ on the transverse axis; then $OTPN$ is a rectangle. Choose $X$ so that $OAXB$ is a rectangle. By proposition 2.1.4, $X$ lies on an asymptote, and $\triangle OAX \sim \triangle QTO$. Therefore $TQ = TO \cdot \frac{OA}{OB} = PN \cdot \frac{OA}{OB}$, and hence

$$\begin{aligned} PQ \cdot PQ' &= (PT - TQ)(PT + TQ) = PT^2 - TQ^2 \\ &= ON^2 - PN^2 \cdot \frac{OA^2}{OB^2} = ON^2 - (ON^2 - OA^2) = OA^2, \end{aligned}$$

where the penultimate equality follows from proposition 2.1.3. □

[7] In the real Euclidean plane such points exist only when $e > \sqrt{2}$. In the complex Euclidean plane they always exist.

**Proposition 2.1.15.** *Through points $P,Q$ on $\alpha$ or on its conjugate hyperbola draw two parallel lines. If they meet the asymptotes at $R,R'$ and $S,S'$, respectively, then $PR \cdot PR' = QS \cdot QS'$.*

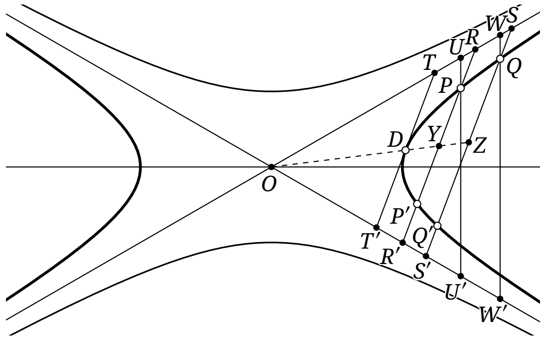


FIGURE 2.9

**Proof.** We treat the case $P,Q \in \alpha$; the proof for the conjugate hyperbola is entirely analogous. As in Figure 2.9, through $P,Q$ draw the lines parallel to the conjugate axis, meeting the asymptotes at $U,U'$ and $W,W'$, respectively. Similar triangles give $\frac{PR}{QS} = \frac{PU}{QW}$ and $\frac{PR'}{QS'} = \frac{PU'}{QW'}$. Consequently,

$$\frac{PR \cdot PR'}{QS \cdot QS'} = \frac{PU \cdot PU'}{QW \cdot QW'} = \frac{OB^2}{OB^2} = 1,$$

where the penultimate equality uses proposition 2.1.14. □

**Proposition 2.1.16.** *Let a secant of $\alpha$ or of its conjugate hyperbola meet the two asymptotes at $R,R'$ and meet $\alpha$ at $P,P'$. Then $PR = P'R'$. If the tangent to $\alpha$ at $D$ meets the asymptotes at $T,T'$, then $DT = DT'$.*

**Proof.** It suffices to consider $\alpha$, as in Figure 2.9. By proposition 2.1.15, $PR \cdot PR' = P'R \cdot P'R'$, from which $PR = P'R'$ follows. Translate a chord $PP'$ parallel to the tangent until $P,P'$ tend to $D$; in the limit it becomes the tangent and yields $DT = DT'$. □

After the relevant material of chapter 3 has been studied, we shall give another proof of this property by affine transformations in section 3.3.

We now introduce diameters of a hyperbola and their basic properties.

**Proposition 2.1.17.** *The locus of the midpoints of a family of parallel chords of a hyperbola is a line through its center.* This line is called a *diameter* of the hyperbola, and its intersections with the hyperbola are the *endpoints of the diameter*. *The tangents to the hyperbola at those endpoints are parallel to the chords of the family.*

**Proof.** As in Figure 2.9, let $PP',QQ'$ be two parallel chords, with midpoints $Y,Z$. Suppose the lines $PP',QQ'$ meet the asymptotes at $R,R'$ and $S,S'$, respectively. By proposition 2.1.16, $Y,Z$ are also the midpoints of $RR',SS'$. Since $RR' \parallel SS'$, similarity shows that $O,Y,Z$ are collinear. Now let $P,P'$ tend to an endpoint $D$ of the diameter. Their chord tends to the tangent at $D$, which is therefore parallel to $QQ'$. □

**Proposition 2.1.18.** *The tangents to a hyperbola at the endpoints of a chord meet on the diameter that bisects the chord.*

**Proof.** Take two parallel chords $AB,A'B'$ of the hyperbola, and put $AA' \cap BB' = P$. Similar triangles show that $P$ lies on the line joining the midpoints of $AB$ and $A'B'$, that is, on the diameter bisecting these chords. Let $A'B'$ tend to $AB$. The lines $AA',BB'$ tend to the tangents at the endpoints of $AB$, proving the assertion. □

**Remark.** In the preceding arguments involving tangents we used the following idea. Let $A$ be a point of a curve $\gamma$, and let $A' \neq A$ be another point of $\gamma$. As $A'$ tends to $A$ along $\gamma$, the line $AA'$ tends to the tangent to $\gamma$ at $A$. This is, in fact, the definition of the tangent to a general curve at a point.

We shall repeatedly use the same limiting idea for conics: a tangent may be regarded as a line whose two intersections with the conic coincide. Thus, when no ambiguity can result, for a point $P$ on a conic $\alpha$ we shall write $PP$ for the tangent to $\alpha$ at $P$.

When we speak below of the *length* of a diameter of a hyperbola, we mean the distance between its two endpoints. A *radius* of the hyperbola is the segment from the center to one endpoint of a diameter. If a diameter does not meet the hyperbola, its “length” means the part cut from it by the conjugate hyperbola, and its corresponding radius is the segment from the center to one of those intersections.[8] With this convention we have the following result.

**Proposition 2.1.19.** *Through a point $P$ on $\alpha$ or its conjugate hyperbola draw a line $l$ meeting the two asymptotes at $Q,Q'$. If $OD$ is a radius parallel to $l$, then $OD^2 = PQ \cdot PQ'$.*

**Proof.** Apply proposition 2.1.15 to the points $P,D$, using the definition of a radius. □

We next introduce conjugate diameters.

**Proposition 2.1.20.** *Let $l_1,l_2$ be two diameters of a hyperbola. The chords parallel to $l_1$ are bisected by $l_2$ if and only if the chords parallel to $l_2$ are bisected by $l_1$.* When this holds, $l_1,l_2$ are called a pair of *conjugate diameters*; each is the conjugate diameter of the other with respect to this hyperbola.

**Proof.** As in Figure 2.10, suppose the diameter $l_1 = OP$ bisects the chord $XY$, and let $Z$ be the reflection of $Y$ in $O$. Then $OP \parallel XZ$. If the diameter $l_2 \parallel XY$, it plainly bisects $XZ$, as required. □

**Remark.** The radii corresponding to a pair of conjugate diameters are likewise called *conjugate radii*. Of the two conjugate diameters of a hyperbola, one meets the hyperbola and the other does not. For example, if $OP$ above meets $\alpha$, then the line $OQ$ parallel to $XY$ cannot meet $\alpha$.

**Corollary 2.1.21.** *The tangent to a hyperbola at an endpoint of a diameter is parallel to the conjugate diameter.*

**Proof.** In the preceding proposition, let a chord tend to the tangent. □

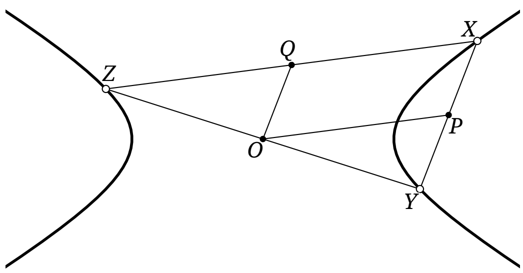


FIGURE 2.10

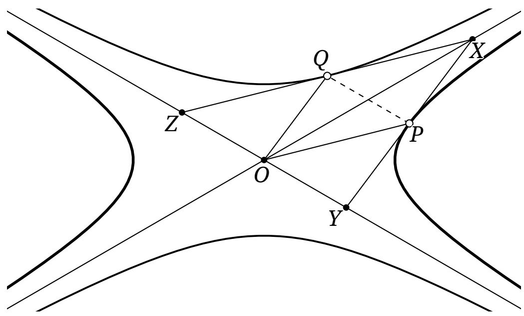


FIGURE 2.11

[8]Over the complex plane a diameter that does not meet the real hyperbola has two imaginary intersections with it, so its length and radius may instead be defined directly, although the length is imaginary. We shall occasionally do this later; in the present chapter we work over the real plane and use the convention just stated.

**Proposition 2.1.22.** *A pair of conjugate radii of a hyperbola is also a pair of conjugate radii of its conjugate hyperbola. Moreover, a pair of conjugate radii of two conjugate hyperbolas, together with the tangents at their noncentral endpoints, forms a parallelogram. One diagonal of this parallelogram lies on one asymptote, and the other is parallel to the other asymptote.*

**Proof.** As in Figure 2.11, let $P \in \alpha$, and let the tangent to $\alpha$ at $P$ meet the two asymptotes at $X, Y$. If $OQ$ is the conjugate radius of $OP$ with respect to $\alpha$, then $OQ \parallel XY$. Let $XQ$ meet the other asymptote at $Z$.

By proposition 2.1.19, $OQ^2 = PX \cdot PY$, while proposition 2.1.16 gives $PX = PY$. Thus $OQ = PX$, and $XPOQ$ is a parallelogram. Hence $XQ \parallel OP$. Since $P$ is the midpoint of $XY$, it follows that $O$ is the midpoint of $YZ$; consequently $Q$ is the midpoint of $XZ$, and $QP$ is parallel to the asymptote $YZ$.

By proposition 2.1.16 and the method of coincidence, $XZ$ is tangent to the conjugate hyperbola $\beta$. Conversely, if $OP'$ is the conjugate radius of $OQ$ with respect to $\beta$, the same argument gives $QP' \parallel YZ$, so $P = P'$. Thus $OQ, OP$ are also conjugate radii of $\beta$, proving the first assertion; the preceding construction then proves the second. $\square$

**Proposition 2.1.23.** *Take the segment joining the noncentral endpoints of a pair of conjugate radii of a hyperbola as one diagonal of a rectangle whose sides are parallel to the axes. The other two vertices of the rectangle lie on one asymptote.*

**Proof.** As in Figure 2.12, let $OP, OQ$ be conjugate radii. Draw $QS \parallel OB$ and $PT \parallel OB$ to meet one asymptote at $S, T$. It remains to prove that $PTQS$ is a rectangle. Let $PQ$ and $AB$ meet the asymptote at $X, Y$, respectively.

The radii $OA, OB$ are themselves conjugate. By proposition 2.1.22, both $PQ$ and $AB$ are parallel to an asymptote. Since $QS \parallel OB$, $\triangle QXS \sim \triangle BYO$. Again by proposition 2.1.22, $QP = 2QX$ and $BA = 2BY$; hence $\frac{QP}{BA} = \frac{QX}{BY} = \frac{QS}{BO}$. Together with $QS \parallel BO$ and $QP \parallel BA$, this gives $\triangle QSP \sim \triangle BOA$, so $\angle QSP = 90^\circ$. Similarly, $\angle QTP = 90^\circ$, and therefore $QSPT$ is a rectangle. $\square$

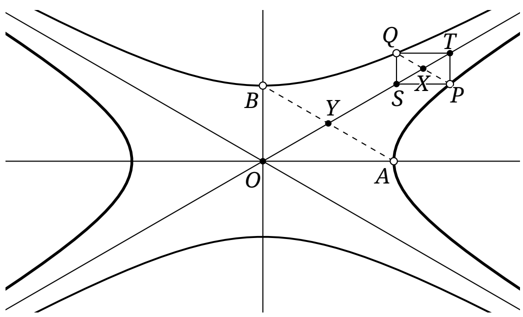


FIGURE 2.12

**Proposition 2.1.24.** *Let $P$ be any point of $\alpha$, and let $OQ$ be the conjugate radius of $OP$. Then $OP^2 - OQ^2 = OA^2 - OB^2$.*

**Proof.** As in Figure 2.13, draw $PT \parallel OB$ and $QT \parallel OA$, meeting at $T$; then $T$ lies on an asymptote. Put $M = PT \cap OA$ and $N = QT \cap OB$. The Pythagorean theorem gives

$$OP^2 - OQ^2 = (OM^2 + PM^2) - (ON^2 + NQ^2) = (NT^2 - NQ^2) - (MT^2 - MP^2).$$

Let $T'$ be the reflection of $T$ in $OA$. It lies on the other asymptote, so proposition 2.1.14 gives

$$MT^2 - MP^2 = (MT + MP)(MT - MP) = PT' \cdot PT = OB^2.$$

Similarly, $NT^2 - NQ^2 = OA^2$, so $OP^2 - OQ^2 = OA^2 - OB^2$. $\square$

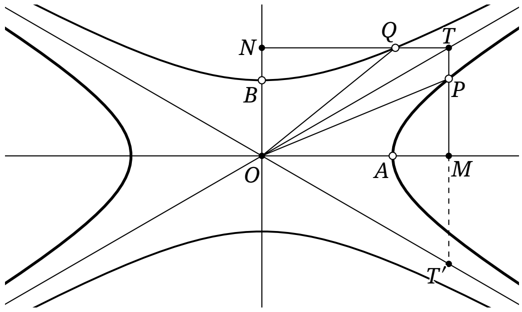


FIGURE 2.13

**Proposition 2.1.25.** *Let $P$ be any point of $\alpha$, and let $OQ$ be the conjugate radius of $OP$. Then $OQ^2 = F'P \cdot FP$.*

**Proof.** The median-length formula gives

$$OP^2 + \frac{1}{4}FF'^2 = \frac{1}{2}(FP^2 + F'P^2) = \frac{1}{2}(F'P - FP)^2 + F'P \cdot FP = \frac{1}{2}AA'^2 + F'P \cdot FP,$$

where the last equality uses the first definition of a hyperbola. Hence

$$F'P \cdot FP = OP^2 + \frac{1}{4}FF'^2 - \frac{1}{2}AA'^2 = OP^2 + OF^2 - 2OA^2 = OP^2 + OB^2 - OA^2 = OQ^2,$$

where the last two equalities use propositions 2.1.7 and 2.1.24, respectively. □

---

## Exercises

**Exercise 2.6.** A hyperbola has been drawn. Construct its foci and directrices with ruler and compass.

**Exercise 2.7.** If a hyperbola has eccentricity $e$, find the eccentricity of its conjugate hyperbola.

**Exercise 2.8.** Prove that a pair of conjugate radii of an equilateral hyperbola are symmetric about an asymptote.

**Exercise 2.9.** Let $\alpha$ be a hyperbola, let $n$ be a fixed line, and let $l$ be an asymptote of $\alpha$. A variable line $m \parallel n$ meets $\alpha$ at $P, Q$. Prove that $\operatorname{dist}(P,l) \cdot \operatorname{dist}(Q,l) = \text{const}$.

**Exercise 2.10.** Let $l_1, l_2$ be the asymptotes of a hyperbola $\alpha$, let $O$ be its center, and let $P \in \alpha$. Prove that:

(1) the parallel through $P$ to $l_2$ meets $l_1$ at $A$, and the parallel through $P$ to $l_1$ meets $l_2$ at $B$; the area $S_{PAOB}$ of the parallelogram $PAOB$ is constant;
(2) the tangent to $\alpha$ at $P$ meets $l_1, l_2$ at $A, B$, and the area $S_{\triangle AOB}$ is constant.

**Exercise 2.11.** Let $O$ be the center of a hyperbola $\alpha$. Points $P, Q$ lie on $\alpha$ and its conjugate hyperbola, respectively, and satisfy $PO \perp QO$. Prove that $\frac{1}{OP^2} - \frac{1}{OQ^2} = \text{const}$.

**Exercise 2.12.** A chord $PP'$ of a hyperbola with center $O$ is bisected at $M$ by a diameter $OM$. The ray $OM$ meets the hyperbola at $Q$ and the tangent at $P$ at $R$. Prove that $OM \cdot OR = OQ^2$.

**Exercise 2.13.** Let $\alpha$ be a fixed hyperbola. A variable point $A$ lies on one branch, and variable points $B,C$ lie on the other, with the centroid $G$ of $\triangle ABC$ also on $\alpha$. Prove that $S_{\triangle ABC}$ is minimal if and only if the line $AG$ passes through the center of $\alpha$.

## 2.2 Elementary Properties of the Ellipse

### 2.2.1 Basic Properties of the Ellipse

We now consider the ellipse.

**Definition 2.2.1.** An *ellipse* is the locus $\alpha$ of points for which the ratio of the distance from a fixed point $F$ to the distance from a fixed line $l$ is a constant $e<1$. The point $F$ is called a *focus* of $\alpha$, and the line $l$ its *directrix*.

Draw through $F$ the perpendicular to $l$, and below write $X$ for its foot. The ellipse $\alpha$ is plainly symmetric about $FX$. There are two points $A,A'$ of $FX$ on the ellipse, on the same side of $l$. They are the *major-axis vertices*, and the midpoint $O$ of $AA'$ is the *center* of the ellipse. The perpendicular to $AA'$ through $O$ meets $\alpha$ at $B,B'$, the *minor-axis vertices*, as in Figure 2.14. The circle with diameter $AA'$ is the *major auxiliary circle* of the ellipse, often simply its *auxiliary circle*; the circle with diameter $BB'$ is its *minor auxiliary circle*.

For the remainder of this discussion retain this notation, taking $A$ as the left major-axis vertex, called the *left vertex*, and $A'$ as the right major-axis vertex, called the *right vertex*. Similarly, take $B$ and $B'$ as the *upper vertex* and *lower vertex*, respectively.[9]

As in the hyperbolic case, we build up the properties of the ellipse by a sequence of propositions. Many proofs are analogous to those already given for the hyperbola and are omitted; some of the figures retain the auxiliary lines that indicate the corresponding arguments.

**Proposition 2.2.2.** *If the extension of a chord $PP'$ of an ellipse meets the directrix $l$ at $K$, then $FK$ bisects the exterior angle of $\angle PFP'$, as in Figure 2.14.*

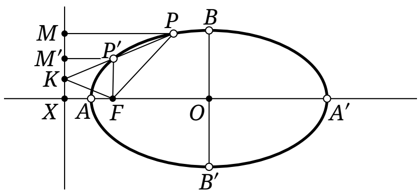


FIGURE 2.14

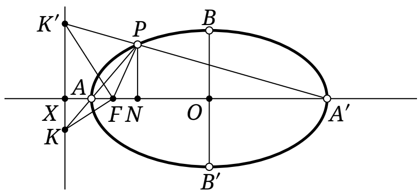


FIGURE 2.15

**Proposition 2.2.3.** *Let $N$ be the projection on the major axis of a point $P$ on an ellipse $\alpha$. Then, as in Figure 2.15, $\frac{PN^2}{NA\cdot NA'}=\frac{OB^2}{OA^2}$ and $OB<OA$.*

**Proposition 2.2.4.** *The ellipse $\alpha$ is symmetric about the minor axis $BB'$. It therefore has a second focus $F'$ and corresponding directrix $l'$.*

Below, write $X'$ for the intersection of the other directrix $l'$ with $AA'$.

[9]As for the hyperbola, these names refer only to orientation and apply when the major axis is horizontal. If it is vertical, the two major-axis vertices are instead called upper and lower, and the two minor-axis vertices left and right.

**Proposition 2.2.5.** *For the ellipse $\alpha$,*
*(1)* $FA = e \cdot AX$*;*
*(2)* $OA = e \cdot OX$*;*
*(3)* $OF = e \cdot OA$*;*
*(4)* $OF \cdot OX = OA^2$.

**Proof.** The four assertions are proved in the same way as parts (1)–(4) of proposition 2.1.7. ☐

**Proposition 2.2.6.** *For every point $P \in \alpha$, $PF + PF' = AA'$, as in Figure 2.16.*

**Proof.** The proof is analogous to that of proposition 2.1.8. ☐

**Corollary 2.2.7.** *For the ellipse $\alpha$,*[10]
*(1)* $OB^2 + OF^2 = OA^2$*;*
*(2)* $FX = \frac{OB^2}{OF}$.

**Proof.** For (1), apply proposition 2.2.6 at a minor-axis vertex. Part (2) is analogous to part (6) of proposition 2.1.7. ☐

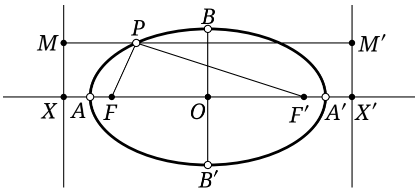


FIGURE 2.16

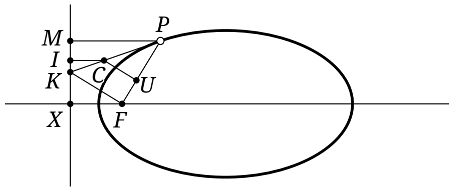


FIGURE 2.17

This establishes the basic properties of the ellipse. It also has the following properties.

**Proposition 2.2.8.** *If the tangent to an ellipse at $P$ meets the directrix $l$ at $K$, then $KF \perp FP$.*

**Corollary 2.2.9.** *The tangents to an ellipse at the endpoints of a chord $PP'$ through the focus $F$ meet on the directrix $l$.*

**Proposition 2.2.10** (Adams property)**.** *Let $C$ lie on the tangent to $\alpha$ at $P$, and draw $CI \perp l$ and $CU \perp PF$, with respective feet $I, U$. Then $FU = e \cdot CI$, as in Figure 2.17.*

---

## Exercises

**Exercise 2.14.** Prove propositions 2.2.2 to 2.2.6, 2.2.8 and 2.2.10 and corollary 2.2.9.

**Exercise 2.15.** At the end of section 2.1.1 we proved that every point of a hyperbola other than $A$,$A'$ lies outside $A$,$A'$. Give the analogous proof that every point of an ellipse other than $A$,$A'$ lies inside $A$,$A'$ in the same axial sense.

---

[10]Combining this with proposition 2.2.5, in the customary notation, if $a, b, c$ are respectively the semimajor axis, the semiminor axis, and the semifocal distance, then $c^2 = a^2 - b^2$, the eccentricity is $e = \frac{c}{a}$, and the focal parameter is $p = \frac{b^2}{c}$.

**Exercise 2.16.** Fix a point $M$ on the major axis of an ellipse, and let $F$ be one of its foci. As $P$ varies on the ellipse, draw through $M$ the perpendicular to the tangent at $P$, meeting the line $PF$ at $K$. Determine the locus of $K$.

**Exercise 2.17.** Let $a,b$ be respectively the semimajor and semiminor axes of an ellipse $\alpha$, and let $F$ be one of its foci. Prove that among all chords of $\alpha$ through $F$, the shortest is perpendicular to the major axis and has length $\frac{2b^2}{a}$. This chord is called a *latus rectum* of $\alpha$.

**Exercise 2.18.** Let $O$ be the center of an ellipse $\alpha$, and let $A,B \in \alpha$ satisfy $OA \perp OB$. Prove that:
(1) $\frac{1}{OA^2} + \frac{1}{OB^2} = \text{const}$;
(2) the line $AB$ envelops a fixed circle.[11]

### 2.2.2 Auxiliary Circle and Diameters

For the hyperbola we obtained the properties of diameters by using the asymptotes. For the ellipse we use the auxiliary circle instead. Its key property is the following.

**Proposition 2.2.11.** *Let $P \in \alpha$, let $N$ be its projection on $AA'$, and extend $NP$ to $Q$ so that $\frac{\overline{QN}}{\overline{PN}} = \left|\frac{OA}{OB}\right|$. Then $OQ = OA$; that is, $Q$ lies on the auxiliary circle of $\alpha$, as in Figure 2.18.*

**Proof.** By proposition 2.2.3,

$$QN^2 = PN^2 \cdot \frac{OA^2}{OB^2} = \left[\frac{OB^2}{OA^2}(AN \cdot A'N)\right] \cdot \frac{OA^2}{OB^2} = AN \cdot A'N.$$

The right-triangle altitude theorem gives $\angle AQA' = 90°$, so $Q$ lies on the circle centered at $O$ with radius $OA$. □

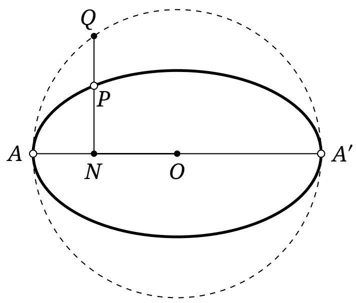


FIGURE 2.18

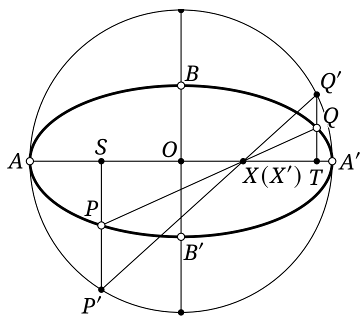


FIGURE 2.19

We shall often use this construction. Accordingly, for the remainder of this chapter call $Q$ the *point corresponding to $P$ on the auxiliary circle*. In this construction the directed angle $\overrightarrow{\measuredangle} A'OQ$ is called the *eccentric angle*, or *parameter angle*, of $Q$.[12] Indeed, on the ellipse in standard form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a > b > 0),$$

a point $P$ may be written $(a\cos\theta, b\sin\theta)$, and the geometric angle corresponding to $\theta$ is $\overrightarrow{\measuredangle} A'OQ$. One may also define an eccentric angle for a hyperbola: it is the angle $\angle HOQ$ occurring in exercise 2.2,

[11] By exercise 2.5, a hyperbola has the analogous property. This circle is called the *inner director circle* of a central conic.

[12] One may instead take $\overrightarrow{\measuredangle} AOQ$ as the parameter angle, but the chosen major-axis vertex must remain fixed within a configuration; the right-hand vertex is usually chosen.

with a suitable orientation. That orientation is less simple to specify and of little use here, so we shall not pursue it.

The auxiliary circle has the following frequently used properties.

**Proposition 2.2.12.** *Let $P,Q$ be points of an ellipse $\alpha$, and let $P',Q'$ be their corresponding points on the auxiliary circle. Put $\phi = \measuredangle(AA',PQ)$ and $\phi' = \measuredangle(AA',P'Q')$. Then:*

*(a)* $PQ \cap P'Q' \in AA'$;

*(b) the tangent to $\alpha$ at $P$ and the tangent to the auxiliary circle at $P'$ meet on $AA'$;*

*(c)* $\frac{\tan\phi'}{\tan\phi} = \frac{OA}{OB}$;

*(d)* $\theta_P + \theta_Q = 2\phi' + \pi$, *where $\theta_P, \theta_Q$ are the parameter angles of $P,Q$.*[13]

**Proof.** As in Figure 2.19, let $PP'$ and $QQ'$ meet $AA'$ at $S,T$, and let $PQ$ and $P'Q'$ meet $AA'$ at $X,X'$, respectively.

For (a), $\triangle PSX \sim \triangle QTX$ gives $\frac{\overline{SX}}{\overline{TX}} = \frac{\overline{PS}}{\overline{QT}}$. Likewise, $\frac{\overline{SX'}}{\overline{TX'}} = \frac{\overline{P'S}}{\overline{Q'T}}$. But proposition 2.2.11 gives $\frac{\overline{PS}}{\overline{QT}} = \frac{\overline{P'S}}{\overline{Q'T}}$, so $\frac{\overline{SX}}{\overline{TX}} = \frac{\overline{SX'}}{\overline{TX'}}$. Thus $X = X'$.

For (b), let $Q \longrightarrow P$ in (a). Then $Q' \longrightarrow P'$, while $PQ$ and $P'Q'$ tend to the corresponding tangents.

For (c), the lines $PQ, P'Q'$ meet at a point $X \in AA'$, and proposition 2.2.11 gives $\frac{\tan\phi'}{\tan\phi} = \frac{\overline{Q'T}}{\overline{QT}} = \frac{OA}{OB}$.

For (d), consider the case in which $X$ lies on the segment $PQ$; the other configurations are analogous. Observe that $\overrightarrow{\measuredangle} OP'X = -\overrightarrow{\measuredangle} OQ'X$. Hence

$$\begin{aligned}\theta_P + \theta_Q &= \overrightarrow{\measuredangle} A'OP' + \overrightarrow{\measuredangle} A'OQ' = (\overrightarrow{\measuredangle} A'XP' - \overrightarrow{\measuredangle} OP'X) + (\overrightarrow{\measuredangle} A'XQ' - \overrightarrow{\measuredangle} OQ'X)\\ &= \overrightarrow{\measuredangle} A'XP' + \overrightarrow{\measuredangle} A'XQ' = 2\measuredangle AXQ' + \pi = 2\phi' + \pi. \qquad \square\end{aligned}$$

The auxiliary circle now gives an elementary construction of diameters.

**Proposition 2.2.13.** *The locus of the midpoints of a family of parallel chords of an ellipse is a line through its center.* This line is a *diameter*, and its intersections with the ellipse are the *endpoints of the diameter*. *The tangents to the ellipse at those endpoints are parallel to the chords of the family.*

**Proof.** In Figure 2.20, the chords $PQ \parallel RS$. We prove that their midpoints $M,N$ and the center $O$ of the ellipse are collinear; this proves the first assertion, and the assertion about tangents follows by taking a limit.

Let $P',Q',R',S'$ be the points corresponding to $P,Q,R,S$ on the auxiliary circle. Through $M,N$ draw parallels to the minor axis, meeting $P'Q'$ and $R'S'$ at $M',N'$. The intercept theorem shows that $M',N'$ are the midpoints of $P'Q', R'S'$, respectively.

By proposition 2.2.12, the lines $PQ, P'Q'$ meet the major axis at the same point $X$, and $SR, S'R'$ meet it at the same point $Y$. Moreover,

$$\frac{\tan\measuredangle Q'XA'}{\tan\measuredangle R'YA'} = \frac{\tan\measuredangle QXA' \cdot (OA/OB)}{\tan\measuredangle RYA' \cdot (OA/OB)} = 1.$$

Thus $P'Q' \parallel R'S'$. The perpendicular-bisector property of a circle then shows that $M',N',O$ are collinear.

Since $PP' \parallel MM'$,

$$\frac{\operatorname{dist}(M',AA')}{\operatorname{dist}(M,AA')} = \frac{\operatorname{dist}(P',AA')}{\operatorname{dist}(P,AA')} = \frac{OA}{OB}.$$

The analogous relation holds for $N,N'$, so $\frac{\operatorname{dist}(M',AA')}{\operatorname{dist}(M,AA')} = \frac{\operatorname{dist}(N',AA')}{\operatorname{dist}(N,AA')}$. It follows that $M,O,N$ are collinear. $\square$

This use of the auxiliary circle is essentially the same as the parallel projection used in section 1.4, and as the affine transformations to be introduced in chapter 3.

[13] The angles $\theta_P, \theta_Q$ are directed modulo $360°$, whereas $\phi'$ is directed modulo $180°$. The equality is nevertheless well defined, because multiplication by 2 makes $2\phi'$ well defined modulo $360°$.

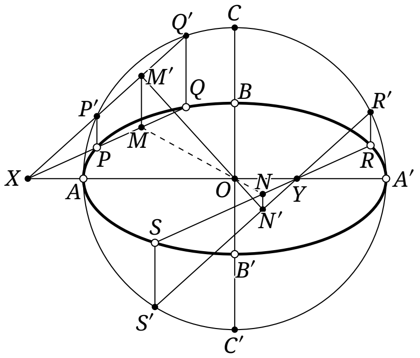


FIGURE 2.20

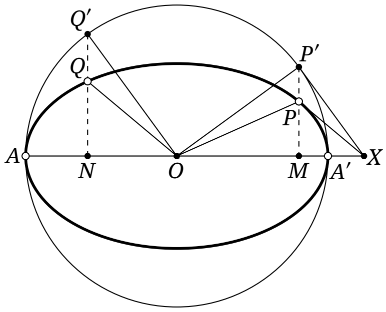


FIGURE 2.21

The next property of diameters has a proof analogous to that of proposition 2.1.18 for the hyperbola.

**Proposition 2.2.14.** *The tangents to an ellipse at the endpoints of a chord meet on the diameter that bisects the chord.*

When we speak below of the length of a diameter of an ellipse, we mean the distance between its endpoints. A *radius* of an ellipse is the segment from the center to one endpoint of a diameter. We now introduce conjugate diameters.

**Proposition 2.2.15.** *Let $l_1, l_2$ be two diameters of an ellipse. The chords parallel to $l_1$ are bisected by $l_2$ if and only if the chords parallel to $l_2$ are bisected by $l_1$.* When this holds, $l_1, l_2$ form a pair of *conjugate diameters*; each is the conjugate diameter of the other with respect to this ellipse.

**Proof.** The proof is entirely analogous to that of proposition 2.1.20 for the hyperbola, using a parallelogram. □

**Remark.** The radii corresponding to a pair of conjugate diameters are likewise called *conjugate radii*.

**Corollary 2.2.16.** *The tangent to an ellipse at an endpoint of a diameter is parallel to the conjugate diameter.*

**Proof.** Let the chord in the preceding proposition tend to the tangent. □

**Proposition 2.2.17.** *Let $OP, OQ$ be conjugate radii of $\alpha$. Through $P, Q$ draw parallels to the minor axis, meeting the auxiliary circle at $P', Q'$, with $P, P'$ on the same side of the major axis and likewise $Q, Q'$ on the same side. Then $P'O \perp Q'O$.*

**Proof.** As in Figure 2.21, draw the tangent to the ellipse at $P$ and the tangent to the auxiliary circle at $P'$. By proposition 2.2.12, they meet at a point $X \in AA'$. Let $M, N$ be the respective projections of $P, Q$ on $AA'$; then $P, P', M$ are collinear, as are $Q, Q', N$.

The property of conjugate diameters gives $PX \parallel QO$, while $PM \parallel QN$. Hence $\triangle QNO \sim \triangle PMX$. Together with proposition 2.2.11, this yields $\frac{P'M}{Q'N} = \frac{PM}{QN} = \frac{MX}{NO}$. Thus $\triangle Q'NO \sim \triangle P'MX$, so $Q'O \parallel P'X$. Since the line at $P'$ is tangent to the auxiliary circle, $P'X \perp P'O$, and we obtain $P'O \perp Q'O$. □

**Proposition 2.2.18.** *For every pair of conjugate radii of an ellipse, the sum of the squares of their lengths is* $OA^2 + OB^2$.

**Proof.** Let $OP, OQ$ be conjugate radii, let $P', Q'$ be the corresponding points on the auxiliary circle, and let $M, N$ be the projections of $P, Q$ on the major axis, as in Figure 2.21. We have $OQ' = OP'$, and proposition 2.2.17 gives $Q'O \perp OP'$. Since $NQ', MP' \perp MN$, it follows that $\triangle Q'ON \cong \triangle OMP'$, and hence $MO = NQ'$. Therefore

$$NO^2 + MO^2 = NO^2 + NQ'^2 = OQ'^2 = OA^2.$$

Consequently,

$$\begin{aligned} OP^2 + OQ^2 &= (OM^2 + PM^2) + (ON^2 + QN^2) = OM^2 + ON^2 + \frac{OB^2}{OA^2}(NQ'^2 + MP'^2) \\ &= OM^2 + ON^2 + \frac{OB^2}{OA^2}(OQ'^2 + OP'^2 - NO^2 - OM^2) \\ &= OA^2 + \frac{OB^2}{OA^2}(2OA^2 - OA^2) = OA^2 + OB^2, \end{aligned}$$

where the second equality uses proposition 2.2.11. ☐

**Proposition 2.2.19.** *Let $P$ be any point of $\alpha$, and let $OQ$ be the conjugate radius of $OP$. Then $OQ^2 = F'P \cdot FP$.*

**Proof.** The median-length formula gives

$$OP^2 + \frac{1}{4}FF'^2 = \frac{1}{2}(FP^2 + F'P^2) = \frac{1}{2}(F'P + FP)^2 - F'P \cdot FP = \frac{1}{2}AA'^2 - F'P \cdot FP,$$

where the last equality uses the first definition of an ellipse. Hence

$$F'P \cdot FP = \frac{1}{2}AA'^2 - \frac{1}{4}FF'^2 - OP^2 = 2OA^2 - OF^2 - OP^2 = OA^2 + OB^2 - OP^2 = OQ^2,$$

where the last two equalities use corollary 2.2.7 and proposition 2.2.18, respectively. ☐

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## Exercises

**Exercise 2.19.** Prove propositions 2.2.14 and 2.2.15.

**Exercise 2.20.** A chord $PP'$ of an ellipse with center $O$ is bisected at $M$ by a diameter $OM$. The ray $OM$ meets the ellipse at $Q$ and the tangent at $P$ at $R$. Prove that $OM \cdot OR = OQ^2$.

**Exercise 2.21.** Let $O$ be the center of an ellipse $\alpha$, and let $A, B \in \alpha$. The tangent at $A$ is parallel to $OB$. Prove that:

(1) the tangent at $B$ is parallel to $OA$;
(2) $OA^2 + OB^2 = \text{const}$.

**Exercise 2.22.** Let a central conic $\alpha$ have center $O$ and focus $F$. Let $P \in \alpha$, let $OQ$ be the conjugate radius of $OP$, and put $PF \cap OQ = Y$. Prove that $PY$ is always equal to the semi-principal axis.

**Exercise 2.23.** The endpoints $A, B$ of a rigid light rod move in two ray-shaped grooves $OX, OY$, respectively. Prove that the locus of a point $P$ dividing $AB$ in a fixed ratio is part of an ellipse.[14]

[14] A *point of fixed division ratio* of $A, B$ means a point $P$ on the line $AB$ for which a fixed real number $\lambda$ satisfies $\frac{\overline{AP}}{\overline{BP}} = \lambda$.

## 2.3 Elementary Properties of the Parabola

We now turn to the parabola.

**Definition 2.3.1.** A *parabola* is the locus $\alpha$ of points equidistant from a fixed point $F$ and a fixed line $l$. The point $F$ is called the *focus* of $\alpha$, and the line $l$ its *directrix*.

Draw $FX \perp l$, with $X \in l$. The parabola $\alpha$ is plainly symmetric about $FX$. The midpoint $A$ of $FX$ lies on the parabola and is called its *vertex*; the line $FX$ is its *axis*, as in Figure 2.22. For the remainder of this section we retain this notation.

**Proposition 2.3.2.** *Let $N$ be the projection on the axis of a point $P \in \alpha$. Then, as in Figure 2.22, $PN^2 = 4AF \cdot AN$.*

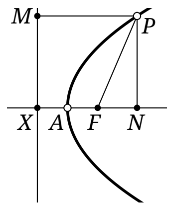


FIGURE 2.22

**Proof.** Let $M$ be the projection of $P$ on the directrix. Then

$$PN^2 = PF^2 - FN^2 = PM^2 - (AN - AF)^2 = (AN + AF)^2 - (AN - AF)^2 = 4AF \cdot AN.$$

□

**Proposition 2.3.3.** *Let the line through $F$ perpendicular to the axis meet $\alpha$ at $L, L'$. Then $LF = L'F = 2AF$.*

**Proof.** Set $N = F$ in proposition 2.3.2. □

The next two propositions are analogous to the hyperbolic results propositions 2.1.2 and 2.1.11, respectively; their proofs are left to the reader.

**Proposition 2.3.4.** *If the line containing a chord $PP'$ of $\alpha$ meets $l$ at $K$, then $FK$ bisects the exterior angle of $\angle PFP'$.*

**Proposition 2.3.5.** *Let $C$ lie on the tangent to $\alpha$ at $P$. Draw $CI \perp l$ and $CU \perp FP$, with respective feet $I, U$. Then $FU = CI$.*

The structure of a parabola is comparatively simple, and in chapter 1 its properties were also developed by elementary geometry. We therefore do not repeat a longer discussion here.

We shall now prove the properties of parabolic diameters by elementary methods. Later they will also be proved by affine transformations in exercise 3.25, and by projective geometry in theorem 6.4.8.

**Proposition 2.3.6.** *The locus of the midpoints of a family of parallel chords of $\alpha$ is a line parallel to the axis.* This line is called a *diameter*, and its intersection with the parabola is the *endpoint of the diameter*. *The tangent to $\alpha$ at that endpoint is parallel to the chords of the family.*

**Proof.** In Figure 2.23, $QQ'$ is a chord and $RR' \parallel QQ'$ is tangent to $\alpha$ at $P$; here $R, R'$ are the intersections of $RR'$ with the tangents to $\alpha$ at $Q, Q'$, respectively. It is enough to prove that $PV$ is parallel to the axis.

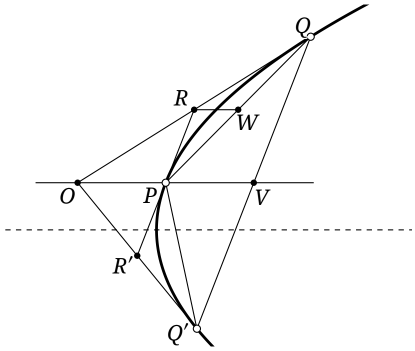


FIGURE 2.23

Through $P$ draw the line $OV$ parallel to the axis, meeting $QQ'$ at $V$ and $QR$ at $O$. Let $W$ be the midpoint of $PQ$. By claim 1.5.6, $RW \parallel OV$, so $R$ is the midpoint of $OQ$. Together with $PR \parallel QV$, this shows that $P$ is the midpoint of $OV$. Similarly, if the tangent at $Q'$ meets $OV$ at $O'$, then $P$ is the midpoint of $O'V$; hence $O' = O$.

Now let $V'$ be the midpoint of $QQ'$. Again by claim 1.5.6, $V'O$ is parallel to the axis. Since $OV$ is also parallel to the axis, $V = V'$. Thus $V$ is the midpoint of $QQ'$, and the result follows. □

The preceding argument also gives several properties of diameters.

**Proposition 2.3.7.** *Suppose a line $l_Q$ parallel to the axis bisects a chord $QQ'$ of $\alpha$. If $l_Q$ meets the tangent at $Q$ at $O$, the chord $QQ'$ at $V$, and $\alpha$ at $P$, then $OP = PV$.*

**Corollary 2.3.8.** *Draw the normal to $\alpha$ at $P$, meeting the axis at $V$, and let $U$ be the projection of $P$ on the axis. Then $UV = 2AF$.*

**Proof.** Let the tangent at $P$ meet the axis at $W$. By symmetry, the diameter corresponding to the chord containing $PU$ is the axis itself, so proposition 2.3.7 gives $UA = WA$. The right-triangle altitude theorem and proposition 2.3.2 now give

$$UV = \frac{PU^2}{UW} = \frac{PU^2}{2AU} = \frac{4AF \cdot AU}{2AU} = 2AF. \qquad \square$$

**Proposition 2.3.9.** *The tangents to $\alpha$ at the endpoints of a chord meet on the diameter that bisects the chord.*

We may now state the property of diameters uniformly for all three conics.

**Theorem 2.3.10** (Properties of diameters)**.** *The locus of the midpoints of a family of parallel chords of a conic is a line, called a diameter of the conic. The tangent at an endpoint of this diameter is parallel to the chords of the family, and the tangents at the endpoints of any chord in the family meet on the diameter. A diameter of a central conic passes through its center; a diameter of a parabola is parallel to its axis.*

We next give some properties peculiar to diameters of a parabola.

**Proposition 2.3.11.** *Let a diameter $PV$ of $\alpha$ bisect a chord $QQ'$ at $V$ and meet $\alpha$ at $P$. Then $QV^2 = 4FP \cdot PV$.*

For the proof we first establish a lemma.

**Lemma 2.3.12.** *Let $P, Q \in \alpha$, and let the tangents at $P, Q$ meet at $R$. Through $P$ draw the line parallel to the axis, meeting $QR$ at $O$. Then*

$$\triangle FQR \overset{+}{\sim} \triangle FRP \overset{+}{\sim} \triangle ROP.$$

**Proof.** The similarity $\triangle FQR \overset{+}{\sim} \triangle FRP$ is theorem 1.5.9. It remains to prove the other similarity.

Let $OQ$ meet the axis of $\alpha$ at $T$. From $\triangle FQR \overset{+}{\sim} \triangle FRP$, $\angle FRP = \angle FQR = \angle FTR = \angle POR$, where the middle equality uses lemma 1.5.3. On the other hand, lemma 1.5.1 says that $PR$ bisects $\angle OPF$, so $\angle FPR = \angle OPR$. Hence $\triangle FRP \sim \triangle ROP$. □

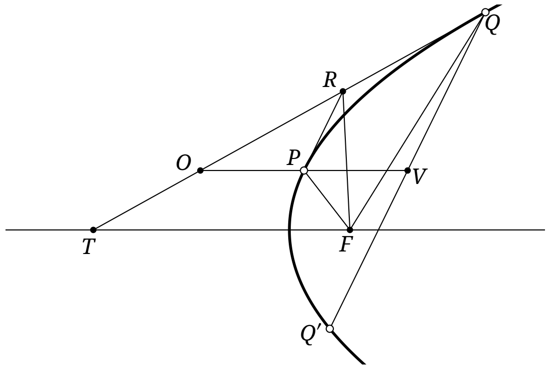


FIGURE 2.24

**Proof of proposition 2.3.11.** As in Figure 2.24, let the diameter $PV$ meet the tangent at $Q$ at $O$, let $OQ$ meet the axis at $T$, and let the tangent at $P$ meet $OQ$ at $R$. By lemma 2.3.12, $\triangle FRP \overset{+}{\sim} \triangle ROP$, and hence $PR^2 = FP \cdot PO$. By proposition 2.3.7, $P$ bisects $OV$, and consequently $QV = 2PR$. Therefore

$$QV^2 = 4PR^2 = 4FP \cdot PO = 4FP \cdot PV. \qquad \square$$

Lemma 2.3.12 also yields the following conclusion.

**Proposition 2.3.13.** *Through a point $V$ of a chord $QQ'$ of $\alpha$, draw the line parallel to the axis, meeting $\alpha$ at $P$ and the tangent at $Q$ at $O$. If the tangent at $P$ meets $OQ$ at $R$, then $OR = QR$.*

**Proof.** As in the proof of proposition 2.3.11, draw through $V$ the parallel to the axis, meeting $\alpha$ at $P$ and the tangent at $Q$ at $O$. Let the tangent at $P$ meet $OQ$ at $R$. We still have $\triangle FQR \sim \triangle FRP \sim \triangle ROP$. Thus $OR = \frac{FR \cdot RP}{PF} = QR$. □

**Proposition 2.3.14.** *A diameter $SX$ of $\alpha$ bisects a chord $AB$ at $X$ and meets $\alpha$ at $S$. Let $P \in AB$, and draw $PU \parallel SX$ to meet $\alpha$ at $U$. Then $AP \cdot BP = 4SF \cdot UP$.*

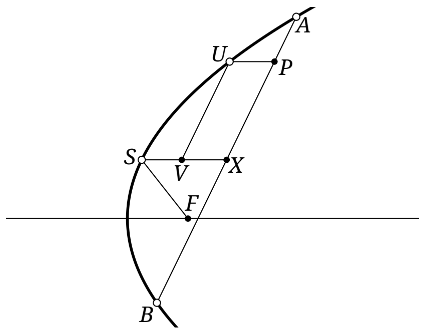


FIGURE 2.25

**Proof.** As in Figure 2.25, through $P$ draw the parallel to the axis, meeting the parabola at $U$, and through $U$ draw the parallel to $AB$, meeting $SX$ at $V$. Since $UV \parallel AX$, the diameter $SX$ also bisects the chord of the parabola corresponding to the line $UV$. Hence, by proposition 2.3.11,

$$\begin{aligned} AP\cdot BP &= (AX-XP)(AX+XP) = AX^2 - PX^2 \\ &= AX^2 - UV^2 = 4SF\cdot SX - 4SF\cdot SV = 4SF\cdot VX = 4SF\cdot UP. \end{aligned}$$ □

Although conjugate diameters were defined for ellipses and hyperbolas, there is no corresponding definition for a parabola.

---

### Exercises

**Exercise 2.24.** Prove propositions 2.3.4 and 2.3.5.

**Exercise 2.25.** A parabola has been drawn. Construct its focus and directrix with ruler and compass.

**Exercise 2.26.** Let a parabola $\alpha$ have focal parameter $p$ and focus $F$. Prove that among all chords of $\alpha$ through $F$, the shortest is perpendicular to the axis and has length $2p$. This chord is called a *latus rectum* of $\alpha$.

**Exercise 2.27.** Let $XY$ be a chord of a parabola $\alpha$, with midpoint $M$. Let $Q$ be the projection of $M$ on the axis, and let the line $XY$ meet the axis at $P$. Prove that $\frac{MQ^2}{PQ} = p$, where $p$ is the focal parameter of $\alpha$.

**Exercise 2.28.** Let $F$ be the focus of a parabola, and let one of its diameters meet the directrix at $P$. Prove that $PF$ is perpendicular to every chord bisected by this diameter.

**Exercise 2.29.** Let $A$ be the vertex of a parabola. Points $P, Q$ on the parabola satisfy $PA \perp QA$. Prove that the line $PQ$ passes through a fixed point.

**Exercise 2.30.** A parabola $\alpha$, a point $P \notin \alpha$, and three collinear points $C, D, E$ are given. Construct with straightedge and compass a chord $AB$ of the parabola such that the line $AB$ passes through $P$ and $\frac{\overline{PA}}{\overline{PB}} = \frac{CE}{DE}$.

**Exercise 2.31.** Through a point $V$ of a chord $QQ'$ of $\alpha$, draw the parallel to the axis, meeting the tangent at $Q$ at $O$ and $\alpha$ at $P$. Prove that $\frac{QV}{Q'V} = \frac{OP}{VP}$.

## 2.4 The Power Theorem for Conics

We now establish a single power theorem common to all three types of conic.

**Theorem 2.4.1** (Power theorem for conics)**.** *Let $\alpha$ be a conic, and let two lines $l_1, l_2$ through $P$ meet $\alpha$ at $A, B$ and $C, D$, respectively.*[15] *Then $\frac{PA \cdot PB}{PC \cdot PD}$ depends only on the directions of $l_1, l_2$.*[16] *In particular, for a central conic this ratio is the ratio of the squares of the lengths of radii parallel to the two lines.*

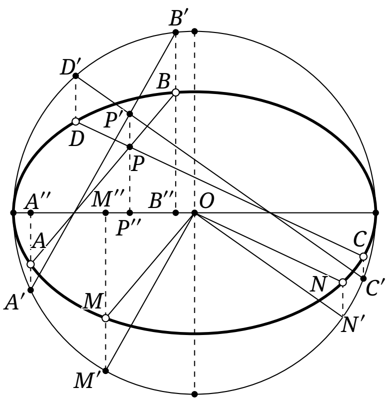


FIGURE 2.26

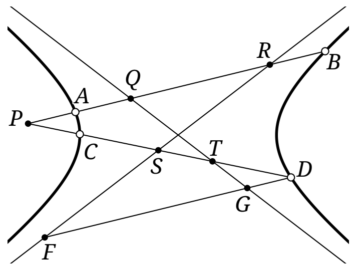


FIGURE 2.27

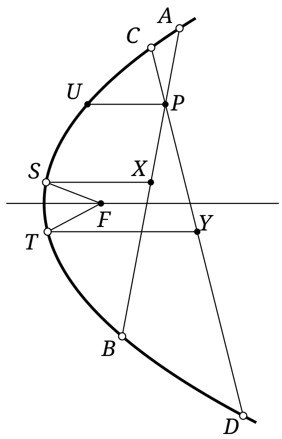


FIGURE 2.28

**Proof.** We consider the ellipse, hyperbola, and parabola in turn.

*1. The ellipse.* In Figure 2.26, draw through $A, B, C, D$ the parallels to the minor axis. Let them meet the auxiliary circle at $A', B', C', D'$ and the major axis at $A'', B'', C'', D''$. Put $A'B' \cap C'D' = P'$, and through $P'$ draw the parallel to the minor axis, meeting the major axis at $P''$. Through the center $O$ draw the parallels to $AB, CD$, meeting the ellipse at $M, N$. Through $M, N$ draw parallels to the minor axis, meeting the auxiliary circle at $M', N'$ and the major axis at $M'', N''$.

As in the proof of proposition 2.2.13, we have $A'B' \parallel OM'$ and $C'D' \parallel ON'$. Similar triangles give

$$\frac{A'P' \cdot B'P'}{OM'^2} = \frac{A''P'' \cdot B''P''}{OM''^2} = \frac{AP \cdot BP}{OM^2},$$

and similarly

$$\frac{C'P' \cdot D'P'}{ON'^2} = \frac{CP \cdot DP}{ON^2}.$$

The circle power theorem gives

$$\frac{A'P' \cdot B'P'}{C'P' \cdot D'P'} \cdot \frac{ON'^2}{OM'^2} = 1.$$

Therefore

$$\frac{AP \cdot BP}{CP \cdot DP} = \frac{OM^2}{ON^2}.$$

The lengths $OM, ON$ depend only on the directions of $AB, CD$, since they are radii parallel to those lines.

*2. The hyperbola.* In Figure 2.27, let $AB$ meet the two asymptotes at $Q, R$, and let $CD$ meet them at $S, T$. Through $D$ draw the parallel to $AB$, meeting the asymptotes at $F, G$.

Since $AB \parallel FG$, similar triangles give

$$\frac{FD}{SD} \cdot \frac{GD}{TD} = \frac{PR}{PS} \cdot \frac{PQ}{PT}.$$

Together with proposition 2.1.15, this yields

$$\frac{AQ \cdot AR}{SD \cdot TD} = \frac{FD \cdot GD}{SD \cdot TD} = \frac{PQ \cdot PR}{PS \cdot PT}.$$

---

[15]As explained at the end of section 2.1, a tangent is included as a limiting secant: if, for example, $l_1$ is tangent to $\alpha$ at $A$, then $B$ is regarded as coinciding with $A$. Whenever a secant of a curve degenerates into a tangent below, we shall understand it in this way without further comment.

[16]Equivalently, if $l_1, l_2$ are respectively parallel to fixed lines $m, n$, the ratio is constant.

The properties of proportions therefore give

$$\frac{FD\cdot GD}{SD\cdot TD}=\frac{PQ\cdot PR-AQ\cdot AR}{PS\cdot PT-SD\cdot TD}.$$

Let $E$ be the midpoint of $QR$. By proposition 2.1.16, it is also the midpoint of $AB$, so

$$PQ\cdot PR=(PE-QE)(PE+QE)=PE^2-QE^2,$$

$$PA\cdot PB=(PE-AE)(PE+BE)=PE^2-AE^2.$$

Subtracting gives

$$PQ\cdot PR-PA\cdot PB=AE^2-QE^2=(AE-QE)(AE+QE)=AQ\cdot AR,$$

and hence $PQ\cdot PR-AQ\cdot AR=PA\cdot PB$. Similarly, $PS\cdot PT-DS\cdot DT=PC\cdot PD$. Consequently,

$$\frac{PA\cdot PB}{PC\cdot PD}=\frac{PQ\cdot PR-AQ\cdot AR}{PS\cdot PT-DS\cdot DT}=\frac{FD\cdot GD}{SD\cdot TD}.$$

The right-hand side depends only on the directions of $CD$ and $FD$. Thus, when the directions of $AB$ and $CD$ are fixed, $\frac{PA\cdot PB}{PC\cdot PD}=\text{const}$.

In particular, if $AB,CD$ are parallel to radii $OM,ON$, respectively, then proposition 2.1.19 gives

$$\frac{PA\cdot PB}{PC\cdot PD}=\frac{OM^2}{ON^2}.$$

*3. The parabola.* We prove the case in which $P$ is inside the parabola; the other cases are analogous. In Figure 2.28, let $SX$ be the diameter corresponding to $AB$, meeting the parabola at $S$ and $AB$ at $X$. Through $P$ draw the parallel to the axis, meeting the parabola at $U$. Let $TY$ be the diameter corresponding to $CD$, meeting the parabola at $T$ and $CD$ at $Y$. By proposition 2.3.14,

$$\frac{AP\cdot BP}{CP\cdot DP}=\frac{4SF\cdot UP}{4TF\cdot UP}=\frac{SF}{TF}.$$

The positions of $S,T$ depend only on the directions of $AB,CD$, by the property of diameters. □

**Corollary 2.4.2.** *If four points $A,B,C,D$ of a conic $\alpha$ are concyclic, then one of the two angle bisectors of the lines $AB,CD$ is parallel to the principal axis of $\alpha$, and the other is perpendicular to it.*

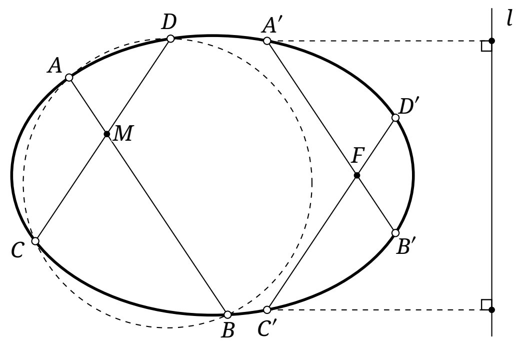


FIGURE 2.29

**Proof.** Put $M=AB\cap CD$. The circle power theorem gives $\frac{\overline{AM}\cdot\overline{BM}}{\overline{CM}\cdot\overline{DM}}=1$. Translate $AB,CD$ parallel to themselves until they pass through a focus $F$ of $\alpha$, meeting $\alpha$ at $A',B'$ and $C',D'$, respectively. The elliptic case is shown in Figure 2.29.

By the power theorem for conics,

$$\frac{\overline{A'F}\cdot\overline{B'F}}{\overline{C'F}\cdot\overline{D'F}}=\frac{\overline{AM}\cdot\overline{BM}}{\overline{CM}\cdot\overline{DM}}=1.$$

Thus

$$\overline{A'F}\cdot\overline{FB'}=\overline{C'F}\cdot\overline{FD'}:=s. \tag{♣}$$

On the other hand, exercise 1.25 gives[17]

$$\frac{1}{\overline{FA'}} - \frac{1}{\overline{FB'}} = \frac{1}{\overline{FC'}} - \frac{1}{\overline{FD'}}. \tag{$\spadesuit$}$$

Putting the fractions over common denominators and using ($\clubsuit$), we obtain

$$\overline{A'F} + \overline{FB'} = \overline{C'F} + \overline{FD'} := t. \tag{$\diamond$}$$

By the formulas of Vieta (Viète), $\overline{A'F}, \overline{FB'}$ are the roots of $x^2 - tx + s = 0$, and so are $\overline{C'F}, \overline{FD'}$. Hence either $\overline{A'F} = \overline{C'F}$, $\overline{FB'} = \overline{FD'}$, or $\overline{A'F} = \overline{FD'}$, $\overline{C'F} = \overline{FB'}$.

Suppose without loss of generality that the first case holds, and let $l$ be the directrix corresponding to $F$. Then $\frac{\operatorname{dist}(A',l)}{\operatorname{dist}(C',l)} = \frac{A'F}{C'F} = 1$. Thus $A', C'$ are symmetric about the principal axis, and so are $A'F, C'F$. Since $AB, CD$ are parallel to $A'F, C'F$, one bisector of the angle between them is parallel to the principal axis and the other perpendicular to it. □

**Remark.** The preceding proof is somewhat lengthy, although the result is easy to visualize. Alternatively, translate $AB, CD$ parallel to themselves until they are tangent to $\alpha$, at $A', C'$, say, and let the two tangents meet at $M'$. The power theorem for conics gives

$$\frac{A'M' \cdot A'M'}{C'M' \cdot C'M'} = \frac{AM \cdot BM}{CM \cdot DM} = 1,$$

so $A'M' = C'M'$. Symmetry suggests that $A'M'$ and $C'M'$ are symmetric about the axis of $\alpha$, which again proves the result. The step asserting this symmetry is not quite rigorous, however, and that is why the preceding argument was used.

The power theorem has the following useful special case.

**Corollary 2.4.3.** *Let a diameter of a central conic with center $O$ meet the conic at $A, A'$, and suppose the diameter $AA'$ bisects a chord $PP'$ at $Q$. If $OB$ is the conjugate radius of $OA$, then* $\frac{PQ^2}{QA \cdot QA'} = \frac{OB^2}{OA^2}$.

**Proof.** By the definition of conjugate radii, $OB \parallel PP'$. The result follows from the power theorem for conics. □

We can now give further characterizations of four concyclic points on a conic.

**Theorem 2.4.4.** *Four points on an ellipse are concyclic if and only if the sum of their parameter angles is zero.*[18]

**Proof.** Let $A, B, C, D$ lie on an ellipse, let $A', B', C', D'$ be their corresponding points on the auxiliary circle, let their parameter angles be $\theta_A, \theta_B, \theta_C, \theta_D$, and let $l$ be the principal axis. Then

$$A, B, C, D \text{ are concyclic} \iff \measuredangle(l, AB) + \measuredangle(l, CD) = 0$$
$$\iff \measuredangle(l, A'B') + \measuredangle(l, C'D') = 0 \iff \theta_A + \theta_B + \theta_C + \theta_D = 0.$$

The first equivalence uses corollary 2.4.2; the second and third use proposition 2.2.12. □

**Theorem 2.4.5.** *Four points on a hyperbola are concyclic if and only if the product of their directed distances from one asymptote is $\frac{a^4b^4}{c^4}$, where $a, b, c$ are the semi-transverse axis, semi-conjugate axis, and semifocal distance, respectively.*

[17] The formula in exercise 1.25 contains absolute values. By reversing the chosen positive directions of directed segments where necessary, it can always be put in the following form.

[18] Parameter angles are directed modulo 360°.

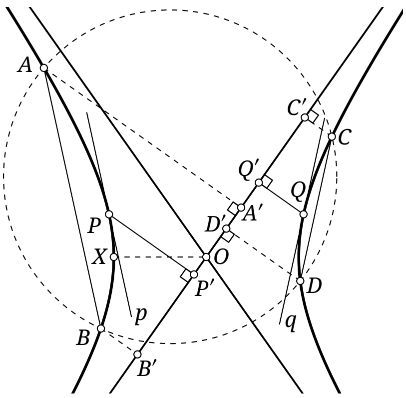


FIGURE 2.30

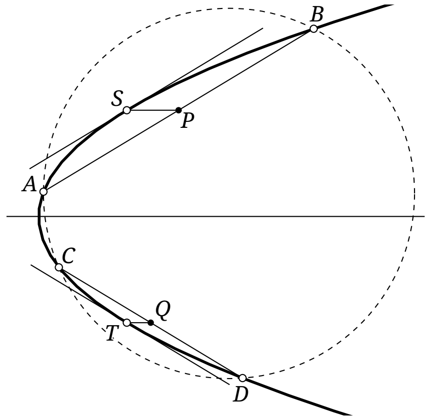


FIGURE 2.31

**Proof.** We first prove necessity. As in Figure 2.30, suppose that $A,B,C,D$ are concyclic points of the hyperbola, and let $A',B',C',D'$ be their respective projections on an asymptote $l$. We must show that

$$\overline{AA'}\cdot\overline{BB'}\cdot\overline{CC'}\cdot\overline{DD'}=\frac{a^4b^4}{c^4}.$$

At least two of the four points $A,B,C,D$ lie on the same branch of the hyperbola. Without loss of generality, let these be $A,B$; a tangent $p$ parallel to $AB$ therefore exists. By corollary 2.4.2, one bisector of the angle formed by the lines $AB$ and $CD$ is parallel to the transverse axis. Reflecting $p$ in the transverse or conjugate axis therefore gives a tangent $q$ parallel to $CD$.

We consider the case in which $p,q$ are symmetric about the conjugate axis; the case of the transverse axis is analogous. Let $p,q$ touch the hyperbola at $P,Q$, respectively, and let $P',Q'$ be the respective projections of $P,Q$ on $l$. By exercise 2.9, $\overline{AA'}\cdot\overline{BB'}=PP'^2$ and $\overline{CC'}\cdot\overline{DD'}=QQ'^2$.

Since $p,q$ are symmetric about the conjugate axis, $PQ$ is always parallel to the transverse axis $OX$, where $O$ is the center and $X$ is the left vertex. By exercise 2.9, $PP'\cdot QQ'=\operatorname{dist}^2(X,l)$. Consequently,

$$\overline{AA'}\cdot\overline{BB'}\cdot\overline{CC'}\cdot\overline{DD'}=\operatorname{dist}^4(X,l).$$

By proposition 2.1.4, we have $\tan\angle(XO,l)=\frac{b}{a}$. By proposition 2.1.7(5), $\sin\angle(XO,l)=\frac{b}{c}$. Thus $\operatorname{dist}(X,l)=OX\sin\angle(XO,l)=a\cdot\frac{b}{c}$, proving necessity.

For sufficiency, we do not yet know that $A,B,C,D$ are concyclic. The given product of directed distances is positive, and an asymptote separates the two branches of the hyperbola. Each branch therefore contains an even number of the four points. Without loss of generality, let $A,B$ lie on the same branch and $C,D$ lie on the same branch. We may therefore still choose tangents $p,q$ parallel to $AB,CD$ and take all the projections on $l$ as before, with

$$\overline{AA'}\cdot\overline{BB'}\cdot\overline{CC'}\cdot\overline{DD'}=(PP'\cdot QQ')^2.$$

Suppose that $P$ is fixed. Once the product of the directed distances is known, $QQ'$ is also known. On a branch of the hyperbola, however, the distance of a point from $l$ varies monotonically as the point moves along the branch. Such a point $Q$ can therefore only be the reflection of $P$ in the transverse or conjugate axis. Thus $p,q$ must be symmetric about the transverse or conjugate axis, and corollary 2.4.2 shows that the four points are concyclic. □

**Theorem 2.4.6.** *Four points on a parabola are concyclic if and only if the sum of their directed distances from the axis of the parabola is zero.*

**Proof.** We prove necessity only. As in Figure 2.31, let $A,B,C,D$ be four concyclic points on the parabola. We must prove that the sum of their directed distances from the axis is zero. Choose tangents parallel to $AB,CD$, touching the parabola at $S,T$, respectively, and let $l$ be the axis of the parabola.

Let $P,Q$ be the respective midpoints of $AB,CD$. By proposition 2.3.6, $\operatorname{dist}_\pm(A,l)+\operatorname{dist}_\pm(B,l)=2\operatorname{dist}_\pm(P,l)$, and $\operatorname{dist}_\pm(C,l)+\operatorname{dist}_\pm(D,l)=2\operatorname{dist}_\pm(Q,l)$. By corollary 2.4.2, the directed angles made by the tangents at $S,T$ with the axis are opposites (why?). They are therefore symmetric about the axis, and $\operatorname{dist}_\pm(P,l)+\operatorname{dist}_\pm(Q,l)=0$, proving the assertion. □

---

### Exercises

**Exercise 2.32.** Prove the sufficiency in theorem 2.4.6.

**Exercise 2.33.** In Cartesian coordinates, let the principal axis of a conic $\alpha$ be parallel to the $x$-axis. Prove that four points $A,B,C,D\in\alpha$ are concyclic if and only if $k_{AB}+k_{CD}=0$.

**Exercise 2.34.** Prove that the centroid of a triangle inscribed in a parabola lies on the axis of the parabola if and only if the circumcircle of the triangle passes through the vertex of the parabola.

**Exercise 2.35.** In Cartesian coordinates, let the principal axis of a conic $\alpha$ be parallel to the $x$-axis, and fix $P\in\alpha$. Variable points $A,B\in\alpha$ satisfy $k_{PA}+k_{PB}=0$. Prove that the slope of $AB$ is constant.

**Exercise 2.36.** Let a hyperbola with center $O$ be given, and let $P$ be any point of the plane. A line $l$ through $P$ meets the hyperbola at $A,B$. Through $P$ draw a parallel to one asymptote, meeting the hyperbola and the other asymptote at $C,D$, respectively. Prove that $\frac{PA\cdot PB}{PC\cdot OD}$ depends only on the direction of $l$.

**Exercise 2.37.** Let an ellipse $\alpha$ have center $O$, and let $l$ be one of the bisectors of the angle between its major and minor axes. A variable line $a\parallel l$ meets $\alpha$ at $A,B$. Prove that the circumcenter of $\triangle ABO$ moves on an equilateral hyperbola.

## 2.5 Circles of Curvature*

We now introduce the circles of curvature of conics. We begin with the general definition.

**Definition 2.5.1.** Let $\gamma$ be a smooth curve and $P\in\gamma$, as in Figure 2.32. The line $n$ through $P$ perpendicular to the tangent to $\gamma$ at $P$ is the *normal* to $\gamma$ at $P$. Choose $A,B\in\gamma$ and let both points tend to $P$ along $\gamma$. The circle $\lim_{A,B\to P}\odot(ABP)$ is the *osculating circle*, or *circle of curvature*, of $\gamma$ at $P$. Its radius $r$ is the *radius of curvature*, its center $O$ the *center of curvature*, and $1/r$ the *curvature* of $\gamma$ at $P$.

**Theorem 2.5.2.** *The osculating circle of a curve at a point is tangent to the curve there, and its center lies on the normal at that point.*

**Proof.** This is immediate from the definition. □

We now study the curvature of conics.

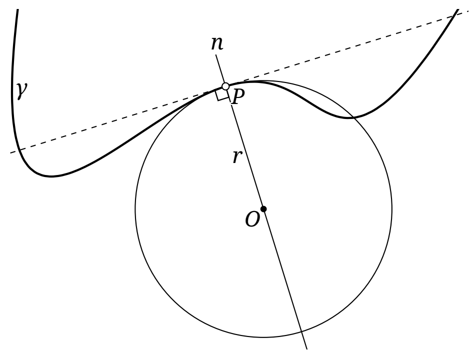


FIGURE 2.32

**Proposition 2.5.3.** *Let $l$ be the tangent to a conic $\alpha$ at $P$, and suppose the osculating circle at $P$ meets $\alpha$ again at $Q$.*[19] *Of the two angle bisectors of $l$ and $PQ$, one is parallel to the axis of $\alpha$ and the other perpendicular to it.*

**Proof.** First regard the point of osculation as three distinct points $P, A, B$, and then take the limit. Since $l = \lim_{A,B\to P} AB$, the assertion follows from corollary 2.4.2. □

We first consider the curvature of a parabola.

**Lemma 2.5.4.** *Let $F$ be the focus of a parabola $\alpha$, and let $P \in \alpha$. The chord of the osculating circle at $P$ that passes through $P$ and is parallel to the axis of $\alpha$ has length $4FP$.*

**Proof.** As in Figure 2.33, choose $Q \in \alpha$, $Q \neq P$, and draw through $Q$ the parallel to the axis, meeting the tangent at $P$ at $U$. Let $\omega$ be the circle through $Q$ tangent to $PU$ at $P$. As $Q \longrightarrow P$, the circle $\omega$ tends to the osculating circle at $P$.

Let $UQ$ meet $\omega$ again at $W$, and complete the parallelogram $UPVQ$. Since $PU$ is tangent to $\alpha$ at $P$ and $QV \parallel UP$, the diameter property shows that the chord corresponding to $QV$ is bisected by the diameter $PV$. By proposition 2.3.11, $QV^2 = 4PF \cdot PV$. The tangent–secant theorem gives

$$UW = \frac{UP^2}{UQ} = \frac{QV^2}{PV} = 4PF.$$

As $Q \longrightarrow P$, we have $U \longrightarrow P$, and therefore $\lim_{Q\to P} UW = 4PF$ is the required chord length. □

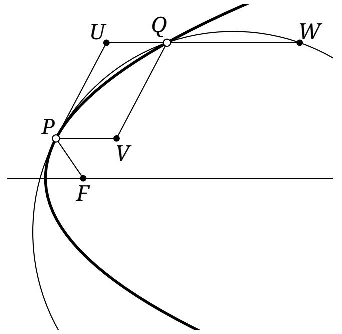


FIGURE 2.33

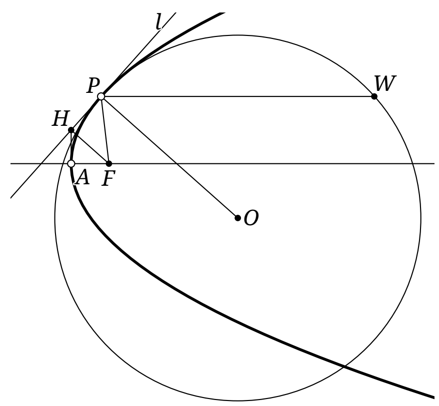


FIGURE 2.34

[19] We have already explained that, in the real Euclidean plane, two conics have at most four intersections, with tangencies counted with multiplicity. By the definition of an osculating circle, it is the limiting case of a circle through three points of the curve; hence its point of tangency with the conic has multiplicity three—or, in a special case, four—and the osculating circle and the conic may therefore have one further intersection.

**Proposition 2.5.5.** *Let a parabola $\alpha$ have focus $F$ and vertex $A$. For $P \in \alpha$, let $l$ be the tangent at $P$ and let the osculating circle be $\odot(O, OP)$. Then*

$$OP = \frac{2PF^2}{\operatorname{dist}(F,l)} = \sqrt{\frac{4PF^3}{AF}}.$$

**Proof.** As in Figure 2.34, let $A$ be the vertex of $\alpha$, let $H$ be the projection of $F$ on $l$, and let the line through $P$ parallel to the axis meet $\odot O$ again at $W$. By lemma 2.5.4, $PW = 4PF$. Since $OP \perp l$,

$$OP = \frac{PW}{2\cos\angle OPW} = \frac{4PF}{2\sin\angle(PW,l)} = \frac{2PF}{\sin\angle HPF} = \frac{2PF}{HF/PF} = \frac{2PF^2}{\operatorname{dist}(F,l)},$$

where the third equality uses the optical property of the parabola.

By lemma 2.3.12, $\triangle FAH \sim \triangle FHP$, so $\frac{FH}{FA} = \frac{FP}{FH}$. Thus $HF = \sqrt{AF \cdot PF}$, and hence

$$OP = \frac{2PF^2}{HF} = \sqrt{\frac{4PF^3}{AF}}.$$ □

We next consider osculating circles of ellipses and hyperbolas.

**Lemma 2.5.6.** *Let $O$ be the center of a central conic $\alpha$. For $P \in \alpha$, let $OS$ be the conjugate radius of $OP$. The chord of the osculating circle at $P$ passing through $P$ and $O$ has length $\frac{2OS^2}{OP}$.*

**Proof.** We treat the ellipse shown in Figure 2.35; the hyperbolic case is entirely analogous. Choose $Q \in \alpha$, $Q \neq P$, and through $Q$ draw the parallel to $OP$, meeting the tangent at $P$ at $U$. Let $\omega$ be the circle through $Q$ tangent to $PU$ at $P$. As $Q \to P$, the circle $\omega$ tends to the osculating circle at $P$.

Let $UQ$ meet $\omega$ again at $W$, and complete the parallelogram $UPVQ$. Since $PU$ is tangent to $\alpha$ at $P$ and $QV \parallel UP$, the diameter property says that the chord corresponding to $QV$ is bisected by the diameter $PV$. Let $P'$ be the other intersection of $OP$ with $\alpha$. By corollary 2.4.3, $QV^2 = (VP \cdot VP') \cdot \frac{OS^2}{OP^2}$. The tangent–secant theorem gives

$$UW = \frac{UP^2}{UQ} = \frac{QV^2}{VP} = VP' \cdot \frac{OS^2}{OP^2}.$$

As $Q \to P$, both $U, V \to P$, and hence

$$\lim_{Q \to P} UW = PP' \cdot \frac{OS^2}{OP^2} = \frac{2OS^2}{OP},$$

which is the required chord length. □

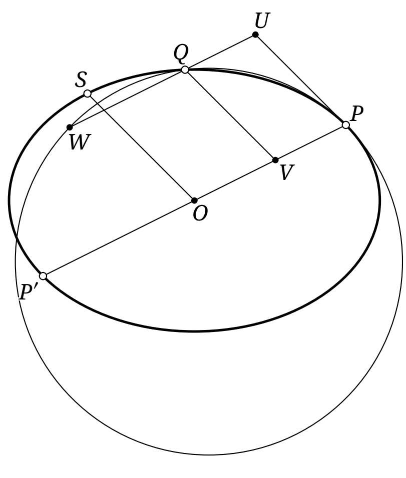

FIGURE 2.35

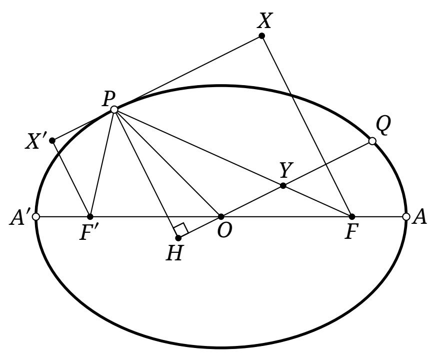

FIGURE 2.36

**Lemma 2.5.7.** *Let $O$ be the center of a central conic $\alpha$, let $P \in \alpha$, and let $OQ$ be the conjugate radius of $OP$. If the normal at $P$ meets $OQ$ at $H$, then $PH \cdot OQ = AO \cdot BO$, where $A, B$ are respectively a principal-axis vertex and a secondary-axis vertex.*

**Proof.** We treat the ellipse shown in Figure 2.36; the hyperbolic case is entirely analogous. Let $A, A'$ be the major-axis vertices and $F, F'$ the foci. Let $X, X'$ be the respective projections of $F, F'$ on the tangent at $P$. By the definition of conjugate diameters and the diameter property, $XX' \parallel OQ$. Put $Y = PF \cap OQ$.

The optical property gives $\angle F'PX' = \angle FPX$, and therefore $\triangle F'PX' \sim \triangle FPX$. Thus $\frac{F'X'}{F'P} = \frac{FX}{FP}$. The same optical property gives $\angle F'PH = \angle FPH$, while $X'F' \parallel PH$, so $\angle X'F'P = \angle HPF$. Hence $\triangle F'X'P \sim \triangle PHY$, and $\frac{PH}{PY} = \frac{F'X'}{F'P}$. Consequently,

$$\frac{PH^2}{OA^2} = \frac{PH^2}{PY^2} = \frac{F'X'^2}{F'P^2} = \frac{F'X' \cdot FX}{F'P \cdot FP} = \frac{|OA^2 - OF^2|}{OQ^2} = \frac{OB^2}{OQ^2},$$

where the first equality uses exercise 2.22, the fourth uses propositions 1.3.7 and 2.2.19, and the last uses corollary 2.2.7. Taking nonnegative square roots and rearranging gives $PH \cdot OQ = OA \cdot OB$. $\square$

**Proposition 2.5.8.** *Let $O$ be the center of a central conic $\alpha$. For $P \in \alpha$, let $OQ$ be the conjugate radius of $OP$, and let $O'$ be the center of the osculating circle at $P$. Then*

$$O'P = \frac{OQ^2}{\operatorname{dist}(P, OQ)} = \frac{OQ^3}{OA \cdot OB},$$

*where $A, B$ are respectively a principal-axis vertex and a secondary-axis vertex.*

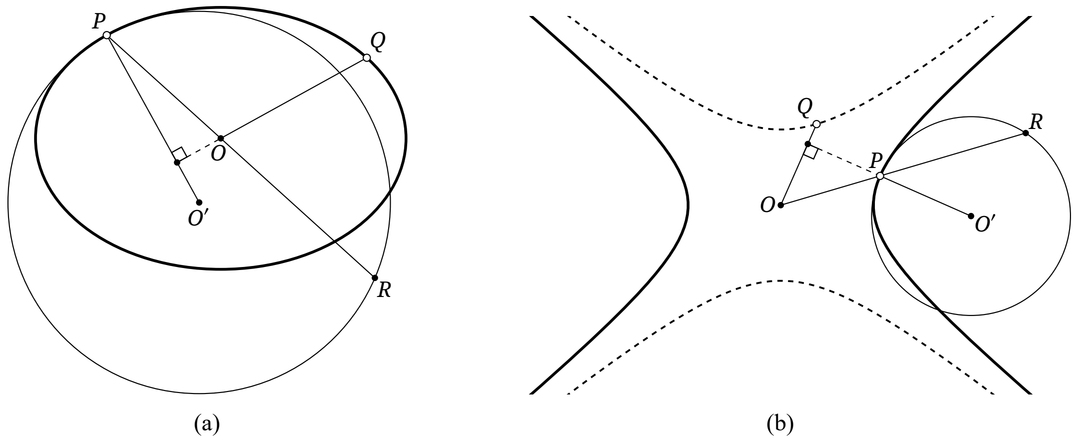


FIGURE 2.37

**Proof.** As in Figure 2.37, let $OP$ meet $\odot O'$ again at $R$. By lemma 2.5.6, $PR = \frac{2OQ^2}{OP}$. Therefore

$$O'P = \frac{PR}{2\cos\angle O'PR} = \frac{OQ^2}{OP}\sec\angle O'PR = \frac{OQ^2}{OP} \cdot \frac{OP}{\operatorname{dist}(P, OQ)} = \frac{OQ^2}{\operatorname{dist}(P, OQ)} = \frac{OQ^3}{OA \cdot OB},$$

where the last equality follows from lemma 2.5.7. $\square$

For all three conics the radius of curvature has the following uniform expression.

**Proposition 2.5.9.** *Let $F$ be a focus of a conic $\alpha$, and let $P \in \alpha$. If $r$ is the radius of the osculating circle at $P$, and $\theta$ is the angle between the normal at $P$ and $PF$, then $r\cos^3\theta = ep$, where $e$ is the eccentricity and $p$ the focal parameter of $\alpha$.*

**Proof.** First consider a central conic. Let $O$ be its center and let $a,b,c$ be its semi-principal axis, semi-secondary axis, and semifocal distance, respectively. Let $OQ$ be the conjugate radius of $OP$, and put $Y = PF \cap OQ$. By exercise 2.22, $PY = a$, and $\cos\theta = \frac{\operatorname{dist}(P,OQ)}{PY} = \frac{\operatorname{dist}(P,OQ)}{a}$. Thus, by proposition 2.5.8 and lemma 2.5.7,

$$r\cos^3\theta = \frac{OQ^3}{ab}\cdot\left(\frac{\operatorname{dist}(P,OQ)}{a}\right)^3 = \frac{OQ^3}{ab}\cdot\left(\frac{b}{OQ}\right)^3 = \frac{b^2}{a} = \frac{c}{a}\cdot\frac{b^2}{c} = ep.$$

For a parabola, let $l$ be the tangent at $P$. Since $\angle(FP,l)+\theta = 90°$, $\cos\theta = \sin\angle(FP,l) = \frac{\operatorname{dist}(F,l)}{FP}$. The proof of proposition 2.5.5 gave $\operatorname{dist}(F,l) = \sqrt{AF\cdot PF}$, so $\cos\theta = \sqrt{\frac{AF}{PF}}$. Hence

$$r\cos^3\theta = \sqrt{\frac{4PF^3}{AF}}\cdot\left(\sqrt{\frac{AF}{PF}}\right)^3 = 2AF = p = ep. \qquad \square$$

Isaac Newton famously derived the law of universal gravitation from Johannes Kepler's laws by a geometric argument. We can make a similar derivation, though not by precisely Newton's method. Suppose a particle $P$ of mass $m$ moves under a central force about a force center $S$ along a conic $\alpha$, with $S$ as a focus. If its speed at $P$ is $v$, its velocity is tangent to $\alpha$. Let the angle between the normal at $P$ and the position vector $\boldsymbol{\rho} = \overrightarrow{SP}$ be $\theta$, and let the central force be $\boldsymbol{F} = F(\rho)\,\hat{\boldsymbol{\rho}}$, where $\hat{\boldsymbol{\rho}}$ is the radial unit vector. Applying Newton's law of motion in the normal direction gives $F\cos\theta = \frac{mv^2}{r}$, where $r$ is the radius of curvature of $\alpha$ at $P$. Conservation of angular momentum gives $L = mv\rho\cos\theta = \text{const}$. Therefore $F = \frac{L^2}{m\rho^2}\frac{1}{r\cos^3\theta}$. By proposition 2.5.9, $F = \frac{L^2}{mep\rho^2}$, which is the inverse-square law.

There is also a uniform construction of the center of curvature. We first prove a lemma.

**Lemma 2.5.10.** *Let $F$ be a focus of a conic $\alpha$ and let $P \in \alpha$. Suppose the normal at $P$ meets the principal axis at $N$, and let $K$ be the projection of $N$ on $PF$. Then $KP = ep$, where $e$ is the eccentricity and $p$ the focal parameter of $\alpha$.*

**Proof.** We treat the ellipse shown in Figure 2.38; the hyperbolic and parabolic cases are entirely analogous. Let $l$ be the directrix, let $S$ be the projection of $P$ on $l$, and let the tangent at $P$ meet $l$ at $T$. By proposition 2.2.8, $TF \perp FP$, so $S,T,F,P$ are concyclic and $\angle PSF = \angle PTF$. Since $TP \perp PN$ and $TF \perp PF$, we also have $\angle FPN = \angle PTF$. Thus $\angle PSF = \angle FPN$.

Also $SP \parallel FN$, so $\angle SPF = \angle PFN$, and therefore $\triangle SPF \sim \triangle PFN$. Let $L$ be the projection of $F$ on $PS$. The points $K,L$ correspond under this similarity, and hence $PK = SL\cdot\frac{PF}{PS} = pe$. $\square$

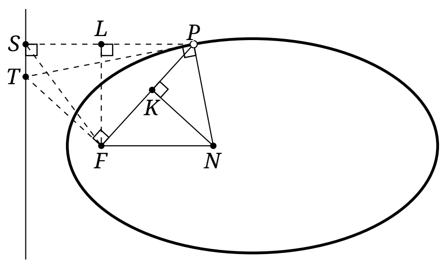


FIGURE 2.38

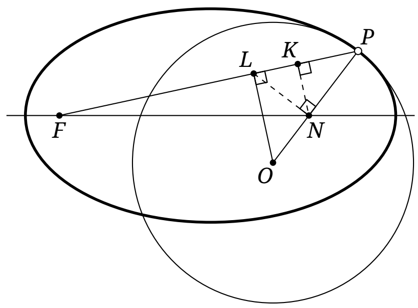


FIGURE 2.39

**Proposition 2.5.11.** *Let $F$ be a focus of a conic $\alpha$ and $P \in \alpha$. Suppose the normal at $P$ meets the principal axis at $N$. Draw $NL \perp PN$ to meet $PF$ at $L$, and draw $LO \perp FP$ to meet $PN$ at $O$. Then $O$ is the center of curvature at $P$.*

**Proof.** The elliptic case is shown in Figure 2.39. Let $K$ be the projection of $N$ on $FP$, and put $\angle FPN = \theta$. Then

$$OP = LP\sec\theta = NP\sec^2\theta = KP\sec^3\theta = ep\sec^3\theta,$$

where the last equality is lemma 2.5.10. The result now follows from proposition 2.5.9. □

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## Exercises

**Exercise 2.38.** Given a conic $\alpha$ and a point $P \in \alpha$, construct the center of curvature at $P$ with straight-edge and compass.

**Exercise 2.39.** Let $\alpha$ be a central conic. Prove that the tangents at the four endpoints of any pair of conjugate diameters enclose a parallelogram whose area is constant.[20]

**Exercise 2.40.** A conic has eccentricity $e$ and focal parameter $p$. Find its curvature at a principal-axis vertex. If it is an ellipse, also find the curvature at a minor-axis vertex.

**Exercise 2.41*.** Suppose the trajectory of a particle in a central force field is a central conic whose center is the force center. Determine the form of the force field.

**Exercise 2.42.** Let an equilateral hyperbola $\alpha$ have center $O$. For $A \in \alpha$, draw $AC \perp OA$ to meet $\alpha$ again at $C$, and choose $D$ so that $4\overline{AD} = \overline{AC}$. Prove that the center of curvature of $\alpha$ at $A$ is $\mathfrak{r}_D(O)$, the reflection of $O$ in $D$.

**Exercise 2.43.** Let an ellipse $\alpha$ have semimajor axis $a$, semiminor axis $b$, and center $O$, and let $P$ lie on its minor axis. Determine the range of $OP$ for which every circle centered at $P$ has fewer than four distinct intersections with $\alpha$.

**Exercise 2.44.** Let $A,B,C,D$ be points of a conic $\alpha$, and let $A',B',C',D'$ be the respective second intersections with $\alpha$ of the osculating circles at $A,B,C,D$. Prove that:
(1) if $\alpha$ is not an ellipse, then $A,B,C,D$ are concyclic if and only if $A',B',C',D'$ are concyclic;
(2) if $\alpha$ is an ellipse, then the concyclicity of $A',B',C',D'$ is necessary but not sufficient for the concyclicity of $A,B,C,D$.

**Exercise 2.45.** Let a conic $\alpha$ have eccentricity $e \in \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{5}}{2}\right)$. As $P$ varies on $\alpha$, let $d$ be the length of the chord cut from the normal at $P$ by $\alpha$. Prove that $d$ is minimal if and only if the center of curvature at $P$ also lies on $\alpha$.

**Exercise 2.46*.** A rectangle $ABCD$ has three of its vertices on a conic of eccentricity $e$ and focal parameter $p$. Find a lower bound for the minimum perimeter of the rectangle.

[20]For a diameter that does not meet a hyperbola, take the tangents to the conjugate hyperbola at its two intersections with that diameter.

**Chapter 3**

# Foundations of Projective Geometry

This chapter introduces the basic concepts of projective geometry for use in later applications.

## 3.1 Projective Transformations and Their Basic Properties

We begin with projective transformations in general.

### 3.1.1 Basic Concepts of Projective Geometry

Before proceeding, readers should review the basic notions of sets and maps. Those unfamiliar with them may consult appendix A; only section A.1.1 is needed for what follows.

We first introduce affine and projective planes.

**Definition 3.1.1** (Affine plane)**.**

(a) To every line in a Euclidean plane, adjoin a new element, called its *point at infinity*, that is, a point at infinity on the line. Below we write $\infty_l$ for the point at infinity on a line $l$; the original points of the line are correspondingly called *finite points*. Every original Euclidean line has exactly one point at infinity, and two parallel lines meet at the same point at infinity.

(b) All points at infinity in a plane form a line, called the *line at infinity*; below we write it as $l_\infty$. A line other than the line at infinity is called a *finite line*.

(c) A Euclidean line after its point at infinity has been adjoined is called an *affine line*. A Euclidean plane after the line at infinity has been adjoined is called an *affine plane* $\mathbb{A}^2$.

**Remark.** The following comments clarify the definition.

(a) One may wonder why a line is assigned only one point at infinity even though points on it can tend to infinity in two directions. If the two directions gave two distinct points at infinity, parallel lines would meet at two points, in sharp contrast with the Euclidean property that two nonparallel lines have a unique intersection. Identifying the two directions makes parallel lines meet at a unique point as well. Thus any two affine lines have exactly one point of intersection, a structure much more compatible with the original Euclidean plane and convenient for further discussion.

(b) The convention that all points at infinity form a line is consistent: because each original Euclidean line contains exactly one such point, the locus of the points at infinity meets every line in exactly one point.

(c) The preceding definition implicitly starts from a real Euclidean plane, so the resulting object is more precisely a *real affine plane*. One may instead adjoin a line at infinity to a complex Euclidean plane, obtaining a *complex affine plane* $\mathbb{C}\mathbb{A}^2$. Both are affine planes, but unless stated otherwise $\mathbb{A}^2$ will mean the real affine plane.[1] Points of the complex affine plane are classified as *real points* and *imaginary points*. The real points are the real points of the corresponding Euclidean plane and the points at infinity of its real lines; all others are imaginary. Lines are similarly classified as *real lines* and *imaginary lines*, a real line being one that contains infinitely many real points and all other lines being imaginary. As in the complex Euclidean plane, an imaginary line contains at most one real point.

[1] When it is important to emphasize the real ground field, one may write $\mathbb{R}\mathbb{A}^2$.

(d) More generally, one defines an $n$-dimensional *affine space* $\mathbb{A}^n$ in the same way: adjoin one point at infinity to every line in an $n$-dimensional space and let all such points form an $(n-1)$-dimensional hyperplane. For example, in three-dimensional affine space the points at infinity form a plane. This produces the $n$-dimensional affine space $\mathbb{A}^n$. Two parallel lines in $\mathbb{A}^n$ still meet at the same point at infinity.

The notion of a projective plane now follows from that of an affine plane.

**Definition 3.1.2.** If finite points and the point at infinity of an affine line are treated on equal terms, without distinction, the line is called a *projective line*. Likewise, if the finite and infinite elements of an affine plane $\mathbb{A}^2$ are treated on equal terms, it becomes a *projective plane* $\mathbb{P}^2$.

**Remark.**

(a) Starting from a real affine plane gives a *real projective plane* $\mathbb{RP}^2$. Starting from a complex affine plane gives a *complex projective plane* $\mathbb{CP}^2$, whose points and lines may again be classified as real or imaginary.

(b) Both real and complex projective planes are projective planes. Unless otherwise stated, "projective plane" will mean a real projective plane.[2] As with affine spaces, the notion extends to an $n$-dimensional *projective space* $\mathbb{P}^n$. In particular, a projective line is the one-dimensional projective space $\mathbb{P}^1$.

(c) In later applications we shall not always distinguish sharply between the projective and affine planes. For example, a configuration may first be treated projectively and then its finite and infinite elements separated for further argument. No difficulty results from this practice within the scope of this book.

(d) The definitions of affine and projective planes used here follow [**3**] and are intended to give beginners a simplified point of entry. More standard definitions will appear later, for example in section 10.5.1; this makes no difference to the later purely geometric applications. For the moment it is enough to understand elements at infinity.

The following Euclidean model helps visualize affine and projective planes.[3]

In figure 3.1, let $\mathbb{S}^2$ be the unit sphere centered at $O$, with equator the great circle $c$. Identify each antipodal pair on $c$ and regard the pair as one point at infinity, such as $\infty_l$ in the figure. Regard each point of the lower hemisphere $S$, apart from $c$, as a finite point.

The equator $c$ then represents the line at infinity $l_\infty$. A semicircle of a great circle in the lower hemisphere represents a finite line $l$, while its two intersections with $c$ are identified as the single point $\infty_l$.

This gives a Euclidean model of the affine plane. Let the plane $\pi$ be tangent to $S$ at its south pole $S_0$. For $A \in S$, put $OA \cap \pi = A'$. The map $\varphi\colon S \to \pi$, $A \mapsto A'$, is a bijection and sends the equator, the line at infinity of $S$, to the line at infinity of $\pi$.

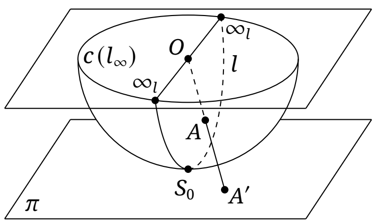


FIGURE 3.1

This model still singles out the equator from the other points of $S$. To put all points on the same footing, consider the whole unit sphere $\mathbb{S}^2$ and identify every antipodal pair. Denote the resulting object by $\mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$.

After any great circle has been chosen as equator, antipodal identification makes the upper and lower open hemispheres equivalent, so that only the lower hemisphere need be considered. Antipodal points of the equator are likewise identified, and the affine model above is recovered.

[2]When the real ground field is to be emphasized, one may write $\mathbb{RP}^2$.

[3]This supplementary discussion may be skipped; it is not needed for the material that follows.

The map $\varphi$ and its inverse $\varphi^{-1}$ are intuitively continuous. The discussion may therefore be summarized formally as follows.

**Theorem 3.1.3.** *The real projective plane admits a homeomorphism* $\mathbb{RP}^2 \cong \mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$.

In rough terms, identifying antipodal points of $\mathbb{S}^2$ gives the *quotient space* $\mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$, which is homeomorphic to $\mathbb{RP}^2$.[4] A precise definition of homeomorphism is given in the appendix. Informally, two spaces are homeomorphic if one can be continuously deformed into the other; more precisely, there is a bijection whose map and inverse are both continuous. Since our purpose here is geometric intuition, we construct the bijection geometrically without proving continuity rigorously.

To speak of a homeomorphism one must in fact first give $\mathbb{RP}^2$ a topology. It is usually defined from three-dimensional Euclidean space $\mathbb{R}^3$ by the quotient topology.

**Theorem 3.1.4.** *Fix a point $O$ in $\mathbb{R}^3$, give $\mathbb{R}^3 \setminus \{O\}$ the subspace topology, and define an equivalence relation $\sim$ on this space by $P \sim Q \Longleftrightarrow Q \in OP$. Then*

$$(\mathbb{R}^3 \setminus \{O\})/\sim \ \cong \mathbb{RP}^2.$$

**Proof.** This statement is more naturally taken as the definition of the topology on $\mathbb{RP}^2$; it is the most direct way to define the real projective plane as a topological space. Here we only explain geometrically why it gives the model obtained by adjoining points at infinity. By the preceding model, it suffices to show

$$\mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x}) \cong (\mathbb{R}^3 \setminus \{O\})/\sim.$$

Embed $\mathbb{S}^2$ in $\mathbb{R}^3$ with center $O$. Every line through $O$ meets $\mathbb{S}^2$ in an antipodal pair. Define

$$f: \mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x}) \cong (\mathbb{R}^3 \setminus \{O\})/\sim, \qquad [P] \mapsto [P].$$

The class $[P]$ on the sphere consists of $P$ and its antipode; its image is the class $[P] = OP \setminus \{O\}$ in the second quotient. Thus $f$ simply associates an antipodal pair with the line through it and $O$, and both $f$ and $f^{-1}$ are intuitively continuous. In keeping with the geometric purpose of this chapter, we omit a formal proof of continuity.

Equivalently, one may model $\mathbb{RP}^2$ by the set of all lines through $O$ in $\mathbb{R}^3$. Each such line represents a point of $\mathbb{RP}^2$, and the plane through $O$ determined by two such lines represents their joining line. Fix a plane $\pi$ through $O$. Its lines represent the points at infinity of $\mathbb{RP}^2$, and the intersection of any plane through $O$ with $\pi$ represents the point at infinity on the corresponding projective line. $\square$

The relevant topological notions are also discussed in the appendix. They are not otherwise needed here and may be skipped by readers unfamiliar with them.

We can now define a perspectivity.

**Definition 3.1.5.** Let $\pi, \pi'$ be two distinct projective planes, and let $f: \pi \to \pi'$ be a bijection taking points of $\pi$ to points of $\pi'$. If, for every $P \in \pi$, the line joining $P$ to $f(P)$ passes through a fixed point $O$ outside both planes, then $f$ is called a *perspectivity*, or *perspective correspondence*. The point $O$ is its *perspector*, or *perspective center*. If $\pi \cap \pi' = l$, the line $l$ is its *perspectrix*, or *perspective axis*.

Every point of the perspective axis is fixed. A perspectivity also preserves lines, and is therefore a special projective correspondence.

A perspectivity is precisely the central projection of section 1.4. In figure 3.2, points $A, B, C, D, E$ of $\pi$ are sent, under the perspectivity with center $O$, to $A', B', C', D', E'$ of $\pi'$. Since $D$ lies on the perspective axis $l$, we have $D' = D$. Since $EO \parallel \pi'$, the point $E'$ is at infinity in $\pi'$. Such a point $E$ is called a *vanishing point* of $\pi$ under this perspectivity, and all vanishing points of $\pi$ form the *vanishing line*. The plane through $O$ parallel to $\pi'$ meets $\pi$ precisely in this vanishing line $l_0$.

The position of a conic relative to the line at infinity can be described by perspectivity.

**Proposition 3.1.6.** *For a conic:*

*(1) an ellipse, and in particular a circle, is disjoint from the line at infinity and has no point at infinity;*

[4] The construction is called a quotient space of $\mathbb{S}^2$.

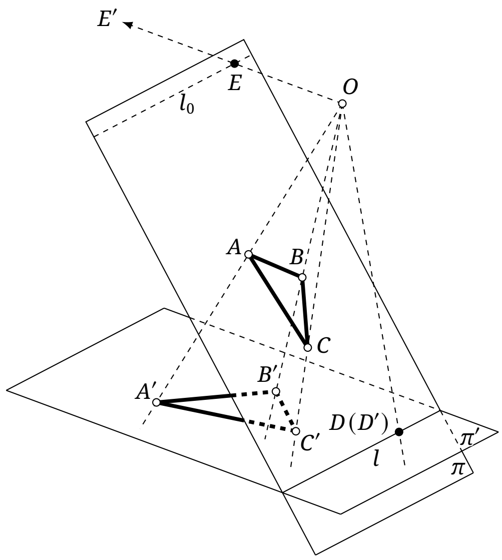


FIGURE 3.2

*(2) a hyperbola meets the line at infinity in two distinct points, at which its two asymptotes are tangent to it;*
*(3) a parabola is tangent to the line at infinity and has one point at infinity, lying on its axis.*

**Proof.** For (1), in figure 1.12a of section 1.4, central projection—that is, a perspectivity $f$—takes the circle $\omega$ in the base plane of the cone to an ellipse $\alpha$ in $\pi$. A point of $\pi$ at infinity would have its preimage on the vanishing line of the plane of $\omega$. The plane through $S$ parallel to $\pi$ meets the plane of $\omega$ in a line outside $\omega$. Thus $\omega$ contains no preimage of a point of $\pi$ at infinity, and $\alpha$ has no point at infinity.

For (2), in figure 1.12b of section 1.4, the perspectivity $f$ takes the circle $\omega$ in the base plane of the cone to a hyperbola $\alpha$ in $\pi$. Find the vanishing line in the plane of $\omega$ as in (1): the plane through $S$ parallel to $\pi$ meets the plane of $\omega$ in a line cutting $\omega$ at two distinct points. These are the two preimages of points at infinity of $\pi$, so the image hyperbola $\alpha$ has two distinct points at infinity.

For (3), in figure 1.13 of section 1.4, let the perspectivity $f$ carry the circle $\omega$ in the base plane of the cone to a parabola $\alpha$ in $\pi$. To find the vanishing line in the plane of $\omega$, note that the cone $S-\omega$ now has exactly one generator $l_0$ parallel to $\pi$. Hence the plane through $S$ parallel to $\pi$ meets the plane of $\omega$ in a line tangent to $\omega$ at $P=l_0\cap\omega$. Thus the vanishing line is tangent to $\omega$ at $P$. The image conic $\alpha$ is therefore tangent to the line at infinity and has one point at infinity. By symmetry $l_0$ is parallel to the axis of $\alpha$, so $f(P)$ is the point at infinity of that axis. Thus the point at infinity on the axis of $\alpha$ is precisely the point at infinity on $\alpha$. $\square$

**Remark.** The proposition was stated in the real projective plane. In the complex projective plane, the equation of an ellipse or circle is quadratic and that of a finite line is linear. The fundamental theorem of algebra therefore gives two intersections, with tangency counted twice. Letting the finite line tend to the line at infinity shows that the latter also meets an ellipse or circle twice, though at imaginary points.

The proposition also has an intuitive interpretation. For a hyperbola, let a point approach one asymptote and tend to infinity. Its tangent approaches that asymptote, so each asymptote is tangent to the hyperbola at one point at infinity, giving two points at infinity in all. For a parabola, let a point move farther and farther along the curve. Its tangent gradually becomes parallel to the axis: the point at infinity on the axis thus approaches the tangent at the moving point. When the moving point reaches infinity, it becomes that point at infinity on the axis. Whichever direction along the parabola the point follows, it tends to the same point at infinity on the axis. The line at infinity therefore has only one intersection with the parabola, or two coincident intersections, and is tangent to it.

We next define projective correspondences and transformations.

**Definition 3.1.7.** Let $\pi$ and $\pi'$ be projective planes, possibly the same. A bijection $f : \pi \to \pi'$ taking points of $\pi$ to points of $\pi'$ is a *projective correspondence* if it is a composition of finitely many perspectivities. When $\pi = \pi'$, it is called a *projective transformation*.

The definition immediately gives the following result.

**Theorem 3.1.8.** *The composition of projective correspondences is a projective correspondence. Every projective correspondence is invertible, and its inverse is a projective correspondence. The composition of projective transformations is a projective transformation. Every projective transformation is invertible, and its inverse is a projective transformation.*

**Proof.** A projective correspondence is a composition of perspectivities. Each perspectivity is a bijection and hence invertible, with inverse again a perspectivity. The inverse of the composition is therefore also a composition of perspectivities, hence a projective correspondence. A projective transformation is simply a projective correspondence from a plane to itself, so its inverse is again a projective transformation. □

It follows that all projective transformations of a projective plane $\pi$ form a group under composition.[5] Transformation groups illustrate the entry of algebraic ideas into geometry and the passage from elementary to higher geometry.

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## Exercises

**Exercise 3.1.** For each of the following sets $G$, determine whether $(G,\circ)$ is a group, where $\circ$ denotes composition of transformations.
(1) all rotations of the Euclidean plane;
(2) all translations of the Euclidean plane;
(3) all similarities of the Euclidean plane;
(4) all homotheties of the Euclidean plane;
(5) all spiral similarities of the Euclidean plane;
(6) all projective transformations of a projective plane, with $\circ$ again denoting composition.
Among the sets in (1)–(5) that are groups, determine every subgroup relation.

**Exercise 3.2.** Must a composition of perspectivities be a perspectivity? Give a reason for your answer.

**Exercise 3.3.** A perspectivity takes $\triangle ABC$ in a projective plane $\pi$ to $\triangle A'B'C'$ in $\pi'$. Put $P = BC \cap B'C'$, $Q = CA \cap C'A'$, and $R = AB \cap A'B'$. Must $P, Q, R$ be collinear? Give a reason for your answer.

**Exercise 3.4.** Let $f : \pi \to \pi'$ be a perspectivity with center $O$, and let $l$ be the vanishing line in $\pi$. Prove that:
(1) two lines meeting on $l$ are carried by $f$ to parallel lines;
(2) if $P, Q \in l$ are fixed and $R$ is arbitrary in $\pi$, then $\angle f(P)f(R)f(Q)$ is constant;
(3) if $p$ is a fixed line in $\pi$ and $f$ varies with $O$, then the line $p' = f(p)$ passes through a fixed point for every choice of $O$.

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[5] Readers unfamiliar with groups may consult the appendix; only section A.3.1 is needed here.

**Exercise 3.5*.** Give Euclidean models of the affine and projective lines. Prove, or explain geometrically, that

$$\mathbb{P}^1 \cong \mathbb{S}^1/(\boldsymbol{x} \sim -\boldsymbol{x}) \cong \mathbb{S}^1,$$

where $\mathbb{S}^1$ is a circle in the Euclidean plane.

**Exercise 3.6**.** Let $O \in \mathbb{RP}^2$. Prove, or explain geometrically, that $\mathbb{RP}^2 \setminus \{O\} \cong M$, where $M$ is the open *Möbius band*. Geometrically, start with a rectangular strip without the edges of one pair of opposite sides, and glue the other pair together with reversed orientation. Precisely,

$$M = ([0,1] \times (-1,1))/((0,y) \sim (1,-y)).$$

**Exercise 3.7**.** As with $\mathbb{RP}^2$, the complex projective plane can be viewed as a quotient of complex Euclidean space. Fix $O \in \mathbb{C}^3$ and define on $\mathbb{C}^3 \setminus \{O\}$ the equivalence relation $\sim$ $P \sim Q \Longleftrightarrow Q \in OP$. Then topologically $\mathbb{CP}^2 \cong (\mathbb{C}^3 \setminus \{O\})/\sim$.

Embed the five-sphere $\mathbb{S}^5$ in $\mathbb{R}^6$ with center $O$. Fix six distinct pairwise perpendicular lines $l_1, l_2, \ldots, l_6$ through $O$. Let $\pi_1, \pi_2, \pi_3$ be the planes determined respectively by $(l_1, l_2)$, $(l_3, l_4)$, and $(l_5, l_6)$, and choose a positive orientation for angles in each plane. Given $P \in \mathbb{S}^5$ and an angle $\theta$, let $P_i$ be the projection of $P$ onto $\pi_i$, rotate $P_i$ about $O$ through $\theta$ in the chosen direction to obtain $P_i'$, and define $Q_P(\theta)$ by

$$\overrightarrow{OQ_P(\theta)} = \overrightarrow{OP_1'} + \overrightarrow{OP_2'} + \overrightarrow{OP_3'}.$$

(1) For fixed $P$, prove that the locus of $Q_P(\theta)$ as $\theta$ varies is a circle in $\mathbb{S}^5$ centered at $O$.
(2) Denote the circle corresponding to $P$ in (1) by $C_P$. Define an equivalence relation $\backsim$ on $\mathbb{S}^5$ by $P \backsim P' \Longleftrightarrow P' \in C_P$. Prove, or explain geometrically, that $\mathbb{CP}^2 \cong \mathbb{S}^5/\backsim$.

### 3.1.2 Basic Properties of Projective Correspondences

Projective correspondences have the following basic properties.

**Theorem 3.1.9.** *Under a projective correspondence, points remain points and lines remain lines; collinear points remain collinear, and concurrent lines remain concurrent.*

**Proof.** These properties are required by the definition of a projective correspondence. □

**Theorem 3.1.10.** *Under a projective correspondence, intersecting curves remain intersecting, disjoint curves remain disjoint, and tangent curves remain tangent.*

**Proof.** If two curves have a common point, the image of that point lies on the images of both curves, so intersecting curves remain intersecting. If two disjoint curves had intersecting images, invertibility and the preceding observation would imply that the original curves intersect, a contradiction. Thus disjoint curves remain disjoint.

It remains to prove the assertion about tangency. Suppose curves $\alpha$ and $\beta$ are carried to $\alpha'$ and $\beta'$ and are tangent at $O$. Choose nearby points $A \in \alpha$ and $B \in \beta$, with images $A', B'$, and let the image of $O$ be $O'$. As $A \to O$, the line $AO$ tends to the tangent to $\alpha$ at $O$. By continuity of the projective correspondence, $A' \to O'$, and $A'O'$ tends to the tangent to $\alpha'$ at $O'$. Similarly, as $B \to O$, the line $BO$ tends to the tangent to $\beta$ at $O$, while $B'O'$ tends to the tangent to $\beta'$ at $O'$. Since $\alpha$ and $\beta$ are tangent, $AO$ and $BO$ tend to the same line, and so do $A'O'$ and $B'O'$. Hence $\alpha'$ and $\beta'$ are tangent at $O'$.

Equivalently, first suppose that a curve $\alpha$ is tangent to a line $l$ at $A$, and choose a nearby point $B \in \alpha$. The corresponding curve $\alpha'$ contains the corresponding points $A', B'$. As $B \to A$, the secant $AB$ tends to $l$. Correspondingly $B' \to A'$, so $A'B'$ tends to the tangent to $\alpha'$ at $A'$; this limiting line is the image of $l$. Tangency of two curves is handled in the same way. □

**Theorem 3.1.11.** *A projective correspondence carries a nondegenerate conic to a nondegenerate conic and a degenerate conic to a degenerate conic, though its type may change. In particular, every nondegenerate conic can be carried to a circle by a projective correspondence.*

**Proof.** By the central-projection discussion in section 1.4, circles, ellipses, hyperbolas, and parabolas can be transformed into one another by central projection. A central projection is a perspectivity, so definition 3.1.7 shows that a conic remains a conic under a projective correspondence.

A degenerate conic is a pair of lines; its image is therefore again a pair of lines and hence a degenerate conic. Finally, to carry a nondegenerate conic to a circle, simply reverse one of the central projections in section 1.4 (the projections in section 1.4 carry a circle to a conic, so here one uses the inverse). □

Projective correspondences can also be characterized as follows.

**Theorem 3.1.12.** *Let $\pi$ and $\pi'$ be projective planes, possibly coincident. Every bijection $f: \pi \to \pi'$ that carries collinear points to collinear points is a projective correspondence, and these are all the projective correspondences from $\pi$ to $\pi'$. In particular, when $\pi = \pi'$, they are precisely the projective transformations of $\pi$.*

**Remark.** The proof below may appear to require continuity of both $f$ and $f^{-1}$: for $P \in \pi$, if $P' \to P$ in $\pi$, one seems to need $f(P') \to f(P)$ in $\pi'$.[6] We do not emphasize this for two reasons. First, this book is chiefly concerned with geometric configurations, and continuity of a map is understood unless otherwise stated. Second, the theorem in fact needs no prior continuity assumption: preservation of lines alone suffices. A complete argument is given in example 10.6.34 of section 10.6; it uses the uniqueness of an automorphism of the real field.

**Proof.** [7] By theorem 3.1.9, every projective correspondence preserves lines. We prove the converse.

**A. Uniqueness from four points.** We first show that a line-preserving transformation is uniquely determined by four points, no three collinear. Suppose $f: \pi \to \pi'$ takes $A, B, C, D$ to $A', B', C', D'$, with no three points in either quadruple collinear. We prove existence and uniqueness of such an $f$. For this proof put $E = AB \cap CD$ and $F = AD \cap BC$, and write $f(E) = E'$ and $f(F) = F'$. Line preservation forces $E'$ and $F'$ to be the corresponding intersections of opposite sides of $A'B'C'D'$, so they are uniquely determined whenever $f$ exists.

First consider existence. Since a bijective line-preserving map is invertible and compositions of line-preserving maps preserve lines, it is enough to carry an arbitrary quadrangle $ABCD$ to a unit square $A'B'C'D'$. Choose a perspectivity $g: \pi \to \pi'$ with center $O$ so that $OE, OF \parallel \pi'$ and $OE \perp OF$. Then $EF$ is the vanishing line, $g(EF)$ is the line at infinity of $\pi'$, and $g(E), g(F)$ are points at infinity in perpendicular directions. Thus the quadrangle $ABCD$ becomes a rectangle under $g$. Stretch this rectangle in the directions of its sides[8] to obtain a square. Both steps preserve lines, and together they give a line-preserving map that takes $ABCD$ to a unit square.

For uniqueness, again use the facts that compositions of line-preserving transformations preserve lines and that these transformations are invertible. It is enough to show that a line-preserving map from a unit square $ABCD$ to an arbitrary quadrangle $A'B'C'D'$ is unique. Here $E$ and $F$ are the two relevant points at infinity. In figure 3.3a, let $BD \cap AC = Z_{11}$, and let $FZ_{11}$ meet $AB, CD$ at $Y_{11}, X_{11}$. Because $F$ is the point at infinity in the direction $BC$, we have $X_{11}Y_{11} \parallel BC$, so $Y_{11}$ and $X_{11}$ are the midpoints of $AB$ and $CD$.

Repeat the construction in the two smaller rectangles $AY_{11}X_{11}D$ and $BY_{11}X_{11}C$. Let $Z_{21} = DY_{11} \cap AX_{11}$, and let $FZ_{21}$ meet $AB, CD$ at $Y_{21}, X_{21}$. Likewise let $BX_{11} \cap CY_{11} = Z_{22}$, and let $FZ_{22}$ meet $AB, CD$ at $Y_{22}, X_{22}$. The points $Y_{21}, Y_{22}$ and $X_{21}, X_{22}$ quarter the segments $AB$ and $CD$. Repeating the construction in the four smaller rectangles $AY_{21}X_{21}D$, $Y_{21}Y_{11}X_{11}X_{21}$, $Y_{22}Y_{11}X_{11}X_{22}$,

[6] This limit can be formulated rigorously using the topology introduced in the smaller-print discussion above.

[7] This proof is rather lengthy and has little bearing on the later applications; it may be omitted.

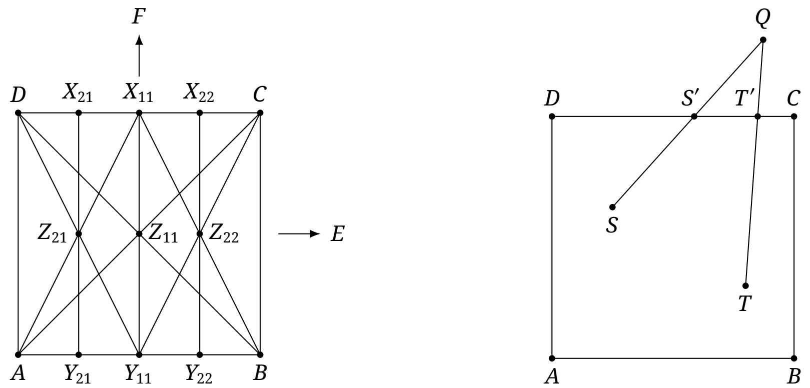
[8] This familiar transformation is discussed again in definition 3.3.11.

and $BY_{22}X_{22}C$ gives the points dividing $AB$ and $CD$ into eighths. Continued repetition gives their $2^n$-division points, where $n$ is a positive integer tending to infinity.

By the preceding construction, every point on the segments $AB, CD$ can be approximated by these $2^n$-division points. Since $f$ preserves lines and is continuous, every point $P'$ on the image segments $A'B', C'D'$ can be approximated by the analogous construction, with each point in the preceding steps replaced by its image under $f$. Thus the corresponding image $f(P)$ must satisfy the preceding conditions. Passing to the limit makes the construction determine $f(P)$ uniquely; hence $f(P)$ is unique. The same argument applies to $BC$ and $DA$, so the image of every boundary point of the square is uniquely determined.

Now choose $M \in CD$ and $N \in BC$, and put $L = FM \cap EN$. Since $f(M), f(N), f(E), f(F)$ are uniquely determined, line preservation gives $f(L) = f(M)f(F) \cap f(E)f(N)$, so $f(L)$ is determined. As $M$ and $N$ vary, $L$ ranges over the interior of the square. Thus the image of every point on or inside $ABCD$ is determined.

Finally, in figure 3.3b, choose $S', T' \in CD$ and $S, T$ inside the square, and put $Q = SS' \cap TT'$. The points $f(S), f(S'), f(T), f(T')$ have already been determined, and hence $f(Q) = f(S)f(S') \cap f(T)f(T')$. As $S', T'$ vary on $CD$ and $S, T$ approach $CD$ from within the square, $Q$ ranges over the region above $CD$ apart from $CD$ itself. The image of every point there is therefore determined. The same reasoning applies below $AB$, to the left of $AD$, and to the right of $BC$. Thus $f$ is uniquely determined on all of $\pi$.




(a) (b)

FIGURE 3.3

**B. Construction from four corresponding points.** It remains to show that, once the images of four points, no three collinear, are known, the required projective correspondence can be constructed. It then follows that every line-preserving transformation is a projective correspondence. Let $A, B, C, D \in \pi$ correspond to $A', B', C', D' \in \pi'$. We construct a composition of perspectivities taking the four points respectively to their prescribed images.

Assume first that the two quadruples are in general position. Otherwise, project the first quadruple by an auxiliary perspectivity to a general auxiliary plane and continue to denote the resulting points by $A, B, C, D$. Here "general position" merely excludes the finitely many incidence conditions that make adjacent auxiliary planes coincide, make the two lines used to determine a perspector coincide, or place a perspector in its source or image plane. The auxiliary planes and centers range over infinitely many positions, so all these degeneracies can be avoided simultaneously. Also assume $A' \notin \pi$.

*1.* Put $E = AB \cap CD$ and $E' = A'B' \cap C'D'$. Choose a general plane $\pi_1$ through $A'$ but not $A$, and a general point $O_1 \in AA' \setminus \{A, A'\}$. Then $O_1 \notin \pi \cup \pi_1$. Let $g_1 : \pi \to \pi_1$ be the perspectivity with center $O_1$, and denote the images of $B, C, D, E$ by $B_1, C_1, D_1, E_1$. Since $O_1 \in AA'$, we have $g_1(A) = A'$. Since $A, B, E$ are collinear, so are $A', B_1, E_1$. On the other hand, $A', B', E'$ are also collinear. Thus $B_1B'$ and $E_1E'$ lie in the plane determined by the intersecting lines $A'B_1$ and $A'B'$, and we may put $O_2 = B_1B' \cap E_1E'$.

*2.* Next choose a general plane $\pi_2$ through $A'B'$ such that $\pi_2 \neq \pi_1, \pi'$ and $O_2 \notin \pi_1 \cup \pi_2$. Let $g_2 : \pi_1 \to \pi_2$ be the perspectivity with center $O_2$. Since $O_2 \in B_1B' \cap E_1E'$, we have $g_2(B_1) = B'$, $g_2(E_1) = E'$, and, because $A' \in \pi_1 \cap \pi_2$, $g_2(A') = A'$. Put $C_2 = g_2(C_1)$ and $D_2 = g_2(D_1)$. Since $E \in CD$, we have $E_1 \in C_1D_1$ and hence $E' \in C_2D_2$. But $E' \in C'D'$ as well, so $C_2D_2$ and $C'D'$ meet at $E'$. The four points $C_2, D_2, C', D'$ are coplanar, and the lines $C_2C'$ and $D_2D'$ therefore meet.
*3.* Put $O_3 = C_2C' \cap D_2D'$, and let $g_3 : \pi_2 \to \pi'$ be the perspectivity with center $O_3$. Since $O_3 \in C_2C' \cap D_2D'$, we have $g_3(C_2) = C'$ and $g_3(D_2) = D'$. Since $\pi_2 \cap \pi' = A'B'$, the map $g_3$ fixes every point of $A'B'$; in particular, $g_3(A') = A'$ and $g_3(B') = B'$. The three perspectivities successively send the quadruple as follows:

$$(A,B,C,D) \to (A',B_1,C_1,D_1) \to (A',B',C_2,D_2) \to (A',B',C',D').$$

Thus $g_3 \circ g_2 \circ g_1$ is the desired projective correspondence. □

The theorem allows a projective correspondence of a real projective plane to be defined as a continuous line-preserving transformation. This statement is specific to $\mathbb{RP}^2$. In the complex projective plane $\mathbb{CP}^2$, not every continuous line-preserving transformation is projective, chiefly because the two choices i and −i play equivalent roles in the definitions; see the remarks after corollary 3.2.6 and definition 3.5.29 and section 10.6.2.

The proof also gives the following consequence.

**Theorem 3.1.13.** *Four points, no three collinear, uniquely determine a projective correspondence between two projective planes. More precisely, let $\pi$ and $\pi'$ be projective planes. If $A,B,C,D$ are four such points in $\pi$ and $A',B',C',D'$ are four such prescribed images in $\pi'$, then there is a unique projective correspondence $f$ with those images.*

Although theorem 3.1.12 was stated strictly only for the real projective plane, this four-point result holds also in the complex projective plane. A proof by one-dimensional projective correspondences will be given in section 3.5.1.

Projective transformations provide convenient proofs of many results.

**Theorem 3.1.14.** *Five points determine a conic in a projective plane. More precisely, through five points, no three collinear, there passes exactly one nondegenerate conic; through five points, no four collinear, there passes exactly one conic when degenerate conics are allowed.*

**Proof.** Among five points no four collinear, there are four of which no three are collinear. Carry them by a projective transformation to $(1,1), (1,-1), (-1,1), (-1,-1)$ in a rectangular coordinate system, and let the fifth point go to $(a,b)$. Solving the quadratic equation (1.1) shows that the conic through the five transformed points may, and must, have the equation

$$c(b^2-1)x^2 - c(a^2-1)y^2 + c(a^2-b^2) = 0,$$

where $c$ is any nonzero constant. The numbers $a,b$ cannot both equal $\pm 1$, and different nonzero values of $c$ define the same curve. Hence exactly one, possibly degenerate, conic passes through the transformed five points. By theorem 3.1.11, exactly one, possibly degenerate, conic passes through the original five points.

If no three of the five points are collinear, the pigeonhole principle shows that two lines cannot together contain all five points. Thus through five points, no three collinear, there passes exactly one nondegenerate conic. □

**Remark.** A purely geometric treatment of this result is given in section 6.2.1.

In what follows, write $\mathfrak{C}(ABCDE)$ for the conic through $A,B,C,D,E$.

**Example 3.1.15.** In figure 3.4, the lines $P_1P_2, Q_1Q_2, R_1R_2$ are concurrent at $S$; $P_1, Q_1, R_1$ lie on $l_1$, while $P_2, Q_2, R_2$ lie on $l_2$, and $l_1 \cap l_2 = O$. Put $X = Q_1R_2 \cap Q_2R_1$, $Y = R_1P_2 \cap P_1R_2$, and $Z = P_1Q_2 \cap P_2Q_1$. Prove that $O, X, Y, Z$ are collinear.

**Solution.** Apply a projective transformation carrying $S$ and $O$ to points at infinity in perpendicular directions, and use primes for transformed points; the resulting configuration is shown in figure 3.5. Since $P_1'P_2', Q_1'Q_2', R_1'R_2'$ pass through the same point at infinity $S'$, we have $P_1'P_2' \parallel Q_1'Q_2' \parallel R_1'R_2'$. Likewise $l_1' \parallel l_2'$, and these two lines are perpendicular to the preceding three.

The points $X', Y', Z'$ therefore all lie on the axis of symmetry of $l_1', l_2'$. They are collinear on a line parallel to $l_1'$ and $l_2'$, and this line passes through $O'$. By theorem 3.1.9, the original points $O, X, Y, Z$ are collinear. □

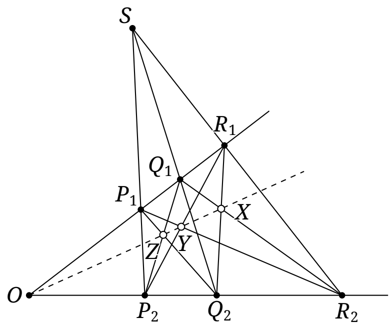



FIGURE 3.4

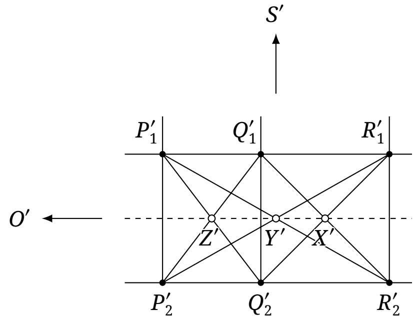



FIGURE 3.5

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## Exercises

**Exercise 3.8.** Let $OX, OY, OZ$ be three fixed lines, and let $A, B$ be fixed points whose joining line $AB$ passes through $O$. A point $R$ varies on the line $OZ$, and $P = RA \cap OX$, $Q = RB \cap OY$. Prove that the line $PQ$ passes through a fixed point of the line $AB$.

**Exercise 3.9.** Let $A, B$ be fixed points and $l$ a fixed line. Choose arbitrary $P, Q \in l$, and put $L = AP \cap BQ$, $M = AQ \cap BP$. Prove that the line $LM$ passes through a fixed point of the line $AB$.

**Exercise 3.10** (Pappus's theorem (Πάππος))**.** Let $A_1, B_1, C_1$ lie on the line $l_1$ and $A_2, B_2, C_2$ on the line $l_2$. Put $T_1 = A_1B_2 \cap A_2B_1$, $T_2 = B_1C_2 \cap B_2C_1$, and $T_3 = C_1A_2 \cap C_2A_1$. Prove that $T_1, T_2, T_3$ are collinear.

**Exercise 3.11.** Let $A, B, C$ be noncollinear, and suppose a conic $\alpha$ is tangent to the lines $BC, CA, AB$ at $A_1, B_1, C_1$, respectively. Prove that $AA_1, BB_1, CC_1$ are concurrent.

# 3.2 Cross-Ratios and Harmonic Ratios

## 3.2.1 Invariants of Projective Transformations

We now consider invariants of projective transformations.[9] We begin with two elementary notions.

[9] From this section onward we shall not always distinguish projective correspondences from projective transformations, since no ambiguity arises in applications; the intended meaning is clear from context.

**Definition 3.2.1** (Ranges and pencils)**.**

(1) A *range of points* is a set of points $A,B,C,\dots$ on a line $l$. It may be written $l(A,B,C,\dots)$; if these points are respectively the intersections of $a,b,c,\dots$ with $l$, it may also be written $l(a,b,c,\dots)$. The line $l$ is the *base* of the range.[10]

(2) A *pencil of lines* is a set of lines $a,b,c,\dots$ through one point $O$. It may be written $O(a,b,c,\dots)$; if the lines pass respectively through $A,B,C,\dots$, it may also be written $O(A,B,C,\dots)$. Their common point $O$ is the *vertex* of the pencil.

The definitions of ranges and pencils immediately give the following result.

**Theorem 3.2.2.** *A projective transformation carries a range of points to a range of points and a pencil of lines to a pencil of lines.*

We next introduce the cross-ratio, the invariant that we shall need.

**Definition 3.2.3** (Cross-ratio)**.**

(1) For three points $A,B,C$ in a range, define their *simple ratio* by $(A,B,C) := \frac{\overline{AC}}{\overline{BC}}$. Here $A,B$ are the *base points* and $C$ is the *division point*. For four points $A,B,C,D$, their *cross-ratio* is
$$(A,B;C,D) := \frac{(A,B,C)}{(A,B,D)} = \frac{\overline{AC}}{\overline{BC}} \div \frac{\overline{AD}}{\overline{BD}}.$$
The points $A,B$ are the *pair of base points*, and $C,D$ the *pair of division points*.

(2) For three lines $a,b,c$ in a pencil, define their *simple ratio* by $(a,b,c) := \frac{\sin\measuredangle(a,c)}{\sin\measuredangle(b,c)}$. Here $a,b$ are the *base lines* and $c$ is the *division line*. For four lines $a,b,c,d$, their cross-ratio is
$$(a,b;c,d) := \frac{(a,b,c)}{(a,b,d)} = \frac{\sin\measuredangle(a,c)}{\sin\measuredangle(b,c)} \div \frac{\sin\measuredangle(a,d)}{\sin\measuredangle(b,d)}.$$
The lines $a,b$ are the *pair of base lines*, and $c,d$ the *pair of division lines*.

**Remark.** When no ambiguity results, the simple ratios $(A,B,C)$ and $(a,b,c)$ may be abbreviated to $(ABC)$ and $(abc)$, and the cross-ratios $(A,B;C,D)$ and $(a,b;c,d)$ to $(AB,CD)$ and $(ab,cd)$. If the four lines $a,b,c,d$ of a pencil with vertex $O$ pass through $A,B,C,D$, their cross-ratio $(a,b;c,d)$ may also be written $O(A,B;C,D)$ or even $O(AB,CD)$.

Quantities involving elements at infinity are defined by limits.

**Proposition 3.2.4.** *In a range $l(A,B,C,D)$, suppose $A,B$ are finite and $D$ is the point at infinity. Then $(A,B,D) = 1$ and $(A,B;C,D) = (A,B,C)$.*

**Proof.** Indeed,
$$(A,B,D) = \lim_{\substack{C' \in AB,\\ |AC'| \to +\infty}} \frac{\overline{AC'}}{\overline{BC'}} = 1,$$
and hence $(A,B;C,D) = \frac{(A,B,C)}{(A,B,D)} = (A,B,C)$. □

**Remark.** To compute the cross-ratio of a pencil containing the line at infinity, one usually converts it by theorem 3.2.5 to the cross-ratio of a range. The simple ratio of a pencil containing the line at infinity is not well defined: an angle between a finite line and the line at infinity is not naturally defined, nor can the quantity be converted to the simple ratio of a range.

The next theorem establishes that the cross-ratio is an invariant of projective transformations.

[10] When no ambiguity results, $l$ may be omitted from the notation. The word "range" imposes no restriction on the number of points: $A,B,C,D$ on $l$ may form a range, and so may all points of $l$. The same convention applies to pencils below.

**Theorem 3.2.5.** *Let $l_1, l_2$ be two lines. As in figure 3.6, let four lines $a,b,c,d$ through a point $P$ meet $l_1$ at $A,B,C,D$ and $l_2$ at $A',B',C',D'$, respectively. Then*

$$(A,B;C,D) = (a,b;c,d) = (A',B';C',D'). \tag{3.1}$$

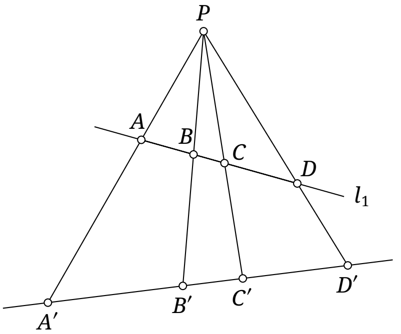


FIGURE 3.6

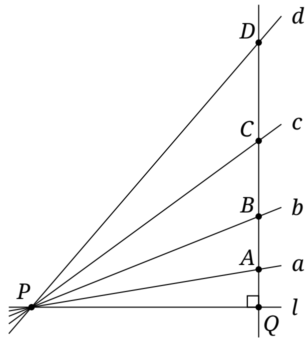


FIGURE 3.7

**Proof.** By the generalized angle-bisector theorem,

$$(A,B;C,D) = \frac{\overline{AC}}{\overline{BC}} \div \frac{\overline{AD}}{\overline{BD}} = \frac{PA \sin\measuredangle(a,c)}{PB \sin\measuredangle(b,c)} \div \frac{PA \sin\measuredangle(a,d)}{PB \sin\measuredangle(b,d)} = (a,b;c,d).$$

Likewise $(A',B';C',D') = (a,b;c,d)$. □

**Remark.** The simple ratio of a pencil cannot in general be converted directly into that of a range, since

$$(A,B,C) = \frac{PA \sin\measuredangle(a,c)}{PB \sin\measuredangle(b,c)} = \frac{PA}{PB}(a,b,c).$$

This also explains why a pencil containing the line at infinity cannot be handled through the simple ratio of a range. The simple ratio of a pencil is introduced mainly for formal symmetry and has little use in applications.

**Corollary 3.2.6.** *Projective transformations preserve cross-ratios. Thus if a range $(A,B,C,D)$ is carried to $(A',B',C',D')$, then $(A,B;C,D) = (A',B';C',D')$; and if a pencil $(a,b,c,d)$ is carried to $(a',b',c',d')$, then $(a,b;c,d) = (a',b';c',d')$.*

**Proof.** In theorem 3.2.5, regard $l_1$ as a line in $\pi$, $l_2$ as a line in $\pi'$, and $P$ as the center of a perspectivity $f : \pi \to \pi'$. Then $f(A) = A'$, $f(B) = B'$, $f(C) = C'$, and $f(D) = D'$. Thus theorem 3.2.5 proves invariance of the cross-ratio of a range under a perspectivity. Theorem 3.2.5 also converts the cross-ratio of a pencil into that of a range, proving invariance of the cross-ratio of a pencil under a perspectivity as well. A projective transformation is a composition of perspectivities, so the result follows. □

**Remark.** Accordingly, a projective correspondence may also be defined as a line-preserving transformation that preserves the cross-ratio of a range. Unlike the formulation in theorem 3.1.12—see the remark following theorem 3.1.12—our earlier discussion tacitly required the map and its inverse to be continuous.

This also explains why theorem 3.1.12 holds strictly only over the real projective plane. Complex conjugation of every point[11] conjugates all lengths and therefore the cross-ratio, which generally changes. Hence it is not projective, even though it is continuous and preserves lines. If one imposes the stronger condition of analyticity, discussed in section A.9, complex conjugation is excluded and only the identity remains.

[11] In coordinates, simply conjugate every coordinate. A geometric interpretation is given in the remark after definition 3.5.29.

The cross-ratio of a pencil has the following useful form.

**Theorem 3.2.7.** *Let $a,b,c,d$ be four lines in a pencil with vertex $P$. Let $l$ be a line through $P$, and let $k_a,k_b,k_c,k_d$ be the tangents of its angles with these four lines: $k_i = \tan(l,i)$ for $i = a,b,c,d$, with* $\tan 90° = \infty$.[12] *Then*

$$(a,b;c,d) = \frac{k_c - k_a}{k_c - k_b} \div \frac{k_d - k_a}{k_d - k_b}.$$

**Proof.** As in figure 3.7, choose any point $Q$ on $l$ other than $P$ and draw through $Q$ the perpendicular to $l$, meeting $a,b,c,d$ at $A,B,C,D$. Put $PQ = s$. Then $QA,QB,QC,QD$ are respectively $sk_a,sk_b,sk_c,sk_d$, and hence

$$(ab,cd) = (AB,CD) = \frac{\overline{AC}}{\overline{BC}} \div \frac{\overline{AD}}{\overline{BD}} = \frac{sk_c - sk_a}{sk_c - sk_b} \div \frac{sk_d - sk_a}{sk_d - sk_b} = \frac{k_c - k_a}{k_c - k_b} \div \frac{k_d - k_a}{k_d - k_b}. \qquad \square$$

The analogous cotangent form follows in exactly the same way; we omit its proof.

**Theorem 3.2.8.** *Let $a,b,c,d$ be four lines in a pencil with vertex $P$. For any line $l$ through $P$, write $k_a,k_b,k_c,k_d$ for the cotangents of the angles from $l$ to these four lines, with* $\cot 0° = \infty$.[13] *Then*

$$(a,b;c,d) = \frac{k_c - k_a}{k_c - k_b} \div \frac{k_d - k_a}{k_d - k_b}.$$

Projective invariants now give efficient proofs of further results, such as the following theorem.

**Theorem 3.2.9** (Carnot's theorem)**.** *Let $A,B,C$ be noncollinear, with $A_1,A_2 \in BC$, $B_1,B_2 \in AC$, and $C_1,C_2 \in AB$. The six points $A_1,A_2,B_1,B_2,C_1,C_2$ lie on one conic if and only if*

$$\frac{\overline{BA_1} \cdot \overline{BA_2}}{\overline{CA_1} \cdot \overline{CA_2}} \cdot \frac{\overline{CB_1} \cdot \overline{CB_2}}{\overline{AB_1} \cdot \overline{AB_2}} \cdot \frac{\overline{AC_1} \cdot \overline{AC_2}}{\overline{BC_1} \cdot \overline{BC_2}} = 1. \tag{3.2}$$

The idea is to carry the common conic projectively to a circle, for which the assertion is easy. Since the left-hand side of (3.2) is not visibly a product of cross-ratios, we first construct another projective invariant.

**Lemma 3.2.10.** *Given points $A,B,C$, let $A_1 \in BC$, $B_1 \in AC$, and $C_1 \in AB$. The quantity*

$$s = \frac{\overline{BA_1}}{\overline{CA_1}} \cdot \frac{\overline{CB_1}}{\overline{AB_1}} \cdot \frac{\overline{AC_1}}{\overline{BC_1}}$$

*is invariant under projective transformations.*

**Proof.** Since a projective transformation is a composition of perspectivities, it is enough to prove invariance under a perspectivity. Let $f : \pi \to \pi'$ have center $P$, use primes for images, for example $f(A) = A'$, and denote the transformed value of $s$ by $s'$. As in theorem 3.2.5, the generalized angle-bisector theorem gives

$$s = \frac{PB \sin\measuredangle BPA_1}{PC \sin\measuredangle CPA_1} \cdot \frac{PC \sin\measuredangle CPB_1}{PA \sin\measuredangle APB_1} \cdot \frac{PA \sin\measuredangle APC_1}{PB \sin\measuredangle BPC_1} = \frac{\sin\measuredangle BPA_1 \cdot \sin\measuredangle CPB_1 \cdot \sin\measuredangle APC_1}{\sin\measuredangle CPA_1 \cdot \sin\measuredangle APB_1 \cdot \sin\measuredangle BPC_1}.$$

Similarly,

$$s' = \frac{\sin\measuredangle B'PA_1' \cdot \sin\measuredangle C'PB_1' \cdot \sin\measuredangle A'PC_1'}{\sin\measuredangle C'PA_1' \cdot \sin\measuredangle A'PB_1' \cdot \sin\measuredangle B'PC_1'}.$$

The triples $B,B',P$ and $A_1,A_1',P$ are collinear, so $\measuredangle BPA_1 = \measuredangle B'PA_1'$. The other corresponding angles are likewise equal, and hence $s = s'$. $\square$

We can now prove Carnot's theorem using the projective invariant just constructed.

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[12] Here $\infty$ may mean either positive or negative infinity; the formula shows that the effect is the same. This corresponds to approaching the unique point at infinity of a line from either direction. The same convention applies below.

[13] The *cotangent*, denoted by cot, is the reciprocal of the tangent.

**Proof of Theorem 3.2.9.** First suppose the six points lie on a conic $\alpha$. Carry $\alpha$ projectively to a circle. The circle case of (3.2) follows at once from the power theorem. By lemma 3.2.10, under a projective transformation

$$\text{LHS of (3.2)} = \left(\frac{\overline{BA_1}}{\overline{CA_1}} \cdot \frac{\overline{CB_1}}{\overline{AB_1}} \cdot \frac{\overline{AC_1}}{\overline{BC_1}}\right) \cdot \left(\frac{\overline{BA_2}}{\overline{CA_2}} \cdot \frac{\overline{CB_2}}{\overline{AB_2}} \cdot \frac{\overline{AC_2}}{\overline{BC_2}}\right) = \text{const.}$$

Thus the equation holds for every $\alpha$.

Conversely, by theorem 3.1.14, let $\alpha$ be the conic through $A_1, A_2, B_1, B_2, C_1$, and let its other intersection with $AB$ be $C_2'$. Then $C_2'$ satisfies the condition imposed on $C_2$ by (3.2). For fixed $A_1, A_2, B_1, B_2, C_1$, the condition in (3.2) determines $C_2$ uniquely. Hence $C_2 = C_2'$, as required. □

**Alternative proof of the necessity in Theorem 3.2.9.** Suppose the six points lie on a conic $\alpha$. By the power theorem for conics, the left-hand side of (3.2), written as

$$\text{LHS} = \frac{\overline{AC_1} \cdot \overline{AC_2}}{\overline{AB_1} \cdot \overline{AB_2}} \cdot \frac{\overline{BA_1} \cdot \overline{BA_2}}{\overline{BC_1} \cdot \overline{BC_2}} \cdot \frac{\overline{CB_1} \cdot \overline{CB_2}}{\overline{CA_1} \cdot \overline{CA_2}},$$

depends only on the directions of $AB, BC, CA$. Translate these three lines separately until each is tangent to $\alpha$. Then $A_2 = A_1$, $B_2 = B_1$, and $C_2 = C_1$. By exercise 3.11, the lines $AA_1, BB_1, CC_1$ are concurrent, and Ceva's theorem yields

$$\text{LHS of (3.2)} = \left(\frac{\overline{BA_1}}{\overline{CA_1}} \cdot \frac{\overline{CB_1}}{\overline{AB_1}} \cdot \frac{\overline{AC_1}}{\overline{BC_1}}\right)^2 = (-1)^2 = 1.$$

□

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## Exercises

**Exercise 3.12.**

(1) Prove theorem 3.2.8.

(2) In a rectangular coordinate system, four lines $a, b, c, d$ of a pencil have slopes $k_a, k_b, k_c, k_d$. Prove that

$$(a,b;c,d) = \frac{k_a - k_c}{k_b - k_c} \div \frac{k_a - k_d}{k_b - k_d}.$$

**Exercise 3.13.** Use projective transformations to prove Ceva's theorem and Menelaus' theorem.

**Exercise 3.14.** Let $A_1, A_2, \ldots, A_{n-1}, A_n$ be $n$ points in the plane, and let $B_1, B_2, \ldots, B_{n-1}, B_n$ be another $n$ points lying respectively on the lines $A_1A_2, A_2A_3, \ldots, A_{n-1}A_n, A_nA_1$. Prove that

$$\frac{\overline{A_1B_1}}{\overline{A_2B_1}} \cdot \frac{\overline{A_2B_2}}{\overline{A_3B_2}} \cdots \frac{\overline{A_{n-1}B_{n-1}}}{\overline{A_nB_{n-1}}} \cdot \frac{\overline{A_nB_n}}{\overline{A_1B_n}}$$

is invariant under projective transformations.[14]

**Exercise 3.15.** Find the indicated cross-ratios in a rectangular coordinate system.

(1) For $P_1(3,1)$, $P_2(7,5)$, $Q_1(6,4)$, $Q_2(9,7)$, find $(P_1, P_2; Q_1, Q_2)$.

(2) For $l_1: 2x - y + 1 = 0$, $l_2: 3x + y - 2 = 0$, $l_3: 7x - y = 0$, and $l_4: 5x - 1 = 0$, find $(l_1, l_2; l_3, l_4)$.

### 3.2.2 Properties of the Cross-Ratio

The cross-ratio is fundamental in projective geometry. We record several of its properties.

---

[14] Invariance of the cross-ratio is the case $n = 2$, and lemma 3.2.10 is the case $n = 3$.

**Proposition 3.2.11.** *For four points $P_1,P_2,P_3,P_4$ in a range:*

*(1)* $(P_3,P_4;P_1,P_2) = (P_1,P_2;P_3,P_4)$*;*
*(2)* $(P_2,P_1;P_3,P_4) = (P_1,P_2;P_4,P_3) = \frac{1}{(P_1,P_2;P_3,P_4)}$*;*
*(3)* $(P_2,P_1;P_4,P_3) = (P_1,P_2;P_3,P_4)$*;*
*(4)* $(P_1,P_3;P_2,P_4) = (P_4,P_2;P_3,P_1) = 1-(P_1,P_2;P_3,P_4)$*.*

**Proof.** For (1), the definition gives

$$(P_1P_2,P_3P_4) = \frac{\overline{P_1P_3}}{\overline{P_2P_3}} \div \frac{\overline{P_1P_4}}{\overline{P_2P_4}} = \frac{\overline{P_1P_3}\cdot\overline{P_2P_4}}{\overline{P_2P_3}\cdot\overline{P_1P_4}},$$
$$(P_3P_4,P_1P_2) = \frac{\overline{P_3P_1}}{\overline{P_4P_1}} \div \frac{\overline{P_3P_2}}{\overline{P_4P_2}} = \frac{\overline{P_3P_1}\cdot\overline{P_4P_2}}{\overline{P_4P_1}\cdot\overline{P_3P_2}} = \frac{\overline{P_1P_3}\cdot\overline{P_2P_4}}{\overline{P_2P_3}\cdot\overline{P_1P_4}} = (P_1P_2,P_3P_4).$$

For (2), the definition gives

$$(P_1P_2,P_4P_3) = \frac{(P_1P_2P_4)}{(P_1P_2P_3)} = \left(\frac{(P_1P_2P_3)}{(P_1P_2P_4)}\right)^{-1} = \frac{1}{(P_1P_2,P_3P_4)},$$

and

$$(P_2P_1,P_3P_4) = (P_3P_4,P_2P_1) = \frac{1}{(P_3P_4,P_1P_2)} = \frac{1}{(P_1P_2,P_3P_4)}.$$

Here the first and last equalities use (1).

Assertion (3) now follows from (2):

$$(P_2P_1,P_4P_3) = \frac{1}{(P_1P_2,P_4P_3)} = (P_1P_2,P_3P_4).$$

For (4), expanding the definition gives

$$\begin{aligned}(P_1P_3,P_2P_4) &= \frac{\overline{P_1P_2}\cdot\overline{P_3P_4}}{\overline{P_3P_2}\cdot\overline{P_1P_4}} = \frac{(\overline{P_1P_3}+\overline{P_3P_2})(\overline{P_3P_2}+\overline{P_2P_4})}{\overline{P_1P_4}\cdot\overline{P_3P_2}}\\ &= \frac{\overline{P_1P_3}\cdot\overline{P_2P_4}+\overline{P_3P_2}(\overline{P_1P_3}+\overline{P_3P_2}+\overline{P_2P_4})}{\overline{P_1P_4}\cdot\overline{P_3P_2}}\\ &= \frac{\overline{P_1P_3}\cdot\overline{P_2P_4}-\overline{P_2P_3}\cdot\overline{P_1P_4}}{-\overline{P_2P_3}\cdot\overline{P_1P_4}} = -\frac{\overline{P_1P_3}\cdot\overline{P_2P_4}}{\overline{P_2P_3}\cdot\overline{P_1P_4}}+1 = 1-(P_1P_2,P_3P_4).\end{aligned}$$

Also,

$$(P_4P_2,P_3P_1) = (P_3P_1,P_4P_2) = (P_1P_3,P_2P_4) = 1-(P_1P_2,P_3P_4).$$

The first two equalities use (1) and (3), respectively. □

**Proposition 3.2.12.** *For four lines $l_1,l_2,l_3,l_4$ in a pencil:*

*(1)* $(l_3,l_4;l_1,l_2) = (l_1,l_2;l_3,l_4)$*;*
*(2)* $(l_2,l_1;l_3,l_4) = (l_1,l_2;l_4,l_3) = \frac{1}{(l_1,l_2;l_3,l_4)}$*;*
*(3)* $(l_2,l_1;l_4,l_3) = (l_1,l_2;l_3,l_4)$*;*
*(4)* $(l_1,l_3;l_2,l_4) = (l_4,l_2;l_3,l_1) = 1-(l_1,l_2;l_3,l_4)$*.*

**Proof.** Let a line $l$ meet $l_1,l_2,l_3,l_4$ at $P_1,P_2,P_3,P_4$, respectively. The cross-ratio of the lines equals the corresponding cross-ratio of these points, and the result follows from proposition 3.2.11. □

**Proposition 3.2.13.**

*(1) If $A,B,C,D,E$ are collinear, then*

$$(A,B;C,D)(A,B;D,E)(A,B;E,C) = 1.$$

*(2) If $a,b,c,d,e$ are concurrent, then*

$$(a,b;c,d)(a,b;d,e)(a,b;e,c) = 1.$$

**Proof.** For five points,

$$(A,B;C,D)\,(A,B;D,E)\,(A,B;E,C) = \frac{(A,B,C)}{(A,B,D)} \cdot \frac{(A,B,D)}{(A,B,E)} \cdot \frac{(A,B,E)}{(A,B,C)} = 1.$$

The proof for five lines is analogous. □

**Proposition 3.2.14.** *Let $P,Q,A,B,C$ be five points such that no three are collinear, except possibly $A,B,C$. Then*

$$(PQ,PA;PB,PC) = (QP,QA;QB,QC) \quad \Longleftrightarrow \quad A,B,C \textit{ are collinear}.$$

**Proof.** Suppose first that $A,B,C$ are collinear, and let $PQ$ meet their line at $D$. Then

$$(PQ,PA;PB,PC) = (D,A;B,C) = (QP,QA;QB,QC).$$

Conversely, let $C'$ be $AB\cap PC$, let $C''$ be $AB\cap QC$, and put $K = AB\cap PQ$. The hypothesis gives $(K,A;B,C') = (K,A;B,C'')$, so $C' = C''$. It follows that $C' = C'' = C$, and hence $A,B,C$ are collinear. □

We now turn to the special cross-ratio called harmonic.

**Definition 3.2.15** (Harmonic conjugacy)**.**

(1) For four points $P_1,P_2,P_3,P_4$ in a range, if $(P_1,P_2;P_3,P_4) = -1$, the pair $P_3,P_4$ is said to *separate harmonically* the pair $P_1,P_2$. Equivalently, the two pairs are *harmonic conjugates*; we also say, more briefly, that $P_1,P_2,P_3,P_4$ are harmonic conjugates. The four points form a *harmonic range*, and $P_4$ is the *fourth harmonic point* of $P_1,P_2,P_3$.
(2) For four lines $l_1,l_2,l_3,l_4$ in a pencil, if $(l_1,l_2;l_3,l_4) = -1$, the pair $l_3,l_4$ is said to separate harmonically the pair $l_1,l_2$. Equivalently, the two pairs are harmonic conjugates; we also say, more briefly, that $l_1,l_2,l_3,l_4$ are harmonic conjugates. The four lines form a *harmonic pencil*, and $l_4$ is the *fourth harmonic line* of $l_1,l_2,l_3$.
(3) In either case the cross-ratio $-1$ is called the *harmonic ratio*.

**Remark.** By proposition 3.2.11, if $(\alpha\beta,\gamma\delta) = -1$, then

$$(\beta\alpha,\gamma\delta) = (\alpha\beta,\delta\gamma) = (\gamma\delta,\alpha\beta) = -1.$$

Thus the two base points or lines are interchangeable, as are the two division points or lines. The base pair and division pair are also interchangeable as wholes.

Harmonic ranges and pencils have the following properties.

**Proposition 3.2.16** (Projective characterization of the midpoint)**.** *Let $\infty_{AB}$ be the point at infinity of $AB$. A point $M$ is the midpoint of $AB$ if and only if $A,B,M,\infty_{AB}$ are harmonic conjugates.*

**Proof.** Since $(AB,M\infty_{AB}) = (ABM)$,

$$M \text{ is the midpoint of } AB \Longleftrightarrow \frac{\overline{AM}}{\overline{BM}} = -1 \Longleftrightarrow (ABM) = -1.$$

This is equivalent to $A,B,M,\infty_{AB}$ being harmonic conjugates. □

**Proposition 3.2.17.** *As in figure 3.8, let $A,B,C,D$ be collinear, with $C$ on the segment $AB$, and let $P$ lie off the line $AB$. Any two of the following conditions imply the other two:*

*(1) $A,B,C,D$ form a harmonic range;*
*(2) $CP$ bisects $\angle APB$;*
*(3) $DP$ bisects the exterior angle of $\angle APB$;*
*(4) $CP \perp DP$.*

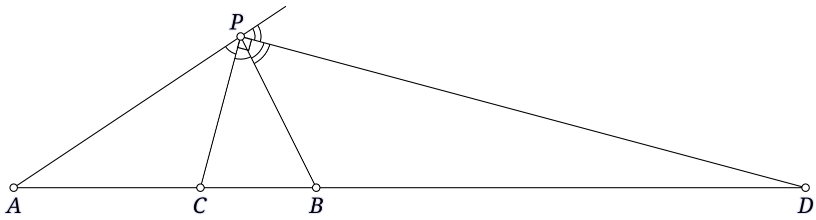


FIGURE 3.8

**Proof.** Conditions (2) and (3) give $\measuredangle APC = -\measuredangle BPC$ and $\measuredangle APD = \measuredangle BPD$. Hence

$$(A,B;C,D) = (PA,PB;PC,PD) = \frac{\sin\measuredangle APC}{\sin\measuredangle BPC} \div \frac{\sin\measuredangle APD}{\sin\measuredangle BPD} = (-1) \div 1 = -1.$$

The two angle bisectors are plainly perpendicular, so (1) and (4) follow.

Assume (1) and (3). Let the internal angle bisector of $\measuredangle APB$ meet $AB$ at $C'$. Then $A,B,C',D$ are harmonic. With $A,B,D$ fixed, the point $C$ satisfying $(AB,CD) = -1$ is unique, so $C = C'$. This proves (2), and then (4) follows from (2) and (3).

Assume (1) and (2). Let the external angle bisector of $\measuredangle APB$ meet $AB$ at $D'$. Then $A,B,C,D'$ are harmonic conjugates. With $A,B,C$ fixed, the point $D$ satisfying $(AB,CD) = -1$ is unique, so $D = D'$. This proves (3), and (2) together with (3) gives (4).

Conditions (2) and (4) plainly imply (3), and then (1). Similarly, (3) and (4) imply (2), and then (1).

Finally assume (1) and (4). Since $CP \perp DP$, we have $\measuredangle APD = \measuredangle APC + 90°$ and $\measuredangle BPD = \measuredangle BPC + 90°$. Therefore

$$\begin{aligned}-1 = (A,B;C,D) = (PA,PB;PC,PD) &= \frac{\sin\measuredangle APC}{\sin\measuredangle BPC} \div \frac{\sin\measuredangle APD}{\sin\measuredangle BPD}\\ &= \frac{\sin\measuredangle APC}{\sin\measuredangle BPC} \div \frac{\sin(\measuredangle APC + 90°)}{\sin(\measuredangle BPC + 90°)} = \frac{\tan\measuredangle APC}{\tan\measuredangle BPC}.\end{aligned}$$

Thus $\measuredangle APC = -\measuredangle BPC$, proving (2), and (2) together with (4) gives (3). □

**Proposition 3.2.18.** *Let $A,B,C,D$ be collinear and let $O$ be the midpoint of $AC$. The following nine conditions are equivalent:*

*(1) $A,C,B,D$ form a harmonic range;*
*(2) $\overline{AB}\cdot\overline{CD} = \overline{BC}\cdot\overline{AD}$;*
*(3) $\overline{AC}\cdot\overline{BD} = 2\overline{AB}\cdot\overline{CD}$;*
*(4) $\overline{AC}\cdot\overline{BD} = 2\overline{BC}\cdot\overline{AD}$;*
*(5) $\overline{OA}^2 = \overline{OC}^2 = \overline{OB}\cdot\overline{OD}$;*
*(6) $\overline{DA}\cdot\overline{DC} = \overline{DB}\cdot\overline{DO}$;*
*(7) $\overline{BA}\cdot\overline{BC} = \overline{BD}\cdot\overline{BO}$;*
*(8) $\dfrac{2}{\overline{BD}} = \dfrac{1}{\overline{BA}} + \dfrac{1}{\overline{BC}}$;*
*(9) $\dfrac{2}{\overline{DB}} = \dfrac{1}{\overline{DA}} + \dfrac{1}{\overline{DC}}$.*

**Proof.** Expanding the definition of cross-ratio and using proposition 3.2.11 proves the equivalence of (1) with (2), (3), and (4). Direct calculation with directed segments proves the equivalence with (5)–(9); the details are left to the reader.

There is also a geometric proof of the connection between (1) and (5)–(9). Because it is closely related to poles and polars, we defer it to section 3.6.1. □

## Exercises

**Exercise 3.16.** Prove that if $A,B,C,D$ are four distinct points of a range, then $(A,B;C,D) \neq 1$.

**Exercise 3.17.** For four points $P_1,P_2,P_3,P_4$ in a range, suppose $(P_1,P_2;P_3,P_4) = \lambda$, and let $i,j,k,l$ be a permutation of 1,2,3,4.

(1) Find every possible value of $(P_i,P_j;P_k,P_l)$.
(2) Repeat the problem when $\lambda$ is the harmonic ratio.

**Exercise 3.18.** Let $A,B,C,D,E,F$ be six collinear points. Prove that:

(1) $(A,B;C,D)(A,B;E,F) = (A,B;C,F)(A,B;E,D)$;
(2) if $(A,B;C,D) = (B,C;D,A)$, then $(A,C;B,D) = -1$.

**Exercise 3.19.** Prove proposition 3.2.18 by calculation.

# 3.3 Affine Transformations and Their Basic Properties

## 3.3.1 Basic Concepts of Affine Transformations

We now study a special kind of projective transformation: the affine transformation. We begin with a perspective affine correspondence, which is a special kind of perspectivity.

**Definition 3.3.1.** Let $\pi$ and $\pi'$ be two distinct affine planes. A bijection $f: \pi \to \pi'$ taking points of $\pi$ to points of $\pi'$ is called a *perspective affine correspondence* if, for all $P,Q \in \pi$, the lines joining $P$ to $f(P)$ and $Q$ to $f(Q)$ are parallel. If $\pi \cap \pi' = l$, the line $l$ is called the *affine axis* of the correspondence.

It follows immediately that every point on the affine axis is fixed by a perspective affine correspondence.

**Theorem 3.3.2.** *A perspective affine correspondence is a special affine transformation.*

**Proof.** By definition, a perspective affine correspondence preserves lines. We show that a perspective affine correspondence $f: \pi \to \pi'$ preserves the line at infinity. By definition, there is a finite line $l$ such that, for every point of $\pi$, the line joining it to its image in $\pi'$ is parallel to $l$.

Suppose, to the contrary, that $f$ maps a point at infinity $P$ of $\pi$ to a finite point $P'$ of $\pi'$. Since $PP' \parallel l$, the line $PP'$ also passes through the point at infinity $O$ of $l$. A line has only one point at infinity, so $O = P$. This would mean that $l$ passes through the point at infinity $P$ of $\pi$. For every point $Q$, the line $Qf(Q)$ would then also pass through this point at infinity of $\pi$, and hence would lie in $\pi$. Thus $f$ could map the points of $\pi$ only to the intersection $\pi \cap \pi'$, contradicting bijectivity.

The same argument shows that $f$ cannot map a finite point of $\pi$ to a point at infinity of $\pi'$. □

A perspective affine correspondence is precisely the parallel projection introduced in section 1.4. Figure 3.9 gives an example: the points $A,B,C,D$ of $\pi$ correspond respectively to $A',B',C',D'$ of $\pi'$. Since $D$ lies on the affine axis, $D' = D$.

The following observation is immediate.

**Theorem 3.3.3.** *A perspective affine correspondence is a perspectivity whose center lies at infinity.*

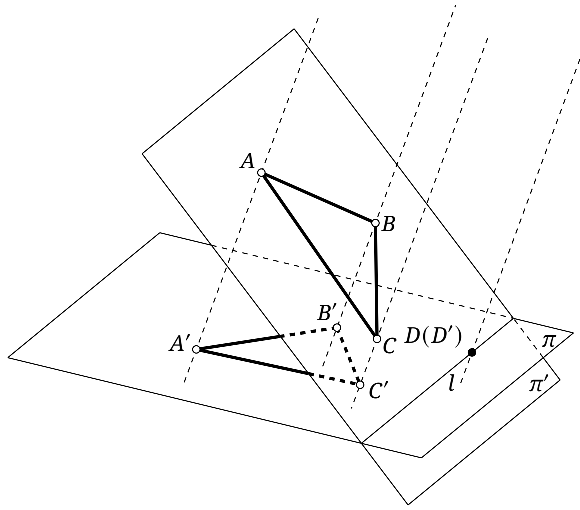


FIGURE 3.9

**Proof.** When the center of a perspectivity lies at infinity, the lines joining points to their images are mutually parallel, which is exactly a perspective affine correspondence. □

Proceeding as for projective correspondences, we define general affine correspondences as follows.

**Definition 3.3.4.** Let $\pi$ and $\pi'$ be two affine planes, which may coincide. A bijection $f: \pi \to \pi'$ taking points of $\pi$ to points of $\pi'$ is called an *affine correspondence* if it is a finite composition of perspective affine correspondences. In particular, if $\pi = \pi'$, it is called an *affine transformation*.

As in the projective case, the affine transformations of a plane form a group.

**Theorem 3.3.5.** *The composite of affine correspondences is an affine correspondence. Every affine correspondence is invertible, and its inverse is an affine correspondence. Likewise, the composite of affine transformations is an affine transformation, and every affine transformation has an inverse that is again an affine transformation.*

**Proof.** The proof is entirely analogous to that of theorem 3.1.8, on observing that every perspective affine correspondence is invertible and that its inverse is again a perspective affine correspondence. □

An affine correspondence or transformation is a special projective correspondence or transformation, so several properties follow at once.

**Theorem 3.3.6.** *Under an affine correspondence or transformation, points remain points, lines remain lines, ranges of points remain ranges, and pencils of lines remain pencils. Collinear points remain collinear, and concurrent lines remain concurrent.*

**Theorem 3.3.7.** *Under an affine correspondence or transformation, intersecting curves remain intersecting, disjoint curves remain disjoint, and tangent curves remain tangent.*

The preceding notion may equivalently be introduced by the following definition.

**Definition 3.3.8.** Let $\pi$ and $\pi'$ be two affine planes, which may coincide. A bijection $f: \pi \to \pi'$ taking points of $\pi$ to points of $\pi'$ is an affine correspondence if it satisfies both of the following conditions:

(1) collinear points have collinear images;
(2) $f$ preserves $l_\infty$: points on the line at infinity are mapped to points on the new line at infinity.

If $\pi = \pi'$, the affine correspondence is called an affine transformation.

Affine correspondences also admit the following characterization.

**Theorem 3.3.9.** *An affine correspondence or transformation is a projective correspondence or transformation that preserves the line at infinity: the line at infinity is carried to the line at infinity. Conversely, a projective correspondence or transformation preserving the line at infinity is an affine correspondence or transformation.*

**Proof.** The center of a perspective affine correspondence lies at infinity, so such a correspondence plainly preserves the line at infinity. It remains to prove that every projective correspondence preserving the line at infinity is an affine correspondence.

First, such a projective correspondence is uniquely determined by the images of at most three noncollinear finite points. Suppose the noncollinear finite points $A, B, C$ of $\pi$ have images $A', B', C'$ under such a correspondence $f: \pi \to \pi'$. Since $f$ preserves both the line at infinity and collinearity, $f(\infty_{AB}) = \infty_{A'B'}$; similarly, $f(\infty_{BC}) = \infty_{B'C'}$. Thus the images are known for the four points $A, C, \infty_{AB}, \infty_{BC}$, no three of which are collinear. A projective correspondence is uniquely determined by the images of four such points, so there is at most one possible $f$.

We next show that this $f$ is a finite composition of perspective affine correspondences. We may assume that $\pi \neq \pi'$ and that none of the pairs $A, A'$, $B, B'$, $C, C'$ consists of coincident points. Otherwise, first translate $\pi$ to a plane $\pi^*$ for which these conditions hold. The lines joining corresponding points before and after this translation are mutually parallel, and translation preserves lines, so this preliminary translation is itself a perspective affine correspondence.

Choose a plane $\pi_1$ through $A'$ that is distinct from both $\pi$ and $\pi'$, and that contains none of $B, C, B', C'$. Construct a perspective affine correspondence $f_1: \pi \to \pi_1$ whose point–image lines are parallel to $AA'$. Then $f_1(A) = A'$. Denote the images of $B, C$ by $B_1, C_1$.

Next choose a plane $\pi_2$ through $A', B'$ that is distinct from both $\pi_1$ and $\pi'$, and contains neither $B_1$ nor $C_1$. Construct a perspective affine correspondence $f_2: \pi_1 \to \pi_2$ whose point–image lines are parallel to $B_1B'$. Then $f_2(B_1) = B'$. Since $A' \in \pi_1 \cap \pi_2$, the point $A'$ lies on the affine axis and is fixed by $f_2$. Denote the image of $C_1$ by $C_2$.

Finally, construct a perspective affine correspondence $f_3: \pi_2 \to \pi'$ whose point–image lines are parallel to $C_2C'$. Then $f_3(C_2) = C'$. Since $A', B' \in \pi_2 \cap \pi'$, we have $f_3(A') = A'$ and $f_3(B') = B'$. Consequently $f = f_3 \circ f_2 \circ f_1$ has the required action. $\square$

Thus an affine correspondence may equivalently be defined as a projective correspondence that preserves the line at infinity. The proof also establishes the following result.

**Theorem 3.3.10.** *Three noncollinear finite points uniquely determine an affine correspondence. More precisely, let $\pi, \pi'$ be two planes, let $A, B, C$ be noncollinear finite points of $\pi$, and let $A', B', C'$ be their images under an affine correspondence $f: \pi \to \pi'$. Then $f$ is uniquely determined.*

We conclude with a special affine transformation.

**Definition 3.3.11.** Fix a line $l$, a point $O \in l$, and a nonzero scalar $a$. For each point $P$ of the plane, choose $P'$ so that the length of the projection onto $l$ of the directed segment $\overline{OP'}$[15] is $a$ times the length of the projection onto $l$ of $\overline{OP}$, while the projections of $\overline{OP'}$ and $\overline{OP}$ in the direction perpendicular to $l$ have equal lengths. This defines a transformation $f: P \mapsto P'$ fixing $O$, called a *stretch* with *center* $O$, direction $l$, and *stretch factor* $a$.

[15]For a line $l$ and points $P, Q$, let $P', Q'$ be the projections of $P, Q$ onto $l$. The length $P'Q'$ is called the length of the projection of $PQ$ onto $l$; for a directed segment, $\overline{P'Q'}$ is the length of the projection of $\overline{PQ}$ onto $l$.

**Theorem 3.3.12.** *A stretch is an affine transformation.*

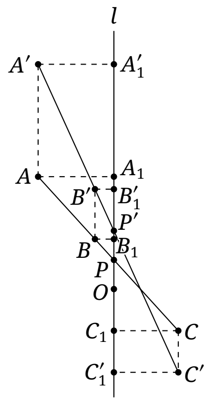


FIGURE 3.10

**Proof.** Let $A,B,C$ be collinear, and consider a stretch with center $O$, direction $l$, and factor $a$. Let the common line of $A,B,C$ meet $l$ at $P$. In this proof, primes denote images under the stretch, and a subscript 1 denotes projection onto $l$, as in figure 3.10.

By definition, the projections of $\overline{OA}$ and $\overline{OA'}$ in the direction perpendicular to $l$ have the same length, and hence $\overline{AA_1} = \overline{A'A_1'}$. Similarly, $\overline{BB_1} = \overline{B'B_1'}$, and $P' \in l$. On the other hand,

$$\overline{P'A_1'} = \overline{OA_1'} - \overline{OP'} = a\overline{OA_1} - a\overline{OP} = a\overline{PA_1},$$

and similarly $\overline{P'B_1'} = a\overline{PB_1}$.

Since $A,B,P$ are collinear, $\frac{\overline{AA_1}}{\overline{BB_1}} = \frac{\overline{PA_1}}{\overline{PB_1}}$. After the stretch,

$$\frac{\overline{P'A_1'}}{\overline{P'B_1'}} = \frac{\overline{PA_1}}{\overline{PB_1}} = \frac{\overline{AA_1}}{\overline{BB_1}} = \frac{\overline{A'A_1'}}{\overline{B'B_1'}},$$

so $A',B',P'$ are collinear. The same argument shows that $A',C',P'$ are collinear, and hence $A',B',C'$ are collinear.

Thus a stretch preserves lines and is therefore a projective transformation. It plainly preserves the line at infinity, so it is an affine transformation. □

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## Exercises

**Exercise 3.20.** Is a composite of perspective affine correspondences necessarily a perspective affine correspondence? Prove the assertion if it is true, and give a counterexample if it is false.

**Exercise 3.21.**

(1) Is a composite of stretches necessarily a stretch? Justify your answer.
(2) Can every affine transformation be expressed as a composite of stretches? Justify your answer.

**Exercise 3.22.** Which of the following are affine transformations, and which are projective transformations?

(1) a translation;
(2) a similarity;
(3) a homothety;
(4) a rotation;

(5) a stretch.

### 3.3.2 Properties and Applications of Affine Transformations

We now study properties specific to affine correspondences and transformations. General projective correspondences and transformations need not possess these properties.

**Theorem 3.3.13.** *Under a real affine correspondence, a conic remains a conic. More precisely, an ellipse is carried to an ellipse or a circle, and a circle to a circle or an ellipse; moreover, some affine correspondence carries any given ellipse to a circle. A hyperbola remains a hyperbola, and a parabola remains a parabola.*

**Proof.** By proposition 3.1.6, together with the fact that an affine transformation sends points at infinity to points at infinity, an ellipse or circle remains an ellipse or circle, a hyperbola remains a hyperbola, and a parabola remains a parabola.

To carry an ellipse to a circle, reverse the parallel projection of a circle to an ellipse described in section 1.4 (the parallel projection in section 1.4 carries a circle to an ellipse, so here one uses the inverse). □

**Remark.** The theorem shows that circles and ellipses can be transformed into one another by affine correspondences: they are "projectively equivalent." Thus, within projective geometry, circles and ellipses have much in common. In what follows, especially when projective geometry is applied to conics, we will often treat a circle as a kind of ellipse without further comment. The context will make clear whether the word "ellipse" includes circles.

The theorem above is stated for real affine planes. In a complex affine plane, ellipses, circles, and hyperbolas all meet the line at infinity and can be transformed into one another by complex affine correspondences. A parabola, however, is always tangent to the line at infinity and therefore remains a parabola even under a complex affine correspondence. An example of a complex affine correspondence appears at the end of this subsection.

**Theorem 3.3.14.** *Affine correspondences preserve parallelism.*

**Proof.** An affine correspondence sends points at infinity to points at infinity, so lines meeting at a point at infinity are carried to lines that again meet at a point at infinity. □

**Theorem 3.3.15.** *An affine correspondence preserves the simple ratio of a range.*

**Proof.** Suppose the range $(A,B,C)$ is carried to $(A',B',C')$, and the point at infinity $D$ of $AB$ is carried to $D'$. Then $D'$ is also a point at infinity, and

$$(ABC) = (AB,CD) = (A'B',C'D') = (A'B'C').$$

□

**Remark.** It is essential to note that an affine transformation does not preserve the simple ratio of a pencil. As emphasized earlier, although the cross-ratio of a pencil can be converted into the cross-ratio of a range, the simple ratio of a pencil cannot be converted directly into the simple ratio of a range.

**Corollary 3.3.16.** *An affine correspondence preserves ratios of the lengths of parallel or collinear segments.*

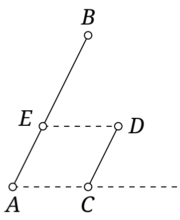


FIGURE 3.11

**Proof.** Preservation of the simple ratio of a range proves the assertion for collinear segments. It remains to consider parallel segments. Suppose, as in figure 3.11, that $AB \parallel CD$; we show that $\frac{\overline{AB}}{\overline{CD}}$ is invariant.

Through $D$ draw $DE \parallel AC$, meeting $AB$ at $E$. Then $AE \parallel CD$ and $\overline{AE} = \overline{CD}$. Since $ED \parallel AC$ and $AE \parallel CD$, and since an affine correspondence preserves parallelism, the parallelogram $ACDE$ is carried to a parallelogram. Hence the relations $AE \parallel CD$ and $\overline{AE} = \overline{CD}$ are preserved. Moreover, $\frac{\overline{AB}}{\overline{AE}}$ is invariant because $A, B, E$ are collinear. It follows that $\frac{\overline{AB}}{\overline{CD}}$ is invariant. □

**Theorem 3.3.17.** *An affine correspondence preserves the ratio of the areas of corresponding figures. More precisely, suppose figures $\mathscr{A}, \mathscr{B}$ have areas $S_1, S_2$ and are carried to figures $\mathscr{A}', \mathscr{B}'$ with areas $S_1', S_2'$. Then $\frac{S_1'}{S_2'} = \frac{S_1}{S_2}$.*

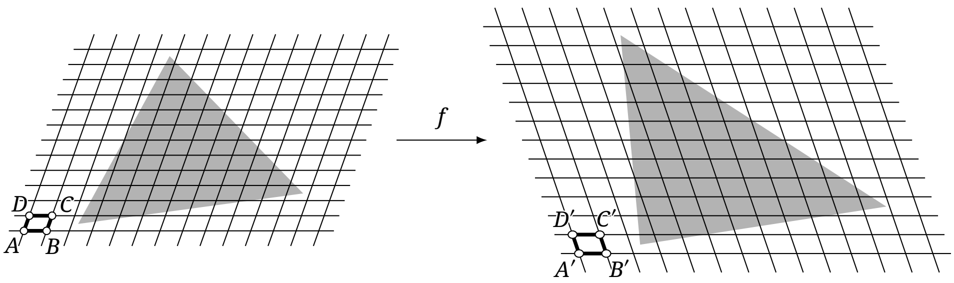


FIGURE 3.12

**Proof.** Consider an affine correspondence $f: \pi \to \pi'$. In $\pi$, choose a parallelogram $ABCD$ as in figure 3.12. Translates of infinitely many copies of this parallelogram tile $\pi$ without overlap or gaps. Under $f$, it is carried to a parallelogram $A'B'C'D'$, whose translates similarly tile $\pi'$.

As $ABCD$ tends to an infinitesimal parallelogram, a figure $\mathscr{A}$ in $\pi$ can be subdivided into translates of $ABCD$, while the corresponding figure $\mathscr{A}'$ is subdivided in the corresponding way into translates of $A'B'C'D'$. As the parallelograms become infinitesimal, the two subdivisions correspond under the affine transformation. Thus $\mathscr{A}$ is divided into the same number of translates of $ABCD$ as $\mathscr{A}'$ is into translates of $A'B'C'D'$. Therefore

$$\frac{S_1}{S_1'} = \lim_{\substack{|AB| \to 0 \\ |AD| \to 0}} \frac{S_{ABCD}}{S_{A'B'C'D'}}.$$

For any other corresponding pair $\mathscr{B}, \mathscr{B}'$, the same argument gives

$$\frac{S_2}{S_2'} = \lim_{\substack{|AB| \to 0 \\ |AD| \to 0}} \frac{S_{ABCD}}{S_{A'B'C'D'}}.$$

Thus $\frac{S_1}{S_1'} = \frac{S_2}{S_2'}$, that is, $\frac{S_1'}{S_2'} = \frac{S_1}{S_2}$. □

**Remark.** A completely rigorous proof would require calculus. Since these notes assume only a secondary-school background, we have presented the proof in terms of subdivisions; its underlying idea is the same as that used in calculus.

Affine transformations can also give short proofs of geometric results, as the following examples illustrate. From this point onward, as in the discussion of projective transformations, we will not distinguish strictly between affine transformations and affine correspondences unless necessary.

**Example 3.3.18.** In a trapezoid $ABCD$ with $AB \parallel CD$, let $M_1, M_2$ be the midpoints of $AB, CD$, respectively, and put $P = AC \cap BD$, $Q = AD \cap BC$. Prove that $P, Q, M_1, M_2$ are collinear.

**Solution.** Parallelism, midpoints, and collinearity are preserved by affine transformations. We may therefore apply an affine transformation carrying $ABCD$ to an isosceles trapezoid with legs $BC$ and $AD$. This can be done by carrying $\triangle M_1CD$ to an isosceles triangle with base $CD$. For an isosceles trapezoid the result is immediate. □

**Example 3.3.19.** In a rectangular coordinate system, find the area of the ellipse $\alpha: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b > 0)$.

**Solution.** Apply the stretch with center the origin $O$, direction the $y$-axis, and factor $\frac{a}{b}$. It lengthens the $y$-axis by the factor $\frac{a}{b}$ and carries the ellipse to the circle $\alpha': x^2 + y^2 = a^2$. The right vertex $A(a,0)$, upper vertex $B(0,b)$, and origin $O(0,0)$ are carried respectively to $A'(a,0)$, $B'(0,a)$, and $O'(0,0)$. By theorem 3.3.17,

$$\frac{S_\alpha}{S_{\triangle AOB}} = \frac{S_{\alpha'}}{S_{\triangle A'O'B'}} = \frac{\pi a^2}{a^2/2} = 2\pi.$$

Consequently,

$$S_\alpha = 2\pi \cdot S_{\triangle AOB} = 2\pi \cdot \frac{ab}{2} = \pi ab.$$

□

**Example 3.3.20.** As in figure 3.13, a line $l$ meets a hyperbola $\alpha$ at $P, Q$ and its two asymptotes at $X, Y$. Use an affine transformation to prove that $PX = QY$.

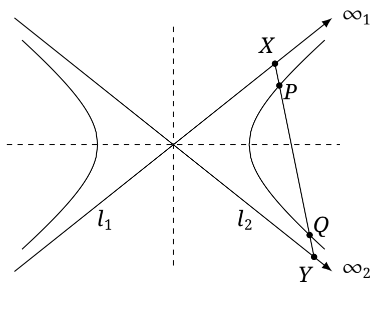


FIGURE 3.13

**Solution.** We show that there is an affine correspondence after which one of the axes of the hyperbola perpendicularly bisects the image of $PQ$. Symmetry then gives $XP = YQ$ in the transformed figure, and invariance of ratios of collinear segments gives the same equality in the original figure.

Let $\infty_1, \infty_2$ be the points at infinity on the two asymptotes. Let $\pi$ be the original plane, and choose a plane $\pi'$ with $P, Q \in \pi \cap \pi'$. Consider a projective correspondence $f: \pi \to \pi'$ with $f(P) = P$ and $f(Q) = Q$, and write $f(\infty_1) = \infty_1'$ and $f(\infty_2) = \infty_2'$. For an axis of the transformed hyperbola to perpendicularly bisect $PQ$, it is enough to choose $\infty_1', \infty_2'$ so that one bisector of the angle formed by two lines in their respective directions is perpendicular to $PQ$. Such points at infinity are easily chosen.

Let $S=\infty_1\infty_1'\cap\infty_2\infty_2'$. All four points lie on the plane at infinity, so $S$ also lies there. It now suffices to take $f$ to be the perspectivity with center $S$. A perspectivity whose center lies at infinity is a perspective affine correspondence, completing the proof. □

Over a complex affine plane, a hyperbola can be carried to a circle by a complex affine transformation. For the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $(a,b>0)$, apply the stretch with center the origin $O$, direction the $y$-axis, and factor $\frac{a}{b}\mathrm{i}$. In other words, stretch the $y$-axis by the complex factor $\frac{a}{b}\mathrm{i}$; the hyperbola becomes the circle $x^2+y^2=a^2$. A stretch is, of course, an affine transformation.

Complex affine transformations require some care. They still preserve ratios of parallel or collinear directed segments. [16] Area, however, has no suitable definition for imaginary figures, so the invariance of area ratios is no longer available.

The coordinate description above may appear insufficiently geometric. Every circle or ellipse meets $l_\infty$ in two imaginary points, whereas a hyperbola meets it in two real points. Thus a stretch perpendicular to the transverse axis can carry a hyperbola to a circle, except that its factor is imaginary. One construction is to fix the two transverse-axis vertices, take one imaginary intersection of the hyperbola's conjugate axis with the curve, and carry it to a real point on that same axis whose distance from the center is the semi-transverse axis. These three point–image pairs determine an affine transformation, which is a stretch fixing the transverse axis. Since affine transformations preserve conics, and in view of the original symmetry, the resulting conic must be a circle. The following is an example.

**Example 3.3.21.** Prove that, for a central conic of eccentricity $e$, the product of the tangents of the directed angles from a pair of conjugate diameters to its principal axis is $(e^2-1)$.

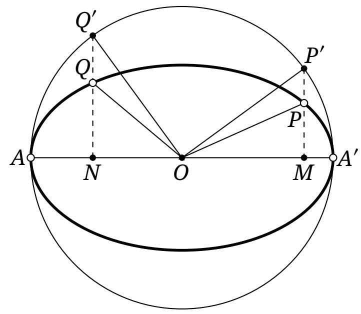


FIGURE 3.14

**Solution.** First suppose the conic is an ellipse $\alpha$ with semimajor and semiminor axes $a,b$. Stretch in the direction of the minor axis, with center the ellipse's center $O$ and factor $\frac{a}{b}$. The ellipse is carried to its auxiliary circle $\omega$. Let $OP,OQ$ be a pair of conjugate radii of the original ellipse, with $P,Q$ on the ellipse, and denote their images by $OP',OQ'$. Let $PP',QQ'$ meet the major axis at $M,N$, respectively. Then $PP',QQ'\perp MN$, and by proposition 2.2.17, $OP'\perp OQ'$, as shown in figure 3.14.

[16]Even in a complex plane, this ratio need not be defined metrically; it can be defined by vectors. For example, if $A,B,C$ are collinear and $\overrightarrow{AC}=\lambda\overrightarrow{BC}$, then $(ABC)=\lambda$. Cross-ratios of ranges in a complex plane are defined just as naturally.

The stretch gives $\frac{\overline{P'M}}{\overline{PM}} = \frac{\overline{Q'N}}{\overline{QN}} = \frac{a}{b}$. Consequently,

$$\begin{aligned}\tan\measuredangle MOP\cdot\tan\measuredangle MOQ &= \frac{\overline{PM}}{\overline{OM}}\cdot\frac{\overline{QN}}{\overline{ON}} = \frac{b^2}{a^2}\cdot\frac{\overline{P'M}}{\overline{OM}}\cdot\frac{\overline{Q'N}}{\overline{ON}}\\ &= \frac{b^2}{a^2}\cdot\tan\measuredangle MOP'\cdot\tan\measuredangle MOQ' = \frac{b^2}{a^2}\cdot\tan\measuredangle MOP'\cdot\tan(90^\circ+\measuredangle MOP')\\ &= -\frac{b^2}{a^2}\cdot\tan\measuredangle MOP'\cdot\cot\measuredangle MOP' = -\frac{b^2}{a^2} = e^2-1.\end{aligned}$$

This proves the assertion for an ellipse. For a hyperbola the argument is entirely analogous, except that the stretch factor is $\frac{a}{b}\mathrm{i}$. This factor comes from the coordinate equation. To avoid that calculation, note geometrically that a complex stretch carries the hyperbola to a circle, even though its factor has not yet been found. Repeating the elliptic argument already shows that the product of tangents is constant.

We may therefore consider a special case. Let one diameter tend to an asymptote. By corollary 2.1.21, its conjugate diameter tends to the same asymptote. Hence proposition 2.1.4 shows that the product of tangents is $[(\pm)\frac{b}{a}]^2 = \frac{b^2}{a^2} = e^2-1$, where $a,b$ are respectively the semi-transverse and semi-conjugate axes. ☐

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## Exercises

**Exercise 3.23.** In a trapezoid $ABCD$ with $AB\parallel CD$, put $P=AC\cap BD$ and $Q=AD\cap BC$. Prove that the line $PQ$ bisects both $AB$ and $CD$.

**Exercise 3.24.** Let $\alpha$ be a central conic of eccentricity $e$ and center $O$. For a variable point $P\in\alpha$, let $n$ be the normal at $P$. Prove that the ratio of the tangents of the directed angles from $OP$ and $n$, respectively, to the principal axis of $\alpha$ is $(1-e^2)$.

**Exercise 3.25.** Use an affine transformation to prove that the locus of the midpoints of a family of parallel chords of a parabola is a line parallel to its axis.

**Exercise 3.26.** In a rectangular coordinate system, let $A,B$ be two points of the ellipse $\alpha: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a>b>0)$ symmetric about the origin, and let $C$ be any point of $\alpha$ distinct from $A,B$. Prove that $k_{AC}k_{BC} = -\frac{b^2}{a^2}$ whenever the slopes of both $AC$ and $BC$ exist.

**Exercise 3.27.** For a given ellipse, find the ratio of the greatest possible area of an inscribed triangle to the area of the ellipse. Prove also that an inscribed triangle has greatest area if and only if its centroid coincides with the center of the ellipse.

**Exercise 3.28.** Let $\Gamma$ be a hyperbola and $l$ a line, and choose $P_0\in\Gamma$. Recursively construct $P_n$ as follows: through $P_{n-1}$ draw the line parallel to $l$, meeting $\Gamma$ again at $Q_n$, and let $P_n,Q_n$ be symmetric about the conjugate axis of $\Gamma$. Prove that $S_{\triangle P_{n-1}P_nP_{n+1}}$ is constant.

**Exercise 3.29.**

(1) Let $F$ be a focus of a conic $\alpha$, and let a variable chord $AB$ of $\alpha$ pass through $F$. Prove that the locus of the midpoint of $AB$ is a conic with the same eccentricity as $\alpha$.

(2) Use part (1) to prove that, for a central conic $\alpha$ with focus $F$, the circle having any chord of $\alpha$ through $F$ as diameter is tangent to two fixed circles. What happens when $\alpha$ is a parabola?

**Exercise 3.30.** Two diameters of an ellipse are called *conjugate diameters* if the tangent at either endpoint of one is parallel to the other diameter. Given an ellipse $C$, prove that all circles whose centers lie on $C$ and which are tangent to a pair of conjugate diameters of $C$ have the same radius.

**Exercise 3.31* (Napoleon–Barlotti theorem** (Napoléon)**).** Let $P$ be an $n$-gon. On each of its $n$ sides, construct externally a regular $n$-gon, and join in order the centers of these $n$ regular polygons to form an $n$-gon $P'$. Prove that $P'$ is regular if and only if some affine transformation carries $P$ to a regular $n$-gon.

## 3.4 Important Theorems in Projective Geometry

The preceding sections introduced the most important basic concepts of projective geometry. We now proceed to some of its deeper results, beginning with several fundamental theorems.

**Theorem 3.4.1** (Pappus' theorem)**.** *As in figure 3.15, let $A_1, B_1, C_1$ lie on a line $l_1$, and let $A_2, B_2, C_2$ lie on a line $l_2$. Put $T_3 = A_1B_2 \cap A_2B_1$, $T_1 = B_1C_2 \cap B_2C_1$, and $T_2 = C_1A_2 \cap C_2A_1$. Then $T_1, T_2, T_3$ are collinear.*

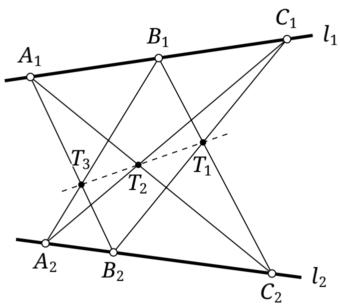


FIGURE 3.15

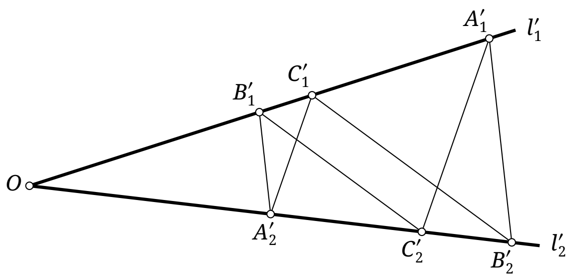


FIGURE 3.16

**Proof.** Apply a projective transformation taking $T_1, T_2$ to infinity, and use primes for the transformed points, as in figure 3.16. Then $A_1'C_2' \parallel A_2'C_1'$ and $B_1'C_2' \parallel B_2'C_1'$. We prove $A_2'B_1' \parallel A_1'B_2'$. It follows that $T_3'$ is also at infinity, so $T_1', T_2', T_3'$ are collinear; hence so are $T_1, T_2, T_3$.

Let $O = l_1' \cap l_2'$. If $O$ is at infinity, then the segments $B_1'C_1'$ and $B_2'C_2'$ are parallel and equal, as are $A_1'C_1'$ and $A_2'C_2'$. Therefore $A_1'B_1'$ and $A_2'B_2'$ are also parallel and equal. Thus $A_1'B_2' \parallel A_2'B_1'$ is immediate.

If $O$ is finite, then

$$\frac{\overline{B_1'A_1'}}{\overline{A_2'B_2'}} = \frac{\overline{B_1'C_1'}+\overline{C_1'A_1'}}{\overline{A_2'C_2'}+\overline{C_2'B_2'}} = \frac{\overline{B_1'C_1'}+\dfrac{\overline{A_2'C_2'}}{\overline{OA_2'}}(\overline{OB_1'}+\overline{B_1'C_1'})}{\overline{A_2'C_2'}+\dfrac{\overline{B_1'C_1'}}{\overline{OB_1'}}(\overline{OA_2'}+\overline{A_2'C_2'})}$$
$$= \frac{\overline{OB_1'}}{\overline{OA_2'}} \cdot \frac{\overline{OA_2'}\cdot\overline{B_1'C_1'}+\overline{A_2'C_2'}\cdot\overline{OB_1'}+\overline{A_2'C_2'}\cdot\overline{B_1'C_1'}}{\overline{OB_1'}\cdot\overline{A_2'C_2'}+\overline{B_1'C_1'}\cdot\overline{OA_2'}+\overline{B_1'C_1'}\cdot\overline{A_2'C_2'}} = \frac{\overline{OB_1'}}{\overline{OA_2'}}.$$

The second equality uses $C_1'A_2' \parallel A_1'C_2'$ and $C_2'B_1' \parallel B_2'C_1'$. The final relation gives $A_1'B_2' \parallel A_2'B_1'$. □

**Theorem 3.4.2** (Desargues' theorem)**.** *As in figure 3.17, if the lines joining corresponding vertices of two triangles are concurrent, then the intersections of corresponding sides are collinear. More precisely, for* $\triangle A_1B_1C_1$ *and* $\triangle A_2B_2C_2$*, put* $P=B_1C_1\cap B_2C_2$*,* $Q=A_1C_1\cap A_2C_2$*, and* $R=A_1B_1\cap A_2B_2$*. If* $A_1A_2, B_1B_2, C_1C_2$ *are concurrent at* $S$*, then* $P,Q,R$ *are collinear.*

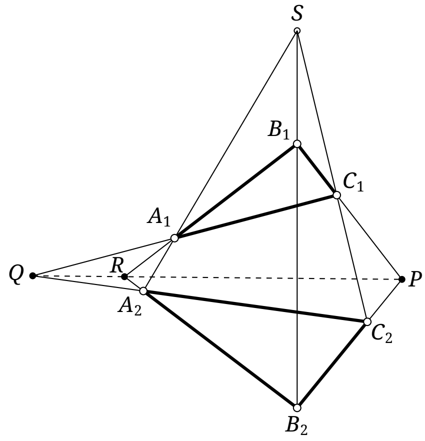


FIGURE 3.17

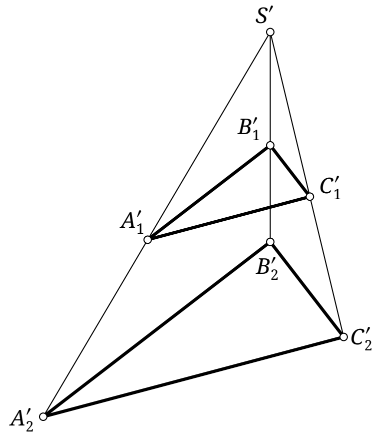


FIGURE 3.18

**Proof.** Apply a projective transformation taking $P,Q$ to infinity, and use primes for the transformed points, as in figure 3.18. Then $B_1'C_1' \parallel B_2'C_2'$ and $A_1'C_1' \parallel A_2'C_2'$. We prove $A_1'B_1' \parallel A_2'B_2'$. This places $R'$ at infinity, so $P',Q',R'$ are collinear and consequently so are $P,Q,R$.

If $S'$ is at infinity, $\triangle A_2'B_2'C_2'$ is a translate of $\triangle A_1'B_1'C_1'$, so $A_1'B_1' \parallel A_2'B_2'$ is immediate. If $S'$ is finite, then

$$\frac{\overline{S'A_1'}}{\overline{S'A_2'}} = \frac{\overline{S'C_1'}}{\overline{S'C_2'}} = \frac{\overline{S'B_1'}}{\overline{S'B_2'}},$$

where the two equalities follow respectively from $A_1'C_1' \parallel A_2'C_2'$ and $B_1'C_1' \parallel B_2'C_2'$. Hence $A_1'B_1' \parallel A_2'B_2'$. □

**Theorem 3.4.3** (Converse of Desargues' theorem)**.** *If the intersections of corresponding sides of two triangles are collinear, then the lines joining corresponding vertices are concurrent. More precisely, for* $\triangle A_1B_1C_1$ *and* $\triangle A_2B_2C_2$*, put* $P=B_1C_1\cap B_2C_2$*,* $Q=A_1C_1\cap A_2C_2$*, and* $R=A_1B_1\cap A_2B_2$*. If* $P,Q,R$ *are collinear, then* $A_1A_2, B_1B_2, C_1C_2$ *are concurrent.*

**Proof.** As in the proof of theorem 3.4.2, transform $PQR$ into the line at infinity; the result then follows at once. Alternatively, apply Desargues' theorem to $\triangle A_1RA_2$ and $\triangle C_1PC_2$. Since $P,Q,R$ are collinear, the lines $A_1C_1, RP, A_2C_2$ are concurrent. Hence $B_1, B_2, (A_1A_2\cap C_1C_2)$ are collinear, which says precisely that $A_1A_2, B_1B_2, C_1C_2$ are concurrent. □

Desargues' theorem motivates the following terminology.

**Definition 3.4.4.** Two triangles satisfying the hypotheses of Desargues' theorem are called *perspective triangles*. The intersection $S$ of the lines joining corresponding vertices is their *perspector*, or *center of perspectivity*; the line $PQR$ through the intersections of corresponding sides is their *perspective axis*, also called their *perspectrix*.

The center and axis just defined should not be confused with the center and axis of a perspectivity introduced earlier, despite sharing the same names. The intended meaning will be clear from context.

We next consider several theorems about circles. The forms of Pascal's, Brianchon's, and Newton's theorems given below are in fact restricted forms; their general versions will be presented later.

**Lemma 3.4.5.** *Given a circle and a line $l$ disjoint from it, there is a projective transformation that carries the circle to a circle and $l$ to the line at infinity.*

**Proof.** First use a projective transformation to carry $l$ to the line at infinity. The circle then becomes an ellipse, since it does not meet the line at infinity. An affine transformation can carry this ellipse to a circle. ☐

**Theorem 3.4.6** (Pascal's theorem)**.** *As in figure 3.19, the intersections of the three pairs of principal opposite sides of a hexagon inscribed in a circle are collinear.*[17] *More precisely, let $ABCDEF$ be inscribed in a circle, and put $P = AB \cap DE$, $Q = BC \cap EF$, and $R = CD \cap FA$. Then $P,Q,R$ are collinear.*

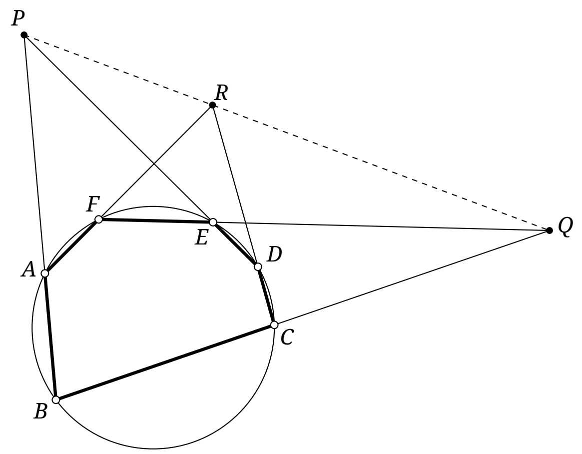


FIGURE 3.19

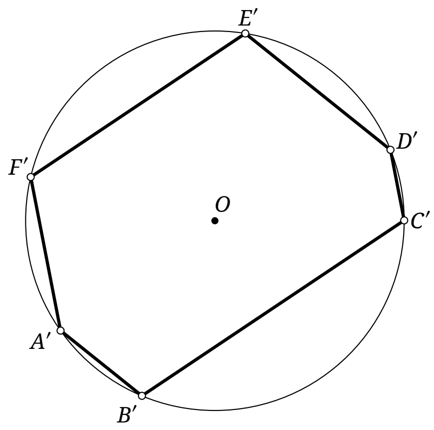


FIGURE 3.20

**Proof.** By lemma 3.4.5, there is a projective transformation preserving the circle and carrying $P,Q$ to infinity. Use primes for the transformed points and let $O$ be the center of the circle, as in figure 3.20. Then $A'B' \parallel D'E'$ and $B'C' \parallel E'F'$.

The first parallelism gives $\widehat{A'B'C'D'E'} = \widehat{B'A'F'E'D'}$, and the second gives $\widehat{C'D'E'} = \widehat{B'A'F'}$. Consequently,

$$\widehat{A'B'C'} = \widehat{A'B'C'D'E'} - \widehat{C'D'E'} = \widehat{B'A'F'E'D'} - \widehat{B'A'F'} = \widehat{F'E'D'}.$$

Thus $C'D' \parallel F'A'$, so the image of $R$ is also at infinity. Hence $P,Q,R$ were collinear before the transformation. ☐

This proof assumes that $ABCDEF$ is a strictly convex hexagon, which ensures that $PQ$ does not meet the circle and permits the use of lemma 3.4.5. If complex affine transformations are admitted, however, the line $l$ in lemma 3.4.5 always has two intersections with the circle unless it is tangent to the circle. After carrying $l$ to the line at infinity, a complex affine transformation can always restore the conic to a circle. The theorem then holds for six points $A,B,C,D,E,F$ in general position. Exercise 0.43 already gave a proof using a homothety; later we shall give another proof for the general case without introducing the complex projective plane.

If two adjacent vertices in Pascal's hexagon tend to coincide, their joining line tends to the tangent at the limiting point. Degenerate forms of Pascal's theorem can therefore be written for "hexagons" with coincident vertices. The following is the case of one pair of coincident adjacent vertices.

[17] Here principal opposite sides means sides separated by two intervening sides.

**Proposition 3.4.7.** *As in figure 3.21, let $ABCDE$ be a pentagon inscribed in a circle. The tangent at $A$ meets the line $CD$ at $P$; put $Q = AB \cap DE$ and $R = BC \cap EA$. Then $P, Q, R$ are collinear.*

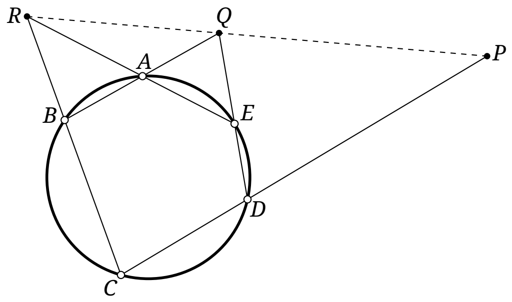


FIGURE 3.21

**Lemma 3.4.8.** *Given a circle and a point $C$ in its interior, there is a projective transformation that carries the circle to a circle and $C$ to the center of the resulting circle.*

**Proof.** We shall prove this later. □

**Theorem 3.4.9** (Brianchon's theorem). *As in figure 3.22, the three principal diagonals of a hexagon circumscribed about a circle are concurrent.*[18] *More precisely, let $ABCDEF$ be a hexagon inscribed in a circle. Let the tangents at $A, B$ meet at $P$, those at $B, C$ at $Q$, those at $C, D$ at $R$, those at $D, E$ at $S$, those at $E, F$ at $T$, and those at $F, A$ at $U$. Then $PS, QT, RU$ are concurrent.*

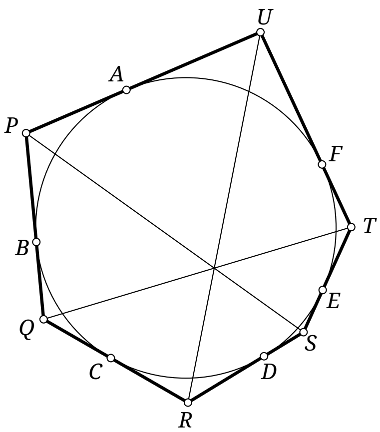


FIGURE 3.22

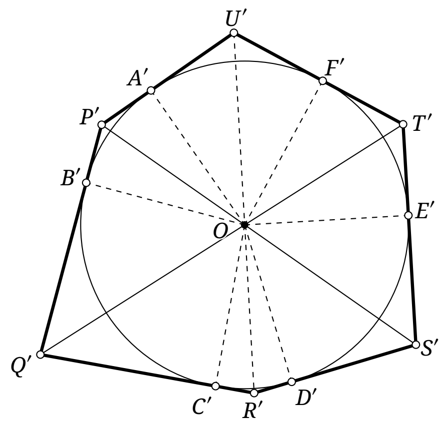


FIGURE 3.23

**Proof.** By lemma 3.4.8, apply a projective transformation preserving the circle and carrying $PS \cap QT$ to its center $O$. Use primes for the transformed points, as in figure 3.23. It is enough to prove that $U', O, R'$ are collinear. This makes $U, R, (PS \cap QT)$ collinear before the transformation, which proves the assertion.

Tangency gives $\angle A'OP' = \angle B'OP'$ and $\angle E'OS' = \angle D'OS'$. Since $P'S'$ passes through $O$,

$$\angle A'OE' = 180^\circ - \angle A'OP' - \angle E'OS' = 180^\circ - \angle B'OP' - \angle D'OS' = \angle B'OD'.$$

Similarly, because $Q'T'$ passes through $O$, $\angle E'OC' = \angle F'OB'$.

[18] A principal diagonal joins two vertices separated by two intervening vertices.

It follows that

$$\angle A'OC' = 360^\circ - \angle A'OE' - \angle E'OC' = 360^\circ - \angle B'OD' - \angle F'OB' = \angle F'OD'.$$

On the other hand, tangency gives $\angle A'OU' = \angle F'OU'$ and $\angle C'OR' = \angle D'OR'$. Therefore

$$\angle U'OR' = \angle U'OA' + \angle A'OC' + \angle C'OR' = \frac{360^\circ}{2} = 180^\circ.$$

Thus $U', O, R'$ are collinear, as required. □

As with the proof of Pascal's theorem, this argument assumes that $ABCDEF$ is strictly convex, so that $O$ lies inside the circle and lemma 3.4.8 applies. We shall see that the proof of lemma 3.4.8 uses the duality principle and therefore requires $C$ to be inside the circle. In the complex projective plane, however, as in lemma 3.4.5, the analogous result requires only that $C$ not lie on the circle. Brianchon's theorem consequently holds for six points $A, B, C, D, E, F$ in general position. A proof of the general case not using the complex projective plane will be given later.

Brianchon's theorem also has degenerate forms. When two tangents coalesce, their intersection tends to the point of tangency. Newton's theorem below is the case in which two pairs of tangents coalesce.

**Proposition 3.4.10** (Newton's theorem)**.** *As in figure 3.24, the two diagonals of a quadrilateral circumscribed about a circle and the two lines joining opposite points of tangency are concurrent.*

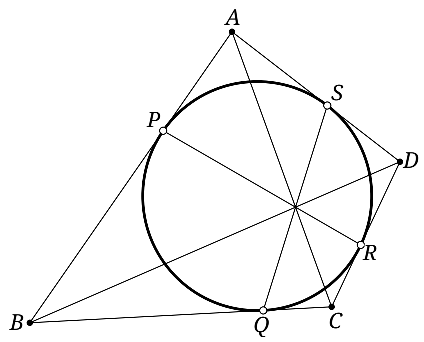


FIGURE 3.24

**Proof.** In figure 3.24, let $ABCD$ be circumscribed about the circle, with points of tangency $P, Q, R, S$. Apply Brianchon's theorem to the six tangents $AB, AB, BC, CD, CD, DA$, where two pairs coincide. The lines $PR, BD, AC$ are concurrent. The same argument shows that $QS$ passes through their common point. □

---

## Exercises

**Exercise 3.32.** Let $A, B, C$ be noncollinear, and let $P$ vary on a fixed line through $C$. Put $X = AP \cap BC$ and $Y = AC \cap BP$. Prove that $XY$ passes through a fixed point.

**Exercise 3.33.** For the various cases in which points of tangency or tangent lines coincide, state and draw the corresponding degenerate forms of Pascal's and Brianchon's theorems.

**Exercise 3.34.** Choose $A', B', C'$ on the sidelines $BC, CA, AB$ of $\triangle ABC$, respectively. Put $F = B'C' \cap BC$, $G = C'A' \cap CA$, $H = A'B' \cap AB$, $M = FH \cap BB'$, and $N = FG \cap CC'$. Prove that $MG, NH, BC$ are concurrent.

**Exercise 3.35** (Converse of Pascal's theorem)**.** Suppose five vertices of a hexagon $ABCDEF$ lie on a circle $\omega$. If the intersections of its three pairs of principal opposite sides are collinear, prove that its sixth vertex also lies on $\omega$.

**Exercise 3.36.** The tangent to a hyperbola at $A$ meets its asymptotes at $A_1,A_2$, and the tangent at $B$ meets them at $B_1,B_2$. Prove that $A_1B_2,A_2B_1,AB$ are concurrent.

**Exercise 3.37.** Through a fixed point $C$ not on a circle, draw four chords $A_1B_1,A_2B_2,A_3B_3,A_4B_4$. Put $D=A_1A_2\cap A_3A_4$ and $E=B_1B_2\cap B_3B_4$. Prove that $C,D,E$ are collinear.

## 3.5 One-Dimensional Projective Transformations

We now study more closely the properties of ranges of points and pencils of lines in projective geometry.

### 3.5.1 Projective Correspondences of One-Dimensional Fundamental Forms

The term *one-dimensional fundamental form* means either a range of points or a pencil of lines. The following definition describes the correspondences between such forms.

**Definition 3.5.1** (Perspectivity)**.** There are three kinds of perspectivity between one-dimensional fundamental forms.

(1) As in figure 3.25, joining the points of a range $(A,B,C,D)$ to a point $P$ outside its base line gives a perspectivity between the range $(A,B,C,D)$ and the pencil $(PA,PB,PC,PD)$, written $(A,B,C,D)\overline{\wedge}(PA,PB,PC,PD)$. If $P$ is at infinity, this is more specifically called a *perspective affine correspondence*.

(2) As in figure 3.25, consider the ranges $(A,B,C,D)$ and $(A',B',C',D')$. If the lines joining corresponding points are concurrent at $P$, this gives a correspondence between $(A,B,C,D)$ and $(A',B',C',D')$. The two ranges are called *perspective ranges*, and we write

$$(A,B,C,D)\overset{(P)}{\overline{\wedge}}(A',B',C',D').$$

The point $P$ is the *center of perspectivity*.

(3) Consider the pencils $(a,b,c,d)$ and $(a',b',c',d')$ in figure 3.26. Suppose the intersections of corresponding lines all lie on a line $l$. This gives a correspondence between the pencils $(a,b,c,d)$ and $(a',b',c',d')$. The two pencils are called *perspective pencils*, and we write

$$(a,b,c,d)\overset{(l)}{\overline{\wedge}}(a',b',c',d').$$

The line $l$ is the *perspective axis*.

If two one-dimensional fundamental forms $[\pi]$ and $[\pi']$ can be related by a composition of perspectivities—that is, if there are $n$ fundamental forms $[\pi_1],\dots,[\pi_n]$ such that

$$[\pi]\overline{\wedge}[\pi_1]\overline{\wedge}\cdots\overline{\wedge}[\pi'],$$

then $[\pi]$ and $[\pi']$ are said to be in *projective correspondence*, written $[\pi]\overline{\wedge}[\pi']$. If the intermediate forms $[\pi_1],\dots,[\pi_n]$ can be chosen so that every perspectivity is a perspective affine correspondence, then $[\pi]$ and $[\pi']$ are said to be in *affine correspondence*.

This definition closely parallels that of a projective correspondence of planes. In particular, a projective or affine transformation on a line can be regarded as the restriction to that line of a projective or affine transformation of planes.

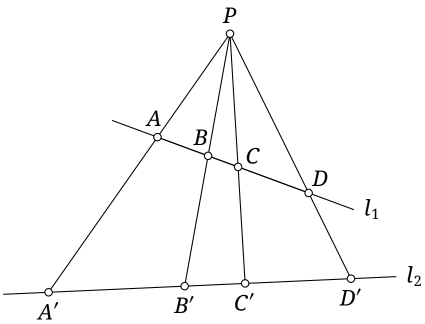


FIGURE 3.25

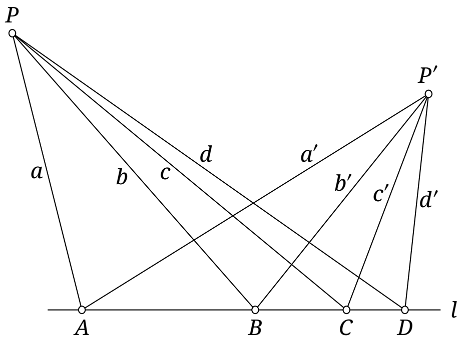


FIGURE 3.26

Indeed, let $f: \pi \to \pi'$ be a perspectivity with center $O$. For any line $l \in \pi$, put $f(l) = l'$. The restriction $f|_l$ gives a perspectivity $l \to l'$, written $l \overset{(O)}{\barwedge} l'$. If $O$ is at infinity, then $f$ is a perspective affine correspondence and $f|_l$ gives a perspective affine correspondence $l \to l'$. Thus theorem 3.3.9 yields the following.

**Theorem 3.5.2.** *A perspectivity $l \to l'$ is an affine correspondence if and only if it preserves the point at infinity; that is, the point at infinity on $l$ corresponds to the point at infinity on $l'$.*

Our principal concern will be general perspectivities and projective correspondences. The definition immediately gives the next theorem.

**Theorem 3.5.3.** *A projective correspondence between one-dimensional fundamental forms preserves cross-ratios. Thus, if $(\alpha,\beta,\gamma,\delta)$ and $(\alpha',\beta',\gamma',\delta')$ are ranges or pencils with $(\alpha,\beta,\gamma,\delta) \barwedge (\alpha',\beta',\gamma',\delta')$, then*

$$(\alpha,\beta;\gamma,\delta) = (\alpha',\beta';\gamma',\delta').$$

**Proof.** This is an immediate consequence of theorem 3.2.5. □

We next study projective correspondences and perspectivities between such forms.

**Theorem 3.5.4** (von Staudt's theorem)**.** *Three pairs of corresponding elements determine a projective correspondence between two one-dimensional fundamental forms.*

**Proof.** Let $\alpha,\beta,\gamma$ and their images $\alpha',\beta',\gamma'$ under a projective correspondence of one-dimensional fundamental forms be given. We prove that they determine a unique projective correspondence. First we prove existence.

Suppose first that both triples are ranges of points. A pencil can always be converted into a range by a perspectivity, so this is sufficient. Let the base lines $l, l'$ of the ranges $l(\alpha,\beta,\gamma)$ and $l'(\alpha',\beta',\gamma')$ be distinct. If necessary, translate the first range; a translation is itself a perspectivity with center at infinity. Choose an arbitrary point of $\alpha\alpha'$ as a center of perspectivity, and choose through $\alpha'$ a line $l''$ distinct from $l$ and $l'$. The resulting perspectivity $g: l \to l''$ satisfies $g(\alpha) = \alpha'$.

Now let $h: l'' \to l'$ be the perspectivity whose center is the intersection of the line joining $\beta'$ to $g(\beta)$ with the line joining $\gamma'$ to $g(\gamma)$. Then $h(g(\beta)) = \beta'$ and $h(g(\gamma)) = \gamma'$. Moreover, $\alpha' = l' \cap l''$, so $h(\alpha') = \alpha'$. Hence $h \circ g$ is a projective correspondence having the prescribed three pairs.

For uniqueness, if a fourth element $\delta$ has image $\delta'$, invariance of the cross-ratio requires

$$(\alpha,\beta;\gamma,\delta) = (\alpha',\beta';\gamma',\delta').$$

This equation determines $\delta'$ uniquely. □

**Corollary 3.5.5.** *Every projective correspondence between one-dimensional fundamental forms is a composition of perspectivities. Moreover, every cross-ratio-preserving map between one-dimensional fundamental forms is a projective correspondence.*[19]

**Proof.** The first assertion was established in the construction used in the proof of theorem 3.5.4. For the converse, a cross-ratio-preserving map is uniquely determined by three pairs of corresponding elements. Indeed, once $\alpha,\beta,\gamma$ and their images $\alpha',\beta',\gamma'$ under the cross-ratio-preserving map $f$ are given, the equation

$$(\alpha,\beta;\gamma,\delta)=(\alpha',\beta';\gamma',f(\delta))$$

uniquely determines $f(\delta)$ for every $\delta$ and hence uniquely determines $f$; one can verify that this map satisfies the condition of preserving cross-ratios. By theorem 3.5.4, the three prescribed pairs also determine a composition of perspectivities $f'$. Since each perspectivity preserves cross-ratios, $f'$ does also. Uniqueness now gives $f=f'$. ☐

The preceding theorem also supplies another proof of theorem 3.1.13.

**Revisit** (theorem 3.1.13)**.** *Four points, no three collinear, uniquely determine a projective correspondence.*

**Alternative proof of theorem 3.1.13.** Let $A,B,C,D$ and their prescribed images $A',B',C',D'$ be given. Existence was constructed in the proof of theorem 3.1.12; we prove uniqueness.

Put $O=AB\cap CD$. Since a projective correspondence preserves lines, necessarily $O':=f(O)=A'B'\cap C'D'$. Thus the images of the three points $A,B,O$ on $AB$ are known, and theorem 3.5.4 uniquely determines the image under $f$ of every point of $AB$. The same argument applies to $CD$.

Let $P$ be any point on neither $AB$ nor $CD$. If $P\notin AC$, put $S=CP\cap AB$ and $T=AP\cap CD$. The points $S':=f(S)$ and $T':=f(T)$ are uniquely determined, and preservation of lines gives $f(P)=C'S'\cap A'T'$. If $P\in AC$, use $S=DP\cap AB$ instead. Thus $f(P)$ is unique in every case. ☐

**Theorem 3.5.6.** *A projective correspondence between two ranges is a perspectivity if and only if the intersection of their base lines is a fixed point of the correspondence.*

**Proof.** Necessity is immediate. Conversely, suppose a projective correspondence $f$ takes the range $(A,B,C,X)$ on $l$ to $(A',B',C',X)$ on $l'$, where $X=l\cap l'$ and $f(X)=X$. We prove $(A,B,C,X)\overline{\wedge}(A',B',C',X)$. Let $S=AA'\cap BB'$. Projection from $S$ defines a perspectivity $g$ from $l$ to $l'$. The three points $X,A,B$ have the same respective images $X,A',B'$ under $f$ and $g$. By theorem 3.5.4, three pairs determine a projective correspondence, so $f=g$ and $f$ is a perspectivity. ☐

The following analogous statement is left to the reader to prove.

**Theorem 3.5.7.** *A projective correspondence between two pencils is a perspectivity if and only if the line joining their vertices is a fixed line of the correspondence.*

Theorems 3.5.6 and 3.5.7 are dual: interchanging points and lines transforms either statement into the other. This relationship will be studied in detail in the next section.

We now give some applications of projective correspondences between one-dimensional fundamental forms.

**Theorem 3.5.8.** *Let a line meet the sidelines $P_2P_3,P_3P_1,P_1P_2$ of $\triangle P_1P_2P_3$ at $Q_1,Q_2,Q_3$, respectively. Choose $R_1,R_2,R_3$ on the lines $P_2P_3,P_3P_1,P_1P_2$, respectively, and put $(P_2,P_3;R_1,Q_1)=k_1$, $(P_3,P_1;R_2,Q_2)=k_2$, and $(P_1,P_2;R_3,Q_3)=k_3$. Then:*

*(1) $R_1,R_2,R_3$ are collinear if and only if $k_1k_2k_3=1$;*

[19] A map $f:[\pi]\to[\pi']$ is said to preserve cross-ratios if, for all $\alpha,\beta,\gamma,\delta\in[\pi]$, $(\alpha,\beta;\gamma,\delta)=(f(\alpha),f(\beta);f(\gamma),f(\delta))$. Some authors therefore define a one-dimensional projective correspondence as a cross-ratio-preserving map between one-dimensional fundamental forms.

*(2)* $P_1R_1, P_2R_2, P_3R_3$ *are concurrent if and only if* $k_1k_2k_3 = -1$.

**Proof.** For (1), let $T = R_3Q_1 \cap P_1P_3$, as in figure 3.27. Suppose first that $R_1, R_2, R_3$ are collinear. Then

$$(P_2, P_3; R_1, Q_1) \overset{(R_3)}{\overline{\overline{\wedge}}} (P_1, P_3; R_2, T).$$

Applying proposition 3.2.13 to $P_1, P_3, R_2, T, Q_2$ gives

$$(P_1, P_3; R_2, T)(P_3, P_1; R_2, Q_2)(P_1, P_3; T, Q_2) = 1.$$

Here $(P_3, P_1; R_2, Q_2) = (P_1, P_3; Q_2, R_2)$. Invariance of cross-ratios under a perspectivity shows that this equation is equivalent to $k_1k_2k_3 = 1$.

Conversely, the equation $k_1k_2k_3 = 1$ gives

$$(P_2, P_3; R_1, Q_1) = (P_1, P_3; R_2, T).$$

By theorem 3.5.6, this projective correspondence is a perspectivity, so the lines joining corresponding points are concurrent. Hence $R_1, R_2, R_3$ are collinear.

For (2), let $R = R_3P_3 \cap R_2P_2$ and $T = P_1R \cap Q_1Q_2$, as in figure 3.28. If $P_1R_1, P_2R_2, P_3R_3$ are concurrent, apply part (1) to $\triangle P_1P_2R_1$ and $\triangle P_1P_3R_1$. This gives

$$\begin{cases} (P_1, P_2; R_3, Q_3)(P_2, R_1; P_3, Q_1)(R_1, P_1; R, T) = 1, \\ (P_1, P_3; R_2, Q_2)(P_3, R_1; P_2, Q_1)(R_1, P_1; R, T) = 1. \end{cases}$$

Dividing the first equation by the second and expanding the cross-ratios gives $k_1k_2k_3 = -1$.

Conversely, let $R' = P_1R \cap P_2P_3$. The same computation, now starting from $k_1k_2k_3 = -1$, gives

$$(P_2, P_3; R_1, Q_1) = (P_2, P_3; R', Q_1).$$

The definition of the cross-ratio and the method of coincidence give $R' = R_1$, and the three lines are concurrent. □

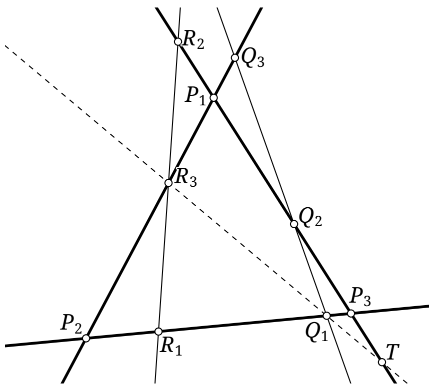


FIGURE 3.27

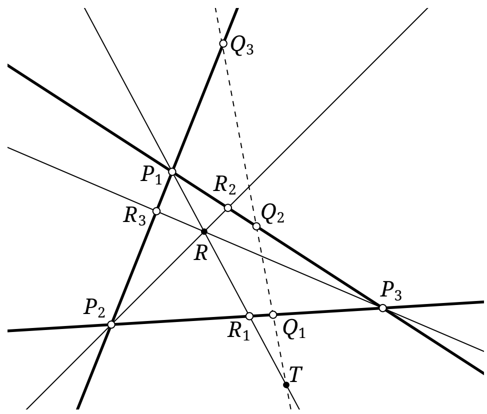


FIGURE 3.28

The preceding theorem gives a new proof of Menelaus' and Ceva's theorems.

**Alternative proofs of Theorems 0.1.14 and 0.1.15.** Take $Q_1Q_2Q_3$ in theorem 3.5.8 to be the line at infinity. Parts (1) and (2) then become Menelaus' theorem and Ceva's theorem, respectively. □

**Example 3.5.9.** Prove Pappus' theorem using projective correspondences of one-dimensional fundamental forms.

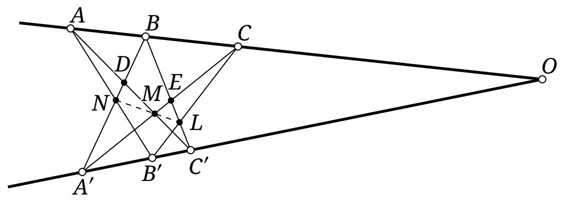


FIGURE 3.29

**Solution.** In figure 3.29, the points $A,B,C$ are collinear, as are $A',B',C'$, and the two base lines meet at $O$. Put $N=AB'\cap A'B$, $M=AC'\cap A'C$, and $L=BC'\cap B'C$. We prove that $N,M,L$ are collinear. Let $D=AC'\cap A'B$ and $E=A'C\cap BC'$. Observe that

$$(B,D,N,A')\overset{(A)}{\barwedge}(O,C',B',A')\overset{(C)}{\barwedge}(B,C',L,E).$$

Hence $(B,D,N,A')\barwedge(B,C',L,E)$. The intersection $B$ of the two base lines is fixed by this projective correspondence, so theorem 3.5.6 gives $(B,D,N,A')\barwedge(B,C',L,E)$. The lines joining corresponding points are therefore concurrent, which gives the collinearity of $N,M,L$. ☐

We conclude by introducing complete quadrilaterals and complete quadrangles, and proving their properties by means of projective correspondences of one-dimensional fundamental forms.

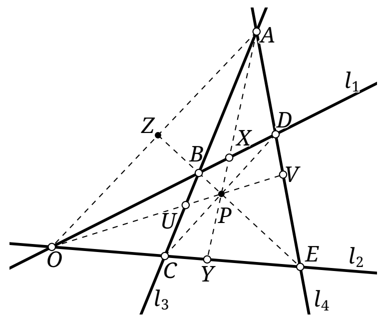


FIGURE 3.30

**Definition 3.5.10.**

(1) Four lines in a projective plane, no three concurrent, together with their six points of intersection form a *complete quadrilateral*. The six points are its *vertices*, and the four lines its *sides*. Two vertices not lying on a common side are *opposite vertices*; their joining line is a *diagonal*. The three diagonals form the *diagonal trilateral*.

(2) Four points in a projective plane, no three collinear, together with the six lines joining them form a *complete quadrangle*. The four points are its *vertices*, and the six lines its *sides*. Two sides having no common vertex are *opposite sides*; their intersection is a *diagonal point*. The three diagonal points form the *diagonal triangle*.[20]

In figure 3.30, the lines $l_1,l_2,l_3,l_4$ and their intersections $A,B,C,D,E,O$ form a complete quadrilateral, whose diagonals are $AO,CD,BE$. The points $B,C,D,E$ and the lines $BD,BC,CE,DE,EB,CD$ form a complete quadrangle, whose diagonal points are $A,O,P$ and whose diagonal triangle is $\triangle AOP$.

These notions extend to complete $n$-laterals and complete $n$-gons. Given $n$ lines in a plane, no three concurrent, the lines together with their $\mathrm{C}_n^2=\frac{n(n-1)}{2}$ pairwise intersections form a *complete n-lateral*; the $n$ lines are its sides and the $\frac{n(n-1)}{2}$ points its vertices. Dually, given $n$ points in a plane, no three collinear, the points together with the $\mathrm{C}_n^2=\frac{n(n-1)}{2}$ lines joining pairs of them form a *complete n-gon*; the $n$ points are its vertices and the $\frac{n(n-1)}{2}$ lines its sides.

[20] The expression "complete quadrilateral" is sometimes used ambiguously; in most such cases it denotes the four-line configuration, and the intended meaning is clear from context.

There are corresponding "simple" configurations. Given $n$ lines $l_1, l_2, \dots, l_n$ in a plane, no three concurrent, construct their successive intersections, namely the $n$ points $(l_1 \cap l_2)$, $(l_2 \cap l_3)$, ..., $(l_{n-1} \cap l_n)$, $(l_n \cap l_1)$. These $n$ lines and $n$ points form a *simple n-lateral*; the $n$ lines are its sides and the $n$ points are its vertices. Given $n$ points $A_1, A_2, \dots, A_n$ in a plane, no three collinear, join them successively to obtain the $n$ lines $A_1A_2, A_2A_3, \dots, A_{n-1}A_n, A_nA_1$. These $n$ points and $n$ lines form a *simple n-gon*; the $n$ points are its vertices and the $n$ lines are its sides.

For $n = 3$, the simple triangle, simple trilateral, complete triangle, and complete trilateral are the same configuration; below we simply call it a triangle.

We also fix the following terminology. A complete $n$-gon is inscribed in a conic when all its vertices lie on the conic; a complete $n$-lateral is circumscribed about a conic when all its sides are tangent to the conic. The notation $\{A_1, A_2, \dots, A_n\}$, or $A_1A_2 \cdots A_n$, denotes the complete $n$-gon with vertices $A_1, A_2, \dots, A_n$; similarly, $\{l_1, l_2, \dots, l_n\}$, or $l_1 l_2 \cdots l_n$, denotes the complete $n$-lateral with sides $l_1, l_2, \dots, l_n$.

**Theorem 3.5.11** (Harmonic property of complete quadrangles and quadrilaterals). *In the complete quadrilateral formed by the lines $l_1, l_2, l_3, l_4$ in figure 3.30, let the two diagonals $BE, CD$ meet at $P$. Then the four lines $l_1, l_2, OA, OP$ form a harmonic pencil.*

**Proof.** Put $U = AC \cap OP$, $V = AE \cap OP$, $X = OD \cap AP$, $Y = OE \cap AP$, and $Z = BE \cap AO$. Then

$$\begin{aligned}(l_1, l_2, OA, OP) \barwedge (D, E, A, V) \overset{(O)}{\barwedge} (X, Y, A, P) \overset{(O)}{\barwedge} (B, E, Z, P) \\ \overset{(A)}{\barwedge} (U, V, O, P) \overset{(C)}{\barwedge} (A, V, E, D) \barwedge (OA, OP, l_2, l_1).\end{aligned}$$

Consequently,

$$(l_1, l_2; OA, OP) = (D, E; A, V) = (A, V; E, D).$$

By proposition 3.2.11, however, $(A, V; E, D) = 1/(D, E; A, V)$. Thus the equality requires the cross-ratio to be $-1$; it cannot be 1 by exercise 3.16. Hence the four lines form a harmonic pencil. ☐

## Exercises

**Exercise 3.38.** Given three lines $a, b, c$ in one pencil, construct using only a straightedge a line $d$ such that $(a, b; c, d) = -1$.

**Exercise 3.39.** Prove Desargues' theorem using one-dimensional projective transformations.

**Exercise 3.40.** The points $B, A, A', B'$ occur in this order on a line $l$, and $AA'$ and $BB'$ have the same midpoint $M$. Under a one-dimensional projective correspondence, use a superscript $*$ for corresponding elements. Suppose the images occur on $l^*$ in the order $B^*, A^*, M^*, A'^*, B'^*$, and suppose $A^*A'^*$ and $B^*B'^*$ have the same midpoint $M'$. Prove that $M' = M^*$.

**Exercise 3.41.** Given $\triangle ABC$ and a point $P$ in the plane, put $A_1 = PA \cap BC$, $B_1 = PB \cap AC$, $C_1 = PC \cap AB$, and $A_2 = AA_1 \cap B_1C_1$. Find $(A, A_2; A_1, P)$.

**Exercise 3.42.** Choose collinear points $P, Q, R$ on the sidelines $BC, CA, AB$ of $\triangle ABC$, respectively. Suppose $P, Q, R$ are fixed and $B, C$ lie on respective fixed lines. Prove that $A$ also lies on a fixed line.

**Exercise 3.43.** Let $A, B, C, P, Q, R, X$ belong to one range, with $A, B, C$ distinct. If $(A, B, C, P, Q, R) \barwedge (B, C, A, Q, R, X)$, prove that $X = P$.

**Exercise 3.44.** Let ranges $[P]$ and $[P']$ be in projective correspondence $[P]\barwedge[P']$. If $P_r,P'_r$ and $P_s,P'_s$ are two pairs of corresponding points, prove that the intersection of $P_rP'_s$ and $P_sP'_r$ lies on a fixed line.

### 3.5.2 One-Dimensional Projective Transformations and Involutions

Projective correspondences of one-dimensional fundamental forms lead to a notion of projective transformation.

**Definition 3.5.12.** Two one-dimensional fundamental forms are called *superposed* if they are ranges on the same base line or pencils with the same vertex. A projective correspondence between two superposed fundamental forms is a *one-dimensional projective transformation*; an affine correspondence between them is a *one-dimensional affine transformation*.[21]

This definition parallels the earlier, two-dimensional definition. The affine case follows immediately from theorem 3.5.2.

**Theorem 3.5.13.** *A one-dimensional projective transformation on a line $l$ is affine if and only if it fixes the point at infinity of $l$.*

One-dimensional affine transformations will play little role, so we concentrate on general one-dimensional projective transformations. These have the following readily established property.

**Proposition 3.5.14.**
*A one-dimensional projective transformation preserves cross-ratios of corresponding quadruples. Conversely, a cross-ratio-preserving map between superposed one-dimensional fundamental forms is a one-dimensional projective transformation.*

**Proof.** The first assertion follows because this is a special projective correspondence between one-dimensional fundamental forms, and such correspondences preserve cross-ratios. Conversely, corollary 3.5.5 says that a cross-ratio-preserving map between such forms is a projective correspondence; superposition makes it a one-dimensional projective transformation. ☐

The criterion by preservation of cross-ratios can be simplified as follows.

**Proposition 3.5.15.** *Let $X$ be a one-dimensional fundamental form, let $\alpha,\beta,\gamma\in X$ be three prescribed elements, and suppose a map $f: X\to X$ takes them to $\alpha',\beta',\gamma'$, respectively. Then $f$ is projective if and only if, for every $\xi\in X$, writing $\xi'=f(\xi)$, one has*

$$(\alpha\beta,\gamma\xi)=(\alpha'\beta',\gamma'\xi').$$

**Proof.** Necessity is immediate. For sufficiency it is enough to consider ranges, since pencils can be carried projectively to ranges. For arbitrary points $\xi_1,\xi_2,\xi_3,\xi_4$, use primes for their images under $f$. We have

$$\begin{aligned}(\alpha\beta\xi_1)-(\alpha\beta\xi_2)&=\frac{\overline{\alpha\xi_1}}{\overline{\beta\xi_1}}-\frac{\overline{\alpha\xi_2}}{\overline{\beta\xi_2}}=\frac{\overline{\alpha\xi_1}\cdot\overline{\beta\xi_2}-\overline{\beta\xi_1}\cdot\overline{\alpha\xi_2}}{\overline{\beta\xi_1}\cdot\overline{\beta\xi_2}}\\&=\frac{\overline{\alpha\xi_1}\cdot(\overline{\beta\alpha}+\overline{\alpha\xi_2})-(\overline{\alpha\xi_1}+\overline{\beta\alpha})\cdot\overline{\alpha\xi_2}}{\overline{\beta\xi_1}\cdot\overline{\beta\xi_2}}=\frac{(\overline{\alpha\xi_1}-\overline{\alpha\xi_2})\cdot\overline{\beta\alpha}}{\overline{\beta\xi_1}\cdot\overline{\beta\xi_2}}=\frac{\overline{\xi_1\xi_2}\cdot\overline{\alpha\beta}}{\overline{\beta\xi_1}\cdot\overline{\beta\xi_2}}.\end{aligned}$$

[21] A projective transformation between two ranges on the same line $l$ is called a projective transformation on $l$; one between two pencils with the same vertex $P$ is called a projective transformation at $P$. Here "one-dimensional" is a modifier in the terminology of this definition, so, when there is no danger of ambiguity, we simply say projective transformation or affine transformation.

Therefore

$$(\xi_1\xi_2\xi_3) = \frac{\overline{\xi_1\xi_3}}{\overline{\xi_2\xi_3}} = \left( [(\alpha\beta\xi_1)-(\alpha\beta\xi_3)] \frac{\overline{\beta\xi_1}\cdot\overline{\beta\xi_3}}{\overline{\alpha\beta}} \right) \div \left( [(\alpha\beta\xi_2)-(\alpha\beta\xi_3)] \frac{\overline{\beta\xi_2}\cdot\overline{\beta\xi_3}}{\overline{\alpha\beta}} \right)$$
$$= \frac{(\alpha\beta\xi_1)-(\alpha\beta\xi_3)}{(\alpha\beta\xi_2)-(\alpha\beta\xi_3)} \frac{\overline{\beta\xi_1}}{\overline{\beta\xi_2}}.$$

Hence

$$(\xi_1\xi_2,\xi_3\xi_4) = \frac{(\xi_1\xi_2\xi_3)}{(\xi_1\xi_2\xi_4)} = \frac{(\alpha\beta\xi_1)-(\alpha\beta\xi_3)}{(\alpha\beta\xi_2)-(\alpha\beta\xi_3)} \div \frac{(\alpha\beta\xi_1)-(\alpha\beta\xi_4)}{(\alpha\beta\xi_2)-(\alpha\beta\xi_4)},$$

and similarly

$$(\xi_1'\xi_2',\xi_3'\xi_4') = \frac{(\alpha'\beta'\xi_1')-(\alpha'\beta'\xi_3')}{(\alpha'\beta'\xi_2')-(\alpha'\beta'\xi_3')} \div \frac{(\alpha'\beta'\xi_1')-(\alpha'\beta'\xi_4')}{(\alpha'\beta'\xi_2')-(\alpha'\beta'\xi_4')}.$$

The hypothesis gives

$$\frac{(\alpha'\beta'\xi_i')}{(\alpha\beta\xi_i)} = \frac{(\alpha\beta\gamma)}{(\alpha\beta\xi_i)} \div \frac{(\alpha'\beta'\gamma')}{(\alpha'\beta'\xi_i')} \times \frac{(\alpha'\beta'\gamma')}{(\alpha\beta\gamma)} = \frac{(\alpha\beta,\gamma\xi_i)}{(\alpha'\beta',\gamma'\xi_i')} \times \frac{(\alpha'\beta'\gamma')}{(\alpha\beta\gamma)} = \text{const.}$$

Thus $(\xi_1'\xi_2',\xi_3'\xi_4') = (\xi_1\xi_2,\xi_3\xi_4)$ for all four elements, so $f$ is projective. □

We next ask how a projective transformation is determined.

**Theorem 3.5.16** (Fundamental theorem for projectivities on a line)**.** *Three pairs of corresponding elements on a one-dimensional fundamental form uniquely determine a one-dimensional projective transformation.*

**Proof.** This follows immediately from theorem 3.5.4. □

**Theorem 3.5.17.** *A nonidentity one-dimensional projective transformation has at most two fixed elements.*[22]

**Proof.** If such a transformation had three fixed elements, then it and the identity id would have three prescribed pairs in common. By theorem 3.5.16, the transformation would be id. □

This leads to the following definition.

**Definition 3.5.18.** A nonidentity one-dimensional projective transformation is classified by its number of fixed elements:

(1) it is *hyperbolic* if it has exactly two fixed elements;
(2) it is *parabolic* if it has exactly one fixed element;
(3) it is *elliptic* if it has no fixed element.

**Remark.** It is enough to consider a projective transformation between superposed ranges.[23] Suppose its three pairs are $A,B,C$ and $A',B',C'$, and let $D$ be a fixed point. Put $\overline{AD}=x$. The equality $(AB,CD) = (A'B',C'D)$ becomes

$$\frac{\overline{AC}}{\overline{BC}} \div \frac{x}{x-\overline{AB}} = \frac{\overline{A'C'}}{\overline{B'C'}} \div \frac{x-\overline{AA'}}{x-\overline{AB'}}. \tag{$*$}$$

After simplification, this is a quadratic equation in $x$.[24] Two distinct real roots, one repeated real root, and no real roots correspond respectively to the three cases in definition 3.5.18.

---

[22] A *fixed element*, also called a *self-corresponding element*, is an element left unchanged by the transformation.

[23] For superposed pencils, intersect every line of the pencil with a line $l$ not through the common vertex, thereby converting the pencils to ranges.

[24] If $A=A'$, $B=B'$, and $C=C'$, equation $(*)$ is an identity. If the quadratic coefficient in $(*)$ vanishes, one root may be taken to be infinity, while the remaining linear equation supplies the other root; the equation can still be treated as having two roots.

The definition in definition 3.5.18 is formulated over the real projective plane. Over the complex projective plane, $x$ ranges over $\mathbb{C}\cup\{\infty\}$ and always has two solutions, which may coincide or be imaginary. Thus a one-dimensional projective transformation over the complex projective plane has exactly two fixed elements, counted with multiplicity. The three cases in definition 3.5.18 become, respectively, two distinct real fixed elements, a repeated fixed element, and two distinct imaginary fixed elements.

The next theorem describes the fixed elements of a projective transformation.

**Theorem 3.5.19.** *Let a nonparabolic one-dimensional projective transformation have fixed elements $\mu,\nu$, which may be imaginary. For every element $\alpha$ and its image $\alpha'$ under the transformation, $(\alpha\alpha',\mu\nu)=\mathrm{const}$.*

**Proof.** Suppose $\alpha,\beta$ have images $\alpha',\beta'$. Invariance of the cross-ratio gives

$$\frac{(\mu\nu\alpha)}{(\mu\nu\beta)}=(\mu\nu,\alpha\beta)=(\mu\nu,\alpha'\beta')=\frac{(\mu\nu\alpha')}{(\mu\nu\beta')},$$

and consequently

$$(\mu\nu,\alpha\alpha')=\frac{(\mu\nu\alpha)}{(\mu\nu\alpha')}=\frac{(\mu\nu\beta)}{(\mu\nu\beta')}=(\mu\nu,\beta\beta').$$

By proposition 3.2.11, this is equivalent to $(\alpha\alpha',\mu\nu)=(\beta\beta',\mu\nu)$. □

The converse is also true; its proof is left as an exercise.

**Theorem 3.5.20.** *Let $\mu,\nu$ be distinct elements of a one-dimensional fundamental form. Suppose a bijection $f$ of this form to itself fixes $\mu,\nu$ and takes every element $\alpha$ to an element $\alpha'$ satisfying $(\mu\nu,\alpha\alpha')=k$, where $k$ is a given number. Then $f$ is projective.*

We now study a special projective transformation: an involution in one dimension.

**Definition 3.5.21.** An *involutory transformation*, or simply an *involution*, is a projective transformation $f\neq\mathrm{id}$ satisfying $f\circ f=\mathrm{id}$. If $f$ is one-dimensional, it is called a *one-dimensional involution*.[25]

**Remark.** Conventions differ as to whether the identity is counted as an involution. Here it is excluded.

One-dimensional involutions have the following important property.

**Theorem 3.5.22.** *Two pairs of corresponding elements uniquely determine a one-dimensional involution. More precisely, suppose elements $\alpha,\beta$ and their images $\alpha',\beta'$ are prescribed.*[26] *Then there exists a unique involution $f$ with these pairs.*

**Proof.** Let $f:[\pi]\to[\pi]$ be a transformation on a line $l$ or at a point $P$, with prescribed pairs $\alpha,\alpha'$ and $\beta,\beta'$, so that $f(\alpha')=\alpha$ and $f(\beta')=\beta$. Let $\gamma\in[\pi]$ be arbitrary.
*1.* If $\alpha=\alpha'$ and $\beta=\beta'$ do not both hold, assume without loss of generality that $\alpha\neq\alpha'$. An involution must satisfy

$$(\alpha,\alpha';\beta,\gamma)=(\alpha',\alpha;\beta',f(\gamma)).$$

This rule determines $f(\gamma)$ uniquely. It preserves cross-ratios, hence defines a projective transformation, and direct verification gives $f(f(\gamma))=\gamma$ for every $\gamma$. Thus it defines a unique involution.
*2.* If both $\alpha$ and $\beta$ are fixed, an involution must satisfy

$$(\alpha,\beta;\gamma,f(\gamma))=(\alpha,\beta;f(\gamma),\gamma).$$

[25] An involution between ranges on the same line $l$ is also called an involution on $l$; one between pencils with the same vertex $P$ is called an involution at $P$.

[26] The two pairs must not coincide. Since an involution composed with itself is the identity, in addition to $\alpha\neq\beta$ and $\alpha'\neq\beta'$, one must require $\alpha\neq\beta'$ and $\beta\neq\alpha'$.

There are only two possibilities: $f(\gamma)=\gamma$, or $\alpha,\beta,\gamma,f(\gamma)$ form a harmonic range or pencil. The first would give the identity and is excluded. The harmonic condition uniquely determines $f(\gamma)$; again the resulting map preserves cross-ratios and satisfies $f(f(\gamma))=\gamma$ for every $\gamma$. Thus it uniquely determines an involution. □

**Corollary 3.5.23.** *Given two distinct fixed elements of an involution, two further elements form a corresponding pair if and only if they are harmonic conjugates with respect to the fixed elements.*

**Proof.** This is the second case in the proof of theorem 3.5.22. □

As in the preceding discussion of one-dimensional projective transformations, an involution cannot have three fixed points, since it would then be the identity. This leads to the following definition.

**Definition 3.5.24.** Over the real projective plane, an involution on a line $l$ or at a point $P$ is *hyperbolic* if it has two fixed elements, and *elliptic* if it has none. It cannot have exactly one fixed element.

**Remark.** As in the remark following definition 3.5.18, setting $f(\gamma)=\gamma$ in

$$(\alpha,\alpha';\beta,\gamma)=(\alpha',\alpha;\beta',f(\gamma))$$

gives a quadratic equation. Thus, over the complex projective plane, every involution has two fixed elements. The two cases above correspond to two distinct real fixed elements and two distinct imaginary fixed elements.

Here is why a single fixed element, or equivalently a repeated fixed element, is impossible. Suppose an involution $f$ has only one fixed element $\xi$. Choose $\alpha\neq\xi$ and write $f(\alpha)=\alpha'$. Construct the fourth harmonic element $\xi'$ such that $(\alpha\alpha',\xi\xi')=-1$; clearly $\xi'\neq\xi$. Since $f$ is an involution and preserves cross-ratios, we have

$$-1=(f(\alpha)f(\alpha'),f(\xi)f(\xi'))=(\alpha'\alpha,f(\xi)f(\xi'))=(\alpha\alpha',\xi f(\xi')).$$

The last equality uses proposition 3.2.11 and $f(\xi)=\xi$. Uniqueness of the fourth harmonic element gives $f(\xi')=\xi'$. Since $\xi$ is the only fixed element, $\xi=\xi'$, a contradiction. □

**Theorem 3.5.25** (Desargues' involution theorem)**.** *If a line meets the three pairs of opposite sides of a complete quadrangle, the resulting three pairs of intersection points are corresponding pairs of one involution on that line.*

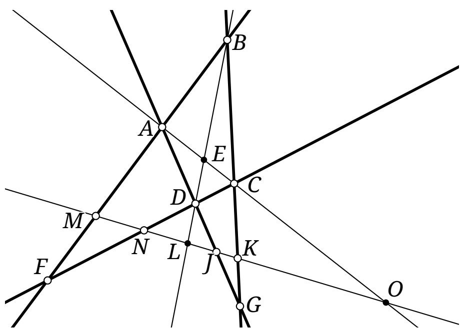


FIGURE 3.31

**Proof.** Let a line $l$ meet the three pairs of opposite sides of the complete quadrangle $A,B,C,D$ in the pairs $M,N$; $J,K$; and $L,O$, respectively, as in figure 3.31. It is enough to prove $(L,O;N,J)=(O,L;M,K)$. But

$$(L,O,N,J)\overset{(D)}{\overline{\overline{\wedge}}}(E,O,C,A)\overset{(B)}{\overline{\overline{\wedge}}}(L,O,K,M),$$

and $(L,O;K,M)=(O,L;M,K)$, which proves the assertion. □

Finally, involutions allow perpendicularity to be defined projectively.

**Definition 3.5.26** (Projective definition of perpendicularity)**.** Fix an elliptic *absolute involution* on the line at infinity of the plane.[27] Two lines are called *perpendicular* if their points at infinity are a corresponding pair of the absolute involution. Over the complex projective plane, this involution has two imaginary fixed points at infinity, called the *circular points*.

We show that this agrees with the usual definition of perpendicularity. Under the usual definition, consider all lines through a point $P$ and the map $g$ taking each such line $l$ to the line $g(l)$ through $P$ perpendicular to $l$. It is a one-dimensional projective transformation: it rotates all lines through 90°, so angles between corresponding lines and hence cross-ratios are preserved. Moreover, $a \perp b$ implies $b \perp a$, so $g(g(a)) = a$; thus $g$ is an involution.

Intersecting every line through $P$ with the line at infinity $l_\infty$ induces an involution $f$ on $l_\infty$, because perspectivities preserve cross-ratios. A pair of corresponding lines gives a pair of corresponding points of $f$.

For another point $P'$, define $g'$ similarly. If a line $l$ through $P$ and a line $l'$ through $P'$ meet the line at infinity at the same point, then $l \parallel l'$, and therefore $g(l) \parallel g'(l')$. Hence $g(l)$ and $g'(l')$ also meet the line at infinity at the same point. Thus every choice of $P$ induces the same involution $f$.

The map $g$ plainly has no fixed line, so it is elliptic, and the induced involution $f$ is elliptic as well. Choosing any finite point $P$ therefore determines an elliptic involution $f$ on the line at infinity, and this one involution determines all perpendicularity relations. □

According to definition 3.5.26, the choice of an absolute involution may appear arbitrary, and indeed perpendicularity depends on that choice. This is precisely the distinction between a projective plane and an ordinary Euclidean plane. Projective transformations do not preserve angles, so perpendicularity is not a purely projective notion; it involves metric structure. The elements of an abstract projective plane have equal status, and perpendicularity cannot be defined without additional data. Choosing an absolute involution amounts to adding metric structure to the projective plane. Since its elements have equal status, there is freedom in how this metric structure is added, and different choices can yield different metric structures. Conversely, a given Euclidean plane determines a unique such elliptic involution.

Combining corollary 3.5.23 and definition 3.5.26 gives an equivalent formulation.

**Theorem 3.5.27** (Projective characterization of perpendicularity)**.** *Two lines are perpendicular if and only if their points at infinity and the two circular points form a harmonic range.*

The following example applies the projective definition of perpendicularity.

**Example 3.5.28.** Give a projective proof that, for every triangle, its orthocenter, circumcenter, and centroid are collinear.

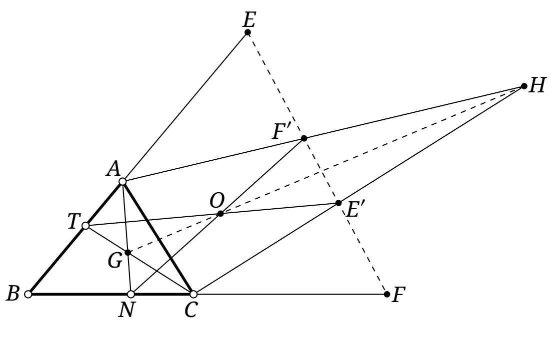


Figure 3.32

[27]Here "absolute" means chosen in advance: it is additional structure fixed on the projective plane, and all geometric relations are subsequently considered relative to it. Once chosen, the involution remains fixed.

**Solution.** Refer to the schematic figure 3.32.[28] Let $E,F$ be the points at infinity on the sides $BA,BC$ of $\triangle ABC$, respectively, and let $E',F'$ be their respective partners under the absolute involution on the line at infinity. By the projective definition of perpendicularity, $BA \perp CE'$ and $BC \perp AF'$. Thus $H = CE' \cap AF'$ is the orthocenter of $\triangle ABC$.

Let $T,N$ be the midpoints of $BA,BC$, respectively. Then $G = CT \cap AN$ is the centroid. Since $BA \perp TE'$ and $BC \perp NF'$, the lines $TE',NF'$ are perpendicular bisectors of $BA,BC$, and $O = TE' \cap NF'$ is the circumcenter.

Because $TN$ is a midline of $\triangle ABC$, the lines $TN$ and $AC$ meet at a point at infinity. Thus $TN,AC,E'F'$ are concurrent. Applying Desargues' theorem to $\triangle NAF'$ and $\triangle TCE'$ shows that the intersections $G,O,H$ of corresponding sides are collinear. □

---

## Exercises

**Exercise 3.45.** Prove theorem 3.5.20.

**Exercise 3.46.** Do all involutions on a fixed line (or at a fixed point), together with the identity, form a group? Give a reason.

**Exercise 3.47.** Let $f$ be a parabolic projective transformation between two superposed ranges, and let $E$ be its fixed point. Prove that, for every point $P$ of the range, $(P,f^2(P);f(P),E) = -1$.

**Exercise 3.48.** Suppose a one-dimensional projective transformation $f$ has a nonfixed pair $\alpha,\alpha'$ satisfying the involution condition, namely $f(\alpha) = \alpha', f(\alpha') = \alpha$, and $\alpha \neq \alpha'$. Prove that $f$ is an involution.

**Exercise 3.49.** Let the six points $A,B,C,A',B',C'$ be collinear and satisfy $(AA',BC) = (BB',CA) = (CC',AB) = -1$. Prove that $A,B,C$ and $A',B',C'$ form three corresponding pairs of a single involution.

**Exercise 3.50.** Let $P$ be a point in the plane of $\triangle ABC$, and let $l$ be a line through $P$. Reflect the lines $PA,PB,PC$ in $l$; let their images meet the lines $BC,CA,AB$ at $A',B',C'$, respectively. Prove that $A',B',C'$ are collinear.

**Exercise 3.51.** The triangles $\triangle ABC$ and $\triangle A'B'C'$ are centrally symmetric. Through $A',B',C'$ draw lines $l_1,l_2,l_3$, respectively, with $l_1 \parallel l_2 \parallel l_3$. Prove that $(l_1 \cap BC)$, $(l_2 \cap CA)$, and $(l_3 \cap AB)$ are collinear.

**Exercise 3.52*.** Prove that every one-dimensional projective transformation is a composition of at most two one-dimensional involutions.

**Exercise 3.53.** Prove that the two circular points are the points at infinity on the two lines of slopes $+\mathrm{i}$ and $-\mathrm{i}$ in a Cartesian coordinate system.

**Exercise 3.54* (Laguerre's theorem).** Let $I,J$ be the circular points, and let $l_1,l_2$ meet $l_\infty$ at $P,Q$, respectively. Put $\mu = (PQ,IJ)$. Prove that $\measuredangle(l_1,l_2) = \frac{1}{2\mathrm{i}} \ln \mu$.
*Hint.* Use exercise 3.53.

---

[28] The figure is only schematic and, of course, cannot represent the actual configuration.

### 3.5.3 Introduction of Imaginary Elements*

Imaginary elements were introduced algebraically earlier. We now complete the picture by introducing imaginary points and lines in a projective plane by purely geometric means. This is included only for logical completeness; readers not interested in the construction may omit this subsection.

Our aim is to define hypothetical imaginary elements—imaginary points and imaginary lines—on a real projective plane, and to explain why, from a purely projective viewpoint, they can be treated on the same footing as real elements. The earlier algebraic construction enlarged the scalar field from $\mathbb{R}$ to $\mathbb{C}$. Here instead we adjoin an element imagined not to exist in the real projective plane.[29]

An elliptic involution on a line has no real fixed point but has two imaginary fixed points. This suggests the following geometric definition.

**Definition 3.5.29.** Let $A$,$B$,$A'$,$B'$ occur in this order on a line $l$ of a real projective plane. They determine an elliptic involution $f$ on $l$, with $(A,A')$ and $(B,B')$ as corresponding pairs. Imagine that $f$ has two fixed points $C,C'$, which are uniquely determined by $A,B,A',B'$. These are the two *imaginary points* on $l$, denoted respectively by $(ABA'B')$ and $(AB'A'B)$. Either is called the *conjugate imaginary point* of the other, and the two are said to be *conjugate*.

**Remark.** The reader can readily prove that two corresponding pairs of an elliptic involution necessarily interlace: their order may be $A,B,A',B'$ or $A,B',A',B$, but not $A,A',B',B$ or $A,A',B,B'$. The order of the four points must therefore be respected in the notation. In a hyperbolic involution, by contrast, if $(A,A')$ and $(B,B')$ are corresponding pairs, one pair of corresponding points is adjacent; possible orders include $B',A',A,B$ and $A,B,B',A'$.

The two imaginary points are distinguished by the two orders, but a cyclic permutation of all four entries in the clockwise direction still denotes the same point; for example, $(AB'A'B) = (BAB'A') = (B'A'BA)$. This gives a geometric definition of conjugate points. From this viewpoint, the complex-conjugation transformation mentioned in the remark following corollary 3.2.6 interchanges the two conjugate imaginary points while fixing every real point. The definition and all subsequent properties are unchanged because the two points have exactly symmetric roles. The "imaginary distance" to a real point, obtained by solving the condition that the imaginary point be a fixed point of an elliptic involution, is replaced by its complex conjugate; hence cross-ratios change. This also supplements the remark following theorem 3.1.12.

Imaginary lines are defined dually.

**Definition 3.5.30.** Let $P$ be a point of a real projective plane, and let $a,b,a',b'$ be four lines through $P$, occurring in this cyclic order, clockwise or counterclockwise. They determine an elliptic involution $f$ at $P$, with $(a,a')$ and $(b,b')$ as corresponding pairs. Imagine that $f$ has two fixed lines $c,c'$, uniquely determined by $a,b,a',b'$. These are the two *imaginary lines* through $P$, denoted respectively by $(aba'b')$ and $(ab'a'b)$. Either is the *conjugate imaginary line* of the other, and the two are said to be conjugate.

For example, let $(a,a')$ and $(b,b')$ be two pairs of perpendicular lines through one point, meeting the line at infinity in $(A,A')$ and $(B,B')$, respectively. By the preceding definition, $(ABA'B')$ and $(AB'A'B)$ are the two circular points.

To place imaginary and real elements on a comparable footing, we next define incidence between an imaginary point and an imaginary line.

[29] Thus imaginary elements could in fact be considered entirely within the real projective plane. After this subsection, however, we shall continue to distinguish a "real projective plane," in which imaginary elements are not considered, from a "complex projective plane," in which they are.

**Definition 3.5.31.** Let $l$ be an imaginary line and $P$ an imaginary point of a real projective plane. We say that *$P$ lies on $l$* if there are four real lines $a,b,a',b'$ in a pencil and four real points $A,B,A',B'$ in a range such that $A,B,A',B'$ lie respectively on $a,b,a',b'$, and $l=(aba'b')$, $P=(ABA'B')$.

This definition is natural and agrees with the earlier algebraic one. Under the perspectivity

$$(a,b,a',b')\overline{\wedge}(A,B,A',B'),$$

the intersections with $AB$ of the imaginary fixed lines of the involution determined by $(a,a')$ and $(b,b')$ are exactly the imaginary fixed points of the involution determined by $(A,A')$ and $(B,B')$.

The definition immediately yields the following facts, which were also clear from the algebraic viewpoint.

**Theorem 3.5.32.** *An imaginary line contains exactly one real point, and an imaginary point lies on exactly one real line.*

**Proof.** We prove the first assertion; the second is dual. An imaginary line cannot contain two real points, since the line joining two real points is real. On the other hand, the pencil $(a,b,a',b')$ defining an imaginary line has a real vertex, and the imaginary line passes through that vertex. □

The same definitions allow us to treat further incidence questions.

**Theorem 3.5.33.** *In a projective plane, two imaginary points not lying on the same real line determine a unique line, which is imaginary. Two imaginary lines not passing through the same real point meet in a unique point, which is imaginary.*

**Proof.** We first show that, for any two imaginary points, a real point collinear with them can be constructed. This establishes the existence of a collinear real point; it does not presuppose the existence or uniqueness of the imaginary line.

Let the two imaginary points lie on distinct real lines $m,n$, and write $M=(ABA'B')$ and $N=(CDC'D')$. Put $S=m\cap n$. Let $i$ be the elliptic involution on $m$ determined by the pairs $(A,A')$, $(B,B')$, and let $j$ be the involution on $n$ determined by $(C,C')$, $(D,D')$.

Because $A,B,A',B'$ interlace, the circles $\odot(AA')$ and $\odot(BB')$ have two real intersections; choose one and call it $E$. Taking each line through $E$ to its perpendicular is an involution. Thus, for $X\in m$, let $EX'\perp EX$ meet $m$ at $X'$. The resulting involution on $m$ has $EA\perp EA'$ and $EB\perp EB'$, so its pairs include $(A,A')$ and $(B,B')$; it is therefore $i$. Similarly, choose $F\in\odot(CC')\cap\odot(DD')$. For a variable point $X$ on $n$, draw $FX'\perp FX$ to meet $n$ at $X'$. This again defines an involution on $n$, namely $j$.

Put $U=i(S)$ and $V=j(S)$. Then $\angle SEU=90°$. Let its internal and external bisectors meet $m$ at $P,P_1$, respectively. Then $PE\perp P_1E$, so $P_1=i(P)$, and proposition 3.2.17 gives $(SU,PP_1)=-1$. Similarly, let the internal and external bisectors of the right angle $SFV$ meet $n$ at $Q,Q_1$; then $(SV,QQ_1)=-1$. The pairs $(S,U)$ and $(P,P_1)$ also determine $i$, so, after interchanging $P,P_1$ if necessary, we may write $M=(SPUP_1)$. Likewise, take $N=(SQVQ_1)$.

Let $f: m\to n$ be the projective correspondence taking $S,U,P$ to $S,V,Q$, respectively. Since $(SU,PP_1)=(SV,QQ_1)=-1$, invariance of cross-ratios gives $f(P_1)=Q_1$. As $f$ fixes $S$, theorem 3.5.6 shows that it is a perspectivity, with center $O=PQ\cap UV$. The imaginary line $((OS)(OP)(OU)(OP_1))$ passes by definition through the imaginary points $M,N$ and contains the real point $O$.

It remains to prove uniqueness and to show that the real point must be $O$. Suppose another imaginary line through $M,N$ has real point $O'$, unique by theorem 3.5.32. The perspectivity $f'$ from $m$ to $n$ with center $O'$ transports the involution defining the imaginary line to the involutions $i,j$ induced on $m,n$. Thus $f'$ fixes $S$ and carries corresponding pairs of $i$ to corresponding pairs of $j$. Hence it takes $U$ to $V$ and takes $P$ to $Q$, while fixing $S$. Consequently $f'=f$ and $O'=O$. This proves uniqueness. The assertion concerning two imaginary lines is dual. □

**Alternative proof.** Write the two imaginary points as $(ABA'B')$ and $(CDC'D')$. It is enough to choose $A,B,A',B',C,D,C',D'$ so that the ranges $(A,B,A',B')$ and $(C,D,C',D')$ are perspective, and to show that their center of perspectivity $O$ is independent of these choices. The required line is then $((OA)(OB)(OA')(OB'))$. The details are left to the reader. □

We have now defined imaginary elements directly on a real projective plane and may treat them just like real elements. For example, perspectivities and projective correspondences involving imaginary elements may be defined in the same way as for real elements. Some details may not have been treated fully carefully; these are left to the reader to consider and complete. This is, after all, an introductory course, and we do not intend to construct the entire theory with complete rigor from the axioms. In section 1.2 we also defined objects such as imaginary circles and imaginary ellipses algebraically. At present we cannot yet give purely geometric definitions of them. Once the projective definition of conics, including circles, has been developed in chapter 6, however, their introduction is natural, since we have already shown that imaginary elements may be introduced directly into projective relations.

Thus, when imaginary elements are admitted, the following statements can be explained geometrically: a nondegenerate conic and a line always have two intersections counted with multiplicity, and two nondegenerate conics always have four intersections counted with multiplicity. For an intuitive picture, consider two congruent ellipses with the same center and with perpendicular major axes. They plainly have four common points. Deform the ellipses continuously. If the number of distinct intersections in the real projective plane decreases, two intersection points must approach each other. When they coincide, the number of distinct intersections decreases by one and the two conics are tangent there, but the number counted with multiplicity is unchanged. If the conics continue to move from tangency until they separate there, the number of intersections in the real projective plane decreases by two, but they may be regarded as meeting at two imaginary points. Similarly, starting with an ellipse and its major axis gives an intuitive picture of why they always have two intersections. This is only an intuitive argument, not a rigorous proof. For reasons of space we shall not pursue a rigorous treatment here; the reader may rely on the preceding intuition or simply regard the statements as known facts.

We finish with another representation of imaginary elements. Begin with points on a real line $l$, and choose an imaginary point $X$ not on $l$. For $P \in l$, define $f(P) = P$ if $P$ is real; if $P$ is imaginary, define $f(P)$ to be the unique real point on $PX$. In this way, the real point $f(P)$ represents $P$.

Let $l'$ be the unique real line through $X$. Then $f$ is a bijection from $l$, including its imaginary points, to $(\mathbb{RP}^2 \setminus l') \cup l$. Indeed, on the real points of $l$, the map is plainly a bijection to the real points of $l$. If $P$ is imaginary, then $f(P)$ is unique and lies on neither $l$ nor $l'$; otherwise $P$ would be real. Conversely, for a point $f(P)$ of $\mathbb{RP}^2$ on neither $l$ nor $l'$, $Xf(P)$ meets $l$ at an imaginary point $P$; a real intersection would make $PX$ a real line, a contradiction.

Thus every line, with its imaginary points included, may be pictured as represented by another real plane "embedded" at that line. Dually, the lines through a point, including imaginary lines, may be represented by the lines of a real plane.

---

## Exercises

**Exercise 3.55.** Given imaginary points $M = (ABA'B')$ and $N = (CDC'D')$, construct with straightedge and compass the real point on $MN$.

**Exercise 3.56.** Let $m,n$ be lines in a plane and $X$ a point outside both. Using the final representation of imaginary elements above, investigate when the ranges on $m,n$, including imaginary points, are perspective.

**Exercise 3.57.** Prove the part of theorem 3.5.33 concerning the intersection of two imaginary lines.

**Exercise 3.58.** Let the sixteen points $A_i, B_i, A_i', B_i'$ $(i = 1,2,3,4)$ be collinear, and let $C_i = (A_i B_i A_i' B_i')$ be imaginary points. Express the cross-ratio $(C_1 C_2, C_3 C_4)$ in terms of the sixteen given points.

## 3.6 Polarity with Respect to a Circle

### 3.6.1 Polarity and Its Basic Properties

We begin with the special case of polarity with respect to a circle. The general construction will be introduced later.

**Definition 3.6.1** (Polarity with respect to a circle)**.** Let a circle have center $O$ and radius $r$. Its *polar correspondence*, or polar transformation, called its *polarity* for short, takes each point $A \neq O$ in the plane to a line $a$ perpendicular to $OA$, with foot $A'$ satisfying $\overline{OA} \cdot \overline{OA'} = r^2$; the same polarity takes the line $a$ to the point $A$. The line $a$ is the *polar* of $A$ with respect to the circle $O$, and $A$ is the *pole* of $a$ with respect to the circle $O$.

The definition immediately gives the following result.

**Theorem 3.6.2.** *With respect to a circle of center $O$, the polar of $O$ is the line at infinity. The polar of a point on the line at infinity is a diameter of the circle, perpendicular to every finite line through that point at infinity.*

**Proof.** In definition 3.6.1, take the respective limits $\overline{OA} \to 0$ and $\overline{OA} \to \infty$. □

**Theorem 3.6.3.** *The polar of a point on a circle is the tangent to the circle at that point.*

**Proof.** By definition 3.6.1, if $A$ lies on the circle, then $\overline{OA} = r$ and the defining equation gives $\overline{OA'} = r$, so $A' = A$. The line through $A$ perpendicular to $OA$ is the tangent at $A$. □

**Theorem 3.6.4.** *From a point $A$ outside $\odot O$, draw the two tangents to the circle. The line joining their points of tangency is the polar $a$ of $A$.*

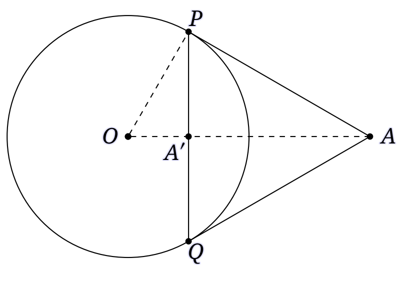


FIGURE 3.33

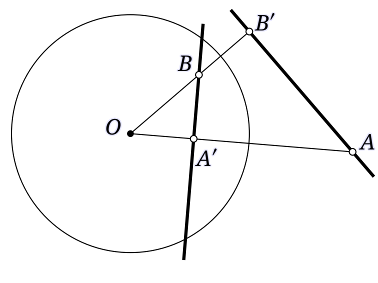


FIGURE 3.34

**Proof.** Let the points of tangency be $P, Q$, and let $A' = PQ \cap AO$, as in figure 3.33. Clearly $PQ \perp OA$. It remains to prove $\overline{OA'} \cdot \overline{OA} = r^2 = PO^2$, which is the right-triangle altitude theorem applied to $\triangle OPA$. □

**Theorem 3.6.5** (Polarity principle)**.** *For polarity with respect to a circle, if the polar of $A$ passes through $B$, then the polar of $B$ passes through $A$.*

**Proof.** Choose $B$ on the polar of $A$, as in figure 3.34. Let the polar of $A$ meet $OA$ at $A'$, and let the polar of $B$ meet $OB$ at $B'$. We show that $AB'$ is the polar of $B$.

By the definition of polarity, $\overline{OA'}\cdot\overline{OA}=\overline{OB'}\cdot\overline{OB}$. Hence $\triangle AB'O \sim \triangle BA'O$, so $\angle AB'O = \angle BA'O = 90^\circ$. Together with $\overline{OB}\cdot\overline{OB'}=r^2$, this says that $AB'$ is the polar of $B$. ☐

**Corollary 3.6.6.** *With respect to a fixed circle, the pole of a line is the common point of the polars of all points on the line. The polar of a point is the line containing the poles of all lines through that point.*

**Corollary 3.6.7.** *The polar of a point $A$ inside $\odot O$ can be constructed as follows. Draw a chord $PQ$ through $A$, and let the tangents at $P,Q$ meet at $S$. Draw a second chord $P'Q'$ through $A$, and let the tangents at $P',Q'$ meet at $S'$. Then $SS'$ is the polar of $A$.*

The polarity principle gives another proof of theorem 3.6.4.

**Alternative proof of Theorem 3.6.4.** In figure 3.33, the polars of $P,Q$ are the tangents at those points. The polar of $P$ passes through $A$, so the polar of $A$ passes through $P$; similarly it passes through $Q$. Thus $PQ$ is the polar of $A$. ☐

**Theorem 3.6.8.** *The cross-ratio of four points in a range equals the cross-ratio of their four corresponding polars.*

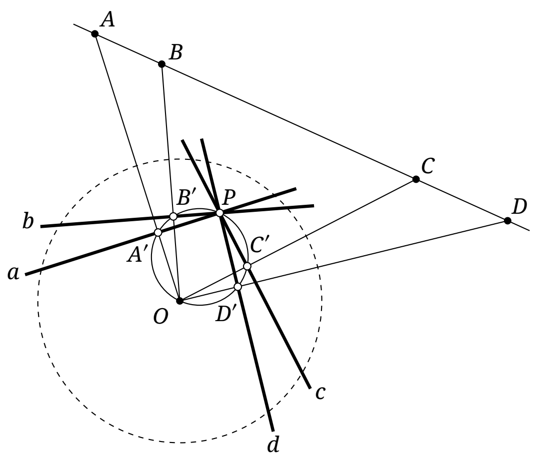


FIGURE 3.35

**Proof.** Let $A,B,C,D$ be four collinear points and let their polars with respect to $\odot O$ be $a,b,c,d$, respectively, as in figure 3.35. Let these polars meet $OA,OB,OC,OD$ at $A',B',C',D'$, respectively. If $P$ is the pole of $AD$, then corollary 3.6.6 shows that $a,b,c,d$ all pass through $P$.

By definition, $PA' \perp A'O$, and the analogous relations hold for $B',C',D'$. Thus $A',B',C',D'$ lie on $\odot(OP)$, and

$$(A,B;C,D)=(AO,BO;CO,DO)=(PA',PB';PC',PD').$$

For the last equality, compare the pencils $(OA,OB,OC,OD)$ and $(PA',PB',PC',PD')$. The angle between each line of the first and its corresponding line of the second is $90^\circ$. The angle between any two lines of the first pencil therefore equals the angle between the corresponding two lines of the second, so the two pencils have the same cross-ratio. ☐

We next consider the behavior of poles and polars under projective transformations.

**Theorem 3.6.9.** *If a projective transformation preserves a circle, it preserves polarity with respect to that circle. More precisely, let $p$ be the polar of $P$ with respect to a circle $\omega$, and let $f$ preserve $\omega$. Then $f(p)$ is the polar of $f(P)$ with respect to $\omega$, and $f(P)$ is the pole of $f(p)$ with respect to $\omega$.*

**Proof.** By theorems 3.6.3 and 3.6.4 and corollary 3.6.7, the polar of any point with respect to $\omega$ can be constructed using only tangents, joins of points, and intersections of lines. Projective transformations preserve all these incidence and tangency relations. Applying the same construction after the transformation therefore produces the polar of a point in the same way. Thus projective transformations preserve the polarity relation. □

This invariance yields the following important result.

**Theorem 3.6.10** (Harmonic property of poles and polars)**.** *As in figure 3.36, let $A,B$ lie on a circle $\omega$, let $P,Q \in AB$, and let $p$ be the polar of $P$. Then*

$$Q \in p \quad \Longleftrightarrow \quad (PQ,AB) = -1.$$

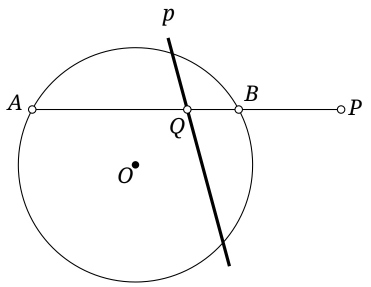


FIGURE 3.36

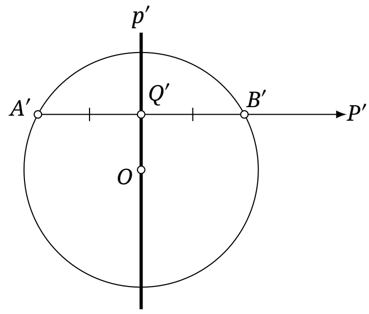


FIGURE 3.37

**Proof.** By the polarity principle and the invariance of the harmonic relation under interchange of the two base points, the statement is unchanged when $P,Q$ are interchanged. At least one of $P,Q$ lies outside the circle; without loss of generality, let this be $P$.

First prove the forward implication. Choose a projective transformation $f$ preserving $\omega$ and taking $P$ to a point at infinity; lemma 3.4.5 gives such a transformation. Use primes for the corresponding points or lines under $f$, as in figure 3.37. By theorem 3.6.9, $p'$ is still the polar of $P'$. By theorem 3.6.2, $p'$ passes through the center $O$ and is perpendicular to $A'B'$, so $p'$ is the perpendicular bisector of $A'B'$. Hence $Q'$ is the midpoint of $A'B'$, and $(P'Q',A'B') = -1$. Correspondingly, before the projective transformation, $(PQ,AB) = -1$.

The reverse implication is entirely similar. Apply the same projective transformation. The relation $(P'Q',A'B') = -1$ similarly implies that $Q'$ is the midpoint of $A'B'$, so $OQ' \perp A'B'$. Thus $OQ'$ is the polar of $P'$, and hence $Q' \in p'$. □

**Proposition 3.6.11.** *For a complete quadrangle inscribed in a circle, the polar of each vertex of its diagonal triangle is the opposite side of that triangle. Thus, if $A,B,C$ are the vertices of the diagonal triangle, as in figure 3.38, their polars are $BC,CA,AB$, respectively.*

**Proof.** By lemma 3.4.5, apply a projective transformation preserving the circle and taking $AB$ to the line at infinity. Use primes for the transformed elements. Now $A',B'$ lie at infinity, so the four vertices of the inscribed complete quadrangle form a parallelogram. An inscribed parallelogram is a rectangle; hence $C'$ is the center of the circle and its polar is $A'B'$. Since $A'C' \perp B'C'$, the polar of $A'$ is $B'C'$ and the polar of $B'$ is $A'C'$. □

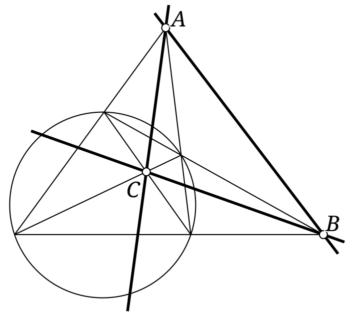


FIGURE 3.38

**Proposition 3.6.12.** *Suppose the polars of the vertices $A,B,C$ of $\triangle ABC$ with respect to a circle $\omega$ are $EF,FD,DE$, respectively, and form $\triangle DEF$. Then:*

*(1) the polars of $D,E,F$ form $\triangle ABC$;*
*(2) $\triangle ABC$ and $\triangle DEF$ are perspective.*

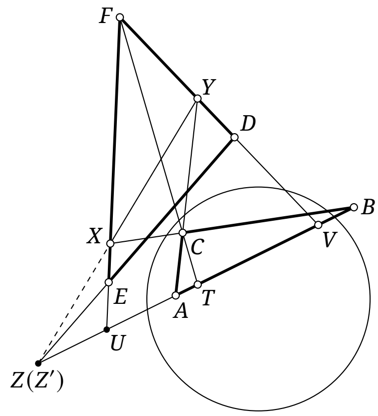


FIGURE 3.39

**Proof.** For (1), the polarity principle shows that the polars of $D,E,F$ are $BC,CA,AB$, respectively, and the conclusion is immediate.

For (2), let $f$ send each point $P$ on the line $AB$ to its polar. Then $f(P)$ passes through $F$, and theorem 3.6.8 shows that $f$ is a projective correspondence. Let $g$ send each line $p$ through $F$ to the point $p \cap AB$ on $AB$. The map $g$ is also a projective correspondence, so $h := g \circ f$ gives a projectivity of the line $AB$. For a point $P$ on $AB$, the definition puts $h(P)$ on the polar of $P$. By the polarity principle, the polar of $h(P)$ passes through $P$; it also passes through $F$. Hence $h(h(P)) = FP \cap AB = P$, so $h$ is an involution of $AB$.

Put $BC \cap EF = X$, $AC \cap DF = Y$, and $DE \cap AB = Z$. Let the lines $XY,EF,CF,DF$ meet $AB$ at $Z',U,T,V$, respectively, as in figure 3.39. By definition, $h(A) = U$ and $h(B) = V$. The polar $DE$ of $C$ passes through $Z$, and the polar $AB$ of $F$ also passes through $Z$. The polarity principle therefore shows that the polar of $Z$ passes through both $C$ and $F$: thus $CF$ is the polar of $Z$, and $h(Z) = T$. Consequently, $(A,U)$, $(B,V)$, and $(Z,T)$ are three pairs of corresponding points of $h$.

Apply Desargues' involution theorem to the complete quadrangle $CXFY$ and the line $AB$. The three pairs of opposite sides of the complete quadrangle meet $AB$ in the pairs $(A,U)$, $(B,V)$, and $(Z',T)$. Since two pairs of corresponding points determine an involution, we must have $Z' = Z$. Thus $X,Y,Z$ are collinear, and Desargues' theorem shows that $\triangle ABC$ and $\triangle DEF$ are perspective. $\square$

Here is an application of polarity.

**Example 3.6.13.** Prove example 4.4.23 using poles and polars.

**Solution.** Put $K = PG_a \cap G_bG_c$ and $T = G_bG_c \cap BC$. Consider polarity with respect to $\odot I$. The polar $G_bG_c$ of $A$ passes through $T$, and the polar $BC$ of $G_a$ also passes through $T$. Hence the polar of $T$ passes through $A, G_a$ and is therefore $AG_a$.

By theorem 3.6.10, $(KT, G_cG_b) = -1$. Moreover,

$$(PG_a, PT, PB, PC) \overline{\wedge} (G_a, T, B, C) \overset{(A)}{\overline{\wedge}} (K, T, G_c, G_b).$$

Thus the pencil $P(G_a, T, B, C)$ is harmonic. Since $\angle G_aPT = 90°$, proposition 3.2.17 shows that $G_aP$ bisects $\angle BPC$, and therefore $\angle G_cPB = \angle G_bPC$. Also $\angle BG_cP = \angle CG_bP$, so $\triangle PG_cB \sim \triangle PG_bC$. □

The polar construction gives a more geometric proof of conditions (5)–(9) in proposition 3.2.18. We restate the relevant result.

**Theorem 3.6.14.** *Given four collinear points $A, B, C, D$, let $\omega = \odot(AC)$ have center $O$. The following conditions are equivalent:*

*(1) $B$ lies on the polar of $D$;*
*(2) $D$ lies on the polar of $B$;*
*(3) $(AC, BD) = -1$;*
*(4) $\overline{OA}^2 = \overline{OC}^2 = \overline{OB} \cdot \overline{OD}$;*
*(5) $\overline{DA} \cdot \overline{DC} = \overline{DB} \cdot \overline{DO}$;*
*(6) $\overline{BA} \cdot \overline{BC} = \overline{BD} \cdot \overline{BO}$;*
*(7) $\dfrac{2}{\overline{DB}} = \dfrac{1}{\overline{DA}} + \dfrac{1}{\overline{DC}}$;*
*(8) $\dfrac{2}{\overline{BD}} = \dfrac{1}{\overline{BA}} + \dfrac{1}{\overline{BC}}$.*

**Proof.** The equivalence of (1), (2), and (3) is theorem 3.6.10; the equivalence of (3) with (4)–(8) appears as conditions (1) and (5)–(9) of proposition 3.2.18. We give a geometric proof of the implication from (1) to (4)–(8).

As in figure 3.40, draw through $B$ the perpendicular to $AC$, meeting $\omega$ at one of its intersections $P$, and assume (1). We seek to prove (4)–(8). By definition 3.6.1, $BP$ is the polar of $D$, so $PD$ is tangent to $\omega$. The right-triangle altitude theorem gives

$$\overline{OB} \cdot \overline{OD} = \overline{OP}^2 = \overline{OA}^2 = \overline{OC}^2,$$

which is (4). The secant–tangent theorem and the right-triangle altitude theorem give

$$\overline{DA} \cdot \overline{DC} = \overline{DP}^2 = \overline{DB} \cdot \overline{DO},$$

which is (5). Hence

$$\frac{1}{\overline{DA}} + \frac{1}{\overline{DC}} = \frac{\overline{DA} + \overline{DC}}{\overline{DA} \cdot \overline{DC}} = \frac{2\overline{DO}}{\overline{DB} \cdot \overline{DO}} = \frac{2}{\overline{DB}},$$

where the second equality uses (5). This proves (7). Finally, $B, D$ have symmetric roles, so (5) and (7) imply (6) and (8), respectively. □

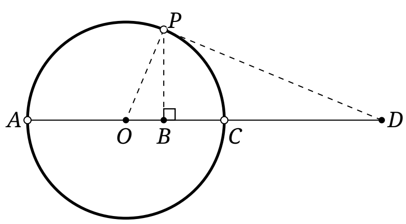


FIGURE 3.40

## Exercises

**Exercise 3.59.** Let $ABCD$ be inscribed in a circle $\omega$, and put $X=AC\cap BD$, $Y=AB\cap CD$. Prove that the tangents to $\omega$ at $B,C$ meet at a point of the line $XY$.

**Exercise 3.60.** Let the polar of $P$ with respect to $\omega$ meet $\omega$ at $A,B$, and let an arbitrary line through $P$ meet $\omega$ at $C,D$. Prove that:
(1) for every $Q\in\omega$, $(QA,QB;QC,QD)=-1$;
(2) $AC\cdot BD=AD\cdot BC$;
(3) if $M$ is the midpoint of the segment $AB$, then the line $AB$ bisects $\angle CMD$.

**Exercise 3.61.** Let $\triangle ABC$ be inscribed in $\omega$, and let the tangents at $A,B,C$ form $\triangle A'B'C'$. For an arbitrary point $S$, let the lines $AS,BS,CS$ meet the lines $BC,CA,AB$ at $A_1,B_1,C_1$, respectively. Prove that $\triangle A_1B_1C_1$ and $\triangle A'B'C'$ are perspective.

**Exercise 3.62.** Let $A$ be the right-angle vertex of $\triangle ABC$. Points $D,E$ lie on the segments $AB,AC$, respectively, and $B,C,E,D$ are concyclic. Put $G=BE\cap CD$ and $H=ED\cap CB$. Let $HI\perp AG$ at $I$, and put $M=ID\cap AH$. If $BG=BH$, prove that $GM\perp MH$.

### 3.6.2 Circles and the Duality Principle

Polarity takes points to lines and lines to points, revealing a duality between them. This leads to the duality principle. We first define what is meant by duality.

**Definition 3.6.15.** Consider a configuration, or a proposition, in a projective plane built only from the elements, relations, and operations in table 1. The configuration, or proposition, obtained by interchanging the two entries in each row is called its *dual configuration*, or its *dual proposition*.

TABLE 1

| | | |
|---|---|---|
| 1 | point | line |
| 2 | draw a line through a point | choose a point on a line |
| 3 | a point lies on a line | a line passes through a point |
| 4 | a point does not lie on a line | a line does not pass through a point |
| 5 | join two points | intersect two lines |
| 6 | $n$ points are collinear | $n$ lines are concurrent |
| 7 | four points have cross-ratio $t$; in particular, four points form a harmonic range | four lines have cross-ratio $t$; in particular, four lines form a harmonic pencil |

**Example 3.6.16.** *A simple triangle and a simple trilateral are dual configurations; in other words, the dual of a triangle is again a triangle.*

Indeed, a simple triangle begins with three points $A,B,C$, while a simple trilateral begins dually with three lines $a,b,c$. Joining $AB,BC,CA$ is dual to taking the intersections $a\cap b,b\cap c,c\cap a$.

**Example 3.6.17.** *A simple n-gon and a simple n-lateral are dual configurations, and a complete n-gon and a complete n-lateral are dual configurations.* In other words, the dual of an $n$-sided polygon is again an $n$-sided polygon. The proof is the same as in example 3.6.16.

**Example 3.6.18.** *Desargues' theorem and its converse are dual propositions.*

Desargues' theorem says that, if the lines joining corresponding vertices of two triangles are concurrent, then the intersections of corresponding sides are collinear. Its converse says that, if the intersections of corresponding sides are collinear, then the lines joining corresponding vertices are concurrent. The following rows match the corresponding elements and operations:

| *Desargues' theorem* | *Its converse* |
|---|---|
| triangles | triangles |
| corresponding vertices | corresponding sides |
| joining lines | intersection points |
| concurrent | collinear |
| corresponding sides | corresponding vertices |
| intersection points | joining lines |
| collinear | concurrent |

Thus the two statements are dual.

For dual propositions, we have the following result.

**Theorem 3.6.19** (Duality principle). *If a proposition of projective geometry is true, then its dual proposition is also true.*

**Proof.** Apply polarity with respect to a circle to the construction in the original proposition. By the preceding properties of polarity, the paired elements, relations, and operations in rows 1–7 of definition 3.6.15 are interchanged. □

So far we have considered the most basic dualities. Because the polarity used here is taken with respect to a circle, the further dualities in table 2 may be added to table 1; the duality principle remains valid.

TABLE 2

| | | |
|---|---|---|
| 8 | circle | circle |
| 9 | center of a circle | line at infinity |
| 10 | a point lies on a circle | a line is tangent to the circle |
| 11 | a point does not lie on a circle | a line is not tangent to the circle |
| 12 | a point is the pole of a line with respect to the circle | a line is the polar of a point with respect to the circle |

The assertion that the dual of a circle is a circle applies only when the configuration or proposition contains a single circle. It fails in general for several circles. The duality principle here is implemented by polarity with respect to one chosen circle; any other circle must then be transformed by that polarity, and its polar reciprocal with respect to the chosen circle is no longer a circle.[30] Thus proposition 0.4.10 and the later theorem 4.5.7 may look dual, but they are not directly a pair of dual propositions under this principle.

**Example 3.6.20.** Brianchon's theorem and Pascal's theorem are dual propositions. Brianchon's theorem states that the principal diagonals of a hexagon circumscribed about a circle are concurrent; Pascal's theorem states that the intersections of the principal opposite sides of a hexagon inscribed in a circle are collinear.

A corresponding comparison is

[30]The nature of the resulting figure will become clear in chapter 6.

| *Brianchon's theorem* | *Pascal's theorem* |
|---|---|
| circle | circle |
| circumscribed hexagon | inscribed hexagon |
| principal diagonals | intersections of principal opposite sides |
| concurrent | collinear |

A circumscribed hexagon has six sidelines tangent to the circle, so its dual has six vertices on the circle and is an inscribed hexagon. A principal diagonal joins vertices separated by two intervening vertices; dually, one takes the intersection of the corresponding two sidelines, giving an intersection of principal opposite sides.

**Example 3.6.21.** Lemma 3.4.8 is the dual of lemma 3.4.5.

**Remark.** The preceding propositions contain a relation of the form "there exists a projective transformation." This relation also has a dual, namely the same existence assertion, as follows at once from polarity with respect to a circle.

Here is an application of the duality principle.

**Proposition 3.6.22.** *For a complete quadrilateral circumscribed about a circle, the polar of each vertex of its diagonal trilateral is the opposite side of that trilateral. Thus, if $A,B,C$ are the vertices of the diagonal trilateral, as in figure 3.41, their polars are $BC,CA,AB$, respectively.*

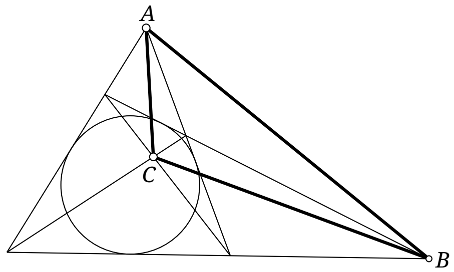


FIGURE 3.41

**Proof.** This is the dual of proposition 3.6.11, so it holds. ☐

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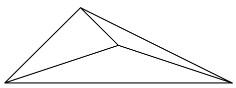
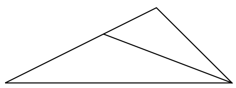
## Exercises

**Exercise 3.63.** Draw the dual of each of the following configurations:

(1)



(2)



**Exercise 3.64.** State the dual of each of the following propositions:

(1) exercise 3.32;
(2) Pappus' theorem and Newton's theorem;
(3) theorem 3.6.10.

**Exercise 3.65.**

(1) At first sight, the dual of "there is exactly one circle through three noncollinear points" should be "there is exactly one circle tangent to three nonconcurrent lines." The latter statement is false. What has gone wrong?

(2) Can the assertion "every projective transformation between two superposed ranges is a composition of at most two one-dimensional involutions" have a dual proposition? Explain.

**Exercise 3.66.** Let $(a,b,c)$ and $(a',b',c')$ be pencils with distinct vertices, and put $C_1 = a\cap b'$, $C_2 = b\cap a'$, $A_1 = b\cap c'$, $A_2 = c\cap b'$, $B_1 = c\cap a'$, and $B_2 = a\cap c'$. Use the duality principle to prove that $A_1A_2, B_1B_2, C_1C_2$ are concurrent.

**Exercise 3.67*.** The duality principle extends to projective spaces of arbitrary dimension. This exercise considers duality in three-dimensional projective space $\mathbb{P}^3$.

(1) Begin, as in definition 3.6.1, with polarity with respect to a sphere. For a sphere of radius $r$ centered at $O$ and a point $P$ in space, choose $P'$ on the ray $OP$ so that $\overline{OP'}\cdot\overline{OP} = r^2$. The plane $\pi$ through $P'$ perpendicular to $OP$ is called the *polar plane* of $P$, and $P$ is the *pole* of $\pi$. Following the discussion of section 3.6.1, prove:
   (i) if $P$ lies on the sphere, its polar plane is the tangent plane there;
   (ii) if $P$ lies outside the sphere, the plane containing the points of tangency of all tangents from $P$ is its polar plane;
   (iii) *Polarity principle:* if the polar plane of $A$ passes through $B$, then the polar plane of $B$ passes through $A$;
   (iv) as $P$ varies on a fixed line $p$, a fixed line $p'$ lies in every polar plane of $P$; call $p'$ the polar line of $p$.

(2) Next consider the duality principle in $\mathbb{P}^3$. Note that it should follow from the properties of polarity. Complete the following table of dualities.

TABLE 3

| | | |
|---|---|---|
| 1 | point | plane |
| 2 | line | line |
| 3 | draw a plane through a point | choose a point in a plane |
| 4 | draw a plane through a line | (i) ______ |
| 5 | draw the plane through three points | (ii) ______ |
| 6 | (iii) ______ | choose a line in a plane |
| 7 | (iv) ______ | intersect two planes in a line |
| 8 | $n$ points lie in one plane | (v) ______ |
| 9 | (vi) ______ | $n$ planes have a common line |
| 10 | (vii) ______ | $n$ lines lie in one plane |

(3) Apply the results of part (2):
   (i) construct the dual in $\mathbb{P}^3$ of a complete quadrangle in a plane;
   (ii) state the dual in $\mathbb{P}^3$ of Desargues' theorem;
   (iii) use duality in $\mathbb{P}^3$ to prove the following. Let $\alpha,\beta,\gamma,\delta$ be four distinct planes. If the intersection line of $\alpha,\beta$ is coplanar with that of $\gamma,\delta$, then the four planes have a common point; moreover, the intersection lines of $\alpha,\gamma$ and $\beta,\delta$ are coplanar, and so are the intersection lines of $\alpha,\delta$ and $\beta,\gamma$.

(4) Interested readers may investigate the duality principle in general $\mathbb{P}^n$.

## Chapter 4

# Foundations of Modern Euclidean Geometry

## 4.1 Inversion

### 4.1.1 Inversion and Its Basic Properties

**Definition 4.1.1** (Inversion)**.** An *inversion* in a circle $\omega$ with center $O$ and radius $r$ sends a point $A$ of the plane to the point $A'$ on the line $OA$ satisfying $\overline{OA'} \cdot \overline{OA} = r^2$; the inverse of a point at infinity is $O$. The number $r^2$ is called the *power of the inversion*, $\omega$ its *circle of inversion*, and $O$ its *center*.

In what follows, write $\mathfrak{i}_{\odot O}$ for this inversion, abbreviated to $\mathfrak{i}$ when there is no ambiguity. In particular, the inverse of $P$ in the circumcircle of $\triangle ABC$ is denoted by $\mathbf{i}_{\triangle ABC}P$, or simply by $\mathbf{i}P$ when no confusion can arise.

**Remark.** One may in fact also allow $r \in \mathbb{I}$, so that the power of an inversion may be any nonzero real number. When $r \in \mathbb{I}$, the circle is imaginary, but the inverse point $A'$ is still defined by $\overline{OA'} \cdot \overline{OA} = r^2 < 0$; the points $A$ and $A'$ then lie on opposite sides of $O$. Inversion in the imaginary circle with center $O$ and radius $r \in \mathbb{I}$ may be constructed by first inverting in the circle with center $O$ and radius $|r|$, and then applying the central reflection in $O$.

Inversion has the following basic properties.

**Theorem 4.1.2.** *An inversion $\mathfrak{i}$ is invertible, and its inverse is itself: $\mathfrak{i}^{-1} = \mathfrak{i}$.*

**Proof.** This is immediate from the definition. □

**Theorem 4.1.3.** *An inversion fixes every line through its center $O$ as a set.*

**Proof.** This is immediate from the definition. □

**Remark.** The image of the center $O$ is not defined, but it may be interpreted as a limit. If a curve $\gamma$ passes through $O$, let a point $P$ on $\gamma$ tend to $O$ along $\gamma$. The limiting image of $O$ on $\gamma$ is a point at infinity. For different curves $\gamma$, this corresponding point at infinity need not be the same. In particular, on a line through $O$, inversion sends $O$, in this limiting sense, to the point at infinity of that line.

**Lemma 4.1.4.** *Suppose that an inversion centered at $O$ sends points $A, B$, with $A, B, O$ noncollinear, to $A', B'$, respectively. Then $\triangle AOB \eqsim \triangle B'OA'$, and the four points $A', B', A, B$ are concyclic.*

**Proof.** By the definition of inversion, $OA \cdot OA' = OB \cdot OB'$. Both the similarity and the concyclicity follow. □

**Theorem 4.1.5.** *An inversion sends a circle not passing through its center $O$ to a circle that does not pass through $O$.*

**Proof.** As in Figure 4.1, let the line joining the center of the given circle to the center of inversion $O$ meet the circle at $A, D$, and let $B$ be any point of the circle. Suppose that inversion sends $A, B, D$ to $A', B', D'$, respectively.

By lemma 4.1.4, $\triangle OAB \sim \triangle OB'A'$ and $\triangle ODB \sim \triangle OB'D'$. Hence

$$\angle D'B'A' = \angle OB'A' - \angle OB'D' = \angle OAB - \angle ODB = \angle ABD = 90^\circ.$$

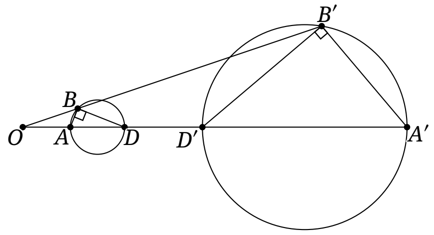



FIGURE 4.1

Thus $B'$ lies on the circle with diameter $A'D'$, and the original circle with diameter $AD$ is sent to the circle with diameter $A'D'$. ☐

**Theorem 4.1.6.** *An inversion centered at $O$ sends a circle $\omega$ through $O$ to a line $l$ not through $O$. The point of $\omega$ antipodal to $O$ is sent to the orthogonal projection of $O$ on $l$. Conversely, it sends a line $l$ not through $O$ to a circle $\omega$ through $O$, and the orthogonal projection of $O$ on $l$ is sent to the point of $\omega$ antipodal to $O$.*

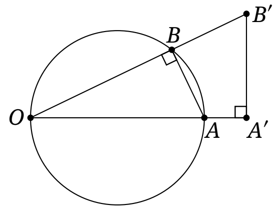



FIGURE 4.2

**Proof.** In Figure 4.2, let $A$ be the point of the circle antipodal to its point $O$, and let $A'$ be the inverse of $A$. For a point $B$ of the circle, write $B'$ for its inverse. By lemma 4.1.4, $\triangle AOB \sim \triangle B'OA'$, so $\angle B'A'O = \angle ABO = 90°$. Thus $B'$ lies on the line through $A'$ perpendicular to $OA$. The inverse of the circle through $O$ is therefore a line not through $O$, and the image $A' = f(A)$ is the orthogonal projection of $O$ on $A'B'$. The converse follows in the same way. ☐

**Theorem 4.1.7.** *Curves that are tangent before inversion remain tangent after inversion; disjoint curves remain disjoint, and intersecting curves remain intersecting.*[1]

**Proof.** The proof is straightforward and may be completed by imitating the proof of theorem 3.1.10; the details are left to the reader. ☐

**Theorem 4.1.8.** *Inversion preserves angles: the angle of intersection of two curves is the same before and after inversion.*

**Proof.** Suppose that the two curves meet at $B$, and choose nearby points $A, C$ on the respective curves, in the configuration of Figure 4.3. Let $A', B', C'$ be the images of $A, B, C$. By lemma 4.1.4, $\triangle OCB \sim \triangle OB'C'$ and $\triangle OAB \sim \triangle OB'A'$. Consequently,

$$\begin{aligned}\angle C'B'A' = \angle A'B'C' &= \angle A'OC' + \angle B'A'O + \angle B'C'O \\ &= \angle AOC + \angle ABO + \angle CBO = \angle AOC + \angle ABC.\end{aligned}$$

Let $A$ and $C$ tend to $B$ along the two curves. Then $AB, BC$ tend to the tangents of the original curves, while $A'B', B'C'$ tend to the tangents of their images at $B'$. Since $\angle AOC \to 0$, the angles $\angle ABC$ and $\angle A'B'C'$ are equal. ☐

[1]Here and below, straight lines are included among curves.

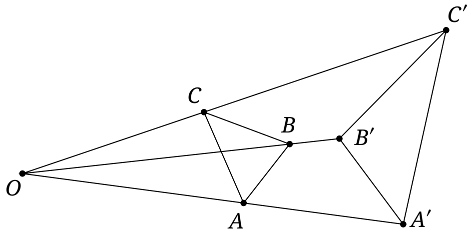



FIGURE 4.3

**Remark.** More precisely, the proof gives $\measuredangle ABC = -\measuredangle A'B'C'$. Thus, in the language of directed angles, theorem 4.1.8 says that inversion changes the sign of the angle of intersection.

**Theorem 4.1.9.** *If $\odot O_1$ and $\odot O$ are orthogonal, inversion in $\odot O$ leaves $\odot O_1$ invariant. Conversely, if $\odot O_1$ is invariant under inversion in $\odot O$, then the two circles are orthogonal.*

**Proof.** Let $\odot O_1$ and $\odot O$ meet at $X, Y$. The points $X, Y$ are fixed by the inversion, so the image $\odot O_1'$ of $\odot O_1$ also passes through $X, Y$.

If the circles are orthogonal, the tangents to $\odot O_1$ at $X, Y$ both pass through $O$. After inversion, the tangents to $\odot O_1'$ at $X, Y$ also both pass through $O$, which forces $\odot O_1' = \odot O_1$.

If the circles are not orthogonal, the tangent directions at $X, Y$ change, because inversion reverses directed angles.[2] Thus $\odot O_1' \neq \odot O_1$. Invariance therefore implies orthogonality as well. □

**Theorem 4.1.10.** *As in Figure 4.4, let $P, Q$ be inverse points with respect to $\odot O$, and let the segment $PQ$ meet the circle at $R$. For every point $A$ on the circle but not on the line $PQ$, the line $AR$ bisects $\angle PAQ$, and $\frac{AP}{AQ} = \frac{PR}{QR}$.*

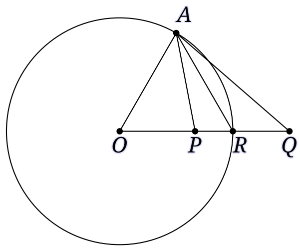



FIGURE 4.4

**Proof.** The definition of inversion gives $OA^2 = OP \cdot OQ$, and hence $\triangle OAP \sim \triangle QOA$. Thus $\angle OAP = \angle OQA$. Since $OA = OR$, we also have $\angle OAR = \angle ORA$, and therefore

$$\angle PAR = \angle OAR - \angle OAP = \angle ORA - \angle Q = \angle QAR.$$

The angle-bisector theorem gives $\frac{AP}{AQ} = \frac{PR}{QR}$. □

We next record several properties related to projective geometry.

**Theorem 4.1.11.** *Given $\odot O$ and a point $P$, let $Q$ be the intersection of the line $OP$ with the polar $p$ of $P$. Then $OQ \perp p$ and $Q = \mathrm{i}_{\odot O} P$.*

[2] In the orthogonal case, $\measuredangle -90^\circ$ and $\measuredangle 90^\circ$ are equivalent, so the tangent direction does not change.

**Proof.** This is immediate from the definitions of polar and inversion. □

**Corollary 4.1.12.** *If $P,Q$ are inverse with respect to $\odot O$, and $PQ$ meets $\odot O$ at $M,N$, then $P,Q,M,N$ form a harmonic range.*

**Proof.** This follows from the preceding result and theorem 3.6.10. □

**Theorem 4.1.13.** *Let $l$ be a line through $O$. Inversion in $\odot O$ preserves the cross-ratio of points on $l$, and hence restricts to a projectivity of $l$. Moreover, this projectivity is an involution.*

**Proof.** Let the radius of $\odot O$ be $r$. For four points $A,B,C,D$ on $l$, use primes for their inverse points. Then

$$\begin{aligned}(A'B',C'D') &= \frac{\overline{A'C'}}{\overline{B'C'}} \div \frac{\overline{A'D'}}{\overline{B'D'}} = \frac{\overline{OC'}-\overline{OA'}}{\overline{OC'}-\overline{OB'}} \div \frac{\overline{OD'}-\overline{OA'}}{\overline{OD'}-\overline{OB'}} \\ &= \frac{r^2/\overline{OC}-r^2/\overline{OA}}{r^2/\overline{OC}-r^2/\overline{OB}} \div \frac{r^2/\overline{OD}-r^2/\overline{OA}}{r^2/\overline{OD}-r^2/\overline{OB}} = \frac{\overline{OB}\cdot(\overline{OA}-\overline{OC})}{\overline{OA}\cdot(\overline{OB}-\overline{OC})} \div \frac{\overline{OB}\cdot(\overline{OA}-\overline{OD})}{\overline{OA}\cdot(\overline{OB}-\overline{OD})} \\ &= \frac{-\overline{OB}\cdot\overline{AC}}{-\overline{OA}\cdot\overline{BC}} \div \frac{-\overline{OB}\cdot\overline{AD}}{-\overline{OA}\cdot\overline{BD}} = \frac{\overline{AC}}{\overline{BC}} \div \frac{\overline{AD}}{\overline{BD}} = (AB,CD).\end{aligned}$$

Thus the map preserves cross-ratio and, by proposition 3.5.14, is a one-dimensional projectivity. By theorem 4.1.2, its square is the identity, so it is an involution. □

The preceding theorem yields the following consequence.

**Proposition 4.1.14.** *Suppose that an inversion $f$ centered at $P$ sends a circle $\omega$ with center $Q$ to a circle $\omega'$ with center $S$, and sends $Q$ to $Q'$. Then $P,Q'$ are inverse with respect to $\omega'$.*

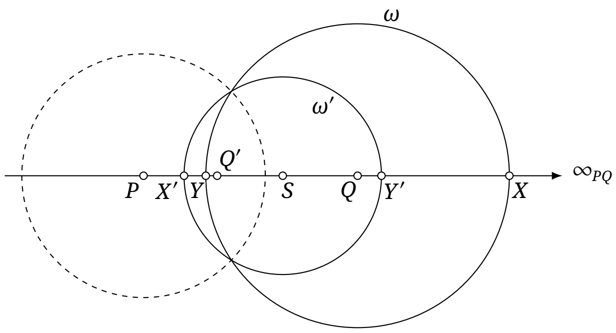



FIGURE 4.5

**Proof.** Let $PQ$ meet $\omega$ at $X,Y$ and $\omega'$ at $X',Y'$. Plainly $X'=f(X)$ and $Y'=f(Y)$. If $\infty_{PQ}$ is the point at infinity of $PQ$, then $f(\infty_{PQ})=P$. By theorem 4.1.13, $(X'Y',PQ')=(XY,\infty_{PQ}Q)=-1$. By implication (1)⇒(5) of proposition 3.2.18, $\overline{SP}\cdot\overline{SQ'}=r^2_{\omega'}$; hence $P,Q'$ are inverse with respect to $\omega'$. □

Finally, inversion does not in general send a conic to a conic. The following exercises provide examples.

## Exercises

**Exercise 4.1.** Prove that the image of the circumcircle of a triangle under inversion in its incircle is the nine-point circle of the contact triangle.

**Exercise 4.2.** Given $\triangle ABC$, let $\odot A$ have radius $\sqrt{AB \cdot AC}$, and let $l$ be the bisector of $\angle BAC$. Prove that under the transformation $\mathsf{f}_l \circ \mathsf{i}_{\odot A}$:

(1) the points $B, C$ are interchanged;
(2) the incenter of $\triangle ABC$ is sent to the $A$-excenter.

**Exercise 4.3.** Given $\odot O$, let $\omega$ be a circle not passing through $O$. Prove that the center of $\mathsf{i}_{\odot O}(\omega)$ is $\mathsf{i}_{\odot O}(\mathsf{i}_{\omega}(O))$.

**Exercise 4.4.** Given points $A, B$ and a circle $\omega$, prove that the following conditions are equivalent:

(1) $A, B$ are inverse with respect to $\omega$;
(2) $\omega$ is an Apollonius circle of $A, B$;
(3) both the line $AB$ and the circle $\odot AB$ are orthogonal to $\omega$.

**Exercise 4.5.** Let $\odot O$ have radius $R$, and let $\odot P$ be a circle of radius $r$ not passing through $O$. If $OP = d$, prove that the radius of $\mathsf{i}_{\odot O}(\odot P)$ is $\left|\frac{rR^2}{d^2 - r^2}\right|$.

**Exercise 4.6.** Define inversion in three-dimensional space as follows: inversion in the sphere with center $O$ and radius $r$ sends a point $A$ to the point $A'$ on the line $OA$ satisfying $\overline{OA'} \cdot \overline{OA} = r^2$. Determine the images of lines, circles, planes, and spheres under spatial inversion.

**Exercise 4.7*.** Let two fixed points $A, B$ in the plane be separated by $2a$, where $a > 0$. The locus of a moving point $P$ satisfying $PA \cdot PB = a^2$ is called a *Bernoulli lemniscate*. Prove that inversion in the auxiliary circle of an equilateral hyperbola sends the hyperbola to a Bernoulli lemniscate, as in Figure 4.6.

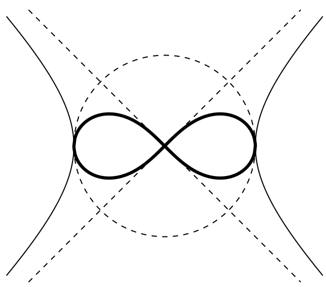


FIGURE 4.6

**Exercise 4.8*.** Suppose that all of space is filled with a medium whose refractive index is spherically symmetric about a point $O$, and that the refractive index $n$ at distance $\rho$ from $O$ is

$$n(\rho) = \frac{n_0 a^2}{a^2 + \rho^2},$$

where $n_0, a$ are positive constants. Prove that:

(1) every ray traveling in the medium follows a circle or a straight line;
(2) every point $A$ is imaged at $\mathsf{f}_O(\mathsf{i}_{\odot(O,a)}(A))$;

(3) if a spherical cavity centered at a point is hollowed out of the medium, then every ray emitted from the center of the cavity passes through another common point in the medium.

### 4.1.2 Applications of Inversion

This subsection concerns applications of inversion. Inversion allows us to prove certain results in plane geometry quickly. We begin with two famous theorems in plane geometry that admit quick proofs by inversion.

**Theorem 4.1.15** (Ptolemy's theorem (Πτολεμαῖος))**.** *For every quadrilateral ABCD,*

$$AB \cdot CD + AD \cdot BC \geq AC \cdot BD,$$

*with equality if and only if $A, B, C, D$ are concyclic.*

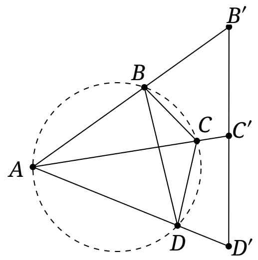



FIGURE 4.7

**Proof.** As in Figure 4.7, invert about $A$ with unit radius 1, and denote the images of $B, C, D$ by $B', C', D'$. By lemma 4.1.4, $C'D' = \frac{CD}{AC \cdot AD}$, $D'B' = \frac{DB}{AD \cdot AB}$, and $B'C' = \frac{BC}{AB \cdot AC}$. Observe that $B'C' + D'C' \geq B'D'$; substituting the expressions for $B'C', D'C', B'D'$ proves the inequality. Equality holds precisely when $B', C', D'$ are collinear, which by theorem 4.1.6 is equivalent to the concyclicity of $A, B, C, D$. □

**Corollary 4.1.16** (Three-chord lemma)**.** *Let $AB, AC, AD$ be three chords of one circle, with $AD$ lying between $AB$ and $AC$. Then*

$$AB \sin\angle CAD + AC \sin\angle BAD = AD \sin\angle BAC.$$

**Proof.** By the sine rule, $\sin\angle CAD = \frac{CD}{2R}$, $\sin\angle BAD = \frac{BD}{2R}$, and $\sin\angle BAC = \frac{BC}{2R}$. The assertion is therefore an immediate consequence of Ptolemy's theorem. □

**Theorem 4.1.17** (Casey's theorem)**.** *Let $\odot O_1, \odot O_2, \odot O_3, \odot O_4$, in this cyclic order, be internally tangent to $\odot O$. If $t_{ij}$ denotes the length of a direct common tangent to $\odot O_i$ and $\odot O_j$, then*

$$t_{12}t_{34} + t_{23}t_{14} = t_{13}t_{24}.$$

*See Figure 4.8.*[3]

We first prove a lemma.

**Lemma 4.1.18.** *Let $\odot_1, \odot_2$ have radii $r_1, r_2$, respectively, and a direct common tangent of length $t$. Then $\frac{t}{\sqrt{r_1 r_2}}$ is invariant under inversion.*

[3]When all four circles degenerate to points, this is Ptolemy's theorem.

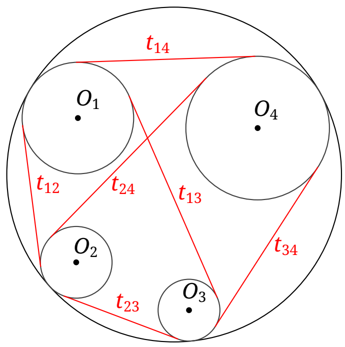



FIGURE 4.8

**Proof.** Let the distance between the centers be $d$. If the circles intersect, let $k$ be the cosine of their angle of intersection. The cosine rule gives $k = \frac{r_1^2+r_2^2-d^2}{2r_1r_2}$. Geometrically, $t^2 = d^2-(r_1-r_2)^2$, so $\frac{t}{\sqrt{r_1r_2}} = \sqrt{2(1-k)}$. By theorem 4.1.8, inversion preserves $k$, and hence $\frac{t}{\sqrt{r_1r_2}}$ is also invariant.

When the circles are disjoint, one may either continue the argument directly to the complex plane or use the spatial method discussed in section 4.5. □

We now return to the proof of Casey's theorem.

**Proof of Theorem 4.1.17.** Let $r_i$ be the radius of $\odot O_i$. Choose a point $R$ on $\odot O$ and invert in an arbitrary circle centered at $R$. The image of $\odot O$ is a line $l$, and the four image circles are tangent to $l$. Let their points of tangency, in order, be $A_1,A_2,A_3,A_4$, and use primes for lengths after inversion. Then $t'_{ij} = A_iA_j$. Since $A_1,A_2,A_3,A_4$ are collinear, one immediately verifies

$$A_1A_2 \cdot A_3A_4 + A_2A_3 \cdot A_1A_4 = A_1A_3 \cdot A_2A_4.$$

This may also be viewed as Ptolemy's theorem for a generalized circle of infinite radius. Hence $t'_{12}t'_{34} + t'_{23}t'_{14} = t'_{13}t'_{24}$. Equivalently,

$$\frac{t'_{12}}{\sqrt{r'_1r'_2}} \cdot \frac{t'_{34}}{\sqrt{r'_3r'_4}} + \frac{t'_{23}}{\sqrt{r'_2r'_3}} \cdot \frac{t'_{14}}{\sqrt{r'_1r'_4}} = \frac{t'_{13}}{\sqrt{r'_1r'_3}} \cdot \frac{t'_{24}}{\sqrt{r'_2r'_4}}.$$

By lemma 4.1.18, the corresponding relation before inversion is

$$\frac{t_{12}}{\sqrt{r_1r_2}} \cdot \frac{t_{34}}{\sqrt{r_3r_4}} + \frac{t_{23}}{\sqrt{r_2r_3}} \cdot \frac{t_{14}}{\sqrt{r_1r_4}} = \frac{t_{13}}{\sqrt{r_1r_3}} \cdot \frac{t_{24}}{\sqrt{r_2r_4}},$$

which is precisely the required equality. □

**Remark.** Casey's theorem admits the following extensions.

(1) Let $\odot O_1, \odot O_2, \odot O_3, \odot O_4$, in this order, each be tangent to $\odot O$. When $\odot O_i$ and $\odot O_j$ are both internally tangent to $\odot O$, or both externally tangent to it, let $t_{ij}$ be the length of their direct common tangent. When one is internally and the other externally tangent to $\odot O$, let $t_{ij}$ be the length of their transverse common tangent. Then $t_{12}t_{34} + t_{23}t_{14} = t_{13}t_{24}$.

(2) The converse also holds: if $t_{12}t_{34} + t_{23}t_{14} = t_{13}t_{24}$, the four circles $O_1,O_2,O_3,O_4$ are tangent to a common circle.[4] If it is unclear whether $O_1O_2O_3O_4$ is a quadrilateral in the usual sense—for instance, if $O_1O_2$ and $O_3O_4$ cross—the equality may be written

$$|t_{12}t_{34} \pm t_{23}t_{14}| = t_{13}t_{24}$$

for a suitable choice of sign.

[4]The tangent lengths here are understood in the extended sense of part (1); whether each tangency is internal or external must be determined from the configuration and the explanation in part (1).

(3) The theorem remains valid when one or more circles degenerate to points. A common tangent length between a point and a circle then means the length of a tangent drawn from the point, while that between two points means their distance. In the real Euclidean plane a point inside a circle has no real tangent. One may nevertheless work in the complex Euclidean plane. By the secant–tangent theorem, if any secant through a point $P$ outside a circle $\odot O$ meets the circle at $A,B$, the length of a tangent from $P$ to $\odot O$ is $\sqrt{\overline{PA}\cdot\overline{PB}}$. By continuation, when $P$ is inside $\odot O$ and $AB$ is any chord through $P$, its tangent length may be taken as $\sqrt{\overline{PA}\cdot\overline{PB}}$.[5]

We give two further applications of inversion, the first involving a conic.

**Example 4.1.19.** Given a parabola, prove that if the orthocenter of one of its inscribed triangles is the focus, then the incircle of that triangle is fixed.

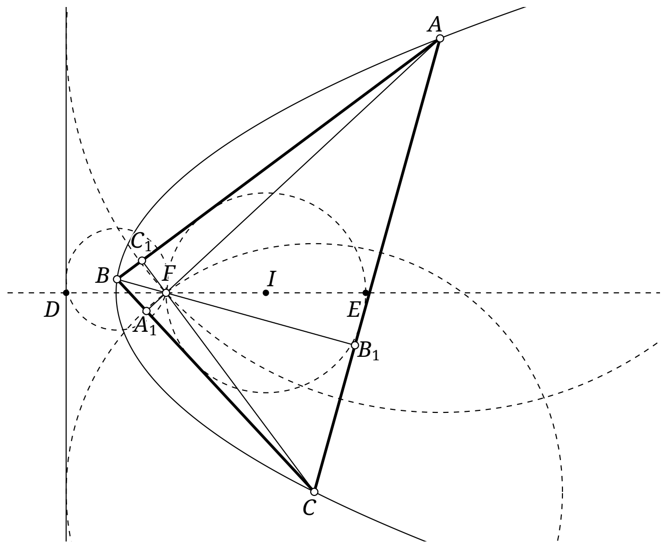



FIGURE 4.9

**Solution.** By lemma 0.5.22,

$$AF\cdot FA_1 = BF\cdot FB_1 = CF\cdot FC_1 = t,$$

where $t = 4R^2\cos A\cos B\cos C$ and $R$ is the radius of $\odot(ABC)$. On the other hand, by the definition of a parabola, the three circles $\odot(A,AF)$, $\odot(B,BF)$, and $\odot(C,CF)$ are tangent to the directrix, as in Figure 4.9.

Invert about $F$ with radius $\sqrt{-2t}$.[6] By theorem 4.1.6, and because the points antipodal to $F$ on $\odot A,\odot B,\odot C$ are sent to $A_1,B_1,C_1$, respectively, $\mathfrak{i}(\odot A)=BC$, $\mathfrak{i}(\odot B)=AC$, and $\mathfrak{i}(\odot C)=AB$. Before inversion, the directrix $l$ is tangent to these three circles, so $\mathfrak{i}(l)$ is tangent to $AB,BC,CA$. Thus $\mathfrak{i}(l)$ is the incircle $\odot I$ of $\triangle ABC$. Moreover, theorem 4.1.6 gives $F\in\mathfrak{i}(l)$, so the orthocenter $F$ lies on $\odot I$.

Let the inradius be $r$. By proposition 0.6.18, $FI^2 = 2r^2 - t$. Since $FI = r$, we have $r^2 = t$. Let $D$ be the intersection of $l$ with the axis of the parabola, and let $E$ be the other intersection of $\odot I$ with the axis. The points $D,E$ are inverse, so $DF\cdot FE = 2t = 2r^2$. Substitution of $EF = 2r$ gives $DF = r$. Thus $\odot I$ is fixed: its center lies on the axis at a distance from the focus equal to the distance from the focus to the directrix, and its radius is that same distance. □

[5]It is then imaginary; a direct calculation in the complex plane gives the same expression.

[6]This is an example of an inversion with imaginary radius.

The next example concerns a triangle.

**Example 4.1.20.** In $\triangle ABC$, as in Figure 4.10, let the bisector of $\angle A$ meet the segment $BC$ at $D$, and let $N$ be a point of the segment $BC$ distinct from $D$. Among the intersections of the perpendicular bisector of the segment $AD$ with $\odot(ABN)$ and $\odot(ACN)$, choose the two farthest apart and call them $X, Y$. Prove that $X, Y, N, D$ are concyclic.

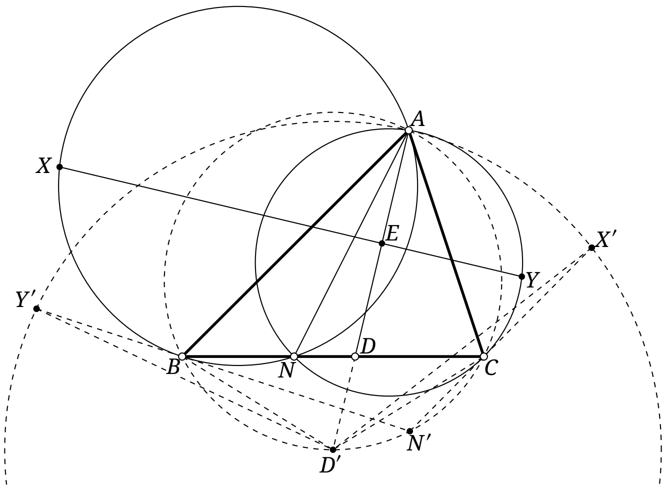



FIGURE 4.10

**Solution.** Let $\omega$ be the circle centered at $A$ with radius $r = \sqrt{AB \cdot AC}$, and set $f = \mathsf{f}_{AD} \circ \mathsf{i}_{\omega}$. Then $f(B) = C, f(C) = B,$[7] $f(BC) = \odot(ABC)$, and $f(AD) = AD$. Since $D = BC \cap AD$, its image $D' := f(D)$ is the other intersection of $AD$ with $\odot(ABC)$. Similarly, $N' := f(N)$ is the other intersection of $\mathsf{f}_{AD}(AN)$ with $\odot(ABC)$. The circles $\odot(ABN)$ and $\odot(ACN)$ pass through the center of inversion $A$ and through $N$; hence $f$ sends them respectively to the lines $N'C$ and $N'B$.

Let $AD \cap XY = E$. Then $\overline{Af(E)} \cdot \overline{AE} = \overline{AD} \cdot \overline{AD'}$, and $\overline{AD} = 2\overline{AE}$, so $\overline{Af(E)} = 2\overline{AD'}$. Since $AE \perp XY$ at $E$, the point $f(E)$ is antipodal to $A$ on the circle $f(XY)$. Thus $f(XY)$ is the circle $\odot D'$ centered at $D'$ and passing through $A$. The points $X' := f(X)$ and $Y' := f(Y)$ are therefore the intersections of the rays $N'C$ and $N'B$, respectively, with $\odot D'$.

Because $AD$ bisects $\angle BAC$, the point $D'$ is the midpoint of the arc $\widehat{BD'C}$, and hence $BD' = CD'$. Since $B, C, D', N'$ are concyclic, $\angle D'BN' = \angle D'CN'$, that is, $\angle Y'BD' = \angle X'CD'$. Together with $Y'D' = X'D'$, this gives $\triangle Y'BD' \cong \triangle X'CD'$. Thus $\triangle X'CD'$ is obtained by rotating $\triangle Y'BD'$ about $D'$. By proposition 0.1.5, $\measuredangle(Y'D', X'D') = \measuredangle(Y'B, X'C)$, that is, $\angle Y'D'X' = \angle Y'N'X'$. Hence $X', Y', N', D'$ are concyclic, and so are their inverse images $X, Y, N, D$. □

These examples show that the main point in using inversion is to choose its circle so that a complicated configuration becomes simple. The preceding example uses the same construction as exercise 4.2: inversion followed by a reflection. We shall call it an *axial-reflection inversion*.[8] It interchanges two vertices of a triangle and therefore has many useful properties. We shall use it again when discussing pseudocircles.

[7] This is also the assertion of exercise 4.2.

[8] It is sometimes called a $\sqrt{bc}$-inversion. If the reflection is instead in the bisector of $\angle ABC$ and is composed with an inversion centered at $B$, the corresponding name is $\sqrt{ac}$-inversion; similarly one obtains $\sqrt{ab}$-inversion.

## Exercises

**Exercise 4.9.** Let $\triangle A'B'C'$ be the orthic triangle of an acute triangle $\triangle ABC$. A circle $\omega$ is externally tangent to each of $\odot(B'C'A)$, $\odot(C'BA')$, and $\odot(A'CB')$. Prove that $\omega$ is also tangent to $\odot(ABC)$.

**Exercise 4.10.** Given $\triangle ABC$, let $B'$ be the reflection of $B$ in $AC$ and $C'$ the reflection of $C$ in $AB$. If $\odot(ABB') \cap \odot(ACC') = \{A, P\}$, prove that the line $AP$ passes through the circumcenter of $\triangle ABC$.

**Exercise 4.11.** For a given triangle, prove that the incenter and circumcenter of the original triangle are collinear with the orthocenter of its contact triangle. *Hint.* Invert in the incircle.

**Exercise 4.12.** Let $D$ be a point on the bisector of $\angle A$ of $\triangle ABC$, and let $\triangle DEF$ be the circumcircle mid-arc triangle of $\triangle ABC$. Let $ED, FD$ meet $\mathscr{C}$ again at $G, H$, respectively. On the rays $AB, AC$ choose $M, N$ so that $A, I, G, M$ and $A, I, H, N$ are respectively concyclic, where $I$ is the incenter of $\triangle ABC$. Prove that $BC \parallel MN$.

**Exercise 4.13.** Given three circles in the plane, construct with straightedge and compass a circle tangent to all three.

**Exercise 4.14.** Let $ABCDEFG$ be a regular heptagon. Prove that $\frac{1}{AB} = \frac{1}{AC} + \frac{1}{AD}$.

**Exercise 4.15* (Descartes' circle theorem).** Let $\odot O_1, \odot O_2, \odot O_3, \odot O_4$ be pairwise externally tangent, with radii $r_1, r_2, r_3, r_4$. Prove that

$$\sum_{i=1}^{4} \frac{2}{r_i^2} = \left( \sum_{i=1}^{4} \frac{1}{r_i} \right)^2.$$

**Exercise 4.16.** Prove that an isocenter of a triangle inscribed in a conic lies on the principal axis of the conic if and only if the circumcircle of the triangle is tangent to the auxiliary circle of the conic.[9] *Hint.* Use Casey's theorem.

# 4.2 Simson Lines

## 4.2.1 Simson and Steiner Lines

We begin with the theorems of Simson and Steiner and their associated lines. Although elementary, these results are important and will be used substantially later. Simson's theorem introduces the following line.

**Theorem 4.2.1** (Simson's theorem)**.** *The orthogonal projections of a point on the three sidelines of a triangle are collinear if and only if the point lies on the circumcircle of the triangle.* When the three projections are collinear, their line is called the *Simson line* of the point with respect to the triangle. For $P \in \mathscr{C}_{\triangle ABC}$, in what follows write $\mathscr{S}_{\triangle ABC}(P)$ for this line, abbreviated to $\mathscr{S}(P)$ when there is no ambiguity.

[9] Recall that, in order to state certain properties uniformly for the three kinds of conic, we also call the tangent to a parabola at its vertex its "auxiliary circle"; the principal axis of a parabola is its axis.

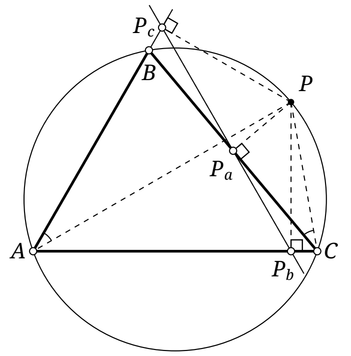



FIGURE 4.11

**Proof.** As in Figure 4.11, let $P_a, P_b, P_c$ be the projections of $P$ on the three sidelines of $\triangle ABC$. The points $P, C, P_b, P_a$ are concyclic, so $\measuredangle PP_bP_a = \measuredangle PCP_a = \measuredangle PCB$. Similarly, $\measuredangle PP_bP_c = \measuredangle PAB$. Therefore

$$\begin{aligned} P_a, P_b, P_c \text{ are collinear} &\iff \measuredangle PP_bP_a = \measuredangle PP_bP_c \\ &\iff \measuredangle PCB = \measuredangle PAB \iff P, A, B, C \text{ are concyclic.} \end{aligned}$$ $\square$

Simson lines have several further properties.

**Lemma 4.2.2.** *Let $P \in \mathscr{C}_{\triangle ABC}$, and let the perpendicular from $P$ to $AC$ meet the circumcircle again at $B'$. Then $BB' \parallel \mathscr{S}(P)$.*

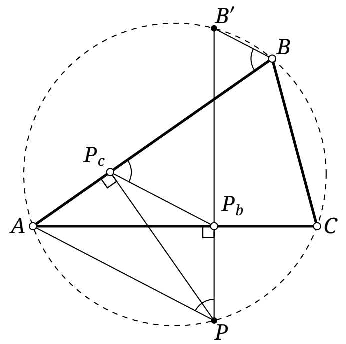



FIGURE 4.12

**Proof.** In Figure 4.12, let $P_b, P_c$ be the projections of $P$ on $AC, AB$. Then $\mathscr{S}(P) = P_bP_c$. We have $\measuredangle ABB' = \measuredangle APB' = \measuredangle BP_cP_b$, where the last equality follows from the concyclicity of $A, P, P_b, P_c$. Hence $BB' \parallel P_bP_c = \mathscr{S}(P)$. $\square$

**Proposition 4.2.3.** *Let $O$ be the circumcenter of $\triangle ABC$. If $P_1, P_2 \in \mathscr{C}_{\triangle ABC}$, then*[10]

$$2\measuredangle(\mathscr{S}(P_1), \mathscr{S}(P_2)) = \overrightarrow{\measuredangle} P_2OP_1.$$

**Proof.** By lemma 4.2.2, when $P$ moves counterclockwise through an angle $\theta$ about the center of the circumcircle, the corresponding point $B'$ moves clockwise through $\theta$. The inscribed-angle theorem shows that $BB'$ rotates clockwise through $\frac{\theta}{2}$ about $B$. Since $\mathscr{S}(P) \parallel BB'$, the line $\mathscr{S}(P)$ rotates clockwise through $\frac{\theta}{2}$, proving the proposition. $\square$

[10]Equivalently, when a point moves uniformly in angle around the circumcircle, its Simson line rotates in the opposite direction at half the angular speed. The angle on the right is directed modulo 360°, whereas the angle on the left is directed modulo 180°; multiplication by 2 makes the equality meaningful modulo 360°.

**Lemma 4.2.4.** *Let $O$ be the circumcenter of $\triangle ABC$, and let $A', P \in \mathscr{C}$. Then*

$$\measuredangle(\mathscr{S}_{\triangle ABC}(P), \mathscr{S}_{\triangle A'BC}(P)) = \frac{1}{2}\overrightarrow{\measuredangle}AOA'.$$

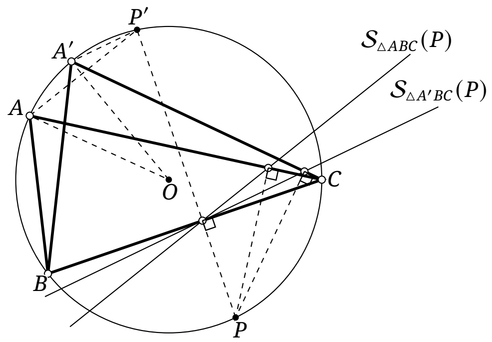



FIGURE 4.13

**Proof.** In Figure 4.13, draw $PP' \perp BC$, meeting $\odot O$ at $P'$. By lemma 4.2.2, $\mathscr{S}_{\triangle ABC}(P) \parallel AP'$ and $\mathscr{S}_{\triangle A'BC}(P) \parallel A'P'$. Also, $\measuredangle(AP', A'P') = \measuredangle AP'A' = \overrightarrow{\measuredangle}AOA'/2$. □

**Theorem 4.2.5.** *If $\triangle ABC$ and $\triangle A'B'C'$ have the same circumcircle $\mathscr{C}$ with center $O$, then for every $P \in \mathscr{C}$,*

$$\measuredangle(\mathscr{S}_{\triangle ABC}(P), \mathscr{S}_{\triangle A'B'C'}(P)) = \frac{1}{2}(\overrightarrow{\measuredangle}AOA' + \overrightarrow{\measuredangle}BOB' + \overrightarrow{\measuredangle}COC').$$

**Proof.** Observe that

$$\begin{aligned}\text{LHS} = {} & \measuredangle(\mathscr{S}_{\triangle ABC}(P), \mathscr{S}_{\triangle A'BC}(P)) + \measuredangle(\mathscr{S}_{\triangle A'BC}(P), \mathscr{S}_{\triangle A'B'C}(P)) \\ & + \measuredangle(\mathscr{S}_{\triangle A'B'C}(P), \mathscr{S}_{\triangle A'B'C'}(P)).\end{aligned}$$

Apply lemma 4.2.4 to the three terms. □

**Corollary 4.2.6.** *Suppose that $\triangle ABC$ and $\triangle A'B'C'$ have the same circumcircle $\mathscr{C}$. If, for one point $P \in \mathscr{C}$, the Simson lines of $P$ with respect to the two triangles meet at angle $\theta$, then the Simson lines of every point of $\mathscr{C}$ with respect to the two triangles also meet at angle $\theta$.*

**Proof.** By theorem 4.2.5, the angle is independent of the position of $P$. □

The following theorem of Steiner is closely related to Simson's theorem and introduces the Steiner line.

**Theorem 4.2.7** (Steiner's theorem)**.** *For a point $P$ on the circumcircle of $\triangle ABC$, the Simson line of $P$ bisects the segment $PH$, where $H$ is the orthocenter of the triangle.*

**Proof.** In Figure 4.14, lemma 0.5.21 shows that the reflection $H'$ of $H$ in $AC$ lies on the circumcircle. Since $PB'$ and $BH'$ are both perpendicular to $AC$, the quadrilateral $BH'PB'$ is a trapezoid; being inscribed in $\mathscr{C}$, it is isosceles.

Reflect $P$ in $AC$ to obtain $P'$. Then $P'H \parallel BB'$, and lemma 4.2.2 gives $BB' \parallel \mathscr{S}(P)$. Thus $\mathscr{S}(P) \parallel P'H$. Since $P_b$ is the midpoint of $PP'$, the line $\mathscr{S}(P)$ is a midline of $\triangle P'HP$, so $\mathscr{S}(P)$ bisects the segment $PH$. □

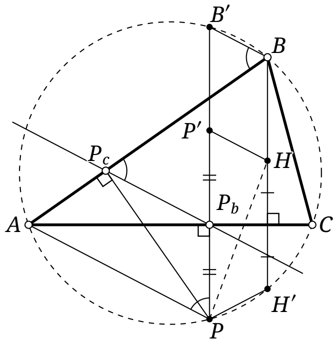



FIGURE 4.14

**Theorem 4.2.8.** *For $P$ on the circumcircle of $\triangle ABC$, the reflections of $P$ in the three sidelines are collinear on a line through the orthocenter $H$, as in Figure 4.15. This line is $\mathfrak{h}_{P,2}(\mathscr{S}(P))$.* It is called the *Steiner line* of $P$ with respect to $\triangle ABC$. In what follows, write it as $\mathscr{W}_{\triangle ABC}(P)$, abbreviated to $\mathscr{W}(P)$ when no ambiguity can arise.

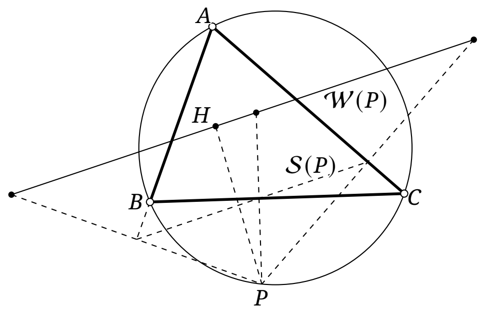



FIGURE 4.15

**Proof.** Apply the homothety $\mathfrak{h}_{P,2}$ to the three projections in Simson's theorem. Their images are the three reflections, and hence are collinear. Steiner's theorem shows that their line passes through $H$. □

**Theorem 4.2.9.** *Let $l$ be a line through the orthocenter $H$ of $\triangle ABC$. The reflections of $l$ in the three sidelines are concurrent at a point $P$ of $\odot(ABC)$, as in Figure 4.16.* The point $P$ is called the *anti-Steiner point* of $l$ with respect to $\triangle ABC$. In what follows, write $\mathscr{W}^{-1}_{\triangle ABC}(l)$ for it, abbreviated to $\mathscr{W}^{-1}(l)$ when no ambiguity can arise.[11]

**Proof.** If $\mathscr{W}(P)=l$, the three reflected lines plainly meet at $P$. The method of coincidence now proves the theorem. □

Steiner's theorem also gives the following result.

**Proposition 4.2.10.** *For $\triangle ABC$, let $P$ lie on $\mathscr{C}$, let $H$ be the orthocenter, and let $M_a,M_b,M_c$ be the side midpoints. If $K=PH\cap\mathscr{S}(P)$, then $K$ lies on $\mathscr{C}_{\triangle M_aM_bM_c}$, that is, $\mathscr{N}$. Moreover, $\mathscr{S}_{\triangle M_aM_bM_c}(K)\perp\mathscr{S}(P)$.*

[11]For a point $P$, the point $\mathscr{W}^{-1}(PH)$ is also sometimes called the anti-Steiner point of $P$.

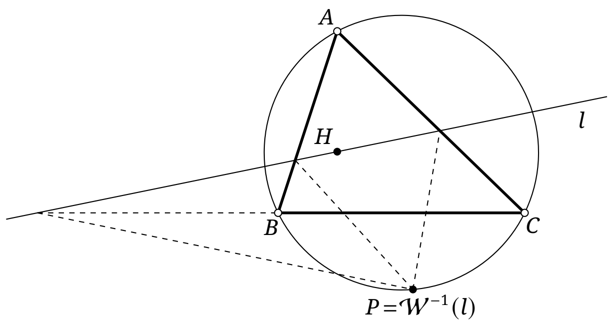



FIGURE 4.16

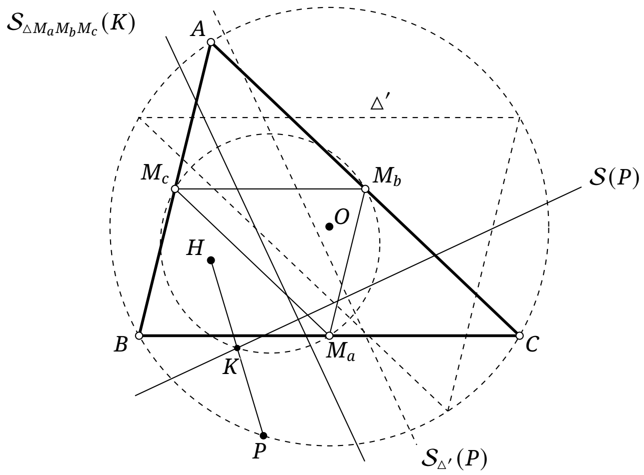



FIGURE 4.17

**Proof.** By Steiner's theorem, $K$ is the midpoint of $PH$, as in Figure 4.17. Under the homothety $\mathfrak{h}_{H,1/2}$, the point $P$ is sent to $K$, so $K$ plainly lies on $\mathscr{C}_{\triangle M_aM_bM_c}$. By lemma 0.5.20, the antipodal triangle $\triangle'$ of $\triangle ABC$ is homothetic to $\triangle M_aM_bM_c$ about $H$ in the ratio $2:1$, and $P,K$ are corresponding points. Hence $\mathscr{S}_{\triangle'}(P) \parallel \mathscr{S}_{\triangle M_aM_bM_c}(K)$. By theorem 4.2.5 and the definition of the antipodal triangle, $\mathscr{S}_{\triangle'}(P) \perp \mathscr{S}(P)$, so $\mathscr{S}(P) \perp \mathscr{S}_{\triangle M_aM_bM_c}(K)$. $\square$

We conclude with several properties of parabolas that follow from the theorems of Simson and Steiner.

**Theorem 4.2.11.** *The circumcircle of a triangle circumscribed about a parabola passes through the focus of the parabola.*

**Proof.** In Figure 4.18, claim 1.5.2 shows that the three projections of the focus $F$ on the lines $AB, BC, CA$ are collinear. Simson's theorem therefore gives $F \in \mathscr{C}$. $\square$

**Theorem 4.2.12.** *The orthocenter of a triangle circumscribed about a parabola lies on the directrix.*

**Proof.** In Figure 4.18, theorem 4.2.11 applies, and $\mathscr{S}(F)$ is the tangent to the parabola at its vertex. The directrix is $l = \mathfrak{h}_{F,2}(\mathscr{S}(F))$. By Steiner's theorem $\mathscr{S}(F)$ bisects the segment joining the focus $F$ to the orthocenter $H$ of $\triangle ABC$, so $H \in l$. $\square$

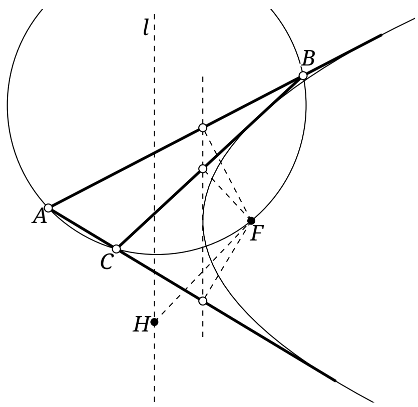



FIGURE 4.18

**Proposition 4.2.13.** *The directrix of a parabola is the Steiner line of its focus with respect to every triangle circumscribed about the parabola.*

**Proof.** By theorem 4.2.11, the Simson line of the focus with respect to a circumscribed triangle is the tangent at the vertex. The directrix and the tangent at the vertex are homothetic about the focus in the ratio $2:1$. By theorem 4.2.8, the directrix is therefore the Steiner line of the focus. □

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## Exercises

**Exercise 4.17.** Given $\triangle ABC$ and a point $P$ of $\mathscr{C}$, construct with straightedge and compass a line $l$ such that the parabola with focus $P$ and directrix $l$ is tangent to the lines $AB, BC, CA$.

**Exercise 4.18.** Let $AB < BC$ in $\triangle ABC$. Let $H$ be the orthocenter, $M$ the midpoint of the segment $BC$, and $N$ the midpoint of the arc $\widehat{BAC}$ of $\mathscr{C}$. A point $D$ on the side $CA$ is such that $MD$ is the perpendicular bisector of $HN$. Prove that $CD = DA + AB$.

**Exercise 4.19.** Given $\triangle ABC$ and $P_1, P_2, P_3 \in \mathscr{C}$, prove that the three lines $\mathscr{S}(P_1), \mathscr{S}(P_2), \mathscr{S}(P_3)$ are concurrent if and only if $\measuredangle ABP_1 + \measuredangle BCP_2 + \measuredangle CAP_3 = 0$.

**Exercise 4.20.** Let $H$ be the orthocenter of $\triangle ABC$ and let $P \in \mathscr{C}$. The perpendicular bisector of the segment $HP$ meets $CA, AB$ at $Q, R$, respectively. Prove that $A, P, Q, R$ are concyclic.

**Exercise 4.21.** A parabola is tangent to both sides of a fixed angle. Find the locus of the midpoint of the segment cut from a moving tangent by the two sides of the angle.

**Exercise 4.22.** The tangents at two points $P, Q$ of a parabola with focus $F$ meet at $A$. If $O$ is the circumcenter of $\triangle APQ$, prove that $AF \perp FO$.

**Exercise 4.23** (Droz-Farny theorem). Let a line $l_1$ through the orthocenter $H$ of $\triangle ABC$ meet the corresponding sidelines at $A_1, B_1, C_1$. Let the line $l_2$ through $H$ perpendicular to $l_1$ meet those sidelines at $A_2, B_2, C_2$. Prove that the midpoints $A_0, B_0, C_0$ of $A_1A_2, B_1B_2, C_1C_2$ are collinear.

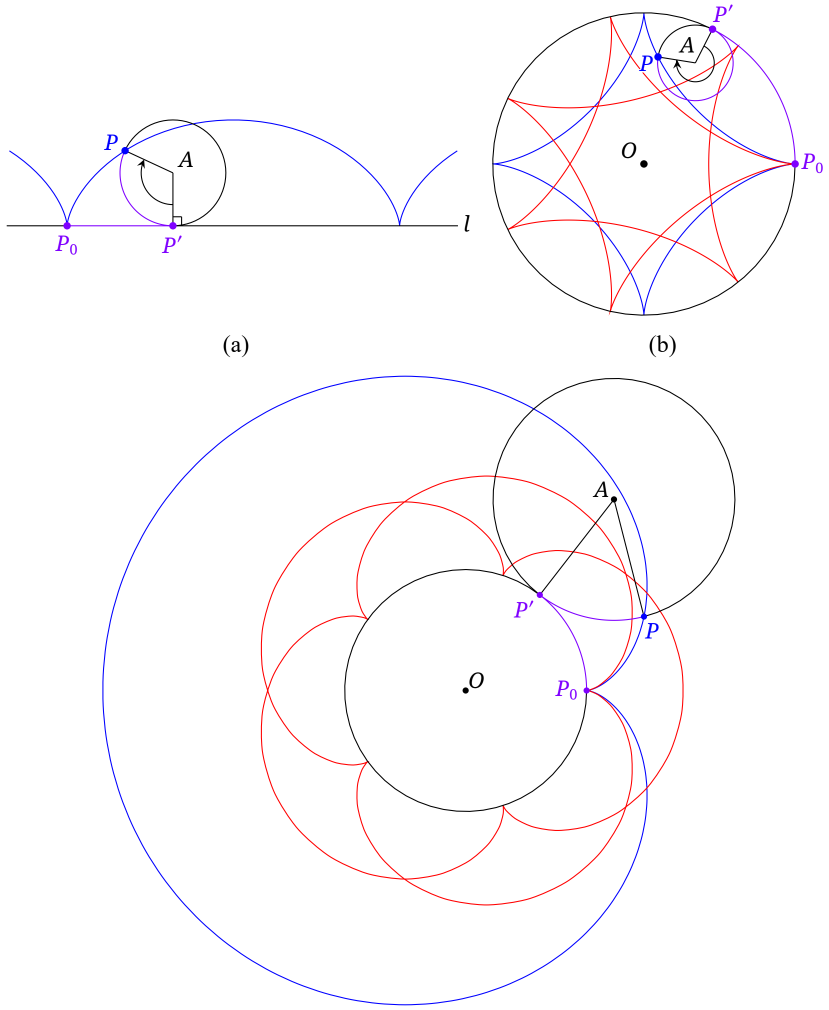
**Exercise 4.24.** Let $ABCD$ be a tangential quadrilateral. A line $l$ through $A$ meets the segment $BC$ at $M$ and the line $CD$ at $N$. If $I_1, I_2, I_3$ are the incenters of $\triangle ABM$, $\triangle MNC$, and $\triangle NDA$, respectively, prove that the orthocenter of $\triangle I_1 I_2 I_3$ lies on $l$.

### 4.2.2 Cycloids and the Steiner Deltoid*

This optional subsection concerns cycloidal curves. They are not part of the standard course associated with this book and will scarcely be used later. They are included both to introduce the celebrated Steiner deltoid and to emphasize that classical plane geometry is not restricted to points, lines, circles, and conics.

**Definition 4.2.14.** Let $\odot A$ be a circle of fixed radius and let $P$ be a point fixed on $\odot A$.

(1) When $\odot A$ rolls without slipping along a fixed line $l$, the locus of $P$ is a *cycloid*.
(2) When $\odot A$ rolls without slipping inside a fixed circle $\odot O$, the locus of $P$ is a *hypocycloid*. If $r_{\odot O} = 3r_{\odot A}$, it is a *deltoid*; if $r_{\odot O} = 4r_{\odot A}$, it is an *astroid*, also called a *tetracuspid*.
(3) When $\odot A$ rolls without slipping outside a fixed circle $\odot O$, the locus of $P$ is an *epicycloid*. In particular, when $r_{\odot A} = r_{\odot O}$, the epicycloid is a *cardioid*.




(a) (b)

(c)

FIGURE 4.19

Figure 4.19 shows several cycloidal curves.[12]

The blue curve in Figure 4.19a is a cycloid generated as $\odot A$ rolls along the line $l$. Initially $P$ coincides with $P_0$. When the point of contact of $\odot A$ with $l$ has reached $P'$, the point $P$ has reached the position shown. Rolling without slipping means that the arc $\overset{\frown}{P'P}$ on $\odot A$ and the segment $P_0P'$ on $l$—the two purple pieces in the figure—have equal lengths.

The blue and red curves in Figure 4.19b are hypocycloids. The blue one is an astroid, generated by $\odot A$ rolling inside $\odot O$ with $r_{\odot O}/r_{\odot A}=4$. Initially $P=P_0$; when the point of contact of the two circles has reached $P'$, the point $P$ is in the position shown. Rolling without slipping requires the arc $\overset{\frown}{P'P}$ on $\odot A$ and the arc $\overset{\frown}{P_0P'}$ on $\odot O$—the two purple pieces—to have equal lengths. For the red hypocycloid, the ratio of the radii of $\odot O$ and $\odot A$ is $7/2$. The blue curve in Figure 4.21 below is a deltoid.

The blue and red curves in Figure 4.19c are epicycloids. The blue one is a cardioid, generated by $\odot A$ rolling outside $\odot O$ with $r_{\odot O}/r_{\odot A}=1$. Initially $P=P_0$; when the point of contact of the circles has reached $P'$, the point $P$ is in the position shown. Rolling without slipping requires the arc $\overset{\frown}{P'P}$ on $\odot A$ and the arc $\overset{\frown}{P_0P'}$ on $\odot O$—the two purple pieces—to have equal lengths. For the red epicycloid, the ratio of the radii of $\odot O$ and $\odot A$ is $5/2$.

In the hypocycloidal case, if $r_{\odot O}=2r_{\odot A}$, the point $P$ of $\odot A$ moves back and forth along a diameter segment of $\odot O$. The hypocycloid then degenerates into two coincident segments.[13] This configuration is called a *Tusi couple*.[14]

The figures also show that a cycloidal curve has a cusp whenever the fixed point on the rolling circle is simultaneously at the point of contact with the fixed line or circle; the curve is not differentiable there. Thus a deltoid has three cusps, an astroid four, and a cardioid one.

Our main aim in this subsection is to prove that the Simson lines envelop a deltoid. An *envelope* is a curve tangent to every member of a family of moving curves or lines. We first establish several properties of the deltoid; other hypocycloids and epicycloids can be studied similarly.

**Proposition 4.2.15.** *Let $\odot A$, whose radius is one third that of the fixed circle $\odot O$, roll inside $\odot O$, and suppose that a point $D$ on $\odot A$ traces a deltoid. Let $B$ be an intersection of the deltoid with $\odot O$. If, at a particular position, $\odot A$ is tangent to $\odot O$ at $C$, then $\measuredangle CAD=3\measuredangle COB$.*

**Proof.** Since the rolling is without slipping, the arcs $\overset{\frown}{CD}$ and $\overset{\frown}{CB}$ have equal lengths. Their radii are in the ratio $1:3$, so their central angles are in the ratio $3:1$. □

**Proposition 4.2.16.** *Let $M$ be fixed on a circle $\omega$, and let $Q,R$ be moving points of $\omega$ such that $\measuredangle RQM=-2\measuredangle QRM$. Then the lines $QR$ envelop a deltoid.*

**Proof.** As in Figure 4.20, let $O$ be the center of $\omega$, construct the circle $\gamma=\odot(O,3OM)$, and choose $B\in\gamma$ so that $\overrightarrow{OB}=3\overrightarrow{MO}$. Extend $OQ$ to $A$ so that $Q$ is the midpoint of $AO$, and construct $\odot(A,AQ)$. This circle is tangent to $\gamma$ at a point $C$, with $C,A,Q,O$ collinear. On $\odot A$ choose $D$ so that the length of $\overset{\frown}{CD}$, not necessarily the minor arc, equals that of $\overset{\frown}{CB}$.

By proposition 4.2.15, the point $D$ moves on a deltoid $\alpha$ inscribed in $\gamma$ and having $B$ as a cusp, and $\measuredangle CAD=3\measuredangle AOB$. The angle relations give $\measuredangle CAD=3\measuredangle COB=-3\measuredangle MOQ=\measuredangle QOR$, so $AD\parallel OR$, and hence $D,Q,R$ are collinear.

[12]The word "cycloid" is sometimes used collectively for cycloids, hypocycloids, and epicycloids; the intended meaning is normally clear from context.

[13]The outward and return motions of $P$ each trace a segment, and those two traces coincide. A proof is left to the reader.

[14]The full name of Naṣīr is Naṣīr al-Dīn Muḥammad ibn Muḥammad ibn al-Ḥasan al-Ṭūsī (نصير الدين محمد بن محمد بن الحسن الطوسي). In an Arabic name, *ibn* (or *bin*) means "son of," whereas *al-Ṭūsī* is a geographical family designation. By Arabic convention, using the latter alone as the person's name is not especially appropriate, although "Tusi couple" is the established English name. In this book the author follows his own preference and refers to him as Naṣīr.

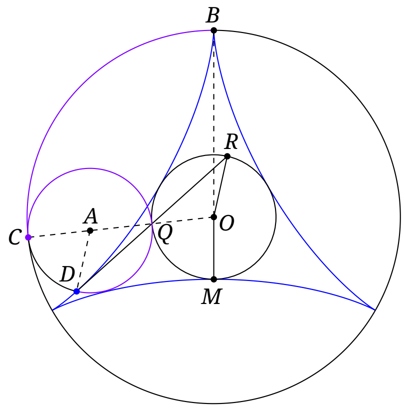



FIGURE 4.20

At this instant $C$ is the instantaneous center of rotation of the rolling circle $\odot A$. Thus the tangent to $\alpha$ at $D$ is perpendicular to $CD$. Since $QR \perp CD$, the line $QR$ is this tangent. As $Q$ is arbitrary, all the lines $QR$ are tangents of $\alpha$; and because $Q$ and the position of $\odot A$ correspond one-to-one, they give all tangents of $\alpha$. □

To prove that the envelope of the Simson lines is a deltoid, we need several lemmas about Simson lines. Throughout the rest of this subsection, for $\triangle ABC$, let $H$ and $O$ denote its orthocenter and circumcenter, respectively.

**Lemma 4.2.17.** *Given $\triangle ABC$, there is a unique equilateral triangle $\triangle XYZ$ inscribed in $\mathscr{C}$ such that, for every $P \in \mathscr{C}$, $\mathscr{S}_{\triangle ABC}(P) \parallel \mathscr{S}_{\triangle XYZ}(P)$.*

**Proof.** By theorem 4.2.5, the required condition is

$$\begin{aligned} 0 &= \vec{\measuredangle} AOX + \vec{\measuredangle} BOY + \vec{\measuredangle} COZ \\ &= \vec{\measuredangle} AOX + (\vec{\measuredangle} AOX + \vec{\measuredangle} XOY + \vec{\measuredangle} BOA) + (\vec{\measuredangle} AOX + \vec{\measuredangle} XOZ + \vec{\measuredangle} COA) \\ &= 3\vec{\measuredangle} AOX + (\vec{\measuredangle} XOY + \vec{\measuredangle} XOZ) + (\vec{\measuredangle} BOA + \vec{\measuredangle} COA) = 3\vec{\measuredangle} AOX + 0 + (\vec{\measuredangle} BOA + \vec{\measuredangle} COA). \end{aligned}$$

Thus $3\vec{\measuredangle} AOX = \alpha$, where $\alpha$ is a fixed directed angle. This identity is modulo $360°$, so

$$\vec{\measuredangle} AOX = \left(\frac{\alpha + k\cdot 2\pi}{3}\right) = \frac{\alpha}{3} + \frac{2\pi}{3}k \qquad (k \in \mathbb{Z})$$

gives all the solutions. They differ by integral multiples of $120°$, which merely select different vertices of the same equilateral triangle. Hence all solutions determine one and the same triangle $\triangle XYZ$, up to an immaterial relabeling of its vertices. □

**Lemma 4.2.18.** *Given $\triangle ABC$, let $\triangle XYZ$ be the equilateral triangle supplied by lemma 4.2.17, and let $\triangle UVW$ be its antipodal triangle. Then $\mathscr{S}_{\triangle ABC}(K)$ is tangent to $\mathscr{N}$ if and only if $K \in \{U, V, W\}$.*

**Proof.** Refer to Figure 4.21. We first prove that $\mathscr{S}(U)$ is tangent to $\mathscr{N}$ when $K = U$. By theorem 4.2.5, $\mathscr{S}_{\triangle XYZ}(U) \perp \mathscr{S}_{\triangle UVW}(U)$, while $\mathscr{S}(U) \parallel \mathscr{S}_{\triangle XYZ}(U)$. Hence $\mathscr{S}(U) \perp \mathscr{S}_{\triangle UVW}(U)$.

Now $\mathscr{S}_{\triangle UVW}(U) = UO$, and the tangent $l$ at $U$ is perpendicular to $UO$. Thus $\mathscr{S}(U) \parallel l$. Let $M$ be the midpoint of $UH$. Steiner's theorem gives $M \in \mathscr{S}(U)$. The lines $\mathscr{S}(U)$ and $l$ therefore correspond under the homothety of center $H$ and ratio $1:2$, so $\mathscr{S}(U)$ is tangent to $\mathscr{N}$. Similarly, $\mathscr{S}(V)$ and $\mathscr{S}(W)$ are also tangent to $\mathscr{N}$.

Next we prove that tangency fails when $K \neq U, V, W$. Let $\triangle KK'K''$ be the equilateral triangle inscribed in $\mathscr{C}$, and let $\triangle'$ be its antipodal triangle. If $\mathscr{S}(K)$ were tangent to $\mathscr{N}$, reversing the preceding argument would give $\mathscr{S}(K) \parallel \mathscr{S}_{\triangle'}(K)$. But $\triangle' \neq \triangle XYZ$, contrary to lemma 4.2.17. □

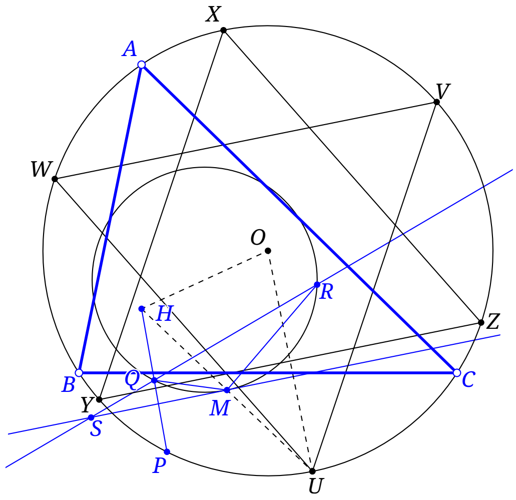



FIGURE 4.21

**Lemma 4.2.19.** *Given $\triangle ABC$, choose $\triangle UVW$ as in lemma 4.2.18, and let $H$ be the orthocenter. Let $M$ be the midpoint of $HU$, let $P \in \mathscr{C}$, and suppose $\mathscr{S}(P) \cap \mathscr{N} = \{Q, R\}$, where $Q$ is the midpoint of $HP$. Then $\measuredangle RQM = 2\measuredangle MRQ$.*

**Proof.** Continue with the notation of the proof of lemma 4.2.18, and put $S = QR \cap \mathscr{S}(U)$, as in Figure 4.21. By proposition 4.2.3, $\measuredangle MSQ = -\overrightarrow{\measuredangle} UOP/2 = -\measuredangle UAP$. The circles $\mathscr{C}$ and $\mathscr{N}$ are homothetic about $H$ in the ratio $2:1$, so inscribed angles subtending corresponding arcs are equal; hence $\measuredangle PAU = \measuredangle QRM$. Together with $\measuredangle QMS = \measuredangle QRM$ from the tangent–chord theorem, this yields $\measuredangle QSM = \measuredangle SMQ$. Therefore $\measuredangle RQM = \measuredangle RSM + \measuredangle SMQ = 2\measuredangle MRQ$. □

**Theorem 4.2.20.** *Given $\triangle ABC$, as $P$ moves on $\mathscr{C}$, the line $\mathscr{S}(P)$ envelops a deltoid.* This deltoid is called the *Steiner deltoid* of $\triangle ABC$.

**Proof.** Let $\mathscr{S}(P) \cap \mathscr{N} = \{R, Q\}$, with $Q$ the midpoint of $HP$. Choose $\triangle UVW$ as in lemma 4.2.18, and let $M$ be the midpoint of $UH$. By lemma 4.2.19, $\measuredangle RQM = -2\measuredangle QRM$. By proposition 4.2.16, the lines $QR$ envelop a deltoid. □

Figure 4.22 displays the envelope of the Simson lines.

---

## Exercises

**Exercise 4.25.** Prove that triangles with the same circumcircle have congruent Steiner deltoids.

**Exercise 4.26.** Let the ratio of the radii of $\odot O$ and $\odot A$ be $k > 0$. When $\odot A$ rolls without slipping inside, respectively outside, $\odot O$, the locus of a point of $\odot A$ is a hypocycloid, respectively an epicycloid, $\alpha$. Prove that $\alpha$ is closed if and only if $k \in \mathbb{Q}$. If $k = p/q$, where $p, q$ are relatively prime positive integers, prove that $\alpha$ has $p$ cusps.

**Exercise 4.27.** As $\odot A$ rolls without slipping along a line $l$, a point $P$ of $\odot A$ traces a cycloid $\alpha$. For $Q \in \alpha$, let $h$ be its distance from $l$, and let $\theta$ be the angle between the tangent to $\alpha$ at $Q$ and $l$. Prove that $\frac{\cos\theta}{\sqrt{h}} = \text{const}$.

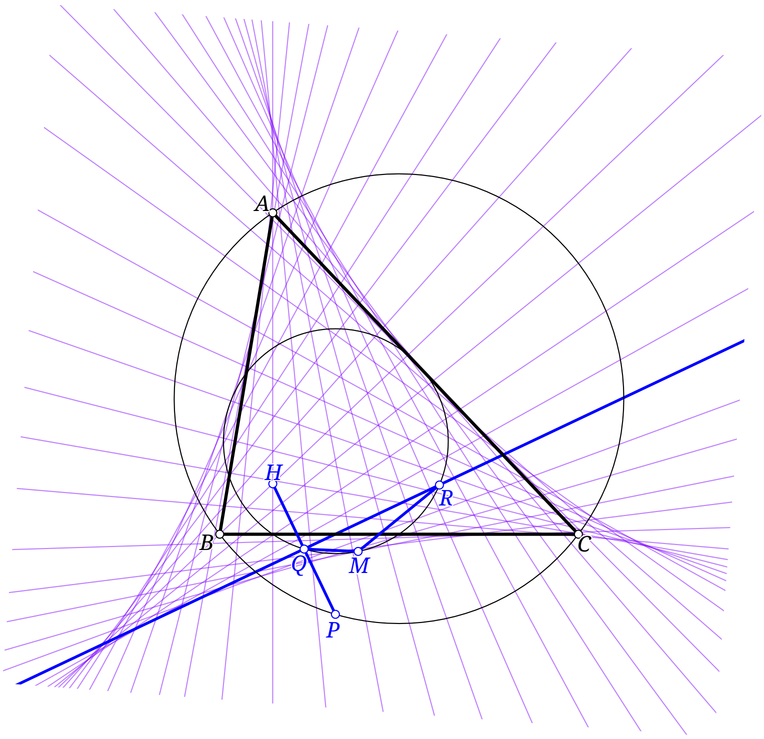



FIGURE 4.22

**Exercise 4.28.** Let $A$ be fixed on $\odot O$, and let $B$ move on $\odot O$. If $C$ is the reflection of $A$ in $OB$, prove that the lines $BC$ envelop a cardioid.

**Exercise 4.29.** Let the fixed lines $OP, OQ$ be perpendicular. Points $A, B$ move on $OP, OQ$, respectively, while $AB$ remains constant. Prove that the lines $AB$ envelop an astroid.

**Exercise 4.30.** Let $P$ be fixed on a circle $\omega$, and let $Q$ move on $\omega$. Denote by $l_Q$ the reflection of the line $PQ$ in the tangent to $\omega$ at $Q$. Prove that the lines $l_Q$ envelop a cardioid.[15]

**Exercise 4.31.** This exercise studies properties of the Tusi couple. Suppose the radii of $\odot O$ and $\odot A$ are in the ratio $2:1$, and $\odot A$ rolls without slipping inside $\odot O$. Prove that:
(1) a fixed point $P$ of $\odot A$ moves on a segment;
(2) a point $P'$ fixed relative to $\odot A$ and lying inside it moves on an ellipse.

### 4.2.3 Generalizations of Simson's Theorem*

#### Oblique Simson Lines

**Theorem 4.2.21.** *Fix a triangle $\triangle ABC$ and an angle $\alpha$. For $P \in \mathscr{C}$, choose points $D, E, F$ on the lines $BC, CA, AB$, respectively, so that $\measuredangle(PD, BC) = \measuredangle(PE, CA) = \measuredangle(PF, AB) = \alpha$. Then $D, E, F$ are collinear, as in Figure 4.23.* Their common line is called the *$\alpha$-oblique Simson line*, or the *$\alpha$-Carnot line*, of $P$ with respect to $\triangle ABC$; below we write it as $\mathscr{S}(P, \alpha)$.

[15]When drinking milk or coffee, one may sometimes notice a heart-shaped pattern of light on its surface, especially when the light source is near the rim of the cup. This exercise partially explains that phenomenon.

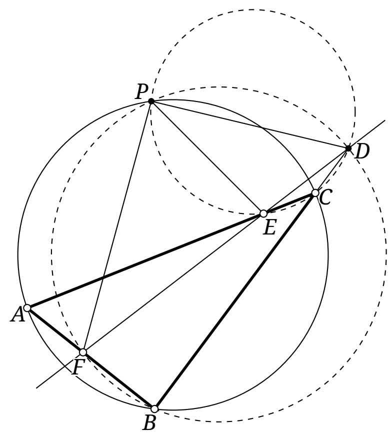



FIGURE 4.23

**Proof.** The proof is similar to that of Simson's theorem. The quadruples $P,D,B,F$ and $P,D,E,C$ are concyclic, and hence $\measuredangle PDE = \measuredangle PCE = \measuredangle PBA = \measuredangle PDF$. Thus $D,E,F$ are collinear. □

The $\alpha$-Simson lines have the following properties.

**Proposition 4.2.22.** *Fix $\triangle ABC$ and $P \in \mathscr{C}$. As $\alpha$ varies, the lines $\mathscr{S}(P,\alpha)$ envelop a parabola.*

**Proof.** Put $l_\alpha = \mathscr{S}(P,\alpha)$, and let $l_\alpha$ meet the three sidelines at $D,E,F$. Construct a parabola $\gamma$ tangent to $AB,BC,CA,l_\alpha$. By theorem 4.2.11, the circles $\odot(ABC),\odot(DBF),\odot(DEC)$ meet at the focus of the parabola; their common point is precisely $P$. Thus the focus of $\gamma$ is $P$. There is only one parabola with prescribed focus $P$ tangent to the three fixed lines $AB,BC,CA$; see exercise 4.17. Consequently every line $l_\alpha$ is tangent to this same parabola $\gamma$. □

**Proposition 4.2.23.** *Fix $\triangle ABC$ and an angle $\alpha$. If $P$ moves uniformly around $\mathscr{C}$, then its $\alpha$-Carnot line rotates in the opposite direction at half the angular speed.*

**Proof.** This follows at once by the argument of proposition 4.2.3. □

**Proposition 4.2.24.** *Let $O$ be the circumcenter of $\triangle ABC$. Given angles $\alpha_1,\alpha_2$, if $P_1,P_2 \in \mathscr{C}$, then*

$$\measuredangle(\mathscr{S}(P_1,\alpha_1),\mathscr{S}(P_2,\alpha_2)) = \alpha_1 - \alpha_2 - \frac{\overrightarrow{\measuredangle} P_1OP_2}{2}.$$

**Proof.** We have

$$\measuredangle(\mathscr{S}(P_1,\alpha_1),\mathscr{S}(P_2,\alpha_2)) = \measuredangle(\mathscr{S}(P_1,\alpha_1),\mathscr{S}(P_1,\alpha_2)) + \measuredangle(\mathscr{S}(P_1,\alpha_2),\mathscr{S}(P_2,\alpha_2)).$$

The first angle is plainly $\alpha_1 - \alpha_2$, and by proposition 4.2.23 the second is $-\frac{1}{2}\overrightarrow{\measuredangle} P_1OP_2$. □

Like ordinary Simson lines, the $\alpha$-Simson lines envelop a deltoid.

**Proposition 4.2.25.** *Fix $\triangle ABC$ and a nonzero angle $\alpha$. As $P$ moves on $\mathscr{C}$, the lines $\mathscr{S}(P,\alpha)$ envelop a deltoid.* We may call it the *$\alpha$-Steiner deltoid*.

We shall not give a complete proof, but briefly describe its idea. The $\alpha$-Simson line has many properties analogous to those of the ordinary Simson line. Generalizing a few key facts suffices. In the preceding proof for the ordinary Simson lines, the crucial ingredients were theorem 4.2.5 and Steiner's theorem. Taking $\alpha = 90^\circ$ as an intermediate case, proposition 4.2.24 readily gives the required generalization of theorem 4.2.5. On the other hand, Steiner's theorem says that the homothety $\mathfrak{h}_{H,1/2}$ sends $\mathscr{C}$ to $\mathscr{N}$ and sends every $P \in \mathscr{C}$ to a point of $\mathscr{S}(P)$. The following proposition is a generalization of Steiner's theorem. With these two generalizations, the preceding argument, after an appropriate rotation, proves that all $\alpha$-Simson lines envelop a deltoid.

**Proposition 4.2.26.** *Fix $\triangle ABC$ and a nonzero angle $\alpha$, and let $H$ be the orthocenter. There exist a fixed circle $\odot K$ and a spiral similarity $\mathfrak{s}$ centered at $H$ such that*

*(1) $\mathfrak{s}(\mathscr{C}) = \odot K$;*
*(2) $\mathfrak{s}(P) \in \mathscr{S}(P, \alpha)$ for every $P \in \mathscr{C}$.*

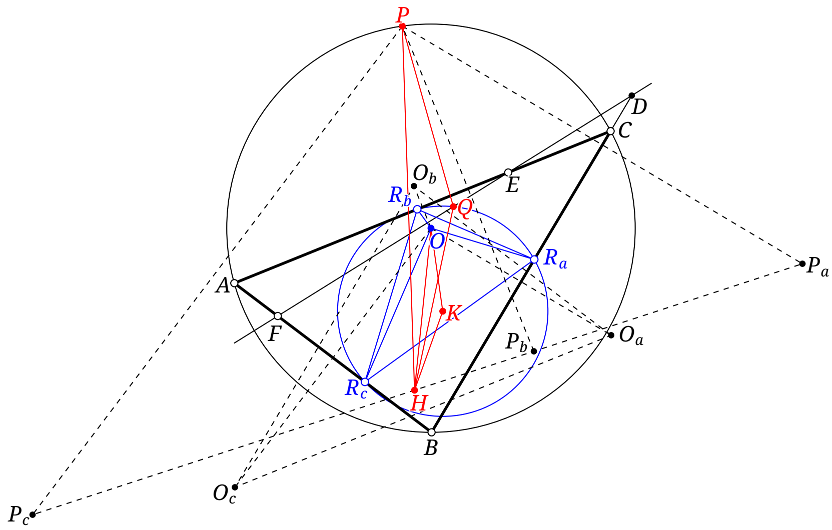



FIGURE 4.24

**Proof.** As in Figure 4.24, let $O$ be the circumcenter of $\triangle ABC$. Choose $R_a \in BC$, $R_b \in CA$, and $R_c \in AB$ so that $\measuredangle(OR_a, BC) = \measuredangle(OR_b, CA) = \measuredangle(OR_c, AB) = \alpha$. The quadruples $O, R_a, R_b, C$, $O, R_b, R_c, A$, and $O, R_c, R_a, B$ are concyclic. Computing the corresponding angles gives $\triangle R_aR_bR_c \cup O \overset{+}{\sim} \triangle ABC \cup H$.

Let $\odot K = \odot(R_aR_bR_c)$. For every $P \in \mathscr{C}$, define $\mathfrak{s}(P)$ to be the point $Q$ for which $\triangle HPQ \overset{+}{\sim} \triangle HOK$. Since $\frac{HP}{HQ} = \frac{HO}{HK}$ and $\measuredangle QHP$ are both constant, $\mathfrak{s}$ is indeed a spiral similarity. We verify the two properties.

Let $\triangle O_aO_bO_c$ be the reflection triangle of $O$. A reflection triangle and a pedal triangle are homothetic in the ratio $2:1$, while the pedal triangle of $O$ is the medial triangle, directly similar to $\triangle ABC$. Hence $\triangle O_aO_bO_c \cong \triangle ABC \overset{+}{\sim} \triangle R_aR_bR_c$. By proposition 0.5.28, $H$ is the circumcenter of $\triangle O_aO_bO_c$, whereas $K$ is the circumcenter of $\triangle R_aR_bR_c$. It follows that $\triangle O_aO_bO_c \cup H \overset{+}{\sim} \triangle R_aR_bR_c \cup K$.

Thus $\triangle HOO_b \overset{+}{\sim} \triangle KOR_b$, and hence $\triangle KOH \overset{+}{\sim} \triangle R_bOO_b$. Since $OR_b = O_bR_b$, we obtain $HK = OK$, so $K$ lies on the perpendicular bisector of $OH$.

From $\triangle HPQ \overset{+}{\sim} \triangle HOK$ we have $\triangle KHQ \overset{+}{\sim} \triangle OHP$, and therefore $\frac{KQ}{OP} = \frac{HK}{HO} = \frac{OK}{OH} = \frac{KR_b}{HO_b}$, where the last equality follows from $\triangle HOO_b \overset{+}{\sim} \triangle KOR_b$. Moreover, $\triangle ABC \cup O \cong \triangle O_aO_bO_c \cup H$ gives $OP = HO_a$. Thus $KQ = KR_b$, so $Q \in \odot K$, proving (1).

Let $\mathscr{S}(P, \alpha)$ meet the three sidelines at $D, E, F$, and let $P_a, P_b, P_c$ be the reflections of $P$ in them. One readily observes that

$$\triangle PFP_c \overset{+}{\sim} \triangle PDP_a \overset{+}{\sim} \triangle PEP_b \overset{+}{\sim} \triangle OR_bO_b \overset{+}{\sim} \triangle OKH \overset{+}{\sim} \triangle PQH.$$

Consequently $P_aP_bP_c \cup H \overset{+}{\sim} DEF \cup Q$. Since $P_a, P_b, P_c, H$ are collinear on $\mathscr{W}(P)$, the four points $D, E, F, Q$ are collinear. This proves (2). □

**Theorems of Seimiya (清宮) and Turner**

**Theorem 4.2.27** (Seimiya's theorem)**.** *Let $P,Q$ be two points on the circumcircle of $\triangle ABC$, and let $L,M,N$ be the reflections of $P$ in the three sidelines. If $QL,QM,QN$ meet $BC,CA,AB$ at $X,Y,Z$, respectively, then $X,Y,Z$ are collinear.*

This is another generalization of Simson's theorem; the latter is recovered when $P=Q$.

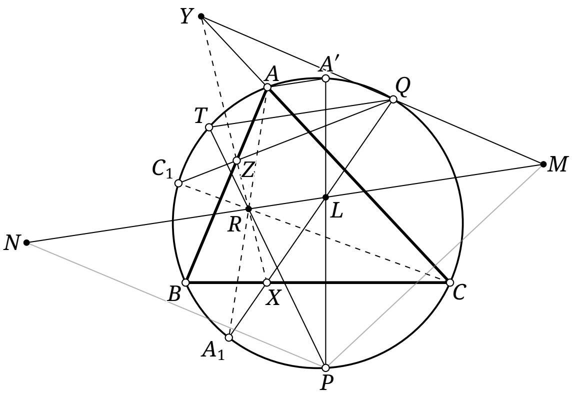



FIGURE 4.25

**Proof.** In Figure 4.25, the points $M,L,N$ lie on $\mathscr{W}(P)$. Put $PL\cap\mathscr{C}=\{P,A'\}$. Since $\mathscr{W}(P)\parallel\mathscr{S}(P)$, lemma 4.2.2 gives $AA'\parallel\mathscr{W}(P)$. Draw $QT\parallel\mathscr{W}(P)$, with $T$ the second intersection with $\mathscr{C}$, and put $R=TP\cap\mathscr{W}(P)$. Let $QX\cap\mathscr{C}=\{A_1,Q\}$.

Apply Pascal's theorem to the six points $Q,A_1,A,A',P,T$ on $\mathscr{C}$.[16] Since $QA_1\cap A'P=L$ and $AA'\cap QT=\infty_{\mathscr{W}(P)}$, the points $A_1A\cap PT$, $L$, and $\infty_{\mathscr{W}(P)}$ are collinear. Thus the three lines $A_1A,PT,\mathscr{W}(P)$ are concurrent, and $A,A_1,R$ are collinear.

Similarly, let $QZ$ meet $\mathscr{C}$ again at $C_1$. Then $C,C_1,R$ are collinear. Apply Pascal's theorem to $A,A_1,Q,C_1,C,B$. We have $AA_1\cap C_1C=R$, $A_1Q\cap CB=X$, and $QC_1\cap BA=Z$; hence $R,X,Z$ are collinear. In the same way $R,Y,Z$ are collinear, and therefore so are $X,Y,Z$. □

**Theorem 4.2.28** (Turner's theorem)**.** *Given $\triangle ABC$ and a point $P$, put $Q=\mathbf{i}P$, where* **i** *denotes inversion in the circumcircle. Let $L,M,N$ be the reflections of $P$ in the three sidelines. If $QL,QM,QN$ meet $BC,CA,AB$ at $X,Y,Z$, respectively, then $X,Y,Z$ are collinear.*

Turner's theorem is also a generalization of Simson's theorem, and again reduces to it when $P=Q$.

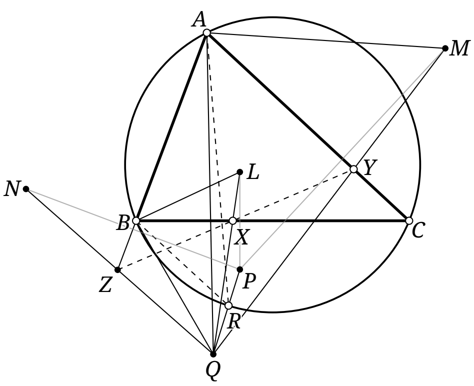



FIGURE 4.26

[16] The statement of theorem 3.4.6 is given only for a convex inscribed hexagon, but the discussion following it explains that the theorem holds in all configurations.

**Proof.** In Figure 4.26, let the segment $PQ$ meet $\mathscr{C}$ at $R$. Then theorem 4.1.10 shows that $AR$ bisects $\angle PAQ$, while reflection shows that $AC$ bisects $\angle PAM$. Thus $2\measuredangle RAC = \measuredangle QAM$. Similarly, $2\measuredangle RBC = \measuredangle QBL$. Since $R \in \mathscr{C}$, we have $\measuredangle RAC = \measuredangle RBC$, and therefore $\measuredangle RBL = \measuredangle RAM$.

By theorem 4.1.10, $\frac{BQ}{BP} = \frac{QR}{PR} = \frac{AQ}{AP}$. Reflection gives $BP = BL$ and $AP = AM$, whence $\frac{BQ}{BL} = \frac{AQ}{AM}$. Thus $\triangle QBL \overset{\pm}{\sim} \triangle QAM$, so $\measuredangle LQB = \measuredangle MQA$. Let $l$ be the angle bisector of $\angle AQL$. In each of the pairs $(QB, QM)$ and $(QA, QL)$, the two lines are reflections of one another in $l$.

Similarly, $QC$ and $QN$ are reflections of one another in $l$. Applying the result of exercise 3.50 to $Q$ and $\triangle ABC$ proves that $X, Y, Z$ are collinear. □

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### Exercises

**Exercise 4.32.** Prove proposition 4.2.25.

**Exercise 4.33.** Fix $\triangle ABC$ and a nonzero angle $\alpha$. Let $P_1, P_2, P_3 \in \mathscr{C}$, put $s_i = \mathscr{S}(P_i, \alpha)$, and set $\beta = \overrightarrow{\measuredangle} AOP_1 + \overrightarrow{\measuredangle} BOP_2 + \overrightarrow{\measuredangle} COP_3 - 2\alpha$. Prove that:
(1) if $\beta = 0$, then $s_1, s_2, s_3$ are concurrent;
(2) if $\beta \neq 0$, then $s_1, s_2, s_3$ are not concurrent, and the $\beta$-Steiner deltoid of the triangle formed by these lines is the $\alpha$-Steiner deltoid of $\triangle ABC$.

**Exercise 4.34.** Fix $\triangle ABC$ and a diameter $m$ of $\mathscr{C}$. Let moving points $P, Q$ of $\mathscr{C}$ remain symmetric about $m$. Let $L, M, N$ be the reflections of $P$ in the three sidelines, and let $QL, QM, QN$ meet them at $X, Y, Z$, respectively. By Seimiya's theorem these points are collinear; denote their line by $l(P)$. Prove that, as $P$ moves, $l(P)$ passes through a fixed point of $\mathscr{N}$.

**Exercise 4.35.** Fix $\triangle ABC$ and a diameter $m$ of $\mathscr{C}$. Let $P$ move on $m$ and put $Q = \mathbf{i}P$. Let $L, M, N$ be the reflections of $P$ in the three sidelines, and let $QL, QM, QN$ meet them at $X, Y, Z$, respectively. By Turner's theorem these points are collinear; denote their line by $l(P)$. Prove that, as $P$ moves, $l(P)$ passes through a fixed point of $\mathscr{N}$.

## 4.3 Cevian and Pedal Triangles

This section introduces two important concepts in plane geometry: cevian triangles and pedal triangles.

### Cevian Triangles

**Definition 4.3.1.** Given $\triangle ABC$ and a point $P$, put $A_1 = AP \cap BC$, $B_1 = BP \cap AC$, and $C_1 = CP \cap AB$, as in Figure 4.27. The triangle $\triangle A_1B_1C_1$ is the *cevian triangle* of $P$ with respect to $\triangle ABC$; the lines $AA_1, BB_1, CC_1$ are the three *cevians* of $P$; and the circumcircle of $\triangle A_1B_1C_1$ is the *cevian circle* of $P$. Conversely, $\triangle ABC$ is the *anticevian triangle* of $P$ with respect to $\triangle A_1B_1C_1$.

The cevian triangle of a point and the reference triangle are plainly perspective, and therefore have a perspective axis. This permits the following definition, although we shall not yet use it.

**Definition 4.3.2.** Given $\triangle ABC$ and a point $P$, let $l$ be the perspective axis of $\triangle ABC$ and the cevian triangle of $P$. The line $l$ is the *trilinear polar* of $P$ with respect to $\triangle ABC$, and $P$ is the *trilinear pole* of $l$.

The definition makes the construction of a cevian triangle immediate. The next proposition gives a construction of an anticevian triangle.

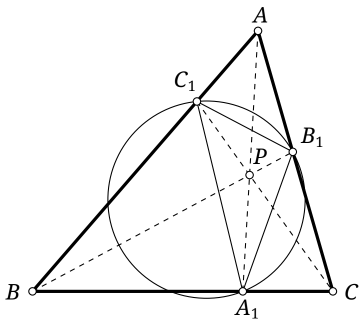



FIGURE 4.27

**Proposition 4.3.3.** *The anticevian triangle of P with respect to $\triangle ABC$ can be constructed as follows. Construct the cevian triangle $\triangle A_1B_1C_1$ of P, and put $M = A_1B_1 \cap AB$, $A' = AA_1 \cap CM$, $B' = BB_1 \cap CA'$, and $C' = CC_1 \cap BA'$. Then $\triangle A'B'C'$ is the required anticevian triangle.*

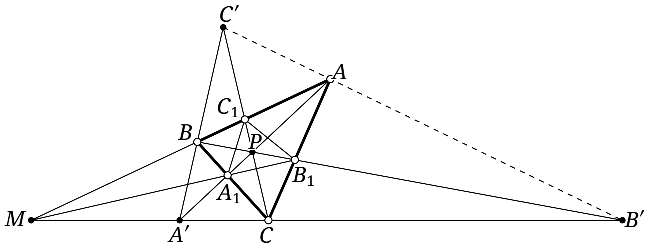



FIGURE 4.28

**Proof.** In Figure 4.28, it remains only to show that $C', B', A$ are collinear. Apply Desargues' theorem to $\triangle MBA'$ and $\triangle B_1CP$, whose center of perspectivity is $A_1$. ☐

Projective geometry gives the following important property of cevian triangles.

**Theorem 4.3.4** (Harmonic property of a cevian triangle)**.** *Let the cevian triangle of P with respect to $\triangle ABC$ be $\triangle A_1B_1C_1$.*[17] *Let the sides of $\triangle A_1B_1C_1$ meet the corresponding cevians at $A_2, B_2, C_2$. Then $(A, P, A_2, A_1)$, $(B, P, B_1, B_2)$, and $(C, P, C_1, C_2)$ are harmonic ranges, as in Figure 4.29.*

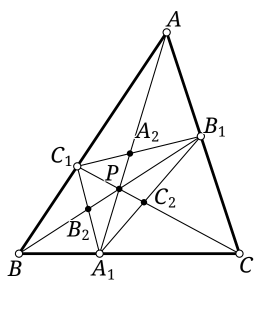



FIGURE 4.29

[17]As noted earlier, the reference triangle may be omitted when no ambiguity can result. Thus "the cevian triangle of P" here means the cevian triangle of P with respect to $\triangle ABC$; the same convention will be used below.

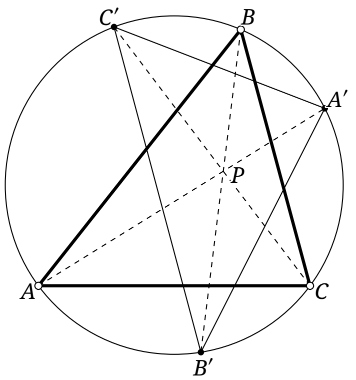
**Proof.** For the complete quadrangle $ABCP$, the harmonic property of a complete quadrangle, theorem 3.5.11, shows that $B_1C, B_1B, B_1A_1, B_1C_1$ form a harmonic pencil.[18] Since $B_1(C,B,A_1,C_1) \overline{\overline{\wedge}} (P,A,A_1,A_2)$, the points $P,A,A_1,A_2$ form a harmonic range. The other two statements follow in the same way. □

There is also a circumcevian triangle.

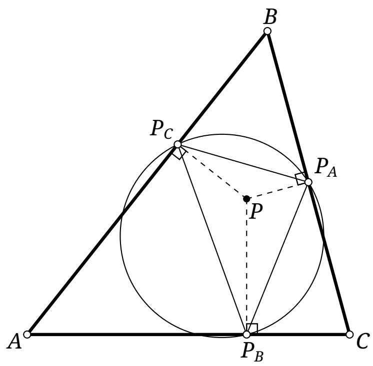
**Definition 4.3.5.** Let $P$ lie inside $\triangle ABC$, put $AP \cap \odot(ABC) = \{A, A'\}$, and define $B', C'$ similarly. The triangle $\triangle A'B'C'$, shown in Figure 4.30a, is the *circumcevian triangle* of $P$ with respect to $\triangle ABC$.

### Pedal Triangles

**Definition 4.3.6.** Given $\triangle ABC$ and a point $P$, drop perpendiculars from $P$ to the three sidelines. The triangle $\triangle P_AP_BP_C$ formed by their feet, as in Figure 4.30b, is the *pedal triangle* of $P$ with respect to $\triangle ABC$, and its circumcircle is the *pedal circle* of $P$. Conversely, $\triangle ABC$ is the *antipedal triangle* of $P$ with respect to $\triangle P_AP_BP_C$.




(a)




(b)

FIGURE 4.30

Not every point of the plane has a nondegenerate pedal triangle in the traditional sense. Simson's theorem gives the precise exception.

**Theorem 4.3.7.** *The pedal triangle of a point $P$ with respect to $\triangle ABC$ is degenerate, its three vertices being collinear, if and only if $P \in \mathscr{C}$.*

The definition of a homothety also immediately yields the following fact.

**Theorem 4.3.8.** *The reflection triangle and pedal triangle of a point with respect to a given triangle are homothetic about that point in the ratio* $2:1$.

The following important theorem connects pedal and cevian triangles.

**Theorem 4.3.9.** *The pedal triangle and circumcevian triangle of $P$ with respect to $\triangle ABC$ are directly similar, as in Figure 4.31.*

**Proof.** In the configuration of Figure 4.31, let $\triangle P_AP_BP_C$ be the pedal triangle and $\triangle A'B'C'$ the circumcevian triangle. Then $\measuredangle AA'C' = \measuredangle ACC' = \measuredangle P_BP_AP$, where the last equality follows from the concyclicity of $P, P_A, C, P_B$. Similarly, $\measuredangle AA'B' = \measuredangle P_CP_AP$, and hence $\measuredangle P_CP_AP_B = \measuredangle C'A'B'$. The other two pairs of corresponding angles are equal in the same way, so $\triangle P_AP_BP_C \overset{+}{\sim} \triangle A'B'C'$. □

[18]The four points $A, B, C, P$ form a concave quadrilateral here, but the harmonic property of a complete quadrangle still holds, since projective theorems do not depend on this positional relation. Some readers may initially find such an application unfamiliar, but it is standard and should quickly become natural.

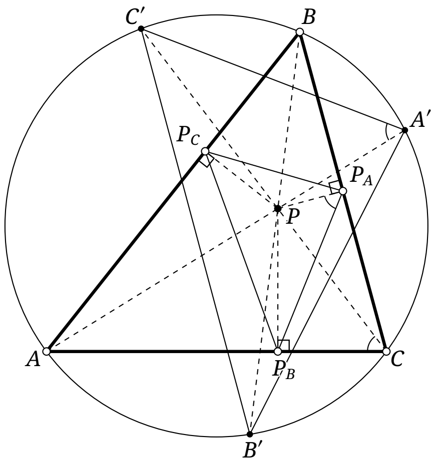



FIGURE 4.31

**Theorem 4.3.10.** *If two points are inverse in the circumcircle of $\triangle ABC$, then their circumcevian triangles with respect to $\triangle ABC$ are oppositely similar.*

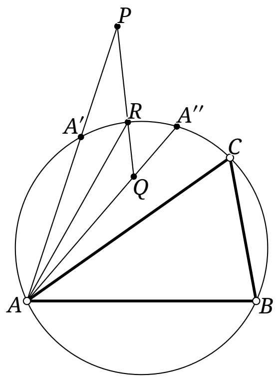



FIGURE 4.32

**Proof.** As in Figure 4.32, let $P,Q$ be inverse points, let their circumcevian triangles be $\triangle A'B'C'$ and $\triangle A''B''C''$, and let the segment $PQ$ meet $\odot(ABC)$ at $R$. By theorem 4.1.10, $\angle PAR = \angle QAR$, so $\widehat{A'R} = \widehat{A''R}$. Thus $A'$ and $A''$ are reflections of each other in $PQ$. The same holds for $B',B''$ and for $C',C''$. Hence the two circumcevian triangles are reflections of each other in $PQ$ and are oppositely similar. □

**Corollary 4.3.11.** *If two points are inverse in the circumcircle of $\triangle ABC$, then their pedal triangles with respect to $\triangle ABC$ are oppositely similar.*

**Proof.** This follows from theorems 4.3.9 and 4.3.10. □

---

## Exercises

**Exercise 4.36.** Prove that the trilinear pole of the line at infinity with respect to $\triangle ABC$ is the centroid of $\triangle ABC$.

**Exercise 4.37.** Given $\triangle ABC$ and two points $P,Q$, prove that the six vertices of their cevian triangles lie on a conic. This conic is called the *bicevian conic* of $P,Q$ with respect to $\triangle ABC$.

**Exercise 4.38.** Given $\triangle ABC$ and a point $P$, let the cevian circle of $P$ meet the three sidelines again, apart from the three vertices of the cevian triangle, at $A',B',C'$, respectively. Prove that $AA',BB',CC'$ are concurrent at a point $P'$. The point $P'$ is the *cyclocevian conjugate* of $P$ with respect to $\triangle ABC$, and the transformation $P \mapsto P'$ is *cyclocevian conjugation* with respect to $\triangle ABC$.

**Exercise 4.39.** Let $H$ be the orthocenter of $\triangle ABC$. Points $P,Q$, neither on $\mathscr{C}$, lie on the same side of $H$ on a line through $H$. If $HP > HQ$, prove that the perimeter of the pedal triangle of $P$ is greater than that of the pedal triangle of $Q$.

**Exercise 4.40.** Given $\triangle ABC$, construct with straightedge and compass a point $S$ whose pedal triangle with respect to $\triangle ABC$ is equilateral.

**Exercise 4.41** (Boutte's theorem)**.** Prove that the reflection triangle of a triangle and the pedal triangle of its nine-point center are homothetic about the centroid in the ratio $4:1$.

**Exercise 4.42.** Given $\triangle ABC$ and a point $P$, let $R$ and $O$ be the circumradius and circumcenter. Prove that the area of the pedal triangle of $P$ is $\frac{|PO^2-R^2|}{4R^2} \cdot S_{\triangle ABC}$.

**Exercise 4.43.** Given $\triangle ABC$ and a point $P$, theorem 4.3.9 shows that the pedal triangle and circumcevian triangle of $P$ are related by a spiral similarity. Let $\theta$ be the directed angle of rotation of the spiral similarity that sends the circumcevian triangle to the pedal triangle. Prove that:
(1) $\measuredangle PBA + \measuredangle PCB + \measuredangle PAC = 90^\circ + \theta$;
(2) $|\theta|$ equals the angle of intersection of $\mathscr{N}$ and the pedal circle of $P$.

## 4.4 Isogonal and Isotomic Conjugacy

This section concerns two special relations associated with a triangle: isogonal and isotomic conjugacy.

### 4.4.1 Isogonal Conjugacy and Its Basic Properties

**Definition 4.4.1.** Let $P$ be a point distinct from the vertices of $\triangle ABC$. Reflect the lines $PA,PB,PC$ in the bisectors of $\angle A, \angle B, \angle C$, respectively. The three reflected lines meet at a point $P'$, which may be a point at infinity, as in Figure 4.33. The point $P'$ is the *isogonal conjugate* of $P$ with respect to $\triangle ABC$, and the transformation $P \mapsto P'$ is *isogonal conjugation*. The lines $AP',BP',CP'$ are the *isogonal lines* of $AP,BP,CP$ in the corresponding vertex angles.

If $P'$ exists, it is plainly unique; its existence is guaranteed by theorem 4.4.2. We shall write $\mathbf{g}_{\triangle ABC}P$ for the isogonal conjugate of $P$ with respect to $\triangle ABC$, and simply $\mathbf{g}P$ when no ambiguity can result.

**Theorem 4.4.2.** *For a point $P$ and a triangle $\triangle ABC$, the circumcenter of the reflection triangle of $P$ is $P' = \mathbf{g}P$.*

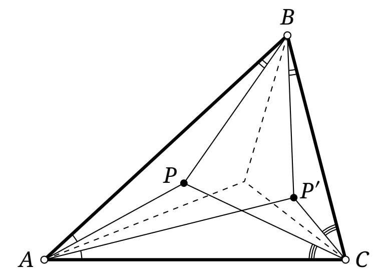



FIGURE 4.33

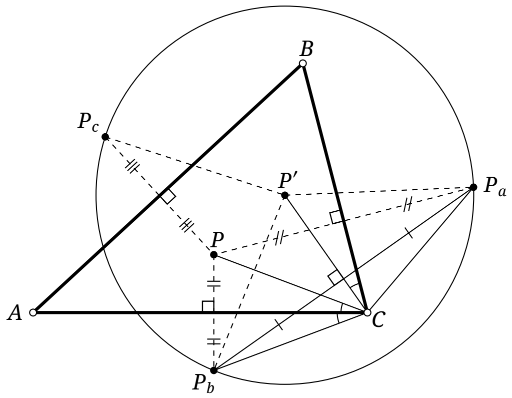



FIGURE 4.34

**Proof.** First suppose $P$ lies inside $\triangle ABC$, and let $\triangle P_aP_bP_c$ be its reflection triangle, as in Figure 4.34. Let $P'$ be the center of $\odot(P_aP_bP_c)$. Then $\angle P_aCP' = \angle P_aCP_b/2 = \angle BCA$, and consequently

$$\angle BCP' = \angle P_aCP' - \angle BCP_a = \angle ACB - \angle BCP = \angle ACP.$$

The other two pairs of angles are equal in the same way, so $P' = \mathbf{g}P$.

The same argument applies when $P$ lies outside the triangle. If $P \in \odot(ABC)$, however, $\triangle P_aP_bP_c$ degenerates into $\mathscr{W}(P)$, and $P'$ is not defined in the ordinary sense. We may regard $\mathscr{W}(P)$ as a circle whose radius tends to infinity; its center is then the point at infinity in the direction perpendicular to $\mathscr{W}(P)$. Since $\mathscr{W}(P) \parallel \mathscr{S}(P)$, this is precisely the required $P'$. This description uses a limiting argument; a direct proof appears in part (4) of theorem 4.4.3. □

The next theorem gives the fundamental properties of isogonal conjugation.

**Theorem 4.4.3.** *Isogonal conjugation with respect to $\triangle ABC$ has the following properties.*

*(1) If $P$ does not lie on a sideline of the triangle, then $\mathbf{g}(\mathbf{g}P) = P$; equivalently, $\mathbf{g}^2 = \mathrm{id}$.*[19]
*(2) The isogonal conjugate of a point on a sideline is the opposite vertex.*
*(3) Exactly four points of the plane are fixed by isogonal conjugation: the incenter and the three excenters.*
*(4) If $P \in \odot(ABC)$, then $\mathbf{g}P$ is the point at infinity in the direction perpendicular to $\mathscr{S}(P)$.*

**Proof.** Parts (1)–(3) are immediate, and part (4) was discussed in the proof of theorem 4.4.2. We give another proof of (4).

In the configuration of Figure 4.35, let $P_b, P_c$ be the projections of $P$ on $AC, AB$, respectively, and let the reflection of $AP$ in the bisector of $\angle BAC$ meet $\mathscr{S}(P) = P_bP_c$ at $X$. Since $A, P_c, P_b, P$ are concyclic,

$$\angle AP_cP_b = 180^\circ - \angle APP_b = 90^\circ + \angle PAP_b = 90^\circ + \angle XAP_c.$$

[19]Thus $P$ and $\mathbf{g}P$ play symmetric roles under isogonal conjugation. They may be said to be *isogonally conjugate*, or to form a pair of isogonal conjugates.

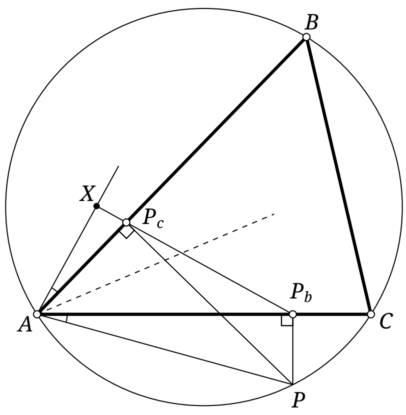



FIGURE 4.35

Thus $\angle AXP_b = \angle AP_cP_b - \angle XAP_c = 90°$. The reflections of $PB$ and $PC$ in the corresponding angle bisectors are likewise perpendicular to $\mathscr{S}(P)$, proving (4). ☐

Theorem 4.4.2 has the following consequences.

**Corollary 4.4.4.** *If $P,P'$ are isogonal conjugates with respect to $\triangle ABC$, then the center of the pedal circle of $P'$ is the midpoint of $PP'$, and its radius is half the circumradius of the reflection triangle of $P$.*

**Proof.** In the proof of theorem 4.4.2, apply the homothety $\mathfrak{h}_{P',1/2}$ to $\odot(P_aP_bP_c)$. ☐

**Corollary 4.4.5.** *For a fixed triangle, two distinct points have the same pedal circle if and only if they are isogonal conjugates. In that case the center of their common pedal circle is their midpoint.*

**Proof.** If $P,P'$ are isogonal conjugates, corollary 4.4.4 shows that both pedal circles have the midpoint of $PP'$ as center. The reflection triangle and pedal triangle of any point are homothetic in the ratio $2:1$, so the radius of a pedal circle is half the circumradius of the corresponding reflection triangle. Another application of corollary 4.4.4 therefore shows that the two radii are equal.

Conversely, suppose $P$ and $Q$ have the same pedal circle, and put $P' = \mathbf{g}P$. Then $P'$ has that pedal circle as well. By Dirichlet's principle (the pigeonhole principle),[20] at least two vertices of the pedal triangle of $Q$ coincide with vertices of the pedal triangle of $P$, or at least two coincide with vertices of that of $P'$, since the circle meets each sideline in at most two points. Two perpendicular feet determine $Q$, so $Q = P$ or $Q = P'$. ☐

**Corollary 4.4.6.** *The circumcenter and orthocenter of a triangle are isogonal conjugates.*

**Proof.** Their pedal circles are both the nine-point circle.[21] ☐

The following construction also produces the isogonal conjugate.

**Proposition 4.4.7.** *Let $P$ lie inside $\triangle ABC$. From $A,B,C$, draw perpendiculars to the corresponding sides of the pedal triangle of $P$. These three perpendiculars meet at the isogonal conjugate $P'$ of $P$, as in Figure 4.36.*

[20] If $n$ objects are placed in $k$ classes, some class contains at least $\lceil \frac{n}{k} \rceil$ objects.

[21] This can also be proved directly using only elementary geometry, without corollary 4.4.5; the details are left to the reader.

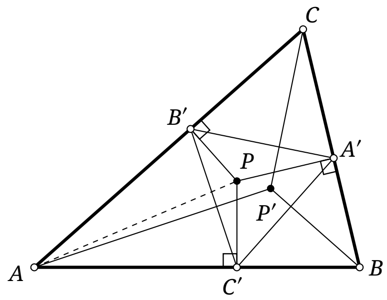



FIGURE 4.36

**Proof.** Put $P' = \mathbf{g}P$. Since $P, B', A, C'$ are concyclic, we have $\angle PC'B' = \angle PAB' = \angle P'AB$, and hence $\angle(C'B', AP') = \angle(AC', PC') = 90°$. The other two perpendiculars are obtained similarly, so their common point is $P'$. ☐

Isogonal conjugates also satisfy the following metric properties.

**Proposition 4.4.8.** *Let $AA_1, AA_2$ be isogonal lines in $\angle A$ of $\triangle ABC$, meeting the side $BC$ at $A_1, A_2$, respectively. Then*

$$\frac{AB^2}{AC^2} = \frac{BA_1}{CA_1} \cdot \frac{BA_2}{CA_2}.$$

**Proof.** The generalized angle-bisector theorem gives $\frac{AB \sin\measuredangle BAA_1}{AC \sin\measuredangle CAA_1} = \frac{\overline{BA_1}}{\overline{CA_1}}$, and similarly $\frac{AB \sin\measuredangle BAA_2}{AC \sin\measuredangle CAA_2} = \frac{\overline{BA_2}}{\overline{CA_2}}$. Multiply the two identities and use $\measuredangle BAA_2 = -\measuredangle CAA_1$ and $\measuredangle CAA_2 = -\measuredangle BAA_1$. This gives $\frac{AB^2}{AC^2} = \frac{BA_1}{CA_1} \cdot \frac{BA_2}{CA_2}$. ☐

**Theorem 4.4.9.** *Given $\triangle ABC$, if the directed distances of a point $P$ from $BC, CA, AB$ are in the ratio $\alpha : \beta : \gamma$, then those of $\mathbf{g}P$ are in the ratio $\alpha^{-1} : \beta^{-1} : \gamma^{-1}$.*

**Proof.** By the subtended-angle formula,

$$\begin{aligned}
\operatorname{dist}(\mathbf{g}P, AB) : \operatorname{dist}(\mathbf{g}P, AC) &= \overline{A\mathbf{g}P} \sin\measuredangle BA\mathbf{g}P : \overline{A\mathbf{g}P} \sin\measuredangle \mathbf{g}PAC \\
&= \sin\measuredangle BA\mathbf{g}P : \sin\measuredangle \mathbf{g}PAC = \sin\measuredangle PAC : \sin\measuredangle BAP \\
&= \overline{AP} \sin\measuredangle PAC : \overline{AP} \sin\measuredangle BAP = \beta : \gamma = \gamma^{-1} : \beta^{-1}.
\end{aligned}$$

The ratios involving the remaining side follow similarly. ☐

**Proposition 4.4.10.** *Let $R, O$ be the circumradius and circumcenter of $\triangle ABC$. Given a point $P$, put $Q = \mathbf{g}P$, and let $\triangle P_A P_B P_C$ and $\triangle Q_A Q_B Q_C$ be the pedal triangles of $P, Q$. Then*

$$R^2 - OP^2 = 2R \cdot \frac{\overline{PP_B} \cdot \overline{PP_C}}{\overline{QQ_A}}.$$

**Proof.** As in Figure 4.37, extend $AP, AQ$ to meet the circumcircle again at $U, V$, respectively. Since $P$ and $Q$ are isogonal conjugates, $\angle BAV = \angle UAC$, and hence the corresponding chords satisfy $BV = CU$. Since $\angle BUP = \angle QVB$ and

$$\angle BPU = \angle ABP + \angle BAP = \angle QBC + \angle QAC = \angle QBC + \angle VBC = \angle QBV,$$

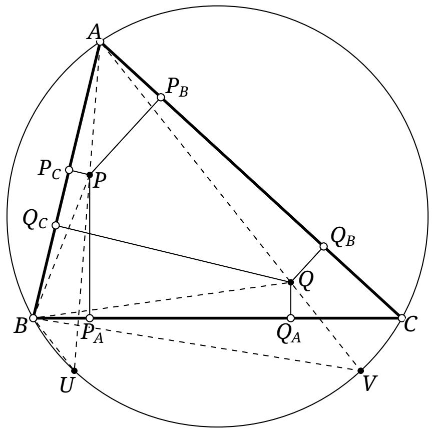



FIGURE 4.37

we have $\triangle BPU \sim \triangle QBV$. Therefore

$$\begin{aligned} R^2 - OP^2 &= AP \cdot PU = AP \cdot \frac{BV \cdot BP}{BQ} = AP \cdot BV \cdot \frac{PP_C}{QQ_A} \\ &= AP \cdot CU \cdot \frac{PP_C}{QQ_A} = AP \cdot 2R \sin\angle UAC \cdot \frac{PP_C}{QQ_A} = 2R \cdot \frac{PP_B \cdot PP_C}{QQ_A}. \end{aligned}$$

The successive equalities use, in order, the power-of-a-point theorem, $\triangle BPU \sim \triangle QBV$, $\triangle BPP_C \sim \triangle BQQ_A$, the equal-chord relation $BV = CU$, the sine rule, and $PP_B = AP \sin\angle UAC$. □

**Remark.** Taking $P = Q = I$, the incenter, gives the Euler–Chapple formula theorem 0.6.20, $OI^2 = R^2 - 2Rr$, where $r$ is the inradius.

**Proposition 4.4.11.** *Let $Q = \mathbf{g}P$. Suppose $AP, AQ$ meet the circumcircle of $\triangle ABC$ again at $D, E$, respectively, and put $K = AQ \cap BC$. Then*

$$\frac{\overline{AP}}{\overline{PD}} = \frac{\overline{QK}}{\overline{KE}}.$$

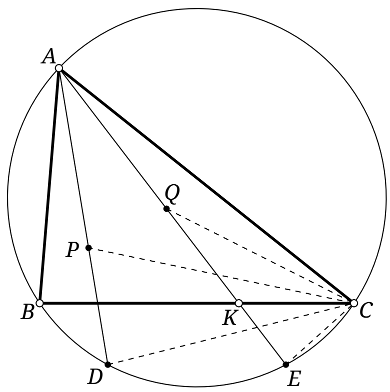



FIGURE 4.38

**Proof.** As in Figure 4.38, the argument of proposition 4.4.10 gives $\triangle PDC \sim \triangle CEQ$. Hence $\frac{PD}{DC} = \frac{CE}{EQ} \Rightarrow PD \cdot EQ = DC \cdot CE$. Also $\angle ADC = \angle CEK$ and $\angle DAC = \angle BEA = \angle ECK$, so $\triangle ADC \sim \triangle CEK$. Thus $\frac{AD}{DC} = \frac{CE}{EK} \Rightarrow AD \cdot EK = DC \cdot CE$. It follows that $AD \cdot EK = PD \cdot EQ \Rightarrow \frac{AD}{PD} = \frac{EQ}{EK}$. Consequently,

$$\frac{AP}{PD} = \frac{AD - PD}{PD} = \frac{AD}{PD} - 1 = \frac{EQ}{EK} - 1 = \frac{EQ - EK}{EK} = \frac{QK}{KE}.$$ □

We conclude this subsection with an application of a pair of isogonally conjugate points: the celebrated theorem of Morley. It has many proofs; the one given here uses such a pair.

**Theorem 4.4.12** (Morley's theorem)**.** *The intersections of adjacent angle trisectors of a triangle form an equilateral triangle. In Figure 4.39, $\triangle A_1B_1C_1$ is equilateral.*

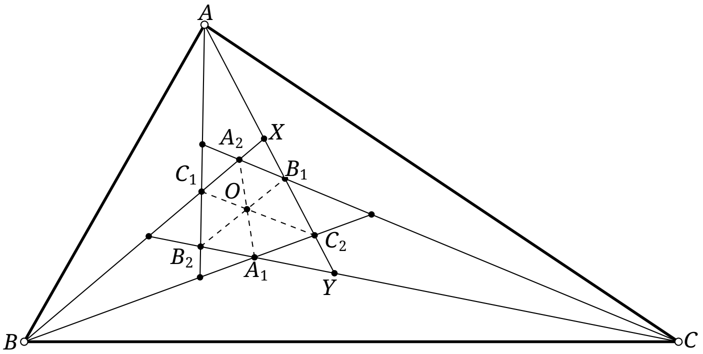



FIGURE 4.39

**Proof.** Let the six angle trisectors meet at $A_1, A_2, B_1, B_2, C_1, C_2$, as in Figure 4.39. The points $A_1, A_2$ are isogonal conjugates by the trisector construction. Hence

$$\begin{aligned} C_2(B,C_1,A_2,X) &\overset{(BX)}{\doublebarwedge} A(B,C_1,A_2,X) \barwedge A(C,Y,A_1,B_2) \\ &\overset{(CB_2)}{\doublebarwedge} B_1(C,Y,A_1,B_2) \barwedge B_1(A_1,B_2,C,Y). \end{aligned}$$

The second projectivity follows because reflecting all lines in an angle bisector preserves cross-ratios, and the last uses part (1) of proposition 3.2.11.[22]

In the projectivity of pencils $C_2(B,C_1,A_2,X) \barwedge B_1(A_1,B_2,C,Y)$, the line joining the vertices, $C_2B_1$, corresponds to itself. By theorem 3.5.7, the projectivity is a perspectivity, and the intersections of corresponding lines are collinear. Thus $A_1 = C_2B \cap B_1A_1$, $O = C_2C_1 \cap B_1B_2$, and $A_2 = C_2A_2 \cap B_1C$ are collinear. Therefore $C_1C_2, B_1B_2, A_1A_2$ are concurrent.

The point $C_1$ is the incenter of $\triangle ABC_2$, so $\angle AC_1C_2 = 90^\circ + \frac{\angle ABC_2}{2} = 90^\circ + \frac{\angle ABC}{3}$. The point $B_1$ is the incenter of $\triangle AB_2C$, so

$$\angle AB_2B_1 = 90^\circ - \frac{\angle B_2AC}{2} - \frac{\angle B_2CA}{2} = 90^\circ - \frac{\angle BAC}{3} - \frac{\angle BCA}{3}.$$

It follows that

$$\angle B_2OC_1 = \angle AC_1C_2 - \angle AB_2B_1 = \frac{\angle ABC + \angle BCA + \angle CAB}{3} = 60^\circ.$$

Similarly,

$$\angle C_1OA_2 = \angle A_2OB_1 = \angle B_1OC_2 = \angle C_2OA_1 = \angle A_1OB_2 = 60^\circ.$$

Finally, $A_1$ is the incenter of $\triangle BA_2C$, whence $\angle BA_2A_1 = \angle CA_2A_1$. Together with $\angle C_1OA_2 = \angle B_1OA_2$, this gives $\triangle C_1A_2O \cong \triangle B_1A_2O$, and hence $C_1O = B_1O$. Similarly $A_1O = B_1O$, so $\triangle A_1B_1C_1$ is equilateral with center $O$. □

The equilateral triangle $\triangle A_1B_1C_1$ in Morley's theorem is the *first Morley triangle* of $\triangle ABC$, usually called simply its *Morley triangle*. Its center is the *first Morley center*, usually the *Morley center*. The two trisectors at each of the three vertices have six pairwise intersections in all. The triangle formed by the three intersections other than the vertices of the Morley triangle is the *first Morley adjunct triangle*, usually the *Morley adjunct triangle*; in Figure 4.39 it is $\triangle A_2B_2C_2$.

[22] Recall that a cross-ratio-preserving transformation between one-dimensional fundamental forms is a one-dimensional projectivity.

The proof also gives the following fact.

**Proposition 4.4.13.** *The Morley and Morley adjunct triangles of a given triangle are perspective, with the Morley center as their center of perspectivity.*

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## Exercises

**Exercise 4.44.** Given $\triangle ABC$ and points $P,Q$, prove that $P,Q$ are isogonal conjugates if and only if $\measuredangle CPA + \measuredangle CQA = \measuredangle CBA$ and $\measuredangle APB + \measuredangle AQB = \measuredangle ACB$.

**Exercise 4.45.** The image of the incircle of an equilateral triangle under isogonal conjugation is a deltoid whose cusps are the three vertices of the triangle.

**Exercise 4.46*.** Given $\triangle ABC$, prove that:
(1) there is a unique triangle $\triangle XYZ$ with the same circumcircle as $\triangle ABC$ such that the line joining each vertex of $\triangle XYZ$ to its isogonal conjugate is tangent to the circumcircle; this triangle is called the *circumtangential triangle* of $\triangle ABC$;
(2) the circumtangential triangle, the triangle formed by the three cusps of the Steiner deltoid, and the Morley triangle of $\triangle ABC$ are pairwise homothetic.

**Exercise 4.47.** Given $\triangle ABC$ and a point $P$, prove that the pedal triangle of $\mathbf{g}P$ is homothetic to the antipedal triangle of $P$ with respect to its circumcevian triangle.

**Exercise 4.48.** Fix $\triangle ABC$ and $\alpha \in (0, 90^\circ)$. For points $P,Q$, choose $P_1,Q_1$ on the line $BC$, $P_2,Q_2$ on the line $CA$, and $P_3,Q_3$ on the line $AB$ so that

$$\begin{aligned}\measuredangle(BC,PP_1) &= \measuredangle(CA,PP_2) = \measuredangle(AB,PP_3) = \alpha \\ &= -\measuredangle(BC,QQ_1) = -\measuredangle(CA,QQ_2) = -\measuredangle(AB,QQ_3).\end{aligned}$$

Prove that:
(1) $P,Q$ are isogonal conjugates if and only if $P_1,P_2,P_3,Q_1,Q_2,Q_3$ lie on one circle $\omega$;
(2) if $P,Q$ are isogonal conjugates, then the center $R$ of $\omega$ lies on the perpendicular bisector of $PQ$, and $\angle RPQ = 90^\circ - \alpha$.

**Exercise 4.49.** Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$. Let $D$ be the second intersection of the line $AP$ with $\mathscr{C}$, and let $T = DQ \cap BC$. Prove that $PT \parallel AQ$.

**Exercise 4.50.** Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$. Let $D,E$ be the second intersections of the lines $BP,CQ$ with $\mathscr{C}$, respectively, and let $DE$ meet $AB,AC$ at $K,L$. Prove that $PK \parallel AB$ and $QL \parallel AC$.

**Exercise 4.51.** Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$. Let $D,E$ be the second intersections of the lines $AP,AQ$ with $\mathscr{C}$, respectively. For $F \in \mathscr{C}$, put $L = EF \cap BC$. Prove that $\triangle FPD \stackrel{-}{\sim} \triangle QLE$.

**Exercise 4.52.** Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$. Let $D$ be the second intersection of the line $AQ$ with $\mathscr{C}$, put $Q' = \mathfrak{r}_{BC}(Q)$, and let $Q'D$ meet $\mathscr{C}$ at $D,E$. Prove that $AE \perp OP$, where $O$ is the circumcenter of $\triangle ABC$.

### 4.4.2 Isogonal Conjugacy and Conics

We first examine the relation between isogonal conjugacy and conics.

**Theorem 4.4.14.** *For points $P, P'$ not lying on the sidelines of a given triangle, $P' = \mathbf{g}P$ if and only if there is a conic with foci $P, P'$ tangent to all three sidelines. In the limiting case in which $P'$ tends to infinity, the conic may be regarded as a parabola, and $P'$ as its point at infinity.*[23]

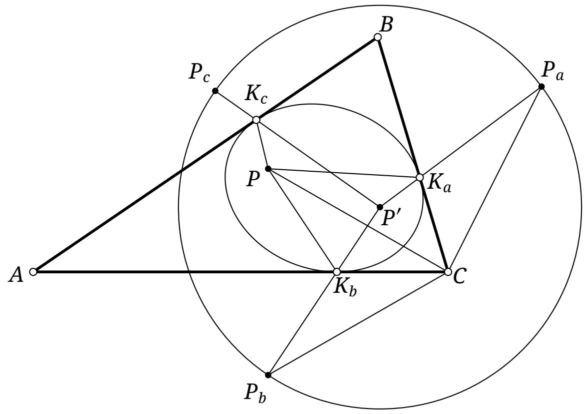


FIGURE 4.40

**Proof.** We prove the forward implication in the configuration of Figure 4.40; the other configurations are analogous. Let $\triangle P_aP_bP_c$ be the reflection triangle of $P$, and put $K_a = P'P_a \cap BC$, with $K_b, K_c$ defined similarly. The figure gives $\angle PK_aB = \angle P'K_aC$, and analogously at the other two sides. Moreover, for each $i \in \{a,b,c\}$, $PK_i + P'K_i = P_iK_i + P'K_i = P'P_i$, and the three final quantities are equal. The optical property of a conic therefore proves the assertion. □

Figure 4.41 shows how the position of $P$ determines the type of conic obtained. Points in the unshaded region give ellipses, points in the shaded region give hyperbolas, and points on the circumcircle give parabolas. Why?

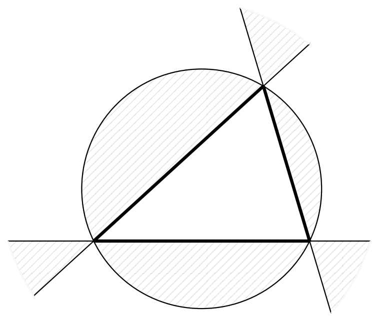

FIGURE 4.41

**Corollary 4.4.15.** *Let $P, Q$ be isogonal conjugates with respect to $\triangle ABC$, and let $\alpha$ be the conic with foci $P, Q$ tangent to its three sidelines. Then the auxiliary circle of $\alpha$ is the common pedal circle of $P, Q$.*

[23] Treating one focus of a parabola as a point at infinity does lead to some correct conclusions, including its optical property, so the following proof also applies to a parabola. This viewpoint is not valid in every context, however, and must be used with care.

**Proof.** This follows directly from theorem 1.3.5, since $P,Q$ are the foci. □

**Corollary 4.4.16.** *Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$, and let $\triangle P_AP_BP_C$ and $\triangle Q_AQ_BQ_C$ be their pedal triangles. Then*

$$\overline{PP_A}\cdot\overline{QQ_A}=\overline{PP_B}\cdot\overline{QQ_B}=\overline{PP_C}\cdot\overline{QQ_C}.$$

**Proof.** Together with proposition 1.3.7, this is an immediate consequence of corollary 4.4.15. □

A notable special case of theorem 4.4.14 occurs when an ellipse tangent to the three sidelines has its center at the centroid. Its further properties are most conveniently described in the complex plane, where the point $(a,b)$ is represented by the complex number $a+\mathrm{i}b$.

**Theorem 4.4.17** (Marden's theorem)**.** *Let the vertices of $\triangle ABC$ be represented by the complex numbers $z_a,z_b,z_c$, and put $P(z)=(z-z_a)(z-z_b)(z-z_c)$. If $p,q$ are the two roots of $P'(z)$, then the corresponding points $P,Q$ are isogonal conjugates. There is an ellipse tangent to the three sidelines with foci $P,Q$, center the centroid $G$ of the triangle, and points of tangency the three side midpoints.* This ellipse is the *Steiner inellipse*, or *inscribed Steiner ellipse*, of $\triangle ABC$.

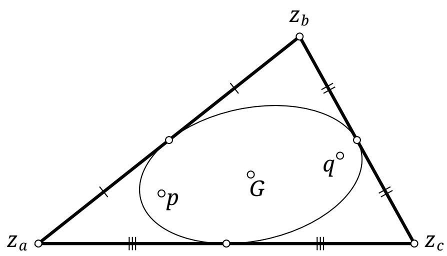


FIGURE 4.42

**Proof.** We show that $\angle BAP=\angle CAQ$. Without loss of generality take $z_a=0$ and let $\overrightarrow{AC}$ point along the positive real axis. Then $P(z)=z^3-(z_b+z_c)z^2+z_cz_bz$ and $P'(z)=3z^2-2(z_b+z_c)z+z_bz_c$. Vieta's formulas give $pq=\frac{z_bz_c}{3}$, so $pq$ and $z_bz_c$ have the same argument. But $\arg(pq)=\arg(p)+\arg(q)=\angle PAC+\angle QAC$, while $\arg(z_bz_c)=\arg(z_b)+\arg(z_c)=\angle BAC+0$. Hence $\angle QAC=\angle BAC-\angle PAC=\angle PAB$, proving that $P$ and $Q$ are isogonally conjugate.

Again by Vieta's formulas, the complex coordinate of the center $G$ is $g=\frac{p+q}{2}=\frac{z_a+z_b+z_c}{3}$, so $G$ is the centroid.

Finally, apply an affine transformation taking an equilateral triangle to the triangle in the statement. Its incircle becomes an inscribed ellipse of the image triangle, its center becomes the centroid, and its three points of tangency become the side midpoints. A conic with a prescribed center tangent to three prescribed lines is unique, as will be proved in corollary 7.4.21. This completes the proof. □

**Remark.** More generally, for every complex polynomial $P(z)$, the centroid of its zeros, counted with multiplicity here and below, coincides with the centroid of the zeros of $P'(z)$. Translate the coordinate system so that the former centroid is 0. The coefficient of the second-highest power in $P(z)$ is then 0, and the corresponding coefficient in $P'(z)$ is also 0. Vieta's formulas now show that the centroid of the zeros of $P'(z)$ is 0 as well.

The preceding discussion of isogonal conjugacy for a triangle extends to polygons.

**Definition 4.4.18.** Let $A_1A_2\cdots A_n$ be an $n$-gon and let $P$ be a point of the plane. For each $i\in\{1,2,\ldots,n\}$, reflect the line $PA_i$ in the bisector of $\angle A_i$. If the resulting $n$ lines are concurrent at $P'$, then $P'$ is the *isogonal conjugate* of $P$ with respect to the polygon. Below we write $P'=\mathbf{g}_{A_1A_2\cdots A_n}P$.

Properties of isogonal conjugates of a triangle also extend to polygons. The following three results generalize corollary 4.4.5, theorem 4.4.14, and exercise 4.48, respectively.

**Theorem 4.4.19.** *Let $\mathscr{A}$ be a polygon and $P$ a point. The isogonal conjugate $\mathbf{g}_{\mathscr{A}}P$ exists if and only if the projections of $P$ on all sidelines of $\mathscr{A}$ lie on one circle $\omega$. In that case the projections of $\mathbf{g}_{\mathscr{A}}P$ on those sidelines also lie on $\omega$, whose center is the midpoint of $P$ and $\mathbf{g}_{\mathscr{A}}P$.*

**Theorem 4.4.20.** *Let $\mathscr{A}$ be a polygon and $P$ a point. The isogonal conjugate $\mathbf{g}_{\mathscr{A}}P$ exists if and only if there is a conic $\alpha$ with one focus at $P$ tangent to every sideline of $\mathscr{A}$. In that case $\mathbf{g}_{\mathscr{A}}P$ is the other focus of $\alpha$.*

**Remark.** It follows that a general pentagon, which we shall prove determines a unique conic tangent to its five sidelines in theorem 4.5.10, has only one pair of isogonal conjugates. Apart from special cases, a polygon with more sides has no isogonal conjugates. For a quadrilateral, the points possessing isogonal conjugates in fact form a circumcubic.

**Theorem 4.4.21.** *Let $\mathscr{A}$ be a polygon with $n$ sides, let $\alpha \in (0,90^\circ)$, and let $P$ be a point. On each sideline choose a point so that the directed angle from the sideline to the line joining that point to $P$ is $\alpha$. Then $Q = \mathbf{g}_{\mathscr{A}}P$ exists if and only if the $n$ chosen points lie on one circle $\omega$. In that case, on each sideline choose similarly a point so that the directed angle from the sideline to its joining line with $Q$ is $-\alpha$. These $n$ points also lie on $\omega$. Its center $R$ lies on the perpendicular bisector of $PQ$, and $\angle RPQ = 90^\circ - \alpha$.*

The proofs of these three generalized theorems are left to the reader.

Finally, a pair of isogonally conjugate points yields a proof of Pascal's theorem in its general form.

**Theorem 4.4.22** (Pascal's theorem)**.** *Let $A,B,C,D,E,F$ be any six points on a conic, and put $X = AB \cap DE$, $Y = BC \cap EF$, and $Z = CD \cap FA$. Then $X,Y,Z$ are collinear.*

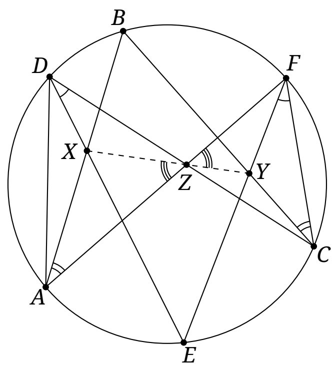


FIGURE 4.43

**Proof.** A projective transformation takes the conic to a circle, so it is enough to consider that case. We prove the configuration in Figure 4.43; the others are analogous.

We have $\triangle ADZ \stackrel{-}{\sim} \triangle CFZ$. Let the similarity that sends $\triangle ADZ$ to $\triangle CFZ$ send $X$ to $X'$. Then $\angle X'FZ = \angle XDZ = \angle YFC$ and, similarly, $\angle X'CZ = \angle YCF$. Hence $Y,X'$ are isogonal conjugates with respect to $\triangle ZFC$. Therefore $\angle AZX = \angle CZX' = \angle FZY$, and $X,Y,Z$ are collinear. $\square$

Unlike the earlier version of Pascal's theorem, which assumed that $ABCDEF$ was a convex hexagon in the ordinary sense, this general form applies to six points in arbitrary positions. Recall also that Pascal's theorem applies to degenerate cases in which some pairs of the six points tend to coincide.

Here is an application.

**Example 4.4.23.** Let $N_B$ be the midpoint of the arc $\widehat{ABC}$ of the circumcircle of $\triangle ABC$, and let $N_C$ be the midpoint of $\widehat{ACB}$. Let $\triangle I_aI_bI_c$ be the excentral triangle, and put $X = AC \cap N_BN_C$. Prove that $I_c, X, M_c$ are collinear, where $M_c$ is the midpoint of $AB$.

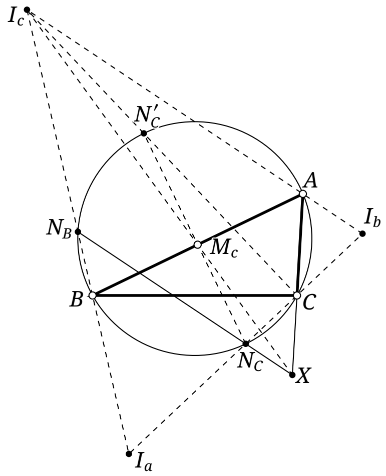


FIGURE 4.44

**Solution.** In Figure 4.44, let $N'_C$ be the antipode of $N_C$ on the circumcircle. Then $N'_CN_C$ passes through $M_c$. By theorem 0.6.3 and proposition 0.6.8, the triples $I_c, N_B, B$ and $I_c, N'_C, C$ are collinear. Apply Pascal's theorem to the hexagon $CABN_BN_CN'_C$. Since $CA \cap N_BN_C = X$, $AB \cap N_CN'_C = M_c$, and $BN_B \cap N'_CC = I_c$, the points $I_c, M_c, X$ are collinear. $\square$

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## Exercises

**Exercise 4.53.** Prove that the Steiner inellipse and the circumscribed Steiner ellipse of a triangle are homothetic, and find the ratio of homothety.

**Exercise 4.54.** Prove that among all ellipses inscribed in a triangle, the Steiner inellipse has greatest area, and find the ratio of the areas of the triangle and this ellipse.

**Exercise 4.55.** If $P, Q$ are isogonal conjugates with respect to a quadrilateral, prove that they are also isogonal conjugates with respect to every triangle formed by three of the four lines bounding the quadrilateral.

**Exercise 4.56.** Let $O, H$ be the circumcenter and orthocenter of $\triangle ABC$, and let $D$ be the projection of $A$ on $BC$. Draw $DE \perp OD$. Prove that $O, H$ are isogonal conjugates with respect to the quadrilateral bounded by $AB, BC, CA, DE$.

**Exercise 4.57.** Prove theorems 4.4.19 to 4.4.21.

**Exercise 4.58.** Prove that a point $P$ has an isogonal conjugate with respect to a quadrilateral $ABCD$ if and only if $\measuredangle APD = \measuredangle CPB$.

**Exercise 4.59.** Let $I$ be the incenter of $\triangle ABC$. The circle $\odot(B,I,C)$ meets the incircle at $P,Q$. Let $E,F$ be the projections of $I$ on $AB,AC$, and put $S = EF \cap AI$. Prove that the ellipse with foci $P,Q$ passing through $S$ is inscribed in $\triangle ABC$.

**Exercise 4.60.** In an acute triangle $\triangle ABC$ with $AB > AC$, let $O,H$ be the circumcenter and orthocenter, and let $P$ lie on the arc $\widehat{BC}$ of $\mathscr{C}$. Draw $HD \parallel AP$ with $D \in BC$, and let $U$ be the midpoint of $HP$. Prove that $\angle ODU = \angle HDC$.

**Exercise 4.61.** Suppose that $ABCD$ and $A'B'C'D'$ are both inscribed in one conic. Put $P = AB \cap A'B'$, $Q = BC \cap B'C'$, $R = CD \cap C'D'$, and $S = DA \cap D'A'$. If $P,Q,R$ are collinear, prove that $S$ lies on their line as well.

**Exercise 4.62.** Let $P{-}ABC$ be a tetrahedron. Its insphere touches the base $ABC$ at $X$, and its $P$-exsphere touches that base at $Y$. Prove that $X,Y$ are isogonal conjugates with respect to $\triangle ABC$.

**Exercise 4.63.** Let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $P$ move on $\mathscr{C}$. The lines $PM_a,PM_b,PM_c$ meet $\mathscr{C}$ again at $A',B',C'$, respectively. Prove that the area of the triangle bounded by $AA',BB',CC'$ is independent of the position of $P$.

*Hint.* Use Pascal's theorem.

**Exercise 4.64.** In the Droz-Farny theorem of exercise 4.23, fix $\triangle ABC$ and let $l_1,l_2$ vary. Find the envelope of the line $\overline{A_0B_0C_0}$.

### 4.4.3 Isotomic Conjugacy

Isotomic conjugacy is analogous to isogonal conjugacy.

**Definition 4.4.24.** Let $P$ be a point distinct from the vertices of $\triangle ABC$, and let $\triangle A_1B_1C_1$ be its cevian triangle. Reflect $A_1$ in the midpoint of $BC$ to obtain $A_2$, and define $B_2,C_2$ similarly. The lines $AA_2,BB_2,CC_2$ are concurrent at a point $P'$, which may be at infinity, as in Figure 4.45. The point $P'$ is the *isotomic conjugate* of $P$ with respect to $\triangle ABC$, and the transformation $P \mapsto P'$ is *isotomic conjugation*. The lines $AA_2,BB_2,CC_2$ are the *isotomic lines* of $AA_1,BB_1,CC_1$ in $\angle A,\angle B,\angle C$ of $\triangle ABC$, respectively.

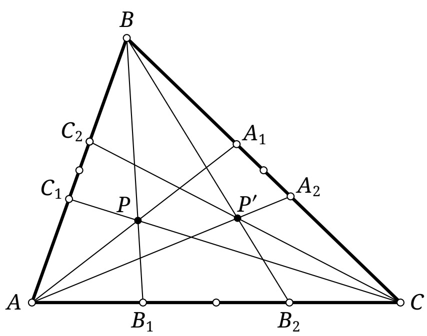



FIGURE 4.45

Apply an affine transformation taking the triangle to an equilateral triangle. Since an affine transformation preserves ratios of collinear segments, it follows that the isotomic conjugacy of points is preserved under affine transformations; in the equilateral image triangle, these pairs are also isogonally conjugate. The existence and uniqueness of the isotomic conjugate therefore follow from those of the isogonal conjugate. They may also be proved directly from Ceva's theorem.

We shall write $\mathbf{t}_{\triangle ABC}P$ for the isotomic conjugate of $P$ with respect to $\triangle ABC$, and simply $\mathbf{t}P$ when there is no ambiguity.

**Theorem 4.4.25.** *Isotomic conjugation with respect to $\triangle ABC$ has the following properties.*

*(1) If $P$ does not lie on a sideline, then $\mathbf{t}(\mathbf{t}P) = P$; equivalently, $\mathbf{t}^2 = \mathrm{id}$.*[24]

*(2) The isotomic conjugate of a point on a sideline is the opposite vertex.*

*(3) Exactly four points are fixed: the centroid, and the reflections of the three vertices in the midpoints of their opposite sides; equivalently, the centroid and the three vertices of its anticevian triangle.*

*(4) Affine transformations preserve pairs of isotomic conjugates, with respect to the image of $\triangle ABC$.*

*(5) The locus of points whose isotomic conjugates are at infinity is a circumscribed ellipse of $\triangle ABC$.*[25] *Every affine transformation taking $\triangle ABC$ to an equilateral triangle takes this ellipse to the circumcircle of the image triangle. It is homothetic to the Steiner inellipse about the centroid in the ratio* $2:1$. It is called the *circumscribed Steiner ellipse* of $\triangle ABC$.

**Proof.** The assertions are straightforward. For (5), map the triangle affinely to an equilateral one. Isotomic conjugacy is preserved, and for an equilateral triangle it coincides with isogonal conjugacy. The points whose isogonal conjugates are at infinity lie on the circumcircle, proving the assertion. □

**Theorem 4.4.26.** *Given $\triangle ABC$, suppose the directed distances of $P$ from $BC, CA, AB$ are in the ratio $\alpha:\beta:\gamma$. Then the corresponding distances of $\mathbf{t}P$ are in the ratio $(a^2\alpha)^{-1}:(b^2\beta)^{-1}:(c^2\gamma)^{-1}$, where $a = BC$, $b = CA$, and $c = AB$.*

**Proof.** We have

$$\begin{aligned}
\mathrm{dist}(\mathbf{t}P, AB):\mathrm{dist}(\mathbf{t}P, AC) &= \frac{[\triangle AB\mathbf{t}P]}{c} : \frac{[\triangle AC\mathbf{t}P]}{b} \\
&= \frac{[\triangle ACP]}{c} : \frac{[\triangle ABP]}{b} = (c[\triangle ABP])^{-1} : (b[\triangle ACP])^{-1} \\
&= [c\cdot c\,\mathrm{dist}(P, AB)]^{-1} : [b\cdot b\,\mathrm{dist}(P, AC)]^{-1} = (c^2\gamma)^{-1} : (b^2\beta)^{-1}.
\end{aligned}$$

The other ratios follow similarly. □

In general, isotomic conjugates have fewer of the elegant properties enjoyed by isogonal conjugates.

## Exercises

**Exercise 4.65.** Prove that among all ellipses circumscribed about a triangle, its circumscribed Steiner ellipse has least area, and find the ratio of the areas of that ellipse and the triangle.

[24] Thus $P$ and $\mathbf{t}P$ play symmetric roles. They may be said to be isotomically conjugate, or to form a pair of isotomic conjugates.

[25] Strictly speaking, the three vertices are to be excluded.

**Exercise 4.66.** Let $G$ be the centroid of $\triangle ABC$. Points $P_1, P_2$ are isotomic conjugates, and the centroids of their cevian triangles are $G_1, G_2$. Prove that $G$ is the midpoint of $G_1G_2$.

**Exercise 4.67.** For $\triangle ABC$, let $H, O, G$ be its orthocenter, circumcenter, and centroid, respectively, and put $H' = \mathbf{t}H$. Prove that:

(1) $H'$ is the center of perspectivity of $\triangle ABC$ and the pedal triangle of $\mathfrak{f}_O(H)$;
(2) if $H_a$ is the projection of $A$ on $BC$, $A'$ is the midpoint of $H_aA$, and $B', C'$ are defined similarly, then $\triangle A'B'C'$ and the cevian triangle of $H'$ have center of perspectivity $G$;
(3) the reflection triangle of $G$ with respect to the medial triangle of $\triangle ABC$ and $\triangle ABC$ have center of perspectivity $H'$.

**Exercise 4.68.** Let $I$ be the incenter of $\triangle ABC$, put $I' = \mathbf{t}I$, and let $I^* = \mathbf{a}I'$. Prove that:

(1) the distances from $I'$ to the three sidelines are inversely proportional to the squares of the corresponding side lengths;
(2) the line through $I^*$ parallel to $BC$ is cut by $AB, AC$ in a segment $a$. Similarly, the parallels through $I^*$ to $CA, AB$ give segments $b, c$. The lengths of $a, b, c$ are equal. Thus $I'$ is called the *congruent-parallelians point*.

**Exercise 4.69** (Grinberg's theorem** (Гринберг)**).** For a triangle, prove the identity of transformations

$$\text{cyclocevian conjugation} = \mathbf{t} \circ \mathbf{a} \circ \mathbf{g} \circ \mathbf{c} \circ \mathbf{t}.$$

## 4.5 Power of a Point and Radical Axes

This section introduces two important tools of plane geometry: power of a point and the radical axis.

**Definition 4.5.1.** Let a circle with center $O$ have radius $r$. For a point $P$ of the plane, the number $OP^2 - r^2$ is the *power* of $P$ with respect to the circle, denoted by $\operatorname{pow}_{\odot O}(P)$.

**Proposition 4.5.2.** *The locus of points $P$ having a fixed power $k$ with respect to a circle $\odot O$ is a circle concentric with $\odot O$.*

**Proof.** This is immediate. □

**Theorem 4.5.3** (Power of a point theorem)**.** *Let $P$ be fixed and let an arbitrary line through $P$ meet a circle $\odot O$ at $X, Y$; in the tangent case, $X = Y$ is the point of contact. Then*

$$\overline{XP} \cdot \overline{YP} = \operatorname{pow}_{\odot O}(P).$$

**Proof.** Let the line $OP$ meet $\odot O$ at $A, B$. Then

$$\overline{XP} \cdot \overline{YP} = \overline{AP} \cdot \overline{BP} = (OP + r_{\odot O})(OP - r_{\odot O}) = \operatorname{pow}_{\odot O}(P).$$

□

**Proposition 4.5.4.** *If $\odot O_1$ and $\odot O_2$ are orthogonal, then* $\operatorname{pow}_{\odot O_1}(O_2) = r^2_{\odot O_2}$.

**Proof.** Orthogonality gives $r^2_{\odot O_1} + r^2_{\odot O_2} = O_1O_2^2$. Therefore LHS $= O_1O_2^2 - r^2_{\odot O_1} = r^2_{\odot O_2}$. □

**Theorem 4.5.5.** *For two nonconcentric circles, the locus of points having equal powers with respect to the two circles is a line.* This line is called their *radical axis*.

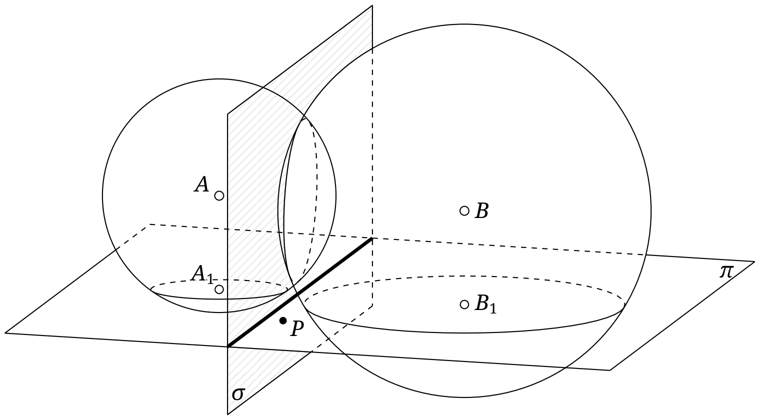



FIGURE 4.46

**Proof.** First suppose the circles meet at $X, Y$. These two points plainly have equal powers. For an arbitrary point $P$, let $XP$ meet the circles again at $Y_1, Y_2$. By the power-of-a-point theorem, the two powers are $\overline{PX}\cdot\overline{PY_1}$ and $\overline{PX}\cdot\overline{PY_2}$. They are equal if and only if $PY_1 = PY_2$, equivalently $Y_1 = Y_2 = Y$. Thus the locus is the line $XY$.

If the two circles are tangent at $X$, every point $P$ on their common tangent $l$ at $X$ has power $PX^2$ with respect to both. If $P' \notin l$, let $XP'$ meet the circles again at $Y_1, Y_2$. Its powers are $\overline{P'Y_1}\cdot\overline{P'X}$ and $\overline{P'Y_2}\cdot\overline{P'X}$, which are unequal because $Y_1 \neq Y_2$. Hence the locus is $l$.

In the complex Euclidean plane, two disjoint circles may be regarded as meeting at two imaginary points, and the result follows as in the first case. We also give an argument not using the complex Euclidean plane.

Let the disjoint circles be $\odot A_1, \odot B_1$. Lift the configuration to three-space, regarding them as the sections by a plane $\pi$ of two intersecting spheres with centers $A, B$, chosen so that $AB \parallel \pi$, as in Figure 4.46. As in the planar case, define the power of a point with respect to a sphere as the square of its distance from the center minus the square of the sphere's radius. The points of equal power with respect to the two spheres form the plane $\sigma$ containing their circle of intersection. For every $P \in \pi$, $\mathrm{pow}_{\odot A_1}(P) = \mathrm{pow}_{\text{sphere } A}(P)$ and $\mathrm{pow}_{\odot B_1}(P) = \mathrm{pow}_{\text{sphere } B}(P)$. Since $AA_1 = BB_1$, equality of the powers with respect to the spheres is equivalent to equality with respect to the circles. The required locus is therefore the line $\sigma \cap \pi$. $\square$

**Remark.** The disjoint case also has a planar proof. For $\odot A, \odot B$, there is a point $P_0$ on the line $AB$ having equal powers. For any $P$, the power-of-a-point theorem gives

$$\mathrm{pow}_{\odot A}(P) = \mathrm{pow}_{\odot B}(P) \iff AP_0^2 - BP_0^2 = AP^2 - BP^2.$$

By theorem 0.2.10, the locus is the line through $P_0$ perpendicular to $AB$.

This planar method, however, is difficult to extend directly to some later theorems involving powers of circles, whereas the three-dimensional method extends readily.

The radical axis has the following immediate property.

**Proposition 4.5.6.** *The radical axis of two intersecting circles is the line containing their common chord. The radical axis of two tangent circles is their common tangent at the point of contact.*

**Remark.** In the complex Euclidean plane, two disjoint circles meet at two imaginary points, whose joining line is their radical axis. Tangent circles may be viewed as the limiting case in which two intersections coalesce.

**Proof.** This follows from the proof of theorem 4.5.5. □

**Theorem 4.5.7** (Radical-center theorem)**.** *Let three circles have noncollinear centers, and let each pair have a radical axis. The three radical axes are concurrent.* Their common point is the *radical center* of the circles.[26]

**Proof.** For circles $\odot O_1, \odot O_2, \odot O_3$, let $l_{ij}$ be the radical axis of $\odot O_i$ and $\odot O_j$. The point $l_{12} \cap l_{23}$ has the same power with respect to all three circles, and therefore lies on $l_{13}$. □

**Remark.** When the three circles intersect pairwise, the theorem says that their three common chords are concurrent. A spatial proof is immediate. Take three pairwise intersecting spheres with centers $A, B, C$ and a common point $P$. Viewed orthogonally from above the plane determined by $A, B, C$, the spheres appear as circles $\odot A_1, \odot B_1, \odot C_1$, their pairwise circles of intersection appear as the three common chords, and $P$ appears as their common point.

We give two applications of the radical axis.

**Theorem 4.5.8** (Davis's theorem)**.** *On the lines $BC, CA, AB$ of $\triangle ABC$, choose respectively $A_1, A_2$; $B_1, B_2$; and $C_1, C_2$. If each of the three quadruples $B_1, B_2, C_1, C_2$, $C_1, C_2, A_1, A_2$, and $A_1, A_2, B_1, B_2$ is concyclic, then all six points lie on one circle.*

**Proof.** Let the circles through $B_1, B_2, C_1, C_2$, through $C_1, C_2, A_1, A_2$, and through $A_1, A_2, B_1, B_2$ be $\omega_1, \omega_2, \omega_3$, respectively. Suppose, to the contrary, that the six points are not concyclic. Then $\omega_1, \omega_2, \omega_3$ are distinct. The radical axis of $\omega_2, \omega_3$ is $A_1A_2$; that of $\omega_3, \omega_1$ is $B_1B_2$; and that of $\omega_1, \omega_2$ is $C_1C_2$. The three lines $A_1A_2, B_1B_2, C_1C_2$ are plainly not concurrent, contradicting the radical-center theorem. □

The second application is Brianchon's theorem in its general form.

**Theorem 4.5.9** (Brianchon's theorem)**.** *Let six lines $l_i$ $(i = 1, \ldots, 6)$ be tangent to one conic, and put $A_{ij} = l_i \cap l_j$. Then $A_{12}A_{45}$, $A_{23}A_{56}$, and $A_{34}A_{61}$ are concurrent.*

**Proof.** A projective transformation takes the conic to a circle, so it suffices to consider that case. We treat the configuration of Figure 4.47; the others are analogous. Suppose $ABCDEF$ is circumscribed about $\omega$. We must prove that $AD, BE, CF$ are concurrent.

Construct a circle $\omega_1$ tangent to the lines $AB$ and $DE$, and construct $\omega_2, \omega_3$ similarly, choosing them so that the length of a direct common tangent from $\omega$ to each $\omega_i$ is the same number $a$. The tangent lengths from $B$ to $\omega_1, \omega_2$ are then equal, so $B$ has equal powers with respect to those circles. The same is true of $E$, and hence $BE$ is their radical axis.

Similarly, $AD$ and $CF$ are the radical axes of $\omega_1, \omega_3$ and $\omega_2, \omega_3$, respectively. The radical-center theorem proves their concurrence. □

As with the weaker version introduced earlier, Brianchon's theorem also applies when some pairs of tangent lines tend to coincide. Here is one application.

[26] This result is sometimes also called Monge's theorem. If the three centers are collinear, the three radical axes are parallel. Thus in the projective plane the restriction on the centers may be omitted.

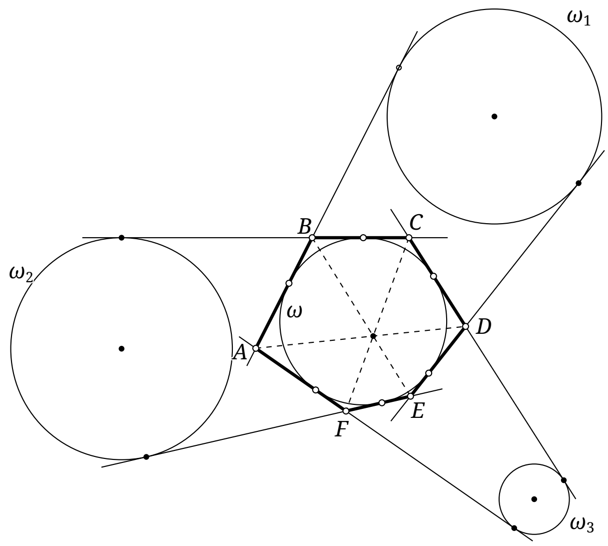



FIGURE 4.47

**Theorem 4.5.10.** *Given five lines, no three concurrent, there is a unique nondegenerate conic tangent to all five.*

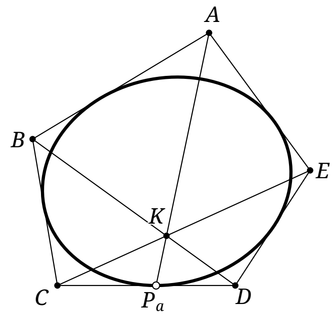



FIGURE 4.48

**Proof.** The limiting form of Brianchon's theorem constructs the point of tangency on each line. Let the five tangents form the pentagon $ABCDE$, as in Figure 4.48, and put $K = BD \cap CE$ and $P_a = AK \cap CD$. Apply Brianchon's theorem to the six lines $AB, BC, CP_a, P_aD, DE, EA$. If a conic $\alpha$ satisfying the conditions exists, its point of tangency with $CD$ must be $P_a$.

Construct similarly the points of tangency $P_b, P_c, P_d, P_e$ on $DE, EA, AB, BC$. Any admissible $\alpha$ must pass through these five points, and hence $\alpha = \mathfrak{C}(P_aP_bP_cP_dP_e)$, the unique conic satisfying the conditions. □

In analogy with the notation used earlier, below we write $\mathfrak{C}(l_1l_2l_3l_4l_5)$ for the conic tangent to the five lines $l_1, l_2, l_3, l_4, l_5$.

---

## Exercises

**Exercise 4.70.** Given two circles, construct their radical axis with straightedge and compass. How many methods can you find?

**Exercise 4.71.** Let $\omega_1, \omega_2$ be circles, neither passing through the center of the other, and let their radical axis be $l$. Put $\omega = \mathfrak{i}_{\omega_1}(\omega_2)$. Prove that the radical axes of $\omega, \omega_1$ and of $\omega, \omega_2$ are both $l$.

**Exercise 4.72.** The circle $\odot O_2$ is internally tangent to $\odot O_1$ at $T$ and is also tangent to the chord $AB$ of $\odot O_1$ at $P$. Let $TP$ meet $\odot O_1$ at $T, M$. Prove that $\operatorname{pow}_{\odot O_2}(M) = MA^2 = MB^2$.

**Exercise 4.73.** Let $C \in \odot(AB)$, and let $D$ be the projection of $C$ on $AB$. A circle $\odot P$ is tangent to the segments $CD, DB$ and internally tangent to the arc $\widehat{BC}$. Prove that $AC = AG$, where $G$ is the point at which $\odot P$ touches $DB$.

**Exercise 4.74.** In $\triangle ABC$, let $D, E$ lie on the segments $AC, AB$, respectively. The circles $\odot(BD)$ and $\odot(CE)$ meet at $P, Q$. If $H$ is the orthocenter of $\triangle ABC$, prove that $P, Q, H$ are collinear.

**Exercise 4.75.** Circles $\odot O_1, \odot O_2$ of unequal radii meet at $P, Q$ and are internally tangent to $\odot O$ at $A, B$, respectively. Prove that $OP \perp PQ$ if and only if $A, B, Q$ are collinear.

**Exercise 4.76.** Let $AD$ be the altitude to the hypotenuse $BC$ of the right triangle $\triangle ABC$, and let $T$ be the midpoint of $AD$. Through $T$ draw a parallel to $BC$, meeting $AB, AC$ at $A_1, A_2$; a parallel to $CA$, meeting $BC, BA$ at $B_1, B_2$; and a parallel to $AB$, meeting $CA, CB$ at $C_1, C_2$. Prove that the six points $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic.

**Exercise 4.77.** Let $ABCD$ be inscribed in $\odot O$, and put $J = AC \cap BD$, $E = AB \cap CD$, and $F = AD \cap BC$. Prove that:

(1) $\operatorname{pow}_{\odot O}(E) + \operatorname{pow}_{\odot O}(J) = EJ^2$;
*Hint.* If $\odot(JCD) \cap EJ = \{J, L\}$, then $E, B, L, D$ are concyclic.

(2) $OJ \perp EF$.

**Exercise 4.78.** Given five tangents to a conic, construct their five points of tangency using only an unmarked straightedge.

**Exercise 4.79** (Linearity of power of a point)**.** For two fixed circles $\odot O_1, \odot O_2$ in the plane, define $f : P \mapsto (\operatorname{pow}_{\odot O_1}(P) - \operatorname{pow}_{\odot O_2}(P))$. Prove that $f$ is a linear function[27]; that is,

$$f(P) = 2\overrightarrow{O_1O_2} \cdot \overrightarrow{OP} + c,$$

where $O$ is fixed and $c$ is constant.

Also prove the following two common forms. If $A, B, C$ are collinear, then

$$\overline{AB} \cdot f(C) + \overline{BC} \cdot f(A) + \overline{CA} \cdot f(B) = 0.$$

In vector form, for a fixed point $O$ and three points $A, B, C$, if $\overrightarrow{OC} = k\overrightarrow{OA} + (1-k)\overrightarrow{OB}$, then $f(C) = kf(A) + (1-k)f(B)$.

## 4.6 Pencils of Circles

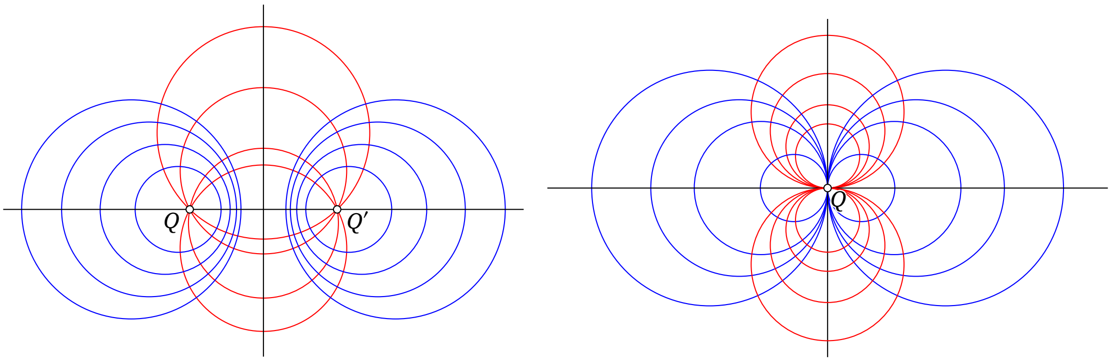
We begin with the following terminology.

[27] Here "linear function" does not mean a so-called "linear map" of the form $\boldsymbol{v} \mapsto c\boldsymbol{v}$; rather, it denotes a property analogous to that of a function of degree one, and the precise formula is given explicitly here.

**Definition 4.6.1.**

a. If the centers of three circles are collinear, their three pairwise radical axes are either parallel or coincident. When all three coincide, the circles are said to be *coaxial*.
b. Given two circles, the set of all circles coaxial with them, including the two original circles, is a *pencil of circles*. Its members have a common radical axis, called the *radical axis* or *axis of equal power* of the pencil.
c. If the two generating circles meet at two distinct points, all circles of the pencil pass through those points, and the pencil is *hyperbolic*. If the generating circles are tangent, all circles of the pencil are tangent at their common point, and the pencil is *parabolic*. If the generating circles are disjoint, all circles in the pencil are pairwise disjoint, and the pencil is *elliptic*.
d. An elliptic pencil contains two degenerate point circles.[28] Their points are the *limit points* of the pencil. A parabolic pencil contains one point circle, namely its common point of tangency, which is likewise called its limit point.

The red circles in Figure 4.49a form a hyperbolic pencil. The blue circles in figure 4.49a form an elliptic pencil, with limit points $Q,Q'$. In Figure 4.49b, the red and blue circles form two parabolic pencils, both having $Q$ as their limit point.




(a) (b)

FIGURE 4.49

**Remark.** The preceding definitions were made in the real Euclidean plane. In the complex Euclidean plane, two disjoint circles meet at two imaginary points. Accordingly, two distinct points of the complex Euclidean plane generate a pencil, elliptic or hyperbolic: the set of all circles through them. Its radical axis is their joining line.

In the limiting case when the two points coalesce, “two coincident points generate a parabolic pencil.” Its circles are mutually tangent at that double point. To determine the pencil completely, one must additionally specify a line through the double point; this is their common tangent and also their radical axis. Thus the pencil consists of the circles through the double point tangent to the prescribed line.[29]

If the radii of a sequence of circles in a pencil tend to infinity while their centers tend to infinity, the limiting generalized circle is the radical axis. It is therefore sometimes useful to regard the radical axis itself as a member of the pencil.

Ordinary concentric circles have no radical axis in the Euclidean plane. In the projective plane, however, their radical axis may be regarded as the line at infinity, as is readily verified by taking limits of nonconcentric circles. In this limiting sense a family of concentric circles is also a pencil, but it has neither an axis of centers nor the conjugate pencil defined below.

[28]The word “circle” in a pencil may include a point circle, treated as a circle of radius zero. Thus the power of $P$ with respect to the point circle $O$ is $\text{pow}_O(P) = OP^2$.

[29]For convenience we usually use the slightly imprecise phrase “two coincident points generate a parabolic pencil.” We may then say uniformly that two possibly coincident points in the complex Euclidean plane determine a pencil.

We now establish some basic properties.

**Proposition 4.6.2.** *The centers of all circles in a pencil $\mathscr{O}$ form a line perpendicular to its radical axis.* This line is the *axis of centers* of $\mathscr{O}$.

**Proof.** This is immediate. □

**Proposition 4.6.3.** *Let $P$ lie on the radical axis of a pencil $\mathscr{O}$, and suppose $P$ lies outside the circles concerned.*[30] *Then:*

*(1) the tangent lengths from $P$ to all circles in $\mathscr{O}$ are equal;*
*(2) the circle $\odot P$ centered at $P$ with radius equal to this tangent length is orthogonal to every circle in $\mathscr{O}$;*
*(3) as $P$ moves on the radical axis of $\mathscr{O}$, the circles $\odot P$ themselves form a pencil $\mathscr{O}'$.*

The pencil $\mathscr{O}'$ is the *conjugate pencil* or *associated pencil* of $\mathscr{O}$.

**Proof.** For (1), $P$ has the same power with respect to every circle in the pencil, and the power-of-a-point theorem identifies that power with the square of the tangent length. Part (2) follows from (1) and proposition 4.5.4.

For (3), first suppose $\mathscr{O}$ is elliptic, with limit points $Q,Q'$. Regard them as point circles. The tangent length from $P$ to any circle of $\mathscr{O}$ equals $PQ=PQ'$. Thus $\odot P$ passes through $Q,Q'$, and all such circles form a hyperbolic pencil.

If $\mathscr{O}$ is parabolic, its circles have a common point of tangency $Q$. Each $\odot P$ passes through $Q$ and is tangent to the axis of centers of $\mathscr{O}$, so these circles form a parabolic pencil.

Finally, suppose $\mathscr{O}$ is hyperbolic, with common points $Q,Q'$, and let $M$ be the midpoint of $QQ'$. The squared radius $r^2_{\odot P}$ is the square of the tangent length from $P$ to a circle of $\mathscr{O}$, and hence

$$r^2_{\odot P} = PQ\cdot PQ' = (PM+MQ)(PM-MQ) = PM^2 - MQ^2.$$

Consequently $\operatorname{pow}_{\odot P}(M) = MP^2 - (PM^2 - MQ^2) = MQ^2$, independent of $P$. Since $P$ moves on the line $QQ'$, take the line $l$ through $M$ perpendicular to $QQ'$, namely the axis of centers of $\mathscr{O}$. Then $l$ is the common radical axis of the circles $\odot P$. □

The following partial converse is left to the reader.

**Proposition 4.6.4.** *Let $\alpha,\beta$ be two circles of a pencil $\mathscr{O}$. If a circle $\omega$ is orthogonal to both, then it is orthogonal to every circle of $\mathscr{O}$ and belongs to the conjugate pencil of $\mathscr{O}$.*

The conjugate pencil has the following properties.

**Proposition 4.6.5.** *The conjugate of an elliptic pencil is the hyperbolic pencil through its two limit points. The conjugate of a hyperbolic pencil is the elliptic pencil having its two common points as limit points. The conjugate of a parabolic pencil is the parabolic pencil having the same limit point and tangent to the original axis of centers.*

**Proposition 4.6.6.** *The radical axis of the conjugate pencil is the axis of centers of the original pencil, and the axis of centers of the conjugate pencil is the radical axis of the original pencil.*

**Proposition 4.6.7.** *The conjugate of the conjugate pencil is the original pencil.*

There is an interesting physical interpretation. Consider two parallel, uniform, infinitely long straight wires carrying equal and opposite linear charge densities. In a plane perpendicular to the wires, the electrostatic equipotentials form an elliptic pencil whose limit points are the intersections of the wires with that plane; the electric field lines form its conjugate pencil.

[30] In the complex Euclidean plane, tangents may also be drawn from a point inside a circle; see the remark following Casey's theorem.

Figure 4.49a shows a conjugate elliptic and hyperbolic pair; Figure 4.49b shows a conjugate pair of parabolic pencils.

We next consider inversion of pencils of circles.

**Proposition 4.6.8.** *Inversion in a circle sends a pencil of circles to another pencil, including a concentric pencil, or to a pencil of lines. In particular, if the circle of inversion belongs to the original pencil, the image pencil coincides with the original pencil.*

**Proof.** The circles of a hyperbolic pencil have two common points. If the center of inversion is neither, their images form the pencil through the images of those common points. If the center of inversion is one of them, their images are the lines through the image of the other common point. If the circle of inversion is a member of the pencil, both common points are fixed, and the image circles still pass through them.

The parabolic case is analogous. The elliptic case follows either by passing to the complex Euclidean plane or by a lifting argument like that used in theorem 4.5.5.[31] □

For an elliptic pencil we have the more precise result below.

**Proposition 4.6.9.** *Let $\mathscr{O}$ be an elliptic pencil.*

*(1) Inversion in any circle of $\mathscr{O}$ interchanges its two limit points.*
*(2) An inversion whose circle is centered at one limit point sends $\mathscr{O}$ to a family of concentric circles centered at the image of the other limit point.*

**Proof.** For (1), regard the limit points as point circles. By proposition 4.6.8, the pencil is invariant under inversion, so the image of either point circle is the other point circle in the pencil.

For (2), inversion centered at one limit point sends the conjugate pencil to a pencil of lines whose vertex is the image of the other limit point; all those lines pass through that image. Since inversion is conformal, every image circle of the original pencil is orthogonal to every line of this pencil. These can only be the concentric circles centered at the image of the other limit point. □

**Corollary 4.6.10.** *For two disjoint nonconcentric circles, there is an inversion that sends them to concentric circles.*

**Proof.** Choose the center of inversion at a limit point of the elliptic pencil generated by the two circles. □

The following theorem may be viewed as a generalization of the radical axis.

**Theorem 4.6.11.** *For two circles $\omega_1, \omega_2$, the locus of points at which the ratio of the two powers is a fixed number different from* 1 *is a circle in the pencil generated by $\omega_1, \omega_2$, as in Figure 4.50. Conversely, on any circle of that pencil the ratio of the powers with respect to $\omega_1, \omega_2$ is a fixed number different from* 1.

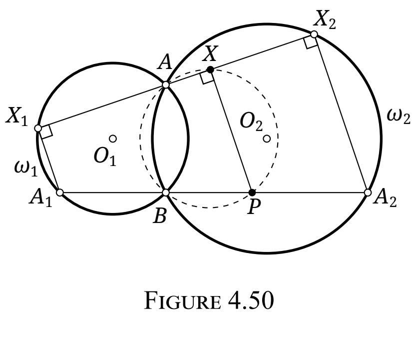



FIGURE 4.50

[31]Inversion may be defined similarly in three-dimensional space; see exercise 4.6.

**Proof.** We prove only the first assertion, and first assume that the two circles intersect. The disjoint case follows by passing to the complex Euclidean plane or by the lifting argument used earlier; the tangent case is the limit in which the two intersections coalesce.

Let $O_i$ be the center of $\omega_i$, let the circles meet at $A,B$, and let $A_i$ be the antipode of $A$ on $\omega_i$. Since $A_1B\perp AB$ and $A_2B\perp AB$, the points $A_1,A_2,B$ are collinear. Suppose $X$ has fixed power ratio $k$, and let $XA$ meet $\omega_i$ again at $X_i$. By the power-of-a-point theorem, $k=\frac{\overline{XX_1}\cdot\overline{XA}}{\overline{XX_2}\cdot\overline{XA}}=\frac{\overline{XX_1}}{\overline{XX_2}}$. Since $AA_1,AA_2$ are diameters, $X_i$ is the projection of $A_i$ on $AX$. Choose $P$ on the line $A_1A_2$ so that $\frac{\overline{PA_1}}{\overline{PA_2}}=k$. Proportionality of intercepts gives $PX\perp XA$, and hence $X\in\odot(AP)$. Also $AB\perp BP$, so $B\in\odot(AP)$. Thus $\odot(AP)$ passes through $A,B$ and belongs to the pencil generated by $\omega_1,\omega_2$. $\square$

The final result concerns polarity.

**Proposition 4.6.12.** *Let $P$ lie outside a pencil of circles $\mathscr{O}$. The polars of $P$ with respect to the distinct circles of $\mathscr{O}$ are concurrent.*

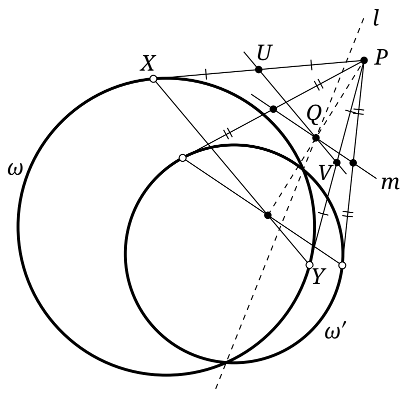



FIGURE 4.51

**Proof.** Let $l$ be the radical axis of $\mathscr{O}$, as in Figure 4.51. Draw tangents from $P$ to a circle $\omega\in\mathscr{O}$, touching it at $X,Y$. Then $XY$ is the polar of $P$. Let $U,V$ be the midpoints of $PX,PY$. Regarding $P$ as a point circle, the power-of-a-point theorem gives $\operatorname{pow}_\omega(U)=UX^2=UP^2=\operatorname{pow}_P(U)$, and similarly $\operatorname{pow}_\omega(V)=\operatorname{pow}_P(V)$. Thus $UV$ is the radical axis of $\omega$ and the point circle $P$.

It follows that the polar of $P$ with respect to $\omega$ is the image under the homothety $\mathfrak{h}_{P,2}$ of the radical axis of $P$ and $\omega$. Let $UV$ be the radical axis of the point circle $P$ and $\omega$, and let $UV$ meet $l$ at $Q$. For every other $\omega'\in\mathscr{O}$, let $m$ be the radical axis of the point circle $P$ and $\omega'$. By the radical-center theorem, $m$ passes through $Q$. Hence all the polars pass through $\mathfrak{h}_{P,2}(Q)$. $\square$

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## Exercises

**Exercise 4.80.** For a parabolic pencil, prove that:

(1) inversion centered at its limit point sends the pencil to a family of parallel lines;
(2) all pairwise external centers of similitude of its circles coincide.

**Exercise 4.81.**

(1) Prove propositions 4.6.4 to 4.6.7.
(2) Prove the disjoint-circle case of theorem 4.6.11 by a three-dimensional argument.

**Exercise 4.82.** Two nested circles have radii $R_1, R_2$ and distance $d > 0$ between their centers. Inversion in a circle of radius $r$ sends them to concentric circles. Find the radii of the image circles.

**Exercise 4.83.** Let circles $\alpha, \beta, \gamma$ belong to one pencil. A chord $AB$ of $\alpha$ is tangent to $\beta$. The tangent lengths from $A, B$ to $\gamma$ are $a, b$, respectively. Prove that $\frac{a+b}{AB} = \text{const}$.

**Exercise 4.84.** Let $\mathscr{O}$ be a pencil. For points $X, Y$, suppose that the polars of $X$ with respect to the circles of $\mathscr{O}$ meet at $X'$, and those of $Y$ meet at $Y'$. Put $Z = XY \cap X'Y'$ and $Z' = XY' \cap X'Y$. Prove that the polars of $Z$ with respect to the circles of $\mathscr{O}$ meet at $Z'$.

**Exercise 4.85.** Let $ABCD$ be inscribed in a circle $\omega_1$. Suppose the segments $AC, BD$ are tangent to a circle $\omega_2$ at $X, Y$, while the lines $AB, CD$ are tangent to another circle $\omega$ at $U, V$. Prove that $\omega$ can be chosen so that $X, Y, U, V$ are collinear and $\omega_1, \omega_2, \omega$ belong to one pencil.

## 4.7 Closure Theorems

We now apply the theory of pencils of circles to several beautiful results known as closure theorems. The meaning of “closure” will become clear from the four theorems below.

For two circles $\omega_1, \omega_2$, let $\mathscr{M}_0(\omega_1, \omega_2)$ be the family of circles tangent to both in the same sense—externally to both or internally to both—and let $\mathscr{M}_1(\omega_1, \omega_2)$ be the family of circles tangent internally to one and externally to the other.

### Steiner’s Porism

**Theorem 4.7.1** (Steiner’s porism)**.** *Let $\omega_1, \omega_2$ be disjoint circles and let $k \in \{0, 1\}$. Starting with some $\alpha_1 \in \mathscr{M}_k(\omega_1, \omega_2)$, choose successively $\alpha_2, \ldots, \alpha_n \in \mathscr{M}_k(\omega_1, \omega_2)$ so that $\alpha_i$ and $\alpha_{i+1}$ are tangent for $i = 1, 2, \ldots, n-1$. If, for one choice of $\alpha_1$, the last circle $\alpha_n$ is also tangent to $\alpha_1$, then infinitely many such chains $(\alpha_1, \ldots, \alpha_n)$ exist, and every member of $\mathscr{M}_k(\omega_1, \omega_2)$ may be chosen as $\alpha_1$.*

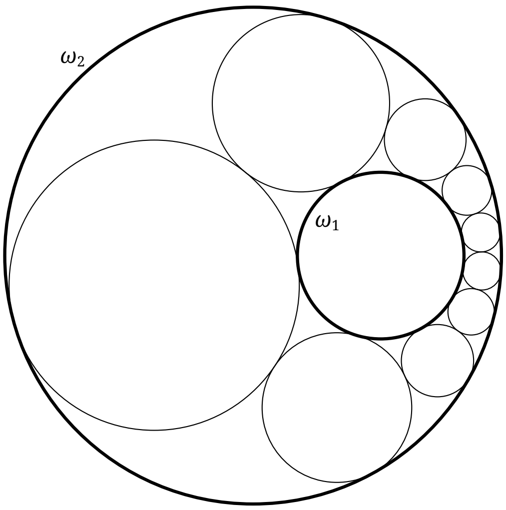


FIGURE 4.52

**Proof.** By corollary 4.6.10, an inversion takes $\omega_1, \omega_2$ to concentric circles. In that configuration the assertion is immediate. $\square$

Figure 4.52 illustrates the case $k=1$ and $n=9$. [32] A chain $(\alpha_1,\ldots,\alpha_n)$ satisfying the hypotheses of the theorem is said to satisfy the *Steiner closure condition*, and is called a *Steiner chain*.

**Poncelet's Closure Theorem**

We next use pencils of circles to prove the circular case of Poncelet's celebrated theorem. Its general form will be established later.

**Theorem 4.7.2** (Poncelet's theorem)**.** *Let $\omega_1,\omega_2$ be circles and let $A_0\in\omega_1$. Draw a tangent $A_0A_1$ to $\omega_2$, meeting $\omega_1$ again at $A_1$; draw the other tangent $A_1A_2\neq A_0A_1$ to $\omega_2$, meeting $\omega_1$ again at $A_2$; and continue in this way until the tangent $A_{n-1}A_n\neq A_{n-2}A_{n-1}$ meets $\omega_1$ again at $A_n$. If $A_n=A_0$ for one initial point $A_0$, then infinitely many closed chains $(A_0,\ldots,A_n)$ exist, and every point of $\omega_1$ outside $\omega_2$ may be used as $A_0$.*[33]

This is a special case of the following theorem; the proof given here is adapted from [**6**].

**Theorem 4.7.3** (Poncelet's closure theorem)**.** *Let the circles $\omega_0,\omega_1,\ldots,\omega_n$ belong to one pencil; apart from $\omega_0$, any two or more of them may coincide. Starting from $A_0\in\omega_0$, draw a tangent $A_0A_1$ to $\omega_1$ that meets $\omega_0$ again at $A_1$, then a tangent $A_1A_2$ to $\omega_2$ that meets $\omega_0$ again at $A_2$, and so on, finally drawing a tangent $A_{n-1}A_n$ to $\omega_n$ that meets $\omega_0$ again at $A_n$.*

*If $A_n=A_0$ for one initial point $A_0$, then infinitely many such closed chains $(A_0,\ldots,A_n)$ exist, and every point of $\omega_0$ for which the successive construction is possible may be chosen as $A_0$.*[34]

Figure 4.53 illustrates theorem 4.7.3 for $n=4$. An $n$-gon $A_0A_1\cdots A_{n-1}$ satisfying its hypotheses is said to satisfy the *Poncelet closure condition*.

We first prove the following lemma.

**Lemma 4.7.4.** *Let $A_0,A_1,B_0,B_1$ lie on a circle $\omega_0$, and suppose $A_0A_1,B_0B_1$ are tangent to a circle $\omega_1$ at $X,Y$, respectively, as in figure 4.54. Then there is a circle $\omega'$ in the pencil containing $\omega_0,\omega_1$ that is tangent to both $A_0B_0$ and $A_1B_1$, with both points of tangency on the line $XY$.*

**Proof.** Put $Z=A_0A_1\cap B_0B_1$, $P=XY\cap A_0B_0$, and $Q=XY\cap A_1B_1$. The triangle $XZY$ is isosceles, so $\angle ZYX=\angle ZXY$. Also $\angle B_1A_1A_0=\angle B_1B_0A_0$, and hence $\triangle XQA_1\sim\triangle YPB_0$. Thus $\angle PQA_1=\angle QPB_0$, and there is a circle $\omega'$ tangent to $A_1B_1$ at $Q$ and to $A_0B_0$ at $P$.

The powers of $A_0,A_1,B_0,B_1$ with respect to $\omega_1$ and $\omega'$ have the respective ratios $\frac{A_0X^2}{A_0P^2}$, $\frac{A_1X^2}{A_1Q^2}$, $\frac{B_0Y^2}{B_0P^2}$, and $\frac{B_1Y^2}{B_1Q^2}$. From $\triangle XQA_1\sim\triangle YPB_0$ we obtain $\frac{A_1X^2}{A_1Q^2}=\frac{B_0Y^2}{B_0P^2}$. Similarly, $\triangle B_1QY\sim\triangle A_0PX$, because $\angle A_1B_1B_0=\angle A_1A_0B_0$, and therefore $\frac{A_0X^2}{A_0P^2}=\frac{B_1Y^2}{B_1Q^2}$. Finally, the sine rule gives

$$\frac{A_0X}{A_0P}=\frac{\sin\angle A_0PX}{\sin\angle A_0XP}=\frac{\sin\angle B_0PY}{\sin\angle B_0YP}=\frac{B_0Y}{B_0P}.$$

[32] In the electronic edition the author has embedded an animation to illustrate this closure theorem: as $\alpha_1$ varies, a family of circles satisfying the conditions continues to exist. To view it, use a JavaScript-enabled PDF viewer such as Adobe Acrobat Reader, and click the figure to start playback. The animation merely gives a more vivid view of this beautiful theorem, and inability to play it does not affect the text. The same kind of animation appears later in figures 4.53, 4.56 and 4.59; other viewers display the first frame.

[33] The condition that $A_0$ lie outside $\omega_2$ ensures that the required real tangents exist. Over the complex Euclidean plane this restriction may be omitted.

[34] If $A_{i-1}$ lies inside $\omega_i$, a real tangent $A_{i-1}A_i$ cannot be drawn. The assertion means that, whenever the successive tangent construction produces all the points $A_1,A_2,\ldots,A_n$, its final point $A_n$ must coincide with $A_0$. Over the complex Euclidean plane, tangents can always be drawn, so this restriction may be omitted. There are also two tangent choices at each step. They are to be chosen as one continuous branch: as $A_0$ varies continuously on $\omega_0$, all the points $A_i$ must vary continuously on $\omega_0$.

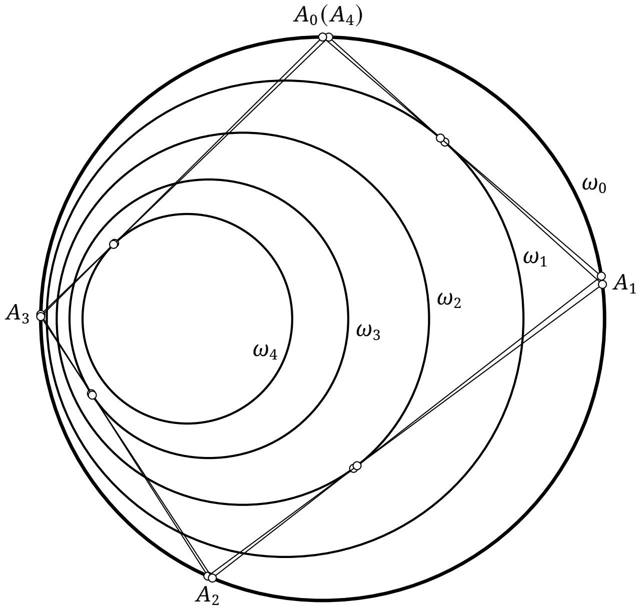


FIGURE 4.53

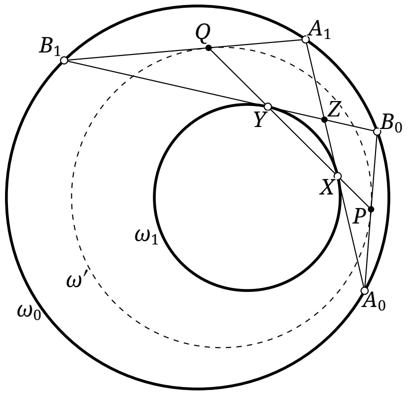


FIGURE 4.54

Consequently,

$$\frac{\operatorname{pow}_{\omega_1}(A_0)}{\operatorname{pow}_{\omega'}(A_0)} = \frac{\operatorname{pow}_{\omega_1}(A_1)}{\operatorname{pow}_{\omega'}(A_1)} = \frac{\operatorname{pow}_{\omega_1}(B_0)}{\operatorname{pow}_{\omega'}(B_0)} = \frac{\operatorname{pow}_{\omega_1}(B_1)}{\operatorname{pow}_{\omega'}(B_1)}.$$

By theorem 4.6.11, the four points therefore lie on a circle in the pencil generated by $\omega_1, \omega'$. That circle is $\omega_0$, so $\omega_0, \omega_1, \omega'$ belong to one pencil. □

**Proof of theorem 4.7.3.** Suppose the $n$-gon $A_0 A_1 \cdots A_{n-1}$ satisfies the Poncelet closure condition. Choose another point $B_0 \in \omega_0$. Draw a tangent $B_0 B_1$ to $\omega_1$, meeting $\omega_0$ again at $B_1$; then draw a tangent $B_1 B_2$ to $\omega_2$, meeting $\omega_0$ again at $B_2$; continue in this way, finally drawing a tangent $B_{n-1} B_n$ to $\omega_n$ that meets $\omega_0$ again at $B_n$.[35] We prove that $B_n = B_0$.

For the fixed pair $A_0, B_0$, all the lines $A_i B_i$ are tangent to one circle $\omega'$, and $\omega_0, \omega_i, \omega'$ belong to one pencil. Indeed, lemma 4.7.4 gives a circle $\omega'$ in the pencil tangent to $A_0 B_0, A_1 B_1$. Applying the lemma to $A_1 B_1, A_2 B_2$ gives another such circle $\omega''$. The required member of the pencil tangent to $A_1 B_1$ in the prescribed configuration is unique,[36] so $\omega'' = \omega'$. Repetition shows that $A_3 B_3, A_4 B_4, \ldots, A_n B_n$ are tangent to the same circle $\omega'$.

[35] At each step choose the tangent with the same orientation as the corresponding side of $A_0 A_1 \cdots A_{n-1}$: as $B_0$ tends continuously to $A_0$, the points $B_1, \ldots, B_n$ are to tend continuously to $A_1, \ldots, A_n$.

[36] A pencil may contain two circles tangent to a given line, but their relative positions select only one of them here.

Since $A_n = A_0$, the two tangents from $A_0 = A_n$ to $\omega'$ coincide when chosen on the same branch. Their second intersections with $\omega_0$ are $B_0$ and $B_n$, respectively, and hence $B_n = B_0$. □

Poncelet's theorem says that a polygon inscribed in one circle and circumscribed about another may be "rotated" about the inner circle. During this motion each diagonal envelops a fixed circle, possibly a degenerate point circle, coaxial with the inscribed and circumscribed circles. Figure 4.55 illustrates this phenomenon.

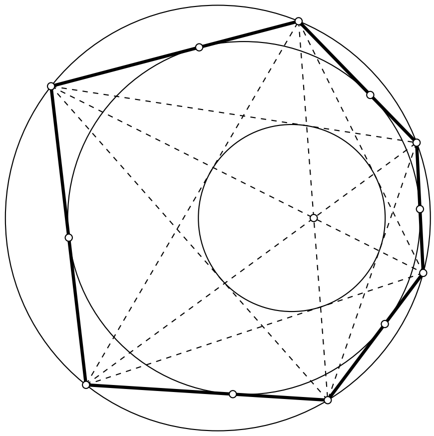

FIGURE 4.55

**Emch's Closure Theorem***

We next introduce Emch's closure theorem.

**Theorem 4.7.5** (Emch's theorem). *Let $\omega_1, \omega_2, \delta$ be three circles in the plane, any of which may degenerate to a point circle, and let $k \in \{0,1\}$. Starting with $\alpha_1 \in \mathscr{M}_k(\omega_1, \omega_2)$, choose successively $\alpha_2, \dots, \alpha_n \in \mathscr{M}_k(\omega_1, \omega_2)$ so that one of the two intersection points of $\alpha_i$ and $\alpha_{i+1}$ lies on $\delta$ for $i = 1, 2, \dots, n-1$.*

*If, for one choice of $\alpha_1$, one intersection point of $\alpha_n$ and $\alpha_1$ also lies on $\delta$, then infinitely many such chains $(\alpha_1, \dots, \alpha_n)$ exist, and every member of $\mathscr{M}_k(\omega_1, \omega_2)$ may be chosen as $\alpha_1$.*

Figure 4.56 illustrates a case of Emch's theorem with $k = 1$ and $n = 12$. We prove directly its generalization, the Great Emch theorem; the proof given here is adapted from [**9**].

Poncelet's theorem may be viewed as a limiting case of Emch's theorem. Keep the center of the outer circle $\omega_1$ fixed and let its radius tend to infinity. At the same time, keep fixed the intersections of the $\alpha_i$ with $\delta$, while letting the centers and radii of the $\alpha_i$ tend to infinity. The arcs of the $\alpha_i$ cut off by $\delta$ then tend to line segments, and Emch's theorem becomes Poncelet's theorem for $\delta$ and $\omega_2$.

The following theorem generalizes Emch's theorem.

**Theorem 4.7.6** (Great Emch theorem). *Let $\mathscr{O}_1, \mathscr{O}_2$ be two pencils of circles with a common circle $\delta$. Let $\alpha_1, \dots, \alpha_n \in \mathscr{O}_1$ and $\beta_1, \dots, \beta_n \in \mathscr{O}_2$, and suppose, for every $i$, that $\delta$ lies between $\alpha_i$ and $\beta_i$. Choose $k_1, \dots, k_n \in \{0,1\}$ and put $\mathscr{M}_{(i)} = \mathscr{M}_{k_i}(\alpha_i, \beta_i)$. Starting with $\omega_1 \in \mathscr{M}_{(1)}$, choose circles $\omega_2, \dots, \omega_n$ with $\omega_i \in \mathscr{M}_{(i)}$ so that, for $i = 1, 2, \dots, n-1$, one intersection point of $\omega_i$ and $\omega_{i+1}$ lies on $\delta$.*

*If, for one choice of $\omega_1$, an intersection point of $\omega_n$ and $\omega_1$ also lies on $\delta$, then infinitely many such chains $(\omega_1, \dots, \omega_n)$ exist, and every member of $\mathscr{M}_{(1)}$ may be chosen as $\omega_1$.*[37]

The relation of this theorem to Emch's theorem is the same as that of Poncelet's closure theorem to Poncelet's theorem. Its proof is due to Protasov (Протасов), and it is therefore also called *Protasov's theorem*.

[37]Once $\omega_{i-1}$ is fixed, there may be two choices for $\omega_i$. As in the footnote to theorem 4.7.3, choose one continuous branch: when $\omega_1$ varies continuously, every chosen $\omega_i$ must vary continuously.

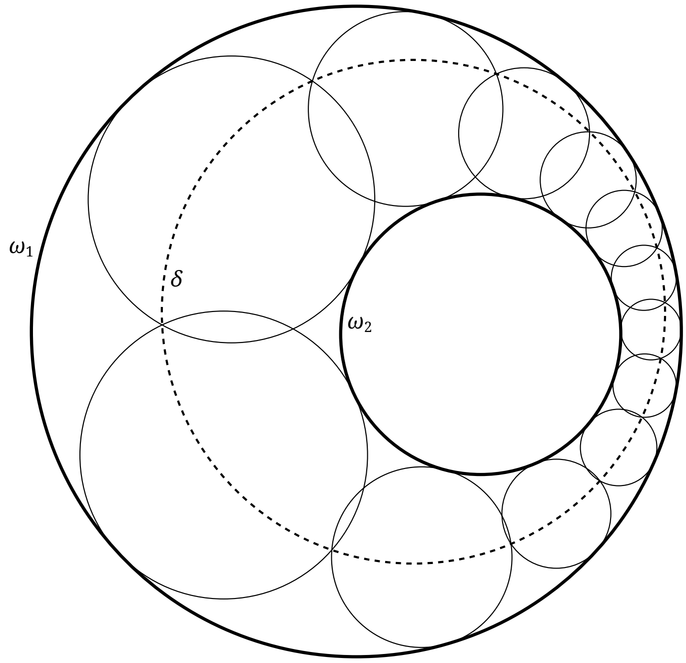


FIGURE 4.56

We begin with two lemmas.

**Lemma 4.7.7.** *Let circles $\alpha, \beta, \gamma$ belong to one pencil $\mathscr{O}$, and suppose the chord $AC$ of $\alpha$ is tangent to $\beta$. Let a circle $\omega$ pass through $A, C$ and be tangent to $\gamma$ in a prescribed sense, internal or external. Then, as $\omega$ varies, it is tangent to a fixed circle $\xi$ concentric with $\alpha$, and $\omega \in \mathscr{M}_0(\xi, \gamma)$.*

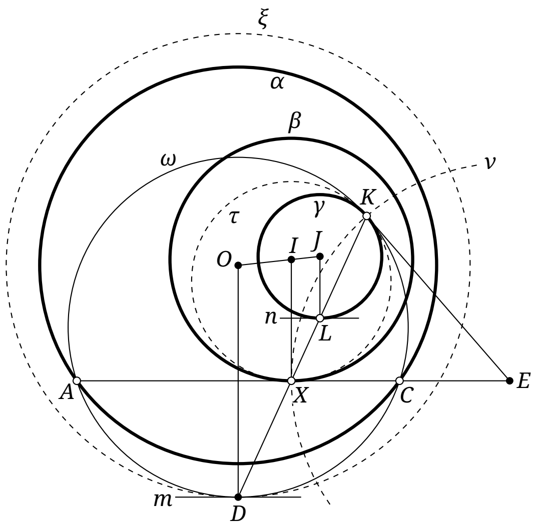


FIGURE 4.57

**Proof.** Consider the case in which $\omega$ is internally tangent to $\gamma$, as in figure 4.57. Let $K$ be their point of tangency, and let their common tangent at $K$ meet $AC$ at $E$. The secant–tangent theorem applied to $\omega$ gives $KE^2 = AE \cdot CE = \mathrm{pow}_\alpha(E)$, where the last equality is the power-of-a-point theorem.

Consider the circle $\nu = \odot(E, EK)$. By proposition 4.5.4, it is orthogonal to $\alpha$; it is plainly orthogonal to $\gamma$ as well. Hence proposition 4.6.4 shows that $\nu$ is orthogonal to every circle of $\mathscr{O}$, in particular to $\beta$. Thus $\nu$ passes through the point $X$ at which $AC$ is tangent to $\beta$, and $KE = XE$. There is therefore a circle $\tau$ tangent to $KE$ at $K$ and to $AE$ at $X$.

The three circles $\omega, \tau, \gamma$ are tangent at $K$, so $K$ is their common external center of similitude. Let $KX$ meet $\omega, \gamma$ again at $D, L$, respectively. Then $D, X, L$ are corresponding points under the homotheties among $\omega, \tau, \gamma$. Consequently, the tangent $m$ to $\omega$ at $D$, the tangent $AC$ to $\tau$ at $X$, and the tangent $n$ to $\gamma$ at $L$ are parallel.

Let $O, I, J$ be the centers of $\alpha, \beta, \gamma$, respectively. We have $JL \perp n$ and $IX \perp AC$. Also $AC \parallel m$ implies that $D$ is the midpoint of $\widehat{AC}$, so $OD \perp m$. Hence $OD \parallel IX \parallel JL$. It follows that $\frac{OD-JL}{OD-IX} = \frac{OJ}{OI}$, and therefore $OD = \frac{IX \cdot OJ - JL \cdot OI}{OJ - OI} = \text{const}$. Thus $\omega$ is internally tangent to the fixed circle $\xi = \odot(O, OD)$, as required. $\square$

The converse is also true.

**Lemma 4.7.8.** *Let $\alpha, \gamma, \xi$ be circles with $\alpha$ and $\xi$ concentric. A circle $\omega \in \mathscr{M}_0(\gamma, \xi)$ meets $\alpha$ at $A, C$. Then, for every such $\omega$, the chord $AC$ is tangent to one fixed circle $\beta$, and $\alpha, \beta, \gamma$ belong to one pencil.*

**Proof.** For one choice of $\omega$, the pencil generated by $\alpha, \gamma$ contains a unique circle $\beta$ tangent to $AC$. By lemma 4.7.7, $\omega$ is tangent to a fixed circle $\xi'$ concentric with $\alpha$, and $\omega \in \mathscr{M}_0(\xi, \gamma)$. But only one circle concentric with $\alpha$ can be tangent to $\omega$ in the prescribed sense, so $\xi' = \xi$.

Conversely, if $\beta$ is fixed and the chord $AC$ in lemma 4.7.7 ranges over all possible positions, the corresponding circles $\omega$ range over $\mathscr{M}_0(\xi, \gamma)$. Hence, as $\omega$ ranges over that family in the present lemma, every chord $AC$ is tangent to the same circle $\beta$. $\square$

**Proof of theorem 4.7.6.** By proposition 4.6.9(2), an inversion takes $\mathscr{O}_1$ to a family of concentric circles. Let $X_i$ be the common point of $\omega_i, \omega_{i+1}, \delta$. By lemma 4.7.8, each line $X_iX_{i+1}$ is tangent to a circle $\beta_i'$ in $\mathscr{O}_2$. Poncelet's closure theorem now proves the result. $\square$

Figure 4.58 illustrates the great Emch theorem for $n = 6$ and $k_i = 1$ for every $i$.

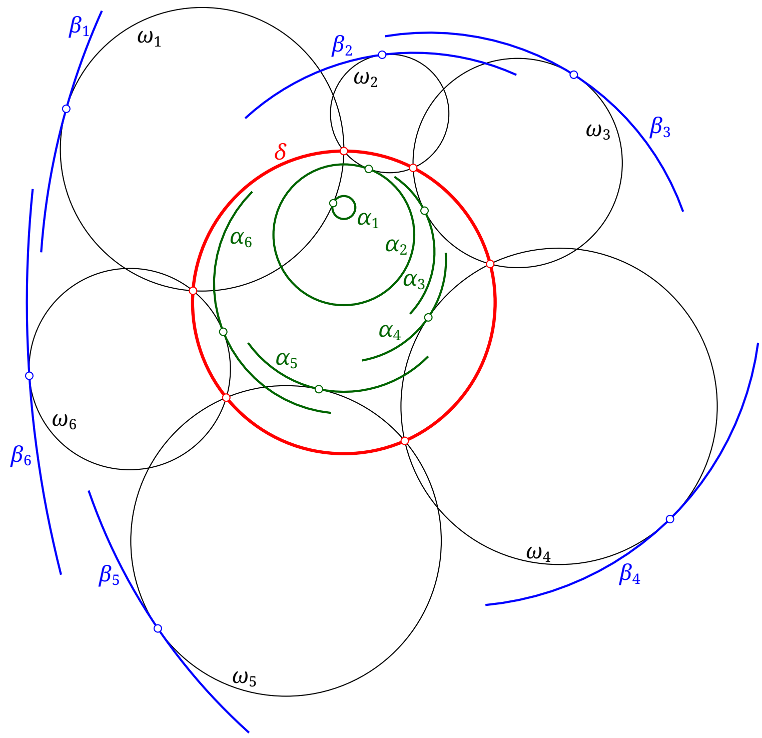


FIGURE 4.58

## The Zigzag Porism*

The final closure theorem, the zigzag porism, follows immediately from Emch's theorem.

**Theorem 4.7.9** (Zigzag porism)**.** *Let $\omega_1,\omega_2$ be two circles and let $d>0$ be fixed. Starting from $A_0\in\omega_1$, choose successively the $2n$ distinct points $B_1,A_1,B_2,A_2,\dots,B_n,A_n$ so that, for $i=1,2,\dots,n$, $A_i\in\omega_1$, $B_i\in\omega_2$, and $A_{i-1}B_i=B_iA_i=d$. If $A_n=A_0$ for one initial point $A_0$, then infinitely many such closed zigzags $(A_0,\dots,A_n;B_1,\dots,B_n)$ exist. Every point $A_0$ of $\omega_1$ may be used as the initial point, provided the circle centered at $A_0$ with radius $d$ meets $\omega_2$ in two distinct points.*[38]

**Proof.** At each $A_i$, draw the circle $\alpha_i=\odot(A_i,d)$. Adjacent circles $\alpha_i,\alpha_{i+1}$ have an intersection point on $\omega_2$. Every circle with center on $\omega_1$ and radius $d$ is tangent to the two fixed circles $\gamma_1,\gamma_2$ concentric with $\omega_1$ and having radii $r_{\omega_1}+d$ and $|r_{\omega_1}-d|$, respectively. Hence there is a $k\in\{0,1\}$ such that $\alpha_i\in\mathscr{M}_k(\gamma_1,\gamma_2)$ for every $i$. The result is therefore Emch's theorem. □

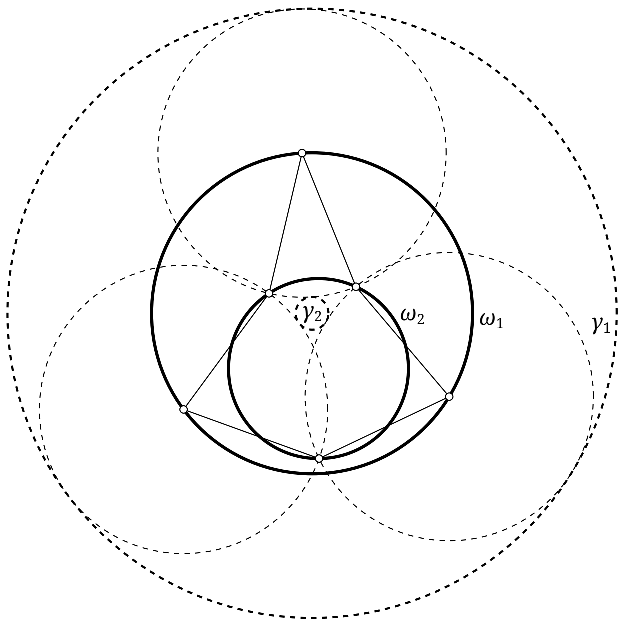


FIGURE 4.59

Figure 4.59 illustrates a case with $n=3$.

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## Exercises

**Exercise 4.86.** Prove that the centers of the circles in a Steiner chain lie on one central conic.

**Exercise 4.87.** Formulate the three-dimensional generalization of Steiner's porism.

**Exercise 4.88.** Let $\omega_1,\omega_2$ be nested circles, and suppose there is a quadrilateral inscribed in $\omega_1$ and circumscribed about $\omega_2$. Prove that the diagonals of every such quadrilateral meet at one fixed point, and that this point is a limit point of the elliptic pencil determined by $\omega_1,\omega_2$.

**Exercise 4.89.** Let $\omega_1,\omega_2$ be nested circles, and suppose there is a hexagon inscribed in $\omega_1$ and circumscribed about $\omega_2$. For any such hexagon $ABCDEF$, let its six points of tangency with $\omega_2$ form the hexagon $A'B'C'D'E'F'$. Prove that the six lines $AD$, $BE$, $CF$, $A'D'$, $B'E'$, $C'F'$ are concurrent at a fixed point.

[38] In an intuitive, if not entirely rigorous, picture, a flea makes successive jumps of length $d$ back and forth between two circles without retracing a jump. If it eventually returns to its starting point from one position, it does so from every admissible starting position.

**Exercise 4.90**.** Suppose the closure condition in Emch's theorem is satisfied, and consider a chain of circles in $\mathscr{M}_k(\omega_1,\omega_2)$. Thus, for $i=1,2,\dots,n$, one intersection point of $\alpha_i$ and $\alpha_{i+1}$ lies on $\delta$, where $\alpha_{n+1}=\alpha_1$. Prove that their other intersection points all lie on one fixed circle.

The following result is due to Avksentyev (Авксентьев).

**Exercise 4.91** (Avksentyev's theorem).** Prove the following "closure theorem."[39]

Let $\alpha,\gamma,\eta$ be coaxial circles, each pair being nested, and let $\xi$ be a circle concentric with $\alpha$. A circle $\delta\in\mathscr{M}_0(\xi,\eta)$ meets $\alpha$ at $A,B$. Through $A,B$, draw one tangent to $\gamma$ from each point, and let these tangents meet $\delta$ again at $A',B'$, respectively. If $A'=B'$ for one choice of $\delta$, then for every $\delta\in\mathscr{M}_0(\xi,\eta)$ the tangents can be chosen so that $A'=B'$.

*Hint.* Use exercises 0.70 and 4.83. The later result proposition 5.6.11 may also be used without proof.

## 4.8 Basic Properties of Complete Quadrilaterals

We conclude the chapter with several elementary geometric properties of a complete quadrilateral. We begin with the following fact.

**Lemma 4.8.1.** *There is a unique parabola tangent to all four sides of a complete quadrilateral; that is, every complete quadrilateral has a unique inscribed parabola.*

**Proof.** By theorem 4.5.10, the four sides of the complete quadrilateral together with $l_\infty$ determine a unique conic tangent to all five lines. By proposition 3.1.6, this conic is a parabola. □

This lemma makes several further results immediate.

**Theorem 4.8.2** (Miquel's theorem)**.** *The circumcircles of the four triangles determined by the four lines of a complete quadrilateral pass through one point.* This point is the *Miquel point* of the complete quadrilateral.

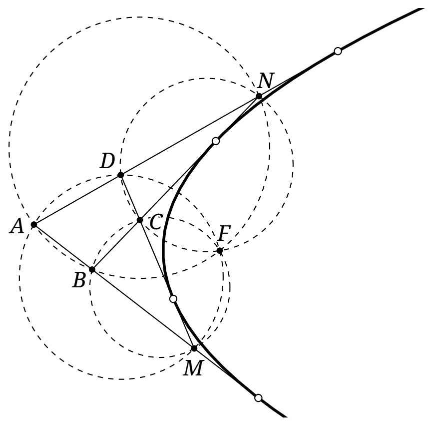


FIGURE 4.60

**Proof.** As in figure 4.60, there is a unique parabola tangent to the four lines. By theorem 4.2.11, all four circumcircles pass through its focus. □

[39] Its closure is somewhat different from that in the four preceding theorems. Avksentyev, however, used this name when publishing the result, and we retain it.

**Remark.** The expressions "the Miquel point of the complete quadrangle $ABCD$," "the Miquel point of the complete quadrilateral $ABCD$," and "the Miquel point of the quadrilateral $ABCD$" all mean the Miquel point of the complete quadrilateral formed by the four lines $AB$, $BC$, $CD$, and $DA$. The same convention will be used for its Gauss line (Gauß), and for its Aubert line, and related objects.

The result also has a proof not involving a parabola.

**Alternative proof.** Let the four lines meet at $A,B,C,D,M,N$, as in figure 4.60. Suppose the circles $\odot(DCN)$ and $\odot(ADM)$ meet again at $F$. Then $\measuredangle FCN = \measuredangle FDN = \measuredangle FDA = \measuredangle FMA$, so $F \in \odot(CBM)$. The same argument gives $F \in \odot(ABN)$. □

The first proof immediately gives the following property.

**Theorem 4.8.3.** *The Miquel point of a complete quadrilateral is the focus of its inscribed parabola.*

The Miquel point leads at once to a Simson line.

**Theorem 4.8.4.** *The projections of the Miquel point of a complete quadrilateral on its four sides are collinear.* Their common line is the *Simson line* of the complete quadrilateral.

**Proof.** Take the inscribed parabola of the complete quadrilateral. By theorem 4.8.3 and claim 1.5.2, the four projections are collinear on the tangent to the parabola at its vertex, its "auxiliary circle." □

**Alternative proof.** Apply Miquel's theorem and Simson's theorem. □

The first proof also gives the following characterization.

**Proposition 4.8.5.** *The Simson line of a complete quadrilateral is the "auxiliary circle" of its inscribed parabola.*

We next introduce the line associated with Gauss.

**Theorem 4.8.6** (Newton's theorem)**.** *The midpoints of the three diagonals of a complete quadrilateral are collinear.* Their common line is called the *Gauss line*, the *Newton line*, or the *Newton–Gauss line* of the complete quadrilateral.

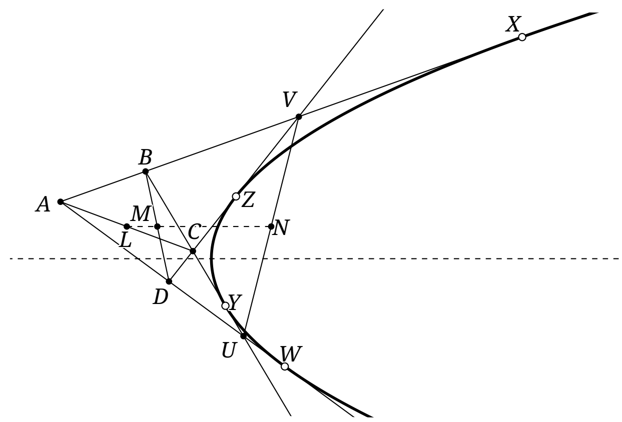


FIGURE 4.61

**Proof.** Let the inscribed parabola $\alpha$ of the complete quadrilateral $\{AB,BC,CD,DA\}$ touch its four sides at $X,Y,Z,W$, and let $L,M,N$ be the midpoints of its three diagonals, as in figure 4.61. Denote by $y_P$ the directed distance of a point $P$ from the axis of the parabola, positive above the axis. By claim 1.5.5,

$$y_N = \frac{y_V + y_U}{2} = \frac{1}{2}\left(\frac{y_X + y_Z}{2} + \frac{y_Y + y_W}{2}\right) = \frac{y_X + y_Y + y_Z + y_W}{4}.$$

The same expression is obtained for $y_L$ and $y_M$, so the three midpoints are collinear. $\square$

The proof gives another useful property.

**Proposition 4.8.7.** *The Gauss line of a complete quadrilateral is parallel to the axis of its inscribed parabola.*

The following area characterization of the Gauss line is due to Léon Anne.

**Theorem 4.8.8** (Léon Anne's theorem)**.** *Let $l$ be the Gauss line of a quadrilateral $ABCD$. For a point $P$ in the plane,*

$$P \in l \quad \Longleftrightarrow \quad [\triangle PAB] + [\triangle PCD] = [\triangle PBC] + [\triangle PDA] = \frac{S_{ABCD}}{2}.$$

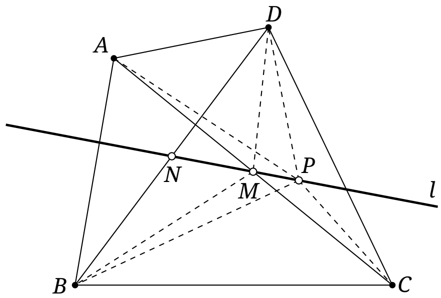


FIGURE 4.62

**Proof.** As in figure 4.62, let $M,N$ be the midpoints of $AC,BD$, respectively, so that $l = MN$. The identity $[\triangle PAB] + [\triangle PCD] + [\triangle PBC] + [\triangle PDA] = S_{ABCD}$ shows that it suffices to prove $P \in l \Longleftrightarrow [\triangle PBC] + [\triangle PDA] = \frac{S_{ABCD}}{2}$.

For any point $P$ in the plane, directed-area decomposition gives

$$\begin{aligned}
[\triangle PBC] + [\triangle PDA] &= [\triangle MBC] + [\triangle MDA] + [\triangle MPC] + [\triangle DMP] - [\triangle BMP] - [\triangle MPA] \\
&= \frac{S_{\triangle ABC}}{2} + \frac{S_{\triangle ADC}}{2} + ([\triangle MPC] - [\triangle MPA]) + ([\triangle DMP] - [\triangle BMP]) \\
&= \frac{S_{ABCD}}{2} + 0 + ([\triangle DMP] - [\triangle BMP]).
\end{aligned}$$

Thus it remains only to prove $P \in l \Longleftrightarrow [\triangle DMP] = [\triangle BMP]$. Let $N' = PM \cap DB$. By lemma 0.1.13, the last area equality holds if and only if $N'$ is the midpoint of $BD$, that is, if and only if $N' = N$, or equivalently if and only if $N,M,P$ are collinear. This is precisely $P \in l$. $\square$

**Remark.** The proof shows that the points satisfying this area equality lie on a line $l$, and that the midpoints of both diagonals of $ABCD$ lie on $l$. Similarly, the midpoint of the two intersection points of the pairs of opposite sides of $ABCD$ also lies on $l$. This gives a more elementary proof of Newton's theorem.

We now turn to the Aubert line.

**Theorem 4.8.9.** *The orthocenters of the four triangles determined by a complete quadrilateral are collinear.* Their common line is the *Aubert line*, also called the *ortholine*, of the complete quadrilateral. *It has the following properties:*

*(1) it is perpendicular to the Gauss line;*
*(2) it is parallel to the Simson line;*
*(3) its distance from the Miquel point is twice the distance from the Simson line to the Miquel point.*

**Proof.** Consider the parabola tangent to the four lines. By theorem 4.2.12, the four orthocenters are collinear on its directrix. The remaining three properties follow from theorem 4.8.3 and propositions 4.8.5 and 4.8.7. ☐

Here is a more elementary proof of the collinearity. The verification of the three further properties is left to the reader.

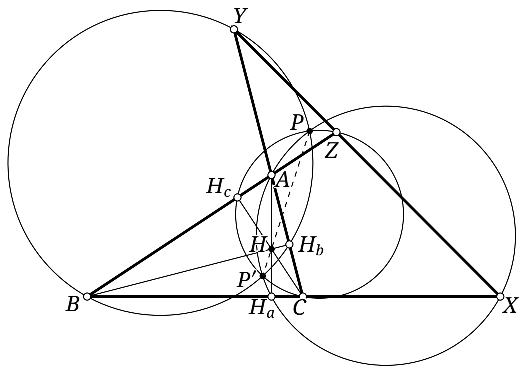


FIGURE 4.63

**Alternative proof.** As in figure 4.63, the triangle $ABC$ and the line $\overline{XYZ}$ form the complete quadrilateral $ACXZ$. Take the three circles having its diagonals as diameters. Define $H_a$ by $\odot(AX) \cap BC = \{X, H_a\}$, and define $H_b, H_c$ cyclically. Then $AH_a \perp BC$, and $AH_a, BH_b, CH_c$ meet at the orthocenter $H$ of $\triangle ABC$.

By lemma 0.5.22, $AH \cdot HH_a = CH \cdot HH_c = BH \cdot HH_b$. The power-of-a-point theorem therefore shows that $H$ has equal powers with respect to the three circles. The orthocenters of the other three triangles have the same property. Thus all four orthocenters lie on the common radical axis of the three circles and are collinear. ☐

The first proof gives a further characterization.

**Proposition 4.8.10.** *The Aubert line of a complete quadrilateral is the directrix of its inscribed parabola.*

The second proof gives the following important property.

**Theorem 4.8.11** (Gauss–Bodenmiller theorem)**.** *The three circles having the diagonals of a complete quadrilateral as diameters are coaxial, and their common radical axis is the Aubert line.*

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## Exercises

**Exercise 4.92.** Our proof of Newton's theorem used a parabola. Find a proof that does not use conics.

**Exercise 4.93.** In the parallelogram $ACGD$, choose $B, E$ on the sides $AC, AD$, respectively, and put $F = BD \cap CE$. Prove that the Gauss line of the quadrilateral $ABFE$ is parallel to $GF$.

**Exercise 4.94.** Prove that the reflections of the Miquel point of a complete quadrilateral in its four sides are collinear on the Aubert line.

**Exercise 4.95.** Prove that the four circumcenters of the triangles determined by a complete quadrilateral and its Miquel point lie on one circle, called the *Miquel circle*.

**Exercise 4.96** (Five-circle theorem)**.** Prove that the circumcircles of the five triangles at the tips of a pentagram meet pairwise in five further points, distinct from the intersection points already present in the pentagram, and that these five points are concyclic. In figure 4.64, these are the five open points on the dashed circle.

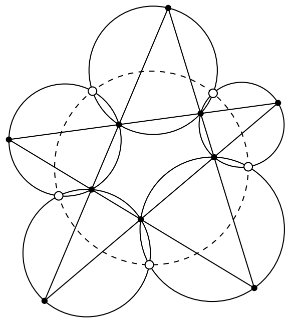

FIGURE 4.64

**Exercise 4.97.** Let $l$ be the Gauss line of a quadrilateral $ABCD$. A point $P$ inside $ABCD$ satisfies $\angle APD + \angle CPB = 180°$. Prove that the line joining the orthocenters of $\triangle APD$ and $\triangle CPB$ is perpendicular to $l$.

**Exercise 4.98.** Let $A, B, C, D$ lie on a circle $\omega$. Prove that the radical axis of $\omega$ and the circle through the four circumcenters of the triangles of the complete quadrilateral $ABCD$ is the Aubert line of that complete quadrilateral.

## Chapter 5

# Selected Triangle Centers

This chapter introduces a selection of triangle centers and some of their properties. The Encyclopedia of Triangle Centers (ETC)[1] catalogs and numbers tens of thousands of triangle centers; here we discuss only a number of the best known.

For clarity, the following table lists the triangle centers used in this book, the symbols adopted for them, their ETC numbers, and the sections in which they first occur. Some centers are introduced only in later chapters. Where the ETC supplies an established name, the table uses it; where the ETC gives no name, the table gives a defining description instead. A convenient local descriptor used elsewhere in the text is not thereby presented as an official ETC name.

When the text uses a convenient book shorthand, it is set in smaller type within the same name-or-description cell.

Symbols consisting of two letters, such as *Fe* and *Ni*, are already sufficiently distinctive. Once such a point has been defined, we may therefore use its symbol without restating its meaning whenever the reference triangle is clear.

TABLE 1. Triangle centers used in this book

| ETC | Name or defining description | Symbol | Location |
|---|---|---|---|
| 1 | Incenter | $I$ | 0.6.1 |
| 2 | Centroid | $G$ | 0.5.1 |
| 3 | Circumcenter | $O$ | 0.5.1 |
| 4 | Orthocenter | $H$ | 0.5.2 |
| 5 | Nine-point center | $Ni$ | 0.5.3 |
| 6 | Symmedian point (Lemoine point) | $L$ | 5.3.1 |
| 7 | Gergonne point | $Ge$ | 5.2 |
| 8 | Nagel point | $Na$ | 5.2 |
| 9 | Mittenpunkt | $Mi$ | 5.3.2 |
| 10 | Spieker center | $Sp$ | 5.2 |
| 11 | Feuerbach point | $Fe$ | 5.1 |
| 12 | Harmonic conjugate of the Feuerbach point with respect to the incenter and nine-point center<br>BOOK SHORTHAND: *Feuerbach perspector* | $Fp$ | 5.1 |
| 13 | First isogonic center (first Fermat–Torricelli point) | $T_1$ | 5.9.1 |
| 14 | Second isogonic center (second Fermat–Torricelli point) | $T_2$ | 5.9.1 |
| 15 | First isodynamic point (first Apollonius point) | $Ap_1$ | 5.8 |
| 16 | Second isodynamic point (second Apollonius point) | $Ap_2$ | 5.8 |
| 17 | First Napoleon point | $Np_1$ | 5.9.1 |
| 18 | Second Napoleon point | $Np_2$ | 5.9.1 |
| 19 | Clawson point | $Cl$ | 5.5 |
| 22 | Exeter point | $Et$ | 5.3.2 |
| 25 | Homothetic center of the orthic and tangential triangles | $Ht$ | 5.3.2 |
| 26 | Circumcenter of the tangential triangle<br>BOOK SHORTHAND: *tangential circumcenter* | $To$ | 5.3.2 |
| 30 | Euler point at infinity | $Eu$ | 7.6.3 |

*Continued on next page*

[1] See `https://faculty.evansville.edu/ck6/encyclopedia/ETC.html`.

*Table 1 continued*

| ETC | Name or defining description | Symbol | Location |
|---|---|---|---|
| 33 | Perspector of the orthic and intangents triangles | *Ic* | 5.5 |
| | BOOK SHORTHAND: *inner Clawson point* | | |
| 39 | Brocard midpoint | $Br_{\mathrm{m}}$ | 5.7.2 |
| 40 | Bevan point | *Be* | 5.4 |
| 51 | Centroid of the orthic triangle | *Gh* | 7.4.3 |
| | BOOK SHORTHAND: *orthocentroid point* | | |
| 54 | Kosnita point (Coșniță) | *Ko* | 5.1 |
| 55 | Internal similicenter of the circumcircle and incircle | *In* | 5.2 |
| 56 | External similicenter of the circumcircle and incircle | *Ex* | 5.2 |
| 57 | Isogonal conjugate of the Mittenpunkt | *Ta* | 5.4 |
| | BOOK SHORTHAND: *tangential perspector* | | |
| 69 | Symmedian point of the anticomplementary triangle | *Al* | 5.3.2 |
| | BOOK SHORTHAND: *anticomplementary Lemoine point* | | |
| 75 | Isotomic conjugate of the incenter | *Ti* | 7.5.2 |
| 76 | Third Brocard point | $Br_3$ | 5.8 |
| 485 | Vecten point | $Vc_1$ | 5.9.1 |
| 486 | Inner Vecten point | $Vc_2$ | 5.9.1 |
| — | Excenter | $I_{a,b,c}$ | 0.6.2 |
| — | First Brocard point | $Br_1$ | 5.7.1 |
| — | Second Brocard point | $Br_2$ | 5.7.1 |

## 5.1 The Nine-Point Circle and the Feuerbach Point

We have already introduced the nine-point circle. We now investigate several further properties, beginning with the isogonal conjugate of its center *Ni*.

**Theorem 5.1.1** (Kosnita's theorem)**.** *Let $O$ be the circumcenter of $\triangle ABC$, and let $\triangle O_A O_B O_C$ be the Carnot triangle of $O$. Then $\triangle ABC$ and $\triangle O_A O_B O_C$ are perspective from a point Ko, and Ko* $=$ **g***Ni.* The point *Ko* is called the *Kosnita point*.

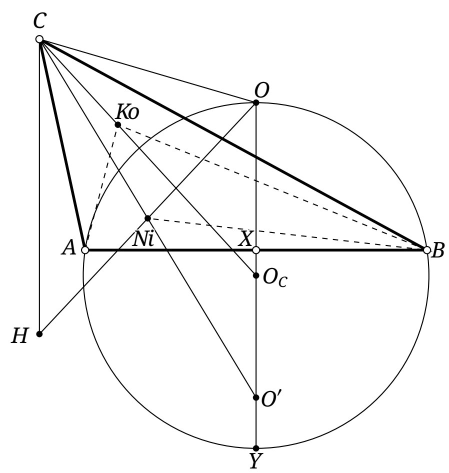


FIGURE 5.1

**Proof.** Write $O$ for the circumcenter and $H$ for the orthocenter. Let $O' = \mathfrak{f}_{AB}(O)$ be the reflection of $O$ in $AB$, set $OO' \cap AB = X$, and let $Y$ be the point of $\odot O_C$ antipodal to $O$, as in Figure 5.1. By lemma 0.5.18, $CHO'O$ is a parallelogram. Since $Ni$ is the midpoint of its diagonal $HO$, the points $C, Ni, O'$ are collinear. Moreover, by the right-triangle altitude theorem,

$$OC^2 = OA^2 = OX \cdot OY = (2OX) \cdot \frac{OY}{2} = OO' \cdot OO_C.$$

Hence $\triangle COO_C \sim \triangle O'OC$, and therefore $\angle O_C CO = \angle CO'O = \angle HCNi$. Since $H$ and $O$ are isogonally conjugate, $\angle HCA = \angle OCB$, so $\angle ACNi = \angle BCO_C$.

Similarly, $\angle BANi = \angle CAO_A$ and $\angle ABNi = \angle CBO_B$. The existence theorem for isogonal conjugates now proves that $Ko$ exists. □

Another celebrated result concerning the nine-point circle is Feuerbach's theorem.

**Theorem 5.1.2** (Feuerbach's theorem)**.** *The nine-point circle of a triangle is tangent to its incircle and to each of its three excircles.*[2]

**Proof.** As in Figure 5.2, let the side lengths of $\triangle ABC$ be $a, b, c$, let $\triangle M_a M_b M_c$ be its medial triangle, and let $\odot I$ be its incircle. Consider the point circles $\odot M_a, \odot M_b, \odot M_c$, degenerated to the points $M_a, M_b, M_c$, together with $\odot I$. If $t_{ij}$ denotes the length of a common tangent to $\odot M_i$ and $\odot M_j$, then $t_{M_aM_b} = \frac{c}{2}$, $t_{M_bM_c} = \frac{a}{2}$, $t_{M_cM_a} = \frac{b}{2}$, $t_{M_aI} = \frac{|b-c|}{2}$, $t_{M_bI} = \frac{|a-c|}{2}$, and $t_{M_cI} = \frac{|b-a|}{2}$. With a suitable choice of sign,

$$\left| t_{M_aM_b} t_{M_cI} \pm t_{M_bM_c} t_{M_aI} \right| = t_{M_cM_a} t_{M_bI}.$$

The converse of Casey's theorem therefore shows that the four circles are tangent to one circle. Since the first three are point circles, that common tangent circle is precisely $\odot(M_aM_bM_c) = \mathcal{N}$. □

**Corollary 5.1.3.** *If the circumradius and inradius of $\triangle ABC$ are $R$ and $r$, and its incenter is $I$, then $INi = \frac{R}{2} - r$.*

**Proof.** The circles $\mathcal{N}$ and $\mathcal{I}$ are tangent, and their relative positions show that the tangency is internal. Since $r_{\mathcal{N}} = \frac{R}{2}$, the conclusion follows. □

This gives a trigonometry-free proof of proposition 0.6.18: if $H$ and $I$ are the orthocenter and incenter of $\triangle ABC$, then $HI^2 = 2r^2 - 2R\rho$, where $r = r_{\mathcal{I}}$, $R = r_{\mathcal{C}}$, and $\rho = \rho_{\triangle ABC}$.

**Proof of proposition 0.6.18.** The Euler–Chapple formula gives $OI^2 = R^2 - 2Rr$; exercise 0.73 gives $OH^2 = R^2 - 4R\rho$; and corollary 5.1.3 gives $INi = \frac{R}{2} - r$. Since $INi$ is a median of $\triangle IOH$, the median-length formula yields

$$HI^2 = 2NiI^2 + \frac{1}{2}OH^2 - OI^2 = 2r^2 - 2R\rho.$$

□

The point of tangency $Fe$ of a triangle's Euler circle and incircle is its *Feuerbach point*. The three points where the Euler circle touches the excircles are the *excentral Feuerbach points*, and the triangle they form is the *Feuerbach triangle*.

**Proposition 5.1.4.** *For $\triangle ABC$, the Feuerbach triangle $\triangle Fe_A Fe_B Fe_C$ is perspective with $\triangle ABC$, as in Figure 5.2.* For convenience in this book, we call its center of perspectivity $Fp$ the *Feuerbach perspector*.

[2] For an equilateral triangle the nine-point circle and the incircle coincide; this may be regarded as a limiting case of tangency.

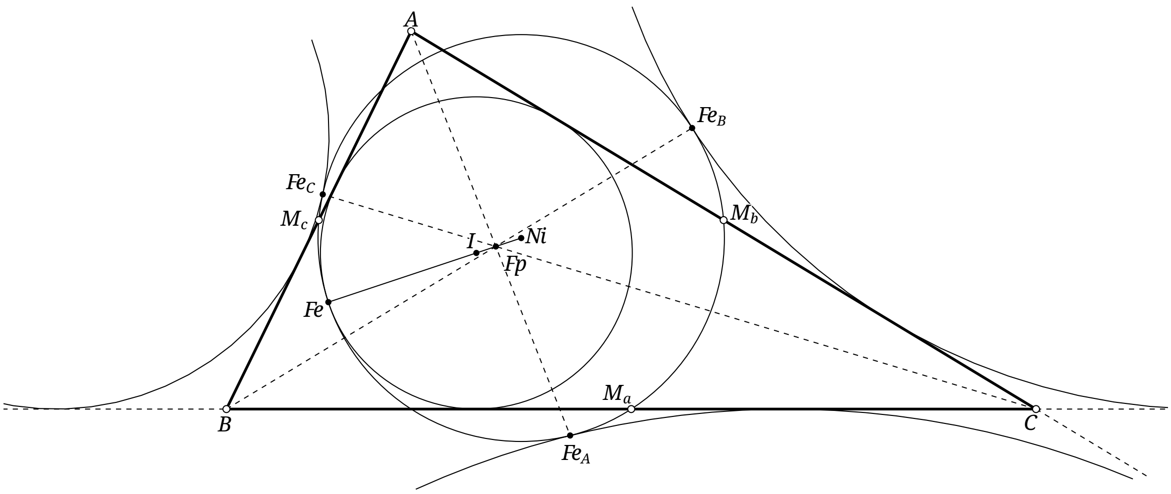


FIGURE 5.2

**Proof.** The nine-point circle and the $A$-excircle touch at $Fe_A$, so $Fe_A$ is their internal similicenter. The point $A$ is the external similicenter of the $A$-excircle and the incircle. Hence proposition 0.4.10 shows that $AFe_A$ passes through the internal similicenter $Fp$ of the nine-point circle and the incircle; their external similicenter is $Fe$. Similarly, $Fp \in BFe_B$ and $Fp \in CFe_C$. □

The proof also gives the following fact.

**Proposition 5.1.5.** *The Feuerbach perspector Fp of a triangle is the internal similicenter of the nine-point circle and the incircle.*

The Feuerbach perspector has one further useful property.

**Proposition 5.1.6.** *For any triangle, its incenter I and the points Ni, Fe, Fp are collinear and form a harmonic range, as in Figure 5.2.*

**Proof.** Since the nine-point circle and the incircle touch at $Fe$, the points $Ni, I, Fe$ are collinear. By proposition 5.1.5, so are $Ni, I, Fp$; hence all four points are collinear. Since $Fp$ is the internal similicenter, $\frac{\overline{FpNi}}{\overline{FpI}} = -\frac{r_{\odot Ni}}{r_{\odot I}} = -\frac{\overline{FeNi}}{\overline{FeI}}$. Thus $(Fp, Fe; Ni, I) = -1$. □

---

## Exercises

**Exercise 5.1.** Prove that when the Kosnita point $Ko$ of a triangle is a point at infinity, its reflection triangle degenerates into a line.

**Exercise 5.2.** Prove that the circumcenter of a triangle's reflection triangle is the reflection of the original circumcenter in $Ko$.
*Hint.* Boutte's theorem.

**Exercise 5.3.** For $\triangle ABC$, let $O$ be the circumcenter and let $\triangle A'B'C'$ be the reflection triangle. Prove that $\odot(AOA')$, $\odot(BOB')$, and $\odot(COC')$ are coaxial and that $Ko$ lies on their common radical axis.

**Exercise 5.4.** Suppose that $\triangle ABC$ satisfies $AB+AC=2BC$. Let $AFe\cap BC=P$, and let $S$ be the midpoint of the arc $\widehat{BC}$ of $\mathscr{C}$. Prove that $AFe=PS$.

**Exercise 5.5.** Let $I$ be the incenter of $\triangle ABC$, let $M_a$ be the midpoint of $BC$, and let $\triangle DEF$ be the intouch triangle. Set $FE\cap BC=Q$ and $AFe\cap DI=P$. Prove that $M_aI$ is the radical axis of $\odot(IPFe)$ and $\odot(IDQ)$.

**Exercise 5.6.** Let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $H$ be the orthocenter of $\triangle ABC$. Draw $M_aD\perp AI$, meeting $AH$ at $D$; draw $M_bE\perp BI$, meeting $BH$ at $E$; and draw $M_cF\perp CI$, meeting $CH$ at $F$. Prove that $Fe\in\odot(DEF)$.

**Exercise 5.7.** Let $G$ be the centroid of $\triangle ABC$, and let $K$ be an intersection of $FeG$ with $\mathscr{C}$. Prove that, among the segments $AK,BK,CK$, the length of the longest equals the sum of the lengths of the other two.

## 5.2 The Gergonne and Nagel Points

We now introduce the celebrated Gergonne and Nagel points.

**Definition 5.2.1.** For a triangle $\triangle ABC$, the center of perspectivity $Ge$ of the triangle and its intouch triangle is called the *Gergonne point*. The center of perspectivity $Na$ of the triangle and its extouch triangle is called the *Nagel point*.

We first prove that $Ge$ and $Na$ exist. Let the side lengths of $\triangle ABC$ be $a,b,c$, and put $p=\frac{a+b+c}{2}$. By proposition 0.6.13,

$$\frac{CG_a}{BG_a}\cdot\frac{BG_c}{AG_c}\cdot\frac{AG_b}{CG_b}=\frac{p-c}{p-b}\cdot\frac{p-b}{p-a}\cdot\frac{p-a}{p-c}=1,$$
$$\frac{CN_a}{BN_a}\cdot\frac{BN_c}{AN_c}\cdot\frac{AN_b}{CN_b}=\frac{p-b}{p-c}\cdot\frac{p-a}{p-b}\cdot\frac{p-c}{p-a}=1.$$

Ceva's theorem therefore proves the existence of both points. The existence of $Ge$ also has a projective proof: keep the incircle fixed and move the intersection of $AG_a$ and $BG_b$ to the center of the incircle.

We record several important properties.

**Proposition 5.2.2.** *The line joining a vertex of $\triangle ABC$ to $Na$ divides the triangle's perimeter into two equal parts. Thus, if $ANa\cap BC=D$, then $AB+BD=AC+CD$; the analogous statement holds at the other vertices.*

**Proof.** This follows immediately from proposition 0.6.13. □

**Proposition 5.2.3.** *For every triangle, $Ge=\mathbf{t}Na$.*

**Proof.** This follows directly from proposition 0.6.13 and the definition of isotomic conjugation. □

We next study the isogonal conjugates of $Ge$ and $Na$. Introduce two more triangle centers: the internal similicenter $In$ and external similicenter $Ex$ of the incircle and circumcircle. When no ambiguity is possible, we call them simply the triangle's *internal similicenter* and *external similicenter*.

The definition gives the following property at once.

**Proposition 5.2.4.** *If $I$ and $O$ are the incenter and circumcenter of a triangle, then $O$,$In$,$I$,$Ex$ form a harmonic range.*

**Proof.** The two homotheties give $\frac{\overline{InO}}{\overline{InI}} = -\frac{R}{r}$ and $\frac{\overline{ExO}}{\overline{ExI}} = \frac{R}{r}$. Hence $(OIn, IEx) = -1$. □

The next proposition relates these similicenters to the isogonal conjugates of $Ge$ and $Na$.

**Proposition 5.2.5.** *For every triangle, $In = \mathbf{g}Ge$ and $Ex = \mathbf{g}Na$.*

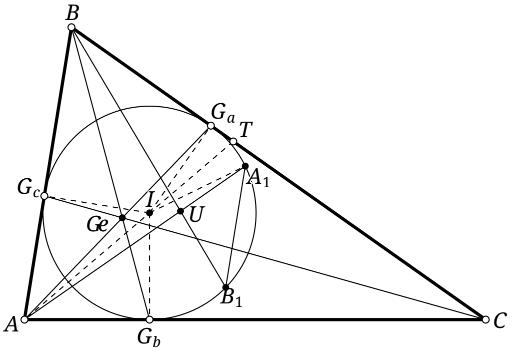


FIGURE 5.3

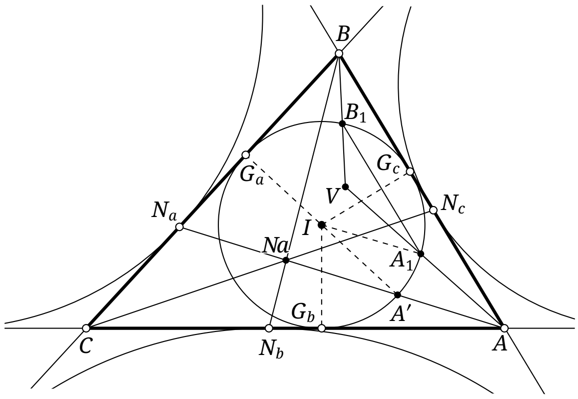


FIGURE 5.4

**Proof.** First prove $In = \mathbf{g}Ge$. In Figure 5.3, let $I$ be the incenter, put $U = \mathbf{g}Ge$, and let $\triangle G_aG_bG_c$ be the intouch triangle. On the lines $AU, BU, CU$, let the farther intersections with $\mathscr{I}$ be $A_1, B_1, C_1$, respectively, and let the bisector of $\angle BAC$ meet $BC$ at $T$. Then

$$\angle G_aIA_1 = 2\angle G_aIT = 2(\angle AIC - 90^\circ) = 2\left(\angle ABC + \frac{\angle BAC}{2} - 90^\circ\right),$$

$$\begin{aligned}\angle G_cIA_1 &= \angle G_aIG_c + \angle G_aIA_1\\ &= (180^\circ - \angle ABC) + 2\left(\angle ABC + \frac{\angle BAC}{2} - 90^\circ\right) = \angle BAC + \angle ABC.\end{aligned}$$

Similarly, $\angle G_cIB_1 = \angle BAC + \angle ABC$, so $AB \parallel A_1B_1$. In the same way, $A_1C_1 \parallel AC$ and $B_1C_1 \parallel BC$. Thus $\triangle ABC$ and $\triangle A_1B_1C_1$ are homothetic with center $U$. The circles $\odot(ABC) = \mathscr{C}$ and $\odot(A_1B_1C_1) = \mathscr{I}$ are corresponding figures under this homothety, so $\mathscr{C}$ and $\mathscr{I}$ are homothetic about $U$. Their relative positions show that $U$ is the internal similicenter $In$.

It remains to prove $Ex = \mathbf{g}Na$. In Figure 5.4, let $I$ be the incenter, put $V = \mathbf{g}Na$, and let $\triangle G_aG_bG_c$ and $\triangle N_aN_bN_c$ be the intouch and extouch triangles. Let the segments $VA, VB, VC$ meet $\mathscr{I}$ at $A_1, B_1, C_1$, respectively. We need the following lemma. □

**Lemma 5.2.6.** *Suppose that the A-excircle and incircle of $\triangle ABC$ touch $BC$ at $N_a$ and $G_a$, respectively. The point $A'$ of the incircle antipodal to $G_a$ lies on the line $AN_a$.*

**Proof.** As in Figure 5.5, draw $XY$ through $A'$ parallel to $CB$, meeting the sides $AB$,$AC$ at $Y$,$X$, respectively. Then $\mathscr{I}_{\triangle ABC} = \mathscr{I}_{a\triangle XYA}$. Consider the homothety centered at $A$ that sends $\triangle XYA$ to $\triangle CBA$. It sends $\mathscr{I}_{\triangle ABC} = \mathscr{I}_{a\triangle XYA}$ to $\mathscr{I}_{a\triangle BCA}$. The points $A'$,$N_a$ at which corresponding circles touch corresponding sides are corresponding points under the homothety. Hence $A$,$A'$,$N_a$ are collinear. □

We return to the proposition.

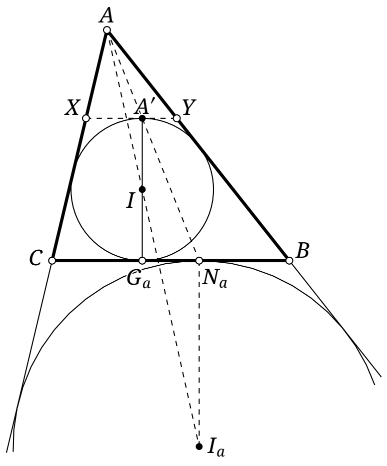


FIGURE 5.5

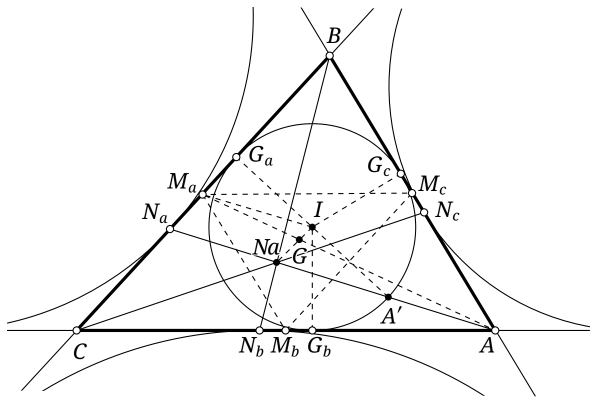


FIGURE 5.6

**Completion of the proof of proposition 5.2.5.** By lemma 5.2.6, the intersection $A'$ of the segment $N_aA$ with $\mathscr{I}$ nearer $A$ is antipodal to $G_a$ on $\mathscr{I}$. Consequently,

$$\begin{aligned}\angle A'IA_1 &= 2\angle A'VA = 2(\angle N_aA'G_a - \angle A'AI) = 2(90^\circ - \angle G_aN_aA - \angle A'AI)\\ &= 2(90^\circ - \angle BCA - \angle CAN_a - \angle N_aAI) = 2\left(90^\circ - \angle BCA - \frac{\angle CAB}{2}\right),\\ \angle G_cIA_1 &= 180^\circ - \angle G_aIG_c - \angle A'IA_1 = \angle ABC - 2\left(90^\circ - \angle BCA - \frac{\angle CAB}{2}\right)\\ &= \angle ABC + \angle CAB + 2\angle BCA - 180^\circ = \angle BCA.\end{aligned}$$

Similarly $\angle G_cIB_1 = \angle BCA$, whence $A_1B_1 \parallel AB$. The other two pairs of corresponding sides are also parallel. There is therefore a homothety centered at $V$ that sends $\triangle ABC$ to $\triangle A_1B_1C_1$. The circles $\odot(ABC) = \mathscr{C}$ and $\odot(A_1B_1C_1) = \mathscr{I}$ are corresponding figures under it, so $\mathscr{C}$ and $\mathscr{I}$ are homothetic about $V$. Their relative positions show that $V$ is the external similicenter *Ex*. □

The Nagel point also has the following interesting property.

**Proposition 5.2.7.** *For every triangle, its incenter $I$, centroid $G$, and Nagel point Na are collinear, and $I = \mathbf{c}Na$.* Their common line is called the *Nagel line*.

**Proof.** In Figure 5.6, for $\triangle ABC$, let $\triangle G_aG_bG_c$, $\triangle N_aN_bN_c$, and $\triangle M_aM_bM_c$ be its intouch, extouch, and medial triangles, respectively. By lemma 5.2.6, the point $A'$ antipodal to $G_a$ lies on *ANa*.

Because *Na* and *Ge* are isotomically conjugate, $M_a$ is the midpoint of $N_aG_a$; and $I$ is the midpoint of $A'G_a$. Hence $M_aI \parallel ANa$.

Consider the homothety between the medial triangle $\triangle M_aM_bM_c$ and the original triangle $\triangle ABC$. Since $M_a, A$ are corresponding points and $M_aI \parallel ANa$, the lines $M_aI, ANa$ correspond under this homothety. Thus $M_aI$ passes through the Nagel point of $\triangle M_aM_bM_c$. Likewise, $M_bI, M_cI$ pass through the Nagel point of $\triangle M_aM_bM_c$. Their common point $I$ is therefore the Nagel point of $\triangle M_aM_bM_c$.

Thus $I, Na$ are corresponding points under the homothety between $\triangle M_aM_bM_c$ and $\triangle ABC$, and $I = \mathbf{c}Na$. □

**Corollary 5.2.8.** *The incenter of a triangle is the Nagel point of its medial triangle.*

We conclude the section with another center related to the Nagel point.

**Definition 5.2.9.** The incenter $Sp$ of the medial triangle of $\triangle ABC$ is called the *Spieker center* of $\triangle ABC$. The incircle of the medial triangle is called the *Spieker circle*.

The definition immediately gives the following result.

**Proposition 5.2.10.** *The Spieker center of a triangle is the complement of its incenter.*

**Proof.** Apply the homothety between the triangle and its medial triangle. □

The next propositions describe the connection between the Spieker center and the Nagel point.

**Proposition 5.2.11.** *The Spieker center of a triangle is the midpoint of its Nagel point and incenter.*

**Proof.** This follows immediately from propositions 5.2.7 and 5.2.10. □

**Proposition 5.2.12.** *The Nagel point and centroid of a triangle are, respectively, the external and internal similicenters of the Spieker circle and the incircle.*

**Proof.** The homothety between the triangle and its medial triangle shows that the radii of the Spieker circle and incircle are in the ratio $1:2$. If $G$ and $I$ are the centroid and incenter, then $\frac{\overline{GI}}{\overline{GSp}} = -2$ and $\frac{\overline{NaI}}{\overline{NaSp}} = 2$, which proves the assertion. □

**Proposition 5.2.13.** *In Figure 5.7, let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $A_1$,$A_2$,$A_3$ be the midpoints of the segments $NaA$,$NaB$,$NaC$. The Spieker circle is the incircle of $\triangle A_1A_2A_3$, and $\triangle A_1A_2A_3$ is centrally symmetric to $\triangle M_aM_bM_c$ about $Sp$. The six points where the Spieker circle touches the sides of these two triangles form three pairs of antipodal points on the circle.*

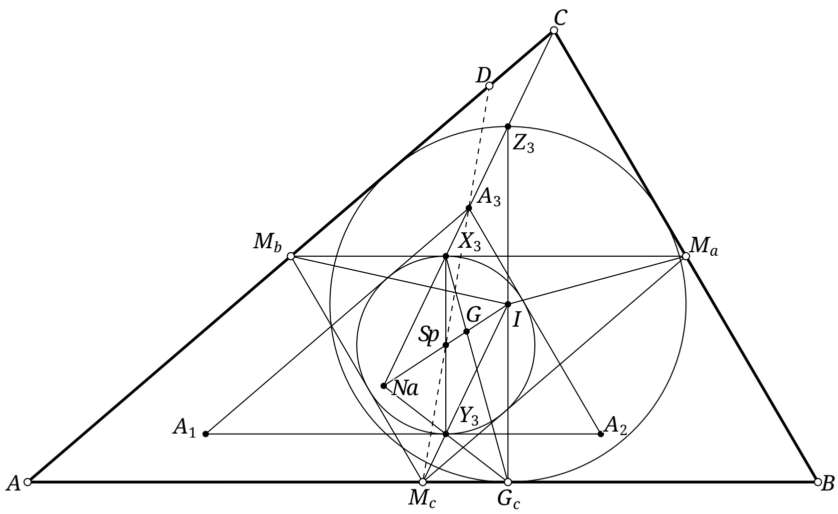


FIGURE 5.7

**Proof.** The Nagel point $Na$ is the external similicenter of the Spieker circle and the incircle, with ratio $\frac{1}{2}$, while $\mathfrak{h}_{Na,1/2}(\triangle ABC) = \triangle A_1A_2A_3$. Thus the Spieker circle is the incircle of $\triangle A_1A_2A_3$.

Moreover, $A_1A_2 \parallel M_aM_b$, and both lines are tangent to the Spieker circle. Their points of tangency $X_3, Y_3$ are therefore antipodal. The other two pairs follow similarly, and the asserted central symmetry is immediate. □

**Proposition 5.2.14.** *The line joining a vertex of a triangle to its Nagel point meets the corresponding side of the medial triangle at that side's point of tangency with the Spieker circle. Thus, in Figure 5.7, if $\triangle M_aM_bM_c$ is the medial triangle of $\triangle ABC$, then $X_3 = NaC \cap M_aM_b$ is the point where the Spieker circle touches $M_aM_b$.*

**Proof.** The line $CNa$ bisects the perimeter of $\triangle ABC$. Since $\triangle CM_bM_a$ and $\triangle CAB$ are homothetic about $C$, it also bisects the perimeter of $\triangle CM_aM_b$. Put $M_aM_b \cap NaC = X_3'$. Then $CM_b + M_bX_3' = CM_a + M_aX_3'$. The quadrilateral $CM_bM_cM_a$ is a parallelogram, so this equation becomes $M_bM_c + M_aX_3' = M_aM_c + M_bX_3'$. By proposition 0.6.13, the point of tangency $X_3$ of the Spieker circle with $M_aM_b$ satisfies $M_bM_c + M_aX_3 = M_aM_c + M_bX_3$. Hence $X_3 = X_3'$. □

**Proposition 5.2.15.** *The line joining the midpoint of a side of a triangle to the Spieker center bisects the triangle's perimeter.*

**Proof.** In the configuration of Figure 5.7, let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$. We prove that $\mathcal{S}p M_c$ bisects the perimeter of $\triangle ABC$. Put $M_c\mathcal{S}p \cap AC = D$, and through $B$ draw $BE \parallel M_c\mathcal{S}p$, meeting the extension of $AC$ at $E$; the point $E$ is not shown in the figure.

The triangles $\triangle M_aM_bM_c$ and $\triangle ABC$ are homothetic, and $M_c\mathcal{S}p$ is an angle bisector of $\triangle M_aM_bM_c$. Hence it is parallel to the bisector of $\angle ACB$, as is $BE$. It follows that $CB = CE$.[3] Since $M_cD$ is a midline of $\triangle ABE$,

$$AM_c + AD = BM_c + ED = BM_c + DC + CE = BM_c + DC + CB.$$

□

---

## Exercises

**Exercise 5.8.** A thin wire of uniform linear density is bent into a triangle. Prove that its center of mass is the Spieker center of the triangle.

**Exercise 5.9.** The incircle $\mathcal{I}$ of $\triangle ABC$ touches $BC$ at $D$. Let $Ge^*$ and $Na^*$ be the Gergonne and Nagel points of $\triangle ABD$. Prove that $(BGe^* \cap CNa^*) \in \mathcal{I}$.

**Exercise 5.10.** Let $H$ be the orthocenter of $\triangle ABC$, and let $\triangle A'B'C'$ be its south-pole triangle. The triangle formed by $\mathsf{f}_{BC}(A')$, $\mathsf{f}_{CA}(B')$, and $\mathsf{f}_{AB}(C')$ is called the *Fuhrmann triangle*, and its circumcircle the *Fuhrmann circle*. Prove that $HNa$ is a diameter of the Fuhrmann circle.

**Exercise 5.11.** Let $I$ be the incenter of $\triangle ABC$, let $\triangle M_aM_bM_c$ be its medial triangle, and let $\triangle G_aG_bG_c$ be its intouch triangle. Choose $\triangle A_1A_2A_3$ so that its sides are respectively parallel to those of $\triangle ABC$ and its incircle is the Spieker circle. Prove that the lines $A_1A_2$, $M_cI$, and $NaG_c$ are concurrent at the point $Y_3$ where $A_1A_2$ touches the Spieker circle.

**Exercise 5.12.** For any triangle, prove that $In$ is the pole, with respect to $\mathcal{C}$, of the trilinear polar of the incenter.

**Exercise 5.13.** Prove that the tangential triangle of the circumcevian triangle of the incenter is perspective with the original triangle, with center of perspectivity $Ex$.

---

[3] Indeed, if the bisector of $\angle ACB$ meets $AB$ at $T$, then $\angle CBE = \angle BCT = \angle ACT = \angle CEB$, where the first and last equalities follow from $CT \parallel BE$.

**Exercise 5.14.** For $\triangle ABC$, let $A', B', C'$ be the respective inverses of $A, B, C$ under an inversion centered at *Na*. Let $\triangle PQR$ be the antipedal triangle of *Na* with respect to $\triangle A'B'C'$, and let $\triangle XYZ$ be the cevian triangle of *Na* with respect to $\triangle PQR$. Prove that *Na* is the circumcenter of $\triangle XYZ$.

**Exercise 5.15.** A variable triangle has fixed incenter $I$ and circumcenter $O$. Prove that the locus of *Sp* is a circle centered at the midpoint of $IO$.

**Exercise 5.16*.** For $\triangle ABC$, let $I, H, O$ be its incenter, orthocenter, and circumcenter, respectively. Prove that *Ge*, $I$, and $\mathfrak{f}_O(H)$ are collinear.

## 5.3 The Lemoine Point

### 5.3.1 The Lemoine Point

We now introduce the important Lemoine point.

**Definition 5.3.1.** In Figure 5.8, let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$. Reflect the medians $AM_a, BM_b, CM_c$ in the respective angle bisectors at $A, B, C$, obtaining $AA', BB', CC'$. These three lines are the *symmedians* of $\triangle ABC$. They are concurrent; their common point is called the *symmedian point*, the *Lemoine point*, or the *isogonal conjugate of the centroid*.

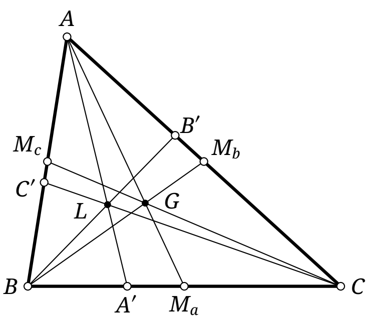

FIGURE 5.8

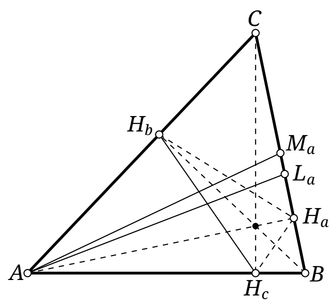

FIGURE 5.9

The definition immediately gives the following result.

**Proposition 5.3.2.** *The Lemoine point and centroid of a triangle are isogonally conjugate.*

We first establish some properties of symmedians.

**Proposition 5.3.3.** *A symmedian divides the opposite side in the squared ratio of the adjacent sides. More precisely, if the symmedian $AA'$ of $\triangle ABC$ meets $BC$ at $A'$, then $\frac{BA'}{CA'} = \left(\frac{AB}{AC}\right)^2$.*

**Proof.** Let the median $AM$ meet $BC$ at $M$, so that $BM = CM$, and apply proposition 4.4.8. □

**Proposition 5.3.4.** *A symmedian of a triangle bisects the corresponding side of its orthic triangle. Thus, if $\triangle H_aH_bH_c$ is the orthic triangle of $\triangle ABC$, then the symmedian $AL_a$ bisects $H_bH_c$, as in Figure 5.9.*

**Proof.** Let $AM_a$ be a median of $\triangle ABC$. The points $C,B,H_c,H_b$ are concyclic, and hence $\measuredangle AH_bH_c = -\measuredangle ABC$ and $\measuredangle AH_cH_b = -\measuredangle ACB$. Thus $\triangle AH_bH_c \eqsim \triangle ABC$. Since $\angle L_aAB = \angle M_aAC$, the lines $AL_a$ and $AM_a$ correspond under this opposite similarity. As $AM_a$ is a median of $\triangle ABC$, $AL_a$ is a median of $\triangle AH_bH_c$. □

We next give several important characterizations of the Lemoine point.

**Theorem 5.3.5** (Grebe's first theorem)**.** *The ratio of the [directed] distances from the Lemoine point to the three sides of a triangle equals the ratio of the corresponding side lengths. Conversely, any point whose [directed] distances have that ratio is the Lemoine point.*

**Proof.** Apply proposition 0.5.3 and theorem 4.4.9. □

**Theorem 5.3.6** (Grebe's second theorem)**.** *Among all points of the plane, the Lemoine point of $\triangle ABC$ minimizes the sum of the squares of the distances to the three sides.*

**Proof.** Let the distances from $P$ to the three sides be $d_a,d_b,d_c$, and let the side lengths be $a,b,c$. A minimizing point plainly lies inside the triangle. By the Cauchy inequality,[4]

$$(a^2+b^2+c^2)(d_a^2+d_b^2+d_c^2) \geq (ad_a+bd_b+cd_c)^2 = 4S^2_{\triangle ABC}.$$

The equality condition, together with theorem 5.3.5, identifies the minimizer as the Lemoine point. □

**Lemma 5.3.7.** *The line joining a vertex of a triangle to the pole, with respect to its circumcircle, of the opposite side is a symmedian.*

**Proof.** For $\triangle ABC$ in Figure 5.10, let $A_1$ be the pole of $BC$ with respect to $\mathscr{C}$. It is the inverse in $\mathscr{C}$ of the midpoint $M_a$ of $BC$. Let $P$ be the midpoint of the arc $\widehat{BC}$. Then $M_a,P,A_1$ are collinear, and theorem 4.1.10 shows that $AP$ bisects $\angle M_aAA_1$. But $AP$ also bisects $\angle BAC$. Hence $AM_a$ and $AA_1$ are isogonal lines at $A$. □

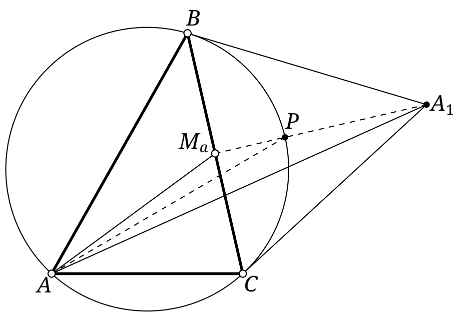


FIGURE 5.10

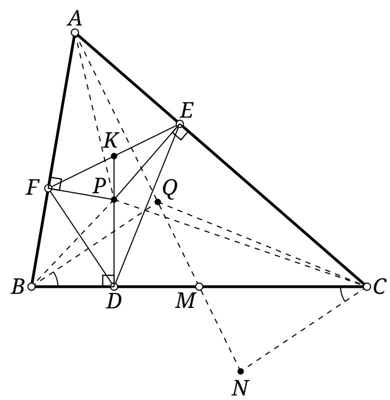


FIGURE 5.11

[4] The Cauchy inequality states that $(\sum_{i=1}^n a_i^2)(\sum_{i=1}^n b_i^2) \geq (\sum_{i=1}^n a_ib_i)^2$. If some $b_i \neq 0$, equality holds precisely when all ratios $\frac{a_i}{b_i}$ with $b_i \neq 0$ are equal and $a_i = 0$ whenever $b_i = 0$. Geometrically, in $\mathbb{R}^n$ it says that the product of the squared magnitudes of two vectors is at least the square of their inner product.

**Proposition 5.3.8.** *The tangential triangle of a triangle is the anticevian triangle of its Lemoine point.*

**Proof.** This is immediate from lemma 5.3.7. □

**Corollary 5.3.9.** *A projectivity that leaves a triangle's circumcircle invariant sends its Lemoine point to the Lemoine point of the image triangle.*

**Proof.** By proposition 5.3.8, the Lemoine point is constructed as the center of perspectivity of the triangle and its tangential triangle. The operations in this construction—drawing tangents and taking a center of perspectivity— are preserved by projectivities. If the circumcircle is invariant, the image of the Lemoine point is therefore the Lemoine point of the image triangle. □

**Corollary 5.3.10.** *The Gergonne point Ge of a triangle is the Lemoine point of its intouch triangle.*

**Proof.** This follows from the definition of the intouch triangle and proposition 5.3.8. □

Finally, the Lemoine point has the following important property.

**Theorem 5.3.11** (Lemoine's theorem)**.** *A point is the centroid of its own pedal triangle with respect to $\triangle ABC$ if and only if it is the Lemoine point of $\triangle ABC$.*

**Proof.** In Figure 5.11, let $P,Q$ be isogonally conjugate points, and let $\triangle DEF$ be the pedal triangle of $P$. Put $AQ \cap BC = M$, draw $CN \parallel BQ$ with $N \in AM$, and put $DP \cap EF = K$. Using $P = \mathbf{g}Q$ in both chains, followed by the concyclicity of $P,D,B,F$ and $P,E,A,F$ in the first, and of $P,D,B,F$ and $P,D,C,E$ in the second, gives

$$\angle CNQ = \angle BQM = \angle QBA + \angle QAB = \angle PBC + \angle PAC = \angle PFE + \angle PFD = \angle DFE,$$
$$\angle NCQ = \angle NCB + \angle QCB = \angle QBC + \angle QCB = \angle PBA + \angle PCA = \angle PDF + \angle PDE = \angle FDE.$$

Consequently $\triangle CNQ \sim \triangle DFE$. Moreover, using $P = \mathbf{g}Q$ and then the concyclicity of $P,D,B,F$, we obtain $\angle NCM = \angle QBC = \angle PBA = \angle PDF = \angle FDK$, so $CM$ and $DK$ are corresponding lines of this similarity, and $M,K$ are corresponding points.

For sufficiency, let $P$ be the Lemoine point, so $Q$ is the centroid and $M$ is the midpoint of $BC$. The quadrilateral $BNCQ$ is a parallelogram; hence $CM$ is a median of $\triangle CNQ$, and $DK$ is a median of $\triangle DEF$. Similarly, $FP$ and $EP$ are the other two medians of $\triangle DEF$.

Conversely, suppose that $P$ is the centroid of $\triangle DEF$. Then $K$ is the midpoint of $EF$, so $M$ is the midpoint of $NQ$. The quadrilateral $BNCQ$ is therefore a parallelogram, and $M$ is the midpoint of $BC$. Thus $AM$ is a median of $\triangle ABC$; similarly $BQ$ and $CQ$ are the other two medians. Hence $Q$ is the centroid and $P$ the Lemoine point. □

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## Exercises

**Exercise 5.17.** Let $G,L$ be the centroid and Lemoine point of $\triangle ABC$, and let $H'$ be the orthocenter of the pedal triangle of $L$. Prove that $G,L,H'$ are collinear and that $GL = LH'$.

**Exercise 5.18.** Given $\triangle ABC$, let points $D,E,F$ vary on the lines $BC,CA,AB$, respectively. Prove that if $DE^2 + EF^2 + FD^2$ attains its minimum, then $\triangle DEF$ is the pedal triangle of the Lemoine point $L$.

**Exercise 5.19.** Let $\triangle H_aH_bH_c$ be the orthic triangle of $\triangle ABC$, let $M$ be the midpoint of $BC$, and put $AM \cap H_bH_c = X$. Prove that the line $XH_a$ passes through the Lemoine point $L$ of $\triangle ABC$.

**Exercise 5.20.** Let $L$ be the Lemoine point of $\triangle ABC$.

(1) Through $L$ draw a line parallel to $BC$, meeting $AB,AC$ at $D_1,D_2$, respectively. Through $L$ draw a line parallel to $CA$, meeting $BC,BA$ at $E_1,E_2$, respectively; through $L$ draw a line parallel to $AB$, meeting $AC,BC$ at $F_1,F_2$, respectively. Prove that the six points $D_1,D_2,E_1,E_2,F_1,F_2$ are concyclic. Their circle is the *first Lemoine circle* of $\triangle ABC$.

(2) Through $L$ draw an antiparallel to $BC$, meeting $AB,AC$ at $D_1,D_2$, respectively. Through $L$ draw an antiparallel to $CA$, meeting $BC,BA$ at $E_1,E_2$, respectively; through $L$ draw an antiparallel to $AB$, meeting $AC,BC$ at $F_1,F_2$, respectively. Prove that the six points $D_1,D_2,E_1,E_2,F_1,F_2$ are concyclic. Their circle is the *second Lemoine circle* of $\triangle ABC$.

**Exercise 5.21.** Let $\triangle DEF$ be the medial triangle of $\triangle ABC$. Suppose $\odot(AED) \cap \odot(CEF) = \{P,E\}$, $\odot(BDE) \cap \odot(CDF) = \{Q,D\}$, and $PE \cap DQ = K$. Prove that $CK$ is a symmedian of $\triangle ABC$.

### 5.3.2 Applications of the Lemoine Point

We apply the preceding results to several related triangle centers.

#### The Mittenpunkt

**Theorem 5.3.12.** *The excentral triangle and medial triangle of a triangle are perspective.* Their center of perspectivity *Mi* is called the *Mittenpunkt*.[5]

**Proof.** The orthic triangle of the excentral triangle is the original triangle. Applying proposition 5.3.4 to the excentral triangle shows that the lines joining corresponding vertices of the excentral and medial triangles are the symmedians of the excentral triangle. They therefore concur at its Lemoine point. □

The proof gives the following result.

**Proposition 5.3.13.** *The Mittenpunkt of a triangle is the Lemoine point of its excentral triangle.*

We shall continue our study of the Mittenpunkt in the next section.

#### The Anticomplementary Lemoine Point

**Definition 5.3.14.** For brevity in this book, the Lemoine point $Al$ of the anticomplementary triangle of a triangle is called its *anticomplementary Lemoine point*.

**Proposition 5.3.15.** *If $H$ is the orthocenter of $\triangle ABC$, then $H = \mathbf{t}Al$.*

**Proof.** In Figure 5.12, let $\triangle A'B'C'$ be the anticomplementary triangle of $\triangle ABC$, and let the tangents to its circumcircle at $B',C'$ meet at $K$. By lemma 5.3.7, $Al \in KA'$. Draw $A'X \perp BC$ at $X$ and $AY \perp BC$ at $Y$, put $A'X \cap B'C' = Z$, and put $KA' \cap B'C' = W$. If $H$ is the orthocenter of $\triangle ABC$, then $H \in AY$. Let $M$ be the midpoint of $BC$; clearly $A,M,A'$ are collinear.

The tangential triangle of $\triangle A'B'C'$ is the anticevian triangle of $Al$ with respect to $\triangle A'B'C'$. Hence theorem 4.3.4 gives the harmonic range $(A',W,Al,K)$. By proposition 3.2.11,

$$(A',K;Al,W) = (A',K;W,Al)^{-1} = \frac{1}{1-(A',W;Al,K)} = \frac{1}{2}.$$

[5]The name is German: *mitten* means "in the middle" and *Punkt* means "point." It is not a personal name, and the word already includes the sense of "point."

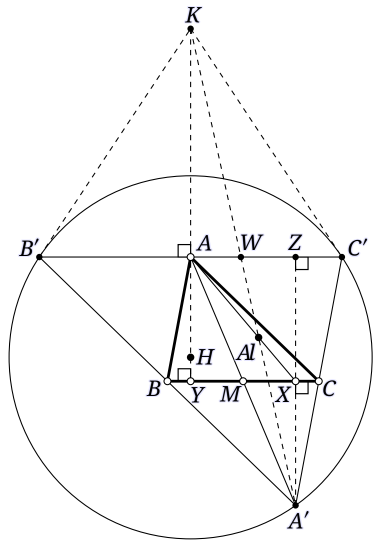


FIGURE 5.12

Therefore
$$\frac{AlA'}{AlK} = \frac{WA'}{2WK} = \frac{A'Z}{2AK} = \frac{A'X}{AK}.$$
Thus $X, Al, A$ are collinear. Since $X, Y$ are symmetric about $M$, the same argument cyclically shows that $Al$ and $H$ are isotomically conjugate. □

### The Homothety Center of the Orthic and Tangential Triangles

The following proposition introduces this center.

**Proposition 5.3.16.** *The tangential triangle and orthic triangle of a triangle are homothetic.* We denote their homothety center by $Ht$.

**Proof.** Let $\triangle L^aL^bL^c$ and $\triangle H_aH_bH_c$ be the tangential and orthic triangles of $\triangle ABC$, and let $H, O$ be its orthocenter and circumcenter. Clearly $L^bL^c \perp AO$. Since $H = \mathbf{g}O$, proposition 4.4.7 also gives $H_bH_c \perp AO$. Thus $L^bL^c \parallel H_bH_c$, and the other pairs of corresponding sides are parallel as well. Hence the two triangles are homothetic. □

**Proposition 5.3.17.** *For every triangle,* $Ht = \mathbf{g}Al$.

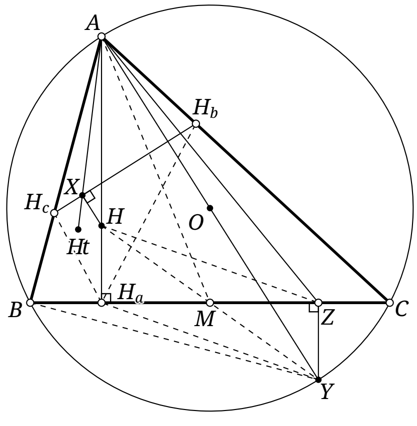


FIGURE 5.13

**Proof.** In Figure 5.13, let $\triangle H_aH_bH_c$ be the orthic triangle of $\triangle ABC$, and draw $HX \perp H_bH_c$ at $X$. The point $H$ is the incenter of the orthic triangle, and $X$ is the point where its incircle touches $H_bH_c$. The circumcenter $O$ is the incenter of the tangential triangle, and $A$ is the point where its incircle touches the corresponding side. The homothety therefore gives $Ht, X, A$ collinear.

Let $Y$ be antipodal to $A$ on the circumcircle, and draw $YZ \perp BC$ at $Z$. By lemma 0.5.16, $HBYC$ is a parallelogram. Hence $H, Y$ are symmetric about the midpoint $M$ of $BC$, as are $H_a, Z$, and $H_aH \parallel YZ$ and $H_aH = YZ$. By lemma 0.5.22 and proposition 0.6.17, $HX \cdot AY = AH \cdot H_aH = AH \cdot YZ$, and hence $\frac{XH}{ZY} = \frac{AH}{AY}$. Moreover $XH \parallel AY$ and $AH \parallel YZ$, so $\angle XHA = \angle ZYA$. Thus $\triangle AXH \sim \triangle AZY$. It follows that $\angle XAH = \angle ZAY$. Since $\angle BAH = \angle OAC$, we have $\angle BAX = \angle CAZ$: the lines $AHt$ and $AZ$ are isogonal at $A$. Cyclically, $Ht = \mathbf{gt}H$. Since $Al = \mathbf{t}H$, we obtain $Ht = \mathbf{g}Al$. □

**Proposition 5.3.18.** *For every triangle, $Ht \in \mathscr{C}$.*

**Proof.** If the triangle is acute, the incenter of its tangential triangle is its circumcenter $O$, and the incenter of its orthic triangle is its orthocenter $H$. The homothety therefore shows that $O, H, Ht$ are collinear. If the triangle is obtuse, replace the incenters in this argument by the corresponding excenters. The right-triangle case remains valid when understood as a limit. □

### The Exeter Point

Before introducing the Exeter point, call the circumcevian triangle of the centroid the *circummedial triangle*. The following theorem introduces the Exeter point.

**Theorem 5.3.19.** *The tangential triangle and circummedial triangle of a triangle are perspective.* Their center of perspectivity $Et$ is called the *Exeter point*.

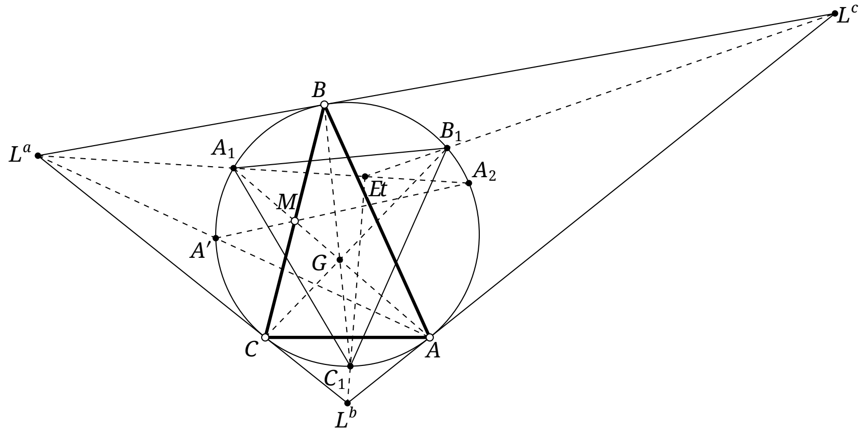


FIGURE 5.14

**Proof.** In Figure 5.14, let $\triangle L^aL^bL^c$ be the tangential triangle of $\triangle ABC$, and let $A'$ be the second intersection of $AL^a$ with $\mathscr{C}$. The line isogonal to $AL^a$ in $\angle CL^aB$ meets $\mathscr{C}$ at $A_1, A_2$, where $A_1$ and $A$ lie on opposite sides of the line $BC$.

Put $AA_1 \cap A'A_2 = M$. By proposition 3.6.11, $M$ lies on $BC$, the polar of $L^a$ with respect to $\mathscr{C}$. On the other hand, $L^aA'$ and $L^aA_1$ are isogonal in $\angle BL^aC$, so $M$ lies on its angle bisector. Thus $M$ is the midpoint of $BC$.

Since $AM$ passes through the centroid $G$, $A_1$ is a vertex of the circummedial triangle. Construct its other vertices $B_1, C_1$ cyclically, giving the other two pairs of isogonal lines. By proposition 5.3.8, the lines $AL^a, BL^b, CL^c$ concur at the Lemoine point $L$. The three corresponding isogonal lines $L^aA_1, L^bB_1, L^cC_1$ therefore concur at $Et = \mathbf{g}_{\triangle L^aL^bL^c}L$. □

The proof also gives the following proposition.

**Proposition 5.3.20.** *The Exeter point $Et$ and the Lemoine point $L$ of a triangle are isogonally conjugate with respect to the tangential triangle.*

The Exeter point also lies on the Euler line.

**Proposition 5.3.21.** *For every triangle, $Et \in \mathscr{E}$.*

For the proof, introduce one further center: the circumcenter $To$ of the tangential triangle, which for brevity in this book we call the *tangential circumcenter*.

**Lemma 5.3.22.** *For every triangle, $To \in \mathscr{E}$.*

**Proof.** Consider the homothety between the tangential and orthic triangles, whose center is $Ht$. Their circumcenters are $To$ and $Ni$, respectively, so $Ht, To, Ni$ are collinear. Since $Ht, Ni \in \mathscr{E}$, also $To \in \mathscr{E}$. □

**Proof of proposition 5.3.21.** The Lemoine point $L$ of $\triangle ABC$ is the Gergonne point of the tangential triangle, while $Et$ is its isogonal conjugate. Thus $Et$ is the internal similicenter of the tangential triangle. The circumcenter and incenter of that triangle are $To$ and the original circumcenter $O$, respectively, so $Et \in OTo$. By lemma 5.3.22, both $O$ and $To$ lie on $\mathscr{E}$, and hence so does $Et$. □

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## Exercises

**Exercise 5.22.** Prove that the medial triangle of a triangle's excentral triangle is perspective with the original triangle, with center of perspectivity $Mi$.

**Exercise 5.23.** Prove the following generalization of the Exeter point. Given $\triangle ABC$ and a point $P$, the circumcevian triangle of $P$ is perspective with the tangential triangle of $\triangle ABC$. The center of perspectivity $P'$ is called the *Exeter transform* of $P$.

**Exercise 5.24.** Let $\triangle A'B'C'$ be the anticomplementary triangle of $\triangle ABC$, and let $A'', B'', C''$ be the orthogonal projections of $A', B', C'$ on $BC, CA, AB$, respectively. Prove that $\triangle ABC$ and $\triangle A''B''C''$ are perspective from $Al$.

**Exercise 5.25.** For $\triangle ABC$, prove that the pole, with respect to $\mathscr{C}$, of the trilinear polar of its orthocenter is $Ht$.

**Exercise 5.26.** If $O$ is the circumcenter of $\triangle ABC$, prove that $Ni, To, O, Ht$ form a harmonic range.

**Exercise 5.27.** Let $G, O$ be the centroid and circumcenter of $\triangle ABC$. Prove that $G, Et, O, Ht$ form a harmonic range.

**Exercise 5.28*.** Prove that for every triangle, $Mi = \mathbf{c}Ge$.

**Exercise 5.29*.** Prove that for every triangle, $Al, Ge, Na$ are collinear.

## 5.4 The Mittenpunkt and the Tangential Perspector

Section 5.3 introduced the Mittenpunkt. In this section we focus on the Mittenpunkt and its isogonal conjugate, called here the tangential perspector. We first define the tangential perspector and derive its related properties; later we prove that it is isogonally conjugate to the Mittenpunkt.

By proposition 0.6.7, the excentral and intouch triangles are homothetic. This allows the following definition.

**Definition 5.4.1.** In Figure 5.15, the homothety center *Ta* of a triangle's excentral and intouch triangles is called its *tangential perspector*.

The homothety immediately gives some elementary properties.

**Proposition 5.4.2.** *In Figure 5.15, let $\triangle G_aG_bG_c$ be the intouch triangle of $\triangle ABC$, let $N$ be the midpoint of the arc $\widehat{BAC}$, let $S$ be the midpoint of $G_bG_c$, and let $P$ be the foot of the perpendicular from $G_a$ to $G_bG_c$. Then $N,S,Ta$ are collinear, and $A,P,Ta$ are collinear.*

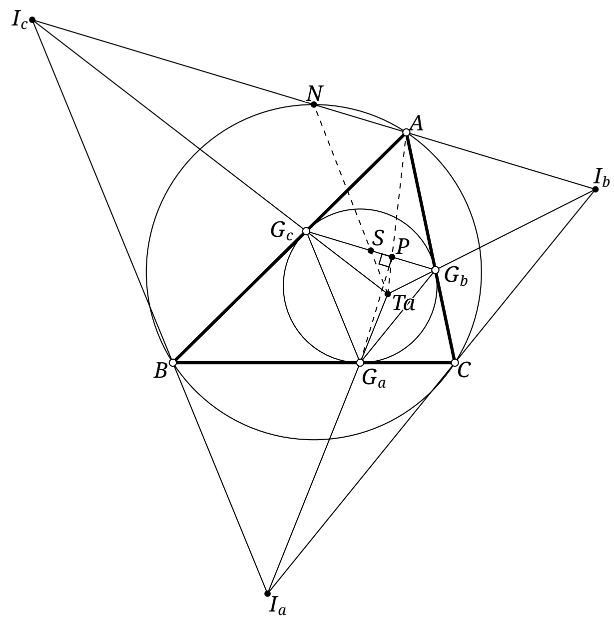


FIGURE 5.15

**Proof.** Let $\triangle I_aI_bI_c$ be the excentral triangle. Since $N$ is the midpoint of $I_bI_c$, the points $N,S$ correspond under the homothety from $\triangle I_aI_bI_c$ to $\triangle G_aG_bG_c$. Thus $N,S,Ta$ are collinear. The incenter $I$ is the orthocenter of the excentral triangle, so $I_aA \perp I_bI_c$. Hence $A,P$ are another pair of corresponding points, and $A,P,Ta$ are collinear. □

The same homothety yields a particularly attractive result.

**Proposition 5.4.3.** *If $G$ is the centroid of a triangle, then $Mi,Ta,G,Ge$ are collinear, and $Mi = \mathbf{c}Ge$; see Figure 5.16.*

**Proof.** Retain the notation of proposition 5.4.2. By corollary 5.3.10 and proposition 5.3.13, *Ge* and *Mi* are the Lemoine points of $\triangle G_aG_bG_c$ and $\triangle I_aI_bI_c$, respectively. Since the homothety center of these two triangles is *Ta*, the points $Mi,Ge,Ta$ are collinear.

Let $\triangle M_aM_bM_c$ be the medial triangle, as in Figure 5.16. Under the homothety from $\triangle I_aI_bI_c$ to $\triangle G_aG_bG_c$, the points $Mi,Ge$ correspond, and hence $I_aMi \parallel G_aGe$, or equivalently $I_aMi \parallel AGe$. By the definition of $Mi$, $M_a \in I_aMi$, so $M_aMi \parallel AG_a$. Under the homothety from $\triangle ABC$ to $\triangle M_aM_bM_c$,

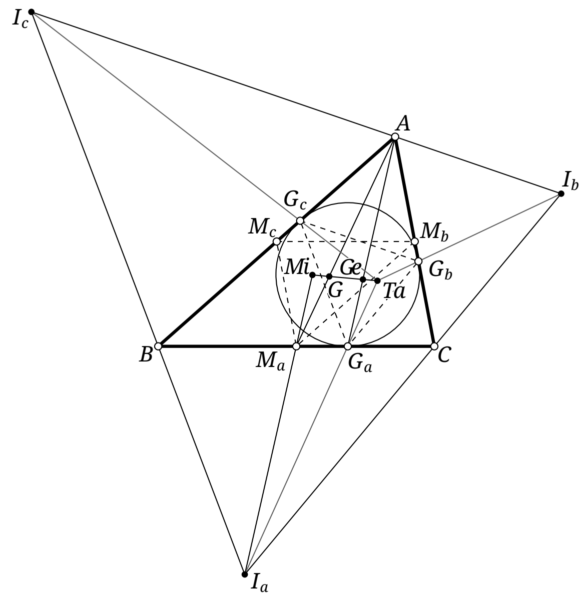


FIGURE 5.16

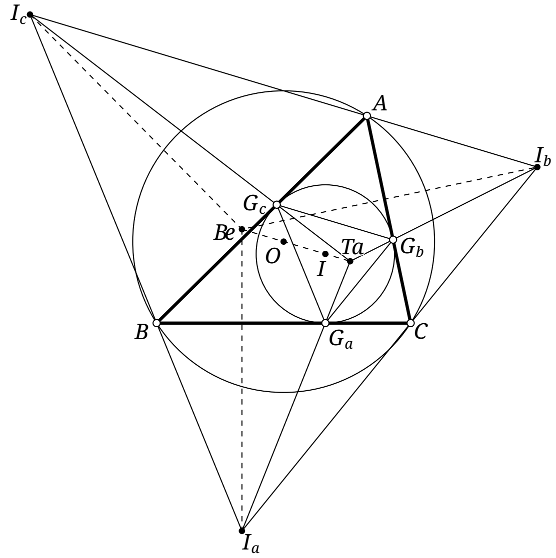


FIGURE 5.17

$M_a$,$A$ correspond; hence $M_aMi$ and $AGe$ are corresponding lines. Therefore $M_aMi$ passes through the Gergonne point of $\triangle M_aM_bM_c$. Likewise, $M_cMi$ and $M_bMi$ pass through the Gergonne point of $\triangle M_aM_bM_c$. Thus $Mi$ is the Gergonne point of $\triangle M_aM_bM_c$.

Thus $Mi$ and $Ge$ correspond under the homothety from $\triangle ABC$ to $\triangle M_aM_bM_c$, whose center is $G$. Hence $Mi, G, Ge$ are collinear and $Mi = \mathbf{c}Ge$. □

The proof also gives a small additional property.

**Proposition 5.4.4.** *The Mittenpunkt of a triangle is the Gergonne point of its medial triangle.*

Before proceeding, we introduce the Bevan point.

**Proposition 5.4.5.** *Let $\triangle I_aI_bI_c$ be the excentral triangle of $\triangle ABC$. Through $I_a$ draw the line $l_a$ perpendicular to $BC$, and define $l_b, l_c$ cyclically. The three lines are concurrent at a point $Be$; see Figure 5.17.* This point is called the *Bevan point*.

**Proof.** The incenter $I$ is the orthocenter of the excentral triangle, and $\triangle ABC$ is its pedal triangle. By proposition 4.4.7, the stated construction gives the point isogonally conjugate to $I$ with respect to $\triangle I_aI_bI_c$, proving concurrency. □

The proof yields an important characterization.

**Proposition 5.4.6.** *The Bevan point of a triangle is the circumcenter of its excentral triangle.*

**Proof.** The proof of proposition 5.4.5 showed that the incenter and the Bevan point are isogonally conjugate with respect to the excentral triangle. The incenter is the orthocenter of that triangle, so the result follows from corollary 4.4.6. □

We can now prove the following property of the tangential perspector.

**Proposition 5.4.7.** *If $I, O$ are the incenter and circumcenter of a triangle, then $Ta, I, O, Be$ are collinear, and $O$ is the midpoint of $BeI$.* Their common line is called the *OI line*, or the *Bevan line*; see Figure 5.17.

**Proof.** With respect to the excentral triangle, the points $Be, O, I$ are, respectively, its circumcenter, nine-point center, and orthocenter. Hence they are collinear and $O$ is the midpoint of $BeI$. Moreover, $Be$ and $I$ are the circumcenters of the excentral and intouch triangles. Their homothety therefore gives $Be, I, Ta$ collinear. □

**Proposition 5.4.8.** *If $H, I$ are the orthocenter and incenter of $\triangle ABC$, then $Be, Sp, Mi, H$ are collinear. Their common line is parallel to $IGe$, and $Sp$ is the midpoint of $HBe$.*

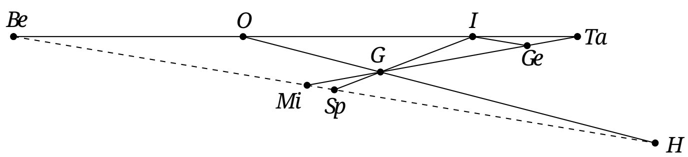


FIGURE 5.18

**Proof.** In Figure 5.18, for $\triangle ABC$, let $I, O, G, H$ be its incenter, circumcenter, centroid, and orthocenter, respectively. The Euler-line theorem gives $\frac{\overline{OG}}{\overline{GH}} = \frac{1}{2}$. By proposition 5.2.7, $\frac{\overline{IG}}{\overline{GSp}} = 2$; by proposition 5.4.7, $Be, O, I, Ta$ are collinear and $\overline{BeO} = \overline{OI}$; and by proposition 5.4.3, $Mi, G, Ge, Ta$ are collinear with $\frac{\overline{GGe}}{\overline{GMi}} = -2$.

The circumcenters of the excentral and intouch triangles are $Be$ and $I$, while their Lemoine points are $Mi$ and $Ge$. Their homothety therefore gives $BeMi \parallel IGe$. Since $\frac{IG}{GSp} = \frac{GeG}{GMi} = 2$, we also have $MiSp \parallel IGe$, and hence $Be, Mi, Sp$ are collinear.

Apply the converse of Menelaus's theorem to $\triangle IOG$ and the points $Be, Sp, H$. Since

$$\frac{IBe}{BeO} \cdot \frac{OH}{HG} \cdot \frac{GSp}{SpI} = 2 \times \frac{3}{2} \times \frac{1}{3} = 1,$$

the points $Be, Sp, H$ are collinear. Finally, Menelaus's theorem in $\triangle IBeSp$ with the transversal $\overline{OGH}$[6] gives

$$1 = \frac{IO}{OBe}\cdot\frac{BeH}{HSp}\cdot\frac{SpG}{GI} = 1\cdot\frac{BeH}{HSp}\cdot\frac{1}{2}.$$

Thus $Sp$ is the midpoint of $BeH$. □

We finish by proving that the Mittenpunkt and tangential perspector are isogonally conjugate. We first need a lemma.

**Lemma 5.4.9.** *In Figure 5.19, let $\triangle G_aG_bG_c$ be the intouch triangle and $\triangle I_aI_bI_c$ the excentral triangle of $\triangle ABC$, whose incenter is $I$. Let $N$ be the midpoint of the arc $\widehat{BAC}$, and put $TaN \cap G_aI = T$. Then*

*(1) $\angle TAB = \angle TaAC$;*

*(2) $I_c, T, G_b$ are collinear.*

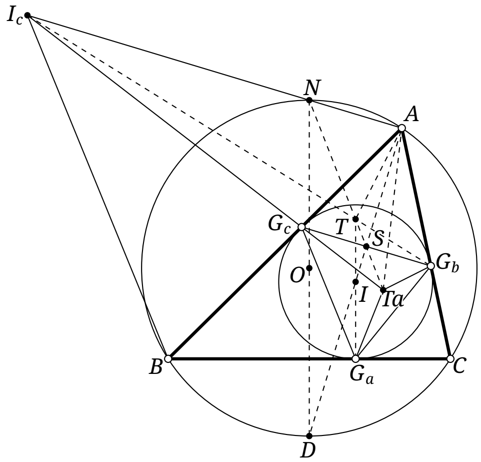


FIGURE 5.19

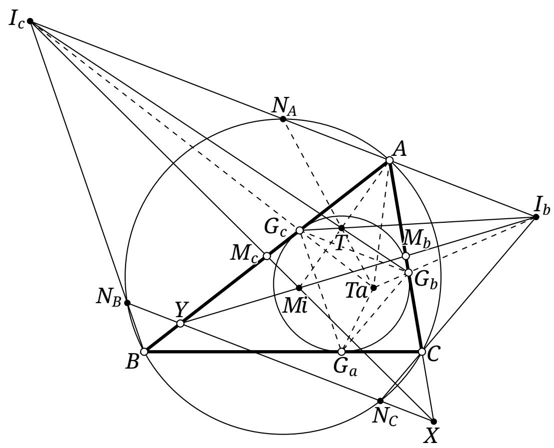


FIGURE 5.20

**Proof.** Retain the notation of proposition 5.4.2. Since $AG_c = AG_b$, the line $AS$ passes through the incenter $I$. By proposition 0.6.8, its intersection $D$ with $\odot(ABC)$ is antipodal to $N$.

The circumcenter $O$ is the midpoint of $ND$, so $(D,N;O,\infty_{DN}) = -1$, where $\infty_{DN}$ is the point at infinity of $DN$. By proposition 5.4.7, $O, I, Ta$ are collinear, so

$$(D,N,O,\infty_{DN}) \overset{(I)}{\doublebarwedge} (S,N,Ta,T).$$

Hence $(S,N;Ta,T) = -1$. Since $NA \perp AS$, proposition 3.2.17 shows that $AS$ bisects $\angle TATa$. It also bisects $\angle BAC$, and therefore $\angle TAB = \angle TaAC$.

For the second assertion, $S$ is the midpoint of $G_bG_c$, so $(G_c,G_b;S,\infty_{G_cG_b}) = -1$, where $\infty_{G_cG_b}$ is the point at infinity of $G_cG_b$. Put $I_cG_b \cap NTa = T'$. Since $I_bI_c \parallel G_bG_c$,

$$(G_c,G_b,S,\infty_{G_cG_b}) \overset{(I_c)}{\doublebarwedge} (Ta,T',S,N),$$

and hence $(Ta,T';S,N) = -1$. But also $-1 = (S,N;Ta,T) = (Ta,T;S,N)$. Uniqueness of the harmonic conjugate gives $T = T'$, so $I_c, T, G_b$ are collinear. □

[6] For three points, the overline does not denote a directed segment; it emphasizes that the three points lie on one line.

**Theorem 5.4.10.** *The Mittenpunkt and tangential perspector of a triangle are isogonally conjugate.*

**Proof.** In Figure 5.20, let $\triangle N_A N_B N_C$ be the complementary arc-midpoint triangle, $\triangle I_a I_b I_c$ the excentral triangle, and $\triangle G_a G_b G_c$ the intouch triangle of $\triangle ABC$. Let $N_B N_C$ meet the lines $AB$,$AC$ at $Y$,$X$. By example 4.4.23, $I_c$,$M_c$,$X$ and $I_b$,$M_b$,$Y$ are collinear.

By definition, $I_c M_c$ and $I_b M_b$ pass through $Mi$. Put $G_a I \cap N_A Ta = T$. By lemma 5.4.9, $I_c$,$T$,$G_b$ are collinear, and similarly $G_c$,$T$,$I_b$ are collinear. Apply Desargues's theorem to $\triangle I_b G_c Y$ and $\triangle I_c G_b X$. The circumcircle of $\triangle ABC$ is the nine-point circle of the excentral triangle, so $N_B N_C$ is a midline of $\triangle I_a I_b I_c$ and is parallel to $I_b I_c$. Thus the three lines joining corresponding vertices, $I_b I_c$,$G_c G_b$,$YX$, meet at one point at infinity. Desargues's theorem says that the intersections of corresponding sides are collinear; hence $Mi$,$T$,$A$ are collinear.

Now lemma 5.4.9 gives $\angle MiAB = \angle TaAC$. The two cyclic analogues show that $Mi$ and $Ta$ are isogonally conjugate. □

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## Exercises

**Exercise 5.30.** Prove that the tangential perspector of a triangle is

(1) the center of perspectivity of the complementary arc-midpoint triangle and the medial triangle of the intouch triangle;
(2) the center of perspectivity of the triangle and the orthic triangle of its intouch triangle.

**Exercise 5.31.** Prove that the Bevan point is the center of perspectivity of the extouch and excentral triangles.

**Exercise 5.32.** Prove that the Bevan point is the midpoint of the Nagel point and the reflection of the orthocenter in the circumcenter.

**Exercise 5.33.** Let $N$ be the midpoint of the arc $\widehat{BAC}$ of the circumcircle of $\triangle ABC$, and let $I$ be the incenter. Prove that $ATa$, $NI$, and $BC$ are concurrent.

**Exercise 5.34.** Let $I,O$ be the incenter and circumcenter of $\triangle ABC$, and let $H'$ be the orthocenter of the tangential triangle. The points $I,O,H'$ are collinear. Let $K$ be such that $O,H',I,K$ form a harmonic range. Prove that

(1) $In,H',I,Be$ form a harmonic range;
(2) $Ex,H',I,Ta$ form a harmonic range;
(3) $I,K,Be,Ta$ form a harmonic range.

**Exercise 5.35.** Let $\triangle DEF$ be the intouch triangle of $\triangle ABC$. Put $EF \cap BC = K$, let $AD$ meet $\mathscr{I}$ at $D,R$, let $\odot(BRC)$ meet $\mathscr{I}$ at $R,J$, and put $DJ \cap AK = P$. If $M$ is the midpoint of $BC$ and $PM \cap EF = Q$, prove that $D,Q,Ta$ are collinear.

**Exercise 5.36.** Prove that the trilinear polar of $Ta$ is the radical axis of $\mathscr{C}$ and $\mathscr{I}$.

## 5.5 The Clawson Point*

This optional section introduces the Clawson point. It is related to the Bevan point and to the internal and external similicenters discussed above, but is not needed substantially later in the book.

We begin with two auxiliary triangles.

**Definition 5.5.1.** In Figure 5.21a, the *intangents triangle* $\triangle A_1B_1C_1$ of $\triangle ABC$ has as its sides the three transverse common tangents, other than $AB,BC,CA$, of $\mathcal{I}$ with $\mathcal{I}_a,\mathcal{I}_b,\mathcal{I}_c$, respectively. In Figure 5.21b, the *extangents triangle* $\triangle A_2B_2C_2$ has as its sides the three direct common tangents, other than $AB,BC,CA$, of the pairs among $\mathcal{I}_a,\mathcal{I}_b,\mathcal{I}_c$.

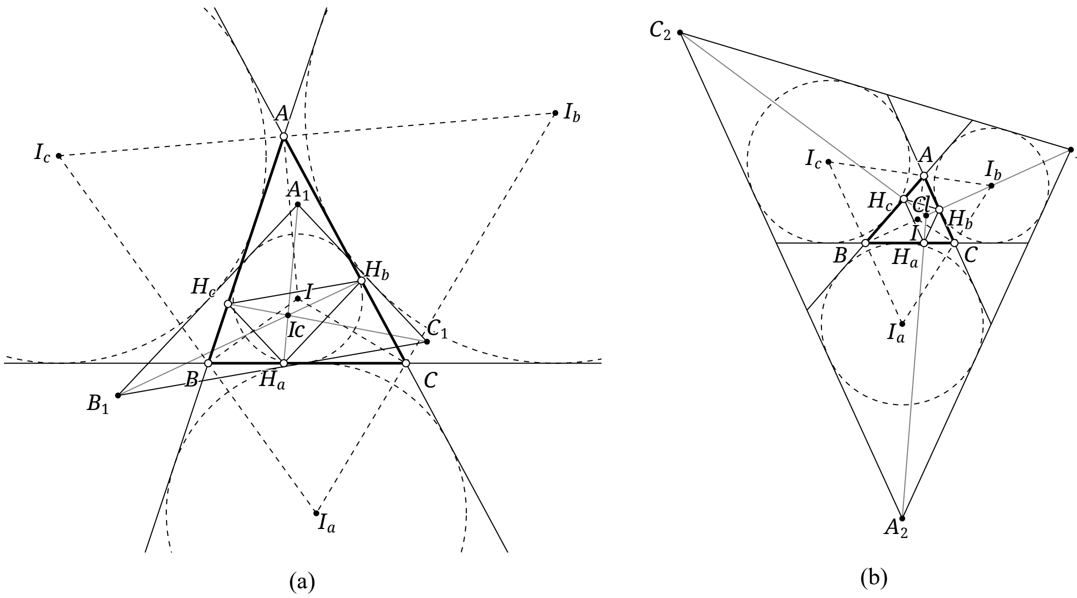


FIGURE 5.21

If $I$ and $I_a,I_b,I_c$ are the incenter and excenters of $\triangle ABC$, then the sides $B_1C_1,C_1A_1,A_1B_1$ of the intangents triangle are the reflections of $BC,CA,AB$ in $II_a,II_b,II_c$, respectively. The sides $B_2C_2,C_2A_2,A_2B_2$ of the extangents triangle are the reflections of $BC,CA,AB$ in $I_bI_c,I_cI_a,I_aI_b$, respectively.

Throughout this section, $I,H$ denote the incenter and orthocenter; $\triangle H_aH_bH_c$, $\triangle A_1B_1C_1$, $\triangle A_2B_2C_2$, $\triangle I_aI_bI_c$, $\triangle L^aL^bL^c$, $\triangle G_aG_bG_c$, and $\triangle X_aX_bX_c$ denote, respectively, the orthic, intangents, extangents, excentral, tangential, intouch, and incenter-cevian triangles of $\triangle ABC$. Its inradius and circumradius are $r,R$.

The following theorem introduces the Clawson points.

**Theorem 5.5.2.** *The triangles $\triangle A_2B_2C_2$ and $\triangle H_aH_bH_c$ are homothetic with center Cl, and $\triangle A_1B_1C_1$ and $\triangle H_aH_bH_c$ are homothetic with center Ic.* The point *Cl* is the *Clawson point*; for convenience in this book, *Ic* is called the *inner Clawson point* of $\triangle ABC$.

**Proof.** Consider the configuration in Figure 5.21; the other cases are similar. For $\triangle H_aH_bH_c$ and $\triangle A_1B_1C_1$,

$$\begin{aligned}\angle(A_1B_1,AB) &= 180^\circ - 2\angle(AB,CI_c) = 180^\circ - 2(180^\circ - \angle CBA - \angle ICB)\\ &= -180^\circ + 2\left(\angle CBA + \frac{180^\circ - \angle CBA - \angle BAC}{2}\right) = \angle CBA - \angle BAC,\\ \angle(H_aH_b,AB) &= \angle(H_aH_b,BH_b) - \angle(BH_b,AB) = (90^\circ - \angle BAC) - \angle H_cCB\\ &= 90^\circ - \angle BAC - (90^\circ - \angle CBA) = \angle CBA - \angle BAC.\end{aligned}$$

Thus $A_1B_1 \parallel H_aH_b$, and the other corresponding sides are parallel as well. The two triangles are homothetic.

Similarly, for $\triangle H_aH_bH_c$ and $\triangle A_2B_2C_2$,

$$\begin{aligned}\angle(A_2B_2,AB) &= 2\angle(I_bI_a,AB) = 2(\angle AI_aI_b - \angle BAI_a)\\ &= 2(\angle IBC - \angle BAI) = 2\left(\frac{\angle CBA}{2} - \frac{\angle BAC}{2}\right) = \angle CBA - \angle BAC,\end{aligned}$$

where the second line uses the cyclic quadrilateral $BICI_a$. Hence $A_2B_2 \parallel H_aH_b$, and cyclically the remaining corresponding sides are parallel. These triangles are also homothetic. ☐

We now establish some properties of the Clawson points.

**Proposition 5.5.3.** *For every triangle, $H,I,Ic$ are collinear.*

We first need a lemma.

**Lemma 5.5.4.** *For $\triangle ABC$, the point $I$ is also the incenter of $\triangle A_1B_1C_1$, and $A_1I \perp BC$, $B_1I \perp CA$, $C_1I \perp AB$.*

**Proof.** It is immediate that $I$ is the incenter. For the perpendicularities, use Figure 5.22. Let $\triangle G'_aG'_bG'_c$ be the intouch triangle of $\triangle A_1B_1C_1$, and put $Z = BC \cap A_1B_1$, $Y = BC \cap A_1C_1$.

Since $\mathsf{f}_{II_c}(AC) = BC$ and $\mathsf{f}_{II_c}(AB) = A_1B_1$,

$$\mathsf{f}_{II_c}(A) = \mathsf{f}_{II_c}(AC \cap AB) = BC \cap A_1B_1 = Z.$$

Thus the lines $AB$ and $ZA_1$ are symmetric about $II_c$. The points $G_c, G'_c$ are also symmetric about $II_c$, so $ZG'_c = AG_c$. Similarly, $YG'_b = AG_b$. Therefore

$$A_1Z = A_1G'_c + ZG'_c = A_1G'_c + AG_c = A_1G'_b + AG_b = A_1G'_b + YG'_b = A_1Y.$$

As $A_1I$ bisects $\angle ZA_1Y$, it follows that $A_1I \perp BC$. The other two perpendicularities follow cyclically. ☐

**Proof of proposition 5.5.3.** The points $H,I$ are the incenters of the orthic and intangents triangles, respectively. The homothety between these two triangles therefore gives $H,I,Ic$ collinear. ☐

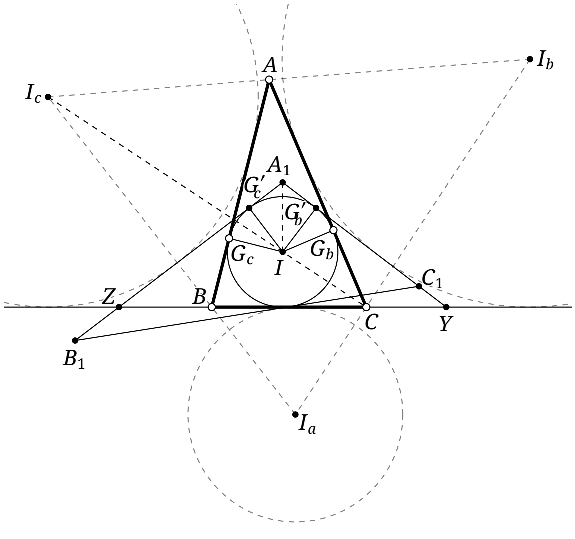


FIGURE 5.22

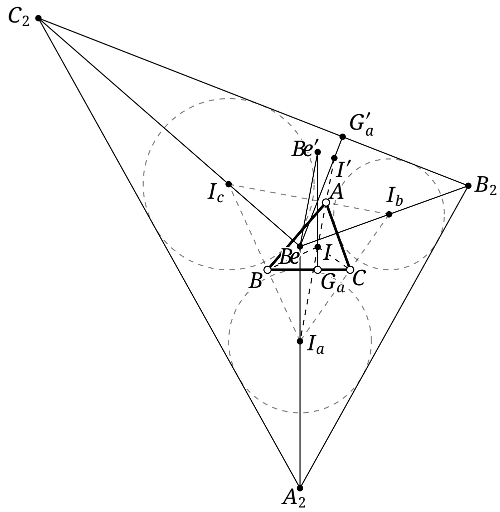


FIGURE 5.23

**Proposition 5.5.5.** *For every triangle, $H, Mi, Sp, Cl, Be$ are collinear.*

We again begin with a lemma.

**Lemma 5.5.6.** *For $\triangle ABC$, the Bevan point $Be$ is the incenter of $\triangle A_2B_2C_2$, and $A_2Be \perp BC$, $B_2Be \perp CA$, $C_2Be \perp AB$.*

**Proof.** The lines $A_2I_a, B_2I_b, C_2I_c$ are plainly the angle bisectors of $\triangle A_2B_2C_2$. As in the proof of lemma 5.5.4, one obtains $A_2I_a \perp BC$, $B_2I_b \perp CA$, and $C_2I_c \perp AB$. By proposition 5.4.5, the three lines concur at the Bevan point. □

**Proof of proposition 5.5.5.** We already know that $H, Mi, Sp, Be$ are collinear. The points $Be, H$ are the incenters of the extangents and orthic triangles. Their homothety gives $Be, H, Cl$ collinear as well. □

**Proposition 5.5.7.** *The triangles $\triangle L^aL^bL^c$, $\triangle A_1B_1C_1$, and $\triangle A_2B_2C_2$ have the common homothety center $In$. Each of them is also perspective with $\triangle X_aX_bX_c$ from the common center $In$.*

We require two lemmas.

**Lemma 5.5.8.** *The inradius of $\triangle A_2B_2C_2$ is $2R + r$.*

**Proof.** In Figure 5.23, let $Be'$ be the reflection of $Be$ in $I_bI_c$. The points $Be, I$ are the circumcenter and orthocenter of $\triangle I_aI_bI_c$. By lemma 0.5.18, $Be'I \parallel I_aBe$ and $Be'I = I_aBe$. Since $I_aBe \perp BC$, also $Be'G_a \perp BC$.

Let $\triangle G'_aG'_bG'_c$ be the intouch triangle of $\triangle A_2B_2C_2$. We have $\mathfrak{f}_{I_bI_c}(Be) = Be'$ and $\mathfrak{f}_{I_bI_c}(B_2C_2) = BC$. Since $G'_a, G_a$ are the orthogonal projections of $Be, Be'$ on $B_2C_2, BC$, respectively, $\mathfrak{f}_{I_bI_c}(G'_a) = G_a$. Put $I' := \mathfrak{f}_{I_bI_c}(I) \in BeG'_a$.

Because $I$ is the orthocenter of $\triangle I_aI_bI_c$, lemma 0.5.19 gives $I' \in \odot(I_aI_bI_c)$. The circle $\odot(ABC)$ is the nine-point circle of the excentral triangle, and $Be$ is the circumcenter of $\triangle I_aI_bI_c$. Thus $BeI' = r_{\odot(I_aI_bI_c)} = 2r_{\odot(ABC)} = 2R$. Consequently the inradius of $\triangle A_2B_2C_2$ is

$$BeG'_a = Be'G_a = G_aI + IBe' = G_aI + I'Be = 2R + r.$$

□

**Lemma 5.5.9.** *For $\triangle A_1B_1C_1$ and $\triangle A_2B_2C_2$, let $C_1A_1, A_1B_1, C_2A_2, A_2B_2$ meet $BC$ at $Y, Z, Y', Z'$, respectively. Then $I_b$ is the circumcenter of $\triangle AYZ'$, and $I_c$ is the circumcenter of $\triangle AY'Z$.*

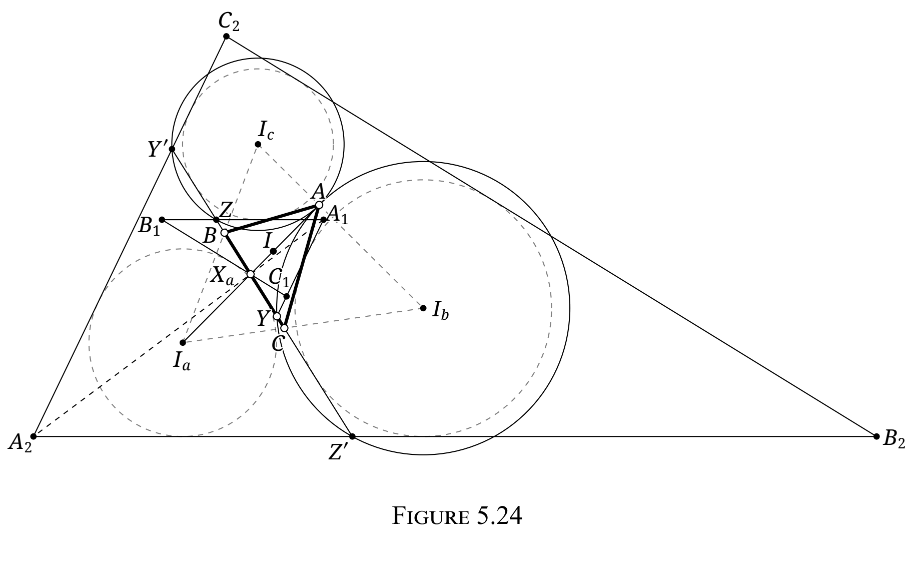


FIGURE 5.24

**Proof.** Since $A_2B_2 = \mathfrak{f}_{I_aI_b}(AB)$ and $BC = \mathfrak{f}_{I_aI_b}(AC)$,

$$\mathfrak{f}_{I_aI_b}(A) = \mathfrak{f}_{I_aI_b}(AB \cap AC) = A_2B_2 \cap BC = Z'.$$

Thus $I_aI_b$ is the perpendicular bisector of $Z'A$. Similarly, the perpendicular bisectors of $AY, ZA, AY'$ are $II_b, II_c, I_cI_a$, respectively, proving the claim. □

**Proof of proposition 5.5.7.** The triangles $\triangle L^aL^bL^c$ and $\triangle H_aH_bH_c$ are homothetic. Together with theorem 5.5.2, this shows that $\triangle L^aL^bL^c$, $\triangle A_1B_1C_1$, and $\triangle A_2B_2C_2$ are pairwise homothetic. Let their inradii be $r_L, r_1, r_2$.

Let $r, R$ be the inradius and circumradius, respectively. The internal similicenter is *In*, and proposition 5.4.7 says that *Be*, $I$ are symmetric about $O$. Therefore $\overline{InO} = \frac{\overline{IO}}{r+R} \cdot R$, $\overline{InI} = -\frac{\overline{IO}}{r+R} \cdot r$, and $\overline{InBe} = \overline{InO} + \overline{IO} = \frac{\overline{IO}}{r+R} \cdot (2R+r)$. The incenters of $\triangle L^aL^bL^c$, $\triangle A_1B_1C_1$, and $\triangle A_2B_2C_2$ are $O, I, Be$, respectively. By lemma 5.5.8, $\frac{r_L}{r_2} = \frac{R}{2R+r} = \frac{\overline{InO}}{\overline{InBe}}$ and $\frac{r_1}{r_2} = \frac{r}{2R+r} = -\frac{\overline{InI}}{\overline{InBe}}$. Thus all three triangles have the common homothety center *In*; in particular $L^a, A_1, A_2, In$ are collinear.

It remains to prove the perspectivities with $\triangle X_aX_bX_c$. In Figure 5.24, the line of centers of $\odot(AYZ')$ and $\odot(AY'Z)$ is $I_bI_c$ by lemma 5.5.9. Since $A$ is a common point of the two circles, the line through $A$ perpendicular to $I_bI_c$, namely $AI$, is their radical axis. As $X_a \in AI$, $\operatorname{pow}_{\odot(AYZ')}(X_a) = \operatorname{pow}_{\odot(AY'Z)}(X_a)$. Equivalently, $\overline{X_aY} \cdot \overline{X_aZ'} = \overline{X_aY'} \cdot \overline{X_aZ}$, and hence $\frac{\overline{X_aY}}{\overline{X_aZ}} = \frac{\overline{X_aY'}}{\overline{X_aZ'}}$. The isosceles triangles $\triangle A_1ZY$ and $\triangle A_2Z'Y'$ have equal vertex angles, so $\triangle A_1ZY \sim \triangle A_2Z'Y'$. In fact they are homothetic. By the definition of homothety, their center of homothety is $X'_a = BC \cap A_1A_2$, and the homothety gives $\frac{\overline{X'_aY}}{\overline{X'_aZ}} = \frac{\overline{X'_aY'}}{\overline{X'_aZ'}}$. Hence $X'_a = X_a$. Therefore $A_1, A_2, X_a$ are collinear. Combining this with the collinearity obtained earlier shows that $L^a, A_1, A_2, X_a, In$ are collinear. The two cyclic analogues then show that *In* is their common center of perspectivity. □

**Corollary 5.5.10.** *For every triangle, Cl, Ic, Ht, In are collinear.*

**Proof.** The proof is left as an exercise. □

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## Exercises

**Exercise 5.37.** Prove corollary 5.5.10.

**Exercise 5.38.** The external similicenter *Ex* of a triangle is the homothety center of its tangential triangle and the central reflection of its intangents triangle about its incenter. Prove this assertion.

**Exercise 5.39.** Prove that for an acute triangle the area of its extangents triangle is at least 100 times the area of its orthic triangle.

**Exercise 5.40.** Prove that the distances from *Cl* to the three sides are proportional to the tangents of the corresponding opposite angles.

**Exercise 5.41**.** Given $\triangle ABC$ and points $U, V$, let the trilinear polar of $U$ meet the sidelines of $\triangle ABC$ at $U_1, U_2, U_3$. Let $\triangle V_1V_2V_3$ be the circumcevian triangle of $V$, and let the lines $U_1V_1, U_2V_2, U_3V_3$ meet $\odot(ABC)$ again at $W_1, W_2, W_3$, respectively.

(1) *Jerabek's theorem.* Prove that $\triangle ABC$ and $\triangle W_1W_2W_3$ are perspective from a point $W$.

(2) By part (1), define the transformation $\varsigma$ by $\varsigma(\triangle ABC, U, V) = W$. For $\triangle ABC$, suppose $O, I, H$ are its circumcenter, incenter, and orthocenter, respectively, and $I_a, I_b, I_c$ are its excenters. Prove that
$$\varsigma(\triangle ABC, I, O) = \varsigma(\triangle I_a I_b I_c, H, O) = Cl.$$

## 5.6 Pseudocircles*

We now discuss a family of circles called *pseudocircles*. In general, a *pseudotangent circle* is tangent to two lines and to another circle, whereas a *pseudocircumcircle* passes through two prescribed points and is tangent to another circle. These circles are not themselves triangle centers, but they are closely connected with centers such as *In*, *Ex*, *Ta* and occur frequently in olympiad problems. This optional section is not used later in the systematic development; uninterested readers may omit it.

We begin with two useful lemmas.

**Lemma 5.6.1.** *In Figure 5.25, let $T$ be a common point of two circles. A chord $AB$ of the larger circle touches the smaller circle at $S$, and the line $ST$ meets the larger circle again at $M$. The following are equivalent:*
*(1) $T$ is a point of tangency of the two circles;*
*(2) $ST$ bisects $\angle ATB$;*
*(3) $M$ is the midpoint of the arc $\widehat{AMB}$ of the larger circle;*
*(4) the tangent to the larger circle at $M$ is parallel to $AB$.*

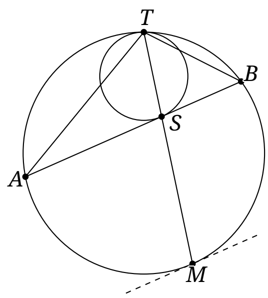


FIGURE 5.25

**Proof.** The equivalence of (2), (3), and (4) is immediate. We prove the equivalence of (1) and (4).

If (1) holds, $T$ is the external similicenter of the two circles, and $S, M$ are corresponding points. The tangent to the smaller circle at $S$ is therefore parallel to the tangent to the larger circle at $M$, proving (4).

Conversely, assume (4). Let $\omega$ be the circle tangent to the larger circle at $T$ and tangent to $AB$ at $S'$. The first part gives $T, S', M$ collinear, and hence $S = S'$. There is a unique circle through $T$ tangent to the segment $AB$ at $S$, so $\omega$ is the smaller circle and (1) holds. □

**Lemma 5.6.2.** *Fix $\triangle ABC$ and consider the axial-reflection inversion centered at $A$. Two points $P, Q$ are interchanged by this transformation if and only if $\triangle ABP \overset{\pm}{\sim} \triangle AQC$. In particular:*
*(1) $B, C$ are interchanged; $AB, AC$ are interchanged as lines; and $\mathscr{C}$ is sent to $BC$;*
*(2) the incenter $I$ and the $A$-excenter $I_a$ are interchanged;*
*(3) the circumcenter $O$ and $\mathfrak{f}_{BC}(A)$ are interchanged.*

**Proof.** By the definition of axial-reflection inversion, if $P,Q$ are interchanged, then $AB\cdot AC=AP\cdot AQ$, while $AP,AQ$ are isogonal at $A$. Hence $\triangle ABP \stackrel{\pm}{\sim} \triangle AQC$. The converse is identical.

Part (1) is immediate. For (2), the corresponding angles of $\triangle ABI$ and $\triangle AI_aC$ are equal, as a direct calculation shows. Hence $\triangle ABI \stackrel{\pm}{\sim} \triangle AI_aC$, so the incenter $I$ and the $A$-excenter $I_a$ are interchanged. For (3), put $A'=\mathfrak{f}_{BC}(A)$. Since $AA'\perp BC$, the lines $AA'$ and $AO$ are isogonal at $A$ because the orthocenter and circumcenter are isogonally conjugate. If $R$ is the circumradius, the sine rule gives

$$AA'\cdot AO=2AB\sin B\cdot R=2AB\sin B\cdot\frac{AC}{2\sin B}=AB\cdot AC.$$

Thus $A'$ and $O$ are interchanged. □

### 5.6.1 Mixtilinear Circumcircles

We first introduce the mixtilinear circumcircles of a triangle.

**Definition 5.6.3.** Given $\triangle ABC$, its $A$-*mixtilinear circumcircle* is the circle through $B,C$ internally tangent to $\mathcal{I}$; it is denoted by $\omega_A$ in Figure 5.26. The $B$- and $C$-mixtilinear circumcircles are defined cyclically, and the three are collectively called the *mixtilinear circumcircles* of the triangle.

The $A$-mixtilinear circumcircle plainly exists and is unique. We now give its principal properties.

**Proposition 5.6.4.** *Let the A-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at P, let $\mathcal{I}$ touch BC at D, and let $I_a$ be the A-excenter. Then $P,D,I_a$ are collinear.*

**Proof.** Instead define $P$ as the second intersection of $I_aD$ with $\mathcal{I}$; we prove that $\odot(BCP)$ is tangent to $\mathcal{I}$.

In Figure 5.26, let $M,N$ be the midpoints of $DI_a,PD$, and let $D'$ be the projection of $I_a$ on $BC$. Write $I$ for the incenter. Clearly $IN\perp NI_a$, while $IB\perp BI_a$ and $IC\perp CI_a$. Thus $B,C,N,I_a,I$ are concyclic. The power theorem gives $DB\cdot DC=DI_a\cdot DN=DM\cdot DP$, so $M,B,C,P$ are concyclic. By proposition 0.6.13, $D'B=DC$; because $M$ is the midpoint of $I_aD$ and $I_aD'\perp BC$, $M$ is the midpoint of the arc $\widehat{BC}$ of $\omega_A$.

Let a circle $\mathcal{I}'$ internally tangent to $\omega_A$ at $P$ touch $BC$ at $D_0$. By lemma 5.6.1, $PD_0$ passes through $M$, whence $D_0=PM\cap BC=D$. Therefore $\mathcal{I}'=\mathcal{I}$. □

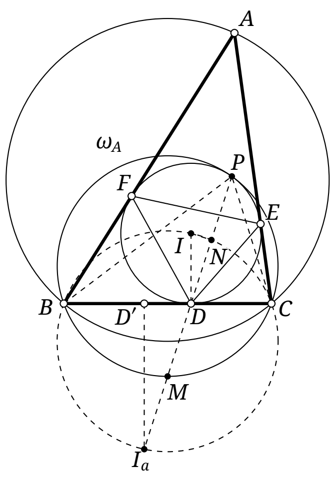


FIGURE 5.26

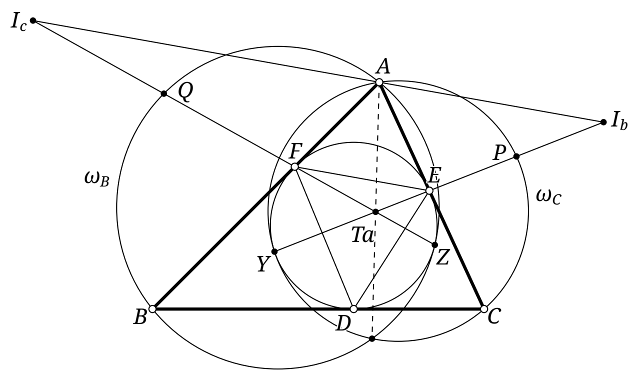


FIGURE 5.27

**Corollary 5.6.5.** *Let the A-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch its incircle at P, let the incircle touch BC at D, and let $I_a$ be its A-excenter. Then $PI_a$ bisects $\angle BPC$.*

**Proof.** Combine proposition 5.6.4 and lemma 5.6.1. ☐

**Corollary 5.6.6.** *Let the A-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch its incircle at P. Let the incircle touch BC at D, and let $I_a$ be the A-excenter. Let the segment $PI_a$ meet $\omega_A$ again at $M$. Then $M$ is the midpoint of $DI_a$ and also the midpoint of the arc $\widehat{BC}$ of $\omega_A$.*

**Proof.** This is the point $M$ constructed in the proof of proposition 5.6.4, where both assertions were established. ☐

**Proposition 5.6.7.** *The radical center of the three mixtilinear circumcircles of a triangle is Ta.*

**Proof.** Let $\omega_A, \omega_B, \omega_C$ be the three mixtilinear circumcircles of $\triangle ABC$, and let them touch $\mathcal{I}$ at $X, Y, Z$. Let $\triangle I_aI_bI_c$ and $\triangle DEF$ be its excentral and intouch triangles. By definition 5.4.1, $I_bI_c \parallel EF$ and $Ta = I_bE \cap I_cF$.

By proposition 5.6.4, $Y \in I_bE$ and $Z \in I_cF$. The four points $Y, Z, E, F$ lie on $\mathcal{I}$, so Reim's theorem gives $Y, Z, I_b, I_c$ concyclic. Since $Ta = YI_b \cap ZI_c$, the power theorem yields $\overline{TaY} \cdot \overline{TaI_b} = \overline{TaZ} \cdot \overline{TaI_c}$. Let $I_bE$ meet the arc $\widehat{AC}$ of $\omega_B$ at $P$, and let $I_cF$ meet the arc $\widehat{AB}$ of $\omega_C$ at $Q$. By corollary 5.6.6, $P, Q$ are the midpoints of $I_bE, I_cF$. As $EF \parallel I_bI_c$ by proposition 0.6.7, $\frac{\overline{PTa}}{\overline{ETa}} = \frac{\overline{QTa}}{\overline{FTa}}$, and therefore $\overline{TaY} \cdot \overline{TaP} = \overline{TaZ} \cdot \overline{TaQ}$. Thus $\operatorname{pow}_{\omega_B}(Ta) = \operatorname{pow}_{\omega_C}(Ta)$, so $Ta$ lies on the radical axis of $\omega_B, \omega_C$. Cyclically, it lies on the other radical axes as well, and hence is the radical center. ☐

**Proposition 5.6.8.** *Given $\triangle ABC$, as in Figure 5.28, let the A-mixtilinear circumcircle $\omega_A$ touch $\mathcal{I}$ at P and meet AC again at Q. Let $\mathcal{I}$ touch BC, AC at D, E, and let T be the midpoint of DE. Then:*

*(1) $\triangle PBD \sim \triangle PTE$;*
*(2) $\triangle PDT \sim \triangle PEQ$;*
*(3) $P, T, D, B$ are concyclic, as are $P, T, E, Q$;*
*(4) $\triangle PBT \sim \triangle PDE \sim \triangle PTQ$;*
*(5) $\triangle QET \sim \triangle TDB$.*

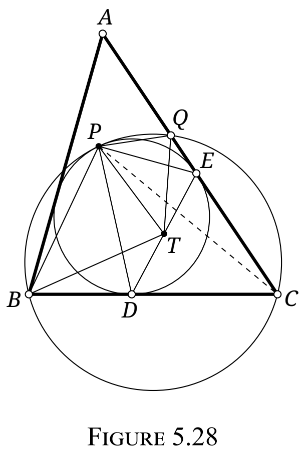

FIGURE 5.28

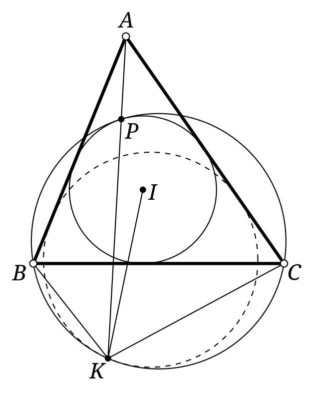

FIGURE 5.29

**Proof.** (1) We have $\angle BPD = \angle CPD = \angle TPE$, where the first equality follows from lemma 5.6.1 and the second from lemma 5.3.7. The tangent–chord theorem gives $\angle BDP = \angle TEP$. Hence $\triangle PBD \sim \triangle PTE$.

(2) Similarly, $\angle QPE = \angle CPE = \angle TPD$, by lemma 5.6.1 and lemma 5.3.7, respectively, while the tangent–chord theorem gives $\angle PEQ = \angle PDT$. Hence $\triangle PDT \sim \triangle PEQ$.

(3) Part (1) gives $\angle PBD = \angle PTE$, so $P,B,D,T$ are concyclic. Part (2) gives $\angle PTD = \angle PQE$, so $P,T,E,Q$ are concyclic.

(4) We have $\angle PTB = \angle PDB = \angle PED$, by (3) and (1), respectively. Part (3) also gives $\angle PBT = \angle PDE$, so $\triangle PBT \sim \triangle PDE$. Similarly, $\triangle PDE \sim \triangle PTQ$.

(5) We have $\angle TQE = \angle TPE = \angle BPD = \angle BTD$, by (3), (1), and (3), respectively. Also, $\angle QET = \angle BDT$ because $\triangle ECD$ is isosceles. Hence $\triangle QET \sim \triangle TDB$. □

**Proposition 5.6.9.** *Given $\triangle ABC$, let the A-mixtilinear circumcircle $\omega_A$ touch $\mathcal{I}$ at P, and let AP meet $\omega_A$ again at K. If I is the incenter, then KI bisects $\angle BKC$.*

**Proof.** In Figure 5.29, let $\omega \neq \mathcal{I}$ be the other circle tangent to $AB,AC$, and $\omega_A$. By propositions 0.4.8 and 0.4.9, the external similicenter of $\mathcal{I}$ and $\omega_A$ is $P$; that of $\mathcal{I}$ and $\omega$ is $A$; and that of $\omega$ and $\omega_A$ is their point of tangency $K'$. By proposition 0.4.10, $A,P,K'$ are collinear, so $K = K'$. The next proposition completes the proof. □

**Proposition 5.6.10.** *Let a circle $\omega_0$ touch the rays $AB,AC$ at $Y,X$. Let a circle $\omega$ pass through $B,C$ and be internally tangent to $\omega_0$ at K, with A,K on opposite sides of XY. If I is the incenter of $\triangle ABC$, then KI bisects $\angle BKC$.*

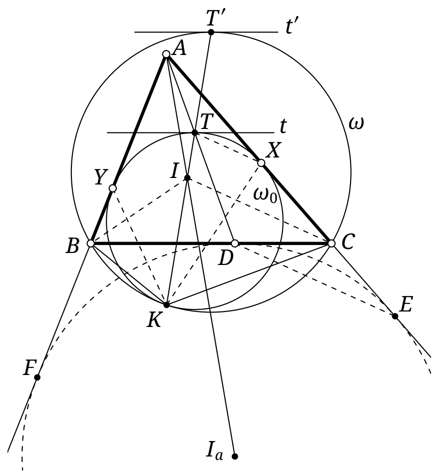


FIGURE 5.30

**Proof.** Let the $A$-excircle $\mathcal{I}_a$ touch the three sidelines at $D,E,F$. Let $T$ be the intersection of $AD$ with $\omega_0$ nearer $A$, and redefine $K$ as the second intersection of $IT$ with $\omega_0$. We prove that $\odot(BKC)$ is tangent to $\omega_0$ and that $KI$ bisects $\angle BKC$, as in Figure 5.30.

The point $A$ is the external similicenter of $\omega_0$ and $\mathcal{I}_a$, with $T,D$ and $X,E$ corresponding. Thus $TX \parallel DE$. Also $IC \parallel DE$, since both lines are perpendicular to $I_aC$; hence $TX \parallel IC$. The tangent–chord theorem gives the first equality in $\angle CXK = \angle XTK = \angle CIK$, so $I,X,C,K$ are concyclic.[7] Similarly $Y,I,B,K$ are concyclic. Therefore

$$\angle CKI = \angle IXA = \angle IYA = \angle BKI,$$

where the middle equality follows from $\triangle IXA \cong \triangle IYA$.

[7]This may be viewed as a degenerate form of Reim's theorem: regard $X$ as a repeated point, so that $XX$ is the tangent $AC$ to $\omega_0$ at $X$. From $T,K,X,X$ concyclic, Reim's theorem gives $I,X,C,K$ concyclic.

Since $T,D$ correspond under the homothety between $\omega_0$ and $\mathcal{I}_a$, $BC$ is parallel to the tangent $t$ to $\omega_0$ at $T$. Extend $KT$ to meet $\omega$ at $T'$. Since $KT'$ bisects $\angle BKC$, the tangent $t'$ to $\omega$ at $T'$ is parallel to $t$. By lemma 5.6.1, $\omega$ and $\omega_0$ are tangent at $K$. □

The preceding proposition generalizes proposition 5.6.9. The following analogue of corollary 5.6.5 can be proved in the same way; the details are left to the reader.

**Proposition 5.6.11.** *Given $\triangle ABC$, let a circle $\omega_0$ touch the rays $AB,AC$ at $Y,X$, respectively. Let a circle $\omega$ pass through $B,C$ and touch $\omega_0$ internally at $K$, with $A,K$ on the same side of $XY$. Then $KI_a$ bisects $\angle BKC$, where $I_a$ is the $A$-excenter of $\triangle ABC$.*

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## Exercises

**Exercise 5.42.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at $P$, let $\mathcal{I}$ touch $BC$ at $D$, and let $H_a$ be the projection of $A$ on $BC$. Prove that $PD$ bisects $AH_a$.

**Exercise 5.43.** Let $I,J$ be the incenter and $A$-excenter of $\triangle ABC$. Suppose $A,X,Y$ are collinear and $B,C,X,Y$ are concyclic. Prove that

$$\measuredangle BXJ + \measuredangle CXJ = 0 \quad \Longleftrightarrow \quad \measuredangle BYI + \measuredangle CYI = 0.$$

**Exercise 5.44.** Prove that the triangle formed by the three points of tangency of the mixtilinear circumcircles with $\mathcal{I}$ is perspective with the original triangle.

**Exercise 5.45.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at $P$. Put $AP \cap BC = X$, let $AP$ meet $\omega_A$ again at $K$, let $I$ be the incenter, and let $D$ be the point where $\mathcal{I}$ touches $BC$. Prove that:

(1) $P,I,D,K$ are concyclic;
(2) $\triangle PBK \sim \triangle CDK$ and $\triangle PCD \sim \triangle BDK$;
(3) $\dfrac{BD}{CD} = \sqrt{\dfrac{BK}{CK}} = \sqrt[3]{\dfrac{BX}{CX}}$.

**Exercise 5.46.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at $P$. Put $AP \cap BC = X$, let $AP$ meet $\omega_A$ again at $K$, let $I$ be the incenter, and let $D$ be the point where $\mathcal{I}_a$ touches $BC$. Prove that $\angle BID + \angle APC = 180^\circ$.

**Exercise 5.47.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at $P$, and let $AP$ meet $\omega_A$ again at $K$. If $\triangle DEF$ is the intouch triangle of $\triangle ABC$, $AD \cap EF = X$, and $I$ is the incenter, prove that $IK$ bisects $XD$.

**Exercise 5.48*.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathcal{I}$ at $P$, let $K$ be the projection of $P$ on $BC$, and let $D$ be the point where $\mathcal{I}$ touches $BC$. Prove that the midpoints of the segments $BC$, $PK$, and $GeD$ are collinear.

### 5.6.2 Mixtilinear Incircles and Excircles

We next consider the mixtilinear incircles and excircles of a triangle.

#### Mixtilinear Incircles

**Definition 5.6.12.** Given $\triangle ABC$, its $A$-*mixtilinear incircle* is the circle tangent to the rays $AB,AC$ and internally tangent to $\mathscr{C}$; it is denoted by $\omega_A$ in Figure 5.31. The $B$- and $C$-mixtilinear incircles are defined cyclically, and the three are collectively called the *mixtilinear incircles* of the triangle.

Its existence is clear from the definition. The following are its main properties.

**Theorem 5.6.13** (Mannheim's theorem)**.** *In Figure 5.31, let the $A$-mixtilinear incircle $\omega_A$ of $\triangle ABC$ touch $AB,AC$ at $D,E$. Then the midpoint of $DE$ is the incenter $I$ of $\triangle ABC$.*

**Proof.** It suffices to prove that $D,E,I$ are collinear, for $I$ lies on the bisector of $\angle A$.

Apply the axial-reflection inversion centered at $A$. The lines $AB,AC$ and the circle $\mathscr{C}$, all tangent to $\omega_A$, are sent to $AC,AB,BC$, respectively. Since tangency is preserved, $\omega_A$ is sent to the $A$-excircle $\mathscr{I}_a$.[8] The points $D,E$ are sent to the points $D',E'$ where $\mathscr{I}_a$ touches $AC,AB$, while $I$ is sent to the $A$-excenter $I_a$. The points $A,I_a,D',E'$ are concyclic, so their inverse images $I,D,E$ are collinear. ☐

**Proposition 5.6.14.** *Given $\triangle ABC$, as in Figure 5.31, let its $A$-mixtilinear incircle $\omega_A$ touch $AB,AC$ at $D,E$ and $\mathscr{C}$ at $T_a$. If $I$ is the incenter, then $B,T_a,I,D$ are concyclic, and so are $C,T_a,I,E$.*

**Proof.** Under the axial-reflection inversion centered at $A$, the circle $\mathscr{C}$ is sent to $BC$, $\omega_A$ to the $A$-excircle $\mathscr{I}_a$, $T_a$ to the point $T'_a$ where $\mathscr{I}_a$ touches $BC$, and $D$ to the point $D'$ where $\mathscr{I}_a$ touches $AC$. The points $I,B$ are sent to the $A$-excenter $I_a$ and the point $C$, respectively. Since $I_a,C,D',T'_a$ are concyclic, their inverse images $I,B,D,T_a$ are concyclic. The other assertion follows similarly. ☐

**Corollary 5.6.15.** *As in figure 5.31, let the $A$-, $B$-, and $C$-mixtilinear incircles $\omega_A,\omega_B,\omega_C$ touch $\mathscr{C}$ at $T_a,T_b,T_c$, respectively, and let $\triangle N_aN_bN_c$ be the complementary arc-midpoint triangle of $\triangle ABC$. Then $\triangle T_aT_bT_c$ and $\triangle N_aN_bN_c$ are perspective from the incenter $I$.*

**Proof.** Suppose the $A$-mixtilinear incircle touches $AB,AC$ at $D,E$, and let $I$ be the incenter of $\triangle ABC$. By proposition 5.6.14,

$$\measuredangle CT_aI = \measuredangle CEI = -\measuredangle BDI = -\measuredangle BT_aI.$$

Thus $IT_a$ bisects $\angle BT_aC$, and $T_a,I,N_a$ are collinear. The other two collinearities follow cyclically. ☐

**Proposition 5.6.16.** *Given $\triangle ABC$, as in Figure 5.31, let the $A$-mixtilinear incircle $\omega_A$ touch $\mathscr{C}$ at $T_a$, and let the $A$-excircle $\mathscr{I}_a$ touch $BC$ at $T'_a$. Then $AT_a$ and $AT'_a$ are isogonal lines at $A$.*

**Proof.** The axial-reflection inversion centered at $A$ interchanges $T_a$ and $T'_a$. ☐

**Corollary 5.6.17.** *The triangle formed by the three points where the mixtilinear incircles touch $\mathscr{C}$ is perspective with $\triangle ABC$ from $Ex$.*

**Proof.** In proposition 5.6.16, the line $AT'_a$ passes through $Na$. Since $Ex = \mathbf{g}Na$, its isogonal line $AT_a$ passes through $Ex$. The other two vertex lines do so cyclically. ☐

### Mixtilinear Excircles

We next introduce the mixtilinear excircles.

[8] Consider why the image is the excircle rather than the incircle.

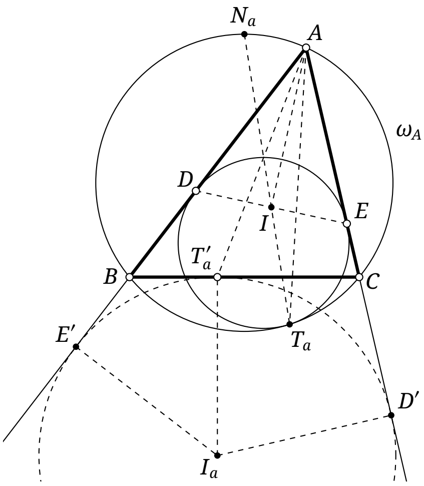


FIGURE 5.31

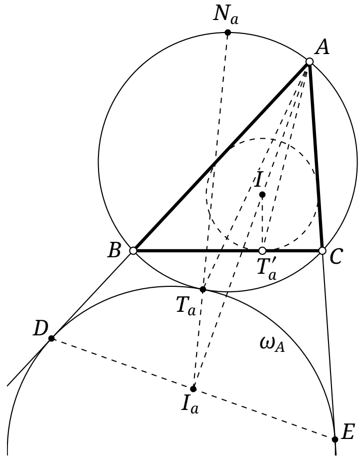


FIGURE 5.32

**Definition 5.6.18.** Given $\triangle ABC$, its *A-mixtilinear excircle* is the circle tangent to the rays $AB$,$AC$ and externally tangent to $\mathscr{C}$; it is denoted by $\omega_A$ in Figure 5.32. The $B$- and $C$-mixtilinear excircles are defined cyclically, and the three are collectively called the *mixtilinear excircles* of the triangle.

The properties of the mixtilinear excircle closely parallel those of the mixtilinear incircle. We list them, leaving the analogous proofs to the reader.

**Theorem 5.6.19** (Mannheim's theorem)**.** *Given $\triangle ABC$, as in Figure 5.32, let the A-mixtilinear excircle touch the rays $AB$,$AC$ at $D$,$E$. Then the midpoint of $DE$ is the A-excenter $I_a$.*

**Proposition 5.6.20.** *Let $\triangle ABC$ have A-excenter $I_a$. As in figure 5.32, let the A-mixtilinear excircle touch the rays $AB$,$AC$ at $D$,$E$ and $\mathscr{C}$ at $T_a$. Then $B$,$T_a$,$I_a$,$D$ are concyclic, and so are $C$,$T_a$,$I_a$,$E$.*

**Proposition 5.6.21.** *Given $\triangle ABC$, as in figure 5.32, let the A-mixtilinear excircle touch $\mathscr{C}$ at $T_a$, and let $N_a$ be the midpoint of the arc $\widehat{BAC}$ of $\mathscr{C}$. Then the line $T_aN_a$ passes through the A-excenter $I_a$.*[9]

**Proposition 5.6.22.** *Given $\triangle ABC$, use its incircle $\mathscr{I}$ and circumcircle $\mathscr{C}$. As in figure 5.32, let the A-mixtilinear excircle touch $\mathscr{C}$ at $T_a$, and let $\mathscr{I}$ touch $BC$ at $T'_a$. Then $AT_a$ and $AT'_a$ are isogonal lines at $A$.*

**Corollary 5.6.23.** *The triangle formed by the three points where the mixtilinear excircles touch $\mathscr{C}$ is perspective with $\triangle ABC$ from In.*

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## Exercises

**Exercise 5.49.** Prove theorem 5.6.19, propositions 5.6.20 to 5.6.22, and corollary 5.6.23.

**Exercise 5.50.** Let the $A$-mixtilinear incircle of $\triangle ABC$ touch $\mathscr{C}$ at $T$ and $AB$,$AC$ at $F$,$E$. Let the segment $AT$ meet the mixtilinear incircle again at $P$, and let $I$ be the incenter. Prove that:

(1) $\triangle TBI \sim \triangle TFE \sim \triangle TIC$;

[9]This is analogous to corollary 5.6.15, although corollary 5.6.15 has a perspectivity and the present result does not. Nevertheless, corollary 5.6.15 includes the conclusion that $N_aT_a$ passes through $I$, which is analogous to the present conclusion.

(2) the quadrilaterals satisfy $TBFI \sim TFPE \sim TIEC$;

(3) $\dfrac{BF}{CE} = \dfrac{TB}{TC} = \left(\dfrac{IB}{IC}\right)^2$;

(4) $BF \cdot CE = \frac{EF^2}{4}$.

**Exercise 5.51.** Let the $A$-mixtilinear excircle of $\triangle ABC$ touch $\mathscr{C}$ at $T$ and the rays $AB,AC$ at $F,E$. Let the ray $AT$ meet it again at $P$, and let $I_a$ be the $A$-excenter. Prove that:

(1) the quadrilaterals satisfy $TBFI_a \sim TFPE \sim TI_aEC$;

(2) $\dfrac{BF}{CE} = \dfrac{TB}{TC} = \left(\dfrac{I_aB}{I_aC}\right)^2$;

(3) $BF \cdot CE = \frac{EF^2}{4}$.

**Exercise 5.52.** Let $\omega_A, \omega_B, \omega_C$ be the three mixtilinear incircles of $\triangle ABC$, and let $I,O$ be its incenter and circumcenter.

(1) Let $L$ be the midpoint of the arc $\widehat{BC}$ of $\mathscr{C}$, let $\mathscr{I}$ touch $BC$ at $D$, and let $X$ be the midpoint of $ID$. Prove that $XL$ is the radical axis of $\omega_B, \omega_C$, and that $X$ is the radical center of $\mathscr{I}, \omega_B, \omega_C$.

(2) Let $P$ be the radical center of $\omega_A, \omega_B, \omega_C$. Prove that $P \in OI$ and $\frac{\overline{PO}}{\overline{PI}} = \frac{2R}{r}$, where $R,r$ are the radii of $\mathscr{C}, \mathscr{I}$, respectively.

**Exercise 5.53.** Let $\omega_A, \omega_B, \omega_C$ be the three mixtilinear excircles of $\triangle ABC$, let $I,O$ be its incenter and circumcenter, and let $r,R$ be its inradius and circumradius. Prove that:

(1) the radical center $X$ of $\mathscr{I}, \omega_B, \omega_C$ satisfies $IX \perp BC$ and $IX = 2R - \frac{r}{2}$;

(2) the radical center $P$ of $\omega_A, \omega_B, \omega_C$ lies on the line $OI$ and $\frac{\overline{PO}}{\overline{PI}} = \frac{2R}{4R-r}$.

### 5.6.3 Mixtilinear Circles of a Quadrilateral

We conclude with the quadrilateral generalization. Many of the preceding triangle results are degenerations of results about quadrilateral mixtilinear circles. The central theorem below resolves many configurations of this kind.

**Theorem 5.6.24** (Sawayama's theorem (沢山))**.** *In Figure 5.33, let ABCD be inscribed in a circle $\omega_0$. Let $\omega$ be internally tangent to $\omega_0$, tangent to $AC,BD$ at $X,Y$, and intersect the segment $AD$. If $I_1,I_2$ are the incenters of $\triangle BAD$ and $\triangle CAD$, then $I_1,I_2,X,Y$ are collinear.*

**Proof.** By lemma 4.7.4, there is a circle $\omega'$ tangent to both $\omega_0$ and $\omega$ at their common point $T$, and tangent to $AB,CD$. Let its points of tangency with those lines be $P,Q$; then $P,Q,X,Y$ are collinear.

For $\triangle ABD$, apply the axial-reflection inversion centered at $B$. It interchanges $AB,DB$, sends $AD$ to the circumcircle of $\triangle ABD$, and sends $T$ to a point $T^*$ on $AD$. The circles $\omega,\omega'$ are sent to circles $\omega^*,\omega'^*$ tangent to $AD$ at $T^*$. The points $P,Y$ are sent to $P^*,Y^*$ on the rays $BD,BA$, where $P^*,Y^*$ are the points at which $\omega'^*,\omega^*$ touch those rays. By lemma 5.6.2, $I_1$ is sent to the $B$-excenter $I_1^*$ of $\triangle ABD$; see Figure 5.34.

The centers $O_1,O_2$ of $\omega^*,\omega'^*$ lie on $AI_1^*,DI_1^*$, respectively, and we have $\triangle T^*I_1^*D \cong \triangle P^*I_1^*D$ and $\triangle T^*I_1^*A \cong \triangle Y^*I_1^*A$. Hence $\measuredangle BP^*I_1^* = -\measuredangle I_1^*T^*A = \measuredangle BY^*I_1^*$, so $B,Y^*,I_1^*,P^*$ are concyclic. Before inversion this says that $Y,I_1,P$ are collinear. Similarly $X,I_2,Q$ are collinear, proving the theorem. $\square$

This is the basic form of Sawayama's theorem. Letting $B$ and $C$ coalesce gives Mannheim's theorem. The preceding formulation imposes particular incidence and tangency positions.[10] The following version covers general positions.

[10] For example, the points of tangency are required to lie on segments rather than arbitrary lines, and the two circles are internally rather than externally tangent.

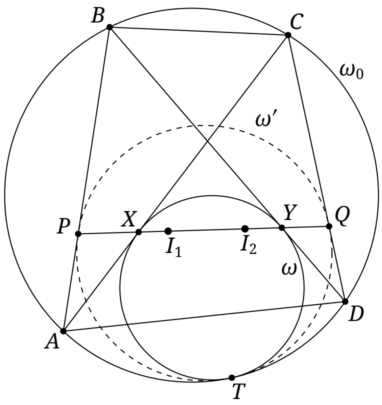


FIGURE 5.33

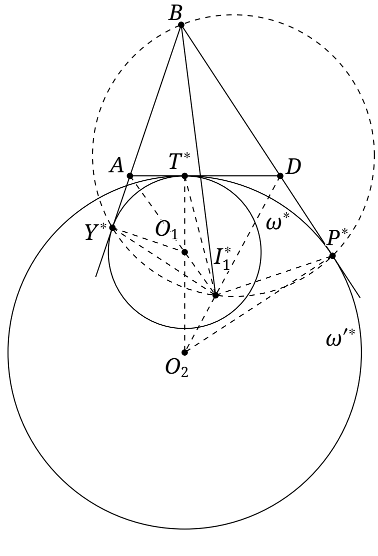


FIGURE 5.34

**Theorem 5.6.25** (Sawayama's theorem)**.** *Let $A,B,C,D$ lie on $\odot O$. A circle $\odot P$ is tangent to $\odot O$ and touches the lines $AC,DB$ at $X,Y$. Let $I_1$ be the intersection of the bisectors of $\angle ABD$ and $\angle YDA$, and let $I_2$ be the intersection of the bisectors of $\angle ACD$ and $\angle XAD$. Then $I_1,I_2,X,Y$ are collinear.*

The same configuration gives a further result.[11]

**Proposition 5.6.26.** *In Figure 5.33, let $ABCD$ be a quadrilateral inscribed in a circle $\omega_0$, and let a circle $\omega$ touch $\omega_0$ internally at $T$, touch $AC,BD$ at $X,Y$, and intersect the segment $AD$. Put $O = AC \cap BD$, and let $I,I_1,I_2$ be the incenters of $\triangle OAD$, $\triangle BAD$, and $\triangle CAD$, respectively. Then:*
*(1) $A,D,I_1,I_2$ are concyclic;*
*(2) $A,X,I_1,I,T$ are concyclic, and $D,Y,I_2,I,T$ are concyclic.*

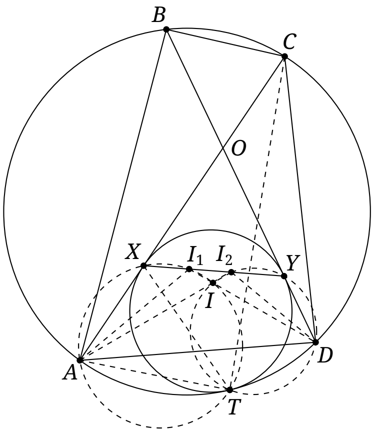


FIGURE 5.35

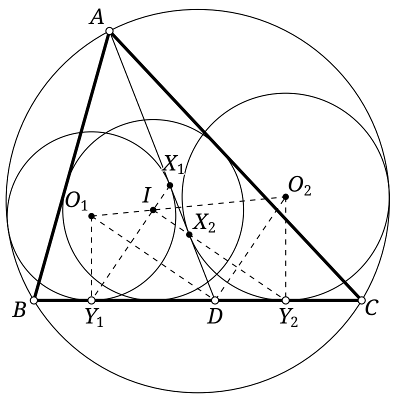


FIGURE 5.36

**Proof.** (1) Because $I_1,I_2$ are the respective incenters,

$$\angle AI_1D = 90^\circ + \frac{\angle ABD}{2} = 90^\circ + \frac{\angle ACD}{2} = \angle AI_2D.$$

Thus $A,I_1,I_2,D$ are concyclic.

[11]We state it for the positions in theorem 5.6.24; the reader may formulate the general-position version.

(2) Clearly $I_1,I,D$ are collinear, and Sawayama's theorem gives $X,I_1,I_2$ collinear. By part (1), $\angle I_2I_1I=\angle I_2AD=\angle I_2AX$, so $A,X,I_1,I$ are concyclic. Furthermore,

$$\angle XI_1A=\angle I_2DA=\frac{\angle ADC}{2}=\frac{\angle ATC}{2}=\angle ATX,$$

where the first equality follows from part (1), and the last from lemma 5.6.1. Thus $A,X,I_1,T$ are concyclic, and hence $A,X,I_1,I,T$ are concyclic. The other five-point circle follows similarly. $\square$

We give one application: Thébault's third theorem.

**Theorem 5.6.27** (Thébault's third theorem)**.** *In Figure 5.36, let $D$ be a point on side $BC$ of $\triangle ABC$, and let $I$ be its incenter. A circle $\odot O_1$ is tangent to the segments $AD,BD$ and internally tangent to $\mathscr{C}$; a circle $\odot O_2$ is tangent to the segments $AD,CD$ and internally tangent to $\mathscr{C}$. Then $O_1,O_2,I$ are collinear.*

**Proof.** Let $\odot O_1$ touch $AD,BD$ at $X_1,Y_1$, and let $\odot O_2$ touch $AD,CD$ at $X_2,Y_2$. The points $O_1,O_2$ lie on the bisectors of $\angle ADB,\angle ADC$, so $O_1D\perp O_2D$. Also $X_1Y_1\perp DO_1$ and $X_2Y_2\perp DO_2$; hence $O_1D\parallel X_2Y_2$ and $O_2D\parallel X_1Y_1$.

Moreover $O_1Y_1\parallel O_2Y_2$, and Sawayama's theorem gives $I=X_1Y_1\cap X_2Y_2$. Apply Pappus's theorem to the collinear points $\infty_{O_1Y_1},\infty_{X_2Y_2},\infty_{X_1Y_1}$ and the collinear points $D,Y_1,Y_2$. It follows that $O_1,O_2,I$ are collinear. $\square$

---

### Exercises

**Exercise 5.54.** Use Sawayama's theorem to prove propositions 5.6.10 and 5.6.11.

**Exercise 5.55.** The statement of proposition 5.6.26 and Thébault's third theorem impose particular relative positions on the lines and circles. Formulate both results for general positions.

**Exercise 5.56.** Let $D$ be a point on side $BC$ of $\triangle ABC$, and let $I$ be the incenter of $\triangle ABC$. Let $\odot O_1$ be tangent to the segments $AD,BD$ and internally tangent to $\mathscr{C}$, and let $\odot O_2$ be tangent to the segments $AD,CD$ and internally tangent to $\mathscr{C}$. Put $\theta=\frac{1}{2}\angle ADB$. Prove that:

(1) $\frac{IO_1}{IO_2}=\tan^2\theta$;

(2) $r_{\odot O_1}\cos^2\theta+r_{\odot O_2}\sin^2\theta=r_{\mathscr{I}}$.

**Exercise 5.57.** Let the $A$-mixtilinear circumcircle $\omega_A$ of $\triangle ABC$ touch $\mathscr{T}$ at $P$. Let $I$ be the incenter, $D$ its projection on $BC$, and $X,Y$ the second intersections of $\omega_A$ with $AB,AC$, respectively. If $J$ is the $A$-excenter of $\triangle AXY$, prove that $P,J,I,D$ are concyclic.

## 5.7 The Brocard Points

We have already introduced the triangle centers *Ge*, *Na*, *Mi*, *Ta*, *Be*, *In*, *Ex*, together with the related Clawson point and mixtilinear circles; these objects are closely interwoven. Beginning with this section, we turn to the Brocard points, the Apollonius point, and the Torricelli point, another group of intimately related triangle centers.

### 5.7.1 The Brocard Points

**Definition 5.7.1.** In Figures 5.37a and 5.37b, there is a point $Br_1$ inside $\triangle ABC$ such that

$$\angle BABr_1 = \angle CBBr_1 = \angle ACBr_1,$$

and a point $Br_2$ inside $\triangle ABC$ such that

$$\angle ABBr_2 = \angle BCBr_2 = \angle CABr_2.$$

The point $Br_1$ is the *first Brocard point*, and $Br_2$ is the *second Brocard point*; together they are called the *Brocard points*.

We prove existence and uniqueness by giving a construction of $Br_1$.

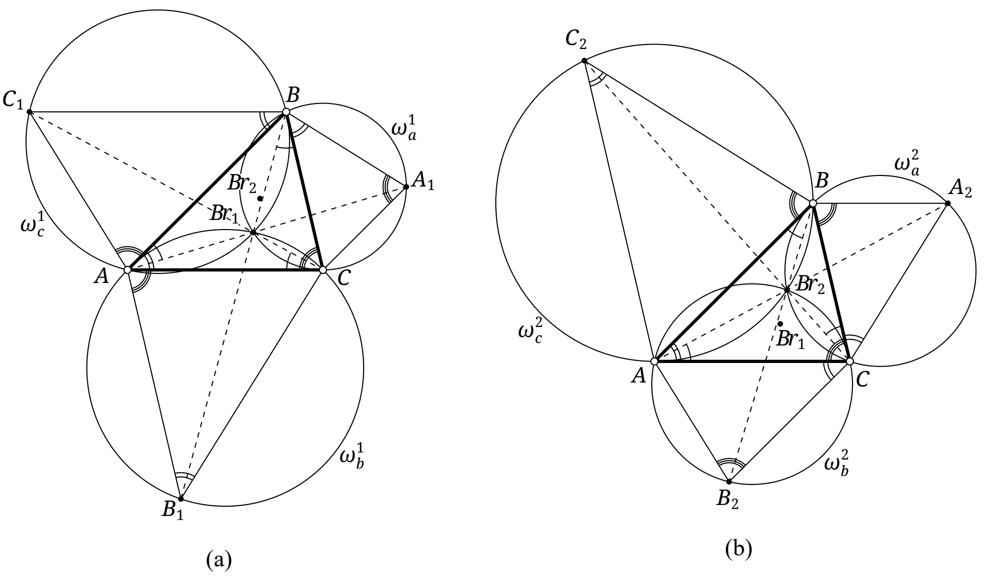


FIGURE 5.37

In Figure 5.37a, construct externally on $BC, CA, AB$ the triangles $\triangle BCA_1, \triangle B_1CA, \triangle BC_1A$, each oppositely similar to $\triangle ABC$, and denote their circumcircles by $\omega_a^1, \omega_b^1, \omega_c^1$, respectively. We show that these three circles have a common point satisfying the defining condition for the first Brocard point.

Let $Br_1$ be the second intersection of $\omega_a^1$ and $\omega_b^1$, the first being $C$. Then

$$\begin{aligned}\angle ABr_1B &= 2\pi - \angle ABr_1C - \angle BBr_1C = \angle BA_1C + \angle AB_1C \\ &= \angle BAC + \angle BCA = \pi - \angle B = \pi - \angle AC_1B,\end{aligned}$$

so $Br_1 \in \omega_c^1$. Moreover,

$$\angle CBBr_1 = \pi - \angle BBr_1C - \angle BCBr_1 = \angle BA_1C - \angle BCBr_1 = \angle BCA - \angle BCBr_1 = \angle ACBr_1.$$

Similarly, $\angle CBBr_1 = \angle BABr_1$, and hence $Br_1$ is the first Brocard point.

For uniqueness, suppose that $Br_1'$ is another first Brocard point. The same calculation gives $\angle BBr_1'A = 180° - \angle ABC$, so $Br_1' \in \omega_c^1$; similarly it lies on $\omega_a^1$ and $\omega_b^1$. Thus $Br_1' = Br_1$. The existence and uniqueness of $Br_2$ follow in the same way. □

We next record several interesting properties of the Brocard points.

**Proposition 5.7.2.** *The first and second Brocard points of a triangle are isogonally conjugate.*

**Proof.** For $\triangle ABC$, the definition of isogonal conjugation shows that $\mathbf{g}Br_1$ satisfies the defining conditions for $Br_2$. Since $Br_2$ is unique, $\mathbf{g}Br_1 = Br_2$. □

**Corollary 5.7.3.** *For $\triangle ABC$,*

$$\angle BABr_1 = \angle CBBr_1 = \angle ACBr_1 = \angle ABBr_2 = \angle BCBr_2 = \angle CABr_2 := \varpi.$$

The angle $\varpi$ is called the *Brocard angle.* [12]

**Corollary 5.7.4.** *For $\triangle ABC$, there is an inscribed ellipse whose foci are $Br_1, Br_2$.* It is called the *Brocard ellipse.*

**Proof.** Apply theorem 4.4.14. ☐

**Proposition 5.7.5.** *In Figure 5.37a, construct externally $\triangle BCA_1, \triangle B_1CA, \triangle BC_1A$, each oppositely similar to $\triangle ABC$. Then:*

*(1) the circumcircles of $\triangle BCA_1, \triangle B_1CA, \triangle BC_1A$ have the common point $Br_1$;*
*(2) the lines $AA_1, BB_1, CC_1$ concur at $Br_1$.*

**Proof.** The first assertion was proved while establishing the existence and uniqueness of $Br_1$. For the second, angle calculation gives

$$\angle ABr_1A_1 = \angle ABr_1C + \angle CBr_1A_1 = 180^\circ - \angle AB_1C + \angle CBA_1 = 180^\circ - \angle BAC + \angle BAC = 180^\circ.$$

Hence $Br_1 \in AA_1$. Similarly, $Br_1 \in BB_1, CC_1$. ☐

**Proposition 5.7.6.** *In Figure 5.37b, construct externally $\triangle CA_2B, \triangle CAB_2, \triangle C_2AB$, each oppositely similar to $\triangle ABC$. Then:*

*(1) the circumcircles of $\triangle BCA_2, \triangle B_2CA, \triangle BC_2A$ have the common point $Br_2$;*
*(2) the lines $AA_2, BB_2, CC_2$ concur at $Br_2$.*

**Proof.** This is entirely analogous to proposition 5.7.5; the details are left to the reader. ☐

**Proposition 5.7.7.** *In $\triangle ABC$,*

$$\cot\varpi = \frac{AB^2 + BC^2 + CA^2}{4S_{\triangle ABC}} = \cot A + \cot B + \cot C.$$

**Proof.** Applying corollary 0.2.7 to $\triangle ABr_1B$ gives $\cot\varpi = \frac{AB^2 + Br_1A^2 - Br_1B^2}{4S_{\triangle ABBr_1}}$, and there are two cyclic analogues. Hence, by the elementary property of proportions, using corollary 0.2.7 for the last equality, [13]

$$\cot\varpi = \frac{\sum_{\text{cyc}}(AB^2 + Br_1A^2 - Br_1B^2)}{\sum_{\text{cyc}} 4S_{\triangle ABBr_1}} = \frac{AB^2 + BC^2 + CA^2}{4S_{\triangle ABC}} = \sum_{\text{cyc}} \frac{CA^2 + AB^2 - BC^2}{4S_{\triangle ABC}} = \sum_{\text{cyc}} \cot A. \quad ☐$$

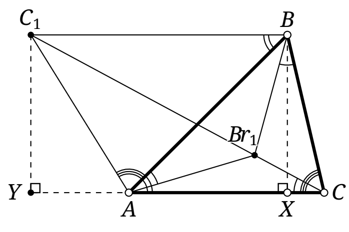


FIGURE 5.38

[12] The Brocard angle of $\triangle ABC$ will be denoted by $\varpi_{\triangle ABC}$, or simply by $\varpi$ when no ambiguity can arise.

[13] The subscript cyc indicates cyclic summation, here over the cyclic permutations of $A, B, C$. Thus, for example, $\sum_{\text{cyc}} 4S_{\triangle ABBr_1} = 4S_{\triangle ABBr_1} + 4S_{\triangle BCBr_1} + 4S_{\triangle CABr_1}$.

**Alternative proof.** In Figure 5.38, construct externally $\triangle BC_1A \stackrel{-}{\sim} \triangle ABC$. Let $X, Y$ be the orthogonal projections of $B, C_1$ onto $AC$. Since $AC \parallel BC_1$, we have $BX = C_1Y$ and $\angle ABC = \angle BC_1A = \angle C_1AY$. By proposition 5.7.5, $Br_1 \in CC_1$. Therefore

$$\begin{aligned}\cot A + \cot B + \cot C &= \cot\angle CAB + \cot\angle C_1AY + \cot\angle ACB \\ &= \frac{AX}{BX} + \frac{AY}{C_1Y} + \frac{CX}{BX} = \frac{AX+AY+CX}{C_1Y} = \frac{CY}{C_1Y} = \cot\angle C_1CY = \cot\varpi. \qquad \square\end{aligned}$$

The identity $\cot\varpi = \frac{AB^2+BC^2+CA^2}{4S_{\triangle ABC}}$ then follows from corollary 0.2.7, exactly as in the first proof.

**Proposition 5.7.8.** *The pedal triangle and the circumcevian triangle of either Brocard point of $\triangle ABC$ are similar to $\triangle ABC$.* [14]

**Proof.** In Figure 5.39, let $\triangle A'B'C'$ be the pedal triangle of $Br_1$. The points $A, B', Br_1, C'$ are concyclic, as are $A', C, B', Br_1$. Hence $\angle C'B'Br_1 = \angle C'ABr_1 = \angle ACBr_1$ and $\angle A'B'Br_1 = \angle A'CBr_1$, so $\angle A'B'C' = \angle C$. Similarly, $\angle B'A'C' = \angle B$ and $\angle A'C'B' = \angle A$. Thus $\triangle ABC \stackrel{+}{\sim} \triangle C'A'B'$. By theorem 4.3.9, the circumcevian triangle of $Br_1$ is also similar to the original triangle. The proof for $Br_2$ is identical. $\square$

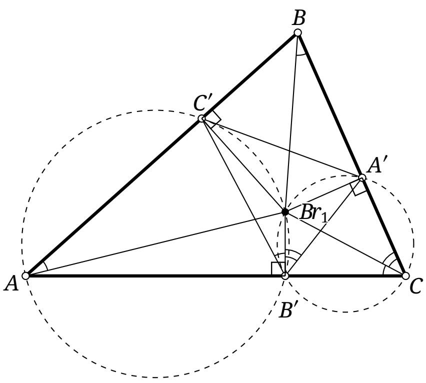


FIGURE 5.39

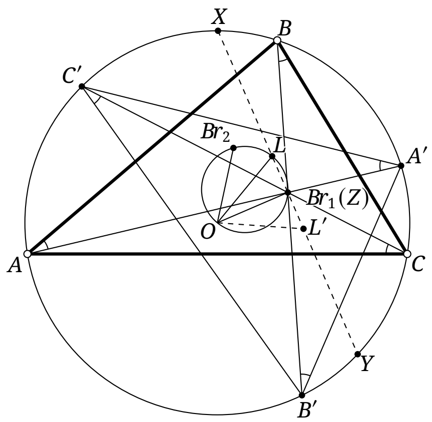


FIGURE 5.40

**Proposition 5.7.9.** *If $O$ is the circumcenter of $\triangle ABC$, then $OBr_1 = OBr_2$ and $\angle Br_1OBr_2 = 2\varpi$.*

**Proof.** In Figure 5.40, let $\triangle A'B'C'$ be the circumcevian triangle of $Br_1$. Consider the rotation $f := \mathfrak{r}_{O,-2\varpi}$. Since $\angle AOC' = 2\angle ACC' = 2\varpi$, we have $f(A) = C'$; similarly, $f(B) = A'$ and $f(C) = B'$. Thus $f(\triangle ABC) = \triangle A'B'C'$.

The point $f(Br_2)$ is the second Brocard point of $\triangle A'B'C'$. On the other hand, $\angle C'A'Br_1 = \angle Br_1CA = \varpi$, and likewise $\angle B'C'Br_1 = \angle A'B'Br_1 = \varpi$. Hence $Br_1$ itself is the second Brocard point of $\triangle A'B'C'$. By uniqueness, $f(Br_2) = Br_1$, proving the assertion. $\square$

**Proposition 5.7.10.** *If $O$ and $R$ are the circumcenter and circumradius of $\triangle ABC$, then*

$$OBr_1 = OBr_2 = R\sqrt{1 - 4\sin^2\varpi}.$$

[14] The order of corresponding vertices is not the usual one. If the pedal triangle of $Br_1$ is $\triangle A'B'C'$, for example, the correct correspondence is $\triangle ABC \sim \triangle C'A'B'$, not $\triangle ABC \sim \triangle A'B'C'$.

**Proof.** In Figure 5.40, the triangle $\triangle C'A'B'$ is obtained from $\triangle ABC$ by rotating clockwise through $2\varpi$ about $O$. Hence

$$\angle BBr_1A' \stackrel{\mathrm{m}}{=} \frac{\widehat{BA'}+\widehat{AB'}}{2} = \frac{\widehat{AB'}}{2}+\frac{\widehat{AC'}}{2} = \frac{\widehat{C'B'}}{2} \stackrel{\mathrm{m}}{=} \angle Br_1CB',$$

$$\angle Br_1BA' \stackrel{\mathrm{m}}{=} \frac{\widehat{A'B'}}{2} = \frac{\widehat{BC}}{2} \stackrel{\mathrm{m}}{=} \angle Br_1B'C.$$

Thus $\triangle Br_1B'C \sim \triangle A'BBr_1$. By the power-of-a-point theorem,

$$-\mathrm{pow}_{\mathscr{C}}(Br_1) = R^2 - OBr_1^2 = BBr_1 \cdot B'Br_1 = BA' \cdot B'C = BA'^2 = (2R\sin\varpi)^2.$$

Here $BBr_1 \cdot B'Br_1 = BA' \cdot B'C$ follows from the preceding similarity, and the last equality follows from the sine rule. Therefore $OBr_1 = R\sqrt{1-4\sin^2\varpi}$. □

**Proposition 5.7.11.** *Let $O$ be the circumcenter of a triangle. Its Lemoine point lies on the circumcircle of $\triangle OBr_1Br_2$ and is antipodal to $O$ on that circle.*

**Proof.** In Figure 5.40, let $\triangle A'B'C'$ be the circumcevian triangle of $Br_1$. Let $O$ be the circumcenter of $\triangle ABC$, and let $L, L'$ be the Lemoine points of $\triangle ABC, \triangle A'B'C'$, respectively. Under the rotation $f := \mathrm{r}_{O,-2\varpi}$ we have $f(\triangle ABC) = \triangle A'B'C'$, and hence $f(L) = L'$. If $LL' \cap \odot O = X, Y$, then $LX = L'Y$; thus $XY$ and $LL'$ have the same midpoint, say $Z$.

By lemma 3.4.8, there is a projectivity $g$ preserving $\mathscr{C} = \odot(\triangle ABC)$ and satisfying $g(Br_1) = O$. Denote images under $g$ by asterisks. The triangles $\triangle A^*B^*C^*$ and $\triangle A'^*B'^*C'^*$ are centrally symmetric about $O$, so their Lemoine points and $O$ are collinear. By corollary 5.3.9, $g$ preserves Lemoine points; therefore $L, L', Br_1$ were collinear before the transformation.

Now $X^*, Y^*$ are the intersections of $L^*L'^*$ with $\mathscr{C}$, and $L^*, L'^*$ are centrally symmetric about $O$. Hence $X^*, Y^*$ are also centrally symmetric about $O$, and exercise 3.40 gives $Z^* = O$. Since also $Br_1^* = O$, we obtain $Z = Br_1$. Thus $Br_1$ is the midpoint of $LL'$. Because $OL = OL'$, we have $OBr_1 \perp LL'$, and consequently $OL$ is a diameter of $\odot(OBr_1Br_2)$. □

**Proposition 5.7.12.** *The contact triangle of the Brocard ellipse with the sides of a triangle is perspective with the original triangle; the center of perspectivity is the Lemoine point.*

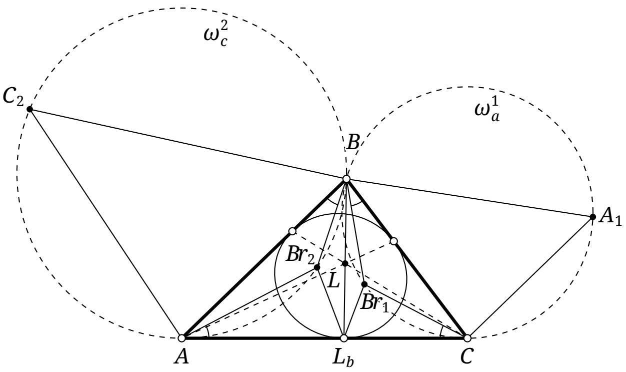


FIGURE 5.41

**Proof.** In Figure 5.41, let $L$ be the Lemoine point of $\triangle ABC$ and put $BL \cap AC = L_b$. Construct externally $\triangle BCA_1, \triangle C_2AB$, each oppositely similar to $\triangle ABC$, and denote their circumcircles by $\omega_a^1, \omega_c^2$. Then $Br_1 \in \omega_a^1$ and $Br_2 \in \omega_c^2$.

Since $\angle ABBr_2 = \angle CBBr_1$, the sine rule gives

$$\frac{CBr_1}{ABr_2} = \frac{2r_{\omega_a^1}\sin\angle ABBr_2}{2r_{\omega_c^2}\sin\angle CBBr_1} = \frac{r_{\omega_a^1}}{r_{\omega_c^2}}.$$

The ratio of the two radii is the similarity ratio of $\triangle BCA_1$ and $\triangle C_2AB$, and hence

$$\frac{CBr_1}{ABr_2} = \frac{BC}{AC_2} = \frac{BC}{AB} \cdot \frac{AB}{AC_2} = \left(\frac{BC}{AB}\right)^2 = \frac{CL_b}{AL_b},$$

where the last equality is proposition 5.3.3. Together with $\angle Br_2AC = \angle Br_1CA$, this yields $\triangle CL_bBr_1 \overset{-}{\sim} \triangle AL_bBr_2$. Thus $\angle Br_2L_bA = \angle Br_1L_bC$, and the optical property of an ellipse shows that $L_b$ is the point where the Brocard ellipse touches $AC$. The other two sides are analogous, proving both the perspectivity and its center. $\square$

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## Exercises

**Exercise 5.58.** Prove that $\varpi \leq 30°$.

**Exercise 5.59.** Prove that the pedal triangles of the two Brocard points of a triangle are congruent.

**Exercise 5.60.** Consider the following extension of the Brocard point to a quadrilateral. Prove that a convex quadrilateral $A_1A_2A_3A_4$ has a point $P$ in its interior such that

$$\angle PA_1A_2 = \angle PA_2A_3 = \angle PA_3A_4 = \angle PA_4A_1$$

if and only if the quadrilateral is cyclic and $A_1A_2 \cdot A_3A_4 = A_1A_4 \cdot A_2A_3$.

**Exercise 5.61.** Let $G, L$ be the centroid and Lemoine point of $\triangle ABC$. Prove that $ABr_1, BL, CG$ are concurrent, and that $ABr_2, BG, CL$ are concurrent.

**Exercise 5.62.** Suppose that $\triangle ABC \overset{+}{\sim} \triangle A'B'C'$ and $AA' \parallel BB' \parallel CC'$. Let $Br_1$ be the first Brocard point of $\triangle ABC$ and $Br_2'$ the second Brocard point of $\triangle A'B'C'$. Prove that $Br_1Br_2' \parallel AA'$.

**Exercise 5.63*.** For $\triangle ABC$, prove:

(1) (Abī Khuzām) $\varpi^3 \leq (A-\varpi)(B-\varpi)(C-\varpi)$;
(2) (Yff's conjecture)[15] $8\varpi^3 \leq ABC$.

**Exercise 5.64*.** Prove that the eccentricity of the Brocard ellipse is $\sqrt{1-4\sin^2\varpi}$.

### 5.7.2 Brocard Triangles

We now study several further objects associated with the Brocard points, in particular the Brocard triangles. We begin with some definitions.

**Definition 5.7.13.** Let $L, O$ be the Lemoine point and circumcenter of $\triangle ABC$. The line $OL$ is the *Brocard axis* of $\triangle ABC$, and the circle $\odot(OL)$ with diameter $OL$ is its *Brocard circle*.

The midpoint $Br_{\mathrm{m}}$ of $Br_1Br_2$ is the *Brocard midpoint*. The point $Br_3$ inside the triangle whose distances from the three sides are inversely proportional to the cubes of the corresponding side lengths is the *third Brocard point*. [16]

We already know that $Br_1, Br_2$ lie on the Brocard circle and are mirror images of one another in the Brocard axis. The following additional properties of $Br_3$ and $Br_{\mathrm{m}}$ are immediate.

[15] Yff's conjecture was proved by Abī Khuzām.

[16] That is, $\mathrm{dist}(Br_3, BC) : \mathrm{dist}(Br_3, CA) : \mathrm{dist}(Br_3, AB) = BC^{-3} : CA^{-3} : AB^{-3}$.

**Proposition 5.7.14.** *The Brocard midpoint of a triangle is the center of its Brocard ellipse.*

**Proof.** This is immediate. □

**Proposition 5.7.15.** *The Brocard midpoint of a triangle lies on its Brocard axis.*

**Proof.** The points $Br_1, Br_2$ are mirror images in the Brocard axis, and $Br_{\mathrm{m}}$ is their midpoint. □

**Proposition 5.7.16.** *The Lemoine point and the third Brocard point of a triangle are isotomically conjugate.*

**Proof.** Combine theorems 4.4.26 and 5.3.5 with the definition of $Br_3$. □

**Proposition 5.7.17.** *In a triangle, the complement of the third Brocard point is the Brocard midpoint.*

**Proof.** This result will be proved after section 7.4.2; it follows immediately from corollary 7.4.20 below. □

We turn to the Brocard triangles. The following theorem introduces the first one.

**Theorem 5.7.18.** *For $\triangle ABC$, put $B_a = BBr_1 \cap CBr_2$, $B_b = CBr_1 \cap ABr_2$, and $B_c = ABr_1 \cap BBr_2$. Then $\triangle B_aB_bB_c$ is inscribed in the Brocard circle.* It is called the *first Brocard triangle* of $\triangle ABC$.

**Proof.** In Figure 5.42,

$$\measuredangle Br_2B_bBr_1 = \measuredangle ACB_b + \measuredangle B_bAC = 2\varpi = \angle Br_1OBr_2,$$

where the last equality follows from proposition 5.7.9. Hence $B_b, O, Br_1, Br_2$ are concyclic, so $B_b$ lies on the Brocard circle. The proofs for $B_a, B_c$ are identical. □

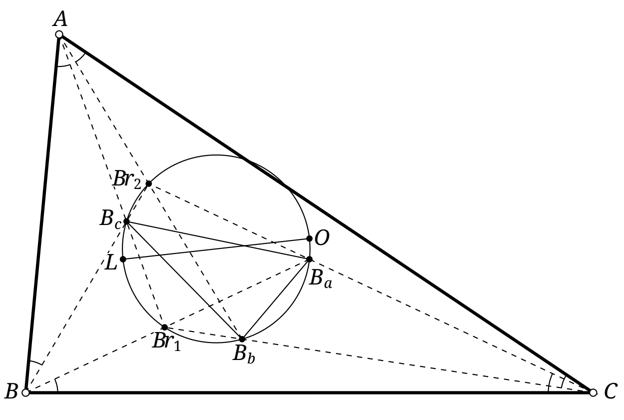


FIGURE 5.42

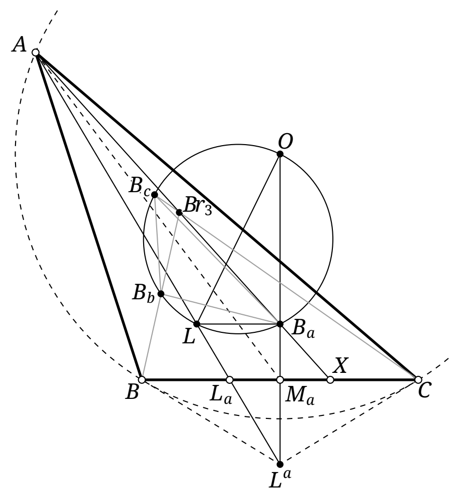


FIGURE 5.43

The first Brocard triangle has the following properties.

**Proposition 5.7.19.** *The first Brocard triangle is oppositely similar to the original triangle.*

**Proof.** Let $\triangle B_aB_bB_c$ be the first Brocard triangle of $\triangle ABC$, as in Figure 5.42. Then

$$\measuredangle B_bB_aB_c = \measuredangle B_bBr_1B_c = -\measuredangle Br_1AC - \measuredangle ACBr_1 = -(\measuredangle BAC - \varpi) - \varpi = -\measuredangle BAC,$$

where the first equality is theorem 5.7.18. The other two corresponding angles are handled similarly, and hence $\triangle B_aB_bB_c \mathrel{\overset{-}{\sim}} \triangle ABC$. $\square$

**Proposition 5.7.20.** *A triangle and its first Brocard triangle are perspective; their center of perspectivity is the third Brocard point.*

**Proof.** Let $\triangle B_aB_bB_c$ be the first Brocard triangle of $\triangle ABC$. Write $L, O$ for the Lemoine point and circumcenter, $M_a$ for the midpoint of $BC$, and put $AB_a \cap BC = X$ and $AL \cap BC = L_a$. Let $\triangle L^aL^bL^c$ be the tangential triangle, as in Figure 5.43.

By the definition of the first Brocard triangle, $\triangle BB_aC$ is isosceles with base angles $\varpi$, so $B_a \in OM_a$. By proposition 5.7.11, $LB_a \perp OB_a$; since $OB_a \perp BC$, it follows that $LB_a \parallel BC$.

The triangle $\triangle L^aL^bL^c$ is the anticevian triangle of $L$, and theorem 4.3.4 gives $(A, L_a; L, L^a) = -1$. Projecting from $B_a$ gives

$$(A, L_a, L, L^a) \overset{(B_a)}{\barwedge} (X, L_a, \infty_{BC}, M_a),$$

and hence $(X, L_a; \infty_{BC}, M_a) = -1$. Therefore $M_a$ is the midpoint of $XL_a$, so $\mathbf{t}L \in AX$, and hence $\mathbf{t}L \in AB_a$. Cyclically, $\mathbf{t}L \in BB_b, CB_c$. Since proposition 5.7.16 gives $Br_3 = \mathbf{t}L$, the stated perspectivity follows. $\square$

The next theorem introduces the second Brocard triangle.

**Theorem 5.7.21.** *For $\triangle ABC$, construct the following six circles: $\omega_a^1$ through $A, C$ tangent to $AB$; $\omega_a^2$ through $A, B$ tangent to $AC$; $\omega_b^1$ through $B, A$ tangent to $BC$; $\omega_b^2$ through $B, C$ tangent to $BA$; $\omega_c^1$ through $C, B$ tangent to $CA$; and $\omega_c^2$ through $C, A$ tangent to $CB$. Let $\omega_a^1 \cap \omega_a^2 = A, C_a$, and define $C_b, C_c$ cyclically. Then $\triangle C_aC_bC_c$ is inscribed in the Brocard circle.* It is called the *second Brocard triangle* of $\triangle ABC$.

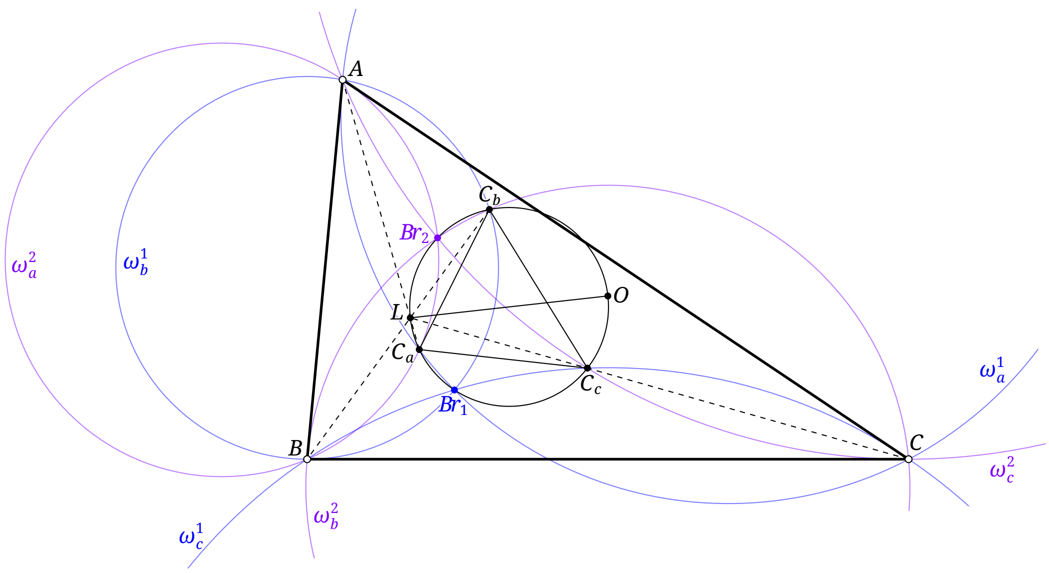


FIGURE 5.44

**Proof.** In Figure 5.44, the circles $\omega_a^1, \omega_b^1, \omega_c^1, \omega_a^2, \omega_b^2, \omega_c^2$ are precisely those in Figures 5.37a and 5.37b; the circles $\omega_a^1, \omega_b^1, \omega_c^1$ pass through $Br_1$, while $\omega_a^2, \omega_b^2, \omega_c^2$ pass through $Br_2$. Therefore

$$\begin{aligned}\measuredangle Br_1C_aBr_2 &= \measuredangle C_aBr_1C + \measuredangle BBr_2C_a + \measuredangle(CBr_1, BBr_2) = \measuredangle C_aAC + \measuredangle BAC_a + \measuredangle(CBr_1, BBr_2)\\ &= \measuredangle BAC + \measuredangle(CBr_1, BBr_2) = -\measuredangle ACBr_1 - \measuredangle Br_2BA = -2\varpi = \measuredangle Br_1OBr_2,\end{aligned}$$

where the final equality uses proposition 5.7.9. Hence $C_a$ lies on the Brocard circle. The other two vertices follow cyclically. □

The second Brocard triangle has the following property.

**Proposition 5.7.22.** *A triangle and its second Brocard triangle are perspective; their center of perspectivity is the Lemoine point.*

**Proof.** Use the notation of theorem 5.7.21 and Figure 5.44. By theorem 5.7.21, $O, Br_1, Br_2, C_a$ are concyclic; at the same time $Br_2, B, A, C_a$ lie on $\omega_a^2$. Thus

$$\measuredangle OC_aA = \measuredangle OC_aBr_2 + \measuredangle Br_2C_aA = \measuredangle OBr_1Br_2 + \measuredangle Br_2BA.$$

By proposition 5.7.9, $\measuredangle OBr_1Br_2 = 90° - \varpi$, while $\measuredangle Br_2BA = \varpi$. Hence $\measuredangle OC_aA = 90°$. On the other hand, proposition 5.7.11 gives $\angle OC_aL = 90°$, so $A, C_a, L$ are collinear. The other two collinearities follow cyclically, proving the result. □

**Proposition 5.7.23.** *The first and second Brocard triangles of a triangle are perspective; their center of perspectivity is the centroid of the original triangle.*

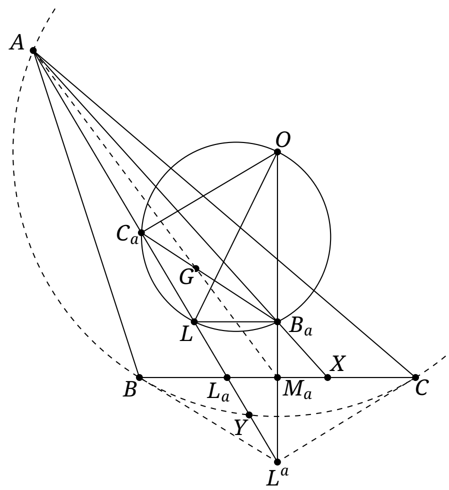


FIGURE 5.45

**Proof.** Continue with the notation in the proof of proposition 5.7.20. In Figure 5.45, let $\triangle C_aC_bC_c$ be the second Brocard triangle, set $AM_a \cap C_aB_a = G$, and let $AL$ meet $\mathscr{C}$ at $A, Y$. By propositions 5.7.11 and 5.7.22, the point $C_a$ is the intersection of $AL$ with the Brocard circle, and $OC_a \perp AL$.

Menelaus's theorem applied to $\triangle L_aL^aM_a$ with transversal $\overline{AB_aX}$ gives

$$\frac{L_aX}{XM_a} \cdot \frac{M_aB_a}{B_aL^a} \cdot \frac{L^aA}{AL_a} = 1.$$

Since $M_a$ is the midpoint of $L_aX$, we have $\frac{L_aX}{XM_a} = 2$, and therefore $\frac{M_aB_a}{B_aL^a} = \frac{AL_a}{2AL^a}$.

By theorem 4.3.4, $(A, L_a; L, L^a) = -1$. Since the polar of $L^a$ with respect to $\odot(ABC)$ is $BC$, theorem 3.6.10 gives $(A, Y; L, L^a) = -1$. Moreover, $OC_a \perp AL$ shows that $C_a$ is the midpoint of $AY$. These three conditions determine the relative positions of $A,C_a,L,L_a,Y,L^a$ on $AL$; a direct calculation of segment ratios gives $AL_a \cdot L^aC_a = AL^a \cdot C_aA$. Applying Menelaus's theorem to $\triangle AL^aM_a$ with transversal $\overline{C_aGB_a}$ now yields

$$1 = \frac{AG}{GM_a} \cdot \frac{M_aB_a}{B_aL^a} \cdot \frac{L^aC_a}{C_aA} = \frac{AG}{GM_a} \cdot \frac{AL_a}{2AL^a} \cdot \frac{L^aC_a}{C_aA} = \frac{AG}{2GM_a}.$$

Thus $\frac{AG}{GM_a} = 2$, so $G$ is the centroid. Hence the centroid of the triangle lies on $B_aC_a$. Similarly, it also lies on $B_bC_b, B_cC_c$. The triangles are therefore perspective, with the centroid as their center of perspectivity. □

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## Exercises

**Exercise 5.65.** Let $H$ be the orthocenter of $\triangle ABC$. Prove that $H, Al, Br_3$ are collinear.

**Exercise 5.66.** Prove that the orthocenter of the orthic triangle lies on the Brocard axis.

**Exercise 5.67.** Let the side lengths of $\triangle ABC$ be $a,b,c$. For $i = 1,2,3$, compute

$$[\triangle Br_iBC] : [\triangle Br_iCA] : [\triangle Br_iAB].$$

What pattern do you find? Use it to conjecture why $Br_3$ is called the third Brocard point.

**Exercise 5.68.** Prove that a triangle and its first Brocard triangle have the same centroid.

**Exercise 5.69.** For a triangle, prove that the pedal triangle of $Ni$ and the medial triangle of the first Brocard triangle are perspective, with center of perspectivity $Ni$.

**Exercise 5.70.** Let $L$ be the Lemoine point of $\triangle ABC$, and let $AL \cap \mathscr{C} = A, P$. Prove that the midpoint of $AP$ is a vertex of the second Brocard triangle.

**Exercise 5.71.** Let $\triangle B_aB_bB_c$ be the first Brocard triangle of $\triangle ABC$, and define $E_i = \mathbf{g}B_i$ for $i = a,b,c$. The triangle $\triangle E_aE_bE_c$ is called the *third Brocard triangle* of $\triangle ABC$.

(1) Prove that the third Brocard triangle is perspective with the original triangle and that the distances from the center of perspectivity to the three sides are proportional to the cubes of the corresponding side lengths. This center is therefore called the *third power point*.
(2) Prove that the third power point and the third Brocard point are isogonally conjugate.
(3) Let $B'_a, B'_b, B'_c$ be the inverses of $B_a, B_b, B_c$ in $\mathscr{C}$. Prove that $\triangle B'_aB'_bB'_c$ is perspective with $\triangle ABC$ and that their center of perspectivity is the third power point.

**Exercise 5.72.** Let $\triangle C_aC_bC_c$ be the second Brocard triangle of $\triangle ABC$, and define $D_i = \mathbf{g}C_i$ for $i = a,b,c$. The triangle $\triangle D_aD_bD_c$ is called the *fourth Brocard triangle*, or the *D-triangle*, of $\triangle ABC$.

(1) Prove that $\omega_b^1 \cap \omega_c^2 = A, D_a$, with the circles defined as in theorem 5.7.21.
(2) Let $G$ be the centroid of $\triangle ABC$. Prove that $G$ is the center of perspectivity of $\triangle ABC$ and $\triangle D_aD_bD_c$.
(3) Prove that the circle with diameter joining the orthocenter and centroid of $\triangle ABC$—the *orthocentroidal circle*—is the circumcircle of $\triangle D_aD_bD_c$.
(4) Prove that $\triangle D_aD_bD_c \overset{\pm}{\sim} \triangle ABC$.
(5) Prove that $\triangle D_aD_bD_c$ and $\triangle ABC$ have the same Lemoine point.

**Exercise 5.73.** For $\triangle ABC$, choose $D$ so that $\triangle BCD$ is equilateral and $A,D$ lie on the same side of $BC$. Let the perpendicular bisector of $AD$ meet the line $BC$ at $J$, and let $AJ \cap \mathscr{C} = A,T$. Prove that $\mathscr{S}(T)$ is parallel to the Brocard axis of $\triangle ABC$.

**Exercise 5.74.** Let $\triangle XYZ$ be the first Brocard triangle of $\triangle ABC$, and let $BX,CX$ meet $YZ$ at $P,Q$, respectively. Through $P$ draw the perpendicular to $AC$, and through $Q$ the perpendicular to $AB$; let these two lines meet at $T$. Prove that $TB = TC$.

## 5.8 The Apollonius Points

We now introduce the celebrated Apollonius points.

Recall the Apollonius circle from theorem 0.3.7. Given $\triangle ABC$, the locus of points $P$ satisfying $\frac{PA}{PB} = \frac{AC}{BC}$ is an Apollonius circle for the pair $A,B$; the intersections of the internal and external angle bisectors of $\angle ACB$ with the line $AB$ are a pair of antipodal points on it. We call this circle the *$C$-Apollonius circle* of $\triangle ABC$. The $A$- and $B$-Apollonius circles are defined cyclically, and the three are collectively called the *Apollonius circles* of the triangle.

When $CA = CB$, the locus of points $P$ satisfying $\frac{PA}{PB} = \frac{AC}{BC}$ is the perpendicular bisector of $AB$, rather than a circle in the ordinary sense. In the limiting sense it may be regarded as a circle whose center is $\infty_{AB}$ and whose radius is infinite; we continue to call it the $C$-Apollonius circle of $\triangle ABC$.

The Apollonius circles yield the following theorem, which also defines the Apollonius points.

**Theorem 5.8.1.** *For a given triangle $\triangle ABC$, there are exactly two points $P$ satisfying*

$$PA \cdot BC = PB \cdot CA = PC \cdot AB.$$

They are called the *Apollonius points*, or the *isodynamic points*, of $\triangle ABC$.

**Proof.** If the triangle is scalene, assume without loss of generality that $BC < AC < AB$. The condition $PA \cdot BC = PB \cdot CA$ is equivalent to $\frac{PA}{PB} = \frac{CA}{CB}$, so $P$ lies on the $C$-Apollonius circle of $\triangle ABC$. Likewise, $PA \cdot BC = PC \cdot AB$ is equivalent to $\frac{PA}{PC} = \frac{AB}{BC}$, so $P$ lies on the $B$-Apollonius circle of $\triangle ABC$.

Since $BC < AC$, the definition of an Apollonius circle shows that $B$ lies inside the $C$-Apollonius circle. Similarly, $BC < AB$ shows that $C$ lies inside the $B$-Apollonius circle. Each of the $B$- and $C$-Apollonius circles thus has a point lying inside the other, so they must intersect strictly at two distinct points.

When $P$ lies on both the $B$- and $C$-Apollonius circles of $\triangle ABC$, it already satisfies $PA \cdot BC = PB \cdot CA = PC \cdot AB$. Thus it also satisfies $\frac{PB}{PC} = \frac{AB}{AC}$, so it lies on the $A$-Apollonius circle of $\triangle ABC$ as well.

If the triangle is isosceles but not equilateral, assume without loss of generality that $AC = BC$. If $AC = BC < AB$, both the $A$- and $B$-Apollonius circles contain $C$ in their interiors, while they respectively do not contain the points $B,A$ on the other circle. Hence these two circles also intersect strictly at two distinct points, which are the two Apollonius points. If $AC = BC > AB$, the $A$- and $B$-Apollonius circles contain the points $B,A$ of the other circle, respectively, in their interiors; they again have two distinct intersections, which are the Apollonius points.

Thus, in the nonequilateral case, $P$ is one of the two distinct common points of the three Apollonius circles of $\triangle ABC$.

In fact, this proposition should require that $\triangle ABC$ is not equilateral. For an equilateral triangle, the three Apollonius circles degenerate into the perpendicular bisectors of the sides and have only one common point, the center of the triangle. The equilateral case is so special that we generally do not consider it; hence we shall not explicitly state its exclusion in what follows. $\square$

The Apollonius circles have the following properties.

**Proposition 5.8.2.** *The three Apollonius circles of* $\triangle ABC$*:*

(1) *are coaxial, and their centers are collinear;*
(2) *have two common points* $Ap_1, Ap_2$*, precisely the two Apollonius points of* $\triangle ABC$*;*
(3) *are all orthogonal to* $\mathscr{C}$.

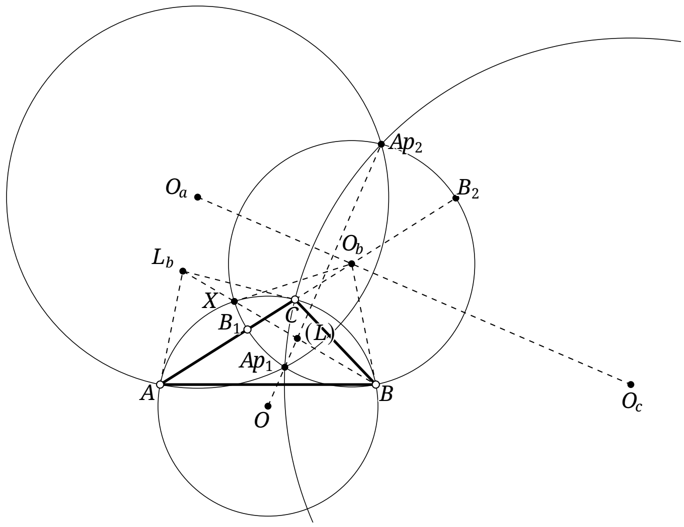


FIGURE 5.46

**Proof.** The second assertion was established in the proof of theorem 5.8.1, and the coaxiality and collinearity in the first assertion follow at once. It remains to prove orthogonality. Consider the $B$-Apollonius circle $\odot O_b$; it suffices to show that $OB \perp O_b B$.

In the configuration of Figure 5.46, let $AC \cap \odot O_b = B_1, B_2$. The lines $BB_1, BB_2$ are the internal and external angle bisectors of $\angle ABC$. Therefore

$$\begin{aligned}\angle O_b BC &= \angle O_b BB_1 - \angle CBB_1 = \angle O_b B_1 B - \angle CBB_1\\ &= \angle O_b B_1 B - \angle ABB_1 = \angle CAB = \frac{\angle COB}{2} = \frac{\pi - 2\angle CBO}{2} = \frac{\pi}{2} - \angle CBO.\end{aligned}$$

Hence $\angle OBO_b = 90^\circ$. □

**Corollary 5.8.3.** *The two Apollonius points of* $\triangle ABC$ *are inverses of each other in* $\mathscr{C}$.

**Proof.** Since the circumcircle is orthogonal to each Apollonius circle, theorem 4.1.9 shows that inversion in the circumcircle preserves all three Apollonius circles. An Apollonius point changes under this inversion, so it can only be sent to the other common point of the three circles, namely the other Apollonius point. □

**Corollary 5.8.4.** *Of the two Apollonius points of* $\triangle ABC$*, one lies inside* $\mathscr{C}$ *and the other outside it.*

In view of corollary 5.8.4, the Apollonius point $Ap_1$ inside $\mathscr{C}$ is called the *first Apollonius point*, the *first isodynamic point*, or the *positive isodynamic point*. The point $Ap_2$ outside $\mathscr{C}$ is called the *second Apollonius point*, the *second isodynamic point*, or the *negative isodynamic point*.

The Apollonius points have the following properties.

**Proposition 5.8.5.** *The two Apollonius points* $Ap_1, Ap_2$ *of* $\triangle ABC$ *lie on the Brocard axis. Moreover, the Lemoine point is the inverse in* $\mathscr{C}$ *of the midpoint of* $Ap_1 Ap_2$.

**Proof.** Let $\odot O$ be the circumcircle of $\triangle ABC$. By corollary 5.8.3, the points $O, Ap_1, Ap_2$ are collinear. It remains to prove that the Lemoine point $L$ lies on this line and has the stated inversion relation.

We first show that the polars, with respect to $\odot O$, of the centers of the three Apollonius circles are the three symmedians. Consider the $B$-Apollonius circle $\odot O_b$, as in Figure 5.46. Let the tangents to $\odot O$ at $A, C$ meet at $L_b$. By lemma 5.3.7, $BL_b$ is a symmedian of $\triangle ABC$. The polar of $L_b$ passes through $O_b$, so the principle of pole and polar says that the polar of $O_b$ passes through $L_b$. On the other hand, because $\odot O$ and $\odot O_b$ are orthogonal, $O_bB$ is tangent to $\odot O$ at $B$; hence the polar of $O_b$ also passes through $B$. It is therefore the symmedian $BL_b$. Consequently, $BL_b$ also passes through the other intersection $X$ of $\odot O_b$ and $\odot O$.

By proposition 5.8.2, the three centers $O_a, O_b, O_c$ are collinear, while their polars are the three symmedians. Thus the pole of $O_aO_c$ is the common point of the symmedians, namely $L$. The line $O_aO_c$ plainly perpendicularly bisects $Ap_1Ap_2$. It follows from theorem 4.1.11 that $L$ is the inverse of the midpoint of $Ap_1Ap_2$. Since $O, Ap_1, Ap_2$ are collinear, we also have $L \in Ap_1Ap_2$. □

**Corollary 5.8.6.** *If $O, L$ are the circumcenter and Lemoine point of $\triangle ABC$, then $O, L, Ap_1, Ap_2$ form a harmonic range.*

**Proof.** Let $M$ be the midpoint of $Ap_1Ap_2$. By proposition 5.8.5, $OL \cdot OM = OAp_1 \cdot OAp_2$. Apply proposition 3.2.18. □

**Proposition 5.8.7.** *The pedal triangle of either Apollonius point of a triangle is equilateral.*

**Proof.** The case of $Ap_1$ is shown in Figure 5.47. Let $R$ be the circumradius of $\triangle ABC$, let $Ap$ be either Apollonius point, and let $\triangle A'B'C'$ be its pedal triangle. The points $A', Ap, B', C$ lie on the circle $\odot(ApC)$ with diameter $ApC$. Hence $A'B' = ApC \sin\angle ACB = ApC \cdot \frac{AB}{2R}$. Cyclically, $A'C' = ApB \cdot \frac{AC}{2R}$ and $B'C' = ApA \cdot \frac{BC}{2R}$. Since $ApC \cdot AB = ApB \cdot AC = ApA \cdot BC$, we obtain $A'B' = A'C' = B'C'$, so the pedal triangle is equilateral. □

**Proposition 5.8.8.** *If $L$ is the Lemoine point of $\triangle ABC$, then $\angle ApBrL = 60°$.* [17]

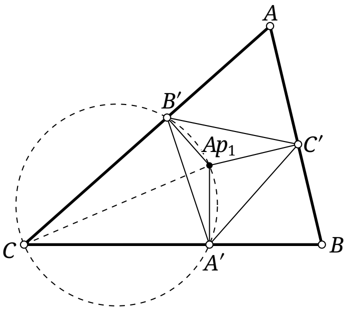


FIGURE 5.47

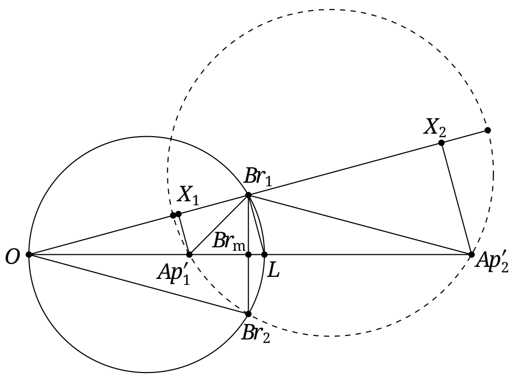


FIGURE 5.48

**Proof.** Let $O, L$ be the circumcenter and Lemoine point of $\triangle ABC$. By propositions 5.7.9 and 5.7.11, the points $O, Br_1, Br_2, L$ are concyclic, $OL$ is a diameter of their circle, $Br_1, Br_2$ are symmetric about $OL$, and $Br_1Br_2 \cap OL = Br_m$, as in Figure 5.48.

Consider $\triangle OBr_1Br_2$, and denote its Apollonius points by $Ap_1', Ap_2'$. The $O$-Apollonius circle of this triangle degenerates into the line $OL$. Hence $Ap_1', Ap_2'$ are the intersections of $OL$ with the $Br_2$-Apollonius circle of $\triangle OBr_1Br_2$.

[17]Here $Ap$ may be either $Ap_1$ or $Ap_2$, and $Br$ may be either $Br_1$ or $Br_2$—but not $Br_3$. The subscripts are therefore suppressed.

We have
$$\frac{Ap_1'O}{Ap_1'Br_1}=\frac{OBr_2}{Br_1Br_2}=\frac{OBr_1}{2Br_mBr_1}.$$
Let $X_1,X_2$ be the projections of $Ap_1',Ap_2'$ onto $OBr_1$. Since $\triangle OAp_1'X_1\sim\triangle OBr_1Br_m$, $\frac{OBr_1}{Br_mBr_1}=\frac{OAp_1'}{X_1Ap_1'}$. Consequently $Br_1Ap_1'=2Ap_1'X_1$, so $\angle OBr_1Ap_1'=30°$. Since $OBr_1\perp LBr_1$, we obtain $\angle Ap_1'Br_1L=60°$. Similarly, $\angle Ap_2'Br_1L=60°$, and symmetry in $Br_1,Br_2$ gives $\angle Ap'BrL=60°$. It remains only to prove proposition 5.8.9. ☐

**Proposition 5.8.9.** *If $O$ is the circumcenter of $\triangle ABC$, then $\triangle OBr_1Br_2$ and $\triangle ABC$ have the same Apollonius points.*

**Proof.** Retain the notation from the proof of proposition 5.8.8, as in Figure 5.48. By corollary 5.8.3, the points $O,L,Ap_1',Ap_2'$ form a harmonic range.

By propositions 5.7.9 and 5.7.11, $\angle Br_1OL=\varpi$. We have already shown that $\angle OBr_1Ap_1'=30°$ and $\angle OBr_1Ap_2'=150°$. Hence $\angle Br_1Ap_1'Ap_2'=30°+\varpi$ and $\angle Br_1Ap_2'Ap_1'=30°-\varpi$. Applying the sine rule to $\triangle OBr_1Ap_1'$ and $\triangle OBr_2Ap_2'$ gives
$$OAp_1'=\frac{\sin 30°}{\sin(30°+\varpi)}OBr_1,\qquad OAp_2'=\frac{\sin 30°}{\sin(30°-\varpi)}OBr_1.$$
Consequently,
$$OAp_1'\cdot OAp_2'=\frac{\sin^2 30°}{\sin(30°-\varpi)\sin(30°+\varpi)}OBr_1^2=\frac{OBr_1^2}{1-4\sin^2\varpi}=R^2,$$
where the last equality is proposition 5.7.10 and $R$ is the circumradius of $\triangle ABC$. Thus $Ap_1'=\mathbf{i}Ap_2'$.

Once $O,L$ are fixed, the two conditions $Ap_1'=\mathbf{i}Ap_2'$ and $(O,L;Ap_1',Ap_2')=-1$ determine $Ap_1',Ap_2'$ uniquely. By corollaries 5.8.3 and 5.8.6, the Apollonius points $Ap_1,Ap_2$ of $\triangle ABC$ satisfy the same two conditions. Hence $Ap_1'=Ap_1$ and $Ap_2'=Ap_2$. ☐

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## Exercises

**Exercise 5.75.** Prove that the pedal triangle of any point on any one of a triangle's three Apollonius circles is isosceles.

**Exercise 5.76.** Prove that a point whose circumcevian triangle is equilateral is an Apollonius point of the original triangle.

**Exercise 5.77.** Let $\omega$ be a circle centered at an Apollonius point $Ap$ of $\triangle ABC$. Under the general inversion $\mathfrak{i}_\omega$, let $A,B,C$ be sent to $A',B',C'$, respectively. Prove that $\triangle A'B'C'$ is equilateral.

**Exercise 5.78.** The circle through the two Apollonius points and the centroid of a triangle is called its *Parry circle*. Prove that:

(1) the Parry circle is orthogonal to both the circumcircle and the Brocard circle;
(2) the inverse of the centroid in the circumcircle lies on the Parry circle;
(3) * The Lemoine point lies on the radical axis of the circumcircle and the Parry circle.

# 5.9 The Torricelli Points

## 5.9.1 The Torricelli Points

We next introduce the Torricelli points, which are closely related to the Apollonius points.

**Theorem 5.9.1.** *In Figure 5.49a, construct externally on the three sides of* $\triangle ABC$ *the equilateral triangles* $\triangle BCA_1$*,* $\triangle ACB_1$*, and* $\triangle ABC_1$*. Then the three lines* $AA_1$*,* $BB_1$*, and* $CC_1$ *concur at a point* $T_1$*.*

*In Figure 5.49b, construct the equilateral triangles* $\triangle BCA_2, \triangle ACB_2, \triangle ABC_2$ *inward on the three sides of* $\triangle ABC$*. Then the three lines* $AA_2$*,* $BB_2$*, and* $CC_2$ *concur at a point* $T_2$*.*

The point $T_1$ is the *first Torricelli point*, and $T_2$ is the *second Torricelli point*; together they are the *Torricelli points* of $\triangle ABC$.

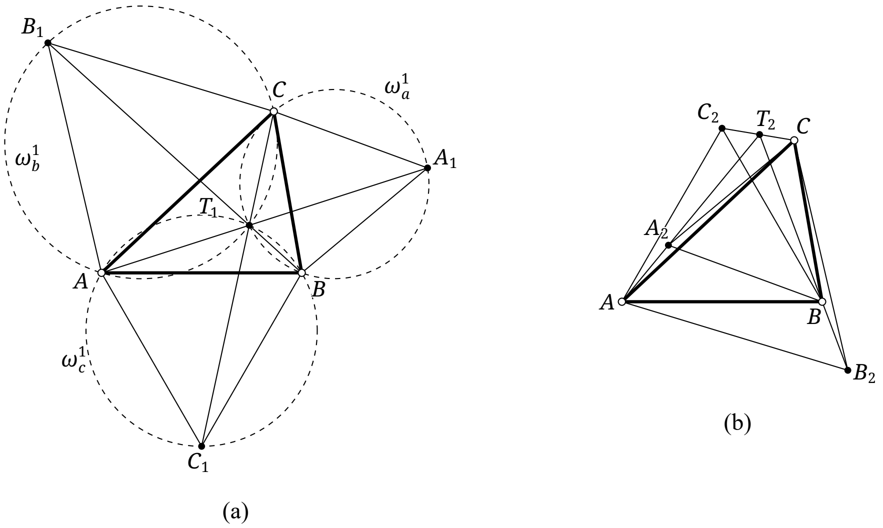


FIGURE 5.49

**Proof.** Consider the outward construction in Figure 5.49a. Let $\omega_a^1, \omega_b^1, \omega_c^1$ be the circumcircles of $\triangle BCA_1$, $\triangle ACB_1$, and $\triangle ABC_1$, respectively, and let $T_1$ be the second intersection of $\omega_b^1$ and $\omega_c^1$, the first being $A$. Cyclicity gives $\angle CT_1A = 180° - \angle AB_1C = 120°$, and similarly $\angle BT_1A = 120°$. Thus $\angle CT_1B = 120°$, so $T_1, C, A_1, B$ are concyclic and $T_1 \in \omega_a^1$. Hence $T_1$ is the common point of the three circles.

Now $\angle AT_1C = 120°$ and $\angle CT_1A_1 = \angle CBA_1 = 60°$, so $A, T_1, A_1$ are collinear. The other two collinearities follow cyclically. The inward construction is analogous and is left to the reader. □

The construction in this proof also gives the following characterization, proposition 5.9.2, of $T_1$ and $T_2$.

**Proposition 5.9.2.** *Each side of a triangle subtends at either Torricelli point an angle of* $120°$ *or* $60°$*. Conversely, a point at which all three sides subtend angles of* $120°$ *or* $60°$ *is a Torricelli point.* [18]

Accordingly, $T_1$ is also called the *first isogonic point* or the *positive isogonic point*, while $T_2$ is called the *second isogonic point* or the *negative isogonic point*. Together they are the *isogonic points* of $\triangle ABC$.

The first Torricelli point has the following fundamental property.

---

[18] For $\triangle ABC$ in Figure 5.49, for example, $\angle AT_1C = \angle BT_1C = \angle BT_1A = 120°$, while $\angle AT_2B = \angle BT_2C = 60°$ and $\angle AT_2C = 120°$. In particular, if all three angles of the triangle are less than $120°$, every side subtends an angle of $120°$ at $T_1$.

**Proposition 5.9.3.** *If every angle of a triangle is less than* $120^\circ$*, the sum of the distances from its first Torricelli point to the three vertices is minimal.*

**Proof.** In Figure 5.50, let $X$ be any point of the plane and apply the rotation $\mathfrak{r}_{A,60^\circ}$. Denote images by primes. Since $B'X' = BX$,

$$XA + XB + XC = X'X + X'B' + XC \geq CB',$$

with equality when $X = T_1$. □

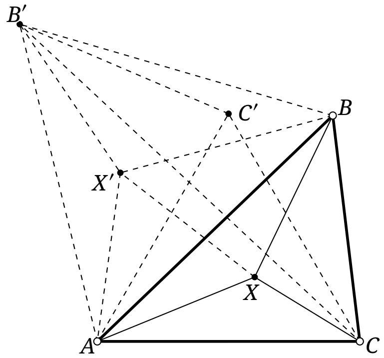


FIGURE 5.50

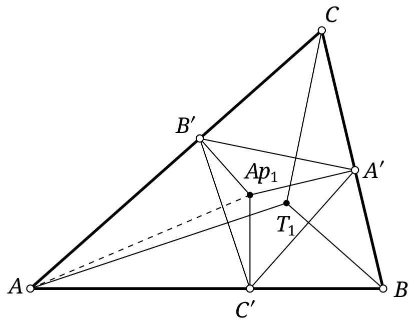


FIGURE 5.51

This property was first proved by Fermat. Thus $T_1$ is also called the *first Fermat point*, and correspondingly $T_2$ the *second Fermat point*; together they are the *Fermat points*. [19]

The equilateral-triangle construction in Figure 5.49 also gives the following result.

**Proposition 5.9.4.** *Construct outwardly, or inwardly, an equilateral triangle on each side of* $\triangle ABC$*, and denote the three new vertices by* $A_1, B_1, C_1$*. Then* $AA_1 = BB_1 = CC_1$*.*

**Proof.** We have $\triangle ABA_1 \cong \triangle C_1BC$, and hence $AA_1 = CC_1$. Cyclically, $BB_1 = AA_1$. □

The relation between Torricelli points and isogonal conjugation is as follows.

**Proposition 5.9.5.** *For every triangle,* $T_1 = \mathbf{g}\mathit{Ap}_1$ *and* $T_2 = \mathbf{g}\mathit{Ap}_2$*.*

**Proof.** For $\triangle ABC$, let $\triangle A'B'C'$ be the pedal triangle of an Apollonius point $\mathit{Ap}$. Through $A,B,C$ draw the perpendiculars to $B'C', C'A', A'B'$, respectively. By proposition 4.4.7, they concur at $T := \mathbf{g}\mathit{Ap}$. We show that $T$ is a Torricelli point; the case $T_1, \mathit{Ap}_1$ is illustrated in Figure 5.51.

Since $AT \perp B'C'$ and $BT \perp C'A'$, $\measuredangle(AT,BT) = \measuredangle(B'C',A'C') = 60^\circ$, where the last equality follows from proposition 5.8.7. Thus $\angle ATB$ is $60^\circ$ or $120^\circ$. The other two sides are analogous, so proposition 5.9.2 makes $T$ a Torricelli point. The relative positions show that $\mathit{Ap}_1$ corresponds to $T_1$ and $\mathit{Ap}_2$ to $T_2$. □

We conclude this subsection with a generalization of the Torricelli-point construction.

**Theorem 5.9.6.** *Given* $\triangle ABC$ *and a point P, let* $f_A$ *be the involution in the pencil at A that fixes PA and interchanges* $AB,AC$*. Let* $f_B$ *be the involution in the pencil at B that fixes PB and interchanges* $BC,BA$*, and let* $f_C$ *be the involution in the pencil at C that fixes PC and interchanges* $CA,CB$*. Let* $a,a'$*;* $b,b'$*; and* $c,c'$ *be corresponding pairs of lines under* $f_A, f_B, f_C$*, and set* $A' = b' \cap c$*,* $B' = c' \cap a$*, and* $C' = a' \cap b$*. Then* $\triangle ABC$ *and* $\triangle A'B'C'$ *are perspective.*

Before proving the theorem, we record several common special cases.

[19]In some sources, "the Fermat point" means only the first Fermat point; the intended meaning is normally clear from context.

**Theorem 5.9.7.** *Given $\triangle ABC$, let $a,a'$; $b,b'$; and $c,c'$ be pairs of isogonal lines at $A,B,C$, respectively, and define $A' = b' \cap c$, $B' = c' \cap a$, and $C' = a' \cap b$. Then $\triangle ABC$ and $\triangle A'B'C'$ are perspective.*

**Proof.** In theorem 5.9.6, take $P = I$, the incenter. Then $f_A$ is reflection in $AI$: it fixes $AI$, interchanges $AB,AC$, and preserves cross-ratios of lines because reflection reverses every directed angle. Similarly, $f_B$ and $f_C$ are reflections in $BI$ and $CI$. □

**Example 5.9.8.** A triangle and its Morley triangle are perspective. Their center of perspectivity is called the *first Morley–Taylor–Marr center*, or the *second Morley center*.

The following is another special case of theorem 5.9.7.

**Theorem 5.9.9.** *In Figure 5.52, construct externally on the three sides of $\triangle ABC$ the isosceles triangles $\triangle A'BC, \triangle AB'C, \triangle ABC'$, each with base angle $\phi$. Then $AA', BB', CC'$ concur at a point $X(\phi)$.*

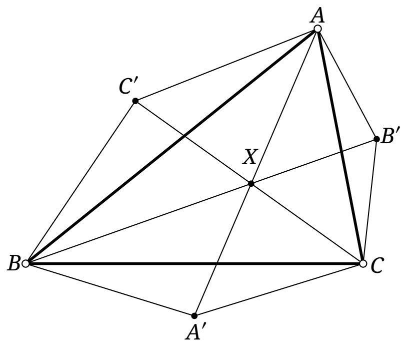


FIGURE 5.52

**Remark.** The angle $\phi$ is understood in the generalized sense. Write $\phi \equiv \theta \pmod{180^\circ}$, where $\theta \in (-90^\circ, 90^\circ]$, and set $X(\phi) = X(\theta)$. If $0 < \theta < 90^\circ$, use the ordinary outward construction. If $-90^\circ < \theta < 0$, construct inwardly with base angle $-\theta$. At $\theta = 90^\circ$, the limiting construction gives $X(90^\circ)$ as the orthocenter of $\triangle ABC$; at $\theta = 0$, the three isosceles triangles degenerate, $A', B', C'$ become the side midpoints, and $X(0)$ is the centroid.

**Example 5.9.10.** In theorem 5.9.9, $X(0)$ is the centroid, $X(90^\circ)$ the orthocenter, $X(60^\circ) = T_1$, $X(-60^\circ) = T_2$, and $X(-\varpi) = Br_3$.

**Example 5.9.11.** Construct outwardly, or inwardly, an equilateral triangle on each side of $\triangle ABC$. The triangle formed by the centers of these three equilateral triangles is perspective with $\triangle ABC$. The respective centers of perspectivity $Np_1, Np_2$ are called the *first Napoleon point* and the *second Napoleon point*. [20]

**Proof.** Take $\phi = \pm 30^\circ$ in theorem 5.9.9. □

**Example 5.9.12.** Construct outwardly, or inwardly, a square on each side of $\triangle ABC$. The triangle formed by the centers of the squares is perspective with $\triangle ABC$. The respective centers of perspectivity $Vc_1, Vc_2$ are the *outer Vecten point*[21] and the *inner Vecten point*, respectively.

[20]Together $Np_1, Np_2$ are called the *Napoleon points* of $\triangle ABC$.

[21]This may be abbreviated to *Vecten point*.

**Proof.** Take $\phi = \pm 45^\circ$ in theorem 5.9.9. □

We now prove theorem 5.9.6.

**Proof of theorem 5.9.6.** Apply a projectivity sending $A,B$ to points at infinity in perpendicular directions, sending $\angle ACB$ to a right angle, and sending $P$ to a point on its angle bisector. Thus $AB = l_\infty$, as in Figure 5.53. Let $P_1, P_2$ be the projections of $P$ onto $BC, AC$, respectively, and put $PP_1 = PP_2 := p$.

We have $a \parallel a' \parallel PP_1 \parallel AC$. Let the distance from the moving line $a$ to $AC$ be $u$. Consider the transformation $f'_A$ that sends $a$ to the line $a'$ on the same side of $AC$ and at distance $\frac{p^2}{u}$ from $AC$. It fixes $PP_1$, interchanges $AC, AB$, and is an involution in the pencil at $A$. [22] The two corresponding pairs $(AB, AC)$ and $(AP, AP)$ determine the involution uniquely, so $f_A = f'_A$.

Let the distance from the moving line $b$ to $BC$ be $v$. Similarly, $f_B$ sends $b$ to the line $b'$ at distance $\frac{p^2}{v}$ from $BC$. For $f_C$, note that $CP$ bisects the right angle $ACB$. Reflection of a line $c$ in $CP$ therefore satisfies the defining conditions. Write $\tan \measuredangle(BC, c) = w$; then $\tan \measuredangle(BC, c') = \frac{1}{w}$.

Let the projections of $A', B', C'$ onto $BC$ and $AC$ be, respectively, $A'_1, A'_2$; $B'_1, B'_2$; and $C'_1, C'_2$. For $A'$, the preceding description gives $CA'_1 = \frac{v}{w}$ and $A'A'_1 = v$; for $B'$, it gives $CB'_1 = \frac{p^2}{u}$ and $B'B'_1 = \frac{p^2}{uw}$; and for $C'$, it gives $CC'_1 = u$ and $C'C'_1 = \frac{p^2}{v}$. The line $B'B'_2$ is $BB'$, and $A'A'_1$ is $AA'$. Put $AA' \cap BB' = X$. Then $XA'_1 = B'B'_1 = \frac{p^2}{uw}$. Since $\frac{XA'_1}{CA'_1} = \frac{p^2}{uv} = \frac{C'C'_1}{CC'_1}$, the point $X$ lies on $CC'$, proving the perspectivity. □

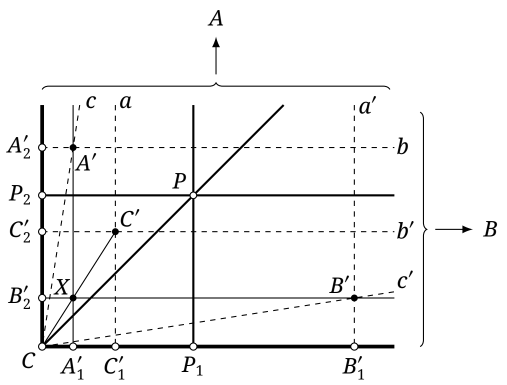


FIGURE 5.53

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## Exercises

**Exercise 5.79.** For a general triangle, determine when the angles subtended by its sides at each Torricelli point in proposition 5.9.2 are 120° and when they are 60°.

**Exercise 5.80.** Prove that the antipedal triangle of either Torricelli point is equilateral.

**Exercise 5.81.** Prove that $T_1$ lies on the circumcircle of the reflection triangle of $T_2$, and that $T_2$ lies on the circumcircle of the reflection triangle of $T_1$.

[22] One way to see the involution is to compare $\mathrm{dist}(a, AC) \cdot \mathrm{dist}(a', AC) = p^2$ with the defining metric relation for an inversion and use theorem 4.1.13; alternatively, one may verify directly that cross-ratios are preserved and that the square of the transformation is the identity.

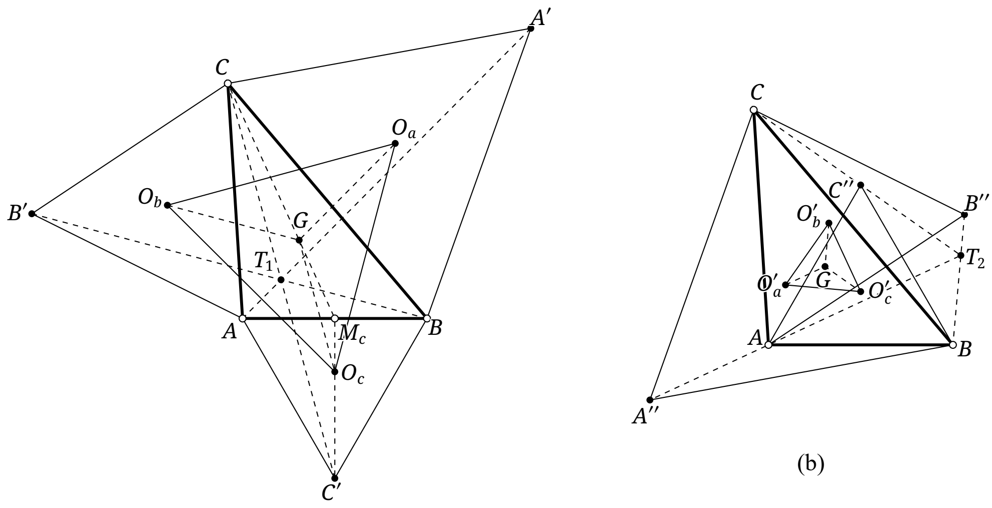
**Exercise 5.82.** Let $I,O$ be the incenter and circumcenter of $\triangle ABC$, and let $I'$ be the incenter of the pedal triangle of either Torricelli point $T$. Prove that $TI' \parallel OI$.

**Exercise 5.83*.** Investigate the following generalization of the Fermat point. Given $\triangle ABC$ and real numbers $a,b,c$, determine the position of a point $P$ at which $a|PA|+b|PB|+c|PC|$ is extremal.

**Exercise 5.84.** Prove that:
(1) a triangle and its first Morley adjunct triangle are perspective; their center of perspectivity is called the *second Morley–Taylor–Marr center*;
(2) ** The first and second Morley centers and the Morley center of a triangle are collinear.

### 5.9.2 Napoleon Triangles

We now introduce the Napoleon triangles, a construction related to the Torricelli points, and use them to establish further properties of those points.




(a) (b)

FIGURE 5.54

**Theorem 5.9.13** (Napoleon's theorem). *In Figure 5.54, construct outwardly, or inwardly, an equilateral triangle on each side of* $\triangle ABC$. *The centers of the three equilateral triangles form an equilateral triangle whose center is the centroid of* $\triangle ABC$. This new triangle is called the *outer Napoleon triangle*, or respectively the *inner Napoleon triangle*.

**Proof.** We prove the outward case; the inward case is analogous. In Figure 5.54a, let $A',B',C'$ be the third vertices of the three equilateral triangles, let their centers be $O_a,O_b,O_c$, and let $G$ and $M_c$ be the centroid of $\triangle ABC$ and the midpoint of $AB$. Let $T_1$ be the first Torricelli point.

Since $O_c$ is the center of the equilateral triangle $\triangle ABC'$, $\frac{M_cO_c}{M_cC'}=\frac{1}{3}$. Since $G$ is the centroid, $\frac{GM_c}{CM_c}=\frac{1}{3}$. Thus $\triangle GM_cO_c \sim \triangle CM_cC'$, and consequently $GO_c \parallel CC'$ and $GO_c=\frac{1}{3}CC'$. Cyclically, $GO_a \parallel AA'$ and $GO_a=\frac{1}{3}AA'$, while $GO_b \parallel BB'$ and $GO_b=\frac{1}{3}BB'$. By proposition 5.9.4, $GO_a=GO_b=GO_c$, so $G$ is the circumcenter of $\triangle O_aO_bO_c$.

Moreover, $GO_a \parallel AA'$ and $GO_b \parallel BB'$, whence $\angle O_aGO_b=\angle A'T_1B'=120°$, where the last equality follows from proposition 5.9.2. Cyclically, $\angle O_bGO_c=\angle O_cGO_a=120°$. Since $G$ is already the circumcenter, $\triangle O_aO_bO_c$ is equilateral and has center $G$. $\square$

**Remark.** The proof above uses a property of the Torricelli point. The assertion that $\triangle O_aO_bO_c$ is equilateral may also be viewed as a special case of Echols's second theorem.

In the language of Napoleon triangles, example 5.9.11 becomes the following statement.

**Theorem 5.9.14.** *A triangle and its outer Napoleon triangle are perspective from $Np_1$; the triangle and its inner Napoleon triangle are perspective from $Np_2$.*

We first establish some properties of Napoleon triangles.

**Proposition 5.9.15.** *For a given triangle, the pedal triangle of the first Apollonius point is homothetic to the outer Napoleon triangle; the pedal triangle of the second Apollonius point is homothetic to the inner Napoleon triangle. In both cases the center of homothety is the Lemoine point.*

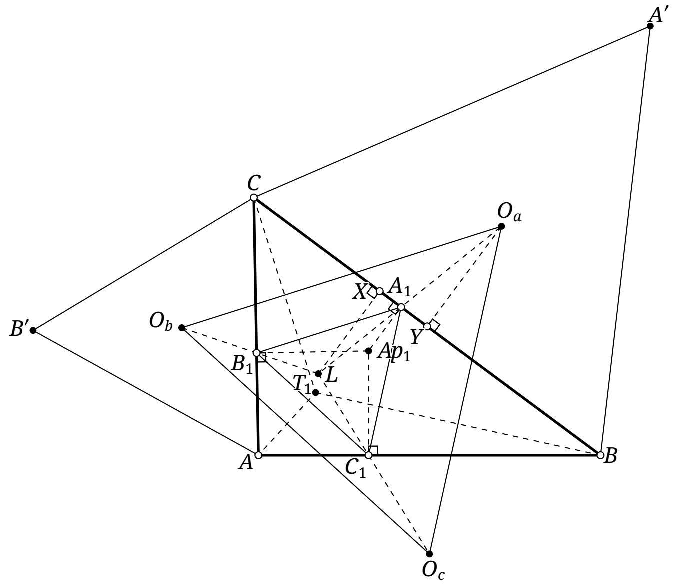



FIGURE 5.55

**Proof.** We prove only the assertion for the outer Napoleon triangle. In Figure 5.55, let $\triangle O_aO_bO_c$ be that triangle and let $\triangle A_1B_1C_1$ be the pedal triangle of $Ap_1$. Since $Ap_1 = \mathbf{g}T_1$ by proposition 5.9.5, proposition 4.4.7 gives $CT_1 \perp A_1B_1$.

On the other hand, $O_b, O_a$ are the centers of $\odot(AB'C)$ and $\odot(A'BC)$, respectively. By proposition 5.9.2, these two circles meet at $C, T_1$, and hence $CT_1 \perp O_aO_b$. Thus $A_1B_1 \parallel O_aO_b$, with two cyclic analogues. The triangles $\triangle O_aO_bO_c$ and $\triangle A_1B_1C_1$ are therefore homothetic. Let $L$ be their center of homothety and let $k = \frac{LO_a}{LA_1}$ be the ratio. We show that $L$ is the Lemoine point.

Let $X, Y$ be the projections of $L, O_a$ onto $BC$. Then

$$k - 1 = \frac{A_1O_a}{A_1L} = \frac{O_aY}{LX} = \frac{\frac{\sqrt{3}}{6}BC}{\operatorname{dist}(L, BC)}.$$

The cyclic analogues give

$$\frac{\operatorname{dist}(L, BC)}{BC} = \frac{\operatorname{dist}(L, AB)}{AB} = \frac{\operatorname{dist}(L, AC)}{AC}.$$

By theorem 5.3.5, $L$ is the Lemoine point. $\square$

**Proposition 5.9.16.** *The first and second Torricelli points of a triangle lie, respectively, on the circumcircles of its inner and outer Napoleon triangles.*

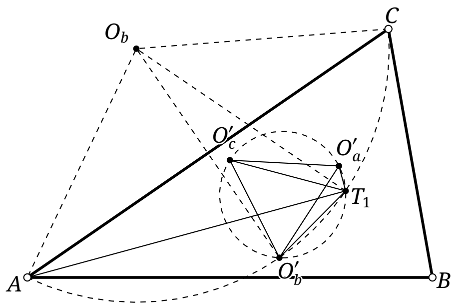



FIGURE 5.56

**Proof.** Let $\triangle O'_aO'_bO'_c$ be the inner Napoleon triangle of $\triangle ABC$. We prove that $T_1 \in \odot(O'_aO'_bO'_c)$; the assertion for $T_2$ is analogous.

In Figure 5.56, let $O_b$ be the reflection of $O'_b$ in the line $AC$. The triangles $\triangle AO_bO'_b$ and $\triangle CO_bO'_b$ are equilateral. Hence $A, C, O'_b$ lie on a circle centered at $O_b$. Since $\angle AT_1C = 120°$, the point $T_1$ lies on the same circle, so $A, C, T_1, O'_b$ are concyclic and $\angle AT_1O'_b = \angle ACO'_b = 30°$. Similarly, $\angle AT_1O'_c = 30°$, and therefore $\angle O'_cT_1O'_b = 60° = \angle O'_cO'_aO'_b$. Thus $O'_a, O'_b, O'_c, T_1$ are concyclic. □

**Proposition 5.9.17.** *Let $r_1, r_2$ be the circumradii of the outer and inner Napoleon triangles of $\triangle ABC$. Then*

$$\frac{r_1}{r_2} = \frac{AAp_2}{AAp_1} = \frac{BAp_2}{BAp_1} = \frac{CAp_2}{CAp_1}.$$

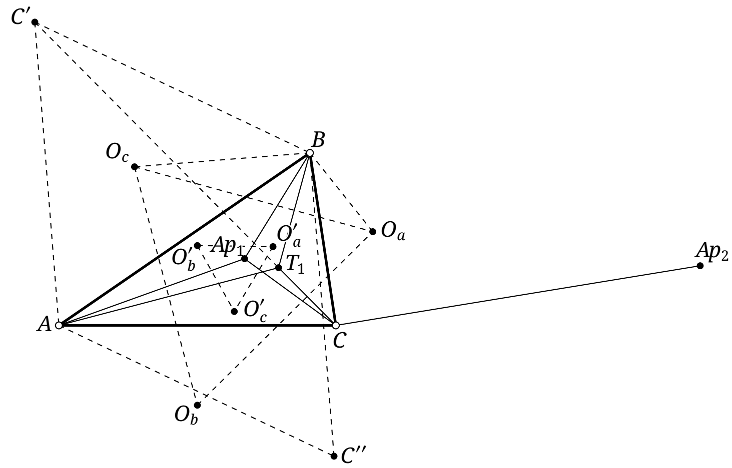



FIGURE 5.57

**Proof.** We prove only $\frac{r_1}{r_2} = \frac{CAp_2}{CAp_1}$. Let $\triangle O'_aO'_bO'_c$ and $\triangle O_aO_bO_c$ be the inner and outer Napoleon triangles. Let $C', C''$ be the third vertices of the equilateral triangles constructed outwardly and inwardly on $AB$, respectively, as in Figure 5.57.

The points $A, B, C', T_1$ are concyclic, $C', T_1, C$ are collinear, and $Ap_1 = \mathbf{g}T_1$. Hence $\angle CAAp_1 = \angle BAT_1 = \angle CC'B$, while $\angle ACAp_1 = \angle C'CB$. Thus $\triangle ACAp_1 \sim \triangle C'CB$, and $\frac{AC}{C'C} = \frac{CAp_1}{CB}$. Similarly, $\frac{AC}{C''C} = \frac{CAp_2}{CB}$, and therefore $\frac{CAp_1}{CAp_2} = \frac{CC''}{CC'}$. We also have $\angle C'BO_c = \angle O_cBA = \angle O_aBC = 30°$

and $\angle C'BC = \angle O_cBO_a$. Also $\frac{BO_c}{BC'} = \frac{BO_a}{BC} = \frac{1}{\sqrt{3}}$, so $\triangle O_cO_aB \sim \triangle C'CB$. Hence $\frac{O_aO_c}{CC'} = \frac{O_aB}{CB} = \frac{1}{\sqrt{3}}$. Similarly, $\frac{O'_aO'_c}{CC''} = \frac{1}{\sqrt{3}}$. Consequently

$$\frac{r_1}{r_2} = \frac{O_aO_c}{O'_aO'_c} = \frac{CC'}{CC''} = \frac{CAp_2}{CAp_1}.$$

□

We next use Napoleon triangles to derive several further properties of the Torricelli points.

**Proposition 5.9.18.** *Let $L,G$ be the Lemoine point and centroid of $\triangle ABC$. The line $LG$ bisects both $Ap_1T_1$ and $Ap_2T_2$.*

**Proof.** In Figure 5.58, let $Q$ be the midpoint of $T_1Ap_1$. For $\triangle ABC$, let $O,G,L$ denote its circumcenter, centroid, and Lemoine point, respectively. Let $\triangle A_1B_1C_1$ be the pedal triangle of $Ap_1$, and $\triangle O_aO_bO_c$ the outer Napoleon triangle.

Since $T_1 = \mathbf{g}Ap_1$, corollary 4.4.5 shows that $Q$ is the circumcenter of $\triangle A_1B_1C_1$. This triangle is equilateral, so $Q$ is its center. By proposition 5.9.15, the triangles $\triangle O_aO_bO_c$ and $\triangle A_1B_1C_1$ are homothetic from $L$, while theorem 5.9.13 says that $G$ is the center of $\triangle O_aO_bO_c$. Thus $Q,G$ correspond under this homothety and $Q,G,L$ are collinear. □

**Proposition 5.9.19.** *For every triangle, $Ap_1T_1 \parallel Ap_2T_2 \parallel \mathscr{E}$.*

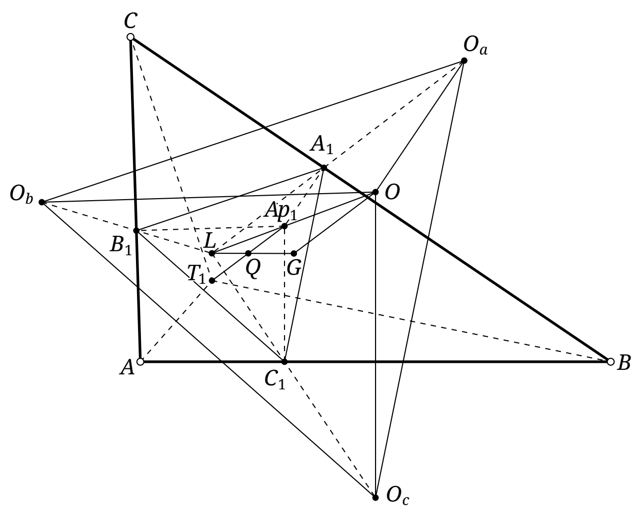



FIGURE 5.58

**Proof.** Continue with the notation of the proof of proposition 5.9.18. Both $O$ and $O_a$ lie on the perpendicular bisector of $BC$, so $OO_a \perp BC$ and $A_1Ap_1 \parallel OO_a$. Cyclically, $B_1Ap_1 \parallel OO_b$ and $C_1Ap_1 \parallel OO_c$. Thus $Ap_1, O$ are also corresponding points of the homothety in that proof. It follows that $QAp_1 \parallel GO$, and hence $Ap_1T_1 \parallel \mathscr{E}$. The proof for $Ap_2T_2$ is analogous. □

**Remark.** This proof gives another proof that $O,Ap_1,Ap_2,L$ are collinear. The points $O,Ap_1$ correspond under the homothety between the outer Napoleon triangle and the pedal triangle of $Ap_1$, so $O,L,Ap_1$ are collinear; similarly, $O,L,Ap_2$ are collinear.

**Proposition 5.9.20.** *If $L$ is the Lemoine point of $\triangle ABC$, then $L, T_1, T_2$ are collinear. More precisely, $L = T_1T_2 \cap Ap_1Ap_2$.*

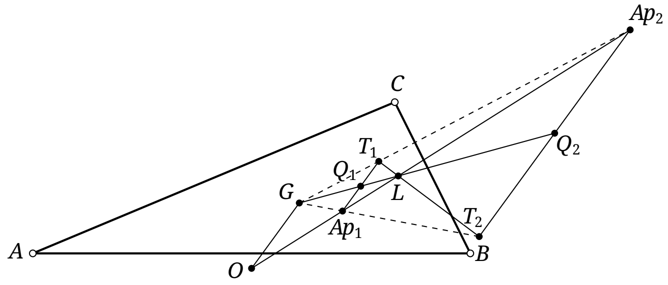



FIGURE 5.59

**Proof.** As in Figure 5.59, let $Q_1, Q_2$ be the midpoints of $Ap_1T_1, Ap_2T_2$, and let $G$ be the centroid. By proposition 5.9.18, $G, Q_1, L$ are collinear; similarly $G, Q_2, L$ are collinear. Hence $G, Q_1, Q_2, L$ lie on one line.

By proposition 5.9.19, $Ap_1T_1 \parallel Ap_2T_2$. Applying example 3.3.18, the intersection of $Ap_1Ap_2$ and $T_1T_2$ lies on $Q_1Q_2$. Thus $T_1T_2, Ap_1Ap_2, GL$ are concurrent. The latter two lines already meet at $L$ by proposition 5.8.5, so the common point is $L$. □

**Proposition 5.9.21.** *If $G$ is the centroid of $\triangle ABC$, then $G = Ap_1T_2 \cap Ap_2T_1$.*

**Proof.** Continue with the notation of proposition 5.9.20, and let $O$ be the circumcenter. By corollary 5.8.6, $(Ap_1Ap_2, LO) = -1$. Since $Q_1$ is the midpoint of $T_1Ap_1$, $(Ap_1T_1, Q_1\infty_{Ap_1T_1}) = -1$. Hence

$$(Ap_1, Ap_2, L, O) \barwedge (Ap_1, T_1, Q_1, \infty_{Ap_1T_1}).$$

The point $Ap_1$ is fixed by this projectivity, so theorem 3.5.6 says that the correspondence is a perspectivity. Its joins of corresponding points are therefore concurrent. Since $G \in LQ_1$ and, by proposition 5.9.19, $G \in O\infty_{Ap_1T_1}$, we obtain $G \in Ap_2T_1$. Similarly, $G \in Ap_1T_2$. □

**Theorem 5.9.22** (Lester's theorem)**.** *For every triangle, its two Torricelli points, circumcenter, and nine-point center are concyclic.* Their circle is called the *Lester circle.*

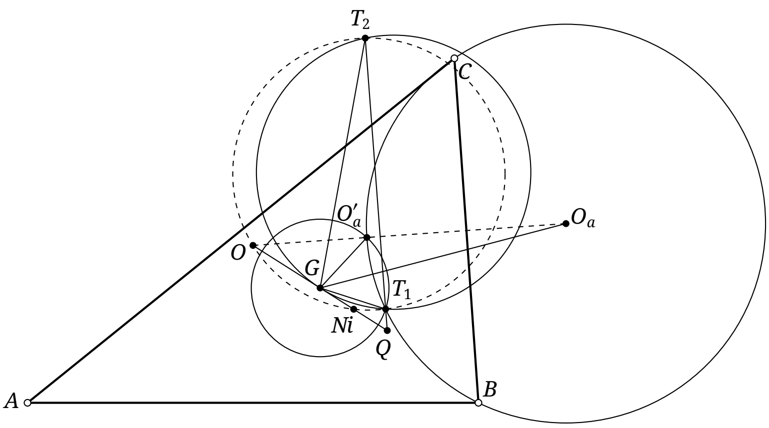



FIGURE 5.60

**Proof.** Let $G$ be the centroid of $\triangle ABC$, and let $\triangle O'_aO'_bO'_c$ and $\triangle O_aO_bO_c$ be the inner and outer Napoleon triangles, as in Figure 5.60. By proposition 5.9.16, the points $T_1, O'_a, O'_b, O'_c$ lie on a circle $\omega_1$, whose center is $G$ by Napoleon's theorem. Also $\angle BT_1C = \angle BO'_aC = 120°$, so $C, B, O'_a, T_1$ lie on a circle $\omega_2$. Since $O_aC = O_aO'_a = O_aB$, this circle is centered at $O_a$.

The common chord of $\omega_1, \omega_2$ is $O'_aT_1$, so $O'_a, T_1$ are symmetric about the line of centers $GO_a$. Similarly, $T_2, O_a$ are symmetric about $GO'_a$.

Apply example 0.3.9 to $\triangle GT_1T_2$. Let the tangent to $\odot(GT_1T_2)$ at $G$ meet $T_1T_2$ at $Q$, and let $O$ be the reflection of $Q$ in $G$. Then $O \in O_aO'_a$, so $O$ lies on the perpendicular bisector of $BC$. Cyclically it lies on the perpendicular bisectors of $AB$ and $AC$; hence $O$ is the circumcenter of $\triangle ABC$. By theorem 0.5.25, the nine-point center $Ni$ is the midpoint of $GQ$.

Finally, power of a point with respect to $\odot(GT_1T_2)$ gives

$$QT_1 \cdot QT_2 = QG^2 = 2QG \cdot \frac{QG}{2} = QNi \cdot QO.$$

Therefore $O, Ni, T_1, T_2$ are concyclic. $\square$

We finish by listing several further properties of the Napoleon points.

**Proposition 5.9.23.** *If $L$ is the Lemoine point of $\triangle ABC$, then $L \in Np_1Np_2$.*

**Proposition 5.9.24.** *If $O$ is the circumcenter of $\triangle ABC$, then $O = Np_1T_1 \cap Np_2T_2$.*

**Proposition 5.9.25.** *For every triangle, $Ni = Np_2T_1 \cap Np_1T_2$.*

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## Exercises

**Exercise 5.85.** For $\triangle ABC$, let $O_1, O_2$ be the centers of the pedal circles of $T_1, T_2$, respectively. Prove that $O_1T_1 \parallel O_2T_2$.

**Exercise 5.86.** Given $\triangle ABC$, let $D, E, F$ lie on the segments $BC, CA, AB$, respectively, and suppose that $\triangle DEF$ is equilateral. Determine the locus of its center.

**Exercise 5.87.** Prove that the line joining the two Torricelli points bisects the segment joining the orthocenter and centroid.

**Exercise 5.88.** Prove that the Euler line of the cevian triangle of $T_1$ is parallel to the Euler line of the original triangle.

**Exercise 5.89.** Given $\triangle ABC$, suppose that rotating the segment $AB$ counterclockwise through 60° makes it coincide with the segment $AC$. Let $D$ vary in the plane, and rotate the segment $AD$ counterclockwise through 90° to obtain the segment $AE$. Choose $M, N$ on the segments $BD, CD$, respectively, so that $MN \parallel AD$. Prove that there is a fixed point $F$, independent of $D$, such that $EF \parallel \mathscr{E}_{\triangle DMN}$ always holds.

**Exercise 5.90.** Prove proposition 5.9.23.

**Exercise 5.91.** For $\triangle ABC$, let $D, E, F$ lie on the segments $BC, CA, AB$, respectively, and satisfy $\frac{BD}{DC} = \frac{CE}{EA} = \frac{AF}{FB}$. Let $G$ be the centroid of $\triangle ABC$, and let $T_1, T'_1$ be the first Torricelli points of $\triangle ABC, \triangle DEF$, respectively. Prove that $\angle GT'_1T_1 = 2\angle GT_1T'_1$.

Part II

# Advanced Conic Geometry

*Conics and Geometric Transformations*

The preceding part introduced the foundations of plane geometry. Building on that material, this part develops plane geometry further by synthetic methods, with particular emphasis on questions connected with conics. We investigate conics through projective geometry, study geometric transformations associated with triangles, and examine important special conics and their applications.

# Chapter 6

# Basic Projective Properties of Conics

Chapter 3 introduced the fundamental concepts of projective geometry. In this chapter we use projective methods to study conics.

## 6.1 Poles and Polars

Since a projectivity can send a conic to a circle, many properties previously established for circles extend directly to arbitrary conics. The theory of poles and polars from section 3.6 is a first example.

**Definition 6.1.1** (Polarity)**.** Let $\alpha$ be a conic.

(1) Given a point $A$, choose a projectivity $f$ sending $\alpha$ to a circle $\omega$ and $A$ to $A'$. Let $a'$ be the polar of $A'$ with respect to $\omega$, and let $a = f^{-1}(a')$. The line $a$ is the *polar* of $A$ with respect to $\alpha$, and $A$ is the *pole* of $a$. We write $a = \mathfrak{p}_\alpha(A)$ and $A = \mathfrak{p}_\alpha^{-1}(a)$; when no ambiguity can arise, we abbreviate these to $a = \mathfrak{p}(A)$ and $A = \mathfrak{p}^{-1}(a)$.

(2) The correspondence that sends a point $A$ to its polar $a$ and a line $a$ to its pole $A$ is called the *polar correspondence* with respect to $\alpha$. It gives a one-to-one transformation between points and lines, called the *polarity* with respect to $\alpha$.

There are infinitely many ways to send $\alpha$ to a circle. Nevertheless, $\mathfrak{p}_\alpha(A)$ is independent of this choice, as will follow from theorem 6.1.3.

**Theorem 6.1.2.** *Projectivities preserve pole–polar relations. If $a$ is the polar of $A$ with respect to a conic $\alpha$, and a projectivity sends $a, A, \alpha$ to $a', A', \alpha'$, respectively, then $a'$ is the polar of $A'$ with respect to $\alpha'$.*

**Proof.** Send $\alpha$ and $\alpha'$ projectively to the same circle and apply the definition. ☐

Sending the conic to a circle also gives the following constructions.

**Theorem 6.1.3.** *For a conic $\alpha$ and a point $A$, its polar may be constructed as follows.*

*(1) If $A \in \alpha$, then $\mathfrak{p}(A)$ is the tangent to $\alpha$ at $A$.* [1]

*(2) If $A$ is exterior to $\alpha$, draw the two tangents from $A$ to $\alpha$. The line joining their points of tangency is $\mathfrak{p}(A)$.*

*(3) If $A$ is interior to $\alpha$, draw a chord $PQ$ through $A$ and let the tangents at $P, Q$ meet at $S$. Draw another chord $P'Q'$ through $A$ and let the tangents at $P', Q'$ meet at $S'$. Then $SS' = \mathfrak{p}(A)$.*

**Proof.** The three assertions are the projective extensions of theorems 3.6.3 and 3.6.4 and corollary 3.6.7, respectively. ☐

**Remark.** In the second assertion, "exterior" means that two real tangents can be drawn from $A$; this is a statement in the real projective plane. In the complex projective plane, two tangents can also be drawn from an interior point, as is easily proved analytically. These tangents are imaginary lines, and the same assertion remains valid.

[1] Thus $\mathfrak{p}_\alpha(A)$ may denote the tangent at $A$ when $A \in \alpha$. When we wish to emphasize tangency, however, we write $\mathfrak{t}_\alpha(A)$, or simply $\mathfrak{t}(A)$. The same notation will also be used for tangents to general curves.

These constructions show directly that every point has a uniquely determined polar. The following statements are the corresponding extensions of theorems 3.6.5, 3.6.8 and 3.6.10, corollary 3.6.6, and proposition 3.6.12.

**Theorem 6.1.4** (La Hire's theorem, or the polarity principle)**.** *For a polarity with respect to a conic, if the polar of $A$ passes through $B$, then the polar of $B$ passes through $A$.*

**Theorem 6.1.5.** *With respect to a fixed conic, the pole of a line is the common point of the polars of all points on that line. Dually, the polar of a point is the line containing the poles of all lines through that point.*

**Theorem 6.1.6.** *With respect to a fixed conic, the polars of four collinear points form four concurrent lines, and the cross-ratio of the four points equals the cross-ratio of the corresponding polars.*

**Theorem 6.1.7** (Harmonic property of poles and polars)**.** *Let $A,B$ lie on a conic $\alpha$, let $P,Q \in AB$, and let $p$ be the polar of $P$. Then $Q \in p \iff (PQ,AB) = -1$.*

**Theorem 6.1.8** (Chasles's theorem)**.** *Suppose that the polars of the vertices $A,B,C$ of $\triangle ABC$ with respect to a conic $\alpha$ are the sides $EF,FD,DE$ of a triangle $\triangle DEF$. Then:*

*(1) the polars of $D,E,F$ are the sides of $\triangle ABC$; thus $\triangle ABC$* and *$\triangle DEF$* are called *polar triangles* of one another with respect to $\alpha$;

*(2) the two triangles are perspective.* Their center of perspectivity is called the *perspector* of $\alpha$ with respect to $\triangle ABC$, or of $\triangle ABC$ with respect to $\alpha$.

Two further observations are useful. First, theorem 6.1.7 gives an equivalent purely projective definition.

**Definition 6.1.9** (Projective definition of pole and polar)**.** Given a conic $\alpha$ and a point $P$, let a variable secant $l$ through $P$ meet $\alpha$ in two points. On $l$ choose $Q$ so that $P,Q$ harmonically separate those two intersections. Then $Q$ moves on a fixed line, called the *polar* of $P$; the point $P$ is called the *pole* of that line.

Second, the polarity principle leads to the following definition.

**Definition 6.1.10.** If the pole of one line with respect to a conic lies on another line, then the pole of the latter also lies on the former; the two lines are called *conjugate* with respect to the conic. If the polar of one point passes through another point, then the polar of the latter also passes through the former; the two points are called *conjugate* with respect to the conic.

**Theorem 6.1.11.** *The line joining two conjugate points meets the conic in a pair of points harmonically separated by the conjugate pair.*

**Proof.** Let $A,B$ be conjugate and let $AB$ meet the conic at $P,Q$. Then $B = \mathfrak{p}(A) \cap PQ$, so $(A,B;P,Q) = -1$. □

Dually, we have the following result; its proof is left to the reader.

**Theorem 6.1.12.** *From the intersection of two conjugate lines, draw the two tangents to the conic. These tangents are harmonically separated by the conjugate pair.*

**Definition 6.1.13.** A triangle is *self-polar* with respect to a conic if each vertex is the pole of the opposite side; it is then called a *self-polar triangle*.

Thus propositions 3.6.11 and 3.6.22 extend to the following results.

**Theorem 6.1.14** (Brocard's theorem)**.** *The diagonal triangle of a complete quadrangle inscribed in a conic is self-polar with respect to that conic.*

**Remark.** The same construction is meaningful when the conic degenerates into two intersecting lines $l_1 \cup l_2$. In Figure 3.30, draw through $A$ two secants $\overline{ABC}$ and $\overline{ADE}$ of this degenerate conic, and put $BE \cap CD = P$ and $BD \cap CE = O = l_1 \cap l_2$. The theorem says that the "polar" of $A$ is $PO$. This gives another interpretation of theorem 3.5.11 in terms of a degenerate conic.

**Theorem 6.1.15.** *The diagonal trilateral of a complete quadrilateral circumscribed about a conic is self-polar with respect to that conic.*

The definition of a self-polar triangle also yields a useful object in triangle geometry.

**Theorem 6.1.16.** *Given $\triangle ABC$ with orthocenter $H$, there is a circle $\omega$ centered at $H$ such that $\triangle ABC$ is self-polar with respect to $\omega$, and the circumcircle $\mathscr{C}$ and nine-point circle $\mathscr{N}$ are inverse in $\omega$. Its radius satisfies*

$$r_\omega^2 = \overline{AH} \cdot \overline{H_aH} = \overline{BH} \cdot \overline{H_bH} = \overline{CH} \cdot \overline{H_cH} = -4R^2 \cos A \cos B \cos C = -2R\rho_{\triangle ABC},$$

*where $R$ is the circumradius and $\triangle H_aH_bH_c$ is the orthic triangle.* The circle $\omega$ is called the *polar circle* of $\triangle ABC$, denoted by $\mathscr{P}_{\triangle ABC}$, or simply $\mathscr{P}$.

**Proof.** Construct the circle centered at $H$ with radius $r_\omega$. By the definition of a self-polar triangle, lemma 0.5.22 immediately implies that $\triangle ABC$ is self-polar with respect to this circle. Under the general inversion $\mathfrak{i}_{\mathscr{P}}$, the vertices of $\triangle ABC$ and $\triangle H_aH_bH_c$ correspond, so their circumcircles $\mathscr{C}, \mathscr{N}$ are inverse circles. □

**Remark.** For an obtuse, acute, or right triangle, respectively, its polar circle is a real circle, an imaginary circle, or a point circle.

Poles and polars of a conic have one further important special property.

**Theorem 6.1.17.** *The polar of a focus of a conic is the corresponding directrix.*

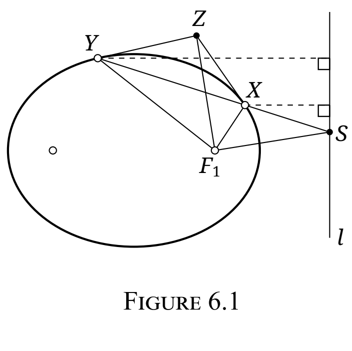


FIGURE 6.1

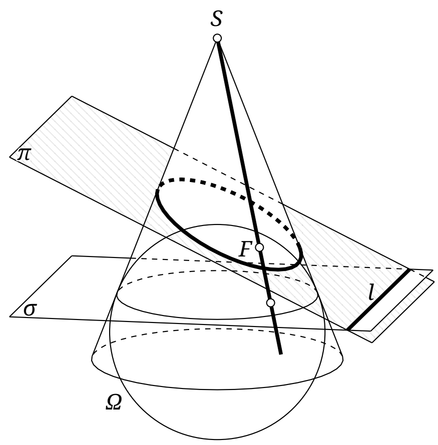


FIGURE 6.2

**Proof.** The parabolic case follows from theorem 1.5.7. We prove the elliptic case; the hyperbolic case is analogous. In Figure 6.1, let $F_1$ be a focus of the ellipse and let $l$ be its polar. We show that $l$ is the corresponding directrix.

Let an arbitrary secant $XY$ meet $l$ at $S$, and let the tangents at $X,Y$ meet at $Z$. By the polarity principle, $ZF_1 = \mathfrak{p}(S)$. Hence theorem 1.3.3 gives $ZF_1 \perp SF_1$, while theorem 1.3.8 says that $ZF_1$ bisects $\angle XF_1Y$. Thus $F_1S$ is the external angle bisector of $\angle YF_1X$. By the external angle-bisector theorem, $\frac{F_1X}{F_1Y} = \frac{SX}{SY} = \frac{\operatorname{dist}(X,l)}{\operatorname{dist}(Y,l)}$, so $l$ is the directrix. □

**Alternative proof.** There is an analogous polarity principle in three-dimensional space: points and planes are mutually polar, while the polar of a line is a line; compare exercise 3.67. We may therefore use the two Dandelin spheres.

In Figure 6.2, let the polar planes of $S,F$ with respect to the sphere $\Omega$ be $\sigma,\pi$. The polar line of $SF$ is their line of intersection, namely the directrix $l$; recall section 1.4. The pole of $l$ with respect to the circle $\sigma \cap \Omega$ is plainly $SF \cap \sigma$. Central projection of these elements onto $\pi$ proves the assertion. □

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## Exercises

**Exercise 6.1.** Prove theorem 6.1.12.

**Exercise 6.2.** Given a conic $\alpha$ and a point $A$, draw two lines through $A$ meeting $\alpha$ at $X_1,X_2$ and $Y_1,Y_2$. Let the tangents at $X_1,X_2$ meet at $X$, and those at $Y_1,Y_2$ meet at $Y$. Prove that $XY = \mathfrak{p}(A)$.

**Exercise 6.3** (Converse of Brocard's theorem)**.** If $\triangle ABC$ is self-polar with respect to a conic $\alpha$, prove that it is the diagonal triangle of some complete quadrangle inscribed in $\alpha$.

**Exercise 6.4.** Given a parabola $\alpha$ and a point $Q$, let $N$ be the projection of $Q$ onto the axis. Through $Q$ draw the perpendicular to its polar, meeting the axis at $L$. Prove that $NL$ equals the focal parameter.

**Exercise 6.5.** Let $P,Q$ lie on an ellipse $\alpha$, and let $Q'$ be the reflection of $Q$ in its major axis. Suppose the lines $PQ,PQ'$ meet the major axis at $M,N$. Prove that the length of the semimajor axis equals the tangent length from the center of $\alpha$ to $\odot(MN)$.

**Exercise 6.6.** Let $F,l$ be a corresponding focus and directrix of an ellipse $\alpha$. A chord $PQ$ passes through $F$. Choose $F',P',Q' \in l$ so that $PP' \parallel QQ' \parallel FF'$. Prove that $S^2_{\triangle PQF'} = 4S_{\triangle PP'F'}S_{\triangle QQ'F'}$.

**Exercise 6.7.** Let $l_1,l_2$ be conjugate diameters of a central conic $\alpha$[2] Prove that the tangents at the intersections of $l_1$ with $\alpha$ are parallel to $l_2$.

**Exercise 6.8.** Prove that the polar circles of the four triangles of a complete quadrilateral are coaxial.

**Exercise 6.9.** Let $F$ be a focus of an ellipse $\alpha$, and let a line through $F$ meet $\alpha$ at $A,B$. Through $A,B$ draw perpendicular lines meeting $\alpha$ again at $P,Q$, respectively. Prove that $\angle APF = \angle BQF$.
*Hint.* Use the projective definition of an angle bisector.

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[2]That is, diameters that are conjugate to one another.

**Exercise 6.10* (Hesse's theorem).** Let $A,A'$ and $B,B'$ be two pairs of conjugate points with respect to a conic $\alpha$. Put $C = AB \cap A'B'$ and $C' = AB' \cap A'B$. Prove that $C,C'$ are conjugate.

**Exercise 6.11* (Gaskin's theorem).** Prove that the circumcircle of a self-polar triangle of a conic is orthogonal to the director circle of the conic.

## 6.2 Quadratic Point Ranges

In this section we develop a projective theory of conics by means of quadratic point ranges.

### 6.2.1 Basic Concepts

The following theorem motivates the definition.

**Theorem 6.2.1.** *Let $A_1,A_2,A_3,A_4$ be four points on a conic $\alpha$. If $P$ is any point of $\alpha$, then $(PA_1,PA_2;PA_3,PA_4)$ is independent of the choice of $P$.*

**Proof.** A projectivity sends $\alpha$ to a circle and preserves cross-ratios. For a circle, the inscribed-angle theorem shows that every directed angle $\measuredangle A_iPA_j$ is independent of $P$. By the definition of cross-ratio, $(PA_1,PA_2;PA_3,PA_4)$ is therefore independent of the choice of $P$. □

**Remark.** The point $P$ may coincide with one of $A_1,A_2,A_3,A_4$; the corresponding join is then interpreted in the limiting sense as the tangent. Thus, if $P = A_1$, the line $PA_1$ means $\mathrm{t}_\alpha(A_1)$.

**Definition 6.2.2** (Quadratic point range)**.**

(1) A *quadratic point range* is a collection of points $A,B,C,\ldots$ on a conic $\alpha$, denoted by $\alpha(A,B,C,\ldots)$. [3]

(2) For four points $A,B,C,D$ on $\alpha$, define their *cross-ratio* by
$$\alpha(A,B;C,D) := (AP,BP;CP,DP),$$
where $P$ is any point of $\alpha$; this is well defined by theorem 6.2.1. When no ambiguity can arise, we may write $\alpha(AB,CD)$, $(A,B;C,D)$, or $(AB,CD)$.

(3) If $\alpha(AB,CD) = -1$, the range $\alpha(A,B,C,D)$ is a *harmonic quadratic range*; the quadrangle $ACBD$ is then called a *harmonic quadrangle*. [4]

We begin with some elementary properties.

**Proposition 6.2.3.** *If $ABCD$ is a cyclic harmonic quadrangle, then $AB\cdot CD = BC\cdot AD$.*

**Proof.** Choose any point $P$ on the circumcircle of $ABCD$. Then $(PA,PC;PB,PD) = -1$. By the definition of cross-ratio, $\sin\measuredangle BPA\cdot\sin\measuredangle DPC = -\sin\measuredangle BPC\cdot\sin\measuredangle DPA$. The sine rule now gives $BA\cdot DC = BC\cdot DA$. □

[3] The symbol $\alpha$ may be suppressed when the supporting conic is clear. As with an ordinary point range, the term imposes no restriction on the number of points.

[4] The order matters: saying that the quadrangle $ABCD$ is harmonic means that $(A,C,B,D)$ is a harmonic quadratic range.

**Proposition 6.2.4.** *Suppose an inversion sends a line $a_1$ to a line or circle $a_2$, and sends $A,B,C,D \in a_1$ to $A',B',C',D'$, respectively. Then $a_1(AB,CD) = a_2(A'B',C'D')$.*

**Proof.** If $a_1,a_2$ are both lines, the assertion follows from theorem 4.1.13. If $a_1$ is a line and $a_2$ a circle, it is essentially theorem 3.6.8. In Figure 3.35, inversion in $\odot O$ sends $A,B,C,D$ to $A',B',C',D'$, and

$$(AB,CD) = (PA',PB';PC',PD') = (A',B';C',D').$$

□

**Proposition 6.2.5** (Harmonic quadratic ranges)**.** *If $A,B,P,Q$ form a harmonic quadratic range on a conic $\alpha$, then $\mathfrak{p}^{-1}(PQ) \in AB$.*

**Proof.** Let $R = \mathfrak{p}^{-1}(PQ) = \mathrm{t}(P) \cap \mathrm{t}(Q)$. Suppose $AR \cap \alpha = A,B'$ and $AR \cap PQ = T$. The harmonic property of poles and polars gives $(A,B';R,T) = -1$, hence $(PA,PB';PR,PT) = -1$. The hypothesis gives $(A,B;P,Q) = -1$, and hence $(PA,PB;PP,PQ) = -1$. Note that $PT = PQ$, while $PP = \mathrm{t}(P) = PR$ denotes the tangent to $\alpha$ at $P$. □

**Proposition 6.2.6.** *Let the polar of $P$ with respect to a conic $\alpha$ meet $\alpha$ at $M,N$, and let any line through $P$ meet $\alpha$ at $A,B$. Then $A,B,M,N$ form a harmonic quadratic range on $\alpha$.*

**Proof.** By theorem 6.1.3, $PM$ is tangent to $\alpha$ at $M$. By theorem 6.1.7, the intersections with $AB$ of $MA,MB,MM,MN$, where $MM = \mathrm{t}(M)$, form a harmonic range. Thus $(MA,MB;MM,MN) = -1$, and hence $(A,B;M,N) = -1$. □

The cross-ratio of a quadratic range permits a purely projective definition of a conic.

**Definition 6.2.7** (Projective definition of a conic)**.**

(1) Let a projectivity $f$ send the pencil at $A$ to the pencil at $B$. The locus of the intersection of a line of the first pencil with its image under $f$ is a *conic*, or *quadratic curve*. If $f$ is a perspectivity, the conic is *degenerate*.

(2) A nondegenerate conic is an *ellipse* if it does not meet the line at infinity, a *parabola* if it is tangent to the line at infinity, and a *hyperbola* if it meets the line at infinity in two points. [5]

**Remark.** The construction above does not appear to produce a double line. In the limiting sense, however, a double line is obtained when two distinct lines of a degenerate conic tend to coincidence.

We verify that this projective definition is equivalent to the earlier one. If $f$ is a perspectivity, it has a corresponding line fixed by $f$, which can only be $AB$. By theorem 3.5.7, the intersections of all other pairs of corresponding lines lie on one line $l$. Every point of $AB$ also qualifies as an intersection because $f(AB) = AB$. The conic therefore degenerates into $l \cup AB$.

Conversely, if $f$ is not a perspectivity, theorem 3.5.7 shows that the locus cannot be a line. Suppose three points $C,D,E$, distinct from $A,B$, are known to lie on the locus. If a further point $F$ also lies on it, then preservation of cross-ratio requires

$$(AD,AC;AE,AF) = (BD,BC;BE,BF).$$

Let $\alpha$ be the conic through $A,B,C,D,E$ furnished by the earlier definition; its existence is guaranteed by theorem 3.1.14. If $F \notin \alpha$, put $AF \cap \alpha = A,F'$. Then

$$(AD,AC;AE,AF) = (AD,AC;AE,AF') = (BD,BC;BE,BF'),$$

Since $F \notin \alpha$, however, $BF \neq BF'$, and hence $(BD,BC;BE,BF') \neq (BD,BC;BE,BF)$, a contradiction. Thus $F \in \alpha$.

[5] Here a circle is included among the ellipses. As elsewhere, the unqualified term "conic" usually means a nondegenerate conic, though the context will make clear when degenerate conics are included.

Finally, once the global projective definition of a conic is accepted, the classification in the second part is equivalent to the earlier metric classification by proposition 3.6.22. This proves the equivalence of the definitions. ☐

**Proposition 6.2.8.** *In projective definition A, the conic generated by a projectivity between the pencils at A and B passes through A and B.*

**Proof.** This was proved in the preceding discussion. ☐

Here are some applications of the projective definition.

**Theorem 6.2.9.** *Given $\triangle ABC$ and a fixed point $Z$, let a variable line $l$ through $Z$ meet $BC$,$AC$ at $A'$,$B'$, and put $Q = AA' \cap BB'$. Then the locus of $Q$ is a conic through $A$,$B$,$C$ tangent to both $AZ$ and $BZ$.*

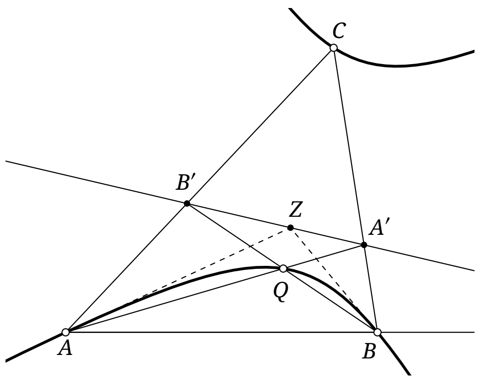


FIGURE 6.3

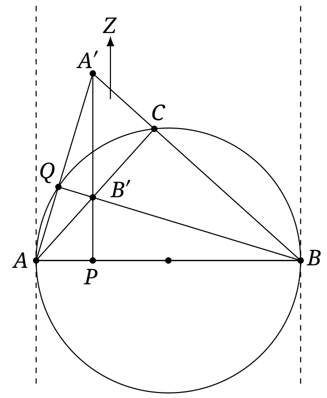


FIGURE 6.4

**Proof.** As $A'$ moves on the line $BC$, clearly $[AA'] \overset{(BC)}{\barwedge} [A'Z] \overset{(AC)}{\barwedge} [B'B]$.[6] Thus $[AA'] \barwedge [BB']$ gives a projectivity from the pencil at $A$ to the pencil at $B$. By the projective definition, the locus of $AA' \cap BB'$ is a conic $\alpha$ through $A$,$B$.

When $l = CZ$, we have $Q = C$, so $C \in \alpha$. As $A'$ tends to $AZ \cap BC$, the point $Q$ tends to $A$ and $QA$ tends to the tangent to $\alpha$ there; hence $AZ$ is that tangent. Similarly, $BZ$ is tangent at $B$. ☐

We can also prove the result without using the projective definition of a conic.

**Alternative proof.** As in Figure 6.4, apply a projectivity sending $\triangle ABC$ to an isosceles right triangle with right angle at $C$, and sending $Z$ to the point at infinity in the direction perpendicular to $AB$. This can be done because a projectivity is determined by four corresponding points. Then $A'B' \perp AB$, so $B'$ is the orthocenter of $\triangle A'AB$. Hence $AA' \perp BB'$, and $Q$ lies on the circumcircle of $\triangle ABC$, which is plainly tangent to $AZ$ and $BZ$. ☐

**Example 6.2.10.** Given $\triangle ABC$ and real numbers $u,v,w$, for a variable point $P$ write $[\triangle PBC] = \lambda$, $[\triangle PCA] = \mu$, and $[\triangle PAB] = \nu$. The locus of points satisfying $\frac{u}{\lambda} + \frac{v}{\mu} + \frac{w}{\nu} = 0$ is a conic through $A,B,C$.

[6] Brackets around an element denote the range or pencil generated as that point or line varies. For example, $[AA'] \overset{(BC)}{\barwedge} [A'Z]$ means that, as $A'$ varies, the pencil at $A$ generated by $AA'$ and the pencil at $Z$ generated by $A'Z$ are in projective correspondence. The lines $AA'$ and $A'Z$ are representative elements of the two pencils: for every $A'$, the line $AA'$ corresponds to the line $A'Z$.

**Proof.** Let the side lengths of $\triangle ABC$ be $a,b,c$, put $\measuredangle ACB=\alpha$, $\measuredangle ABC=\beta$, and write $\measuredangle PCA=\theta$, $\measuredangle PBA=\phi$. Then

$$\frac{\mu}{\lambda}=\frac{\frac{1}{2}PC\cdot b\sin\theta}{\frac{1}{2}PC\cdot a\sin(\alpha-\theta)}=\frac{b\sin\theta}{a\sin(\alpha-\theta)},\quad \frac{\nu}{\lambda}=\frac{\frac{1}{2}PB\cdot c\sin\phi}{\frac{1}{2}PB\cdot a\sin(\beta-\phi)}=\frac{c\sin\phi}{a\sin(\beta-\phi)}.$$

Thus the required relation is

$$\begin{aligned}0=u+\frac{v}{\mu/\lambda}+\frac{w}{\nu/\lambda}&=u+\frac{av}{b}\frac{\sin(\alpha-\theta)}{\sin\theta}+\frac{aw}{c}\frac{\sin(\beta-\phi)}{\sin\phi}\\&=\frac{av}{b}\sin\alpha\cot\theta-\frac{av}{b}\cos\alpha+\frac{aw}{c}\sin\beta\cot\phi-\frac{aw}{c}\cos\beta+u.\end{aligned}$$

There are therefore fixed constants $U,V$ such that $\cot\phi=U\cot\theta+V$. A direct calculation using theorem 3.2.8 shows that the map $[PB]\to[PC]$ from the pencil at $B$ to the pencil at $C$ preserves the cross-ratios of corresponding elements, and is therefore a projectivity. [7] The projective definition now shows that the locus is a conic through $B,C$; similarly, it also passes through $A$. ☐

Finally, the converse of theorem 6.2.1 gives another projective generation of a conic. Its proof is left to the reader.

**Theorem 6.2.11.** *Let $A,B,C,D$ be four points, no three collinear. The locus of points $P$ for which $(PA,PB;PC,PD)$ has a prescribed constant value is a conic through $A,B,C,D$.*

**Remark.** The literal definition omits $A,B,C,D$ from the locus, but all four belong to it in the limiting sense.

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## Exercises

**Exercise 6.12.** Prove Pascal's theorem using quadratic point ranges.

**Exercise 6.13.** Prove that five lines, no three concurrent, are tangent to a unique conic.

**Exercise 6.14.** The variable lines $AB,BC,CA$ pass through fixed points $P,Q,R$, respectively, while $A,B$ move on fixed lines $l_1,l_2$. Prove that the locus of $C$ is a conic. When does it degenerate?

**Exercise 6.15.** Let $I$ be the incenter of $\triangle ABC$ and $\triangle DEF$ its intouch triangle. Choose $D',E',F'$ on the lines $DI,EI,FI$, respectively, so that $\frac{\overline{ID}}{\overline{ID'}}=\frac{\overline{IE}}{\overline{IE'}}=\frac{\overline{IF}}{\overline{IF'}}$. Prove that the lines $AD'$, $BE'$, and $CF'$ are concurrent at a point $Y$, and that $Y$ moves on a conic through the six points $A,B,C,I,Ge,Na$.

### 6.2.2 Projectivities and Involutions of Quadratic Ranges

Just as an ordinary point range admits perspectivities, projectivities, and involutions, so does a quadratic point range on a conic.

**Definition 6.2.12.** Let $\alpha$ be a conic.

(1) Given a quadratic range $\alpha(A,B,C,\dots)$ and a point $O\in\alpha$, joining $O$ to the members of the range gives a pencil $O(A,B,C,\dots)$. This is a *perspectivity* between the quadratic range and the pencil, written

$$\alpha(A,B,C,\dots)\overline{\wedge}O(A,B,C,\dots).$$

[7] One may instead use the formula for cross-ratio-preserving maps in section 6.6.2.

(2) Suppose that two quadratic ranges $\alpha(A,B,C,\ldots)$ and $\alpha(A',B',C',\ldots)$ satisfy
$$O(A,B,C,\ldots)\barwedge O(A',B',C',\ldots)$$
for some $O\in\alpha$. They are then said to be in *projective correspondence*, written
$$\alpha(A,B,C,\ldots)\overset{(O)}{\barwedge}\alpha(A',B',C',\ldots).$$
The point $O$ is a center of this projectivity. This *projective correspondence* sends points of $\alpha$ to points of $\alpha$ and is called a *projectivity*, or a projective transformation, of $\alpha$; the conic $\alpha$ is its *base conic*.

**Remark.** The center $O$ may be omitted from the notation. Indeed, if the displayed projectivity holds at one $O\in\alpha$, then for every $P\in\alpha$,
$$P(AB,CD)=O(AB,CD)=O(A'B',C'D')=P(A'B',C'D')$$
by theorem 6.2.1. Consequently, $\alpha(A,B,C,D)\overset{(P)}{\barwedge}\alpha(A',B',C',D')$ also holds. Thus we may omit the center $(O)$ and simply write $\alpha(A,B,C,D)\barwedge\alpha(A',B',C',D')$.

The definition immediately gives the following property.

**Theorem 6.2.13.** *Corresponding elements of a quadratic range and a pencil in perspective have equal cross-ratios.*

As for one-dimensional projectivities, the following results hold; the proofs are left to the reader.

**Theorem 6.2.14.** *A projectivity of a conic preserves cross-ratios. Conversely, every cross-ratio-preserving correspondence between two quadratic ranges on a conic is a projectivity.*

**Theorem 6.2.15.** *The composite of two projectivities of a conic is again a projectivity.*

**Theorem 6.2.16.** *A projectivity between two quadratic ranges on a conic is uniquely determined by three pairs of corresponding points.*

Projectivities of quadratic ranges have a useful special property.

**Proposition 6.2.17.** *If $\alpha(A,B,C,D)\barwedge\alpha(A',B',C',D')$, then $A'(A,B,C,D)\doublebarwedge A(A',B',C',D')$.*

**Proof.** We have the chain
$$A'(A,B,C,D)\doublebarwedge\alpha(A,B,C,D)\barwedge\alpha(A',B',C',D')\doublebarwedge A(A',B',C',D').$$
The join $AA'$ of the two vertices corresponds to itself. The projectivity between the two pencils is therefore a perspectivity; by theorem 3.5.7, the intersections of all remaining pairs of corresponding lines are collinear. □

**Theorem 6.2.18.** *Let $f$ be a projectivity between quadratic ranges on a conic $\alpha$. For every $A,B\in\alpha$, the intersection of the lines $Af(B),Bf(A)$ lies on a fixed line.* This line is called the *projective axis* of $f$.

**Proof.** Write $\alpha(A_1,A_2,A_3,\ldots)\barwedge\alpha(A'_1,A'_2,A'_3,\ldots)$. By proposition 6.2.17, for each fixed $i$ the pencils satisfy $A'_i(A_1,A_2,\ldots)\doublebarwedge A_i(A'_1,A'_2,\ldots)$; call this perspectivity $f_i$. By theorem 3.5.7, for every $j$ the point $A'_iA_j\cap A_iA'_j$ lies on the axis of $f_i$.

It remains to show that these axes are independent of $i$. Put $X=A'_1A_2\cap A_1A'_2$, $Y=A'_2A_3\cap A_2A'_3$, and $Z=A'_3A_1\cap A_3A'_1$. The axis of $f_1$ is $XZ$ and that of $f_2$ is $XY$. Pascal's theorem makes $X,Y,Z$ collinear, so the two axes coincide. The other $f_i$ are analogous. □

**Corollary 6.2.19.** *Let $f$ be a projective transformation of a conic $\alpha$. The intersections of the projective axis of $f$ with $\alpha$ are fixed points of $f$.*

As in dimension one, projectivities of a conic are classified by their fixed points. With no real fixed point they are *elliptic*; with exactly one, *parabolic*; and with two, *hyperbolic*. These cases correspond to the projective axis being exterior, tangent, or secant to the conic.

We now turn to involutions.

**Definition 6.2.20.** A projectivity $f$ of a conic $\alpha$ is an *involutory projectivity*, or simply an *involution*, if $f \neq \mathrm{id}$ and $f \circ f = \mathrm{id}$. Its projective axis is then called its *axis of involution*.

The following analogues of the one-dimensional theory are left to the reader.

**Theorem 6.2.21.** *An involution on a conic is uniquely determined by two pairs of corresponding points.*

**Theorem 6.2.22.** *If an involution $f$ on a conic $\alpha$ has fixed points $X,Y$ and $f(P) = P'$, then $P,P',X,Y$ form a harmonic quadratic range.*

Involutions on a conic also have the following properties.

**Proposition 6.2.23.** *The tangents at a pair of corresponding points of an involution on a conic $\alpha$ meet on its axis of involution.*

**Proof.** Suppose $\alpha(A,B,C,D) \barwedge \alpha(A',B',C',D')$ is an involution. Then $\alpha(A,B,C,A') \barwedge \alpha(A',B',C',A)$. By proposition 6.2.17, $A'(A,B,C,A') \barwedge A(A',B',C',A)$. Consequently the limiting lines $AA$ and $A'A'$—that is, the tangents at $A$ and $A'$—meet on the projective axis. □

**Proposition 6.2.24.** *The fixed points of an involution on a conic $\alpha$ are the intersections of its axis with $\alpha$.*

**Proof.** This is corollary 6.2.19. □

Accordingly, an involution is *elliptic* if it has no real fixed points and *hyperbolic* if it has two. These cases correspond to an axis exterior or secant to the conic. There is no parabolic involution with exactly one fixed point. For if an involution $f$ on $\alpha$ had the unique fixed point $T$, its axis would be the tangent $\mathfrak{t}(T)$. Given any $P$ on that tangent, the two tangents from $P$ to $\alpha$ would touch at $T$ and some $P'$, making $T,P'$ a corresponding pair. As $P$ is arbitrary, $T$ would correspond to every point of $\alpha$, contradicting bijectivity.

**Proposition 6.2.25.** *For a given conic $\alpha$, an axis of involution determines a unique involution.*

**Proof.** Over the complex projective plane, let the axis meet $\alpha$ at $C,D$. By proposition 6.2.17, these are the fixed points of the involution, and two pairs of corresponding points determine an involution. □

**Proposition 6.2.26.** *The joins of corresponding points of an involution $f$ on a conic $\alpha$ pass through a fixed point: the pole, with respect to $\alpha$, of the axis of involution.* This point is called the *center of involution*.

**Proof.** Let $A,A'$ be corresponding points of the involution. Then $\mathfrak{t}(A) \cap \mathfrak{t}(A')$ lies on its axis; that is, $\mathfrak{p}^{-1}(AA')$ lies on the axis. The polarity principle therefore says that $AA'$ passes through the pole of that axis. □

The axis cannot be tangent to the conic; equivalently, the center of involution cannot lie on the conic, for otherwise no projective correspondence would result.

**Proposition 6.2.27.** *A fixed point not on a conic determines a unique involution having that point as center. Equivalently, if the joins of corresponding points of two quadratic ranges pass through a fixed point, their correspondence is an involution.*

**Proof.** Take the polar of the fixed point as the axis and apply proposition 6.2.25. □

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## Exercises

**Exercise 6.16.** Prove theorems 6.2.15, 6.2.16, 6.2.21 and 6.2.22.

**Exercise 6.17.** Let $A, B$ be fixed points of a conic $\alpha$, and let $l$ be a fixed line. For a variable point $P \in \alpha$, put $R = PA \cap l$, and let $BR$ meet $\alpha$ again at $Q$. Prove that $f : [P] \to [Q]$ is a projectivity of $\alpha$.

**Exercise 6.18.** Suppose the quadratic ranges $(A, B, C, D)$ and $(A', B', C', D')$ on a conic $\alpha$ are projectively corresponding, and the positions of their three corresponding pairs $(A, A'), (B, B'), (C, C')$ are known. Given $D$, construct its corresponding point $D'$ as follows: put $AB' \cap A'B = B_0$, $AC' \cap A'C = C_0$, and $A'D \cap B_0C_0 = D_0$. Prove that $D'$ is the second intersection of $AD_0$ with $\alpha$.

**Exercise 6.19.** Let $\alpha$ be a conic and let $A, B \notin \alpha$. For variable $P \in \alpha$, let $PA$ meet $\alpha$ again at $P'$, and let $P'B$ meet it again at $P''$. Of the maps $f_1 : [P] \to [P']$, $f_2 : [P'] \to [P'']$, and $f_3 : [P] \to [P'']$, which are involutions of $\alpha$?

**Exercise 6.20.** Let conics $\alpha_1, \alpha_2$ meet at $A$, and let $O$ be a fixed point exterior to $\alpha_2$. A variable chord $PQ$ of $\alpha_2$ passes through $O$. Let the lines $AP, AQ$ meet $\alpha_1$ again at $P', Q'$. Prove that the line $P'Q'$ passes through a fixed point.

## 6.3 Fundamental Theorems on Conics

We now collect several important theorems of projective geometry concerning conics. The most fundamental are Pascal's and Brianchon's theorems.

**Theorem 6.3.1** (Pascal's theorem)**.** *For any six points $X_1, X_2, X_3, X_4, X_5, X_6$ on a conic, the three points $X_1X_2 \cap X_4X_5$, $X_2X_3 \cap X_5X_6$, and $X_3X_4 \cap X_6X_1$ are collinear.*

**Theorem 6.3.2** (Brianchon's theorem)**.** *Let six lines $l_i$, $i = 1, \dots, 6$, be tangent to one conic, and put $A_{ij} = l_i \cap l_j$. Then the lines $A_{12}A_{45}$, $A_{23}A_{56}$, and $A_{34}A_{61}$ are concurrent.*

We have already proved these theorems using homotheties, projectivities, isogonal conjugacy, and radical axes. Here is a projective-geometric proof of Pascal's theorem using quadratic point ranges. Brianchon's theorem cannot be proved projectively by quadratic point ranges; after sections 6.5 and 6.6, it will emerge as the dual statement.

**Proof of theorem 6.3.1.** In Figure 6.5, let the conic be $\alpha$ and denote the three points in the assertion by $A, B, C$, in that order. Put $P = X_5X_6 \cap X_1X_2$ and $Q = X_1X_6 \cap X_4X_5$. Then

$$(X_5, B, P, X_6) \doublebarwedge X_2 (X_5, B, P, X_6) \barwedge \alpha (X_5, X_3, X_1, X_6) \doublebarwedge X_4 (X_5, X_3, X_1, X_6) \doublebarwedge (Q, C, X_1, X_6).$$

Thus $(X_5, B, P, X_6) \barwedge (Q, C, X_1, X_6)$. This projectivity has the fixed point $X_6$, so by theorem 3.5.6 it is a perspectivity. The joins of corresponding points $X_5Q, BC, PX_1$ are concurrent. Since $A = PX_1 \cap X_5Q$, the points $A, B, C$ are collinear. □

The two theorems also apply in limiting cases where two points, or two tangents, coalesce. Figure 6.6 shows a degeneration of Brianchon's theorem in which the lines containing the sides $CP_a, DP_a$ of the circumscribed hexagon $ABCP_aDE$ coincide and $P_a$ is the point of tangency. It follows that $AP_a, EC, BD$ concur at $K$. [8]

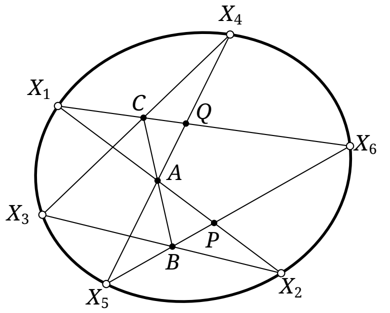


FIGURE 6.5

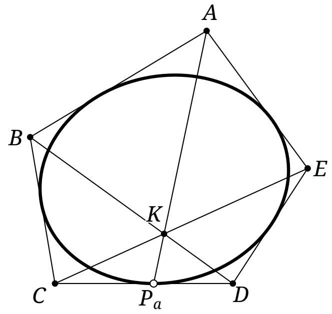


FIGURE 6.6

We now prove the converses.

**Theorem 6.3.3** (Converse of Pascal's theorem)**.** *For six points $X_1,X_2,X_3,X_4,X_5,X_6$, put $A=X_1X_2 \cap X_4X_5$, $B=X_2X_3 \cap X_5X_6$, and $C=X_3X_4 \cap X_6X_1$. If $A,B,C$ are collinear, then $X_1,X_2,X_3,X_4,X_5,X_6$ lie on one conic.*

**Proof.** By theorem 3.1.14, the conic $\alpha = \mathfrak{C}(X_1X_2X_3X_4X_5)$ is uniquely determined. Put again $A = X_1X_2 \cap X_4X_5$, $B=X_2X_3 \cap X_5X_6$, and $C=X_3X_4 \cap X_6X_1$, and let the second intersection of $\alpha$ with $BX_5$ be $Y$. Pascal's theorem gives $X_3X_4 \cap X_1Y \in AB$, so $X_3X_4 \cap X_1Y = C$. Hence $Y = X_6$. □

**Theorem 6.3.4** (Converse of Brianchon's theorem)**.** *Let $l_1,l_2,l_3,l_4,l_5,l_6$ be six lines and put $l_i \cap l_j = A_{ij}$. If $A_{12}A_{45}, A_{23}A_{56}, A_{34}A_{61}$ are concurrent, then the six lines are tangent to one conic.*

**Proof.** The argument is similar to the proof of the converse of Pascal's theorem, with "five points determine a conic" replaced by "five lines determine a conic," namely theorem 4.5.10. The details are left to the reader. □

Pascal's theorem gives a straightedge-only construction of arbitrarily many further points of a conic from five known points. Dually, Brianchon's theorem gives a straightedge-only construction of arbitrarily many further tangents from five known tangents. We leave the explicit constructions to the reader.

Here is an application of Brianchon's theorem.

**Example 6.3.5.** Use Brianchon's theorem to prove theorem 4.2.12: the orthocenter of a triangle circumscribed about a parabola lies on the directrix of the parabola.

**Proof.** In Figure 6.7, let $H$ be the orthocenter of a triangle $ABC$ circumscribed about the parabola $\alpha$. Draw a tangent $l_1$ through $H$ and the line $l_2$ through $H$ perpendicular to $l_1$. It suffices to prove that $l_2$ is also tangent to $\alpha$, for theorem 1.5.7 will then place $H$ on the directrix.

Let $Y,X$ be the points at infinity on $AC, l_2$, and put $P = AB \cap l_1$. Consider the six-line figure $\{BP, PH, HX, XY, YC, CB\}$. Its principal diagonals $BX, PY, HC$ concur, since they are the three altitudes of $\triangle HBP$. By the converse of Brianchon's theorem, a conic is tangent to $AB, BC, CA, l_1, l_2, XY$. The five lines $AB, BC, CA, l_1, XY$ already determine that conic as $\alpha$, so $l_2$ is tangent to $\alpha$. □

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[8] Let $AP, BP$ be the two tangents from $P$ to a conic, touching at $A, B$. As $P$ approaches the conic, $A, B, P$ tend to the same point, while $AP$ and $BP$ coalesce into the tangent there. Thus the intersection of two coalescing tangents becomes their point of tangency.

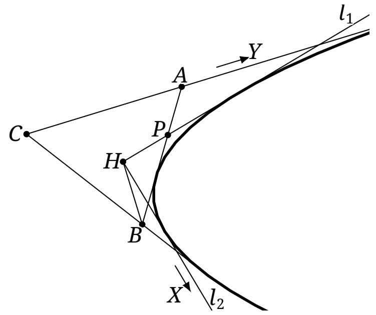


FIGURE 6.7

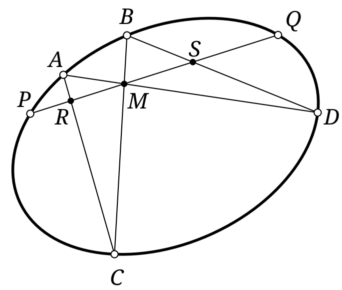


FIGURE 6.8

Another celebrated projective theorem on conics is Candy's theorem.

**Theorem 6.3.6** (Candy's theorem)**.** *In Figure 6.8, let $P,Q$ lie on a conic $\alpha$ and let $M \in PQ$. Draw chords $AD,BC$ through $M$, and put $R = AC \cap PQ$, $S = BD \cap PQ$. Then $\frac{1}{MP} + \frac{1}{MQ} = \frac{1}{MR} + \frac{1}{MS}$.*

**Proof.** The chain

$$(P,R,M,Q) \overset{=}{\wedge} C(P,R,M,Q) \overset{=}{\wedge} \alpha(P,A,B,Q) \overset{=}{\wedge} D(P,A,B,Q) \overset{=}{\wedge} (P,M,S,Q)$$

gives $(PR,MQ) = (PM,SQ)$. Expanding the two cross-ratios proves the identity. $\square$

When $M$ is the midpoint of the chord, this specializes to the following theorem, named for the shape of its configuration.

**Theorem 6.3.7** (Butterfly theorem)**.** *Let $M$ be the midpoint of a chord $PQ$ of a conic $\alpha$. Through $M$ draw chords $AD,BC$, and put $R = AC \cap PQ$, $S = BD \cap PQ$. Then $MR = MS$.*

We next record two further results.

**Theorem 6.3.8.** *In Figure 6.9, let $\triangle A_1B_1C_1$ and $\triangle A_2B_2C_2$ be the anticevian triangles of $D_1,D_2$ with respect to $\triangle A'B'C'$. Then the points $A_1,B_1,C_1,D_1,A_2,B_2,C_2,D_2$ lie on one conic.* This conic is called the *double-anticevian conic* of $D_1,D_2$ with respect to $\triangle A'B'C'$.

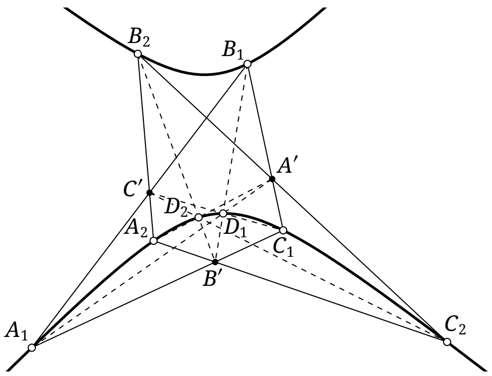


FIGURE 6.9

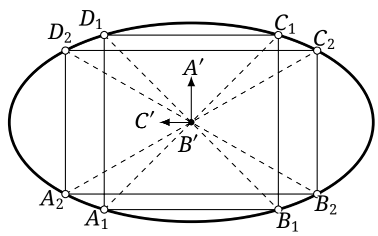


FIGURE 6.10

**Proof.** Let $\alpha = \mathfrak{C}(A_1B_1C_1D_1D_2)$. Brocard's theorem applied to the complete quadrangle $A_1B_1C_1D_1$ shows that $\triangle A'B'C'$ is self-polar with respect to $\alpha$, so $B'C' = \mathfrak{p}_\alpha(A')$. Let $A'D_2$ meet $\alpha$ again at $A_2'$. Then $B'(D_2,A_2',C',A')$ is a harmonic pencil. On the other hand, the complete quadrangle $A_2B_2C_2D_2$ makes $B'(D_2,A_2,C',A')$ a harmonic pencil. Hence $B'A_2' = B'A_2$, so $A_2' = A_2$ and $A_2 \in \alpha$. Cyclically, $B_2,C_2 \in \alpha$. $\square$

**Alternative proof.** In Figure 6.10, use a projectivity to send $A_1B_1C_1D_1$ to a square. The points $A', C'$ become the points at infinity in two perpendicular directions, $A_2B_2C_2D_2$ becomes a rectangle with sides parallel to those of the square, and $B'$ becomes the common center of $A_1B_1C_1D_1$ and $A_2B_2C_2D_2$. By symmetry, the points $A_1, B_1, C_1, D_1, A_2, B_2, C_2, D_2$ lie on one conic. $\square$

The last theorem does not itself mention a conic, but poles and polars make its proof especially simple.

**Theorem 6.3.9.** *Given $\triangle ABC$ and fixed points $P, Q$, let $\triangle A_1B_1C_1$ and $\triangle A_2B_2C_2$ be their respective cevian triangles. Put $C_3 = CC_1 \cap A_2B_2$ and $C_4 = CC_2 \cap A_1B_1$, and define $A_3, A_4$ and $B_3, B_4$ similarly. Then the six lines $A_1A_4, A_2A_3, B_1B_4, B_2B_3, C_1C_4, C_2C_3$ are concurrent.* Their common point is called the *crosspoint* of $P, Q$ with respect to $\triangle ABC$.

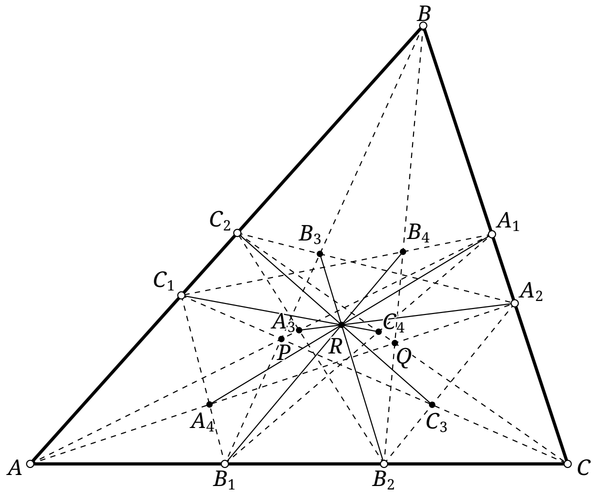


FIGURE 6.11

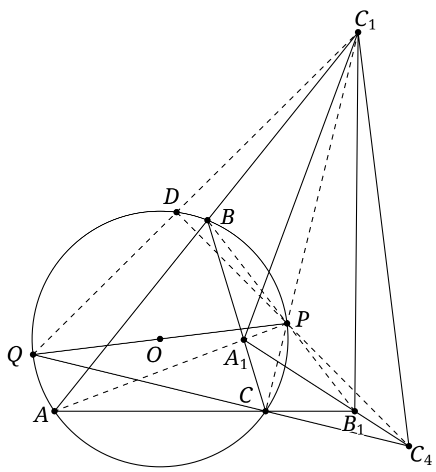


FIGURE 6.12

**Proof.** Let $\alpha = \mathfrak{C}(ABCPQ)$ and apply a projectivity sending $\alpha$ to a circle with diameter $PQ$. In Figure 6.12, Brocard's theorem for the complete quadrangle $ABCP$ gives $\mathfrak{p}_{\odot(ABC)}(C_1) = A_1B_1$. Let $QC_1$ meet the circle again at $D$. In $\triangle QPC_1$, the lines $PD, QC$ are altitudes because $PQ$ is a diameter. Let them meet at $C_4'$, the orthocenter.

Brocard's theorem applied to the complete quadrangle $QCPD$ gives $C_4' \in \mathfrak{p}(C_1)$, and hence $C_4' \in A_1B_1$. Since also $C_4' \in CQ$, we have $C_4' = C_4$. Thus $C_4$ is the orthocenter of $\triangle QPC_1$, so $C_1C_4 \perp PQ$, that is, $\mathfrak{p}^{-1}(PQ) \in C_1C_4$. The other five lines are analogous; after the projectivity they are all parallel in the direction perpendicular to $PQ$ and concur at the corresponding point at infinity. $\square$

The proof also gives the following description.

**Theorem 6.3.10.** *The crosspoint of $P, Q$ with respect to $\triangle ABC$ is $\mathfrak{p}^{-1}_{\mathfrak{C}(ABCPQ)}(PQ)$.*

**Proof.** In the transformed configuration of theorem 6.3.9, the common point of the six lines is the point at infinity perpendicular to $PQ$. This is exactly the intersection of the tangents at $P, Q$ to $\odot(ABC)$, hence the pole of $PQ$. Thus, after the projectivity, the crosspoint is $R = \mathfrak{p}^{-1}_{\odot(ABC)}(PQ)$. Consequently, before the projectivity it is $R = \mathfrak{p}^{-1}_{\mathfrak{C}(ABCPQ)}(PQ)$, as claimed. $\square$

## Exercises

**Exercise 6.21.**

(1) Prove the converse of Brianchon's theorem.
(2) Given five tangents to a conic, construct further tangents using only a straightedge.
(3) Given five points on a conic, construct further points of the conic and the tangents at the five given points using only a straightedge.

**Exercise 6.22.** Prove that a circumconic of $\triangle ABC$ is a rectangular hyperbola if and only if it passes through the orthocenter of $\triangle ABC$.

**Exercise 6.23.** In Figure 6.13, two conics have four common tangents and four common points. Prove that the intersection of the diagonals of the quadrilateral formed by the common tangents coincides with the intersection of the diagonals of the quadrilateral formed by the four common points.

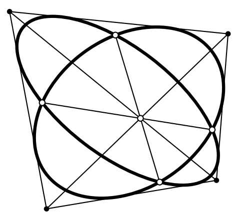

FIGURE 6.13

**Exercise 6.24.** Prove that two triangles $\triangle ABC, \triangle DEF$ are inscribed in one conic if and only if they are circumscribed about one conic.

**Exercise 6.25** (Hexagon theorem)**.** A conic $\alpha$ meets the sides $AB, BC, CA$ of $\triangle ABC$ at $C_1, C_2; A_1, A_2;$ and $B_1, B_2$, respectively. Let $A_3, B_3, C_3$ be the poles of $A_1A_2, B_1B_2, C_1C_2$ with respect to $\alpha$. Prove that $\triangle ABC$ and $\triangle A_3B_3C_3$ are perspective.

**Exercise 6.26.** A parabola touches the corresponding sides of $\triangle ABC$ at $A', B', C'$. Prove that the line through $C'$ parallel to the axis of the parabola meets $A'B'$ on the median of $\triangle ABC$ through $C$.

**Exercise 6.27.** Let fixed points $P, Q$ lie on the principal axis $l$ of a conic $\alpha$. Variable points $A, B, C, D \in \alpha$ satisfy $P \in AB, CD$ and $Q \in BC$. Prove that $\frac{\tan\measuredangle(l,BC)}{\tan\measuredangle(l,AD)}$ is constant.

**Exercise 6.28** (Quadrilateral butterfly theorem)**.** In a convex quadrilateral $ACBD$, the line $CD$ bisects $AB$ at $P$. Choose $F, E, G, H$ on the segments $DB, BC, CA, AD$, respectively, so that $EH$ and $GF$ pass through $P$. If $GH, EF$ meet $AB$ at $U, V$, prove that $P$ is the midpoint of $UV$.

**Exercise 6.29.** Let $H, G$ be the orthocenter and centroid of $\triangle ABC$. Prove that $\mathcal{S}p \in \mathfrak{C}(ABCGH)$.
*Hint.* Use exercise 6.22.

**Exercise 6.30.** Let $R$ be the crosspoint of $P,Q$ with respect to $\triangle ABC$, and put $U = BP \cap CQ$, $V = BQ \cap CP$. Prove that $R \in UV$.

**Exercise 6.31*.** Let $G$ be the centroid of $\triangle ABC$. For any point $P$, prove that the crosspoint of $P,G$ is $\mathbf{ct}P$.

## 6.4 Conics from a Projective Viewpoint

We have already met many important structures associated with conics, using several different methods. We now build the theory systematically in projective terms, treating such elements as centers, diameters, axes, asymptotes, and vertices.

Throughout this section we work in the complex projective plane, so every line meets every conic in two points, possibly coincident. The conics themselves remain real: algebraically, the coefficients in equation (1.1) are real. The complex projective plane is introduced only to facilitate the discussion.

### 6.4.1 Affine Properties of Conics

By the affine properties of a conic we mean those involving its center, diameters, and asymptotes, since all are tied to the line at infinity. We now reinterpret the earlier metric definitions projectively.

#### Centers and Diameters

**Definition 6.4.1** (Projective definitions of center and diameter)**.** The *center* of a conic is the pole of the line at infinity. A finite polar of a point at infinity is called a *diameter* of the conic.

This agrees with the earlier definition. For a central conic $\alpha$, the metric center $O$ is characterized by the property that whenever a line $l$ through $O$ meets $\alpha$ at $A,B$, the point $O$ is the midpoint of $AB$. Under the projective definition, $\mathfrak{p}(O) = l_\infty$. Let a line $l$ through $O$ meet $\alpha$ at $A,B$. Then theorem 6.1.7 gives $(AB, O\infty_l) = -1$, which says exactly that $O$ is the midpoint of $AB$.

We did not previously assign a center to a parabola. By proposition 3.1.6, a parabola is tangent to the line at infinity at the point at infinity of its axis, and the present definition makes that point its center.

For diameters, under the earlier definition and by propositions 2.1.17, 2.2.13 and 2.3.6, a diameter is the locus of the midpoints of a family of parallel chords. Equivalently, it is a line through the center.[9] Under the projective definition, the polarity principle likewise says that a diameter is a line through the center. Thus the definitions agree. □

The basic facts may be summarized without further proof.

**Theorem 6.4.2.** *A central conic has a unique finite center. A parabola has a unique center, the point at infinity on its axis.*

**Theorem 6.4.3.** *Every diameter of a conic passes through its center. The center is the midpoint of the segment cut from the diameter by the conic.*

**Remark.** In the real projective plane, the second sentence applies when the diameter has real intersections with the conic; a tangent counts as two coincident intersections and the segment then has length zero. In the complex projective plane every line has two intersections, so the statement is unconditional.

[9] In the earlier terminology, a diameter of a parabola was a line parallel to its axis; the present definition also assigns the point at infinity on the axis as the center of the parabola.

**Theorem 6.4.4** (Diameter theorem)**.** *The locus of the midpoints of a family of parallel chords of a conic is a line, which is a diameter of the conic. The tangents at the endpoints of that diameter are parallel to the chords, and the tangents at the endpoints of any chord in the family meet on the same diameter.*

This is precisely theorem 2.3.10; its projective proof is immediate and is left to the reader.

**Definition 6.4.5.** Two diameters of a conic are *conjugate diameters* if they are conjugate lines with respect to the conic.

This agrees with the elementary definition in chapter 2; the more explicit verification is theorem 6.4.8 below.

**Theorem 6.4.6.** *For a conic, the diameter conjugate to a given diameter is the polar of the point at infinity on the given diameter.*

**Proof.** The polar of that point at infinity is itself a diameter, and the two lines are conjugate by construction. ☐

**Theorem 6.4.7.** *The tangents at the finite endpoints of a diameter are parallel to its conjugate diameter.* [10]

**Proof.** By the polarity principle, the tangent at an endpoint of a diameter passes through the pole of that diameter. By the definition of conjugate diameters, the pole lies on the conjugate diameter and is a point at infinity, giving the required parallelism. This is also corollary 2.1.21. ☐

**Theorem 6.4.8.** *For a central conic, every chord parallel to one diameter is bisected by the conjugate diameter.*

**Proof.** Let $l$ be a diameter of a central conic $\alpha$, let a chord $AB$ be parallel to $l$, and let $M$ be its midpoint. Since $(AB, M\infty_{AB}) = -1$, we have $\infty_{AB} \in \mathfrak{p}(M)$. Thus $\infty_l \in \mathfrak{p}(M)$, and the polarity principle gives $M \in \mathfrak{p}(\infty_l)$, the diameter conjugate to $l$. ☐

**Example 6.4.9.** If a parallelogram is inscribed in a central conic $\alpha$, prove that its diagonals are diameters of $\alpha$ and that its two pairs of opposite sides are parallel to a pair of conjugate diameters.

**Proof.** Let the parallelogram $ABCD$ be inscribed in a central conic $\alpha$, and put $O = AC \cap BD$, $X = AB \cap CD$, and $Y = AD \cap BC$. Then $X, Y$ are points at infinity. Brocard's theorem gives $O = \mathfrak{p}^{-1}(XY)$, so $O$ is the center and $AC, BD$ are diameters. It also gives $\mathfrak{p}(X) = YO$ and $\mathfrak{p}(Y) = XO$; since $O$ is the center, $YO, XO$ are conjugate diameters. Since $X$ is at infinity, $AB \parallel CD \parallel XO$ and $AD \parallel BC \parallel YO$. ☐

### Asymptotes

**Definition 6.4.10.** A tangent to a conic at one of its intersections with the line at infinity is called an *asymptote*, provided the tangent is not itself the line at infinity.

This agrees with the traditional definition. Let $P$ be a point on a curve $\gamma$ with tangent $l_P$, and let $Q$ be a point of $\gamma$ at infinity with tangent $l$. As $P \to Q$, we have $l_P \to l$, so $\operatorname{dist}(P, l) \to 0$. We may also see the definition by sending a hyperbola perspectively to a circle and examining the images of its asymptotes.

[10] The endpoints are the finite intersections of the diameter with the conic; for a parabola, the intersection at infinity is excluded here.

Refer to Figure 1.16 in section 1.4. Under a perspectivity $f$, a circle $\omega$ in a section plane $\sigma$ of a cone through $K$ is sent to a hyperbola $\alpha$ in a plane $\pi$. In figure 1.16, the generator $SQ$ is parallel to an asymptote $OP$, and the generator $SQ'$ is parallel to the other asymptote $OP'$.

Let $SQ,SQ'$ meet $\sigma$, and hence $\omega$, at $T,T'$. Since these two generators are parallel to $\pi$, the line $TT'$ is the vanishing line of $\sigma$ under $f$, while the points at infinity on $OP,OP'$ are sent to $T,T'$ by $f^{-1}$. If $SO\cap\sigma=O'$, then $O'=f^{-1}(O)$. We need only show that $O'T,O'T'$ are tangents to $\omega$.

The vertical section $CSD$ of the cone is shown in Figure 6.14. Let $K,K'$ be its intersections with $\omega$; then $KK'$ is a diameter of $\omega$, and $R=TT'\cap KK'$. The points $A,B$ are the vertices of $\alpha$ on the generators $CD,CB$, and $O$ is the midpoint of $AB$. In this section, $SR\|AB$ and the two lines meet at the same point $\infty_0$ at infinity. Hence $(A,B,O,\infty_0)\overset{(S)}{\barwedge}(K,K',O',R)$, so $(KK',O'R)=(AB,O\infty_0)=-1$. Since $KK'\perp TT'$, the line $TT'$ is the polar of $O'$ with respect to $\omega$. Therefore $O'T,O'T'$ are tangents, as required. □

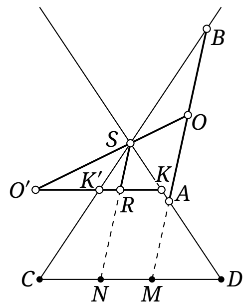


FIGURE 6.14

Under this definition, an ellipse also has asymptotes: it is not tangent to the line at infinity, so it has two imaginary intersections with that line and hence two imaginary asymptotes. A parabola is tangent to the line at infinity, whose tangent there is again the line at infinity and therefore is excluded by the definition. Thus:

**Theorem 6.4.11.** *A parabola has no asymptote; a hyperbola has two real asymptotes; an ellipse, including a circle, has two imaginary asymptotes.*

**Theorem 6.4.12.** *The two asymptotes of a central conic meet at its center and harmonically separate every pair of conjugate diameters.*

**Proof.** Let $t,t'$ be the asymptotes of a central conic $\alpha$, touching it at the points at infinity $\infty_t,\infty_{t'}$. Their intersection is the pole of $l_\infty=\infty_t\infty_{t'}$, namely the center $O$ of $\alpha$.

Let $l,l'$ be conjugate diameters. Then $\infty_{l'}\in\mathfrak{p}(\infty_l)$. Since $l_\infty$ contains $\infty_l$ and meets $\alpha$ at $\infty_t,\infty_{t'}$, theorem 6.1.7 gives $(\infty_l\infty_{l'},\infty_t\infty_{t'})=-1$, hence $(ll',tt')=-1$. □

**Example 6.4.13.** Use projective geometry to prove the following for a hyperbola.

(1) The triangle formed by its two asymptotes and any tangent has constant area.
(2) The point of tangency bisects the segment cut from the tangent by the two asymptotes.
(3) Through any point of the hyperbola draw lines parallel to the two asymptotes. The parallelogram bounded by these lines and the asymptotes has constant area.

**Proof.** Let $l_1,l_2$ be the asymptotes of a hyperbola $\alpha$, meeting at its center $O$. Let tangents $l_3,l_4$ meet $l_1,l_2$ at $A,B$ and $C,D$, respectively, with $l_3$ tangent at $M$.

For the first assertion, put $P=AD\cap BC$. Apply the dual of Brocard's theorem, theorem 6.1.15, to the complete quadrilateral $l_1l_2l_3l_4$. The line $\mathfrak{p}(P)$ passes through the center $O$, so $P\in\mathfrak{p}(O)=l_\infty$. Hence $AD\|BC$, and $S_{\triangle ADB}=S_{\triangle ADC}$. Adding $S_{\triangle AOD}$ to both sides gives $S_{\triangle ABO}=S_{\triangle CDO}$. Since $l_3,l_4$ were arbitrary, the area is constant.

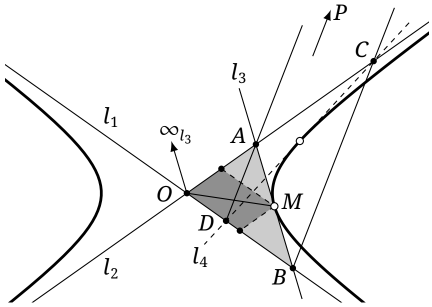


FIGURE 6.15

For the second assertion, theorem 6.4.7 says that $O\infty_{l_3}$ and $OM$ are conjugate diameters. By theorem 6.4.12, $O(AB,M\infty_{l_3}) = (l_1,l_2;OM,O\infty_{l_3}) = -1$. Thus $M$ is the midpoint of $AB$. The third assertion now follows because the parallelogram has half the area of the triangle in the first assertion. □

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## Exercises

**Exercise 6.32.** Give a projective proof that the point of tangency of a hyperbola tangent is the midpoint of the segment cut from that tangent by the asymptotes.

**Exercise 6.33.** Let $l_1,l_2$ be the asymptotes of a hyperbola $\alpha$, and let tangents $t_1,t_2$ meet $l_1,l_2$ at $A_1,B_1$ and $A_2,B_2$, respectively. Prove projectively that $A_1B_2 \parallel A_2B_1$.

**Exercise 6.34.** If a parallelogram is inscribed in a central conic, prove that its diagonals are diameters and its pairs of opposite sides are parallel to a pair of conjugate diameters.

**Exercise 6.35.** Prove that the locus of the midpoints of chords of a conic through a fixed point is a conic of the same type. [11]

**Exercise 6.36.** Let $P,Q$ be conjugate points with respect to a central conic $\alpha$. If the midpoint $A$ of $PQ$ lies on $\alpha$, prove that $PQ$ is parallel to an asymptote of $\alpha$.

**Exercise 6.37.** Prove that:
(1) conjugate diameters of a rectangular hyperbola are symmetric about its asymptotes;
(2) the angle of intersection of two concentric rectangular hyperbolas is twice the angle between their asymptotes.

### 6.4.2 Circles and the Circular Points

Five points determine a conic, whereas three determine a circle. What are the two remaining points? The answer is that every circle passes through the two circular points. This gives a projective definition of a circle.

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[11] Thus an ellipse, hyperbola, or parabola gives, respectively, an ellipse, hyperbola, or parabola.

**Definition 6.4.14** (Projective definition of a circle)**.** A *circle* is a nondegenerate conic through the two circular points.

To compare this definition with the familiar metric one, we first derive a basic consequence.

**Theorem 6.4.15.** *Every pair of conjugate diameters of a circle is perpendicular.*

**Proof.** The intersections of a circle with the line at infinity are the circular points $I,J$. Let conjugate diameters meet the line at infinity at $P,Q$, and let $O$ be the center. By theorem 6.4.12, $OI,OJ$ are the two asymptotes of the circle and $(OP,OQ;OI,OJ) = -1$. The projective definition of perpendicularity, theorem 3.5.27, now gives $OP \perp OQ$. □

Conversely, a conic whose every pair of conjugate diameters is perpendicular is a circle in the ordinary metric sense. The projective and metric definitions therefore agree.

**Theorem 6.4.16.** *Let two lines $l_1,l_2$ avoid the circular points $I,J$, and let them meet $l_\infty$ at $A,B$, respectively. The angle $\measuredangle(l_1,l_2)$ and the cross-ratio $(AB,IJ)$ determine one another uniquely.*

**Proof.** Let $O = l_1 \cap l_2$. Take another pair $l_1',l_2'$ avoiding the circular points, meeting at $O'$, and meeting $l_\infty$ at $A',B'$, respectively. Put $P = l_1 \cap l_1'$ and $Q = l_2 \cap l_2'$. Then $\measuredangle(l_1,l_2) = \measuredangle POQ$ and $\measuredangle(l_1',l_2') = \measuredangle PO'Q$. We must prove $\measuredangle POQ = \measuredangle PO'Q \Longleftrightarrow (AB,IJ) = (A'B',IJ)$.

First suppose $\measuredangle POQ = \measuredangle PO'Q$. Then $P,Q,O,O'$ lie on a circle $\omega$, which also passes through $I,J$. Consequently

$$l_\infty(A,B,I,J) \overline{\wedge} O(A,B,I,J) \overline{\wedge} \omega(P,Q,I,J) \overline{\wedge} O'(A',B',I,J) \overline{\wedge} l_\infty(A',B',I,J),$$

and hence $(AB,IJ) = (A'B',IJ)$. The proof of the converse is similar. □

This gives a projective characterization of angle. In the preceding theorem, if $l_1$ tends to a line through a circular point $I$, then for every $l_2$ not through $J$ we have $(AB,IJ) = (IB,IJ) = 0$. Thus the angle $\measuredangle(l_1,l_2)$ and this cross-ratio are not in one-to-one correspondence. If $l_2$ also tends to a line through $J$, the corresponding cross-ratio $(IJ,IJ)$ is undefined. Yet repeating the proof of theorem 6.4.16 suggests that it should hold for all $l_1,l_2$, even lines through the circular points. The only explanation is that we cannot reasonably define the angle between a line through a circular point and another line; equivalently, their angle does not have a definite value. We therefore call a line through a circular point an *isotropic line*.

The relation can be made quantitative.

**Theorem 6.4.17** (Laguerre's theorem)**.** *Let two lines $l_1,l_2$ avoid the circular points $I,J$, and let them meet $l_\infty$ at $A,B$, respectively. Then $\measuredangle(l_1,l_2) = \frac{1}{2\mathrm{i}}\ln(AB,IJ)$.*[12]

**Proof.** For arbitrary lines, write their directed angle as $\theta(l_1,l_2)$, so

$$\theta(l_1,l_2) + \theta(l_2,l_3) = \theta(l_1,l_3).$$

Let $\Theta(l_1,l_2)$ be the cross-ratio in which the two points at infinity of $l_1,l_2$ are separated by $I,J$. By proposition 3.2.13,

$$\ln\Theta(l_1,l_2) + \ln\Theta(l_2,l_3) = \ln[\Theta(l_1,l_2)\Theta(l_2,l_3)] = \ln\Theta(l_1,l_3).$$

By theorem 6.4.16, a fixed function $g$ satisfies $\ln\Theta(l_1,l_2) = g(\theta(l_1,l_2))$. Since $\theta(l_1,l_2)$ can take any value $a$ and $\theta(l_2,l_3)$ any value $b$, the two equations give $g(a+b) = g(a) + g(b)$ for all $a,b$. This is the standard *Cauchy functional equation*. In geometry we require $g(x)$ to be continuous, and hence $g(x) = Cx$ for a constant $C$.[13] Thus $\theta(l_1,l_2) = C^{-1}\ln\Theta(l_1,l_2)$.

[12] Here we use the logarithm and inverse trigonometric functions over the complex numbers; see the discussion of complex functions in Appendix A.

[13] The solution of this equation is explained in detail in section A.9.2.

When $\theta(l_1,l_2)=90^\circ=\pi/2$, the projective definition of perpendicularity requires $\Theta(l_1,l_2)=-1$. Since $\ln(-1)=\mathrm{i}\pi(1+2k)$, $k\in\mathbb{Z}$,[14] and directed angles differing by $\pi$ are equivalent here,[15]

$$C\left\{\frac{\pi}{2}+n\pi\,|\,n\in\mathbb{Z}\right\}=\{\mathrm{i}\pi(1+2k)\,|\,k\in\mathbb{Z}\}.$$

Hence $C=\pm2\mathrm{i}$, and

$$\theta(l_1,l_2)=\pm\frac{1}{2\mathrm{i}}\ln\Theta(l_1,l_2).$$

□

The proof has not quite returned to the required formula $\theta(l_1,l_2)=\frac{1}{2\mathrm{i}}\ln\Theta(l_1,l_2)$, since a possible sign remains. This sign comes from the choice of a positive sense for angles. Until now the two circular points $I,J$ have been treated as equivalent; the theorem shows that interchanging them negates the angle. Their ordering therefore fixes its positive sense. The usual convention that counterclockwise angles are positive is also a choice, so the sign does not invalidate the proof. In general, it is enough to choose one orientation consistently; the internal ordering of the circular points need not be considered explicitly.

For real angles $\theta$, which suffice for our purposes, the Cauchy equation requires continuity of $g$. This is a natural geometric requirement. For complex angles, however, the stronger condition of analyticity is needed: continuity alone allows $g(z)=a\mathrm{Re}(z)+b\mathrm{Im}(z)$. See section A.9 for the detailed derivation concerning the Cauchy equation.

Although continuity is natural geometrically, analyticity is less so. We therefore give a new calculation of the function that avoids any continuity assumption.

**Alternative proof.** Let $l$ be any line through $O=l_1\cap l_2$, and put $k_1=\tan\measuredangle(l,l_1)$, $k_2=\tan\measuredangle(l,l_2)$, $a=\tan\measuredangle(l,OI)$, and $b=\tan\measuredangle(l,OJ)$. By theorem 3.2.7,[16]

$$(AB,IJ)=(IJ,AB)=\frac{a-k_1}{b-k_1}\div\frac{a-k_2}{b-k_2}.$$

By the projective definition of perpendicularity in definition 3.5.26, when $k_1=0$ and $k_2=\infty$ we have $l_1\perp l_2$, so $-1=(AB,IJ)=\frac{a}{b}$, and hence $b=-a$. When $\measuredangle(l,l_1)=45^\circ$ and $\measuredangle(l,l_2)=135^\circ$, we have $k_1=1,k_2=-1$, and

$$-1=(AB,IJ)=\frac{a-1}{-a-1}\div\frac{a+1}{-a+1}\quad\Rightarrow\quad a^2=-1\quad\Rightarrow\quad a=\pm\mathrm{i}.$$

Thus $a=\pm\mathrm{i}$ and $b=\mp\mathrm{i}$.

For a general angle, choose $l=l_1$. Then $k_1=0$ and $k_2=\tan\measuredangle(l_1,l_2)$, while

$$(AB,IJ)=\frac{a}{b}\div\frac{a-k_2}{b-k_2}=\frac{\pm\mathrm{i}+k_2}{\pm\mathrm{i}-k_2}.$$

Writing $\theta=\measuredangle(l,l_2)$ and $\Theta=(AB,IJ)$, we obtain

$$\theta=\arctan k_2=\mp\arctan\left(\frac{1-\Theta}{1+\Theta}\mathrm{i}\right)=\mp\frac{1}{2\mathrm{i}}\ln\Theta.$$

□

Again a possible sign separates this result from the required formula $\theta=\frac{1}{2\mathrm{i}}\ln\Theta$. For that formula to agree with the usual counterclockwise-positive convention, we must choose $a=-\mathrm{i}$, $b=+\mathrm{i}$. This fixes the positions of the two circular points on the line at infinity. The calculation therefore also proves the following result.

[14]This is a multivalued function over the complex number field; see section A.9.

[15]For general angles, reduction modulo $2\pi$ is the safest convention. In this cross-ratio definition, however, angles differing by $\pi$ give the same cross ratio, and the directed angle used in the definition is itself taken modulo $\pi$.

[16]For the complex-function facts used here, see section A.9.

**Theorem 6.4.18.** *Choose any reference line l. The tangents of the directed "angles" from l to the lines through the first circular point I and the second circular point J are, respectively,* $-\mathrm{i}$ *and* $+\mathrm{i}$.

**Remark.** In complex analysis, $\arctan x = \frac{1}{2\mathrm{i}} \ln \frac{1+\mathrm{i}x}{1-\mathrm{i}x}$. At $x = \pm\mathrm{i}$ the inverse tangent has no corresponding angle, explaining algebraically why lines through $I,J$ are directionless isotropic lines. Thus the phrase "tangent of the directed angle" in the theorem is only formal. It can be defined without the angle itself: for a line $l'$ through $I$ or $J$, choose $P \in l'$, let $Q$ be its projection onto $l$, and put $O = l \cap l'$; the tangent is $\frac{\overline{QP}}{\overline{OQ}}$.

Although this value is fixed, it is indeed independent of the reference line $l$; see the exercises.

Here are two applications of the circular points.

**Theorem 6.4.19** (Three-conics theorem). *As in figure 6.16, if three conics have two common points, the three chords joining their remaining pairwise intersections are concurrent. More precisely, let the conics* $\alpha_1, \alpha_2, \alpha_3$ *in* $\mathbb{CP}^2$ *have common points* $P, Q$. *Let* $A_3, B_3$ *be the other intersections of* $\alpha_1, \alpha_2$, *let* $A_1, B_1$ *be the other intersections of* $\alpha_2, \alpha_3$, *and let* $A_2, B_2$ *be the other intersections of* $\alpha_3, \alpha_1$. *Then* $A_1B_1, A_2B_2, A_3B_3$ *are concurrent.*

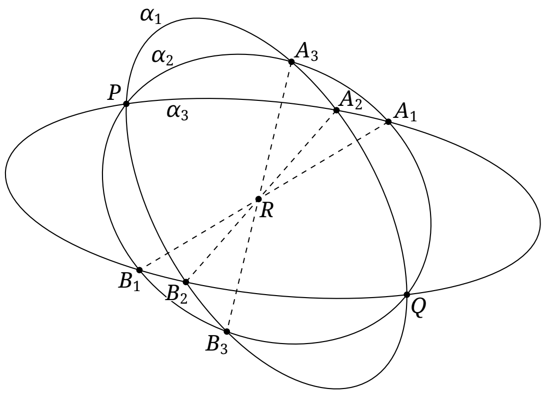


FIGURE 6.16

**Proof.** Apply a projectivity sending $P,Q$ to the circular points. The three conics become circles, and the assertion becomes the radical-center theorem. [17] □

**Example 6.4.20.** Give a purely geometric proof of theorem 3.1.14.

**Proof.** Among five points, no four collinear, choose two points $I,J$: if no three are collinear, choose any two; if three are collinear, choose the other two. Call the remaining points $A,B,C$. Apply a projectivity sending $I,J$ to the circular points. Every nondegenerate conic through the two circular points is a circle. If $A,B,C$ are not collinear, exactly one circle passes through them, and it also passes through the two circular points. Thus exactly one nondegenerate conic passes through the five points after the projectivity, and the same is true before it.

If $A,B,C$ are collinear, no circle passes through them. Thus no nondegenerate conic passes through the five points after the projectivity. By the pigeonhole principle, however, exactly one pair of lines passes through the five points: one through the images of $A,B,C$, the other through the images of $I,J$, namely $l_\infty$. Hence there is again a unique conic through the five points, and it is degenerate. □

[17] In the complex Euclidean plane, continuation from the case of two intersecting circles shows that their radical axis is their common chord.

**Remark.** This argument completes the purely geometric foundation of the theory of conics. Its use of the projective definition of a circle to prove theorem 3.1.14 is not circular reasoning.

The projective definition also explains the earlier convention of regarding a line as a generalized circle with center and radius at infinity. For three collinear points $A,B,C$ and the circular points $I,J$, the conic through all $A,B,C,I,J$ is the degenerate conic $\overline{ABC} \cup l_\infty$. Since it passes through $I,J$, it retains circle-like properties. Strictly speaking the degenerate circle is the union of the finite line and $l_\infty$, though ignoring the latter generally has no effect on the geometry.

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## Exercises

**Exercise 6.38.** Give a projective proof of the perpendicular-diameter theorem: the diameter of a circle perpendicular to a nondiametral chord bisects that chord.

**Exercise 6.39.** If $\tan\theta = \pm \mathrm{i}$, prove from the tangent addition formula that $\tan(\theta+\phi) = \pm\mathrm{i}$ for every $\phi$. Hence show that the positions of $I,J$ in theorem 6.4.18 are independent of the reference line $l$. Although $\theta$ itself is not well defined, its formal tangent remains valid.

**Exercise 6.40.** Prove that if all four sides of a quadrilateral are isotropic lines, then its diagonals perpendicularly bisect one another.

**Exercise 6.41.** Prove that an affine transformation is a similarity if and only if it preserves the two circular points.

### 6.4.3 Metric Properties of Conics

This section develops the metric properties of conics projectively, through the circular points.

#### Projective Definitions of Axes and Vertices

We begin with projective definitions of the axes and vertices of a conic.

**Definition 6.4.21** (Projective definitions of an axis and a vertex)**.** An *axis* of a conic is a diameter that bisects a family of chords perpendicular to it.[18] A finite intersection of an axis with the conic is a *vertex*.

This definition makes the conic symmetric about each of its axes and is therefore equivalent to the earlier definition.

**Theorem 6.4.22.** *A parabola has a unique axis and a unique vertex. Its axis is the polar of the fourth harmonic point of the two circular points and the point at infinity of the parabola.*

**Proof.** Let the line at infinity $l_\infty$ be tangent to the parabola at $P_\infty$, and let $Q_\infty$ be the fourth harmonic point of $P_\infty$ with respect to $I,J$. The polar of $Q_\infty$ passes through $P_\infty$; denote its other intersection with the parabola by $V$. Since $(I,J;P_\infty,Q_\infty) = -1$, we have $VP_\infty \perp VQ_\infty$.

The line $VP_\infty$, being the polar of the point at infinity $Q_\infty$, is a diameter. Let an arbitrary line through $Q_\infty$ meet the parabola at $M_1,M_2$ and $VP_\infty$ at $M$. Since $VP_\infty$ is the polar of $Q_\infty$, we have $(Q_\infty,M;M_1,M_2) = -1$. Thus $M$ is the midpoint of $M_1M_2$. Since $M_1M_2 \parallel VQ_\infty$ and $VQ_\infty \perp VP_\infty$, we have $M_1M_2 \perp VP_\infty$. Hence $VP_\infty$ satisfies the definition of an axis. Every diameter of the parabola passes through its uniquely determined point $P_\infty$; hence the axis is unique. □

[18]Both bisection and perpendicularity have projective definitions; see proposition 3.2.16 and theorem 3.5.27. Thus this is a projective definition.

**Theorem 6.4.23.** *Every central conic other than a circle has exactly one pair of axes and four vertices. Its axes are the internal and external angle bisectors of its two asymptotes.*

**Proof.** Let the line at infinity $l_\infty$ meet the conic at $P_\infty, Q_\infty$. The tangents $t_1, t_2$ there are the asymptotes and meet at the center $C$. Let the internal and external angle bisectors of $\angle P_\infty C Q_\infty$ be $a,b$, meeting the line at infinity at $A_\infty, B_\infty$. Since $a \perp b$, we have $(t_1,t_2;a,b) = -1$, or equivalently $(P_\infty, Q_\infty; A_\infty, B_\infty) = -1$. Thus $a,b$ are conjugate diameters and hence axes, because each diameter bisects the chords conjugate to it.

Suppose another pair of axes $a', b'$ existed. Axes occur in conjugate pairs, and $a' \perp b'$, so $(a',b';CI,CJ) = -1$, just as $(a,b;CI,CJ) = -1$. The involution interchanging $a$ with $b$ and $a'$ with $b'$ sends each diameter to its conjugate diameter. Its invariant lines $CI, CJ$ would therefore be the asymptotes. Then $I,J$ would lie on the conic, forcing the conic to be a circle, a contradiction. □

The following metric property of conjugate diameters, previously stated in example 3.3.21, is restated here without another projective proof.

**Proposition 6.4.24.** *For a pair of conjugate diameters of a central conic, the product of the tangents of their directed angles with the principal axis is $(e^2 - 1)$, where $e$ is the eccentricity.*

### Foci and Directrices

We now treat foci and directrices projectively.

**Definition 6.4.25** (Projective definitions of a focus and a directrix)**.** Draw the tangents to a conic from the two circular points. The finite pairwise intersections of the four tangents are the *foci* of the conic. The polar of a focus with respect to the conic is its *directrix*.

Before showing that this agrees with the traditional definition, we derive some projective properties of the foci.

**Theorem 6.4.26.** *The center of a circle is its focus, and its directrix is the line at infinity.*

**Proof.** A circle passes through the circular points, so its tangents there are its asymptotes. They meet at the center of the circle. The center is therefore a focus, whose polar is the line at infinity. □

**Theorem 6.4.27.** *A parabola has one focus, lying on its axis. Its directrix is perpendicular to the axis, and the points of contact of its isotropic tangents lie on the directrix.*[19]

**Proof.** Because the parabola is tangent to the line at infinity $l_\infty$, it has only two finite isotropic tangents. Their intersection is the focus $F$, so the parabola has one focus and one directrix.

As in figure 6.17,[20] let the two isotropic tangents be $IA, JB$, with points of contact $A,B$, where $I,J$ are the circular points. Then $AB$ is the directrix; let $Q_\infty = AB \cap l_\infty$, and let the parabola touch $l_\infty$ at $P_\infty$. Apply Brianchon's theorem to the six-line hexagon $\{IA, AF, FB, BJ, JP_\infty, P_\infty I\}$, which has degenerated to a triangle. It follows that $FP_\infty, IB, AJ$ concur at a point $C$. Hence, by theorem 3.5.11, $(P_\infty, Q_\infty; I, J) = -1$. Equivalently, $(D, Q_\infty; A, B) = -1$, where $D = AB \cap FP_\infty$. Thus $FP_\infty$ is the polar of $Q_\infty$. Moreover $FP_\infty \perp FQ_\infty$, so $FP_\infty$ perpendicularly bisects a family of parallel chords through the direction $Q_\infty$. By the definition of an axis, $FP_\infty$ is the axis. Therefore $F$ lies on the axis and $AB$ is perpendicular to the axis, since $AB \parallel FQ_\infty$. □

[19] An *isotropic tangent* is, as the name suggests, a tangent that is also an isotropic line: here, a finite tangent through a circular point.

[20] A figure in the complex projective plane cannot be drawn literally; this and the analogous figures below are only schematics intended to aid visualization.

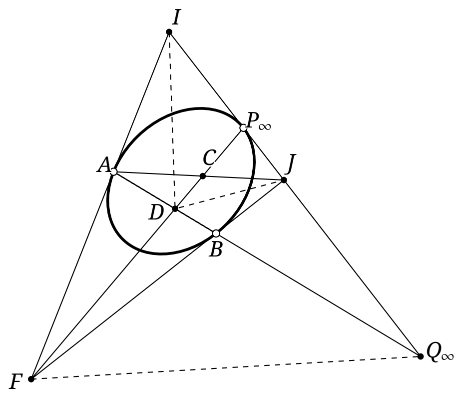


FIGURE 6.17

**Theorem 6.4.28.** *Every real central conic other than a circle has four foci, two real and two imaginary, one pair on each axis. It has four corresponding directrices, two real and two imaginary, each pair perpendicular to the relevant axis.*

**Proof.** The four isotropic tangents drawn from $I,J$ to the central conic meet in four points $F,F',G,G'$. The lines joining their points of contact in the appropriate pairs, $MM',NN',MN,M'N'$, are the four directrices; the ellipse case is shown in figure 6.18.

The complete quadrilateral $FG'F'G$ is circumscribed about the conic. By theorem 6.1.15, its diagonal triangle $\triangle CP_\infty Q_\infty$ is self-polar. Hence $C = \mathfrak{p}^{-1}(P_\infty Q_\infty)$ is the center. Also $FQ_\infty = \mathfrak{p}(P_\infty)$, and therefore $(P_\infty, Q_\infty; I, J) = -1$. Thus $GG' \perp FF'$. Both pass through $C$, so they are the two axes and the foci lie on them.

Since $FF'$ passes through the pole $F$ of $MM'$, the line $MM'$ passes through the pole $P_\infty$ of $FF'$. Similarly, $NN'$ passes through $P_\infty$. Hence the polars of $F,F'$, namely their directrices, are parallel to the axis $GG'$ and perpendicular to the axis $FF'$. Similarly, the directrices corresponding to $G,G'$ are perpendicular to the axis $GG'$.

An imaginary line contains at most one real point, so $F,G$ cannot both be real, and $F',G'$ cannot both be real. The four isotropic tangents form two pairs of conjugate complex lines, and conjugate complex lines meet at a real point. Thus two of the four foci are real and two are imaginary; the same holds for their corresponding directrices. □

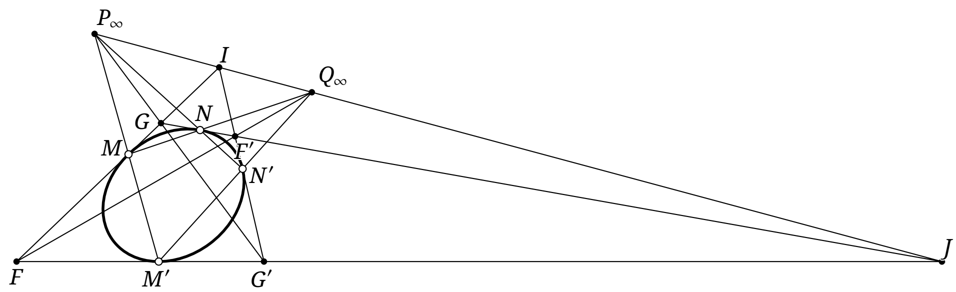


FIGURE 6.18

Here are two applications of the projective definition of a focus.

**Example 6.4.29.** Prove projectively that conjugate lines through the same focus of a conic are perpendicular.

**Solution.** Let $F$ be the focus and $I,J$ the circular points. Then $FI,FJ$ are tangent to the conic. By theorem 6.1.12, the two conjugate lines are harmonically separated by $FI,FJ$; theorem 3.5.27 now gives the result. □

**Example 6.4.30.** Let $F,l$ be a corresponding focus and directrix of a conic $\alpha$. The pole of a chord $PQ$ is $T$, and $R = PQ \cap l$. Prove projectively that $FT,FR$ are the internal and external angle bisectors of $\angle PFQ$.

**Solution.** Put $C = PQ \cap FT$. Since $R \in l = \mathfrak{p}(F)$ and $R \in PQ = \mathfrak{p}(T)$, the principle of polarity gives $\mathfrak{p}(R) = TF$. Hence $-1 = (TP,TQ;TR,TC) = (FP,FQ;FR,FC)$. By theorem 6.1.7, $R \in \mathfrak{p}(C)$, so $FR,FC$ are conjugate lines. By example 6.4.29, $FR \perp FC$, and therefore $FR \perp FT$. Since $F(P,Q,R,C)$ is a harmonic pencil, proposition 3.2.17 proves the assertion. □

We can now compare the projective and traditional definitions. Let $F,l$ be a corresponding focus and directrix of a conic $\alpha$. For any chord $PQ$, put $R = PQ \cap l$. By example 6.4.30, $FR$ is an external angle bisector of $\angle PFQ$. Thus $\frac{FP}{FQ} = \frac{PR}{QR} = \frac{\operatorname{dist}(P,l)}{\operatorname{dist}(Q,l)}$. Therefore $\alpha$ is the locus of points whose distances from a fixed point $F$ and a fixed line $l$ have constant ratio. This is precisely the second definition of a conic, so the two definitions agree.

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## Exercises

**Exercise 6.42.** Prove projectively that the axes of a central conic are the internal and external angle bisectors of its asymptotes.

**Exercise 6.43.** For a central conic, determine the relation between the eccentricities associated with its real focus–directrix pairs and those associated with its imaginary focus–directrix pairs.

**Exercise 6.44.** A fixed conic $\alpha$ is tangent to two fixed lines $t_1,t_2$. A variable tangent $m$ meets them at $M_1,M_2$, and $F$ is a focus of $\alpha$. Prove projectively that $\angle M_1FM_2$ is constant.
*Hint.* If four fixed tangents $l_1,l_2,l_3,l_4$ to a conic meet a variable tangent $l$ at $A_1,A_2,A_3,A_4$, then $(A_1A_2,A_3A_4)$ is independent of $l$.

**Exercise 6.45.** Prove projectively that the circumcircle of every triangle circumscribed about a parabola passes through the focus of the parabola.

**Exercise 6.46.** Give a projective proof of Monge's circle theorem.

# 6.5 Polarity and Duality

## 6.5.1 Basic Concepts of Polarity and Duality

As with the polarity discussed in section 3.6, the theory of poles and polars of a conic defines a *polar transformation*, usually called a *polarity*: with respect to a conic, it sends each point of the plane to its polar and each line to its pole. We now extend this operation to arbitrary curves.

**Definition 6.5.1.** Under a polarity with respect to a conic, the *polar curve* of a smooth curve[21] is either the locus of the poles of all its tangents or, equivalently, the curve enveloped by the polars of all its points.[22]

We verify the equivalence of the two constructions. Let $A,B$ lie on a curve $\gamma$, whose tangents there are $a,b$. Under the polarity, let $A,B$ correspond to the lines $a',b'$, and let $a,b$ correspond to the points $A',B'$. The polarity principle gives $A'\in a'$ and $B'\in b'$. As $B\to A$, the chord $AB$ tends to $a$, while $a\cap b$ tends to $A$. Consequently

$$A'=\mathfrak{p}^{-1}(a)=\mathfrak{p}^{-1}\left(\lim_{B\to A}AB\right)=\lim_{B\to A}\mathfrak{p}^{-1}(AB)=\lim_{B\to A}(a'\cap b').$$

Here we have used continuity of the polarity: taking the limit before or after applying the polarity gives the same result. We shall not treat this continuity rigorously in this course, but it is intuitively easy to understand.

Let $\gamma'$ be the envelope obtained from the polars of the points of $\gamma$. The lines $a',b'$ are tangent to $\gamma'$ at, say, $A'',B''$. As $B\to A$, $b'\to a'$, and $\lim_{B\to A}(a'\cap b')=A''$. Hence $A'=A''$. Thus the points obtained as poles of the tangents of $\gamma$ are exactly the points of contact of the envelope $\gamma'$, proving the equivalence.

**Theorem 6.5.2.** *Let $\mathfrak{d}$ be a polarity with respect to a conic. For every curve $\gamma$, $\mathfrak{d}(\mathfrak{d}(\gamma))=\gamma$.*

**Proof.** Regard $\mathfrak{d}(\gamma)$ as the locus of the poles of the tangents of $\gamma$. Applying $\mathfrak{d}$ again gives the envelope of the polars of those poles, which is the envelope of the original tangents of $\gamma$. That envelope is $\gamma$ itself. □

We write $\mathfrak{d}_\alpha$ for polarity with respect to a conic $\alpha$. The three symbols $\mathfrak{t}$, $\mathfrak{p}$, and $\mathfrak{d}$ have distinct roles. The symbol $\mathfrak{t}$ emphasizes a tangent: if $P\in\alpha$, then $\mathfrak{t}_\alpha(P)=\mathfrak{p}_\alpha(P)$, but $\mathfrak{t}_\alpha$ may also be used for a general curve rather than a conic. The symbol $\mathfrak{p}$ emphasizes the pole–polar correspondence and does not act on an entire curve; moreover, when a line $a$ is the input, $\mathfrak{p}^{-1}(a)$ denotes its pole, because $\mathfrak{p}$ is reserved for sending a point to its polar. By contrast $\mathfrak{d}(a)$ denotes the pole of $a$ directly, because $\mathfrak{d}$ is the polarity transformation interchanging points and lines. The latter notation also foregrounds the transformational viewpoint.

We next determine the polar curve of one conic with respect to another.

**Theorem 6.5.3.** *The polar curve $\alpha$ of a circle $\omega_1$ with respect to another circle $\omega$ is a conic. It is an ellipse, parabola, or hyperbola according as the center of $\omega$ lies inside, on, or outside $\omega_1$.*

**Proof.** Let $\omega$ have center $O$ and let $\omega_1$ have center $O_1$. Suppose first that $O$ lies inside $\omega_1$, as in figure 6.19, and put $\omega_2=\mathfrak{i}_\omega(\omega_1)$. For $X\in\omega_2$, let $p(X)$ be the line through $X$ perpendicular to $OX$. Then $p(X)=\mathfrak{p}(\mathfrak{i}_\omega(X))$. As $X$ traverses $\omega_2$, the envelope of $p(X)$ is therefore the polar curve of $\omega_1$.

By theorem 1.3.6, these lines envelop the ellipse having $\omega_2$ as auxiliary circle and having foci $O$ and $\mathfrak{f}_{O_2}(O)=O'$, where $O_2$ is the center of $\omega_2$. This proves the case in question. The cases in which $O$ lies outside or on $\omega_1$ have similar proofs, which we leave to the reader. □

The construction admits the following sharper description.

**Proposition 6.5.4.** *Let $\omega,\omega_1$ be circles with centers $O,O_1$. Then $\mathfrak{d}_\omega(\omega_1)$ is the conic having $O$ and $\mathfrak{p}_\omega(O_1)$ as a corresponding focus and directrix, and $\mathfrak{i}_\omega(\omega_1)$ as its auxiliary circle.*[23]

[21]Here *smooth* means sufficiently regular for the construction: in particular, the curve has suitable continuity and a unique tangent at each point. A nondegenerate conic has these properties. A curve pieced together from a curved arc and a straight segment need not: all tangents along the straight segment have the same pole, producing pathologies under polarity. Unless otherwise stated, all curves considered here have the required regularity.

[22]If every member of a family of variable lines is tangent to a curve, the family is said to *envelop* the curve, and the curve is their *envelope*.

[23]By convention, the tangent to a parabola at its vertex is also called its "auxiliary circle"; it may be viewed as a limiting circle of an ellipse.

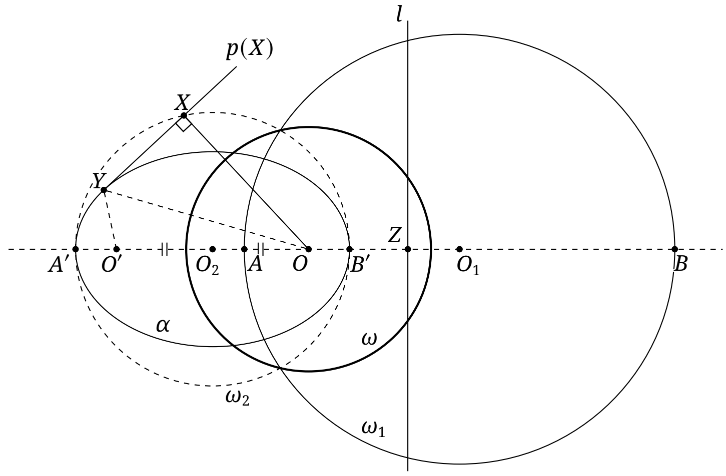


FIGURE 6.19

**Proof.** The proof of theorem 6.5.3 already shows that $O$ is a focus and that $\mathfrak{i}_\omega(\omega_1)$ is the auxiliary circle. We need only prove that $\mathfrak{p}(O_1)$ is the corresponding directrix. Put $\alpha = \mathfrak{d}_\omega(\omega_1)$, $\omega_2 = \mathfrak{i}_\omega(\omega_1)$, $l = \mathfrak{p}_\omega(O_1)$, and $l \cap OO_1 = Z$. Let $A, B$ be the intersections of $\omega_1$ with $OO_1$, and $A', B'$ those of $\omega_2$ with $OO_1$, as in figure 6.19.

Clearly $(AB, O_1\infty_{OO_1}) = -1$. Under $\mathfrak{i}_\omega$, the points $A, B, \infty_{OO_1}$ go to $A', B', O$, respectively, while $\mathfrak{i}_\omega(O_1) = Z$ by theorem 4.1.11. Hence theorem 4.1.13 gives $(A'B', ZO) = -1$. By definition 6.1.9, $Z \in \mathfrak{p}_\alpha(O)$. Since $O$ lies on an axis of $\alpha$, symmetry gives $\mathfrak{p}_\alpha(O) \perp OO_1$. Thus $l = \mathfrak{p}_\alpha(O)$; because $O$ is a focus of $\alpha$, theorem 6.1.17 completes the proof. ☐

Reversing the preceding polarity yields the following result.

**Theorem 6.5.5.** *Let $\alpha$ be a conic and $F$ one of its foci. A polarity with respect to a circle centered at $F$ sends $\alpha$ to a circle, namely the inverse of an auxiliary circle of $\alpha$ in that circle of polarity.*

Polarity with respect to a circle is an especially useful case and is called a *standard polarity*. A standard polarity centered at $P$ means polarity with respect to some circle centered at $P$.

**Theorem 6.5.6.** *The polar curve of one conic $\alpha_1$ with respect to another conic $\alpha$ is a conic.*

**Proof.** One proof combines theorem 6.5.3 with a projectivity. Send two distinct (possibly imaginary) intersections of the conics to the circular points; the assertion then reduces to theorem 6.5.3. If all four intersections coalesce, regard the configuration as a limiting case.[24]

Without complex projectivities, however, it is not always possible to send two conics to two circles simultaneously. We therefore give a second proof. Choose five points $X_1, X_2, X_3, X_4, X_5$ on $\alpha_1$, and let their polars with respect to $\alpha$ be $x_1, x_2, x_3, x_4, x_5$. These five lines determine a unique conic $\beta$ tangent to all of them. For any point $X \in \alpha_1$, let $x$ be its polar. Apply Pascal's theorem to $X_1, X_2, X_3, X_4, X_5, X$. By the polarity principle, $x_1, x_2, x_3, x_4, x_5, x$ satisfy the hypothesis of the converse of Brianchon's theorem. Thus $x$ is also tangent to $\beta$, proving that the polar curve is $\beta$. ☐

The earlier duality for circles now extends to arbitrary conics. Elements corresponding under polarity with respect to a conic are dual. By theorem 6.5.6, replacing "circle" by "conic" in tables 1 and 2, with a conic dual to a conic, gives the following general definition.

[24]Because conics are quadratic curves, two conics in the complex projective plane have four intersections, possibly coincident. These cases will be examined in detail later.

**Definition 6.5.7.** In the projective plane, suppose a figure or proposition is constructed only from the elements, relations, and operations in the following table. The figure or proposition obtained by interchanging the two entries in every row is its *dual figure* or *dual proposition*.[25]

TABLE 1

| | | |
|---|---|---|
| 1 | point | line |
| 2 | draw a line through a point | choose a point on a line |
| 3 | a point lies on a line | a line passes through a point |
| 4 | a point does not lie on a line | a line does not pass through a point |
| 5 | join two points | intersect two lines |
| 6 | $n$ points are collinear | $n$ lines are concurrent |
| 7 | four points have cross ratio $t$; in particular, they form a harmonic range | four lines have cross ratio $t$; in particular, they form a harmonic pencil |
| 8 | conic | conic |
| 9 | a point lies on a conic | a line is tangent to a conic |
| 10 | a point does not lie on a conic | a line is not tangent to a conic |
| 11 | a point is the pole of a line with respect to a conic | a line is the polar of a point with respect to a conic |

**Theorem 6.5.8** (Principle of duality)**.** *If a proposition of projective geometry is true, then its dual proposition is also true.*

As before, a polarity transforms a proposition into its dual by interchanging the two columns of table 1. This is the *generalized principle of polarity*, usually shortened to the *principle of polarity*. For example, a tangent to a conic is sent to a point on the polar curve of that conic, corresponding to row 9 of table 1. Historically, the principle of polarity has sometimes also been called the *principle of duality*.

Here are some examples.

**Example 6.5.9.** "There is a unique conic through five given points" is dual to "There is a unique conic tangent to five given lines."

**Example 6.5.10.** Pascal's theorem and Brianchon's theorem are dual; their converses are dual as well.

**Example 6.5.11.** Theorems 6.1.14 and 6.1.15 are dual.

**Example 6.5.12.** Prove that if two complete quadrilaterals have the same three diagonal lines, then a conic is tangent to all eight lines forming the two quadrilaterals.

**Solution.** Regard $A_iB_iC_iD_i$ $(i=1,2)$ in theorem 6.3.8 as complete quadrangles and $A', B', C'$ as their three diagonal points. That result says: if two complete quadrangles have the same three diagonal points, their eight vertices lie on a conic. The present assertion is its dual. ☐

Two applications of example 6.5.12 follow.

[25]Previously a proposition could contain only one circle, since one polarity does not in general preserve several circles simultaneously. Here several conics are allowed because the polar curve of any conic with respect to a conic is again a conic.

**Theorem 6.5.13** (Emelyanov's theorem (Емельянов)). *The Euler circle of the diagonal triangle of a complete quadrilateral passes through its Miquel point.*

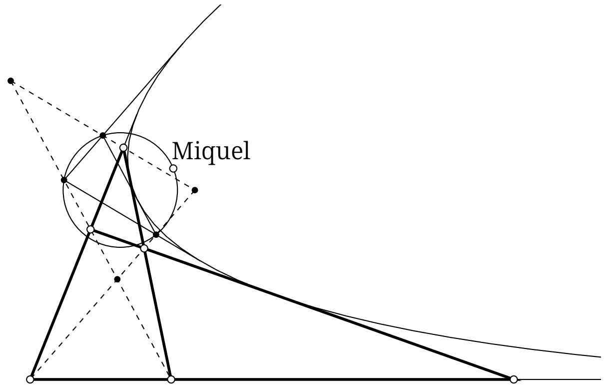


FIGURE 6.20

**Proof.** Let the original complete quadrilateral be $\mathfrak{a}$, and let $\mathfrak{b}$ be the complete quadrilateral formed by the three midlines of its diagonal triangle together with $l_\infty$. The two complete quadrilaterals have the same diagonal triangle. By example 6.5.12, a parabola is tangent to all eight of their lines. The conic tangent to the four lines of $\mathfrak{a}$ and to $l_\infty$ is necessarily the inscribed parabola $\alpha$ of $\mathfrak{a}$. By theorem 4.2.11, the circumcircle of the medial triangle of the diagonal triangle—its Euler circle—passes through the focus of $\alpha$. The conclusion now follows from theorem 4.8.3. ☐

**Proposition 6.5.14.** *The center of the circumcircle of the diagonal triangle of a complete quadrilateral lies on its Aubert line.*

**Proof.** The orthocenter of a medial triangle is the circumcenter of the original triangle. The proof now follows by the same method as theorem 6.5.13. ☐

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## Exercises

**Exercise 6.47.** Prove the hyperbolic and parabolic cases of theorem 6.5.3.

**Exercise 6.48.** Let $C$ be the center of a conic $\beta$. Suppose $\beta$ is the polar curve of a circle $\alpha$ with respect to a circle $\omega$. Prove that the polar of $C$ with respect to $\omega$ coincides with the polar of the center of $\omega$ with respect to $\alpha$.

**Exercise 6.49.** Prove that the polar curve of a rectangular hyperbola with respect to its auxiliary circle is the hyperbola itself.

**Exercise 6.50.** State the duals of the following results:
(1) propositions 6.2.5 and 6.2.6;
(2) Chasles's theorem, the hexagon theorem, and the three-conics theorem.

**Exercise 6.51.** Given $\triangle ABC$ and a point $P$ on its Euler circle, find a parabola $\alpha$ such that, for every tangent $l$ to $\alpha$, the Miquel point of the complete quadrilateral formed by $l$ and the three sides of the cevian triangle of the trilinear pole of $l$ is $P$.

**Exercise 6.52**.** In the real projective plane, construct a continuous bijection $j\colon X \mapsto j(X)$ from points to lines (continuity may be assumed here, although it is in fact unnecessary) satisfying the polarity principle $A \in j(B) \Longleftrightarrow B \in j(A)$. Prove that if $\{A \mid A \in j(A)\}$ is nonempty, it defines a conic and $j$ gives the polarity with respect to this conic.

### 6.5.2 Applications of Standard Polarity

Standard polarity is especially useful because polarity with respect to a circle has the following metric property.

**Theorem 6.5.15.** *Let $\omega$ be a circle with center $O$. For any points $A, B$, $\measuredangle AOB = \measuredangle(\mathfrak{p}_\omega(A), \mathfrak{p}_\omega(B))$.*

**Proof.** Since $\mathfrak{p}(A) \perp OA$ and $\mathfrak{p}(B) \perp OB$, the assertion follows immediately. □

Thus standard polarity preserves angle relations as well as projective ones. Our first application is the property in exercise 6.22, stated here as the following characterization of rectangular hyperbolas.

**Theorem 6.5.16.** *A conic $\alpha$ through the vertices of $\triangle ABC$ is a rectangular hyperbola if and only if it also passes through the orthocenter $H$ of the triangle.*

We first prove a lemma.

**Lemma 6.5.17.** *Let $H$ be the orthocenter of $\triangle ABC$. Under a standard polarity centered at $H$, let $\triangle A'B'C'$ be the polar triangle of $\triangle ABC$. Then the two triangles are homothetic, and $H$ is also the orthocenter of $\triangle A'B'C'$.*

**Proof.** As in figure 6.21, $H$ is the orthocenter of $\triangle ABC$, so $AH \perp BC$. By the definition of polarity with respect to a circle, $B'C' = \mathfrak{d}(A) \perp AH$. Hence $B'C' \parallel BC$. The other two pairs of sides are parallel in the same way, so $\triangle A'B'C'$ and $\triangle ABC$ are homothetic.

Moreover $A' = \mathfrak{d}(BC)$, whence $A'H \perp BC$ and $A, H, A'$ are collinear. Thus $A'H \perp BC$, and together with $B'C' \parallel BC$ this gives $A'H \perp B'C'$. The other two altitudes are analogous, proving that $H$ is the orthocenter of $\triangle A'B'C'$. □

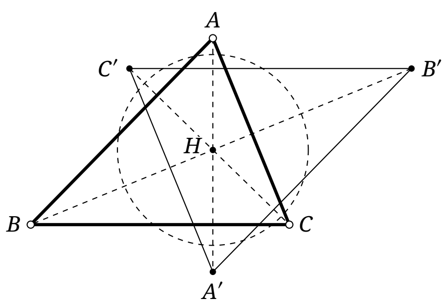


FIGURE 6.21

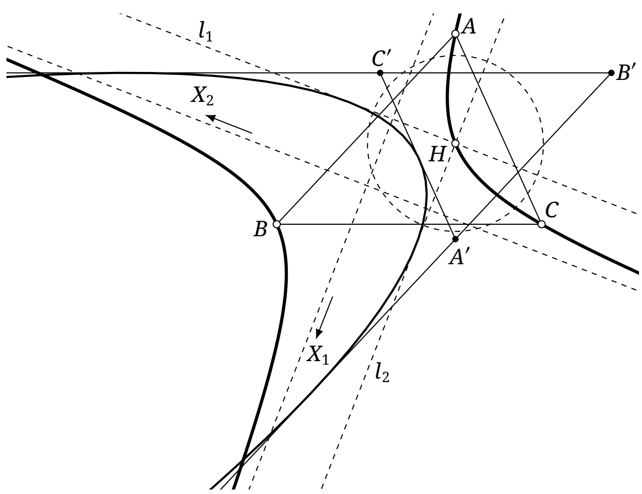


FIGURE 6.22

**Proof of theorem 6.5.16.** First suppose that $\alpha$ passes through the orthocenter $H$ of $\triangle ABC$, as in figure 6.22. Apply a standard polarity $\mathfrak{d}$ centered at $H$ and put $\beta := \mathfrak{d}(\alpha)$. By the polarity principle, $\mathfrak{d}(H) = l_\infty$ is tangent to $\beta$, so $\beta$ is a parabola.

Since $A,B,C \in \alpha$, the polar triangle $\triangle A'B'C'$ is circumscribed about $\beta$. By lemma 6.5.17, $H$ is its orthocenter; hence theorem 4.2.12 shows that $H$ lies on the directrix of $\beta$.

Draw the two tangents $l_1,l_2$ to $\beta$ through $H$, and let their inverse images under the polarity be $X_1,X_2$. Then $X_1,X_2 \in \alpha$. Since $l_1,l_2$ pass through the center of the polarity, $X_1,X_2$ lie at infinity. By theorem 1.5.7, $l_1 \perp l_2$, and theorem 6.5.15 therefore gives $HX_1 \perp HX_2$. Thus $\alpha$ contains the two points at infinity in perpendicular directions and is a rectangular hyperbola. The proof of the converse is similar. □

The orthotransversal is another application, although it does not itself involve a conic.

**Theorem 6.5.18.** *Given $\triangle ABC$ and a point $P$, draw through $P$ the perpendiculars to $AP,BP,CP$, meeting $BC,CA,AB$ at $X,Y,Z$, respectively. Then $X,Y,Z$ are collinear.* Their common line is the *orthotransversal* of $P$ with respect to $\triangle ABC$.

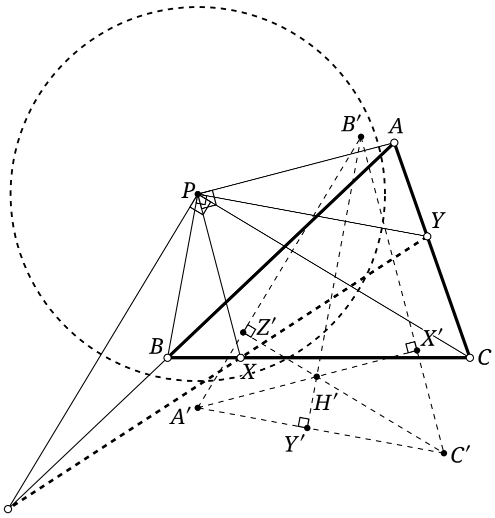


FIGURE 6.23

**Proof.** Apply a standard polarity $\mathfrak{d}$ centered at $P$, and let $\triangle A'B'C'$ be the polar triangle of $\triangle ABC$, as in figure 6.23. Since $\mathfrak{d}(B) = A'C'$, we have $PB \perp A'C'$. The line $PY$ passes through the center of the polarity, so its pole is the point at infinity in the perpendicular direction, namely $\mathfrak{d}(PY) = \infty_{PB}$.

The polarity principle gives $\mathfrak{d}(Y) = \mathfrak{d}(PY)\mathfrak{d}(AC) = \infty_{PB}B'$. Since $PB \perp A'C'$, we have $\mathfrak{d}(Y) \perp A'C'$. Similarly, $\mathfrak{d}(X)$ and $\mathfrak{d}(Z)$ are perpendicular to $B'C'$ and $A'B'$, respectively. Thus $\mathfrak{d}(X)$, $\mathfrak{d}(Y)$, and $\mathfrak{d}(Z)$ concur at the orthocenter of $\triangle A'B'C'$. By polarity, the original points $X,Y,Z$ are collinear. □

We next introduce the Frégier point.

**Theorem 6.5.19** (Frégier's theorem)**.** *Let $P$ be a point on a conic $\alpha$. All chords of $\alpha$ that subtend a right angle at $P$ pass through a fixed point.* This point is called the *Frégier point* of $P$ with respect to $\alpha$.

**Proof.** Apply a standard polarity $\mathfrak{d}$ centered at $P$. Then $\mathfrak{d}(P) = l_\infty$. Put $\beta = \mathfrak{d}(\alpha)$; since $\beta$ is tangent to $l_\infty$, it is a parabola.

Let $AB$ be a chord of $\alpha$ with $AP \perp BP$. Because $AP$ passes through the center of the polarity, its pole lies at infinity in the direction perpendicular to $AP$, namely $\mathfrak{d}(AP) = \infty_{BP}$; likewise $\mathfrak{d}(BP) = \infty_{AP}$. Since $A \in AP$ and $B \in BP$, the polarity principle shows that $\mathfrak{d}(A),\mathfrak{d}(B)$ are the finite tangents to $\beta$ through $\infty_{BP},\infty_{AP}$, respectively. They are therefore perpendicular. By theorem 1.5.7, $\mathfrak{d}(A) \cap \mathfrak{d}(B)$ moves on a fixed line, the directrix of $\beta$. Taking inverse polars shows that $AB$ passes through a fixed point. □

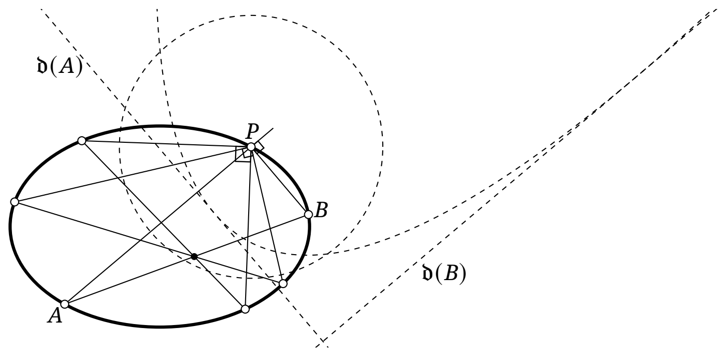


FIGURE 6.24

**Alternative proof.** For $A \in \alpha$, rotate $AP$ through 90° about $P$, and let the resulting line meet $\alpha$ again at $B$. The map $f\colon [AP] \to [BP]$ preserves angles between pairs of lines and hence preserves cross-ratios; it is therefore a projectivity of the pencil at $P$. Each map in $[A] \barwedge [AP] \barwedge [BP] \barwedge [B]$ is projective, so $g\colon [A] \to [B]$ is a projectivity of $\alpha$. Clearly $g(B) = A$, and hence $g^2 = \mathrm{id}$. Thus $g$ is an involution, and the joins of corresponding points $A, B$ pass through its center. □

We next locate the Frégier point. For a conic $\alpha$, let $x$ be its principal axis, and consider two special chords $AB$. First choose $AP \parallel x$ and $BP \perp x$; then $AB$ is a diameter and $\mathfrak{f}_x(P) \in AB$. Next let $B \to P$; the line $BP$ tends to $\mathfrak{t}(P)$, while $AP$ becomes the line perpendicular to $\mathfrak{t}(P)$. The intersection of these two special positions of $AB$ is the Frégier point $P'$.

**Proposition 6.5.20.** *For a central conic, the locus of the Frégier points of its points is a conic homothetic to the original conic about its center, with ratio $\frac{e^2}{2-e^2}$, where e is the eccentricity of the original conic.*

**Proof.** Consider first an ellipse, as in figure 6.25. Let $Q$ be the reflection of $P \in \alpha$ in the major axis; then the major axis perpendicularly bisects $PQ$ at $T$. Let $\mathfrak{t}(P)$ meet the major axis at $Z$. The line through $P$ perpendicular to $\mathfrak{t}(P)$ meets the major axis at $X$. Put $P' = OQ \cap PX$. Then $P'$ is the Frégier point of $P$.

Apply the affine transformation fixing the major axis and sending the ellipse to a circle; let $P$ go to $Y$. If $a, b, c$ are the semimajor axis, semiminor axis, and semifocal distance, respectively, then $YT = \frac{a}{b}PT$, while $O, X, T, Z$ remain fixed and $YZ$ is tangent to the circle. Since $OY \perp YZ$, the right-triangle altitude theorem applied to $\triangle OYZ$ gives $OT \cdot ZT = YT^2 = \frac{a^2}{b^2}PT^2$. Also $PX \perp PZ$, so the same theorem gives $XT \cdot ZT = PT^2$. Therefore $\frac{OT}{XT} = \frac{a^2}{b^2}$ and $\frac{TX}{XO} = \frac{b^2}{a^2-b^2} = \frac{a^2-c^2}{c^2} = \frac{1-e^2}{e^2}$. Menelaus's theorem in $\triangle OQT$ with transversal $\overline{PXP'}$ gives

$$1 = \frac{P'O}{P'Q} \cdot \frac{QP}{PT} \cdot \frac{TX}{XO} = \frac{P'O}{P'Q} \cdot 2 \cdot \frac{1-e^2}{e^2}.$$

Thus $\frac{P'O}{P'Q} = \frac{e^2}{2-2e^2}$, and hence

$$\frac{OP'}{OQ} = \frac{e^2}{2-2e^2+e^2} = \frac{e^2}{2-e^2} = \text{const.}$$

The hyperbolic case follows similarly, using a complex affine transformation.[26] □

[26]Although lengths in the complex plane can involve a choice of square root, ratios of parallel or collinear directed segments require no extra structure: if $\overrightarrow{AB} = \lambda\overrightarrow{AC}$, define $\frac{AB}{AC} = \lambda$.

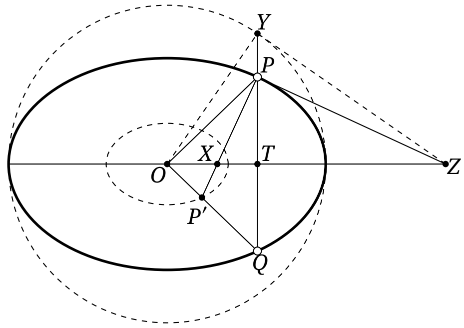


FIGURE 6.25

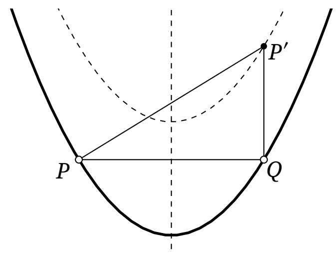


FIGURE 6.26

**Proposition 6.5.21.** *As shown in figure 6.26, the locus of the Frégier points of a parabola is the parabola obtained by translating the original one along its principal axis, in the direction of opening, through twice the focal parameter.*

**Proof.** For a point $P$ on a parabola $\alpha$, reflect $P$ in the axis to $Q$. The line through $Q$ parallel to the axis meets the normal at $P$ at the Frégier point $P'$. By corollary 2.3.8, $P'Q$ is twice the focal parameter. □

Frégier's theorem has the following extension.

**Theorem 6.5.22.** *Let $P$ lie on a fixed conic $\alpha$, and let $\phi$ be a fixed acute angle. The chords of $\alpha$ subtending either $\phi$ or $180° - \phi$ at $P$ envelop a conic.*

**Proof.** Use the standard polarity centered at $P$ as in the first proof of Frégier's theorem. For a qualifying chord $AB$, the angle between $\flat(A)$ and $\flat(B)$ is now $\phi$ or $180° - \phi$, rather than $90°$. By theorem 1.5.8, their intersection $\flat(A) \cap \flat(B)$ lies on a conic. The polarity principle then shows that the original chords $AB$ envelop a conic. □

This result can be generalized further.

**Theorem 6.5.23.** *The joins of corresponding points under a noninvolutory projectivity of a fixed conic envelop a conic.*

**Proof.** Let $f$ be a noninvolutory projectivity of a conic $\alpha$, and fix $P \in \alpha$. For a variable $A$, the perspectivities $[A] \doublebarwedge [AP]$ and $[f(A)] \doublebarwedge [f(A)P]$, together with the projectivity $[A] \barwedge [f(A)]$, induce a projectivity $g\colon [PA] \to [Pf(A)]$ of the pencil at $P$. Since $f$ is not an involution, neither is $g$.

If $g$ is not parabolic, let $m,n$ be its possibly imaginary invariant lines. By theorem 3.5.19, for every $A$ we have $(PA, Pf(A); m,n) = \text{const}$. Apply a projectivity making $m,n$ pass through the two circular points. By theorem 6.4.16, the directed angle between $PA$ and $Pf(A)$ is then a fixed angle other than $90°$; otherwise $g$ would be an involution. By theorem 6.5.22, the lines $Af(A)$ envelop a conic. Thus $Af(A)$ also envelops a conic before the projectivity. If $g$ is parabolic, take a limit of the nonparabolic case. □

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## Exercises

**Exercise 6.53.** What curve is enveloped by all rhombi inscribed in a fixed ellipse?

**Exercise 6.54.** For what eccentricity does the Frégier point of every point of a conic lie at infinity?

**Exercise 6.55.** Let the asymptotes of a hyperbola centered at $O$ form an angle $\theta$. If $P'$ is the Frégier point of $P$ on the hyperbola, prove that $\frac{OP}{OP'} = \cos\theta$.

**Exercise 6.56.** Given five points $A,B,C,D,P$, prove that the orthotransversals of $P$ with respect to $\triangle ABC, \triangle BCD, \triangle CDA, \triangle DAB$ concur at the Frégier point of $P$ with respect to $\mathfrak{C}(ABCDP)$.

**Exercise 6.57.** Let $P$ be a point not on a conic $\alpha$. Variable perpendicular lines $m,n$ through $P$ have respective intersection points $A,B$ with $\alpha$, one chosen on each line. If $Q$ is the orthogonal projection of $P$ onto $AB$, prove that the locus of $Q$ is a circle.

**Exercise 6.58.** Given a triangle, prove that the following four lines are concurrent: the trilinear polar of a point, its orthotransversal with respect to the given triangle, its orthotransversal with respect to its cevian triangle, and its orthotransversal with respect to its anticevian triangle.

**Exercise 6.59.** Let $P,Q$ be fixed points and $A$ a variable point on a conic $\alpha$. The lines $AP,AQ$ meet $\alpha$ again at $M,N$, respectively. Prove that $MN$ envelops a conic.

**Exercise 6.60.** Let $\triangle ABC$ be inscribed in a parabola $\alpha$ and have orthocenter equal to the focus $F$ of $\alpha$. Construct a triangle $\triangle A'B'C'$ circumscribed about $\alpha$ so that its three points of contact are $A,B,C$. Let $l$ be the tangent to $\alpha$ at its vertex. Prove that:
(1) $\odot(ABC)$ is tangent to $l$;
(2) the orthocenter of $\triangle A'B'C'$ is fixed;
(3) if $T$ is the point of tangency in part 1 and $O$ is the circumcenter of $\triangle A'B'C'$, then $O$ moves on a fixed line and $OT \perp l$.

**Exercise 6.61**.** Triangles $\triangle ABC$ and $\triangle A'B'C'$ have the following property: the perpendiculars from $A,B,C$ to $B'C',C'A',A'B'$ concur at $Q_1$. Prove that:
(1) the perpendiculars from $A',B',C'$ to $BC,CA,AB$ concur at $Q_2$;
(2) if the two triangles are also perspective from a point $P$, then $Q_1,Q_2,P$ are collinear (Sondat's theorem).

## 6.6 Quadratic Line Ranges

### 6.6.1 Quadratic Line Ranges

Duality turns quadratic point ranges into quadratic line ranges. We begin with the dual of theorem 6.2.1.

**Theorem 6.6.1.** *Let $l_1,l_2,l_3,l_4$ be four tangents to a conic $\alpha$, and let $l$ be any tangent to $\alpha$. Let $l_1,l_2,l_3,l_4$ meet $l$ at $A_1,A_2,A_3,A_4$, respectively. Then $(A_1A_2,A_3A_4)$ is independent of the choice of $l$.*

**Remark.** The tangent $l$ may coincide with one of $l_1,l_2,l_3,l_4$. For example, if $l = l_1$, the limiting intersection $l \cap l_1$ is the point of contact of $l_1$ with $\alpha$.

**Definition 6.6.2** (Quadratic line range)**.**
(1) A *quadratic line range* is a collection of tangents $a,b,c,\ldots$ to a conic $\alpha$, denoted $\alpha(a,b,c,\ldots)$.[27]

[27] When no ambiguity can arise, the symbol $\alpha$ specifying the conic of the quadratic line range may be omitted. As with a quadratic point range, the term imposes no restriction on the number of members under consideration.

(2) Given four tangents $a,b,c,d$ to $\alpha$, define their *cross ratio* by

$$\alpha(a,b;c,d) := (a \cap p, b \cap p; c \cap p, d \cap p),$$

where $p$ is any tangent to $\alpha$. By theorem 6.6.1, this is well defined. It may be abbreviated as $\alpha(ab,cd)$, or, when unambiguous, as $(a,b;c,d)$ or $(ab,cd)$.

(3) If $\alpha(ab,cd) = -1$, then $\alpha(a,b,c,d)$ is a *harmonic quadratic line range* on $\alpha$.

Quadratic point ranges and quadratic line ranges are dual; in particular, harmonic quadratic point ranges and harmonic quadratic line ranges are dual. They also have the following relationship.

**Theorem 6.6.3.** *Let $A,B,C,D$ lie on a conic $\alpha$, and let the tangents there be $a,b,c,d$. Then $\alpha(AB,CD) = \alpha(ab,cd)$.*

**Proof.** Apply polarity with respect to $\alpha$. □

In analogy with projective definition A, we can define a conic using a projective correspondence between point ranges.

**Definition 6.6.4** (Projective definition of a conic)**.** Let a projectivity $f$ send the point range on a line $l_1$ to the point range on a line $l_2$. The joins of corresponding points envelop a conic tangent to $l_1, l_2$. This conic is nondegenerate if and only if $f$ is not a perspectivity.

**Remark.** The degeneration here is not the usual degeneration of a conic into two lines. In the limiting case where $f$ is a perspectivity, the envelope degenerates into two points: $l_1 \cap l_2$ and the center of perspectivity.

The distinction comes from two dual ways of regarding a conic. All earlier definitions, including projective definition A, regard it as a set of points: a *curve of order two*. Definition B regards it as the envelope of a set of tangent lines: a *curve of class two*. A degenerate curve of order two is a pair of lines; dually, a degenerate curve of class two is a pair of points. Accordingly, definitions 6.2.7 and 6.6.4 are also called the *projective definition of a curve of order two* and the *projective definition of a curve of class two*, respectively.

For a nondegenerate conic the two viewpoints produce the same geometric object and ordinarily need not be distinguished.[28] Their algebraic distinction will become clearer after homogeneous point and line coordinates are introduced.

Thus the following rows may be added to table 1.

TABLE 2

| | | |
|---|---|---|
| 12 | curve of order two; in particular, a degenerate curve of order two | curve of class two; in particular, a degenerate curve of class two |
| 13 | $n$ points lie on one conic (of order two) | $n$ lines are tangent to one conic (of class two) |
| 14 | four members of a quadratic point range have cross ratio $t$; in particular, they form a harmonic quadratic point range | four members of a quadratic line range have cross ratio $t$; in particular, they form a harmonic quadratic line range |

We now specialize projective definition B to parabolas.

[28] Degenerate cases may also be treated by limits. Unless stated otherwise, a degenerate conic will mean a degenerate curve of order two, namely two possibly coincident lines. A curve of class two degenerated into two points will be called a *degenerate curve of class two*.

**Theorem 6.6.5.** *Let $l_1, l_2$ be intersecting lines. Suppose $B$ moves uniformly on $l_1$ and $C$ moves uniformly, possibly at a different speed, on $l_2$, and the two points do not pass through $l_1 \cap l_2$ simultaneously. Then the line $BC$ envelops a parabola tangent to $l_1, l_2$.*

**Proof.** Uniform motion makes $f : [B] \to [C]$ cross-ratio preserving, hence projective. Projective definition B shows that $BC$ envelops a conic. The two points reach infinity simultaneously, so the conic is tangent to $l_\infty$ and is therefore a parabola. □

The converse takes the following form.

**Theorem 6.6.6.** *The tangents to a parabola at fixed points $X, Y$ meet at $A$. Let $P$ vary on the parabola, and let the tangent at $P$ meet $AX, AY$ at $B, C$. Then $B$ moves uniformly on $AX$ if and only if $C$ moves uniformly on $AY$.*

**Proof.** Let $B'$ on $AX$ and $C'$ on $AY$ move uniformly so that $C' = A$ when $B' = X$, and $C' = Y$ when $B' = A$. By theorem 6.6.5, the lines $B'C'$ envelop a parabola $\alpha$.

As $B'C' \to AX$, both lines are tangent to $\alpha$, so their intersection $B' = B'C' \cap AX$ tends to the point of contact of $AX$ with $\alpha$; during this limit $B' \to X$. Thus $AX$ is tangent to $\alpha$ at $X$, and similarly $AY$ is tangent at $Y$.

The conditions of tangency to $AX$ at $X$, to $AY$ at $Y$, and to $l_\infty$ determine a unique conic.[29] By identification, $\alpha$ is the given parabola. Since the envelope of $B'C'$ is $\alpha$, every tangent $BC$ gives the same correspondence between $B, C$ as that between $B', C'$, which proves the result. □

**Alternative proof.** As in figure 6.27, let $F$ be the focus. By theorem 4.2.11, the points $A, B, C, F$ are concyclic. Choose the positive directions of $\overline{AB}$ and $\overline{AC}$ to agree with $\overline{AX}$ and $\overline{AY}$, respectively. The three-chord theorem gives

$$\overline{AB} \sin \measuredangle FAC + \overline{AC} \sin \measuredangle BAF = AF \sin \measuredangle BAC.$$

Differentiating with respect to time yields

$$\sin \measuredangle FAC \cdot \frac{\mathrm{d}}{\mathrm{d}t}\overline{AB} + \sin \measuredangle BAF \cdot \frac{\mathrm{d}}{\mathrm{d}t}\overline{AC} = 0.$$

The angles $\measuredangle FAC, \measuredangle BAF$ are fixed, so either velocity is constant exactly when the other is. □

In these theorems, the special feature of uniform motion determines the parabola. We can use this to prove several properties of parabolas.

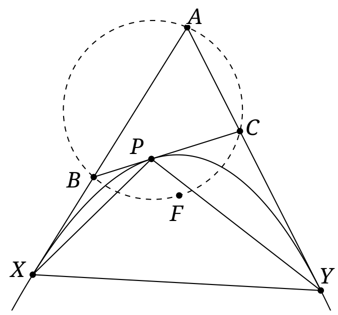


FIGURE 6.27

[29]This is the limiting case of tangency to the five lines consisting of the coincident pair $AX, AX$, the coincident pair $AY, AY$, and $l_\infty$. To interpret the first pair, start with two lines $AX, A'X'$ and let $A'X' \to AX$ while keeping their intersection at $X$. The second pair is interpreted similarly.

**Theorem 6.6.7.** *Let the tangents to a parabola at $X,Y$ meet at $A$. For a point $P$ on the parabola, let its tangent meet $AX,AY$ at $B,C$, respectively. Then $\frac{CP}{PB}=\frac{YC}{CA}=\frac{AB}{BX}$.*

**Proof.** The tangents $AX,AY,BC$ induce a projectivity between the ranges on $AX$ and $AY$, namely $f: A\mapsto Y,\ B\mapsto C,\ X\mapsto A$. Since the corresponding points $B,C$ move uniformly, we have $\frac{AB}{BX}=\frac{YC}{CA}$. The same argument for the fractional-linear projectivity between the ranges on $BC$ and $AX$ gives $\frac{CP}{PB}=\frac{AB}{BX}$. □

**Theorem 6.6.8.** *Let the tangents to a parabola at $X,Y$ meet at $A$. For a point $P$ on the parabola, let its tangent meet $AX,AY$ at $B,C$, respectively. Then $S_{\triangle PXY}=2S_{\triangle ABC}$.*

**Proof.** Put $S_{\triangle ABC}=S$ and, by theorem 6.6.7, set $t=\frac{CP}{PB}=\frac{YC}{CA}=\frac{AB}{BX}$. Then

$$\begin{aligned}
S_{\triangle PCY}&=\frac{CP}{CB}\cdot\frac{CY}{CA}\cdot S=\frac{t}{1+t}\cdot t\cdot S,\\
S_{\triangle PBX}&=\frac{BP}{BC}\cdot\frac{BX}{BA}\cdot S=\frac{1}{1+t}\cdot\frac{1}{t}\cdot S,\\
S_{\triangle AXY}&=\frac{AX}{AB}\cdot\frac{AY}{AC}\cdot S=\frac{1+t}{t}\cdot(1+t)\cdot S,\\
S_{\triangle PXY}&=S_{\triangle AXY}-S_{\triangle PCY}-S_{\triangle PBX}-S=2S.
\end{aligned}$$

□

We now dualize the definitions of projectivities associated with quadratic point ranges.

**Definition 6.6.9.**

(1) Let $\alpha(a,b,c,\ldots)$ be a quadratic line range and $l$ any tangent to $\alpha$. Intersecting its members with $l$ gives the point range $l(a,b,c,\ldots)$. This is a perspectivity

$$\alpha(a,b,c,\ldots)\overline{\wedge}\, l(a,b,c,\ldots).$$

(2) Given a conic $\alpha$ and a quadratic line range $\alpha(a,b,c,\ldots)$, let $A,B,C,\ldots$ be its points of contact with $\alpha$. These points form a quadratic point range on $\alpha$, and

$$\alpha(a,b,c,\ldots)\overline{\overline{\wedge}}\,\alpha(A,B,C,\ldots)$$

is a perspectivity between a quadratic line range and a quadratic point range.

(3) Two quadratic line ranges $\alpha(a,b,c,\ldots)$ and $\alpha(a',b',c',\ldots)$ are *projective* if, for a tangent $l$ to $\alpha$,

$$l(a,b,c,\ldots)\overline{\wedge}\, l(a',b',c',\ldots).$$

We write this projectivity as

$$\alpha(a,b,c,\ldots)\overset{(l)}{\wedge}\alpha(a',b',c',\ldots).$$

The tangent $l$ is called a center of this projectivity. In fact, if $\alpha(a,b,c,\ldots)\overset{(l)}{\wedge}\alpha(a',b',c',\ldots)$, then for every tangent $l'$ to $\alpha$, $\alpha(a,b,c,\ldots)\overset{(l')}{\wedge}\alpha(a',b',c',\ldots)$ also holds. Thus we may simply write

$$\alpha(a,b,c,\ldots)\overline{\wedge}\,\alpha(a',b',c',\ldots).$$

The map sending $\alpha(a,b,c,\ldots)$ to $\alpha(a',b',c',\ldots)$ is called a *projectivity*, or *projective transformation*, of quadratic line ranges on $\alpha$; the conic $\alpha$ is its *base conic*.

**Remark.** Projectivities of quadratic point ranges occur more often, so "a projectivity on the conic $\alpha$" ordinarily refers to that pointwise case. A projectivity of quadratic line ranges may instead be called a projectivity on the curve of class two $\alpha$, since such a curve is regarded as an envelope of lines. Correspondingly, a projectivity on the curve of order two $\alpha$ means one between quadratic point ranges. When context removes ambiguity, we may say simply projectivity without specifying point or line ranges. For example, the term at the end of part 3 of the preceding definition clearly refers to quadratic line ranges. The same convention applies to involutions on a conic discussed below.

The following properties of these perspectivities follow directly from their definitions.

**Theorem 6.6.10.** *Corresponding elements have equal cross-ratios under a perspectivity between a quadratic line range and a point range, and under a perspectivity between a quadratic line range and a quadratic point range.*

The next results are dual to their counterparts for quadratic point ranges, so their proofs are omitted.

**Theorem 6.6.11.** *A projectivity on a curve of class two preserves cross-ratios. Conversely, every cross-ratio-preserving correspondence between two quadratic line ranges on such a curve is a projectivity.*

**Theorem 6.6.12.** *The composite of projectivities on a curve of class two is a projectivity.*

**Theorem 6.6.13.** *A projectivity between two quadratic line ranges on a conic $\alpha$ is completely determined by three pairs of corresponding lines.*

**Proposition 6.6.14.** *Let $a,b,c,d$ and $a',b',c',d'$ be two ordered sets of tangents to a conic $\alpha$. If $\alpha(a,b,c,d) \barwedge \alpha(a',b',c',d')$, then $a'(a,b,c,d) \doublebarwedge a(a',b',c',d')$.*

**Theorem 6.6.15.** *Let $f$ be a projectivity of quadratic line ranges on a conic $\alpha$. For any $a,b \in \mathbf{T}\alpha$, the line joining $a \cap f(b)$ and $b \cap f(a)$ passes through a fixed point.* This point is called the *projective center* of $f$. [30]

**Corollary 6.6.16.** *The two tangents from the projective center of a projectivity $f$ on a curve of class two $\alpha$ are precisely the invariant lines of $f$.*

Projectivities on a curve of class two are classified by their number of real invariant lines. They are *elliptic*, *parabolic*, or *hyperbolic* when they have zero, one, or two invariant lines, respectively. These cases give zero, one, or two distinct tangents from the projective center to the conic; equivalently, the center lies inside, on, or outside the conic. We next consider involutions on a curve of class two.

**Definition 6.6.17.** Let $f$ be a projectivity between quadratic line ranges on a conic $\alpha$. If $f \neq \mathrm{id}$ and $f \circ f = \mathrm{id}$, then $f$ is an *involutory projectivity*, or simply an *involution*, of quadratic line ranges on $\alpha$. Its projective center is the *center of the involution*.

**Theorem 6.6.18.** *An involution between quadratic line ranges on a conic $\alpha$ is completely determined by two pairs of corresponding lines.*

**Theorem 6.6.19.** *Let an involution $f$ on a curve of class two $\alpha$ have invariant lines $m,n$, and let $p,p'$ correspond under $f$. Then $p,p',m,n$ form a harmonic quadratic line range.*

[30] We write $\mathbf{T}\gamma$ for the set of all tangents to a curve $\gamma$. As a circle $\omega$ shrinks to a point $P$, the set $\mathbf{T}\omega$ tends to the pencil of all lines through $P$; we also write $\mathbf{T}P$ for that pencil.

**Proposition 6.6.20.** *For an involution on a curve of class two $\alpha$, the line joining the points of contact of any pair of corresponding tangents passes through the center of the involution.*

**Proposition 6.6.21.** *The invariant lines of an involution on a curve of class two $\alpha$ are the two tangents to $\alpha$ from its center.*

Such an involution is *elliptic* or *hyperbolic* according as it has no real invariant lines or two invariant lines; equivalently, its center lies inside or outside the conic. It cannot be parabolic, with just one invariant line; that is, the center of an involution cannot lie on the conic.

**Proposition 6.6.22.** *For a conic $\alpha$, a chosen center of involution determines an involution of quadratic line ranges on $\alpha$.*

**Proposition 6.6.23.** *The intersections of corresponding lines under an involution $f$ on a curve of class two $\alpha$ lie on a fixed line. This line is the polar, with respect to $\alpha$, of the center of the involution.* It is called the *axis of the involution*.

The axis of such an involution cannot be tangent to the conic.

Dualizing theorem 6.5.23 gives the following result.

**Theorem 6.6.24.** *Under a noninvolutory projectivity between quadratic line ranges on a fixed conic, the intersections of corresponding lines trace a conic.*

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## Exercises

**Exercise 6.62.** Given $\triangle ABC$ and a line $l$, let $P$ vary on $l$, and put $E = AP \cap BC$, $F = BP \cap CA$. Prove that $EF$ envelops a conic tangent to all three sides of $\triangle ABC$.

**Exercise 6.63.** Given an equilateral triangle, prove that the cevian triangle of a point on its circumcircle has constant area.

**Exercise 6.64.** Given $\triangle ABC$, let $P \in \mathscr{C}$ and put $\mathfrak{f}_{BC}(P) = P_A$. Draw $P_AQ_A \perp \mathscr{W}(P)$ with $Q_A \in BC$, and define $Q_B, Q_C$ cyclically. Prove that $S_{\triangle Q_AQ_BQ_C}$ is constant.

**Exercise 6.65.** On the perpendicular bisectors of the three sides of $\triangle ABC$, choose $A', B', C'$ so that $\triangle BA'C \overset{\pm}{\sim} \triangle CB'A \overset{\pm}{\sim} \triangle AC'B$. Prove that $\triangle A'B'C'$ and $\triangle ABC$ are perspective and that their perspective axis envelops a conic.

### 6.6.2 Sets Endowed with a Cross Ratio

We have encountered four kinds of *one-parameter set*, meaning a set whose elements can be described by one parameter: a point range, a line pencil, a quadratic point range on a conic, and a quadratic line range on a conic. For a point range, fix one point and use directed distance from it as a parameter. For a line pencil, fix one line and use directed angle from it. Part 1 of definition 6.2.12 and part 1 of definition 6.6.9 give one-to-one correspondences of quadratic point and line ranges with pencils and point ranges, respectively. Thus these too can be described by one parameter. This motivates an abstract definition.

**Definition 6.6.25.** Let a one-parameter set $X$ have elements described by a parameter $t \in \mathbb{F} \cup \{\infty\}$.[31] A *set endowed with a cross ratio* is a pair $(X,f)$, where

$$f : X \times X \times X \times X \to \mathbb{F} \cup \{\infty\}, \quad (X_1,X_2,X_3,X_4) \mapsto \frac{t_3 - t_1}{t_3 - t_2} \div \frac{t_4 - t_1}{t_4 - t_2}, \tag{6.1}$$

and $t_i$ is the parameter of $X_i$. The value of $f$ is the *cross ratio* of $X_1,X_2,X_3,X_4$. When $f$ is understood, the pair is denoted simply by $X$.

For $X_1,X_2,X_3,X_4$ in the same set $X$ endowed with a cross ratio, write their cross ratio as $X(X_1,X_2;X_3,X_4)$. An alternative notation is $X(X_1X_2,X_3X_4)$. When no ambiguity arises, it may also be written as $(X_1,X_2;X_3,X_4)$ or $(X_1X_2,X_3X_4)$. The elements $X_1,X_2,X_3,X_4$ are *harmonic* if $(X_1X_2,X_3X_4) = -1$.[32]

**Example 6.6.26.** The four preceding constructions are sets endowed with a cross ratio.

(1) For the point range on a line $l$, the underlying set is $l$ itself.[33] Fix $O \in l$ and parameterize $P \in l$ by $\overline{OP}$.[34] Then $l(P_1P_2,P_3P_4) = \frac{\overline{P_1P_3}}{\overline{P_2P_3}} \div \frac{\overline{P_1P_4}}{\overline{P_2P_4}}$.
(2) For the line pencil at $P$, the underlying set is $\mathbf{T}P$, the set of lines through $P$. Fix $l \in \mathbf{T}P$ and parameterize $a \in \mathbf{T}P$ by $k_a := \tan\measuredangle(l,a)$. Then $\mathbf{T}P(ab,cd) = \frac{k_c - k_a}{k_c - k_b} \div \frac{k_d - k_a}{k_d - k_b}$. This agrees with definition 3.2.3 by exercise 3.17. Choose any line $l_0$ not through $P$. For $a \in \mathbf{T}P$, the intersections $l_0 \cap a$ form a point range on $l_0$. By theorem 3.2.5, using $\overline{(l_0 \cap l)(l_0 \cap a)}$ as the parameter of $a$ gives an equivalent cross-ratio expression.[35]
(3) For the quadratic point range on a conic $\alpha$ regarded as a curve of order two, the underlying set is $\alpha$. Fix $O \in \alpha$ and give $P \in \alpha$ the parameter of the line $OP$ in the pencil at $O$. Then $\alpha(P_1P_2,P_3P_4) = \mathbf{T}O(OP_1,OP_2;OP_3,OP_4)$.
(4) For the quadratic line range on a conic $\alpha$ regarded as a curve of class two, the underlying set is $\mathbf{T}\alpha$. Fix a tangent $l$ to $\alpha$. For every $a \in \mathbf{T}\alpha$, use the parameter of the point $a \cap l$ in the point range on $l$. Then $\mathbf{T}\alpha(ab,cd) = l(a \cap l, b \cap l; c \cap l, d \cap l)$.

The familiar identities for cross-ratios of point ranges and line pencils extend immediately from proposition 3.2.11 (or proposition 3.2.12) and proposition 3.2.13.

**Theorem 6.6.27.** *For $\xi_1,\xi_2,\xi_3,\xi_4$ in a set $X$ endowed with a cross ratio,*

*(1)* $(\xi_3,\xi_4;\xi_1,\xi_2) = (\xi_1,\xi_2;\xi_3,\xi_4)$.
*(2)* $(\xi_2,\xi_1;\xi_3,\xi_4) = (\xi_1,\xi_2;\xi_4,\xi_3) = \frac{1}{(\xi_1,\xi_2;\xi_3,\xi_4)}$.
*(3)* $(\xi_2,\xi_1;\xi_4,\xi_3) = (\xi_1,\xi_2;\xi_3,\xi_4)$.
*(4)* $(\xi_1,\xi_3;\xi_2,\xi_4) = (\xi_4,\xi_2;\xi_3,\xi_1) = 1 - (\xi_1,\xi_2;\xi_3,\xi_4)$.

**Proposition 6.6.28.** *If $\xi_1,\xi_2,\xi_3,\xi_4,\xi_5$ belong to the same set endowed with a cross ratio, then* $(\xi_1\xi_2,\xi_3\xi_4)(\xi_1\xi_2,\xi_4\xi_5)(\xi_1\xi_2,\xi_5\xi_3) = 1$.

We can now define projectivities and involutions uniformly.

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[31] Here $\mathbb{F}$ is a *number field*; see definition A.4.31. In our applications it is usually $\mathbb{R}$ or $\mathbb{C}$, corresponding to the real or complex projective plane. Here and throughout this book, *number ring* and *number field* are used literally for a ring or field consisting of complex numbers; in particular, *number field* does not have its usual narrower number-theoretic meaning. As before, the number field will be left implicit unless the distinction matters.

[32] The expression $\frac{t_3-t_1}{t_3-t_2} \div \frac{t_4-t_1}{t_4-t_2}$ is also called the *cross ratio* of the numbers $t_1,t_2,t_3,t_4$ and is denoted by $(t_1,t_2;t_3,t_4)$ or $(t_1t_2,t_3t_4)$.

[33] A line is itself a set of points, which is why it can serve directly as the underlying set; the same observation applies to the conic in item 3.

[34] The parameter ranges over $\mathbb{C} \cup \{\infty\}$ in $\mathbb{CP}^2$ and over $\mathbb{R} \cup \{\infty\}$ in $\mathbb{RP}^2$; the same convention applies below.

[35] By definition this cross ratio of lines is written $\mathbf{T}P(ab,cd)$. The symbol $\mathbf{T}$ may be omitted, as in the earlier notation; the same convention applies to quadratic line ranges in item 4.

**Definition 6.6.29.**

(1) Let $X_1, X_2$ be sets endowed with cross-ratios over the same number field $\mathbb{F}$. A bijection $\varphi: X_1 \to X_2$ is a *projectivity* if it preserves every cross ratio:
$$X_1(P_1P_2, P_3P_4) = X_2(\varphi(P_1)\varphi(P_2), \varphi(P_3)\varphi(P_4)) \quad \text{for all } P_1, P_2, P_3, P_4 \in X_1.$$
If $\varphi$ maps the elements $\xi_1, \xi_2, \xi_3, \ldots$ of $X_1$ to $\zeta_1, \zeta_2, \zeta_3, \ldots$, respectively, write
$$X_1(\xi_1, \xi_2, \xi_3, \ldots) \barwedge X_2(\zeta_1, \zeta_2, \zeta_3, \ldots).$$
When the meaning is clear, this may be abbreviated to $(\xi_1, \xi_2, \xi_3, \ldots) \barwedge (\zeta_1, \zeta_2, \zeta_3, \ldots)$; the elements $(\xi_1, \xi_2, \xi_3, \ldots)$ and $(\zeta_1, \zeta_2, \zeta_3, \ldots)$ are then said to be in projective correspondence. Denote the collection of all projectivities from $X_1$ to $X_2$ by $\mathrm{Prj}(X_1, X_2)$.[36]

(2) A transformation $\psi \in \mathrm{Prj}(X, X)$ is a *projective transformation* of $X$; write $\mathrm{Prj}(X) = \mathrm{Prj}(X, X)$.

(3) If $\psi \in \mathrm{Prj}(X)$ satisfies $\psi^2 = \mathrm{id}_X$ but $\psi \neq \mathrm{id}_X$, it is an *involutory transformation*, or simply an *involution*, of $X$.[37]

(4) These correspondences, transformations, and involutions are all defined on one-parameter sets and are therefore called *one-dimensional projectivities* , *one-dimensional projective transformations* , and *one-dimensional involutions* , respectively.

The earlier definitions for the four concrete kinds of sets agree with this definition. The following results follow directly from it.

**Theorem 6.6.30.** *Let $X_1, X_2, X_3$ be sets endowed with cross-ratios over the same number field $\mathbb{F}$. Then:*

*(1)* $\mathrm{id}_{X_1}$ *is a projective transformation;*

*(2) if $\varphi \in \mathrm{Prj}(X_1, X_2)$ and $\psi \in \mathrm{Prj}(X_2, X_3)$, then $\psi \circ \varphi \in \mathrm{Prj}(X_1, X_3)$;*

*(3) if $\varphi \in \mathrm{Prj}(X_1, X_2)$, then $\varphi^{-1} \in \mathrm{Prj}(X_2, X_1)$.*

**Corollary 6.6.31.** *For every set X endowed with a cross ratio,* $\mathrm{Prj}(X)$ *is a group.*

**Example 6.6.32.** Let $\alpha$ be a conic, let $l \in \mathbf{T}\alpha$ be fixed, and let $O$ be fixed. For variable $P \in \alpha$, let $l_P$ be the tangent at $P$, put $P' = l_P \cap l$, and draw $OP'$. Then:

(1) $\varphi: \alpha \to \mathbf{T}\alpha$, $P \mapsto l_P$, is a projectivity from the quadratic point range on $\alpha$ to the quadratic line range on $\alpha$, because theorem 6.6.3 shows that it preserves cross-ratios;

(2) $\psi: \mathbf{T}\alpha \to l$, $l_P \mapsto P'$, is a projectivity from the quadratic line range on $\alpha$ to the point range on $l$; it preserves cross-ratios;

(3) $\chi: l \to \mathbf{T}O$ is a projectivity from the point range on $l$ to the line pencil at $O$; it preserves cross-ratios;

(4) $\tau: \alpha \to \mathbf{T}O$ is a projectivity from the quadratic point range on $\alpha$ to the line pencil at $O$, because $\tau = \chi \circ \psi \circ \varphi$ and theorem 6.6.30 applies. It preserves cross-ratios.

Thus the unified definition of projectivities between sets endowed with cross-ratios extends the meaning of our earlier definitions. For example, the earlier definitions specified the correspondences in items 2 and 3, whereas we now also have those in items 1 and 4.

We now give unified properties of projective correspondences, projective transformations, and involutions. Since every element of each set can be described by one parameter, their structures are similar and the results extend directly from section 3.5.2. Many proofs will therefore not be repeated.

[36] Since projectivities preserve the algebraic structure given by cross-ratios, some authors borrow the appendix notation Aut, Hom, Iso for homomorphisms and isomorphisms to denote sets of projectivities or projective transformations. These symbols may, however, have different meanings in different settings. For instance, in a two-dimensional plane, $\mathrm{Aut}(\pi)$ sometimes denotes all projective transformations of $\pi$, and sometimes all transformations that preserve lines. The notation $\mathrm{Prj}(\pi)$ for all projective transformations of $\pi$ is unambiguous, so we use it here, although it is not standard.

[37] Some authors include the identity among the involutions; this book does not.

**Theorem 6.6.33.** *Three pairs of corresponding elements determine a projectivity. More precisely, let $X_1, X_2$ be sets endowed with cross-ratios over a number field $\mathbb{F}$ and let $\varphi \in \mathrm{Prj}(X_1, X_2)$. The images of three distinct elements $\xi_1, \xi_2, \xi_3 \in X_1$ determine $\varphi$ uniquely.*

**Theorem 6.6.34.** *Let $X$ be a set endowed with a cross ratio over a number field $\mathbb{F}$ and let $\varphi \in \mathrm{Prj}(X)$.*

*(1) The images under $\varphi$ of three distinct elements $\xi_1, \xi_2, \xi_3 \in X$ determine $\varphi$ uniquely. This is the statement that three pairs of corresponding elements determine a one-dimensional projective transformation.*
*(2) If three distinct elements $\xi_1, \xi_2, \xi_3 \in X$ are invariant under $\varphi$, then $\varphi = \mathrm{id}_X$. This is the statement that three invariant elements determine a one-dimensional identity projectivity.*
*(3) If $\varphi \neq \mathrm{id}_X$, it has at most two invariant elements.*
   *(a) Over $\mathbb{C}$, it always has two invariant elements, possibly coincident.*
   *(b)* Over $\mathbb{R}$, it is called *elliptic*, *parabolic*, or *hyperbolic* according as it has zero, exactly one, or two invariant elements.

**Theorem 6.6.35.** *Let $X$ be a set endowed with a cross ratio over a number field $\mathbb{F}$ and let $\varphi$ be an involution of $X$.*

*(1) Two distinct pairs of elements corresponding under $\varphi$ determine it uniquely. This is the statement that two pairs of corresponding elements determine a one-dimensional involution.*
*(2) The involution has at most two invariant elements.*
   *(a) Over $\mathbb{C}$, it always has two distinct invariant elements.*
   *(b) Over $\mathbb{R}$, it has either two distinct invariant elements or none.* With no invariant element it is called *elliptic*; with two distinct invariant elements it is called *hyperbolic*.

**Theorem 6.6.36.** *Let $X$ be endowed with a cross ratio, and let a bijection $\varphi: X \to X$ satisfy $\varphi(\xi_1) = \xi_1$, $\varphi(\xi_2) = \xi_2$ for two distinct elements $\xi_1, \xi_2 \in X$. For any $\zeta \in X$, write $\varphi(\zeta) = \zeta'$.*

*(1) $\varphi \in \mathrm{Prj}(X)$ if and only if a fixed scalar $k$ satisfies $(\xi_1\xi_2, \zeta\zeta') = k$ for every $\zeta$.*
*(2) It is an involution if and only if $(\xi_1\xi_2, \zeta\zeta') = -1$ for every $\zeta$.*

**Theorem 6.6.37.** *Let $X$ and $X'$ be sets endowed with cross-ratios. Suppose $\xi_1, \xi_2, \xi_3 \in X$ correspond under $\varphi: X \to X'$ to $\xi'_1, \xi'_2, \xi'_3 \in X'$. Then $\varphi$ is projective if and only if, for every $\xi \in X$, writing $\xi' = \varphi(\xi)$, one has $(\xi_1\xi_2, \xi_3\xi) = (\xi'_1\xi'_2, \xi'_3\xi')$.*

Here is another concrete application.

**Example 6.6.38.** Let $\alpha$ be a conic, $l$ a line, $P \in \alpha$, and $k$ a constant. A variable line $p \in \mathbf{T}P$ satisfies $p \cap \alpha = \{P, P'\}$ and $p \cap l = Q$. Choose $R \in p$ so that $(PP', QR) = k$. Prove that the locus of $R$ is a conic or a line.

**Proof.** Choose a fixed line $p_0$ through $P$, meeting $\alpha$ again at $P'_0$ and $l$ at $Q_0$, and choose $R_0 \in p_0$ so that $(PP'_0, Q_0R_0) = k$. Equality of cross-ratios gives $p(P, P', Q, R) \barwedge (P, P'_0, Q_0, R_0)$. By theorem 3.5.6, the lines $P'_0P'$, $Q_0Q$, and $R_0R$ concur at a point $S$. Hence the following three maps are projective:

$$\varphi: \mathbf{T}P \to \alpha, \quad p \mapsto P'; \quad \psi: \alpha \to l, \quad P' \mapsto (S = P'P'_0 \cap l); \quad \chi: l \to \mathbf{T}R_0, \quad S \mapsto R_0S.$$

By theorem 6.6.30(2), $\tau = \chi \circ \psi \circ \varphi \in \mathrm{Prj}(\mathbf{T}P, \mathbf{T}R_0)$. Since $R = p \cap \tau(p)$, projective definition A of a conic shows that the locus of $R$ is a conic, with a line as the degenerate case. □

We conclude by describing the possible forms of one-dimensional projectivities.

**Theorem 6.6.39.** *Let the elements of $X, Y$ be described, respectively, by single parameters $s, t \in \mathbb{F} \cup \{\infty\}$, and denote them by $\xi(s), \eta(t)$. Then*

$$\mathrm{Prj}(X, Y) = \left\{ \varphi: \xi(s) \mapsto \eta(t) \;\middle|\; t = \frac{As + B}{Cs + D},\ A, B, C, D \in \mathbb{F},\ AD \neq BC \right\}. \tag{6.2}$$

**Proof.** Suppose three elements $\xi(s_1)$, $\xi(s_2)$, $\xi(s_3)$ and their corresponding elements $\eta(t_1)$, $\eta(t_2)$, $\eta(t_3)$ under a projectivity $\varphi$ are given. For any $\xi(s)$, let its corresponding element under $\varphi$ be $\eta(t)$. Since a projectivity preserves cross-ratios,

$$\frac{s_3-s_1}{s_3-s_2}\div\frac{s-s_1}{s-s_2}=\frac{t_3-t_1}{t_3-t_2}\div\frac{t-t_1}{t-t_2}.$$

Solving for $t$ gives $t=\frac{As+B}{Cs+D}$, with $AD\neq BC$ lest $t$ be constant. Conversely, every transformation of this form preserves cross ratios. □

**Corollary 6.6.40.** *Let the elements of $X,Y$ be described, respectively, by single parameters $s,t\in\mathbb{F}\cup\{\infty\}$, and denote them by $\xi(s),\eta(t)$. Let a projectivity be described by $\varphi:\ \xi(s)\mapsto\eta(f(s))$, where $f(0)=0$ and $f(\infty)=\infty$. Then there is a constant $c$ distinct from $0$ and $\infty$ such that $f(s)=cs$ for every $s$. Consequently, $\frac{s_1-s_3}{s_2-s_3}=\frac{f(s_1)-f(s_3)}{f(s_2)-f(s_3)}$ for all $s_1,s_2,s_3$.*

**Proof.** In theorem 6.6.39, substitute $s=0,t=0$ and $s=\infty,t=\infty$. This gives $B=C=0$, so $f(s)=As/D$, with $AD\neq 0$. □

**Theorem 6.6.41.** *Let the elements of a set $X$ endowed with a cross ratio be described by a single parameter $t\in\mathbb{F}\cup\{\infty\}$, and denote them by $\xi(t)$. Then*

$$\operatorname{Prj}(X)=\left\{\varphi:\ \xi(t)\mapsto\xi\left(\frac{At+B}{Ct+D}\right)\ \middle|\ A,B,C,D\in\mathbb{F},\ AD\neq BC\right\}. \tag{6.3}$$

*In particular, the involutions of $X$ are*

$$\left\{\varphi:\ \xi(t)\mapsto\xi\left(-\frac{At+B}{Ct+A}\right)\ \middle|\ A,B,C\in\mathbb{F},\ A^2\neq BC\right\}. \tag{6.4}$$

**Proof.** Suppose three elements $\xi(t_1),\xi(t_2),\xi(t_3)$ and their corresponding elements $\xi(t_1'),\xi(t_2'),\xi(t_3')$ under a projectivity $\varphi$ are given. For any $\xi(t)$, let its corresponding element under $\varphi$ be $\xi(t')$. Preservation of cross ratio gives

$$\frac{t_3-t_1}{t_3-t_2}\div\frac{t-t_1}{t-t_2}=\frac{t_3'-t_1'}{t_3'-t_2'}\div\frac{t'-t_1'}{t'-t_2'}.$$

It follows that $t'=\frac{At+B}{Ct+D}$ with $AD\neq BC$, since otherwise $t'$ would be constant; conversely, every transformation of this form preserves cross-ratios.

Now put $f(t)=\frac{At+B}{Ct+D}$, so that $t'=f(t)$. An involution must satisfy $f(f(t))=t$ and $f\neq\mathrm{id}$. Direct calculation gives precisely the conditions on $A,B,C,D$ in (6.4); conversely, every $f(t)$ satisfying those conditions is an involution. □

Define

$$\mathrm{PGL}(2,\mathbb{F}):=\left\{\varphi:\ \mathbb{F}\cup\{\infty\}\to\mathbb{F}\cup\{\infty\},\quad t\mapsto\frac{At+B}{Ct+D}\ \middle|\ A,B,C,D\in\mathbb{F},\ AD\neq BC\right\}.$$

Under composition this is the two-dimensional *projective general linear group*; write $\mathrm{PGL}(2)$ when the number field is understood. A map of the form $t\mapsto\frac{At+B}{Ct+D}$ is a *fractional linear transformation*. Equation (6.3) gives the following correspondence. For every $\varphi:\ t\mapsto\frac{At+B}{Ct+D}$ in $\mathrm{PGL}(2)$, associate the projectivity $\varphi:\ \xi(t)\mapsto\xi\left(\frac{At+B}{Ct+D}\right)$ of $X$. This is a bijection that respects composition, and hence a group isomorphism between $\mathrm{PGL}(2)$ and $\operatorname{Prj}(X)$.

**Theorem 6.6.42.** *For every set $X$ endowed with a cross ratio,* $\operatorname{Prj}(X)\cong\mathrm{PGL}(2)$.

Every such $X$ is projectively isomorphic to a line by matching its parameter, and a projective line is the one-dimensional projective space $\mathbb{P}^1$. Hence the statement can also be written as follows.

**Theorem 6.6.43.** *Over the same number field,* $\mathrm{Prj}(\mathbb{P}^1) \cong \mathrm{PGL}(2)$.

More conventionally, the *projective general linear group* is defined by

$$\mathrm{PGL}(n,\mathbb{F}) := \mathrm{GL}(n,\mathbb{F})/\mathrm{Z}(\mathrm{GL}(n,\mathbb{F})),$$

where $\mathrm{GL}(n,\mathbb{F})$ is the $n$-dimensional general linear group and

$$\mathrm{Z}(\mathrm{GL}(n,\mathbb{F})) = \{r\mathbf{I}_n \mid r \in \mathbb{F}, r \neq 0\}.$$

Here $\mathbf{I}_n$ is the $n \times n$ identity matrix. Readers may skip this small-print passage for now and consult section A.3.2 for the algebraic notions used here.

Denote the two-dimensional projective general linear group defined by (6.3) by $\mathrm{PGL}^*(2)$. The two definitions look different, but we can show that $\mathrm{PGL}^*(2)$ and $\mathrm{PGL}(2)$ are isomorphic and hence essentially the same. The following map $\sigma$ gives the group-homomorphism relation $\mathrm{GL}(2) \sim \mathrm{PGL}^*(2)$:

$$\sigma : \mathrm{GL}(2) \longrightarrow \mathrm{PGL}^*(2), \qquad \begin{bmatrix} A & B \\ C & D \end{bmatrix} \mapsto \left(\varphi : t \mapsto \frac{At+B}{Ct+D}\right)$$

The image transformation $\varphi$ is the identity if and only if $B = C = 0$ and $A = D \neq 0$. Hence the kernel of $\sigma$ is $\ker\sigma = \{r\mathbf{I}_2 \mid r \in \mathbb{F}, r \neq 0\} = \mathrm{Z}(\mathrm{GL}(2))$. The fundamental homomorphism theorem therefore gives $\mathrm{PGL}^*(2) \cong \mathrm{GL}(2)/\mathrm{Z}(\mathrm{GL}(2)) = \mathrm{PGL}(2)$.

As a further example, adjoin a single *point at infinity* $\infty$ to the complex plane $\mathbb{C}$, approached along any direction. It may be viewed as a complex number of infinite modulus and indeterminate argument. The result is the *extended complex plane* $\hat{\mathbb{C}}$. Unlike the real affine plane, which has an entire line of points at infinity, it has just one.

**Example 6.6.44.** The points of the extended complex plane form the one-parameter set $\hat{\mathbb{C}}$. For four points $P_1, P_2, P_3, P_4$ corresponding to complex numbers $z_1, z_2, z_3, z_4$, respectively, define

$$(P_1P_2, P_3P_4) = \frac{z_3 - z_1}{z_3 - z_2} \div \frac{z_4 - z_1}{z_4 - z_2}.$$

This makes $\hat{\mathbb{C}}$ a set endowed with a cross ratio over $\mathbb{C}$, with the following properties.

(1) Write $P(z)$ when the point $P$ corresponds to the complex number $z$. The elements of $\mathrm{Prj}(\hat{\mathbb{C}})$ have the form

$$\varphi : \mathbb{C} \cup \{\infty\} \longrightarrow \mathbb{C} \cup \{\infty\}, \quad z \longmapsto \frac{az+b}{cz+d}, \qquad ad \neq bc.$$

They are the *Möbius transformations*; $\mathrm{Prj}(\hat{\mathbb{C}})$ is the *Möbius group* and $\mathrm{Prj}(\hat{\mathbb{C}}) \cong \mathrm{PGL}(2,\mathbb{C})$.

(2) Four points $P_1, P_2, P_3, P_4$ are collinear or concyclic if and only if $(P_1P_2, P_3P_4) \in \mathbb{R}$.

(3) Every composition of translations, rotations, direct similarities, and complex inversions is a Möbius transformation, and every Möbius transformation has such a decomposition.[38] Möbius transformations send collinear or concyclic points to collinear or concyclic points. A real inversion alone and a reflection in an axis are not Möbius transformations because each involves complex conjugation.[39]

The proofs are left as exercises.

**Example 6.6.45.** Given $\triangle ABC$, successively perform an axial-reflection inversion at $A$, then at the new vertex $B$, and then at the new vertex $C$. Prove that the composite of the three transformations is the identity.[40]

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[38] A *real inversion* centered at the origin in the extended complex plane has the form $z \mapsto \frac{k}{\bar{z}}$, where $k$ is a constant. The extra conjugation is the distinction here: a *complex inversion* means an ordinary inversion in the complex plane, usually centered at the origin, followed by complex conjugation. To distinguish the earlier inversions from these, we may call them real inversions.

[39] If real inversions and axis reflections are also allowed, the transformations have one of the two forms $\varphi : z \mapsto \frac{az+b}{cz+d}$ and $\varphi : z \mapsto \frac{a\bar{z}+b}{c\bar{z}+d}$, where $ad \neq bc$. Both are projective semilinear transformations; see chapter 10.

[40] Each transformation is taken at the corresponding vertex of the triangle produced by the preceding step; the three transformations are not fixed in advance.

**Solution.** Although neither a real inversion nor an axis reflection alone is Möbius, each applies complex conjugation, so their conjugations cancel in the composite called an axial-reflection inversion. Thus each of the three steps, and hence their composite, is Möbius. The composite fixes $A, B, C$. Since three corresponding points determine a one-dimensional projective transformation and $\mathrm{Prj}(\hat{\mathbb{C}}) \cong \mathrm{PGL}(2, \mathbb{C})$, it must be the identity. □

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## Exercises

**Exercise 6.66.**
(1) Prove the properties in example 6.6.44.
(2) Determine the form of an involution of $\hat{\mathbb{C}}$ and describe how it may be decomposed.

**Exercise 6.67.** Let $\alpha$ be a conic. Prove that $\varphi \in \mathrm{Prj}(\alpha)$ if and only if there is a fixed line $l$ such that, for all $P, Q \in \alpha$, $P\varphi(Q) \cap Q\varphi P \in l$.

**Exercise 6.68.** Let $\alpha$ be a conic and $\varphi \in \mathrm{Prj}(\alpha)$. Prove that there is a fixed conic or fixed point $\alpha'$ such that, for every $P \in \alpha$, $P\varphi P \in \mathbf{T}\alpha'$.

**Exercise 6.69.** Let $\alpha$ be a conic in a Cartesian plane and $P$ a fixed point on it. A variable line $l$ meets $\alpha$ at $A, B$, and the slopes of the lines $PA, PB$ are $k_1, k_2$. Prove that the line $AB$ passes through a fixed, possibly infinite, point if and only if constants $a, b, c$ exist such that $a(k_1 + k_2) + b(k_1 k_2) + c = 0$ identically.

**Exercise 6.70*.** A pencil of circles is also a one-parameter set. Endow it with a cross ratio and investigate its properties.

**Exercise 6.71.** Given $\triangle ABC$, let a variable line $l$ pass through a fixed point at infinity $X$. Let $n$ be the Gauss line of the complete quadrilateral $\{AB, BC, CA, l\}$.[41] Prove that $n$ envelops a parabola $\alpha$ inscribed in the medial triangle of $\triangle ABC$, that the point of $\alpha$ at infinity is $X$, and that the point of contact of $\alpha$ with $n$ is $l \cap n$.

# 6.7 Pencils of Conics

Pencils of conics generalize the pencils of circles introduced earlier.

## 6.7.1 Definition and Classification

In the complex projective plane, two conics have four intersections counted with multiplicity. For example, two disjoint circles have two finite imaginary intersections, whose join is their radical axis, and the two circular points, which are imaginary points at infinity. Since five points determine a conic, we are led to the following definition.

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[41] The midpoints of the three pairs of opposite vertices of a complete quadrilateral are collinear; their line is its Gauss line. It will be studied further later.

**Definition 6.7.1.** Given four possibly coincident points $A,B,C,D$ in a plane,[42] the set of all conics, including degenerate conics, through $A,B,C,D$ is a *pencil of conics*. It is denoted $\mathrm{pen}(A,B,C,D)$ or $\mathrm{pen}(ABCD)$. If the four points are the intersections of conics $\gamma_1,\gamma_2$, it may also be denoted $\mathrm{pen}(\gamma_1,\gamma_2)$.

The definition allows the four base points $A,B,C,D$ of $\mathscr{P}=\mathrm{pen}(ABCD)$ to coincide. Coincident points alone tell us only that the curves of $\mathscr{P}$ have a common point of tangency, without giving further information about that point. We must therefore also specify the directions in which the points approach one another. To do so, suppose $\mathscr{P}$ contains a nondegenerate conic $\alpha_0$, and suppose, for example, that $A=B$. Keep $A,C,D$ fixed on $\alpha_0$, choose $B'\in\alpha_0$, and take $\lim_{B'\to A}\mathrm{pen}(AB'CD)$ to obtain $\mathrm{pen}(ABCD)$.[43] This manner of approach is appropriate because $\alpha_0\in\mathscr{P}$ remains in $\mathrm{pen}(AB'CD)$ throughout the process. Choosing another nondegenerate conic in $\mathscr{P}$ as the reference does not change the result.

Thus the present definition indeed generalizes the earlier notion of a pencil of circles.

**Proposition 6.7.2.** *Every pencil of circles in the sense of definition 4.6.1 is a pencil of conics in the present sense.*

**Remark.** The term pencil of circles here includes a family of concentric circles. Strictly speaking, degenerate members should also be included in definition 4.6.1; see the discussion below. Nevertheless, we shall generally refer to a pencil of circles directly as a pencil of conics.

**Proof.** Let $I,J$ be the circular points. Every circle passes through $I,J$. Choose two circles $\omega_1,\omega_2$ in the pencil. In the definition in definition 4.6.1, their radical axis passes through the two intersections in $\omega_1\cap\omega_2$, whether real or imaginary. For all the circles to have a common radical axis, they must all pass through the two intersections in $\omega_1\cap\omega_2$. Together with $I,J$, these are the four base points of a pencil of conics. Parabolic pencils and families of concentric circles follow as limiting cases. □

The number and pattern of coincident base points give five types.

(1) If $A,B,C,D$ are distinct, $\mathscr{P}$ is a *nondegenerate pencil of conics*. It contains the three degenerate conics $AB\cup CD$, $AC\cup BD$, and $AD\cup BC$. This is the most common case; figure 6.28 is an example. Elliptic and hyperbolic pencils of circles are also examples: their four base points are the two possibly imaginary points generating the circle pencil and the two circular points.

(2) If exactly two base points coincide, say $A=B$, then $\mathscr{P}$ is a *single-contact pencil of conics*. All its members are tangent at the double base point $A(B)$. Choosing $\alpha_0\in\mathscr{P}$ and letting $B$ approach $A$ along $\alpha_0$ is equivalent, at this order of contact, to letting $B$ approach along the common tangent $a=\mathrm{t}_{\alpha_0}(A)$. It therefore suffices to specify that $B$ approaches $A$ along a given line $a$. Taking $B\to A$ in the three degenerate conics in case 1 gives the two degenerate members of this pencil: $a\cup CD$ and $AC\cup AD$. Figure 6.29 gives an example, and a parabolic pencil of circles is of this type.

(3) If $A=B$ and $C=D$, with $A\neq C$, then $\mathscr{P}$ is a *double-contact pencil of conics*. As in case 2, it suffices to specify the directions $a,c$ of $B\to A$ and $D\to C$, where both $a$ and $c$ are lines. This makes $\mathscr{P}$ a limiting case of case 1. Its members have common points of tangency $A(B),C(D)$, with common tangents $a,c$, respectively.

    Taking the limit $B\to A$, $C\to D$ in case 1 gives the two degenerate members $a\cup c$ and $AC\cup AC$. Here $AC\cup AC$ denotes a pair of coincident lines.[44] Figure 6.30 is an example. The parabolas with Cartesian equations $y=ax^2$, with parameter $a$, form such a pencil with double base points at the origin and at the point at infinity of the $y$-axis. A family of concentric circles gives another, whose double base points are the circular points.

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[42] When points coincide, their manner of approach must also be specified; this is made precise below.

[43] If more than two points coincide, use the analogous construction. For example, if $A,B,C$ coincide, keep $A,D$ fixed, choose $B',C'$ on $\alpha_0$, and let $B',C'\to A$.

[44] The symbol $\cup$ here denotes union with multiplicity. It is not quite set-theoretic union, which collapses repeated elements; the intended meaning is clear from context.

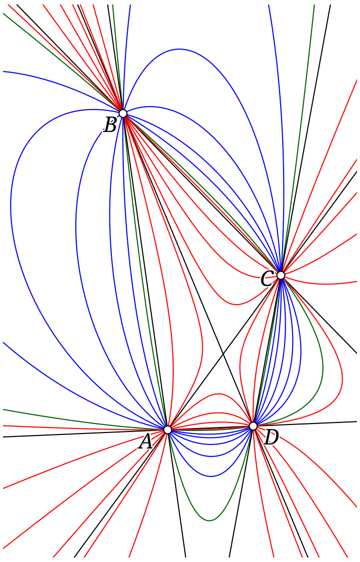


FIGURE 6.28

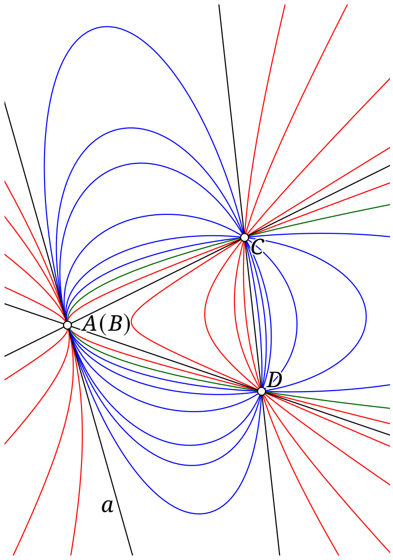


FIGURE 6.29

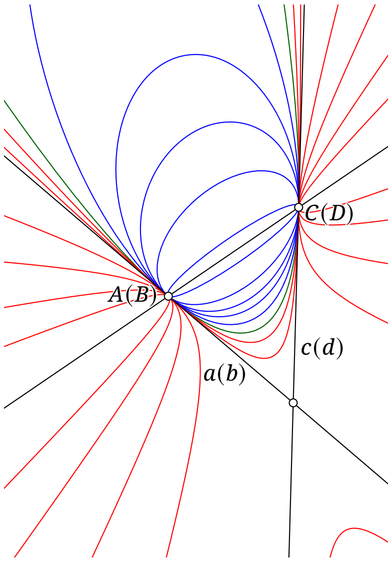


FIGURE 6.30

(4) If exactly three base points coincide, say $A = B = C \neq D$, then $\mathscr{P}$ is an *osculating pencil of conics*. Its members have triple contact at $A(B)(C)$. The approach of $B, C$ cannot be specified merely along the common tangent: that would keep $A, B, C$ collinear, unlike approach along $\alpha_0$.

Since three points determine a circle, there is a circle through $A, B, C$. As $B, C \to A$ along $\alpha_0 \in \mathscr{P}$, the circle $\odot(ABC)$ tends to a fixed circle $\omega$, the *circle of curvature* of $\alpha_0$ at $A$. The other curves in $\mathscr{P}$ have the same circle of curvature $\omega$ at $A$. Thus it suffices to specify $\omega$ and take $B, C \to A$ along $\omega$. All members of $\mathscr{P}$ share the tangent $a$ at $A$ and osculate $\omega$ there.

Taking $B, C \to A$ in case 1 shows that the sole degenerate member of $\mathscr{P}$ is $a \cup AD$. Figure 6.31 is an example.[45]

(5) If all four base points coincide, $\mathscr{P}$ is a *hyperosculating pencil of conics*. As in case 4, there are still a line $a$ and a circle $\omega$ such that every curve in $\mathscr{P}$ is tangent to $a$ at $A$ and osculates $\omega$ at $A(B)(C)(D)$. However, we can no longer use $\omega$ alone, as in case 4, to specify how $B, C, D$ approach $A$: when a curve $\alpha_0 \in \mathscr{P}$ is used as the reference, $A, B, C, D$ need not be concyclic. In this case, the manner of approach must be specified using a curve $\alpha_0$ of $\mathscr{P}$.

[45] Two curves with contact of multiplicity three are said to *osculate*.

The only degenerate member is $a \cup a$, a pair of coincident lines. Figure 6.32 is an example. All parabolas with Cartesian equations $y = x^2 + a$, where $a$ is the parameter, form a hyperosculating pencil whose common point of tangency is the point at infinity of the $y$-axis.[46]

The last four types are collectively called *degenerate pencils of conics*.

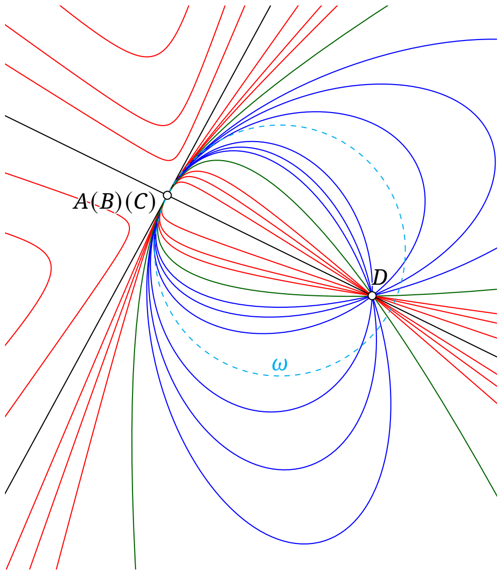


FIGURE 6.31

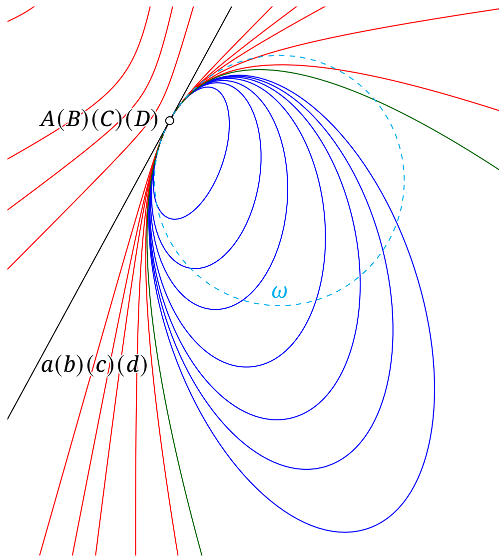


FIGURE 6.32

The dual construction is as follows.[47]

**Definition 6.7.3.** Given four possibly coincident lines $a, b, c, d$, the set of all conics tangent to $a, b, c, d$ is a *dual pencil of conics*, denoted $\mathrm{pen}(a, b, c, d)$ or $\mathrm{pen}(abcd)$. If these are the common tangents of conics $\gamma_1, \gamma_2$, the notation $\mathrm{pen}_{\mathrm{T}}(\gamma_1, \gamma_2)$ uses a subscript to distinguish the dual pencil.

**Remark.** A dual pencil of conics is generally not a pencil of conics in the sense defined earlier, because its conics often have no common points. More precisely, regarded as curves tangent to the given lines, its members are curves of class two. Thus an ordinary pencil may also be called a *pencil of curves of order two*, and a dual pencil a *pencil of curves of class two*. Their nondegenerate members are all ordinary conics. An ordinary pencil has members degenerated into pairs of lines; dually, a dual pencil has members degenerated into pairs of points. Except when the degenerate members matter, both families are often called *pencils of conics*, the intended meaning being clear from context.

Coincident lines in a dual pencil are interpreted by limits in the same way. Let $\mathcal{Q} = \mathrm{pen}(abcd)$ contain a nondegenerate conic $\alpha_0$, and suppose $a = b$. Then $\alpha_0$ is tangent to $a, c, d$. Hold $a, c, d$ fixed and let a tangent $b'$ to $\alpha_0$ tend to $b$. Then $\lim_{b' \to b} \mathrm{pen}(ab'cd) = \mathrm{pen}(abcd)$. The members of $\mathcal{Q}$ now share a common point, namely the point of contact of $a(b)$ with all the curves.

We can also classify dual pencils. Many of their properties follow by duality from those of ordinary pencils, so we omit detailed explanations.

(1) If $a, b, c, d$ are distinct, $\mathcal{Q}$ is a *nondegenerate dual pencil of conics*. Its degenerate members of class two are $(a \cap b) \cup (c \cap d)$, $(a \cap c) \cup (b \cap d)$, and $(a \cap d) \cup (b \cap c)$. See figure 6.33.

(2) If exactly two lines coincide, say $a = b$, choose a point $A \in a$ and let $b \to a$ through $A$. The result is a *single-tangent dual pencil of conics*. Every member passes through $A$ and has double contact there. Its degenerate members of class two are $A \cup (c \cap d)$ and $(a \cap c) \cup (a \cap d)$. Figure 6.34 gives an example.

[46] Two curves with contact of multiplicity four are said to *hyperosculate*.

[47] Two conics in $\mathbb{CP}^2$ have four possibly coincident intersections; dually, they also have four possibly coincident common tangents.

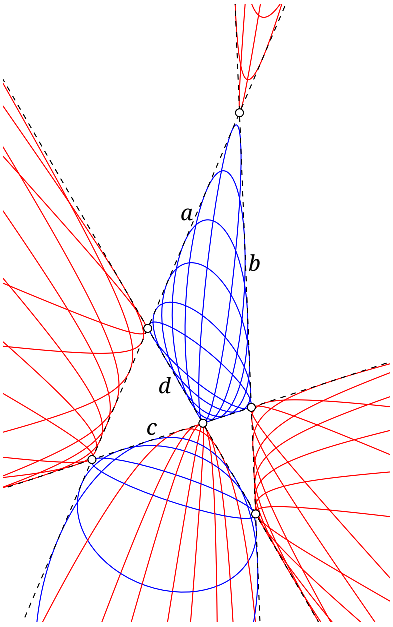


FIGURE 6.33

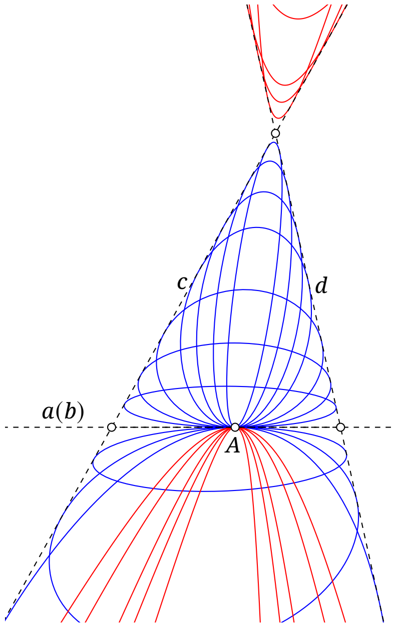


FIGURE 6.34

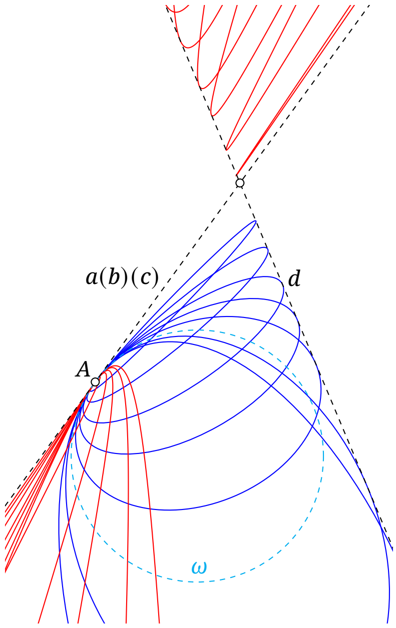


FIGURE 6.35

(3) If $a=b\neq c=d$, choose $A\in a$ and $C\in c$, let $b\to a$ while always passing through $A$, and let $d\to c$ while always passing through $C$. Taking this limit gives a *double-tangent dual pencil of conics*. Every member passes through $A, C$ and has double contact at both. Apart from their degenerate members, this family is essentially the same as a double-contact pencil; see also figure 6.30. Its degenerate members of class two are $A\cup C$ and the double point $(a\cap c)\cup(a\cap c)$.

(4) If $a=b=c\neq d$, the result is an *osculating dual pencil of conics*. As in the discussion of the intersection of coincident tangents in case 2, there is a point $A$ on $a(b)(c)$ through which every member of $\mathscr{Q}$ passes, and $A$ is their common point of contact of multiplicity three. The curves therefore have properties at $A$ analogous to those of an osculating pencil at its point of osculation: there is a circle $\omega$ through $A$ that every member of $\mathscr{Q}$ osculates at $A$. To specify how $b, c$ approach $a$, it suffices to specify $\omega$ and require $b, c$ to remain tangent to $\omega$ while tending to $a$. The unique degenerate member of class two is $A\cup(a\cap d)$. See figure 6.35.

(5) If $a = b = c = d$, the result is a *hyperosculating dual pencil of conics*. As in case 4, there is a point $A$ on $a(b)(c)(d)$ through which every member of $\mathscr{Q}$ passes, and $A$ is their common point of contact of multiplicity four. There is also a circle $\omega$ through $A$ that all the curves of $\mathscr{Q}$ osculate at $A$.

There is no simpler way to specify how $b,c,d$ approach $a$: one must choose a curve $\alpha_0 \in \mathscr{Q}$ and require $b,c,d$ to remain tangent to $\alpha_0$ as they approach $a$.

As with double-tangent dual pencils, hyperosculating dual pencils and hyperosculating ordinary pencils are essentially the same, apart from their degenerate members; see also figure 6.32. The unique degenerate member of class two in the dual pencil is $A \cup A$, a pair of coincident points.

The last four types are collectively called *degenerate dual pencils of conics*.

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## Exercises

**Exercise 6.72.** Prove that polarity with respect to any nondegenerate member of a double-tangent or double-contact pencil sends each nondegenerate member of that pencil to another nondegenerate member of the same pencil.

**Exercise 6.73.** Can an osculating dual pencil contain two distinct circles? Justify your answer. What about a hyperosculating dual pencil?

**Exercise 6.74.** Prove that:
(1) a family of concentric circles belongs to one double-contact pencil;
(2) all parabolas with Cartesian equations $y = ax^2$, with parameter $a$, belong to one double-contact pencil;
(3) all parabolas with Cartesian equations $y = x^2 + a$, with parameter $a$, belong to one hyperosculating pencil;
(4) conics with one common focus and the corresponding common directrix belong to one double-contact pencil;
(5) conics with the same two foci belong to one dual pencil of conics.

**Exercise 6.75.** In Cartesian coordinates, find the equations of the conics in each pencil:
(1) $A(0,0)$, $D(0,1)$, $B = C = A$, with $B,C \to A$ along the circle $\odot(D,AD)$; find $\text{pen}(ABCD)$;
(2) $a\colon\ y = 0$, $c\colon\ y = 1 + x$, $d\colon\ y = 1 - x$, and $b = a$, with $b \to a$ through the origin; find $\text{pen}(abcd)$;
(3) $a\colon\ y = 0$, $d\colon\ x = 0$, and $b = c = a$, with $b,c \to a$ while remaining tangent to $x^2 + (y-1)^2 = 1$; find $\text{pen}(abcd)$.

### 6.7.2 Basic Properties of Pencils of Conics

We discuss some elementary properties of pencils of conics, beginning with a property of dual pencils.

**Theorem 6.7.4.** *If four lines $a,b,c,d$ form a quadrilateral that is not a parallelogram, then the locus of the centers of all conics in the dual pencil* $\text{pen}(abcd)$ *is the Gauss line of the quadrilateral formed by $a,b,c,d$.*

**Proof.** As in figure 6.36a, let a conic $\alpha$ be tangent to all four sides of a quadrilateral $ABCD$, and let $O$ be its center. Reflect $ABCD$ in $O$ to obtain $A'B'C'D'$. By symmetry, the lines $A'B', B'C', C'D', D'A'$ are also tangent to $\alpha$. Put $U = BC \cap B'C'$ and $V = AD \cap A'D'$. Symmetry gives $BC \parallel B'C'$ and $AD \parallel A'D'$, so $U, V \in l_\infty$. Let $M,N$ be the midpoints of $AC,BD$, respectively. Then $MN$ is the Gauss line.

Apply Brianchon's theorem to the hexagon $ABUC'D'V$ circumscribed about $\alpha$. Its principal diagonals $AC', BD', UV$ are concurrent. Since $UV$ is the line at infinity, $AC' \parallel BD'$.

Since $O$ is the midpoint of $CC'$, the segment $OM$ is a midline of $\triangle CAC'$, so $OM \parallel AC'$. Similarly, $ON$ is a midline of $\triangle DBD'$, so $ON \parallel BD'$. Together with $AC' \parallel BD'$, this shows that $M, N, O$ are collinear, as required.

The parabolic case is understood by a limit. Conversely, given a point on the Gauss line, reversing the construction and applying the converse of Brianchon's theorem shows that it can serve as the center of a conic. □

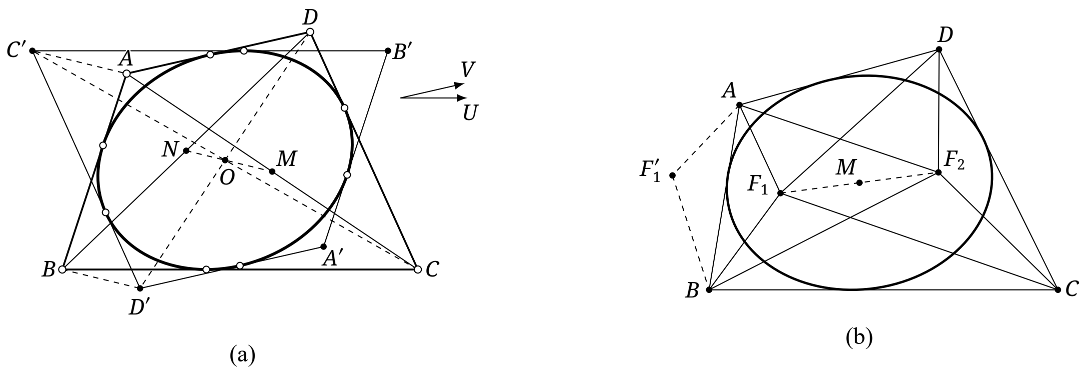


FIGURE 6.36

The proposition also has a more elementary proof. The following argument comes from [**6**].

**Alternative proof.** First consider central conics; the parabolic case follows by a limit. Let $F_1, F_2$ be the foci of a conic inscribed in the quadrilateral, and let $M$ be their midpoint, hence the center of the conic. Reorder the four vertices if necessary so that $ABCD$ is an ordinary quadrilateral, and consider the convex case shown in figure 6.36b; the other cases are similar.

Reflect $F_1$ in the line $AB$ to $F_1'$. Then

$$[\triangle F_1AB] + [\triangle F_2AB] = S_{F_1'AF_2B} = \frac{1}{2}(AF_1' \cdot AF_2 \sin\angle F_1'AF_2 + BF_1' \cdot BF_2 \sin\angle F_1'BF_2).$$

By theorem 4.4.20, $F_1, F_2$ are isogonally conjugate with respect to the quadrilateral $ABCD$. Therefore

$$\angle F_1'AF_2 = \angle BAF_2 + \angle BAF_1' = \angle F_1AB + \angle F_2AB = \angle BAD,$$

and similarly $\angle F_1'BF_2 = \angle ABC$. Hence

$$\begin{aligned}[\triangle F_1AB] + [\triangle F_2AB] &= \frac{1}{2}(AF_1 \cdot AF_2 \sin\angle BAD + BF_1 \cdot BF_2 \sin\angle ABC)\\ &= \frac{1}{2}\sum_{X=A,B} XF_1 \cdot XF_2 \sin\angle X.\end{aligned}$$

An analogous calculation gives an expression for $[\triangle F_1CD] + [\triangle F_2CD]$. Thus

$$\begin{aligned}\frac{1}{2}\sum_{X=A,B,C,D} XF_1 \cdot XF_2 \sin\angle X &= [\triangle F_1AB] + [\triangle F_1CD] + [\triangle F_2AB] + [\triangle F_2CD]\\ &= 2([\triangle BMA] + [\triangle CMD]),\end{aligned}$$

where the last equality follows from exercise 0.9. The same left-hand side also equals $2([\triangle BMC] + [\triangle AMD])$. Thus $[\triangle BMA] + [\triangle CMD] = [\triangle BMC] + [\triangle AMD]$, and Léon Anne's theorem, theorem 4.8.8, proves that $M$ lies on the Gauss line. □

**Corollary 6.7.5** (Monge's theorem)**.** *If a quadrilateral has an incircle, its center lies on the Gauss line of the quadrilateral.*

**Proof.** This is the circle case of the preceding theorem. □

**Corollary 6.7.6.** *Let $l$ be a line distinct from the three sides of the diagonal triangle of a complete quadrilateral $\{a,b,c,d\}$. As $\alpha$ varies over* $\mathrm{pen}(abcd)$*, the pole $\mathfrak{p}_\alpha^{-1}(l)$ lies on a fixed line.*

**Proof.** Apply a projectivity sending $l$ to $l_\infty$. Since $l$ is distinct from the sides of the diagonal triangle, the transformed lines $a,b,c,d$ do not form a parallelogram. The pole of $l_\infty$ with respect to a conic is its center, so theorem 6.7.4 applies. □

**Corollary 6.7.7.** *Let $ABCD$ be a complete quadrangle and let $P$ be a point other than the three vertices of its diagonal triangle. As $\alpha$ varies over* $\mathrm{pen}(ABCD)$*, the polar $\mathfrak{p}_\alpha(P)$ passes through a fixed point.*

**Proof.** This is the dual of corollary 6.7.6. □

**Alternative proof.** Send two of $A,B,C,D$ to the circular points; the result becomes proposition 4.6.12. □

The following theorem of Hesse is an application of corollary 6.7.7.

**Theorem 6.7.8** (Hesse's theorem)**.** *For a conic, if two of the three pairs of opposite vertices of a complete quadrilateral are conjugate, then the third pair is conjugate as well.*

**Proof.** Let $X,X'$ and $Y,Y'$ be conjugate pairs with respect to a conic $\alpha$, and put $Z = XY \cap X'Y'$ and $Z' = X'Y \cap XY'$. We prove that $Z,Z'$ are conjugate. Let $XX' \cap \alpha = A,B$ and $YY' \cap \alpha = C,D$, and let $\mathscr{W} = \mathrm{pen}(ABCD)$. By theorem 6.1.11, $(A,B;X,X') = (C,D;Y,Y') = -1$. Thus $X,X'$ and $Y,Y'$ are conjugate with respect to every member of $\mathscr{W}$.

Choose $\beta \in \mathscr{W}$ so that $\mathfrak{p}_\beta(X) = X'Y'$. The polarity principle gives $X \in \mathfrak{p}_\beta(Y')$, and conjugacy of $Y,Y'$ gives $Y \in \mathfrak{p}_\beta(Y')$. Hence $\mathfrak{p}_\beta(Y') = XY$. By the polarity principle, the polar of $Z$ contains $X,Y'$, so $\mathfrak{p}_\beta(Z) = XY'$. Similarly, choose $\gamma \in \mathscr{W}$ with $\mathfrak{p}_\gamma(Z) = X'Y$.

By corollary 6.7.7, there is a fixed point $Z''$ conjugate to $Z$ for every member of $\mathscr{W}$. The preceding two choices force $Z'' = X'Y \cap XY' = Z'$. Since $\alpha \in \mathscr{W}$, the points $Z,Z'$ are conjugate with respect to $\alpha$. □

We next turn from the centers of a dual pencil to those of an ordinary pencil.

**Theorem 6.7.9.** *Let $ABCD$ be a complete quadrangle. The locus of the centers of all conics in* $\mathrm{pen}(ABCD)$ *is a conic through the midpoints of the six sides of $ABCD$ and its three diagonal points.* This conic is the *nine-point conic* of the complete quadrangle $ABCD$, also called its *nine-point quadratic curve*.

**Proof.** Let $K,L,M,N$ be the midpoints of $AB,BC,CD,DA$. Each can be the center of a conic through $A,B,C,D$.[48]

Let $\alpha \in \mathrm{pen}(ABCD)$ have center $O$, and let the poles of $OK,OL,OM,ON$ with respect to $\alpha$ be $K',L',M',N'$. By theorem 6.1.6, $O(KL,MN) = (K'L',M'N')$. The pole of the diameter $OK$ is the point at infinity in its *conjugate direction*.[49] Since $K$ is the midpoint of $AB$, theorem 6.4.8 gives $K' = \infty_{AB}$. The same argument applies to $L',M',N'$. Therefore $O(KL,MN)$ is the cross ratio of the

[48] For example, by symmetry, the points $A,B,C,D$ together with the reflections of $C,D$ in $K$ lie on a conic.

[49] Here the conjugate direction means the direction of the corresponding conjugate diameter.

points at infinity of $AB, BC, CD, DA$, independent of $\alpha$. By theorem 6.2.11, the locus of $O$ is a conic $\beta$.

Repeating the argument that $K, L, M, N \in \beta$ for the other side midpoints shows that all six side midpoints of $ABCD$ lie on $\beta$. In the limiting sense, the center of a conic degenerated into two lines is their intersection. The three degenerate members $AB \cup CD$, $AC \cup BD$, and $AD \cup BC$ therefore show that the three diagonal points also lie on $\beta$. ☐

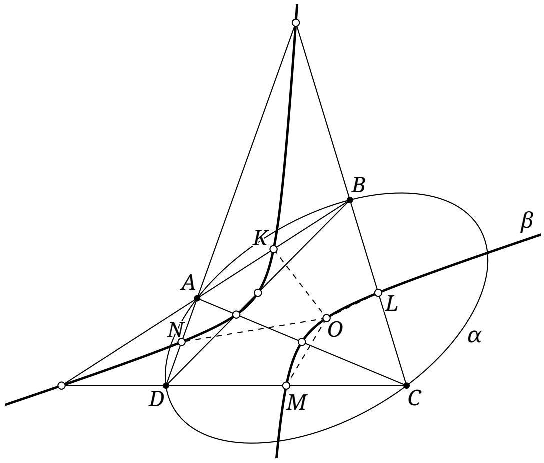


FIGURE 6.37

**Corollary 6.7.10.** *Given a complete quadrangle ABCD and a line l, the poles of l with respect to the conics in* pen$(ABCD)$ *trace a conic through the three diagonal points of ABCD.*

**Proof.** Send $l$ to $l_\infty$ by a projectivity. Its poles become the centers of the conics, and theorem 6.7.9 applies. ☐

**Corollary 6.7.11.** *Given a complete quadrilateral* $\{a, b, c, d\}$ *and a point P, the polars of P with respect to the conics in* pen$(abcd)$ *envelop a conic tangent to the three sides of the diagonal triangle of* $\{a, b, c, d\}$.

**Proof.** This is the dual of corollary 6.7.10. ☐

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## Exercises

**Exercise 6.76.** Prove that the centroid of a complete quadrangle $ABCD$ is the center of its nine-point conic.

**Exercise 6.77.** Prove that the nine-point conic of a complete quadrangle is a rectangular hyperbola if and only if the quadrangle is cyclic.

**Exercise 6.78.** Let $T$ be a point inside a convex quadrilateral $ABCD$ such that $\text{dist}(T, AB) = \text{dist}(T, CD)$ and $\text{dist}(T, BC) = \text{dist}(T, AD)$. Prove that $T$ lies on the Gauss line of $ABCD$ if and only if its vertices lie on a circle, its sidelines are tangent to a circle, or it is a proper trapezoid or a parallelogram.

**Exercise 6.79.** Let $A,B$ lie inside $\angle POQ$. Points $X,Y$ lie on the rays $OP,OQ$, respectively, and are such that the exterior angles of $\angle AXB$ and $\angle AYB$ are bisected by $OP$ and $OQ$, respectively. Let $C,Z$ be the midpoints of $AB,XY$.

(1) If $\angle O$ is a right angle, prove that $C,Z,O$ are collinear.
(2) If $\angle O$ is not a right angle, prove that $O \in CZ$ if and only if the polygonal paths $AXB$ and $AYB$ have equal length.

**Exercise 6.80.** Fix a triangle. If two of the three pairs of opposite vertices of a complete quadrilateral are isogonally conjugate (respectively, isotomically conjugate), prove that the remaining pair is likewise isogonally conjugate (respectively, isotomically conjugate).

*Hint.* Try using pencils of conics to represent isogonally conjugate points and isotomically conjugate points.

**Exercise 6.81.** Let $\mathscr{P}$ be a pencil of conics, and let $P$ and $l$ be a variable point and a variable line in the plane. The polars of $P$ with respect to the curves in $\mathscr{P}$ form a pencil with vertex $Q$, and the poles of $l$ with respect to those curves trace a conic $\gamma$. Fix four members $\alpha_1,\alpha_2,\alpha_3,\alpha_4$ of $\mathscr{P}$.

(1) Prove that the cross ratio of the four lines $\mathfrak{p}_{\alpha_1}(P)$, $\mathfrak{p}_{\alpha_2}(P)$, $\mathfrak{p}_{\alpha_3}(P)$, $\mathfrak{p}_{\alpha_4}(P)$ in the pencil at $Q$ is a constant $k_1$ independent of $P$.
(2) Prove that the cross ratio of the four poles $\mathfrak{p}_{\alpha_1}^{-1}(l)$, $\mathfrak{p}_{\alpha_2}^{-1}(l)$, $\mathfrak{p}_{\alpha_3}^{-1}(l)$, $\mathfrak{p}_{\alpha_4}^{-1}(l)$ in the quadratic point range on $\gamma$ is a constant $k_2$ independent of $l$.
(3) Prove that $k_1 = k_2$.
(4) State the dual assertion.

### 6.7.3 Desargues' Involution Theorem

Desargues' involution theorem links pencils of conics with involutions.

**Theorem 6.7.12** (Desargues' involution theorem)**.** *Given four points $A,B,C,D$ and a line $l$, the pencil $\mathscr{P} = \mathrm{pen}(ABCD)$ completely determines an involution on $l$: the two possibly coincident intersections of $l$ with each conic in $\mathscr{P}$ form a corresponding pair.*

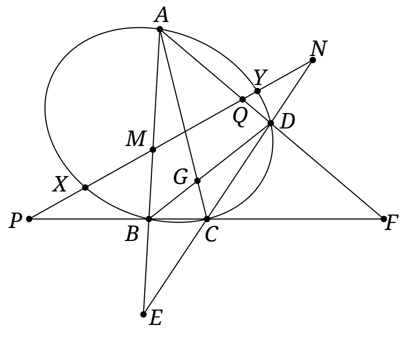


FIGURE 6.38

**Proof.** As in figure 6.38, let $l$ meet $AB,CD,BC,AD$ at $M,N,P,Q$, respectively, and let it meet a conic $\alpha \in \mathscr{P}$ at $X,Y$. The pencil contains the degenerate conics $AB \cup CD$ and $BC \cup DA$, so $M,N$ and $P,Q$ are two corresponding pairs. Two such pairs determine an involution. It remains to show $(P,N;X,Y) = (Q,M;Y,X)$. But

$$(P,N,X,Y) \doublebarwedge C(B,D,X,Y) \doublebarwedge \alpha(B,D,X,Y) \doublebarwedge A(B,D,X,Y) \doublebarwedge (M,Q,X,Y),$$

and therefore $(P,N;X,Y) = (M,Q;X,Y) = (Q,M;Y,X)$. $\square$

**Remark.** The version of Desargues' involution theorem in section 3.5.2 is a weakening obtained by considering only the three degenerate members of the pencil. The intersections of the three pairs of opposite sides of the complete quadrangle with $l$ form three corresponding pairs of the involution: $AB \cap l, CD \cap l$; $AC \cap l, BD \cap l$; and $AD \cap l, BC \cap l$. All three pairs belong to the same involution.

Conversely, if $X, Y$ correspond in this involution, some member of $\mathscr{P}$ passes through both. Indeed, the conic $\mathfrak{C}(ABCDX)$ belongs to $\mathscr{P}$, and its other intersection with $l$ is the mate of $X$. This is the second sense in which the pencil "completely determines" the involution.

**Corollary 6.7.13.** *At most two conics in a pencil are tangent to a given line.*

**Proof.** By Desargues' involution theorem, the pencil induces an involution on the line. A point where a conic of the pencil is tangent to this line is an invariant point of the involution, and an involution has at most two such points. □

The dual statement is the following.

**Theorem 6.7.14** (Plücker's theorem; dual Desargues involution theorem). *Given four lines $a,b,c,d$ and a point $P$, the dual pencil $\mathscr{Q} = \operatorname{pen}(abcd)$ completely determines an involution on the pencil of lines through $P$: the two possibly coincident tangents from $P$ to each conic in $\mathscr{Q}$ form a corresponding pair.*

**Remark.** The three degenerate members of class two in $\mathscr{Q}$ show in particular that the joins of $P$ to the three pairs of opposite vertices of the complete quadrilateral $\{a,b,c,d\}$ form three corresponding pairs of this involution. Conversely, if $m,n$ correspond, some member of $\mathscr{Q}$ is tangent to both.

We give several applications of Desargues' theorem. First, Desargues' involution theorem proves the four-conics theorem.

**Theorem 6.7.15** (Four-conics theorem). *As in figure 6.39, three conics have twelve pairwise intersections. Suppose two points can be chosen from the four intersections of each pair so that the six chosen points lie on one conic. Then the three lines joining the two remaining intersections of each pair are concurrent.*

*More explicitly, let $\alpha_1, \alpha_2, \alpha_3$ be three conics in $\mathbb{CP}^2$, with $\alpha_2 \cap \alpha_3 = P_1, Q_1, P_1', Q_1'$, $\alpha_3 \cap \alpha_1 = P_2, Q_2, P_2', Q_2'$, and $\alpha_1 \cap \alpha_2 = P_3, Q_3, P_3', Q_3'$. If $P_1, Q_1, P_2, Q_2, P_3, Q_3$ lie on a conic $\alpha_0$, then $P_1'Q_1', P_2'Q_2', P_3'Q_3'$ are concurrent.*

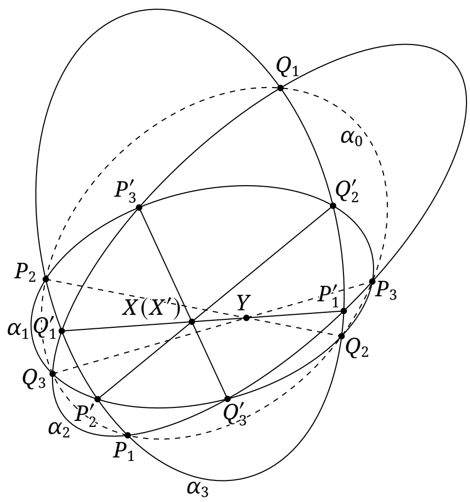


FIGURE 6.39

**Proof.** Put $X = P_2'Q_2' \cap P_1'Q_1'$ and $X' = P_3'Q_3' \cap P_1'Q_1'$. It suffices to show $X = X'$. Apply the three-conics theorem to $\alpha_0, \alpha_2, \alpha_3$; the lines $P_1'Q_1'$, $P_2Q_2$, and $P_3Q_3$ concur at a point $Y$.

Consider the conics $\alpha_1, \alpha_3$ and the degenerate conic $P_2'Q_2' \cup P_2Q_2$ in the pencil $\operatorname{pen}(P_2Q_2P_2'Q_2')$, together with the line $P_1'Q_1'$. Desargues' involution theorem shows that the pairs $M,N$, $Q_1',P_1'$, and $X,Y$ belong to one involution, where $M,N = \alpha_1 \cap P_1'Q_1'$. Repeating the argument with the analogous pencil $\operatorname{pen}(P_3Q_3P_3'Q_3')$ and the line $P_1'Q_1'$ gives the pairs $M,N$, $Q_1',P_1'$, and $X',Y$ in another involution. Since two corresponding pairs determine an involution and these two involutions already have two corresponding pairs in common, $X' = X$. $\square$

The four-conics theorem also has a converse, whose proof is left to the reader.

**Proposition 6.7.16** (Converse of the four-conics theorem). *Let $\alpha_1, \alpha_2, \alpha_3$ be three conics, and denote their pairwise intersections by $\alpha_2 \cap \alpha_3 = P_1,Q_1,P_1',Q_1'$, $\alpha_3 \cap \alpha_1 = P_2,Q_2,P_2',Q_2'$, and $\alpha_1 \cap \alpha_2 = P_3,Q_3,P_3',Q_3'$. Suppose the four triples of lines $P_1Q_1,P_2Q_2,P_3'Q_3'$; $P_1Q_1,P_3Q_3,P_2'Q_2'$; $P_2Q_2,P_3Q_3,P_1'Q_1'$; and $P_1'Q_1',P_2'Q_2',P_3'Q_3'$ are concurrent at $X_1,X_2,X_3,X_4$, respectively. Then $P_1,Q_1,P_2,Q_2,P_3,Q_3$ lie on a conic.*

We next give two applications of the dual form.

**Example 6.7.17.** Let the four sidelines of a quadrilateral $ABCD$ be tangent to a circle $\odot O$, and let $X$ be the orthogonal projection of $O$ on $BD$. Prove that the lines $AX$ and $CX$ are symmetric with respect to $BD$.

**Solution.** Draw the two tangents $u,v$ from $X$ to $\odot O$; they may be imaginary. Apply Plücker's theorem at $X$ to the complete quadrilateral formed by $AB,BC,CD,DA$. The pairs $(XA,XC)$, $(XB,XD)$, and $(u,v)$ belong to one involution $g$ on the pencil at $X$.

On the same pencil, let $f$ be reflection in $OX$. It preserves cross-ratios and its square is the identity, so it is an involution. Moreover, it takes $XB$ to $XD$ and $u$ to $v$. Since two corresponding pairs determine an involution, $f = g$. Now $B,X,D$ are collinear, and $f$ plainly leaves $XO$ invariant; hence $XO$ and $BD$ are the invariant lines of this involution. By corollary 3.5.23, $X(A,C;B,O) = -1$. Since $XB \perp XO$, proposition 3.2.17 shows that $XB$ and $XO$ are the two angle bisectors determined by $AX$ and $CX$. Thus $AX$ and $CX$ are symmetric with respect to $BD$. $\square$

**Example 6.7.18.** Let a complete quadrangle be inscribed in a central conic, and suppose two of its diagonal lines pass respectively through the two real foci. Prove that its diagonal triangle is right-angled, with right-angle vertex at the intersection of those two diagonal lines.

**Solution.** Let $I,J$ be the circular points. Draw the tangents $l_1,l_2$ from $I$ and $l_3,l_4$ from $J$ to the central conic $\alpha$. By the projective definition of a focus, the four pairwise intersections of these tangents other than $I,J$ form two pairs of foci. Let the real foci, the two intended in the statement, be $F_1 = l_1 \cap l_3$ and $F_2 = l_2 \cap l_4$. Let the diagonal lines $AB,CD$ of the inscribed complete quadrangle $ABCD$ pass through $F_1,F_2$, respectively, and meet at $E$; let $F,G$ be the other two diagonal points.

The harmonic property of a complete quadrangle gives $E(FG,F_1F_2) = -1$. Put $FG \cap \alpha = M,N$. Brocard's theorem gives $F \in \mathfrak{p}(G)$, and the harmonic property of poles and polars yields $E(FG,MN) = -1$. Define an involution $f$ on the pencil at $E$ by $(EF,EG;l,f(l)) = -1$ for every line $l$ through $E$. Its invariant lines are $EF,EG$, while $(EA,EC)$ and $(EM,EN)$ are corresponding pairs.

Brocard's theorem also gives $\mathfrak{p}(E) = FG$, so $EM,EN$ are tangents to $\alpha$. Apply Plücker's theorem to $\alpha$ and the complete quadrilateral $l_1l_2l_3l_4$. The pairs $(EF_1,EF_2)$, $(EI,EJ)$, and $(EM,EN)$ belong to an involution $g$ on the pencil at $E$. Since two pairs determine an involution, $f = g$. Hence $E(FG,IJ) = -1$, and the projective definition of perpendicularity gives $EF \perp EG$. $\square$

The final example is Poncelet's celebrated closure theorem; the proof here follows [**6**].

**Theorem 6.7.19** (Poncelet's closure theorem)**.** *Let a pencil $\mathscr{F}$ contain conics $\alpha_0, \alpha_1, \dots, \alpha_n$. From $A_0 \in \alpha_0$, draw a tangent to $\alpha_1$ and let its other intersection with $\alpha_0$ be $A_1$; from $A_1$, draw a tangent to $\alpha_2$ and let the other intersection be $A_2$; continue in this way. If the construction closes, $A_n = A_0$, for one starting point, then the analogous construction from every $B_0 \in \alpha_0$ also closes, with $B_n = B_0$.*[50]

We first need a lemma.

**Lemma 6.7.20.** *Let the conics $\alpha_0, \alpha_1, \alpha_2, \dots$ belong to one pencil $\mathscr{P}$, as in figure 6.40.*

*(1) Suppose $A, B, C \in \alpha_0$, the line $AB$ is tangent to $\alpha_1$ at $K$, and $AC$ is tangent to $\alpha_2$ at $L$. Then there is $D \in \alpha_0$ such that:*
   - *(a) $\alpha_1$ is tangent to $CD$ at $KL \cap CD$, and $\alpha_2$ is tangent to $BD$ at $KL \cap BD$;*
   - *(b) some $\alpha_j \in \mathscr{P}$ is tangent to $AD$ at $AD \cap KL$ and to $BC$ at $BC \cap KL$.*

*(2) Suppose $A, B, C, D \in \alpha_0$, and $\alpha_1$ is tangent to $AB$ at $K$ and to $CD$ at $M$. Then some $\alpha_j \in \mathscr{P}$ is tangent to $AC$ at $KM \cap AC$ and to $BD$ at $BD \cap KM$.*

**Proof.** For part 1, let $\alpha_1 \cap KL = K, M$ and put $\mathscr{F} = \mathrm{pen}(\alpha_1, CM \cup AB)$. Desargues' involution theorem gives an involution on $AC$ from $\mathscr{P}$: the two possibly imaginary intersections of $\alpha_1$ with $AC$ form one pair, and the intersections $A, C$ of $\alpha_0$ with $AC$ form another. The pencil $\mathscr{F}$ also induces an involution on $AC$: the two intersections of $\alpha_1$ with $AC$ form one pair, and the intersections $A, C$ of $CM \cup AB$ with $AC$ form another. Since two corresponding pairs determine an involution, $\mathscr{F}, \mathscr{P}$ induce the same involution $f$ on $AC$.

Since $\alpha_2 \in \mathscr{P}$ is tangent to $AC$ at $L$, the point $L$ is invariant under $f$. Therefore a member $\gamma \in \mathscr{F}$ has a double intersection with $AC$ at $L$. Three of the four intersections of $\alpha_1$ and $AB \cup CM$ are $K, K, M$: the point of tangency $K$ is a double intersection. The members of $\mathscr{F}$ are tangent to $AB$ at $K$. Thus $\gamma$ passes through $M$, has a double intersection with $AB$ at $K$, and is determined by $K, K, L, L, M$. It must be the double line $KL \cup KL$.

Its intersection $M$ with $CM$ is also double. Hence $\mathscr{F} = \mathrm{pen}(KKMM)$ is a double-contact pencil, and $\alpha_1$ is tangent to $CM$ at $M$. Let $C, D = CM \cap \alpha_0$. Applying the same argument to $B, C, D$ shows that $\alpha_2$ is tangent to $BD$ at $N = BD \cap KL$, proving part 1(a).

Let $J = AD \cap NL$ and choose $\alpha_j \in \mathscr{P}$ through $J$. The same argument shows that it is tangent to $AD$ at $J$ and likewise tangent to $BC$ at $BC \cap KL$. This proves part 1(b); the reader is asked to complete the details. Part 2 has a proof similar to part 1 and is also left as an exercise. □

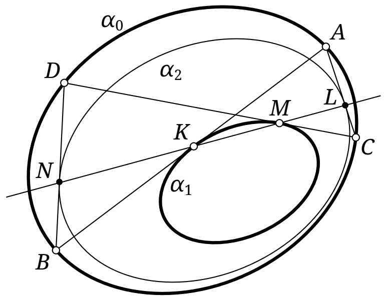


FIGURE 6.40

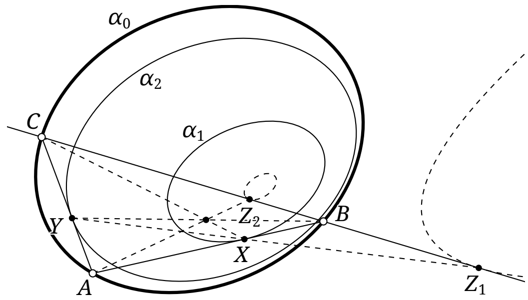


FIGURE 6.41

**Claim 6.7.21.** *Let $\alpha_0, \alpha_1, \alpha_2, \dots$ belong to a pencil $\mathscr{F}$. Suppose $A, B, C \in \alpha_0$, $AB$ is tangent to $\alpha_1$ at $X$, and $AC$ is tangent to $\alpha_2$ at $Y$. There are at most two points $Z \in BC$ at which some $\alpha_j \in \mathscr{F}$ is tangent to $BC$. One has $Z, X, Y$ collinear; for the other, $CX, BY, AZ$ are concurrent.*[51]

[50] As in the circle-pencil version in section 4.7, the direction of the tangent chosen at each step is part of the construction.

[51] The two points may coincide, hence "at most two."

**Proof.** There are at most two such points by corollary 6.7.13. Denote the one collinear with $X,Y$ by $Z_1$; it exists by lemma 6.7.20. Let $Z_2$ be defined by concurrence of $CX,BY,AZ_2$. We show that some $\alpha_j$ is tangent to $BC$ at $Z_2$.

For the complete quadrangle $BXYC$, theorem 3.5.11 gives $(BC,Z_1Z_2)=-1$. The pencil $\mathscr{F}$ induces an involution on $BC$, in which $B,C$ form a pair and $Z_1$ is invariant. If $Z_2'$ is the mate of $Z_2$, then $-1=(BC,Z_1Z_2)=(CB,Z_1Z_2')$. But proposition 3.2.11 also gives $(CB,Z_1Z_2)=(BC,Z_1Z_2)^{-1}=-1$. Consequently, $(CB,Z_1Z_2')=(CB,Z_1Z_2)$, so $Z_2'=Z_2$. Thus $Z_2$ is invariant, and Desargues' involution theorem gives the desired tangent. □

**Proof of theorem 6.7.19.** We use induction, beginning with $n=3$. Let $A_0A_1,A_1A_2,A_2A_0$ be tangent to $\alpha_1,\alpha_2,\alpha_3$ at $X_1,X_2,X_3$, respectively; these three points are not collinear. From an arbitrary $B_0\in\alpha_0$, draw $B_0B_1$ tangent to $\alpha_1$ at $Y_1$, choosing the same tangent direction as for $A_0A_1$, and let it meet $\alpha_0$ again at $B_1$.

By lemma 6.7.20(2), some $\alpha'\in\mathscr{F}$ is tangent to $A_0B_0,A_1B_1$ at their intersections $Z_0,Z_1$ with $X_1Y_1$. By lemma 6.7.20(1), there is $B_2\in\alpha_0$ such that $\alpha_2$ is tangent to $B_1B_2$ and $\alpha'$ to $B_2A_2$, at the intersections $Y_2,Z_2$ of these lines with $Z_1X_2$. Another application of lemma 6.7.20(2) gives $\alpha''\in\mathscr{F}$ tangent to $A_2A_0,B_2B_0$ at their intersections $K,Y_3$ with $Z_2Z_0$.

Apply claim 6.7.21 to $A_1,A_0,A_2$ and the line $A_0A_2$. If $K\neq X_3$, then, since $X_1,X_2,X_3$ are not collinear, the existence of $\alpha''$ tangent to $A_0A_2$ at $K$ would force $X_1,X_2,K$ to be collinear. But Desargues' theorem applied to $\triangle A_0A_1A_2$ and $\triangle Z_0Z_1Z_2$ shows that they are not collinear, since the joins of corresponding vertices are not concurrent. Thus $K=X_3$ and $\alpha''=\alpha_3$. The case $n=3$ is proved.

Assume the result for $n=k-1$; we prove it for $n=k$. As in the construction of $\alpha''$ when $n=3$, choose $\beta\in\mathscr{F}$ tangent to $A_0A_2$ and $B_0B_2$. The $(k-1)$-gon $A_0A_2\cdots A_{n-1}$ has successive sides tangent to $\beta,\alpha_3,\dots,\alpha_n$. The induction hypothesis completes the proof. □

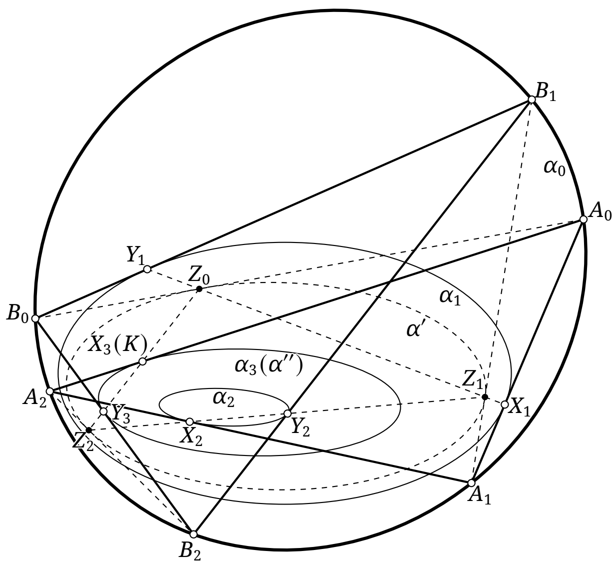


FIGURE 6.42

The lemma and Poncelet's theorem are conic analogues of lemma 4.7.4 and theorem 4.7.3. One could therefore send the pencil of conics to a pencil of circles by a projectivity, interpreting multiple base points by limits. The argument above instead illustrates the use of Desargues' involution theorem.

Poncelet's theorem says that, once one such polygon $A_0\cdots A_n$ exists, it may be "rotated" while its sides remain tangent to the corresponding conics. This invariant hints at a group structure on a conic.

## Exercises

**Exercise 6.82.**

(1) Complete the proof of lemma 6.7.20.
(2) Prove the converse of the four-conics theorem, and state the duals of the four-conics theorem and its converse.
(3) State the duals of lemma 6.7.20 and claim 6.7.21 and of Poncelet's closure theorem.

**Exercise 6.83.** Given $A,B,C,D$ and a line $l$ through none of them, Desargues' involution theorem induces an involution on $l$ from $\mathrm{pen}(ABCD)$. For a point $P \in l$, construct its mate using a straightedge alone.

**Exercise 6.84.** Given a pencil of circles $\mathscr{W}$ and a line $l$, Desargues' involution theorem gives the involution induced by $\mathscr{W}$ on $l$. If $Q$ is the intersection of $l$ with the radical axis of $\mathscr{W}$, prove that the involution is an inversion centered at $Q$.

**Exercise 6.85.** Three circles are pairwise disjoint. Their pairwise transverse common tangents form a hexagon. Prove that this hexagon is circumscribed about a conic.

**Exercise 6.86.** Let an ellipse $\alpha$ have center $O$, semimajor axis $a$, semiminor axis $b$, and semifocal distance $c$. Draw the two lines $l_1,l_2$ parallel to the major axis and at distance $\frac{ab}{c}$ from it. A line $l$ not through $O$ meets $\alpha$ at $A,B$ and meets $l_1,l_2$ at $C,D$, respectively. Prove that $\angle AOC = \angle BOD$.

**Exercise 6.87.** Let $\triangle ABC$ and $\triangle DEF$ both be inscribed in a conic $\Gamma$ and circumscribed about a conic $\gamma$. Suppose $EF,BC$ touch $\gamma$ at $L,K$, and let $AL,DK$ meet $\Gamma$ again at $N,M$. Prove that $AM,EF,BC,DN$ are concurrent.

**Exercise 6.88.** Let an ellipse $\alpha$ have foci $F_1,F_2$, and let $\odot O = \odot(F_1F_2)$. A variable point $P \in \odot O$ determines $PF_1 \cap \alpha = A,B$ and $PF_2 \cap \alpha = C,D$, with $A,C$ on one side of the major axis and $B,D$ on the other. If $E = AD \cap BC$, prove that $E$ moves on two fixed lines.

**Exercise 6.89.** For $\triangle ABC$, let the two direct common tangents of its circumcircle $\mathscr{C}$ and its $A$-excircle $\mathscr{I}_a$ meet the line $BC$ at $P,Q$. Prove that $\angle PAB = \angle CAQ$.

**Exercise 6.90.** Let $A$ vary on a conic $\alpha_0$. The two tangents from $A$ to a conic $\alpha_1$ meet $\alpha_0$ again at $B,C$. If the line $BC$ passes through a fixed point, what condition must $\alpha_0,\alpha_1$ satisfy?

**Exercise 6.91.** A variable triangle $\triangle ABC$ has fixed incircle and fixed $A$-pseudocircumcircle. Determine the locus of $A$.

**Exercise 6.92.** Let $A,B$ be fixed and $P$ vary on a conic $\alpha$. The lines $PA,PB$ meet $\alpha$ again at $X,Y$. Prove that $XY$ envelops a conic $\beta$, that $\beta$ is tangent to $\alpha$ at two points, and that those two points are collinear with $A,B$.

**Exercise 6.93.** For $\triangle ABC$, let $M \in \mathscr{C}$. The two tangents from $M$ to the incircle $\mathscr{I}$ meet the line $BC$ at $X_1,X_2$. The $A$-pseudoincircle is tangent to $\mathscr{C}$ at $T$. Prove that $X_1,X_2,M,T$ are concyclic.

**Exercise 6.94.** An inscribed conic $\alpha$ of $\triangle ABC$ touches $BC$ at $D$. From an arbitrary $X \in \odot(ABC)$ draw the two tangents to $\alpha$, meeting $BC$ at $U, V$. Prove that $\odot(XUV)$ passes through a fixed point $T \in \odot(ABC)$, and that $T$ is the Miquel point of the degenerate quadrilateral $ABDC$.[52]

**Exercise 6.95* (Ivory's theorem).** Let ellipses $\alpha_1, \alpha_2$ and hyperbolas $\beta_1, \beta_2$ have the same two foci. Let $\beta_1$ meet $\alpha_1, \alpha_2$ at $A, B$, and $\beta_2$ meet $\alpha_1, \alpha_2$ at $C, D$, respectively. Suppose that $A, B, C, D$ all lie on the same side of the line joining the two foci and also on the same side of the perpendicular bisector of the segment joining them. Prove that $AD = BC$.

[52]The Miquel point of this degenerate quadrilateral may be characterized by a limit. Besides $T \in \odot(ABC)$, one has $\measuredangle BDT = \measuredangle ACT$ and $\measuredangle CDT = \measuredangle ABT$. For the first equality, take $C'$ near $C$ and put $B' = C'D \cap AB$. The definition gives $T, D, B, B'$ concyclic, so $\measuredangle BDT = \measuredangle B'BT = \measuredangle ACT$; the last equality uses the cyclicity of $A, B, C, T$. The other relation is analogous.

## Chapter 7

# Triangles and Geometric Transformations

Building on the preceding material, this chapter develops the geometry of triangles associated with geometric transformations, including orthopoles, isoconjugation, trilinear polarity, and binary operations.

## 7.1 Orthopoles

### 7.1.1 Orthopoles and Their Basic Properties

This subsection studies orthopoles. For convenience, throughout the subsection, for $\triangle ABC$, let $O$ and $H$ denote its circumcenter and orthocenter, respectively, and write its medial triangle as $\triangle \equiv \triangle MNL$. Plainly, $O$ is the orthocenter of $\triangle LMN$, while $\mathscr{N}$ is the circumcircle of $\triangle MNL$.

Before introducing orthopoles, we first record the following lemma.

**Lemma 7.1.1.** *For $\triangle ABC$, let $D \in \mathscr{N}$, and let $D'$ be the reflection of $D$ in $MN$. Then $AD' \perp OD'$.*

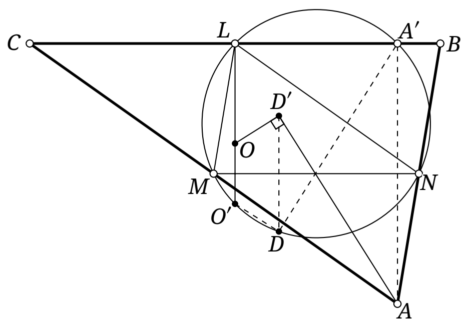


FIGURE 7.1

**Proof.** As shown in figure 7.1, let $O'$ and $A'$ be the reflections of $O$ and $A$, respectively, in $MN$. Then $A'$ is the orthogonal projection of $A$ on $BC$, so $A' \in \mathscr{N}$. Since $O$ is the orthocenter of $\triangle LMN$, lemma 0.5.21 gives $O' \in \mathscr{N}$. Clearly $O'L \perp MN$ and $A'L \parallel MN$, whence $O'L \perp A'L$. Thus $A'O'$ is a diameter of $\mathscr{N}$, and hence $A'D \perp O'D$. Reflection in $MN$ now gives $AD' \perp OD'$. □

**Theorem 7.1.2.** *Given $\triangle ABC$ and a line $l$, let $D,E,F$ be the orthogonal projections of $A,B,C$, respectively, on $l$. The perpendicular to $BC$ through $D$, the perpendicular to $CA$ through $E$, and the perpendicular to $AB$ through $F$ are concurrent at a point $K$.* This point is called the *orthopole* of $l$ with respect to $\triangle ABC$. Below we write it as $\mathfrak{o}_{\triangle ABC}(l)$, or simply $\mathfrak{o}(l)$ when there is no ambiguity.

**Proof.** As shown in figure 7.2, translate $l$ in the direction perpendicular to it until it becomes a line $l'$ through $O$. Denote the orthogonal projections of $A,B,C$ on $l'$ by $D',E',F'$, respectively.

We first prove the existence of the orthopole of a line through $O$, hence of the orthopole of $l'$. Put $K' = \mathscr{W}_{\triangle}^{-1}(l')$, and let $D'',E'',F''$ be the reflections of $K'$ in $MN,NL,LM$, respectively. By lemma 7.1.1, $D''A, E''B, F''C \perp l'$. By the method of coincidence, $K'$ is therefore $\mathfrak{o}(l')$.

The line $l$ is a translate of $l'$. Consequently, the three perpendiculars in the statement are parallel to $D'K', E'K', F'K'$, respectively. They are therefore concurrent, and their point of concurrence is obtained from $K'$ by the same translation. □

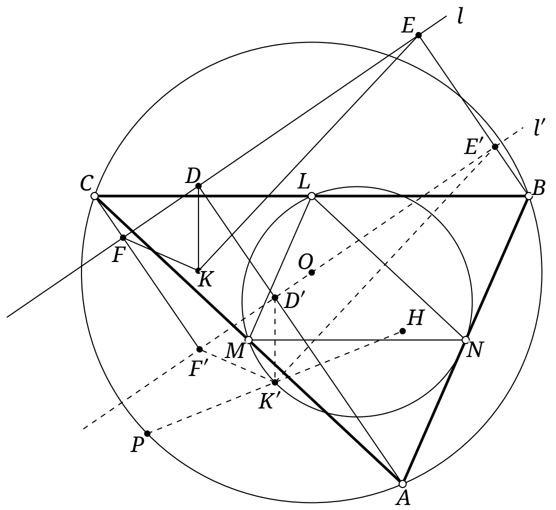


FIGURE 7.2

The proof immediately gives several properties of orthopoles.

**Theorem 7.1.3.** *For $\triangle ABC$, the orthopole of a line through the circumcenter lies on the nine-point circle and is the inverse Steiner point of that line with respect to the medial triangle.*

**Corollary 7.1.4.** *Let a parabola be inscribed in the medial triangle of a triangle. The orthopole, with respect to the original triangle, of the parabola's directrix is its focus.*

**Proof.** Combine the preceding theorem with proposition 4.2.13. □

**Theorem 7.1.5.** *For $\triangle ABC$, suppose that $l'$ is obtained by translating $l$ in the direction perpendicular to $l$. The same translation carries $\mathfrak{o}(l)$ to $\mathfrak{o}(l')$.*

**Proposition 7.1.6.** *With respect to a given triangle, the locus of the orthopoles of a family of parallel lines is part of a line perpendicular to them. This locus line is the Simson line of a point on the circumcircle of the triangle.*

**Proof.** By theorem 7.1.5, the orthopoles lie on a line perpendicular to the given parallel lines. It remains to prove the second assertion.

In the notation of figure 7.2, let $l$ be a line and put $K = \mathfrak{o}(l)$. Translate $l$ perpendicularly to itself until it passes through $O$, and denote the resulting line by $l'$. Then $K' := \mathfrak{o}(l') \in \mathscr{N}$. Extend $HK'$ to meet $\mathscr{C}$ at $P$. The homothety between $\mathscr{N}$ and $\mathscr{C}$ shows that $K'$ is the midpoint of $PH$. We shall prove that $KK' = \mathscr{S}(P)$, which completes the proof.

By lemma 0.5.20, the antipodal triangle $\triangle'$ of $\triangle ABC$ is homothetic to $\triangle$ about $H$ in the ratio $2:1$. Hence $\mathscr{S}_\triangle(K')$ and $l_P := \mathscr{S}_{\triangle'}(P)$ are homothetic about $H$ in the ratio $1:2$, so $l_P \parallel \mathscr{S}_\triangle(K')$. Since $l' = \mathscr{W}_\triangle(K') \parallel \mathscr{S}_\triangle(K')$, we also have $l' \parallel l_P$.

By theorem 4.2.5 and the definition of the antipodal triangle, $l_P \perp \mathscr{S}(P)$; hence $l' \perp \mathscr{S}(P)$. Since $KK' \perp l'$ and $K'$ is the midpoint of $PH$, it follows that $KK' = \mathscr{S}(P)$. [1] □

[1] By Steiner's theorem, the intersection of $\mathscr{S}(P)$ with $PH$ is the midpoint of $PH$, and $K'$ has this property.

**Proposition 7.1.7.** *The Steiner line of a point on the circumcircle of a triangle passes through the orthopole of the Simson line of that point.*

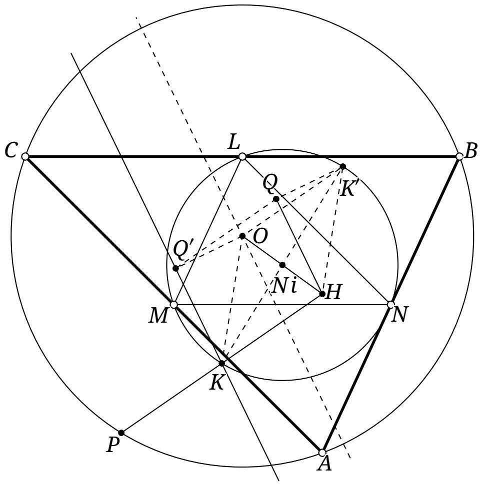


FIGURE 7.3

**Proof.** As shown in figure 7.3, let $P \in \mathscr{C}$, let $K$ be the midpoint of $PH$, and note that $K$ lies on both $l := \mathscr{S}(P)$ and $\mathscr{N}$. Put $Q = \mathfrak{o}(l)$. Since the Simson and Steiner lines are parallel, it suffices to prove that $QH \parallel l$.

Let $K'$ be the point of $\mathscr{N}$ antipodal to $K$. By lemma 1.5.3, $\mathscr{S}_\triangle(K) \perp \mathscr{S}_\triangle(K')$. Put $l' = \mathscr{W}_\triangle(K')$. Then $l' \parallel \mathscr{S}_\triangle(K')$, while proposition 4.2.10 gives $\mathscr{S}_\triangle(K) \perp \mathscr{S}(P)$. Thus $l' \parallel l$.

The line $l'$ passes through the orthocenter $O$ of $\triangle$. Hence theorem 7.1.3 gives $K' = \mathfrak{o}(l')$. Let $Q'$ be the projection of $O$ on $l$. By theorem 7.1.5, $OQ'$ is parallel and equal in length to $K'Q$, and therefore $OK'$ is parallel and equal in length to $QQ'$. Since $K, K'$ and $O, H$ are both symmetric about $Ni$, $OK'$ is parallel and equal in length to $KH$. Consequently $QQ'$ is parallel and equal in length to $KH$, and hence $Q'K$ is parallel and equal in length to $QH$. □

**Proposition 7.1.8.** *Let $H$ and $T$ be the orthocenters of two triangles $\triangle ABC$ and $\triangle PQR$ having a common circumcircle $\mathscr{C}$, and suppose that $\overrightarrow{\measuredangle} AOP + \overrightarrow{\measuredangle} BOQ + \overrightarrow{\measuredangle} COR = 0$. Let $l_A, l_B, l_C$ be the Simson lines of $A, B, C$ with respect to $\triangle PQR$, and let $l_P, l_Q, l_R$ be the Simson lines of $P, Q, R$ with respect to $\triangle ABC$. If $S$ is the midpoint of $HT$, then:*

*(1) the six lines $l_A, l_B, l_C, l_P, l_Q, l_R$ pass through $S$;*
*(2) the orthopoles of $PQ, QR, RP$ with respect to $\triangle ABC$ and those of $AB, BC, CA$ with respect to $\triangle PQR$ are all $S$.*

**Proof.** (1) By theorem 4.2.5, $l_P \parallel \mathscr{S}_{\triangle PQR}(P) = PT$. Steiner's theorem says that $l_P$ passes through the midpoint of $PH$, so it also passes through the midpoint $S$ of $HT$. The other five Simson lines pass through $S$ in the same way.

(2) Let $X$ be the projection of $A$ on $PQ$. Then $X \in l_A$, and part (1) gives $S \in l_A$; thus $l_A = XS$. By theorem 4.2.5, $l_A \parallel \mathscr{S}_{\triangle ABC}(A) = AH$, while $AH \perp BC$. Consequently $l_A \perp BC$. Similarly, if $Y, Z$ are the projections of $B, C$ on $PQ$, then $YS \perp AC$ and $ZS \perp AB$. The definition of the orthopole therefore gives $\mathfrak{o}_{\triangle ABC}(PQ) = S$. □

**Corollary 7.1.9.** *The orthopole of a line with respect to a triangle is the intersection of the Simson lines of the two points in which that line meets the circumcircle.*

**Proof.** In proposition 7.1.8, the point $S$ is the orthopole of $PQ$ with respect to $\triangle ABC$, and it is also $l_P \cap l_Q$. □

**Corollary 7.1.10.** *The locus of the orthopoles of the variable lines through a fixed point on a triangle's circumcircle is part of the Simson line of that point.*

---

## Exercises

**Exercise 7.1.** Given $\triangle ABC$ and two points $P, Q \in \mathscr{C}$, let $L, M, N$ be the reflections of $P$ in the three sidelines of the triangle. The lines $QL, QM, QN$ meet $BC, CA, AB$ at $X, Y, Z$, respectively; Seimiya's theorem gives that $X, Y, Z$ are collinear. Prove that the orthopole of the perpendicular bisector of $PQ$ lies on the line $\overline{XYZ}$.

**Exercise 7.2.** Given $\triangle ABC$ and a line $l$, let $\theta_a, \theta_b, \theta_c$ be the directed angles from the three sidelines of $\triangle ABC$ to $l$, and let $R$ be the radius of $\mathscr{C}$. Prove that $\operatorname{dist}(\mathfrak{o}(l), l) = 2R|\cos\theta_a \cos\theta_b \cos\theta_c|$.

**Exercise 7.3.** Any three of the four lines of a complete quadrilateral form a triangle, so there are four such triangles. Prove that the orthopoles of an arbitrary line with respect to these four triangles are collinear.

**Exercise 7.4.** Given $\triangle ABC$ and a line $l$, let $O$ be the circumcenter and let $P$ vary on $l$. Write $\triangle XYZ$ for the pedal triangle of $P$. Put $W = AO \cap l$, let $B', C'$ be the projections of $W$ on $AB, AC$, respectively, and put $E = B'C' \cap YZ$. Prove that $\mathfrak{o}(l) \in XE$.

**Exercise 7.5.** Given $\triangle ABC$ and a line $l$, reflect $l$ in the three sidelines of the medial triangle of $\triangle ABC$, and let the three reflected lines form a triangle $\triangle_l$. Prove that $\mathfrak{o}(l)$ lies on one of the incircles or excircles of $\triangle_l$.

**Exercise 7.6.** Given a circle $\omega$, a point $O \in \omega$, and a positive number $a$, let $l$ be a variable line through $O$ and let its second intersection with $\omega$ be $P$. On $l$, choose $Q$ so that $|QP| = a$. The locus of $Q$ is called a *limaçon of Pascal*, or simply a *limaçon*.

Now let $\triangle ABC$ and a point $P$ be given, and let $l$ vary through $P$. Prove that the locus of $\mathfrak{f}_l(\mathfrak{o}(l))$ is a limaçon.

### 7.1.2 Orthopoles and Conics*

We now study properties relating orthopoles and conics. Although the first theorem does not itself involve conics, our proof uses them.

**Theorem 7.1.11** (Lemoyne's theorem). *Given $\triangle ABC$ and a line $l$, let $O$ be the circumcenter and put $S = \mathfrak{o}(l)$. For every $P \in l$, the power of $S$ with respect to the pedal circle of $P$ is constant, and its value is $-2\operatorname{dist}_\pm(S, l) \cdot \operatorname{dist}_\pm(O, l)$.*

**Proof.** Let $Q$ be the center of the pedal circle $\omega$ of $P$, let $L$ be an intersection of $\omega$ with $l$, and draw through $L$ the line $t$ perpendicular to $l$, as in figure 7.4. By corollary 4.4.15, there is a conic $\gamma$, with foci $P$ and $\mathbf{g}P$, which is inscribed in $\triangle ABC$ and has $\omega$ as its outer auxiliary circle. Its center is $Q$, and theorem 1.3.5 says that $t$ is tangent to $\gamma$. By theorem 6.7.4, the Gauss line $n$ of the complete quadrilateral $\{t, BC, CA, AB\}$ passes through $Q$.

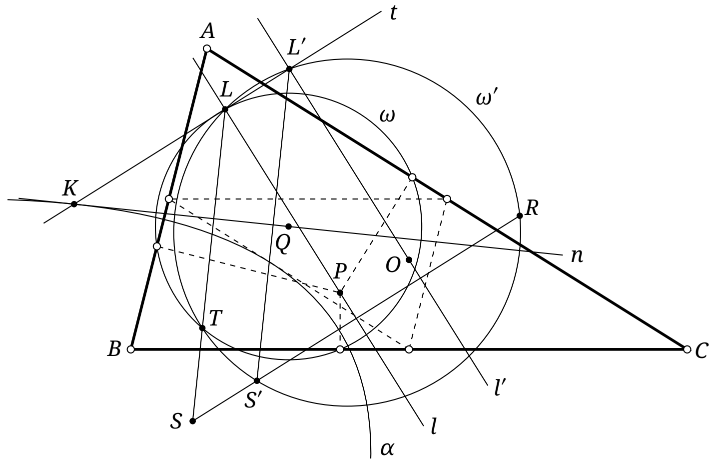


FIGURE 7.4

Put $K = t \cap n$. By exercise 6.71, the line $n$ envelops a parabola $\alpha$ inscribed in the medial triangle of $\triangle ABC$; its point of contact with $\alpha$ is $K$, and its axis is parallel to $t$, so its directrix is parallel to $l$. By theorem 4.2.12, the orthocenter $O$ of the medial triangle lies on that directrix. Thus the line $l'$ through $O$ parallel to $l$ is the directrix of $\alpha$, and corollary 7.1.4 gives that $S' := \mathfrak{o}(l')$ is the focus of $\alpha$.

Put $L' = l' \cap t$. By lemma 1.5.1, the tangent $n$ to $\alpha$ perpendicularly bisects $S'L'$. By theorem 7.1.5, $LL' \parallel SS'$ and $\overline{LL'} = \overline{SS'} = -\operatorname{dist}_{\pm}(O,l)$. Hence $LS \parallel L'S'$, so $LS \perp n$ and $T := \mathfrak{f}_n(L) \in \omega$. The quadrilateral $LTS'L'$ is therefore an isosceles trapezoid and has a circumcircle $\omega'$.

Let $R$ be the second intersection of $SS'$ with $\omega'$. The lines $t, l, l', SR$ bound a rectangle, while $L, L', R, S'$ are concyclic. It follows that $\overline{S'R} = 2\operatorname{dist}_{\pm}(S',l) + \overline{LL'} = 2\operatorname{dist}_{\pm}(S',l) + \overline{SS'}$, and consequently $\overline{SR} = \overline{SS'} + \overline{S'R} = 2(\operatorname{dist}_{\pm}(S',l) + \overline{SS'}) = 2\operatorname{dist}_{\pm}(S,l)$. The power-of-a-point theorem now gives

$$\operatorname{pow}_{\omega}(S) = \overline{ST} \cdot \overline{SL} = \overline{SR} \cdot \overline{SS'} = -2\operatorname{dist}_{\pm}(S,l) \cdot \operatorname{dist}_{\pm}(O,l).$$

□

**Corollary 7.1.12** (Griffiths's theorem)**.** *Given a triangle, the pedal circles of all points on a line through its circumcenter pass through the orthopole of that line.* This point is called the *Griffiths point* of the line.

**Proof.** By theorem 7.1.3, the orthopole of a line through the circumcenter lies on the nine-point circle. The pedal circle of the circumcenter is the nine-point circle, so the orthopole has zero power with respect to it. By theorem 7.1.11, the power of the orthopole with respect to the pedal circle of an arbitrary point of the line has the same constant value; it is therefore zero, and all these pedal circles pass through the orthopole. □

Theorems theorem 7.1.5 and corollary 7.1.10 give the loci of the orthopoles of the variable lines through, respectively, a point at infinity and a point on the circumcircle. Before treating the general case, we need a lemma.

**Lemma 7.1.13.** *Given $\triangle ABC$ and a line $l$, let $\theta_a, \theta_b, \theta_c$ be the directed angles from the three sidelines of the triangle to $l$. If $P$ varies on $l$ and $\triangle P_aP_bP_c$ is its pedal triangle, then there is a unique point $S$ for which $[\triangle SP_bP_c] : [\triangle SP_cP_a] : [\triangle SP_aP_b]$ is constant. This point is $S = \mathfrak{o}(l)$, and the constant ratio is $a\sec\theta_a : b\sec\theta_b : c\sec\theta_c$,*[2] *where $a, b, c$ are the side lengths of $\triangle ABC$.*

[2] The function sec, called the *secant*, is the reciprocal of the cosine.

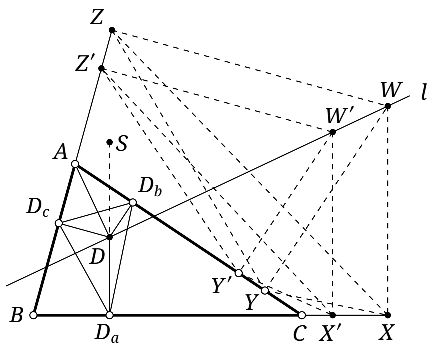


FIGURE 7.5

**Proof.** As shown in figure 7.5, choose any two points $W, W' \in l$, and denote their pedal triangles by $\triangle XYZ$ and $\triangle X'Y'Z'$. Write $[\triangle SYZ]:[\triangle SZX]:[\triangle SXY] = \lambda:\mu:\nu$. By theorem 0.1.12, $\lambda\overrightarrow{SX} + \mu\overrightarrow{SY} + \nu\overrightarrow{SZ} = \mathbf{0}$. If we also require $\lambda:\mu:\nu = [\triangle SY'Z']:[\triangle SZ'X']:[\triangle SX'Y']$, then $\lambda\overrightarrow{SX'} + \mu\overrightarrow{SY'} + \nu\overrightarrow{SZ'} = \mathbf{0}$. Subtraction gives

$$\mathbf{0} = \lambda\overrightarrow{XX'} + \mu\overrightarrow{YY'} + \nu\overrightarrow{ZZ'} = \lambda\frac{WW'\cos\theta_a}{a}\cdot\overrightarrow{BC} + \mu\frac{WW'\cos\theta_b}{b}\cdot\overrightarrow{CA} + \nu\frac{WW'\cos\theta_c}{c}\cdot\overrightarrow{AB}.$$

Thus the unique solution is $\lambda:\mu:\nu = a\sec\theta_a : b\sec\theta_b : c\sec\theta_c$. It remains to show that $S = \mathfrak{o}(l)$.

Let $D$ be the projection of $A$ on $l$, and let $\triangle D_aD_bD_c$ be the pedal triangle of $D$. Then $\measuredangle D_aDD_c = \measuredangle D_aBD_c$, $\measuredangle D_aDD_b = \measuredangle D_aCD_b$, $\measuredangle D_cDA = \measuredangle(AB,l) = \theta_c$, and $\measuredangle D_bDA = \measuredangle(AC,l) = \theta_b$. Hence

$$\begin{aligned}\frac{[\triangle DD_aD_c]}{[\triangle DD_aD_b]} &= \frac{\frac{1}{2}DD_c\cdot DD_a\sin\angle B}{\frac{1}{2}DD_b\cdot DD_a\sin\angle C} = \frac{DD_c}{DD_b}\cdot\frac{\sin\angle B}{\sin\angle C}\\ &= \frac{AD\cos\measuredangle ADD_c}{AD\cos\measuredangle ADD_b}\cdot\frac{b}{c} = \frac{\cos\theta_c}{\cos\theta_b}\cdot\frac{b}{c} = \frac{[\triangle SD_aD_c]}{[\triangle SD_aD_b]}.\end{aligned}$$

By lemma 0.1.13, we obtain $S \in DD_a$. Similarly, let $E, F$ be the projections of $B, C$ on $l$, and let $E_b, F_c$ be the projections of $E, F$ on $CA, AB$, respectively. Then $S \in EE_b$ and $S \in FF_c$. By the definition of the orthopole, $S = \mathfrak{o}(l)$. □

**Theorem 7.1.14.** *Given $\triangle ABC$ and a fixed finite point $P$ not on its circumcircle, the locus of the orthopole $Q$ of a variable line through $P$ is a circumellipse of the pedal triangle of $P$.* It is called the *orthopolar ellipse* of $P$.

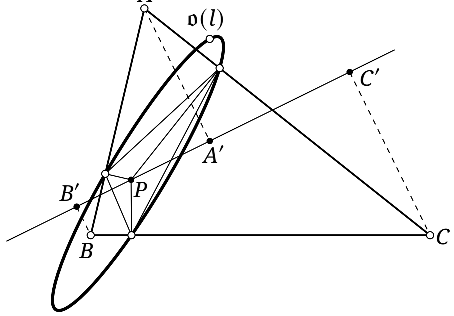


FIGURE 7.6

**Proof.** Let $\triangle P_aP_bP_c$ be the pedal triangle of $P$, let $l$ be a line through $P$, and let $\theta_a, \theta_b, \theta_c$ be the directed angles from the three sidelines of $\triangle ABC$ to $l$. If $A', B', C'$ are the projections of $A, B, C$ on $l$, respectively, then

$$0 = \overline{B'C'} + \overline{C'A'} + \overline{A'B'} = a\cos\theta_a + b\cos\theta_b + c\cos\theta_c. \qquad (\clubsuit)$$

For $Q = \mathfrak{o}(l)$, denote the directed areas by $\lambda = [\triangle P_b P_c Q]$, $\mu = [\triangle P_c P_a Q]$, and $\nu = [\triangle P_a P_b Q]$. By lemma 7.1.13,

$$\frac{1}{\lambda} : \frac{1}{\mu} : \frac{1}{\nu} = \frac{\cos\theta_a}{a} : \frac{\cos\theta_b}{b} : \frac{\cos\theta_c}{c}. \tag{$\diamondsuit$}$$

Combining (♣) and (◇) gives

$$\frac{a^2}{\lambda} + \frac{b^2}{\mu} + \frac{c^2}{\nu} = 0. \tag{$\spadesuit$}$$

It follows from example 6.2.10 that the locus of $Q$ is a circumconic $\alpha$ of $\triangle P_a P_b P_c$. Suppose that $Q$ could be a point at infinity. The area of $\triangle P_a P_b P_c$ would then be negligible relative to $\lambda, \mu, \nu$, and hence $\lambda + \mu + \nu = 0$. Together with (♠), a straightforward calculation shows that this system has no real solution for $\lambda, \mu, \nu$. Thus $\alpha$ has no point at infinity and is therefore an ellipse. □

We next examine several properties of the orthopolar ellipse.

**Theorem 7.1.15.** *Given $\triangle ABC$ and a point $P$, let $H$ be the orthocenter of the triangle and let $K$ be the midpoint of $HP$. The orthopolar ellipse $\gamma$ of $P$ is centered at $K$. Moreover, the orthopoles corresponding to any two perpendicular lines through $P$ are diametrically opposite points of $\gamma$.*[3]

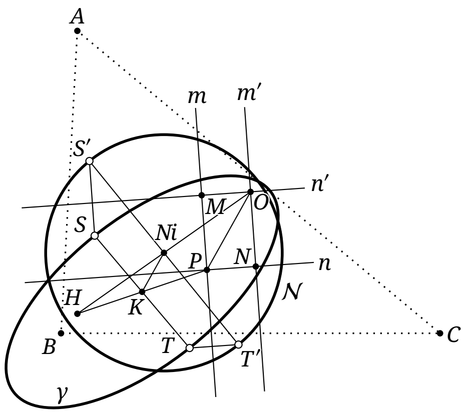


FIGURE 7.7

**Proof.** As in figure 7.7, let $O$ be the circumcenter. Let $m, n$ be two perpendicular lines through $P$, and let $m', n'$ be the lines through $O$ parallel to $m, n$, respectively. Denote the orthopoles of these four lines by $S, T, S', T'$. The four lines form the rectangle $MONP$, also shown in figure 7.7. By theorem 7.1.5, the segments $SS'$, $PM$, and $NO$ are parallel and equal in length; likewise, $TT'$, $PN$, and $MO$ are parallel and equal in length.

By theorem 7.1.3 and proposition 4.2.3, and because a Steiner line is parallel to the corresponding Simson line, $S'$ and $T'$ are diametrically opposite points of $\mathcal{N}$. Now redefine $K$ as the midpoint of $ST$. Since $Ni$ is the midpoint of $HO$, we have

$$\begin{aligned} 2\overrightarrow{PK} &= (\overrightarrow{PNi} + \overrightarrow{NiS'} - \overrightarrow{SS'} - \overrightarrow{KS}) + (\overrightarrow{PNi} + \overrightarrow{NiT'} - \overrightarrow{TT'} - \overrightarrow{KT}) \\ &= 2\overrightarrow{PNi} + (\overrightarrow{NiS'} + \overrightarrow{NiT'}) - (\overrightarrow{SS'} + \overrightarrow{TT'}) - (\overrightarrow{KS} + \overrightarrow{KT}) \\ &= 2\overrightarrow{PNi} + \mathbf{0} - (\overrightarrow{PN} + \overrightarrow{NO}) - \mathbf{0} = 2\overrightarrow{PNi} - \overrightarrow{PO} = 2\overrightarrow{PNi} + (\overrightarrow{NiO} + \overrightarrow{NiH}) - \overrightarrow{PO} \\ &= (\overrightarrow{PNi} + \overrightarrow{NiO}) + (\overrightarrow{PNi} + \overrightarrow{NiH}) - \overrightarrow{PO} = \overrightarrow{PO} + \overrightarrow{PH} - \overrightarrow{PO} = \overrightarrow{PH}. \end{aligned}$$

Thus $K$ is the midpoint of $HP$. Since $m$ and $n$ were arbitrary, $\gamma$ has infinitely many chords with midpoint $K$; hence $K$ is its center. □

[3] More generally, two points of a central conic that are symmetric about its center are also called *diametrically opposite*.

**Exercise 7.7.** Given $\triangle ABC$, determine a line such that the pedal circle of every point on it is tangent to $\mathscr{I}$.

**Exercise 7.8.** Given $\triangle ABC$ with circumcenter $O$, let $P$ be a point, let $\triangle DEF$ be its pedal triangle, and let $\gamma$ be its orthopolar ellipse. Put $OP \cap \mathscr{C} = Q,R$. Complete the following tasks.

(1) Let $B',C',E',F'$ be the projections of $B,C,E,F$, respectively, on the Simson line $\mathscr{S}(Q)$. Prove that $\frac{\overline{E'F'}}{\overline{B'C'}} = \frac{\overline{RP}}{\overline{QR}}$.
(2) Using part (1), prove that there is an affine transformation of the plane that maps $\triangle ABC$ to $\triangle DEF$ and $\mathscr{C}$ to $\gamma$.
(3) Using part (2), prove the following:
    (a) the two axes of $\gamma$ are parallel to $\mathscr{S}(R)$ and $\mathscr{S}(Q)$, respectively;
    (b) the lengths of the major and minor axes of $\gamma$ are $|r_{\mathscr{C}} \pm OP|$, respectively, where $r_{\mathscr{C}}$ is the radius of $\mathscr{C}$;
    (c) the inner and outer auxiliary circles of $\gamma$ are tangent to $\mathscr{N}$.

**Exercise 7.9*.** Given $\triangle ABC$, find a point $X$ whose orthopolar ellipse is an inellipse of $\triangle ABC$.

**Exercise 7.10*.** Given $\triangle ABC$, let $\gamma_0$ be its Simson deltoid, and let $\gamma(P)$ be the orthopolar ellipse of a point $P$. Prove that:

(1) $\mathfrak{d}(\mathbf{T}\mathscr{C}) = \gamma_0$;
(2) $\gamma_0$ and $\gamma(P)$ are always tangent.

# 7.2 Isoconjugation

In this section we study isoconjugation.

Many results in the sections on isoconjugation, trilinear polarity, and binary functions can be obtained directly by symbolic reasoning, without constructing a mental picture. Accordingly, we shall not illustrate many of them.[4]

## 7.2.1 Basic Definition of Isoconjugation

**Theorem 7.2.1.** *Let $\triangle ABC$ and fixed points $X,X' \notin AB \cup BC \cup CA$ be given. For every point $P \notin AB \cup BC \cup CA$, there is a unique point $P'$ satisfying all three conditions:*

*(1) $A(PX,BC) = A(P'X',CB)$;*
*(2) $B(PX,CA) = B(P'X',AC)$;*
*(3) $C(PX,AB) = C(P'X',BA)$.*

*This defines a map*

$$\mathbf{j}: \mathbb{P}^2 \setminus (AB \cup BC \cup CA) \longrightarrow \mathbb{P}^2 \setminus (AB \cup BC \cup CA), \quad P \longmapsto P'.$$

The map $\mathbf{j}$ is called an *isoconjugation* of $\triangle ABC$, and $P'$ is called the *isoconjugate* of $P$.

**Proof.** As in figure 7.8, draw through $B$ the line $l_B$ such that $B(PX,CA) = (l_B, BX'; BA, BC)$. Construct $l_C$ analogously and put $P' = l_B \cap l_C$. Conditions (2) and (3) hold by construction; we prove (1).

Consider the pencil of curves of class two $\mathscr{Q} = \mathrm{pen}(PX, PX', P'X, P'X')$. By Plücker's theorem it induces an involution $g$ in the pencil at $B$, with $g(BX) = BX'$ and $g(BP) = BP'$. Since $B(PX,CA) = B(P'X',AC)$, we also have $g(BC) = BA$.

[4] In fact, the author recommends trying to solve every proposition by reasoning from the known theorems without relying on a diagram.

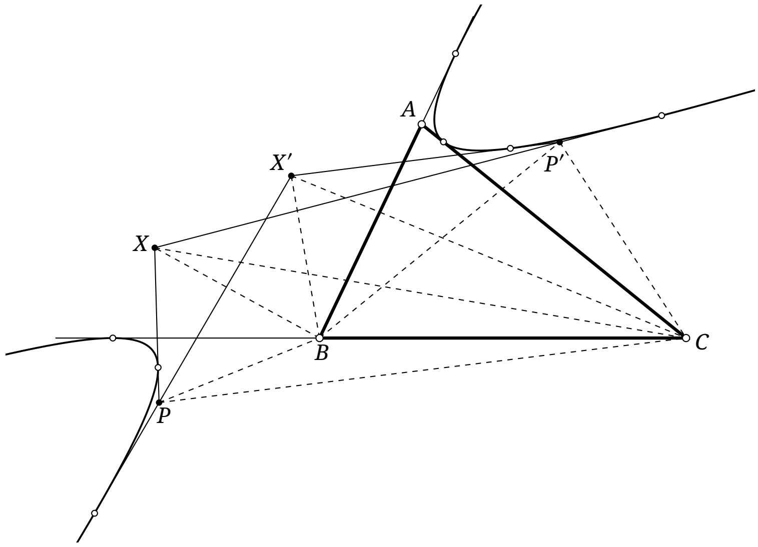


FIGURE 7.8

Plücker's theorem therefore yields a conic $\alpha$ tangent to all six lines $PX, PX', P'X, P'X', BA, BC$. The line $CA$ is tangent to $\alpha$ by the same argument. Applying Plücker's theorem once more shows that the three pairs $(AB,AC)$, $(AX,AX')$, and $(AP,AP')$ belong to one involution. Hence $A(PX,BC) = A(P'X',CB)$, as required. □

**Remark.** The proof shows that any two of the three conditions in theorem 7.2.1 imply the third.

For a point on $BC$, $CA$, or $AB$, other than $A,B,C$, one may take a limit and define its isoconjugate to be $A$, $B$, or $C$, respectively. A limit does not, however, assign an isoconjugate to any vertex: approaching a vertex in different directions makes the isoconjugate approach different points of the opposite side. Once the direction of approach is prescribed, the corresponding point of the opposite side is determined.

The definition immediately gives the following result.

**Proposition 7.2.2.** *If* $\mathbf{j}$ *is an isoconjugation of* $\triangle ABC$*, then* $\mathbf{j}^2 = \mathrm{id}$*. Thus, whenever* $P' = \mathbf{j}P$*, one also has* $P = \mathbf{j}P'$. The points $P$ and $P'$ are said to be *isoconjugate* under $\mathbf{j}$ with respect to $\triangle ABC$.

**Theorem 7.2.3.** *Let* $\mathbf{j}$ *be an isoconjugation of* $\triangle ABC$*, and suppose that* $\mathbf{j}X = X$. Such a point $X$ is called a *fixed point* of $\mathbf{j}$. *Then the three vertices of the anticevian triangle of $X$ are also fixed points of* $\mathbf{j}$.

**Proof.** Let $\triangle X^aX^bX^c$ be the anticevian triangle of $X$. By theorem 4.3.4, $B(X^aX,AC) = -1 = B(X^aX,CA)$, and similarly $C(X^aX,BA) = C(X^aX,AB)$. Hence $\mathbf{j}X^a = X^a$; the other two vertices follow cyclically. □

**Corollary 7.2.4.** *An isoconjugation has four fixed points; the anticevian triangle of any one of them has the other three as its vertices.*

**Proof.** By the following theorem 7.2.6, a contradiction argument readily shows that an isoconjugation has at most four fixed points, and the result follows. □

Isoconjugation generalizes both isogonal and isotomic conjugation.

**Theorem 7.2.5.** *For a triangle, isogonal conjugation is the isoconjugation whose fixed points are the incenter and the three excenters, while isotomic conjugation is the isoconjugation whose fixed points are the centroid and the three vertices of the anticomplemental triangle.*

The fixed points of an isoconjugation may be imaginary; see the exercises. Given a point $X$, setting $X' = X$ in theorem 7.2.1 gives an isoconjugation having $X$ as a fixed point. We denote it by $\mathbf{j}^X_{\triangle ABC}$, or simply by $\mathbf{j}^X$ when the triangle is clear.

**Theorem 7.2.6.** *Given $\triangle ABC$ and a point $X$, let $P' = \mathbf{j}^X P$ and let $\triangle X^a X^b X^c$ be the anticevian triangle of $X$. Then, for every $P$, $(AP, AP'; AX, X^b X^c) = -1$. Here $X^b X^c$ passes through $A$, so the expression is a cross-ratio in the pencil at $A$.*

**Proof.** By definition, $AP$ and $AP'$ are corresponding lines of the involution $f$ in the pencil at $A$ that interchanges $AB$ and $AC$ and fixes $AX$. The fixed lines of $f$ are $AX$ and $X^b X^c$, so the assertion follows from corollary 3.5.23. ☐

**Theorem 7.2.7.** *If $\mathbf{j}$ is an isoconjugation of $\triangle ABC$, then for any four points $P, Q, R, S$, $A(PQ, RS) = A(\mathbf{j}P\mathbf{j}Q, \mathbf{j}R\mathbf{j}S)$.*

**Proof.** The correspondence $A[P] \to A[\mathbf{j}(P)]$ induced by the isoconjugation is an involution in the pencil at $A$. It is therefore a projectivity and preserves cross-ratios. ☐

**Corollary 7.2.8.** *If $f$ is an isoconjugation of $\triangle ABC$, then for all $P, Q$, $A(PQ, BC) = A(f(P)f(Q), CB)$.*

**Proof.** In the involution induced by an isoconjugation in the pencil at $A$, the lines $AB$ and $AC$ correspond by definition. ☐

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## Exercises

**Exercise 7.11.** Given a triangle and an isoconjugation, prove that the isoconjugate image of the line at infinity is a circumconic of the triangle.

**Exercise 7.12.** Given $\triangle ABC$ and points $P, P', Q$, construct with straightedge and compass a point $Q'$ such that $(P, P')$ and $(Q, Q')$ are two corresponding pairs under one isoconjugation.

**Exercise 7.13.** Given $\triangle ABC$, let $I, G, L$ be its incenter, centroid, and Lemoine point. Prove that the pairs $(L, \mathbf{t}I)$, $(I, G)$, and $(Ge, Mi)$ belong to one isoconjugation.

**Exercise 7.14.** Given $\triangle ABC$ and a point $P$ not on its sidelines, let $Q$ be the pole, with respect to the polar circle of $\triangle ABC$, of the trilinear polar of $P$. The point $Q$ is called the *polar conjugate* of $P$ with respect to $\triangle ABC$. Prove that polar conjugacy is an isoconjugation and that its fixed points are imaginary.

### 7.2.2 Isoconjugation and Conics

#### Isoconjugation and Conics

We now examine the relation between isoconjugation and conics. The following characterization gives its pencil definition.

**Theorem 7.2.9** (Pencil definition of isoconjugation)**.** *Given $\triangle ABC$ and a point $X$, let $\triangle X^aX^bX^c$ be the anticevian triangle of $X$, and let $\mathscr{W} = \mathrm{pen}(XX^aX^bX^c)$ be the pencil of conics through those four points. For any two points $P,P'$, $P' = \mathbf{j}^X P$ if and only if $P$ and $P'$ are conjugate with respect to every conic in $\mathscr{W}$.*

**Proof.** Suppose $P' = \mathbf{j}^X P$. We prove that $P$ and $P'$ are conjugate with respect to every $\alpha \in \mathscr{W}$. Put $PP' \cap \alpha = U,V$. By the pole–polar properties, it is enough to show that $(PP',UV) = -1$.

By Desargues's involution theorem, $\mathscr{W}$ induces an involution $g$ on $PP'$. Four of its corresponding pairs are $(PP' \cap AX, PP' \cap X^bX^c)$, $(PP' \cap BX, PP' \cap X^cX^a)$, $(PP' \cap CX, PP' \cap X^aX^b)$, and $(U,V)$. As in the proof of theorem 7.2.6, $g$ is precisely the involution satisfying $(Qg(Q),PP') = -1$. Hence $P,P',U,V$ form a harmonic range.

The converse follows by reversing the argument. □

Thus theorem 7.2.9 gives another construction of isoconjugation. We shall also say that this is the isoconjugation induced on $\triangle ABC$ by the pencil $\mathscr{W}$.

**Theorem 7.2.10.** *Given $\triangle ABC$ and points $X,P$, let $\triangle X^aX^bX^c$ be the anticevian triangle of $X$. Then $P\mathbf{j}^X P \in \mathbf{T}\mathfrak{C}(PXX^aX^bX^c)$.*

**Proof.** Put $\alpha = \mathfrak{C}(PXX^aX^bX^c)$. By the pencil definition, the polar of $P$ with respect to $\alpha$ passes through $\mathbf{j}^X P$. Since $P \in \alpha$, that polar is the tangent $P\mathbf{j}^X P$. □

The following is the isoconjugate-line property of a complete quadrilateral.

**Theorem 7.2.11** (Isoconjugate-line property of a complete quadrilateral)**.** *Suppose that two of the three pairs of opposite vertices of a complete quadrilateral are corresponding pairs under an isoconjugation of a triangle. Then the remaining pair is also a corresponding pair. Explicitly, let $X,X'$ and $Y,Y'$ be two corresponding pairs under an isoconjugation $\mathbf{j}$ of $\triangle ABC$, and set $Z = XY \cap X'Y'$ and $Z' = XY' \cap X'Y$. Then $Z' = \mathbf{j}Z$.*

**Proof.** By the pencil definition of isoconjugation, this is an immediate consequence of Hesse's theorem. □

**Alternative proof.** Consider the complete quadrilateral of the four lines $\{XY, XY', X'Y, X'Y'\}$. Its three pairs of opposite vertices are $(X,X')$, $(Y,Y')$, and $(Z,Z')$. By the dual of Desargues' involution theorem, theorem 3.5.25, the three pairs of lines $(AX,AX')$, $(AY,AY')$, and $(AZ,AZ')$ are corresponding pairs of one involution on the pencil at $A$. On the other hand, by the definition of isoconjugation, $\mathbf{j}$ induces on that pencil an involution having $(AX,AX')$ and $(AY,AY')$ as two corresponding pairs. Two corresponding pairs determine an involution, so the two involutions are the same. Hence $Z,Z'$ satisfy the condition for isoconjugation at $A$. The same argument at $B$ and $C$ gives $Z' = \mathbf{j}Z$. □

**Corollary 7.2.12** (Isogonal- and isotomic-line property of a complete quadrilateral)**.** *If two of the three pairs of opposite vertices of a complete quadrilateral are corresponding pairs under the isogonal, respectively isotomic, conjugation of a triangle, then so is the remaining pair.*

Here is one application.

**Example 7.2.13.** The isogonal case gives another proof of proposition 5.9.21: the lines $Ap_1T_2$ and $Ap_2T_1$ of a triangle meet at its centroid. By propositions 5.9.19 and 5.9.20, $Ap_1$ and $T_1$ are isogonal conjugates, as are $T_2$ and $Ap_2$. Hence corollary 7.2.12 shows that $Ap_1T_2 \cap Ap_2T_1$ and $Ap_1Ap_2 \cap T_1T_2$ are isogonal conjugates. The point $Ap_1Ap_2 \cap T_1T_2$ is the Lemoine point $L$, whose isogonal conjugate is the centroid $G$. Therefore $G = Ap_1T_2 \cap Ap_2T_1$.

For a set of points $\gamma$ and an isoconjugation $\mathbf{j}$, define $\mathbf{j}(\gamma)=\{\mathbf{j}P\,|\,P\in\gamma\}$. This is the *isoconjugate image* of $\gamma$.[5] As for inversion and polarity, the following basic incidence property holds; we omit the analogous proof.

**Theorem 7.2.14.** *Given $\triangle ABC$ and an isoconjugation $\mathbf{j}$, consider curves in the projective plane after their portions on AB, BC, and CA have been removed. Under $\mathbf{j}$, tangent curves remain tangent, intersecting curves remain intersecting, and disjoint curves remain disjoint.*

In particular, for a line $l$, we can study its isoconjugate image $\mathbf{j}(l)$.

**Theorem 7.2.15.** *The isoconjugate image of a line $l$ that passes through no vertex of $\triangle ABC$ is a circumconic of the triangle. Conversely, the isoconjugate image of such a circumconic is a line.*

**Proof.** Use the pencil definition, and suppose that $\mathbf{j}$ is induced by a conic pencil $\mathscr{W}$. Let $\beta$ be the locus of the poles of $l$ with respect to all conics in $\mathscr{W}$. For any $P'\in\beta$, corollary 6.7.10 shows that $P'$ is the pole of $l$ with respect to some $\alpha\in\mathscr{W}$. The polars of $P'$ with respect to all conics of $\mathscr{W}$ pass through a fixed point $P=\mathbf{j}P'$. Since $l=\mathfrak{p}_\alpha(P')$, the point $P$ lies on $l$. Thus the isoconjugate image of $\beta$ is $l$, and conversely the image of $l$ is $\beta$. □

**Example 7.2.16.** Use isoconjugation to prove the property of a rectangular hyperbola in theorem 8.1.1: a circumconic of a triangle is a rectangular hyperbola if and only if it passes through the orthocenter.

**Solution.** First prove necessity. Let $\alpha$ be a circumrectangular hyperbola of $\triangle ABC$. Then $l=\mathbf{g}(\alpha)$ is a line. By theorem 4.4.3, the isogonal conjugates $M,N$ of the two points at infinity on $\alpha$ lie on $\odot(ABC)$, and their Simson lines are perpendicular. Together with proposition 4.2.3, this shows that $M,N$ are diametrically opposite points of $\odot(ABC)$, so $l$ passes through the circumcenter $O$. Since the isogonal conjugate of the orthocenter $H$ is $O$, the point $H$ lies on $\alpha$. The converse is the same argument in reverse. □

For a line through a vertex, the image can be characterized as follows.

**Theorem 7.2.17.** *Suppose that the isoconjugate image of a line $l$ with respect to $\triangle ABC$ is a conic $\alpha$. Let $l_A$ be the tangent to $\alpha$ at $A$, and put $D=l\cap BC$. Then the isoconjugate image of $AD$ is $l_A$.*

**Proof.** Take in theorem 7.2.15 the limit as the line approaches one through the vertex $A$. In the limiting sense, $A$ and its opposite side $BC$ are isoconjugate. The image conic therefore degenerates into two lines, one of which is $BC$. After that component is removed, the other component is the isoconjugate image of the line through $A$; equivalently, the image of $l_A$ is a line.

The isoconjugate of $D$ is $A$. As $P$ tends to $D$ along $l$, its isoconjugate $P'$ tends to $A$ along $\alpha$, hence along $l_A$. Since the image of $AD$ is a line, that line must be $l_A$. □

**Corollary 7.2.18.** *For a given triangle, the isoconjugate image of a line through a vertex is again a line through that vertex.*

**Theorem 7.2.19.** *Given $\triangle ABC$ and a point $X$, let $l$ be a line through $X$ that passes through no vertex of the triangle. Then $\mathbf{j}^X(l)$ is the conic $\alpha$ through $A,B,C,X$ that is tangent to $l$ at $X$.*

**Proof.** It is clear that $\mathbf{j}^X(l)$ passes through $A,B,C,X$. It remains to prove tangency.

Suppose $\mathbf{j}^X(l)$ met $l$ at a point $X'\neq X$. Then $\mathbf{j}^X(X')$ would lie on both $l$ and $\mathbf{j}^X(l)$, so it would equal either $X$ or $X'$. But $\mathbf{j}^X(X)=X\neq X'$, while $\mathbf{j}^X(X')\neq X'$: otherwise $X'$ would be a fixed point and

[5] If a curve $\gamma$ passes through a vertex $A$, the value at $A$ is understood as the limit $\mathbf{j}A=\lim_{P\in\gamma,P\to A}\mathbf{j}P$.

$l$ would pass through a vertex of the triangle. This contradiction proves that $\mathbf{j}^X(l)$ is tangent to $l$ at $X$. □

Finally, we record an important projective property.

**Theorem 7.2.20.** *Given $\triangle ABC$ and an isoconjugation $\mathbf{j}$, let $\gamma,\gamma'$ each be either a line or a circumconic of the triangle. If $\mathbf{j}(\gamma)=\gamma'$, then $\mathbf{j}:\gamma\to\gamma'$ is a projectivity.*

**Proof.** For a variable point $P$ on $\gamma$, the map between the two pencils $[AP]\to[A\mathbf{j}P]$ is a projectivity by definition. When $\gamma$ and $\gamma'$ are lines or circumconics, moreover, $A[AP]\barwedge\gamma[P]$ and $A[A\mathbf{j}P]\barwedge\gamma'[\mathbf{j}P]$. Consequently $\gamma[P]\barwedge\gamma'[\mathbf{j}P]$. □

### An Introduction to Polygonal Isoconjugation*

In definition 4.4.18 we extended isogonal conjugation to polygons. General isoconjugation extends in the same way, as we now briefly explain. Let an $n$-gon $A_1A_2\cdots A_n$ and points $X,X'$ not lying on any of its sidelines be given. Throughout, subscripts are read modulo $n$, so that $A_{i+n}=A_i$. For a point $P$ in the plane, suppose that there is a point $Q$ such that $A_i(PX,A_{i+1}A_{i-1})=A_i(QX',A_{i-1}A_{i+1})$ for $i=1,\dots,n$. Then $P$ and $Q$ are said to be *isoconjugate* with respect to the polygon when $X,X'$ are prescribed as a corresponding pair, and the resulting transformation $P\mapsto Q$ is called the corresponding *isoconjugation*.

Points in special positions may be included by taking limits.

Clearly, $X,X'$ themselves form a corresponding pair under this isoconjugation. As in the case of a triangle, the three pairs of lines $(A_iP,A_iQ)$, $(A_iA_{i-1},A_iA_{i+1})$, and $(A_iX,A_iX')$ are corresponding pairs of one involution in the pencil at $A_i$. Moreover, if $A_i(PX,A_{i+1}A_{i-1})=A_i(QX',A_{i-1}A_{i+1})$ holds for $i=1,\dots,n-1$, then the condition for $i=n$ follows automatically.

Since the two conditions for $i=1$ and $i=2$ already determine $Q$ uniquely, isoconjugates with respect to a polygon need not always exist. For a general quadrilateral, the points that possess isoconjugates trace a curve; for a general $n$-gon with $n\geq 6$, there need not be any corresponding pair other than $(X,X')$. The pentagonal case is exceptional: a general pentagon has further corresponding pairs, but only finitely many, as follows.

The five sidelines of a general pentagon are tangent to a conic $\gamma$. Draw the two tangents $u_1,u_2$ from $X$ to $\gamma$ and the two tangents $v_1,v_2$ from $X'$ to $\gamma$, and put $U=u_1\cap v_1$, $U'=u_2\cap v_2$, $V=u_1\cap v_2$, and $V'=u_2\cap v_1$. The four tangent lines $u_1,u_2,v_1,v_2$ determine a complete quadrilateral whose pairs of opposite vertices are precisely $(X,X')$, $(U,U')$, and $(V,V')$. The dual of Desargues' involution theorem, theorem 3.5.25, now shows that $(U,U')$ and $(V,V')$ are also corresponding pairs under this isoconjugation.

Polygonal isoconjugation can be related to isoconjugations of triangles.

**Proposition 7.2.21.** *Let $\mathbf{j}$ be the isoconjugation of an $n$-gon $A_1A_2\cdots A_n$ for which $(X,X')$ is a corresponding pair. Put $B_i=A_iA_{i-1}\cap A_{i+1}A_{i+2}$, again reading subscripts modulo $n$, and let $\mathbf{j}_i$ be the isoconjugation of $\triangle A_iA_{i+1}B_i$ for which $(X,X')$ is a corresponding pair. Then, for points $P,Q$,*

$$Q=\mathbf{j}P \quad\Longleftrightarrow\quad Q=\mathbf{j}_1P=\mathbf{j}_2P=\cdots=\mathbf{j}_{n-2}P.$$

**Proof.** Suppose that $Q=\mathbf{j}P$. At the vertices $A_i,A_{i+1}$ of $\triangle A_iA_{i+1}B_i$, the points $P,Q$ satisfy

$$A_i(PX,A_{i+1}A_{i-1})=A_i(QX',A_{i-1}A_{i+1}),\quad A_{i+1}(PX,A_{i+2}A_i)=A_{i+1}(QX',A_iA_{i+2}).$$

Existence and uniqueness in the definition of triangular isoconjugation therefore give $Q=\mathbf{j}_iP$.

Conversely, suppose that $Q=\mathbf{j}_iP$ for $i=1,\dots,n-2$. Then $A_i(PX,A_{i+1}A_{i-1})=A_i(QX',A_{i-1}A_{i+1})$ holds for $i=1,\dots,n-1$; as in the triangular case, the relation for $i=n$ follows as well. □

Under the hypotheses of this proposition, polygonal isoconjugation also has a pencil definition. Let $\mathscr{P}_i$ be the pencil of conics determined by the four fixed points of $\mathbf{j}_i$. The pencil definition for a triangle shows that $Q = \mathbf{j}P$ if and only if $P,Q$ are conjugate with respect to every conic in each of $\mathscr{P}_1,\dots,\mathscr{P}_{n-2}$. Hesse's theorem then gives the following result.

**Theorem 7.2.22.** *Given a polygon and an isoconjugation on it, suppose that $P,P'$ and $Q,Q'$ are two isoconjugate pairs. Then $PQ \cap P'Q'$ and $PQ' \cap P'Q$ form another corresponding pair under the same isoconjugation.*

As before, this result can alternatively be proved from the dual of Desargues' involution theorem.

We introduced isogonal conjugation for polygons in section 4.4.2. It is related to polygonal isoconjugation as follows.

**Proposition 7.2.23.** *The isoconjugation of a polygon for which the circular points $I,J$ form a corresponding pair is precisely isogonal conjugation.*

**Proof.** Let $(P,Q)$ and $(I,J)$ be two corresponding pairs under the isoconjugation, and denote three consecutive vertices of the polygon by $B,A,C$. We are given $A(PI,BC) = A(QJ,CB)$ and must prove that $AP,AQ$ are symmetric with respect to the angle bisectors of $\angle BAC$.

The corresponding pairs $(AI,AJ)$ and $(AB,AC)$ determine an involution in the pencil at $A$, and the cross-ratio condition shows that $(AP,AQ)$ is also a corresponding pair of this involution. Let its fixed lines be $m,n$. By corollary 3.5.23, $(AI,AJ;m,n) = -1$, so $m \perp n$. Similarly $(AB,AC;m,n) = -1$, and proposition 3.2.17 shows that $m,n$ are the two angle bisectors of $\angle BAC$. Reflection in the bisectors of $\angle BAC$ is also an involution in the pencil at $A$ and fixes $m,n$, so it coincides with the preceding involution. Hence $AP$ and $AQ$ are isogonal, as required. $\square$

Thus, when $X \neq X'$, the preceding construction specializes to isogonal conjugation on replacing $X,X'$ by the circular points. In section 4.4.2 we noted that the locus of points of a quadrilateral that possess isogonal conjugates is a circumcubic. Likewise, for a fixed isoconjugation of a quadrilateral, the locus of points that possess isoconjugates is a cubic curve. It is circumscribed about the quadrilateral because, by definition, the isoconjugate of a vertex is undefined but may be regarded as an arbitrary point; consequently, every vertex is added to the locus as a limiting point.

The discussion above gives a simple property of this cubic: if $P,P'$ and $Q,Q'$ are two isoconjugate pairs on it, then $PQ \cap P'Q'$ and $PQ' \cap P'Q$ also lie on it. In particular, on passing to the limiting case, the tangents at $P,P'$ meet at the third intersection of $PP'$ with the cubic.

Figure 7.9 depicts the locus $\gamma$ of points of a quadrilateral $ABCD$ that possess isogonal conjugates. Put $E = AB \cap CD$ and $F = AD \cap BC$. Proposition 7.2.21 shows immediately that the isoconjugation of $ABCD$ can also be regarded as the corresponding isoconjugation of the quadrilateral $AECF$; hence $\gamma$ also passes through $E,F$. By theorem 4.4.20, every pair of isogonal conjugates of a quadrilateral is the pair of foci of one of its inconics. Accordingly, two inconic ellipses of the quadrilateral are shown in the figure. Their pairs of foci $(P,P')$ and $(Q,Q')$ are two pairs of isogonal conjugates. Put $R = PQ \cap P'Q'$ and $R' = PQ' \cap P'Q$; the points $R,R'$ are also shown.

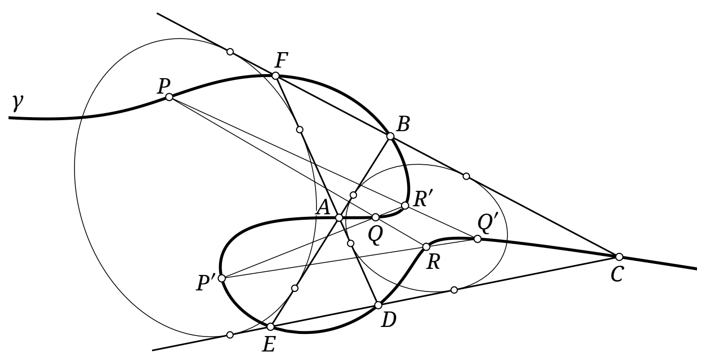


FIGURE 7.9

## Exercises

**Exercise 7.15*.** Fill in the details of the proof of proposition 7.2.21, and use the dual of Desargues' involution theorem to prove theorem 7.2.22.

**Exercise 7.16*.** For a fixed isoconjugation of a quadrilateral, let $\gamma$ be the curve consisting of the points that possess isoconjugates. Let $l$ be a line in general position that passes through none of the six vertices of the complete quadrilateral determined by the four sidelines. Using the preceding results, without presupposing any knowledge of cubic curves, prove that, in the complex projective plane, $l$ and $\gamma$ have exactly three intersections, counted with multiplicity.

**Exercise 7.17.** Given $\triangle ABC$, two isoconjugations $f, g$, and a fixed line $l$, let $P$ vary on $l$. Prove that the line $f(P)g(P)$ passes through a fixed point.

**Exercise 7.18.** Given $\triangle ABC$ and a point $X$, let $\triangle X_A X_B X_C$ be the anticevian triangle of $X$. If $\alpha \in \operatorname{pen}(XX_A X_B X_C)$ and $P \in \alpha$, prove that $P\mathbf{j}^X P \in \mathbf{T}\alpha$.

**Exercise 7.19.** Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be four fixed conics in one pencil. Prove that, for every point $P$ of the plane, the polars of $P$ with respect to $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ belong to one pencil, and that their cross-ratio is constant.

**Exercise 7.20.** Given $\triangle ABC$, suppose $(P,Q)$ and $(R,S)$ are two pairs of isogonal conjugates, and let $H$ be the orthocenter. Let $\triangle P_a P_b P_c$ and $\triangle R_a R_b R_c$ be the pedal triangles of $P$ and $R$. If $P_bP_c \cap R_bR_c = H$, prove that $BC \parallel QS$.

**Exercise 7.21.** Given $\triangle ABC$, let $\triangle DEF$ be its orthic triangle. The perpendicular bisector of $BC$ meets $EF$ at $Q$, and put $S = \mathbf{g}Q$. The line through $A$ parallel to $BC$ meets $\mathscr{C}$ again at $T$. Prove that $S, H, T$ are collinear.

**Exercise 7.22**.** Consider the isogonal conjugation $\mathbf{g}$ of a quadrilateral $ABCD$, and let $\Gamma$ be the locus of points that possess isogonal conjugates. For $U, V \in \Gamma$, exercise 7.16 gives a third intersection of $UV$ with $\Gamma$; in this problem denote it by $U * V$. When the two points coincide, use the tangent, and count intersections with multiplicity. All special configurations are understood by taking limits. The following parts should not use the fact that $\Gamma$ is a cubic.

(1) Choose two pairs of isogonal conjugates $U, U' = \mathbf{g}U$ and $V, V' = \mathbf{g}V$ on $\Gamma$. Prove that $\Gamma$ is also the isogonal-conjugacy locus of the quadrilateral $UVU'V'$. In particular, for a general point $Z$, $Z \in \Gamma \iff \measuredangle UZV + \measuredangle U'ZV' = 0$.
(2) Fix two general points $U, V \in \Gamma$, and construct an opposite similarity transformation $f$ such that $f(U) = U'$ and $f(V) = V'$. Let $\omega$ vary over the circles through $U, V$, and let $W, Z$ be the intersections of $\omega$ and $f(\omega)$ other than the circular points. Prove that the finite points of $\omega \cap \Gamma$ other than $U, V$ are precisely $W, Z$, and that, as $\omega$ varies, the line $WZ$ always passes through a fixed point of $\Gamma$. Denote this point by $T_{UV}$.
(3) Let $N$ be the point at infinity on the axis of the parabola inscribed in the complete quadrilateral $ABCD$. For two fixed general points $U, V \in \Gamma$, prove that $N \in \Gamma$ and $T_{UV} = N * (U * V)$.
(4) For four general points $P, Q, R, S$ on $\Gamma$, prove that $(P * Q) * (R * S) = (P * R) * (Q * S)$.
(5) Let the nine intersections $P_{ij} = l_i \cap m_j$ of six lines $l_1, l_2, l_3, m_1, m_2, m_3$ be distinct. If eight of them lie on $\Gamma$, prove that the ninth also lies on $\Gamma$.

## 7.3 Trilinear Polarity

### 7.3.1 Trilinear Polars and Trilinear Poles

Recall the following definitions.

**Definition 7.3.1.** Given $\triangle ABC$, the perspective axis $l$ of the cevian triangle of a point $P$ and $\triangle ABC$ is called the *trilinear polar* of $P$ with respect to $\triangle ABC$; in turn, $P$ is called the *trilinear pole* of $l$ with respect to $\triangle ABC$. We write $\mathscr{T}_{\triangle ABC}(P)$, or simply $\mathscr{T}(P)$ when there is no ambiguity, for the trilinear polar of $P$. The notation $\mathscr{T}^{-1}$ denotes the trilinear-pole operation.

**Proposition 7.3.2.** *For $\triangle ABC$ and a point $P$, as in figure 7.10, let $A_1 = AP \cap BC$ and $A_2 = \mathscr{T}(P) \cap BC$. Then $A_1, A_2, B, C$ form a harmonic range.*

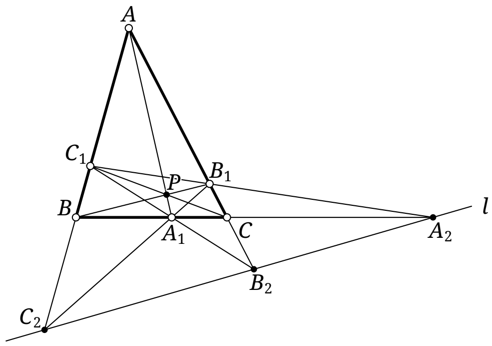


FIGURE 7.10

**Proof.** Let $\triangle A_1B_1C_1$ be the cevian triangle of $P$. Apply the harmonic property of a complete quadrangle to $CB_1C_1B$. □

**Corollary 7.3.3.** *Given $\triangle ABC$ and a line $l$ passing through no vertex, its trilinear pole can be constructed as follows; see figure 7.10.*

*(1) Put $A_2 = l \cap BC$, and choose $A_1$ so that $(A_1A_2, BC) = -1$. Define $B_1, B_2$ and $C_1, C_2$ cyclically.*
*(2) The lines $AA_1, BB_1, CC_1$ concur at the trilinear pole $P$ of $l$.*

The construction also shows that every line through no vertex has a unique trilinear pole.
We now derive several properties involving isoconjugation.

**Proposition 7.3.4.** *Let $\mathbf{j}$ be an isoconjugation of $\triangle ABC$, with $\mathbf{j}P = P'$. If $\mathscr{T}(P)$ and $\mathscr{T}(P')$ meet $BC$ at $A_1$ and $A_1'$, respectively, then the isoconjugate image of the line $AA_1$ is $AA_1'$.*

**Proof.** By proposition 7.3.2, $A(P'A_1', CB) = -1 = A(PA_1, BC)$. By the definition of isoconjugation, for every point $Q$ on $AA_1$, $\mathbf{j}Q \in AA_1'$. Together with corollary 7.2.18, this proves the assertion. □

**Definition 7.3.5.** Let $\mathbf{j}$ be an isoconjugation of $\triangle ABC$, and let $P, P'$ be an isoconjugate pair under it. Each of the lines $\mathscr{T}(P)$ and $\mathscr{T}(P')$ is called the *isoconjugate line* of the other; together they form an *isoconjugate pair of lines*. Equivalently, the isoconjugate of a line is the trilinear polar of the isoconjugate of its trilinear pole. In the special cases $\mathbf{j} = \mathbf{g}$ and $\mathbf{j} = \mathbf{t}$, these are called *isogonal lines* and *isotomic lines*, respectively.

**Proposition 7.3.6.** *Given $\triangle ABC$ and a line $l$, form the complete quadrilateral from $l$ and the three sidelines. Its Gauss line is the complemental image of the isotomic line of $l$.*[6]

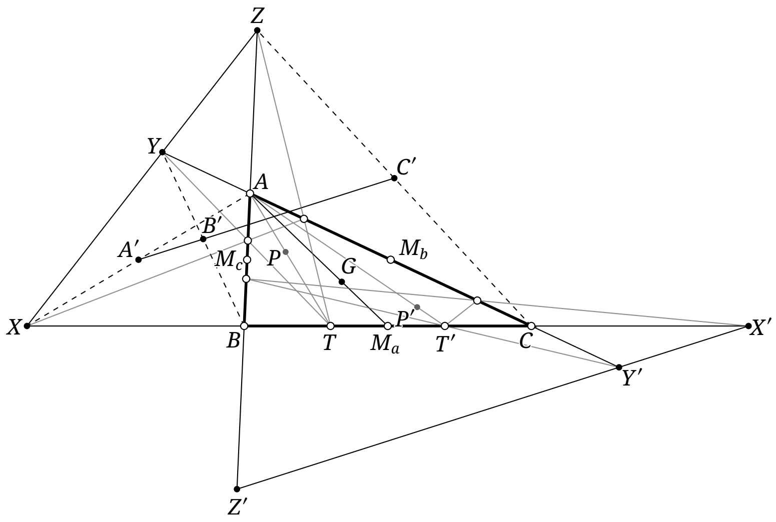


FIGURE 7.11

**Proof.** As in figure 7.11, let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $G$ be the centroid. Suppose $l$ meets the three sides at $X,Y,Z$, while its isotomic line $l'$ meets them at $X',Y',Z'$. Let $P,P'$ be the trilinear poles of $l,l'$. Then $P'=\mathbf{t}P$. Let $A',B',C'$ be the midpoints of the three diagonals of the complete quadrilateral; $\overline{A'B'C'}$ is the Gauss line.

Put $T=AP\cap BC$ and $T'=AP'\cap BC$. By proposition 7.3.2, $(BC,XT)=-1=(BC,X'T')=(CB,X'T')$. Thus $(B,C,X,T)\barwedge(C,B,X',T')$. Since $T$ and $T'$ are symmetric about $M_a$ by isotomic conjugacy, and $B,C$ are also symmetric about $M_a$, this projectivity is the half-turn about $M_a$. Hence $X$ and $X'$ are symmetric about $M_a$.

The point $M_a$ is the midpoint of $XX'$, and the centroid property gives $\overline{AG}=2\overline{GM_a}$. Hence $G$ is also the centroid of $\triangle AXX'$. Since $A'$ is the midpoint of $AX$, the points $A',G,X'$ are collinear and $\overline{GX'}=2\overline{A'G}$. The same holds cyclically for $B',C'$, so the result follows from the homothety $\mathfrak{h}_{G,-2}$. $\square$

We shall also need the following reciprocity.

**Theorem 7.3.7.** *Let $\mathbf{j}$ be an isoconjugation of $\triangle ABC$. For any two points $P,Q$, $\mathbf{j}Q\in\mathscr{T}(P)\iff\mathbf{j}P\in\mathscr{T}(Q)$.*

**Proof.** Let $\mathscr{T}(P)$ meet $BC,CA,AB$ at $A_1,B_1,C_1$, and let $\mathscr{T}(Q)$ meet them at $A_1',B_1',C_1'$. Put $P'=\mathbf{j}P$ and $Q'=\mathbf{j}Q$, and let $K=\mathscr{T}(Q')\cap CA$.

On $AC$, take the involution $g$ defined by $(U,g(U);A,C)=-1$ for $U\in AC$. Its fixed points are $A,C$. By proposition 7.3.4, $g(BQ'\cap AC)=K$ and $g(BP\cap AC)=B_1$. Consequently $(K,BP\cap AC,A,C)\barwedge(BQ'\cap AC,B_1,A,C)$, and hence $B(KP,AC)=B(Q'B_1,AC)=B(Q'B_1,C_1C)$, where the second equality uses the collinearity of $A,B,C_1$.

Since $\mathbf{j}P=P'$ and, by proposition 7.3.4, $\mathbf{j}(BK)=BB_1'$, corollary 7.2.8 gives $B(KP,AC)=B(B_1'P',CA)=B(B_1'P',CC_1')$, where the second equality uses the collinearity of $C_1',B,A$. Therefore $B(Q'B_1,C_1C)=B(B_1'P',CC_1')$. Similarly, $C(Q'B_1,C_1B)=C(C_1'B,P'B_1')=C(B_1'P',BC_1')$. It follows that

[6] This means the image under the complement map. For a figure $\gamma$, its complemental image is defined precisely by $\mathbf{c}\gamma:=\{\mathbf{c}P\,|\,P\in\gamma\}$.

$$B(Q'B_1, C_1C) = C(Q'B_1, C_1B) \Longleftrightarrow B(B'_1P', CC'_1) = C(B'_1P', BC'_1).$$

By proposition 3.2.14, this says precisely that $Q', B_1, C_1$ are collinear if and only if $B'_1, P', C'_1$ are collinear. □

We conclude the subsection with several notable trilinear polars.

**Theorem 7.3.8.** *The trilinear polar of the centroid of a triangle is the line at infinity.*

**Proof.** This is immediate. □

**Definition 7.3.9.** The *orthic axis* of a triangle is the trilinear polar of its orthocenter. We denote the orthic axis of $\triangle ABC$ by $\mathscr{A}_{\triangle ABC}$, or simply by $\mathscr{A}$ when there is no ambiguity.

**Theorem 7.3.10.** *The orthic axis of a triangle is the common radical axis of the circumcircle, the nine-point circle, and the polar circle, and it is perpendicular to the Euler line.*

**Proof.** Let the orthic axis $\mathscr{A}$ meet the three sidelines at $A', B', C'$, and let $\triangle H_aH_bH_c$ be the orthic triangle of $\triangle ABC$. The points $H_b, H_c, B, C$ are concyclic, and $\mathscr{N} = \odot(H_aH_bH_c)$. Hence

$$\mathrm{pow}_{\mathscr{C}}(A') = \overline{A'B} \cdot \overline{A'C} = \overline{A'H_b} \cdot \overline{A'H_c} = \mathrm{pow}_{\mathscr{N}}(A').$$

Thus $A'$ lies on the radical axis of $\mathscr{C}$ and $\mathscr{N}$; cyclically, so do $B'$ and $C'$. Therefore $\mathscr{A}$ is that radical axis. By theorem 6.1.16, $\mathscr{C}$ and $\mathscr{N}$ are inverse with respect to the polar circle $\mathscr{P}$, so all three circles have the same radical axis. Their centers lie on the Euler line, which is therefore perpendicular to $\mathscr{A}$. □

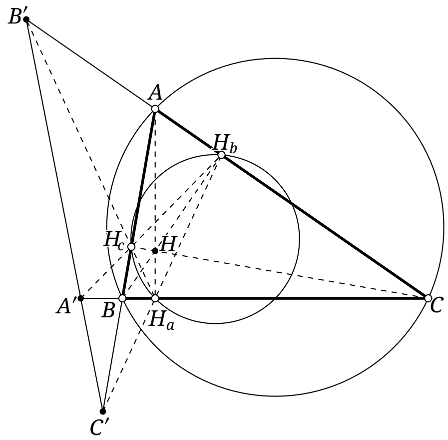


FIGURE 7.12

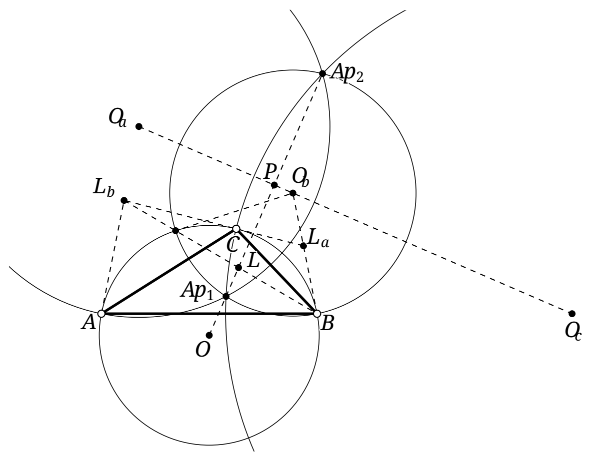


FIGURE 7.13

**Definition 7.3.11.** The *Lemoine axis* of a triangle is the trilinear polar of its Lemoine point.

**Theorem 7.3.12.** *The Lemoine axis of a triangle is perpendicular to the Brocard axis and is each of the following:*

*(1) the axis of centers of the three Apollonius circles;*
*(2) the perspective axis of the triangle and its tangential triangle;*
*(3) the polar of the Lemoine point with respect to the circumcircle.*

**Proof.** As in figure 7.13, consult section 5.8 and the related results in figure 5.46. Construct the three Apollonius circles of $\triangle ABC$; their centers $O_a, O_b, O_c$ are collinear. Moreover, $BL_b = \mathfrak{p}_{\mathscr{C}}(O_b)$, where $\triangle L_aL_bL_c$ is the tangential triangle. The harmonic pole–polar property gives $B(AC, L_bO_b) = -1$. By lemma 5.3.7, $L \in BL_b$, so $B(AC, LO_b) = -1$. Hence proposition 7.3.2 yields $O_b = \mathscr{T}(L) \cap AC$. Cyclically, $O_a, O_c$ are the other two intersections of $\mathscr{T}(L)$ with the sidelines. Thus the line $\overline{O_aO_bO_c}$ is the Lemoine axis, proving (1).

By proposition 5.8.2, the Apollonius circle $\odot O_b$ is orthogonal to $\mathscr{C}$, so $O_bB$ is tangent to $\mathscr{C}$. Hence $O_b = L_aL_c \cap AC$. Cyclically, $O_a, O_c$ are the other intersections of corresponding sides of $\triangle L_aL_bL_c$ and $\triangle ABC$. This proves (2).

Let $Ap_1, Ap_2$ be the two Apollonius points. The line $\overline{OAp_1Ap_2}$ is the Brocard axis and contains $L$; it is plainly perpendicular to the Lemoine axis $\overline{O_aO_bO_c}$.

Let the Brocard and Lemoine axes meet at $P$. Then $P$ is the midpoint of $Ap_1Ap_2$. By proposition 5.8.5, $L = \mathbf{i}P$, and therefore the Lemoine axis is $\mathfrak{p}_{\mathscr{C}}(L)$. This proves (3). □

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## Exercises

**Exercise 7.23.** Let $G$ be the centroid of $\triangle ABC$, and let $\triangle P_1P_2P_3$ be the cevian triangle of a point $P$. Put $X = \mathbf{a}P$. Let $T_1, T_2, T_3$ be the midpoints of $AP_1, BP_2, CP_3$, respectively. Set $A' = T_2T_3 \cap BC$, and define $B', C'$ cyclically. Prove that $A', B', C'$ are collinear and that $\mathscr{T}(X) = \overline{A'B'C'}$.

**Exercise 7.24.** Prove that the orthic axis of a triangle is

(1) the perspective axis of its medial and tangential triangles;
(2) the common radical axis of its orthocentroidal circle,[7] its circumcircle, and the circumcircle of its tangential triangle.

**Exercise 7.25.** Prove that the Lemoine axis of a triangle is the radical axis of its circumcircle and Brocard circle.

**Exercise 7.26.** Prove that the statements "two points are isoconjugate" and "two lines are isoconjugate" are dual.

**Exercise 7.27.** Define the trilinear polar of the incenter of a triangle to be its *antiorthic axis*. Prove that

(1) the antiorthic axis of the triangle is the orthic axis of its excentral triangle;
(2) the antiorthic axis is the common perspective axis of the excentral triangle, the extangents triangle, the Feuerbach triangle, and the original triangle.

### 7.3.2 Trilinear Polarity and Conics

As preparation, we first introduce the perspector of an inconic. Here and below, "inconic" means only that the conic is tangent to the three sidelines; it need not lie inside the triangle.

**Theorem 7.3.13.** *The contact triangle of an inconic of a triangle is perspective with the original triangle, as in figure 7.14.* The center of this perspectivity is called the *perspector*, or *Brianchon point*, of the inconic with respect to the triangle.

[7]The circle with diameter joining the centroid and orthocenter of a triangle is the *orthocentroidal circle* of the triangle.

**Proof.** This is a special case of Chasles's theorem. It also follows directly from Brianchon's theorem in the case where three pairs of tangents coincide. □

**Proposition 7.3.14.** *Given $\triangle ABC$ and a point $P$ not on a sideline, there is a unique inconic $\alpha$ whose perspector is $P$.*

**Proof.** Four points determine a projective transformation of the plane. Hence there is a unique projective transformation fixing the three vertices of the triangle and carrying $P$ to its Gergonne point $Ge$. Under this transformation $\alpha$ exists and becomes the incircle. Its uniqueness may also be seen from the fact that the perspector uniquely determines the three contact points: the conic is thereby prescribed through six points counted with multiplicity, three pairs being coincident. □

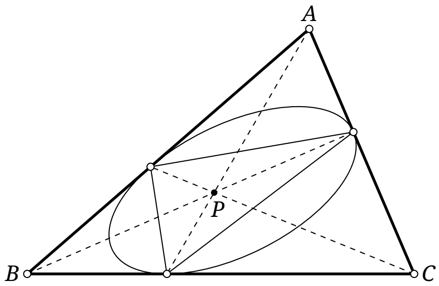


FIGURE 7.14

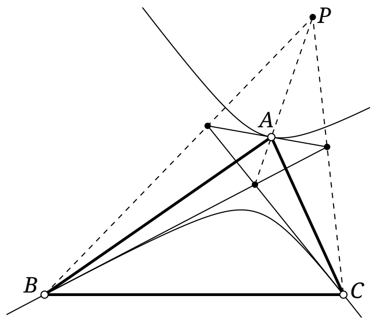


FIGURE 7.15

The analogous definition applies to a circumconic.

**Theorem 7.3.15.** *The triangle formed by the tangents to a circumconic at the three vertices is perspective with the original triangle, as in figure 7.15.* The center of this perspectivity is called the *perspector* of the circumconic with respect to the triangle.

**Proof.** This is a special case of Chasles's theorem, or equivalently the dual of theorem 7.3.13. □

**Proposition 7.3.16.** *Given $\triangle ABC$ and a point $P$ not on a sideline, there is a unique circumconic $\alpha$ whose perspector is $P$.*

**Proof.** Apply a projective transformation that fixes the three vertices and carries $P$ to the Lemoine point. The conic $\alpha$ then exists and becomes the circumcircle. Uniqueness also follows because the perspector uniquely determines the three tangents—the sidelines of the anticevian triangle of $P$—so the conic is specified by six tangent lines counted with multiplicity, three pairs being coincident. □

For inconics we have the following result.

**Theorem 7.3.17.** *Given $\triangle ABC$, let $P$ vary and let $\alpha$ be the inconic with perspector $X$. Then $\mathscr{T}(P) \in \mathbf{T}\alpha \iff P \in \mathscr{T}(X)$. Moreover, as $P$ varies on $\mathscr{T}(X)$, the correspondence $\mathscr{T}(X)[P] \to \alpha[\mathscr{T}(P)]$ is a projectivity.*

**Proof.** We prove only the forward implication; the converse is left to the reader. Let $\triangle X_aX_bX_c$ be the cevian triangle of $X$, whose vertices are the contact points of $\alpha$ with the three sidelines. Let $\mathscr{T}(X)$ meet them at $A_1', B_1', C_1'$. Let $\triangle A_2B_2C_2$ be the cevian triangle of $P$, and let $\mathscr{T}(P)$ meet the sidelines at $A_1, B_1, C_1$, as in figure 7.16.

Apply Newton's theorem to the quadrilateral $BC_1B_1C$. Then $BB_1, CC_1, X_bX_c$ concur, and therefore

$$(B_1, X_b; A, C) = (B, X_c; A, C_1). \qquad (*)$$

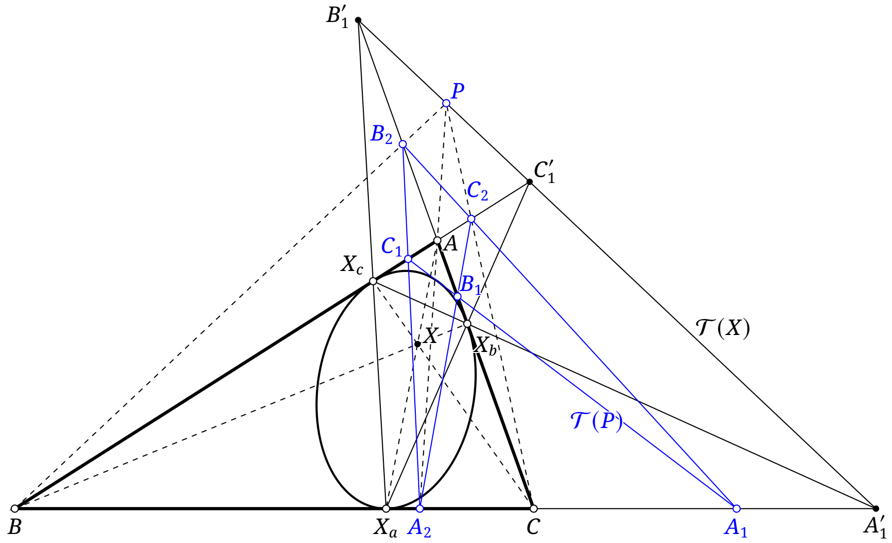


FIGURE 7.16

On $AC$, take the involution $g$ with fixed points $A,C$. By proposition 7.3.2, $g(B_2)=B_1$ and $g(B'_1)=X_b$. Hence

$$B(PB'_1,C'_1C)=B(B_2B'_1,AC)=B(B_1X_b,AC)=C(BX_c,AC_1)=C(C_1A,X_cB),$$

where $(*)$ gives the third equality. An analogous involution on $AB$ gives $C(PB'_1,C'_1B)=C(C_1A,X_cB)$. Thus $B(PB'_1,C'_1C)=C(PB'_1,C'_1B)$, and proposition 3.2.14 shows that $P,B'_1,C'_1$ are collinear.

It remains to prove projectivity. Since $AC[B_2]\barwedge\mathcal{T}(X)[P]\barwedge AB[C_2]$, we have $AC[B_2]\barwedge AB[C_2]$. Therefore the line $B_2C_2$ envelopes a conic $\gamma$ tangent to $AB$ and $AC$. Taking $P=A'_1$ shows that $\gamma$ is also tangent to $BC$. We obtain the chain

$$\mathcal{T}(X)[P]\barwedge AC[B_2]\barwedge\gamma[B_2C_2]\barwedge BC[A_1]\barwedge\alpha[\overline{A_1B_1C_1}],$$

which proves the assertion. □

**Theorem 7.3.18.** *Let* $\mathbf{j}$ *be an isoconjugation of* $\triangle ABC$*. Let* $\alpha$ *be the inconic with perspector* $X$*, and let* $\beta=\mathbf{j}(\mathcal{T}(X))$ *be the circumconic. For any point* $P$*,* $P\in\alpha\iff\mathcal{T}(\mathbf{j}P)\in\mathbf{T}\beta$*.*

**Proof.** If $Q\in\mathcal{T}(X)$, then theorem 7.3.17 says that $\mathcal{T}(Q)\in\mathbf{T}\alpha$. Let the circumconic be $\gamma=\mathbf{j}(\mathcal{T}(\mathbf{j}P))$. If $Q\in\gamma$, then $\mathbf{j}Q\in\mathcal{T}(\mathbf{j}P)$, and theorem 7.3.7 gives $P\in\mathcal{T}(Q)$. Thus, whenever $Q\in\gamma\cap\mathcal{T}(X)$, the line $\mathcal{T}(Q)$ is tangent to $\alpha$ and passes through $P$.

By theorem 7.2.14, applying isoconjugation shows that the tangency of $\mathcal{T}(\mathbf{j}P)$ to $\beta=\mathbf{j}(\mathcal{T}(X))$ is equivalent to the tangency of $\gamma$ to $\mathcal{T}(X)$, that is, to the coincidence of their two intersection points. The preceding paragraph shows that, if the two intersection points coincide, their trilinear polars are both tangent to $\alpha$ and pass through $P$. Thus the two tangents to $\alpha$ through $P$ coincide, which means $P\in\alpha$. Hence the coincidence of the two intersections of $\gamma$ and $\mathcal{T}(X)$ is equivalent to $P\in\alpha$, proving the assertion. □

Similarly, we can study the relation between circumconics and trilinear polarity.

**Theorem 7.3.19.** *Given* $\triangle ABC$*, let* $P$ *vary and let* $\alpha$ *be the circumconic with perspector* $X$*. Then* $P\in\alpha\iff X\in\mathcal{T}(P)$*. Moreover, as* $P$ *varies on* $\alpha$*, the correspondence* $\alpha[P]\to X[\mathcal{T}(P)]$ *is a projectivity.*

**Proof.** Restate theorem 7.3.17 as follows. Given $\triangle ABC$, let $l$ be a variable line and let $\alpha$ be an inconic with perspective axis $n$, namely the perspective axis of its contact triangle and the original triangle; it is the trilinear polar of the perspector. Then $l\in\mathbf{T}\alpha\iff\mathcal{T}^{-1}(l)\in n$, and, as $l\in\mathbf{T}\alpha$ varies, the correspondence $\alpha[l]\to n[\mathcal{T}^{-1}(l)]$ is a projectivity. The theorem is exactly the dual of this statement;

note that the dual of constructing the trilinear polar of a point is constructing the trilinear pole of a line. ☐

**Corollary 7.3.20.** *Given $\triangle ABC$, an isoconjugation* $\mathbf{j}$ *with respect to it, and a point $P$, the circumconic* $\mathbf{j}(\mathscr{T}(P))$ *has perspector* $\mathbf{j}P$.

**Proof.** Equivalently, suppose $\alpha$ is a circumconic with perspector $\mathbf{j}P$. We must show that, for every $P'$, $P' \in \alpha \Longleftrightarrow \mathbf{j}P' \in \mathscr{T}(P)$. By theorem 7.3.7, $\mathbf{j}P' \in \mathscr{T}(P) \Longleftrightarrow \mathbf{j}P \in \mathscr{T}(P')$. The assertion now follows from theorem 7.3.19. ☐

We now give examples in which trilinear polarity converts a collinearity or concurrency question into a more accessible one.

**Example 7.3.21.** Let $G$ be the centroid of $\triangle ABC$. Fix points $X, T$ and a directed angle $\theta$, and put $f = \mathbf{j}^X$. For a point $P$, write $f(G) = G'$ and $f(P) = P'$. Prove that the locus of the points $P$ satisfying $\measuredangle(TP, \mathscr{T}(P')) = \theta$ is a conic through $G'$ and $T$.

**Solution.** Through $T$, draw $l_0$ so that $\measuredangle(TP, l_0) = \theta$. The condition $\measuredangle(TP, \mathscr{T}(P')) = \theta$ requires $l_0 \parallel \mathscr{T}(P')$, or equivalently that $l_\infty, l_0, \mathscr{T}(P')$ are concurrent. Put $Q' = \mathscr{T}^{-1}(l_0)$. Since $\mathscr{T}^{-1}(l_\infty) = G$, theorem 7.3.19 shows that $A, B, C, Q', P', G$ lie on one conic. If $f(Q') = Q$, then theorem 7.2.15 shows that $Q, P, G'$ are collinear.

Now choose a line $l$ through $T$. The point $P$ on $l$ satisfying the condition is the intersection of $l$ and $QG'$. As $l$ varies, we have $T \in l_0 = \mathscr{T}(Q')$. By theorem 7.3.19, $Q'$ varies on a fixed circumconic of $\triangle ABC$, and by theorem 7.2.15, $Q$ varies on a fixed line $\sigma$.

As $l$ rotates about $T$:

(1) $l$ and $l_0$ rotate through the same angle, so the correspondence $[l] \to [l_0]$ preserves cross-ratios of corresponding elements and is projective;
(2) because $l_0 = \mathscr{T}(Q')$, proposition 7.3.2 says that $l_0$ and $AQ'$ cut $BC$ harmonically.[8] Thus $[(l_0 \cap BC)] \to [AQ' \cap BC]$ is the involution on $BC$ with fixed points $B, C$, and $[l_0] \to [AQ']$ is projective;
(3) by theorem 7.2.20, $[Q'] \to [Q]$ is projective, and hence so is the chain $[AQ'] \to [Q] \to [G'Q]$.

Therefore the pencil $[l]$ at $T$ and the pencil $[G'Q]$ at $G'$ are projective. Since $P = l \cap G'Q$, the projective definition of a conic shows that the locus of $P$ is a conic through $T$ and $G'$. ☐

**Corollary 7.3.22.** *Let $G, H$ be the centroid and orthocenter of $\triangle ABC$, and let $f$ be an isoconjugation. For a point $P$, write $f(G) = G'$ and $f(P) = P'$. Then $\mathscr{T}(P') \perp HP$ if and only if $A, B, C, G', H, P$ lie on a conic.*

**Proof.** Set $\theta = 90^\circ$ and $T = H$ in example 7.3.21. One readily checks that $P = A, B, C$ all satisfy $\mathscr{T}(P') \perp HP$.[9] The conic therefore also passes through $A, B, C$. ☐

**Example 7.3.23.** Let $H, G$ be the orthocenter and centroid of $\triangle ABC$. Given a point $P$, let $Q$ be the second intersection of the line $PG$ with the conic $\alpha$ through $A, B, C, H, P$. Then $HQ \perp \mathscr{T}(P)$.

**Solution.** Let $\beta = \mathfrak{C}(ABCGP)$, and let its tangent at $A$ meet $BC$ at $D$. Choose the line $AE$ so that $AE, AD, AB, AC$ form a harmonic pencil. Then $AE = \mathscr{T}(A)$.[10] By theorem 7.3.19, the trilinear polars of $A, G, P$ are concurrent. Since $\mathscr{T}(G) = l_\infty$, it follows that $\mathscr{T}(P) \parallel \mathscr{T}(A) = AE$.

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[8] Their intersections with $BC$, together with $B, C$, form a harmonic range.

[9] Take a limit as the point approaches one of these vertices in some way; the result does not depend on the choice of approach. For example, when $P = A$, regard $f(P)$ as any point of the opposite side $BC$. Then $\mathscr{T}(P') = BC$, so $\mathscr{T}(P') \perp HP$.

[10] Here $\mathscr{T}(A)$ is understood as $\lim_{X \in \beta, X \to A} \mathscr{T}(X)$.

Using the conic $\alpha$ through $H,P,Q,A,B,C$ and the conic $\beta$ through $P,G,A,B,C$, we obtain

$$H(QA,BC)=P(QA,BC)=P(GA,BC)=A(GA,BC)=A(GD,BC),$$

where the coincident line $AA$ in the limiting projection is understood as the tangent $t_\beta(A)$.

Let $g$ be the involution on $BC$ with fixed points $B,C$. Then $g(D)=E$ and $g(AG\cap BC)=\infty_{BC}$, the point at infinity on $BC$. Hence $A(GD,BC)=A(\infty_{BC}E,BC)$, and therefore $H(QA,BC)=A(\infty_{BC}E,BC)=A(E\infty_{BC},CB)$. In the pencils $H(Q,A,B,C)$ and $A(E,\infty_{BC},C,B)$, the lines $A\infty_{BC}$, $AC$, $AB$ are respectively perpendicular to $HA$, $HB$, $HC$. Taking perpendicular lines is a projectivity, and three corresponding pairs determine it. Consequently $HQ\perp AE$. Since $\mathscr{T}(P)\parallel AE$, we have $HQ\perp\mathscr{T}(P)$. □

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## Exercises

**Exercise 7.28.** Prove that the locus of the perspectors of all parabolic inconics of a triangle is its Steiner circumellipse.

**Exercise 7.29.** Given $\triangle ABC$ and the center of one of its inconics $\alpha$, how can one determine the type of $\alpha$?

**Exercise 7.30.** Given $\triangle ABC$ with incenter $I$ and orthocenter $H$, prove that $HI\perp\mathscr{T}(Na)$.

**Exercise 7.31.** Given $\triangle ABC$, let $l$ be a line in the plane and let $P$ vary on $l$. Let $\alpha$ be the circumconic with perspector $P$. Suppose that, for every fixed $l$, the directions of the axes of $\alpha$ remain constant as $P$ varies on $l$. Prove that $l$ passes through a fixed point, and determine that point.

**Exercise 7.32.** Let $G,H$ be the centroid and orthocenter of $\triangle ABC$. For a point $P$ in the plane, prove that $\mathscr{T}(P)\perp\mathscr{E}\iff P\in\mathfrak{C}(ABCGH)$.

**Exercise 7.33.** For $\triangle ABC$, let $G,H,L$ be its centroid, orthocenter, and Lemoine point, respectively. Suppose a point $X$ satisfies $\mathbf{g}X\in OL$. Let $\mathscr{T}(X)$ meet $AB,AC$ at $U,V$, and put $S=\mathscr{W}^{-1}(HX)$. Prove that $A,U,V,S$ are concyclic.

# 7.4 Cevian Geometry

This section introduces notions in triangle geometry related to Ceva's theorem, including cevapoints and crosspoints.

## 7.4.1 Basic Concepts in Cevian Geometry

### Bianticevian Conics

We encountered bianticevian conics in theorem 6.3.8; recall the definition.

**Definition 7.4.1.** Given $\triangle ABC$ and two points $P,Q$, the six vertices of their two anticevian triangles, together with $P,Q$ (eight points in all), lie on one conic; see figure 7.17. It is called the *bianticevian conic* of $P,Q$ with respect to $\triangle ABC$. We denote it by $\mathscr{B}_{\triangle ABC}(P,Q)$, or simply by $\mathscr{B}(P,Q)$ when the reference triangle is clear.

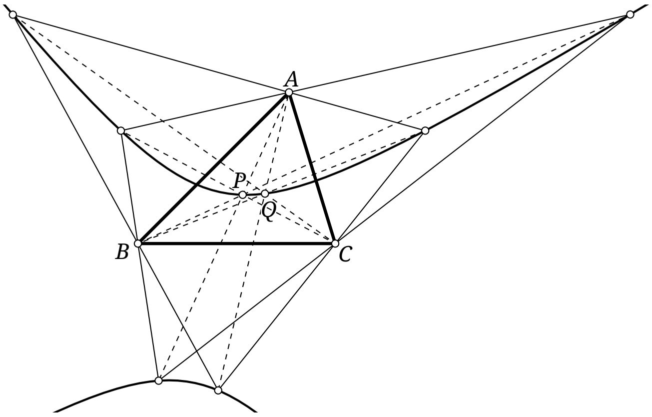


FIGURE 7.17

**Proposition 7.4.2.** *The triangle $\triangle ABC$ is self-polar with respect to each of its bianticevian conics.*

**Proof.** Put $\alpha = \mathscr{B}(P,Q)$. The point $P$ and the three vertices of its anticevian triangle form a complete quadrangle inscribed in $\alpha$. By Brocard's theorem, its diagonal triangle is self-polar with respect to $\alpha$; that diagonal triangle is $\triangle ABC$. ☐

**Theorem 7.4.3.** *Let $\alpha$ be a bianticevian conic of $\triangle ABC$. If $P \in \alpha$, then the anticevian triangle $\triangle P^a P^b P^c$ is inscribed in $\alpha$.*

**Proof.** Suppose $P$ lies on the bianticevian conic $\alpha = \mathscr{B}(X,Y)$, and let $\triangle X^a X^b X^c$ be the anticevian triangle of $X$. Then $\alpha = \mathfrak{C}(X^a X^b X^c XY)$. Consider $\alpha' = \mathscr{B}(X,P)$; then $\alpha' = \mathfrak{C}(X^a X^b X^c XP)$. Since $P \in \alpha$, we must have $\alpha = \alpha'$, as required. ☐

**Theorem 7.4.4.** *Let $\alpha$ be a bianticevian conic of $\triangle ABC$. For any two points $R,S$, $\mathscr{T}(R) = \mathfrak{p}_\alpha(S) \iff \mathscr{T}(S) = \mathfrak{p}_\alpha(R)$.*

**Proof.** We prove the forward implication; the reverse is symmetric. Suppose $\mathscr{T}(R) = \mathfrak{p}(S)$. Choose $X \in \alpha$ and consider $\mathbf{j}^X$. By its pencil construction, $\mathbf{j}^X S \in \mathfrak{p}(S) = \mathscr{T}(R)$. Hence theorem 7.3.7 gives $\mathbf{j}^X R \in \mathscr{T}(S)$. As $X$ varies on $\alpha$, $\mathbf{j}^X(R) \in \mathfrak{p}(R)$ also always holds. Therefore $\mathscr{T}(S) = \mathfrak{p}(R)$. ☐

Here is an application.

**Theorem 7.4.5.** *Given $\triangle ABC$, let the anticevian triangles of $P,Q$ be $\triangle P^a P^b P^c$ and $\triangle Q^a Q^b Q^c$. Set $S_a = P^b Q^b \cap P^c Q^c$, and define $S_b,S_c$ cyclically. Then $\triangle S_a S_b S_c$ is the cevian triangle of $S = \mathscr{T}^{-1}(PQ)$.*

**Proof.** As in figure 7.18, put $\alpha = \mathscr{B}(P,Q)$. Since $P^b P^c \cap Q^b Q^c = A$, Brocard's theorem applied to the complete quadrangle $P^b P^c Q^b Q^c$ on $\alpha$ gives $S_a \in \mathfrak{p}(A) = BC$. Cyclically, $S_b,S_c$ lie on the other two sides of $\triangle ABC$.

Define $S'_a = PQ \cap P^a Q^a$. Since $PP^a \cap QQ^a = A$, Brocard's theorem applied to $PQP^aQ^a$ gives $S'_a \in \mathfrak{p}(A) = BC$. By their definitions, $S_b,S_c \in P^a Q^a$, so $S_b,S_c,S'_a$ are collinear.

Define $S'_b,S'_c$ cyclically. Then $S_c,S_a,S'_b$ are collinear, $S_a,S_b,S'_c$ are collinear, and $S'_b,S'_c$ lie on $CA,AB$, respectively. Since $S'_a,S'_b,S'_c$ lie on $PQ$, the definition of the trilinear polar shows that $\triangle ABC$ and $\triangle S_a S_b S_c$ have perspector $\mathscr{T}^{-1}(PQ)$. ☐

There is likewise a bicevian conic.

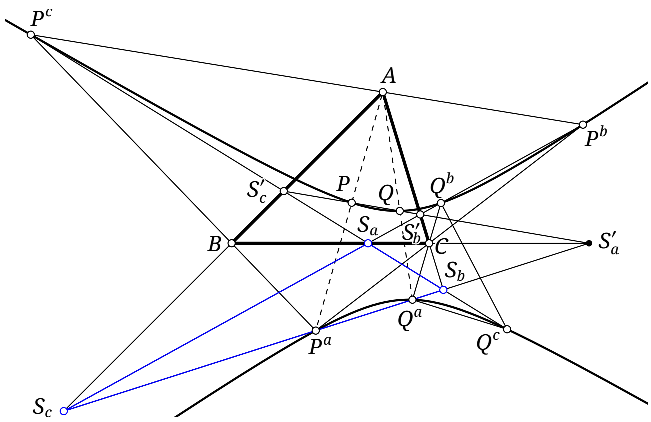


FIGURE 7.18

**Definition 7.4.6.** Given $\triangle ABC$ and points $P, Q$, the six vertices of the cevian triangles of $P, Q$ lie on a conic, called their *bicevian conic* with respect to $\triangle ABC$.

Its existence follows immediately from Ceva's and Carnot's theorems.

### Cevapoints and Crosspoints

The following theorem underlies the definitions that follow.

**Theorem 7.4.7** (Cevian nest theorem)**.** *Suppose $\triangle A_2B_2C_2$ is inscribed in $\triangle A_1B_1C_1$, and $\triangle A_3B_3C_3$ is inscribed in $\triangle A_2B_2C_2$.*[11] *Any two of the following imply the third:*

*(1) $\triangle A_2B_2C_2$ and $\triangle A_3B_3C_3$ have perspector $P$;*
*(2) $\triangle A_3B_3C_3$ and $\triangle A_1B_1C_1$ have perspector $Q$;*
*(3) $\triangle A_1B_1C_1$ and $\triangle A_2B_2C_2$ have perspector $R$.*

When all three conditions hold, the triangles $\triangle A_1B_1C_1, \triangle A_2B_2C_2, \triangle A_3B_3C_3$ form a *Cevian nest*.

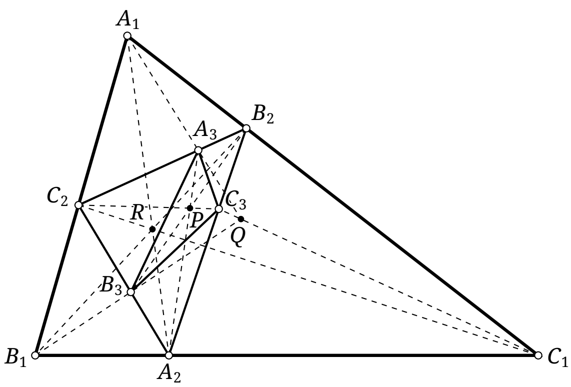


FIGURE 7.19

**Proof.** Applying theorem 6.3.9 to $R, Q$ with respect to $\triangle A_1B_1C_1$ proves $(2),(3) \Rightarrow (1)$. The other two implications follow by the method of coincidence. $\square$

The Cevian nest theorem motivates four binary constructions. The names and symbols below will be used throughout this text.

[11]Here "$\triangle A_2B_2C_2$ is inscribed in $\triangle A_1B_1C_1$" means that $A_2, B_2, C_2$ lie on the lines $B_1C_1, C_1A_1, A_1B_1$, respectively.

**Definition 7.4.8.** Given $\triangle ABC$ and points $P,Q$, define the following.

a. Let $\triangle Q^aQ^bQ^c$ be the anticevian triangle of $Q$, and let $PQ^a,PQ^b,PQ^c$ meet $BC,CA,AB$ at $P_1,P_2,P_3$, respectively. The perspector $R_1$ of $\triangle ABC$ and $\triangle P_1P_2P_3$ is the *cevapoint*, or *Ceva product*, of $P,Q$. We denote it by $\mathrm{Cep}_{\triangle ABC}(P,Q)$ or $(P*Q)_{\triangle ABC}$.
b. Let $\triangle P_aP_bP_c$ be the cevian triangle of $P$, and let $QP_a,QP_b,QP_c$ meet $P_bP_c,P_cP_a,P_aP_b$ at $Q_1,Q_2,Q_3$, respectively. The perspector $R_2$ of $\triangle ABC$ and $\triangle Q_1Q_2Q_3$ is the *crosspoint* of $P,Q$. We denote it by $\mathrm{Crp}_{\triangle ABC}(P,Q)$ or $(P\star Q)_{\triangle ABC}$.
c. The *$Q$-Ceva conjugate* $R_3$ of $P$ is the perspector of the cevian triangle of $P$ and the anticevian triangle of $Q$; equivalently, $R_3$ is called the *Ceva quotient* of $Q$ by $P$. We denote it by $\mathrm{Cej}_{\triangle ABC}(Q,P)$ or $(Q/P)_{\triangle ABC}$.
d. The *$Q$-cross conjugate* $R_4$ of $P$ is the perspector with $\triangle ABC$ of the cevian triangle of $Q$ taken with respect to the cevian triangle of $P$. We denote it by $\mathrm{Crj}_{\triangle ABC}(Q,P)$ or $(Q:P)_{\triangle ABC}$.

The reference triangle may be omitted when no ambiguity results; for example, $R_1=\mathrm{Cep}(P,Q)$ and $R_4=Q:P$.

Although these definitions appear elaborate, the Cevian nest theorem makes their meaning transparent after changing the reference triangle. Suppose $\triangle_1,\triangle_2,\triangle_3$ form a Cevian nest, where $P$ is the perspector of $\triangle_2,\triangle_3$, $Q$ that of $\triangle_3,\triangle_1$, and $R$ that of $\triangle_1,\triangle_2$, as in figure 7.19. Then $P=(Q*R)_{\triangle_2}=(Q\star R)_{\triangle_1}$ and $Q=(R/P)_{\triangle_2}=(P:R)_{\triangle_1}$.

The crosspoint was introduced earlier in section 6.3; that definition plainly agrees with the present one. The preceding theorem 6.3.10 gives the following description.

**Theorem 7.4.9.** *For $\triangle ABC$ and points $P,Q$, the cevapoint $P*Q$ is the intersection of the tangents at $P,Q$ to $\mathscr{B}(P,Q)$. The crosspoint $P\star Q$ is the intersection of the tangents at $P,Q$ to $\mathfrak{C}(ABCPQ)$.*

Thus the cevapoint and crosspoint do not depend on the order of the two arguments.

**Corollary 7.4.10.** *For $\triangle ABC$ and points $P,Q$, $P*Q=Q*P$ and $P\star Q=Q\star P$.*

Ceva conjugates and cross conjugates do depend on order; for example, the Ceva quotient of $P$ by $Q$ must not be confused with the Ceva quotient of $Q$ by $P$.

Changing the reference triangle in the definitions and applying the Cevian nest theorem gives the following relations.

**Theorem 7.4.11.** *Let $\triangle DEF$ be the cevian triangle of $P$ with respect to $\triangle ABC$. For any points $Q,R$:*

*(1) $(P\star Q)_{\triangle ABC}=R$ if and only if $(P*Q)_{\triangle DEF}=R$;*
*(2) $(R/P)_{\triangle ABC}=Q$ if and only if $(R:P)_{\triangle DEF}=Q$;*
*(3) if $(P*Q)_{\triangle ABC}=R$, then $(R/Q)_{\triangle ABC}=P$;*
*(4) if $(P\star Q)_{\triangle ABC}=R$, then $(R:Q)_{\triangle ABC}=P$.*

For a fixed reference triangle, the four operations $\mathrm{Cep}(\cdot,\cdot)$, $\mathrm{Crp}(\cdot,\cdot)$, $\mathrm{Cej}(\cdot,\cdot)$, and $\mathrm{Crj}(\cdot,\cdot)$ may be viewed as binary maps $\mathbb{P}^2\times\mathbb{P}^2\to\mathbb{P}^2$. Their values at points on the sidelines are defined by limits. In other words, they may be regarded as four binary functions.

We give two direct examples.

**Example 7.4.12.** For $\triangle ABC$, suppose $P\star Q=R$. Put $U=BP\cap CQ$ and $V=BQ\cap CP$. Prove that $R\in UV$.

**Solution.** In figure 7.20, apply Pascal's theorem to the hexagon $BPPCQQ$ inscribed in $\mathfrak{C}(ABCPQ)$, interpreting $PP$ and $QQ$ as tangents. By theorem 7.4.9, $PP\cap QQ=R$, and the result follows. $\square$

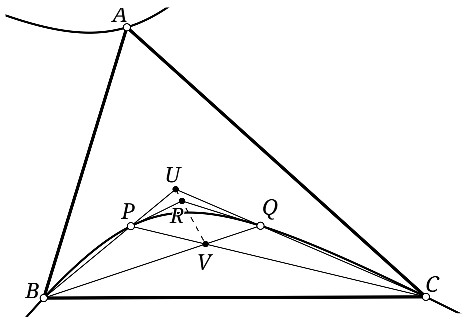


FIGURE 7.20

**Example 7.4.13.** Let the anticevian triangles of $P, Q$ with respect to $\triangle ABC$ be $\triangle P^aP^bP^c$ and $\triangle Q^aQ^bQ^c$. If $R = P * Q$ and $D = \mathscr{T}(R) \cap BC$, prove that $D = P^cQ^b \cap P^bQ^c$.

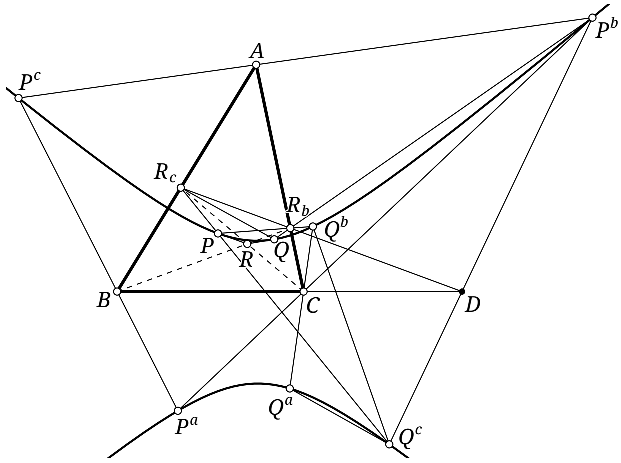


FIGURE 7.21

**Proof.** As in figure 7.21, put $\alpha = \mathscr{B}(P, Q)$ and redefine $D = P^cQ^b \cap P^bQ^c$. We show that $D$ lies on both $BC$ and $\mathscr{T}(R)$.

The complete quadrangle $P^cQ^cP^bQ^b$ is inscribed in $\alpha$, and $A = P^bQ^b \cap P^cQ^c$. Brocard's theorem therefore gives $D \in \mathfrak{p}(A) = BC$.

Put $R_b = PQ^b \cap QP^b$ and $R_c = PQ^c \cap QP^c$. By the Cevian nest theorem, $B, R, R_b$ and $C, R, R_c$ are collinear, and $R_b, R_c$ lie on the corresponding sides of the triangle. Pascal's theorem applied to the hexagon $P^bQP^cQ^bPQ^c$ on $\alpha$ shows that $R_b, R_c, D$ are collinear.

Thus $D = R_bR_c \cap BC$, which is precisely $\mathscr{T}(R) \cap BC$ by the definition of the trilinear polar. $\square$

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## Exercises

**Exercise 7.34.** Prove that the crosspoint of the centroid and orthocenter of a triangle is the Lemoine point.

**Exercise 7.35.** Prove that, for a given triangle, the medial triangle of the cevian triangle of any point in the plane is perspective with the original triangle.

**Exercise 7.36.** Let $P,P'$ be isogonal conjugates with respect to $\triangle ABC$, and let $I_a,I_b,I_c$ be the three corresponding excenters. Put $A' = PI_a \cap BC$, $B' = PI_b \cap CA$, and $C' = PI_c \cap AB$. Prove that the lines $AA', BB', CC', PP'$ are concurrent.

**Exercise 7.37.** Given $\triangle ABC$, let $I$ be its incenter and $I_b, I_c$ its $B$- and $C$-excenters. For a point $P$, put $D = PI_b \cap AC$, $E = PI_c \cap AB$, $F = BD \cap CE$, and $Q = \mathbf{g}F$. Prove that $P,Q,I$ are collinear.

**Exercise 7.38*.** Given $\triangle ABC$ and $U,V \in \mathscr{C}$, prove that the bicevian conic of $U,V$ is a rectangular hyperbola if and only if $Et \in UV$.

### 7.4.2 Basic Properties of Cevian-Geometric Transformations

We first give further characterizations of the cevapoint and crosspoint.

**Theorem 7.4.14.** *Given $\triangle ABC$ and points $P,Q$, let $\alpha$ be the circumconic with perspector $Q$. Then $P * Q = \mathscr{T}^{-1}(\mathfrak{p}_\alpha(P))$.*

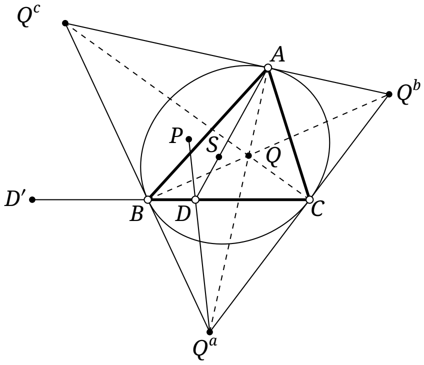


FIGURE 7.22

**Proof.** As in figure 7.22, put $S = P * Q$, and let $\triangle Q^aQ^bQ^c$ be the anticevian triangle of $Q$. By the definition of the perspector, $\alpha$ is an inconic of $\triangle Q^aQ^bQ^c$. By theorem 7.4.11, $S = (P \star Q)_{\triangle Q^aQ^bQ^c}$. The definition of the crosspoint shows that $AS$ and $PQ^a$ meet at a point $D$ of $BC$.

Let $D' = \mathfrak{p}_\alpha(P) \cap BC$. By the polarity principle, $P \in \mathfrak{p}(D')$. Also, $D' \in BC = \mathfrak{p}(Q^a)$, so $Q^a \in \mathfrak{p}(D')$. Hence $\mathfrak{p}(D') = PQ^a$, that is, $D \in \mathfrak{p}(D')$. The harmonic pole–polar property gives $(BC, DD') = -1$. By symmetry and proposition 7.3.2, $\mathscr{T}(S) = \mathfrak{p}(P)$, as required. □

**Corollary 7.4.15.** *Given $\triangle ABC$ and points $P,Q$, let $\alpha$ be the circumconic with perspector $Q$. Then $P/Q = \mathfrak{p}_\alpha^{-1}(\mathscr{T}(P))$.*

**Proof.** If $R = P/Q$, then $P = R * Q = \mathscr{T}^{-1}(\mathfrak{p}_\alpha(R))$, which gives the assertion. □

The crosspoint has an analogous description using an inconic.

**Theorem 7.4.16.** *Let $\alpha$ be an inconic of $\triangle ABC$ with perspector $P$. For any point $Q$ of the plane, $P \star Q = \mathfrak{p}_\alpha^{-1}(\mathscr{T}(Q))$.*

**Proof.** As in figure 7.23, put $R = P \star Q$, and let $\triangle P_aP_bP_c$ be the cevian triangle of $P$; its vertices are the contact points of $\alpha$ with the sidelines. Let $D = \mathscr{T}(Q) \cap BC$, and let the lines $AQ, AD$ meet $P_bP_c$ at $V, U$, respectively. By proposition 7.3.2, $-1 = A(BC, QD) = (P_cP_b, VU)$. The harmonic pole–polar property therefore gives $U \in \mathfrak{p}(V)$.

Now $V \in P_bP_c = \mathfrak{p}(A)$, so the polarity principle gives $A \in \mathfrak{p}(V)$. Together with $U \in \mathfrak{p}(V)$, this yields $\mathfrak{p}(V) = AU = AD$, that is, $D \in \mathfrak{p}(V)$, and therefore $V \in \mathfrak{p}(D)$. Also, $D \in BC = \mathfrak{p}(P_a)$, so $P_a \in \mathfrak{p}(D)$ and hence $\mathfrak{p}(D) = VP_a$. The Cevian nest theorem and the description of the crosspoint give $R \in VP_a$, hence $D \in \mathfrak{p}(R)$.

Repeating the argument with the other two intersections of $\mathscr{T}(Q)$ and the sidelines proves the other two collinearities, and finally gives $\mathfrak{p}(R) = \mathscr{T}(Q)$. □

**Corollary 7.4.17.** *Let $\alpha$ be an inconic of $\triangle ABC$ with perspector $Q$. Then for every point $P$ in the plane, $P : Q = \mathscr{T}^{-1}(\mathfrak{p}_\alpha(P))$.*

**Proof.** If $R = P : Q$, then $P = Q \star R = \mathfrak{p}_\alpha^{-1}(\mathscr{T}(R))$, which gives the assertion. □

**Corollary 7.4.18.** *Let $G$ be the centroid of $\triangle ABC$, and let $\alpha$ be an inconic with perspector $P$. Then the center of $\alpha$ is $P \star G$.*

**Proof.** Set $Q = G$ in theorem 7.4.16. Since $\mathscr{T}(Q) = l_\infty$, the point $P \star G = \mathfrak{p}^{-1}(l_\infty)$ is the center of $\alpha$. □

We next characterize several notable points by these binary operations.

**Theorem 7.4.19.** *For $\triangle ABC$ with centroid $G$ and any point $P$, $P \star G = \mathbf{ct}P$.* The point $\mathbf{ct}P$ is sometimes called the *isotomcomplement* of $P$. When using notation such as $\mathbf{t}, \mathbf{c}, \mathbf{g}$, we omit the composition sign for convenience; for example, $\mathbf{gig}$ denotes $\mathbf{g} \circ \mathbf{i} \circ \mathbf{g}$, that is, $\mathbf{gig}P = \mathbf{g}(\mathbf{i}(\mathbf{g}P))$.

**Proof.** As in figure 7.24, put $P' = \mathbf{t}P$ and $Q = \mathbf{c}P'$. We prove $Q = P \star G$. Let $AP, AP'$ meet $BC$ at $D, D'$, let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and put $F = M_bM_c \cap AD$.

Since $P' = \mathbf{t}P$, $M_a$ is the midpoint of $DD'$. The point $F$ is the midpoint of $AD$, so $FM_a \parallel AD'$. The centroid property gives $\overline{AG} = 2\overline{GM_a}$, while $Q = \mathbf{c}P'$ gives $\overline{GP'} = 2\overline{QG}$. Hence $QM_a \parallel AD'$.

Thus $F, Q, M_a$ are collinear. The two cyclic relations follow in the same way, and the definition of the crosspoint together with the Cevian nest theorem yields $Q = P \star G$. □

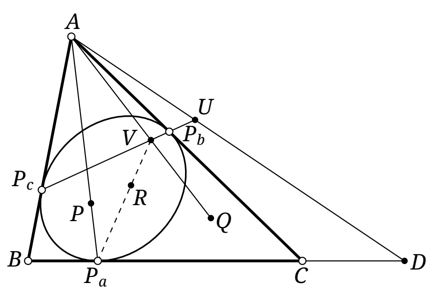


FIGURE 7.23

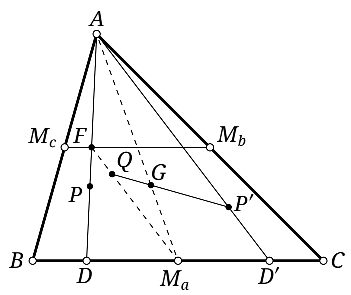


FIGURE 7.24

**Corollary 7.4.20.** *Let $G$ be the centroid of $\triangle ABC$, and let $\alpha$ be an inconic with perspector $P$. Then the center of $\alpha$ is $\mathbf{ct}P$.*

**Proof.** Combine corollary 7.4.18 and theorem 7.4.19. □

**Corollary 7.4.21.** *Given $\triangle ABC$ and a point $P$, there is a unique conic $\alpha$ centered at $P$ and tangent to the three sidelines of $\triangle ABC$.*

**Proof.** By corollary 7.4.20, the perspector of $\alpha$ is $\mathbf{ta}P$, which is uniquely determined. Now apply proposition 7.3.14. ☐

There is also a proof of corollary 7.4.20 that does not use crosspoints.

**Alternative proof of corollary 7.4.20.** Apply an affine transformation taking $\alpha$ to an incircle. If $\alpha$ is a hyperbola, a complex affine transformation may be used; the parabolic case follows as a limit of the central-conic cases. The perspector becomes the Gergonne point $Ge$, and the center becomes the incenter $I$. Affine transformations preserve isotomic conjugacy, the complement relation, and the anticomplement relation. Since $\mathbf{t}Ge = Na$ and $\mathbf{c}Na = I$, the assertion holds before the transformation as well. ☐

The isotomcomplement has another useful property.

**Theorem 7.4.22.** *For $\triangle ABC$ and a point $P$, the point $P \star (\mathbf{ct}P)$ is the centroid of the cevian triangle of $P$.*

**Proof.** Let $\triangle P_aP_bP_c$ be the cevian triangle of $P$. By the definition of a crosspoint, it suffices to prove that the cevian triangle of the centroid of $\triangle P_aP_bP_c$, taken with respect to $\triangle P_aP_bP_c$ (that is, the medial triangle of $\triangle P_aP_bP_c$), is perspective with $\triangle ABC$, with perspector $\mathbf{ct}P$. Put $Q = \mathbf{ct}P$.

Let $K$ be the midpoint of $P_bP_c$. It is enough to show that $A,K,Q$ are collinear; the two cyclic relations then give perspectivity. Let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $V,U$ be the midpoints of $BP_b, CP_c$, respectively. Then $U \in M_aM_b$ and $V \in M_aM_c$, as in figure 7.25. By theorem 7.4.19 and the definition of the crosspoint, noting that $\triangle M_aM_bM_c$ is the cevian triangle of the centroid of $\triangle ABC$, we have $Q = M_bV \cap M_cU$.

Let $E,F$ be the midpoints of $AP_c, AP_b$. Since $F,K,V$ are the midpoints of $AP_b, P_bP_c, BP_b$, respectively, $K \in VF$. Similarly, $K \in EU$, so $K = EU \cap FV$.

Since $E,U$ are the midpoints of $AP_c, CP_c$, respectively, $EU \parallel AC$ and $M_aM_c \parallel AC$, so the three lines $AC, EU, M_aM_c$ pass through one point at infinity. Likewise $AB, FV, M_aM_b$ pass through another. The dual of Pappus's theorem gives the concurrence of $AK, M_cU, M_bV$.[12] Thus $A,K,Q$ are collinear. ☐

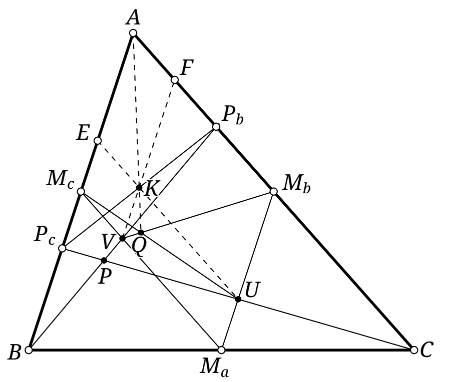


FIGURE 7.25

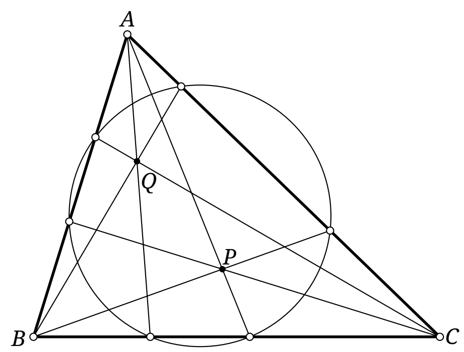


FIGURE 7.26

The preceding theorem 7.4.22 yields a useful characterization of cyclocevian conjugacy, a notion already encountered in exercise 4.38.

**Definition 7.4.23.** Given $\triangle ABC$ and two distinct points $P,Q$, if they have the same cevian circle, then $P,Q$ are called *cyclocevian conjugates* with respect to $\triangle ABC$; see figure 7.26.

For every $P$, existence and uniqueness of its cyclocevian conjugate follow directly from Ceva's and Carnot's theorems.

[12]More explicitly, put $AB = a$, $VF = b$, $M_aM_b = c$, and $AC = b'$, $EU = a'$, $M_aM_c = c'$. Then $AK = (a \cap b')(b \cap a')$, $M_cU = (a \cap c')(c \cap a')$, and $M_bV = (c \cap b')(b \cap c')$.

**Theorem 7.4.24** (Grinberg's theorem)**.** *Cyclocevian conjugation with respect to a triangle is the transformation* **tagct**.

**Proof.** Let $P, Q$ be cyclocevian conjugates with respect to $\triangle ABC$. It is enough to prove that $\mathbf{ct}P$ and $\mathbf{ct}Q$ are isogonal conjugates. As in figure 7.27, write these two points as $P', Q'$, and put $X = AP' \cap P_bP_c$ and $Y = AQ' \cap Q_bQ_c$. By theorem 7.4.22, $X, Y$ are the midpoints of $P_bP_c, Q_bQ_c$, respectively.

Because $P_b, P_c, Q_b, Q_c$ are concyclic, $\triangle AP_bP_c \eqsim \triangle AQ_cQ_b$. The points $X, Y$ correspond under this similarity, so $\measuredangle XAP_b = -\measuredangle YAQ_c$. Thus $AP'$ and $AQ'$ are isogonal in $\angle A$. The two cyclic relations follow in the same way, and hence $Q' = \mathbf{g}P'$. $\square$

**Example 7.4.25.** For a triangle, the cevian circles of its centroid $G$ and orthocenter $H$ are both $\mathscr{N}$, so $G, H$ are cyclocevian conjugates. Moreover, $\mathbf{ct}G = G$, while proposition 5.3.15 gives $\mathbf{ct}H = \mathbf{c}Al = L$. Grinberg's theorem therefore gives $L = \mathbf{g}G$, as expected.

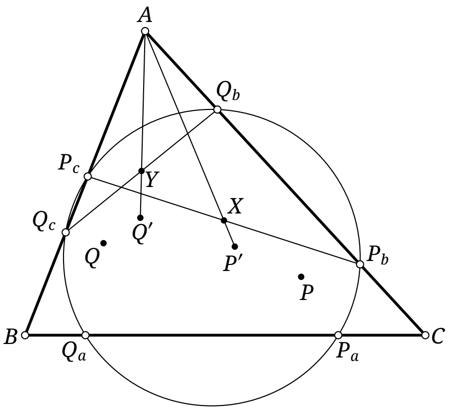


FIGURE 7.27

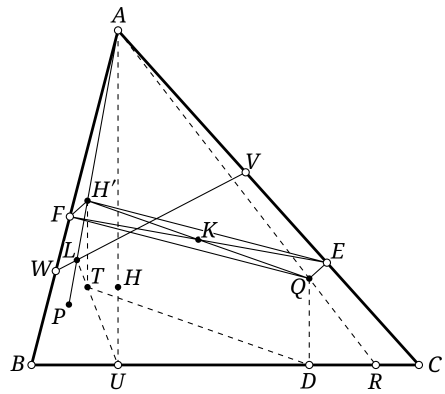


FIGURE 7.28

We next give a result involving the orthocenter.

**Theorem 7.4.26.** *Let $H$ be the orthocenter of $\triangle ABC$. For any point $Q$, the anticomplement of $Q$ with respect to its pedal triangle is $H \star \mathbf{g}Q$, where the crosspoint is taken with respect to $\triangle ABC$.*

**Proof.** As in figure 7.28, let $\triangle DEF$ be the pedal triangle of $Q$, put $T = \mathbf{a}_{\triangle DEF}Q$ and $P = \mathbf{g}Q$, and prove that $T = H \star P$. Let $\triangle UVW$ be the orthic triangle, put $L = AP \cap WV$ and $R = AQ \cap BC$, let $K$ be the midpoint of $EF$, and let $H'$ be the orthocenter of $\triangle AEF$.

By proposition 4.4.7, $H' \in AP$. Since $AQ$ is a diameter of the circumcircle of $\triangle AEF$, claim 0.5.17 gives $\frac{AH'}{AQ} = \cos A = \frac{AV}{AB}$. Together with $\measuredangle WAH' = -\measuredangle CAQ$, this gives $\triangle AVW \cup H' \eqsim \triangle ABC \cup Q$.[13] Hence $\frac{LH'}{LA} = \frac{RQ}{RA}$.

The lines $QE, FH'$ are perpendicular to $AC$, and $QF, EH'$ to $AB$; therefore $FH'EQ$ is a parallelogram and $\overline{H'Q} = 2\overline{KQ}$. Since $K = (\mathbf{c}D)_{\triangle DEF}$ and $Q = (\mathbf{c}T)_{\triangle DEF}$, the segment $DT$ is parallel to $KQ$ and twice as long. Therefore $H'Q$ is parallel and equal in length to $DT$.

Consequently $H'T$ is parallel and equal in length to $QD$, and $QD \parallel AU$. Hence $\frac{H'T}{AU} = \frac{QD}{AU} = \frac{RQ}{RA} = \frac{LH'}{LA}$, which proves that $U, T, L$ are collinear. The two cyclic relations follow in the same way, and the definition of the crosspoint together with the Cevian nest theorem completes the proof. $\square$

[13] Here the symbol $\cup$ means that, under the opposite similarity determined by $\triangle AVW$ and $\triangle ABC$, the points $H'$ and $Q$ also correspond.

## Exercises

**Exercise 7.39.** Find the cyclocevian conjugate of the Gergonne point $Ge$ of a triangle.

**Exercise 7.40.** Prove the previously unproved proposition 5.7.17, namely $\mathbf{c}Br_3 = Br_{\mathrm{m}}$.

**Exercise 7.41.** Let $H, I$ be the orthocenter and incenter of $\triangle ABC$, and let $X, Y, Z$ be the orthocenters of $\triangle BCI, \triangle CAI, \triangle ABI$, respectively. Prove that $\mathbf{g}_{\triangle XYZ} H \in HI$.

**Exercise 7.42.** Given $\triangle ABC$ and points $P, Q$, prove that $(Q \star P) \star P = \mathscr{T}^{-1}_{\triangle_P}(\mathscr{T}(Q))$, where $\triangle_P$ is the cevian triangle of $P$.

### 7.4.3 Further Properties of Cevian-Geometric Transformations

We now study their relation to isoconjugation and the images they produce.

**Theorem 7.4.27.** *Given $\triangle ABC$ and an isoconjugation $\mathbf{j}$ with respect to it, the following identities hold:*

*(1)* $\mathbf{j}(P \star Q) = \mathbf{j}P * \mathbf{j}Q$;
*(2)* $\mathbf{j}(P * Q) = \mathbf{j}P \star \mathbf{j}Q$;
*(3)* $\mathbf{j}(R \diagup P) = \mathbf{j}R : \mathbf{j}P$;
*(4)* $\mathbf{j}(R : P) = \mathbf{j}R \diagup \mathbf{j}P$.

**Proof.** We prove the first identity; the other three follow from it and are left as exercises. Put $R = P \star Q$ and $R' = \mathbf{j}P * \mathbf{j}Q$. We show that $R = \mathbf{j}R'$.

Let $\alpha = \mathscr{B}(\mathbf{j}P, \mathbf{j}Q)$ and $\beta = \mathfrak{C}(ABCPQ)$. As in figure 7.29, put $L = \mathbf{j}P\mathbf{j}Q \cap BC$. By proposition 7.4.2, $\mathfrak{p}_\alpha(A) = BC$, while theorem 7.4.9 gives $\mathfrak{p}_\alpha(R') = \mathbf{j}P\mathbf{j}Q$. The polarity principle therefore yields $\mathfrak{p}_\alpha(L) = AR'$.

The harmonic pole–polar property gives $A(LR', \mathbf{j}Q\mathbf{j}P) = -1$. Applying the isoconjugation gives $A(A\mathbf{j}R', QP) = -1$, where $\mathbf{j}L = A$ and $AA$ is understood as $\mathfrak{t}_\beta(A)$, by theorem 7.2.17.

Let $T$ be the other intersection of $A\mathbf{j}R'$ with $\beta$. Then $A, T, Q, P$ form a harmonic quadratic point range on $\beta$. By proposition 6.2.5, $\mathfrak{p}^{-1}_\beta(PQ) \in AT$, while theorem 7.4.9 gives $R = \mathfrak{p}^{-1}_\beta(PQ)$. Thus $A, R, \mathbf{j}R', T$ are collinear. Similarly, $B, R, \mathbf{j}R'$ are collinear, so $R = \mathbf{j}R'$. $\square$

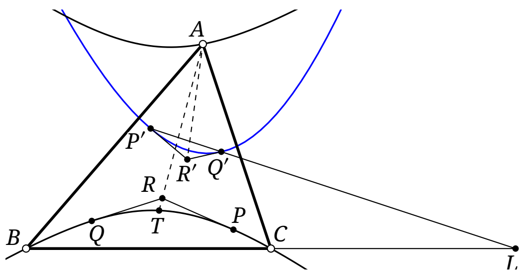


FIGURE 7.29

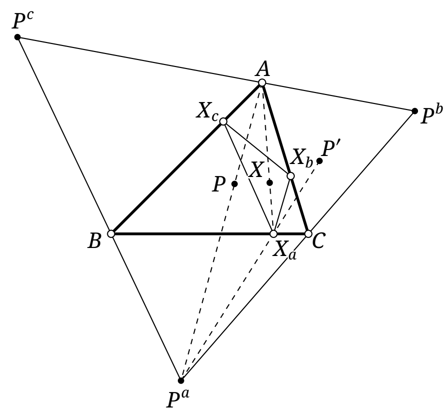


FIGURE 7.30

**Theorem 7.4.28.** *Given $\triangle ABC$ and a point $X$, let $\triangle X_aX_bX_c$ be the cevian triangle of $X$. Then* $\mathrm{Cej}(X,\cdot)=\mathbf{j}^X_{\triangle X_aX_bX_c}$. Here the symbol $\cdot$ in $\mathrm{Cej}(X,\cdot)$ is a placeholder: applying $\mathrm{Cej}(X,\cdot)$ to a point $P$ gives $\mathrm{Cej}(X,P)$. Thus, for every point $P$, $\mathrm{Cej}(X,P)=\mathbf{j}^X_{\triangle X_aX_bX_c}P$.

**Proof.** As in figure 7.30, for a point $P$, put $P'=X/P$. The anticevian triangle of $X$ with respect to $\triangle X_aX_bX_c$ is $\triangle ABC$. By symmetry and theorem 7.2.3, it is enough to prove $X_a(PP',AB)=-1$.

Let $\triangle P^aP^bP^c$ be the anticevian triangle of $P$ with respect to $\triangle ABC$. Since $AX\cap BC=X_a$, the Cevian nest theorem and the definition of the $X$-Ceva conjugate $P'$ of $P$ show that $P',X_a,P^a$ are collinear. Hence $X_a(PP',AB)=X_a(PP^a,AB)=-1$, where the last equality follows from theorem 4.3.4. □

**Corollary 7.4.29.** *Given $\triangle ABC$ and points $X,P$, put $Q=X/P$ and $\mathscr{W}=\mathrm{pen}(ABCX)$. Then $P,Q$ are conjugate with respect to every conic in $\mathscr{W}$.*

The isogonal and isotomic special cases give the following.

**Corollary 7.4.30.** *Let $H$ be the orthocenter of $\triangle ABC$ and $\triangle H_aH_bH_c$ its orthic triangle. Then* $\mathrm{Cej}(H,\cdot)=\mathbf{g}_{\triangle H_aH_bH_c}$.

**Corollary 7.4.31.** *Let $M$ be the centroid of $\triangle ABC$ and $\triangle M_aM_bM_c$ its medial triangle. Then* $\mathrm{Cej}(M,\cdot)=\mathbf{t}_{\triangle M_aM_bM_c}$.

Here is an application of the preceding theorem.

**Theorem 7.4.32.** *Given $\triangle ABC$ and points $X,P$:*
*(1) $P$, $X/P$, and $X\star P$ are collinear;*
*(2) $P$, $P/X$, and $X:P$ are collinear.*

**Proof.** For (1), let $\triangle X_aX_bX_c$ be the cevian triangle of $X$, put $Q=X/P$ and $R=X\star P$. By theorem 7.4.28, $Q=\mathbf{j}^X_{\triangle X_aX_bX_c}P$, and this isoconjugation is induced by the pencil $\mathrm{pen}(ABCX)$. Put $\alpha=\mathfrak{C}(ABCXP)$. By the pencil definition, $Q\in\mathfrak{p}_\alpha(P)$, so $\mathfrak{t}_\alpha(P)=PQ$. By theorem 7.4.9, $PR=\mathfrak{t}_\alpha(P)$; hence $P,Q,R$ are collinear.

For (2), let $\triangle P_aP_bP_c$ be the cevian triangle of $P$, put $Q=P/X$ and $R=X:P$. By theorem 7.4.28, $Q=\mathbf{j}^P_{\triangle P_aP_bP_c}X$. Put $\alpha=\mathfrak{C}(ABCPR)$. The same argument gives $Q\in\mathfrak{p}_\alpha(X)$, while theorem 7.4.9 gives $\mathfrak{p}_\alpha(X)=PR$. Thus $P,Q,R$ are collinear. □

We next study images of lines under Cevian-geometric transformations. For a point $X$ and a line $l$, write $X/l$ or $\mathrm{Cej}(X,l)$ for the image of $l$ under the $X$-Ceva-conjugate map $\mathrm{Cej}(X,\cdot)$; explicitly, it is $\{P\,|\,P=X/Q, Q\in l\}$. The other operation symbols are extended to sets in the same way.

**Theorem 7.4.33.** *Given $\triangle ABC$, a point $X$, and a line $l$:*
*(1) $X/l$ is a conic through the three vertices of the cevian triangle of $X$;*
*(2) $X*l$ is a conic through $A,B,C$;*
*(3) $X\star l$ is a conic through the three vertices of the cevian triangle of $X$;*
*(4) $l/X$ is a conic through the three vertices of the anticevian triangle of $X$;*
*(5) $l:X$ is a conic through $A,B,C$.*
*When $X\in l$, all these loci also pass through $X$.*

**Proof.** When $X\in l$, it is clear that all these loci pass through $X$, since $X*X=X/X=X\star X=X:X=X$.

Part (1) follows immediately from corollary 7.4.29 and theorem 7.2.15.

For (2), as in figure 7.31, let $P$ vary on $l$, put $R=X*P$ and $R_a=AR\cap BC$, and let $\triangle X^aX^bX^c$ be the anticevian triangle of $X$. The definition of the cevapoint and the Cevian nest theorem show that $X^a,R_a,P$ are collinear. As $P$ varies, $A[R]\to[R_a]\to X^a[R_a]\to l[P]$ is a chain of projective correspondences, that is, $A[R]\barwedge l[P]$. Similarly, $B[R]\barwedge l[P]$. Thus $A[R]\to B[R]$ gives a projectivity from the

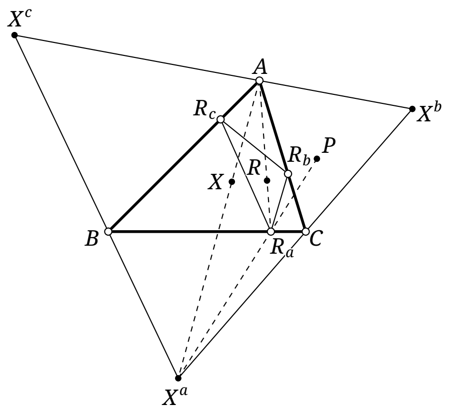


FIGURE 7.31

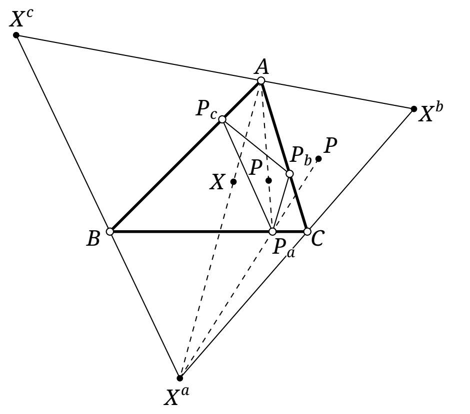


FIGURE 7.32

pencil at $A$ to the pencil at $B$, so the projective definition of a conic shows that the locus of $R$ is a conic through $A,B$. Cyclically, it also passes through $C$.

Part (3) follows by applying part (2) to the cevian triangle of $X$ and using theorem 7.4.11.

For (4), as in figure 7.32, let $P$ vary on $l$, put $Q = P/X$ and $P_a = AP \cap BC$, and let $\triangle X^aX^bX^c$ be the anticevian triangle of $X$. The definition of the $P$-Ceva conjugate $Q$ of $X$ and the Cevian nest theorem show that $X^a, P_a, Q$ are collinear. As $P$ varies, $X^a[Q] \barwedge l[P] \barwedge X^b[Q]$. Hence the locus of $Q$ is a conic through $X^a, X^b$, and cyclically through $X^c$ as well.

Part (5) follows by applying part (4) to the cevian triangle of $X$ and using theorem 7.4.11. □

Here is an application.

**Theorem 7.4.34.** *Given $\triangle ABC$, an isoconjugation $\mathbf{j}$, and points $P,X$, the following are equivalent:*
*(1) $P, \mathbf{j}P, X$ are collinear;*
*(2) $P, X/P, \mathbf{j}X$ are collinear;*
*(3) $X/P$, $\mathbf{j}(X/P)$, and $X$ are collinear.*

**Proof.** First prove the equivalence of (1) and (2). Write $\mathbf{j} = \mathbf{j}^U$, and let $\triangle U^aU^bU^c := \triangle$ be the anticevian triangle of $U$. By theorems 7.4.11 and 7.4.28, the desired equivalence says that $P, (U/P)_\triangle, X$ are collinear if and only if $P, (X:P)_\triangle, (U/X)_\triangle$ are collinear.

Apply $\mathrm{Cep}(P,\cdot)$ and use theorem 7.4.33. The points $P, (U/P)_\triangle, X$ are collinear if and only if $U^a, U^b, U^c, P, U, (P*X)_\triangle$ lie on one conic. Next apply $\mathrm{Cej}(\cdot,X)$ and use theorem 7.4.33 again. The points $U^a, U^b, U^c, P, U, (P*X)_\triangle$ lie on one conic if and only if $(P/X)_\triangle, (U/X)_\triangle, P$ are collinear.

By theorem 7.4.32, the points $(P/X)_\triangle, (X:P)_\triangle, P$ are always collinear. Thus $P, (X:P)_\triangle, (U/X)_\triangle$ are collinear if and only if $(P/X)_\triangle, (U/X)_\triangle, P$ are collinear. We conclude that $P, (U/P)_\triangle, X$ are collinear if and only if $P, (X:P)_\triangle, (U/X)_\triangle$ are collinear, as required.

Replace $P$ by $X/P$ in that equivalence to obtain the equivalence of (2) and (3). □

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## Exercises

**Exercise 7.43.** Prove parts (2)–(4) of theorem 7.4.27.

**Exercise 7.44.** In this text, call the centroid of the orthic triangle of a triangle its *orthocentroid point*, denoted by $Gh$. Given $\triangle ABC$, with Lemoine point $L$ and orthocenter $H$, prove that
(1) $Gh = H \star L$;
(2) $Gh, Ht, L$ are collinear.

**Exercise 7.45.** Let $I,O$ be the incenter and circumcenter of $\triangle ABC$, and let $K$ be the isogonal conjugate of $I$ with respect to the orthic triangle of $\triangle ABC$. Prove that $K \in OI$.

**Exercise 7.46.** Given $\triangle ABC$ and a point $P$, let $\triangle'$ be the anticevian triangle of $P$. If a conic $\alpha$ passes through $P$ and the three vertices of $\triangle'$, prove that $P * \alpha = \mathrm{t}_\alpha(P)$.

**Exercise 7.47.** Given $\triangle ABC$ with circumcenter $O$, orthocenter $H$, and incenter $I$, put $\alpha = \mathfrak{C}(HIABC)$, and let $S$ be the second intersection of $\alpha$ with $OH$. Prove that $S = I * O$.

## 7.5 Barycentric Products

We now introduce barycentric products. Barycentric coordinates serve here as labels for points, and the treatment in this section remains geometric.

### 7.5.1 Basic Notions of Barycentric Coordinates

**Definition 7.5.1.** Given $\triangle ABC$ and a point $P$ of the projective plane, the ratio $u:v:w = [\triangle PBC]:[\triangle PCA]:[\triangle PAB]$ determines $P$ uniquely.[14] The homogeneous triple $[u:v:w]$ is called the *barycentric coordinates* of $P$ with respect to $\triangle ABC$. If $u+v+w \neq 0$, put $S = u+v+w$. The triple $\left(\frac{u}{S}, \frac{v}{S}, \frac{w}{S}\right)$ is called the *normalized barycentric coordinates* of $P$, or simply its barycentric coordinates.

**Remark.** In homogeneous barycentric coordinates $[u:v:w]$, the values $u,v,w$ are not unique; the same ratio $u:v:w$ represents the same point. In normalized barycentric coordinates $(u,v,w)$, however, a triple of values $u,v,w$ corresponds one-to-one to a point $P$, since $u+v+w=1$. Here $P$ must be finite, as the following theorem shows. Barycentric coordinates of points on a sideline or at infinity are obtained in the limiting sense.

Here are some basic examples.

**Theorem 7.5.2.** *For $\triangle ABC$:*

*(1) the barycentric coordinates of $A,B,C$ are respectively $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$;*
*(2) points of $BC,CA,AB$ have barycentric coordinates of the respective forms $[0:v:w]$, $[u:0:w]$, and $[u:v:0]$;*
*(3) the centroid $G$ has barycentric coordinates $[1:1:1]$;*
*(4) $[u:v:w] \in l_\infty$ if and only if $u+v+w=0$.*

**Proof.** Assertions (1), (2), and (3) are immediate; we only explain (4). For every point $P$, $[\triangle PBC] + [\triangle PCA] + [\triangle PAB] = [\triangle ABC]$. When $P$ tends to infinity, $[\triangle ABC]$ remains finite while the three areas on the left become infinite. To obtain finite values of $u,v,w$, we may take them to be $1, \frac{[\triangle PCA]}{[\triangle PBC]}, \frac{[\triangle PAB]}{[\triangle PBC]}$, respectively.[15] Then $u+v+w = \frac{[\triangle ABC]}{[\triangle PBC]} \to 0$. Conversely, if $P$ is finite, then $u+v+w \neq 0$. □

**Definition 7.5.3.** Given $\triangle ABC$ and barycentric coordinates $P[a:b:c]$, $Q[\alpha:\beta:\gamma]$, their *barycentric product* $(P \times Q)_{\triangle ABC}$, usually abbreviated to $P \times Q$, is defined to be $[a\alpha : b\beta : c\gamma]$. Their *barycentric quotient* $(P \div Q)_{\triangle ABC}$, usually abbreviated to $P \div Q$, is defined to be $\left[\frac{a}{\alpha} : \frac{b}{\beta} : \frac{c}{\gamma}\right]$.

The definitions are homogeneous and hence well defined. Clearly $P \times Q = Q \times P$, whereas in general $P \div Q \neq Q \div P$. More precisely, these operations act on barycentric coordinate triples, not only

[14] This direct fact is left to the reader.

[15] These ratios of areas can have finite values.

on points; the distinction will be made explicit below. Since a point has a corresponding barycentric coordinate triple, we freely use $P\times Q$ and $P\div Q$ for the points represented by the resulting triples.

The definitions give the following elementary but useful properties.

**Theorem 7.5.4.** *For a triangle and points $P,Q$, if $R=P\times Q$, then $P=R\div Q$ and $Q=R\div P$.*

**Proof.** This is immediate from the definitions. □

**Theorem 7.5.5.** *Given $\triangle ABC$, let $\mathbb{P}^2_\times := \mathbb{P}^2 \setminus (AB\cup BC\cup CA)$. The set $\mathbb{P}^2_\times$ is an abelian group under the barycentric product $\times$, that is:*

*(1) if $P,Q\in\mathbb{P}^2_\times$, then $P\times Q\in\mathbb{P}^2_\times$;*
*(2) if $P,Q,R\in\mathbb{P}^2_\times$, then $(P\times Q)\times R=P\times(Q\times R)$;*
*(3) there is an identity $\mathbf{1}$ such that, for $P\in\mathbb{P}^2_\times$, $P\times\mathbf{1}=\mathbf{1}\times P=P$; this identity is the centroid $G$.* In barycentric calculations we often write $\mathbf{1}$ for $G$.
*(4) every $P\in\mathbb{P}^2_\times$ has a point $P'$ satisfying $P\times P'=P'\times P=\mathbf{1}$.* We denote this point by $P^{-1}$.
*(5) if $P,Q\in\mathbb{P}^2_\times$, then $P\times Q=Q\times P$.*

**Proof.** All assertions follow directly from the coordinatewise definition. The restriction to $\mathbb{P}^2_\times$, rather than the whole projective plane, excludes points $[u:v:w]$ for which one of $u,v,w$ is zero. □

Thus $\times$ and $\div$ behave like ordinary multiplication and division, with the centroid $G$ playing the role of the number 1. We can therefore use the laws of multiplication and division to handle barycentric products and quotients. Points on a sideline behave in some respects like 0, so expressions involving them require separate care.

The next theorem gives the geometric meaning of the barycentric product.

**Theorem 7.5.6.** *For $\triangle ABC$ and points $P,Q$, if $P\times P'=Q\times Q'$, then $(P,P')$ and $(Q,Q')$ are corresponding pairs under one isoconjugation.*

**Proof.** It is enough to show that, under a fixed isoconjugation $\mathbf{j}$, the product $P\times P'$, where $P'=\mathbf{j}P$, is constant. Equivalently, the ratio

$$([\triangle PBC]\cdot[\triangle P'BC]):([\triangle PCA]\cdot[\triangle P'CA]):([\triangle PAB]\cdot[\triangle P'AB])$$

must be fixed. By cyclic symmetry it suffices to prove $\frac{[\triangle PAB]\cdot[\triangle P'AB]}{[\triangle PCA]\cdot[\triangle P'CA]}=\text{const}$. Put $P_a=AP\cap BC$ and $P'_a=AP'\cap BC$. By lemma 0.1.13, the left side equals $\frac{\overline{BP_a}}{\overline{CP_a}}\cdot\frac{\overline{BP'_a}}{\overline{CP'_a}}$. Let $\infty_a$ be the point at infinity on $BC$. Although $\mathbf{j}\infty_a=A$, the line $A\mathbf{j}\infty_a$ in the calculation below is understood as the limit $\mathbf{j}(A\infty_a)$, as permitted by corollary 7.2.18. Then

$$\begin{aligned}\frac{\overline{BP_a}}{\overline{CP_a}}\cdot\frac{\overline{BP'_a}}{\overline{CP'_a}} &= A(BC,P\infty_a)\cdot A(BC,P'\infty_a)=A(BC,P\infty_a)\cdot A(CB,P\mathbf{j}\infty_a)\\ &= A(BC,P\infty_a)\cdot A(B,C;\mathbf{j}\infty_a,P)=A(B,C;\mathbf{j}\infty_a,\infty_a)=\text{const}.\end{aligned}$$

The second equality uses corollary 7.2.8, and the assertion follows. □

**Corollary 7.5.7.** *Given $\triangle ABC$, if $P,Q$ are a corresponding pair under an isoconjugation $\mathbf{j}$, then, as an operator on points, $\mathbf{j}=P\times Q\div$. In particular, if $X$ is a fixed point, then $\mathbf{j}=X^2\div$.*[16]

[16] Two explanations are useful here. First, powers have their usual meaning: $X^2=X\times X$, $X^3=X\times X\times X$, and $X^{-2}=\mathbf{1}\div X\div X=\mathbf{1}\div X^2$. Second, $P\times Q\div$ is an operator: applied to a point $R$, it gives $P\times Q\div R$. Thus $\mathbf{j}R=P\times Q\div R$ for every $R$ may formally be written as $\mathbf{j}=P\times Q\div$.

**Corollary 7.5.8.** *Given $\triangle ABC$, as operators on points,* $\mathbf{t} = \mathbf{1}^2 \div = \mathbf{1}\div$. *If $I$ is the incenter, then* $\mathbf{g} = I^2 \div$.

**Theorem 7.5.9.** *Given $\triangle ABC$, let $f$ be a projective transformation fixing $A,B,C$. There is a barycentric coordinate $\Phi$ such that, as an operator on points, $f = \Phi\times$.*

**Proof.** Suppose a given point $P$ is mapped to $P'$, and an arbitrary point $Q$ of the plane is mapped to $Q'$. Preservation of cross-ratios gives $A(PQ,BC) = A(P'Q',BC) = A(Q'P',CB)$. By the definition of isoconjugation, $(P,Q')$ and $(Q,P')$ are corresponding pairs under one isoconjugation. Hence $P \times Q' = Q \times P'$, and therefore $Q' = (P' \div P) \times Q$. Take $\Phi = P' \div P$. □

We now refine the preceding definitions. Barycentric coordinates can carry a "dimension," analogous to that of a physical quantity; in this text we call this their *weight*. An ordinary point has barycentric coordinates of weight 1, as does the identity $\mathbf{1}$, which normally represents the centroid $G$. Thus the operations above are, more precisely, operations on barycentric coordinates; a point is represented by barycentric coordinates of weight 1.

If $\Phi,\Psi$ have weights $k,l$, respectively, we assign weights $k+l$ and $k-l$ to $\Phi \times \Psi$ and $\Phi \div \Psi$. Accordingly, an expression written earlier as $R = P \times Q$ actually gives $R$ weight 2, so it is not a point. To balance the weights one may write $R = P \times Q \div \mathbf{1}$. Because $\mathbf{1}$ is the identity, it has no effect on the numerical values of the barycentric coordinates and may therefore be omitted. Writing it explicitly helps clarify the meaning of the expression: in this example, after inserting $\mathbf{1}$, we can read directly that $R$ is the image of the centroid $G$ under the isoconjugation for which $P,Q$ are a corresponding pair.

A second use of weight is that only barycentric coordinates of weight zero and equations whose two sides have equal weight are projectively invariant. Indeed, a point $P$ of weight 1 is mapped to $\Phi \times P$ by the projective transformation on points $f = \Phi\times$.[17] For barycentric coordinates $P_k$ of weight $k$, take barycentric coordinates $P$ with the same numerical values but with weight 1. Then $P_k = P \times \mathbf{1}^{k-1}$, and

$$f(P_k) = f(P) \times f(\mathbf{1})^{k-1} = (\Phi \times P) \times (\Phi \times \mathbf{1})^{k-1} = \Phi^k \times P_k.$$

Thus the projective transformation $f$ actually acts on barycentric coordinates of weight $k$ by the operator $\Phi^k\times$.

For barycentric coordinates of nonzero weight or an equation whose weights are unbalanced, multiplication or division by $\mathbf{1}$ is needed to obtain a projectively invariant expression. For example, under a projective transformation $f$ on points, $R = P \times Q^2$ does not imply $f(R) = f(P) \times f(Q)^2$. Rewriting the equation as $R \times \mathbf{1}^2 = P \times Q^2$ gives weight 3 on both sides, and then $f(R) \times f(G)^2 = f(P) \times f(Q)^2$.

We next relate the complement and anticomplement maps to barycentric coordinates.

**Theorem 7.5.10.** *Given $\triangle ABC$ and a point $P$ with barycentric coordinates $[u:v:w]$:*

*(1)* $\mathbf{c}P = [v+w : w+u : u+v]$*;*

*(2)* $\mathbf{a}P = [v+w-u : w+u-v : u+v-w]$.

**Proof.** We prove the first formula; the analogous proof of the second is left to the reader. Put $Q = \mathbf{c}P$, and choose $u = [\triangle PBC]$, $v = [\triangle PCA]$, $w = [\triangle PAB]$, so that $[\triangle ABC] = u+v+w := S$. If $G$ is the centroid, then $[\triangle GBC] = \frac{S}{3}$. From the definition of the complement, $2\overline{GQ} = \overline{PG}$, and therefore

$$2([\triangle QBC] - [\triangle GBC]) = [\triangle GBC] - [\triangle PBC],$$

$$[\triangle QBC] = \frac{3}{2}[\triangle GBC] - \frac{1}{2}[\triangle PBC] = \frac{S-u}{2} = \frac{v+w}{2}.$$

The other two directed areas follow cyclically, giving

$$\mathbf{c}P = \left[\frac{v+w}{2} : \frac{w+u}{2} : \frac{u+v}{2}\right] = [v+w : w+u : u+v].$$

□

[17] Clearly, $\Phi$ is a barycentric coordinate of weight zero, as can also be seen from the proof of theorem 7.5.9.

The point maps extend naturally to barycentric coordinates of arbitrary weight.

**Definition 7.5.11.** Given $\triangle ABC$ and a barycentric coordinate $P[u:v:w]$, define:
(i) the *complement transformation* $\mathbf{c}_{\triangle ABC}P$, usually abbreviated to $\mathbf{c}P$, as $[v+w:w+u:u+v]$;
(ii) the *anticomplement transformation* $\mathbf{a}_{\triangle ABC}P$, usually abbreviated to $\mathbf{a}P$, as $[v+w-u:w+u-v:u+v-w]$.
Both operations preserve weight.

The definitions are well defined because they depend only on the ratio $u:v:w$, not on the particular values of $u,v,w$. Despite their names, the operations act on arbitrary barycentric coordinates, not only on points of weight 1.

Replacing addition by subtraction in definition (i) gives another operation.

**Definition 7.5.12.** Given $\triangle ABC$ and a barycentric coordinate $P[u:v:w]$, its *difference-point transformation* $\mathbf{d}_{\triangle ABC}P$, usually abbreviated to $\mathbf{d}P$, is defined to be $[v-w:w-u:u-v]$. This operation preserves weight.

We give its geometric meaning. For convenience, let $\mathscr{St}_{\triangle ABC}$ denote the Steiner circumellipse of $\triangle ABC$, abbreviated to $\mathscr{St}$ when no ambiguity arises.

**Theorem 7.5.13.** *For $\triangle ABC$ and a point $P$, $P^{\mathbf{d}} = \mathfrak{p}_{\mathscr{St}}^{-1}(GP) \in l_\infty$, where $G$ is the centroid of $\triangle ABC$.*
For compactness, the symbols $\mathbf{c}$, $\mathbf{a}$, and $\mathbf{d}$ may also be written as superscripts; for example, $P^{\mathbf{d}}$.

**Proof.** The statement is affine invariant, so transform $\triangle ABC$ into an equilateral triangle. Then $\mathscr{St} = \mathscr{C}$, and $G$ is its center. Thus the pole $P' = \mathfrak{p}_{\mathscr{St}}^{-1}(GP)$ is the point at infinity in the direction perpendicular to $GP$.

Write $P[u:v:w]$, with the barycentric coordinates scaled so that $S = u+v+w = [\triangle ABC]$, and take a point $Q$ with $\overline{GQ} = t\overline{GP}$. A calculation analogous to the proof of part (1) of theorem 7.5.10, left to the reader, gives
$$Q = [3u+(t^{-1}-1)S : 3v+(t^{-1}-1)S : 3w+(t^{-1}-1)S].$$
Letting $t\to\infty$ yields the point at infinity on $GP$:
$$P_\infty = [3u-S:3v-S:3w-S] = [2u-v-w:2v-w-u:2w-u-v]. \tag{†}$$

Let $\phi_a,\phi_b,\phi_c$ be the directed angles from the three sidelines to $GP$. From the directed-area formula,
$$\begin{aligned} P_\infty &= \lim_{R\to P_\infty}[BC\cdot CR\sin\measuredangle(BC,CR) : CA\cdot AR\sin\measuredangle(CA,AR) : AB\cdot BR\sin\measuredangle(AB,BR)] \\ &= [\sin\phi_a : \sin\phi_b : \sin\phi_c]. \end{aligned} \tag{$\diamond$}$$
Here $AB=BC=CA$ and $\lim_{R\to P_\infty}(AR:BR:CR) = 1:1:1$, since all three lengths tend to infinity while their pairwise differences remain bounded. Since $GP' \perp GP$, similarly
$$P' = [\sin(\phi_a+90^\circ):\sin(\phi_b+90^\circ):\sin(\phi_c+90^\circ)] = [\cos\phi_a:\cos\phi_b:\cos\phi_c].$$

By (†) and ($\diamond$), write
$$(\sin\phi_a,\sin\phi_b,\sin\phi_c) = \lambda(2u-v-w, 2v-w-u, 2w-u-v).$$
Using (♣) from theorem 7.1.14 gives $\cos\phi_a+\cos\phi_b+\cos\phi_c = 0$. Solving yields $\lambda^{-2} = 4(u^2+v^2+w^2)-4(vw+wu+uv)$, and hence $P' = [v-w:w-u:u-v] = \mathbf{d}P$. □

In this book, the point $\mathbf{d}P$ is called the *difference point* of $P$. The theorem gives the geometric meaning of the difference point. Because the original definition has no explicit geometric interpretation, some calculation is unavoidable. For a more purely geometric treatment, one may instead take $\mathfrak{p}_{\mathscr{St}}^{-1}(GP)$ as its original definition.

The points $P, \mathbf{c}P, \mathbf{a}P$ and the centroid are collinear, so they have the same difference point. A point at infinity is fixed by both the complement and anticomplement maps. Therefore:

**Theorem 7.5.14.** *For* $\triangle ABC$, $\mathbf{c}\circ\mathbf{d}=\mathbf{d}\circ\mathbf{c}=\mathbf{a}\circ\mathbf{d}=\mathbf{d}\circ\mathbf{a}=\mathbf{d}$.

The complement, anticomplement, and difference-point transformations act on barycentric coordinates. Their formulas themselves are regarded as unchanged by a projective transformation, but an expression involving one of these maps is projectively invariant only when the transformed barycentric coordinates have weight zero. The next example illustrates the point.

**Example 7.5.15.** Given $\triangle ABC$ and points $P,Q$, let $R=P\star Q$. Prove that $R=Q\times(\mathbf{c}\mathbf{j}^{\sqrt{Q}}P)$. Here $\mathbf{j}^{\sqrt{Q}}$ abbreviates $\mathbf{j}^{\sqrt{\mathbf{1}\times Q}}$: it is the isoconjugation under which $Q$ corresponds to the centroid $G$, since its fixed point is $\sqrt{Q\times\mathbf{1}}$.

**Solution.** Express the equation entirely in barycentric operations:

$$R=Q\times(\mathbf{j}^{\sqrt{Q}}P)^{\mathbf{c}}=Q\times(Q\div P)^{\mathbf{c}},$$

where corollary 7.5.7 was used. The complement map acts on barycentric coordinates of weight zero, and the whole equation is weight-balanced, so it is projectively invariant. Apply a projective transformation fixing $\triangle ABC$ and carrying $Q$ to the centroid $G$. It remains to prove $P\star G=G\times(G\div P)^{\mathbf{c}}$. But

$$G\times(G\div P)^{\mathbf{c}}=\mathbf{1}\times(\mathbf{1}\div P)^{\mathbf{c}}=(\mathbf{t}P)^{\mathbf{c}}=\mathbf{ct}P,$$

and $P\star G=\mathbf{ct}P$ is precisely theorem 7.4.19. □

The key is projective invariance. Although the complement map alone is not projectively invariant, arranging a weight-zero argument makes the displayed identity invariant and permits the projective reduction.

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## Exercises

**Exercise 7.48.** Given $\triangle ABC$ and points $P,Q,R,S$:
(1) balance the weights in $P^3\times(Q\div R^4)^{\mathbf{a}}=P\times(Q^{-2})^{\mathbf{d}}$;
(2) suppose a projective transformation fixes $A,B,C,S$. Write the image of the equation from part (1) under this transformation.

**Exercise 7.49.** Given $\triangle ABC$ and a barycentric coordinate $P$, prove that
(1) $P\times P^{\mathbf{c}}=(P^{-1})^{\mathbf{c}}$;
(2) $P\times P^{\mathbf{d}}=(P^{-1})^{\mathbf{d}}$.

**Exercise 7.50.** Given $\triangle ABC$ with centroid $G$ and Lemoine point $L$, prove that $Ge\times In=G\times L$.

**Exercise 7.51.** Given $\triangle ABC$ with centroid $G$ and a point $P$, prove that the pairs $(P,\mathbf{c}P)$ and $(G,\mathbf{ct}P)$ belong to one isoconjugation.

### 7.5.2 Barycentric Products and Geometric Transformations

We now use barycentric operations to treat isoconjugation, trilinear polars, and the transformations of Cevian geometry.

#### $X$-Complements, $X$-Anticomplements, and $X$-Difference Points

We first introduce the following notions as an application.

**Definition 7.5.16.** Given $\triangle ABC$, a point $X$, and a point $P$ of the plane, define:

(i) the *$X$-complement* $\mathbf{c}_{X,\triangle ABC}P$ as the image of $P$ under the projective transformation that fixes $X$ and maps $A,B,C$ respectively to the three vertices of the cevian triangle of $X$;

(ii) the *$X$-anticomplement* $\mathbf{a}_{X,\triangle ABC}P$ as the image of $P$ under the projective transformation that fixes $X$ and maps $A,B,C$ respectively to the three vertices of the anticevian triangle of $X$;

(iii) the *$X$-difference point* $\mathbf{d}_{X,\triangle ABC}P$ as the point $\mathfrak{p}^{-1}_{\mathfrak{C}(X,\triangle ABC)}(PX)$, where $\mathfrak{C}(X,\triangle ABC)$ denotes the circumconic of $\triangle ABC$ with perspector $X$, abbreviated to $\mathfrak{C}(X)$ when the reference triangle is clear.

When the reference triangle is clear, we abbreviate these to $\mathbf{c}_X P$, $\mathbf{a}_X P$, and $\mathbf{d}_X P$.

The following theorem is immediate.

**Theorem 7.5.17.** *If $G$ is the centroid of $\triangle ABC$, then $\mathbf{c}_G = \mathbf{c}$, $\mathbf{a}_G = \mathbf{a}$, and $\mathbf{d}_G = \mathbf{d}$.*

Thus these definitions generalize the usual complement, anticomplement, and difference-point maps. They also act on barycentric coordinates of arbitrary weight, preserving that weight, and may be written as superscripts.

By theorem 7.5.17, under the projective transformation $\Phi$ fixing $\triangle ABC$ and taking the centroid $G$ to $X$, the cevian triangle of $X$, the anticevian triangle of $X$, and $\mathfrak{C}(X)$ are respectively the images of the medial triangle, the anticomplemental triangle, and $\mathfrak{C}(G) = \mathscr{St}$. Hence:

**Proposition 7.5.18.** *Given $\triangle ABC$ with centroid $G$ and a point $X$, let $f$ be the projective transformation that fixes $A,B,C$ and takes $G$ to $X$. Then*

*(i)* $\mathbf{c}_X = f \circ \mathbf{c} \circ f^{-1}$;

*(ii)* $\mathbf{a}_X = f \circ \mathbf{a} \circ f^{-1}$;

*(iii)* $\mathbf{d}_X = f \circ \mathbf{d} \circ f^{-1}$.

**Corollary 7.5.19.** *Given $\triangle ABC$ and a point $X$, let $P$ be a barycentric coordinate of weight $k$. Then*

*(i)* $P^{\mathbf{c}_X} = X^k \times (P \div X^k)^{\mathbf{c}}$;

*(ii)* $P^{\mathbf{a}_X} = X^k \times (P \div X^k)^{\mathbf{a}}$;

*(iii)* $P^{\mathbf{d}_X} = X^k \times (P \div X^k)^{\mathbf{d}}$.

**Proof.** The transformation $f$ is represented by the barycentric coordinates of weight zero $\Phi = X \div \mathbf{1}$, and it takes $P$ to $\Phi^k \times P$. Substitution in proposition 7.5.18 gives the three formulas. ☐

### Isoconjugation

The preceding subsection described isoconjugation in barycentric terms. We give some applications.

**Proposition 7.5.20.** *Given $\triangle ABC$ and points $P,P',Q,Q'$, suppose $P \times P' = Q \times Q'$. Then $P,Q$, and $P'Q \cap PQ'$ lie on one circumconic of $\triangle ABC$.*

**Proof.** By theorem 7.5.6, there is an isoconjugation $\mathbf{j}$ such that $P' = \mathbf{j}P$ and $Q' = \mathbf{j}Q$. By theorem 7.2.11, $\mathbf{j}(P'Q \cap PQ') = (PQ \cap P'Q')$. The points $P', Q', (PQ \cap P'Q')$ are collinear, so theorem 7.2.15 proves the assertion. ☐

**Example 7.5.21.** Given $\triangle ABC$ with centroid $G$ and Lemoine point $L$, prove that the lines $GIn$, $LGe$, and $MiTi$ are concurrent. Here $Ti$ denotes the isotomic conjugate of the incenter.

**Solution.** By proposition 5.2.7, the points $I,G,Na$ are collinear. Their isotomic conjugates are $Ti,G,Ge$, respectively, by proposition 5.2.3; hence these three points lie on a circumconic $\alpha$. Since $G = \mathbf{g}L$ and $Ge = \mathbf{g}In$ by proposition 5.2.5, $G \times L = Ge \times In$. By proposition 7.5.20, $GIn$ and $LGe$ meet at a point of $\alpha$. Applying proposition 7.5.20 again, it remains to show $L \times Ti = Ge \times Mi$. We have

$$L \times Ti = L \times (\mathbf{1}^2 \div I) = (L \times \mathbf{1} \div I) \times \mathbf{1} = \mathbf{g}I \times \mathbf{1} = I \times \mathbf{1} = I.$$

Here the first equality uses corollary 7.5.8; the third follows from theorem 7.5.6 and $L = \mathbf{g}G$. Also,

$$Ge \times Mi = Ge \times Ge^{\mathbf{c}} = (\mathbf{1} \div Ge)^{\mathbf{c}} = (\mathbf{t}Ge)^{\mathbf{c}} = \mathbf{c}Na = I.$$

The first, third, fourth, and fifth equalities use propositions 5.2.3, 5.2.7 and 5.4.3 and corollary 7.5.8, respectively. The second equality follows from $P \times P^{\mathbf{c}} = (P^{-1})^{\mathbf{c}}$, which is readily verified from the definitions for every barycentric coordinate $P$. □

For completeness, given $\triangle ABC$ and a point $P$, if the directed distances of $P$ from the three sidelines are in the ratio $[x:y:z]$, then $[x:y:z]$ are its *trilinear coordinates*. Its barycentric coordinates are $[ax:by:cz]$, where $a,b,c$ are the side lengths. Coordinatewise trilinear products and quotients can be defined in the same way, with identity the incenter $I$. They have properties analogous to those of barycentric products and quotients, but the complement, anticomplement, and difference-point operations cannot be defined directly by the preceding method. An $X$-isoconjugation is naturally defined in trilinear coordinates: the trilinear product of each corresponding pair is constant, and the point represented by that constant is $X$. We continue to use barycentric coordinates here, since trilinears are more often used for analytic computations.

### Cevian-Geometric Transformations

The four Cevian operations have the following barycentric formulas.

**Theorem 7.5.22.** *For $\triangle ABC$ and points $P,Q$:*

*(i)* $P * Q = P \div (Q \div P)^{\mathbf{c}}$;
*(ii)* $P \star Q = Q \times (Q \div P)^{\mathbf{c}}$;
*(iii)* $P / Q = Q \times (Q \div P)^{\mathbf{a}}$;
*(iv)* $P : Q = Q \div (P \div Q)^{\mathbf{a}}$.

**Proof.** Identity (ii) was proved in example 7.5.15; we now prove (i). The argument of the complement map has weight zero and both sides have weight 1, so the identity is projectively invariant. Apply a projective transformation fixing $\triangle ABC$ and taking $P$ to the centroid $G$. It remains to prove $Q * G = G \div (Q \div G)^{\mathbf{c}}$. After this transformation, $Q * G$ is the crosspoint of $Q,G$ with respect to the anticomplemental triangle of $\triangle ABC$. Applying the complement map, the homothety $\mathfrak{h}_{G,-1/2}$, gives

$$(Q * G)^{\mathbf{c}} = Q^{\mathbf{c}} \star G^{\mathbf{c}} = Q^{\mathbf{c}} \star G = G \times (G \div Q^{\mathbf{c}})^{\mathbf{c}} = (G^2 \div Q^{\mathbf{c}})^{\mathbf{c}},$$

where the third equality uses (ii). Thus $Q * G = G^2 \div Q^{\mathbf{c}} = G \div (Q \div G)^{\mathbf{c}}$.

For (iii), put $R = P/Q$. By (i), $P = Q * R = Q \div (R \div Q)^{\mathbf{c}}$, which gives $R = Q \times (Q \div P)^{\mathbf{a}}$. Part (iv) is analogous. □

**Corollary 7.5.23.** *For $\triangle ABC$ and points $P,Q$, $P \times Q = (P * Q) \times (P \star Q) = (P/Q) \times (Q:P)$.*

**Example 7.5.24.** Given $\triangle ABC$ with centroid $G$ and orthocenter $H$, determine $G \star H$.

**Solution.** Let $O,L$ be the circumcenter and Lemoine point. Then

$$G \star H = H \times (H \div G)^{\mathbf{c}} = H \times \mathbf{c}H \div G = H \times O \div G = L,$$

because $(H,O)$ and $(G,L)$ are two corresponding pairs under isogonal conjugation. □

**Remark.** Equivalently,

$$G \star H = H \star G = G \times (G \div H)^{\mathbf{c}} = \mathbf{ct}H.$$

Together with $G \star H = L$, this gives $\mathbf{t}H = \mathbf{a}L = Al$, as already known.

### Trilinear Polars

We give several properties relating trilinear polars to barycentric coordinates.

**Theorem 7.5.25.** *For $\triangle ABC$ and any two points $P,Q$, $\mathscr{T}(P)\div P=\mathscr{T}(Q)\div Q$.* Here $\mathscr{T}(P)\div P := \{X\div P|X\in\mathscr{T}(P)\}$, and $\mathscr{T}(Q)\div Q$ is understood similarly. Equivalently, for every $X\in\mathscr{T}(P)$ there is a point $Y\in\mathscr{T}(Q)$ such that $X\div P=Y\div Q$.

**Proof.** The identity is equivalent to $\mathscr{T}(P)\times Q=\mathscr{T}(Q)\times P$. Thus for every $P'\in\mathscr{T}(P)$ there must be a $Q'\in\mathscr{T}(Q)$ such that $(P',Q)$ and $(P,Q')$ belong to one isoconjugation $\mathbf{j}$. This is equivalent to $\mathbf{j}Q\in\mathscr{T}(P)\Longleftrightarrow\mathbf{j}P\in\mathscr{T}(Q)$, which is theorem 7.3.7. ☐

**Theorem 7.5.26.** *For $\triangle ABC$ and any two points $P,Q$, $P\div\mathfrak{C}(P)=\mathscr{T}(Q)\div Q$.*

**Proof.** By corollary 7.3.20, every isoconjugation satisfies $\mathbf{j}(\mathfrak{C}(P))=\mathscr{T}(\mathbf{j}P)$. By theorem 7.5.6, choose $\mathbf{j}=P\times Q\div$. Then $P\times Q\div\mathfrak{C}(P)=\mathscr{T}(Q)$, which proves the assertion. ☐

This gives a property of circumconics.

**Theorem 7.5.27.** *For $\triangle ABC$ and any two points $P,Q$, $\mathfrak{C}(P)\div P=\mathfrak{C}(Q)\div Q$.*

**Proof.** By theorem 7.5.26, $P\div\mathfrak{C}(P)=\mathscr{T}(Q)\div Q$. Setting $P=Q$ in theorem 7.5.26 also gives $Q\div\mathfrak{C}(Q)=\mathscr{T}(Q)\div Q$. Inverting the set quotients proves the result. ☐

**Corollary 7.5.28.** *For $\triangle ABC$ and points $P,Q$, $P\in\mathfrak{C}(Q)\Longleftrightarrow Q\in\mathfrak{C}(P)$.*

**Proof.** This follows by the same argument as theorem 7.5.25. ☐

**Theorem 7.5.29.** *For $\triangle ABC$ and points $P,Q$, $\mathscr{T}(P)\cap\mathscr{T}(Q)=Q\times(Q\div P)^{\mathbf{d}}$.*

**Proof.** By corollary 7.4.15 and theorem 7.5.22,

$$\mathscr{T}(P)=\mathfrak{p}_{\mathfrak{C}(Q)}(P/Q)=\mathfrak{p}_{\mathfrak{C}(Q)}(Q\times(Q\div P)^{\mathbf{a}}).$$

Replacing $P$ by $Q$ in this equation gives $\mathscr{T}(Q)=\mathfrak{p}_{\mathfrak{C}(Q)}(Q/Q)=\mathfrak{p}_{\mathfrak{C}(Q)}(Q)$. By the polarity principle, $\mathscr{T}(P)\cap\mathscr{T}(Q)$ is the pole of the line joining $Q$ and $(Q\times(Q\div P)^{\mathbf{a}})$. The definition of the $X$-difference point and corollary 7.5.19 give

$$\mathscr{T}(P)\cap\mathscr{T}(Q)=\mathbf{d}_Q(Q\times(Q\div P)^{\mathbf{a}})=Q\times(Q\times(Q\div P)^{\mathbf{a}}\div Q)^{\mathbf{d}}=Q\times((Q\div P)^{\mathbf{c}})^{\mathbf{d}}=Q\times(Q\div P)^{\mathbf{d}},$$

where the last equality uses theorem 7.5.14. ☐

**Theorem 7.5.30.** *For $\triangle ABC$ and points $P,Q$, $\mathscr{T}^{-1}(PQ)=Q\div(P\div Q)^{\mathbf{d}}$.*

**Proof.** By the definition of the $X$-difference point and corollary 7.5.19, $PQ=\mathfrak{p}_{\mathfrak{C}(Q)}(\mathbf{d}_QP)$ and $\mathbf{d}_QP=Q\times(P\div Q)^{\mathbf{d}}$. Therefore

$$\mathscr{T}^{-1}(PQ)=(Q\times(P\div Q)^{\mathbf{d}})*Q=Q*(Q\times(P\div Q)^{\mathbf{d}})=Q\div((P\div Q)^{\mathbf{d}})^{\mathbf{c}}=Q\div(P\div Q)^{\mathbf{d}},$$

where the first, third, and fourth equalities use theorems 7.4.14, 7.5.14 and 7.5.22, respectively. ☐

**Corollary 7.5.31.** *Given $\triangle ABC$, three points $P,Q,R$ are collinear if and only if $(Q\div P)^{\mathbf{d}}=(R\div P)^{\mathbf{d}}$.*

**Proof.** The points $P,Q,R$ are collinear if and only if $\mathscr{T}^{-1}(PQ)=\mathscr{T}^{-1}(PR)$; now apply theorem 7.5.30. ☐

The assertion also has a direct proof.

**Alternative proof.** The difference-point map here acts on barycentric coordinates of weight zero, and the equation is balanced. Apply a projective transformation taking $P$ to the centroid; the assertion then follows immediately from the geometric definition of the difference point. ☐

## Exercises

**Exercise 7.52.** Given $\triangle ABC$ with incenter $I$, determine $I \star Mi$.

**Exercise 7.53.** Given $\triangle ABC$ and points $P,Q$, put $S = \mathscr{T}^{-1}(PQ)$ and $T = \mathscr{T}(P) \cap \mathscr{T}(Q)$. Prove that $(P,Q)$ and $(S,T)$ are corresponding pairs under one isoconjugation.

**Exercise 7.54.** Given $\triangle ABC$ and a point $U$, let $\mathbf{j}$ be an isoconjugation. Prove that, for every point $P$:

(1) $\mathscr{T}^{-1}(\mathfrak{p}_{\mathbf{j}(\mathscr{T}(U))}(P)) = \mathbf{j}(U \times (U \div \mathbf{j}P)^{\mathbf{c}})$;

(2) $\mathfrak{p}^{-1}_{\mathbf{j}(\mathscr{T}(U))}(\mathscr{T}(P)) = \mathbf{j}(U \div (\mathbf{j}P \div U)^{\mathbf{a}})$.

**Exercise 7.55.** Given $\triangle ABC$, a circumconic $\alpha$, and a line $l$, put $U = \mathscr{T}^{-1}(l)$ and $R = \mathfrak{p}_\alpha^{-1}(l)$. Prove that $(R \div U) \times (R \div U)^{\mathbf{a}} \div l = \alpha \div U^2$.

**Exercise 7.56.** Given $\triangle ABC$ and a point $P$, prove for every point $Q$ in the plane that $\mathbf{c}_P(\mathscr{T}(\mathbf{j}^P Q)) = \mathscr{T}(\mathbf{j}^P \mathbf{a}_P(Q))$.

# 7.6 Orthotransversals and Orthocorrespondents*

## 7.6.1 Basic Properties of Orthotransversals*

This section develops the properties of orthotransversals. Recall their definition from section 6.5.2.

**Definition 7.6.1.** Given $\triangle ABC$ and a point $P$, draw through $P$ the perpendiculars to $AP,BP,CP$, meeting $BC,CA,AB$ at $X,Y,Z$, respectively. The points $X,Y,Z$ are collinear, and their common line $XYZ$ is called the *orthotransversal* of $P$ with respect to $\triangle ABC$. We denote it by $\mathscr{O}_{\triangle ABC}(P)$, or simply by $\mathscr{O}(P)$ when there is no ambiguity.

The earlier proof used polarity. The Gauss–Bodenmiller theorem provides a second description (theorem 4.8.11). If a line $l$ meets the three sidelines of $ABC$ at $X,Y,Z$, respectively, then the circles $\odot(AX)$, $\odot(BY)$, $\odot(CZ)$ have two common points $P,P'$, and $\overline{XYZ} = \mathscr{O}(P) = \mathscr{O}(P')$. No point other than $P,P'$ has $\overline{XYZ}$ as its orthotransversal; see figure 7.33.

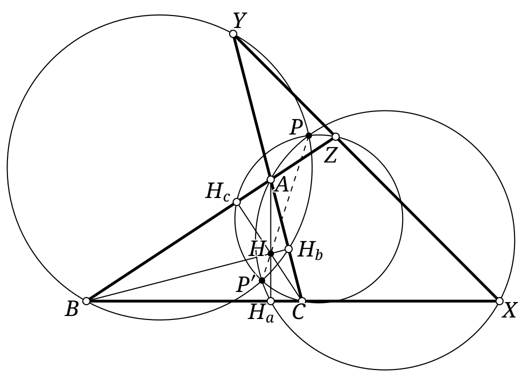


FIGURE 7.33

**Definition 7.6.2.** Given $\triangle ABC$ and a line $l$, there are two points $P,P'$ whose orthotransversal is $l$. They are called the *orthoassociates* of $l$ with respect to $\triangle ABC$; the points $P,P'$ are also said to form an orthoassociate pair with respect to $\triangle ABC$.

The preceding discussion of the Gauss–Bodenmiller theorem gives a construction of the orthoassociates.

**Theorem 7.6.3.** *For a transversal $\overline{XYZ}$ of $\triangle ABC$,*[18] $\mathcal{O}^{-1}(XYZ) = \odot(AX) \cap \odot(BY) \cap \odot(CZ)$.[19]

The Gauss–Bodenmiller theorem also gives the following.

**Theorem 7.6.4.** *Given $\triangle ABC$ and a line $l$, the line through its two orthoassociates is the Aubert line of the complete quadrilateral $\{l,AB,BC,CA\}$.*

We record several elementary properties.

**Theorem 7.6.5.** *The two points of an orthoassociate pair with respect to $\triangle ABC$ are inverse with respect to the polar circle $\mathcal{P}$.*

**Proof.** Refer to figure 4.63. Let $\overline{XYZ}$ be the common orthotransversal and put $\odot(AX) \cap BC = H_a, X$. Then $\mathrm{pow}_{\odot(AX)}(H) = \overline{AH} \cdot \overline{H_aH} = r_{\mathcal{P}}^2$ by theorem 6.1.16. Hence proposition 4.5.4 shows that $\mathcal{P}$ and $\odot(AX)$ are orthogonal, so general inversion $\mathfrak{i}_{\mathcal{P}}$ preserves $\odot(AX)$ by theorem 4.1.9. It likewise preserves $\odot(BY)$ and $\odot(CZ)$, and therefore interchanges their two common points. By theorem 7.6.3, those points are precisely the two orthoassociates of $XYZ$. □

We need one lemma before the next property.

**Lemma 7.6.6.** *Let $\triangle A'B'C'$ be the antipodal triangle of $\triangle ABC$, and let $P$ be a point. Through $P$ draw the line parallel to $BC$, and through $A'$ the line perpendicular to $AP$; let them meet at $D$. Define $E,F$ cyclically. Then $D,E,F$ are collinear, and their line is perpendicular to $HP$.*

**Proof.** As in figure 7.34, let $AP$ meet $\odot(ABC)$ at $A,T$. Then $A',T,D$ are collinear and $\angle ATD = 90°$. Let $H$ be the orthocenter of $\triangle ABC$, and put $D' = AH \cap PD$. Since $\angle AD'D = 90° = \angle ATD$, the points $A,D',T,D$ lie on $\odot(AD)$. Hence $\overline{AP} \cdot \overline{TP} = \overline{D'P} \cdot \overline{DP}$, or $\mathrm{pow}_{\odot(ABC)}(P) = \mathrm{pow}_{\odot(HD)}(P)$ (noting that $D'D$ is a chord of $\odot(HD)$). Cyclically, $\mathrm{pow}_{\odot(ABC)}(P) = \mathrm{pow}_{\odot(HE)}(P) = \mathrm{pow}_{\odot(HF)}(P)$. Thus $P$ has equal power to the three circles $\odot(HD)$, $\odot(HE)$, and $\odot(HF)$. Since they all pass through $H$, their common radical axis is $HP$. Their centers $O_1,O_2,O_3$ are therefore collinear on a line perpendicular to $HP$, and the homothety $\mathfrak{h}_{H,2}$ maps them to $D,E,F$. □

**Theorem 7.6.7.** *Given $\triangle ABC$ and a point $P$, put $P' = \mathbf{g}P$, and let $H'$ be the orthocenter of the pedal triangle of $P'$. Then $P'H' \perp \mathcal{O}(P)$.*

**Proof.** As in figure 7.35, let $\triangle P_1P_2P_3$ be the pedal triangle of $P'$ and let $\odot O'$ be its pedal circle. By corollary 4.4.5, $P$ and $P'$ are symmetric about $O'$. Let $P_1'$ be the point of $\odot O'$ antipodal to $P_1$. Through $P_1'$ draw the perpendicular to $P'P_1$, and through $P'$ the parallel to $P_2P_3$; let them meet at $X'$. Define $Y',Z'$ cyclically. By lemma 7.6.6, $X',Y',Z'$ are collinear and $\overline{X'Y'Z'} \perp P'H'$.

Let $\mathcal{O}(P)$ meet $BC,CA,AB$ at $X,Y,Z$. By proposition 4.4.7, $P_2P_3 \perp AP$. Since $P'X' \parallel P_2P_3$, we have $P'X' \perp AP$. The definition of the orthotransversal also gives $AP \perp PX$, so $PX \parallel P'X'$. Because $P,P'$ are symmetric about $O'$, the lines $PX,P'X'$ correspond under that half-turn.

Also $P_1'X'$ and $BC$ are both perpendicular to $P'P_1$, while $P_1'$ and $P_1$ are symmetric about $O'$. Thus $P_1'X'$ and $BC$ correspond under the same half-turn, forcing $X'$ and $X$ to correspond. Cyclically, $Y',Z'$ correspond to $Y,Z$. Hence $\overline{XYZ} \parallel \overline{X'Y'Z'}$, and consequently $\overline{XYZ} \perp P'H'$. □

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[18]In what follows, a transversal $\overline{XYZ}$ of $\triangle ABC$ is understood to meet its three sidelines at $X,Y,Z$, respectively.

[19]Here $\mathcal{O}^{-1}$ denotes the *orthoassociates* of the transversal.

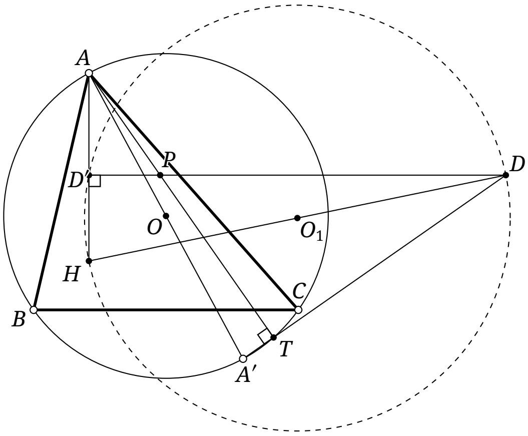


FIGURE 7.34

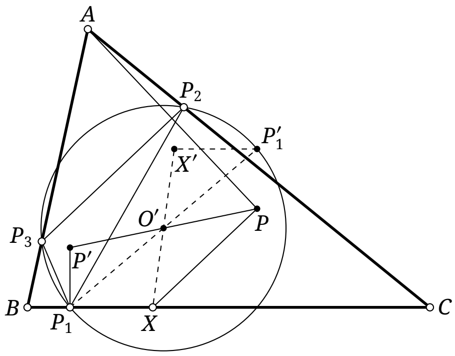


FIGURE 7.35

## Exercises

**Exercise 7.57.** Given $\triangle ABC$ with orthocenter $H$, let $P \in \odot(BC)$ and put $Q = \mathbf{g}P$. Prove that $P,Q,H$ are collinear.

**Exercise 7.58.** Let $\triangle A'B'C'$ be the circumcevian triangle of $P$ with respect to $\triangle ABC$. Prove that $\mathbf{i}(\mathscr{O}_{\triangle ABC}(P) \cap \mathscr{O}_{\triangle A'B'C'}(P)) \in \odot(OP)$.

**Exercise 7.59.** Prove that the two orthoassociates of the Simson line of a point on the circumcircle of a triangle lie on one circumparabola of the triangle.

**Exercise 7.60.** Given points $A,B,C,D$ and another point $P$, prove that the orthotransversals of $P$ with respect to $\triangle ABC, \triangle BCD, \triangle CDA, \triangle DAB$ are concurrent.

### 7.6.2 Further Properties of Orthotransversals*

We continue with relations between orthotransversals and conics.

**Theorem 7.6.8.** *Given $\triangle ABC$ and a point $P$, the tangent at $P$ to any circumrectangular hyperbola through $P$ is perpendicular to $\mathscr{O}(P)$.*

**Proof.** This is the limiting case $Q \to P$ of the following generalization, proposition 7.6.9. □

**Proposition 7.6.9.** *Given $\triangle ABC$, let $P,Q$ be points such that $\alpha = \mathfrak{C}(ABCPQ)$ is a rectangular hyperbola. Through $Q$ draw the perpendiculars to $AP,BP,CP$, meeting $BC,CA,AB$ at $X,Y,Z$, respectively. Then $X,Y,Z$ are collinear and $\overline{XYZ} \perp PQ$.*

**Proof.** As in figure 7.36, put $M = ZQ \cap PB$ and $N = YQ \cap PC$, let $T,S$ be the points at infinity on $ZQ,YQ$, and let $H'$ be the orthocenter of $\triangle PBC$. By theorem 6.5.16, $H' \in \alpha$. The lines $ZQ, BH'$ are both perpendicular to $PC$, so $BH' \parallel ZQ$; similarly $CH' \parallel YQ$. Hence

$$(Z,Q,M,T) \overline{\wedge} B(Z,Q,M,T) \overline{\wedge} \alpha(A,Q,P,H') \overline{\wedge} C(Y,Q,N,S) \overline{\wedge} (Y,Q,N,S).$$

Thus $(ZQ,MT) = (YQ,NS)$, or $\frac{ZQ}{MQ} = \frac{YQ}{NQ}$. Consequently $YZ \parallel MN$.

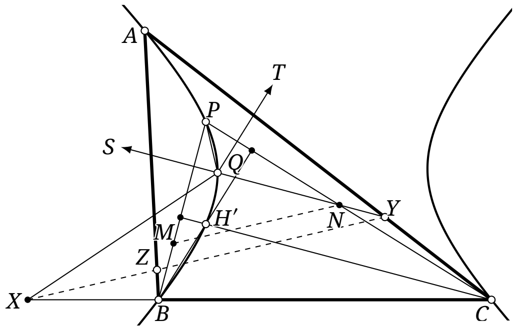


FIGURE 7.36

Since $MQ \perp PN$ and $NQ \perp PM$, the point $Q$ is the orthocenter of $\triangle PMN$. Thus $PQ \perp MN$, and hence $PQ \perp YZ$. The cyclic arguments give $PQ \perp XZ$ and $PQ \perp YX$, so $X,Y,Z$ are collinear and $\overline{XYZ} \perp PQ$. ☐

**Corollary 7.6.10.** *Given $\triangle ABC$ and a point $P$, let $H'$ be the orthocenter of the pedal triangle of $\mathbf{g}P$. Then $H'\mathbf{g}P$ is parallel to the tangent at $P$ to the circumrectangular hyperbola through $P$.*

**Proof.** Combine theorems 7.6.7 and 7.6.8. ☐

**Proposition 7.6.11.** *Given $\triangle ABC$ and a point $P$, let $Q$ be the projection of $P$ on $\mathscr{O}(P)$, and let $\alpha = \mathfrak{C}(ABCPQ)$. Then $\mathscr{O}(P)$ is parallel to an asymptote of $\alpha$.*

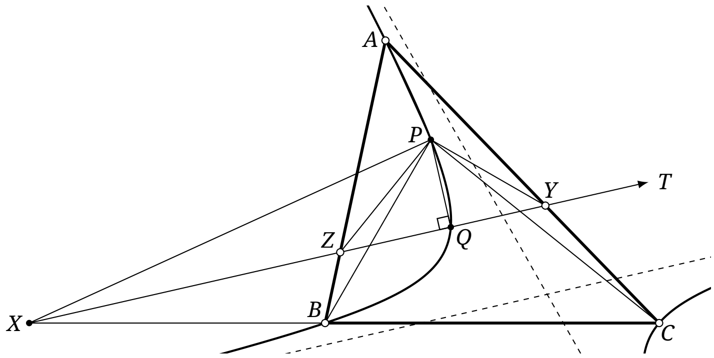


FIGURE 7.37

**Proof.** As in figure 7.37, let $\mathscr{O}(P)$ meet the three sides at $X,Y,Z$, let $T$ be its point at infinity, and put $U = CQ \cap AB$, $V = AQ \cap BC$. The perspectivities $P(Q,Z,Y,X) \overset{(XY)}{\doublebarwedge} A(Q,B,C,X) \overset{(XC)}{\doublebarwedge} Q(A,B,C,T)$ give $P(XY,ZQ) = P(QZ,YX) = Q(AB,CT)$. The rotation $\mathfrak{r}_{P,90^\circ}$ maps the pencil $P(A,B,C,T)$ to $P(X,Y,Z,Q)$, so also $P(AB,CT) = P(XY,ZQ) = Q(AB,CT)$. By the projective definition of a conic, $A,B,C,P,Q,T$ lie on one conic $\alpha$. Its point at infinity $T$ has the direction of $\mathscr{O}(P)$, proving the assertion. ☐

Polarity gives another useful description. We first prove a lemma.

**Lemma 7.6.12.** *Given $\triangle ABC$ and a point $P$, let $\omega_P$ be the pedal circle of $P$, let $\omega$ be any circle centered at $P$, and let $\mathfrak{d}$ be polarity with respect to $\omega$. Let $\triangle A^*B^*C^*$ be the polar triangle of $\triangle ABC$. Then:*

*(1) $\mathfrak{d}(\mathscr{T}(P))$ is the centroid $M^*$ of $\triangle A^*B^*C^*$;*
*(2) $\mathfrak{d}(\mathscr{O}(P))$ is the orthocenter $H^*$ of $\triangle A^*B^*C^*$;*
*(3) the pedal circle $\omega_P$ of $P$ and $\odot(A^*B^*C^*)$ are inverse with respect to $\omega$;*

*(4)* $\mathfrak{d}(\mathfrak{p}_{\omega_P}(P))$ *is the circumcenter* $O^*$ *of* $\triangle A^*B^*C^*$*;*
*(5) the polar triangle* $\triangle P_a^* P_b^* P_c^*$ *of the cevian triangle* $\triangle P_a P_b P_c$ *of* $P$ *is the anticomplemental triangle of* $\triangle A^*B^*C^*$*. The polar triangle* $\triangle P^{a*}P^{b*}P^{c*}$ *of the anticevian triangle* $\triangle P^a P^b P^c$ *of* $P$ *is the medial triangle of* $\triangle A^*B^*C^*$*.*

**Proof.** Parts (1) and (5) are immediate and are left to the reader. We now consider (2), (3), and (4). Part (2) was already obtained in the proof of existence of the orthotransversal in theorem 6.5.18 of section 6.5.2.

For (3), consider the conic $\alpha$ with foci $P, \mathbf{g}P$ tangent to the three sidelines of $\triangle ABC$. By theorem 4.5.7, $\omega_P$ is an auxiliary circle of $\alpha$, so theorem 6.5.5 gives the assertion.

For (4), part (3) shows that the centers of $\odot(A^*B^*C^*)$, $\omega_P$, and $\omega$ are collinear. Therefore $\mathfrak{p}_{\omega_P}(P) \perp O^*P$. Let their intersection be $Z$. The points $P, Z$ are inverse with respect to $\omega_P$; applying the converse of proposition 4.1.14 shows that $Z, O^*$ are inverse with respect to $\omega$. Hence $O^* = \mathfrak{d}(\mathfrak{p}_{\omega_P}(P))$. $\square$

**Theorem 7.6.13.** *If* $P$ *lies on the circumcircle or nine-point circle of* $\triangle ABC$*, then* $\mathscr{O}(P)$ *passes through the circumcenter* $O$*.*

**Proof.** By theorem 6.1.16, the circumcircle and nine-point circle are inverse with respect to $\mathscr{P}$. In view of theorem 7.6.5, it is enough to treat $P \in \odot(ABC)$.

Choose a circle $\omega$ centered at $P$ and consider polarity with respect to it. Abbreviate the polar triangle of $\triangle ABC$ to $\triangle^*$. The circle $\odot(ABC)$ is transformed into an inconic $\alpha$ of $\triangle^*$. By proposition 6.5.4, $\alpha$ is a parabola with focus $P$ and directrix $\mathfrak{d}(O)$. By lemma 7.6.12, $\mathscr{O}(P)$ transforms into the orthocenter $H^*$ of $\triangle^*$. Since theorem 4.2.12 gives $H^* \in \mathfrak{d}(O)$, polarity back yields $O \in \mathscr{O}(P)$. $\square$

There is also a direct proof.

**Alternative proof.** It suffices to treat the circumcircle. As in figure 7.38, let $\mathscr{O}(P)$ meet $BC, CA, AB$ at $X, Y, Z$. Let $PY$ meet $\odot(ABC)$ at $P, Y'$ and $PZ$ meet it at $P, Z'$. By definition, $BP \perp PY$ and $CP \perp PZ$, so $Y', Z'$ are antipodal to $B, C$, respectively. Thus $O = BY' \cap CZ'$. Pascal's theorem applied to the hexagon $Z'PY'BAC$ gives the collinearity of $Z, O, Y$. $\square$

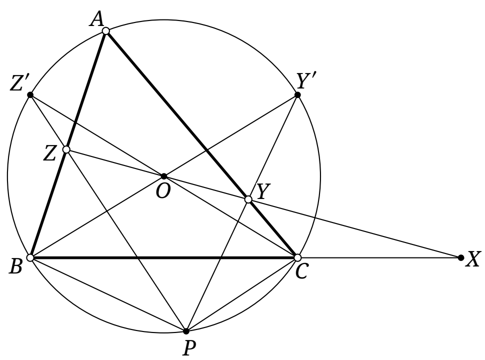


FIGURE 7.38

**Theorem 7.6.14.** *For a triangle and a point* $P$*, the polar of* $P$ *with respect to its pedal circle is the orthotransversal of* $P$ *with respect to its anticevian triangle.*

**Proof.** Apply polarity with respect to a circle centered at $P$. By parts (2) and (5) of lemma 7.6.12, the orthotransversal of $P$ with respect to its anticevian triangle becomes the orthocenter of the medial triangle of the polar triangle of the original triangle, hence the circumcenter of that polar triangle. By part (4) of lemma 7.6.12, the polar of $P$ with respect to its pedal circle also becomes the circumcenter of the polar triangle of the original triangle. The two lines therefore coincide. □

**Theorem 7.6.15.** *For a triangle and a point $P$, the following four lines are concurrent: the trilinear polar of $P$; its orthotransversal; its orthotransversal with respect to its cevian triangle; and its orthotransversal with respect to its anticevian triangle, equivalently its polar with respect to its pedal circle.*

**Proof.** Apply polarity with respect to a circle centered at $P$, and denote the polar triangle of the original triangle by $\triangle^*$. By lemma 7.6.12, the four lines become, respectively, the centroid of $\triangle^*$, the orthocenter of $\triangle^*$, the orthocenter of the anticomplemental triangle of $\triangle^*$, and the orthocenter of the medial triangle of $\triangle^*$. These four points are collinear because $\triangle^*$, the anticomplemental triangle of $\triangle^*$, and the medial triangle of $\triangle^*$ have the centroid of $\triangle^*$ as their common homothety center. Polarity back gives the desired concurrence. □

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## Exercises

**Exercise 7.61.** Prove that the orthotransversal of the incenter of a triangle is perpendicular to the line joining the incenter and circumcenter.

**Exercise 7.62.** Let $\alpha$ be a circumrectangular hyperbola of $\triangle ABC$. Prove that the orthotransversals of two points of $\alpha$ symmetric about its center are parallel.

**Exercise 7.63.** Given $\triangle ABC$, let $P \in \mathscr{C}$. Prove that $\mathscr{W}(P) \cap \mathscr{O}(P) \in \mathfrak{C}(ABCHO)$, where $H, O$ are the orthocenter and circumcenter of $\triangle ABC$, respectively.

**Exercise 7.64.** Let $O, H$ be the circumcenter and orthocenter of an acute triangle $\triangle ABC$. Take $P \in \mathscr{N}$, put $J = BP \cap AH$, and draw through $P$ the perpendicular to $AP$, meeting $BC, CH$ at $Q, K$, respectively. Draw $AL \parallel OQ$ with $L \in BC$. Prove that $J, K, L$ are collinear.

**Exercise 7.65.** Let $O$ be the circumcenter of $\triangle ABC$, take $P \in \mathscr{C}$, and let $\triangle DEF$ be the cevian triangle of $P$. Draw $AT \perp BC$ with $T \in EF$, and draw $AP \perp PS$ with $S \in BC$. Prove that $OS \perp DT$.

### 7.6.3 Orthocorrespondents*

Orthotransversals lead to the following point map.

**Definition 7.6.16.** Given $\triangle ABC$ and a point $P$, the point $\mathscr{T}^{-1}(\mathscr{O}(P))$ is called the *orthocorrespondent* of $P$ with respect to $\triangle ABC$. We denote it by $\mathbf{o}_{\triangle ABC}P$, or simply by $\mathbf{o}P$ when there is no ambiguity.

Unlike the maps $\mathbf{g}, \mathbf{t}, \mathbf{i}, \mathbf{f}$, this map satisfies $\mathbf{o}^2 \neq \mathrm{id}$: if $P' = \mathbf{o}P$, it need not follow that $P = \mathbf{o}P'$. Its inverse operation $\mathbf{o}^{-1}P = \mathscr{O}^{-1}(\mathscr{T}(P))$ gives two points, the two orthoassociates of $\mathscr{T}(P)$. Here is an example of an orthocorrespondent.

**Proposition 7.6.17.** *Each of the two Torricelli points of a triangle is its own orthocorrespondent.*

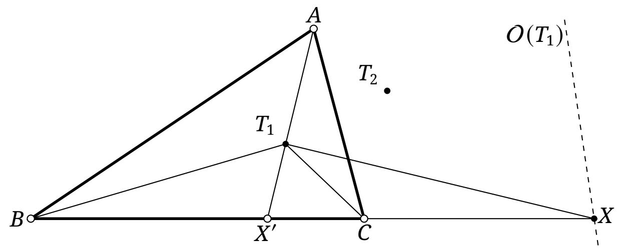


FIGURE 7.39

**Proof.** We prove the assertion for the first Torricelli point; the second is analogous. As in figure 7.39, let the orthotransversal of the first Torricelli point $T_1$ of $\triangle ABC$ meet $BC$ at $X$. By proposition 5.9.2, $\angle BT_1C = \angle AT_1C = 120°$, so $\angle CT_1X = 30°$. Thus $T_1X$ bisects the exterior angle of $\angle BT_1C$, while $AT_1$ bisects its interior angle. By proposition 3.2.17, $A(B,C,X,T_1)$ is a harmonic pencil. The harmonic relations for the pencils with vertices $B,C$ follow in the same way, and proposition 7.3.2 yields $T_1 = \mathbf{o}T_1$. $\square$

We give some further propositions about orthocorrespondents.

**Proposition 7.6.18.** *Suppose $\triangle ABC$ and $\triangle DEF$ have common perspector $P$. Put $Q = \mathbf{o}_{\triangle ABC}P$ and $R = \mathbf{o}_{\triangle DEF}P$. Then $P,Q,R$ are collinear.*

**Proof.** Apply polarity with respect to a circle centered at $P$. The polar of an orthotransversal is the orthocenter of the corresponding polar triangle; duality then says that the trilinear polar of this orthocenter is the polar of the original orthocorrespondent.

Let $\triangle_1, \triangle_2$ be the polar triangles of $\triangle ABC, \triangle DEF$, respectively. Then $\mathfrak{d}(Q) = \mathscr{A}_{\triangle_1}$ and $\mathfrak{d}(R) = \mathscr{A}_{\triangle_2}$. Because $\triangle ABC$ and $\triangle DEF$ have common perspector $P$, their polar triangles $\triangle_1, \triangle_2$ have common perspective axis $\mathfrak{d}(P) = l_\infty$ and are therefore homothetic. Hence $\mathscr{A}_{\triangle_1} \parallel \mathscr{A}_{\triangle_2}$, so $\mathscr{A}_{\triangle_1}, \mathscr{A}_{\triangle_2}, l_\infty$ are concurrent. Before polarity, this is the collinearity of $Q,R,P$. $\square$

**Theorem 7.6.19.** *With respect to a fixed reference triangle, the line joining a point to its orthocorrespondent is parallel to the Euler line of the pedal triangle of its isogonal conjugate.*

**Proof.** For $\triangle ABC$, put $Q = \mathbf{g}P$ and $T = \mathbf{o}P$. Let $\triangle A'B'C'$ be the circumcevian triangle of $P$, and let $\triangle P_aP_bP_c$ be the antipedal triangle of $P$ with respect to $\triangle A'B'C'$, as in figure 7.40. By exercise 4.47, $\triangle P_aP_bP_c$ is homothetic to the pedal triangle of $Q$, so their Euler lines are parallel. It is therefore enough to prove $PT \parallel \mathscr{E}_{\triangle P_aP_bP_c}$.

Let $\mathfrak{i}$ be an inversion centered at $P$ that preserves $\mathscr{C}$. Since each point and its inverse are collinear with $P$, the points $A,B,C$ map to $A',B',C'$. Since $BC$ is a line, $\mathfrak{i}(BC) = \odot(B'C'P)$. Since $PC' \perp P_aP_b$ and $PB' \perp P_aP_c$, we have $P \in \odot(B'PC') = \odot(PP_a)$.

Let $\mathscr{O}(P)$ meet $BC,CA,AB$ at $D,E,F$, and write $D' = \mathfrak{i}(D)$. Since $D \in BC$, we have $D' \in \mathfrak{i}(BC) = \odot(PP_a)$, so $D'$ is the second intersection of $DP$ with $\odot(P_aP)$. Let $H'$ be the orthocenter of $\triangle P_aP_bP_c$. We have $AP \perp PD$ and $AP \perp P_bP_c$, while $P_aH' \perp P_bP_c$; hence $P_aH' \perp PD$. Their intersection on $\odot(P_aP)$ must be $D'$.

Put $T_a = AT \cap BC$ and $T'_a = \mathfrak{i}(T_a)$. Since $T_a \in BC$, inversion gives $T'_a \in \odot(P_aP)$. Inversion preserves the cross-ratio between a line and its image circle, so

$$P_a(T'_aD', C'B') = P(T'_aD', C'B') = P(T_aD, CB) = -1,$$

where the last equality uses $\mathscr{T}(T) = \overline{DEF}$ and proposition 7.3.2. Let $H_a = P_aT'_a \cap P_bP_c$. Then by proposition 7.3.2, $H_a = P_bP_c \cap \mathscr{T}_{\triangle P_aP_bP_c}(H') = P_bP_c \cap \mathscr{A}_{\triangle P_aP_bP_c}$. Defining $H_b, H_c$ cyclically gives $\overline{H_aH_bH_c} = \mathscr{A}_{\triangle P_aP_bP_c}$.

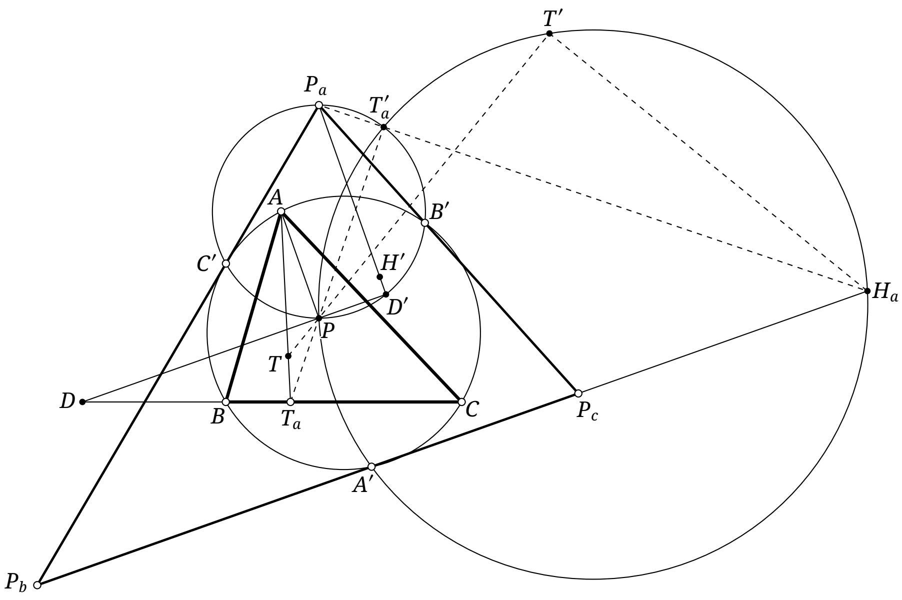


FIGURE 7.40

Finally, put $T' = \mathfrak{i}(T)$. Since $\mathfrak{i}(AT_a) = \odot(PA'T'_a)$ and $T \in AT_a$, we have $T' \in \odot(PA'T'_a)$. The relations $PT'_a \perp H_aT'_a$ and $PA' \perp H_aA'$ give $\odot(PA'T'_a) = \odot(PH_a)$, hence $T' \in \odot(PH_a)$. Therefore $PT' \perp T'H_a$, and cyclically $PT' \perp T'H_b, T'H_c$. Therefore $T', H_a, H_b, H_c$ are collinear, so $T' \in \mathscr{A}_{\triangle P_aP_bP_c}$. By theorem 7.3.10, $\mathscr{E} \perp \mathscr{A}$. Since $P, T, T'$ are collinear, $PT \parallel \mathscr{E}_{\triangle P_aP_bP_c}$. □

**Theorem 7.6.20.** *Given $\triangle ABC$ and a point $P$, let $H$ be the orthocenter, put $Q = \mathbf{g}P$, $T = \mathbf{o}P$, and $S = H/P$, and let $R$ be the isogonal conjugate of $P$ with respect to its anticevian triangle. Then $R, T, Q, S$ are collinear and form a harmonic range.*

We first prove a lemma.

**Lemma 7.6.21.** *Given $\triangle ABC$ and a point $P$, let $\triangle P^*_AP^*_BP^*_C$ be the antipedal triangle of $P$ with respect to its circumcevian triangle. Use primes for the notable objects of $\triangle P^*_AP^*_BP^*_C$; for example, $\mathscr{A}'$, $H'$, and $O'$ denote the orthic axis, orthocenter, and circumcenter of $\triangle P^*_AP^*_BP^*_C$, respectively. Let $\mathfrak{i}$ be an inversion centered at $P$ that preserves $\odot(ABC)$. Use a superscript $^*$ for the image of a point under $\mathfrak{i}$, and put $\mathbf{p} = \mathfrak{h}_{P,2}$. There is a circle $\omega$, centered on $\mathscr{A}'$, that passes through $P$ and is orthogonal to $\mathscr{C}', \mathscr{N}', \mathscr{P}'$. Moreover:*

*(1) if $T = \mathbf{o}P$, then $\mathbf{p}T^* \in \omega$ and $Eu' \in P\mathbf{p}T^*$;*[20]
*(2) if $Q = \mathbf{g}P$, then $\mathbf{p}Q^* \in \omega$ and $O' \in P\mathbf{p}Q^*$;*
*(3) if $R$ is the isogonal conjugate of $P$ with respect to its anticevian triangle, then $\mathbf{p}R^* \in \omega$ and $Ni' \in P\mathbf{p}R^*$;*
*(4) if $H$ is the orthocenter of $\triangle ABC$ and $S = H/P$, then $\mathbf{p}S^* \in \omega$ and $H' \in P\mathbf{p}S^*$.*

**Proof.** By theorem 7.3.10, the circles $\mathscr{C}', \mathscr{N}', \mathscr{P}'$ have common radical axis $\mathscr{A}'$ and form a coaxial pencil. By proposition 4.6.3, there is a circle with center $J \in \mathscr{A}'$ that is orthogonal to $\mathscr{C}', \mathscr{N}', \mathscr{P}'$ and passes through $P$. This proves the existence of $\omega$. We now prove the four remaining assertions.

For (1), the proof of theorem 7.6.19 shows that $T^*$ is the projection of $P$ on $\mathscr{A}'$. Since $J \in \mathscr{A}'$, the perpendicular-diameter theorem gives $\mathbf{p}T^* \in \omega$. The same proof gives $P\mathbf{p}T^* \parallel \mathscr{E}'$, so $Eu' \in P\mathbf{p}T^*$.

For (2), the proof of theorem 7.6.19 identifies the circumcevian triangle of $P$ as $\triangle A^*B^*C^*$. As in figure 7.41, the points $B^*, C^*, P^*_A, P$ are concyclic, as are $B^*, C^*, B, C$. Thus $\measuredangle PP^*_AC^* = \measuredangle PB^*C^* =$

[20]The point at infinity on the Euler line of a triangle is called its *Euler infinity point* and is usually denoted by *Eu*.

$\measuredangle BCC^*$. Together with $C^*P_A^* \perp C^*C$, this gives $PP_A^* \perp BC$. Also, $P_A^*P$ is a diameter of $\mathfrak{i}(BC) = \odot(B^*C^*P)$, so theorem 4.1.6 shows that the inverse image $P_A$ of $P_A^*$ is the projection of $P$ on $BC$.

Let $\triangle X_aX_bX_c$ be the reflection triangle of $P$. Then $P_A$ is the midpoint of $X_aP$. The definition of inversion shows that $X_a^*$ is the midpoint of $PP_A^*$, so $\mathbf{p}X_a^* = P_A^*$ and $\mathbf{p}(\odot(X_a^*X_b^*X_c^*)) = \mathscr{C}'$. By theorem 4.4.2, $Q$ is the center of $\odot(X_aX_bX_c)$. Thus proposition 4.1.14 shows that $Q^*, P$ are inverse with respect to $\odot(X_a^*X_b^*X_c^*)$. Together with $\mathbf{p}(\odot(X_a^*X_b^*X_c^*)) = \mathscr{C}'$, this shows that $P, \mathbf{p}Q^*$ are inverse with respect to $\mathscr{C}'$. Hence $O' \in P\mathbf{p}Q^*$ and $r_{\mathscr{C}'}^2 = \overline{O'P} \cdot \overline{O'\mathbf{p}Q^*}$. Since $\mathscr{C}'$ and $\omega$ are orthogonal, proposition 4.5.4 gives $\operatorname{pow}_\omega(O') = r_{\mathscr{C}'}^2$, and therefore $\operatorname{pow}_\omega(O') = \overline{O'P} \cdot \overline{O'\mathbf{p}Q^*}$. This proves $\mathbf{p}Q^* \in \omega$. □

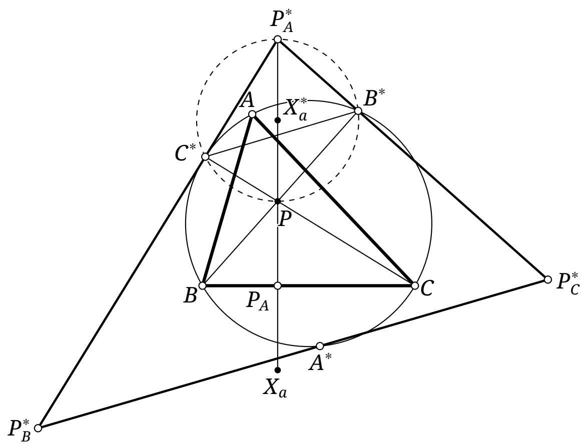


FIGURE 7.41

**Proof of lemma 7.6.21, continued.** For (3), as in figure 7.42, let $\triangle P_aP_bP_c$ and $\triangle P^aP^bP^c$ be the cevian and anticevian triangles of $P$, and let $\triangle P^{A*}P^{B*}P^{C*}$ be the medial triangle of $\triangle P_A^*P_B^*P_C^*$. Denote the inverse images of the three vertices $P^{A*}, P^{B*}, P^{C*}$ under the inversion by $P^A, P^B, P^C$, respectively. Since $AP \perp P_B^*P_C^*$ at $A^*$, theorem 4.1.6 gives $\mathfrak{i}(P_B^*P_C^*) = \odot(AP)$. Let $\infty_a'$ be the point at infinity on $P_B^*P_C^*$; then $\infty_a'^* = P$. Invariance of cross-ratio under inversion gives[21]

$$-1 = (P_B^*P_C^*, P^{A*}\infty_a') = A(P_BP_C, P^AP) = A(P_bB, P^AP).$$

By theorem 4.3.4, $A(P_bB, P^bP) = -1$, so the inverse image $P^A$ of $P^{A*}$ lies on $P^bP^c$. By lemma 4.1.4, $\angle PP^AA = \angle PA^*P^{A*} = 90°$, so $PP^A \perp P^bP^c$. The two cyclic perpendicularities similarly give

$$\mathfrak{i}(\odot(P^AP^BP^C)) = \odot(P^{A*}P^{B*}P^{C*}) = \mathscr{N}'.$$

Since $R = \mathbf{g}_{\triangle P^aP^bP^c}P$, corollary 4.4.4 makes the midpoint $R_0$ of $PR$ the center of $\odot(P^AP^BP^C)$. Hence $\mathbf{p}R_0 = R$; the definition of inversion also gives $\mathbf{p}R^* = R_0^*$. By proposition 4.1.14, the points $R_0^*(=\mathbf{p}R^*)$ and $P$ are inverse with respect to the circle $\mathfrak{i}(\odot(P^AP^BP^C)) = \mathscr{N}'$. Therefore $Ni' \in P\mathbf{p}R^*$ and $r_{\mathscr{N}'}^2 = \overline{Ni'\mathbf{p}R^*} \cdot \overline{Ni'P}$. Since $\mathscr{N}'$ and $\omega$ are orthogonal, $\operatorname{pow}_\omega(Ni') = r_{\mathscr{N}'}^2$, and hence $\operatorname{pow}_\omega(Ni') = \overline{Ni'\mathbf{p}R^*} \cdot \overline{Ni'P}$. Thus $\mathbf{p}R^* \in \omega$. □

**Proof of lemma 7.6.21, concluded.** For (4), theorem 4.3.4 gives $(AP_a, PP^a) = -1$. Let $\infty_{PX_a}$ be the point at infinity on $PX_a$. Then $(\infty_{PX_a}P_A, PX_a) = -1$, so $(A, P_a, P, P^a) \barwedge (\infty_{PX_a}, P_A, P, X_a)$. Let $\triangle H_aH_bH_c$ be the orthic triangle of the original triangle. The lines $A\infty_{PX_a}, P_aP_A$ meet at $H_a$, so theorem 3.5.6 shows that $X_a, H_a, P^a$ are collinear. Thus $\measuredangle PX_aH_a = -\measuredangle P_APH_a$.

Let $\triangle H_a'H_b'H_c'$ be the orthic triangle of $\triangle P_A^*P_B^*P_C^*$. The right angles $\angle AH_aP_A = \angle P_A^*H_a'A^* = \angle H_aP_AP = \angle H_a'A^*P = 90°$ give $\measuredangle PP_A^*H_a' = \angle P_APA^* = -\measuredangle PAH_a$. The definition of inversion also gives $\frac{AP}{P_A^*P} = \frac{P_AP}{A^*P}$, so $APP_AH_a \eqsim P_A^*PA^*H_a'$. Therefore $\measuredangle A^*PH_a' = -\measuredangle P_APH_a$.

[21]As in the discussion of $P_A$ in (2), the points $P_B, P_C$ are the projections of $P$ on $CA, AB$, respectively. This gives the third equality below.

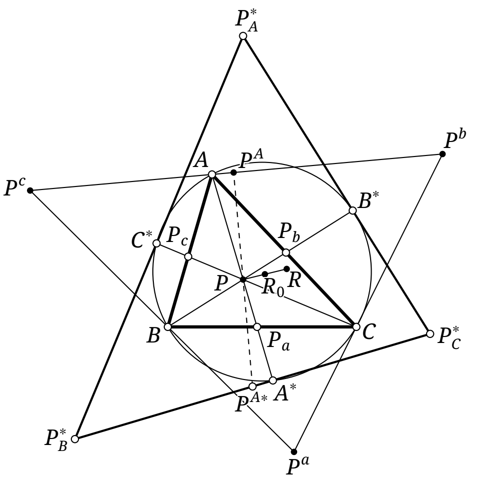


FIGURE 7.42

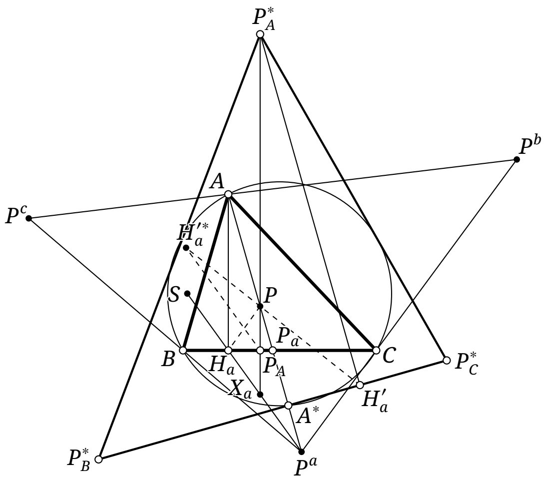


FIGURE 7.43

By lemma 4.1.4, $H'_a, H'^*_a, P^*_A, P_A$ are concyclic. Thus $\measuredangle PP_AH'^*_a = -\measuredangle PH'_aP^*_A = \measuredangle A^*PH'_a$, where the last equality uses $PA^*, P^*_AH'_a \perp P^*_BP^*_C$. Combining the angle relations already proved gives $\measuredangle PP_AH'^*_a = \measuredangle PX_aH_a$, so $H'^*_aP_A \parallel P^aH_a$. Since $P_A$ is the midpoint of $PX_a$, $\mathbf{p}(H'^*_aP_A) = P^aH_a$.

By the definition of the Ceva quotient, $S \in H_aP^a$, and therefore $\mathbf{p}^{-1}(S) \in H'^*_aP_A$. Hence $\mathbf{p}S^* = \mathfrak{i}(\mathbf{p}^{-1}(S)) \in \mathfrak{i}(H'^*_aP_A) = \odot(PH'_aP^*_A)$. Cyclically, $\mathbf{p}S^*$ also lies on $\odot(PH'_bP^*_B)$ and $\odot(PH'_cP^*_C)$. On the other hand, by the definition of the polar circle, the inversion $\mathfrak{i}_{\mathscr{P}'}$ sends $P^*_A, P^*_B, P^*_C$ to $H'_a, H'_b, H'_c$, respectively. Put $\mathfrak{i}_{\mathscr{P}'}(P) = S'$. Then $S'$ lies on $\odot(PH'_aP^*_A)$, $\odot(PH'_bP^*_B)$, and $\odot(PH'_cP^*_C)$, so $S' = \mathbf{p}S^*$. Thus $\mathfrak{i}_{\mathscr{P}'}(P) = \mathbf{p}S^*$. Since $H'$ is the center of $\mathscr{P}'$, we have $H' \in P\mathbf{p}S^*$. Since $\mathscr{P}'$ and $\omega$ are orthogonal, theorem 4.1.9 shows that $\mathfrak{i}_{\mathscr{P}'}$ preserves $\omega$. As $P \in \omega$, it follows that $\mathbf{p}S^* = \mathfrak{i}_{\mathscr{P}'}(P) \in \omega$. □

We now return to theorem 7.6.20.

**Proof of theorem 7.6.20.** In the notation of lemma 7.6.21, the five points $\mathbf{p}Q^*, \mathbf{p}R^*, \mathbf{p}S^*, \mathbf{p}T^*, P$ lie on $\omega$. The lines joining $P$ to the first four points pass through $O', Ni', H', Eu'$, respectively. Hence

$$(\mathbf{p}R^*\mathbf{p}T^*, \mathbf{p}Q^*\mathbf{p}S^*) = P(\mathbf{p}R^*\mathbf{p}T^*, \mathbf{p}Q^*\mathbf{p}S^*) = (Ni'Eu', O'H') = -1,$$

where the last equality is the harmonic form of the Euler-line theorem. By homothety, $R^*, T^*, Q^*, S^*$ also form a harmonic quadratic point range on a circle through $P$. Inversion preserves their cross-ratio; since their circle passes through $P$, before inversion $R, T, Q, S$ are collinear and form a harmonic range. □

We give one further conic property.

**Proposition 7.6.22.** *Given $\triangle ABC$ with centroid $G$ and orthocenter $H$, let $S$ be the projection of $G$ on $AH$. Then $\mathbf{o}(BC) = \mathfrak{C}(ABCGS)$.*[22]

**Proof.** As in figure 7.44, let $A'$ be the point of $\mathscr{C}$ antipodal to $A$, and let $O$ be the circumcenter of $\triangle ABC$. For every $K \in BC$, $\mathscr{O}(K) \perp BC$, so $\mathscr{O}(K)$ passes through the point at infinity $\infty_0$ on $AH$. Since $\mathbf{g}H = O$, corollary 7.2.18 gives $\mathbf{g}(AH) = AO$. By theorem 4.4.3, $\mathbf{g}\infty_0 \in \mathscr{C}$, so $A' = \mathbf{g}\infty_0$. Thus $\mathbf{g}A' = \infty_0 \in \mathscr{O}(K) = \mathscr{T}(\mathbf{o}K)$. By theorem 7.3.7, $\mathbf{go}K \in \mathscr{T}(A')$, so $\mathbf{o}K \in \mathbf{g}(\mathscr{T}(A'))$. Hence $\mathbf{o}(BC) = \mathbf{g}(\mathscr{T}(A'))$, which is a conic $\alpha$ through $A, B, C$.

Take $K$ to be the point at infinity on $BC$. Then $\mathscr{O}(K) = l_\infty$, so $\mathbf{o}K = G$ and $G \in \alpha$. We now prove $S \in \alpha$. As above, for a point $K$ on $BC$, $\mathscr{T}(\mathbf{o}K) \parallel AH$. It is therefore enough to prove $\mathscr{T}(S) \perp BC$. Let $\triangle H_aH_bH_c$ be the cevian triangle of $S$, and let $\mathscr{T}(S)$ meet the three sidelines at $H_1, H_2, H_3$. The

[22]Here $\mathbf{o}(BC)$ is the locus of the orthocorrespondents of the points on the side $BC$.

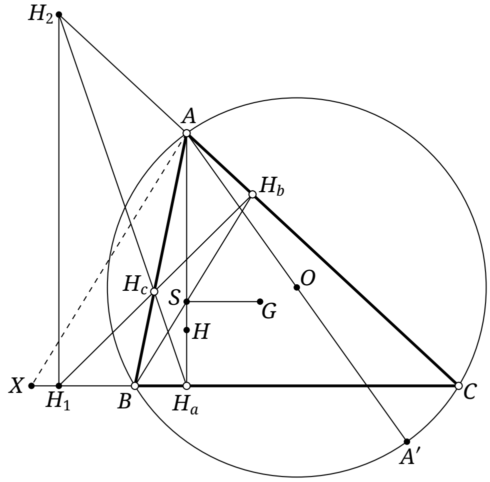


FIGURE 7.44

harmonic property of a complete quadrilateral gives

$$(A,C;H_2,H_b) = -1 \Longrightarrow \frac{AH_2}{CH_2} = \frac{AH_b}{CH_b},$$

$$(H_a,C;H_1,B) = 1-(H_a,H_1;C,B) = 2 \Longrightarrow \frac{H_aH_1}{CH_1} = 2\frac{H_aB}{CB}.$$

Through $A$, draw $AX \parallel BH_b$ meeting $BC$ at $X$. Since $GS \parallel BC$ and $G$ is the centroid, $2 = \frac{AS}{SH_a} = \frac{XB}{BH_a}$. Consequently

$$\frac{H_aH_1}{CH_1} = \frac{XB}{BH_a} \cdot \frac{H_aB}{CB} = \frac{XB}{CB} = \frac{AH_b}{CH_b} = \frac{AH_2}{CH_2}.$$

Thus $\mathscr{T}(S) = H_1H_2 \parallel AH$, as required. □

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## Exercises

**Exercise 7.66.** Reprove theorem 7.6.19 by polarity; this route is substantially shorter.

**Exercise 7.67.** Prove that, for an obtuse triangle, the image of the polar circle under the orthocorrespondent map meets some side of the triangle at two points symmetric about the midpoint of that side.

**Exercise 7.68.** For a triangle, prove that $\mathbf{o}Ap_1 = \mathbf{g}Np_2$ and $\mathbf{o}Ap_2 = \mathbf{g}Np_1$.

**Exercise 7.69.** Let $P,Q$ be isogonal conjugates with respect to $\triangle ABC$, and let $K$ be the perspector of the inconic of $\triangle ABC$ with foci $P,Q$. Prove that $K/P = \mathbf{o}P$.

**Exercise 7.70.** Given $\triangle ABC$ with orthocenter $H$ and circumcenter $O$, let $P$ be a point, and put $Q = \mathbf{o}P$ and $X = HP \cap QG$. Prove that $X \in \mathfrak{C}(ABCGH)$.

## Chapter 8

# Special Conics

This chapter builds on the preceding theory to study conics with special metric properties.

## 8.1 Rectangular Hyperbolas

### 8.1.1 Basic Properties of Rectangular Hyperbolas

We begin with a brief account of the most fundamental properties of a rectangular hyperbola. We have already proved its principal characterization in theorem 6.5.16; we restate it here for emphasis.

**Theorem 8.1.1.** *A circumconic of a triangle is a rectangular hyperbola if and only if it passes through the orthocenter of the triangle.*

**Corollary 8.1.2.** *For a chord $BC$ of a rectangular hyperbola and a point $A$ on the hyperbola, $AB \perp AC$ if and only if the tangent at $A$ is perpendicular to $BC$.*

**Proof.** Take four points $A, A', B, C$ on the rectangular hyperbola. Letting $A' \to A$ turns $AA'$ into the tangent at $A$, and theorem 8.1.1 gives the claim. ☐

We next record several other important properties.

**Theorem 8.1.3.** *The centers of the rectangular circumhyperbolas of $\triangle ABC$ trace the nine-point circle of $\triangle ABC$.*

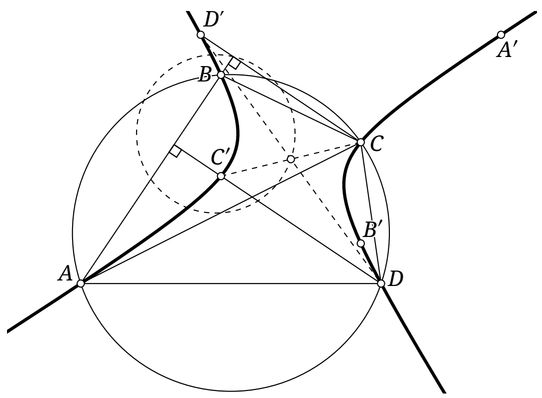


FIGURE 8.1

**Proof.** Let $D$ be the fourth intersection of $\odot(ABC)$ with a rectangular circumhyperbola $\alpha$. Denote the orthocenters of $\triangle BCD, \triangle CDA, \triangle DAB, \triangle ABC$ by $A', B', C', D'$, respectively. All four points lie on $\alpha$, as in figure 8.1.

Let $R$ be the radius of $\odot(ABC)$. By claim 0.5.17, $CD' = 2R|\cos\angle BCA| = 2R|\cos\angle BDA| = DC'$. Thus $CDC'D'$ is a parallelogram, and $C'D'$ is parallel and equal in length to $CD$. The points $D, D'$ and $C, C'$ are symmetric about the midpoints of $CC'$ and $DD'$, respectively.

The analogous statements for $A'$ and $B'$ show that the quadrangles $ABCD$ and $A'B'C'D'$ are centrally symmetric. Since $A', B', C', D'$ all lie on $\alpha$, their center of symmetry is the center $O$ of $\alpha$. Since it is also the midpoint of $DD'$, we have $O \in \mathfrak{h}_{D',1/2}(\odot(ABC)) = \mathcal{N}_{\triangle ABC}$. Conversely, every point of the nine-point circle occurs as the center of a rectangular circumhyperbola; the verification is left to the reader. □

**Theorem 8.1.4.** *Every chord of a rectangular hyperbola subtends equal or supplementary angles at a pair of antipodal points on the hyperbola.*[1] *More precisely, if $A,B$ are antipodal points of a rectangular hyperbola $\alpha$ and $XY$ is a chord, then $\measuredangle XAY = -\measuredangle XBY$.*

**Proof.** As in figure 8.2, let $L,N,M$ be the midpoints of $AX,AY,XY$, respectively. Then $\odot(LMN) = \mathscr{N}_{\triangle AXY}$, so theorem 8.1.3 gives $O \in \odot(LMN)$. Hence $\measuredangle NOL = \measuredangle NML$. The triangles $\triangle LNO$ and $\triangle XYB$ are related by a homothety of ratio $1:2$ centered at $A$, so $\measuredangle NOL = \measuredangle YBX$. In the parallelogram $ANML$, $\measuredangle NML = \measuredangle XAY$. Thus $\measuredangle XAY = \measuredangle YBX$. □

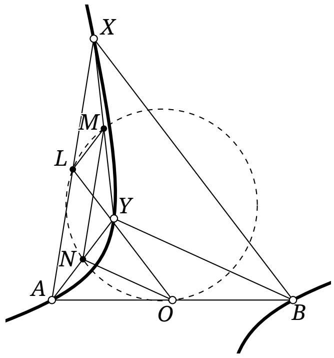


FIGURE 8.2

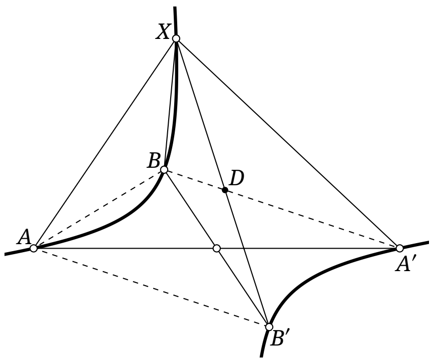


FIGURE 8.3

The converse also holds; its proof is left to the reader.

**Theorem 8.1.5.** *Given three noncollinear points $A,B,C$, the locus of points $X$ satisfying $\measuredangle XAC = -\measuredangle XBC$ is a rectangular hyperbola centered at the midpoint of $AB$.*

This yields another useful property.

**Theorem 8.1.6.** *If $A,B$ are antipodal points of a rectangular hyperbola $\alpha$ and $C \in \alpha$, then the asymptotes of $\alpha$ are parallel to the internal and external bisectors of $\angle BCA$.*

**Proof.** In theorem 8.1.5, let $X$ tend to a point at infinity. □

**Proposition 8.1.7.** *Let $A,A'$ and $B,B'$ be two pairs of antipodal points on a rectangular hyperbola $\alpha$. Then, for every $X \in \alpha$, $\measuredangle AXB = \measuredangle B'XA'$.*

**Proof.** As in figure 8.3, theorem 8.1.4 gives $\measuredangle XAB = \measuredangle BA'X$ and $\measuredangle ABX = \measuredangle XB'A$. Put $D = B'X \cap BA'$. Then

$$\measuredangle AXB = \measuredangle ABX + \measuredangle XAB = \measuredangle XB'A + \measuredangle BA'X = \measuredangle B'DA' + \measuredangle BA'X = \measuredangle B'XA'.$$

□

The angle-transfer property in theorem 8.1.4 is one of the distinctive advantages of rectangular hyperbolas. Apart from circles, other conics do not generally possess it. More precisely, conics of eccentricity $1/2$ or $2$ have certain angle properties as well (see exercise 1.26), but their applications are much less extensive. The following two examples illustrate this property.

[1]Two points of a central conic that are centrally symmetric with respect to its center are also called a pair of antipodal points of the conic.

**Example 8.1.8.** Let $A,B$ be antipodal points of a rectangular hyperbola $\alpha$ with center $O$. A circle $\odot P$ passes through $A,B$, and $C$ is one of the other intersections of $\odot P$ and $\alpha$. Prove that the tangent to $\odot P$ at $C$ is parallel to the tangent to $\alpha$ at $A$; see figure 8.4.

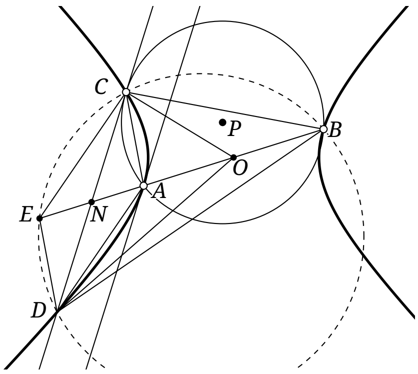


FIGURE 8.4

**Solution.** As in figure 8.4, let the tangent to $\odot P$ at $C$ meet $\alpha$ again at $D$. By the diameter property of a hyperbola, it is enough to show that the line $AB$ bisects $CD$.

Through $C$ draw the line parallel to $AD$, meeting the line $AB$ at $E$. By theorem 8.1.4, $\angle CAD + \angle CBD = 180°$, whereas $\angle ACD + \angle ADC + \angle CAD = 180°$. Hence $\angle ADC + \angle ACD = \angle CBD = \angle CBE + \angle DBE$. Since $CD$ is tangent to $\odot P$, $\angle DCA = \angle CBE$. It follows that $\angle DBE = \angle ADC = \angle DCE$, so $B,C,E,D$ are concyclic. Thus $\angle EDC = \angle CBE = \angle DCA$, whence $ED \parallel AC$. Consequently $ACED$ is a parallelogram, and $BE$ bisects $DC$. □

We now turn to several further special properties.

**Theorem 8.1.9.** *Given $\triangle ABC$ and a point $P$ distinct from its orthocenter, let $I'$ and $I'_a,I'_b,I'_c$ be the incenter and excenters of the cevian triangle of $P$. All four points lie on the rectangular hyperbola through $A,B,C,P$.*

**Proof.** By theorem 6.3.8, the eight points $I',I'_a,I'_b,I'_c,A,B,C,P$ lie on the bianticevian conic of $I'$ and $P$ with respect to the cevian triangle of $P$. Since $I'$ is the orthocenter of $\triangle I'_aI'_bI'_c$, the common conic is a rectangular hyperbola. □

**Theorem 8.1.10.** *Given $\triangle ABC$ and a point $D$, the center $Q$ of the rectangular hyperbola through $A,B,C,D$ lies on the cevian circle of $D$.*

**Proof.** As in figure 8.5, let $\triangle A'B'C'$ be the cevian triangle of $D$ and $\triangle I'_aI'_bI'_c$ its excentral triangle. By theorem 8.1.9, the circle $\odot(A'B'C')$ is the nine-point circle of $\triangle I'_aI'_bI'_c$. But $\odot(A'B'C')$ is precisely the cevian circle of $D$, so the center of the rectangular hyperbola lies on it. □

**Theorem 8.1.11.** *Given $\triangle ABC$ and a point $D$, the center of the rectangular hyperbola through $A,B,C,D$ lies on the pedal circle of $D$.*

**Proof.** As in figure 8.6, let $\triangle A'B'C'$ be the pedal triangle of $D$, let $B_1,C_1$ be the midpoints of $BD,CD$, and let $Q$ be the center of the rectangular hyperbola. By theorem 8.1.3, it is enough to prove $\angle A'C'B' = \angle A'QB'$.

Because $C',D,A',B$ are concyclic, $\angle DC'A' = \angle DBA'$. Since $B_1C_1$ is a midline of $\triangle DBC$, $\angle DBA' = \angle DB_1C_1$. The points $A',D$ are reflections in $B_1C_1$, so $\angle DB_1C_1 = \angle A'B_1C_1$. Finally, $\odot(B_1C_1A')$ is the nine-point circle of $\triangle BCD$, and hence $Q,A',B_1,C_1$ are concyclic. Therefore $\angle A'B_1C_1 = \angle A'QC_1$, and altogether $\angle DC'A' = \angle A'QC_1$.

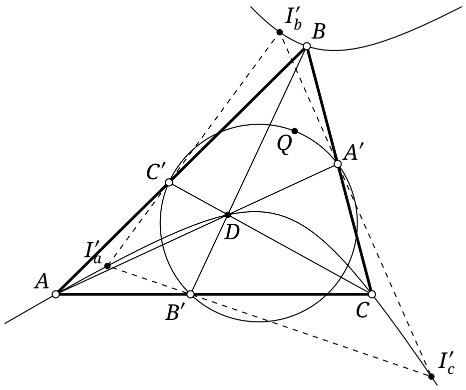


FIGURE 8.5

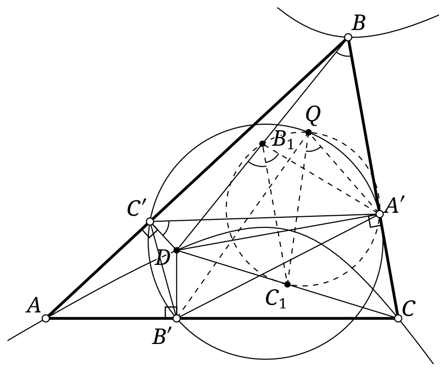


FIGURE 8.6

Likewise $\angle DC'B' = \angle B'QC_1$. Therefore

$$\angle A'C'B' = \angle DC'A' + \angle DC'B' = \angle A'QC_1 + \angle B'QC_1 = \angle A'QB'.$$

□

**Proposition 8.1.12.** *As in figure 8.7, let $\triangle A_2B_2C_2$ be the orthic triangle of $\triangle ABC$, and let $\triangle A_1B_1C_1$ be the cevian triangle of a point $P$. Corresponding sides of these two triangles meet at $A^*,B^*,C^*$. There are infinitely many triangles $\triangle A'B'C'$ such that $A',B',C'$ lie on the lines $BC,CA,AB$, respectively, while the lines $B'C',C'A',A'B'$ pass through $A^*,B^*,C^*$, respectively. Moreover:*

*(1) the lines $AA',BB',CC'$ concur at a point $T$, and the locus of $T$ over all such triangles is the rectangular hyperbola $\alpha$ through $A,B,C,P$;*

*(2) the circumcircles of all such triangles pass through a fixed point $Q$, namely the center of $\alpha$.*

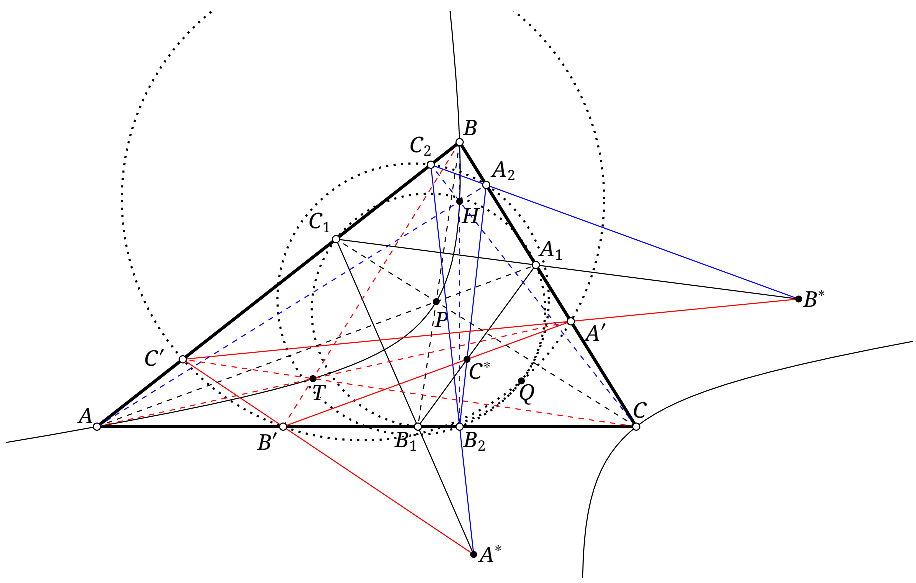


FIGURE 8.7

**Proof.** Let $l$ be any line through $C^*$, meeting $BC,AC$ at $A_c,B_c$, and put $T_c = AA_c \cap BB_c$. Applying theorem 6.2.9 at $C^*$ shows that the locus of $T_c$ is a hyperbola through $A,B,C,P,H$, where $H$ is the orthocenter. It is therefore the rectangular hyperbola $\alpha$.

Similarly, if a line through $A^*$ meets $AC,AB$ at $B_a,C_a$, respectively, the locus of $T_a = BB_a \cap CC_a$ is $\alpha$; the analogous locus of $T_b$ is also $\alpha$. Hence, for any $T \in \alpha$, the cevian triangle $\triangle A'B'C'$ of $T$ satisfies the required incidence conditions. This proves part (1).

The circle $\odot(A'B'C')$ is the cevian circle of $T$. By theorem 8.1.10, it passes through the center $Q$ of the rectangular hyperbola, which is fixed. This proves part (2). ☐

**Theorem 8.1.13.** *Given $\triangle ABC$ and a point $P$, consider the nine circles consisting of the following: the nine-point circles of the four triangles formed by three of $A,B,C,P$; the pedal circle of each one of the four points with respect to the triangle formed by the other three; and the circumcircle of the diagonal triangle of the complete quadrangle $ABCP$. These nine circles pass through one point.* It is called the *Poncelet point* of the complete quadrangle $ABCP$, or the Poncelet point of $P$ with respect to $\triangle ABC$.

**Proof.** The four points $A,B,C,P$ determine a unique rectangular hyperbola. The nine-point circle of any triangle formed by three of the points is the pedal circle of the orthocenter of that triangle, so it passes through the center of the hyperbola. Likewise, the pedal circle of any one of the points with respect to the triangle formed by the remaining three passes through the center of the rectangular hyperbola. Finally, the circumcircle of the diagonal triangle of the complete quadrangle $ABCP$ is the cevian circle of $P$ with respect to $\triangle ABC$, so it too passes through the center. ☐

The proof gives the following characterization.

**Proposition 8.1.14.** *Given $\triangle ABC$ and a point $P$, the Poncelet point of $P$ is the center of the rectangular circumhyperbola through $P$.*

Here is an example concerning the Poncelet point.

**Proposition 8.1.15.** *If $I$ is the incenter of $\triangle ABC$, then the Poncelet point of $ABCI$ is the Feuerbach point $Fe$.*

**Proof.** The pedal circle of $I$ with respect to $\triangle ABC$ is the incircle. Its unique common point with the nine-point circle is $Fe$, which must therefore be the Poncelet point. ☐

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## Exercises

**Exercise 8.1.** Given a quadrilateral $ABCD$, determine the locus of points $P$ for which the four circles $\odot(ABP), \odot(BCP), \odot(CDP), \odot(DAP)$ have equal radii.

**Exercise 8.2.** Let $P,Q$ be antipodal points of a rectangular hyperbola $\alpha$. The circle centered at $P$ with radius $PQ$ meets $\alpha$ at $A,B,C,Q$. Prove that $\triangle ABC$ is equilateral.

**Exercise 8.3.** Let $P$ be the center of a rectangular hyperbola through a cyclic quadrilateral $ABCD$, and let $O$ be the circumcenter. Prove that the line $PO$ passes through the centroid of $ABCD$.

**Exercise 8.4.** Prove that $A,B,C,A',B',C'$ lie on one conic if and only if there is a conic $\alpha$ for which $\triangle ABC$ and $\triangle A'B'C'$ are self-polar triangles.

**Exercise 8.5.** Let $\triangle ABC$ be self-polar with respect to a conic $\alpha$ centered at $O$. Prove that a conic passes through the midpoints of $AB,BC,CA,OA,OB,OC$ and is homothetic to $\alpha$.

**Exercise 8.6.** Given $\triangle ABC$ and a point $P$ in the plane, put $Q = \mathbf{ct}P$. Prove that the cevian circle of $P$ is tangent to $\mathscr{N}$ if and only if the line $Q\mathbf{g}Q$ passes through the Lemoine point.

**Exercise 8.7.** If two tangents can be drawn from the orthocenter of a triangle to its incircle, call them the triangle's two *orthotangents* in this problem. Prove that if one orthotangent cuts the triangle into a smaller triangle and a quadrilateral, then the other orthotangent is parallel to an orthotangent of the smaller triangle.

In particular, the rectangular circumhyperbola through the incenter in the preceding propositions is called the triangle's *Feuerbach hyperbola*. Thus its center is the Feuerbach point. For proposition 8.1.12, [**6**] attributes the case in which $P$ is the incenter to Emelyanov–Emelyanova.

### 8.1.2 Angular Properties of Polarity for a Rectangular Hyperbola

Polarity with respect to a rectangular hyperbola has several special metric features, which we now examine.

**Lemma 8.1.16.** *Let $a$ be the polar of a point $A$ with respect to a rectangular hyperbola $\alpha$ centered at $O$. The internal and external angle bisectors of the lines $OA$ and $a$ are parallel to the two asymptotes of $\alpha$.*

**Proof.** Let $X = \mathfrak{p}^{-1}(OA)$. This is a point at infinity. By the principle of polarity, $X \in a$, so $OX \parallel a$. The lines $OX$ and $OA$ are conjugate diameters and are harmonically separated by the two asymptotes, by theorem 6.4.12. Since the asymptotes of a rectangular hyperbola are perpendicular, proposition 3.2.17 shows that they are the internal and external bisectors of $\angle XOA$. Together with $OX \parallel a$, this proves the claim. ☐

**Corollary 8.1.17.** *Let $a$ be the polar of $A$ with respect to a rectangular hyperbola $\alpha$ centered at $O$, and let $X$ be the projection of $O$ on $a$. Then $OA$ and $OX$ are reflections of one another in the transverse axis (or the conjugate axis) of $\alpha$.*

**Proof.** Let $x$ be the transverse axis and $l$ an asymptote. By lemma 8.1.16, $\measuredangle(l,a) = \measuredangle(OA,l)$. Consequently

$$\begin{aligned}\measuredangle(AO,x) &= \measuredangle(AO,l) + \measuredangle(l,x) = \measuredangle(l,a) + \measuredangle(l,x) = \measuredangle(l,OX) + 90^\circ + \measuredangle(l,x)\\ &= \measuredangle(l,OX) + 90^\circ \pm 45^\circ = \measuredangle(l,OX) \mp 45^\circ = \measuredangle(l,OX) - \measuredangle(l,x) = \measuredangle(x,OX).\end{aligned}$$

The angle between the asymptote and the transverse axis is $45^\circ$ or $-45^\circ$, depending on the chosen asymptote; the final conclusion is the same. Thus $OA$ and $OX$ are symmetric about the transverse axis, and hence also about the conjugate axis. ☐

**Proposition 8.1.18.** *Let $O$ be the center of a rectangular hyperbola. If the polars of two arbitrary points $A,B$ are $a,b$, respectively, then $\measuredangle AOB = \measuredangle(b,a)$.*

**Proof.** Let $l$ be an asymptote. Then lemma 8.1.16 gives $\measuredangle(l,a) = \measuredangle(OA,l)$ and $\measuredangle(l,b) = \measuredangle(OB,l)$. Therefore

$$\measuredangle AOB = \measuredangle(OA,OB) = \measuredangle(OA,l) + \measuredangle(l,OB) = \measuredangle(l,a) + \measuredangle(b,l) = \measuredangle(b,a).$$

☐

**Proposition 8.1.19.** *The polar reciprocal of a rectangular hyperbola with respect to its auxiliary circle is the hyperbola itself.*

**Proof.** Let $A,A'$ be the vertices and $l,l'$ the asymptotes of the rectangular hyperbola $\alpha$. Under this polarity, $\mathfrak{d}(\mathfrak{t}_\alpha(A)) = A$, $\mathfrak{d}(\mathfrak{t}_\alpha(A')) = A'$, $\mathfrak{d}(l) = \infty_{l'}$, and $\mathfrak{d}(l') = \infty_l$. The polar reciprocal of the auxiliary circle is itself, and that circle is tangent to $\alpha$ at $A,A'$. Since polarity preserves tangency, $\mathfrak{d}(\alpha)$ is again tangent to that circle at $A,A'$. Thus the six points $\infty_l,\infty_{l'},A,A,A',A'$ determine $\mathfrak{d}(\alpha)$, which can only be $\alpha$ itself. ☐

**Proposition 8.1.20.** *Let $\omega$ be the auxiliary circle and $x$ the transverse axis of a rectangular hyperbola $\alpha$. Then $\mathfrak{d}_\alpha = \mathfrak{f}_x \circ \mathfrak{d}_\omega$.*

**Proof.** As in figure 8.8, let $A',A$ be the left and right vertices of $\alpha$, respectively, so that $\odot(O,OA)$ is its auxiliary circle. For a point $P$, put $\mathfrak{d}_\alpha(P) = \mathfrak{p}_\alpha(P) = l$ and $\mathfrak{d}_\omega(P) = \mathfrak{p}_\omega(P) = l'$. It is enough to prove that $l$ and $l'$ are reflections in $x$.

Let $X$ be the projection of $P$ on $x$, and let $Y,Y'$ be the intersections of $l,l'$ with $x$, respectively. The projective definition of pole and polar gives $(AA',XY) = -1 = (AA',XY')$, so $Y = Y'$ and the two polars meet on $x$. Let $M,N$ be the projections of $O$ on $l,l'$, respectively. For circle polarity, $O,P,N$ are collinear. By corollary 8.1.17, $OM$ and $ON$ are symmetric about $x$. Since $l,l'$ meet on $x$, the two lines themselves are symmetric about $x$. ☐

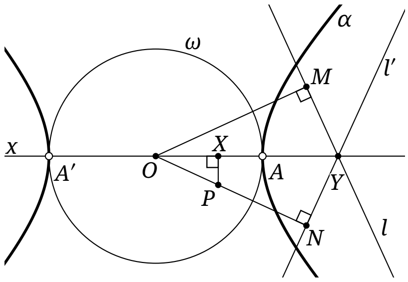


FIGURE 8.8

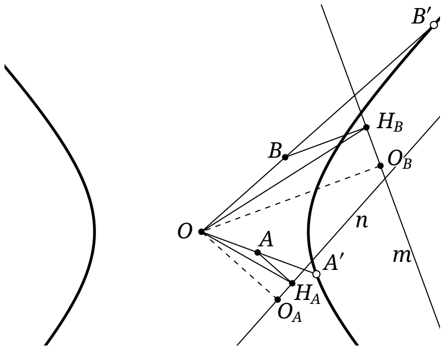


FIGURE 8.9

**Proposition 8.1.21.** *Let $l$ be the polar of $A$ with respect to a rectangular hyperbola $\alpha$ centered at $O$, and let $a$ be its semitransverse axis. Then $OA \cdot \operatorname{dist}(O,l) = a^2$.*

**Proof.** Consider the auxiliary circle $\omega$ of $\alpha$, and put $\mathfrak{p}_\omega(A) = l'$. Clearly $OA \cdot \operatorname{dist}(O,l') = r_\omega^2 = a^2$. By proposition 8.1.20, $l,l'$ are symmetric about the transverse axis, so $OA \cdot \operatorname{dist}(O,l) = a^2$ as well. ☐

Here is a first application.

**Example 8.1.22.** Let $O$ be the center of a rectangular hyperbola, and let $m,n$ be the polars of two arbitrary points $A,B$. Denote the projections of $A,B$ on $n,m$ by $H_A,H_B$, respectively. Prove that $\triangle OAH_A \stackrel{+}{\sim} \triangle OBH_B$.

**Solution.** As in figure 8.9, let $O_A,O_B$ be the projections of $O$ on $n,m$, respectively. By proposition 8.1.18, $\measuredangle O_AOO_B = -\measuredangle(m,n) = \measuredangle AOB$, so $\measuredangle AOO_A = \measuredangle BOO_B$. By proposition 8.1.21, $OA \cdot OO_B = OB \cdot OO_A$, or $\frac{OO_A}{OA} = \frac{OO_B}{OB}$. Thus $\triangle O_AOA \stackrel{+}{\sim} \triangle O_BOB$. Together with $\angle OO_AH_A = \angle O_AH_AA = \angle OO_BH_B = \angle O_BH_BB = 90^\circ$, this gives the similarity of quadrilaterals $OO_AH_AA \stackrel{+}{\sim} OO_BH_BB$, and therefore $\triangle OAH_A \stackrel{+}{\sim} \triangle OBH_B$. ☐

We give some further properties.

**Proposition 8.1.23.** *Let $\alpha$ be a rectangular hyperbola centered at $O$, let $A,B \in \alpha$, let $M$ be the midpoint of $AB$, and put $P = \mathfrak{p}_\alpha^{-1}(AB)$. Then $\triangle OMA \stackrel{-}{\sim} \triangle AMP$ and $\triangle OMB \stackrel{-}{\sim} \triangle BMP$.*

**Proof.** As in figure 8.10, the diameter property gives $O,P,M$ collinear. Since $\mathfrak{p}(P)=AB$, we have $\mathfrak{p}(A)=AP$. Hence proposition 8.1.18 gives $\measuredangle AOP=\measuredangle BAP$, so $\triangle OMA \overset{-}{\sim} \triangle AMP$. The second is analogous. □

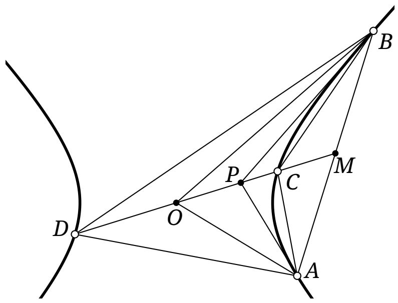


FIGURE 8.10

**Proposition 8.1.24.** *Let $\alpha$ be a rectangular hyperbola with center $O$, let $A,B$ be points on $\alpha$, and let $M$ be the midpoint of $AB$. Let $OM$ meet $\alpha$ at $C,D$. Then $\triangle CMA \overset{-}{\sim} \triangle AMD$ and $\triangle CMB \overset{-}{\sim} \triangle BMD$.*

**Proof.** As in figure 8.10, put $P=\mathfrak{p}^{-1}(AB)$, so $O,M,P$ are collinear. By proposition 8.1.23, $MA^2=MO\cdot MP$. The harmonic property of pole and polar gives $(PM,CD)=-1$; together with proposition 3.2.18, this yields $MC\cdot MD=MP\cdot MO=MA^2=MB^2$. The two asserted similarities follow. □

**Proposition 8.1.25.** *Let $A,A'$ and $B,B'$ be two pairs of conjugate points with respect to a rectangular hyperbola centered at $O$. If $\measuredangle AOB=\measuredangle B'OA'$, then $\triangle AOB \overset{+}{\sim} \triangle B'OA'$.*

**Proof.** As in figure 8.11, let $C=\mathfrak{p}^{-1}(AB)$ and $D=CB'\cap AB$. By the principle of polarity, $C\in\mathfrak{p}(A)$ and $C\in\mathfrak{p}(B)$, so $\mathfrak{p}(B)=B'C$ and $\mathfrak{p}(A)=A'C$. Hence, by proposition 8.1.18, $\measuredangle B'OA'=\measuredangle AOB=\measuredangle(\mathfrak{p}(B),\mathfrak{p}(A))=\measuredangle B'CA'$, and $A',O,B',C$ are concyclic.

Again by proposition 8.1.18, $\measuredangle BOC=\measuredangle(\mathfrak{p}(C),\mathfrak{p}(B))=\measuredangle(AB,B'C)=\measuredangle BDC$, so $O,C,B,D$ are concyclic. It follows that $\measuredangle OA'B'=\measuredangle OCB'=\measuredangle OBA$, so $\triangle AOB \overset{+}{\sim} \triangle B'OA'$. □

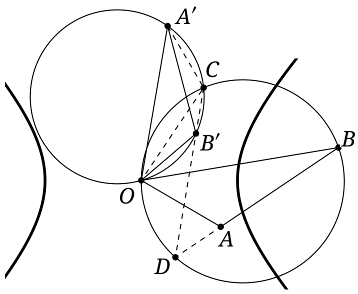


FIGURE 8.11

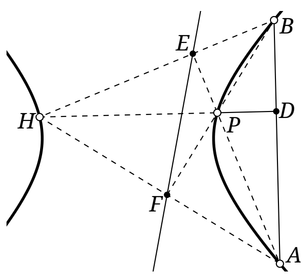


FIGURE 8.12

## Exercises

**Exercise 8.8.** Let $O$ be the center of a rectangular hyperbola, and let $l_A,l_B$ be the polars of $A,B$. If $X,Y$ are the projections of $A,B$ on $l_B,l_A$, respectively, prove that $AX\cdot OY=BY\cdot OX$.

**Exercise 8.9.** Let $A,B,P$ lie on a rectangular hyperbola $\alpha$, let $D$ be the projection of $P$ on $AB$, and put $l = \mathfrak{p}_\alpha(D)$. Prove that $\measuredangle(l,AB) = \measuredangle PBA + \measuredangle PAB$.

**Exercise 8.10.** Given $\triangle ABC$, let $(P,Q)$ and $(R,S)$ be two pairs of isogonal conjugates. Prove that there is a spiral similarity sending the segment $RQ$ to $PS$, with its center on $\mathscr{C}$.

**Exercise 8.11.** Let $D$ be a point on the side $BC$ of $\triangle ABC$, and put $AD \cap \mathscr{C} = A,E$. Draw $EF \parallel BC$, meeting $\mathscr{C}$ again at $F$, and let the line $FD$ meet $\mathscr{C}$ again at $T$. If $I$ is the incenter and $K$ the midpoint of $DI$, prove that $\angle ADI = \angle DTK$.

### 8.1.3 Isogonal Images and Antigonal Conjugation

The isogonal image of a rectangular hyperbola has several special properties, particularly in constructions involving the hyperbola itself.

**Theorem 8.1.26.** *The isogonal image of a line through the circumcenter of a triangle is a rectangular circumhyperbola. Conversely, the isogonal image of a rectangular circumhyperbola is a line through the circumcenter.*

**Proof.** The isogonal conjugate of the circumcenter is the orthocenter, and a circumconic through the orthocenter is a rectangular hyperbola. □

**Proposition 8.1.27.** *Given $\triangle ABC$, let $UV$ be a diameter of $\mathscr{C}$, with $U,V \in \mathscr{C}$. Then $\mathbf{g}(UV)$ is a rectangular hyperbola whose asymptotes are the Simson lines $\mathscr{S}(U)$ and $\mathscr{S}(V)$.*

**Proof.** That the image is a rectangular hyperbola is immediate. By theorem 4.4.3, $\mathbf{g}U$ and $\mathbf{g}V$ are the two points at infinity of $\mathbf{g}(UV)$, and their directions are perpendicular to $\mathscr{S}(U)$ and $\mathscr{S}(V)$, respectively. By lemma 1.5.3, those directions are perpendicular to one another. Hence $\mathbf{g}U \in \mathscr{S}(V)$ and $\mathbf{g}V \in \mathscr{S}(U)$, so the two Simson lines are parallel to the two asymptotes, respectively.

For any point $P$ of $UV$, we have $\mathbf{g}P \in \mathbf{g}(UV)$. By theorem 8.1.11, the center of $\mathbf{g}(UV)$ lies on the pedal circle of $\mathbf{g}P$. By corollary 4.4.5, this is also the pedal circle of $P$. As $P$ tends to $U$ or $V$, its pedal circle degenerates to $\mathscr{S}(U)$ or $\mathscr{S}(V)$, respectively. Thus both Simson lines pass through the center of the hyperbola and are its asymptotes. □

**Theorem 8.1.28.** *With respect to a fixed reference triangle, the center of the isogonal image of a line through the circumcenter is the orthopole of that line.*

**Proof.** In proposition 8.1.27, the intersection of the two asymptotes is the center of the isogonal image. By corollary 7.1.9, that intersection is precisely the orthopole of $UV$. □

**Theorem 8.1.29** (Griffiths's theorem). *Let $l$ be a fixed line through the circumcenter $O$ of $\triangle ABC$. The pedal circles of all points on $l$ pass through a fixed point of the nine-point circle.* This point is called the *Griffiths point* of $l$.

**Proof.** The isogonal image of $l$ is a rectangular hyperbola $\alpha$. By corollary 4.4.5, isogonal conjugates have the same pedal circle. Every pedal circle of a point on $\alpha$ passes through the center of $\alpha$, so the pedal circle of every point on $l$ also passes through the center of $\alpha$. By theorem 8.1.3, the center of $\alpha$ lies on the nine-point circle. □

Together with theorem 8.1.28, this gives the following.

**Corollary 8.1.30.** *With respect to a fixed reference triangle, the Griffiths point of a line through the circumcenter is its orthopole.*

We also obtain the following corollary.

**Corollary 8.1.31** (Fontené's theorem)**.** *Given $\triangle ABC$ with circumcenter $O$, let $P,Q$ be isogonal conjugates. The Poncelet point of $Q$ with respect to $\triangle ABC$ is the orthopole of the line $OP$.*

**Proof.** By Griffiths's theorem, $\mathbf{g}(OP)$ is a rectangular circumhyperbola $\alpha$ whose center is the Griffiths point of $OP$, hence its orthopole. Since $Q = \mathbf{g}P$, the hyperbola $\alpha$ passes through $Q$. By theorem 8.1.13, the Poncelet point of $Q$ is also the center of $\alpha$. □

We now use the preceding properties of isogonal images to study antigonal conjugation.

**Definition 8.1.32.** Given $\triangle ABC$ and a point $P$ different from $A,B,C$ and the orthocenter, points $P',P$ are called *antigonal conjugates* with respect to $\triangle ABC$ if $\measuredangle APB + \measuredangle AP'B = 0$, $\measuredangle BPC + \measuredangle BP'C = 0$, and $\measuredangle CPA + \measuredangle CP'A = 0$. We write the antigonal conjugate of $P$ as $\mathbf{f}_{\triangle ABC}P$, or simply $\mathbf{f}P$ when the reference triangle is clear.

We verify existence and uniqueness. Given $\triangle ABC$ and a point $P$ different from its orthocenter, reflect $\odot(ABP)$ and $\odot(ACP)$ in $AB$ and $AC$, respectively. Unless the reflected circles coincide, they have exactly one common point $P'$ other than $A$, and $\measuredangle APB + \measuredangle AP'B = 0$, $\measuredangle CPA + \measuredangle CP'A = 0$. The identities $\measuredangle APB + \measuredangle BPC + \measuredangle CPA = 0$ and $\measuredangle AP'B + \measuredangle BP'C + \measuredangle CP'A = 0$ also give $\measuredangle BPC + \measuredangle BP'C = 0$. Thus $P'$ is uniquely determined.

We required the reflections of $\odot(ABP), \odot(ACP)$ in $AB,AC$, respectively, to be distinct. If they coincide, their common circle must be the circumcircle $\mathscr{C}$. By lemma 0.5.19, $P$ is then the orthocenter. Thus the antigonal conjugate exists and is unique whenever $P$ is neither the orthocenter nor a vertex of the triangle. The next theorem gives a characterization of antigonal conjugation.

**Theorem 8.1.33.** *Given $\triangle ABC$ and a point $P$, the five points $\mathbf{f}P,P,A,B,C$ lie on a rectangular hyperbola, and $P,\mathbf{f}P$ are antipodal on it.*

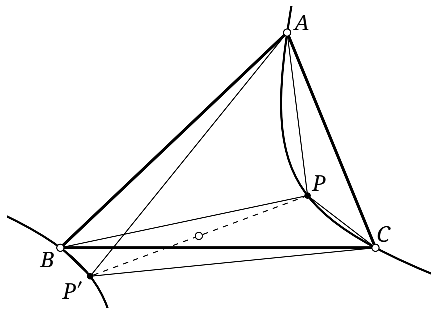


FIGURE 8.13

**Proof.** Draw the rectangular hyperbola $\alpha$ through $A,B,C,P$. By theorem 8.1.4, the point $P'$ antipodal to $P$ on $\alpha$ satisfies the three antigonal relations. Uniqueness gives $P' = \mathbf{f}P$. □

Finally, we use isogonal images to prove an important theorem.

**Theorem 8.1.34.** *Given $\triangle ABC$, $\mathbf{f} = \mathbf{gig}$. Equivalently, the isogonal conjugates of a pair of antigonal conjugates are inverse with respect to the circumcircle.*

**Proof.** For an arbitrary point $P$, consider the rectangular circumhyperbola $\alpha$ through $P$, with center $Q$. Its isogonal image $\mathbf{g}(\alpha)$ is a line $l$ through the circumcenter $O$. Fix $\alpha$ and let $P$ vary on $\alpha$.

By theorem 8.1.33, the correspondence $\alpha[P] \to \alpha[\mathbf{f}P]$ between quadratic point ranges is the half-turn about $Q$, so it is a projectivity and an involution. On the other hand, theorems 4.1.13 and 7.2.20

show that $\alpha[P] \to l[\mathbf{g}P]$, $l[\mathbf{g}P] \to l[\mathbf{ig}P]$, and $l[\mathbf{ig}P] \to \alpha[\mathbf{gig}P]$ are projective correspondences. Thus **gig** is a projectivity. Since $(\mathbf{gig})^2 = \mathrm{id}_\alpha$, the map **gig** is also an involution.

Let $X, Y$ be the points at infinity of $\alpha$. Each is its own antipodal point, so $X, Y$ are fixed by $\mathbf{f}$. By theorem 4.4.3, and because $\mathbf{i}$ fixes every point of $\mathscr{C}$, the points $X, Y$ are also fixed by **gig**. Two fixed points determine an involution, so $\mathbf{f} = \mathbf{gig}$. □

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## Exercises

**Exercise 8.12.**

(1) Determine the antigonal image of the circumcircle of a triangle.

(2) Let $\gamma$ be a curve through the orthocenter $H$ of a triangle. When forming the antigonal image of $\gamma$, interpret $\mathbf{f}H$ as $\lim_{P\in\gamma, P\to H} \mathbf{f}P$. If $\alpha$ is a circumconic tangent to $\gamma$ at $H$, prove that, in this limiting sense, $\mathbf{f}H$ is the fourth intersection of $\alpha$ and $\mathscr{C}$ other than $A, B, C$.

**Exercise 8.13.** Prove $\mathbf{f} = \mathbf{gig}$ without using conics.

**Exercise 8.14.** Let $I$ be the incenter of $\triangle ABC$, and let $P, Q$ be isogonal conjugates. Put $P_a = \mathfrak{i}_{\odot(IBC)}(P)$, and define $P_b, P_c$ cyclically. Prove that $P, P_a, P_b, P_c$ are concyclic.
*Hint.* Invert in the incircle.

**Exercise 8.15.** Let $P, Q$ be antigonal conjugates with respect to $\triangle ABC$. Let $H'$ be the orthocenter of the Carnot triangle of $P$, let $X$ be the circumcenter of $\triangle APQ$, and let $H_1$ be the orthocenter of $\triangle BQC$. Prove that $H'X \perp PH_1$.

**Exercise 8.16*.** Given $\triangle ABC$ with circumcenter $O$, let $P, Q$ be antigonal conjugates. Prove that the Carnot triangles of $P, Q$ have a common orthocenter $H'$, and that the projection of $H'$ on the perpendicular bisector of $PQ$ is the orthopole of $OH'$ with respect to $\triangle ABC$.

# 8.2 Inconics and Circumconics

We now study properties of the inconics and circumconics of a triangle. The important properties of these conics were already developed in section 7.3.2, on trilinear polarity and conics, and section 7.4.2, on the basic properties of Cevian-geometric transformations. Here we add several metric properties, beginning with inconics.

### Metric Properties of Inconics

**Theorem 8.2.1.** *Let a conic $\alpha$ be tangent to the sides of $\triangle ABC$. For a point $X$, the orthotransversal $\mathscr{O}(X)$ is tangent to $\alpha$ if and only if $X$ lies on the Monge circle of $\alpha$.*

**Proof.** As in figure 8.14, suppose $\mathscr{O}(X)$ meets the three sides at $D, E, F$. By definition, $XA \perp XD$, $XB \perp XE$, and $XC \perp XF$. Thus $(XA, XD), (XB, XE), (XC, XF)$ are three corresponding pairs of the same involution of the pencil at $X$, since the projective definition of perpendicularity induces this special involution.

Apply Plücker's theorem to the four lines $AB, BC, CA, \mathscr{O}(X)$. The line $\mathscr{O}(X)$ is tangent to $\alpha$ if and only if the two tangents from $X$ to $\alpha$ form another corresponding pair of this involution, that is, if and only if they are perpendicular. This is precisely the condition that $X$ lie on the Monge circle of $\alpha$. □

**Theorem 8.2.2.** *The Monge circle of every inconic of a triangle is orthogonal to the polar circle $\mathscr{P}$ of the triangle.* For an acute or right triangle, the polar circle is imaginary or a point circle, respectively; the discussion below remains valid.

**Proof.** As in figure 8.14, choose a tangent $l$ of an inconic $\alpha$, meeting $BC, CA, AB$ successively at $D, E, F$. Let $\triangle H_aH_bH_c$ be the orthic triangle and $H$ the orthocenter of $\triangle ABC$. By theorems 7.6.3 and 7.6.4, the circles $\odot(AH_aD)$, $\odot(BH_bE)$, and $\odot(CH_cF)$ are coaxial. Let their two common points be $X, Y$. Then $X, H, Y$ are collinear and $\mathscr{O}(X) = \mathscr{O}(Y) = l$.

By theorem 8.2.1, $X, Y$ lie on the Monge circle $\odot O'$ of $\alpha$. Hence $\odot O'$ is coaxial with the preceding three circles, and

$$HO'^2 - r^2_{\odot O'} = \text{pow}_{\odot O'}(H) = \text{pow}_{\odot(AH_aD)}(H) = \overline{AH} \cdot \overline{H_aH} = r^2_{\mathscr{P}}.$$

The last equality is the definition of the polar circle. Thus $\odot O'$ and $\mathscr{P}$ are orthogonal. □

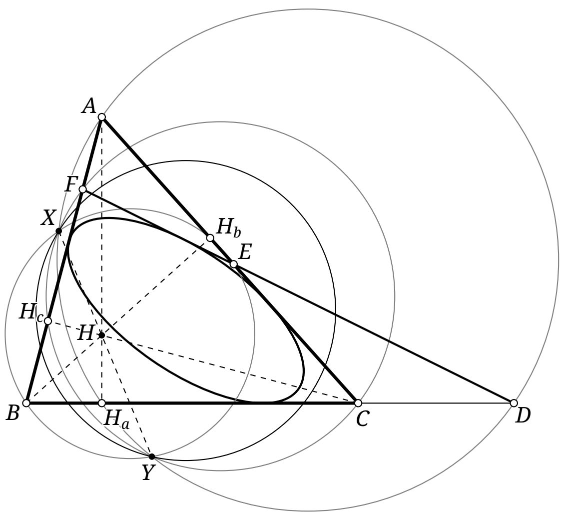


FIGURE 8.14

**Theorem 8.2.3.** *With respect to a fixed reference triangle, as a point moves on the Monge circle of a fixed inconic, its orthocorrespondent lies on a fixed line.*

**Proof.** This follows immediately from the definition of the orthocorrespondent and theorems 7.3.17 and 8.2.1. □

**Theorem 8.2.4.** *For $\triangle ABC$, let $O, H, R$ be its circumcenter, orthocenter, and circumradius, respectively. If an inconic $\alpha$ has foci $P, Q$ and auxiliary-circle radius $r$, then* $4r^2 - HP^2 - HQ^2 = R^2 - OH^2$.

**Proof.** As in figure 8.15, let $\rho$ be the radius of the Monge circle and $O'$ the center of $\alpha$; then $O'$ is also the midpoint of $PQ$. Let $H_a$ be the projection of $H$ on $BC$, and let $H'$ be the reflection of $H$ in $BC$. By lemma 0.5.19, $H' \in \mathscr{C}$. Since the Monge circle is orthogonal to the polar circle by theorem 8.2.2,

$$O'H^2 = \rho^2 + r^2_{\mathscr{P}} = \rho^2 + \overline{AH} \cdot \overline{H_aH} = \rho^2 + \frac{1}{2}\overline{AH} \cdot \overline{H'H} = \rho^2 + \frac{1}{2}\text{pow}_{\mathscr{C}}(H) = \rho^2 + \frac{1}{2}(OH^2 - R^2). \quad (\heartsuit)$$

The second equality uses the definition of the polar circle. On the other hand, the median formula in $\triangle HPQ$ gives

$$O'H^2 = \frac{1}{2}(HP^2 + HQ^2) - \frac{1}{4}PQ^2, \quad (\spadesuit)$$

whereas Monge's circle theorem gives

$$\rho^2 = 2r^2 - \frac{1}{4}PQ^2. \quad (\clubsuit)$$

Combining equations (♥) to (♣) gives $4r^2 - HP^2 - HQ^2 = R^2 - OH^2$. □

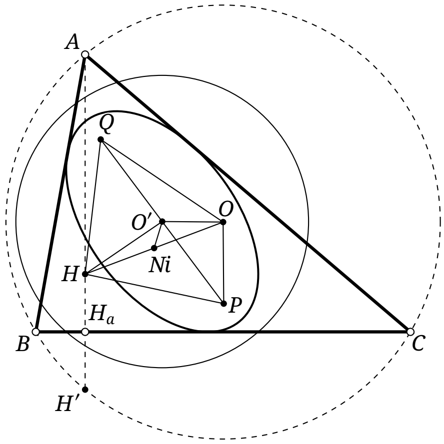


FIGURE 8.15

**Theorem 8.2.5.** *Let $R, O$ be the circumradius and circumcenter of $\triangle ABC$. If a central inconic $\alpha$ has semi-principal axis a, semifocal distance c, and foci $P, Q$, then $(R^2 - OP^2)(R^2 - OQ^2) = 4R^2(a^2 - c^2)$.*

**Proof.** By theorem 4.4.14, $Q = \mathbf{g}P$. Let $\triangle P_AP_BP_C$ and $\triangle Q_AQ_BQ_C$ be the pedal triangles of $P, Q$, respectively. Because $P, Q$ play symmetric roles, as do $A, B, C$, applying proposition 4.4.10, cycling $A, B, C$, and interchanging $P, Q$ yields six relations. Their product is

$$(R^2 - OP^2)^3(R^2 - OQ^2)^3 = (2R)^6 \prod_{\mathrm{cyc}(A,B,C)} (\overline{PP_A} \cdot \overline{QQ_A}).$$

By proposition 1.3.7,

$$\overline{PP_A} \cdot \overline{QQ_A} = \overline{PP_B} \cdot \overline{QQ_B} = \overline{PP_C} \cdot \overline{QQ_C} = a^2 - c^2.$$

Substitution and taking cube roots gives the formula. □

**Theorem 8.2.6** (Aiyar's theorem (ஐயர்)[2]). *Let O and Ni be the circumcenter and nine-point center of $\triangle ABC$, and let R be its circumradius. If an inconic $\alpha$ has center $O'$ and foci $P, Q$, then $OP \cdot OQ = 2R \cdot NiO'$.*

**Proof.** As in figure 8.15, let $a$ and $c$ be the semi-principal axis and semifocal distance of $\alpha$. Applying the median formula to $\triangle OPQ$ gives $O'O^2 = \frac{1}{2}(OP^2 + OQ^2) - c^2$. Since *Ni* is the midpoint of $OH$, a second application of the median formula, now in $\triangle O'HO$, gives

$$O'Ni^2 = \frac{1}{2}(O'H^2 + O'O^2) - \frac{1}{4}OH^2 = \frac{1}{2}\left[O'H^2 + \frac{1}{2}(OP^2 + OQ^2) - c^2\right] - \frac{1}{4}OH^2. \qquad (♡)$$

Rearranging theorem 8.2.5 gives

$$OP^2 + OQ^2 = R^2 - 4(a^2 - c^2) + \frac{OP^2 \cdot OQ^2}{R^2}, \qquad (♤)$$

while equation (♥) in the proof of theorem 8.2.4 gives

$$HO'^2 = (2a^2 - c^2) - \frac{1}{2}(R^2 - OH^2). \qquad (♧)$$

Substituting equations (♤) and (♧) into equation (♡) yields the stated identity. □

[2]That is, V. Ramaswami Aiyar (வி. ராமசுவாமி ஐயர்).

### Metric Properties of Circumconics

We give some further examples concerning circumconics.

**Proposition 8.2.7.** *Let $P$ be one focus of a circumconic $\alpha$ of $\triangle ABC$, and let $Q=\mathbf{g}P$. If $\triangle DEF$ is the pedal triangle of $Q$, then one of its four in/excenters $I$ has the following properties:*

*(1) the principal axis of $\alpha$ is parallel to $QI$;*

*(2) the focal parameter $p$ of $\alpha$ is*

$$p=\frac{QD}{QI}\cdot \operatorname{dist}(P,BC)=\left|\frac{(R^2-OP^2)(R^2-OQ^2)}{4R^2\cdot QI}\right|,$$

*where $O$ is the circumcenter of $\triangle ABC$ and $R=r_{\mathscr{C}}$;*

*(3) the eccentricity of $\alpha$ is $e=\frac{QI}{r'}$, where $r'$ is the radius of the incircle or excircle of $\triangle DEF$ corresponding to $I$.*

*Which in/excenter $I$ is required depends on the type of $\alpha$ and on the positions of $P$ relative to the three sidelines.*

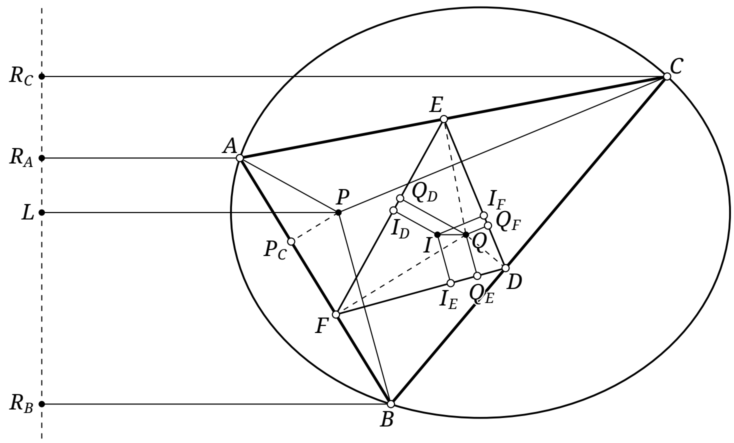


FIGURE 8.16

**Proof.** As in figure 8.16, take the case in which $\alpha$ is an ellipse and $P$ lies inside $\triangle ABC$. We prove that the incenter $I$ of $\triangle DEF$ has the required properties; the other cases are analogous. Let $\triangle P_AP_BP_C$ be the pedal triangle of $P$ with respect to $\triangle ABC$, and let $\triangle I_DI_EI_F$ and $\triangle Q_DQ_EQ_F$ be the pedal triangles of $I,Q$ with respect to $\triangle DEF$, respectively. We first construct $\alpha$.

Construct $APLR_A \overset{-}{\sim} IQQ_DI_D$. Then $AR_A, PL\perp R_AL$. By proposition 4.4.7, $AP\perp EF$, so $AP\parallel QQ_D$. The definition of $L$ gives $\measuredangle APL=-\measuredangle IQQ_D$, so $PL\parallel IQ$. Since $AP\perp EF$ and $AB\perp QF$, we have $\measuredangle BAP=-\measuredangle EFQ$. Thus $\triangle PAP_C \overset{-}{\sim} \triangle QFQ_D$, and similarly $\triangle PBP_C \overset{-}{\sim} \triangle QFQ_E$. Hence

$$PL=\frac{AP}{IQ}\cdot QQ_D=\frac{PP_C\cdot QF}{IQ}=\frac{PB}{IQ}\cdot QQ_E,$$

where the three equalities use $APLR_A \overset{-}{\sim} IQQ_DI_D$, $\triangle PAP_C \overset{-}{\sim} \triangle QFQ_D$, and $\triangle PBP_C \overset{-}{\sim} \triangle QFQ_E$, respectively. Therefore $\triangle BPL \overset{-}{\sim} \triangle IQQ_E$, and similarly $\triangle CPL \overset{-}{\sim} \triangle IQQ_F$.

We may now construct $BPLR_B \overset{-}{\sim} IQQ_EI_E$ and $CPLR_C \overset{-}{\sim} IQQ_FI_F$. The points $R_A,R_B,R_C,L$ are collinear on a line perpendicular to $PL$. Then $\frac{PA}{AR_A}=\frac{QI}{II_D}=\frac{QI}{II_E}=\frac{PB}{R_BB}$, and similarly $\frac{PA}{AR_A}=\frac{PC}{CR_C}$. Hence $A,B,C$ all lie on the conic with focus $P$ and directrix $R_AR_BR_CL$, which is precisely $\alpha$.

We now examine the properties in the statement. (1) By the preceding discussion, $PL$ is the principal axis of $\alpha$, and $PL\parallel QI$.

(2) The preceding discussion gives $p=PL=\frac{PP_C\cdot QF}{QI}$. By corollary 4.4.16, $PP_C\cdot QF=PP_A\cdot QD$, so $p=\frac{PP_A\cdot QD}{QI}$. By symmetry, $p\cdot QI=PP_A\cdot QD=PP_B\cdot QE=PP_C\cdot QF$. Cycling and interchanging

the roles in proposition 4.4.10 gives six equations; multiplying them yields

$$\left|(R^2-OP^2)^3(R^2-OQ^2)^3\right| = (2R)^6\cdot PP_A\cdot QD\cdot PP_B\cdot QE\cdot PP_C\cdot QF = (2R)^6\cdot(p\cdot QI)^3.$$

Thus $p = \left|\frac{(R^2-OP^2)(R^2-OQ^2)}{4R^2\cdot QI}\right|$.

(3) The preceding discussion gives $e = \frac{AP}{AR_A} = \frac{IQ}{II_D} = \frac{QI}{r'}$. □

As an illustration, we use the result to prove example 4.1.19: if $\triangle ABC$ is inscribed in a parabola $\alpha$ and its orthocenter $H$ is the focus, then the incircle $\odot I$ of $\triangle ABC$ is fixed.

**Alternative proof of example 4.1.19.** Let $\mathbf{g}H = O$ be the circumcenter of $\triangle ABC$. Its pedal triangle is the medial triangle; let $I'$ be the incenter of that medial triangle. For $\alpha$, this is the in/excenter required in proposition 8.2.7.

By proposition 8.2.7(1), the axis of $\alpha$ is parallel to $OI'$. The medial and reference triangles are homothetic, with $(O,H)$ and $(I',I)$ as pairs of corresponding points, so $HI$ is parallel to $OI'$ and twice as long. Thus $HI$ is parallel to the axis of $\alpha$, and $I$ lies on that axis. Part (3) of proposition 8.2.7 gives that the inradius of the medial triangle equals $OI'$. Homothety then gives $r_{\odot I} = HI$, and therefore $H \in \odot I$.

Let the line through $H$ perpendicular to the axis meet $\alpha$ at $X,Y$, and let $Z$ be the point at infinity on the axis. Poncelet's porism implies that $\triangle XYZ$ is inscribed in $\alpha$ and tangent to $\odot I$. By proposition 2.3.3, $r_{\odot I} = HX = 2HT$, where $T$ is the vertex of $\alpha$. Hence $\odot I$ is completely determined. □

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**Exercises**

**Exercise 8.17.** Given $\triangle ABC$ with incenter $I$, prove that, for a point $P$, $\mathscr{O}(P) \in \mathbf{T}\mathscr{T} \Longleftrightarrow IP = \sqrt{2}r_{\mathscr{T}}$.

**Exercise 8.18.** In proposition 8.2.7, determine which in/excenter $I$ is required in each case.

**Exercise 8.19.** Under the hypotheses of proposition 8.2.7, prove that the normal to $\alpha$ at $A$ is parallel to $QI_D$, where $I_D$ is the projection of $I$ on $EF$.

**Exercise 8.20.** Given $\triangle ABC$ and an inscribed ellipse $\Gamma$, prove that the inner auxiliary circle of $\Gamma$ is tangent to $\mathscr{N}$ if and only if the circumcenter of $\triangle ABC$ lies on the minor axis of $\Gamma$.

**Exercise 8.21.** Given a conic $\Gamma$ and a circle $\odot I$, suppose there is a triangle $\triangle ABC$ inscribed in $\Gamma$ and circumscribed about $\odot I$. Poncelet's closure theorem gives infinitely many such triangles $\triangle ABC$. Prove that, as $\triangle ABC$ varies, its circumcircle remains tangent to two fixed circles. In particular, if $I$ lies on the principal axis of $\Gamma$, prove that the circumcircle remains tangent to the auxiliary circle of $\Gamma$.

## 8.3 Nine-Point Conics

### 8.3.1 Basic Properties of Nine-Point Conics

Recall theorem 6.7.9, which motivates the following definition.

**Definition 8.3.1.** Given a complete quadrangle $ABCD$, the centers of the conics in the pencil $\mathrm{pen}(ABCD)$ trace a conic through the midpoints of its six sides and the three vertices of its diagonal triangle. This conic is called the *nine-point conic* of the complete quadrangle $ABCD$; see figure 8.17.[3]

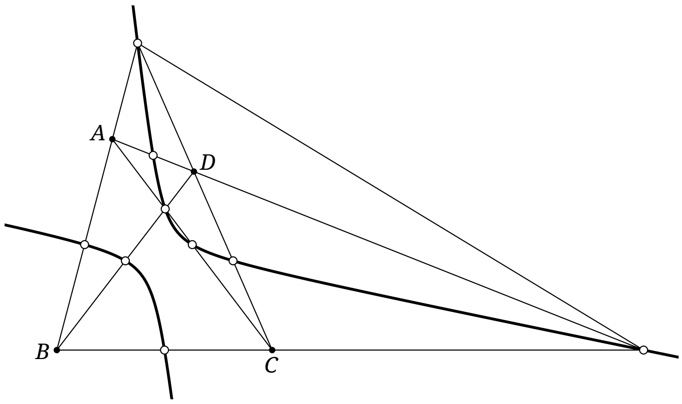


FIGURE 8.17

**Remark.** The generalization of theorem 6.7.9 given by corollary 6.7.10 suggests a broader definition. Given a complete quadrangle $ABCD$ and a line $l$, the poles of $l$ with respect to the conics in $\mathrm{pen}(ABCD)$ trace a conic through the three vertices of the diagonal triangle of $ABCD$. This too may be called the nine-point conic of $l$ with respect to the complete quadrangle $ABCD$. The preceding center-locus definition is the one most often used.

**Example 8.3.2.** If $A,B,C,D$ form an orthocentric system, their nine-point conic is the nine-point circle of any triangle formed by three of the four points. From this viewpoint, theorem 8.1.3 is almost immediate.

We now study the elementary properties of the nine-point conic. The first is part of its definition.

**Theorem 8.3.3.** *The center of every circumconic of a complete quadrangle ABCD lies on its nine-point conic.*

In particular:

**Proposition 8.3.4.** *The Poncelet point of a complete quadrangle lies on its nine-point conic.*

**Proof.** By theorem 8.1.13, the Poncelet point is the center of the rectangular circumhyperbola of the quadrangle, so it lies on the nine-point conic. ☐

The nine-point conic also reveals several distinguished points of a quadrangle.

**Theorem 8.3.5.** *The center of the nine-point conic $\alpha$ of a complete quadrangle ABCD is the centroid M of ABCD.*

**Proof.** The midpoints $M_{AB},M_{BC},M_{CD},M_{DA}$ of $AB,BC,CD,DA$ all lie on the nine-point conic, and plainly form a parallelogram. Hence the center of the parallelogram $M_{AB}M_{BC}M_{CD}M_{DA}$ is the center of $\alpha$. Clearly, the center of this parallelogram is the centroid of the quadrilateral $ABCD$. ☐

[3]As in the remark in section 4.8, one may equivalently speak of the nine-point conic of the corresponding complete quadrilateral $\{AB,BC,CD,DA\}$, or simply of the quadrilateral $ABCD$.

**Theorem 8.3.6.** *Given a complete quadrangle ABCD, let $M_{ij}$ be the midpoint of ij, where $i,j \in \{A,B,C,D\}$. The four circles $\odot(M_{AB}M_{AC}M_{AD})$, $\odot(M_{BC}M_{BD}M_{BA})$, $\odot(M_{CD}M_{CA}M_{CB})$, and $\odot(M_{DA}M_{DB}M_{DC})$ pass through a point Q. The point Q lies on the nine-point conic $\alpha$ of the complete quadrangle ABCD and is antipodal on $\alpha$ to its Poncelet point.* The point $Q$ is called the *Gergonne–Steiner point* of the complete quadrangle $ABCD$.

**Proof.** Let $M$ be the centroid of the quadrilateral $ABCD$. Reflecting the four circles in $M$ turns them into the nine-point circles of $\triangle BCD, \triangle CDA, \triangle DAB, \triangle ABC$, respectively. Those four circles pass through the Poncelet point $P$ of the complete quadrangle, and $P \in \alpha$. Hence the original four circles also pass through a point $Q$, and $P,Q$ are symmetric about $M$. By theorem 8.3.5, $M$ is the center of $\alpha$, so $Q \in \alpha$ and is antipodal to $P$. □

**Proposition 8.3.7.** *Given a complete quadrangle ABCD, for $i,j \in \{A,B,C,D\}$ let $M_{ij}$ denote the midpoint of ij, and put $U = AC \cap BD$, $V = AB \cap CD$, and $W = AD \cap BC$. For its nine-point conic $\alpha$, the tangents at $M_{AC}, M_{BD}$ are parallel to $VW$; the tangents at $M_{AB}, M_{CD}$ are parallel to $WU$; and the tangents at $M_{BC}, M_{AD}$ are parallel to $UV$.*

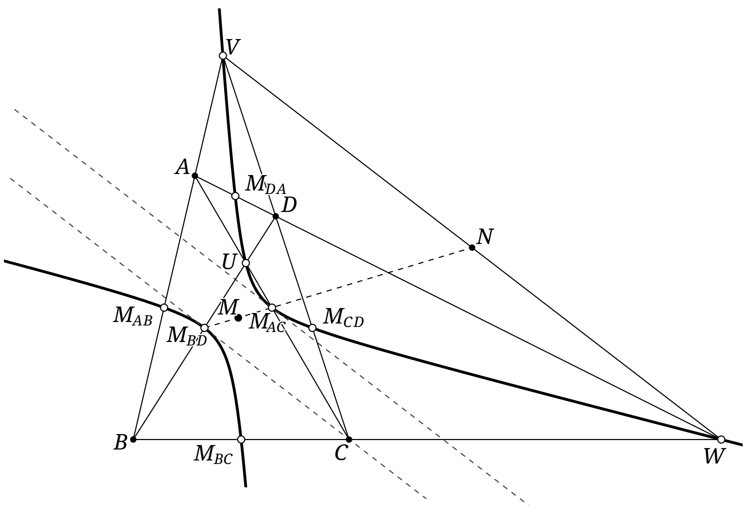


FIGURE 8.18

**Proof.** As in figure 8.18, let $N$ be the midpoint of $VW$ and $M$ the centroid of $ABCD$. By the definition of the Gauss line, $M_{BD}, M_{AC}, N$ are collinear, and their line also passes through $M$. Thus theorem 8.3.5 shows that $\overline{M_{AC}M_{BD}N}$ is a diameter of $\alpha$. By theorem 6.4.4, $VW \parallel \mathrm{t}_\alpha(M_{AC}) \parallel \mathrm{t}_\alpha(M_{BD})$. The other two pairs follow cyclically. □

## Exercises

**Exercise 8.22.** Prove that the centers of all conics through the three vertices and centroid of a triangle trace its Steiner inellipse.

**Exercise 8.23.** Given a complete quadrangle $ABCD$, put $U = AC \cap BD$, $V = AB \cap CD$, $W = AD \cap BC$, $E = AC \cap VW$, and $F = BD \cap VW$. Prove that the tangent at $U$ to the nine-point conic $\alpha$ of $ABCD$ bisects $EF$.

**Exercise 8.24.** Given $\triangle ABC$, a line $l$, and a point $X$, let $\triangle X_aX_bX_c$ be the anticevian triangle of $X$, and let $\alpha_1$ be the nine-point conic of $l$ with respect to the complete quadrangle $XX_aX_bX_c$. Suppose $l$ meets $BC,CA,AB$ at $L_a,L_b,L_c$. There is a conic $\alpha_2$ tangent to $X_aL_a,X_bL_b,X_cL_c$ at $X_a,X_b,X_c$, respectively. Prove that the two direct common tangents of $\alpha_1,\alpha_2$ meet at $X$.

### 8.3.2 Nine-Point Conics and Rectangular Hyperbolas

We now specialize to the nine-point conic of a cyclic quadrangle.

**Theorem 8.3.8.** *The nine-point conic of a quadrilateral $ABCD$ inscribed in a circle $\odot O$ is a rectangular hyperbola through $O$.*

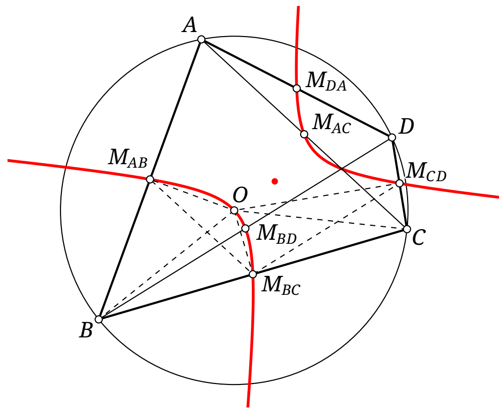


FIGURE 8.19

**Proof.** That the nine-point conic passes through $O$ follows directly from theorem 8.3.3. It remains to show that it is rectangular. For $i,j \in \{A,B,C,D\}$, write $M_{ij}$ for the midpoint of $ij$. The points $M_{BC},M_{AB},O,B$ are concyclic, as are $M_{BC},M_{CD},O,C$. Hence

$$\measuredangle M_{BC}M_{AB}O = \measuredangle M_{BC}BO = \measuredangle OCM_{BC} = \measuredangle OM_{CD}M_{BC}.$$

Similarly, $\measuredangle M_{DA}M_{AB}O = \measuredangle OM_{CD}M_{DA}$. By theorem 8.1.5, the five points $M_{AB},M_{BC},M_{CD},M_{DA},O$ lie on a rectangular hyperbola. $\square$

**Alternative proof.** By corollary 2.4.2, the axis of either parabola in pen$(ABCD)$ is parallel or perpendicular to a bisector of the angle between $AC$ and $BD$. The two parabolas in the pencil therefore have perpendicular axes. Their centers are the two points at infinity in those perpendicular directions. Hence the nine-point conic passes through two points at infinity in perpendicular directions and is a rectangular hyperbola. $\square$

The converse also holds; its proof is left to the reader.

**Theorem 8.3.9.** *If the nine-point conic of a quadrilateral is a rectangular hyperbola, then the quadrilateral is cyclic.*

**Theorem 8.3.10.** *For a quadrilateral $ABCD$ inscribed in a circle $\odot O$, the asymptotes of its nine-point conic are parallel to the internal and external bisectors of the angle between $AC$ and $BD$.*

**Proof.** The alternative proof of theorem 8.3.8 shows that the nine-point conic passes through the points at infinity on the internal and external bisectors of the angle between $AC$ and $BD$. $\square$

**Alternative proof.** Let $\alpha$ be the nine-point conic. From the first proof of theorem 8.3.8 and theorem 8.1.5, the points $M_{AB}, M_{CD}$ are antipodal on $\alpha$. By theorem 8.1.6, the asymptotes of $\alpha$ are parallel to the two angle bisectors of $\angle M_{AB}M_{BC}M_{CD}$. Since $AC \parallel M_{AB}M_{BC}$ and $BD \parallel M_{CD}M_{BC}$, these are the required directions, and the two asymptotes are perpendicular. $\square$

The nine-point conic also gives a projective proof of corollary 2.4.2: if $A,B,C,D$ are concyclic, then the axes of every conic through them are parallel to the internal and external bisectors of the angle between $AC$ and $BD$.

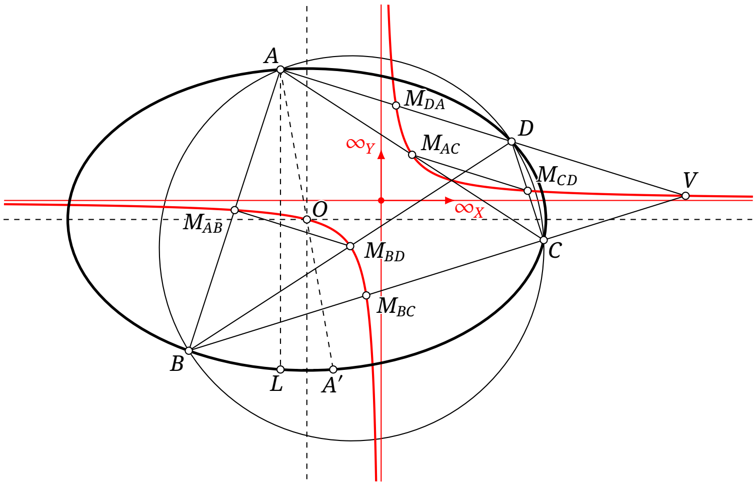


FIGURE 8.20

**Alternative proof of corollary 2.4.2.** The nine-point conic $\alpha$ of $ABCD$ is a rectangular hyperbola. Let $\infty_X, \infty_Y$ be its points at infinity, and put $V = AD \cap BC \in \alpha$. For $i, j \in \{A,B,C,D\}$, write $M_{ij}$ for the midpoint of $ij$. For any conic $\beta$ through $A,B,C,D$, its center $O$ lies on $\alpha$, as in figure 8.20. Let $A'$ be the reflection of $A$ in $O$, so $A' \in \beta$, and let $L$ be the reflection of $A$ in the line $O\infty_X$.

Parallel corresponding lines give

$$A'(B,L,C,D) \barwedge O(M_{AB}, \infty_X, M_{AC}, M_{DA}),$$

and similarly

$$A(B,L,C,D) \barwedge M_{DA}(M_{BD}, \infty_Y, M_{CD}, V).$$

Now $M_{AB}M_{BD} \parallel M_{DA}V \parallel M_{AC}M_{CD}$. Thus the four lines $M_{AB}M_{BD}$, $M_{AC}M_{CD}$, $M_{DA}V$, and $\infty_X\infty_Y$ meet at one point at infinity. By proposition 6.2.27,

$$\alpha(M_{AB}, \infty_X, M_{AC}, M_{DA}) \barwedge \alpha(M_{BD}, \infty_Y, M_{CD}, V),$$

and hence

$$O(M_{AB}, \infty_X, M_{AC}, M_{DA}) \barwedge M_{DA}(M_{BD}, \infty_Y, M_{CD}, V).$$

Therefore $A'(B,L,C,D) \barwedge A(B,L,C,D)$.

The projective definition of a conic now gives $A', B, L, C, D, A \in \beta$. The line $O\infty_X$ is a diameter of $\beta$ and perpendicularly bisects $AL$, so the projective definition of a diameter shows that it is an axis of $\beta$. By theorem 8.3.10, it is parallel to one bisector of the angle between $AC$ and $BD$; the other axis is parallel to the other bisector. $\square$

**Lemma 8.3.11.** *Let a complete quadrangle $ABCD$ be inscribed in the circle $\omega = \odot Q$, and let $\gamma$ be its nine-point conic. If $X \in \gamma$ and $\alpha \in \mathrm{pen}(ABCD)$, then $QX \perp \mathfrak{p}_\alpha(X)$. Conversely, if $\alpha \in \mathrm{pen}(ABCD)$ and a point $X$ satisfies $QX \perp \mathfrak{p}_\alpha(X)$, then $X \in \gamma$.*

**Proof.** Write $\mathscr{A} = \mathrm{pen}(ABCD)$. We first prove the forward implication. By corollary 6.7.7, the polars of $X$ with respect to all conics of $\mathscr{A}$ pass through one point. By the definition of the nine-point conic, $X$ is the center of one member of $\mathscr{A}$, so its polar with respect to that member is $l_\infty$. Hence $\bigcap_{\iota\in\mathscr{A}} \mathfrak{p}_\iota(X) \in l_\infty$, and all the polars of $X$ are parallel. For the circle $\omega = \odot Q$, plainly $\mathfrak{p}_\omega(X) \perp QX$. Since $\mathfrak{p}_\alpha(X) \parallel \mathfrak{p}_\omega(X)$, we obtain $QX \perp \mathfrak{p}_\alpha(X)$.

Conversely, assume $QX \perp \mathfrak{p}_\alpha(X)$. Let $O$ be the center of $\alpha$, and let $A$ be an intersection of $OX$ with $\alpha$. Since $X \in OA$, $\infty_{\mathfrak{t}_\alpha(A)} = \mathfrak{p}_\alpha^{-1}(OA) \in \mathfrak{p}_\alpha(X)$, so $\mathfrak{p}_\alpha(X) \parallel \mathfrak{t}_\alpha(A)$ and $\mathfrak{t}_\alpha(A) \perp QX$. Let $K$ be the point at infinity in the direction perpendicular to $\mathfrak{t}_\alpha(A)$. Then $K \in QX$. As $X$ varies, we have a chain of projectivities

$$[OX] \longrightarrow [\mathfrak{p}_\alpha^{-1}(OX)] = [\infty_{\mathfrak{t}_\alpha(A)}] \longrightarrow [K] \longrightarrow [QX].$$

The first step is projective by theorem 6.1.6. The penultimate step is projective because rotating lines through 90° is a projectivity, so their corresponding intersections with $l_\infty$ also form a projectivity; equivalently, the projective definition of perpendicularity identifies these as corresponding points of the absolute involution on $l_\infty$. Hence the locus of $X$ is a conic. We have already proved that every point of $\gamma$ satisfies the conditions on $X$, so the locus is $\gamma$. □

**Theorem 8.3.12.** *As in figure 8.21, let $\alpha$ be a conic centered at $O$, let $Q$ be a point, and let $\omega$ vary among circles centered at $Q$. Suppose $\omega$ meets $\alpha$ at $A,B,C,D$. The nine-point conic $\gamma$ of $ABCD$ is a fixed rectangular hyperbola, independent of $\omega$. Its asymptotes are parallel to the two axes of $\alpha$, and it passes through $O,Q$. Moreover, the direction conjugate with respect to $\alpha$ to the tangent $\mathfrak{t}_\gamma(O)$[4] is perpendicular to $OQ$, and $\mathfrak{t}_\gamma(Q) \perp \mathfrak{p}_\alpha(Q)$.*

**Proof.** Write $\mathscr{A} = \mathrm{pen}(ABCD)$. By definition, $O,Q \in \gamma$. By theorem 8.3.8, $\gamma$ is a rectangular hyperbola, and corollary 2.4.2 shows that its asymptotes are parallel to the two axes of $\alpha$.

Choose $X \in \gamma$ near $O$. By lemma 8.3.11, $\mathfrak{p}_\alpha(X) \perp XQ$. The direction of $\mathfrak{p}_\alpha(X)$ is conjugate with respect to $\alpha$ to the direction of $OX$. Thus the diameter conjugate to $XO$ is perpendicular to $XQ$. Letting $X \to O$ and using $\mathfrak{p}_\gamma(O) = \mathfrak{t}_\gamma(O)$ shows that its conjugate direction is perpendicular to $OQ$. Thus the tangent at $O$ is independent of $\omega$.

The hyperbola $\gamma$ is now uniquely determined by the two points at infinity on the axes of $\alpha$ and the three points $Q,O,O$, where $O,O$ are coincident and determine the tangent at $O$. Finally, as $X \to Q$, we have $XQ \to \mathfrak{t}_\gamma(Q)$; the relation $XQ \perp \mathfrak{p}_\alpha(X)$ therefore gives $\mathfrak{t}_\gamma(Q) \perp \mathfrak{p}_\alpha(Q)$. □

**Remark.** The result extends directly to a parabola by passage to a limit. Its center $O$ then becomes the point at infinity on the axis. Consequently the condition that the direction conjugate with respect to $\alpha$ to the tangent of $\gamma$ at $O$ is perpendicular to $OQ$ merely repeats that $\gamma$ passes through the point at infinity on the axis, so one piece of determining data is lost.

**Theorem 8.3.13.** *Let $\alpha$ be a conic centered at $O$, and let $Q$ be a fixed point. Let a circle $\omega$ with center $Q$ vary, meeting $\alpha$ at $A,B,C,D$, and let $M$ be the centroid of the complete quadrangle $ABCD$. Then $M$ is fixed. More precisely:*

*i. if $\alpha$ is a central conic of eccentricity $e$, the ratio of the directed distances of $M,Q$ from the secondary axis is $\frac{1}{e^2}$, and the corresponding ratio from the principal axis is $\left(1-\frac{1}{e^2}\right)$;*
*ii. if $\alpha$ is a parabola, then $M$ lies on its axis, and the distance of $M$ from the directrix minus that of $Q$ from the directrix is $-p$, where $p$ is the focal parameter.*

**Proof.** By theorem 8.3.5, $M$ is the center of the nine-point conic of $ABCD$, so it is independent of the choice of $\omega$.

(1) First suppose that $\alpha$ is central. As in figure 8.21, we treat an ellipse; the hyperbolic case is analogous. Denote the directed distances of $Q,M$ from the major and minor axes by $y_Q,x_Q$ and

[4] Since $\mathfrak{t}_\gamma(O)$ passes through the center of $\alpha$, it is plainly a diameter of $\alpha$; here its “conjugate direction” means the direction of the corresponding conjugate diameter.

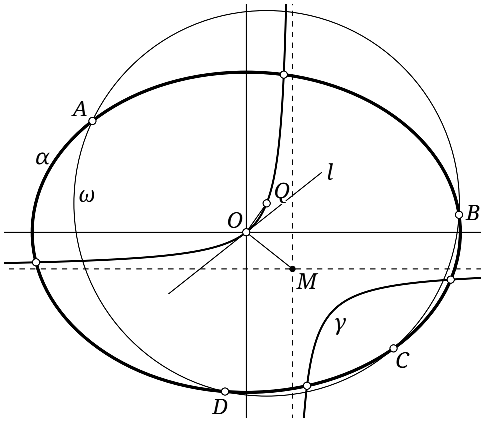


FIGURE 8.21

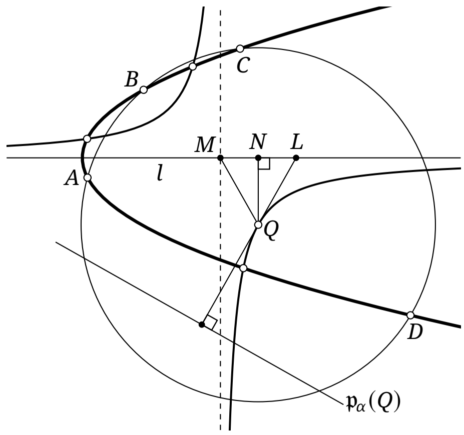


FIGURE 8.22

$y_M, x_M$, respectively. Let $\gamma$ be the nine-point conic of $ABCD$ and $l$ its tangent at $O$. By lemma 8.1.16, the angle bisectors of $l$ and $OM$ are parallel to the asymptotes of $\gamma$, so $l$ and $OM$ are symmetric about the major axis $a$ of $\alpha$.

Let $l'$ be the diameter conjugate to $l$. Since $OQ \perp l'$,

$$\begin{aligned}\frac{y_Q}{x_Q} &= \tan\measuredangle(a,OQ) = \tan(90^\circ + \measuredangle(a,l')) = -\cot\measuredangle(a,l') \\ &= \frac{\tan\measuredangle(a,l)}{1-e^2} = \frac{\tan\measuredangle(a,OM)}{e^2-1} = \frac{y_M/x_M}{e^2-1},\end{aligned}$$

where the fourth equality uses proposition 6.4.24. In addition, example 6.4.13(3) gives $x_M y_M = (x_Q - x_M)(y_Q - y_M)$. Solving these two relations yields $x_M = \frac{x_Q}{e^2}$ and $y_M = \frac{e^2-1}{e^2} y_Q$.

(2) Suppose that $\alpha$ is a parabola with axis $l$. In theorem 8.3.12, consider the case in which the center $O$ lies at infinity. One asymptote of $\gamma$ coincides with $l$, and its center satisfies $M \in l$. Put $L = \mathfrak{t}_\gamma(Q) \cap l$, and let $N$ be the projection of $Q$ on $l$. As in (1), $MN = NL$; see figure 8.22. By theorem 8.3.12, $QL = \mathfrak{t}_\gamma(Q) \perp \mathfrak{p}_\alpha(Q)$, and exercise 6.4 gives $NL = p$. The required difference of distances from the directrix is therefore $-MN = -NL = -p$. □

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## Exercises

**Exercise 8.25.** Prove theorem 8.3.9.

**Exercise 8.26.** In $\triangle ABC$, a circle through $B, C$ meets the segments $AB, AC$ again at $D, E$. Put $F = BE \cap CD$, and let $L, M, N$ be the midpoints of $AF, BC, DE$, respectively. Newton's theorem gives that $L, M, N$ are collinear. Prove that $LM \cdot LN = LA^2$.

**Exercise 8.27.** Let four concyclic points $A, B, C, D$ lie on a parabola. Prove that the centroid $M$ of $ABCD$ lies on the axis of the parabola.

**Exercise 8.28.** Given a parabola $\alpha$ and a point $P$ on its axis, let $\triangle ABC$ be inscribed in $\alpha$ and have incenter $P$. The tangents to $\alpha$ at $A, B, C$ form $\triangle DEF$. Prove that:

(1) the centroid and circumcenter of $\triangle DEF$ each lie on a fixed line;

(2) the Euler line of $\triangle DEF$ passes through a fixed point.

### 8.3.3 Normals to a Conic

#### Normals

Nine-point conics also provide a useful route to properties of conic normals. Recall definition 2.5.1: given a smooth curve $\gamma$ and a point $P$ on it, the line $n$ through $P$ perpendicular to the tangent of $\gamma$ at $P$ is called the *normal* to $\gamma$ at $P$.

**Theorem 8.3.14.** *As in figure 8.23, let the normals at four points $P_1,P_2,P_3,P_4$ of a conic $\alpha$ centered at $O$ concur at $Q$. Then $P_1,P_2,P_3,P_4,O,Q$ lie on a rectangular hyperbola whose asymptotes are parallel to the axes of $\alpha$. It is the nine-point conic of the complete quadrangle formed by the four intersections of $\alpha$ with any circle centered at $Q$.* This fixed hyperbola is called the *Apollonius hyperbola* of $Q$ with respect to $\alpha$.

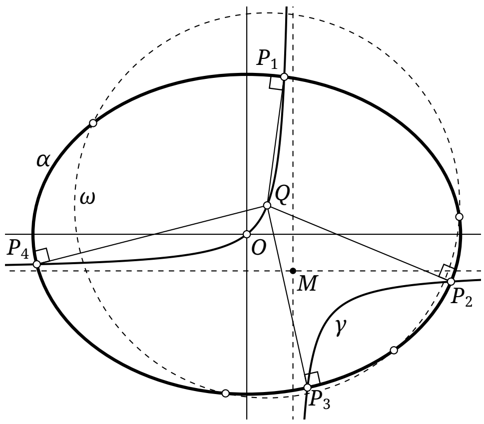



FIGURE 8.23

**Proof.** Choose any circle $\omega$ centered at $Q$ and let its intersections with $\alpha$ be $A,B,C,D$. By theorem 8.3.12, the nine-point conic $\gamma$ of $ABCD$ is fixed and passes through $O,Q$. Since $QP_i \perp \mathfrak{t}_\alpha(P_i) = \mathfrak{p}_\alpha(P_i)$, lemma 8.3.11 gives $P_i \in \gamma$ for all four indices. □

**Theorem 8.3.15.** *Let the normals at four points $P_1,P_2,P_3,P_4$ of a central conic $\alpha$ concur at $Q$. If $P'$ is the reflection of $P_4$ in the center of $\alpha$, then $P_1,P_2,P_3,P'$ are concyclic.* Their circle is called the *Joachimsthal circle*.

**Proof.** We treat an ellipse. Let $\mathscr{A} = \mathrm{pen}(P_1P_2P_3P')$ and let $\gamma$ be the nine-point conic of this complete quadrangle. Keeping the center $O$ fixed, apply an affine stretch in the direction of the minor axis that sends $\alpha$ to a circle $\alpha'$. Let the images of $P_1,P_2,P_3,P',P_4,\gamma$ be $P_1',P_2',P_3',P'',P_4',\gamma'$, respectively. The conic $\gamma'$ is the nine-point conic of $P_1'P_2'P_3'P''$. Since $\alpha'$ is a circle, theorem 8.3.8 shows that $\gamma'$ is a rectangular hyperbola.

By definition, $\gamma'$ contains the midpoints of $P''P_1',P''P_2',P''P_3'$ and also $O$, since $O$ is the center of $\alpha'$. Moreover, $\overline{P_4'O} = \overline{OP''}$. Therefore $\mathfrak{h}_{P'',2}(\gamma')$ is the rectangular hyperbola $\beta'$ through $P_1',P_2',P_3',P_4'$.

Let $\beta$ be the Apollonius hyperbola of $Q$ with respect to $\alpha$. It passes through $P_1,P_2,P_3,P_4$, so its affine image passes through $P_1',P_2',P_3',P_4'$. By theorem 8.3.14, the asymptotes of $\beta$ are parallel to the axes of $\alpha$; the affine stretch preserves those directions and sends $\beta$ to a rectangular hyperbola. Hence its image is $\beta'$.

Since $\mathfrak{h}_{P'',2}(\gamma') = \beta'$, the asymptotes of $\gamma'$ are also parallel to the axes of $\alpha$. Reversing the stretch shows that $\gamma$ is a rectangular hyperbola. By theorem 8.3.9, $P_1,P_2,P_3,P'$ are concyclic. The hyperbolic case follows by a complex affine transformation. □

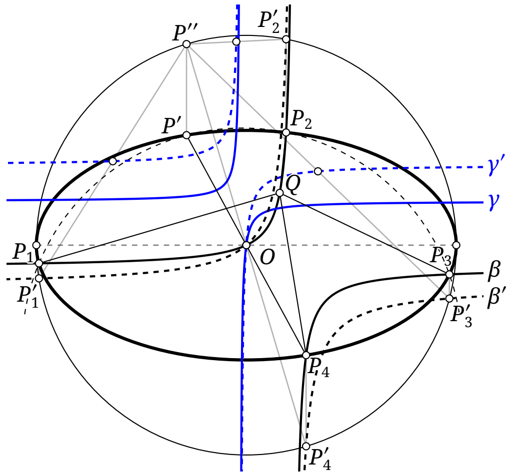



FIGURE 8.24

For a parabola, the limiting fourth "normal" through a given point is the line parallel to the axis: the parabola is tangent to the line at infinity, and this is the normal at that point of tangency. Thus theorem 8.3.15 takes the following form.

**Proposition 8.3.16.** *If the normals at three points of a parabola concur, then their circumcircle also passes through the vertex of the parabola.*

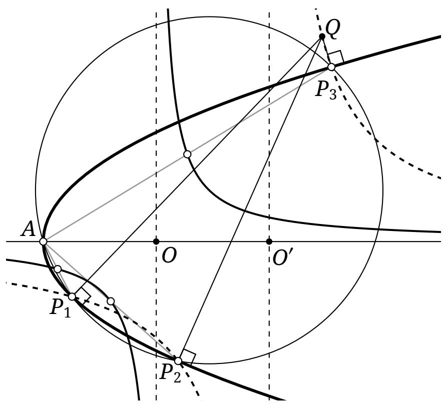



FIGURE 8.25

Here is a direct proof that does not appeal to a limit.

**Proof.** Let $\alpha$ be the parabola and $l$ its axis. Suppose the normals at $P_1,P_2,P_3$ concur. Let the fourth intersection of $\odot(P_1P_2P_3)$ and $\alpha$ be $A$, and put $\mathscr{A} = \mathrm{pen}(AP_1P_2P_3)$. Since $A,P_1,P_2,P_3$ are concyclic, the nine-point conic $\gamma$ of $AP_1P_2P_3$ is a rectangular hyperbola.

Because $\alpha \in \mathscr{A}$, the conic $\gamma$ passes through the center of $\alpha$, namely the point at infinity on $l$. Since $\gamma$ is rectangular, its asymptotes are parallel and perpendicular to $l$. By definition, $\gamma$ passes through the midpoints of $AP_1,AP_2,AP_3$. Hence $\gamma' := \mathfrak{h}_{A,2}(\gamma)$ passes through $P_1,P_2,P_3$ and the two points at infinity in the directions parallel and perpendicular to $l$. Five points determine a conic, so theorem 8.3.14 identifies $\gamma'$ as an Apollonius hyperbola of $\alpha$.

By theorems 8.3.13 and 8.3.14, the center $O'$ of $\gamma'$ lies on $l$. By theorem 8.3.5, the center $O$ of $\gamma$ is the centroid of $AP_1P_2P_3$, and theorem 2.4.6 gives $O \in l$. But $\mathfrak{h}_{A,2}(O) = O'$, so $A,O,O'$ are collinear. Thus $A \in l$ and $A$ is the vertex of the parabola. $\square$

**Evolutes***

We give a few further properties of normals, independent of nine-point conics. First comes the basic terminology.

**Definition 8.3.17** (Evolute)**.** Given a curve $\gamma$, the locus $\gamma'$ of its centers of curvature is called the *evolute* of $\gamma$, and $\gamma$ is called an *involute* of $\gamma'$.

There is an equivalent description.

**Theorem 8.3.18.** *The envelope of the normals to a curve is its evolute.*

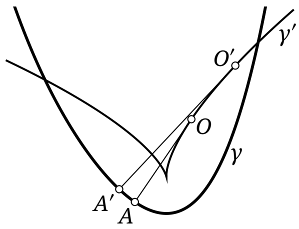



FIGURE 8.26

**Proof.** Because this is not a course in differential geometry, we give an intuitive "proof."[5]

Let $A$ be a point of $\gamma$ with center of curvature $O$, and choose a nearby point $A'$ with center of curvature $O'$, as in figure 8.26. Then $O,O'$ lie on the evolute $\gamma'$. When $A'$ is very close to $A$, as $A$ moves toward $A'$, the intersection of the normals at $A,A'$ becomes the center of curvature $O'$ at $A'$. Since $AO$ is the normal at $A$, as $A \to A'$ we have $O' \in AO$. Thus, as $A$ moves toward $A'$, $O$ moves toward $O'$ in the direction of $OO'$. Consequently $AO$ is tangent to the curve traced by $O$, that is, $AO \in \mathbf{T}\gamma'$. □

The inverse construction has the familiar string interpretation.

**Theorem 8.3.19.** *Let P be fixed on a curve $\gamma'$, and let O move on $\gamma'$. On the tangent* $\mathrm{t}_{\gamma'}(O)$ *choose A so that* $\widehat{PO} + \overline{OA} = \text{const.}$ Here $\widehat{PO}$ denotes arc length along $\gamma'$ and is chosen continuously. There are two possible choices of $A$; require the direction from $O$ to $A$ to agree with the direction of travel along the selected arc from $P$ to $O$. *Then the locus $\gamma$ of A is an involute of $\gamma'$.*

Since differential geometry has not been introduced, we describe this intuitively: wrap an inextensible string around $\gamma'$, fix one end at $P$, and unwind the other end while keeping the free portion taut. The free end traces an involute. For a closed curve the chosen arc may wind around the curve more than once; continuity of its length is the essential point.

**Proof.** Again we argue intuitively. Let $O,O'$ be neighboring points of $\gamma'$, with corresponding points $A,A'$, as in figure 8.26. Because $AO$ is a tangent, $A,O,O'$ become collinear as $O \to O'$. The construction gives $\overline{A'O'} = \overline{AO} + \widehat{OO'} = \overline{AO} + \overline{OO'} = \overline{AO'}$. As $O \to O'$, we have $A \to A'$, so $\angle A'O'A \to 0$. Therefore

$$\angle AA'O' = \angle A'AO' = \frac{180° - \angle A'O'A}{2} \to 90°.$$

Thus $AA' \perp AO'$. But $AA' = \mathrm{t}_\gamma(A)$ in the limit, so $AO$ is the normal to $\gamma$ at $A$, and $\gamma'$ is the envelope of the normals to $\gamma$. □

Our goal here is to determine the form of the evolute of an ellipse; compare section 4.2.2.

[5]This mode of argument is not rigorous by modern mathematical standards, although a physicist would generally accept it completely. Within the purely plane-geometric framework of this course it is nevertheless adequate for our purposes. Mathematicians used similar reasoning when Newton and Leibniz first invented calculus.

**Theorem 8.3.20.** *The evolute of an ellipse is an affine image of an astroid, obtained by a stretch in the direction of the minor axis. The stretch factor equals the ratio of the major axis to the minor axis of the ellipse.*

We use the following description of an astroid.

**Lemma 8.3.21.** *Let $M$ be fixed and $Q,R$ variable on a circle $\omega$, subject to $\measuredangle RQM = -3\measuredangle QRM$. The moving lines $QR$ envelop an astroid. If $O$ is the center of $\omega$ and $K$ is one cusp of the astroid, then $OK = 2OM$ and $\angle KOM = 45°$ or $135°$.*

**Proof.** The proof is identical to that of proposition 4.2.16; we indicate the modified construction from proposition 4.2.16. Let $\omega$ have center $O$. Draw $\gamma = \odot(O, 2OM)$, and choose $B \in \gamma$ so that $\overrightarrow{OB} = 2\overrightarrow{MO}$. Extend $OQ$ to $A$ with $2\overline{QA} = \overline{OQ}$, and draw $\odot(A, AQ)$. This circle is tangent to $\gamma$ at a point $C$, with $C, A, Q, O$ collinear. Choose $D$ on $\odot(A, AQ)$ so that the length of the arc $\widehat{CB}$ equals that of the arc $\widehat{CD}$, which need not be the minor arc. The rest of the proof is exactly as in proposition 4.2.16; the reader is invited to complete it. □

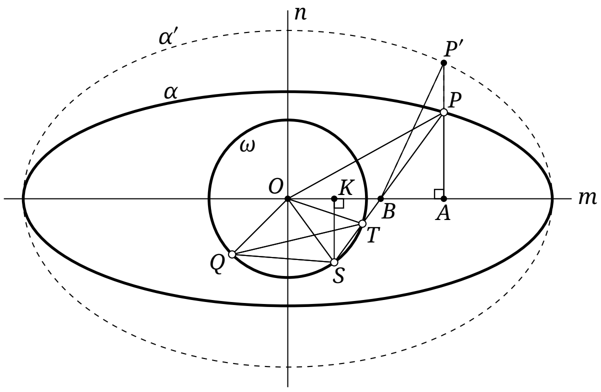



FIGURE 8.27

**Proof of theorem 8.3.20.** As in figure 8.27, draw a circle $\omega$ with perpendicular diameters $m, n$, and let $Q$ be an intersection of $\omega$ with an angle bisector of $m, n$. Let $S, T$ vary on $\omega$ subject to $\measuredangle QST = -3\measuredangle QTS$. By lemma 8.3.21, the lines $ST$ envelop an astroid $\gamma$ whose four cusps all lie on $m$ or $n$. Put $B = ST \cap m$ and $\theta = \angle QTS$. Then $\angle TSQ = 3\theta$ and $\measuredangle QOS = 2\measuredangle QTS = 2\theta$. Moreover,

$$\measuredangle BSO = \measuredangle TSQ - \measuredangle OSQ = 3\theta - \frac{180° - \measuredangle QOS}{2} = 4\theta - 90°,$$
$$\measuredangle SOB = \measuredangle QOB - \measuredangle QOS = 135° - 2\theta,$$
$$\measuredangle OBS = 180° - \measuredangle BSO - \measuredangle SOB = 135° - 2\theta = \measuredangle SOB.$$

Thus $OS = BS$. If $K$ is the projection of $S$ on $m$, then $\overline{OK} = \overline{KB}$.

Choose $A \in m$ so that $\overline{BA} = t\overline{OK} = t\overline{KB}$, where $t$ is constant, and draw $AP \perp m$ with $P \in SB$. Since $\triangle SKB \overset{\pm}{\sim} \triangle PAB$, $\frac{\overline{PA}}{\overline{SK}} = \frac{\overline{AB}}{\overline{KB}} = -t$. Consequently

$$\operatorname{dist}_{\pm}(P, m) = -t \operatorname{dist}_{\pm}(S, m), \quad \operatorname{dist}_{\pm}(P, n) = \overline{AB} + \overline{BO} = (t+2)\overline{KO} = (t+2)\operatorname{dist}_{\pm}(S, n).$$

Thus the locus of $P$ is obtained from the circle-locus of $S$ by stretches of factors $t$ and $t+2$ in the directions of $n$ and $m$. It is an ellipse $\alpha$ with major and minor axes $m, n$, respectively, and axis ratio $k = \frac{t+2}{t}$.

Furthermore,

$$\frac{\tan\measuredangle(m, OP)}{\tan\measuredangle(m, BP)} = \frac{\overline{PA}/\overline{OA}}{\overline{PA}/\overline{BA}} = \frac{\overline{BA}}{\overline{OA}} = \frac{t}{t+2}.$$

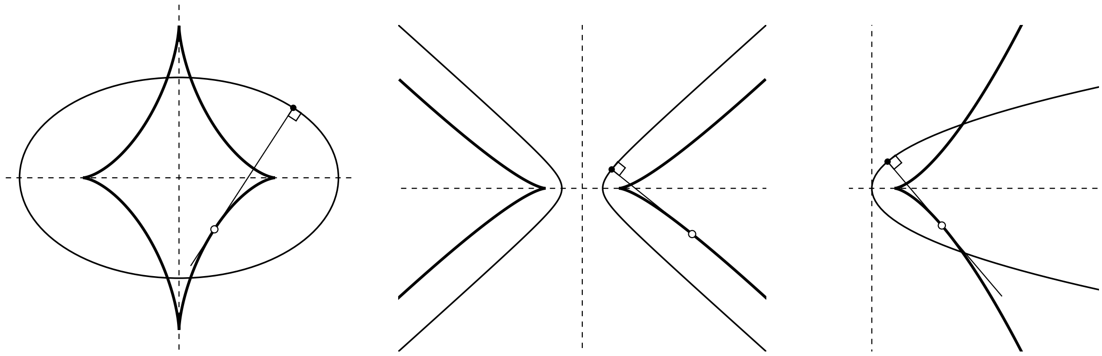
Stretch $\alpha$ in the direction of $n$ by $g=\sqrt{k}$ to obtain an ellipse $\alpha'$, and let $P'$ correspond to $P$. Then $\frac{\tan\measuredangle(m,OP')}{\tan\measuredangle(m,BP')}=\frac{t}{t+2}$. The axis ratio of $\alpha'$ is $k'=k/g=\sqrt{k}$. If $l$ is the normal to $\alpha'$ at $P'$, then exercise 3.24 gives

$$\frac{\tan\measuredangle(m,OP')}{\tan\measuredangle(m,l)}=\frac{1}{k'^2}=\frac{t}{t+2}.$$

Here we used $1-e^2$ equal to the squared ratio of the minor and major axes. Hence $BP'$ is a normal to $\alpha'$.

As $t$ ranges over $(0,+\infty)$, $k'$ ranges over $(1,+\infty)$, so every ellipse shape occurs. Conversely, for any given ellipse $\alpha'$, a circle $\omega$ can be constructed in this way. We have proved that the lines $BP$ envelop the astroid $\gamma$; after the stretch, the normals $BP'$ envelop its image $\gamma'$. The stretch factor is $g=\sqrt{k}=k'$, exactly the axis ratio of $\alpha'$. □

Figure 8.28a shows the evolute of an ellipse. A complex affine transformation gives the evolute of a hyperbola, shown in figure 8.28b, though its geometric meaning is less immediate. The evolute of a parabola is the *semicubical parabola*, shown in figure 8.28c, but it too has no special geometric properties. We do not pursue these two cases here.



(a) (b) (c)

FIGURE 8.28

## Exercises

**Exercise 8.29.** Let $A,B,C$ be points of a parabola $\alpha$ whose normals concur at $P$, and let $G$ be the centroid of $\triangle ABC$. If $d_P,d_G$ are the distances of $P,G$ from the directrix, determine $d_P-2d_G$ in terms of the focal parameter of $\alpha$.

**Exercise 8.30.** Let $P$ be the center of curvature of a conic $\alpha$ at one of its points. Prove that the Apollonius hyperbola of $P$ with respect to $\alpha$ osculates $\alpha$ at that point.

**Exercise 8.31*.** Prove lemma 8.3.21.

**Exercise 8.32*.** Give an intuitive argument, without requiring the full rigor of differential geometry, that at a point where the curvature of a smooth curve has a local extremum, the center of curvature is a cusp of the evolute.

**Exercise 8.33*.**

(1) Find coordinate equations for the evolutes of the three types of conic.
(2) Find a coordinate equation for an involute of a circle.

**Exercise 8.34**.** Let $l$ be a line and let $AB$ be a light inextensible string of fixed length. Initially $B \in l$ and $AB \perp l$. As $B$ moves along $l$, suppose $A$ is drawn after it while the string remains taut. The locus of $A$ is a *tractrix*; equivalently, the segment cut from each tangent to the tractrix by the point of tangency and the line $l$ has constant length. The envelope of the normals to a tractrix is a *catenary*.

Prove geometrically that when a parabola of fixed shape rolls without slipping along a straight line, the locus of its focus is a catenary.

## 8.4 Conics with Common Foci

### 8.4.1 Two Types of Pencils of Conics with Common Foci

We briefly describe two special types of pencils of conics with common foci. In the first, the conics share one focus and its corresponding directrix; in the second, they share both foci.

#### Pencils of Conics with a Common Focus–Directrix Pair

All conics sharing a focus and its corresponding directrix form a double-contact pencil, with the following basic properties.

**Proposition 8.4.1.** *Let $F$ be a point and $l$ a line not through $F$. Let $\mathscr{Q}$ denote the set of all conics for which $F$ and $l$ form a corresponding focus–directrix pair. Then:*

*(1) Given a line $m$ through $F$, let $n$ be the line through $F$ perpendicular to $m$. For every $\alpha \in \mathscr{Q}$, the pole $\mathfrak{p}_\alpha^{-1}(m) = l \cap n$ is fixed.*

*(2) The family $\mathscr{Q}$ is a double-contact pencil of conics, and the double line $l \cup l$ is one of its degenerate members.*

*(3) There is a projectivity carrying $\mathscr{Q}$ to a pencil of concentric circles.*

*(4) Polarity with respect to any nondegenerate $\alpha \in \mathscr{Q}$ preserves the pencil:[6] $\mathfrak{d}_\alpha(\mathscr{Q}) = \mathscr{Q}$.*

**Proof.**

(1) The pole is the intersection of the tangents at the two points of $m \cap \alpha$. The result follows from theorems 1.3.3 and 1.3.4 and claim 1.5.5.

(2) Let $I, J$ be the circular points, and put $I' = FI \cap l$ and $J' = FJ \cap l$. For every $\alpha \in \mathscr{Q}$, the projective definition of a focus says that $FI, FJ$ are tangent to $\alpha$. Since $l$ is the polar of $F$, their points of contact lie on $l$. Thus $FI, FJ$ are tangent to every member at the fixed points $I', J'$, respectively. This is a double-contact pencil, and $I'J' \cup I'J' = l \cup l$ is one of its degenerate members.

(3) By part (2), choose a projectivity $f$ carrying $I', J'$ to the new circular points. Then $f(\mathscr{Q})$ is a pencil of circles and $f(l) = l_\infty$. Projectivities preserve polarity, so, for every $\alpha \in \mathscr{Q}$, $f(F) = \mathfrak{p}_{f(\alpha)}^{-1}[f(l)]$. The pole of the line at infinity with respect to a circle is its center. Consequently every member of $f(\mathscr{Q})$ has the same center $f(F)$, and the transformed pencil is concentric.

(4) Part (3) carries $\mathscr{Q}$ to a concentric pencil of circles, and polarity with respect to any one of those circles preserves the entire concentric pencil. Transforming back proves the assertion.

□

**Corollary 8.4.2.** *Let $\mathscr{Q}$ be a pencil of conics with common focus $F$ and corresponding directrix $l$, and let $\alpha, \beta \in \mathscr{Q}$ be nondegenerate. For any $P \in \beta$, suppose that $\mathfrak{p}_\alpha(P)$ is tangent to $\mathfrak{d}_\alpha(\beta)$ at $Q$. Then $P, Q, F$ are collinear. More generally, the same conclusion holds for any double-contact pencil, with $F$ denoting the common pole of the line through its two fixed points of contact.*

[6] The following equality is equality of the two pencils as sets; it does not assert that every member is individually fixed by the polarity.

**Proof.** Apply a projectivity carrying $\mathcal{Q}$ to a concentric pencil of circles; the assertion is then immediate. □

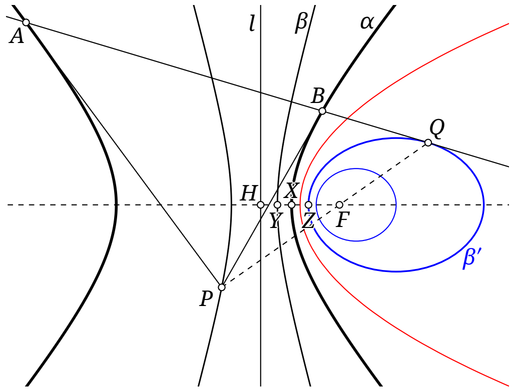



FIGURE 8.29

Figure 8.29 illustrates such a pencil, with common focus $F$ and corresponding directrix $l$. The two black conics are hyperbolas, the two blue members are ellipses, and the red member is a parabola. For the conics $\alpha, \beta$ shown, $\beta'$ is the polar curve of $\beta$ under polarity with respect to $\alpha$. If $P \in \beta$, then $AB = \mathfrak{p}_\alpha(P)$; the figure also shows the collinearity in corollary 8.4.2.

The eccentricities of the members of this pencil satisfy one further relation.

**Proposition 8.4.3.** *Let $\mathcal{Q}$ be a pencil of conics with common focus $F$ and corresponding directrix $l$. Suppose $\beta, \beta' \in \mathcal{Q}$ are polar curves of one another under polarity with respect to $\alpha \in \mathcal{Q}$.*[7] *If $e_1, e_2, e$ are the eccentricities of $\beta, \beta', \alpha$, respectively, then $e_1 e_2 = e^2$.*

**Proof.** Let $H$ be the perpendicular projection of $F$ onto $l$, and let $X, Y, Z$ be the respective intersections of the segment $FH$ with $\alpha, \beta, \beta'$, as in figure 8.29. The line $\mathfrak{p}_\alpha(Y)$ is tangent to $\beta'$ at $Z$; equivalently, $Y, Z$ are conjugate with respect to $\alpha$. Hence

$$\frac{e_1 e_2}{e^2} = \frac{\overline{FY}}{\overline{YH}} \cdot \frac{\overline{FZ}}{\overline{ZH}} \div \left(\frac{\overline{FX}}{\overline{XH}}\right)^2 = \frac{\overline{YF} \cdot \overline{XH}}{\overline{XF} \cdot \overline{YH}} \cdot \frac{\overline{ZF} \cdot \overline{XH}}{\overline{XF} \cdot \overline{ZH}} = (YX, FH)(ZX, FH).$$

By part (3) of proposition 8.4.1, apply a projectivity carrying $\mathcal{Q}$ to a concentric pencil, and mark images by an asterisk. Then $H^* \in l_\infty$ and $F^*$ is the common center. Since projectivities preserve cross-ratios,

$$\frac{e_1 e_2}{e^2} = (Y^*X^*, F^*H^*)(Z^*X^*, F^*H^*) = (Y^*X^*F^*)(Z^*X^*F^*) = \frac{\overline{Y^*F^*} \cdot \overline{Z^*F^*}}{\overline{X^*F^*}^2} = 1.$$

Indeed, polarity is preserved by a projectivity, so $Y^*, Z^*$ are conjugate with respect to the circle $\alpha^*$; their directed distances from its center $F^*$ therefore have product equal to the squared radius $\overline{X^*F^*}^2$. □

### The Pencil of Confocal Conics

We now turn to *confocal conics*: conics having the same two foci are said to be *confocal*. A family of confocal conics is not an ordinary pencil generated by four points; rather, it forms a dual pencil.

**Theorem 8.4.4.** *Given two points, all conics having them as foci have the same center, principal axis, and secondary axis. They form a dual pencil of conics whose degenerate members include the pair of circular points and the pair of foci.*

[7] Part (4) of proposition 8.4.1 ensures that $\mathfrak{d}_\alpha(\beta) \in \mathcal{Q}$.

**Proof.** The common center is plainly the midpoint of the two foci, the common principal axis is their joining line, and the common secondary axis is the perpendicular bisector of that line. The dual-pencil assertion follows at once from the projective definition of a focus. □

For the rest of this section we use the following convenient optical language. Let a line $l$ meet a curve $\gamma$ at $A$, and let $l'$ be the reflection of $l$ in $\mathfrak{t}_\gamma(A)$. We call $l'$ the *reflection* of $l$ at $A$ on $\gamma$. Thus a ray incident along $l$ and reflected from $\gamma$ at $A$ travels along $l'$. We first record two useful elementary properties.

**Proposition 8.4.5.** *For a central conic $\alpha$, two lines that are reflections of one another on $\alpha$ are both tangent to some conic $\beta$ confocal with $\alpha$.*[8] *Conversely, if $\alpha, \beta$ are confocal, the reflection on $\alpha$ of a tangent to $\beta$, taken at an intersection with $\alpha$, is also tangent to $\beta$.*

**Proof.** Given two foci $F_1, F_2$ and a tangent $l$, there is a unique conic with those foci and that tangent. Indeed, put $F_1' = \mathfrak{f}_l(F_1)$. By the optical property, $F_2F_1' \cap l$ is the point of tangency, after which the first definition determines the conic uniquely. It is a hyperbola when $F_1, F_2$ lie on opposite sides of $l$, and an ellipse when they lie on the same side. For the first assertion, choose the confocal conic tangent to one of the two lines; the assertion is then reduced to the converse direction.

Now let $\alpha, \beta$ be confocal with foci $F_1, F_2$, and suppose that $l$ is tangent to $\beta$ and meets $\alpha$ at $A$. We treat the case in which both conics are ellipses, as in figure 8.30; in the other cases only the choice between internal and external angle bisectors changes. Draw the other tangent $l'$ from $A$ to $\beta$. By the little Poncelet theorem, the angle bisectors of $\angle F_1AF_2$ also bisect the angle between $l$ and $l'$. The optical property of $\alpha$ therefore shows that $l, l'$ are reflections of one another in $\mathfrak{t}_\alpha(A)$. Hence $l'$ is the reflection of $l$ at $A$, and uniqueness of a reflected line proves the assertion. □

**Proposition 8.4.6.** *Let $\alpha, \beta$ be confocal conics. Suppose that $l$ is tangent to $\alpha$ at $A$, and put $B = \mathfrak{p}_\beta^{-1}(l)$. Then $AB \perp l$.*

**Proof.** In the dual pencil $\mathscr{P} = \mathrm{pen}_{\mathrm{T}}(\alpha, \beta)$, consider the degenerate member $I \cup J$ formed by the circular points, and put

$$X = \lim_{\substack{\gamma \in \mathscr{P} \\ \gamma \to I \cup J}} \mathfrak{p}_\gamma^{-1}(l).$$

On the one hand, $X \in IJ = l_\infty$. On the other hand, in the limiting case the harmonic property of pole and polar gives $(l \cap IJ, X; I, J) = -1$, and hence $(l, AX; AI, AJ) = -1$. The projective definition of perpendicularity therefore gives $l \perp AX$. By corollary 6.7.6, the points $A, B, X$ are collinear, and consequently $l \perp AB$. □

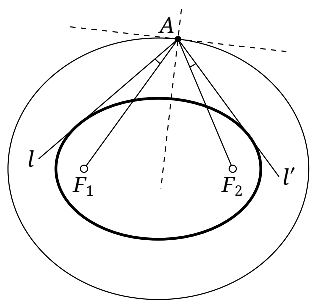



FIGURE 8.30

[8] The conic $\beta$ is nondegenerate provided the two lines do not pass through a focus of $\alpha$; otherwise it is understood in the degenerate sense.

## Exercises

**Exercise 8.35.** Let $\mathscr{Q}$ be a pencil of conics with common focus $F$ and corresponding directrix $l$. Prove that:

(1) if two conics in $\mathscr{Q}$ are polar curves of one another under polarity with respect to a parabola in $\mathscr{Q}$, the product of their eccentricities is 1, and their foci other than $F$ are reflections of one another in $l$;

(2) the auxiliary circles of the conics in $\mathscr{Q}$ form a pencil of circles, every member of which is orthogonal to $\odot(FH)$, where $H$ is the perpendicular projection of $F$ onto $l$.

**Exercise 8.36.** Let $\alpha, \beta$ be distinct nondegenerate conics in a plane $\pi$ with common focus $F$ and corresponding directrix $l$. Prove that the composite $\mathfrak{p}_\beta^{-1} \circ \mathfrak{p}_\alpha$ is a projectivity of $\pi$.

**Exercise 8.37.** In proposition 8.4.6, suppose that $\alpha, \beta$ are ellipses and that $l$ meets $\beta$ at two distinct points $X, Y$. Give two new proofs of the proposition, one using example 6.7.17 and the other using exercise 6.79.

**Exercise 8.38.** Let $\alpha, \beta$ be confocal ellipses. From an exterior point $P$, draw tangents to $\alpha$ at $A$,$B$ and to $\beta$ at $C$,$D$. Prove that the lines $AC$ and $AD$ are reflections of one another in $\mathfrak{t}_\alpha(A)$.

**Exercise 8.39.** An ellipse and a circle are externally tangent, and their two direct common tangents are parallel. Prove that the distance between their centers equals the sum of the semimajor and semiminor axes of the ellipse.

## 8.4.2 Special Properties of Confocal Conics

### Ivory's Theorem

The following is an interesting result about confocal conics.

**Theorem 8.4.7** (Ivory's theorem)**.** *As in figure 8.31, let $\alpha_1, \alpha_2$ be ellipses and $\beta_1, \beta_2$ hyperbolas, all four conics being confocal. Choose $A \in \alpha_1 \cap \beta_1$, $B \in \alpha_1 \cap \beta_2$, $C \in \alpha_2 \cap \beta_2$, and $D \in \alpha_2 \cap \beta_1$, with all four points on the same side of the common principal axis and also on the same side of the common secondary axis. Then $AC = BD$.*

This is often expressed by saying that the diagonals of a curvilinear quadrilateral bounded by two confocal ellipses and two confocal hyperbolas have equal lengths. The formulation above specifies its vertices more precisely. It follows directly from the stronger proposition below.

**Proposition 8.4.8.** *As in figure 8.31, under the hypotheses of theorem 8.4.7, choose any $P$ on the arc $\widehat{AD}$ of $\beta_1$. There is a point $Q$ on the arc $\widehat{AB}$ of $\alpha_1$ with the following property. Let the reflection of $PQ$ at $Q$ on $\alpha_1$ meet the arc $\widehat{BC}$ of $\beta_2$ at $R$, and let the reflection of $QR$ at $R$ on $\beta_2$ meet the arc $\widehat{CD}$ of $\alpha_2$ at $S$. Then $RS$,$SP$ are reflections of one another on $\alpha_2$, and $SP$,$PQ$ are reflections of one another on $\beta_1$. Moreover, $PQ + QR + RS + SP = 2AC = 2BD$, and the lines $PQ$,$QR$,$RS$,$SP$ are tangent to a fixed conic $\gamma$ confocal with $\alpha_1$.*

The central assertion is that the quadrilateral formed by these successively reflected lines inside the curvilinear quadrilateral $ABCD$ has constant perimeter.

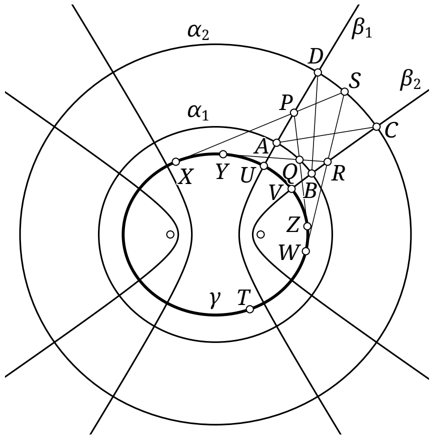



FIGURE 8.31

**Proof.** Let the reflection of $RS$ at $S$ on $\alpha_2$ meet $\beta_1$ at $P'$. By proposition 8.4.5, there is a conic $\gamma$ confocal with $\alpha_1$ to which $PQ, QR, RS, SP'$ are all tangent. Again by proposition 8.4.5, the lines $SP, QP$ are reflections of one another on $\beta_1$ if and only if $P' = P$.

Conversely, fix $\gamma$. Starting from $P$, successively choose $Q, R, S, P'$ so that $PQ, QR, RS, SP'$ are tangent to $\gamma$. Each of these four lines is then the reflection of its predecessor on the corresponding conic. By theorem 8.4.4, the conics $\alpha_1, \alpha_2, \beta_1, \beta_2$ belong to one dual pencil. The dual of Poncelet's closure theorem shows that if $P = P'$ for one starting point, then $P = P'$ for every starting point. Letting $P \to A$ gives $Q \to A$; in this limit, $P = P'$ is equivalent to the coincidence of $P'S$ and $QR$, which in turn is equivalent to $S, R \to C$. Thus, in general, $P = P'$ if and only if $AC$ is tangent to $\gamma$, and this determines $\gamma$. Once the closure condition holds, let $P \to D$. Then $P, S \to D$ and $Q, R \to B$, so $BD$ is also tangent to $\gamma$.

It remains to prove that the perimeter of $PQRS$ is constant; its limiting values are then $2AC$ and $2BD$. We treat the case in which $\gamma$ is an ellipse. Let $X, Y, Z, W$ be the points at which $PS, QR, PQ, SR$, respectively, are tangent to $\gamma$. Let $U$ be the intersection of $\beta_1$ and $\gamma$ nearest $A$, and let $V$ be the intersection of $\beta_2$ and $\gamma$ nearest $B$. On the arc $\widehat{XW}$ of $\gamma$ that contains neither $U$ nor $V$, choose a point $T$; this merely distinguishes the two arcs of $\gamma$ joining the same endpoints.

Apply Graves's theorem at $Q, S$ and MacCullagh's theorem at $P, R$. We obtain

$$QY + QZ + \widehat{YTZ} = \text{const} \equiv k_1, \quad SX + SW + \widehat{XTW} = \text{const} \equiv k_2,$$

$$PZ - PX + \widehat{XU} - \widehat{UZ} = 0, \quad RY - RW + \widehat{WV} - \widehat{YV} = 0.$$

Adding the last three equations and subtracting the first, then rearranging, gives

$$PQ + QR + RS + SP = k_2 - k_1 + 2\widehat{UV}.$$

If $\gamma$ is degenerate, the result is understood by taking a limit. If $\gamma$ is a hyperbola, use theorems 1.3.13 and 1.3.14; the details are left as an exercise. □

### Extremal Perimeter Problems

Another interesting application of confocal conics concerns extremal perimeters.

**Theorem 8.4.9.** *Let $\alpha$ be an ellipse. The perimeter of a convex $n$-gon inscribed in $\alpha$ is maximal if and only if the polygon is circumscribed about an ellipse $\beta$ confocal with $\alpha$.*

**Proof.** Let $A_1A_2 \cdots A_n$ be a convex $n$-gon of maximal perimeter; the case $n = 7$ is shown in figure 8.32. Indices are read cyclically, so for every integer $i$ we set $A_{i+n} = A_i$, and similarly $l_{i+n} = l_i$ and $B_{i+n} = B_i$. Let $l_i$ be the external bisector of $\angle A_{i-1}A_iA_{i+1}$. Then $l_i$ is tangent to $\alpha$ at $A_i$. Otherwise, if its second intersection with $\alpha$ were $B_i$, then $A_{i-1}B_i + B_iA_{i+1} > A_{i-1}A_i + A_iA_{i+1}$, contrary to maximality. Hence the lines $A_{i-1}A_i$ and $A_iA_{i+1}$ are reflections of one another on $\alpha$.

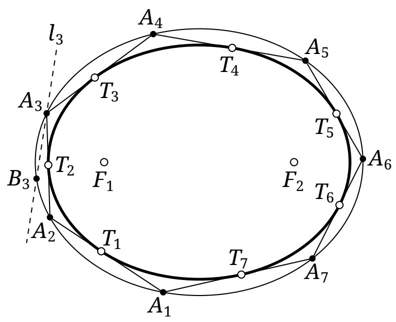



FIGURE 8.32

Let $\beta$ be confocal with $\alpha$ and tangent to $A_1A_2$. By proposition 8.4.5, the line $A_2A_3$ is also tangent to $\beta$; continuing in this way shows that every $A_iA_{i+1}$ is tangent to $\beta$. Conversely, proposition 8.4.5 shows that if a convex polygon is circumscribed about such a $\beta$, it satisfies the necessary condition for maximality established above.

Poncelet's closure theorem supplies infinitely many convex $n$-gons inscribed in $\alpha$ and circumscribed about $\beta$. Moreover, starting from $A_1$ and always drawing the tangent to $\beta$ on a prescribed side to obtain $A_2, A_3, \ldots$, we see that shrinking $\beta$ increases the angle through which one must turn at every step from $A_{i-1}A_i$ to $A_iA_{i+1}$. Thus at most one $\beta$ can satisfy the closure condition.

It remains only to show that all such convex polygons have the same perimeter. Let $T_i$ be the point at which $A_iA_{i+1}$ is tangent to $\beta$. Graves's theorem gives $A_iT_{i-1} + A_iT_i + \overset{\frown}{T_iT_{i+1}T_{i-1}} = \text{const} \equiv C$. Summing over $i = 1, \ldots, n$ and rearranging shows that the sum of the polygon's perimeter and $(n-1)$ times the perimeter of $\beta$ equals the constant $nC$. Thus the polygon's perimeter is constant. □

**Theorem 8.4.10.** *Let $\alpha$ be an ellipse. The perimeter*[9] *of a convex $n$-gon circumscribed about $\alpha$ is minimal if and only if the polygon is inscribed in an ellipse $\beta$ confocal with $\alpha$.*

**Proof.** Let the points of contact of the convex polygon with $\alpha$ be $T_1, \ldots, T_n$, and let $A_i$ be the vertex between $T_{i-1}$ and $T_i$. Again use cyclic indices, so $A_{i+n} = A_i$, $T_{i+n} = T_i$, and similarly for the remaining objects. Fix $T_{i-1}, T_{i+1}$ and ask where $T_i$ must lie for the polygon's perimeter to be minimal. We suppose that $T_i$ lies on the shorter arc between $T_{i-1}$ and $T_{i+1}$. Otherwise the same proof uses an excircle in place of the incircle; verifying this is left to the reader.

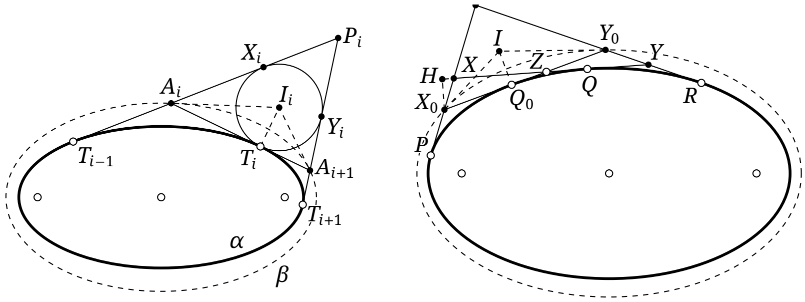
Put $P_i = T_{i-1}A_i \cap T_{i+1}A_{i+1}$; this point is fixed. Let $I_i$ and $\omega_i$ be the incenter and incircle of $\triangle A_iA_{i+1}P_i$, and suppose that $\omega_i$ touches $T_{i-1}A_i, T_{i+1}A_{i+1}, A_iA_{i+1}$ at $X_i, Y_i, Z_i$, respectively. The equal-tangent theorem shows that the length of the corresponding part of the polygon is

$$\overline{T_{i-1}A_iA_{i+1}T_{i+1}} = T_{i-1}X_i + T_{i+1}Y_i = T_{i-1}P_i + T_{i+1}P_i - 2P_iX_i = C - 2P_iX_i,$$

where $C$ is constant. Thus the length decreases as $P_iX_i$ increases. To maximize $P_iX_i$, one must maximize $P_iI_i$. Since $\omega_i$ and $\alpha$ lie on opposite sides of $A_iA_{i+1}$, the maximum occurs when $\omega_i$ is tangent to $\alpha$ at $T_i$. Then $Z_i = T_i$, and $I_iT_i \perp A_iA_{i+1}$.

Draw through $A_i$ a conic $\beta$ confocal with $\alpha$. By proposition 8.4.5, $A_iI_i$ is tangent to $\beta$. Let the other intersection of $A_iA_{i+1}$ with $\beta$ be $A'_{i+1}$, and let the tangent there meet $A_iI_i$ at $I'_i$. Proposition 8.4.6 gives $I'_iT_i \perp A_iA_{i+1}$. Since $A_i, I_i, I'_i$ are collinear, it follows that $I_i = I'_i$, and hence $A_{i+1} = A'_{i+1} \in \beta$. Thus all the vertices $A_i$ lie on $\beta$. The rest of the proof is exactly as for theorem 8.4.9, using Poncelet's closure theorem and Graves's theorem. □

[9] Here, unlike in the more general convention used earlier, every point of tangency is required to lie on the corresponding side segment.




(a) (b)

FIGURE 8.33

**Alternative proof of the local extremal condition.** There is another interesting, though apparently more complicated, way to find the position of $T_i$. For brevity put $X = A_i$, $Y = A_{i+1}$, $P = T_{i-1}$, $Q = T_i$, and $R = T_{i+1}$. Fix a position $Q_0$ of $Q$, and denote the corresponding positions of $X, Y$ by $X_0, Y_0$. Let $\alpha$ and $\beta$ be the angles made by $X_0Y_0$ with $PX_0$ and $RY_0$, respectively, and put $X_0Q_0 = a$, $Y_0Q_0 = b$. Let $\theta$ be the angle through which $XY$ has turned from $X_0Y_0$, and write the length of the broken line as $\overline{PXYR} = \ell(\theta)$. The extremal condition at $Q_0$ is $\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = 0$.

Consider a small rotation $\Delta\theta \ll 1$, and put $Z = X_0Y_0 \cap XY$; then $Q_0Z$ and $QZ$ are also small. Let $H$ be the projection of $X_0$ on $XY$. The triangle $\triangle X_0HZ$ is approximately an isosceles triangle with vertex angle $\angle X_0ZH \to 0$ and base angle $\angle ZHX_0 \to 90°$, and $X_0H \approx (a + Q_0Z)\Delta\theta \approx a\Delta\theta$. Furthermore, $\angle X_0HX \approx 90°$, $HQ \approx HZ \approx X_0Z \approx X_0Q_0$, and $\angle HXX_0 \approx \angle XX_0Q + \Delta\theta \approx \alpha$. It follows that

$$\Delta(PX + XQ) \approx X_0X - HX = \frac{X_0H}{\sin\alpha} - \frac{X_0H}{\tan\alpha} = a\tan\frac{\alpha}{2}\cdot\Delta\theta.$$

Similarly, $\Delta(RY + YQ) = -b\tan\frac{\beta}{2}\cdot\Delta\theta$. Hence

$$0 = \frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \lim_{\Delta\theta\to 0}\frac{\Delta\ell}{\Delta\theta} = \lim_{\Delta\theta\to 0}\frac{\Delta(PX + XQ) + \Delta(RY + YQ)}{\Delta\theta} = a\tan\frac{\alpha}{2} - b\tan\frac{\beta}{2}.$$

Let the bisectors of $\angle(PX_0, X_0Y_0)$ and $\angle(RY_0, X_0Y_0)$ meet at $I$, and let $Q'$ be the projection of $I$ on $X_0Y_0$. Then $IQ' = Q'X_0\tan\frac{\alpha}{2} = Q'Y_0\tan\frac{\beta}{2}$. Comparison with the preceding relation gives $Q' = Q_0$, and hence $IQ_0 \perp X_0Y_0$. As in the first proof, proposition 8.4.6 and the method of coincidence show that $X_0, Y_0$ lie on one conic confocal with the original ellipse. □

For the case $n = 3$ in the preceding two theorems, we may add the following observation. Given a triangle $\triangle ABC$, let $\triangle N_aN_bN_c$ be its extouch triangle. Since $\triangle N_aN_bN_c$ and the original triangle are perspective from *Na*, there is an ellipse tangent to the three sides of $\triangle ABC$ at $N_a, N_b, N_c$. This is called the *Mandart ellipse* of $\triangle ABC$.

Thus a triangle inscribed in an ellipse has maximal perimeter if and only if the Mandart ellipse of the triangle is confocal with the original ellipse. Indeed, proposition 8.4.6 and the optical property show immediately—or the same conclusion follows from the proof of theorem 8.4.10—that at a maximum the tangents at the vertices form the excentral triangle of the inscribed triangle, while the vertices of its extouch triangle are precisely the points at which its sides touch the inner confocal ellipse.

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## Exercises

**Exercise 8.40.** Complete the proof of Ivory's theorem explicitly by using theorems 1.3.13 and 1.3.14.

**Exercise 8.41.** For an ellipse with semimajor axis $a$ and semiminor axis $b$, find the extremal perimeters of its circumscribed triangles, circumscribed quadrilaterals, inscribed triangles, and inscribed quadrilaterals.

**Exercise 8.42* (Ivory's theorem).** In the plane, let $\alpha_1, \alpha_2$ be two thin insulating wires in the shape of confocal ellipses with common center $O$. Suppose that the two wires carry equal total charges and that at $P \in \alpha_i$, for $i = 1, 2$, the linear charge density on $\alpha_i$ is proportional to $\operatorname{dist}(O, \mathrm{t}_{\alpha_i}(P))$. Choose $A_1 \in \alpha_1$ and $A_2 \in \alpha_2$ on the same side of the principal axis and also on the same side of the secondary axis, and suppose that a hyperbola confocal with $\alpha_1$ passes through both points. Prove that the electrostatic potential produced by $\alpha_1$ at $A_2$ equals that produced by $\alpha_2$ at $A_1$.

**Exercise 8.43.** For fixed $n$, prove that the sum of the cosines of the interior angles is constant over all convex $n$-gons inscribed in a given ellipse and circumscribed about another, confocal ellipse.

**Exercise 8.44*.** Prove that the Mandart ellipse of a triangle has perspector *Na*, center *Mi*, and passes through *Fe*.

**Exercise 8.45.** Let $\alpha, \beta$ be confocal ellipses, and suppose that some triangle is inscribed in $\alpha$ and circumscribed about $\beta$. For every such triangle, prove that:

(1) the ratio of its inradius to its circumradius is constant;
(2) its excenters move on a fixed ellipse;
(3) its incenter moves on a fixed ellipse.

Part III

# Axiomatic Projective Geometry

*Projective Nets, Incidence Axioms, and Coordinates*

This part develops an axiomatic treatment of geometry, with axiomatic projective geometry as its principal subject. We begin with projective nets in arbitrary dimension and establish their spans, independence, bases, dimension theory, and fundamental perspectivities. We then formulate the incidence axioms of projective spaces and planes and examine the roles of Desargues' and Pappus' theorems, together with a brief introduction to the axioms of affine and Euclidean geometry. Starting from these axioms, we also construct coordinate structures on a projective plane and use them to study transformations of plane projective geometry.

## Chapter 9

# Basic Concepts of Projective Nets

## General Projective Spaces

This chapter develops the language of "nets" in classical projective geometry. From a modern point of view, a net is a projective space, so the discussion also serves as an introduction to projective spaces of arbitrary dimension. We establish their spans, independence, bases, dimension theory, and fundamental perspectivities. Desargues' and Pappus' properties are treated in the following chapter on axiomatic projective geometry, chapter 10. The modern interpretation of the word "net" is summarized at the end of the chapter.

## 9.1 Definition of a Projective Net

### Projective Nets

We first explain the meanings of an abstract projective "pencil" and "net." We begin with abstract projective points and lines.

**Definition 9.1.1.** An abstract *projective point–line system* consists of a set $\mathscr{P}$ of abstract *projective points*, or simply *points*, and a set $\mathscr{L}$ of abstract *projective lines*, or simply *lines*. [1] They satisfy the following axioms.

(1) Any two distinct projective points $A,B$ determine exactly one projective line, denoted by $AB$.
(2) Every projective line is a set of projective points and contains at least three projective points.
(3) (*Veblen axiom*) If two distinct projective lines have a common projective point, that point is called their *intersection*. If no three of four points $A,B,C,D$ are collinear and $AB$ meets $CD$, then $AC$ meets $BD$.

**Proposition 9.1.2.** *If two distinct lines in a projective point–line system meet, their point of intersection is unique.*

**Proof.** If two projective lines had more than one common projective point, axiom (1) would make them the same projective line, a contradiction. □

**Remark.** We explain the traditional term *pencil*. It is generally used for a one-dimensional projective object consisting of geometric objects of one kind. A "one-dimensional projective object" commonly means a collection of objects of one kind with one degree of freedom, that is, a collection that can be put in bijection with the points of a projective line.[2] For example, the point ranges and pencils of lines considered earlier are pencils. As another example, suppose a pencil of conics is generated by $A,B,C,D$, and choose a line $l$ through $A$ that avoids $B,C,D$. The points $P$ on $l$ then correspond bijectively to the conics $\mathfrak{C}(ABCDP)$ in the pencil. When $P=A$, this is understood as a limit: the conic passes through $A,B,C,D$ and is tangent to $l$ at $A$.

[1] Here "abstract" means that the objects need not be points and lines in the usual geometric sense.

[2] This is a descriptive explanation rather than a rigorous definition. A precise definition will be given for each particular situation when it arises.

**Example 9.1.3.** An ordinary real or complex projective plane is a projective point–line system, with its ordinary points as projective points and its ordinary lines as projective lines. Dually, one may take ordinary lines as abstract projective points and define the intersection of two such lines to be the abstract projective line determined by those two abstract points.

**Example 9.1.4.** Real or complex three-dimensional projective space is a projective point–line system whose projective points and lines are its ordinary points and lines. Here ordinary lines cannot naturally be used as abstract projective points, because it is difficult to say directly what abstract projective line is determined by two skew lines regarded as projective points. Nevertheless, a pencil of lines can still be defined as a pencil: it is the set of all lines through a fixed point and can carry a cross-ratio structure.

A point range, a quadratic point range on a plane conic, and a quadratic line range may likewise be regarded as pencils.

**Example 9.1.5.** Consider the pencils of conics introduced in section 6.7.1. In the complex projective plane, let $\mathscr{P}$ be the set of all conics, including degenerate ones, and regard each conic as a projective point. Let $\mathscr{L}$ be the set of all pencils of conics, each such pencil being a projective line or pencil. The pencil determined by two conics is then the projective line determined by the two corresponding projective points. Thus $\mathscr{P}$ and $\mathscr{L}$ may be regarded as a projective point–line system.

For this example, the Veblen axiom says the following. Given four conics $\gamma_1, \gamma_2, \gamma_3, \gamma_4$, suppose the eight points in $(\gamma_1 \cap \gamma_2) \cup (\gamma_3 \cap \gamma_4)$ lie on a conic; equivalently, $\operatorname{pen}(\gamma_1, \gamma_2)$ and $\operatorname{pen}(\gamma_3, \gamma_4)$ have a common member. One can prove that the eight points in $(\gamma_1 \cap \gamma_3) \cup (\gamma_2 \cap \gamma_4)$ also lie on a conic, or equivalently that $\operatorname{pen}(\gamma_1, \gamma_3)$ and $\operatorname{pen}(\gamma_2, \gamma_4)$ have a common member.

**Definition 9.1.6.**

(1) Let $\mathscr{P}$ be the set of all points of a projective point–line system. A subset $\mathscr{M} \subseteq \mathscr{P}$ is a *projective net*, or simply a *net*, if, for $X, Y \in \mathscr{P}$, the condition $X, Y \in \mathscr{M}$ implies $XY \subseteq \mathscr{M}$. The empty set $\emptyset$ is also regarded as a net, called the *empty net*.

(2) For a net $\mathscr{M}$, if $\mathscr{N} \subseteq \mathscr{M}$ is also a net, then $\mathscr{N}$ is a *subnet* of $\mathscr{M}$. If in addition $\mathscr{N} \neq \mathscr{M}$, it is a *proper subnet*.

(3) Let $\mathscr{N}, \mathscr{N}'$ be nets. A bijection $f: \mathscr{N} \to \mathscr{N}'$ is a *net isomorphism*, or simply an *isomorphism* when there is no ambiguity, if the image of every projective line in $\mathscr{N}$ is a projective line in $\mathscr{N}'$.

**Remark.** A subnet is itself a net; the prefix records only the relative inclusion. By definition, the set of all points of a projective point–line system is also a net.

**Theorem 9.1.7.** *For any family of nets $\mathscr{M}_1, \mathscr{M}_2, \ldots, \mathscr{M}_j$ in a projective point–line system, the intersection $\bigcap_{\alpha=1}^{j} \mathscr{M}_\alpha$ is a net.*

**Proof.** If $X, Y$ lie in the intersection, then $XY \subseteq \mathscr{M}_\alpha$ for every $\alpha$, and hence $XY \subseteq \bigcap_\alpha \mathscr{M}_\alpha$. □

### Spans of Projective Nets

**Definition 9.1.8.** Let $S_1, S_2, \ldots, S_j \subseteq \mathscr{P}$ be sets of projective points, not necessarily nets. The net *spanned* by them is the smallest net containing all these sets; the set $\{S_1, \ldots, S_j\}$ is a *generating set*, or *generating point set*, of that net. In what follows, we use the custom notation net for a net. Thus

$$\operatorname{net}(S_1, \ldots, S_j) := \bigcap_{\mathscr{M} \in \mathscr{F}} \mathscr{M},$$

where

$$\mathscr{F} = \{\mathscr{M} \subseteq \mathscr{P} \mid S_1, \ldots, S_j \subseteq \mathscr{M}, \text{ and } \mathscr{M} \text{ is a net}\}.$$

For nets $\mathscr{M},\mathscr{N}$, their *join* is $\mathscr{M} \vee \mathscr{N} := \operatorname{net}(\mathscr{M},\mathscr{N})$. Joins of more than two nets are defined analogously.

Plainly, the span and join operations for nets satisfy the following properties.

**Proposition 9.1.9.** *For sets $S,T$ of projective points in the same projective point–line system:*
*(1)* $S \subseteq \operatorname{net}(S)$*;*
*(2)* $S \subseteq T \Rightarrow \operatorname{net}(S) \subseteq \operatorname{net}(T)$*;*
*(3)* $\operatorname{net}(\operatorname{net}(S),T) = \operatorname{net}(S,T)$*;*
*(4)* $\operatorname{net}(S,\emptyset) = \operatorname{net}(S)$ *and* $\operatorname{net}(\emptyset) = \emptyset$.

**Proposition 9.1.10.** *Let $\mathscr{L},\mathscr{M},\mathscr{N}$ be nets in the same projective point–line system.*
*(1)* $\mathscr{M} \vee \mathscr{N} = \mathscr{N} \vee \mathscr{M}$ *and* $(\mathscr{M} \vee \mathscr{N}) \vee \mathscr{L} = \mathscr{M} \vee (\mathscr{N} \vee \mathscr{L})$*;*
*(2)* $\mathscr{M} \vee \mathscr{M} = \mathscr{M}$ *and* $\mathscr{M} \vee (\mathscr{M} \cap \mathscr{N}) = \mathscr{M}$*;*
*(3) if nets $\mathscr{M}',\mathscr{N}'$ in the same system satisfy $\mathscr{M}' \supseteq \mathscr{M}$ and $\mathscr{N}' \supseteq \mathscr{N}$, then $\mathscr{M} \vee \mathscr{N} \subseteq \mathscr{M}' \vee \mathscr{N}'$.*

**Theorem 9.1.11.** *Let $f : \mathscr{N} \to \mathscr{N}'$ be a net isomorphism. For every set of points $S \subseteq \mathscr{N}$, $f(\operatorname{net}(S)) = \operatorname{net}(f(S))$. Moreover, $f$ maps nets to nets and subnets to subnets.*

**Proof.** A net is characterized by closure under joining projective points, and a net isomorphism preserves projective lines. It therefore maps nets and subnets to nets and subnets.

Since $S \subseteq \operatorname{net}(S)$, we have $f(S) \subseteq f(\operatorname{net}(S))$ and hence $\operatorname{net}(f(S)) \subseteq \operatorname{net}[f(\operatorname{net}(S))] = f(\operatorname{net}(S))$, where the equality holds because a net isomorphism preserves nets and $f(\operatorname{net}(S))$ is already a net. Hence $\operatorname{net}(f(S)) \subseteq f(\operatorname{net}(S))$. Applying the same argument to the inverse isomorphism $f^{-1} : \mathscr{N}' \to \mathscr{N}$ and the set $f(S)$ gives $\operatorname{net}(S) = \operatorname{net}(f^{-1}[f(S)]) \subseteq f^{-1}(\operatorname{net}[f(S)])$, so $f(\operatorname{net}(S)) \subseteq \operatorname{net}(f(S))$. Thus $f(\operatorname{net}(S)) = \operatorname{net}(f(S))$. □

This construction resembles the span operation in a vector space; see section A.5. In the later axiomatic treatment, chapter 10, a projective point will be represented by homogeneous coordinates and may be viewed as a vector modulo an equivalence relation given by right multiplication. All the constructions above may then be interpreted linearly; for example, the join of two subnets corresponds to the sum of two linear subspaces. The present construction, however, is independent of linear algebra and does not require it as a prerequisite. The following results parallel familiar facts about vector spaces and show how the Veblen axiom guarantees a well-behaved structure.

**Theorem 9.1.12.** *If $\mathscr{M}$ is a nonempty net and $P$ a projective point, then*

$$\operatorname{net}(\mathscr{M},P) = \bigcup_{X \in \mathscr{M}} PX.$$

**Proof.** Put $\mathscr{J} = \bigcup_{X \in \mathscr{M}} PX$. The set $\mathscr{J}$ plainly contains $\mathscr{M}$ and $P$. Since each $X$ on the right belongs to $\mathscr{M}$, it remains only to prove that $\mathscr{J}$ is closed under joining points.

Take $A,B \in \mathscr{J}$ and $Z \in AB$. If $A,B \in \mathscr{M}$, then $AB \subseteq \mathscr{M}$; the conclusion is also immediate if $A = P$ or $B = P$. We therefore consider only the remaining cases.

Suppose first that $A \in \mathscr{M}$ and $B \in PY$, where $Y \in \mathscr{M}$. If $A = Y$, then $AB = AP \subseteq \mathscr{J}$. Otherwise $ZA = AB$ meets $PY$ at $B$, so the Veblen axiom implies that $ZP$ meets $AY$ at a projective point $X$. Since $A,Y \in \mathscr{M}$, we have $AY \subseteq \mathscr{M}$ and hence $X \in \mathscr{M}$. Thus $Z \in PX \subseteq \mathscr{J}$.

Finally suppose that $A \in PX$ and $B \in PY$, with $X,Y \in \mathscr{M}$. If $X = Y$, then $A,B \in PX$ and the assertion is immediate. Assume $X \neq Y$. We may take the points and lines in general position, since the collinear or concurrent degenerate cases follow directly. From $ZA \cap PY = B$, the Veblen axiom gives $ZP \cap AY = T$. Since $TY = AY$ and $TY \cap PX = A$, a second application of the Veblen axiom gives $TP \cap YX = W$. Now $TP = ZP$ and $W \in XY \subseteq \mathscr{M}$, so $Z \in PW \subseteq \mathscr{J}$.

Thus $AB \subseteq \mathscr{J}$ for all $A,B \in \mathscr{J}$, completing the proof. □

**Proposition 9.1.13.** *Let $\mathscr{M}$ be a net and $P,Q$ projective points. If $Q \in \operatorname{net}(\mathscr{M},P)\setminus\mathscr{M}$, then $P \in \operatorname{net}(\mathscr{M},Q)$ and $\operatorname{net}(\mathscr{M},P)=\operatorname{net}(\mathscr{M},Q)$.*

**Proof.** If $\mathscr{M}=\emptyset$, then $\operatorname{net}(\mathscr{M},P)=\{P\}$ and the result is immediate. If $\mathscr{M}\neq\emptyset$, theorem 9.1.12 gives an $X\in\mathscr{M}$ with $Q\in PX$. Since $Q\notin\mathscr{M}$, we have $Q\neq X$; hence $P\in QX\subseteq\operatorname{net}(\mathscr{M},Q)$ and $\operatorname{net}(\mathscr{M},P)\subseteq\operatorname{net}(\mathscr{M},Q)$. Similarly, $\operatorname{net}(\mathscr{M},Q)\subseteq\operatorname{net}(\mathscr{M},P)$, completing the proof. □

**Proposition 9.1.14.** *Let $\mathscr{M}$ be a net and $P_1,P_2,\ldots,P_s$ projective points. If $R\in\operatorname{net}(\mathscr{M},P_1,\ldots,P_s)\setminus\mathscr{M}$, then for some $i\in\{1,\ldots,s\}$,*

$$\operatorname{net}(\mathscr{M},P_1,\ldots,P_s)=\operatorname{net}(\mathscr{M},R,P_1,\ldots,P_{i-1},P_{i+1},\ldots,P_s).$$

**Proof.** Choose the least positive integer $i$ with $R\in\operatorname{net}(\mathscr{M},P_1,\ldots,P_i)$. Put $\mathscr{N}=\operatorname{net}(\mathscr{M},P_1,\ldots,P_{i-1})$, with $\mathscr{N}=\mathscr{M}$ when $i=1$. By minimality, $R\notin\mathscr{N}$ but $R\in\operatorname{net}(\mathscr{N},P_i)$. By proposition 9.1.13, $P_i\in\operatorname{net}(\mathscr{N},R)$, so $R$ may replace $P_i$ in the generating set without changing the generated net. □

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## Exercises

**Exercise 9.1.** Prove propositions 9.1.9 and 9.1.10.

**Exercise 9.2.** Prove that the union $\mathscr{M}\cup\mathscr{N}$ of two nets is a net if and only if $\mathscr{M}\subseteq\mathscr{N}$ or $\mathscr{N}\subseteq\mathscr{M}$.

**Exercise 9.3.** Let $f:\mathscr{N}\to\mathscr{N}'$ be a net isomorphism, let $I$ be an index set, and let $\{\mathscr{M}_i\mid i\in I\}$ be a family of subnets of $\mathscr{N}$. Prove that:
(1) $f\left(\bigcap_{i\in I}\mathscr{M}_i\right)=\bigcap_{i\in I}f(\mathscr{M}_i)$;
(2) the restriction $f|_{\mathscr{M}_i}:\mathscr{M}_i\to f(\mathscr{M}_i)$ is again a net isomorphism.

**Exercise 9.4.** Let $f:\mathscr{N}\to\mathscr{N}'$ be a bijection between two nets, and suppose that three points of $\mathscr{N}$ are collinear if and only if their images are collinear. Prove that $f$ is a net isomorphism. Does the conclusion still hold if "if and only if" is replaced by only the necessary condition, or by only the sufficient condition?

**Exercise 9.5.** For a set $S$ of projective points, put $S_0=S$ and, for $i\in\mathbb{N}$, recursively define

$$S_{i+1}=S_i\cup\Bigg(\bigcup_{\substack{A,B\in S_i\\ A\neq B}}AB\Bigg).$$

Prove that $\bigcup_{i=0}^{\infty}S_i=\operatorname{net}(S)$.

**Exercise 9.6*.** In this problem, a "circle" is also understood to include a degenerate conic through both circular points. Let $\mathscr{C}$ be the set of circles in the complex projective plane. Regard every circle as an abstract projective point and the pencil of circles determined by any two circles as an abstract projective line. Prove that these objects form a projective point–line system.
*Hint.* Consider a projective space containing the plane $\pi$. In this space, choose a sphere $\Sigma$ with center $O$, and choose $N\in\Sigma$ such that $ON\perp\pi$ and $N\notin\pi$. Stereographic projection from $N$ defines $f:\Sigma\to\pi$ by $X\mapsto NX\cap\pi$.[3] For every plane $\Pi$, define $g:\Pi\mapsto f(\Pi\cap\Sigma)$. First show that the range of $g$ is the set of circles in $\pi$. Then show that the family of planes in space containing a fixed line is

[3] When $X=N$, the line $XN$ is naturally understood as the corresponding limit, namely as the tangent line to $\Sigma$ at $N$.

mapped by $g$ to a pencil of circles in $\pi$, and conversely that every pencil of circles in $\pi$ arises from such a family of planes.

## 9.2 Dimension of a Projective Net

### 9.2.1 Bases and Dimension

We continue to study general projective nets. Their bases and dimension can be defined by analogy with those of vector spaces.

**Definition 9.2.1.** Let $S=\{P_1,\ldots,P_n\}$ be a finite set of projective points. The points $P_1,\ldots,P_n$, or the set $S$, are *projectively independent* if $P_i \notin \operatorname{net}(S\setminus\{P_i\})$ for every $P_i \in S$; otherwise they are *projectively dependent*.

If $\mathscr{N}$ is a net and a finite set $S\subseteq\mathscr{N}$ satisfies $\operatorname{net}(S)=\mathscr{N}$, then $S$ is a *generating set*, or *generating point set*, of $\mathscr{N}$. A projectively independent generating set is a *projective basis*, or simply a *basis*, of $\mathscr{N}$.

**Proposition 9.2.2.** *Every finite generating set contains a subset that is a projective basis of the generated net.*

**Proof.** Successively delete from the finite generating set any point generated by the remaining points. The process terminates after finitely many steps. The resulting set still spans the original net, and no one of its points can be deleted; it is therefore projectively independent. ☐

**Theorem 9.2.3** (Steinitz exchange theorem)**.** *Let $P=\{P_1,\ldots,P_r\}$ be a projectively independent set in a net $\mathscr{N}$, and suppose $Q=\{Q_1,\ldots,Q_s\}$ spans $\mathscr{N}$. Then $r\le s$, and one can choose $s-r$ distinct projective points $T=\{T_{r+1},T_{r+2},\ldots,T_s\}\subseteq Q$ such that $\mathscr{N}=\operatorname{net}(P,T)$.*

**Proof.** We use proposition 9.1.14. Begin with $\mathscr{N}=\operatorname{net}(Q_1,\ldots,Q_s)$. Since $P_1\in\mathscr{N}$, proposition 9.1.14 permits $P_1$ to replace one of the $Q_i$; relabeling if necessary, suppose it replaces $Q_1$. Thus $\mathscr{N}=\operatorname{net}(P_1,Q_2,\ldots,Q_s)$. Since $P_2\notin\operatorname{net}(P_1)$, the same proposition permits $P_2$ to replace one of the remaining $Q_i$; suppose it is $Q_2$. Thus $\mathscr{N}=\operatorname{net}(P_1,P_2,Q_3,\ldots,Q_s)$. Projective independence gives $P_3\notin\operatorname{net}(P_1,P_2)$, so $P_3$ may replace one of the remaining $Q_i$. Continuing in this way, we may suppose that the subsequently replaced points are $Q_3,Q_4,\ldots$, in order.

If $r>s$, then once all of $Q_1,\ldots,Q_s$ have been replaced, the definition $\operatorname{net}(Q)=\mathscr{N}$ would force a remaining $P_j$ to lie in $\operatorname{net}(P_1,\ldots,P_{j-1})$, contrary to projective independence. Hence $r\le s$, and at the end of the process $\mathscr{N}=\operatorname{net}(P_1,\ldots,P_r,Q_{r+1},\ldots,Q_s)$. ☐

**Corollary 9.2.4.** *In a net spanned by finitely many points, every projectively independent set can be extended to a projective basis, and every generating set contains a subset that is a basis.*

**Proof.** Apply the Steinitz exchange theorem to an independent set and a finite generating set. The second assertion also follows by deleting redundant points. ☐

If two finite projectively independent sets of respective cardinalities $m,n$ span the same net, the Steinitz exchange theorem gives both $m\le n$ and $n\le m$. Thus $m=n$, and dimension is well defined.

**Definition 9.2.5.** The *dimension* of a nonempty net $\mathscr{N}$ is one less than the number of points in any basis and is denoted by $\dim\mathscr{N}$. A single projective point, viewed as a singleton net, has dimension zero. By convention, the empty net has dimension $-1$. A net of dimension $n$ is called an *n-dimensional net*.

**Remark.** A one-dimensional net is spanned by two projective points and is therefore a projective line. In classical geometry the word "net" originally referred only to the two-dimensional case; it was later extended to arbitrary dimensions. We adopt that broader usage here.

**Example 9.2.6.** Continuing example 9.1.3, take the ordinary points of a projective space as its projective points. A projective plane is then a two-dimensional net, and any three noncollinear points form a basis.

**Example 9.2.7.** Continuing example 9.1.5, the set of all conics through three fixed points may be regarded as a two-dimensional net. Any three conics that do not all pass through a common fourth point form a basis.

**Proof.** Let $\gamma_1, \gamma_2, \gamma_3$ be three conics having three common points but no common fourth point. We show that $\operatorname{net}(\gamma_1, \gamma_2, \gamma_3)$ is the set of conics through the three common points. Every $\gamma \in \operatorname{pen}(\gamma_1, \gamma_2)$ passes through those three points. By theorem 9.1.12, all elements of the conic net are generated by $\operatorname{pen}(\gamma, \gamma_3)$, whose members also pass through the three points.

Conversely, let $\gamma$ be any conic through the three common points. Let $\alpha$ be the conic through those points, the fourth intersection of $\gamma$ with $\gamma_3$, and the fourth intersection of $\gamma_1$ with $\gamma_2$. Then $\alpha \in \operatorname{pen}(\gamma_1, \gamma_2)$ and $\gamma \in \operatorname{pen}(\alpha, \gamma_3)$. □

**Proposition 9.2.8.** *A set $S$ of $r$ points is projectively independent if and only if* $\dim \operatorname{net}(S) = r - 1$.

**Proof.** Necessity is the definition of dimension. Conversely, suppose $\dim \operatorname{net}(S) = r - 1$ but the points are projectively dependent. By proposition 9.2.2, some points can be deleted to obtain a basis $S'$ with $\operatorname{net}(S) = \operatorname{net}(S')$. Then $\dim \operatorname{net}(S) = \dim \operatorname{net}(S') < r - 1$, a contradiction. □

**Proposition 9.2.9.** *If $\mathscr{N}$ is a net and $P$ is a projective point outside $\mathscr{N}$, then* $\dim \operatorname{net}(\mathscr{N}, P) = \dim \mathscr{N} + 1$.

**Proof.** Take a basis $X = \{P_0, \ldots, P_r\}$ of $\mathscr{N}$. Since $P \notin \operatorname{net}(X)$, the set $\{P_0, \ldots, P_r, P\}$ is projectively independent and spans $\operatorname{net}(\mathscr{N}, P)$. □

**Corollary 9.2.10.** *If $\mathscr{M} \subsetneq \mathscr{N}$ are finite-dimensional nets, then* $\dim \mathscr{M} < \dim \mathscr{N}$. *Consequently, a finite-dimensional subnet having the same dimension as its ambient net equals that net.*

**Proof.** Choose $P \in \mathscr{N} \setminus \mathscr{M}$. Then $\operatorname{net}(\mathscr{M}, P) \subseteq \mathscr{N}$, so $\dim \mathscr{N} \geq \dim \operatorname{net}(\mathscr{M}, P) = \dim \mathscr{M} + 1 > \dim \mathscr{M}$. □

**Corollary 9.2.11.** Let $\mathscr{N}$ be a proper subnet of a net $\mathscr{M}$. If $\mathscr{N} \subsetneq \mathscr{M}$ and no subnet $\mathscr{K}$ of $\mathscr{M}$ satisfies $\mathscr{N} \subsetneq \mathscr{K} \subsetneq \mathscr{M}$, then $\mathscr{N}$ is called a *maximal proper subnet*, or a *hypernet* of $\mathscr{M}$. [4] *For a finite-dimensional net $\mathscr{M}$:*

*(1) a proper subnet $\mathscr{N}$ is a hypernet if and only if* $\dim \mathscr{M} = \dim \mathscr{N} + 1$;

*(2) every proper subnet $\mathscr{K} \subseteq \mathscr{M}$ is contained in a hypernet $\mathscr{N} \subseteq \mathscr{M}$.*

**Proof.** For (1), if $\dim \mathscr{N} = \dim \mathscr{M} - 1$, every subnet properly containing $\mathscr{N}$ has dimension at least $\dim \mathscr{M}$ and must therefore be $\mathscr{M}$. Conversely, if $\mathscr{N}$ is a hypernet and $P \in \mathscr{M} \setminus \mathscr{N}$, then $\operatorname{net}(\mathscr{N}, P) = \mathscr{M}$, so $\dim \mathscr{M} = \dim \mathscr{N} + 1$.

For (2), take a basis of the proper subnet $\mathscr{K}$ and extend it to a basis of $\mathscr{M}$. Choose any projective point $P$ from the added part and delete it. Let $\mathscr{N}$ be the net spanned by the remaining basis points. By (1), $\mathscr{N}$ is a hypernet with the required properties. □

[4] The short term "hypernet" is formed by analogy with "hyperplane."

Dimension may equivalently be described in terms of flags.

**Definition 9.2.12.** A chain of strict inclusions among nonempty subnets, $\mathscr{N}_0 \subsetneq \mathscr{N}_1 \subsetneq \cdots \subsetneq \mathscr{N}_k$, is a *flag*. Its *length* is $k$, the number of strict inclusions (one less than the number of subnets). It is a *complete flag* if, for every $i=1,2,\dots,k$, there is no subnet $\mathscr{M}_i$ such that $\mathscr{N}_{i-1} \subsetneq \mathscr{M}_i \subsetneq \mathscr{N}_i$, and there is no nonempty subnet $\mathscr{M}$ such that either $\mathscr{M} \subsetneq \mathscr{N}_0$ or $\mathscr{N}_k \subsetneq \mathscr{M}$. Equivalently, no further subnet can be inserted into the strict inclusion chain. Otherwise the flag is a *partial flag*.

**Corollary 9.2.13.** *Let $\mathscr{N}$ be an $n$-dimensional net. A flag $\mathscr{N}_0 \subsetneq \mathscr{N}_1 \subsetneq \cdots \subsetneq \mathscr{N}_k$ is complete in $\mathscr{N}$ if and only if $k=n$ and $\dim \mathscr{N}_i = i$. Every flag in $\mathscr{N}$ has length at most $n$, and complete flags of length $n$ exist. Equivalently,*

$$\dim \mathscr{N} = \max\{\textit{lengths of strict chains of subnets from a point-net to } \mathscr{N}\}.$$

**Proof.** By corollary 9.2.10, dimensions strictly increase along a strict chain of finite-dimensional subnets. A complete flag must therefore satisfy $\dim \mathscr{N}_i = i$ and $k=n$. By definition, the length of a flag is at most that of a complete flag, and a complete flag has length at most $n$. If $P_0,\dots,P_n$ is a basis of $\mathscr{N}$, then

$$\operatorname{net}(P_0) \subsetneq \operatorname{net}(P_0,P_1) \subsetneq \cdots \subsetneq \operatorname{net}(P_0,\dots,P_n)$$

is a complete flag of length $n$. □

Finally, we give the properties of bases and dimension under net isomorphisms.

**Theorem 9.2.14.** *A net isomorphism maps projectively independent points to projectively independent points, maps a basis of any subnet to a basis of its image, and preserves the dimension of subnets.*

**Proof.** Bases, spans, and the other notions involved are defined in terms of intersections and inclusions of nets. A net isomorphism preserves nets, intersections, and inclusions, so the assertions follow immediately. □

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## Exercises

**Exercise 9.7.** Let $\mathscr{N}$ be an $r$-dimensional net, let $\mathscr{H}$ be one of its hypernets, and let $l \subseteq \mathscr{N}$ be a projective line. Determine $\dim(l \cap \mathscr{H})$.

**Exercise 9.8.** Let $\mathscr{N}$ be a positive-dimensional finite-dimensional net, every one-dimensional subnet of which contains infinitely many projective points. Prove that there is no finite family of proper subnets $\{\mathscr{N}_1,\mathscr{N}_2,\dots,\mathscr{N}_r\}$ such that $\mathscr{N} = \bigcup_{i=1}^r \mathscr{N}_i$.

**Exercise 9.9.** Let $\mathscr{K}$ be a subnet of a net $\mathscr{M}$, with $\dim \mathscr{M} = \dim \mathscr{K} + 2$, and let $l, l' \subseteq \mathscr{M}$ be projective lines disjoint from $\mathscr{K}$. Define

$$f_{l,l'} : l \longrightarrow l', \quad A \longmapsto (\mathscr{K} \vee A) \cap l'.$$

Prove that:

(1) $f_{l,l'}$ is a bijection and $\mathscr{K} \vee A = \mathscr{K} \vee f_{l,l'}(A)$;

(2) if $l'' \subseteq \mathscr{M}$ is another projective line disjoint from $\mathscr{K}$, then $f_{l,l''} = f_{l',l''} \circ f_{l,l'}$.

**Exercise 9.10.** Let $\mathscr{N}$ be a net of dimension at least 3, and let $A,B,C,A',B',C'$ be projective points of $\mathscr{N}$ such that the lines $AA',BB',CC'$ have a common point. Prove that:

(1) $BC$ meets $B'C'$ at a point $P$, $CA$ meets $C'A'$ at a point $Q$, and $AB$ meets $A'B'$ at a point $R$;

(2) $P,Q,R$ are collinear.

**Exercise 9.11.** Determine the dimension of the projective net formed by all circles in the plane, including the degenerate members, in exercise 9.6.

### 9.2.2 The Dimension Formula and Its Applications

**Theorem 9.2.15.** *For nonempty finite-dimensional nets $\mathscr{M}, \mathscr{N}$,*

$$\mathscr{M} \vee \mathscr{N} = \bigcup_{X \in \mathscr{M}, Y \in \mathscr{N}} XY.$$

**Proof.** The union on the right is plainly contained in $\mathscr{M} \vee \mathscr{N}$. It remains to show $\bigcup(XY) \supseteq \mathscr{M} \vee \mathscr{N}$. To this end, take a basis $\{Y_0, \ldots, Y_s\}$ of $\mathscr{N}$, and put $\mathscr{N}_i = \mathrm{net}(Y_0, \ldots, Y_i)$ and $\mathscr{J}_i = \mathrm{net}(\mathscr{M}, \mathscr{N}_i)$. We prove by induction on $i$ that every point of $\mathscr{J}_i$ lies on a line $XY$ with $X \in \mathscr{M}$ and $Y \in \mathscr{N}_i$.

For $i = 0$ this is theorem 9.1.12. Suppose the assertion holds for $i-1$, and take $Z \in \mathscr{J}_i = \mathrm{net}(\mathscr{J}_{i-1}, Y_i)$. If $Z \in \mathscr{J}_{i-1}$, use the induction hypothesis. Otherwise theorem 9.1.12 gives $W \in \mathscr{J}_{i-1}$ such that $Z \in Y_i W$, and the induction hypothesis gives $X \in \mathscr{M}$ and $Y' \in \mathscr{N}_{i-1}$ such that $W \in XY'$. If no three of $Z, Y_i, X, Y'$ are collinear, then $ZY_i \cap XY' = W$, so the Veblen axiom gives a point $Y = ZX \cap Y_i Y'$. Since $Y_i, Y' \in \mathscr{N}_i$, we have $Y \in \mathscr{N}_i$, and therefore $Z \in XY$. The degenerate cases follow directly, completing the induction. □

**Corollary 9.2.16.** *For nonempty finite-dimensional nets $\mathscr{M}, \mathscr{N}$,*

$$\mathscr{M} \vee \mathscr{N} = \bigcup_{X \in \mathscr{M}} \mathrm{net}(X, \mathscr{N}) = \bigcup_{Y \in \mathscr{N}} \mathrm{net}(\mathscr{M}, Y).$$

**Proof.** Combine theorems 9.1.12 and 9.2.15. □

**Theorem 9.2.17** (Modular law)**.** *If $\mathscr{A}, \mathscr{B}, \mathscr{C}$ are finite-dimensional nets with $\mathscr{A} \subseteq \mathscr{C}$, then*

$$\mathscr{A} \vee (\mathscr{B} \cap \mathscr{C}) = (\mathscr{A} \vee \mathscr{B}) \cap \mathscr{C}.$$

**Proof.** The left-hand side is plainly contained in the right-hand side. Conversely, take $Z \in (\mathscr{A} \vee \mathscr{B}) \cap \mathscr{C}$. By theorem 9.2.15, there are $X \in \mathscr{A}$ and $Y \in \mathscr{B}$ such that $Z \in XY$. If $Z = X$, then $Z \in \mathscr{A}$. Otherwise $X \in \mathscr{A} \subseteq \mathscr{C}$ and $Z \in \mathscr{C}$, so the whole projective line $XZ = XY$ lies in $\mathscr{C}$. Hence $Y \in \mathscr{B} \cap \mathscr{C}$, and thus $Z \in \mathscr{A} \vee (\mathscr{B} \cap \mathscr{C})$. Thus the right-hand side of the required equality is contained in the left-hand side, completing the proof. □

**Theorem 9.2.18** (Dimension formula)**.** *For any two finite-dimensional nets $\mathscr{M}, \mathscr{N}$,*

$$\dim(\mathscr{M} \vee \mathscr{N}) = \dim \mathscr{M} + \dim \mathscr{N} - \dim(\mathscr{M} \cap \mathscr{N}).$$

**Proof.** Put $\mathscr{K} = \mathscr{M} \cap \mathscr{N}$. Extend a basis of $\mathscr{K}$ to a basis of $\mathscr{M}$, and denote the added points by $P_1, \ldots, P_a$. Then $a = \dim \mathscr{M} - \dim \mathscr{K}$. Set $\mathscr{M}_i = \mathrm{net}(\mathscr{K}, P_1, \ldots, P_i)$ and $\mathscr{N}_i = \mathrm{net}(\mathscr{N}, P_1, \ldots, P_i)$. If $P_i \in \mathscr{N}_{i-1}$, then $P_i \in \mathscr{M}$ and $P_i \in \mathscr{N} \vee \mathscr{M}_{i-1}$. The modular law gives

$$\mathscr{M} \cap (\mathscr{N} \vee \mathscr{M}_{i-1}) = (\mathscr{M} \cap \mathscr{N}) \vee \mathscr{M}_{i-1} = \mathscr{M}_{i-1},$$

so $P_i \in \mathscr{M}_{i-1}$, contradicting the independence of the chosen basis. Thus $P_i \notin \mathscr{N}_{i-1}$. Starting with $\mathscr{N}$ and adjoining $P_1, P_2, \ldots, P_a$ successively gives $\mathscr{N}_1, \mathscr{N}_2, \ldots, \mathscr{N}_a$. After each $P_i$ is adjoined, the step $\mathscr{N}_{i-1} \to \mathscr{N}_i$ raises the dimension by one. Clearly $\mathscr{N}_a = \mathscr{M} \vee \mathscr{N}$. Hence

$$\dim(\mathscr{M} \vee \mathscr{N}) = \dim \mathscr{N} + a = \dim \mathscr{N} + \dim \mathscr{M} - \dim(\mathscr{M} \cap \mathscr{N}).$$

□

**Corollary 9.2.19.** *The dimension of the join of two disjoint finite-dimensional nets is the sum of their dimensions plus one. That is, if* $\mathscr{M} \cap \mathscr{N} = \emptyset$*, then* $\dim(\mathscr{M} \vee \mathscr{N}) = \dim\mathscr{M} + \dim\mathscr{N} + 1$.

**Corollary 9.2.20.** *Let* $\mathscr{K}$ *be a subnet of a finite-dimensional net* $\mathscr{N}$. *There is a subnet* $\mathscr{H} \subseteq \mathscr{N}$ *such that* $\mathscr{K} \cap \mathscr{H} = \emptyset$, $\mathscr{K} \vee \mathscr{H} = \mathscr{N}$, *and* $\dim\mathscr{H} = \dim\mathscr{N} - \dim\mathscr{K} - 1$. The nets $\mathscr{K}, \mathscr{H}$ are then called *complementary* in $\mathscr{N}$; in what follows, write $\mathscr{K} \dot{\vee} \mathscr{H} = \mathscr{N}$.

**Proof.** By corollary 9.2.4, extend a basis of $\mathscr{K}$ to a basis of $\mathscr{N}$. The net spanned by the new projective points has the properties required of $\mathscr{H}$. The dimension statement follows from the dimension formula. □

**Remark.** If $\mathscr{K} = \mathscr{N}$, take $\mathscr{H} = \emptyset$; this case is included in the corollary.

**Corollary 9.2.21.** *Let* $\mathscr{H}$ *be a hypernet of a* $\mu$*-dimensional net* $\mathscr{N}$, *and let* $\mathscr{M} \subseteq \mathscr{N}$ *be another subnet. If* $\mathscr{M} - \mathscr{H} \neq \emptyset$, *then* $\dim(\mathscr{M} \cap \mathscr{H}) = \dim\mathscr{M} - 1$. *In particular, two distinct hypernets in a* $\mu$*-dimensional net meet in a* $(\mu - 2)$*-dimensional net, and two distinct projective lines in a two-dimensional net meet in a unique projective point.*

**Proof.** Since $\mathscr{M} - \mathscr{H} \neq \emptyset$, we have $\mathscr{M} \vee \mathscr{H} \supsetneq \mathscr{H}$, and corollary 9.2.10 gives $\mathscr{M} \vee \mathscr{H} = \mathscr{N}$. The dimension formula yields

$$\dim(\mathscr{M} \cap \mathscr{H}) = \dim\mathscr{M} + \dim\mathscr{H} - \dim\mathscr{N} = \dim\mathscr{M} + (\mu - 1) - \mu = \dim\mathscr{M} - 1.$$

□

**Proposition 9.2.22.** *Let* $\mathscr{H}$ *be an r-dimensional net with infinitely many points, where* $r \geq 1$, *and let* $F_1, \ldots, F_s \in \mathscr{H}$ *be finitely many points. There is a hypernet* $\mathscr{J}$ *of* $\mathscr{H}$ *such that* $F_i \notin \mathscr{J}$ *for* $i = 1, \ldots, s$.

**Proof.** We use induction on $r$. If $r = 1$, choose as $\mathscr{J}$ any point of $\mathscr{H}$ different from $F_1, \ldots, F_s$.

Assume the result in dimension $r - 1$. Choose $P \in \mathscr{H} \setminus \{F_1, \ldots, F_s\}$. By corollary 9.2.20, there is a hypernet $\mathscr{H}'$ of $\mathscr{H}$ not containing $P$. Put $F_i' = PF_i \cap \mathscr{H}'$. The line $PF_i$ is not contained in $\mathscr{H}'$ and therefore has a unique intersection with it.

By the induction hypothesis, $\mathscr{H}'$ has a hypernet $\mathscr{J}'$ avoiding all the $F_i'$, and $\dim\mathscr{J}' = r - 2$. Set $\mathscr{J} = P \vee \mathscr{J}'$. Since $P \notin \mathscr{J}'$, the net $\mathscr{J}$ is a hypernet of $\mathscr{H}$. By corollary 9.2.21, $\mathscr{J} \cap \mathscr{H}'$ has dimension $r - 2$. It contains the $(r-2)$-dimensional net $\mathscr{J}'$, so $\mathscr{J}' = \mathscr{J} \cap \mathscr{H}'$. If some $F_i$ belonged to $\mathscr{J}$, then $F_i' \in PF_i \subseteq \mathscr{J}$, whence $F_i' \in \mathscr{J} \cap \mathscr{H}' = \mathscr{J}'$, a contradiction. □

**Proposition 9.2.23.** *Let* $U$ *be a projectively independent set and* $U_1, U_2 \subseteq U$. *Then*

$$\operatorname{net}(U_1) \cap \operatorname{net}(U_2) = \operatorname{net}(U_1 \cap U_2), \quad \operatorname{net}(U_1) \vee \operatorname{net}(U_2) = \operatorname{net}(U_1 \cup U_2).$$

*More generally, for any finite family of subsets* $U_1, \ldots, U_r$ *of* $U$,

$$\bigcap_{i=1}^{r} \operatorname{net}(U_i) = \operatorname{net}\left(\bigcap_{i=1}^{r} U_i\right), \quad \bigvee_{i=1}^{r} \operatorname{net}(U_i) = \operatorname{net}\left(\bigcup_{i=1}^{r} U_i\right).$$

**Proof.** The join formula follows directly from the definition of span. For the intersection formula, $U_1 \cap U_2$ is contained in both $\operatorname{net}(U_1)$ and $\operatorname{net}(U_2)$, so $\operatorname{net}(U_1 \cap U_2) \subseteq \operatorname{net}(U_1) \cap \operatorname{net}(U_2)$. Every subset of the projectively independent set $U$ is independent, so the dimension formula gives

$$\dim[\operatorname{net}(U_1) \cap \operatorname{net}(U_2)] = (|U_1| - 1) + (|U_2| - 1) - (|U_1 \cup U_2| - 1) = |U_1 \cap U_2| - 1.$$

Here $|I|$ denotes the cardinality of $I$. This is the dimension of $\operatorname{net}(U_1 \cap U_2)$, so the inclusion gives $\operatorname{net}(U_1) \cap \operatorname{net}(U_2) = \operatorname{net}(U_1 \cap U_2)$. The formulas for more than two subsets follow by induction. □

**Example 9.2.24.** Let $U = \{U_1, \ldots, U_\mu\}$ be a set of $\mu$ projectively independent points, and put $\mathscr{K}_i = \operatorname{net}(U \setminus \{U_i\})$. Prove that:

(1) $\bigcap_{i=1}^{r} \mathscr{K}_i = \operatorname{net}(U_{r+1}, \ldots, U_\mu)$ and $\bigcap_{i=1}^{\mu} \mathscr{K}_i = \emptyset$;
(2) $\bigcap_{i \neq j} \mathscr{K}_i = \{U_j\}$;
(3) if $X \notin \operatorname{net}(U)$, then

$$\bigcap_{i=1}^{r} (\mathscr{K}_i \vee X) = X \vee \operatorname{net}(U_{r+1}, \ldots, U_\mu),$$

and $\bigcap_{i=1}^{\mu} (\mathscr{K}_i \vee X) = \{X\}$.

**Proof.** Both $U$ and $U \cup \{X\}$ are projectively independent. Apply proposition 9.2.23. □

**Example 9.2.25.** Let $\mathscr{H}$ be a $(\mu - 1)$-dimensional net, where $\mu \geq 2$, and let $\mathscr{K}_1, \ldots, \mathscr{K}_\mu$ be hypernets of $\mathscr{H}$. Prove the following.

(1) There is a projectively independent set $U = \{U_1, \ldots, U_\mu\}$ satisfying $\mathscr{K}_i = \operatorname{net}(U \setminus \{U_i\})$ for every $i$ if and only if $\bigcap_{i=1}^{\mu} \mathscr{K}_i = \emptyset$.
(2) Under the condition in (1), $U_i = \bigcap_{j \neq i} \mathscr{K}_j$. Thus $U$ is uniquely determined and $\mathscr{H} = \operatorname{net}(U)$. More generally, for every nonempty $I \subseteq \{1, \ldots, \mu\}$, $\dim \bigcap_{i \in I} \mathscr{K}_i = \mu - |I| - 1$.

**Proof.** Necessity follows at once from example 9.2.24. Conversely, suppose $\bigcap_{i=1}^{\mu} \mathscr{K}_i = \emptyset$. Starting with the $(\mu - 1)$-dimensional net $\mathscr{H}$ and intersecting successively with the $|I|$ hypernets indexed by a nonempty set $I \subseteq \{1, \ldots, \mu\}$, the dimension decreases by at most one at each step. Therefore $\dim \bigcap_{i \in I} \mathscr{K}_i \geq \mu - |I| - 1$. If $\dim \bigcap_{i \in I} \mathscr{K}_i \geq \mu - |I|$ for some $I$, intersecting with the remaining $\mu - |I|$ hypernets would leave an intersection of dimension at least zero, contrary to the hypothesis. Hence $\dim \bigcap_{i \in I} \mathscr{K}_i = \mu - |I| - 1$.

In particular, put $U_i = \bigcap_{j \neq i} \mathscr{K}_j$. Then $U_i$ is a unique projective point, and the empty total intersection implies $U_i \notin \mathscr{K}_i$. For $j \neq i$, we have $U_j \in \mathscr{K}_i$, so $\operatorname{net}(U \setminus \{U_i\}) \subseteq \mathscr{K}_i$. If $U_1, \ldots, U_\mu$ were projectively dependent, some $U_i$ would belong to $\operatorname{net}(U \setminus \{U_i\}) \subseteq \mathscr{K}_i$, a contradiction. Thus the points are projectively independent. Both $\operatorname{net}(U)$ and $\mathscr{H}$ have dimension $\mu - 1$, so $\operatorname{net}(U) = \mathscr{H}$. Moreover, both $\operatorname{net}(U \setminus \{U_i\})$ and $\mathscr{K}_i$ have dimension $\mu - 2$, so they are equal. □

For brevity, we shall henceforth juxtapose generating point sets when there is no ambiguity. Thus, for projective points $A, B, C$, write $AB = \operatorname{net}(A, B) = A \vee B$ and $ABC = \operatorname{net}(A, B, C) = A \vee B \vee C$. If $\mathscr{N}$ is a net, write $A\mathscr{N} = \operatorname{net}(A, \mathscr{N}) = A \vee \mathscr{N}$. This agrees with ordinary geometric notation, in which two juxtaposed points determine a line and three determine a plane.

We shall also identify a singleton with its element when no confusion can arise, especially when writing the intersection of two nets. If $\mathscr{N}_1$ and $\mathscr{N}_2$ have the unique common point $A$, for example, we write $A = \mathscr{N}_1 \cap \mathscr{N}_2$, although the right-hand side is formally the set $\{A\}$. This is analogous to using $l_1 \cap l_2$ for the point of intersection of two lines in synthetic geometry.

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## Exercises

**Exercise 9.12.** Let $\mathscr{K} \subseteq \mathscr{M} \subseteq \mathscr{N}$ be finite-dimensional nets, and let the net $\mathscr{H}$ satisfy $\mathscr{K} \dot{\vee} \mathscr{H} = \mathscr{M}$. Prove that there is a subnet $\mathscr{L} \subseteq \mathscr{N}$ such that $\mathscr{H} \subseteq \mathscr{L}$ and $\mathscr{K} \dot{\vee} \mathscr{L} = \mathscr{N}$.

**Exercise 9.13.** Let $\mathscr{M}, \mathscr{N}$ be nonempty, finitely generated nets such that $\mathscr{M} \cap \mathscr{N} = \emptyset$. Prove that for every $Z \in (\mathscr{M} \vee \mathscr{N}) \setminus (\mathscr{M} \cup \mathscr{N})$ there are unique points $X \in \mathscr{M}$ and $Y \in \mathscr{N}$ such that $Z \in XY$.

**Exercise 9.14.** Let $\mathscr{M}$ be a subnet of $\mathscr{N}$, and define

$$\mathscr{N} / \mathscr{M} := \{\operatorname{net}(X, \mathscr{M}) \mid X \in \mathscr{N} \setminus \mathscr{M}\}.$$

Regard each $\mathrm{net}(X,\mathscr{M})$ as an abstract projective point of $\mathscr{N}/\mathscr{M}$, and define the abstract projective line determined by $\mathrm{net}(X,\mathscr{M})$ and $\mathrm{net}(Y,\mathscr{M})$ to be $\mathrm{net}(X,Y,\mathscr{M})$. Let $\mathscr{A},\mathscr{B}$ also be subnets of $\mathscr{N}$. Prove that:

(1) $\mathscr{N}/\mathscr{M}$ is a net;
(2) $\dim(\mathscr{N}/\mathscr{M}) = \dim\mathscr{N} - \dim\mathscr{M} - 1$;
(3) $\mathrm{net}(\mathscr{A},\mathscr{M})/\mathscr{M}$ is isomorphic to $\mathscr{A}/(\mathscr{A}\cap\mathscr{M})$;
(4) if $\mathscr{M}\subseteq\mathscr{A}\subset\mathscr{N}$, then $(\mathscr{N}/\mathscr{M})/(\mathscr{A}/\mathscr{M})$ is isomorphic to $\mathscr{N}/\mathscr{A}$.

**Exercise 9.15.** Let $\mathscr{N}$ be a $(2r+1)$-dimensional net, where $r\geq 1$, and let $\mathscr{A},\mathscr{B},\mathscr{C}$ be three pairwise disjoint $r$-dimensional subnets.

(1) Prove that $\mathrm{net}(\mathscr{A},\mathscr{B}) = \mathscr{N}$.
(2) Define $\phi:\ \mathscr{A}\to\mathscr{B},\ X\mapsto\mathscr{B}\cap\mathrm{net}(\mathscr{C},X)$. Prove that $\phi$ is a net isomorphism.
(3) Define $\psi:\ \mathscr{A}\to\mathscr{C},\ X\mapsto\mathscr{C}\cap\mathrm{net}(X,\phi(X))$. Prove that $\psi$ is also a net isomorphism.

**Exercise 9.16.** Prove that any three finite-dimensional nets $\mathscr{A},\mathscr{B},\mathscr{C}$ in a projective point–line system satisfy

$$\begin{aligned}&\dim(\mathscr{A}\cap\mathscr{B})+\dim(\mathscr{B}\cap\mathscr{C})+\dim(\mathscr{C}\cap\mathscr{A})+\dim(\mathscr{A}\vee\mathscr{B}\vee\mathscr{C})\\ &\leq\dim(\mathscr{A})+\dim(\mathscr{B})+\dim(\mathscr{C})+\dim(\mathscr{A}\cap\mathscr{B}\cap\mathscr{C}).\end{aligned}$$

Determine the necessary and sufficient condition for equality.

**Exercise 9.17*.** Let an $n$-dimensional net $\mathscr{N}$ contain two complete flags

$$\mathscr{F}_1^i\subsetneq\mathscr{F}_2^i\subsetneq\mathscr{F}_3^i\subsetneq\cdots\subsetneq\mathscr{F}_{n+1}^i\quad(i=1,2).$$

Prove that there are a projective basis $\{P_1,\dots,P_{n+1}\}$ and a permutation $\sigma\in\mathrm{S}_{n+1}$ [5] such that

$$\mathscr{F}_j^1=\mathrm{net}(P_1,\dots,P_j),\quad \mathscr{F}_j^2=\mathrm{net}(P_{\sigma(1)},P_{\sigma(2)},\dots,P_{\sigma(j)}),$$

and $\dim(\mathscr{F}_j^1\cap\mathscr{F}_k^2)+1$ is the number of elements of the set $\{l\in\mathbb{N}_+\mid l\leq k,\ \sigma(l)\leq j\}$.

## 9.3 Pencils of Nets

The dimension theory developed above makes it straightforward to introduce pencils of nets.

**Definition 9.3.1.** Let $\mathscr{K}$ and $\mathscr{H}$ be subnets of a projective net $\mathscr{N}$, with $\mathscr{K}\subseteq\mathscr{H}$ and $\dim\mathscr{K}+2=\dim\mathscr{H}$. The set

$$\mathfrak{B}(\mathscr{K};\mathscr{H}) := \{\mathscr{A}\subseteq\mathscr{N}\mid\mathscr{A}\text{ is a net and }\mathscr{K}\subsetneq\mathscr{A}\subsetneq\mathscr{H}\}$$

is called a *pencil of nets*. The net $\mathscr{K}$ is its *center*, and $\mathscr{H}$ its *spanning net*.

By corollary 9.2.10, every member of $\mathfrak{B}(\mathscr{K};\mathscr{H})$ has dimension $r=(\dim\mathscr{K}+1)$; the pencil is then called an *$r$-dimensional pencil*. Thus the dimension of a pencil refers to the dimension of its members, not to the pencil itself as an $r$-dimensional space. Indeed, propositions 9.3.3 and 9.3.4 show that it has only one degree of freedom and may be put in bijection with the points of a projective line.

In particular, if the center is a zero-dimensional net, that is, a projective point, the pencil is also called a *pencil of lines*. A projective line, regarded as a set of points, may be viewed as a zero-dimensional pencil with center $\emptyset$, since the join of any net with the empty net is the net itself; in this role it is also called a *point range*. [6] If the spanning net is the whole net $\mathscr{N}$, every member is a hypernet of $\mathscr{N}$, and the pencil is called a *pencil of hypernets in $\mathscr{N}$*. This last notion is relative: if $\mathscr{H}$ itself is taken as the ambient net, then $\mathfrak{B}(\mathscr{K};\mathscr{H})$ is a pencil of hypernets in $\mathscr{H}$.

A pencil is said to lie in a net if and only if each member of the pencil is a subnet of that net.

[5] For permutations, see example A.1.13.

[6] This name emphasizes the line as a set of points, although the two terms are often used interchangeably.

**Remark.** A pencil is a set whose constituent nets are its *elements*, not its subsets; membership is therefore denoted by $\in$. A net, on the other hand, is a set of projective points, and each of its subnets is a *subset*, so inclusion is denoted by $\subseteq$. When a projective line is regarded as a zero-dimensional pencil, its member nets are themselves projective points, and the line is consequently the same set whether viewed as a net or as a pencil.

If the members of a pencil are $\mathscr{A}_1,\mathscr{A}_2,\mathscr{A}_3,\ldots$, the pencil may also be denoted by $(\mathscr{A}_1,\mathscr{A}_2,\mathscr{A}_3,\ldots)$. If its center is $\mathscr{K}$ and projective points $A_1,A_2,A_3,\ldots$ satisfy $\mathscr{A}_1=\mathscr{K}A_1$, $\mathscr{A}_2=\mathscr{K}A_2$, $\mathscr{A}_3=\mathscr{K}A_3$, ..., we abbreviate $(\mathscr{K}A_1,\mathscr{K}A_2,\mathscr{K}A_3,\ldots)=\mathscr{K}(A_1,A_2,A_3,\ldots)$. When the center is the empty net it is omitted, so a point range is written $(A_1,A_2,A_3,\ldots)$; in this case the points $A_i$ are collinear.

**Proposition 9.3.2.** *Let $\mathfrak{B}(\mathscr{K};\mathscr{M})$ be a pencil of nets, with center $\mathscr{K}$ and spanning net $\mathscr{M}$, so that $\mathscr{K}\subseteq\mathscr{M}$ and $\dim\mathscr{M}=\dim\mathscr{K}+2$. Any two or more distinct members of $\mathfrak{B}(\mathscr{K};\mathscr{M})$ span $\mathscr{M}$ and have intersection $\mathscr{K}$.*

*Moreover, the union of the projective points in all members of the pencil is exactly its spanning net. Consequently, if a pencil $\mathfrak{M}$ is known to have center $\mathscr{K}$ and the union of the points in all its members is $\mathscr{M}$, then $\dim\mathscr{M}=\dim\mathscr{K}+2$, and $\mathfrak{M}$ is uniquely determined: $\mathfrak{M}=\mathfrak{B}(\mathscr{K};\mathscr{M})$.*

**Proof.** For two distinct members $\mathscr{A}_1,\mathscr{A}_2$ of $\mathfrak{B}(\mathscr{K};\mathscr{M})$, we have $\mathscr{M}\supseteq\mathscr{A}_1\vee\mathscr{A}_2\supsetneq\mathscr{A}_2$. Since $\dim\mathscr{A}_2=\dim\mathscr{M}-1$, corollary 9.2.10 gives $\mathscr{A}_1\vee\mathscr{A}_2=\mathscr{M}$. Adding further distinct members does not change the span, since they are all subnets of $\mathscr{M}$. Also, corollary 9.2.21 gives $\dim(\mathscr{A}_1\cap\mathscr{A}_2)=\dim\mathscr{M}-2=\dim\mathscr{K}$. Because $\mathscr{K}\subseteq\mathscr{A}_1\cap\mathscr{A}_2$, $\mathscr{A}_1\cap\mathscr{A}_2=\mathscr{K}$; intersecting with further members leaves it unchanged.

Now let $P\in\mathscr{M}\setminus\mathscr{K}$. By corollary 9.2.19, $\dim(\mathscr{K}P)=\dim\mathscr{K}+1$, and $\mathscr{K}\subseteq\mathscr{K}P\subseteq\mathscr{M}$; hence $\mathscr{K}P$ is a member of the pencil. Points of $\mathscr{K}$ belong to every member. Conversely, every member is a subnet of $\mathscr{M}$. Thus the union of all points in all members is precisely $\mathscr{M}$. $\square$

**Proposition 9.3.3.** *Let $\mathscr{K}$ be an $(r-1)$-dimensional subnet of a projective net $\mathscr{N}$, and let $l\subseteq\mathscr{N}\setminus\mathscr{K}$ be a projective line. Then the pencil $\mathcal{B}=\mathfrak{B}(\mathscr{K};\mathscr{K}\vee l)$ exists, and:*

*(1) $\mathcal{B}$ is an r-dimensional pencil, while its spanning net $\mathscr{K}\vee l$ has dimension $r+1$;*
*(2) if $\mathcal{B}'=\{\mathscr{K}\vee A|A\in l\}$, then $\mathcal{B}=\mathcal{B}'$;*
*(3) the map $f:l\to\mathcal{B}$, $A\mapsto A\mathscr{K}$, is a bijection.*

**Proof.** By corollary 9.2.19, $\dim(\mathscr{K}l)=(r-1)+1+1=r+1=\dim\mathscr{K}+2$, which proves (1).

For every $A\in l$, corollary 9.2.19 gives $\dim(\mathscr{K}A)=\dim\mathscr{K}+1$; moreover, $\mathscr{K}\subseteq\mathscr{K}A\subseteq\mathscr{K}l$. Thus $\mathscr{K}A$ is a uniquely determined member of $\mathcal{B}$. Conversely, if $\mathscr{H}\in\mathcal{B}$, the inclusion $\mathscr{H}\supseteq\mathscr{K}$ gives $\mathscr{H}l\supseteq\mathscr{K}l$, while the definition of the pencil also gives $\mathscr{H}l\subseteq\mathscr{K}l$. Hence $\mathscr{H}l=\mathscr{K}l$ and $\dim(\mathscr{H}l)=r+1$. The dimension formula yields $\dim(\mathscr{H}\cap l)=r+1-(r+1)=0$, so $\mathscr{H}\cap l$ is a uniquely determined point of $l$. This proves (2) and (3). $\square$

**Remark.** The pencil $\mathfrak{B}(\mathscr{K};\mathscr{K}\vee l)$ in this proposition is abbreviated here to $\mathscr{K}(l)$. Indeed, viewing $l$ as the point range $(A_1,A_2,\ldots)$ gives the earlier notation $\mathscr{K}(A_1,A_2,\ldots)$.

Conversely, a projective line can be used as a transversal of a pencil.

**Proposition 9.3.4.** *Let $\mathfrak{B}(\mathscr{O};\mathscr{M})$ be an r-dimensional pencil, and choose any projective line $l\subseteq\mathscr{M}\setminus\mathscr{O}$; such a line always exists. Then:*

*(1) $\mathscr{M}=\mathrm{net}(\mathscr{O},l)$;*
*(2) the map $f:\mathfrak{B}(\mathscr{O};\mathscr{M})\to l$, $\mathscr{A}\mapsto\mathscr{A}\cap l$, is a bijection.*

**Proof.** Existence of $l$ follows from corollary 9.2.20. Since $\mathscr{O}\cap l=\emptyset$, corollary 9.2.19 gives $\dim(\mathscr{O}l)=\dim\mathscr{O}+2=\dim\mathscr{M}$. Plainly $\mathscr{O}l\subseteq\mathscr{M}$, so $\mathscr{M}=\mathrm{net}(\mathscr{O},l)$, proving (1). Thus $\mathscr{M}=\mathscr{O}l$, and (2) follows from proposition 9.3.3. $\square$

**Corollary 9.3.5.** *Let $\mathscr{N}$ be an n-dimensional net, let $\mathcal{L}$ be a pencil of lines in $\mathscr{N}$ with center $U$, and let $\mathscr{H}$ be a hypernet of $\mathscr{N}$ not containing $U$. Then $l=\{m\cap\mathscr{H}\mid m\in\mathcal{L}\}$ is a projective line in $\mathscr{H}$, and the map $f:\mathcal{L}\to l$, $m\mapsto m\cap\mathscr{H}$, is a bijection.*

**Proof.** The spanning net of $\mathcal{L}$ is a two-dimensional net $\mathscr{A}$. Since $\mathscr{H}$ is a hypernet and $U\notin\mathscr{H}$, corollary 9.2.21 gives $\dim(\mathscr{A}\cap\mathscr{H})=1$, so $\mathscr{A}\cap\mathscr{H}$ is a fixed line. Because the members of $\mathcal{L}$ span $\mathscr{A}$, this line is exactly $l$. By proposition 9.3.4, each $m\in\mathcal{L}$ has a unique intersection point $m\cap\mathscr{H}$. Distinct members cannot have the same point on $l$, since two points determine a line. Hence the map is bijective. □

This completes the basic theory of general projective nets; nets equipped with a cross-ratio structure will be considered next. Readers familiar with linear spaces will recognize an entirely analogous theory, except that the dimension of a net is one less than the number of points in a basis, whereas the dimension of a vector space is the number of vectors in a basis. The words "pencil" and "net" in the preceding sections come from classical projective geometry; terms such as "modular law" and "Steinitz exchange theorem" have been borrowed directly from linear algebra because the corresponding theories are parallel.

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## Exercises

**Exercise 9.18.** Let $\mathscr{K}$ be a subnet of a net $\mathscr{M}$, let $\mathcal{B}=\mathfrak{B}(\mathscr{K};\mathscr{M})$ be a pencil of nets, and let $\mathscr{J}\subseteq\mathscr{M}$ be a subnet. Prove that:

(1) if $\mathscr{J}\subseteq\mathscr{K}$, then every member of $\mathcal{B}$ contains $\mathscr{J}$;
(2) if $\mathscr{K}\subsetneq\mathscr{K}\vee\mathscr{J}\subsetneq\mathscr{M}$, then exactly one member of $\mathcal{B}$ contains $\mathscr{J}$, namely $\mathscr{K}\vee\mathscr{J}$;
(3) if $\mathscr{K}\vee\mathscr{J}=\mathscr{M}$, then no member of $\mathcal{B}$ contains $\mathscr{J}$.

**Exercise 9.19.** Let $r\geq 0$, let $\mathscr{M}$ be an $(r+1)$-dimensional net, and let $\mathscr{K},\mathscr{L}$ be distinct $(r-1)$-dimensional subnets of $\mathscr{M}$. Prove that the pencils $\mathfrak{B}(\mathscr{K};\mathscr{M})$ and $\mathfrak{B}(\mathscr{L};\mathscr{M})$ have a common member if and only if $\dim(\mathscr{K}\vee\mathscr{L})=r$.

**Exercise 9.20.** Let $\mathscr{H}_1,\mathscr{H}_2$ be distinct $r$-dimensional nets. Prove that they belong to a common $r$-dimensional pencil if and only if $\dim(\mathscr{H}_1\cap\mathscr{H}_2)=r-1$, and prove that this pencil is unique.

**Exercise 9.21.** Let $\mathscr{K}$ be a subnet of a net $\mathscr{M}$, with $\dim\mathscr{M}=\dim\mathscr{K}+2$. Define a relation $\sim$ on $\mathscr{M}\setminus\mathscr{K}$ by $P\sim Q \iff \mathscr{K}\vee P=\mathscr{K}\vee Q$. Prove that $\sim$ is an equivalence relation and that its equivalence classes are precisely $\{\mathscr{H}\setminus\mathscr{K}\mid\mathscr{H}\in\mathfrak{B}(\mathscr{K};\mathscr{M})\}$.

**Exercise 9.22*.** Let $\mathscr{N}$ be a $\mu$-dimensional net, let $\mathscr{K}\subseteq\mathscr{N}$ be a $(\mu-2)$-dimensional subnet, and let $\mathscr{I}\subseteq\mathscr{N}$ be a $\nu$-dimensional subnet such that $\mathscr{K}\vee\mathscr{I}=\mathscr{N}$. Put $\mathscr{A}=\mathscr{K}\cap\mathscr{I}$. Prove that $\dim\mathscr{A}=\nu-2$ and that the following maps are mutually inverse bijections:

$$f:\mathfrak{B}(\mathscr{K};\mathscr{N})\to\mathfrak{B}(\mathscr{A};\mathscr{I}),\quad \mathscr{H}\mapsto\mathscr{H}\cap\mathscr{I},$$
$$g:\mathfrak{B}(\mathscr{A};\mathscr{I})\to\mathfrak{B}(\mathscr{K};\mathscr{N}),\quad \mathscr{J}\mapsto\mathscr{K}\vee\mathscr{J}.$$

## 9.4 Nets with a Cross-Ratio Structure

### 9.4.1 Basic Concepts

**Definition**

The projective planes in the preceding examples admit a cross ratio on each of their lines. We now isolate the minimum structure needed to treat cross ratios, perspectivities, and projectivities in a projective net. The cross ratio was defined in another way in section 6.6.2, but that definition applies only to a one-dimensional structure and immediately identifies its values with a number field. We do not yet require so much structure.

**Definition 9.4.1.** Let $\mathscr{N}$ be a net of dimension at least two. A *cross-ratio function $f$* has as its arguments ordered quadruples of projective points on an arbitrary projective line, with the first three points pairwise distinct, and has an abstract set $\Lambda$ as its codomain. The cross ratio of $A,B,C,D$ is denoted by $(A,B;C,D)$, or briefly by $(AB,CD)$. The function is required to satisfy the following conditions.

(1) For every fixed projective line $l$ and every three fixed distinct points $A,B,C \in l$, the map $g\colon l \longrightarrow \Lambda,\quad D \longmapsto (AB,CD)$, is injective.

(2) Let $l,l'$ be any two lines in the same two-dimensional subnet $\mathscr{E} \subseteq \mathscr{N}$, and let $O \in \mathscr{E} \setminus (l \cup l')$. The map $\pi\colon l \longrightarrow l',\quad X \longmapsto OX \cap l'$, preserves the cross ratio of every four points. Thus, for $A,B,C,D \in l$, with $A,B,C$ distinct, $(AB,CD) = (\pi(A)\pi(B),\pi(C)\pi(D))$.

A net equipped with such a function is called a *net with a cross-ratio structure*.

In condition (2), $O \notin l \cup l'$ and corollary 9.2.21 ensure that $\pi$ is well defined. They also give $[(OX \cap l') \vee O] \cap l = X$, so $\pi$ is a bijection.

The map $\pi$ in condition (2) is called a *perspectivity* from $l$ to $l'$, and $O$ is its *center of perspectivity*. We write $l \overset{(O)}{\barwedge} l'$, or, for corresponding points, $(A,B,C,\ldots) \overset{(O)}{\barwedge} (\pi(A),\pi(B),\pi(C),\ldots)$.

A bijection between two lines that preserves the cross ratio of corresponding points is called a *projectivity*. Thus, if $\varphi\colon l \to l'$ is a projectivity, then for $A,B,C,D \in l$, with $A,B,C$ distinct, $(AB,CD) = (\varphi(A),\varphi(B);\varphi(C),\varphi(D))$. We write $l \barwedge l'$, or $(A,B,C,\ldots) \barwedge (\varphi(A),\varphi(B),\varphi(C),\ldots)$. A projectivity from a line to itself is called a *projective transformation*.

A net isomorphism between two nets that preserves cross-ratios on corresponding lines is called a *projective net isomorphism*. Equivalently, a net isomorphism is projective precisely when its restriction to every line is a projectivity.

**Example 9.4.2.** The real projective plane is a net with a cross-ratio structure.

**Theorem 9.4.3.** *A projectivity between two lines is uniquely determined by three distinct pairs of corresponding points. More precisely, let $l,l'$ be two lines. If $A,B,C$ are distinct points of $l$ and $A',B',C'$ are distinct points of $l'$, there is a unique projectivity $l \to l'$ mapping $A,B,C$ to $A',B',C'$, respectively.*

*Every finite composition of perspectivities between lines is a projectivity, and every projectivity between lines is a finite composition of perspectivities.*

*The composition of projectivities is a projectivity, and the inverse of a projectivity is a projectivity.*

**Proof.** The injectivity in condition (1) of definition 9.4.1, together with the requirement that a projectivity preserve cross-ratios, immediately proves uniqueness from three pairs of corresponding points. A composition of perspectivities plainly preserves cross-ratios and is therefore a projectivity. The definition, injectivity of the cross-ratio function, and bijectivity of projectivities likewise show that their compositions and inverses are projectivities. It remains to construct a projectivity as a composition of perspectivities.

Let $l,l'$ be the two lines. Since $\dim(l\vee l')\le\dim l+\dim l'-\dim\emptyset=3$, first suppose $\dim(l\vee l')=3$, so $l\cap l'=\emptyset$. Choose $X\in l$ and $Y\in l'$, and choose three points $A'',B'',C''$ on $XY$. It then suffices to handle the case of intersecting lines, because the desired map is obtained through $(A,B,C)\barwedge(A'',B'',C'')\barwedge(A',B',C')$.

We may therefore assume that $l$ and $l'$ meet and hence lie in a common two-dimensional net $\mathscr{E}$. If $l=l'$, choose a different line $m\subseteq\mathscr{E}$ and a point $O$ lying on neither $l$ nor $m$. Apply the perspectivity $l\overset{(O)}{\barwedge}m$, and then use the construction below to map the three resulting points $A'',B'',C''$ on $m$ to $A',B',C'$ on $l'$ by a composition of perspectivities. Thus it remains to consider $l\ne l'$, in which case the lines have a unique common point.

If one pair of corresponding points coincides, suppose without loss of generality that $A=A'$. This point is the intersection of $l$ and $l'$. Put $O=BB'\cap CC'$. The point $O$ lies on neither $l$ nor $l'$. For example, if $O\in l'$, then $B',O,C'$ are collinear. If $O=B'$, then $C$ would lie on $B'C'=l'$. But $C\in l$ as well, forcing $C=A$, a contradiction; similarly $O\ne C'$. If $O$ is distinct from $B'$ and $C'$, then $B\in OB'\subseteq l'$ and likewise $C\in l'$, forcing $l=BC=l'$, again a contradiction. Hence projection from $O$ gives the required perspectivity.

Finally suppose that none of the three corresponding pairs coincides. Set $m=AC'$. If $A=C'$, then necessarily $A\ne B'$, and one may instead take $m=AB'$; the argument is the same. Choose a point $P\in CC'$ lying on neither $m$ nor $l$ as a center of perspectivity. Then $PA\cap m=A$; put $B''=PB\cap m$. Next take $Q=B''B'\cap AA'$ as a center of perspectivity. As in the preceding case, $Q$ lies on neither $m$ nor $l'$. Since $C'=m\cap l'$, we obtain $(A,B,C)\overset{(P)}{\barwedge}(A,B'',C')\overset{(Q)}{\barwedge}(A',B',C')$, which completes the construction. ☐

**Fundamental Perspectivities**

**Lemma 9.4.4.** *Let $l,l'$ be distinct lines, let a net $\mathscr{N}$ be disjoint from both, and suppose a bijection $f:\ l\to l'$, $A\mapsto A'$, satisfies $A'\in\mathscr{N}\vee A$. Put $\mathscr{P}=l\vee l'$ and $\mathscr{N}_0=\mathscr{N}\cap\mathscr{P}$. Then $\mathscr{N}_0\dot\vee l=\mathscr{P}=\mathscr{N}_0\dot\vee l'$, and $A'\in\mathscr{N}_0\vee A$ for every $A\in l$. Moreover, $0\le\dim\mathscr{N}_0\le 1$.*

**Proof.** Surjectivity of $f$ gives $l'\subseteq\mathscr{N}\vee l$, and hence $\mathscr{P}\subseteq\mathscr{N}\vee l$. By the modular law,

$$l\vee\mathscr{N}_0=l\vee(\mathscr{N}\cap\mathscr{P})=(l\vee\mathscr{N})\cap\mathscr{P}=\mathscr{P}.$$

Since $\mathscr{N}\cap l=\emptyset$ and $\mathscr{N}_0\subseteq\mathscr{N}$, we have $\mathscr{N}_0\cap l=\emptyset$; corollary 9.2.19 therefore gives $\dim\mathscr{N}_0=\dim\mathscr{P}-2$. Similarly, $\mathscr{N}_0\cap l'=\emptyset$, so corollary 9.2.19 gives $\dim(\mathscr{N}_0\vee l')=\dim\mathscr{P}$. Because $\mathscr{N}_0\vee l'\subseteq\mathscr{P}$, we have $\mathscr{N}_0\vee l'=\mathscr{P}$.

For $A\in l$ and $A'=f(A)$, the modular law and $A\in\mathscr{P}$ give

$$(A\vee\mathscr{N})\cap\mathscr{P}=A\vee(\mathscr{N}\cap\mathscr{P})=A\vee\mathscr{N}_0.$$

Since $A'\in A\vee\mathscr{N}$ and $A'\in\mathscr{P}$, it follows that $A'\in\mathscr{N}_0\vee A$. Finally, the dimension formula shows that the join $\mathscr{P}$ of two distinct lines has dimension 2 when they meet and 3 when they are disjoint. Hence $0\le\dim\mathscr{N}_0\le 1$. ☐

**Theorem 9.4.5** (Perspectivities of projective lines)**.** *Let $l,l'$ be distinct lines in an n-dimensional net $\mathscr{E}$ with a cross-ratio structure. Suppose a bijection $f:\ l\to l'$, $A\mapsto A'$, has the following property: there is a nonempty net $\mathscr{N}\subseteq\mathscr{E}\setminus(l\cup l')$ such that $f(A)\in\mathrm{net}(\mathscr{N},A)$ for every $A\in l$. Then $0\le\dim\mathscr{N}\le n-2$, and $f$ is a projectivity.*

**Proof.** The dimension formula gives

$$\dim\mathscr{N}=\dim(\mathscr{N}\vee l)-\dim l+\dim(\mathscr{N}\cap l)\le\dim\mathscr{E}-1+(-1)=n-2.$$

By lemma 9.4.4, replacing $\mathscr{N}$ by $\mathscr{N}\cap(l\vee l')$ does not change the hypothesis. It therefore suffices to consider $\dim\mathscr{N}=0$ and $\dim\mathscr{N}=1$, with $\mathscr{N}\dot\vee l=l\vee l'=\mathscr{N}\dot\vee l'$.

(1) If $\dim \mathscr{N}=0$, write $\mathscr{N}=O$, where $O$ is a projective point. The lines $l,l'$ lie in the two-dimensional net $l\vee l'$, and $A'\in OA$ for every $A\in l$. Consequently $f$ is exactly the perspectivity with center $O$ appearing in definition 9.4.1, and is a projectivity.

(2) If $\dim \mathscr{N}=1$, write $\mathscr{N}=d$, where $d$ is a projective line. Then $\mathscr{P}:=l\vee l'$ is a three-dimensional net and $d\dot{\vee} l=d\dot{\vee} l'=\mathscr{P}$. Choose distinct points $E,E'\in d$, and put $\mathscr{A}=E\vee l$ and $\mathscr{B}=E'\vee l'$. Then $d\vee\mathscr{A}=d\vee\mathscr{B}=\mathscr{A}\vee\mathscr{B}=\mathscr{P}$. Since $E,E'\notin l\cup l'$, proposition 9.2.9 shows that $\mathscr{A}$ and $\mathscr{B}$ are both two-dimensional.

The dimension formula now gives $\dim(d\cap\mathscr{A})=1+2-3=0$. Since $E\in d\cap\mathscr{A}$, this intersection satisfies $d\cap\mathscr{A}=E$; similarly, $d\cap\mathscr{B}=E'$. Also, $\dim(\mathscr{A}\cap\mathscr{B})=2+2-3=1$, so $m=\mathscr{A}\cap\mathscr{B}$ is a line, with $E\notin m$ and $E'\notin m$. There are therefore perspectivities $g\colon\ l\to m$ and $h\colon\ m\to l'$ with centers $E$ and $E'$, respectively.

Take $A\in l$, and write $A'=f(A)$ and $B=g(A)$. Since $A'\in d\vee A$, the two-dimensional nets $d\vee A$ and $\mathscr{B}=E'\vee l'$ contain both $E'$ and $A'$. They are distinct, and the intersection of distinct two-dimensional nets has dimension at most one; their common part is therefore the line $E'A'$. Moreover, $B\in EA\subseteq d\vee A$ and $B\in m\subseteq\mathscr{B}$, so $B\in E'A'$, and hence $h(B)=A'$. Thus $f=h\circ g$, proving that $f$ is a projectivity. $\square$

We may now make the following definitions.

**Definition 9.4.6** (Perspectivities of lines and cross-ratios of pencils)**.** In the notation of theorem 9.4.5, the map $f$ is called a *perspectivity* from $l$ to $l'$, and the lines $l,l'$ are said to be in perspective. The net $\mathscr{N}$ is the *center* of this perspectivity. In the notation already introduced, write $l\overset{(\mathscr{N})}{\barwedge} l'$.

Let $l=(A,B,C,\ldots)$ be a point range, or line, where $A,B,C,\ldots$ are projective points, and suppose $\mathscr{N}\cap l=\emptyset$. The pencil $\mathscr{N}(A,B,C,\ldots)$ is said to be in perspective with the point range $l$, denoted by $\mathscr{N}(A,B,C,\ldots)\,\overline{\barwedge}\,(A,B,C,\ldots)$. On the left, $A,B,C,\ldots$ are only representative points for the corresponding members of the pencil. Thus, if $\mathscr{N}A=\mathscr{N}A'$, $\mathscr{N}B=\mathscr{N}B'$, $\mathscr{N}C=\mathscr{N}C'$, ..., we may also write $\mathscr{N}(A',B',C',\ldots)\,\overline{\barwedge}\,(A,B,C,\ldots)$, even when $A',B',C'$ are not collinear.

Let $\mathcal{M}$ be a pencil with center $\mathscr{O}$, and let $\mathscr{N}_1,\mathscr{N}_2,\mathscr{N}_3,\mathscr{N}_4$ be four of its members. Let $\mathscr{M}$ be the union of the projective points in all members of $\mathcal{M}$, and choose any line $l\subseteq\mathscr{M}\setminus\mathscr{O}$. By proposition 9.3.4, for each $\mathscr{N}_i$ there is a unique point $A_i=l\cap\mathscr{N}_i$. Define the *cross ratio* by $(\mathscr{N}_1\mathscr{N}_2,\mathscr{N}_3\mathscr{N}_4):=(A_1A_2,A_3A_4)$. By theorem 9.4.5, this definition is independent of the choice of $l$.

In what follows, a subscript will, when useful, indicate the point range or the center of the pencil to which the elements belong. Thus a point range may be written $(A,B,C,\ldots)_l$, and the pencil $\mathscr{N}(A,B,C,\ldots)$ may also be written $(\mathscr{N}A,\mathscr{N}B,\mathscr{N}C,\ldots)_{\mathscr{N}}$. The subscript is omitted when there is no ambiguity.

A bijection between two pencils that preserves the cross-ratios of corresponding members is also called a *projectivity*, and is again denoted by $\barwedge$. A projectivity from a pencil to itself is a *projective transformation*.

The cross-ratio function in the initial definition was required only for four points on one line. It has now extended naturally to every pencil, as have the notions of perspectivity and projectivity.

Because a pencil may be put in perspective with a transversal projective line, we immediately obtain the following result.

**Theorem 9.4.7.** *A projectivity between two pencils in a net with a cross-ratio structure is uniquely determined by three pairs of corresponding members. The composition and the inverse of projectivities between pencils are again projectivities.*

As with projective transformations of the projective plane, several basic consequences follow at once.

**Theorem 9.4.8.** *Let $\mathscr{N}$ be a $\mu$-dimensional net, let $\mathscr{K}$ be a subnet of $\mathscr{N}$ of dimension $\mu-2$, and let $\mathscr{I}$ be a subnet of $\mathscr{N}$ of dimension $\nu$, where $1\le\nu\le\mu$ and $\mathscr{K}\vee\mathscr{I}=\mathscr{N}$. Put $\mathscr{A}=\mathscr{K}\cap\mathscr{I}$. Then*

$\dim \mathscr{A} = \nu - 2$, *and the maps*

$$f: \mathfrak{B}(\mathscr{K};\mathscr{N}) \to \mathfrak{B}(\mathscr{A};\mathscr{I}), \quad \mathscr{H} \mapsto \mathscr{H} \cap \mathscr{I},$$
$$g: \mathfrak{B}(\mathscr{A};\mathscr{I}) \to \mathfrak{B}(\mathscr{K};\mathscr{N}), \quad \mathscr{J} \mapsto \mathscr{K} \vee \mathscr{J}$$

*are mutually inverse projectivities.*

In this situation the pencils $\mathfrak{B}(\mathscr{K};\mathscr{N})$ and $\mathfrak{B}(\mathscr{A};\mathscr{I})$ are also said to be in perspective.

**Proof.** The dimension formula gives $\dim \mathscr{A} = (\mu-2) + \nu - \mu = \nu - 2$. If $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$ and $\mathscr{I} \subseteq \mathscr{H}$, then $\mathscr{N} = \mathscr{K} \vee \mathscr{I} \subseteq \mathscr{H} \subsetneq \mathscr{N}$, a contradiction. Thus $\mathscr{I} \nsubseteq \mathscr{H}$. Since there is no subnet strictly between a hypernet and $\mathscr{N}$, we have $\mathscr{H} \vee \mathscr{I} = \mathscr{N}$. The dimension formula therefore gives $\dim(\mathscr{H} \cap \mathscr{I}) = (\mu-1) + \nu - \mu = \nu - 1$. Because $\mathscr{A} \subseteq \mathscr{H} \cap \mathscr{I}$, it follows that $\mathscr{H} \cap \mathscr{I} \in \mathfrak{B}(\mathscr{A};\mathscr{I})$; hence $f$ is well defined.

Conversely, if $\mathscr{J} \in \mathfrak{B}(\mathscr{A};\mathscr{I})$, then $\mathscr{A} = \mathscr{K} \cap \mathscr{I}$ and $\mathscr{A} \subseteq \mathscr{J} \subseteq \mathscr{I}$ imply $\mathscr{K} \cap \mathscr{J} = \mathscr{A}$. Hence $\dim(\mathscr{K} \vee \mathscr{J}) = (\mu-2) + (\nu-1) - (\nu-2) = \mu - 1$, so $\mathscr{K} \vee \mathscr{J} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$ and $g$ is well defined.

Since $\mathscr{J} \subseteq \mathscr{I}$ and $\mathscr{K} \subseteq \mathscr{H}$, the modular law gives

$$(\mathscr{J} \vee \mathscr{K}) \cap \mathscr{I} = \mathscr{J} \vee (\mathscr{K} \cap \mathscr{I}) = \mathscr{J} \vee \mathscr{A} = \mathscr{J},$$
$$\mathscr{K} \vee (\mathscr{I} \cap \mathscr{H}) = (\mathscr{K} \vee \mathscr{I}) \cap \mathscr{H} = \mathscr{N} \cap \mathscr{H} = \mathscr{H}.$$

Thus $f$ and $g$ are mutually inverse bijections.

By corollary 9.2.20, there is a line $d \subseteq \mathscr{I}$ such that $d \dot\vee \mathscr{A} = \mathscr{I}$. Since $d \subseteq \mathscr{I}$, we have

$$d \cap \mathscr{K} = d \cap \mathscr{A} = \emptyset, \quad d \vee \mathscr{K} = (d \vee \mathscr{A}) \vee \mathscr{K} = \mathscr{I} \vee \mathscr{K} = \mathscr{N}.$$

Thus the same line $d$ is a transversal for both pencils in proposition 9.3.4. Moreover, since $d \subseteq \mathscr{I}$, $\mathscr{H} \cap d = (\mathscr{H} \cap \mathscr{I}) \cap d$. By the definition of the cross ratio of a pencil, $f$ preserves cross-ratios; so does its inverse. □

**Corollary 9.4.9.** *Let $\mathscr{N}$ be a $\mu$-dimensional net, let $\mathscr{K}$ be a subnet of $\mathscr{N}$ of dimension $\mu-2$, and let $\mathscr{H}$ be a $(\mu-1)$-dimensional hypernet of $\mathscr{N}$ with $\mathscr{K} \nsubseteq \mathscr{H}$. Put $\mathscr{K}' = \mathscr{K} \cap \mathscr{H}$. Then $f: \mathfrak{B}(\mathscr{K};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{K}';\mathscr{H})$, $\mathscr{M} \longmapsto \mathscr{M} \cap \mathscr{H}$, is a projectivity.*

**Proof.** Since $\mathscr{H}$ is a hypernet and $\mathscr{K} \nsubseteq \mathscr{H}$, the hypotheses give $\mathscr{K} \vee \mathscr{H} = \mathscr{N}$. Apply theorem 9.4.8 with $\nu = \mu - 1$. □

**Theorem 9.4.10** (Perspectivities of equidimensional pencils)**.** *All the objects below are taken in one n-dimensional net with a cross-ratio structure. Let $\mu$ be a positive integer, let $\mathscr{N}, \mathscr{N}'$ be $\mu$-dimensional nets, and let $\mathscr{K} \subseteq \mathscr{N}$ and $\mathscr{K}' \subseteq \mathscr{N}'$ be $(\mu-2)$-dimensional subnets. Each of the following constructions defines a projectivity $f: \mathfrak{B}(\mathscr{K};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{K}';\mathscr{N}')$.*

*(1) Suppose the centers coincide, $\mathscr{K} = \mathscr{K}'$, and there is a fixed net $\mathscr{A}$ such that $\mathscr{A} \nsubseteq \mathscr{K}$, $\mathscr{A} \cap \mathscr{N} = \mathscr{A} \cap \mathscr{N}' = \mathscr{A} \cap \mathscr{K}$, and $\mathscr{A} \vee \mathscr{N} = \mathscr{A} \vee \mathscr{N}'$. For $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$, define $f(\mathscr{H}) = (\mathscr{A} \vee \mathscr{H}) \cap \mathscr{N}'$. Equivalently, $f(\mathscr{H})$ is the unique member of $\mathfrak{B}(\mathscr{K}';\mathscr{N}')$ satisfying $f(\mathscr{H}) \subseteq \mathscr{A} \vee \mathscr{H}$.*

*(2) Suppose the spanning nets coincide, $\mathscr{N} = \mathscr{N}'$, and there is a fixed net $\mathscr{A}$ such that $\mathscr{N} \nsubseteq \mathscr{A}$, $\mathscr{A} \vee \mathscr{K} = \mathscr{A} \vee \mathscr{K}' = \mathscr{A} \vee \mathscr{N}$, and $\mathscr{A} \cap \mathscr{K} = \mathscr{A} \cap \mathscr{K}'$. For $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$, define $f(\mathscr{H}) = \mathscr{K}' \vee (\mathscr{H} \cap \mathscr{A})$. Equivalently, $f(\mathscr{H})$ is the unique member of $\mathfrak{B}(\mathscr{K}';\mathscr{N}')$ satisfying $\mathscr{H} \cap \mathscr{A} = f(\mathscr{H}) \cap \mathscr{A}$.*

In either case $f$ is called a *perspectivity* between the two pencils, and we write $\mathfrak{B}(\mathscr{K};\mathscr{N}) \overset{(\mathscr{A})}{\barwedge} \mathfrak{B}(\mathscr{K}';\mathscr{N}')$. In case (1), $\mathscr{A}$ is called the *center of perspectivity*; in case (2), it is called the *perspective axis*.

**Proof.** For (1), put $\mathscr{B} = \mathscr{A} \vee \mathscr{K}$ and $\mathscr{T} = \mathscr{A} \vee \mathscr{N} = \mathscr{A} \vee \mathscr{N}'$. Since $\mathscr{K} \subseteq \mathscr{N}$, the modular law gives

$$\mathscr{B} \cap \mathscr{N} = (\mathscr{K} \vee \mathscr{A}) \cap \mathscr{N} = \mathscr{K} \vee (\mathscr{A} \cap \mathscr{N}) = \mathscr{K},$$

and similarly $\mathscr{B} \cap \mathscr{N}' = \mathscr{K}$. Also, $\mathscr{B} \vee \mathscr{N} = \mathscr{B} \vee \mathscr{N}' = \mathscr{T}$. The dimension formula yields $\dim \mathscr{T} = \dim \mathscr{B} + \mu - (\mu - 2) = \dim \mathscr{B} + 2$. By theorem 9.4.8, both maps

$$f_1 : \mathfrak{B}(\mathscr{B};\mathscr{T}) \longrightarrow \mathfrak{B}(\mathscr{K};\mathscr{N}), \quad \mathscr{M} \mapsto \mathscr{M} \cap \mathscr{N},$$
$$f_2 : \mathfrak{B}(\mathscr{B};\mathscr{T}) \longrightarrow \mathfrak{B}(\mathscr{K}';\mathscr{N}'), \quad \mathscr{M} \mapsto \mathscr{M} \cap \mathscr{N}'$$

are projectivities. Hence $f = f_2 \circ f_1^{-1}$ is a projectivity. For $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$, the inclusion $\mathscr{K} \subseteq \mathscr{H}$ gives $\mathscr{B} \vee \mathscr{H} = \mathscr{A} \vee \mathscr{H}$, so theorem 9.4.8 gives $f(\mathscr{H}) = (\mathscr{A} \vee \mathscr{H}) \cap \mathscr{N}'$. We have $f(\mathscr{H}) \subseteq \mathscr{A} \vee \mathscr{H}$. If another member $\mathscr{M} \in \mathfrak{B}(\mathscr{K}';\mathscr{N}')$ also satisfies $\mathscr{M} \subseteq \mathscr{A} \vee \mathscr{H}$, then $\mathscr{M} \subseteq (\mathscr{A} \vee \mathscr{H}) \cap \mathscr{N}' = f(\mathscr{H})$. Both sides have dimension $\mu - 1$, so $\mathscr{M} = f(\mathscr{H})$. This proves the equivalent description in (1).

For (2), put $\mathscr{S} = \mathscr{A} \cap \mathscr{N}$ and $\mathscr{R} = \mathscr{A} \cap \mathscr{K} = \mathscr{A} \cap \mathscr{K}'$. Since $\mathscr{K} \subseteq \mathscr{N}$, the modular law gives

$$\mathscr{K} \vee \mathscr{S} = \mathscr{K} \vee (\mathscr{A} \cap \mathscr{N}) = (\mathscr{K} \vee \mathscr{A}) \cap \mathscr{N} = (\mathscr{A} \vee \mathscr{N}) \cap \mathscr{N} = \mathscr{N},$$

and similarly $\mathscr{S} \vee \mathscr{K}' = \mathscr{N}$. Moreover, $\mathscr{S} \cap \mathscr{K} = \mathscr{S} \cap \mathscr{K}' = \mathscr{R}$. The dimension formula gives $\dim \mathscr{R} = \dim \mathscr{S} - 2$. By theorem 9.4.8, the maps

$$f_1 : \mathfrak{B}(\mathscr{K};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{R};\mathscr{S}), \quad \mathscr{H} \mapsto \mathscr{H} \cap \mathscr{S},$$
$$f_2 : \mathfrak{B}(\mathscr{K}';\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{R};\mathscr{S}), \quad \mathscr{M} \mapsto \mathscr{M} \cap \mathscr{S}$$

are projectivities. Therefore $f = f_2^{-1} \circ f_1$ is a projectivity. Since $\mathscr{H} \subseteq \mathscr{N}$, we have $\mathscr{H} \cap \mathscr{S} = \mathscr{H} \cap \mathscr{A}$, and theorem 9.4.8 yields $f(\mathscr{H}) = \mathscr{K}' \vee (\mathscr{H} \cap \mathscr{S}) = \mathscr{K}' \vee (\mathscr{H} \cap \mathscr{A})$. The construction gives $\mathscr{H} \cap \mathscr{A} = f(\mathscr{H}) \cap \mathscr{A}$. If another member $\mathscr{M} \in \mathfrak{B}(\mathscr{K}';\mathscr{N})$ satisfies $\mathscr{M} \cap \mathscr{A} = \mathscr{H} \cap \mathscr{A}$, then $\mathscr{H}, \mathscr{M} \subseteq \mathscr{N}$ implies $\mathscr{M} \cap \mathscr{S} = \mathscr{H} \cap \mathscr{S}$, or $f_2(\mathscr{M}) = f_2(f(\mathscr{H}))$. Injectivity of $f_2$ gives $\mathscr{M} = f(\mathscr{H})$, proving the equivalent description in (2). □

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## Exercises

**Exercise 9.23.** Prove that a nonidentity projective transformation of a projective line in a net with a cross-ratio structure has at most two fixed points.

**Exercise 9.24.** Prove that every projectivity between two pencils in a net with a cross-ratio structure is a composition of finitely many perspectivities.

**Exercise 9.25.** Let $l, l'$ be distinct lines with a unique common point in a net with a cross-ratio structure. Choose distinct points $A, B, C \in l$ and distinct points $A', B', C' \in l'$. By the Veblen axiom, $BC'$ and $B'C$ meet at a point $P$, $CA'$ and $C'A$ meet at a point $Q$, and $AB'$ and $A'B$ meet at a point $R$. Prove that $P, Q, R$ are collinear.

**Exercise 9.26.** Let $\mathscr{N}$ be a three-dimensional net with a cross-ratio structure, and let two-dimensional subnets $\mathscr{E}, \mathscr{E}'$ meet in a projective line $l$. Suppose a net isomorphism $f : \mathscr{E} \to \mathscr{E}'$ restricts to the identity on $l$. Prove that $f$ is a projective net isomorphism.

**Exercise 9.27.** Consider three equidimensional pencils in a net with a cross-ratio structure. Prove that:

(1) if the three pencils have the same center $\mathscr{K}$, and a fixed net $\mathscr{A}$ satisfies the conditions of theorem 9.4.10(1) for every pair of pencils, then, writing $f_{ij}$ for the resulting perspectivity from the $i$th pencil to the $j$th pencil, $f_{02} = f_{12} \circ f_{01}$;

(2) if the three pencils have the same spanning net $\mathscr{N}$, and a fixed net $\mathscr{A}$ satisfies the conditions of theorem 9.4.10(2) for every pair of pencils, then, writing $f_{ij}$ for the resulting perspectivity from the $i$th pencil to the $j$th pencil, $f_{02} = f_{12} \circ f_{01}$.

## 9.4.2 Basic Properties of Perspectivities

**Theorem 9.4.11.** *Two distinct equidimensional pencils have at most one common member. If they have a common member, then a given projectivity between them is a perspectivity if and only if the pencils have either the same center or the same spanning net, and the common member is fixed by the projectivity. In that event, the common member is respectively the intersection of the distinct spanning nets or the join of the distinct centers.*

**Proof.** If two distinct pencils had two different common members, the intersection of those members would be the common center of the pencils, and their join would be the common spanning net. The pencils would therefore be equal, a contradiction. Hence two distinct equidimensional pencils have at most one common member.

Suppose now that the two pencils have a common member $\mathscr{H}_0$, and write the projectivity as $f : \mathfrak{B}(\mathscr{K};\mathscr{N}) \to \mathfrak{B}(\mathscr{K}';\mathscr{N}')$, where $\mathscr{N},\mathscr{N}'$ have dimension $\mu$ and $\mathscr{K},\mathscr{K}'$ have dimension $\mu-2$.

If $f$ is a perspectivity, theorem 9.4.10 says that the pencils have the same center or the same spanning net. In case (1) of that theorem, $\mathscr{H}_0$ itself is the range member satisfying $\mathscr{H}_0 \subseteq \mathscr{A} \vee \mathscr{H}_0$; in case (2), it is the range member satisfying $\mathscr{H}_0 \cap \mathscr{A} = \mathscr{H}_0 \cap \mathscr{A}$. The uniqueness in theorem 9.4.10 gives $f(\mathscr{H}_0) = \mathscr{H}_0$ in either case. We prove the converse in two cases.

(1) Suppose first that $\mathscr{K} = \mathscr{K}'$. Since the pencils are distinct, $\mathscr{N} \neq \mathscr{N}'$. We have $\mathscr{N} \supsetneq \mathscr{N} \cap \mathscr{N}' \supseteq \mathscr{H}_0$, while $\dim \mathscr{N} = \mu = \dim \mathscr{H}_0 + 1$. Thus $\mathscr{N} \cap \mathscr{N}' = \mathscr{H}_0$, and the dimension formula gives $\dim(\mathscr{N} \vee \mathscr{N}') = \mu + \mu - (\mu-1) = \mu+1$. Put $\mathscr{I} = \mathscr{N} \vee \mathscr{N}'$.

Choose two distinct members $\mathscr{H}_1,\mathscr{H}_2$ of the domain pencil, both different from $\mathscr{H}_0$, and, for $i=1,2$, set $\mathscr{M}_i = f(\mathscr{H}_i)$ and $\mathscr{T}_i = \mathscr{H}_i \vee \mathscr{M}_i$. Because $f(\mathscr{H}_0) = \mathscr{H}_0$ and $f$ is bijective, the members $\mathscr{M}_1,\mathscr{M}_2$ are distinct from one another and from $\mathscr{H}_0$. Also, $\mathscr{H}_i \cap \mathscr{H}_0 = \mathscr{M}_i \cap \mathscr{H}_0 = \mathscr{K}$ and $\mathscr{H}_i \cap \mathscr{M}_i \subseteq \mathscr{N} \cap \mathscr{N}' = \mathscr{H}_0$. Hence $\mathscr{H}_i \cap \mathscr{M}_i = \mathscr{K}$ and $\dim \mathscr{T}_i = \mu$.

By the modular law,

$$\mathscr{T}_i \cap \mathscr{N} = (\mathscr{H}_i \vee \mathscr{M}_i) \cap \mathscr{N} = \mathscr{H}_i \vee (\mathscr{M}_i \cap \mathscr{N}) = \mathscr{H}_i \vee \mathscr{K} = \mathscr{H}_i,$$

and similarly $\mathscr{T}_i \cap \mathscr{N}' = \mathscr{M}_i$. Consequently $\mathscr{T}_1 \neq \mathscr{T}_2$. Put $\mathscr{A} = \mathscr{T}_1 \cap \mathscr{T}_2$. The nets $\mathscr{T}_1,\mathscr{T}_2$ are distinct hypernets of the $(\mu+1)$-dimensional net $\mathscr{I}$, so corollary 9.2.21 gives $\dim \mathscr{A} = \mu - 1$. Furthermore,

$$\mathscr{A} \cap \mathscr{N} = (\mathscr{T}_1 \cap \mathscr{N}) \cap (\mathscr{T}_2 \cap \mathscr{N}) = \mathscr{H}_1 \cap \mathscr{H}_2 = \mathscr{K},$$

and similarly $\mathscr{A} \cap \mathscr{N}' = \mathscr{K}$.

By the dimension formula, $\dim(\mathscr{A} \vee \mathscr{N}) = (\mu-1) + \mu - (\mu-2) = \mu+1$; similarly, $\dim(\mathscr{A} \vee \mathscr{N}') = \mu+1$. By the definitions of $\mathscr{A}$ and $\mathscr{T}_i$, $\mathscr{A} \vee \mathscr{N}, \mathscr{A} \vee \mathscr{N}' \subseteq \mathscr{N} \vee \mathscr{N}'$. Since $\dim(\mathscr{N} \vee \mathscr{N}') = \mu+1$, we have $\mathscr{A} \vee \mathscr{N} = \mathscr{A} \vee \mathscr{N}' = \mathscr{I}$. Since $\mathscr{A} \cap \mathscr{N} = \mathscr{K}$ and $\dim \mathscr{A} = \mu - 1 > \mu - 2 = \dim \mathscr{K}$, we have $\mathscr{A} \not\subseteq \mathscr{K}$ and $\mathscr{A} \cap \mathscr{N} = \mathscr{A} \cap \mathscr{N}' = \mathscr{A} \cap \mathscr{K} = \mathscr{K}$. Thus $\mathscr{A}$ satisfies the hypotheses of case (1) in theorem 9.4.10.

For $i=1,2$, we have $\mathscr{A} \cap \mathscr{H}_i = \mathscr{K}$, so $\dim(\mathscr{A} \vee \mathscr{H}_i) = \mu$. Since $\mathscr{A},\mathscr{H}_i \subseteq \mathscr{T}_i$ and $\dim \mathscr{T}_i = \mu$, it follows that $\mathscr{A} \vee \mathscr{H}_i = \mathscr{T}_i$, and therefore $\mathscr{M}_i \subseteq \mathscr{A} \vee \mathscr{H}_i$. The perspectivity defined by $\mathscr{A}$ in theorem 9.4.10(1) agrees with $f$ at $\mathscr{H}_0,\mathscr{H}_1,\mathscr{H}_2$. A projectivity is determined by three pairs of corresponding members, so it is $f$.

(2) Suppose next that $\mathscr{N} = \mathscr{N}'$. Since the pencils are distinct, $\mathscr{K} \neq \mathscr{K}'$, and therefore $\mu \geq 2$. We have $\mathscr{K} \subsetneq \mathscr{K} \vee \mathscr{K}' \subseteq \mathscr{H}_0$, while $\dim \mathscr{H}_0 = \dim \mathscr{K} + 1$. Hence $\mathscr{H}_0 = \mathscr{K} \vee \mathscr{K}'$, and the dimension formula gives $\dim(\mathscr{K} \cap \mathscr{K}') = (\mu-2) + (\mu-2) - (\mu-1) = \mu-3$.

Again choose two distinct domain members $\mathscr{H}_1,\mathscr{H}_2$, both different from $\mathscr{H}_0$, and, for $i=1,2$, put $\mathscr{M}_i = f(\mathscr{H}_i)$ and $\mathscr{J}_i = \mathscr{H}_i \cap \mathscr{M}_i$. Since $\mathscr{H}_0$ is the unique common member, $\mathscr{H}_i \neq \mathscr{M}_i$, and corollary 9.2.21 gives $\dim \mathscr{J}_i = \mu - 2$. Moreover,

$$\mathscr{J}_1 \cap \mathscr{J}_2 = (\mathscr{H}_1 \cap \mathscr{H}_2) \cap (\mathscr{M}_1 \cap \mathscr{M}_2) = \mathscr{K} \cap \mathscr{K}' := \mathscr{R}.$$

Put $\mathscr{A} = \mathscr{J}_1 \vee \mathscr{J}_2$. Then $\dim \mathscr{A} = (\mu-2) + (\mu-2) - (\mu-3) = \mu-1$.

For $i=1,2$, because $\mathscr{K} \subseteq \mathscr{H}_i$ and $\mathscr{K} \subseteq \mathscr{H}_0$, we have

$$\mathscr{K} \cap \mathscr{J}_i = \mathscr{K} \cap \mathscr{M}_i = \mathscr{K} \cap (\mathscr{M}_i \cap \mathscr{H}_0) = \mathscr{K} \cap \mathscr{K}' = \mathscr{R},$$

and similarly $\mathscr{K}' \cap \mathscr{J}_i = \mathscr{R}$. Therefore $\mathscr{H}_i = \mathscr{K} \vee \mathscr{J}_i$ and $\mathscr{M}_i = \mathscr{K}' \vee \mathscr{J}_i$. If $\mathscr{K} \subseteq \mathscr{A}$, then $\mathscr{H}_1, \mathscr{H}_2 \subseteq \mathscr{A}$, and all three nets have dimension $\mu - 1$, forcing $\mathscr{H}_1 = \mathscr{A} = \mathscr{H}_2$, a contradiction. Hence $\mathscr{K} \nsubseteq \mathscr{A}$; similarly, $\mathscr{K}' \nsubseteq \mathscr{A}$. Since $\mathscr{A}$ is a hypernet of $\mathscr{N}$, $\mathscr{A} \vee \mathscr{K} = \mathscr{A} \vee \mathscr{K}' = \mathscr{N}$.

The dimension formula now gives $\dim(\mathscr{A} \cap \mathscr{K}) = (\mu - 1) + (\mu - 2) - \mu = \mu - 3 = \dim \mathscr{R}$. As $\mathscr{R} \subseteq \mathscr{A} \cap \mathscr{K}$, we have $\mathscr{A} \cap \mathscr{K} = \mathscr{R}$; likewise, $\mathscr{A} \cap \mathscr{K} = \mathscr{A} \cap \mathscr{K}' = \mathscr{R}$. Also $\mathscr{A} \subsetneq \mathscr{N}$, and so $\mathscr{N} \nsubseteq \mathscr{A}$ and $\mathscr{A} \vee \mathscr{K} = \mathscr{A} \vee \mathscr{K}' = \mathscr{A} \vee \mathscr{N} = \mathscr{N}$. Thus $\mathscr{A}$ satisfies the hypotheses of case (2) in theorem 9.4.10.

Finally, $\mathscr{A}$ differs from each $\mathscr{H}_i$ and $\mathscr{M}_i$. The intersection of two distinct hypernets has dimension $\mu - 2$, while $\mathscr{J}_i \subseteq \mathscr{A} \cap \mathscr{H}_i$ and $\mathscr{J}_i \subseteq \mathscr{A} \cap \mathscr{M}_i$. Hence $\mathscr{H}_i \cap \mathscr{A} = \mathscr{J}_i = \mathscr{M}_i \cap \mathscr{A}$ for $i = 1, 2$. The perspectivity defined by $\mathscr{A}$ in theorem 9.4.10(2) therefore agrees with $f$ at $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$. Again, uniqueness from three pairs of corresponding members proves that it is $f$. □

In a general perspectivity the center or axis may have relatively large dimension, but it can in fact be replaced by an equivalent object of low dimension.

**Proposition 9.4.12.** *Suppose distinct pencils $\mathcal{B} = \mathfrak{B}(\mathscr{K}; \mathscr{N})$ and $\mathcal{B}' = \mathfrak{B}(\mathscr{K}'; \mathscr{N}')$ are related by a perspectivity $\mathcal{B} \overset{(\mathscr{A})}{\barwedge} \mathcal{B}'$. Then:*

*(1) If $\mathscr{K} = \mathscr{K}'$, put $\mathscr{R} = \mathscr{A} \cap \mathscr{K}$, $\mathscr{P} = \mathscr{N} \vee \mathscr{N}'$, and $\mathscr{A}_0 = \mathscr{A} \cap \mathscr{P}$, and choose $\mathscr{C}$ such that $\mathscr{C} \dot\vee \mathscr{R} = \mathscr{A}_0$. Both $\mathcal{B} \overset{(\mathscr{A}_0)}{\barwedge} \mathcal{B}'$ and $\mathcal{B} \overset{(\mathscr{C})}{\barwedge} \mathcal{B}'$ define the same perspectivity. If $\mathscr{N} \cap \mathscr{N}' = \mathscr{K}$, then* $\dim \mathscr{C} = 1$*; otherwise* $\dim(\mathscr{N} \cap \mathscr{N}') = \dim \mathscr{K} + 1$ *and* $\dim \mathscr{C} = 0$.

*(2) If $\mathscr{N} = \mathscr{N}'$, put $\mathscr{S} = \mathscr{A} \cap \mathscr{N}$ and $\mathscr{R} = \mathscr{A} \cap \mathscr{K} = \mathscr{A} \cap \mathscr{K}'$, and choose $d$ such that $d \dot\vee \mathscr{R} = \mathscr{S}$. Then both $\mathcal{B} \overset{(\mathscr{S})}{\barwedge} \mathcal{B}'$ and $\mathcal{B} \overset{(d)}{\barwedge} \mathcal{B}'$ define the same perspectivity.*

**Proof.** For (1), put $\mathscr{T} = \mathscr{A} \vee \mathscr{N} = \mathscr{A} \vee \mathscr{N}'$. Since $\mathscr{N}, \mathscr{N}' \subseteq \mathscr{T}$, we have $\mathscr{P} \subseteq \mathscr{T}$. Also $\mathscr{N} \subseteq \mathscr{P}$, so the modular law gives

$$\mathscr{A}_0 \vee \mathscr{N} = \mathscr{N} \vee (\mathscr{A} \cap \mathscr{P}) = (\mathscr{N} \vee \mathscr{A}) \cap \mathscr{P} = \mathscr{T} \cap \mathscr{P} = \mathscr{P},$$

and similarly $\mathscr{A}_0 \vee \mathscr{N}' = \mathscr{P}$. The definition gives $\mathscr{A}_0 \cap \mathscr{N} = \mathscr{A} \cap \mathscr{N} = \mathscr{R}$, and likewise $\mathscr{A}_0 \cap \mathscr{N}' = \mathscr{A}_0 \cap \mathscr{K} = \mathscr{R}$.

The pencils are distinct but have the same center, so $\mathscr{N} \neq \mathscr{N}'$. If $\mathscr{A}_0 \subseteq \mathscr{K}$, then $\mathscr{P} = \mathscr{A}_0 \vee \mathscr{N} = \mathscr{N}$, whence $\mathscr{N}' \subseteq \mathscr{N}$; equal dimensions would force $\mathscr{N}' = \mathscr{N}$, a contradiction. Thus $\mathscr{A}_0 \nsubseteq \mathscr{K}$, and $\mathscr{A}_0$ satisfies the hypotheses of theorem 9.4.10(1). For every $\mathscr{H} \in \mathfrak{B}(\mathscr{K}; \mathscr{N})$, we have $\mathscr{H} \subseteq \mathscr{P}$, and the modular law gives

$$(\mathscr{H} \vee \mathscr{A}) \cap \mathscr{P} = \mathscr{H} \vee (\mathscr{A} \cap \mathscr{P}) = \mathscr{H} \vee \mathscr{A}_0.$$

Consequently $(\mathscr{A}_0 \vee \mathscr{H}) \cap \mathscr{N}' = (\mathscr{A} \vee \mathscr{H}) \cap \mathscr{N}'$, so $\mathscr{A}_0$ defines the original perspectivity.

Now choose $\mathscr{C}$ with $\mathscr{C} \dot\vee \mathscr{R} = \mathscr{A}_0$. From $\mathscr{A}_0 \cap \mathscr{N} = \mathscr{A}_0 \cap \mathscr{N}' = \mathscr{A}_0 \cap \mathscr{K} = \mathscr{R}$ we obtain $\mathscr{C} \cap \mathscr{N} = \mathscr{C} \cap \mathscr{N}' = \mathscr{C} \cap \mathscr{K} = \emptyset$. Since $\mathscr{R} \subseteq \mathscr{K} \subseteq \mathscr{N}, \mathscr{N}'$, we also have $\mathscr{C} \vee \mathscr{N} = \mathscr{A}_0 \vee \mathscr{N} = \mathscr{P}$ and $\mathscr{C} \vee \mathscr{N}' = \mathscr{P}$. The fact that $\mathscr{A}_0 \nsubseteq \mathscr{K}$ implies $\mathscr{C} \neq \emptyset$, and hence $\mathscr{C} \nsubseteq \mathscr{K}$. Thus $\mathscr{C}$ also satisfies theorem 9.4.10(1). For every $\mathscr{H} \in \mathfrak{B}(\mathscr{K}; \mathscr{N})$, $\mathscr{R} \subseteq \mathscr{K} \subseteq \mathscr{H}$, so $\mathscr{C} \vee \mathscr{H} = (\mathscr{C} \vee \mathscr{R}) \vee \mathscr{H} = \mathscr{A}_0 \vee \mathscr{H}$. Therefore $\mathscr{C}$ defines the same perspectivity.

Write $\dim \mathscr{N} = \dim \mathscr{N}' = \mu$. The inclusion $\mathscr{K} \subseteq \mathscr{N} \cap \mathscr{N}'$, the equality $\dim \mathscr{K} = \mu - 2$, and the dimensions of the two distinct nets show that $\dim(\mathscr{N} \cap \mathscr{N}')$ is either $\mu - 2$ or $\mu - 1$. The first case is exactly $\mathscr{N} \cap \mathscr{N}' = \mathscr{K}$. Correspondingly, $\dim \mathscr{P}$ is $\mu + 2$ or $\mu + 1$. Because $\mathscr{C} \cap \mathscr{N} = \emptyset$ and $\mathscr{C} \vee \mathscr{N} = \mathscr{P}$, the dimension formula gives

$$\dim \mathscr{C} = \dim \mathscr{P} - \dim \mathscr{N} + \dim \emptyset = \dim \mathscr{P} - \mu - 1,$$

which gives the dimensions asserted in (1).

For (2), the hypotheses of theorem 9.4.10(2) and the modular law give

$$\mathscr{K} \vee \mathscr{S} = \mathscr{K} \vee (\mathscr{A} \cap \mathscr{N}) = (\mathscr{K} \vee \mathscr{A}) \cap \mathscr{N} = \mathscr{N},$$

and similarly $\mathscr{S} \vee \mathscr{K}' = \mathscr{N}$. Moreover, $\mathscr{S} \cap \mathscr{K} = \mathscr{S} \cap \mathscr{K}' = \mathscr{R}$. Since $\mathscr{N} \nsubseteq \mathscr{A}$, we have $\mathscr{N} \nsubseteq \mathscr{S}$; hence $\mathscr{S}$ satisfies the hypotheses of theorem 9.4.10(2). All members of both pencils lie in $\mathscr{N}$, so intersecting them with $\mathscr{A}$ or with $\mathscr{S} = \mathscr{A} \cap \mathscr{N}$ gives the same nets. Thus $\mathscr{S}$ defines the original perspectivity.

Write $s = \dim\mathscr{S}$ and $\mu = \dim\mathscr{N}$. Since $\mathscr{S} \vee \mathscr{K} = \mathscr{N}$, the dimension formula gives $\dim\mathscr{R} = \dim(\mathscr{S} \cap \mathscr{K}) = s + (\mu - 2) - \mu = s - 2$. Choose a line $d \subseteq \mathscr{S}$ such that $d \dot{\vee} \mathscr{R} = \mathscr{S}$. Then $d \cap \mathscr{K} = d \cap \mathscr{K}' = \emptyset$ and $d \vee \mathscr{K} = d \vee \mathscr{K}' = \mathscr{N}$. Because the pencils are distinct but have the same spanning net, $\mathscr{K} \neq \mathscr{K}'$ and $\mu \geq 2$. Hence $\mathscr{N} \nsubseteq d$, and $d$ satisfies the hypotheses of theorem 9.4.10(2).

Take corresponding members $\mathscr{H}, \mathscr{M}$, and put $\mathscr{J} = \mathscr{H} \cap \mathscr{S} = \mathscr{M} \cap \mathscr{S}$. By theorem 9.4.8, $\mathscr{J} \in \mathfrak{B}(\mathscr{R};\mathscr{S})$. Intersecting this pencil with $d$ gives $\mathscr{J} = \mathscr{R} \vee (\mathscr{J} \cap d)$. Since $d \subseteq \mathscr{S}$, $\mathscr{J} \cap d = \mathscr{H} \cap d = \mathscr{M} \cap d$. As $\mathscr{R} \subseteq \mathscr{K}'$, it follows that $\mathscr{M} = \mathscr{K}' \vee \mathscr{J} = \mathscr{K}' \vee (\mathscr{H} \cap d)$. Thus $d$ also defines the original perspectivity. □

Several special cases may now be read off directly.

**Corollary 9.4.13.** *Let $l, l'$ be two lines in an n-dimensional net with a cross-ratio structure, meeting at the unique point X, and let $f: l \to l'$ be a projectivity. Each of the following is necessary and sufficient for f to be a perspectivity:*

*(1) there is a point $O \in (l \vee l') \setminus (l \cup l')$ such that $O, P, f(P)$ are collinear for every $P \in l$; in this case O is the center of perspectivity;*

*(2) $f(X) = X$.*

**Proof.** For $\mu = 1$, the centers in theorem 9.4.10 are $\mathscr{K} = \mathscr{K}' = \emptyset$, and case (1) there is precisely the usual perspectivity of lines. Conditions (1) and (2) follow respectively from proposition 9.4.12 and theorem 9.4.11. □

**Corollary 9.4.14.** *Let two pencils of lines in an n-dimensional net with a cross-ratio structure have distinct centers $X, Y$ and the same two-dimensional spanning net $\mathscr{E}$; that is, the pencils are $\mathfrak{B}(X;\mathscr{E})$ and $\mathfrak{B}(Y;\mathscr{E})$. For a projectivity f between them, each of the following is necessary and sufficient for f to be a perspectivity:*

*(1) there is a line $l \subseteq \mathscr{E} \setminus \{X, Y\}$ such that $m, f(m), l$ have a common point for every $m \in \mathfrak{B}(X;\mathscr{E})$; in this case l is the perspective axis;*

*(2) $f(XY) = XY$.*

**Proof.** This is the case $\mu = 2$ of theorem 9.4.10(2). Conditions (1) and (2) follow respectively from proposition 9.4.12 and theorem 9.4.11. □

**Corollary 9.4.15.** *Let $\mu \geq 2$, and let $\mathfrak{B}(\mathscr{K};\mathscr{N}), \mathfrak{B}(\mathscr{K}';\mathscr{N})$ be two pencils of hypernets in a $\mu$-dimensional net $\mathscr{N}$, and suppose that the join of their centers, $\mathscr{H}_0 = \mathscr{K} \vee \mathscr{K}'$, is also a hypernet. The two pencils have the unique common member $\mathscr{H}_0$. A projectivity f between them is a perspectivity if and only if $f(\mathscr{H}_0) = \mathscr{H}_0$.*

*When f is a perspectivity, its axis may be chosen to be a hypernet $\mathscr{A}$ of $\mathscr{N}$. For every $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$, the three hypernets $\mathscr{A}, \mathscr{H}, f(\mathscr{H})$ then belong to the pencil of hypernets with center $\mathscr{A} \cap \mathscr{H}$. In particular, when $\mu = 2$ this reduces to corollary 9.4.14.*

**Proof.** The net $\mathscr{H}_0$ plainly belongs to both pencils, and by theorem 9.4.11 they have at most one common member; hence it is the unique one. The stated criterion is the case of theorem 9.4.11 in which the two pencils have the same spanning net. In the proof of that theorem the constructed axis $\mathscr{A}$ is a hypernet of $\mathscr{N}$ and satisfies $\mathscr{H} \cap \mathscr{A} = f(\mathscr{H}) \cap \mathscr{A}$. Thus $\mathscr{A}, \mathscr{H}, f(\mathscr{H})$ lie in the pencil of hypernets with center $\mathscr{A} \cap \mathscr{H}$. □

**Proposition 9.4.16.** *Let $\mu \geq 2$, and let $\mathscr{K},\mathscr{L}$ be distinct $(\mu-2)$-dimensional subnets of a $\mu$-dimensional net $\mathscr{N}$. Suppose $\mathscr{H}_0 = \mathscr{K} \vee \mathscr{L}$ is a hypernet of $\mathscr{N}$, and put $\mathscr{R} = \mathscr{K} \cap \mathscr{L}$. Choose a $\nu$-dimensional subnet $\mathscr{S} \subseteq \mathscr{N}$ such that $2 \leq \nu \leq \mu$ and $\mathscr{R} \vee \mathscr{S} = \mathscr{N}$, and put $\mathscr{A} = \mathscr{K} \cap \mathscr{S}$ and $\mathscr{D} = \mathscr{L} \cap \mathscr{S}$. Then:*

*(1) The nets $\mathscr{A},\mathscr{D}$ are distinct $(\nu-2)$-dimensional subnets of $\mathscr{S}$, and $\mathscr{K} = \mathscr{R} \vee \mathscr{A}$, $\mathscr{L} = \mathscr{R} \vee \mathscr{D}$, and $\mathscr{H}_0 \cap \mathscr{S} = \mathscr{A} \vee \mathscr{D}$. Here $\mathscr{A} \vee \mathscr{D}$ is a hypernet of $\mathscr{S}$ and the unique common member of $\mathfrak{B}(\mathscr{A};\mathscr{S})$ and $\mathfrak{B}(\mathscr{D};\mathscr{S})$. By theorem 9.4.8, intersection with $\mathscr{S}$ gives projectivities*

$$g : \mathfrak{B}(\mathscr{K};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{A};\mathscr{S}), \quad \mathscr{H} \mapsto \mathscr{H} \cap \mathscr{S},$$
$$g' : \mathfrak{B}(\mathscr{L};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{D};\mathscr{S}), \quad \mathscr{H} \mapsto \mathscr{H} \cap \mathscr{S},$$

*and $g(\mathscr{H}_0) = \mathscr{A} \vee \mathscr{D} = g'(\mathscr{H}_0)$.*

*(2) For any projectivity $f : \mathfrak{B}(\mathscr{K};\mathscr{N}) \longrightarrow \mathfrak{B}(\mathscr{L};\mathscr{N})$, put $f' = g' \circ f \circ g^{-1}$. Then $f'$ is a projectivity, and $f(\mathscr{H}_0) = \mathscr{H}_0 \iff f'(\mathscr{A} \vee \mathscr{D}) = \mathscr{A} \vee \mathscr{D}$. Consequently $f$ is a perspectivity if and only if $f'$ is a perspectivity.*

*(3) If $f$ in (2) is not a perspectivity, neither is $f'$, and for every $\mathscr{J} \in \mathfrak{B}(\mathscr{A};\mathscr{S})$, $\dim[\mathscr{J} \cap f'(\mathscr{J})] = \nu-2$. In particular, when $\nu = 2$, each pair of corresponding lines has exactly one common point.*

**Proof.** For (1), since $\mathscr{R} \subseteq \mathscr{K},\mathscr{L}$ and $\mathscr{R} \vee \mathscr{S} = \mathscr{N}$, we have $\mathscr{K} \vee \mathscr{S} = \mathscr{L} \vee \mathscr{S} = \mathscr{N}$. Thus theorem 9.4.8 gives $\dim\mathscr{A} = \dim\mathscr{D} = \nu-2$ and the two projectivities in the statement.

Because $\mathscr{H}_0 = \mathscr{K} \vee \mathscr{L}$ is a hypernet of $\mathscr{N}$, the dimension formula gives $\dim\mathscr{R} = (\mu-2)+(\mu-2)-(\mu-1) = \mu-3$. Together with $\mathscr{R} \vee \mathscr{S} = \mathscr{N}$, it also gives $\dim(\mathscr{R} \cap \mathscr{S}) = (\mu-3)+\nu-\mu = \nu-3$. Now $\mathscr{R} \cap \mathscr{A} = \mathscr{R} \cap \mathscr{S}$, so $\dim(\mathscr{R} \vee \mathscr{A}) = (\mu-3)+(\nu-2)-(\nu-3) = \mu-2$. Since $\mathscr{R} \vee \mathscr{A} \subseteq \mathscr{K}$, equality of dimensions gives $\mathscr{K} = \mathscr{R} \vee \mathscr{A}$. Similarly, $\mathscr{L} = \mathscr{R} \vee \mathscr{D}$, and hence $\mathscr{A} \neq \mathscr{D}$.

It follows that $\mathscr{H}_0 = \mathscr{R} \vee \mathscr{A} \vee \mathscr{D}$. Since $\mathscr{A} \vee \mathscr{D} \subseteq \mathscr{S}$ and $\mathscr{R} \cap \mathscr{S} \subseteq \mathscr{A} \vee \mathscr{D}$, the modular law gives

$$\mathscr{A} \vee \mathscr{D} = (\mathscr{A} \vee \mathscr{D}) \vee (\mathscr{R} \cap \mathscr{S}) = [(\mathscr{A} \vee \mathscr{D}) \vee \mathscr{R}] \cap \mathscr{S} = \mathscr{H}_0 \cap \mathscr{S}.$$

Also $\mathscr{R} \subseteq \mathscr{H}_0$ and $\mathscr{R} \vee \mathscr{S} = \mathscr{N}$ imply $\mathscr{H}_0 \vee \mathscr{S} = \mathscr{N}$. Hence $\dim(\mathscr{A} \vee \mathscr{D}) = \dim(\mathscr{H}_0 \cap \mathscr{S}) = (\mu-1)+\nu-\mu = \nu-1$. Thus $\mathscr{A} \vee \mathscr{D}$ is a hypernet of $\mathscr{S}$ and belongs to both induced pencils. By theorem 9.4.11, it is their unique common member. The definitions of $g$ and $g'$ also give $g(\mathscr{H}_0) = \mathscr{A} \vee \mathscr{D} = g'(\mathscr{H}_0)$.

For (2), the map $f'$ is a composition of projectivities and therefore a projectivity. Part (1) gives $g^{-1}(\mathscr{A} \vee \mathscr{D}) = \mathscr{H}_0$, so $f'(\mathscr{A} \vee \mathscr{D}) = g'[f(\mathscr{H}_0)]$. Since $g'(\mathscr{H}_0) = \mathscr{A} \vee \mathscr{D}$ and $g'$ is injective, $f(\mathscr{H}_0) = \mathscr{H}_0$ and $f'(\mathscr{A} \vee \mathscr{D}) = \mathscr{A} \vee \mathscr{D}$ are equivalent. Applying corollary 9.4.15 to $f$ and to $f'$ proves that their perspectivity conditions are equivalent.

For (3), if $f$ is not a perspectivity, then neither is $f'$, so $f'$ does not fix the unique common member $\mathscr{A} \vee \mathscr{D}$. If a corresponding pair of members were equal, that member would be common to the two pencils and would therefore have to be $\mathscr{A} \vee \mathscr{D}$, a contradiction. Conversely, if no corresponding pair is equal, $\mathscr{A} \vee \mathscr{D}$ is not fixed and neither $f'$ nor $f$ is a perspectivity.

Thus, for any $\mathscr{J} \in \mathfrak{B}(\mathscr{A};\mathscr{S})$, the nets $\mathscr{J}$ and $f'(\mathscr{J})$ are distinct hypernets of $\mathscr{S}$. By corollary 9.2.21, $\dim[\mathscr{J} \cap f'(\mathscr{J})] = \nu-2$. When $\nu = 2$, the intersection is zero-dimensional and hence consists of one point. □

We conclude by relating the terminology of this chapter to that of modern projective geometry. The set of all projective points in a projective point–line system is its largest net. In modern language this largest net is a *projective space*, and its nets are the *projective subspaces*. Proper subnets are proper projective subspaces; hypernets are *hyperplanes*; two-dimensional nets are *planes*; and projective lines and projective points are simply the *lines* and *points* of the projective space. More generally, an *n-plane* may mean a projective subspace of arbitrary dimension. This agrees with the traditional use of "net," which originally referred to the two-dimensional case but is also used here for objects of general dimension.

Intersections and joins of nets are respectively intersections and spans, or joins, of projective subspaces; their dimensions are the dimensions of those subspaces. A pencil of nets may be called a pencil of projective subspaces (or, in the broad sense, a pencil of $n$-planes), or, according to the dimension of its members, a point range, pencil of lines, pencil of planes, or pencil of hyperplanes.

There is no single uniform convention for the center and spanning net: the center may be called the axis or center of the pencil, while "spanning space of the pencil" is a natural name for the spanning net. Thus, in modern language, the theory developed in this chapter is synthetic projective geometry in a general axiomatic projective space.

The *Veblen–Young theorem* states that for every $n$-dimensional net $\mathscr{N}$ with $n \geq 3$, there is a division ring $D$ such that $\mathscr{N}$ is isomorphic as a net to the standard projective space $\mathbb{P}^n(D)$. Here the theorem serves only to confirm that the present definitions agree with the modern ones. All later results based on this chapter will use proofs independent of the theorem. Projective spaces over general fields have not yet been introduced; interested readers may consult the later chapter on axiomatic projective geometry, chapter 10, whose development directly yields this result.

In dimension two, the most general net may be non-Desarguesian; see chapter 10. The cross-ratio requirement for a net with a cross-ratio structure, including in dimension two, in fact implies Axiom D of the later axiomatic chapter, chapter 10. That axiom is equivalent to requiring the projective space to be defined over a field. Perspectivities and projectivities in the present sense then arise naturally in projective spaces over fields. No knowledge of division rings or fields is required in this chapter; these comments may be left until chapter 10 is studied.

In particular, add one point at infinity to every line of real or complex Euclidean $n$-space $\mathbb{R}^n$ or $\mathbb{C}^n$, require all points at infinity to lie in one $(n-1)$-dimensional hyperplane, called the *hyperplane at infinity*, and treat finite and infinite points on equal terms. The result is the $n$-dimensional *real projective space* $\mathbb{RP}^n$ or *complex projective space* $\mathbb{CP}^n$. When the underlying field is clear, $\mathbb{P}^n$ denotes an $n$-dimensional projective space. In real and complex projective space, as in plane projective geometry, directed distances may be used to define cross-ratios. This supplies a cross-ratio structure and therefore permits the treatment of perspectivities and projectivities. These concepts and properties will not be repeated when spatial projective geometry is needed later.

## Exercises

**Exercise 9.28.** Let $\varphi: \mathscr{E} \to \mathscr{E}'$ be a projective net isomorphism between nets with cross-ratio structures, and let $f: \mathscr{B} \to \mathscr{B}'$ be a perspectivity between two equidimensional pencils in $\mathscr{E}$. Prove that $\varphi$ maps $\mathscr{B}$ and $\mathscr{B}'$ to pencils in $\mathscr{E}'$ and that $\varphi \circ f \circ \varphi^{-1}$ is again a perspectivity. Determine its center or axis in terms of the center or axis of $f$.

**Exercise 9.29.** In a three-dimensional net $\mathscr{P}$ with a cross-ratio structure, let $k, k'$ be distinct projective lines meeting at $O$, and put $\mathscr{H}_0 = k \vee k'$. Let $\mathscr{E} \subseteq \mathscr{P}$ be a two-dimensional net not containing $O$, and put $X = k \cap \mathscr{E}$ and $Y = k' \cap \mathscr{E}$. Given a projectivity $f: \mathfrak{B}(k;\mathscr{P}) \longrightarrow \mathfrak{B}(k';\mathscr{P})$, define $g: \mathfrak{B}(X;\mathscr{E}) \longrightarrow \mathfrak{B}(Y;\mathscr{E})$, $\quad \mathscr{X} \mapsto f(\mathscr{X} \vee k) \cap \mathscr{E}$. Prove that $g$ is a projectivity and that the following conditions are equivalent:

(1) $f$ is a perspectivity;
(2) $f(\mathscr{H}_0) = \mathscr{H}_0$;
(3) $g(XY) = XY$;
(4) $g$ is a perspectivity.

**Exercise 9.30.** In a three-dimensional net $\mathscr{P}$ with a cross-ratio structure, let $k, k', d$ be three pairwise disjoint lines, and put $\mathscr{B} = \mathfrak{B}(k;\mathscr{P})$ and $\mathscr{B}' = \mathfrak{B}(k';\mathscr{P})$. Define $f_d: \mathscr{B} \longrightarrow \mathscr{B}'$, $\quad \mathscr{H} \mapsto k' \vee (\mathscr{H} \cap d)$. Prove that:

(1) $f_d$ is a perspectivity with axis $d$;
(2) if a projectivity $f: \mathscr{B} \to \mathscr{B}'$ admits three distinct members $\mathscr{H}_i \in \mathscr{B}$ such that each $\mathscr{H}_i \cap f(\mathscr{H}_i)$ meets $d$, then $f = f_d$.

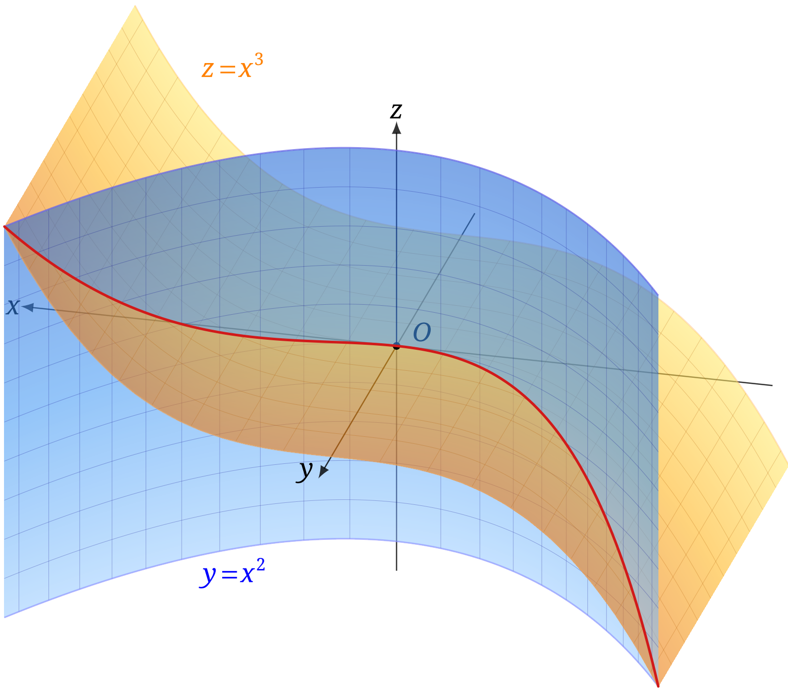



FIGURE 9.1

**Exercise 9.31.** All objects are taken in the same net with a cross-ratio structure. Let $\mu$ be a positive integer, let $\mathscr{N},\mathscr{N}'$ be $\mu$-dimensional nets, and let $\mathscr{K} \subseteq \mathscr{N}$ and $\mathscr{K}' \subseteq \mathscr{N}'$ be $(\mu-2)$-dimensional subnets. Fix a $(\mu-1)$-dimensional net $\mathscr{A} \notin \mathfrak{B}(\mathscr{K};\mathscr{N}) \cup \mathfrak{B}(\mathscr{K}';\mathscr{N}')$, and let $f\colon \mathfrak{B}(\mathscr{K};\mathscr{N}) \to \mathfrak{B}(\mathscr{K}';\mathscr{N}')$ be a bijection. Prove that $f$ is one of the two constructions in theorem 9.4.10 determined by $\mathscr{A}$, with $\mathscr{K} \subsetneq \mathscr{A}$ in case (1) or $\mathscr{A} \subseteq \mathscr{N}$ in case (2), if and only if, for every $\mathscr{H} \in \mathfrak{B}(\mathscr{K};\mathscr{N})$, $\dim(\mathscr{A} \cap \mathscr{H}) = \mu-2$ and $f(\mathscr{H}) \in \mathfrak{B}(\mathscr{A} \cap \mathscr{H};\mathscr{A} \vee \mathscr{H})$.

**Exercise 9.32*.** In real projective three-space, let $X,Y,Z$ be three noncollinear points. The axes $YZ,ZX,XY$ uniquely determine three pencils of planes $\mathscr{A},\mathscr{B},\mathscr{C}$, respectively. Let $f\colon \mathscr{A} \to \mathscr{B}$ and $g\colon \mathscr{A} \to \mathscr{C}$ be projectivities, and suppose none of $f$, $g$, and $g \circ f^{-1}$ is a perspectivity. Prove that for a moving plane $\pi \in \mathscr{A}$, the intersection $\pi \cap f(\pi) \cap g(\pi)$ is always a point and that its locus is a *twisted cubic*. After a projective transformation and removal of the part at infinity, this curve can be represented by $y = x^2$ and $z = x^3$, as shown in figure 9.1.

# Chapter 10

# An Introduction to Axiomatic Plane Geometry*

Before turning to plane geometry in coordinates, we briefly develop an axiomatic system for projective geometry. The axiomatic structure will show that coordinates can grow naturally out of geometry: one need not first define a coordinate system and only afterward give its elements geometric meaning. A full understanding of this chapter requires familiarity with groups, rings, fields, and linear spaces over a general division ring; these topics are developed in detail in sections A.3 to A.5.

## 10.1 An Overview of the Axioms of Projective Geometry

In mathematics, an *axiom* is a basic proposition accepted without proof within a theory and used as a starting point for its development. A mature branch of mathematics generally begins by specifying its fundamental objects and relations, records those relations as axioms, and then builds the theory by rigorous deduction. We now reconstruct projective geometry in this way.

We begin with the following axioms or propositions for a projective plane.

**A1.** Through any two distinct points there passes exactly one line.
**A2.** Any two distinct lines in a projective plane meet.
**A3.** Every projective line contains at least three points.
**A4.** Every projective plane contains three noncollinear points.
**B.** Desargues' theorem holds.
**C.** Pappus' theorem holds.
**D.** The fundamental theorem of one-dimensional projective geometry holds: three pairs of corresponding elements on one-dimensional fundamental forms determine a unique one-dimensional projective transformation. Here projectivities are defined geometrically as in section 3.5: a projective correspondence between one-dimensional fundamental forms is a composition of perspectivities, and a projective correspondence from such a form to itself is a projective transformation.

Axioms A1–A4 are called the *incidence axioms*, or *basic axioms*, of a projective plane; collectively we denote them by A. In particular, A3 requires at least three points on each projective line, one of which may serve as a "point at infinity."[1]

The projective planes studied earlier satisfy A–D; these are the projective planes of classical geometry. Notice also that A2 already implies uniqueness of the intersection of two distinct lines: two distinct common points would, by A1, force the lines to coincide.

It is sometimes useful to regard a plane as embedded in a three-dimensional space. In addition to A1–A4, a three-dimensional projective space satisfies:

**A5.** Through any three noncollinear points there passes exactly one plane.
**A6.** A plane contains the line through each pair of its points.
**A7.** Any two distinct planes have at least two common points.
**A8.** The space contains four noncoplanar points.

[1]A point at infinity is not intrinsically a special kind of point in a projective plane. Historically, however, projective planes arose by adjoining elements at infinity, and A3 reflects that background.

Together, A1–A8 are the *incidence axioms*, or *basic axioms*, of three-dimensional projective space; we denote them by A'. Below, A without a prime continues to mean the plane axioms A1–A4.

The axioms A–D are accepted directly when assumed, but we may also consider planes in which A holds and B does not. To give an overview of the logical relations, we first state:

$$(1)\ \mathrm{A} \not\Rightarrow \mathrm{B};\qquad (2)\ (\mathrm{A}+\mathrm{B}) \not\Rightarrow \mathrm{C};\qquad (3)\ (\mathrm{A}+\mathrm{C}) \Rightarrow \mathrm{B};$$
$$(4)\ (\mathrm{A}+\mathrm{C}) \Longleftrightarrow (\mathrm{A}+\mathrm{D});\qquad (5)\ \mathrm{A}' \Rightarrow \mathrm{B}.$$

Equivalently:

**Theorem 10.1.1.** *For a projective plane:*
*(1) B does not follow from A;*
*(2) even A together with B does not imply C;*
*(3) A together with C implies B;*
*(4) assuming A, propositions C and D are equivalent: A together with C implies D, and conversely A together with D implies C;*
*(5) if the plane is a plane in some three-dimensional projective space, then B holds automatically; in other words, in three-dimensional space, axiom A' implies axiom B.*

**Remark.** The converse of Desargues' theorem in theorem 3.4.3 uses only incidence relations, so its proof remains valid under A. Thus whenever B is assumed, we may also use the converse of Desargues' theorem.

We now justify the assertions in order. Items (1) and (2) require only counterexamples; these will appear below among the non-Desarguesian and non-Pappian projective planes.

For (3), Pappus' theorem implies Desargues' theorem. In figure 10.1, let the lines joining corresponding vertices of $\triangle ABC$ and $\triangle A'B'C'$ be concurrent at $O$, and let the intersections of corresponding sides be $P,Q,R$. We prove their collinearity from A and Pappus' theorem.

Put $S = A'C' \cap AB$. Apply Pappus' theorem first to the two triples $O,C,C'$ and $B,S,A$; the points $T := OS \cap BC$, $U = OA \cap BC'$, and $Q$ are collinear. Apply it next to $O,B,B'$ and $C',A',S$; then $U$, $V := OS \cap B'C'$, and $R$ are collinear. Finally apply it to $B,C',U$ and $V,T,S$. This gives the collinearity of $Q = C'S \cap UT$, $R = BS \cap UV$, and $P = BT \cap C'V$.

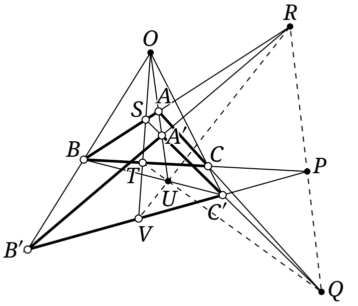



FIGURE 10.1

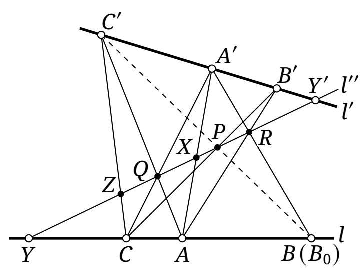



FIGURE 10.2

For (4), we prove the equivalence of Pappus' theorem and the fundamental theorem of one-dimensional projective geometry. First assume D. As in figure 10.2, let $A,B,C \in l$ and $A',B',C' \in l'$. We must prove that $AB' \cap A'B$, $AC' \cap A'C$, and $BC' \cap B'C$ are collinear. Construct $R = AB' \cap A'B$, $Q = AC' \cap A'C$, $P = B'C \cap QR$, and $B_0 = C'P \cap l$; it suffices to prove $B = B_0$. Put $X = AA' \cap QR$, $Z = CC' \cap QR$, $l'' = QR$, $Y = l \cap l''$, and $Y' = l' \cap l''$. We then have the chain of perspectivities

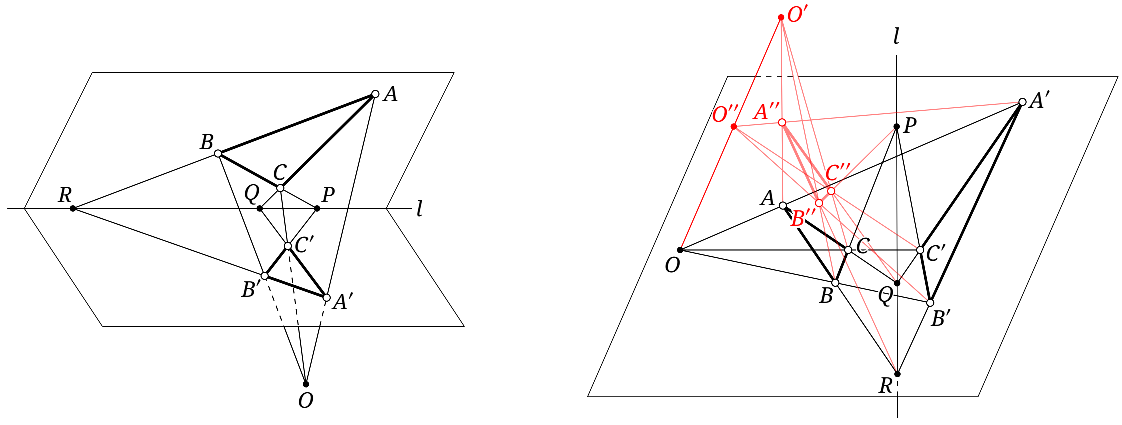
$$l(A,B,C,Y) \overset{(A')}{\barwedge} l''(X,R,Q,Y) \overset{(A)}{\barwedge} l'(A',B',C',Y') \overset{(C)}{\barwedge} l''(Q,P,Z,Y) \overset{(C')}{\barwedge} l(A,B_0,C,Y).$$

Their composition is a projective transformation $f: l \to l$ fixing $A, C, Y$. The identity is also a projective transformation,[2] and D therefore gives $f = \mathrm{id}$. Hence $B = f(B) = B_0$. The converse implication, from Pappus' theorem to D, will be proved after a division-ring structure has been constructed on a projective line.

For (5), we prove Desargues' theorem in three-dimensional projective space. Given $\triangle ABC$ and $\triangle A'B'C'$, suppose $AA', BB', CC'$ concur at $O$. We must prove that the intersections $P = BC \cap B'C'$, $Q = AC \cap A'C'$, and $R = AB \cap A'B'$ exist, and that $P, Q, R$ are collinear.

If the planes $ABC$ and $A'B'C'$ are distinct, let their line of intersection be $l$. Since $A'$ and $B'$ lie in the plane $ABO$, the lines $AB$ and $A'B'$ meet at $R$. Since $R \in AB$ and $AB$ lies in the plane $ABC$, $R$ lies in $ABC$. Similarly it lies in $A'B'C'$, so $R \in l$. Likewise $P, Q \in l$, proving that $P, Q, R$ are collinear.

If the two planes coincide, choose $O'$ outside them and a point $O'' \in OO'$ distinct from $O, O'$. Since $A', O''$ both lie in the plane $AOO'$, the lines $O'A, O''A'$ meet at a point $A''$. Similarly $O'B, O''B'$ meet at $B''$, and $O'C, O''C'$ meet at $C''$. This constructs $\triangle A''B''C''$, whose plane is distinct from $ABC$. The triangles $ABC, A''B''C''$ are perspective from $O'$, and $A'B'C', A''B''C''$ are perspective from $O''$. If $l$ is the intersection of the planes $ABC$ and $A''B''C''$, the preceding case shows that $AB$ meets $A''B''$ at a point $R' \in l$, while $A'B'$ meets $A''B''$ at $R'' \in l$. Both $R', R''$ are the intersection $A''B'' \cap l$, so $AB, A'B'$ meet at the same point of $l$. Thus $R = (AB \cap A'B') \in l$; similarly $P, Q \in l$. $\square$




(a) (b)

FIGURE 10.3

Another possible foundation for projective space uses the *Veblen axiom*, already encountered in chapter 9.

> V. In a projective space, let $A, B, C, D$ be four points with no three collinear. If $AB$ meets $CD$, then $AC$ meets $BD$.

The space may be two-, three-, or higher-dimensional. The plane case holds automatically. We shall treat higher dimensions together later; notice that a three-dimensional subspace of a space satisfying the higher dimensional incidence axioms must itself satisfy A'. The Veblen axiom does not require planes to be defined in advance, yet it produces a structure satisfying the preceding axioms.

**Theorem 10.1.2.** *For a projective space:*

*(1) A1, A2, A5, and A6 imply V.*

*(2) Assume A1, A3, and V. If three noncollinear points $A, B, C$ exist, define the unique projective plane through $A, B, C$ to be the net* $\mathrm{net}(A, B, C)$ *of chapter 9; the resulting plane structure satisfies all the two-dimensional incidence axioms A. If four noncoplanar points $A, B, C, D$ exist,*

[2] Choose a line $l_0 \neq l$ and a point $K$ on neither $l$ nor $l_0$. The composition $l \overset{(K)}{\barwedge} l_0 \overset{(K)}{\barwedge} l$ is the identity, so it satisfies the present geometric definition.

*define their three-dimensional projective space to be* net$(A,B,C,D)$*; then all three-dimensional incidence axioms A' hold.*

**Proof.** For (1), let $A,B,C,D$ be four points with no three collinear and put $O=AB\cap CD$. By A5, $A,B,C$ determine a plane $\pi$. By A6, $O\in AB\subset\pi$, and a second application of A6 gives $D\in CO\subset\pi$. Since $AB$ lies in $\pi$, we also have $O\in\pi$. Thus $A,B,C,D\in\pi$. By A6, $AC,BD\subset\pi$, and A2 shows that $AC,BD$ have a point of intersection.

For (2), A4 and A5 follow from the definition of a plane, while A6 follows from the definition of a net. For A2, take lines $l_1,l_2$ in a plane $\pi$. Lines are one-dimensional nets, and $l_1\vee l_2\subseteq\pi$. The dimension formula gives

$$\dim(l_1\cap l_2)=\dim l_1+\dim l_2-\dim(l_1\vee l_2)\geq 1+1-2=0.$$

Thus $l_1\cap l_2\neq\emptyset$, proving A2 and hence A.

Four noncoplanar points supply A8. It remains to prove A7. If $\pi_1,\pi_2$ are planes in the same three-dimensional net, then $\dim(\pi_1\vee\pi_2)\leq 3$. The dimension formula gives

$$\dim(\pi_1\cap\pi_2)=\dim\pi_1+\dim\pi_2-\dim(\pi_1\vee\pi_2)\geq 2+2-3=1.$$

Their intersection therefore contains at least a line, which proves A7. This establishes all the incidence axioms A'. $\square$

This identifies the earlier notion of a two-dimensional net with a projective plane, and the higher-dimensional case is analogous. In particular, a three-dimensional net is essentially a three-dimensional projective space satisfying A' and no additional prior axioms. The Veblen–Young theorem mentioned earlier then follows from the later discussion. For the present axiomatic treatment, A makes the plane structure explicit and is the more convenient language. The two cross-ratio conditions imposed earlier on a net with a cross-ratio structure amount precisely to defining projectivities as compositions of perspectivities and requiring D.

---

## Exercises

**Exercise 10.1.** Choose suitable axioms of projective geometry. Under those axioms, let $l \overset{(P)}{\doublebarwedge} m \overset{(Q)}{\doublebarwedge} n$, where $l\neq n$, define a projectivity $f: l\to n$, and put $X=l\cap n$. Without using D, prove that $f$ is a perspectivity if and only if either

(a) $l,m,n$ are concurrent, or
(b) $P,Q,X$ are collinear.

**Exercise 10.2.** Choose suitable axioms and prove the following without using D. Suppose $l \overset{(P)}{\doublebarwedge} m \overset{(Q)}{\doublebarwedge} n$, with $l\neq n$ and $X=l\cap n$, induces a projectivity from $l$ to $n$. If $l,m,n$ are not concurrent and $P,Q,X$ are not collinear, then there are a line $m'$, a point $P'\in n$, and a point $Q'\in l$ such that $l \overset{(P')}{\doublebarwedge} m' \overset{(Q')}{\doublebarwedge} n$ induces the same projectivity.

**Exercise 10.3.** Choose suitable axioms and prove that a projectivity between two distinct lines is a perspectivity if and only if it fixes their point of intersection.

**Exercise 10.4*.** Formulate incidence axioms for a general $n$-dimensional projective space.

**Exercise 10.5**.** Assuming A, give a geometric proof that D implies C.[3]

## 10.2 Arithmetic on a Projective Line

Starting from propositions A–D of the preceding section, we now examine the relation between projective geometry and algebraic systems. Despite the word "arithmetic" in the title, the arguments remain synthetic and differ essentially from analytic methods.

**Definition 10.2.1.** Assume A and B.[4] Let $l$ be a projective line in a projective plane. Designate one point of $l$ as its point at infinity $\infty_l$, and choose another point as its *origin*, denoted by $0_l$.[5] We define *addition* on $l$ as follows. For $X, Y \in l$, choose lines $l_x, l_y \neq l$ through $X, Y$, respectively, and a line $l_0 \neq l$ through $0_l$. Put $A = l_x \cap l_0$, $B = l_y \cap l_0$, $C = A\infty_l \cap l_y$, and $D = B\infty_l \cap l_x$, and let $Z = CD \cap l$. Then $Z = X +_l Y$.[6]

The constructions and identities for operations on a line in this section are used whenever the relevant operations are uniquely determined. The same principle applies when the origin or a point at infinity participates in an operation.

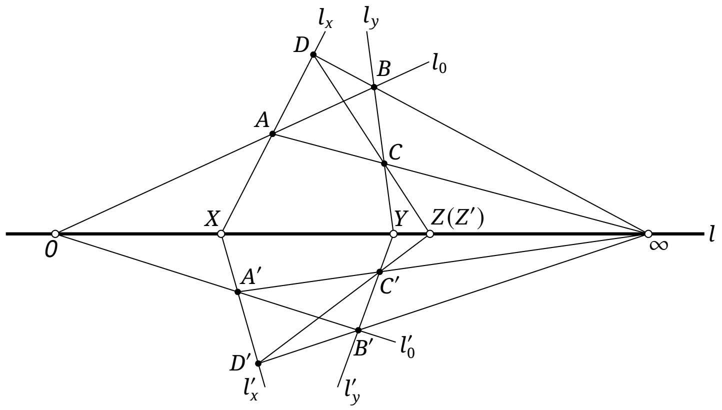



FIGURE 10.4

Assuming A and B, we prove that the sum is unique: for any choices $l_x, l_y, l_0$, the resulting point $Z$ is the same. Indeed, make a second choice $l'_x, l'_y, l'_0$ and define $A', B', C', D'$ analogously, as in figure 10.4.[7] It is enough to prove that $CD$ and $C'D'$ meet $l$ in the same point.

Apply the converse of Desargues' theorem to $\triangle ADB$ and $\triangle A'D'B'$.[8] It follows that $AA', BB', DD'$ are concurrent. Applying the converse to $\triangle ABC$ and $\triangle A'B'C'$ likewise shows that $AA', BB', CC'$ are concurrent. Hence $BB', CC', DD'$ are concurrent, and Desargues' theorem applied to $\triangle BCD$ and $\triangle B'C'D'$ gives the collinearity of $\infty = DB \cap D'B'$, $Y = BC \cap B'C'$, and $Z' := DC \cap D'C'$. Thus $Z' \in l$, so $CD, C'D'$ meet $l$ at the same point, as required. □

[3]There is a purely geometric proof, avoiding the more algebraic division-ring construction used later; see Hartshorne [**14**]. It is omitted here because it is lengthy, is not central to this chapter, and the implication will follow by the later method.

[4]We place the propositions required for a result at the beginning so that its dependence on the axiom system remains explicit.

[5]When no ambiguity can arise, the subscripts are omitted and the points are written simply as $\infty$ and $0$.

[6]When no ambiguity can arise, the addition sign is abbreviated to $+$.

[7]To display the projective incidence structure clearly, the figures in this section draw points at infinity at finite positions.

[8]The listed vertex orders indicate correspondence. Thus this statement means that $AD \cap A'D'$, $AB \cap A'B'$, and $DB \cap D'B'$ are collinear. The same convention is used below, including for direct applications of Desargues' theorem.

We next compare this geometric addition with ordinary addition. Throughout the remainder of the section, the operations are on one fixed line $l$, and $0, \infty \in l$ are fixed. The definition immediately gives $X+Y=Y+X$, the commutative law of addition. The associative law requires proof.

**Theorem 10.2.2** (Associative law of addition)**.** Assuming A and B, *addition on* $l$ *is associative:*

$$(X+Y)+Z=X+(Y+Z) \qquad (X,Y,Z\in l).$$

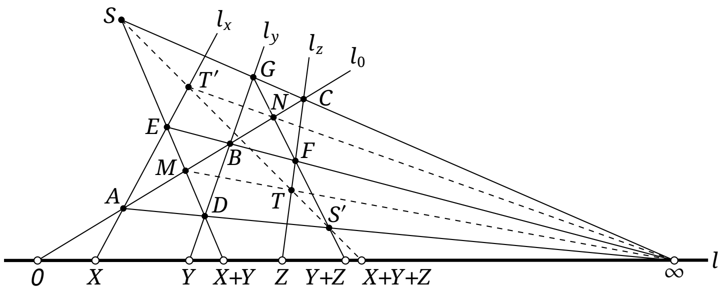



FIGURE 10.5

**Proof.** Choose $X,Y,Z$ and $l_0,l_x,l_y,l_z$ as in figure 10.5; the definition constructs $X+Y$ and $Y+Z$ on $l$. Put $M=ED\cap l_0$, $T=M\infty\cap l_z$, and $S=ED\cap C\infty$. Also put $N=GF\cap l_0$, $T'=N\infty\cap l_x$, and $S'=GF\cap A\infty$. By the definition, $(X+Y)+Z=ST\cap l$ and $X+(Y+Z)=S'T'\cap l$. It remains to show that $S,T,S',T'$ are collinear. Desargues' theorem applied to $\triangle MD\infty$ and $\triangle CGF$ gives the collinearity of $S,S',T$; applied to $\triangle GN\infty$ and $\triangle DAE$, it gives the collinearity of $S',T',S$. The four points therefore lie on one line. □

**Theorem 10.2.3** (Additive identity)**.** Assuming A and B, $0$ *is the* additive identity *on* $l$*; that is,* $0+X=X+0=X$ *for every* $X\in l$.

**Proof.** This follows directly from the definition of addition. □

**Theorem 10.2.4** (Additive inverse)**.** Assuming A and B, *for every* $X\in l\setminus\{\infty\}$ *one can construct its* additive inverse*, denoted by* $-X$*, so that* $X+(-X)=(-X)+X=0$*. Choose a line* $l_x\neq l$ *through* $X$*, and choose two lines through* $0$*, both different from* $l$*, meeting* $l_x$ *at* $A$ *and* $B$*. Put* $C=0A\cap B\infty$ *and* $D=0B\cap A\infty$*. Then* $-X=CD\cap l$*, as shown in figure 10.6.*

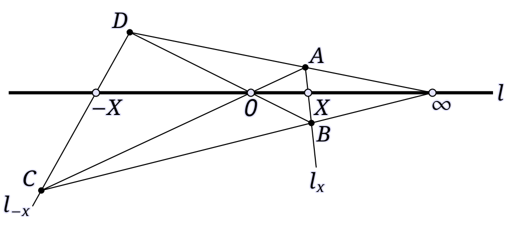



FIGURE 10.6

**Proof.** The assertion follows directly from the definition of addition. □

This theorem allows us to define *subtraction* on $l$, denoted by $-$: for $X,Y\in l$, $X-Y=X+(-Y)$.

We now define multiplication on a projective line.

**Definition 10.2.5.** Assume A and B. Let $0_l$ and $\infty_l$ already be fixed on a projective line $l$, and choose a third point as the *unit point*, denoted by $1_l$.[9] We define *multiplication* on $l$ as follows. For $X,Y \in l$, choose lines $l_x, l_y \neq l$ through $X,Y$, respectively, and a line $l_1 \neq l$ through $1_l$. Put $A = l_x \cap l_1$, $B = l_y \cap l_1$, $C = \infty_l B \cap l_x$, and $D = 0_l A \cap l_y$, and let $Z = CD \cap l$. We denote this operation by $\cdot_l$; then $Z = X \cdot_l Y$.[10]

**Remark.** The order of the factors matters in this definition: commutativity of multiplication does not follow from it alone; in particular, the definition does not imply that $X \cdot Y = Y \cdot X$ for all $X,Y \in l$.

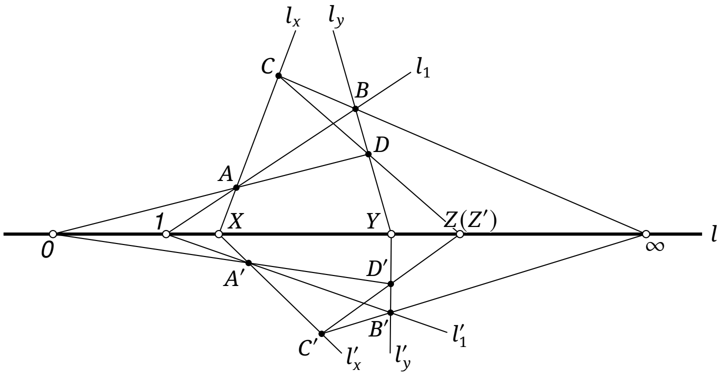



FIGURE 10.7

We prove that the product is unique: the result $Z$ is the same for any auxiliary lines $l_x, l_y, l_1$. For a second choice $l_x', l_y', l_1'$, define $A', B', C', D'$ analogously, as in figure 10.7.[11] It suffices to show that $CD, C'D'$ meet $l$ at the same point. By the converse of Desargues' theorem applied to $\triangle ADB$ and $\triangle A'D'B'$, the lines $AA', BB', DD'$ are concurrent. By the converse applied to $\triangle ABC$ and $\triangle A'B'C'$, the lines $AA', BB', CC'$ are concurrent. Hence $BB', CC', DD'$ are concurrent, and Desargues' theorem applied to $\triangle BCD$ and $\triangle B'C'D'$ shows that[12] $Y = DB \cap D'B'$, $\infty = BC \cap B'C'$, and $Z' := DC \cap D'C'$ are collinear. Thus $Z' \in l$, so $CD, C'D'$ meet $l$ at one point, proving uniqueness. □

In what follows, $0, 1, \infty \in l$ are fixed.

**Theorem 10.2.6** (Associative law of multiplication)**.** Assuming A and B, *multiplication on $l$ is associative:*

$$(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z) \qquad (X,Y,Z \in l).$$

**Proof.** Choose $X,Y,Z$ and $l_1, l_x, l_y, l_z$ as in figure 10.8; the definition constructs $X \cdot Y$ and $Y \cdot Z$. Put $M = ED \cap l_1$, $S = M0 \cap l_z$, and $T = ED \cap C\infty$. Also put $N = GF \cap l_1$, $T' = N\infty \cap l_x$, and $S' = GF \cap A0$. By definition, $(X \cdot Y) \cdot Z = ST \cap l$ and $X \cdot (Y \cdot Z) = S'T' \cap l$. It therefore suffices to prove that $S, T, S', T'$ are collinear. Desargues' theorem applied to $\triangle 0MD$ and $\triangle FCG$ gives the collinearity of $S, T, S'$, while its application to $\triangle GN\infty$ and $\triangle DAE$ gives the collinearity of $S', T', T$. Hence all four points are collinear. □

[9]When no ambiguity can arise, the subscript denoting the line is omitted.

[10]When no ambiguity can arise, this multiplication is abbreviated to $\cdot$.

[11]To display the projective incidence structure clearly, the figures in this section draw points at infinity at finite positions.

[12]The listed vertex orders indicate correspondence. Thus an application to $\triangle ADB$ and $\triangle A'D'B'$ means that $AD \cap A'D'$, $AB \cap A'B'$, and $DB \cap D'B'$ are collinear. The same convention is used below, including for direct applications of Desargues' theorem.

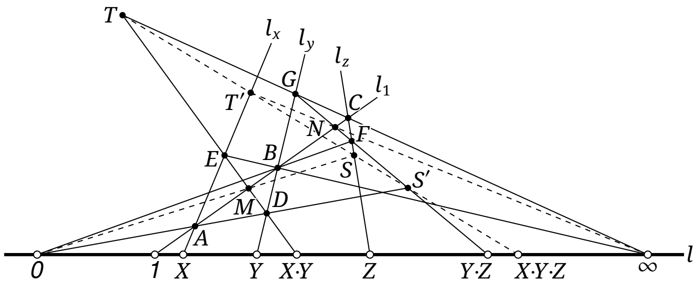



FIGURE 10.8

**Theorem 10.2.7** (Multiplicative zero)**.** Assuming A and B, $\mathit{0}$ *is the multiplicative* zero element*, hence an absorbing element for multiplication:* $\mathit{0}\cdot X = X\cdot \mathit{0} = \mathit{0}$ *for* $X \in l\setminus\{\infty\}$.

**Proof.** This follows directly from the definition of multiplication. ☐

**Theorem 10.2.8** (Multiplicative identity)**.** Assuming A and B, $\mathit{1}$ *is the* multiplicative identity *on* $l$*:* $\mathit{1}\cdot X = X\cdot \mathit{1} = X$ *for every* $X \in l$.

**Proof.** This follows directly from the definition of multiplication. ☐

**Theorem 10.2.9** (Multiplicative inverse)**.** Assuming A and B, *every* $X \in l\setminus\{\mathit{0},\infty\}$ *has a* multiplicative inverse*, denoted by* $X^{-1}$*, satisfying* $X\cdot X^{-1} = X^{-1}\cdot X = \mathit{1}$*. Choose a line* $l_x \neq l$ *through* $X$*, and choose two lines through* $\mathit{1}$*, both different from* $l$*, meeting* $l_x$ *at* $A$ *and* $B$*. Put* $C = \mathit{1}A \cap B\mathit{0}$ *and* $D = \mathit{1}B \cap A\infty$*. Then* $X^{-1} = CD \cap l$*, as in figure 10.9.*

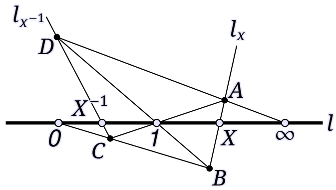



FIGURE 10.9

**Proof.** This follows directly from the definition of multiplication. ☐

**Theorem 10.2.10** (Distributive laws)**.** Assuming A and B, *multiplication distributes over addition on both sides. For all* $X,Y,Z \in l$*, the left distributive law is* $X\cdot(Y+Z) = X\cdot Y + X\cdot Z$*, and the right distributive law is* $(X+Y)\cdot Z = X\cdot Z + Y\cdot Z$.

**Proof.** It is enough to prove, for all $X,Y$, that $X\cdot(Y+\mathit{1}) = X\cdot Y + X$ and $(X+\mathit{1})\cdot Y = X\cdot Y + Y$. Indeed, the case $Z = \mathit{0}$ is immediate. If $Z \neq \mathit{0}$, the first identity gives

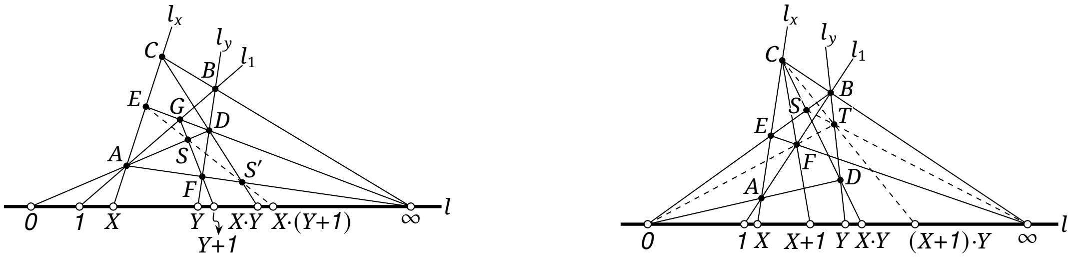
$$X\cdot(Y+Z) = X\cdot(Z\cdot Z^{-1}\cdot Y + Z\cdot \mathit{1}) = X\cdot Z\cdot(Z^{-1}\cdot Y + \mathit{1}) = X\cdot Z\cdot Z^{-1}\cdot Y + X\cdot Z = X\cdot Y + X\cdot Z.$$

The right distributive law $(X+Y)\cdot Z = X\cdot Z + Y\cdot Z$ follows analogously.

We first prove $X\cdot(Y+\mathit{1}) = X\cdot Y + X$. As in figure 10.10a, choose $X,Y$ and the lines $l_x, l_y, l_1$. Multiplication gives $X\cdot Y = CD \cap l$, and addition gives $Y+\mathit{1} = GF \cap l$. Put $S = \mathit{0}A \cap GF$ and $S' = \infty A \cap CD$. Then $ES \cap l = X\cdot(Y+\mathit{1})$ and $ES' \cap l = X\cdot Y + X$. Desargues' theorem applied to $\triangle \infty GF$ and $\triangle CDA$ shows that $E,S,S'$ are collinear. Hence $X\cdot Y + X = X\cdot(Y+\mathit{1})$.

We next prove $(X+\mathit{1})\cdot Y = X\cdot Y + Y$. As in figure 10.10b, choose $X,Y$ and the lines $l_x, l_y, l_1$. Multiplication gives $X\cdot Y = CD \cap l$. Put $E = \mathit{0}B \cap l_x$ and $F = E\infty \cap l_1$; addition then gives $X+\mathit{1} = CF \cap l$. If

$T = 0F \cap l_y$, multiplication gives $CT \cap l = (X+1)\cdot Y$. Finally put $S = CD \cap 0B$. Desargues' theorem applied to $\triangle DCB$ and $\triangle 0EF$ shows that $S, T, \infty$ are collinear. Applying the addition construction to the three lines $CD, BT, 0B$ yields $CT \cap l = X\cdot Y + Y$. □




(a) (b)

FIGURE 10.10

**Theorem 10.2.11** (Commutative law of multiplication)**.** Assuming A and C, *multiplication on* $l$ *is commutative:* $X\cdot Y = Y\cdot X$ *for all* $X, Y \in l$.

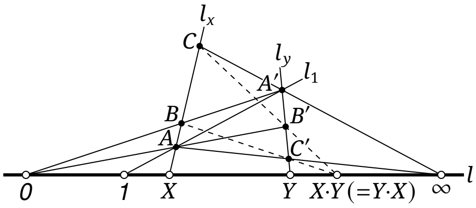



FIGURE 10.11

**Proof.** Choose $X, Y$ and $l_x, l_y, l_1$ as in figure 10.11. Let $l_1$ meet $l_x, l_y$ at $A, A'$, respectively, and put $B' = 0A \cap l_y$, $C = A'\infty \cap l_x$, $B = 0A' \cap l_x$, and $C' = A\infty \cap l_y$. By definition, $CB' \cap l = X\cdot Y$ and $BC' \cap l = Y\cdot X$. Pappus' theorem applied to the collinear triples $A, B, C$ and $A', B', C'$ shows that the points $0 = AB' \cap A'B$, $\infty = AC' \cap A'C$, and $BC' \cap B'C$ are collinear. Hence $B'C$ and $BC'$ meet at the same point of $l$. □

We may now define *division* on $l$, denoted by $/$: for $X, Y \in l$, $X/Y = Y^{-1}\cdot X = X\cdot Y^{-1}$.

The preceding results may be summarized as follows.[13]

**Theorem 10.2.12.** Assuming A and C, *the points of* $l - \{\infty_l\}$*, with the addition and multiplication just defined, form a field. Equivalently,* $l - \{\infty_l\} \cong \mathbb{F}$ *for some field* $\mathbb{F}$.

**Proof.** The field axioms are precisely the properties established above. □

**Remark.** The theorem identifies $l - \{\infty_l\}$ with some field $\mathbb{F}$; it does not determine that field. If the points of the projective plane vary continuously in the usual sense, additional assumptions such as the continuity axioms of section 10.5 lead to $\mathbb{F} \cong \mathbb{R}$ or $\mathbb{F} \cong \mathbb{C}$. Alternatively, if $\infty_l$ is the ordinary point at infinity of a Euclidean line, a direct calculation, left as an exercise, gives

$$\overline{0_l(X+Y)} = \overline{0_lX} + \overline{0_lY}, \quad \overline{0_l(X\cdot Y)}\cdot\overline{0_l1_l} = \overline{0_lX}\cdot\overline{0_lY}.$$

Thus the geometric operations agree with the ordinary ones, and every finite point of $l$ corresponds to a number; in this case as well, $\mathbb{F} \cong \mathbb{R}$ or $\mathbb{F} \cong \mathbb{C}$.

[13] See section A.4.1 of the Appendix.

The logical dependence of the section can now be stated succinctly:

(1) Every result above uses A and B, that is, the incidence axioms and Desargues' theorem.
(2) Commutativity of multiplication, the definition of division, and theorem 10.2.12 additionally use C, that is, Pappus' theorem.

Without commutativity of multiplication, theorem 10.2.12 has the following weaker form, which the reader may verify from the definition of a division ring.

**Theorem 10.2.13.** Assuming A and B, *the points of* $l-\{\infty_l\}$ *form a division ring under the addition and multiplication defined above. Equivalently,* $l-\{\infty_l\}\cong\mathbb{K}$ *for some division ring* $\mathbb{K}$.

---

### Exercises

**Exercise 10.6.** Verify theorems 10.2.3, 10.2.4 and 10.2.7 to 10.2.9 directly from the definitions.

**Exercise 10.7.** Check all the field axioms in theorem 10.2.12. If multiplication is not commutative, verify instead the division-ring assertion in theorem 10.2.13.

**Exercise 10.8.** In a Euclidean model, verify that the geometric addition and multiplication on $l$ agree with the ordinary operations.

**Exercise 10.9.** Assume A and B. Let $ABCD$ be a complete quadrangle whose opposite sides $AC$ and $BD$ meet at $O$. A line $l$ through $O$ meets $AD$ and $BC$ at $M$ and $N$, respectively. If $P=AN\cap CD$ and $Q=CM\cap AB$, prove that $O\in PQ$.

**Exercise 10.10.** In $\mathbb{RP}^2$, let $l,l',l''$ be three mutually parallel lines, and let $A,B,C,D,E$ be given points on $l$. Using only an unmarked straightedge, construct a point $F\in l$ satisfying

$$\overline{AF}=\frac{2\overline{AB}-\overline{DE}}{\overline{BC}}\cdot\overline{AE}+\frac{1}{3}\overline{CD}.$$

## 10.3 One-Dimensional Projective Coordinates

### 10.3.1 One-Dimensional Projective Coordinates and Perspective Affine Transformations

#### One-dimensional projective coordinates

By the preceding section, under the weaker assumptions A and B we have $l-\{\infty_l\}\cong\mathbb{K}$, where $\mathbb{K}$ is a division ring, fixed throughout this section. Once the points $0_l, 1_l, \infty_l$ have been chosen, the geometric addition and multiplication determine this isomorphism up to the corresponding automorphism of $\mathbb{K}$. For $\mathbb{K}=\mathbb{R}$, the four arithmetic operations determine the rational points and the continuity axiom[14] determines the remaining real points, so the identification is unique. For $\mathbb{K}=\mathbb{C}$, continuity fixes the real points and leaves only the equivalent choices i and $-$i; we fix either one. Other division rings are not our main topic, but one can always fix a particular division-ring automorphism. This leads to the following definition.

[14] See the later axiomatic system.

**Definition 10.3.1.** Let $l$ be a projective line. If every point $P \in l - \{\infty_l\}$ corresponds uniquely to an element $p \in \mathbb{K}$, then $p$ is called the *projective coordinate* of $P$, and we write $p = \mathrm{x}_l(P)$. The one-dimensional projective line is denoted by $\mathbb{P}^1(\mathbb{K})$ or $\mathrm{PG}(1, \mathbb{K})$. If three points $A, B, C \in l$ are designated as $0_l, 1_l, \infty_l$, respectively, this choice induces the projective coordinate of every point of $l$. We call $[A, B, C]$ a one-dimensional *projective coordinate system* ; $A, B, C$ are its *origin*, *unit point*, and *point at infinity*, respectively.

To assign a coordinate representation also to $\infty_l$, take $p_1, p_2 \in \mathbb{K}$, not both zero. The expression $[p_1 : p_2]$ is a *homogeneous projective coordinate*, or simply a *homogeneous coordinate*, subject to the following rules.

i. For every $\lambda \in \mathbb{K}^\times$, the representations $[p_1 : p_2]$ and $[p_1\lambda : p_2\lambda]$ represent the same point.[15] Thus we may write $[p_1 : p_2] = [p_1\lambda : p_2\lambda]$.
ii. If $p_2 \neq 0$, put $p = p_1 p_2^{-1}$. Then $[p_1 : p_2] = [p : 1]$, and this representation belongs to the unique point $P$ satisfying $\mathrm{x}_l(P) = p$.
iii. If $p_2 = 0$, the representation denotes $\infty_l$.

When $P$ has homogeneous coordinates $[p_1 : p_2]$, write $\boldsymbol{x}_l(P) = [p_1 : p_2]$. If $\mathrm{x}_l(P) = p$, its *inhomogeneous projective coordinate*, or simply its *inhomogeneous coordinate*, is written $(p)$. Homogeneous and inhomogeneous coordinates represent the same object, so we also write $\boldsymbol{x}_l(P) = (p)$. Thus $\boldsymbol{x}_l(P)$ records a coordinate representation, whereas $\mathrm{x}_l(P)$ is the corresponding element of $\mathbb{K}$. The subscript $l$ indicates that the coordinates are on $l$; it is omitted when no ambiguity can arise.

**Remark.** The bold symbol $\boldsymbol{x}(P)$ denotes a displayed coordinate representation of $P$, homogeneous or inhomogeneous; the ordinary symbol $\mathrm{x}(P)$ denotes the element of $\mathbb{K}$ corresponding to $P$. In one dimension the statements $\boldsymbol{x}(P) = (p)$ and $\mathrm{x}(P) = p$ contain nearly the same numerical information, but in the plane the two notations have genuinely different roles.

Although coordinates have now been introduced, the treatment remains synthetic: a coordinate is initially only a label for a point, obtained from the two geometric operations on a line rather than imported from a Euclidean coordinate system. Here we study the geometric structure produced by these labels.

We now relate the algebraic operations on a line to projective transformations. The coordinate operations in this section are again used whenever their results are uniquely determined. Expressions involving an infinite element are interpreted according to the conventions given below: if cancellation under those conventions yields a determinate value, that value is taken.

First recall the relevant definitions from definitions 3.5.1 and 3.5.12.

**Definition 10.3.2.**
(1) Let $l, l'$ be distinct lines of one projective plane and let $O \notin l \cup l'$. The bijection

$$\varphi : l \longrightarrow l', \qquad A \longmapsto AO \cap l',$$

is a *perspectivity*; $O$ is its *center of perspectivity*.
(2) A finite composite of perspectivities is a *projective correspondence*. A projective correspondence from a line to itself is a *projective transformation* of that line.

The definition of an affine correspondence requires more care. Earlier we defined one as a composite of perspective affine correspondences, while theorem 3.5.2 gave the equivalent description "a projective correspondence that preserves the point at infinity." For a projective line over a general division ring $\mathbb{K}$, with only A and B assumed and C not introduced, the two definitions need not agree. We therefore adopt the following terminology in this part.

[15] When C has not been assumed, $\mathbb{K}$ is only a division ring and multiplication need not be commutative. The reason for defining the scalar $\lambda$ to act on the right will become apparent below.

**Definition 10.3.3.**

(3) An *affine correspondence* is a projective correspondence that maps the point at infinity to the point at infinity. An affine correspondence from a line to itself is an *affine transformation*.

(4) A perspectivity whose center is a point at infinity is a *perspective affine correspondence*. Equivalently, it is a correspondence that is both perspective and affine.

(5) For use below, a self-correspondence obtained as the composite of two perspective affine correspondences is called a *perspective affine transformation*.[16]

We write $\mathrm{Prs}(l,l')$, $\mathrm{Prj}(l,l')$, $\mathrm{Aff}(l,l')$, and $\mathrm{PAff}(l,l')$ for the sets of perspectivities, projective correspondences, affine correspondences, and perspective affine correspondences from $l$ to $l'$, respectively. Likewise $\mathrm{Prj}(l), \mathrm{Aff}(l), \mathrm{PAff}(l)$ denote the corresponding self-transformations of $l$.

For the rest of the section, $\mathit{0}, \mathit{1}, \infty$ and $\mathit{0}', \mathit{1}', \infty'$ denote the origin, unit point, and point at infinity on $l$ and $l'$, respectively; write $\mathrm{x}_{l'}$ also as $\mathrm{x}'$. Because the two lines lie in the same projective plane, $l' - \{\infty_{l'}\} \cong l - \{\infty_l\} \cong \mathbb{K}$.

### One-dimensional perspective affine transformations

We first determine the form of a perspective affine transformation.

**Theorem 10.3.4.** Assuming A and B, *let $\varphi : l \to l'$ be a perspective affine correspondence, so that $\varphi(\infty) = \infty'$. Then:*

*i. $\varphi$ preserves addition,*[17] *$\varphi(X) + \varphi(Y) = \varphi(X+Y)$ for all $X, Y \in l$, if and only if $\varphi(\mathit{0}) = \mathit{0}'$;*

*ii. $\varphi$ preserves multiplication, $\varphi(X) \cdot \varphi(Y) = \varphi(X \cdot Y)$ for all $X, Y \in l$, if and only if $\varphi(\mathit{0}) = \mathit{0}'$ and $\varphi(\mathit{1}) = \mathit{1}'$.*

*Consequently, $\varphi$ induces a division-ring isomorphism $l \setminus \{\infty\} \to l' \setminus \{\infty'\}$ if and only if it preserves the origin, the unit point, and the point at infinity. When it does, the coordinate systems may be chosen so that*[18] *$\mathrm{x}(P) = \mathrm{x}'(\varphi(P))$ for every $P \in l$.*

**Proof.** Since $l$ and $l'$ each form a division ring after removal of their respective points at infinity, necessity follows from the abstract algebraic structure without using specific geometric relations; see exercise A.53. We prove sufficiency. Put $V = l \cap l'$, let $O$ be the center, and choose arbitrary $X, Y \in l$. Write $X' = \varphi(X)$, $Y' = \varphi(Y)$.

For addition, use figure 10.12a. Set $C = \mathit{0}X' \cap OY$, $A = C\infty \cap OX$, and $B = X'\infty \cap OY$, so that $AB \cap l = X + Y$. Similarly set $C' = \mathit{0}'X \cap OY$, $A' = C'\infty' \cap OX$, and $B' = X\infty' \cap OY$, so that $A'B' \cap l' = X' + Y'$. Finally put $P = \mathit{0}'X \cap \mathit{0}X'$, $Q = X'\infty \cap X\infty'$, $R = A\infty \cap A'\infty'$, and $S = AB \cap A'B'$. Desargues' theorem applied to $\triangle X'\mathit{0}\infty$ and $\triangle X\mathit{0}'\infty'$ shows that $P, Q, V$ are collinear. Applied to $\triangle XC'\infty'$ and $\triangle X'C\infty$, it shows that $P, Q, R$ are collinear; and applied to $\triangle AB\infty$ and $\triangle A'B'\infty'$, it shows that $S, R, Q$ are collinear. Thus $V, P, Q, R, S$ are collinear, and in particular $V, Q, S$ are collinear. The converse of Desargues' theorem applied to $\triangle B(X+Y)\infty$ and $\triangle B'(X'+Y')\infty'$ now shows that $BB'$, $(X+Y)(X'+Y')$, and $\infty\infty'$ concur. Since $BB' \cap \infty\infty' = O$, the points $X+Y, X'+Y', O$ are collinear, and we obtain $\varphi(X) + \varphi(Y) = \varphi(X+Y)$.

For multiplication, use figure 10.12b. Define $C = \mathit{1}X' \cap OY$, $A = C\infty \cap OX$, and $B = X'\mathit{0} \cap OY$, and set $C' = \mathit{1}'X \cap OY$, $A' = C'\infty' \cap OX$, and $B' = X\mathit{0}' \cap OY$. Then $AB \cap l = X \cdot Y$ and $A'B' \cap l' = X' \cdot Y'$. Put $P = \mathit{0}'X \cap \mathit{0}X'$, $Q = \mathit{1}X' \cap \mathit{1}'X$, $R = A\infty \cap A'\infty'$, $S = AB \cap A'B'$, and $T = \mathit{0}A \cap \mathit{0}'A'$. Desargues' theorem applied to $\triangle X'\mathit{0}\mathit{1}$ and $\triangle X\mathit{0}'\mathit{1}'$ shows that $P, Q, V$ are collinear. Applied to $\triangle \mathit{1}C\infty$ and $\triangle \mathit{1}'C'\infty'$, it shows that $V, Q, R$ are collinear; and applied to $\triangle \mathit{0}A\infty$ and $\triangle \mathit{0}'A'\infty'$, it shows that $V, T, R$ are collinear. Desargues' theorem applied to $\triangle \mathit{0}AB$ and $\triangle \mathit{0}'A'B'$ shows that $T, P, S$ are

[16]This notion is uncommon and seldom used; it was therefore not introduced in the earlier treatment of classical projective geometry, but is included here as a supplement.

[17]Of course, the addition on the left is $+_{l'}$, whereas that on the right is $+_l$; the intended operations are clear from context. The same convention is used below.

[18]A division-ring isomorphism need not be the identity; the claim is therefore that coordinates *can* be chosen in this way.

collinear. Hence $V,P,Q,R,S,T$ are collinear, and in particular $V,R,S$ are collinear. A final application of the converse of Desargues' theorem to $\triangle A(X\cdot Y)\infty$ and $\triangle A'(X'\cdot Y')\infty'$ shows that $AA'$, $(X\cdot Y)(X'\cdot Y')$, and $\infty\infty'$ concur. Since $AA'\cap\infty\infty'=O$, the points $X\cdot Y, X'\cdot Y', O$ are collinear. Hence $\varphi(X\cdot Y)=\varphi(X)\cdot\varphi(Y)$. □

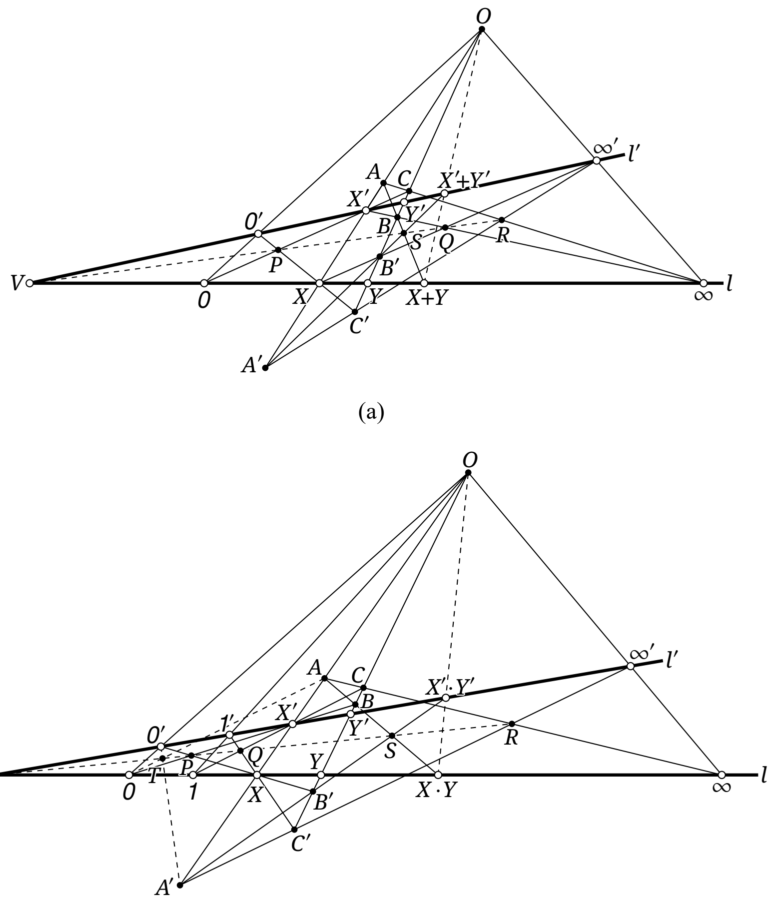



FIGURE 10.12

The general coordinate form is now immediate.

**Theorem 10.3.5.** Assuming A and B, *every* $\varphi\in\mathrm{PAff}(l)$ *has the form*

$$\mathrm{x}(\varphi(P))=a\mathrm{x}(P)+b, \qquad a\neq 0, \tag{10.1}$$

*for all* $P\in l$, *where* $a,b\in\mathbb{K}$.

**Proof.** Let $O,O'$ lie on the line at infinity. Let the perspective affine correspondence $\varphi_1: l\to l'$ have center $O$, and let $\varphi_2: l'\to l$ have center $O'$, so that $\varphi=\varphi_2\circ\varphi_1$. We show that $\mathrm{x}(\varphi(P))=a\mathrm{x}(P)+b$ for every $P\in l$.

The geometric construction is fixed, and its final point $\varphi(P)$ does not depend on how the origin and unit point on $l'$ are chosen. We may therefore take $0'=\varphi_1(0)$, $1'=\varphi_1(1)$, and $\infty'=\varphi_1(\infty)$. By theorem 10.3.4, $\varphi_1$ then induces a division-ring isomorphism. If $P'=OP\cap l'$, then $\mathrm{x}'(P')=\mathrm{x}(P)$.

If $\infty'=\infty$, put $B=\varphi(0)$ and $b=\mathrm{x}(B)$. Use $O'B, OP, 0O$ as the three chosen lines in the addition construction of figure 10.13a. It gives $\varphi(P)=P+B$, hence $\mathrm{x}(\varphi(P))=\mathrm{x}(P)+b$.

If $0'=0$, put $A=\varphi(1)$ and $a=\mathrm{x}(A)$. The multiplication construction in figure 10.13b, with $O'A, OP, 1O$ as the three chosen lines in the definition of multiplication, gives $\varphi(P)=A\cdot P$, hence $\mathrm{x}(\varphi(P))=a\mathrm{x}(P)$.

In the general case, put $l_0 = 0\infty'$. Let $\psi_1 : l \to l_0$ be the perspective affine correspondence with center $O$. It induces a division-ring isomorphism $l\setminus\{\infty\} \to l_0 \setminus \{\infty_{l_0}\}$ when the origin, unit point, and point at infinity on $l_0$ are chosen as $0_0 = \psi_1(0) = 0$, $1_0 = \psi_1(1)$, and $\infty_0 = \psi_1(\infty) = \infty'$, as in figure 10.13c. Let $\psi_2 : l_0 \to l'$ be the perspective affine correspondence with center $O$. Then $\psi_2(0_0) = 0'$, $\psi_2(1_0) = 1'$, and $\psi_2(\infty_0) = \infty'$, so $\psi_2$ also induces a division-ring isomorphism $l_0 \setminus \{\infty_0\} \to l' \setminus \{\infty'\}$.

Now let $\psi'_1 : l \to l_0$ and $\psi'_2 : l_0 \to l'$ be the perspective affine correspondences with center $O'$. Then

$$\varphi = \varphi_2 \circ \varphi_1 = (\psi_1'^{-1} \circ \psi_2'^{-1}) \circ \varphi_1 = (\psi_1'^{-1} \circ \psi_1) \circ \psi_1^{-1} \circ (\psi_2'^{-1} \circ \psi_2) \circ \psi_2^{-1} \circ \varphi_1.$$

The maps $\varphi_1, \psi_1^{-1}, \psi_2^{-1}$ induce division-ring isomorphisms and hence do not change the projective coordinate on the corresponding line. The maps $\psi_1'^{-1} \circ \psi_1$ and $\psi_2'^{-1} \circ \psi_2$ are, respectively, transformations of the second and first special kinds above. Put

$$P' = \varphi_1(P), \qquad P_0 = \psi_2'^{-1}(P') = [(\psi_2'^{-1} \circ \psi_2) \circ \psi_2^{-1}](P').$$

There are $a, b' \in \mathbb{K}$ such that

$$\mathrm{x}_{l_0}(P_0) = \mathrm{x}'(P') + b' = \mathrm{x}(P) + b', \quad \mathrm{x}(\varphi(P)) = a\mathrm{x}_{l_0}(P_0) = a\mathrm{x}(P) + ab'.$$

More explicitly, $b' = \mathrm{x}_{l_0}(\psi_2'^{-1}(0'))$, $a = \mathrm{x}(\psi_1'^{-1}(1_0))$, and $b = ab'$. This proves (10.1). The converse, that every transformation of this form is a perspective affine transformation, is easy to prove: simply reverse the construction in the general case to obtain the corresponding transformation. □

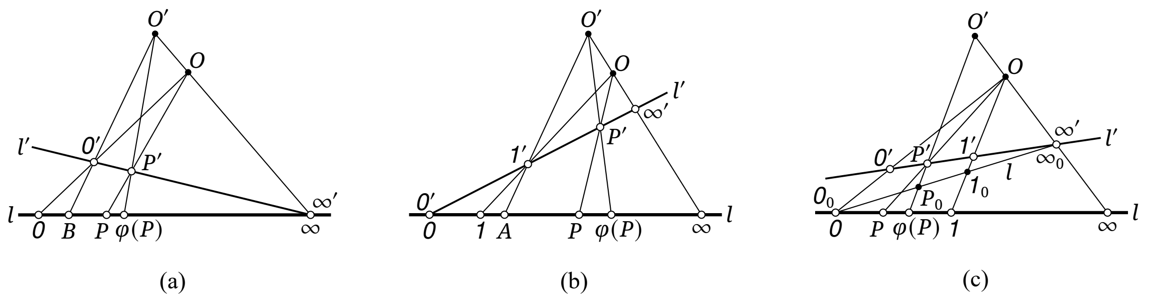



FIGURE 10.13

**Corollary 10.3.6.** Assuming A and B, PAff($l$) *is a group under composition.*

**Proof.** Let $\varphi_i : l \to l, P \mapsto \varphi_i(P)$, satisfy $\mathrm{x}(\varphi_i(P)) = a_i\mathrm{x}(P) + b_i$, where $i = 1,2$. Then

$$\mathrm{x}[\varphi_2(\varphi_1(P))] = a_2[a_1\mathrm{x}(P) + b_1] + b_2 = a_2a_1\mathrm{x}(P) + (a_2b_1 + b_2).$$

This is again a perspective affine transformation, so perspective affine transformations are closed under composition. The inverse of $\varphi_1$ satisfies $\mathrm{x}(\varphi_1^{-1}(P)) = a_1^{-1}\mathrm{x}(P) - a_1^{-1}b_1$, so an inverse exists. Clearly the identity is a perspective affine transformation, and composition of transformations is automatically associative. Thus the perspective affine transformations of the line form a group. □

Another corollary describes the affine transformation obtained by a finite composition of perspective affine correspondences.

**Corollary 10.3.7.** Assuming A and B, *if $\varphi : l \to l$ is a finite composite of perspective affine correspondences, then there exist $a, b \in \mathbb{K}$ such that*

$$\mathrm{x}_l(\varphi(P)) = a\mathrm{x}_l(P) + b \qquad (P \in l).$$

**Proof.** Let the successive intermediate lines be $l_0 = l, l_1, l_2, \ldots$. Choose a point $O$ of the plane and, on every $l_i$, choose $0_i, 1_i, \infty_i$ so that $O, 0_i, 0_l$, $O, 1_i, 1_l$, and $O, \infty_i, \infty_l$ are respectively collinear. For every $i,j$, the perspectivity from $l_i$ to $l_j$ with center $O$ induces a division-ring isomorphism

$$\psi_{ij} : l_i \setminus \{\infty_i\} \longrightarrow l_j \setminus \{\infty_j\}.$$

If an intermediate perspective affine correspondence is $\varphi_i : l_i \to l_{i+1}$, then

$$\varphi_i = \psi_{i,i+1} \circ (\psi_{i+1,i} \circ \varphi_i).$$

The map in parentheses is a perspective affine transformation and has a coordinate relation of the form $a_i \mathrm{x}_{l_i}(P_i) + b_i$, whereas the preceding division-ring isomorphism does not change the coordinate value. Composing all these relations gives $\mathrm{x}_l(\varphi(P)) = a\mathrm{x}_l(P) + b$. ☐

**Corollary 10.3.8.** Assuming A and B, *every affine transformation obtained as a finite composite of perspective affine correspondences is equivalent to a single perspective affine transformation, and hence to a composite of two perspective affine correspondences.*

**Proof.** This follows immediately from theorem 10.3.5 and corollary 10.3.7. ☐

**Definition 10.3.9.** In view of corollary 10.3.8, we use *perspective affine transformation* for either a single such self-transformation or an affine transformation obtained from any finite composite of perspective affine correspondences; the set is denoted by $\mathrm{PAff}(l)$. The group in corollary 10.3.7 is the *one-dimensional perspective affine transformation group* . When $l - \{\infty\} \cong \mathbb{K}$, it is denoted by $\mathrm{AGL}(1, \mathbb{K})$, where the numeral 1 denotes the dimension, or by $\mathrm{AGL}(1)$ when $\mathbb{K}$ is understood.

Here “perspective” means “built from perspective affine correspondences”; it does not assert that the resulting self-transformation is itself a perspectivity. Over a general division ring it need not coincide with every affine transformation defined merely by fixing the point at infinity. The notation PAff emphasizes the geometric construction, whereas AGL emphasizes the algebraic group. In most contexts where the meaning is clear, the notations may be used interchangeably; for a projective line $l$ over $\mathbb{K}$, $\mathrm{PAff}(l) \cong \mathrm{AGL}(1, \mathbb{K})$.

**Corollary 10.3.10.** Assuming A and B, *for three finite points $A, B, C$ on a projective line, with $B \neq C$, define their* simple ratio *by*

$$(ABC) := \mathrm{x}\big((B - C)^{-1} \cdot (A - C)\big).$$

*It is invariant under every perspective affine transformation $\varphi$:*

$$(ABC) = (\varphi(A)\,\varphi(B)\,\varphi(C)).$$

**Proof.** Write $\varphi(P) = X \cdot P + Y$, where $X \in l \setminus \{0, \infty\}$ and $Y \in l \setminus \{\infty\}$. Then

$$\varphi(A) - \varphi(C) = (X \cdot A + Y) - (X \cdot C + Y) = X \cdot (A - C), \quad \varphi(B) - \varphi(C) = X \cdot (B - C),$$

and hence

$$\begin{aligned}(\varphi(A)\varphi(B)\varphi(C)) &= \mathrm{x}\big([\varphi(B) - \varphi(C)]^{-1} \cdot [\varphi(A) - \varphi(C)]\big) \\ &= \mathrm{x}\big((B - C)^{-1} \cdot X^{-1} \cdot X \cdot (A - C)\big) = \mathrm{x}\big((B - C)^{-1} \cdot (A - C)\big).\end{aligned}$$

The corresponding projective coordinate is therefore unchanged. ☐

We also record the following useful consequence.

**Corollary 10.3.11.** Assuming A and B, *suppose coordinate systems on two lines $l, l'$ are chosen so that the joins of points with equal projective coordinates all pass through one point $O$. For every finite composite of perspective affine correspondences $\varphi : l \to l'$, there exist $a, b \in \mathbb{K}$ such that*

$$\mathrm{x}'(\varphi(P)) = a\mathrm{x}(P) + b.$$

**Proof.** Let $\psi: l \to l'$ be the perspectivity with center $O$. Then $\varphi = (\varphi \circ \psi^{-1}) \circ \psi$. The first factor $\varphi \circ \psi^{-1}$ is a perspective affine transformation of $l'$, and the second factor $\psi$ induces the division-ring isomorphism $l - \{\infty\} \cong l' - \{\infty'\}$. Thus the resulting coordinate relation still has the form $\mathrm{x}'(\varphi(P)) = a\mathrm{x}(P) + b$. □

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## Exercises

**Exercise 10.11.** Let $\varphi: l \to l'$ be a perspective affine correspondence in a projective space over a division ring $\mathbb{K}$, with center $O$. Find necessary and sufficient conditions for each of the following:
(1) $\varphi(X - Y) = \varphi(X) - \varphi(Y)$ for all $X, Y \in l$;
(2) $\varphi(X^{-1}) = \varphi(X)^{-1}$ for every $X \in l$.

**Exercise 10.12.** For a prime $p$, find the order of $\mathrm{AGL}(1, \mathbb{F}_p)$.

**Exercise 10.13.** Let $\varphi$ be a perspective affine transformation of a projective line over a division ring $\mathbb{K}$, taking the points with coordinates $0, 1$ to points with coordinates $p, q$, respectively. Determine all possible coordinate formulas for $\varphi$. Is the transformation unique?

### 10.3.2 One-Dimensional Projective Transformations

We first consider the form of a perspective affine transformation, then the forms of a general perspectivity and a projective transformation.

**Lemma 10.3.12.** Assuming A and B, *for a variable point $P$ of a fixed line $l$, each of the following maps $\varphi: l \to l$ is a projective transformation:*
*(1) $P \mapsto A \cdot P$, where $A \in l \setminus \{0, \infty\}$ is fixed;*
*(2) $P \mapsto P \cdot A'$, where $A' \in l \setminus \{0, \infty\}$ is fixed;*
*(3) $P \mapsto P + B$, where $B \in l \setminus \{\infty\}$ is fixed;*
*(4) $P \mapsto P^{-1}$.*

**Proof.** For (1), refer to figure 10.7. Fix the point denoted there by $X$ and let $Y$ vary. Since $l_x, l_y, l_1$ are arbitrary, we may keep $l_x, l_1$, and $B$ fixed. Then only $D$ varies, while $A, B, C$ and the line $0A$ remain fixed. Hence

$$l[Y] \barwedge B[Y] \barwedge (0A)[D] \barwedge C[D] \barwedge l[Z],$$

so $[Y] \barwedge [X \cdot Y]$, which is (1) in the notation of the lemma.[19]

For (2), again refer to figure 10.7, now fixing $Y$ and allowing $X$ to vary. Since $l_x, l_y, l_1$ are arbitrary, we may keep $l_y, l_1$, and $A$ fixed. Then only $C$ varies, while $A, B, D$ and the line $\infty B$ remain fixed. Thus

$$l[X] \barwedge A[X] \barwedge (\infty B)[C] \barwedge D[C] \barwedge l[Z],$$

so $[X] \barwedge [X \cdot Y]$, which proves (2).

For (3), refer to figure 10.4. Fix $Y$, let $X$ vary, and use the freedom in $l_x, l_y, l_0$ to keep $l_y, l_0$, and $A$ fixed. Then only $D$ varies, while $A, B, C$ and the line $\infty B$ remain fixed. Hence

$$l[X] \barwedge A[X] \barwedge (\infty B)[D] \barwedge C[D] \barwedge l[Z],$$

so $[X] \barwedge [X + Y]$, which proves (3).

---

[19] Below, brackets around an element denote the range represented by that element as the moving point or line varies. For example, $B[Y] \barwedge (0A)[D]$ means that the pencil of moving lines $BY$ and the range of moving points $D$ on the base $0A$ are projectively corresponding, with $BY$ and $D$ corresponding for each associated pair of positions.

For (4), refer to figure 10.9. As $X$ varies, the freedom in choosing the two lines through $1$ allows $A$,$D$, and hence the lines $1A$, $1D$, to remain fixed. Therefore

$$l[X] \overset{(A)}{\overline{\wedge}} (1D)[B] \overset{(0)}{\overline{\wedge}} (1A)[C] \overset{(D)}{\overline{\wedge}} l[X^{-1}],$$

which is the projective transformation in (4). □

**Remark.** The map $P \mapsto -P$ is also projective, either directly from figure 10.6 or as a special case of (1), since $-P = (-1) \cdot P$. Right multiplication in (2) is already generated by the other three maps, because

$$P \longmapsto P^{-1} \longmapsto (A')^{-1} \cdot P^{-1} \longmapsto \left((A')^{-1} \cdot P^{-1}\right)^{-1} = (P^{-1})^{-1} \cdot (A'^{-1})^{-1} = P \cdot A'.$$

By theorem A.3.3, the three maps in this chain are respectively (4), (1), and (4): two inversions convert left multiplication into right multiplication.

**Corollary 10.3.13.** Assuming A and B, *for any finite points $A$,$A'$,$B$ on $l$ with $A$,$A' \neq 0$, each of the maps $P \mapsto A \cdot P$, $P \mapsto P \cdot A'$, and $P \mapsto P + B$ is an affine transformation of $l$. The inversion $P \mapsto P^{-1}$ is not affine.*

**Proof.** The definitions of the two operations give

$$A \cdot \infty = \infty \cdot A' = \infty, \qquad \infty + B = B + \infty = \infty$$

for finite nonzero $A$,$A'$ and finite $B$. By contrast, $\infty^{-1} = 0$. □

**Remark.** The zero element of a division ring has no inverse, but here the point at infinity has been adjoined. The earlier geometric construction of the multiplicative inverse gives $\infty^{-1} = 0$ and $0^{-1} = \infty$. Thus, in $\mathbb{K} \cup \{\infty\}$, we additionally stipulate $0^{-1} = \infty$ and $\infty^{-1} = 0$.

Perspective affine transformations previously had only the left-multiplying form $P \mapsto A \cdot P + B$, whereas an affine transformation may have either form $P \mapsto A \cdot P$ or $P \mapsto P \cdot A'$. When C holds, $\mathbb{K}$ is a field and this distinction disappears.

The asymmetry comes from the order chosen in the geometric definition of multiplication. If the clauses in definition 10.2.5 saying that "$\infty_l B$ meets $l_x$ at $C$, and $0_l A$ meets $l_y$ at $D$" are replaced by "$\infty_l A$ meets $l_y$ at $C$, and $0_l B$ meets $l_x$ at $D$," the same arguments develop a parallel right-handed theory, in which a perspective affine transformation has the form $P \mapsto P \cdot A' + B$. The two theories are entirely parallel, so we continue with the present left-multiplying convention rather than repeat the other one. The convention $[p_1 : p_2] = [p_1 \lambda : p_2 \lambda]$ likewise uses right scalar multiplication. Its significance is not yet conspicuous in one dimension but becomes essential in plane coordinates. The preceding lemma gives the general form of a projective transformation.

**Theorem 10.3.14.** Assuming A and B, *every $\varphi \in \mathrm{Prj}(l)$ has the coordinate form*

$$\mathrm{x}(\varphi(P)) = (a\mathrm{x}(P) + b)(c\mathrm{x}(P) + d)^{-1}, \tag{10.2}$$

*for every $P \in l$, where $a,b,c,d \in \mathbb{K}$, with*

$$c = 0 \Longrightarrow ad \neq 0, \qquad c \neq 0 \Longrightarrow b \neq ac^{-1}d.$$

**Proof.** First, let the four finite points $A,B,C,D$ have projective coordinates $a,b,c,d$, respectively, and consider successively

$$\varphi_1 : P \mapsto C \cdot P, \quad \varphi_2 : P \mapsto P + D, \quad \varphi_3 : P \mapsto P^{-1},$$

$$\varphi_4 : P \mapsto (B - A \cdot C^{-1} \cdot D) \cdot P, \qquad \varphi_5 : P \mapsto P + A \cdot C^{-1}.$$

Suppose that $A,B,C,D$ satisfy conditions making all five maps projective transformations. We can then verify that their composite is

$$\begin{aligned}\varphi(P) &= (\varphi_5\circ\varphi_4\circ\varphi_3\circ\varphi_2\circ\varphi_1)(P) = (B-A\cdot C^{-1}\cdot D)\cdot(C\cdot P+D)^{-1}+A\cdot C^{-1}\\ &= (B-A\cdot C^{-1}\cdot D)\cdot(C\cdot P+D)^{-1}+(A\cdot C^{-1})\cdot(C\cdot P+D)\cdot(C\cdot P+D)^{-1}\\ &= (B-A\cdot C^{-1}\cdot D)\cdot(C\cdot P+D)^{-1}+(A\cdot P+A\cdot C^{-1}\cdot D)\cdot(C\cdot P+D)^{-1}\\ &= \left[(B-A\cdot C^{-1}\cdot D)+(A\cdot P+A\cdot C^{-1}\cdot D)\right]\cdot(C\cdot P+D)^{-1} = (A\cdot P+B)\cdot(C\cdot P+D)^{-1}.\end{aligned}$$

Passing through the division-ring isomorphism $l-\{\infty\}\cong\mathbb{K}$ gives the coordinate form. Thus (10.2) is indeed a composite of projective transformations, and so is a projective transformation.

Next we determine the conditions for this expression to be projective. If $C=\mathit{0}$, it reduces directly to $\varphi(P)=(A\cdot P+B)\cdot D^{-1}$. We require $A\neq\mathit{0}$, since otherwise the map is constant, and $D\neq\mathit{0}$, since otherwise it is constantly the point at infinity. Conversely, when $A,D\neq\mathit{0}$, it is a projective transformation. By exercise A.55, the condition that both $A,D$ differ from $\mathit{0}$ is equivalent to $A\cdot D\neq\mathit{0}$, that is, $ad\neq 0$.

If $C\neq\mathit{0}$, then $\varphi_1,\varphi_2,\varphi_3,\varphi_5$ are all projective transformations. Since composites of projective transformations are projective, the condition is equivalent to requiring

$$\varphi_4=\varphi_5^{-1}\circ\varphi\circ\varphi_1^{-1}\circ\varphi_2^{-1}\circ\varphi_3^{-1}$$

to be projective. This requires $B-A\cdot C^{-1}\cdot D\neq\mathit{0}$, or $b\neq ac^{-1}d$.

It remains to show that every one-dimensional projective transformation has the form (10.2). Intuitively, expressions such as $\mathrm{x}(P)^2$ and $\sqrt{\mathrm{x}(P)}$ are not projective transformations, but we cannot yet prove this rigorously. The missing proof will be supplied in section 10.4. □

**Remark.** Although (10.2) places $a,c$ on the left of the variable, the maps in lemma 10.3.12 allow multiplication on either side. Although this formula looks asymmetric, it can be reduced to other forms. For example, in the division ring the identity

$$(xc+d)^{-1}(xa+b)=(a'x+b')(c'x+d')^{-1}$$

holds identically with the following choices: If $c=0$ and $ad\neq 0$, take

$$a'=d^{-1},\quad b'=d^{-1}ba^{-1},\quad c'=0,\quad d'=a^{-1}.$$

If $c\neq 0$ and $q=b-dc^{-1}a\neq 0$, take

$$a'=c^{-1}aq^{-1},\quad b'=c^{-1}+c^{-1}aq^{-1}dc^{-1},\quad c'=q^{-1},\quad d'=q^{-1}dc^{-1}.$$

The reader can verify these identities directly.

At $P=\infty$, direct substitution into $(A\cdot\infty+B)\cdot(C\cdot\infty+D)^{-1}$ would produce the undefined product $\infty\cdot\mathit{0}$. Each step of the earlier factorization into $\varphi_i$ is nevertheless defined, so we adopt the convention

$$(A\cdot\infty+B)\cdot(C\cdot\infty+D)^{-1}=A\cdot C^{-1}.$$

We collect the earlier conventions for calculations involving $\infty$ on $l$, or equivalently on $\mathbb{K}\cup\{\infty\}$.

**Definition 10.3.15** (Operations involving the infinite element)**.** For finite points $A,B,C,D$ and their coordinates $a,b,c,d\in\mathbb{K}$, set

(1) $A+\infty=\infty+A=\infty$, or $a+\infty=\infty+a=\infty$;
(2) for $A\neq\mathit{0}$, $A\cdot\infty=\infty\cdot A=\infty$, or, for $a\neq 0$, $a\infty=\infty a=\infty$;
(3) $\mathit{0}^{-1}=\infty$ and $\infty^{-1}=\mathit{0}$, or $0^{-1}=\infty$ and $\infty^{-1}=0$;
(4) for coefficients that define a projective transformation,

$$(A\cdot\infty+B)\cdot(C\cdot\infty+D)^{-1}=A\cdot C^{-1},$$

or, in coordinates,

$$(a\infty+b)(c\infty+d)^{-1}=ac^{-1}.$$

These are coherent conventions for representing a one-dimensional projective transformation by a single parameter. They do not turn $\mathbb{K}\cup\{\infty\}$ into an ordinary algebraic extension of $\mathbb{K}$.

**Corollary 10.3.16.** Assuming A and B, *let $\varphi$ be a projective transformation of a projective line over $\mathbb{K}$. If the images of the origin, unit point, and point at infinity have projective coordinates $r,s,t$, respectively, then*

$$\mathrm{x}(\varphi(P)) = \left[t(s-t)^{-1}(r-s)\lambda\mathrm{x}(P)+r\lambda\right]\left[(s-t)^{-1}(r-s)\lambda\mathrm{x}(P)+\lambda\right]^{-1}, \tag{10.3}$$

*where $\lambda \in \mathbb{K}^{\times}$.*

**Proof.** Put $p=\mathrm{x}(P)$ and $p'=\mathrm{x}(\varphi(P))$. By theorem 10.3.14, write $p'=f(p)=(ap+b)(cp+d)^{-1}$. The conditions $f(0)=r$, $f(1)=s$, and $f(\infty)=t$ give

$$r=bd^{-1}, \qquad s=(a+b)(c+d)^{-1}, \qquad t=ac^{-1}.$$

The first and last equalities give $a=tc$ and $b=rd$. Substitution into the second gives

$$s=(tc+rd)(c+d)^{-1} \Longrightarrow sc+sd=tc+rd \Longrightarrow (s-t)c=(r-s)d.$$

Taking $d=\lambda\in\mathbb{K}^{\times}$ gives

$$d=\lambda, \quad c=(s-t)^{-1}(r-s)\lambda, \quad b=r\lambda, \quad a=t(s-t)^{-1}(r-s)\lambda,$$

which is (10.3). Direct substitution also verifies that these values give $f(0)=r$, $f(1)=s$, and $f(\infty)=t$. □

**Remark.** If one of $r,s,t,\mathrm{x}(P)$ is $\infty$, use the conventions of definition 10.3.15. For example, when $t=\infty$, formula (10.3) has

$$t(s-t)^{-1}(r-s)=-(r-s), \qquad (s-t)^{-1}(r-s)=0,$$

and therefore reduces to

$$\mathrm{x}(\varphi(P))=\left[-(r-s)\lambda\mathrm{x}(P)+r\lambda\right]\lambda^{-1}=-(r-s)\lambda\mathrm{x}(P)\lambda^{-1}+r.$$

**Corollary 10.3.17.** Assuming A and C, *let $\varphi$ be a projective transformation of $l$, now coordinatized over a field. If the images of the origin, unit point, and point at infinity have coordinates $r,s,t$, respectively, then*

$$\mathrm{x}(\varphi(P))=\frac{t(r-s)\mathrm{x}(P)+r(s-t)}{(r-s)\mathrm{x}(P)+(s-t)}. \tag{10.4}$$

**Proof.** This is corollary 10.3.16, or equivalently (10.3), with commutative multiplication. □

**Corollary 10.3.18.** Assuming A and B, *fix the image coordinates $r,s,t$ of the origin, unit point, and point at infinity in* (10.3). *The parameters $\lambda_1,\lambda_2\in\mathbb{K}^{\times}$ determine the same projective transformation if and only if*

$$\lambda_2^{-1}\lambda_1\in \mathrm{Z}(\mathbb{K})^{\times}.$$

*Equivalently, the distinct transformations are parametrized by the cosets of the quotient group*[20] $\mathbb{K}^{\times}/\mathrm{Z}(\mathbb{K}^{\times})$.

**Proof.** Write $\mathrm{x}(P)=p$, and denote the transformation with parameter $\lambda$ by $\mathrm{x}(\varphi(P))=f_\lambda(p)$. Formula (10.3) is equivalent to

$$f_\lambda(p)=\left[t(s-t)^{-1}(r-s)y+r\right]\left[(s-t)^{-1}(r-s)y+1\right]^{-1}=f_1(y), \qquad y=\lambda p\lambda^{-1}.$$

Define $\imath_\lambda:\mathbb{K}\to\mathbb{K}, p\mapsto \lambda p\lambda^{-1}$. Then $f_\lambda=f_1\circ\imath_\lambda$. Since $f_1$ is bijective,

$$\begin{aligned} f_{\lambda_1}=f_{\lambda_2} &\Longleftrightarrow f_1\circ\imath_{\lambda_1}=f_1\circ\imath_{\lambda_2}\Longleftrightarrow \imath_{\lambda_1}=\imath_{\lambda_2}\\ &\Longleftrightarrow \lambda_1p\lambda_1^{-1}=\lambda_2p\lambda_2^{-1}\quad\text{for every } p\Longleftrightarrow \lambda_2^{-1}\lambda_1p=p\lambda_2^{-1}\lambda_1\quad\text{for every } p.\end{aligned}$$

[20] Although $\mathbb{K}$ is a division ring, $\mathbb{K}^{\times}$ is here regarded as a group under multiplication, not as a ring. Hence this is a quotient group, not a quotient ring.

For $p=\infty$, the last equality holds automatically, because both sides are $\infty$. Thus it is necessary and sufficient that it hold for every $p\in\mathbb{K}$. A sufficient condition is therefore $\lambda_2^{-1}\lambda_1\in Z(\mathbb{K})$. But the allowed parameter values give $\lambda_2^{-1}\lambda_1\neq 0$, so the necessary and sufficient condition is $\lambda_2^{-1}\lambda_1\in Z(\mathbb{K})^\times$. By the definition of a coset, this condition says $\lambda_1\in\lambda_2 Z(\mathbb{K})^\times$. Hence all parameters in one coset give the same transformation, and the equivalence classes are determined by $\mathbb{K}^\times/Z(\mathbb{K})^\times$. □

**Corollary 10.3.19.** *On a projective line over a division ring $\mathbb{K}$, three prescribed pairs of distinct corresponding points determine a unique projective transformation if and only if $\mathbb{K}$ is a field. This is equivalent to the corresponding projective space satisfying C. In other words, propositions C and D are equivalent.*

**Proof.** The points are fixed but the coordinates may be chosen freely, so choose the three original points as the origin, unit point, and point at infinity. By corollary 10.3.18, the image triples determine a unique transformation precisely when, for a given $\lambda\in\mathbb{K}^\times$, every $\mu\in\mathbb{K}^\times$ satisfies $\mu^{-1}\lambda\in Z(\mathbb{K})^\times$. As the products of nonzero $\mu,\lambda$ range over all of $\mathbb{K}^\times$, this is equivalent to $Z(\mathbb{K})^\times=\mathbb{K}^\times$. The zero element also satisfies $0a=a0=0$, so it belongs to $Z(\mathbb{K})$; consequently the condition is equivalent to $Z(\mathbb{K})=\mathbb{K}$, that is, to commutativity of multiplication and hence to $\mathbb{K}$ being a field. □

**Remark.** This corollary shows that C and D are equivalent under A, completing theorem 10.1.1, subject to the promised completion of theorem 10.3.14, namely that every projective transformation has the form (10.2); that proof appears in section 10.4. Conversely, if D is assumed at the outset, the three pairs of corresponding points already determine the transformation, while the field formula (10.4) in theorem 10.3.14 supplies one carrying $0,1,\infty$ to any prescribed triple. Thus theorem 10.3.14 already gives all projective transformations.

The preceding corollary determines the form of an individual projective transformation. The formula in theorem 10.3.14 also immediately yields the following group-theoretic consequence.

**Corollary 10.3.20.** Assuming A and B, $\mathrm{Prj}(l)$ *is a group under composition.* It is called the *one-dimensional projective transformation group* , or the *general projective linear group*; when $l-\{\infty\}\cong\mathbb{K}$, it is denoted by $\mathrm{PGL}(2,\mathbb{K})$, where the numeral 2 corresponds to one dimension, or simply $\mathrm{PGL}(2)$ when $\mathbb{K}$ is understood. *Moreover,*

$$\mathrm{PGL}(2,\mathbb{K})\cong\mathrm{GL}(2,\mathbb{K})/Z(\mathrm{GL}(2,\mathbb{K})). \tag{10.5}$$

**Proof.** The group axioms are immediate, either from the definition or directly from (10.2). We prove the group isomorphism. Denote the group on the right by $\mathrm{PGL}^*(2,\mathbb{K})=\mathrm{GL}(2,\mathbb{K})/Z(\mathrm{GL}(2,\mathbb{K}))$. The following map $\sigma$ gives a group homomorphism $\mathrm{GL}(2)\sim\mathrm{PGL}(2)$:

$$\sigma:\ \mathrm{GL}(2)\longrightarrow\mathrm{PGL}(2),$$
$$\begin{bmatrix}\mathrm{x}(A) & \mathrm{x}(B)\\ \mathrm{x}(C) & \mathrm{x}(D)\end{bmatrix}\mapsto(\varphi:\ P\mapsto(A\cdot P+B)\cdot(C\cdot P+D)^{-1}).$$

Indeed, the conditions $ad\neq 0$ when $c=0$, and $b\neq ac^{-1}d$ when $c\neq 0$, are exactly the conditions for the matrix $\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ to be invertible over a division ring; see example A.5.38.

If $\varphi$ is the identity, then $\varphi(\infty)=A\cdot C^{-1}=\infty$, so $C=\mathit{0}$. Next, $\varphi(\mathit{0})=B\cdot D^{-1}=\mathit{0}$, so $B=\mathit{0}$. Then $\varphi(\mathit{1})=A\cdot D^{-1}=\mathit{1}$, so $A=D\neq\mathit{0}$. Now $\varphi(P)=A\cdot P\cdot A^{-1}$, and we additionally require $A\in Z(l-\{\infty\})$. Thus the identity condition is $B=C=\mathit{0}$ and $A=D\in Z(l-\{\infty\})$.

By theorem A.5.40, the center is $Z(\mathrm{GL}(2))=\{\lambda\mathbf{I}_2\mid\lambda\in Z(\mathbb{K})^\times\}$, exactly corresponding to the above condition $\varphi=\mathrm{id}$. Hence $\ker\sigma=Z(\mathrm{GL}(2))$, and the first isomorphism theorem gives

$$\mathrm{PGL}(2)\cong\mathrm{GL}(2)/Z(\mathrm{GL}(2))=\mathrm{PGL}^*(2).$$

□

**Remark.** Unlike the affine notation, the notation $\mathrm{PGL}(n+1)$ for an $n$-dimensional projective group records its origin in the $(n+1)$-dimensional $\mathrm{GL}(n+1)$. As with AGL, PGL emphasizes the algebraic group and Prj the geometric transformations; in the present setting $\mathrm{Prj}(l) \cong \mathrm{PGL}(2)$, so the notations may be used interchangeably when the meaning is clear.

**Corollary 10.3.21.** Assuming A and B, *every* $\varphi \in \mathrm{Aff}(l)$ *has the form*

$$\mathrm{x}(\varphi(P)) = a\mathrm{x}(P)a' + b, \qquad aa' \neq 0, \tag{10.6}$$

*for every* $P \in l$*, where* $a, a', b \in \mathbb{K}$.

**Proof.** In (10.2), $\mathrm{x}(\varphi(\infty)) = ac^{-1}$, so preserving the point at infinity requires $c = 0$. The form (10.2) then gives $\mathrm{x}(\varphi(P)) = a\mathrm{x}(P)d^{-1} + bd^{-1}$, which can always be written as (10.6). Conversely, every expression in (10.6) plainly has $\mathrm{x}(\varphi(\infty)) = \infty$. ☐

When C holds, the preceding corollary takes the following form over a field.

**Corollary 10.3.22.** Assuming A and C, *over a field every* $\varphi \in \mathrm{Aff}(l)$ *has the form*

$$\mathrm{x}(\varphi(P)) = a\mathrm{x}(P) + b, \qquad a \neq 0, \tag{10.7}$$

*for every* $P \in l$*, where* $a, b \in \mathbb{K}$.

**Proof.** Adding C makes $\mathbb{K}$ a field. Multiplication is then commutative, so the right factor $a'$ in (10.6) can be absorbed into $a$. ☐

**Corollary 10.3.23.** Assuming A and B, $\mathrm{Aff}(l)$ *is a group under composition.* It is the *one-dimensional full affine transformation group* ; when $l - \{\infty\} \cong \mathbb{K}$, it is denoted by $\mathrm{AGL_F}(1, \mathbb{K})$, where 1 denotes a one-dimensional affine transformation, or simply $\mathrm{AGL_F}(1)$ when $\mathbb{K}$ is understood.

**Proof.** This follows either from the definition or directly from the expression in (10.2). The notation Aff emphasizes the geometric structure, while $\mathrm{AGL_F}$ emphasizes the algebraic group; the two can usually be used interchangeably when the meaning is clear. ☐

**Corollary 10.3.24.** *The equality* $\mathrm{AGL_F}(1, \mathbb{K}) = \mathrm{AGL}(1, \mathbb{K})$ *holds if and only if* $\mathbb{K}$ *is a field, equivalently if and only if the corresponding projective space satisfies C.*

**Remark.** In standard algebra, affine transformations are obtained directly from linear spaces (see the later chapters), so the order of multiplication is fixed. The *general affine linear group* in algebra is therefore the perspective affine transformation group constructed geometrically above, and is likewise denoted by AGL. For the same reason, the usual algebraic notion of an affine transformation does not include two-sided multiplication of the form $axa'$. Thus $\mathrm{AGL_F}$ is specific to this geometric approach and is not a widely established notation.

We saw that the simple ratio is invariant under perspective affine transformations. For four points $A, B, C, D$, one can still formally define the *cross-ratio*

$$(AB, CD) = (B - C)^{-1}(A - C)(A - D)^{-1}(B - D).$$

Over a noncommutative division ring, no cross-ratio can be defined so as to be invariant under all full affine and projective transformations. The displayed formal expression is generally not invariant; see the exercises.

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## Exercises

**Exercise 10.14.** Let $A,B,C,D$ be four points of a projective line $l$ over a division ring $\mathbb{K}$. Let $f$ be a projective transformation of $l$, and use primes to denote corresponding points. Prove that there is an $X \in l \setminus \{0,\infty\}$ such that

$$(B-C)^{-1}(A-C)(A-D)^{-1}(B-D) = X^{-1}(B'-C')^{-1}(A'-C')(A'-D')^{-1}(B'-D')X.$$

Deduce that the cross ratio $(B-C)^{-1}(A-C)(A-D)^{-1}(B-D)$ on the left is invariant under projective transformations if and only if $\mathbb{K}$ is a field.

**Exercise 10.15.** For a prime $p$, find the order of $\mathrm{PGL}(2,\mathbb{F}_p)$.

**Exercise 10.16.** List all elements of $\mathrm{AGL}(1)$, $\mathrm{AGL_F}(1)$, and $\mathrm{PGL}(2)$ on the projective line over $\mathbb{F}_3$, and prove that $\mathrm{PGL}(2,\mathbb{F}_3) \cong \mathrm{S}_4$, where $S_4$ is the permutation group.

**Exercise 10.17*.**

(1) Prove that

$$\mathrm{AGL}(1,\mathbb{K}) \cong \mathbb{K}^+ \rtimes \mathbb{K}^\times,$$

where $\mathbb{K}^\times$ acts on $\mathbb{K}^+$ by ordinary left multiplication.

(2*) Prove that

$$\mathrm{AGL_F}(1,\mathbb{K}) \cong \mathbb{K}^+ \rtimes (\mathbb{K}^\times \rtimes \mathrm{Inn}(\mathbb{K})).$$

Use the natural actions: for $\rho \in \mathrm{Inn}(\mathbb{K})$, $u \in \mathbb{K}^\times$, and $b \in \mathbb{K}^+$, $\rho \circ u = \rho(u)$ and $(u,\rho)\circ b = u\rho(b)$. See exercise A.68 for the definition of an inner automorphism.

(3*) Let $\mathbb{H}$ be the quaternion division ring (see example A.4.17) and regard $\mathbb{R}^4$ as the additive group of the four-dimensional real vector space. If $\mathbb{R}_+$ is the multiplicative group of positive reals, prove that

$$\mathrm{AGL_F}(1,\mathbb{H}) \cong \mathbb{R}^4 \rtimes (\mathbb{R}_+ \times \mathrm{SO}(4,\mathbb{R})).$$

**Exercise 10.18*.** Let a group $G$ act on a set $X$, with action denoted by $\circ$.[21] The action is *n-transitive* if, for every two ordered $n$-tuples of distinct elements $(x_1,\dots,x_n)$ and $(y_1,\dots,y_n)$, there is a $g \in G$ such that $g\circ x_i = y_i$ for $i = 1,2,\dots,n$. It is *sharply n-transitive* if that $g$ is unique.

(1) Prove that an $n$-transitive action is sharply $n$-transitive if and only if the pointwise stabilizer of every ordered $n$-tuple of distinct elements is trivial.

(2) Prove that $\mathrm{AGL}(1,\mathbb{K})$ acts sharply 2-transitively on $\mathbb{K}$.[22]

(3) Prove that $\mathrm{AGL_F}(1,\mathbb{K})$ acts 2-transitively on $\mathbb{K}$, and that its action is sharply 2-transitive if and only if $\mathbb{K}$ is a field.

(4) Prove that $\mathrm{PGL}(2,\mathbb{K})$ acts 3-transitively on $\mathbb{P}^1(\mathbb{K})$, and sharply 3-transitively if and only if $\mathbb{K}$ is a field.

**Exercise 10.19*.** Let $\mathrm{PGL}(2,\mathbb{K})$ be the group of projective transformations of the projective line $l$. For $\sigma \in \mathrm{Aut}(\mathbb{K})$, define

$$\phi_\sigma : \mathrm{PG}(1,\mathbb{K}) \to \mathrm{PG}(1,\mathbb{K}), \qquad \boldsymbol{x}^{-1}([x_1 : x_2]) \mapsto \boldsymbol{x}^{-1}([\sigma(x_1) : \sigma(x_2)]).$$

For $g \in \mathrm{PGL}(2,\mathbb{K})$, call $g\circ\phi_\sigma$ a *projective semilinear transformation* of $l$.

(1) Prove that these transformations form the one-dimensional *projective semilinear group*, denoted by $\mathrm{P\Gamma L}(2,\mathbb{K})$.[23]

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[21] For group actions, see section A.3.3.

[22] Here and below, the group acts naturally by applying the indicated transformations directly to the elements of $\mathbb{K}$.

[23] In algebra this is also called the one-dimensional *general projective semilinear group*; by contrast, the PGL group is called the one-dimensional *general projective linear group*.

(2) Prove that $\mathrm{PGL}(2,\mathbb{K}) \trianglelefteq \mathrm{P\Gamma L}(2,\mathbb{K})$, and, when $\mathbb{K}$ is a field, that

$$\mathrm{P\Gamma L}(2,\mathbb{K}) \cong \mathrm{PGL}(2,\mathbb{K}) \rtimes \mathrm{Aut}(\mathbb{K}).$$

**Exercise 10.20***.** Consult the theory of Lie groups. Prove that $\mathrm{PGL}(2,\mathbb{R})$ and $\mathrm{PGL}(2,\mathbb{C})$ are Lie groups, with Lie algebras

$$\mathfrak{pgl}(2,\mathbb{R}) \cong \mathfrak{sl}(2,\mathbb{R}), \qquad \mathfrak{pgl}(2,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C}).$$

## 10.4 Projective Coordinates in the Plane

### Plane projective coordinates

The operations on a projective line now allow us to define projective coordinates in a two-dimensional projective plane.

**Definition 10.4.1.** Write the plane as $\mathbb{P}^2(\mathbb{K})$ or $\mathrm{PG}(2,\mathbb{K})$. Choose two finite lines $x,y$, with intersection $0 = x \cap y$, and use $0$ as the origin on both lines. Choose unit points $1_x \in x$ and $1_y \in y$. Since $x - \{\infty_x\} \cong y - \{\infty_y\} \cong \mathbb{K}$, for any finite point $P$, let $P_x = \infty_y P \cap x$ and $P_y = \infty_x P \cap y$, as in figure 10.14. The points $P_x, P_y$ correspond respectively to definite elements $p_x, p_y \in \mathbb{K}$, namely $p_x = \mathrm{x}_x(P_x)$ and $p_y = \mathrm{x}_y(P_y)$. Thus a finite point $P$ corresponds bijectively to an ordered pair $(p_x, p_y) \in \mathbb{K}^2$.

Below, write $p_x = \mathrm{x}_x(P)$, $p_y = \mathrm{x}_y(P)$, where the subscripts indicate the lines $x,y$; alternatively, write $p_x = \mathrm{x}_1(P)$, $p_y = \mathrm{x}_2(P)$, where the subscripts indicate the first and second coordinates. Write $\boldsymbol{x}(P) = (p_x, p_y)$.[24] This ordered pair is the *inhomogeneous projective coordinate* representation of $P$, or simply its *inhomogeneous coordinate* representation. The resulting assignment is a two-dimensional *projective coordinate system* .

Let $X = \infty_x$, $Y = \infty_y$, $O = 0$, and let $E = \boldsymbol{x}^{-1}(1,1)$.[25] The frame is denoted by $[O,X,Y;E]$. Its four points are the *base points*; $OXY$ is the *coordinate triangle*; $O$ and $E$ are the *origin* and *unit point*; and $OX, OY$ are the *coordinate axes*. Below we use $0$ directly for the origin of the plane projective coordinates.


(a) (b)

FIGURE 10.14

**Remark.** One point was not made explicit in the preceding definition. More precisely, the coordinates on $x,y$ must satisfy a compatibility condition. Besides fixing $0, 1_x, 1_y, \infty_x, \infty_y$, which restricts the division-ring isomorphisms $x - \{\infty_x\} \cong y - \{\infty_y\} \cong \mathbb{K}$, we must require these isomorphisms to have the following property: the joins of points on $x,y$ with the same one-dimensional projective coordinate all pass through a single fixed point, namely $1_x 1_y \cap \infty_x \infty_y$. Equivalently, the division-ring isomorphism is induced by the perspective affine correspondence in theorem 10.3.4.

[24] For example, $\boldsymbol{x}(0) = (0,0)$, $\boldsymbol{x}(1_x) = (1,0)$, and $\boldsymbol{x}(1_y) = (0,1)$. The 0 and 1 on the right denote the zero and unit elements of $\mathbb{K}$, whereas $0, 1$ on the left are point symbols.

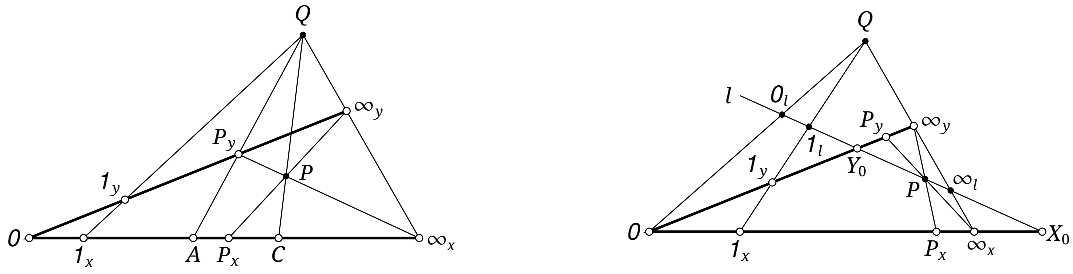
[25] Here the exponent $-1$ again denotes the inverse map: $E$ is the point corresponding to the displayed projective coordinate representation.

Hereafter $x,y$ and the points $0; 1_x, 1_y; \infty_x, \infty_y$ are fixed. We next study the conditions satisfied by points on a line.

**Theorem 10.4.2.** Assuming A and B, *let $l$ be a finite line in a projective plane. There exist $a_x,b_x,a_y,b_y \in \mathbb{K}$, with $a_x,a_y$ not both zero, such that its finite points $P$ have the parametric coordinate equations*

$$\mathrm{x}_x(P) = a_x t + b_x, \qquad \mathrm{x}_y(P) = a_y t + b_y, \tag{10.8}$$

*where $t \in \mathbb{K}$. Thus $P \in l \setminus \{\infty_l\}$ if and only if there is a $t \in \mathbb{K}$ satisfying* (10.8)*. Conversely, every system of this form parametrizes a line.*




(a) (b)

FIGURE 10.15

**Proof.** Put $Q = 1_x 1_y \cap \infty_x \infty_y$. We first show that a line has the asserted parametric form, considering two cases.

Suppose first that $Q \in l$. For any $P \in l - \{Q\}$, put $P_y = P\infty_x \cap y$, $P_x = P\infty_y \cap x$, $C = QP \cap x$, and $A = QP_y \cap x$, as in figure 10.15a. By theorem 10.3.4, $\mathrm{x}_x(A) = \mathrm{x}_y(P_y) = \mathrm{x}_y(P)$. In the definition of addition on the $x$-axis, take $AQ$ and $P_x\infty_y$ as the lines through $A$ and $P_x$, and take $y$ as the line through $0$. The definition then gives $C = A +_x P_x$, and hence

$$\mathrm{x}_x(P) + \mathrm{x}_y(P) = \mathrm{x}_x(C)$$

for every such $P$. In (10.8), choose $a_y = -a_x \neq 0$ and $b_y = \mathrm{x}_x(C) - b_x$. These choices give the preceding equation. If $P \in l - \{Q\}$, the parameter is $t = a_x^{-1}(\mathrm{x}_x(P) - b_x)$. Conversely, for any $t \in \mathbb{K}$, the first equation locates $P_x$; the intersection $\infty_y P_x \cap l$ is then the corresponding point $P$.

Now suppose $Q \notin l$. The perspectivity $\varphi : x \to l$ with center $Q$ induces a division-ring isomorphism $x - \{\infty_x\} \cong l - \{\infty_l\}$. Choose $\varphi(0) = 0_l$, $\varphi(1_x) = 1_l$, and $\varphi(\infty_x) = \infty_l$, and put $t = \mathrm{x}_l(P)$. By corollary 10.3.11, the perspectivity $l \to x$ with center $\infty_y$ has the form $\mathrm{x}_x(P) = a_x t + b_x$ for some $a_x, b_x \in \mathbb{K}$. Similarly, there are $a_y, b_y$ such that $\mathrm{x}_y(P) = a_y t + b_y$. This proves (10.8).

In particular, if $0 \in l$, the origin is fixed, so theorem 10.3.5 gives

$$\mathrm{x}_x(P) = a_x t, \qquad \mathrm{x}_y(P) = a_y t.$$

When $\mathrm{x}_x(P) = 1$, one has $t = a_x^{-1}$ and $\mathrm{x}_y(P) = a_y a_x^{-1}$. Since the $y$-coordinate at the point of a line through the origin whose $x$-coordinate is 1 may take any value, $a_y a_x^{-1}$ may be any element of $\mathbb{K} \cup \{\infty\}$.

We now prove conversely that every system (10.8) represents a line.

If $a_x + a_y = 0$, adding the two equations gives $\mathrm{x}_x(P) + \mathrm{x}_y(P) = b_x + b_y$, which is the first case above with $\mathrm{x}_x(C) = b_x + b_y$.

If $a_x = 0$ and $a_y \neq 0$, the equation reduces to $\mathrm{x}_x(P) = b_x$, which plainly represents the line $(\boldsymbol{x}^{-1}(b_x,0))\infty_y$. Likewise, $a_y = 0$ and $a_x \neq 0$ give the line $(\boldsymbol{x}^{-1}(0,b_y))\infty_x$.

Suppose now that $a_x, a_y, a_x + a_y$ are all nonzero. By (10.8), $\mathrm{x}_x(P) = 0$ corresponds to $t = -a_x^{-1} b_x$, with $\mathrm{x}_y(P) = -a_y a_x^{-1} b_x + b_y$. Similarly, $\mathrm{x}_y(P) = 0$ corresponds to $\mathrm{x}_x(P) = -a_x a_y^{-1} b_y + b_x$. Put $X_0 = \boldsymbol{x}^{-1}(x_0,0)$ and $Y_0 = \boldsymbol{x}^{-1}(0,y_0)$, where

$$x_0 = -a_x a_y^{-1} b_y + b_x, \qquad y_0 = -a_y a_x^{-1} b_x + b_y.$$

If $x_0 = y_0 = 0$, then $X_0 = Y_0 = O$, and

$$a_x^{-1} b_x = a_y^{-1} b_y := r, \qquad k = a_y a_x^{-1},$$

and the given system becomes

$$x = a_x(t + r), \qquad y = k a_x(t + r).$$

Choose a nonzero point $P$ on this set. By the standard form for a line through the origin obtained above, the line $OP$ has a parametrization

$$x = a'_x t, \qquad y = k a'_x t.$$

The change of parameter $t \mapsto a_x^{-1} a'_x t - r$ transforms the given system into this standard form.

Now suppose $x_0, y_0 \neq 0$. Since two points determine a line, it is enough to show that $X_0 Y_0$ has the given parametrization. The perspectivity with center $Q$ induces a division-ring isomorphism, so we may take $\mathrm{x}_l(X_0) = x_0$ and $\mathrm{x}_l(Y_0) = y_0$. For $P \in X_0 Y_0$, put $\mathrm{x}_l(P) = t'$. The perspectivity from $X_0 Y_0$ to the $x$-axis with center $\infty_y$ has the form

$$\mathrm{x}_x(P) = a'_x t' + b'_x.$$

When $P = Y_0$, we have $\mathrm{x}_x(P) = 0$; when $P = X_0$, we have $\mathrm{x}_x(P) = x_0$. These give the two equations

$$0 = a'_x y_0 + b'_x, \qquad x_0 = a'_x x_0 + b'_x.$$

Subtracting the equations eliminates $b'_x$, allowing us to solve for $a'_x$. Substitution back gives $b'_x$; the results are

$$a'_x = x_0(x_0 - y_0)^{-1} = a_x(a_x + a_y)^{-1}, \quad b'_x = -a'_x y_0 = b_x - a_x(a_x + a_y)^{-1}(b_x + b_y).$$

To verify the simplification of $a'_x$, calculate:

$$\begin{aligned}
a'_x = x_0(x_0 - y_0)^{-1} &= (b_x - a_x a_y^{-1} b_y)\left[(b_x - a_x a_y^{-1} b_y) - (b_y - a_y a_x^{-1} b_x)\right]^{-1} \\
&= a_x(a_x^{-1} b_x - a_y^{-1} b_y)\left[a_x(a_x^{-1} b_x - a_y^{-1} b_y) + a_y(a_x^{-1} b_x - a_y^{-1} b_y)\right]^{-1} \\
&= a_x(a_x^{-1} b_x - a_y^{-1} b_y)\left[(a_x + a_y)(a_x^{-1} b_x - a_y^{-1} b_y)\right]^{-1} \\
&= a_x(a_x^{-1} b_x - a_y^{-1} b_y)(a_x^{-1} b_x - a_y^{-1} b_y)^{-1}(a_x + a_y)^{-1} = a_x(a_x + a_y)^{-1}.
\end{aligned}$$

The condition $Q \notin X_0 Y_0$ gives $a_x + a_y \neq 0$. The analogous calculation on the $y$-axis gives

$$a'_y = a_y(a_x + a_y)^{-1}, \quad b'_y = b_y - a_y(a_x + a_y)^{-1}(b_x + b_y).$$

Finally, replacing the parameter $t'$ by $t' = (a_x + a_y)t + (b_x + b_y)$ recovers the original equations in the parameter $t$. Hence the two parametrizations are equivalent. ☐

As $t \to \infty$, the coordinate ratio satisfies $\mathrm{x}_x(P) : \mathrm{x}_y(P) = a_x : a_y$, by definition 10.3.15, independently of $b_x, b_y$.[26] This motivates homogeneous coordinates for all points of the plane.

**Definition 10.4.3.** Let $\pi = \mathbb{P}^2(\mathbb{K})$. For elements $p_1, p_2, p_3 \in \mathbb{K}$, not all zero, the expression $[p_1 : p_2 : p_3]$ is a *homogeneous projective coordinate* representation, or simply a *homogeneous coordinate* representation of a point of $\pi$, with the following meaning.

i. For every $\lambda \in \mathbb{K}^\times$, the representations $[p_1 : p_2 : p_3]$ and $[p_1\lambda : p_2\lambda : p_3\lambda]$ denote the same point; we may write $[p_1 : p_2 : p_3] = [p_1\lambda : p_2\lambda : p_3\lambda]$.
ii. If $p_3 \neq 0$, put $p'_1 = p_1 p_3^{-1}$, $p'_2 = p_2 p_3^{-1}$. Then $[p_1 : p_2 : p_3] = [p'_1 : p'_2 : 1]$ represents the unique point $P$ for which $\boldsymbol{x}(P) = (p'_1, p'_2)$.
iii. If $p_3 = 0$, then $[p_1 : p_2 : p_3]$ represents the point at infinity of a line with parametric equations

$$\mathrm{x}_x(P) = p_1 t + b_x, \qquad \mathrm{x}_y(P) = p_2 t + b_y,$$

where $p_1, p_2$ are not both zero and $b_x, b_y$ are arbitrary.

[26] The ratio of the coefficients of $t$ is also unchanged under a change of the parameter $t$.

Since homogeneous and inhomogeneous coordinates represent the same object, we also write $\boldsymbol{x}(P)=[p_1:p_2:p_3]$ for the homogeneous coordinates of $P$. For convenient component notation, choose one representative at a time and write its entries as $x_1(P), x_2(P), x_3(P)$.[27] Thus

$$\boldsymbol{x}(P)=[x_1(P):x_2(P):x_3(P)].$$

### Lines in a projective plane

The preceding coordinates give a uniform equation for a projective line.

**Theorem 10.4.4.** Assuming A and B, *for every line $l$ in a projective plane there exist $a,b,c\in\mathbb{K}$, not all zero, such that every point $P\in l$ satisfies*

$$ax_1(P)+bx_2(P)+cx_3(P)=0. \tag{10.9}$$

*Conversely, every equation of this form represents a line.*

The right-scalar convention in definition 10.4.3 is essential. In the line equation $ax_1+bx_2+cx_3=0$, replacing $[x_1:x_2:x_3]$ by $[x_1\lambda:x_2\lambda:x_3\lambda]$, $\lambda\neq 0$, gives

$$(ax_1+bx_2+cx_3)\lambda=0,$$

which is equivalent to the original equation. If instead we took $[x_1:x_2:x_3]=[\lambda x_1:\lambda x_2:\lambda x_3]$, the line equation would become

$$a\lambda x_1+b\lambda x_2+c\lambda x_3=0,$$

which is not equivalent over a noncommutative division ring. For consistency, one-dimensional homogeneous projective coordinates also use right scalar equivalence, although the distinction is barely visible there because the corresponding equations are seldom needed. The right factor $\lambda$ reflects the fact that the coefficients $a,b,c$ act on the left. As the proof below shows, this comes from the left coefficient $a$ in the perspective affine formula (10.1). As explained in section 10.3.2, reversing the order in the definition of multiplication produces an entirely parallel theory.

This construction is the *right-module convention*, and the resulting plane is a *right-module projective plane*, sometimes denoted by $\mathrm{PG}_\mathrm{R}(2,\mathbb{K})$ or $\mathbb{P}^2_\mathrm{R}(\mathbb{K})$. Reversing all conventions gives, for $\lambda\neq 0$,

$$[x_1:x_2:x_3]=[\lambda x_1:\lambda x_2:\lambda x_3], \qquad x_1a+x_2b+x_3c=0,$$

the *left-module convention* and a *left-module projective plane*, sometimes written $\mathrm{PG}_\mathrm{L}(2,\mathbb{K})$ or $\mathbb{P}^2_\mathrm{L}(\mathbb{K})$.[28] The two theories are parallel, and a left projective space over $\mathbb{K}$ is a right projective space over the opposite division ring $\mathbb{K}^{\mathrm{op}}$. Below we use the right-module convention and continue to write PG or $\mathbb{P}^n$, without a subscript.

**Proof.** We first use theorem 10.4.2 to derive (10.9) from the parametrization (10.8). For (10.8), a finite point $P\in l-\{\infty_l\}$ has a representative $[p_1:p_2:1]$, where

$$p_1=a_xt+b_x, \qquad p_2=a_yt+b_y.$$

Substitution into the equation (10.9) gives

$$0=ap_1+bp_2+c=(aa_x+ba_y)t+(ab_x+bb_y+c).$$

[27] The three entries of a homogeneous coordinate representation are not uniquely determined; this notation is only a convenient way to select them. In one context, all $x_i(P)$ belonging to a fixed point $P$ may be multiplied on the right by the same nonzero scalar.

[28] There is no universally fixed notation for left- and right-module projective planes. Subscripts L (left) and R (right) may distinguish the two, but many works fix one convention and then omit the subscript. The word "module" is borrowed from the corresponding algebraic structure: although the plane was constructed geometrically here, the result is equivalent to the algebraic left- or right-module projective plane.

For the equation to hold identically, it suffices that each parenthesized expression vanish. Since $a_x, a_y$ are not both zero, suppose without loss of generality that $a_x \neq 0$. Choose any $b \neq 0$, and set

$$a = -ba_y a_x^{-1}, \qquad c = -ab_x - bb_y.$$

Both coefficients in the preceding affine expression then vanish. The point at infinity of the line, represented by $[a_x : a_y : 0]$, satisfies

$$a \cdot a_x + b \cdot a_y + c \cdot 0 = 0.$$

Every point $[p_1 : p_2 : 0]$ on the line at infinity satisfies

$$0 \cdot p_1 + 0 \cdot p_2 + c \cdot 0 = 0,$$

so the line at infinity itself is obtained by $a = b = 0$ and $c \neq 0$.

Conversely, we derive a parametrization of the form (10.8) from a given equation (10.9). If $c = 0$, then $a, b$ cannot both be zero, and the equation reduces to $ap_1 + bp_2 = 0$. Suppose $b \neq 0$; we can then parametrize it by

$$p_1(t) = t, \qquad p_2(t) = -b^{-1}at, \qquad p_3 = 1.$$

If $c \neq 0$ and $a = b = 0$, the equation is the line at infinity. If $a, b$ are not both zero, we may again suppose $b \neq 0$, and parametrize by

$$p_1(t) = t, \qquad p_2(t) = -b^{-1}at - b^{-1}c, \qquad p_3 = 1.$$

Thus every equation (10.9) is a projective line. □

There is also a coordinate description of the span of two points.

**Theorem 10.4.5.** Assuming A and B, *let distinct points $P, Q$ have homogeneous coordinate representations $\boldsymbol{x}(P) = [p_1 : p_2 : p_3], \boldsymbol{x}(Q) = [q_1 : q_2 : q_3]$. Every point $R \in PQ$ has a representation*

$$\boldsymbol{x}(R) = [p_1 a + q_1 b : p_2 a + q_2 b : p_3 a + q_3 b] \tag{10.10}$$

*for some $a, b \in \mathbb{K}$, not both zero. Conversely, every such pair $(a, b)$ gives a point of $PQ$.*

**Proof.** We first show that every point of $PQ$ has the stated form. For convenience, let $\widetilde{\boldsymbol{P}}$ denote a column vector in $\mathbb{K}^3$ representing one choice of homogeneous coordinates of $P$; for example, $\widetilde{\boldsymbol{P}} = [p_1 \;\; p_2 \;\; p_3]^\mathsf{T}$. Write $\langle \widetilde{\boldsymbol{P}} \rangle$ for the homogeneous coordinates corresponding to this column vector. The assertion then becomes $\boldsymbol{x}(R) = \langle \widetilde{\boldsymbol{P}} a + \widetilde{\boldsymbol{Q}} b \rangle$.

If $P, Q$ are both finite, let them correspond to parameters $p, q$ in (10.8). Thus we may take

$$\widetilde{\boldsymbol{P}} = \begin{bmatrix} a_x p + b_x & a_y p + b_y & 1 \end{bmatrix}^\mathsf{T}, \quad \widetilde{\boldsymbol{Q}} = \begin{bmatrix} a_x q + b_x & a_y q + b_y & 1 \end{bmatrix}^\mathsf{T}.$$

Since the points are distinct, $p - q \neq 0$. Taking $a = (p-q)^{-1}(t-q)$ and $b = (p-q)^{-1}(p-t)$, direct calculation gives

$$\langle \widetilde{\boldsymbol{P}} a + \widetilde{\boldsymbol{Q}} b \rangle = [a_x t + b_x : a_y t + b_y : 1], \tag{$*$}$$

which is the finite point $R$ with any parameter $t$. The point at infinity is also obtained, since

$$\langle \widetilde{\boldsymbol{P}}(p-q)^{-1} - \widetilde{\boldsymbol{Q}}(p-q)^{-1} \rangle = [a_x : a_y : 0].$$

If one of the points is at infinity, suppose without loss of generality that $Q \in l_\infty$. The parametric form allows us to take

$$\widetilde{\boldsymbol{P}} = \begin{bmatrix} a_x p + b_x & a_y p + b_y & 1 \end{bmatrix}^\mathsf{T}, \qquad \widetilde{\boldsymbol{Q}} = \begin{bmatrix} a_x & a_y & 0 \end{bmatrix}^\mathsf{T}.$$

Direct calculation gives $\langle \widetilde{\boldsymbol{P}} + \widetilde{\boldsymbol{Q}}(t-p) \rangle = [a_x t + b_x : a_y t + b_y : 1]$, the finite point $R$ for any parameter $t$. The expression $\langle \widetilde{\boldsymbol{P}} \cdot 0 + \widetilde{\boldsymbol{Q}} \cdot 1 \rangle$ plainly gives the point at infinity of the line.

If $P, Q$ are both at infinity, then $PQ = l_\infty$, with $\boldsymbol{x}(P) = [p_1 : p_2 : 0]$ and $\boldsymbol{x}(Q) = [q_1 : q_2 : 0]$. All points at infinity are either of the form $[t : 1 : 0]$, with $t \in \mathbb{K}$, or the point $[1 : 0 : 0]$.

If $p_2,q_2$ are both nonzero, we may take $\widetilde{\boldsymbol{P}}=[p\ \ 1\ \ 0]^{\mathsf{T}}$ and $\widetilde{\boldsymbol{Q}}=[q\ \ 1\ \ 0]^{\mathsf{T}}$. Again $p-q\neq 0$ because the points are distinct. Taking $a=(p-q)^{-1}(r-q)$ and $b=(p-q)^{-1}(p-r)$, calculation gives $\left\langle\widetilde{\boldsymbol{P}}a+\widetilde{\boldsymbol{Q}}b\right\rangle=[r:1:0]$. The remaining point is obtained from

$$\left\langle\widetilde{\boldsymbol{P}}-\widetilde{\boldsymbol{Q}}\right\rangle=[p-q:0:0]=[1:0:0].$$

If one of $p_2,q_2$ is zero, they cannot both be zero because $P,Q$ are distinct. Suppose without loss of generality that $q_2=0$. We may take $\widetilde{\boldsymbol{P}}=[p\ \ 1\ \ 0]^{\mathsf{T}}$ and $\widetilde{\boldsymbol{Q}}=[1\ \ 0\ \ 0]^{\mathsf{T}}$. Calculation gives $\left\langle\widetilde{\boldsymbol{P}}+\widetilde{\boldsymbol{Q}}(r-p)\right\rangle=[r:1:0]$; the remaining point is $\left\langle\widetilde{\boldsymbol{P}}\cdot 0+\widetilde{\boldsymbol{Q}}\cdot 1\right\rangle$ itself.

Conversely, any $a,b$ not both zero give a point of the stated form. Indeed, suppose that $P,Q$ satisfy the line equation in theorem 10.4.4, so that $a_0p_1+b_0p_2+c_0p_3=0$ and $a_0q_1+b_0q_2+c_0q_3=0$. Then we still have

$$\begin{aligned}&a_0(p_1a+q_1b)+b_0(p_2a+q_2b)+c_0(p_3a+q_3b)\\&\quad=(a_0p_1+b_0p_2+c_0p_3)a+(a_0q_1+b_0q_2+c_0q_3)b=0.\end{aligned}$$

□

We can now complete the proof of theorem 10.3.14: every one-dimensional projective transformation has the form (10.2).

**Completion of the proof of theorem 10.3.14.** Let $l,l'$ be lines, let $O,O'$ lie off them, and let $\varphi_1: l\to l'$, $\varphi_2: l'\to l$ be the perspectivities with centers $O,O'$. Consider $\varphi=\varphi_2\circ\varphi_1$. Fix the origin, unit point, and point at infinity $\mathit{0},\mathit{1}_l,\infty_l$ on $l$; we calculate this projective transformation directly in plane coordinates.

Only the coordinates on $l$ matter in the end, so the coordinates off $l$ may be chosen freely. Choose a projective frame so that $l$ is the first coordinate axis and the origin, unit point, and point at infinity on $l$ have representations $[0:0:1],[1:0:1],[1:0:0]$. We may also arrange $\boldsymbol{x}(O)=[1:1:0],\boldsymbol{x}(\infty_{l'})=[0:1:0]$. Indeed this is the frame $[\mathit{0},\infty_l,\infty_{l'};E]$, where $E=\mathit{0}O\cap\mathit{1}_l\infty_{l'}$.[29]

If $O'$ is at infinity, the map is perspective affine and already has the form (10.2). Otherwise write $\boldsymbol{x}(O')=[u:v:1],\boldsymbol{x}(A)=[a:0:1]$, where $A=l\cap l'$. For $Q\in l$, put $q=\mathrm{x}_l(Q)$, so one homogeneous representation is $\boldsymbol{x}(Q)=[q:0:1]$. Let $Q_1=\varphi_1(Q)=OQ\cap l'$ and $Q_2=\varphi_2(Q_1)=O'Q_1\cap l$.

The line $l'=A\infty_{l'}$ has equation $x_1-ax_3=0$, whereas $QO$ has equation $x_1-x_2=qx_3$. Hence $\boldsymbol{x}(Q_1)=(a,a-q)$. Thus $Q_1$ lies on both $l'$ and $OQ$. By theorem 10.4.5, for $\lambda,\mu\in\mathbb{K}$, the homogeneous coordinates of a general point of $O'Q_1$ are

$$\left\langle\widetilde{\boldsymbol{O}'}\lambda+\widetilde{\boldsymbol{Q}_1}\mu\right\rangle=\left\langle\begin{bmatrix}u & v & 1\end{bmatrix}^{\mathsf{T}}\cdot\lambda+\begin{bmatrix}a & (a-q) & 1\end{bmatrix}^{\mathsf{T}}\cdot\mu\right\rangle=[u\lambda+a\mu:v\lambda+(a-q)\mu:\lambda+\mu].$$

For this point to lie on $l$, we require

$$x_2(Q_2)=v\lambda+(a-q)\mu=0\quad\Rightarrow\quad\lambda=v^{-1}(q-a)\mu.$$

Substitution gives the homogeneous coordinate representation

$$\boldsymbol{x}(Q_2)=[a+uv^{-1}(q-a):0:1+v^{-1}(q-a)].$$

Therefore the induced transformation on $l$ satisfies

$$\mathrm{x}_l(\varphi(Q))=\left[uv^{-1}\mathrm{x}_l(Q)+(1-uv^{-1})a\right]\left[v^{-1}\mathrm{x}_l(Q)+(1-v^{-1}a)\right]^{-1}.\qquad(\clubsuit)$$

This is of the form (10.2). A general projective transformation is a finite composite of two-perspectivity maps of the form (♣). As in the proof of corollary 10.3.7, their fractional-linear coordinate expressions (10.2) are closed under composition, completing the proof. □

The same coordinate discussion could be extended to three-dimensional projective space, but we do not pursue it because of space and because the present book is concerned primarily with plane geometry.

[29]The coordinates of $\mathit{0},\infty_l,\infty_{l'}$ are then as displayed. Since the unit point $E$ has coordinates $[1:1:1]$, the line $\mathit{0}E$ has equation $x_1-x_2=0$ and hence point at infinity $[1:1:0]=\boldsymbol{x}(O)$. Likewise, $\mathit{1}_lE$ has equation $x_1-x_3=0$ and therefore passes through $[0:1:0]=\boldsymbol{x}(\infty_{l'})$.

### Exercises

**Exercise 10.21.** In a projective plane over a division ring, with projective frame $[O,X,Y;E]$:
1. Find the inhomogeneous coordinate representations of $O$ and $E$.
2. Write homogeneous coordinate representations of the four base points.
3. Give parametric equations of the form (10.8) for the two coordinate axes and for $OE, EX, EY$.
4. Give equations of the form (10.9) for the three sides of the coordinate triangle. How is the line at infinity represented?

**Exercise 10.22.**
1. Verify $(*)$ directly.
2. Supply the final step in the proof of theorem 10.3.14: prove explicitly that a general projective transformation is a composite of transformations of the form (♣).
3. In (10.8), replace the parameter by $t' = ut + v$, where $u \in \mathbb{K}^\times$ and $v \in \mathbb{K}$. Verify that the point at infinity determined by the new parametrization still has homogeneous coordinate representation $[a_x : a_y : 0]$.

**Exercise 10.23.** Extend the real Euclidean plane to the real projective plane, whose circular points are $I, J$. Let finite points $A, B, C$ form an equilateral triangle with centroid $G$. In the projective frame $[A,B,C;G]$, find homogeneous coordinate representations of $I, J$.

**Exercise 10.24.** Over a field, suppose $X_i$ has homogeneous coordinate representation $[a_i : b_i : c_i]$ for $i = 1,2,3$. Prove that $X_1, X_2, X_3$ are collinear if and only if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

**Exercise 10.25.** In the right-module quaternionic projective plane (see example A.4.17) $\mathbb{P}^2_{\mathrm{R}}(\mathbb{H})$, let the coordinate representations of $A, B, C, D$, respectively, be

$$[1+\mathbf{i} : \mathbf{j}+\mathbf{k} : -2], \quad [\mathbf{i} : \mathbf{k} : 1], \quad [-\mathbf{k} : 2 : 1], \quad [\mathbf{j}+1 : \mathbf{i} : 0].$$

1. Find an equation of the form (10.9) for $AB$, and the homogeneous coordinate representation of its point at infinity.
2. Find the coordinate representation of $AB \cap CD$.
3. How do the answers change in the left-module plane $\mathbb{P}^2_{\mathrm{L}}(\mathbb{H})$ with the same displayed data?

**Exercise 10.26*.** In a right-module projective plane over a general division ring, the points of lines $l_i$ satisfy

$$a_i x_1(P) + b_i x_2(P) + c_i x_3(P) = 0, \qquad i = 1,2.$$

Discuss the possible explicit coordinate representations of $l_1 \cap l_2$.

## 10.5 Axiomatic Plane Geometry*

### 10.5.1 Basic Axioms

The basic incidence axioms A and the relation between B and C have already been introduced. We now examine these axioms more closely.

### Desargues' theorem

The preceding discussion shows that the requirement $l - \{\infty_l\} \cong K$, for a division ring $K$, forces us to adopt Desargues' theorem as an axiom. The following plane satisfies the basic incidence axioms but not Desargues' theorem.

**Example 10.5.1.** In the ordinary Euclidean coordinate plane $\mathbb{R}^2$, consider a broken line $\overline{BAC}$ as in figure 10.16a, where $A$ lies on the $x$-axis and the slope $k_{AC}$ of $AC$ is the cube of the slope $k_{AB}$ of $AB$, that is, $k_{AC} = k_{AB}^3$. Declare the following curves to be the "lines" of a new plane: all ordinary lines parallel to the $x$-axis, all ordinary lines parallel to the $y$-axis, and all broken lines $\overline{BAC}$ satisfying the condition above. Adjoin one common point at infinity to each family of pairwise disjoint lines, and declare all these points at infinity to form a line at infinity. This produces a projective plane.

Verification of the basic axioms is left to the reader. Desargues' theorem, however, does not always hold. In the original $Oxy$-coordinates, take $X(-1,0), Y(-2,0), Z(0,-1)$ and $X'(1,1), Y'(2,1), Z'(0,2)$. In the ordinary Euclidean plane, $\triangle XYZ$ and $\triangle X'Y'Z'$ are clearly centrally symmetric about $Q(0,1/2)$, so $XX', YY', ZZ'$ are concurrent. In the new plane, $ZZ'$ is still the $y$-axis, and neither segment $XX'$ nor segment $YY'$ bends, because each remains on one side of the $x$-axis. Thus $XX', YY', ZZ'$ are concurrent.

The slope of $XZ$ is $-1$, whose cube is again $-1$, so $XZ$ also remains an ordinary line; hence $XZ$ and $X'Z'$ meet at a point at infinity. Likewise, $XY$ and $X'Y'$ are horizontal and meet at a point at infinity. Originally $YZ \parallel Y'Z'$, but $YZ$, whose slope is $-\frac{1}{2}$, bends, whereas the corresponding new line through $Y'$ and $Z'$ has a different point at infinity. Their intersection is therefore finite. The three intersections of corresponding sides are not collinear, so Desargues' theorem fails.

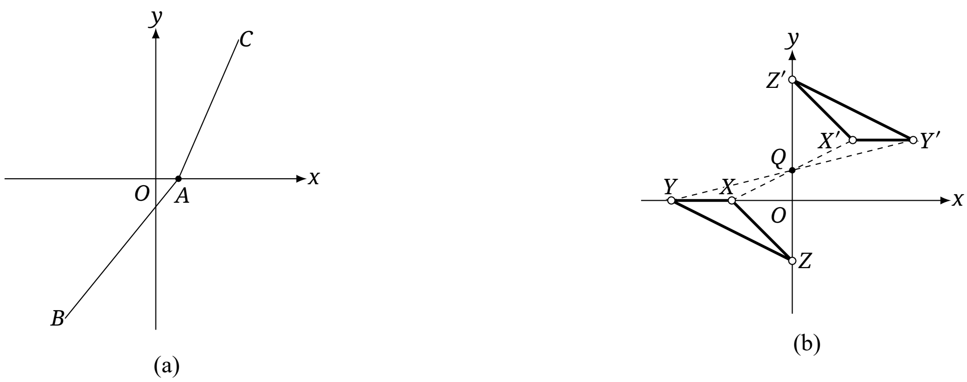



FIGURE 10.16

This example is a variant of the so-called *Moulton plane*; see the exercises. Both constructions replace some ordinary straight lines by bent ones. A projective plane satisfying Desargues' theorem is a *Desarguesian projective plane*; one that does not is a *non-Desarguesian projective plane*. The Moulton plane is the classical example of the latter. In modern algebraic geometry and linear algebra, "projective plane" usually means a Desarguesian plane; in combinatorial, finite, and axiomatic geometry it need not. Unless stated otherwise, projective planes below are Desarguesian.

### Pappus' theorem

Pappus' theorem was used above to prove commutativity of multiplication, and it cannot in fact be derived from Desargues' theorem.

**Example 10.5.2.** Take $K$ to be the quaternion division ring $\mathbb{H}$; see section A.4.1. Pappus' theorem need not hold in the projective plane $\mathbb{HP}^2$ over $\mathbb{H}$. This plane is called the *quaternionic projective plane*.

**Remark.** The most direct verification uses homogeneous coordinates. The reader may check explicitly that Pappus' configuration need not close.

Thus $K$ is a field only when Pappus' theorem is added as an axiom. For example, when $K = \mathbb{R}, \mathbb{C}$, the familiar real and complex projective planes are

$$\mathbb{RP}^2 := \mathbb{P}^2(\mathbb{R}), \qquad \mathbb{CP}^2 := \mathbb{P}^2(\mathbb{C}).$$

Pappus' theorem holds in both planes. A projective plane satisfying Pappus' theorem is a *Pappian projective plane*; otherwise it is *non-Pappian*. Our geometric definition of division on a line used commutativity, so it is unavailable on a line of a non-Pappian projective plane.

**Theorem 10.5.3.** *Every finite Desarguesian projective plane is Pappian.*

**Proof.** Such a plane is coordinatized by a finite division ring. Wedderburn's little theorem says that every finite division ring is a field, and a projective plane over a field is Pappian. □

### The Fano axiom

Besides A–D, the following axiom is sometimes included in an axiomatic treatment of projective planes.

> E. (Fano axiom) The three diagonal points of every complete quadrangle are noncollinear. Equivalently, if no three of $A, B, C, D$ are collinear, then
>
> $$AB \cap CD, \qquad AC \cap BD, \qquad AD \cap BC$$
>
> are noncollinear.

*Fano axiom*

We next determine which coordinatized planes satisfy this axiom.

**Theorem 10.5.4.** *For a projective plane $\mathbb{P}^2(\mathbb{K})$ over a division ring, the Fano axiom fails if and only if* $\operatorname{char}\mathbb{K} = 2$.[30]

**Proof.** For the complete quadrangle $ABCD$, as in figure 10.17a, put $O = AC \cap BD$, $P = AB \cap CD$, and $Q = AD \cap BC$. On the line $l = OP$, let $F = QA \cap l$ and $E = QB \cap l$. Choose $O, E, P$ as the origin, unit point, and point at infinity on $l$. By the earlier result on the additive inverse, $F = -E$, so $\mathrm{x}(F) = \mathrm{x}(-E) = -\mathrm{x}(E) = -1$. Since $E = OP \cap QB$, $F = OP \cap QA$, and $Q = QA \cap QB$, we have

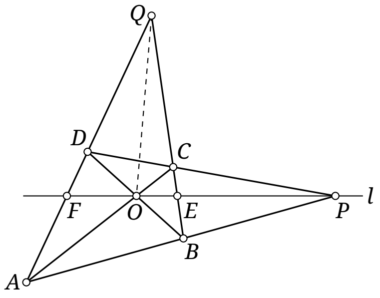
$$\begin{aligned} O, P, Q \text{ collinear} &\iff PQ = OP \iff E = F = Q \\ &\iff \mathrm{x}(E) = \mathrm{x}(F) \iff -1 = 1 \iff 1 + 1 = 0 \iff \operatorname{char}\mathbb{K} = 2. \end{aligned}$$

□

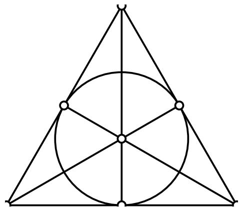
**Example 10.5.5.** Let $K = \mathbb{F}_q$, where $q$ is a positive integral power of a prime; see section A.4.3. This gives the projective plane $\mathrm{PG}(2, q) := \mathbb{P}^2(\mathbb{F}_q)$. Since $\mathbb{F}_q$ is finite, this projective plane has finitely many points. A projective plane with finitely many points is a *finite projective plane*.

For $q = 2$, the plane $\mathbb{P}^2(\mathbb{Z}_2)$ is the *Fano plane*. It has seven points and seven lines, with three points on every line. Figure 10.17b depicts a model in which each drawn segment and the circle represent a line of the Fano plane. Since $\operatorname{char}\mathbb{F}_2 = 2$, the Fano plane does not satisfy the Fano axiom.

[30]The characteristic $\operatorname{char}\mathbb{K}$ is defined in definition A.4.12.




(a)



(b)

In a plane satisfying E we may define harmonic ranges without first introducing division.

**Definition 10.5.6.** Let char $\mathbb{K} \neq 2$, and let $A,B,C$ be collinear on $l$. In the projective coordinate system $[A,C,B]$, the point $D$ with inhomogeneous coordinate $-1$ is the *fourth harmonic point* of $A,B,C$. Equivalently, $A,B,C,D$ form a harmonic range.

Earlier, harmonic ranges were defined through cross-ratios. The definition above extends the notion to projective planes over arbitrary division rings. Since a general cross ratio involves division, it is awkward to define over a noncommutative division ring, but a harmonic range can still be defined.

An axiom system may be chosen freely provided that it is consistent, so one may instead adopt the anti-Fano axiom:

$\overline{\text{E}}$. (Anti-Fano axiom) The three diagonal points of every complete quadrangle are collinear. Equivalently, for four points $A,B,C,D$, no three collinear, the points $AB\cap CD, AC\cap BD, AD\cap BC$ are collinear.

*anti-Fano axiom*

**Corollary 10.5.7.** *A Desarguesian projective plane satisfies the anti-Fano axiom* $\bar{\text{E}}$ *if and only if its coordinating division ring has characteristic* 2*.*

### Basic axioms of an affine plane

An *affine plane* is an incidence structure of points and lines satisfying the following basic axioms.

A1$^0$. Two distinct points determine a unique line.
A2$^0$. There exist three noncollinear points.
F$^0$. If $P \notin l$, there exists a unique line through $P$ disjoint from $l$.

Axiom F$^0$ is the *affine parallel axiom*; two disjoint affine lines are *parallel*. Axioms A1$^0$–A2$^0$ are the *incidence axioms*, collectively denoted by A$^0$.

Affine and projective planes pass naturally into one another. From a projective plane $\pi$, delete a line $l_\infty$, all its points, and the point $l\cap l_\infty$ from every remaining line $l$. The result is an affine plane, and two resulting affine lines are parallel precisely when the corresponding projective lines meet on $l_\infty$. Conversely, adjoin one point at infinity to each parallel class of an affine plane and declare all the new points to form a line $l_\infty$. The result is the *projective completion* of the affine plane. The reader may verify directly that each construction satisfies the basic axioms of the other kind of plane.

Thus an affine plane may be viewed as a projective plane with one line removed. The affine plane over $\mathbb{K}$ is denoted by $\mathbb{A}^2(\mathbb{K})$ or AG$(2,\mathbb{K})$.

**Theorem 10.5.8.** *There are natural identifications*

$$\mathbb{A}^2(\mathbb{K}) \sqcup l_\infty \cong \mathbb{P}^2(\mathbb{K}), \qquad \mathbb{A}^2(\mathbb{K}) \cong \mathbb{P}^2(\mathbb{K}) - l_\infty.$$

*Under this identification, a line $l$ of $\mathbb{P}^2(\mathbb{K})$ with $l \neq l_\infty$ gives the affine line in $\mathbb{P}^2(\mathbb{K})$ $l - (l \cap l_\infty)$. Two affine lines are parallel if and only if the corresponding projective lines meet on $l_\infty$.*

For convenience, one may equivalently regard an affine plane as its projective completion together with a distinguished line at infinity. This was the simplified convention used in chapter 3. As an incidence structure it is then a projective plane with additional data: affine lines are the lines other than $l_\infty$, with their points at infinity removed, and affine transformations are the projective transformations preserving $l_\infty$.

In homogeneous coordinates we may take $l_\infty$ to have equation $x_3 = 0$, in the form of (10.9). Every finite point has a unique inhomogeneous coordinate representation $(x_1,x_2)$, with $(x_1,x_2) = [x_1 : x_2 : 1]$. A point of $\mathbb{A}^2(\mathbb{K})$ can therefore be represented by $(X_1,X_2)$, where $X_1,X_2 \in \mathbb{K}$. These are its *affine coordinates*. A projective line other than $l_\infty$, written $ax_1 + bx_2 + cx_3 = 0$, with $a,b,c \in \mathbb{K}$, therefore becomes the affine line whose coordinate representations satisfy $aX_1 + bX_2 + c = 0$, where $a,b$ are not both zero. Affine geometry is not our main subject, so below we use it only in this projective-completion sense.

### The principle of duality

Interchanging "point" and "line," and replacing "a point lies on a line" by "a line passes through a point," gives the dual forms of the incidence axioms.

A1*. Two distinct lines have a unique point of intersection.
A2*. A line passes through every pair of distinct points.
A3*. At least three lines pass through every point.
A4*. There exist three nonconcurrent lines.

Write A* for A1*–A4*, and denote the duals of B, C, and E by B*, C*, and E*, respectively.

**Theorem 10.5.9.** *For projective planes,* A *is equivalent to* A*. *Assuming* A*, the pairs* B,B*, C,C*, *and* E,E* *are likewise equivalent.*

**Proof.** Assume A. By A2, two distinct lines $l,m$ meet. If they had two distinct common points $P,Q$, A1 would make both equal to the unique line through $P,Q$, so $l = m$, a contradiction. Thus their intersection is unique, proving A1*. A2* is the existence part of A1.

For A3*, fix a point $P$. By A4 choose noncollinear points $A,B,C$. If $P \neq A$, at least one of $AB,AC$ avoids $P$, since otherwise they would have two common points $A,P$. If $P = A$, then $P \notin BC$. Thus some line $l$ avoids $P$. By A3 choose three distinct points $X,Y,Z \in l$. The required three lines are $PX,PY,PZ$. They are distinct: if they coincided, A1 would make $P,X,Y,Z$ collinear, contrary to $P \notin l$. This proves A3*.

For A4*, choose noncollinear $A,B,C$. If $AB,AC,BC$ were concurrent at $P$, then $P \in AB \cap AC$, and A1 would give $P = A$, forcing $A \in BC$, a contradiction. Thus those three lines are nonconcurrent. Conversely, assume A*. Dualize the proof just given by interchanging "point" and "line," "join two points" and "intersect two lines," and "collinear" and "concurrent." This proves the reverse implication; the detailed verification is left as an exercise.

Assuming A, the converse of Desargues' theorem was derived from Desargues' theorem in theorem 3.4.3; this gives B$\Rightarrow$B*, and duality gives the reverse implication. The equivalence of C and C* also follows from the stronger coordinate result below; a synthetic proof is left as an exercise.

Finally, assuming A, suppose E* holds; we prove E, the Fano axiom. For a complete quadrangle $ABCD$, let its diagonal points be $O,P,Q$, as in figure 10.18. In the complete quadrilateral formed by

$AB,BC,CD,DA$, the three diagonal lines are $AC,BD,PQ$. Since $AC\cap BD=O$, axiom E* gives $O\notin PQ$. Hence $O,P,Q$ are noncollinear and E holds. The converse is the dual argument. □

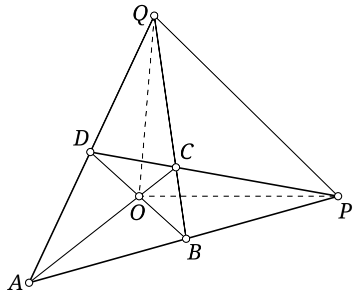



FIGURE 10.18

**Theorem 10.5.10** (Principle of duality). *Let a projective plane satisfy a self-dual axiom system: if all its axioms are collectively denoted by T, then the dual axioms T* also hold. For any statement involving only points, lines, and incidence, form its dual by the operation "*": send each point to a new line, each line to a new point, and reverse incidence, so that $P\in l$ becomes $l^*\in P^*$. The original and dual statements then have the same truth value.*

For Desarguesian and Pappian planes there are more concrete forms.

**Theorem 10.5.11** (Principle of duality). *Let $\mathbb{P}^2(\mathbb{K})$ be a Desarguesian projective plane. For statements involving only points, lines, and incidence, form the dual by the operation "*": replace each point by a new line, each line by a new point, and reverse incidence, so that $P\in l$ becomes $l^*\in P^*$. A statement involving only points, lines, and incidence holds in $\mathbb{P}^2(\mathbb{K})$ if and only if its dual holds in $\mathbb{P}^2(\mathbb{K}^{\mathrm{op}})$, where $\mathbb{K}^{\mathrm{op}}$ is the opposite division ring.*

**Proof.** By theorem 10.4.4, every line has an equation $ax_1+bx_2+cx_3=0$, with $a,b,c$ not all zero, and the replacement $(a,b,c)\to(\lambda a,\lambda b,\lambda c)$, with $\lambda\neq 0$, does not change the line. Send the point $[x:y:z]$ to the line

$$x\circ' x_1+y\circ' x_2+z\circ' x_3=0$$

and the line $ax_1+bx_2+cx_3=0$ to $[a:b:c]$, where $a\circ' b=ba$ is multiplication in $\mathbb{K}^{\mathrm{op}}$; the product on the right is the original product in $\mathbb{K}$. The original incidence condition is $ax+by+cz=0$, while the dual one is

$$0=x\circ' a+y\circ' b+z\circ' c=ax+by+cz.$$

Thus incidence, and consequently every incidence statement, is preserved. □

**Corollary 10.5.12** (Principle of duality). *In a Pappian projective plane, for statements involving only points, lines, and incidence, form the dual by the operation "*": replace each point by a new line, each line by a new point, and reverse incidence, so that $P\in l$ becomes $l^*\in P^*$. The original statement holds if and only if its dual holds.*

**Proof.** The coordinating division ring is a commutative field, hence $\mathbb{K}=\mathbb{K}^{\mathrm{op}}$, so this is immediate from theorem 10.5.11. □

For real and complex projective planes an earlier version of duality was realized by a polarity. The axiomatic argument above both gives another version and extends it to arbitrary division rings. The form in theorem 10.5.10 requires the dual of every axiom satisfied by the plane also to hold. In theorem 10.5.11 and corollary 10.5.12, Desargues' or Pappus' axiom is sufficient for the respective duality principle for incidence statements, even when further axioms are added. An arbitrary additional incidence axiom T need not imply T* under A.

**Example 10.5.13.** The *Hall plane*[31] is a non-Desarguesian plane satisfying A and is not self-dual: interchanging points and lines produces a plane with a different structure. It has 91 points $P_1,\dots,P_{91}$ and 91 lines $l_1,\dots,l_{91}$. Fix incidence indicators $\epsilon_{ij}$, with $\epsilon_{ij}=1$ exactly when $P_i \in l_j$, and $\epsilon_{ij}=0$ otherwise.

The complete structure can be encoded as an additional axiom T: these are exactly the 91 points $P_1,\dots,P_{91}$ and 91 lines $l_1,\dots,l_{91}$, and $P_i \in l_j$ precisely when $\epsilon_{ij}=1$, with $\epsilon$ fixed in advance as above. Because the plane is not self-dual, T* does not hold in the same plane.

Order and continuity axioms will be introduced below only for perspective; they are not a central topic of this course. Despite the variety of planes in this section, the next two chapters use "projective plane" primarily for $\mathbb{RP}^2$, with occasional extensions to $\mathbb{CP}^2$.

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## Exercises

**Exercise 10.27.** Supply the following details:
(1) Verify that the construction in example 10.5.1 satisfies the basic axioms of a projective plane.
(2) Verify that example 10.5.2 is non-Pappian.
(3) Write out explicitly the implication A*⇒A in theorem 10.5.9.

**Exercise 10.28.** Give a purely synthetic proof, without division-ring coordinates, that C and C* are equivalent under A.

**Exercise 10.29.** How many points and lines does $\mathrm{PG}(2,q)$ have? How many points lie on each line?

**Exercise 10.30.** In the polynomial ring $\mathbb{F}_2[x]$, consider the quotient ring $F=\mathbb{F}_2[x]/\langle x^2+x+1\rangle$.[32] Prove that $F$ is a field and determine its number of elements. Does $\mathbb{P}^2(\mathbb{F})$ satisfy the Fano axiom?

**Exercise 10.31.** In a Desarguesian projective plane satisfying the Fano axiom, prove that $A,B,C,D$ is a harmonic range if and only if $B,A,C,D$ is harmonic, and if and only if $A,B,D,C$ is harmonic.

**Exercise 10.32.** If a statement holds in the quaternionic projective plane $\mathbb{HP}^2$, must its dual also hold?

**Exercise 10.33.** Prove that the following construction gives a non-Desarguesian projective plane, called the *Moulton plane*. In the real Euclidean plane $\mathbb{R}^2$, using Cartesian coordinates $Oxy$, declare the new affine lines to be

$$\begin{aligned} &\mathrm{I}: && x=a, \quad a\in\mathbb{R},\\ &\mathrm{II}: && y=mx+b, \quad m\ge 0,\ b\in\mathbb{R},\\ &\mathrm{III}: && y=\begin{cases} mx+b, & x\le 0,\\ 2mx+b, & x\ge 0,\end{cases} \quad m<0,\ b\in\mathbb{R}. \end{aligned}$$

All lines of class I form one parallel class. Two lines of class II, or two of class III, are parallel exactly when their parameters $m$ agree; lines in different classes are not parallel. Adjoin one common point at infinity to every parallel class, and declare all these new points to form a line at infinity.

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[31] Its detailed construction is deferred to an exercise; here the point is its failure of self-duality, so the construction is not developed in the main text.

[32] Angle brackets denote the principal ideal; see exercise A.62.

**Exercise 10.34*.** Formulate the duals of A', B, C, and E in three-dimensional projective space, and investigate their equivalence to the original axioms.

**Exercise 10.35*.** Let $\mathbb{F}_3 = \{0,1,2\}$, adjoin an element $\theta$ satisfying $\theta^2 = 2$, and put $\mathbb{F}_9 = \{a + b\theta \,|\, a,b \in \mathbb{F}_3\}$, with multiplication

$$(a+b\theta)(c+d\theta) = (ac+2bd) + (ad+bc)\theta.$$

Prove that this is the field of order nine and that $\mathbb{F}_9 \cong \mathbb{F}_3[x]/\langle x^2+1\rangle$; see exercise A.62.

Construct a projective plane as follows. Its finite points are the pairs $(x,y)$, with $x,y \in \mathbb{F}_9$. Define two classes of affine lines:

$$\text{I}: \quad L_{m,b} = \{(x, mx+b) \,|\, x \in \mathbb{F}_9\}, \quad m \in \mathbb{F}_9 \setminus \mathbb{F}_3, \quad b \in \mathbb{F}_9;$$

$$\text{II}: \quad B_{u;r,s} = \{(u(a+r\theta), u(b+s\theta)) \,|\, a,b \in \mathbb{F}_3\}, \quad r,s \in \mathbb{F}_3, \quad u \in U,$$

where $U = \{1, \theta, 1+\theta, 1+2\theta\}$. Informally, in the $9 \times 9$ grid of finite points, retain some of the ordinary class-I lines, but replace the lines with slopes $m = 0, 1, 2, \infty$ by class-II lines.

For each $m$, adjoin one common point at infinity $I_m$ to all $L_{m,b}$; for each $u$, adjoin one common point $J_u$ to all $B_{u;r,s}$. Declare all $I_m, J_u$ to form the line at infinity $l_\infty$. This is the *Hall plane* of order nine. Prove that it

(1) satisfies A but is non-Desarguesian;
(2) is not self-dual: interchanging points and lines gives a plane inequivalent to the original;
(3) has 91 points and 91 lines;
(4) satisfies the Fano axiom.

### 10.5.2 Order and Continuity Axioms*

We now sketch several further axioms. Their ideas, rather than all technical proofs, are the point of this optional section.

#### Projective order axioms

For four distinct collinear points $A,B,C,D$, introduce the relations "*separates*" and "*does not separate*." Write $(A,B) \div (C,D)$ and $(A,B) \mathrel{\ddot{-}} (C,D)$, respectively, according as the pair $(C,D)$ separates or does not separate $(A,B)$.

In the real projective line, $(A,B) \div (C,D)$ may be pictured by the cyclic order $A,C,B,D$, with $(AB,CD) < 0$. But $\mathbb{P}^1 \cong \mathbb{S}^1/(\boldsymbol{x} \sim -\boldsymbol{x})$, so a projective line is circle-like, as explained in section 3.1.1, and it is inappropriate to say simply that $C$ lies "between" $A$ and $B$. If $A,C,B,D$ occur in cyclic order, so do $C,B,D,A$ and $B,D,A,C$, for example. This explains why $(C,D) \div (A,B)$ and $(B,A) \div (C,D)$ express the same relation. Formally, separation is governed by the following axioms.

G1. For any three distinct points $A,B,C$ of a line, there is a fourth point $D$ such that $(A,B) \div (C,D)$.
G2. If $(A,B) \div (C,D)$, then $(B,A) \div (C,D)$ and $(C,D) \div (A,B)$. Thus separation is independent of the order within either pair and of the order of the two pairs.
G3. Four distinct collinear points admit exactly one partition into two mutually separating pairs.
G4. If five distinct collinear points satisfy $(C,D) \div (A,B)$ and $(C,E) \div (A,B)$, then $(D,E) \mathrel{\ddot{-}} (A,B)$.
G5. If five distinct collinear points satisfy $(C,D) \mathrel{\ddot{-}} (A,B)$ and $(C,E) \mathrel{\ddot{-}} (A,B)$, then $(D,E) \mathrel{\ddot{-}} (A,B)$.
G6. A perspectivity between lines preserves separation of corresponding point pairs.

These are the *projective order axioms*, collectively denoted by G. A projective plane satisfying G is an *ordered projective plane*.

Once separation is fixed, define the order relation $>$ on $l-\{\infty_l\}$.

**Definition 10.5.14.** On a line $l$ in a projective plane over a division ring, choose $O,E,U$ as origin, unit point, and point at infinity, so the projective coordinate system is $[O,E,U]$. Require:
(1) $E>O$.
(2) The order relation "$<$" satisfies $X>Y$ if and only if $Y<X$.
(3) If $P\in l\setminus\{O,E,U\}$, then $P>O$ when $(E,P)\doteq(O,U)$, and $P<O$ when $(E,P)\div(O,U)$.
(4) With subtraction on the projective line defined as before, for any two points $X,Y$, put $X>Y$ when $X-Y>O$, and $X<Y$ when $X-Y<O$.
These rules determine the order of all points in $l-\{U\}$.

**Proposition 10.5.15.** *For $X,Y,Z\in l\setminus\{U\}$, the preceding order satisfies:*
*(1) exactly one of $X>Y$, $X<Y$, and $X=Y$ holds;*
*(2) if $X>O$ and $Y>O$, then $X+Y>O$;*
*(3) if $X>Y>Z$, then $X>Z$;*
*(4) if $X>Y$, then $X+Z>Y+Z$;*
*(5) if $X>O$ and $Y>O$, then $X\cdot Y>O$;*
*(6) if $X>Y$ and $Z>O$, then $X\cdot Z>Y\cdot Z$.*

**Proof.** For (1), take $P\in l-\{U\}$. If $P=O$, clause (3) of the definition does not apply, so neither $P<O$ nor $P>O$ holds. If $P=E$, then $P>O$, and neither $P=O$ nor $P<O$ holds. If $P\neq O,E$, axiom G3 gives either $(E,P)\div(O,U)$ or $(E,P)\doteq(O,U)$; clause (3) therefore gives either $P>O$ or $P<O$. Since $X-Y\in l-\{U\}$, this argument together with clause (4) proves (1).

For (2), $X>O$, so clause (4) gives $-X=O-X<O$, and hence $(O,U)\div(E,-X)$. The hypothesis gives $(O,U)\doteq(E,Y)$. If $(O,U)\doteq(Y,-X)$, axiom G5 would imply $(O,U)\doteq(E,-X)$, a contradiction. Thus $(O,U)\div(Y,-X)$.

By lemma 10.3.12, with $X$ fixed, the map $P\mapsto P+X$ is a projective transformation, hence a composite of perspectivities. It therefore preserves separation by G6. Thus $(O+X,U+X)\div(X+Y,X-X)$, so $(X,U)\div(X+Y,O)$. Axiom G3 gives $(O,U)\doteq(X,X+Y)$. Together with $(O,U)\doteq(X,E)$, G5 gives $(O,U)\doteq(E,X+Y)$, and hence $X+Y>O$.

For (3), the hypotheses give $X-Y>O$ and $Y-Z>O$, so $X-Z=(X-Y)+(Y-Z)>O$, as required.

For (4), the hypothesis gives $X-Y>O$, so $(X+Z)-(Y+Z)=X-Y>O$, as required.

For (5), the hypothesis gives $(O,U)\doteq(E,X)$. By lemma 10.3.12, $P\mapsto P\cdot X^{-1}$ is a projective transformation and a composite of perspectivities. Axiom G6 therefore gives

$$(O\cdot X^{-1},U\cdot X^{-1})\doteq(E\cdot X^{-1},X\cdot X^{-1}),$$

that is, $(O,U)\doteq(X^{-1},E)$; by G2, $(O,U)\doteq(E,X^{-1})$. The hypothesis also gives $(O,U)\doteq(E,Y)$, so G5 gives $(O,U)\doteq(X^{-1},Y)$. Since $P\mapsto X\cdot P$ is also a projective transformation,

$$(X\cdot O,X\cdot U)\doteq(X\cdot X^{-1},X\cdot Y),$$

that is, $(O,U)\doteq(E,X\cdot Y)$, and hence $X\cdot Y>O$.

For (6), the hypothesis gives $X-Y>O$, so $X\cdot Z-Y\cdot Z=(X-Y)\cdot Z>O$, as required. □

We can now describe the further conditions on $\mathbb{P}^2(\mathbb{K})$ imposed by the order axioms. First introduce an ordered division ring.

**Definition 10.5.16.** A division ring $K$ equipped with a strict total order $<$ (see section A.1.2 for the definition), satisfying irreflexivity, transitivity, and trichotomy, is an *ordered division ring* if, for all $x,y,z\in K$,
(1) (Compatibility with addition.) $x<y$ implies $x+z<y+z$;
(2) (Compatibility with multiplication.) $x>0$ and $y>0$ imply $xy>0$.
The relation $x<y$ may equivalently be written $y>x$. When $K$ is a field, it is called an *ordered field*.

**Theorem 10.5.17.** *The projective plane* $\mathbb{P}^2(\mathbb{K})$ *is ordered if and only if* $\mathbb{K}$ *is an ordered division ring.*

**Proof.** The preceding proposition proves that $\mathbb{K}$ is ordered in an ordered projective plane $\mathbb{P}^2(\mathbb{K})$. We now prove that the projective plane constructed over an ordered division ring $\mathbb{K}$ satisfies the order axioms. For four points $A,B,C,D$ on a line $l$, define $(A,B) \div (C,D)$ to mean that there is a one-dimensional projectivity $h$ with $h(A)=0_l$, $h(B)=\infty_l$, and with one of the elements $\mathrm{x}(h(C)),\mathrm{x}(h(D))$ of $\mathbb{K}$ positive and the other negative.

This is independent of the choice of $h$. Indeed, if $h'$ has the same normalization, then $u=h'h^{-1}$ fixes the origin and infinity. By corollary 10.3.21,

$$\mathrm{x}(u(P))=a\,\mathrm{x}(P)\,a' \qquad (P\in l;\ a,a'\in\mathbb{K}^\times).$$

This expression either preserves all signs simultaneously or reverses all of them, so the fact that the two coordinates have opposite signs is unchanged. The order axioms follow from this definition; their direct verification is left as an exercise. ☐

### The projective continuity axiom

**Theorem 10.5.18.** *Every ordered division ring $K$ is* dense*: for any distinct $a,b\in K$ with $a<b$, there is a $c\in K$ such that $a<c<b$.*

**Proof.** Since $K$ is ordered, its unit element satisfies $1>0$, so it contains $2=1+1>0$. Take $c=2^{-1}(a+b)$; it is easy to verify that this satisfies the required inequalities. ☐

Density does not exclude gaps: the ordered field $\mathbb{Q}$, for example, can still be enlarged to $\mathbb{R}$. We therefore add the following axiom to G.

H. Fix $O,E,U$ on a line and use $[O,E,U]$ to define the order $>$ as above. Partition all points other than $U$ into two classes such that
(1) every point belongs to exactly one class;
(2) each class is nonempty;
(3) every point of the second class is greater than every point of the first.
Then either the first class has a greatest point or the second has a least point.

Axiom H is the *projective continuity axiom*. It is the projective counterpart of a Dedekind cut, one rigorous route to the real numbers. Readers unfamiliar with this construction may simply understand that a line satisfying H has no gaps and skip the details. A line of an ordered projective plane, with its point at infinity removed, is isomorphic to an ordered division ring, so we can consider partitions of that ring directly. For example, splitting $\mathbb{Q}$ into

$$\{q\in\mathbb{Q}\mid q<\sqrt{2}\} \quad \text{and} \quad \{q\in\mathbb{Q}\mid q>\sqrt{2}\}$$

gives two classes neither of which has a least or a greatest element, so $\mathbb{Q}$ fails H and has gaps. The following result is stated without a full proof.

**Theorem 10.5.19.** *If* $\mathbb{P}^2(\mathbb{K})$ *satisfies ABGH, then* $\mathbb{K}\cong\mathbb{R}$*, and the plane is isomorphic to* $\mathbb{R}\mathbb{P}^2$. Such a $\mathbb{K}$ is sometimes called a *complete field*.

### Axioms of an affine plane

An affine plane satisfies the basic axioms $\mathrm{A}^0$ and $\mathrm{F}^0$. We now describe its order and continuity axioms.

On an affine line, separation becomes the relation of one point being *between*, or *not between*, two others. Write $A\overset{*}{C}B$ when $C$ lies between $A$ and $B$. For distinct $A,B$, define the *segment*

$$AB = \{A,B\} \cup \{C \mid A\overset{*}{C}B\}.$$

The points $A,B$ are its *endpoints*, and deleting them leaves the *open segment AB*.

From the projective-completion viewpoint, if $U$ is the point at infinity on the corresponding projective line, then

$$(A,B) \div (C,U) \iff A\overset{*}{C}B.$$

Putting one of the four points in the projective order axioms at infinity leads to the following intrinsic affine axioms.

G1$^0$. If $A\overset{*}{B}C$, then $C\overset{*}{B}A$.
G2$^0$. Given distinct $A,B$ on a line $l$, there is at least one $C \in l$ such that $A\overset{*}{B}C$.
G3$^0$. Among any three distinct points of a line, at most one lies between the other two.
G4$^0$. Let $A,B,C$ be noncollinear. Suppose that a line $l$ contains none of them and that $l \cap AB$ lies between $A$ and $B$. Then $l$ meets exactly one of the two open segments $BC$ and $AC$.

These are the *affine order axioms*, collectively G$^0$. Axiom G4$^0$ is also called *Pasch's axiom*.[33] Viewing the affine plane as a projective plane minus one line derives these from the projective order axioms. More precisely, G1$^0$–G3$^0$ are reformulations of G1–G3, G4$^0$ comes from G6, and the affine forms of G4–G5 follow from G1$^0$–G3$^0$. The details are left as an exercise.

An order on an affine line may now be specified as follows.

Fix points $O,E$ on the line. Require $E > O$. For $P \notin \{O,E\}$, put $P > O$ if either $O\overset{*}{P}E$ or $O\overset{*}{E}P$, and put $P < O$ if $P\overset{*}{O}E$.

This is exactly the affine form of the projective order. It follows that an affine plane satisfying A$^0$B$^0$F$^0$G$^0$ is an affine plane over an ordered division ring.[34]

Order also defines a *ray*. If $P \in l$, the two rays of $l$ with *endpoint P* are

$$\{Q \in l \mid Q \geq P\}, \qquad \{Q \in l \mid Q \leq P\}.$$

Deleting the endpoint gives a *open ray*, also called one *side* of $P$ on $l$.

Segments similarly define the sides of a line. If $A,B \notin l$, they are on the *same side* of $l$ when the open segment $AB$ misses $l$, and on *opposite sides* when it meets $l$. Fixing $A \notin l$, one side of $l$ is the set of all points on the same side as $A$; this set is called a *side* of $l$. Pasch's axiom proves that every line has exactly two sides; the proof is left as an exercise. Each is an *open half-plane*; its union with $l$ is a *closed half-plane*.

The affine continuity axiom is obtained by dropping the qualification that the points under consideration exclude the point at infinity.

H$^0$. On a line with fixed points $O,E$, use G$^0$ to define the order $>$. If all points are partitioned into two nonempty classes, every point belonging to exactly one class, and every point of the second class being greater than every point of the first, then either the first class has a greatest point or the second has a least point.

[33] In many texts, the unqualified phrases "*order axioms*" and "*continuity axiom*" mean their affine forms.
[34] Here B$^0$ denotes the purely affine form of Desargues' theorem, whose statement is not repeated.

This is the *affine continuity axiom*. Every affine plane satisfying $\mathrm{A}^0\mathrm{B}^0\mathrm{F}^0\mathrm{G}^0\mathrm{H}^0$ is isomorphic to the real affine plane.

**Axioms of Euclidean geometry**

In addition to incidence and order, a Euclidean plane requires a metric relation on segments and angles, namely congruence. We use the following axioms.

$\mathrm{A}^0$. The affine incidence axioms.
$\mathrm{B}^0$. The affine form of Desargues' theorem.
F. For every line $l$ and point $A\notin l$, at most one line through $A$ is disjoint from $l$.
$\mathrm{G}^0$. The affine order axioms.
I1. If $A,B\in l$, and $A'$ lies on the same or another line $l'$, then on either prescribed side of $A'$ along $l'$ there is a point $B'$ such that the segment $AB$ is *congruent* to $A'B'$, written $AB\cong A'B'$.
I2. If $AB\cong A''B''$ and $A'B'\cong A''B''$, then $AB\cong A'B'$.
I3. If $A\overset{*}{B}C$, $A'\overset{*}{B}{}'C'$, $AB\cong A'B'$, and $BC\cong B'C'$, then $AC\cong A'C'$.
I4. Two rays $h,k$ with a common endpoint form an *angle*, denoted by $\angle(h,k)$. Given $\angle(h,k)$, a specified half-plane bounded by a line $l'$, and a prescribed ray $h'\subset l'$ with endpoint $O'$, there is exactly one ray $k'$ from $O'$ in that half-plane such that $\angle(h,k)$ is *congruent* to $\angle(h',k')$, written $\angle(h,k)\cong\angle(h',k')$. Also $\angle(h,k)\cong\angle(h,k)$.
I5. For noncollinear $A,B,C$, let $\angle BAC$ denote the angle between the ray with endpoint $A$ through $B$ and the ray with endpoint $A$ through $C$. Given another three distinct noncollinear points $A',B',C'$, if
$$AB\cong A'B',\qquad AC\cong A'C',\qquad \angle BAC\cong\angle B'A'C',$$
then $\angle ABC\cong\angle A'B'C'$ and $\angle ACB\cong\angle A'C'B'$.

Axiom F is the *parallel axiom*; disjoint lines are called *parallel*. Once congruence is available, the existence of parallels follows naturally, so the "unique" formulation in $\mathrm{F}^0$ need not be assumed. Axioms I1–I5 are the *congruence axioms*, collectively I. A plane satisfying $\mathrm{A}^0\mathrm{B}^0\mathrm{F}\mathrm{G}^0\mathrm{I}$ is a *Euclidean plane*.

Although I1 does not state that $B'$ is unique, uniqueness follows. If a second point $C'$ on the prescribed ray also worked, I2 would give $A'B'\cong A'C'$. Suppose $B'\neq C'$, arrange the names so that $B'$ lies between $A'$ and $C'$, and take $D\notin A'B'$.

Axioms I1–I2 imply that every segment is congruent to itself: I1 gives another segment congruent to it, and I2 transfers congruence back to the original segment. Thus $A'D\cong A'D$. The rays $A'B'$ and $A'C'$ coincide; hence I4 gives $\angle B'A'D\cong\angle C'A'D$. Apply I5 to $\triangle A'B'D$ and $\triangle A'C'D$, using $A'B'\cong A'C'$, $A'D\cong A'D$, and $\angle B'A'D\cong\angle C'A'D$, to obtain $\angle A'DB'\cong\angle A'DC'$. Since $B',C'$ are on the same side of $A'D$, the points $D,B',C'$ lie in the same closed half-plane bounded by $A'D$. Together with $\angle A'DB'\cong\angle A'DC'$, I4 makes $B',C'$ lie on the same ray with endpoint $D$. Thus $D,B',C'$ are collinear. But $B',C'$ also lie on $A'B'$, which forces $D\in A'B'$, a contradiction. Hence $B'=C'$. □

We already know that $\mathrm{A}^0\mathrm{B}^0\mathrm{F}^0\mathrm{G}^0$ yields affine geometry over an ordered division ring. The congruence axioms sharpen this description.

**Theorem 10.5.20.** An ordered field $F$ is called a *Pythagorean field* if, for all $x,y\in F$, there is a nonnegative $z\in F$ with $x^2+y^2=z^2$; write $z=\sqrt{x^2+y^2}$. *Every Euclidean plane satisfying $A^0B^0FG^0I$ is an affine plane $\mathbb{A}^2(\mathbb{F})$ over some Pythagorean field $\mathbb{F}$, equipped with a suitable congruence relation.*

**Proof outline.** Because of space, only the main steps are included; interested readers may also consult the relevant literature.

First, I gives the usual elementary results: triangle congruence, existence and uniqueness of right angles, construction of a perpendicular through a point, congruence of right triangles, and the basic theory of isosceles triangles. Lines perpendicular to the same line supply parallel lines; together with F this proves the full affine parallel axiom $F^0$. Hence the plane is $\mathbb{A}^2(\mathbb{K})$ over an ordered division ring.

Second, define the "projection action" of an acute angle $\alpha$ on a segment $m$. Construct a right triangle having hypotenuse $m$ and acute angle $\alpha$ between the hypotenuse and one leg; the length of that leg depends only on $\alpha$ and $m$, by I4–I5. Denote it by $\alpha\circ m$.[35] For any acute angles $\alpha,\beta$ and segment $m$, one must prove

$$\alpha\circ(\beta\circ m)\cong\beta\circ(\alpha\circ m).$$

Use the configuration of figure 10.19a. Rays $a,b,c,d$ start at $O$, with the open rays $b,c,d$ on one side of the line of $a$, and $b,d$ on opposite sides of the line of $c$. Arrange

$$\angle(a,b)\cong\alpha,\quad \angle(a,c)\cong\beta,\quad \angle(c,d)\cong\alpha;$$

then $\angle(b,d)\cong\beta$. If $\alpha\cong\beta$, this is immediate. Otherwise, we may suppose without loss of generality that $b$ lies between $a$ and $c$. Take $P\neq O$ on $a$, draw the lines $PX\perp b$ and $PY\perp c$, with feet $X,Y$, and put $m=OP$. Thus $OX\cong\alpha\circ m$ and $OY\cong\beta\circ m$.

Ordinarily one observes that $O,P,X,Y$ are concyclic and hence $\angle OXY\cong\angle OPY$. Working from the axioms, one can avoid first giving a rigorous definition of a circle: define the midpoint of a segment, take the midpoint $M$ of $OP$, and use congruent triangles to prove that the median to the hypotenuse of a right triangle equals half the hypotenuse. Then $OM\cong PM\cong YM\cong XM$, so $\triangle XMY$ is isosceles, and its base-angle relation proves $\angle OXY\cong\angle OPY$. Let the line $XY$ meet $d$ at $Z$. In $\triangle ZXO$, the angles $\angle OXZ$ and $\angle XOZ$ are known, which shows that $\angle OZX$ is right. Therefore $OZ\cong\alpha\circ(OY)\cong\beta\circ(OX)$, and therefore

$$\alpha\circ(\beta\circ m)\cong\beta\circ(\alpha\circ m).$$


(a) (b)

**Proof outline, continued.** Third, we show that $\mathbb{K}$ is commutative. Use the affine form of Pappus' theorem shown in figure 10.19b. Let $l,l'$ meet at $O$, take $A,B,C\in l$ and $A',B',C'\in l'$, none equal to $O$, and assume

$$CB'\parallel BC',\qquad CA'\parallel AC'.$$

It remains to prove $BA'\parallel AB'$.

Write the segments $OA,OB,OC$ as $a,b,c$, and the lengths of $OA',OB',OC'$ as $a',b',c'$. From $O$, drop a perpendicular to $CB'$; let its relevant acute angles with $l'$ and $l$ be $\lambda,\lambda'$. Define $\mu,\mu'$ similarly from $CA'$, and $\nu,\nu'$ from $BA'$. Uniqueness of right angles and half-planes makes these choices unambiguous. Perpendicularity gives $\lambda\circ b'\cong\lambda'\circ c$ from $CB'$; because $CB'\parallel BC'$, we also obtain $\lambda\circ c'\cong\lambda'\circ b$. Similarly, $CA'$ gives $\mu\circ a'\cong\mu'\circ c$, while $CA'\parallel AC'$ gives $\mu\circ c'\cong\mu'\circ a$. Finally, $BA'$ gives $\nu\circ a'\cong\nu'\circ b$. The last relation gives

$$\lambda'\circ\mu\circ\nu\circ a'\cong\lambda'\circ\mu\circ\nu'\circ b.$$

On the left, commutativity together with $\mu\circ a'\cong\mu'\circ c$ and $\lambda'\circ c\cong\lambda\circ b'$ reduces the expression to $\lambda\circ\mu'\circ\nu\circ b'$. On the right, commutativity together with $\lambda'\circ b\cong\lambda\circ c'$ and $\mu\circ c'\cong\mu'\circ a$ reduces it to $\lambda\circ\mu'\circ\nu'\circ a$. Hence

$$\lambda\circ\mu'\circ\nu\circ b'\cong\lambda\circ\mu'\circ\nu'\circ a.$$

[35] Thus congruent choices $\alpha'\cong\alpha$ and $m'\cong m$ produce congruent legs.

The equal angle actions denoted by $\circ$ can be cancelled, giving $\nu \circ b' \cong \nu' \circ a$, which proves $BA' \parallel AB'$. This weak affine form of Pappus' theorem, together with Desargues' theorem, is equivalent to commutativity of $\mathbb{K}$. Its projective completion satisfies Pappus' theorem.

Finally, in affine coordinates over the resulting field, if $A, B$ have coordinates $(X_1, X_2)$ and $(Y_1, Y_2)$, respectively, define

$$\operatorname{dist}(A,B) = \sqrt{(X_1 - Y_1)^2 + (X_2 - Y_2)^2}$$

and

$$\cos \angle BAC = \frac{\operatorname{dist}(A,B)^2 + \operatorname{dist}(A,C)^2 - \operatorname{dist}(B,C)^2}{2 \operatorname{dist}(A,B) \cdot \operatorname{dist}(A,C)}.$$

Declare

$$AB \cong A'B' \iff \operatorname{dist}(A,B) = \operatorname{dist}(A',B'),$$
$$\angle BAC \cong \angle B'A'C' \iff \cos \angle BAC = \cos \angle B'A'C'.$$

One then verifies that these two functions are well defined and that the resulting system satisfies the preceding axioms. □

**Remark.** If I5 is omitted, strengthen I1 by requiring the point $B'$ to be unique; call the result I1'. Axioms I1' and I2–I4 together form I', the *weak congruence axioms*. Requiring only $\mathrm{A}^0\mathrm{B}^0\mathrm{F}^0\mathrm{G}^0\mathrm{I}'$ may produce geometries over noncommutative division rings, such as a *skew Laurent-series division ring*; we do not develop those examples here.

Since $\mathbb{R}$ is Pythagorean, adjoining continuity immediately gives the following.

**Corollary 10.5.21.** *Every Euclidean plane satisfying $A^0B^0FG^0I$ and the continuity axiom $H^0$ is a Euclidean plane over $\mathbb{R}$, hence the real Euclidean plane.*

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## Exercises

**Exercise 10.36.** Supply the following details:

(1) derive the affine order axioms from the projective order axioms;
(2) state axiom $\mathrm{B}^0$ explicitly;
(3) state explicitly the affine form $\mathrm{C}^0$ of the projective axiom C;
(4) verify directly that the separation relation defined in theorem 10.5.17 satisfies the separation axioms;
(5) prove that every affine line has exactly two sides;
(6*) give a complete proof of theorem 10.5.20.

**Exercise 10.37.** Prove that every ordered projective plane satisfies the Fano axiom; that is, under A, prove G⇒E.

**Exercise 10.38** (Archimedean proposition (Archimedes, Ἀρχιμήδης))**.** Starting from the axioms, prove that the ordered division ring $\mathbb{K}$ of a projective plane satisfying the projective continuity axiom has the following property: for $a, b \in \mathbb{K}$ with $a > 0$, there is $n \in \mathbb{N}$ such that $na \geq b$, where $\geq$ means $>$ or $=$.

**Exercise 10.39.** Use the Archimedean proposition to prove that the ordered division ring coordinatizing a plane satisfying ABGH must be a field.

**Exercise 10.40.** Under $A^0B^0FG^0I$, let $A,B,C$ and $A',B',C'$ be two triples of distinct noncollinear points. Prove that

$$AB \cong A'B', \quad AC \cong A'C', \quad \angle BAC \cong \angle B'A'C' \quad \Longrightarrow \quad BC \cong B'C'.$$

**Exercise 10.41**.** Remove the line at infinity from the Moulton plane of exercise 10.33. Show that, after a suitable metric structure is added, the resulting affine plane fails $B^0$ but satisfies the weak congruence axioms I'.

## 10.6 Two-Dimensional Projective Transformations*

### 10.6.1 Three-Dimensional Projective Coordinates: An Overview

We briefly develop three-dimensional projective space in order to understand two-dimensional projective transformations. Its incidence axioms were given at the beginning of this chapter. A three-dimensional projective space satisfying A' automatically satisfies the Desargues property B, although Pappus' theorem still does not follow from A'. Unless stated otherwise, every result in this section requires only A'.

The operations on projective lines remain valid in space, so every line with its point at infinity removed is isomorphic to a division ring $\mathbb{K}$. Write the space as $\mathrm{PG}(3,\mathbb{K})$ or $\mathbb{P}^3(\mathbb{K})$. Just as the line at infinity is denoted by $l_\infty$, denote the plane at infinity by $\Pi_\infty$, and the part at infinity of a plane $\pi$ by $\infty_\pi$.

**Definition 10.6.1.** Let $\pi,\pi'$ be distinct planes in one three-dimensional projective space.

(1) If $O \notin \pi \cup \pi'$, the bijection

$$\varphi : \pi \longrightarrow \pi', \qquad A \mapsto AO \cap \pi',$$

is a *perspectivity* of planes, with *center of perspectivity* $O$.

(2) A finite composite of perspectivities is a *projectivity*; a projectivity from a plane to itself is a *projective transformation*.

(3) A projectivity carrying points at infinity to points at infinity is an *affine correspondence*; when its source and target are the same plane it is an *affine transformation*.

(4) A perspectivity whose center is a point at infinity is a *perspective affine correspondence*; equivalently, it is both a perspectivity and an affine correspondence. A self-correspondence obtained as a finite composite of these is a *perspective affine transformation*.

Use

$$\mathrm{Prs}(\pi,\pi'), \quad \mathrm{Prj}(\pi,\pi'), \quad \mathrm{Aff}(\pi,\pi'), \quad \mathrm{PAff}(\pi,\pi')$$

for the corresponding sets, and $\mathrm{Prj}(\pi), \mathrm{Aff}(\pi), \mathrm{PAff}(\pi)$ for the three types of transformations of $\pi$.

As in one dimension, the form of a perspective affine transformation can be derived without first constructing spatial coordinates. The proof, however, differs from the one-dimensional argument and relies chiefly on invariants of the transformation.

**Theorem 10.6.2.** *Three distinct noncollinear finite points and three distinct noncollinear image points determine a unique perspective affine transformation of a plane.*

**Proof.** Choose $O,X,Y$ in the source plane. For any finite point $P$, put $P_X = \infty_{OY}P \cap OX, P_Y = \infty_{OX}P \cap OY$. Primes denote images. Once $O',X',Y'$ are fixed, preservation of simple ratios from corollary 10.3.10 determines $P'_X$ and $P'_Y$ through

$$(P'_X O' X') = (P_X O X), \qquad (P'_Y O' Y') = (P_Y O Y).$$

Preservation of collinearity makes $P'$ at most unique, proving uniqueness. The construction in the proof of theorem 3.3.10 proves existence. Although the axiom system had not been introduced there, that proof uses only the basic incidence axioms for points, lines, and planes, and therefore remains valid. □

**Theorem 10.6.3.** *For a plane $\pi$, let $\varphi \in \mathrm{PAff}(\pi)$. There are $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \in \mathrm{GL}(2,\mathbb{K})$ and $c_1, c_2 \in \mathbb{K}$ such that every finite point $P \in \pi$ satisfies*

$$\mathrm{x}_1(\varphi(P)) = a_1\mathrm{x}_1(P) + b_1\mathrm{x}_2(P) + c_1, \quad \mathrm{x}_2(\varphi(P)) = a_2\mathrm{x}_1(P) + b_2\mathrm{x}_2(P) + c_2. \tag{10.11}$$

*Conversely, every transformation having this form is perspective affine.*

**Proof.** Choose projective coordinates. Suppose the origin and the two unit points have image representations

$$\boldsymbol{x}(\varphi(\mathit{0})) = (r_1, r_2), \quad \boldsymbol{x}(\varphi(\mathit{1}_x)) = (s_1, s_2), \quad \boldsymbol{x}(\varphi(\mathit{1}_y)) = (t_1, t_2).$$

Substitution in (10.11) gives the unique formula

$$\begin{aligned} \mathrm{x}_1(\varphi(P)) &= (s_1 - r_1)\mathrm{x}_1(P) + (t_1 - r_1)\mathrm{x}_2(P) + r_1, \\ \mathrm{x}_2(\varphi(P)) &= (s_2 - r_2)\mathrm{x}_1(P) + (t_2 - r_2)\mathrm{x}_2(P) + r_2. \end{aligned} \tag{10.12}$$

By theorem 10.6.2, the perspective affine transformation is at most unique. It is therefore enough to show that each map of the form (10.11) is perspective affine. Denote the map given by this formula by $\varphi$, and verify the required properties.

Substitution of (10.11) into the line parametrization of theorem 10.4.2 shows that the map preserves lines and the line at infinity. In particular, the point $X = \boldsymbol{x}^{-1}(x, 0)$ on the $x$-axis maps to $X' = \boldsymbol{x}^{-1}(a_1x + c_1, a_2x + c_2)$, and $x' = \varphi(x)$ remains a line. Having obtained $X'$, use $\psi \in \mathrm{PAff}(x', x)$ to project $X'$ back to the $x$-axis from a point $Q \in l_\infty$ distinct from $\infty_x, \infty_{x'}$; A3 guarantees such a point. Because $Q \notin x$, one has $x_2(Q) \neq 0$, so choose the homogeneous representation $\boldsymbol{x}(Q) = [q:1:0]$. By theorem 10.4.5, a representative calculation gives

$$\begin{bmatrix} a_1x + c_1 & a_2x + c_2 & 1 \end{bmatrix} - \begin{bmatrix} q & 1 & 0 \end{bmatrix}(a_2x + c_2) = \begin{bmatrix} (a_1 - qa_2)x + (c_1 - qc_2) & 0 & 1 \end{bmatrix}.$$

Thus

$$\boldsymbol{x}(\psi(X')) = ((a_1 - qa_2)x + (c_1 - qc_2), 0).$$

By theorem 10.3.5, $\psi \circ (\varphi|_x) : X \mapsto \psi(X')$ is a perspective affine transformation. The map $\psi$ is itself perspective affine by definition; hence $\varphi|_x$ is a perspective affine correspondence and therefore preserves simple ratios. The same argument shows that $\varphi|_y$ is perspective affine and preserves simple ratios.

For any finite $P$, its projections $P_x = \infty_y P \cap x, P_y = \infty_x P \cap y$ are therefore mapped with the required ratio preservation. Since (10.11) preserves lines,

$$\varphi(P) = \varphi(P_x)\varphi(\infty_y) \cap \varphi(P_y)\varphi(\infty_x).$$

The uniqueness in theorem 10.6.2 completes the proof. □

**Corollary 10.6.4.** *The set* $\mathrm{PAff}(\pi)$*, for* $\pi = \mathbb{P}^2(\mathbb{K})$*, is a transformation group.* It is the *two-dimensional perspective affine transformation group* and is denoted by $\mathrm{AGL}(2,\mathbb{K})$, or $\mathrm{AGL}(2)$ when $\mathbb{K}$ is understood.

The preceding result follows directly from the definitions. As in one dimension, AGL emphasizes the algebraic description, whereas PAff emphasizes the geometric concept. The notations may be interchanged when the meaning is clear, since $\mathrm{PAff}(\pi) \cong \mathrm{AGL}(2)$.

Following the route of section 10.4, we next construct coordinates in $\mathbb{P}^3(\mathbb{K})$. For noncollinear points $A, B, C$, write $ABC$ for their plane. For $P \notin l$, write $Pl = lP$ for the unique plane through $P$ and

$l$.[36] If two lines $l,m$ meet, write $lm$ for their plane.[37] Brackets such as $\underbrace{ABC}$, $\underbrace{Pl}$, or $\underbrace{lm}$ will be used when ambiguity is possible.

**Definition 10.6.5.** Write three-dimensional projective space as $\mathbb{P}^3(\mathbb{K})$ or $\mathrm{PG}(3,\mathbb{K})$. Choose three noncoplanar lines $x,y,z$ through a point $\mathit{0}=x\cap y\cap z$. Use $\mathit{0}$ as the origin on each, and choose unit points $\mathit{1}_x, \mathit{1}_y, \mathit{1}_z$ and points at infinity $\infty_x, \infty_y, \infty_z$. Then

$$x-\{\infty_x\}\cong y-\{\infty_y\}\cong z-\{\infty_z\}\cong \mathbb{K}.$$

For a finite point $P$, define

$$P_x=\underbrace{\infty_y\infty_z P}\cap x, \qquad P_y=\underbrace{\infty_z\infty_x P}\cap y, \qquad P_z=\underbrace{\infty_x\infty_y P}\cap z.$$

The points $P_x,P_y,P_z$ correspond to unique division-ring elements $p_x=\mathrm{x}_x(P_x), p_y=\mathrm{x}_y(P_y), p_z=\mathrm{x}_z(P_z)$. Thus finite points $P$ correspond bijectively to ordered triples $(p_x,p_y,p_z)\in\mathbb{K}^3$. Below, write $p_x=\mathrm{x}_x(P_x)$, $p_y=\mathrm{x}_y(P_y)$, and $p_z=\mathrm{x}_z(P_z)$, where the subscripts indicate the corresponding axes $x,y,z$; alternatively, write $p_x=\mathrm{x}_1(P)$, $p_y=\mathrm{x}_2(P)$, and $p_z=\mathrm{x}_3(P)$, where the subscripts indicate the first, second, and third coordinates. Write $\boldsymbol{x}(P)=(p_x,p_y,p_z)$.[38] This is the *inhomogeneous projective coordinate representation* of $P$, or simply its inhomogeneous coordinate representation. The construction is a three-dimensional *projective coordinate system* .

If $X,Y,Z$ denote the three points at infinity, $O=\mathit{0}$, and $E=\boldsymbol{x}^{-1}(1,1,1)$, write the frame as $[O,X,Y,Z;E]$. These five points are the *base points*; $O$ and $E$ are the *origin* and *unit point*, and $OX,OY,OZ$ are the *coordinate axes*. Below, $\mathit{0}$ denotes the origin of spatial projective coordinates.

**Remark.** As in the plane, one point in the preceding definition needs to be made more precise: the coordinate systems on $x,y,z$ must be compatible. Fixing $\mathit{0}, \mathit{1}_x, \mathit{1}_y, \mathit{1}_z, \infty_x, \infty_y, \infty_z$ restricts the division-ring isomorphisms $x-\{\infty_x\}\cong y-\{\infty_y\}\cong z-\{\infty_z\}\cong\mathbb{K}$. In addition, these isomorphisms must have the following property. Equal one-dimensional coordinates on $x,y$ must join through $\mathit{1}_x\mathit{1}_y\cap\infty_x\infty_y$; those on $y,z$ must join through $\mathit{1}_y\mathit{1}_z\cap\infty_y\infty_z$; and those on $z,x$ must join through $\mathit{1}_z\mathit{1}_x\cap\infty_z\infty_x$. Equivalently, the division-ring isomorphism between every two axes must be induced by a perspective affine correspondence as in theorem 10.3.4.

Hereafter the lines $x,y,z$ and the corresponding points $\mathit{0}; \mathit{1}_x, \mathit{1}_y, \mathit{1}_z; \infty_x, \infty_y, \infty_z$ are fixed. We next study the conditions satisfied by points on a line or a plane.

**Theorem 10.6.6.** *For every finite line $l$ in three-dimensional projective space there are coefficients $a_x,b_x,a_y,b_y,a_z,b_z\in\mathbb{K}$, where $a_x,a_y,a_z$ are not all zero, such that its finite points $P$ are parametrized by*

$$\mathrm{x}_x(P)=a_xt+b_x, \qquad \mathrm{x}_y(P)=a_yt+b_y, \qquad \mathrm{x}_z(P)=a_zt+b_z, \tag{10.13}$$

*where $t$ is a parameter. Thus $P\in l-\{\infty_l\}$ if and only if there is a $t\in\mathbb{K}$ satisfying these equations. Conversely, every system of this form parametrizes a line.*

**Proof.** For $P\in l$, define $P_x=\underbrace{\infty_y\infty_z P}\cap x$ and define $P_y,P_z$ analogously.

[36]Choose two points $X,Y\in l$. Axiom A5 gives the plane $PXY$, and A6 gives $l\subset PXY$. If $X',Y'$ are any other two points of $l$, then A6 gives $X',Y'\in PXY$, hence $P,X',Y'\in PXY$; uniqueness in A5 gives $PX'Y'=PXY$. Thus the plane $Pl$ is well defined.

[37]Let $P=l\cap m$, take $Q\in l\setminus\{P\}$, another point $Q'\in m$, and $R\in l\setminus\{P\}$. Since $P,Q$ lie in the plane $Qm$, A6 gives $PQ=l\subset Qm$; hence $R,P,Q'\in Qm$, and uniqueness in A5 gives $lQ'=RPQ'=Qm$.

[38]For example, $\boldsymbol{x}(\mathit{0})=(0,0,0)$, $\boldsymbol{x}(\mathit{1}_x)=(1,0,0)$, and $\boldsymbol{x}(\mathit{1}_z)=(0,0,1)$. The 0 and 1 on the right denote the zero and unit elements of $\mathbb{K}$, whereas $\mathit{0}, \mathit{1}$ on the left are point symbols.

First suppose, for example, that $\infty_z \in l$ and put $l \cap \underbracket{xy} = A = \boldsymbol{x}^{-1}(m,n,0)$. Incidence axiom A gives $\mathrm{x}_x(P_x) = m$ and $\mathrm{x}_y(P_y) = n$.[39] We may therefore take the parametric form:

$$\mathrm{x}_x(P) = 0 \cdot t + m, \qquad \mathrm{x}_y(P) = 0 \cdot t + n, \qquad \mathrm{x}_z(P) = a_z \cdot t + b_z, \quad a_z \neq 0.$$

The cases involving $\infty_x$ or $\infty_y$ are analogous.

Suppose next that $l$ contains none of the three axis points at infinity. Put

$$l_z = \underbracket{\infty_z l} \cap \underbracket{xy}, \quad l_y = \underbracket{\infty_y l} \cap \underbracket{zx}, \quad l_x = \underbracket{\infty_x l} \cap \underbracket{yz}.$$

The point $\infty_x$, the line $l$, and the line $l_x$ are coplanar, so put $P_{yz} = \infty_x P \cap l_x$. Incidence axiom A gives $P_y = \infty_z P_{yz} \cap y, P_z = \infty_y P_{yz} \cap z$.

Suppose first within this case that $\infty_y \in l_z$. Incidence axiom A shows that $l_z$ and $l_y$ meet the $x$-axis in the same point, namely $P_x$. Thus $\mathrm{x}_x(P_x) = m$ is constant. Applying theorem 10.4.2 to $P_{yz} \in l_x$, and using the two preceding projections, gives

$$\begin{aligned} \mathrm{x}_x(P) &= \mathrm{x}_x(P_x) = 0 \cdot t + m, \\ \mathrm{x}_y(P) &= \mathrm{x}_y(P_y) = \mathrm{x}_y(P_{yz}) = a_y t + b_y, \\ \mathrm{x}_z(P) &= \mathrm{x}_z(P_z) = \mathrm{x}_z(P_{yz}) = a_z t + b_z. \end{aligned}$$

The other corresponding cases are handled in the same way.

It remains to consider the case in which none of $l_x, l_y, l_z$ passes through any of $\infty_x, \infty_y, \infty_z$. Applying theorem 10.4.2 to $l_x$, and using $P_y = \infty_z P_{yz} \cap y$ and $P_z = \infty_y P_{yz} \cap z$, gives a parameter $t$ such that

$$\mathrm{x}_y(P) = \mathrm{x}_y(P_y) = \mathrm{x}_y(P_{yz}) = a_y t + b_y, \quad \mathrm{x}_z(P) = \mathrm{x}_z(P_z) = \mathrm{x}_z(P_{yz}) = a_z t + b_z.$$

Likewise, a second projection gives a parameter $t'$ such that

$$\mathrm{x}_y(P) = a_y' \cdot t' + b_y', \qquad \mathrm{x}_x(P) = a_x' \cdot t' + b_x'.$$

For the common $y$-coordinate, $a_y \cdot t + b_y = a_y' \cdot t' + b_y'$. The hypotheses in this final case ensure that $a_y, a_z, a_y', a_x'$ are all nonzero. Hence

$$t' = ((a_y')^{-1} a_y) \cdot t + (a_y')^{-1}(b_y - b_y')$$

and

$$\mathrm{x}_x(P) = (a_x'(a_y')^{-1} a_y) \cdot t + [a_x'(a_y')^{-1}(b_y - b_y') + b_x'] \equiv a_x t + b_x.$$

Writing the last two coefficients as $a_x, b_x$ proves the required parametric form.

Conversely, the proof that every such parametric system is a finite line follows the proof of theorem 10.4.2. First handle the special cases in which one of $a_x, a_y, a_z$ is zero. In the general case, find the intersections with $\underbracket{xy}, \underbracket{yz}, \underbracket{xz}$, call them $A, B, C$, and verify that the system gives the line $ABC$ obtained by the construction above, up to a change of parameter. We leave these details to the reader. □

As in section 10.4, the preceding result leads to homogeneous coordinates for points at infinity.

**Definition 10.6.7.** In three-dimensional projective space $\Omega = \mathbb{P}^3(\mathbb{K})$, for $p_1, p_2, p_3, p_4 \in \mathbb{K}$ not all zero, the expression $[p_1 : p_2 : p_3 : p_4]$ is a *homogeneous projective coordinate representation* of a point of $\Omega$, or simply a homogeneous coordinate representation, subject to:

i. for every $\lambda \in \mathbb{K}^\times$, the expressions $[p_1 : p_2 : p_3 : p_4]$ and $[p_1\lambda : p_2\lambda : p_3\lambda : p_4\lambda]$ represent the same point, so we may write

$$[p_1 : p_2 : p_3 : p_4] = [p_1\lambda : p_2\lambda : p_3\lambda : p_4\lambda];$$

ii. if $p_4 \neq 0$, put $p_i' = p_i p_4^{-1}$ for $i = 1,2,3$; then $[p_1 : p_2 : p_3 : p_4] = [p_1' : p_2' : p_3' : 1]$, and the quadruple represents the point $P$ with $\boldsymbol{x}(P) = (p_1', p_2', p_3')$;

[39] Indeed, $\underbracket{\infty_y \infty_z P} = \underbracket{l \infty_y}$. Since $A \in l$, we have $A \in \underbracket{\infty_y \infty_z P}$, hence $A\infty_y \subset \underbracket{\infty_y \infty_z P}$. By the definition of plane projective coordinates, $A\infty_y \cap x = \boldsymbol{x}^{-1}(m,0,0)$, which is precisely the intersection of $\underbracket{\infty_y \infty_z P}$ with the $x$-axis; thus $\mathrm{x}_x(P_x) = m$. A reader may visualize the corresponding statement in a real Euclidean space with its points at infinity adjoined, since its basic incidences are the same under A. This example will stand for later elementary spatial incidence assertions using only A, which will not be proved individually.

iii. if $p_4=0$, $[p_1:p_2:p_3:0]$ represents the point at infinity on a line having direction coefficients $p_1,p_2,p_3$, that is, on a line parametrized by

$$\mathrm{x}_x(P)=p_1t+b_x,\quad \mathrm{x}_y(P)=p_2t+b_y,\quad \mathrm{x}_z(P)=p_3t+b_z.$$

Here $p_1,p_2,p_3$ are not all zero, and $b_x,b_y,b_z$ are arbitrary.

Since homogeneous and inhomogeneous coordinates represent the same object, we also write $\boldsymbol{x}(P)=[p_1:p_2:p_3:p_4]$ for the homogeneous coordinates of $P$. When a particular representative has been chosen, denote its entries by the following symbols.[40]

$$x_1(P),x_2(P),x_3(P),x_4(P),\quad \text{so that}\quad \boldsymbol{x}(P)=[x_1(P):x_2(P):x_3(P):x_4(P)].$$

Analogously to theorem 10.4.5, two points determine a line as follows.

**Theorem 10.6.8.** *In three-dimensional projective space, let distinct points $P,Q$ have homogeneous coordinate representations $[p_1:p_2:p_3:p_4]$ and $[q_1:q_2:q_3:q_4]$. A point R lies on PQ if and only if it has a representation*

$$\boldsymbol{x}(R)=[p_1a+q_1b:p_2a+q_2b:p_3a+q_3b:p_4a+q_4b] \tag{10.14}$$

*for some $a,b\in\mathbb{K}$, not both zero.*

**Proof.** The proof is analogous to that of theorem 10.4.5 and is left to the reader. If $P,Q$ are both at infinity, their fourth coordinates $x_4$ are both zero, and the formula gives $x_4(R)=0$. The first three entries then serve as homogeneous coordinates in the two-dimensional plane $\Pi_\infty$, and the resulting line lies in $\Pi_\infty$. □

We can now consider the form of a plane.

**Theorem 10.6.9.** *In three-dimensional projective space, let noncollinear points $P,Q,R$ have homogeneous coordinate representations $[p_1:p_2:p_3:p_4]$, $[q_1:q_2:q_3:q_4]$, and $[r_1:r_2:r_3:r_4]$, respectively. A point T lies in the plane $\underbrace{PQR}$ if and only if it has a representation*

$$\boldsymbol{x}(T)=[p_1a+q_1b+r_1c:p_2a+q_2b+r_2c:p_3a+q_3b+r_3c:p_4a+q_4b+r_4c] \tag{10.15}$$

*for some $a,b,c\in\mathbb{K}$, not all zero.*

**Proof.** Use the convenient notation $\widetilde{\boldsymbol{P}},\langle P\rangle$, and so on, from the proof of theorem 10.4.5.

First suppose that a point has the displayed form. If $\lambda,\mu\in\mathbb{K}$ are not both zero, then

$$S=\left\langle\widetilde{\boldsymbol{P}}\lambda+\widetilde{\boldsymbol{Q}}\mu\right\rangle\in PQ\subset\underbrace{PQR},$$

and, for any $\nu\in\mathbb{K}$,

$$T=\left\langle\widetilde{\boldsymbol{S}}\cdot 1+\widetilde{\boldsymbol{R}}\nu\right\rangle=\left\langle\widetilde{\boldsymbol{P}}\lambda+\widetilde{\boldsymbol{Q}}\mu+\widetilde{\boldsymbol{R}}\nu\right\rangle\in SR\subset\underbrace{PQR}.$$

If $\lambda=\mu=0\neq\nu$, the expression is simply $R$. Thus every displayed combination lies in $\underbrace{PQR}$.

Conversely, let $T\in\underbrace{PQR}$. If $T=R$, take $\lambda=\mu=0\neq\nu$. If $T\neq R$, put $S=TR\cap PQ$. By theorem 10.6.8, there are $\lambda',\mu'\in\mathbb{K}$, not both zero, such that

$$S=\left\langle\widetilde{\boldsymbol{P}}\lambda'+\widetilde{\boldsymbol{Q}}\mu'\right\rangle\in PQ\subset\underbrace{PQR}.$$

Again by theorem 10.6.8, there are $\kappa,\nu\in\mathbb{K}$, not both zero, such that

$$T=\left\langle\widetilde{\boldsymbol{S}}\kappa+\widetilde{\boldsymbol{R}}\nu\right\rangle=\left\langle\widetilde{\boldsymbol{P}}(\lambda'\kappa)+\widetilde{\boldsymbol{Q}}(\mu'\kappa)+\widetilde{\boldsymbol{R}}\nu\right\rangle\in SR\subset\underbrace{PQR}.$$

This is precisely the form asserted in the theorem. □

Two further forms for a plane follow by elementary algebraic rearrangement. We omit the proofs; they can be obtained as in the earlier proofs of the forms of a line.

[40]The four entries of a homogeneous coordinate representation are not uniquely determined; this notation is only a convenient way to select them. In one context, all $x_i(P)$ belonging to a fixed point $P$ may be multiplied on the right by the same nonzero scalar.

**Theorem 10.6.10.** *In three-dimensional projective space, for every finite plane $\pi$, there exist $a_x, a_y, a_z, b_x, b_y, b_z, c_x, c_y, c_z \in \mathbb{K}$ such that*

$$(a_x, a_y, a_z) \neq (0,0,0), \qquad (b_x, b_y, b_z) \neq (0,0,0),$$

*and there is no $\lambda \in \mathbb{K}$ for which*

$$(a_x, a_y, a_z) = (b_x\lambda, b_y\lambda, b_z\lambda).$$

*Equivalently, the right column rank satisfies*

$$\operatorname{rank}^{\mathrm{r}} \begin{bmatrix} a_x & b_x \\ a_y & b_y \\ a_z & b_z \end{bmatrix} = 2.$$

*Its finite points $P$ have the two-parameter representation*

$$\mathrm{x}_x(P) = a_x s + b_x t + c_x, \quad \mathrm{x}_y(P) = a_y s + b_y t + c_y, \quad \mathrm{x}_z(P) = a_z s + b_z t + c_z, \tag{10.16}$$

*where $s, t \in \mathbb{K}$; that is, $P \in \pi - \infty_\pi$ if and only if there exist $s, t \in \mathbb{K}$ satisfying* (10.16). *Conversely, every such system parametrizes a plane.*

**Theorem 10.6.11.** *For every plane $\pi$ in projective space there are $a, b, c, d \in \mathbb{K}$, not all zero, such that its points $P$ satisfy*

$$a x_1(P) + b x_2(P) + c x_3(P) + d x_4(P) = 0. \tag{10.17}$$

*Conversely, every equation of this form represents a plane.*

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## Exercises

**Exercise 10.42.** In three-dimensional projective coordinates:

(1) give the homogeneous coordinate representations of all base points, and the inhomogeneous representations of the finite ones;
(2) give all the preceding forms of the plane $\underline{1_x 1_y 1_z}$;
(3) give all those forms for the line $\underline{1_x 1_y 1_z} \cap \underline{xy}$.

**Exercise 10.43.** Supply the omitted proofs:

(1) prove carefully that every admissible parametrization in theorem 10.6.6 represents a line;
(2) prove theorems 10.6.10 and 10.6.11 in full.

**Exercise 10.44.** Use perspective affine transformations of two-dimensional planes to prove theorem 10.6.10. Do not appeal to the known equation of a line.

**Exercise 10.45.** Count the points, lines, and planes in $\mathrm{PG}(3,q)$.

**Exercise 10.46.** State and prove the principle of duality in three-dimensional projective space.

**Exercise 10.47.** In three-dimensional projective space over a field, let $X_i$ have homogeneous coordinate representation $[a_i : b_i : c_i : d_i]$, $i = 1,2,3,4$. Prove that $X_1, X_2, X_3, X_4$ are coplanar if and only if

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0.$$

**Exercise 10.48.** In the right-module quaternionic projective space (see example A.4.17) $\mathbb{P}^3_{\mathrm{R}}(\mathbb{H})$, the coordinate representations of $A,B,C,D$ are

$$[1+\mathbf{i}:\mathbf{j}+\mathbf{k}:-2:0], \quad [\mathbf{i}:\mathbf{k}:1:3+\mathbf{i}],$$
$$[-\mathbf{k}:2:1:-\mathbf{i}], \quad [\mathbf{j}+1:\mathbf{i}:0:2\mathbf{j}+\mathbf{i}].$$

(1) Find a parametrization of $BC$ in the form (10.13), and the homogeneous coordinate representation of its point at infinity.
(2) Decide whether $AB$ and $CD$ meet and, if so, find a coordinate representation of their intersection.
(3) Decide whether a unique plane passes through $B,C,D$. If so, find its equation in the form (10.17) and parametrize its line at infinity in the form (10.14).
(4) How do the answers change in the left-module space $\mathbb{P}^3_{\mathrm{L}}(\mathbb{H})$ with all other data unchanged?

**Exercise 10.49*.** In a right-module three-dimensional projective space over an arbitrary division ring, suppose the points of planes $\pi_1,\pi_2$ satisfy

$$a_i x_1(P)+b_i x_2(P)+c_i x_3(P)+d_i x_4(P)=0, \qquad i=1,2.$$

Determine an explicit form for their line of intersection.

### 10.6.2 Two-Dimensional Projective Transformations

We now study projective transformations of a plane as induced by perspectivities in three-dimensional space. Perspective affine transformations were treated earlier because their form could be obtained directly; they can of course also be recovered from spatial coordinates.

#### Projective and affine transformations

Successively joining a point to each center of perspectivity and intersecting the target plane gives the following matrix form. The omitted calculation is routine algebra.

**Theorem 10.6.12.** *On a two-dimensional projective plane $\pi$,*

$$\mathrm{Prj}(\pi)=\left\{f: \boldsymbol{x}^{-1}([p_1:p_2:p_3])\mapsto \boldsymbol{x}^{-1}([p_1':p_2':p_3']) \quad \middle| \quad \begin{bmatrix}p_1'\\p_2'\\p_3'\end{bmatrix}=\boldsymbol{A}\begin{bmatrix}p_1\\p_2\\p_3\end{bmatrix}, \quad \boldsymbol{A}\in \mathrm{GL}(3,\mathbb{K})\right\}. \tag{10.18}$$

*For the part whose points are finite both before and after the projective transformation, its form in inhomogeneous projective coordinates is*

$$\begin{aligned} f: \boldsymbol{x}^{-1}(p_1,p_2)\mapsto \boldsymbol{x}^{-1}(&(a_1p_1+b_1p_2+c_1)(a_3p_1+b_3p_2+c_3)^{-1},\\ &(a_2p_1+b_2p_2+c_2)(a_3p_1+b_3p_2+c_3)^{-1}),\end{aligned} \tag{10.19}$$

*where* $\boldsymbol{A}=\begin{bmatrix}a_1&b_1&c_1\\a_2&b_2&c_2\\a_3&b_3&c_3\end{bmatrix}\in \mathrm{GL}(3,\mathbb{K})$.

**Remark.** In one dimension, a few conventions for infinity allow all transformations to be written with inhomogeneous coordinates. In higher dimensions several coordinate directions may become infinite at once, so homogeneous representations cannot be eliminated uniformly.

As in section 10.4, the same calculations give the following group structures; their proofs are omitted.

**Corollary 10.6.13.** *On the plane $\pi=\mathbb{P}^2(\mathbb{K})$, the set* $\mathrm{Prj}(\pi)$ *is a group and*

$$\mathrm{PGL}(3,\mathbb{K})\cong \mathrm{GL}(3,\mathbb{K})/\mathrm{Z}(\mathrm{GL}(3,\mathbb{K})).$$

It is called the *two-dimensional projective transformation group* , or the *projective general linear group*, and is denoted by $\mathrm{PGL}(3,\mathbb{K})$. The number 3 indicates that the group is induced by $\mathrm{GL}(3,\mathbb{K})$; when $\mathbb{K}$ is understood, write $\mathrm{PGL}(3)$.

**Corollary 10.6.14.** *On the plane $\pi = \mathbb{P}^2(\mathbb{K})$, given four source points and four image points, each quadruple in general position (four distinct points with no three collinear), a projective transformation realizing the four correspondences always exists. It is uniquely determined if and only if $\mathbb{K}$ is a field.*

The notations Prj and PGL emphasize the geometric and algebraic descriptions, respectively. We now consider affine transformations, which preserve the points at infinity.

**Corollary 10.6.15.** *On the plane $\pi = \mathbb{P}^2(\mathbb{K})$, an affine transformation $\varphi \in \mathrm{Aff}(\pi)$ has the following form for every finite point $P \in \pi$:*

$$\mathrm{x}_1(\varphi(P)) = [a_1\mathrm{x}_1(P) + b_1\mathrm{x}_2(P)]d + c_1, \quad \mathrm{x}_2(\varphi(P)) = [a_2\mathrm{x}_1(P) + b_2\mathrm{x}_2(P)]d + c_2, \tag{10.20}$$

*where $d \in \mathbb{K}^\times$ and $\boldsymbol{A} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \in \mathrm{GL}(2,\mathbb{K})$. Equivalently,*

$$\begin{bmatrix} \mathrm{x}_1(\varphi(P)) \\ \mathrm{x}_2(\varphi(P)) \end{bmatrix} = \boldsymbol{A} \begin{bmatrix} \mathrm{x}_1(P) \\ \mathrm{x}_2(P) \end{bmatrix} d + \boldsymbol{b}, \qquad \boldsymbol{A} \in \mathrm{GL}(2,\mathbb{K}), \quad \boldsymbol{b} \in \mathbb{K}^2, \quad d \in \mathbb{K}^\times. \tag{10.21}$$

**Proof.** An affine transformation preserves the line at infinity. In (10.18), the condition $p_3 = 0 \Rightarrow p_3' = 0$ is equivalent to $a_3p_1 + b_3p_2 = 0$ for all $p_1, p_2$.[41] Hence $a_3 = b_3 = 0$. Renaming $c_3^{-1}$ as $d$ in (10.19) gives the result. □

**Corollary 10.6.16.** *If C also holds, so that $\pi = \mathbb{P}^2(\mathbb{F})$ is over a field, every affine transformation $\varphi$ has the following form for every finite point $P \in \pi$:*

$$\mathrm{x}_1(\varphi(P)) = a_1\mathrm{x}_1(P) + b_1\mathrm{x}_2(P) + c_1, \quad \mathrm{x}_2(\varphi(P)) = a_2\mathrm{x}_1(P) + b_2\mathrm{x}_2(P) + c_2, \tag{10.22}$$

*where $\boldsymbol{A} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \in \mathrm{GL}(2,\mathbb{F})$. Equivalently, in column form,*

$$\begin{bmatrix} \mathrm{x}_1(\varphi(P)) \\ \mathrm{x}_2(\varphi(P)) \end{bmatrix} = \boldsymbol{A} \begin{bmatrix} \mathrm{x}_1(P) \\ \mathrm{x}_2(P) \end{bmatrix} + \boldsymbol{b}, \quad \boldsymbol{A} \in \mathrm{GL}(2,\mathbb{F}), \quad \boldsymbol{b} \in \mathbb{F}^2.$$

**Corollary 10.6.17.** *On the plane $\pi = \mathbb{P}^2(\mathbb{K})$, the set* $\mathrm{Aff}(\pi)$ *is a group.* It is the *two-dimensional full affine transformation group* and is denoted in this book by $\mathrm{AGL_F}(2,\mathbb{K})$, where 2 indicates the dimension of the affine transformations, or by $\mathrm{AGL_F}(2)$ when $\mathbb{K}$ is understood.

**Corollary 10.6.18.** $\mathrm{AGL_F}(2,\mathbb{K}) = \mathrm{AGL}(2,\mathbb{K})$ *if and only if $\mathbb{K}$ is a field, equivalently, if and only if the corresponding projective plane satisfies C.*

**Remark.** As explained in section 10.4, affine transformations in algebra are derived directly from linear spaces (see the later chapters for details), so the order of multiplication is fixed. Thus the usual algebraic *general affine linear group*, also denoted by AGL, is the perspective affine transformation group defined here geometrically. For the same reason, affine transformations in algebra do not ordinarily include the bilateral form $\boldsymbol{A}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} d$. The notation $\mathrm{AGL_F}$ is specific to this geometric approach and is not a widely established convention.

Comparison with $\mathrm{AGL_F}(1)$ shows that the bilateral multiplication in $axa'$ is not completely symmetric. In two dimensions, we see that the right factor simply allows the whole coordinate column to be multiplied by a single scalar.

[41] The last row of the matrix $\boldsymbol{A}$ in (10.18) is $[a_3 \ \ b_3 \ \ c_3]$, in agreement with the notation in (10.19).

Our definitions of projective, affine, and perspective affine transformations all use three-dimensional space, and hence require B. The notation $\mathrm{AGL_F}$ is more algebraic, whereas Aff expresses the geometric concept. When the meaning is clear, the notations may be used interchangeably, since $\mathrm{AGL_F}(2,\mathbb{K}) \cong \mathrm{Aff}(\pi)$.

### Projective semilinear transformations

We now introduce another kind of transformation of projective space: a projective semilinear transformation. Its effect is not apparent on a one-dimensional projective line, so we discuss it here.

**Definition 10.6.19.** Let $\pi = \mathbb{P}^2(\mathbb{K})$, whose projective transformations form the group $\mathrm{PGL}(3,\mathbb{K})$. For $\sigma \in \mathrm{Aut}(\mathbb{K})$, define

$$\begin{aligned}\phi_\sigma : \mathrm{PG}(2,\mathbb{K}) &\to \mathrm{PG}(2,\mathbb{K}),\\ \boldsymbol{x}^{-1}([x_1 : x_2 : x_3]) &\mapsto \boldsymbol{x}^{-1}([\sigma(x_1) : \sigma(x_2) : \sigma(x_3)]).\end{aligned}$$

For $\sigma \in \mathrm{Aut}(\mathbb{K})$ and $f \in \mathrm{PGL}(3,\mathbb{K})$, the composite $f \circ \phi_\sigma$ is a *projective semilinear transformation* of $\pi$.

**Theorem 10.6.20.** *The map $\phi_\sigma$, and hence every projective semilinear transformation, preserves lines.*

**Proof.** Writing the line equation in the form (10.9) and applying $\sigma$ gives

$$\sigma(ax_1 + bx_2 + cx_3) = \sigma(a)\sigma(x_1) + \sigma(b)\sigma(x_2) + \sigma(c)\sigma(x_3) = 0.$$

Here we used the fact from corollary A.4.38 that a ring isomorphism preserves zero. Every point with coordinates $[x_1 : x_2 : x_3]$ is mapped to a point with coordinates $[\sigma(x_1) : \sigma(x_2) : \sigma(x_3)]$, which satisfies the displayed line equation. ☐

**Theorem 10.6.21.** *The projective semilinear transformations of a projective plane form a group.* It is called the *projective semilinear group*, denoted by $\mathrm{P\Gamma L}(3,\mathbb{K})$ or $\mathrm{P\Gamma L}(3)$ when the division ring is understood; algebraically it is also called the *projective general semilinear group*.

**Proof.** It is enough to verify closure under composition; the remaining group properties are immediate. Use the notation $\phi_\sigma$ from the preceding definition. If $f \in \mathrm{PGL}(3)$ is represented by $\boldsymbol{A} \in \mathrm{GL}(3,\mathbb{K})$ as in (10.18), the composite $\phi_\sigma \circ f \circ \phi_\sigma^{-1}$ has coordinate formula

$$\begin{bmatrix} p_1' \\ p_2' \\ p_3' \end{bmatrix} = \phi_\sigma \left( \boldsymbol{A} \cdot \phi_\sigma^{-1} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \right) = \boldsymbol{A}^\sigma \cdot (\phi_\sigma \circ \phi_\sigma^{-1}) \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \boldsymbol{A}^\sigma \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}.$$

Here $\boldsymbol{A}^\sigma$ is obtained by applying $\sigma$ to every entry of $\boldsymbol{A}$, for example $a_1 \to \sigma(a_1)$. This calculation uses the fact that a division-ring isomorphism preserves addition and multiplication, and readers who do not see it directly can expand the matrices and verify the calculation entry by entry. Thus $\phi_\sigma \circ f \circ \phi_\sigma^{-1} \in \mathrm{PGL}(3)$. For $f, g \in \mathrm{PGL}(3)$ and $\sigma, \rho \in \mathrm{Aut}(\mathbb{K})$,

$$(f \circ \phi_\sigma)(g \circ \phi_\rho) = [f \circ (\phi_\sigma \circ g \circ \phi_\sigma^{-1})] \circ (\phi_\sigma \circ \phi_\rho).$$

Since $\mathrm{Aut}(\mathbb{K})$ is a group, $\phi_\sigma \circ \phi_\rho = \phi_{\sigma\circ\rho}$. The preceding argument gives $f \circ (\phi_\sigma \circ g \circ \phi_\sigma^{-1}) \in \mathrm{PGL}(3)$. Thus the resulting composite is again a projective transformation composed with a transformation $\phi_\bullet$, hence is projective semilinear. ☐

**Corollary 10.6.22.**

$$\mathrm{PGL}(3,\mathbb{K}) \trianglelefteq \mathrm{P\Gamma L}(3,\mathbb{K}).$$

**Proof.** On the projective plane $\pi$, take $\sigma = \mathrm{id}_{\mathbb{K}}$, so that $\phi_\sigma = \mathrm{id}_\pi$. Every projective transformation $f = f \circ \mathrm{id}_\pi$ is therefore projective semilinear. The preceding proof also shows that $\phi_\sigma \circ f \circ \phi_\sigma^{-1}$ is projective for every $\sigma$; hence theorem A.3.36 gives $\mathrm{PGL}(3) \trianglelefteq \mathrm{P\Gamma L}(3)$. □

**Lemma 10.6.23.** *The transformation $\phi_\sigma$ of definition 10.6.19 belongs to* $\mathrm{PGL}(3,\mathbb{K})$ *if and only if* $\sigma \in \mathrm{Inn}(\mathbb{K})$.[42]

**Proof.** If $\sigma \in \mathrm{Inn}(\mathbb{K})$, write $\sigma : x \mapsto uxu^{-1}$. Then $\phi_\sigma$ is the projective transformation with $\boldsymbol{A} = u\mathbf{I}$ in (10.18), because the common right coordinate factor $u^{-1}$ can be canceled in homogeneous coordinates.

Conversely, suppose $\phi_\sigma \in \mathrm{PGL}(3,\mathbb{K})$. By corollary A.4.38, automorphisms preserve the zero and unit elements. Substitute the axis unit points $1_x, 1_y$, the origin $0$, the axis points at infinity $\infty_x, \infty_y$, and the frame unit point $\boldsymbol{x}^{-1}(1,1)$ in (10.18). Elementary algebra then shows that necessarily $\boldsymbol{A} = u\mathbf{I}$, with $u \in \mathbb{K}^\times$. This is precisely $\phi_\sigma$ for $\sigma : x \mapsto uxu^{-1}$. □

**Corollary 10.6.24.** *If $\mathbb{K}$ is a field (the projective plane also satisfies Pappus's theorem C), then*[43]

$$\mathrm{P\Gamma L}(3,\mathbb{K}) \cong \mathrm{PGL}(3,\mathbb{K}) \rtimes \mathrm{Aut}(\mathbb{K}).$$

**Proof.** Taking $f = \mathrm{id}$ gives $\phi_\sigma = \mathrm{id} \circ \phi_\sigma \in \mathrm{P\Gamma L}(3)$. Since the automorphisms $\sigma$ form the transformation group $\mathrm{Aut}(\mathbb{K})$, the maps $\phi_\sigma$ also form a transformation group, which we denote by $H$. Clearly $H \le \mathrm{P\Gamma L}(3)$.

When $\mathbb{K}$ is commutative, the definition of an inner automorphism gives $\mathrm{Inn}(\mathbb{K}) = \{\mathrm{id}_{\mathbb{K}}\}$. By lemma 10.6.23, $\phi_\sigma \in \mathrm{PGL}(3)$ if and only if $\phi_\sigma = \mathrm{id}$, and hence $\mathrm{PGL}(3) \cap H = \{\mathrm{id}\}$.

Clearly $\mathrm{PGL}(3) \cdot H = \mathrm{P\Gamma L}(3)$. The definition of an internal semidirect product in definition A.3.49 therefore gives $\mathrm{P\Gamma L}(3) \cong \mathrm{PGL}(3) \rtimes H$. But $H \cong \mathrm{Aut}(\mathbb{K})$, with isomorphism $\phi_\sigma \mapsto \sigma$, so $\mathrm{P\Gamma L}(3) \cong \mathrm{PGL}(3) \rtimes \mathrm{Aut}(\mathbb{K})$. □

**Remark.** When $\mathbb{K}$ is not a field, we cannot simply write

$$\mathrm{P\Gamma L}(3,\mathbb{K}) \cong \mathrm{PGL}(3,\mathbb{K}) \rtimes (\mathrm{Aut}(\mathbb{K}) / \mathrm{Inn}(\mathbb{K})).$$

Although this removes the overlap of the automorphism-induced transformations $\phi_\sigma$ with projective transformations, a group action cannot in general be defined in this way.

The corresponding affine general semilinear group is defined as follows.

**Definition 10.6.25.** The perspective affine transformations of a plane $\pi$ over $\mathbb{K}$ form the group $\mathrm{AGL}(2,\mathbb{K})$. For $\sigma \in \mathrm{Aut}(\mathbb{K})$, define

$$\begin{aligned} \phi_\sigma : \mathrm{PG}(2,\mathbb{K}) &\longrightarrow \mathrm{PG}(2,\mathbb{K}), \\ \boldsymbol{x}^{-1}([x_1 : x_2 : x_3]) &\longmapsto \boldsymbol{x}^{-1}([\sigma(x_1) : \sigma(x_2) : \sigma(x_3)]). \end{aligned}$$

For $\sigma \in \mathrm{Aut}(\mathbb{K})$ and $f \in \mathrm{AGL}(2,\mathbb{K})$, the composite $f \circ \phi_\sigma$ is called an *affine semilinear transformation* of $\pi$.

The usual definition uses the general affine linear group AGL, rather than $\mathrm{AGL_F}$. This makes no difference here, as the following result shows.

**Theorem 10.6.26.** *Every full affine transformation is affine semilinear.*

[42] See exercises A.42 and A.68 for inner automorphisms.

[43] See section A.3.3 for semidirect products.

**Proof.** For a point $P$, write $\boldsymbol{x}(P) = \begin{bmatrix} \mathrm{x}_1(P) \\ \mathrm{x}_2(P) \end{bmatrix} \in \mathbb{K}^2$. Using the affine form (10.21), we have

$$\boldsymbol{Ax}d + \boldsymbol{b} = (\boldsymbol{A}d)(d^{-1}\boldsymbol{x}d) + \boldsymbol{b}.$$

For $d \in \mathbb{K}^\times$, the map $\sigma\colon \mathbb{K} \to \mathbb{K}$, $x \mapsto d^{-1}xd$, is an automorphism of $\mathbb{K}$. Thus the displayed expression is a perspective affine transformation of the form $\boldsymbol{Ax} + \boldsymbol{b}$, composed with the transformation $\phi_\sigma$ induced by $\sigma$. ☐

Because automorphisms fix zero, $\phi_\sigma$ preserves the line $x_3 = 0$ at infinity. The affine semilinear group is therefore the subgroup of PΓL preserving that line.

**Theorem 10.6.27.** *Affine semilinear transformations preserve lines. All affine semilinear transformations of a plane over* $\mathbb{K}$ *form a group.* This is the *affine semilinear group*, denoted by $\mathrm{A\Gamma L}(2,\mathbb{K})$, or $\mathrm{A\Gamma L}(2)$ when the division ring is understood; algebraically it is the *affine general semilinear group*.

**Theorem 10.6.28.**

$$\mathrm{AGL}(2,\mathbb{K}) \trianglelefteq \mathrm{AGL_F}(2,\mathbb{K}) \trianglelefteq \mathrm{A\Gamma L}(2,\mathbb{K}).$$

**Lemma 10.6.29.** *The map* $\phi_\sigma$ *in definition 10.6.25 belongs to* $\mathrm{AGL}(2,\mathbb{K})$ *if and only if* $\phi_\sigma = \mathrm{id}$.

**Theorem 10.6.30.**

$$\mathrm{A\Gamma L}(2,\mathbb{K}) \cong \mathrm{AGL}(2,\mathbb{K}) \rtimes \mathrm{Aut}(\mathbb{K}).$$

**Remark.** The normal-subgroup assertions follow as in corollary 10.6.22, and the semidirect-product argument is the same as for corollary 10.6.24, using lemma 10.6.29. Let $H$ be the subgroup consisting of all maps $\phi_\sigma$. The common elements of $\mathrm{AGL_F}(2)$ and $H$ need not consist only of $\mathrm{id}_\pi$; it is the intersection of $\mathrm{AGL}(2)$ with $H$ that consists only of $\mathrm{id}_\pi$.

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## Exercises

**Exercise 10.50.** Prove theorems 10.6.12 and 10.6.27 and corollaries 10.6.14 and 10.6.18. Also prove theorems 10.6.28 and 10.6.30 and lemma 10.6.29.

**Exercise 10.51.** Let $\mathbb{K}$ be a division ring.

(1) Prove $\mathrm{AGL_F}(2,\mathbb{K}) \cong \mathrm{AGL}(2,\mathbb{K}) \rtimes \mathrm{Inn}(\mathbb{K})$.

(2) Define the *outer automorphism group* of a group or ring $G$ by $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$. Prove

$$\mathrm{P\Gamma L}(3,\mathbb{K})/\mathrm{PGL}(3,\mathbb{K}) \cong \mathrm{Out}(\mathbb{K}),$$
$$\mathrm{A\Gamma L}(2,\mathbb{K})/\mathrm{AGL_F}(2,\mathbb{K}) \cong \mathrm{Out}(\mathbb{K}).$$

**Exercise 10.52*.** Define the $n$-dimensional *general semilinear group* over $\mathbb{K}$ by

$$\Gamma\mathrm{L}(n,\mathbb{K}) = \{(\boldsymbol{A},\sigma) \mid \boldsymbol{A} \in \mathrm{GL}(n,\mathbb{K}),\ \sigma \in \mathrm{Aut}(\mathbb{K})\},$$

where its action on a vector $\boldsymbol{x} \in \mathbb{K}^n$ is $(\boldsymbol{A},\sigma)\boldsymbol{x} = \boldsymbol{A}\,\sigma(\boldsymbol{x})$, with $\sigma$ applied entrywise. Prove that, when $\mathbb{K}$ is a field,

$$\mathrm{P\Gamma L}(3,\mathbb{K}) \cong \Gamma\mathrm{L}(3,\mathbb{K})/(\mathbb{K}^\times \mathbf{I}_3) := \Gamma\mathrm{L}(3,\mathbb{K})/\{\lambda \mathbf{I}_3 \mid \lambda \in \mathbb{K}^\times\}.$$

What should the kernel be for a general division ring?

**Exercise 10.53.** Determine explicitly all perspective affine, affine, projective, projective semilinear, and affine semilinear transformations of $\mathbb{P}^2(\mathbb{F}_2)$.

**Exercise 10.54.** For a prime $p$, count the elements of the perspective affine, full affine, projective, projective semilinear, and affine semilinear groups of $\mathrm{PG}(2,p)$. Repeat the calculation for $\mathbb{P}^2(\mathbb{F}_9)$, with $\mathbb{F}_9$ as in exercise 10.35.

**Exercise 10.55.** A projective transformation of $\mathbb{P}^2(\mathbb{K})$ sends the four base points $O,X,Y,E$ of the coordinate frame $[O,X,Y;E]$ respectively to $(1_x 1_y \cap l_\infty)$, $1_y$, $E$, and $O$. Determine all possible forms of this projective transformation.

### 10.6.3 Collineations and Correlations

#### Collineations

**Definition 10.6.31.** Let $\pi$ be a projective plane satisfying A; Desargues' theorem need not hold. A bijection of its point set that sends every line, regarded as a set of points, to a line is a *collineation*. The collineations form the *collineation group* $\mathrm{Col}(\pi)$; for $\pi=\mathbb{P}^2(\mathbb{K})$ we also write $\mathrm{Col}(2,\mathbb{K})$.

More generally, for projective planes $\pi,\pi'$, not necessarily over the same division ring, a bijection $f:\pi\to\pi'$ that sends points to points and, for every line regarded as a set of points, has a line as its image set is a *collineation correspondence*. Its set is denoted by $\mathrm{Col}(\pi,\pi')$.

**Theorem 10.6.32** (Fundamental theorem of projective geometry)**.** *Let $\pi=\mathbb{P}^2(\mathbb{K})$ and $\pi'=\mathbb{P}^2(\mathbb{K}')$. For a division-ring isomorphism $\sigma:\mathbb{K}\to\mathbb{K}'$, define*

$$\phi_\sigma:\pi\longrightarrow\pi',\qquad \boldsymbol{x}_\pi^{-1}([x_1:x_2:x_3])\longmapsto\boldsymbol{x}_{\pi'}^{-1}([\sigma(x_1):\sigma(x_2):\sigma(x_3)]).$$

*Then*

$$\mathrm{Col}(\pi,\pi')=\{f\circ\phi_\sigma\,|\,f\in\mathrm{Prj}(\pi'),\ \sigma\in\mathrm{Iso}(\mathbb{K},\mathbb{K}')\}.$$

**Proof.** Let $l,l'$ be any two distinct lines of $\pi'$, with $V=l\cap l'$. Choose distinct points $A,B\in l\setminus\{V\}$ and distinct points $C,D\in l'\setminus\{V\}$. There is a $g\in\mathrm{Prj}(\pi')$ such that $g(0')=V$, $g(1'_x)=A$, $g(\infty'_x)=B$, $g(1'_y)=C$, and $g(\infty'_y)=D$, where the primed symbols $0',1',\infty'$ refer to the coordinate frame in $\pi'$.[44]

Thus, for any $\psi\in\mathrm{Col}(\pi,\pi')$, we can choose $f\in\mathrm{Prj}(\pi')$ so that $\psi'=f\circ\psi$ sends the five points $0,1_x,1_y,\infty_x,\infty_y$ of $\pi$ respectively to $0',1'_x,1'_y,\infty'_x,\infty'_y$ of $\pi'$. Put $O:=1_x1_y\cap\infty_x\infty_y=\boldsymbol{x}^{-1}([1:-1:0])$ and $O':=1'_x1'_y\cap\infty'_x\infty'_y=\boldsymbol{x}_{\pi'}^{-1}([1:-1:0])$. Since $\psi'$ is also a collineation correspondence, $\psi'(O)=O'$. Thus it is enough to treat a $\psi$ that sends $0,1_x,\infty_x,1_y,\infty_y$ respectively to $0',1'_x,\infty'_x,1'_y,\infty'_y$; otherwise one merely restores the extra projective factor.

Let $x=01_x$, $y=01_y$ in $\pi$, and let $x'=0'1'_x$, $y'=0'1'_y$ in $\pi'$. Since $\psi$ sends $0,1_x,1_y$ respectively to $0',1'_x,1'_y$ and sends lines to lines, the projective axioms give $\psi(x)=x'$ and $\psi(y)=y'$. Its restriction to $x\setminus\{\infty_x\}$ is a bijection $x\setminus\{\infty_x\}\to x'\setminus\{\infty'_x\}$, and hence induces a bijection $\sigma:\mathbb{K}\to\mathbb{K}'$.

After its point at infinity is removed, a projective line forms a division ring whose addition and multiplication were constructed entirely from incidences of lines. Since $\psi$ preserves incidence, $\sigma$ preserves both operations: $\sigma(A+B)=\sigma(A)+'\sigma(B)$ and $\sigma(A\cdot B)=\sigma(A)\cdot'\sigma(B)$. Hence $\sigma\in\mathrm{Iso}(\mathbb{K},\mathbb{K}')$.

It remains to show that the normalized $\psi$ equals $\phi_\sigma$. Put $\psi'=\phi_\sigma^{-1}\circ\psi\in\mathrm{Col}(\pi)$. The division-ring isomorphism shows that $\phi_\sigma^{-1}$ sends $0',1'_x,1'_y,\infty'_x,\infty'_y$ respectively back to $0,1_x,1_y,\infty_x,\infty_y$. Hence $\psi'$ fixes every point of $x$, fixes $O$, and satisfies $\psi'(y)=y$. For any $P\in y$, it fixes $OP\cap x$ and $O$, so $\psi'(OP)=OP$, and consequently it fixes the intersection $OP\cap y=P$. Now let $Q$ be any finite point of $\pi$. The map $\psi'$ fixes $\infty_yQ\cap x$ and $\infty_xQ\cap y$, and therefore $\psi'(\infty_yQ)=\infty_yQ$ and $\psi'(\infty_xQ)=\infty_xQ$.

[44] This merely reselects the coordinate frame. Equivalently, $AD\cap BC:=E$ is its unit point. Since projective transformations preserve lines, once $V,B,D,E$ are fixed, $A,C$ are also fixed; a projective transformation realizing the four prescribed pairs exists.

It follows that it fixes the intersection $\infty_y Q \cap \infty_x Q = Q$. Thus the restriction of $\psi'$ to the finite part is the identity. Every point at infinity is the intersection of a finite line with the line at infinity, so the collineation fixes every point at infinity as well. Therefore $\psi' = \mathrm{id}_\pi$, and hence $\psi = \phi_\sigma$, as required. □

The case of a single plane gives the following immediate consequence.

**Theorem 10.6.33** (Fundamental theorem of projective geometry)**.**

$$\mathrm{Col}(2,\mathbb{K}) = \mathrm{P\Gamma L}(3,\mathbb{K}).$$

*Thus every collineation of a projective plane over a division ring is a projective semilinear transformation.*

**Remark.** Our two-dimensional projective transformations were defined through three-dimensional space and therefore require Desargues' theorem. A non-Desarguesian plane therefore has no group $\mathrm{PGL}(3)$ in this sense. The algebraic definition has the same requirement: a plane must admit coordinates over a division ring before the form of a projective transformation can be written down. Thus only collineations can be studied on a non-Desarguesian plane.

For projective spaces of dimension $n \geq 3$ (the incidence axioms in higher dimensions are given later), we can naturally write

$$\mathrm{Col}(n,\mathbb{K}) = \mathrm{P\Gamma L}(n+1,\mathbb{K}).$$

Desargues' theorem holds automatically in dimension three or higher, so in every such projective space the collineation group is the projective semilinear group. This is the general form of the fundamental theorem of projective geometry.

We now consider two concrete examples.

**Example 10.6.34.** For the real projective plane,

$$\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}_\mathbb{R}\}, \qquad \mathrm{Col}(2,\mathbb{R}) = \mathrm{P\Gamma L}(3,\mathbb{R}) = \mathrm{PGL}(3,\mathbb{R}).$$

This proves the assertion in theorem 3.1.12 that every line-preserving transformation of the real projective plane is projective.

Here is an outline of the proof that $\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}_\mathbb{R}\}$. By corollary A.4.38, an automorphism fixes the unit element. Preservation of addition and multiplication then shows first that every integer is fixed, and then that every rational number is fixed. Order is algebraically characterized by

$$a \leq b \quad \Longleftrightarrow \quad b - a = c^2 \quad \text{for some } c \in \mathbb{R}.$$

Automorphisms preserve squares and therefore order. Density of $\mathbb{Q}$ then forces every real number to be fixed.

**Example 10.6.35.** A continuous automorphism $\sigma$ of $\mathbb{C}$ fixes $\mathbb{R}$, because it fixes the dense subfield $\mathbb{Q}$ and preserves limits. Moreover

$$[\sigma(\mathrm{i})]^2 = \sigma(\mathrm{i}^2) = \sigma(-1) = -1,$$

so $\sigma(\mathrm{i}) = \mathrm{i}$ or $-\mathrm{i}$. Thus the only continuous automorphisms are the identity and complex conjugation; if holomorphicity is required, only the identity remains; see section A.9. This is the precise content of the discussion following theorem 3.1.12: over the complex plane, strong analytic hypotheses are needed for the conclusion of theorem 3.1.12.

Without continuity, the *axiom of choice* permits infinitely many highly nonclassical automorphisms, generally preserving neither conjugation nor absolute value; see section A.1.2. Other set-theoretic foundations can change this conclusion. Most mathematicians work with the axiom of choice, but one may deliberately choose another system, such as one with the axiom of determinacy, in which $\mathbb{C}$ has only two automorphisms even without assuming continuity. These set-theoretic axiom systems are included only as an extension and need not be studied in depth. They show that, for continuous infinite coordinate fields typified by $\mathbb{C}$, the higher transformation groups in axiomatic geometry can depend not only on the geometric axioms but also on the surrounding set-theoretic foundation.

### Correlations

**Definition 10.6.36.** Let $\mathscr{P}, \mathscr{L}$ be the point and line sets of a projective plane $\Pi$. A *correlation* is a bijection $\rho: \mathscr{P} \sqcup \mathscr{L} \to \mathscr{P} \sqcup \mathscr{L}$ with

$$\begin{cases} P \in \mathscr{P} \longmapsto \rho(P) \in \mathscr{L}, \\ l \in \mathscr{L} \longmapsto \rho(l) \in \mathscr{P}, \end{cases}$$

and satisfies $P \in l \Rightarrow \rho(l) \in \rho(P)$. Thus it exchanges points and lines and reverses their incidence. The set of correlations is $\mathrm{Cor}(\Pi)$. A correlation $\rho \in \mathrm{Cor}(\Pi)$ satisfying $\rho^2 = \mathrm{id}_\Pi$ is a *polarity*; the set of polarities is $\mathrm{Pol}(\Pi)$.

**Remark.** A correlation is defined on both points and lines, exchanging the two. For a collineation $f \in \mathrm{Col}(\pi)$, it is enough to define the map naturally on the point set: $f(l)$ is then the image set of the points of $l$. Alternatively, one can define a collineation as a bijection $f: \mathscr{P} \sqcup \mathscr{L} \to \mathscr{P} \sqcup \mathscr{L}$ with

$$\begin{cases} P \in \mathscr{P} \longmapsto f(P) \in \mathscr{P}, \\ l \in \mathscr{L} \longmapsto f(l) \in \mathscr{L}, \end{cases} \qquad P \in l \Longrightarrow f(P) \in f(l).$$

This is equivalent to the earlier definition, because the mapped line $f(l)$ is the same whether understood as the image set or as the value of the map. Both correlations and collineations preserve incidence, but a correlation reverses its direction.

The following properties follow directly from the definitions, without using the explicit form of the transformations.

**Proposition 10.6.37.** *For a correlation $\rho$ of a plane $\Pi$, a point $P$ and a line $l$ satisfy*

$$P \in l \quad \Longleftrightarrow \quad \rho(l) \in \rho(P).$$

**Proof.** It suffices to prove $\rho(l) \in \rho(P) \Rightarrow P \in l$. Suppose $\rho(l) \in \rho(P)$ but $P \notin l$, and choose $A \in l$. Since $P \in PA$, we have $\rho(PA) \in \rho(P)$; since $A \in PA$, we have $\rho(PA) \in \rho(A)$. Because $A \neq P$, bijectivity gives $\rho(A) \neq \rho(P)$. Thus $\rho(PA)$ is the intersection $\rho(A) \cap \rho(P)$.

On the other hand, $A \in l$ gives $\rho(l) \in \rho(A)$, so we also have $\rho(l) = \rho(A) \cap \rho(P)$. Uniqueness of the intersection of two lines now gives $\rho(l) = \rho(PA)$, and bijectivity gives $l = PA$, contrary to $P \notin l$. Hence $P \in l$. □

**Proposition 10.6.38** (Principle of polarity)**.** *A correlation $\rho$ of a projective plane to itself satisfies* $\rho^2 = \mathrm{id}$ *if and only if, for every pair of points $P, Q$,*

$$P \in \rho(Q) \quad \Longleftrightarrow \quad Q \in \rho(P).$$

**Proof.** By proposition 10.6.37,

$$P \in \rho(Q) \quad \Longleftrightarrow \quad \rho^2(Q) \in \rho(P).$$

If $\rho^2 = \mathrm{id}$, this gives $P \in \rho(Q) \Leftrightarrow Q \in \rho(P)$. Conversely, suppose this last equivalence holds. Since $P$ is arbitrary, choose two distinct points $P$ on the line $\rho(Q)$. The displayed equivalence and the assumed symmetric incidence show that both corresponding lines $\rho(P)$ contain $Q$ and $\rho^2(Q)$. Their intersection is unique, so $\rho^2(Q) = Q$. For a line $l$, choose distinct $A, B \in l$. proposition 10.6.37 gives $\rho(l) = \rho(A) \cap \rho(B)$, and hence $\rho^2(l) = \rho^2(A)\rho^2(B) = AB = l$. Thus $\rho^2 = \mathrm{id}$. □

As before, for a point $P$, let $\boldsymbol{p}_P$ denote a chosen homogeneous-coordinate column $\begin{bmatrix} x_1(P) \\ x_2(P) \\ x_3(P) \end{bmatrix}$. For a line $l$, let $\boldsymbol{l}_l$ denote the row $[a \ \ b \ \ c]$ of coefficients in its equation (10.9). In a right-module projective plane, the equivalences $\boldsymbol{p}_P \sim \boldsymbol{p}_P \lambda$ and $\boldsymbol{l}_l \sim \lambda \boldsymbol{l}_l$, $\lambda \neq 0$, give representations of the same point or line. For a transformation $\sigma$ of $\mathbb{K}$ and a matrix (or vector) $\boldsymbol{A} = (a_{ij})$, let $\boldsymbol{A}^\sigma$ denote the result of applying $\sigma$ to every entry, so $\boldsymbol{A}^\sigma = (\sigma(a_{ij}))$. The notation immediately gives the following result.

**Proposition 10.6.39.** *In a two-dimensional projective plane, a point $P$ lies on a line $l$ if and only if $\boldsymbol{l}_l \boldsymbol{p}_P = 0$.*

In this section, also define the notation

$$\mathrm{Aut}^-(\mathbb{K}) := \{f : \mathbb{K} \to \mathbb{K} \mid f \text{ is bijective}, \quad f(a+b) = f(a) + f(b), \quad f(ab) = f(b)f(a); \quad \forall a, b \in \mathbb{K}\}$$

for a set of transformations of $\mathbb{K}$. Since $\mathbb{K}$ and $\mathbb{K}^{\mathrm{op}}$ have the same elements and differ only in their multiplication, this set is naturally obtained by pulling back anti-isomorphisms. Define the natural correspondence $\imath : \mathbb{K}^{\mathrm{op}} \to \mathbb{K}$, $x \mapsto x$. Then

$$\mathrm{Aut}^-(\mathbb{K}) = \{\imath \circ \sigma' \mid \sigma' \in \mathrm{Iso}(\mathbb{K}, \mathbb{K}^{\mathrm{op}})\}.$$

Notice that $\imath$ need not be an isomorphism. Elements of $\mathrm{Aut}^-(\mathbb{K})$ are sometimes called *antiautomorphisms*. This notation allows us to describe correlations of a general projective plane.

**Theorem 10.6.40.** *For $\Pi = \mathbb{P}^2(\mathbb{K})$, the correlations are precisely*

$$\mathrm{Cor}(\Pi) = \left\{\rho \mid \boldsymbol{l}_{\rho(P)} = (\boldsymbol{p}_P^\sigma)^\mathsf{T} \boldsymbol{A}, \quad \boldsymbol{p}_{\rho(l)} = \boldsymbol{A}^{-1} (\boldsymbol{l}_l^\sigma)^\mathsf{T}; \quad \boldsymbol{A} \in \mathrm{GL}(3, \mathbb{K}), \quad \sigma \in \mathrm{Aut}^-(\mathbb{K})\right\}. \tag{10.23}$$

*Here $P$ denotes a point and $l$ a line.*

**Proof.** Let $\Pi = \mathbb{P}^2(\mathbb{K})$, and take its dual plane $\Pi^\vee = \mathbb{P}^2(\mathbb{K}^{\mathrm{op}})$. A line $l$ of $\Pi$ naturally serves as a point of $\Pi^\vee$, and a point $P$ as a line, because

$$(P \in l)_\Pi \iff \boldsymbol{l}_l \boldsymbol{p}_P = 0 \iff (\boldsymbol{p}_P)^\mathsf{T} \circ (\boldsymbol{l}_l)^\mathsf{T} = 0 \iff (l \in P)_{\Pi^\vee},$$

where $\circ$ denotes multiplication in the opposite ring. This is the same duality principle as in theorem 10.5.11.

We therefore seek an element of $\mathrm{Col}(\Pi, \Pi^\vee)$. By the fundamental theorem, it is semilinear for an isomorphism $\sigma' \in \mathrm{Iso}(\mathbb{K}, \mathbb{K}^{\mathrm{op}})$ and a matrix $\boldsymbol{A}' \in \mathrm{GL}(3, \mathbb{K}^{\mathrm{op}})$. In the dual plane, its point-coordinate formula is

$$(\boldsymbol{l}_{\rho(P)})^\mathsf{T} = \boldsymbol{A}' \circ \boldsymbol{p}_P^{\sigma'}.$$

The formula still uses matrix multiplication over $\mathbb{K}^{\mathrm{op}}$, and the image is a column representing a point of the dual plane. To return to $\Pi$, use the map $\imath$ defined above and put $\sigma := \imath \circ \sigma' \in \mathrm{Aut}^-(\mathbb{K})$. After identifying the underlying entries through $\imath$, property (c) of theorem A.5.34 gives

$$\boldsymbol{A}' \circ \boldsymbol{p}_P^{\sigma'} = \left[(\boldsymbol{p}_P^\sigma)^\mathsf{T} (\boldsymbol{A}')^\mathsf{T}\right]^\mathsf{T},$$

with the left product taken in $\mathbb{K}^{\mathrm{op}}$ and the right product in $\mathbb{K}$; the numerical entries of $\boldsymbol{A}'$ on the right are unchanged. By theorem A.5.36, $\boldsymbol{A}' \in \mathrm{GL}(3, \mathbb{K}^{\mathrm{op}}) \Leftrightarrow (\boldsymbol{A}')^\mathsf{T} \in \mathrm{GL}(3, \mathbb{K})$. Substitute $\boldsymbol{A} = (\boldsymbol{A}')^\mathsf{T}$. Since the image represents a line in $\Pi$, transpose to obtain

$$\boldsymbol{l}_{\rho(P)} = (\boldsymbol{p}_P^\sigma)^\mathsf{T} \boldsymbol{A}.$$

Once the images of points are fixed, the image of a line is their unique common incident point:

$$\rho(l) = \bigcap_{P \in l} \rho(P).$$

Indeed,

$$\begin{aligned} P \in l &\iff \boldsymbol{l}_l \boldsymbol{p}_P = 0 \iff \sigma(\boldsymbol{l}_l \boldsymbol{p}_P) = 0 \iff (\boldsymbol{p}_P^\sigma)^\mathsf{T} (\boldsymbol{l}_l^\sigma)^\mathsf{T} = 0 \iff (\boldsymbol{p}_P^\sigma)^\mathsf{T} \boldsymbol{A} \boldsymbol{A}^{-1} (\boldsymbol{l}_l^\sigma)^\mathsf{T} = 0 \\ &\iff \boldsymbol{l}_{\rho(P)} [\boldsymbol{A}^{-1} (\boldsymbol{l}_l^\sigma)^\mathsf{T}] = 0 \iff \boldsymbol{l}_{\rho(P)} \boldsymbol{p}_{\rho(l)} = 0 \iff \rho(l) \in \rho(P). \end{aligned}$$

Consequently,

$$\boldsymbol{p}_{\rho(l)} = \boldsymbol{A}^{-1} (\boldsymbol{l}_l^\sigma)^\mathsf{T}.$$

□

**Corollary 10.6.41.** *A projective plane over $\mathbb{K}$ admits a correlation if and only if $\mathbb{K} \cong \mathbb{K}^{\mathrm{op}}$.*

**Proof.** If $\mathbb{K} \ncong \mathbb{K}^{\mathrm{op}}$, then no $\sigma \in \mathrm{Aut}^{-}(\mathbb{K})$ exists; conversely, such a $\sigma$ supplies a correlation through theorem 10.6.40. □

We next give the conditions for a polarity.

**Theorem 10.6.42.** *On the plane $\Pi = \mathbb{P}^2(\mathbb{K})$, a correlation in the form* (10.23) *is a polarity if and only if there is $c \in \mathbb{K}^\times$ such that*

$$\boldsymbol{A}^\sigma c = \boldsymbol{A}^\mathsf{T}, \qquad \sigma^2 : a \mapsto cac^{-1}.$$

**Proof.** First prove necessity. It is enough to impose $\rho^2(P) = P$ for every point, because the action on lines is then determined. From (10.23),

$$\boldsymbol{p}_{\rho^2(P)} = \boldsymbol{A}^{-1}\left\{\left[(\boldsymbol{p}_P^\sigma)^\mathsf{T}\boldsymbol{A}\right]^\sigma\right\}^\mathsf{T} = \boldsymbol{A}^{-1}(\boldsymbol{A}^\sigma)^\mathsf{T}\boldsymbol{p}_P^{\sigma^2}.$$

In the second equality we used property (c) of theorem A.5.34. Notice also that $\sigma$ reverses multiplication, so $(\boldsymbol{AB})^{\sigma\mathsf{T}} = (\boldsymbol{B}^\sigma)^\mathsf{T}(\boldsymbol{A}^\sigma)^\mathsf{T}$. Equality here is understood under the equivalence $\boldsymbol{\mu} \sim \boldsymbol{\mu}\lambda$, $\lambda \neq 0$. Put $\boldsymbol{B} = \boldsymbol{A}^{-1}(\boldsymbol{A}^\sigma)^\mathsf{T}$ and $\sigma^2 = \tau$. Then $\tau \in \mathrm{Aut}(\mathbb{K})$. The condition says that, for every $P$, some $\lambda_P \in \mathbb{K}^\times$ satisfies

$$\boldsymbol{B}\boldsymbol{p}_P^\tau = \boldsymbol{p}_P\lambda_P.$$

For $P \neq Q$, their representatives $\boldsymbol{p}_P, \boldsymbol{p}_Q$ are right-linearly independent in $\mathbb{K}^3$. Their sum $\boldsymbol{p}_P + \boldsymbol{p}_Q \neq \boldsymbol{0}$ therefore represents a point. Applying the relation also to that sum gives

$$(\boldsymbol{p}_P + \boldsymbol{p}_Q)\lambda_{P+Q} = \boldsymbol{B}(\boldsymbol{p}_P + \boldsymbol{p}_Q)^\tau = \boldsymbol{B}\boldsymbol{p}_P^\tau + \boldsymbol{B}\boldsymbol{p}_Q^\tau = \boldsymbol{p}_P\lambda_P + \boldsymbol{p}_Q\lambda_Q.$$

Right linear independence therefore forces $\lambda_{P+Q} = \lambda_P = \lambda_Q$. Hence $\lambda_P$ has one constant value for every point $P$; denote it by $c'$.

For $a \in \mathbb{K}$,

$$\boldsymbol{p}_P ac' = \boldsymbol{B}(\boldsymbol{p}_P a)^\tau = \boldsymbol{B}\boldsymbol{p}_P^\tau \tau(a) = \boldsymbol{p}_P c'\tau(a).$$

Since $\boldsymbol{p}_P \neq \boldsymbol{0}$, at least one component $p_i \neq 0$, and the corresponding component equation is $p_i ac' = p_i c'\tau(a)$. Cancellation gives $ac' = c'\tau(a)$, and hence $\tau(a) = (c')^{-1}ac'$.

We now have $\boldsymbol{B}\boldsymbol{p}_P^\tau = \boldsymbol{p}_P c'$. Substitute successively the standard basis columns

$$\boldsymbol{e}_1 = \begin{bmatrix}1 & 0 & 0\end{bmatrix}^\mathsf{T}, \qquad \boldsymbol{e}_2 = \begin{bmatrix}0 & 1 & 0\end{bmatrix}^\mathsf{T}, \qquad \boldsymbol{e}_3 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}^\mathsf{T}.$$

Since $\tau$ fixes 1, the first substitution gives $\boldsymbol{B}\boldsymbol{e}_1 = \boldsymbol{e}_1 c'$, so $(\boldsymbol{B})_{11} = c'$, $(\boldsymbol{B})_{21} = (\boldsymbol{B})_{31} = 0$. The other two substitutions determine the remaining entries in the same way, and therefore $\boldsymbol{B} = c'\mathbf{I}_3$. With $c = (c')^{-1}$, we obtain $c \in \mathbb{K}^\times$ and

$$\sigma^2 : a \mapsto cac^{-1}, \qquad \boldsymbol{A}^{-1}(\boldsymbol{A}^\sigma)^\mathsf{T} = c^{-1}\mathbf{I}.$$

Thus $\sigma^2 \in \mathrm{Inn}(\mathbb{K})$. Multiplying the matrix equality on the left by $\boldsymbol{A}$ and on the right by $c$ gives $(\boldsymbol{A}^\sigma)^\mathsf{T}c = \boldsymbol{A}$; transposing gives $\boldsymbol{A}^\sigma c = \boldsymbol{A}^\mathsf{T}$. The same calculation in reverse proves sufficiency. □

When $\sigma = \mathrm{id}$, $\mathbb{K}$ must be a field, since otherwise $\mathrm{id} \notin \mathrm{Aut}^{-}(\mathbb{K})$. A correlation $\rho$ then involves only the action of the square matrix $\boldsymbol{A}$. Such a correlation is a *projective correlation*; if it is a polarity, it is called a *projective polarity*. By (10.18) the action of $\boldsymbol{A}$ is a projectivity to the dual plane, and by lemma 10.6.23 every inner automorphism of a field is trivial. Geometrically, over a field a correlation is projective exactly when it preserves the cross-ratios of corresponding elements. For four concurrent lines $a, b, c, d$, define their cross-ratio by intersecting with any transversal $l$:

$$(ab, cd) = (a \cap l, b \cap l; c \cap l, d \cap l).$$

Indeed, a correlation with antiautomorphism $\sigma$ sends a cross-ratio $\lambda$ to $\sigma(\lambda)$, and $\lambda$ ranges over all of $\mathbb{K}$: take a point on an axis, the unit point, the origin, and the point at infinity; their cross ratio is the inhomogeneous coordinate of the chosen point. Thus preservation of every cross ratio forces $\sigma = \mathrm{id}$.

**Corollary 10.6.43.** *For $\Pi = \mathbb{P}^2(\mathbb{F})$ over a field, every projective correlation has the form*

$$\boldsymbol{l}_{\rho(P)} = \boldsymbol{p}_P^{\top}\boldsymbol{A}, \quad \boldsymbol{p}_{\rho(l)} = \boldsymbol{A}^{-1}\boldsymbol{l}_l^{\top}, \qquad \boldsymbol{A} \in \mathrm{GL}(3,\mathbb{F}).$$

*Here $P$ denotes a point and $l$ a line.*

**Corollary 10.6.44.** *On the projective plane $\Pi = \mathbb{P}^2(\mathbb{F})$ over a field, a projective correlation is a projective polarity only if the matrix in corollary 10.6.43 is an invertible symmetric matrix:*

$$\boldsymbol{A}^{\top} = \boldsymbol{A}.$$

**Proof.** Set $\sigma = \mathrm{id}$ in theorem 10.6.42. Then $c\boldsymbol{A}^{\top} = \boldsymbol{A}$ for some $c \in \mathbb{F}^{\times}$, whence

$$\boldsymbol{A} = c\boldsymbol{A}^{\top} = c(c\boldsymbol{A}^{\top})^{\top} = c^2\boldsymbol{A}.$$

Since $\boldsymbol{A}$ is nonzero, $c^2 = 1$, so $c = \pm 1$: indeed, $c^2 = 1 \Leftrightarrow (c+1)(c-1) = 0$, and a product of two nonzero field elements is nonzero, so $c+1 = 0$ or $c-1 = 0$. If $\operatorname{char}\mathbb{F} = 2$, then $1 = -1$ and $\boldsymbol{A}^{\top} = \boldsymbol{A}$. If $\operatorname{char}\mathbb{F} \neq 2$, then $1 \neq -1$, so $\boldsymbol{A}^{\top} = \pm\boldsymbol{A}$, but a skew-symmetric $3\times 3$ matrix is singular; therefore $\boldsymbol{A}^{\top} = \boldsymbol{A}$ by exercise A.93. $\square$

**Remark.** Over a field, one may instead use the polarity principle in place of the equivalent condition $\rho^2 = \mathrm{id}$. Its incidence form is

$$\boldsymbol{x}^{\top}\boldsymbol{A}\boldsymbol{y} = 0 \quad \Longleftrightarrow \quad \boldsymbol{y}^{\top}\boldsymbol{A}\boldsymbol{x} = 0$$

for arbitrary column vectors $\boldsymbol{x},\boldsymbol{y}$. Starting from the form in corollary 10.6.43, the field version of exercise A.21 gives $\boldsymbol{A}^{\top} = \pm\boldsymbol{A}$, and the preceding argument finishes the proof.

For a correlation $\rho$ of a projective plane over a field, a point $P$ satisfying $P \in \rho(P)$ is an *absolute point*. Suppose the projective correlation is represented by $\boldsymbol{A}$. By corollary 10.6.43, $\boldsymbol{l}_{\rho(P)} = \boldsymbol{p}_P^{\top}\boldsymbol{A}$, so

$$P \in \rho(P) \Leftrightarrow \boldsymbol{l}_{\rho(P)}\boldsymbol{p}_P = 0 \Leftrightarrow \boldsymbol{p}_P^{\top}\boldsymbol{A}\boldsymbol{p}_P = 0.$$

Write $\boldsymbol{p}_P = [p_1 \;\; p_2 \;\; p_3]^{\top}$ and $\boldsymbol{A} = (a_{ij})$. Then

$$\boldsymbol{p}_P^{\top}\boldsymbol{A}\boldsymbol{p}_P = \sum_i a_{ii}p_i^2 + \sum_{i<j}(a_{ij} + a_{ji})p_ip_j. \tag{$\bigstar$}$$

This is a quadratic expression in $p_1,p_2,p_3$, and it is not identically zero. Otherwise $a_{ii} = 0$ and $a_{ij} + a_{ji} = 0$, making the odd-dimensional alternating matrix $\boldsymbol{A}$ singular. This is impossible by exercise A.93. Although homogeneous equations of conics have not yet been introduced formally, this already gives an important conclusion.

**Theorem 10.6.45.** *The absolute points of a projective correlation of a projective plane over a field, if any, form a possibly degenerate conic.*

We have not yet formally introduced coordinate equations of conics, but we can explain the idea here; details appear in later chapters. The background discussion in section 1.2, (1.1), gave an equation for conics in the real Euclidean plane. Analogously, the finite part of a conic in a projective plane over a general field can be written

$$a_{11}x^2 + a_{12}xy + a_{22}y^2 + a_{13}x + a_{23}y + a_{33} = 0.$$

Here $(x,y)$ are the inhomogeneous coordinates of a point on the curve. Using homogeneous coordinates $[p_1 : p_2 : p_3]$, with $x = p_1/p_3$ and $y = p_2/p_3$, gives the equation

$$a_{11}p_1^2 + a_{12}p_1p_2 + a_{22}p_2^2 + a_{13}p_1p_3 + a_{23}p_2p_3 + a_{33}p_3^2 = 0. \tag{$\star$}$$

This is exactly the form in ($\bigstar$).

Here ($\star$) is the intrinsic projective form, not merely a formula derived by homogenizing a prior definition. Its coefficients need only be not all zero. In the preceding inhomogeneous form, however, $a_{11},a_{12},a_{22}$ must not all vanish. If they all vanish, the inhomogeneous form degenerates to a line, while the homogeneous form has the additional solution branch $p_3 = 0$. It therefore degenerates to the line $a_{13}p_1 + a_{23}p_2 + a_{33}p_3 = 0$ together with the line at infinity, and is still quadratic. Finally, the coefficients may take general values; there may be no point over $\mathbb{F}$ satisfying the equation. It remains formally quadratic and can be regarded as an imaginary conic.

The following result is the polarity described earlier in chapter 6.

**Theorem 10.6.46.** *Over a field of characteristic different from* 2*, the absolute points of a projective polarity, if any, form a nondegenerate conic. Projective polarities correspond bijectively to nondegenerate conics. Under this correspondence the image of a point is its polar line, and the image of a line is its pole.*[45]

**Proof.** The quadratic equation associated with (☆) can be written

$$\boldsymbol{p}_P^\top \boldsymbol{A}\boldsymbol{p}_P = 0, \qquad \boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12}/2 & a_{13}/2 \\ a_{12}/2 & a_{22} & a_{23}/2 \\ a_{13}/2 & a_{23}/2 & a_{33} \end{bmatrix}.$$

Thus quadratic equations correspond exactly to the symmetric matrices of corollary 10.6.44. The homogeneous-coordinate equivalence $\boldsymbol{A} \sim \lambda\boldsymbol{A}$, $\lambda \neq 0$, gives the same polarity in the correlation formula; likewise, $\boldsymbol{A} \sim \lambda\boldsymbol{A}$ gives the same conic in (☆). Here 2 means $1+1$ over the given field. An invertible symmetric matrix defines a nondegenerate conic. Characteristic 2 is excluded because then $2=0$, so this conclusion does not hold.

We do not define tangents and polars of conics over a general field here. For real or complex projective planes, however, this algebraic polarity agrees with the geometric one of chapter 6. Let $f$ be the polarity and $\gamma$ its absolute conic. If $A \in \gamma$ and $f(A)$ met $\gamma$ again at $B$, then

$$B \in \gamma \Rightarrow B \in f(B), \qquad B \in f(A) \Rightarrow A \in f(B),$$

so $f(B) = AB = f(A)$, contradicting bijectivity. Hence $f(A)$ is tangent at $A$. If $A \notin \gamma$, work in the complex projective plane and let $AP, AQ$ be the two tangents, with points of contact $P, Q$. Since $P \in \gamma$, one has $P \in f(P) = AP$, and $A \in f(P)$; the polarity principle therefore gives $P \in f(A)$. Similarly $Q \in f(A)$, so $f(A) = PQ$, the usual polar. Thus the construction agrees with that of chapter 6. □

**Example 10.6.47.** Since $\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}\}$, every polarity of the real projective plane in the present sense is projective. Thus a possibly degenerate real projective conic may equivalently be defined as the absolute locus of a correlation, and a nondegenerate one as the absolute locus of a polarity in the sense defined in this chapter. The latter is also the description used in exercise 6.52.

The same assertions hold in the complex projective plane after the adjective *projective* is explicitly imposed on the correlation or polarity. The preceding proof of equivalence with the earlier definition uses coordinates. Readers who would prefer a purely geometric proof can find one in the exercises.

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## Exercises

**Exercise 10.56*.**

(1) Prove in detail the assertion of example 10.6.34: $\mathrm{Aut}(\mathbb{R}) = \{\mathrm{id}_{\mathbb{R}}\}$.

(2) Prove in detail the assertion of example 10.6.35: the only continuous automorphisms of $\mathbb{C}$ are the identity and complex conjugation.

**Exercise 10.57*.** In the non-Desarguesian plane $\pi$ of example 10.5.1, define $f_{a,c,\epsilon} : \pi \to \pi$ by sending each point $(x,y)$ to $(ax+c, \epsilon ay)$ in the corresponding real Euclidean coordinate model. Put $G = \{f_{a,c,\epsilon} \mid a > 0,\ c \in \mathbb{R},\ \epsilon = \pm 1\}$, and prove that $G$ is a transformation group of $\pi$ and that $G \leq \mathrm{Col}(\pi)$.

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[45] The conic is meant as a formal homogeneous quadratic equation; its point set over the given field may be empty, just as an imaginary conic has no points in the real projective plane.

**Exercise 10.58.** Do correlations and polarities exist on $\mathbb{P}^2(\mathbb{H})$? If so, give explicit examples and determine their absolute loci.

**Exercise 10.59.** Determine all continuous collineations and polarities of the complex projective plane; they need not be projective. What is the absolute locus of a continuous polarity?

**Exercise 10.60*.** Let $\boldsymbol{A}$ be an invertible $n\times n$ matrix over a division ring $D$, $n\geq 2$, and let $\sigma\in\mathrm{Aut}^-(D)$. Suppose that for all columns $\boldsymbol{x},\boldsymbol{y}\in D^n$,

$$(\boldsymbol{x}^\sigma)^\mathsf{T}\boldsymbol{A}\boldsymbol{y}=0 \quad\Longleftrightarrow\quad (\boldsymbol{y}^\sigma)^\mathsf{T}\boldsymbol{A}\boldsymbol{x}=0.$$

Prove that some $c\in D^\times$ satisfies

$$\sigma^2(a)=cac^{-1}, \qquad \boldsymbol{A}=(\boldsymbol{A}^\sigma)^\mathsf{T}c.$$

**Exercise 10.61.** For a correlation $f$ of a projective plane $\pi$, put $G_f=\{g\in\mathrm{Col}(\pi)\,|\,g\circ f=f\circ g\}$. Prove $G_f\leq\mathrm{Col}(\pi)$. Must $G_f$ be normal?

**Exercise 10.62.** For a projective plane $\pi$, put $G=\mathrm{Col}(\pi)\cup\mathrm{Cor}(\pi)$.
(1) Prove that $G$ is a group, sometimes called the *full projective group*.[46]
(2) Prove $\mathrm{Col}(\pi)\trianglelefteq G$.
(3*) Find the index of $\mathrm{Col}(\pi)$ in $G$; see exercise A.37.
(3) Determine when $G=\mathrm{Col}(\pi)\rtimes\mathbb{Z}_2$.

**Exercise 10.63.** For a projective plane $\pi$, let $G$ be the full projective group of item (2), and let $\rho,\tau\in\mathrm{Pol}(\pi)$. Prove that the smallest subgroup of $G$ containing $\rho,\tau$ has twice the order of the collineation $\rho\circ\tau$ in $G$.

**Exercise 10.64.**
(1) For a prime $q$, count the distinct collineations, correlations, polarities, projective correlations, and projective polarities of $\Pi=\mathbb{P}^2(\mathbb{F}_q)$.
(2) Write all these transformations explicitly for $q=2$.
(3*) Repeat the count for $\pi=\mathbb{P}^2(\mathbb{F}_3[x]/\langle x^2+1\rangle)$; see exercise 10.35.

**Exercise 10.65.** Give a synthetic proof that projective polarities as defined here agree with the earlier definition in chapter 6 for real and complex projective planes.
(1) First prove the following lemma. Let distinct lines $m,n$ meet at $Z$, and choose $X\in m\setminus\{Z\}$, $Y\in n\setminus\{Z\}$. Let $f_1$ be a projectivity from the pencil at $X$ to the range on $m$, and $f_2$ one from the pencil at $Y$ to the range on $n$, with $f_1(XY)=f_2(XY)=Z$. Define a correspondence $g$ from the pencil at $X$ to that at $Y$ as follows. For lines $a,b$ through $X,Y$, respectively, set $g(a)=b$ when $a\cap b$, $f_1(a)$, and $f_2(b)$ are collinear, excluding the singular solution in which $a$ or $b$ is $XY$. Prove that $g$ is a projectivity.
(2) Prove first that the absolute points of a projective correlation lie on a conic. Since a projective correlation preserves cross-ratios, it is enough to prove the following assertion. Let $A_1,\ldots,A_6$ be six points, no three collinear, and let $a_1,\ldots,a_6$ be six lines, no three concurrent, with $A_i\in a_i$. Suppose that for every five distinct indices $i,j,k,l,n\in\{1,\ldots,6\}$,

$$(A_nA_i,A_nA_j;A_nA_k,A_nA_l)=(a_n\cap a_i,a_n\cap a_j;a_n\cap a_k,a_n\cap a_l).$$

Use part (1) to prove that $A_1,\ldots,A_6$ lie on one conic.
(3) Use part (2) to deduce the assertion for a polarity.

[46] Some authors use this name for the collineation group instead.

# Chapter 11

# Epilogue—A History of Geometry

Geometry is one of the oldest and most continuous subjects in the history of mathematics. It first served surveying, architecture, and astronomy; in ancient Greece it was raised into a rigorous deductive theory; and in the modern era it underwent successive turns toward analytic, projective, axiomatic, topological, and algebraic modes of thought. The history of geometry is therefore not merely a chronology of people and theorems. It is also the history of repeated attempts to understand such fundamental questions as what geometric objects are, what geometric methods are, and on what geometric truth depends. Over this long development, some concepts recurred, some methods fell dormant and were later revived, and some objects were renamed in new languages. Geometry changed its appearance many times, yet one persistent question can still be traced through its history: how are we to understand shapes and relations in space?

This chapter aims to sketch, in a reasonably systematic way, the principal lines along which geometry developed from antiquity to the present. We shall focus especially on the ideas that genuinely changed its character: axiomatization, coordinates, invariants of projection, curvature, abstract spaces, and algebraic structure. No chapter of this length can recount every development in every region and branch, nor is the intention to produce a long catalogue of names. Instead, we pause at several turning points to organize widely recognized historical facts, important dates, representative figures, central ideas, and major historical interpretations, so as to give a reasonably complete view of the subject's evolution.

One preliminary qualification is necessary. A "history of geometry" here does not mean a chronological list of all geometric theorems. The preceding chapters have already developed a great deal of plane geometry through its theorems, constructions, and methods. Our concern now is where that material belongs in the history of mathematics and why it acquired special importance at particular times. In other words, this chapter is more concerned with the evolution of geometric ideas than with proving each theorem again. Seeing the continuity between classical plane geometry and modern geometry will also help the reader understand why the apparently scattered objects encountered earlier in the book repeatedly return, and why mathematicians in different centuries assigned them new meanings.

## 11.1 Early Geometry: Measurement and Deduction

Geometry began in practice rather than in theory. As early as the second millennium BCE, the ancient Egyptians had accumulated extensive knowledge of length, area, and slope through resurveying land and constructing pyramids and hydraulic works. The Babylonians of Mesopotamia developed still more refined numerical techniques for calendars, taxation, and engineering. The famous Plimpton 322 tablet, [1] for example, is generally dated to around 1800 BCE. Its contents are related to several Pythagorean triples, showing that mathematicians of the period had, in some sense, mastered constructions of integer solutions to relations such as $a^2+b^2=c^2$. It is important, however, that this knowledge remained largely a body of algorithms or tabular techniques, intended for calculation, measurement, or approximation rather than for the construction of a theory supported by definitions, axioms, and proofs.

This practical background does not diminish the value of ancient geometric knowledge. On the contrary, it reminds us that geometry did not begin as an abstract game detached from reality, but grew out of spatial experience in land, building, astronomy, and craft. Egyptian calculations of area

[1] Plimpton 322 is its modern collection number, taken from the American collector George Arthur Plimpton. The tablet is now held by Columbia University and is generally assigned to the Old Babylonian period.

and volume and problems about slope, together with Mesopotamian tables and measuring techniques, all supplied material for later geometric ideas. What they did not yet constitute was a theory of why the results had to be so. They were more like an extremely effective store of rules, capable of solving many concrete problems but not yet treating their underlying logical structure as an independent object of study. This distinction is essential to understanding the historical significance of Greek geometry.

Modern historians have long debated how much Greek mathematics absorbed from Egypt and Babylonia. A cautious account is that the Greeks did inherit a great deal of practical knowledge from the East, but between the sixth and fourth centuries BCE they accomplished a decisive conceptual shift: they made practical techniques into objects of deductive proof. Thales of Miletus (Θαλῆς ὁ Μιλήσιος), [2] who lived in the sixth century BCE, is traditionally regarded as one of the symbolic figures of this transition. However credible the stories that he measured the height of a pyramid or used an idea of similarity to determine a ship's distance may be, posterity consistently cast him as a forerunner of proof-based geometry. That tradition itself reveals how Greek culture came to understand geometry: it was no longer merely an arithmetical art of measurement, but knowledge that ought to explain why something is true.

The movement from experience to proof did not mean that Greek mathematicians suddenly abandoned practical problems. Many early propositions remained closely connected with measurement, similarity, area, and construction. What changed was the standard by which knowledge was judged. A conclusion was no longer considered sufficient merely because it worked in computation and was reliable in experience; it also had to be placed within a chain of definitions, premises, and inferences. In this sense Greek geometric proof has a special place in the history of civilization. It was not only a mathematical technique, but a new way of organizing knowledge: beginning from a finite set of starting points and joining many conclusions into a whole by reasoning that could be checked.

The wider intellectual setting also helped turn Greek geometry in this direction. Argument, definition, and conceptual distinction gradually became important rational activities in Greek philosophy and education. Mathematics did not develop in isolation, but alongside philosophy, logic, and the growth of scholarly communities. By the time of the Academy of Plato of Athens (Πλάτων ὁ Ἀθηναῖος), [3] geometry was widely regarded as an important discipline for training reason and elevating speculative thought. Aristotle of Stagira (Ἀριστοτέλης ὁ Σταγειρίτης) [4] strengthened the standing of demonstrative knowledge from another direction through his discussions of definition, deduction, and the structure of knowledge. We should not project later formal logic directly onto early geometry; nevertheless, geometry became exemplary in the Greek world in part because it supported, and was supported by, the broader inquiry into what counts as reliable knowledge.

Alongside Thales arose the school of Pythagoras of Samos (Πυθαγόρας ὁ Σάμιος). [5] Living around the turn of the sixth to the fifth century BCE, the Pythagoreans emphasized relations among number and harmony, figure and proportion. The theorem now bearing their name is commonly associated with them, although it had antecedents in Eastern traditions. The Greeks' crucial contribution was not to know the result for the first time, but to place it within deductive argument. In its later standard form, the Pythagorean theorem states that, if a right triangle $ABC$ has $\angle A = 90°$, then

$$AB^2 + AC^2 = BC^2.$$

For Greek mathematics the proposition mattered not merely because it was useful, but because it showed that a relation within a figure could follow necessarily from reasoning.

The importance of the Pythagorean school extended far beyond a single theorem. By linking number, proportion, music, and figure, it made the relation between shape and number a lasting problem. This tendency promoted theories of proportion and similarity, but it also brought a profound crisis: the discovery of incommensurable magnitudes showed that relations among geometric segments could not simply be reduced to ratios of integers. For a tradition centered on the belief that "all

[2] c. 624–c. 546 BCE.

[3] c. 427–347 BCE.

[4] 384–322 BCE.

[5] c. 570–c. 495 BCE.

is number," this was a fundamental challenge. Greek geometry then had to answer a new question: if some geometric magnitudes cannot be adequately described by ordinary integer ratios, can mathematical rigor still be achieved? The later theory of proportion was a systematic response. In the long view, the crisis helped geometry free itself from excessive dependence on a universe of integral ratios and form a more mature theory of magnitude.

The further maturation of Greek geometry depended on extensive work in the fifth and fourth centuries BCE. Hippocrates of Chios (Ἱπποκράτης ὁ Χῖος) [6] studied the areas of lunes in connection with squaring the circle. Theodorus of Cyrene (Θεόδωρος ὁ Κυρηναῖος) and Theaetetus of Athens (Θεαίτητος ὁ Ἀθηναῖος) [7] advanced the study of irrational magnitudes. Eudoxus of Cnidus (Εὔδοξος ὁ Κνίδιος) [8] developed the theory of proportion later incorporated into Euclid's *Elements*, [9] thereby providing a sound basis for treating incommensurable magnitudes. The historical importance of this work is that Greek mathematics did not confine itself to simple figures: it continually confronted such deep problems as how number and shape correspond and whether irrational objects can be admitted into a rigorous theory. From the beginning, then, Greek geometry was not merely the study of figures, but also a foundational training in rigor.

In this process geometry gradually acquired a distinctive synthetic character. The synthetic method does not reject calculation; rather, it begins preferentially from figures, constructions, and relations themselves, placing each proposition within a system in which established conclusions continually yield new ones. The method is especially well suited to expressing equal segments, equal angles, proportion, parallelism, and similarity, and to cultivating a direct grasp of geometric structure. The later rise of analytic geometry radically altered geometric practice, but the classical synthetic method never disappeared, because it preserves structural intuition that an algebraic expression may not display directly. From the standpoint of historical interpretation, the deepest Greek legacy is not only a collection of theorems, but an attitude toward knowledge: a belief that the relations within a figure can explain themselves.

Classical Greek mathematics also produced several celebrated construction problems that were repeatedly revisited: doubling the cube, trisecting an arbitrary angle, and squaring the circle. Strictly speaking, these do not all belong to the core of early elementary geometry, but they played a large historical role. On the one hand, the technical difficulty of the straightedge-and-compass tradition repeatedly forced theory forward. On the other, the problems remind us that Greek geometry was not a static textbook system, but was always accompanied by challenges, setbacks, and unresolved questions. A mature discipline comes into being not merely by arranging known results, but by confronting difficult problems over long periods. Part of the greatness of Greek mathematics is that it did not dismiss these difficulties, but preserved them at the boundaries of theory as motives for later development.

The culmination of this tradition was Euclid of Alexandria (Εὐκλείδης ὁ Ἀλεξανδρεύς), who was active around 300 BCE, and his *Elements*. [10] The greatness of the *Elements* lies not only in its compilation of many earlier results, but still more in its unprecedentedly self-conscious organization: it begins with definitions, postulates, and common notions, and then derives a network of propositions. For the first time, this method made geometry a model for the rational organization of knowledge. The parallel postulate of plane geometry, for example, is often given in later language as follows: through a point outside a line, exactly one line can be drawn parallel to the given line. Whether or not this wording exactly reproduces Euclid's text, its underlying idea is the same: a theoretical system must accept certain starting points, from which the remaining conclusions are to be derived.

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[6] c. 470–c. 421 BCE.

[7] Theodorus of Cyrene lived c. 465–c. 398 BCE; Theaetetus of Athens, c. 417–c. 369 BCE.

[8] His dates are disputed and lie roughly in the first half of the fourth century BCE; common estimates include c. 408–355 and c. 390–c. 340 BCE.

[9] The original Greek title is Στοιχεῖα, meaning "elements" or "fundamental principles"; the Latin title is *Elementa*, and the conventional English title is *Elements*. The work is also referred to simply as the *Elements*.

[10] That is, "Euclid of Alexandria." Very little is known of his life; he is generally thought to have been active around 300 BCE.

The *Elements* remained a standard not simply because it contained many correct propositions, but because it displayed a reproducible scholarly form. For centuries, whether studying geometry or discussing mechanics, optics, astronomy, or even philosophy, writers often idealized the sequence "definitions, then postulates, then proofs." The influence of this form exceeded that of its particular mathematical contents. In many periods, "geometric proof" was almost synonymous with rigorous knowledge. This does not mean that the *Elements* attained the standards of a modern formal system. It continued to rely on spatial intuition, and some conclusions silently assumed continuity, coincidence, and positional relations. Those very features, however, make it a faithful expression of the historical state of classical geometry: rigor had begun, but full formalization had not.

The organization of the *Elements* also deserves historical attention. It is not limited to plane figures, but weaves elementary constructions, proportion, area, number theory, irrational magnitudes, and solid geometry into a whole. For Euclid, "geometry" was therefore never merely a narrow craft of handling plane figures, but a deductive way of organizing mathematics. Some material in the work would now be classified as elementary number theory or solid geometry, but within its classical system these topics belonged to one world sustained jointly by magnitude, ratio, shape, and construction. That unity is difficult for many later elementary textbooks to preserve in full.

The influence of the *Elements* crossed historical periods. In antiquity it summarized the Greek mathematical tradition; in the Middle Ages and Renaissance it became a standard work of mathematical education and scientific training; and in the modern era many philosophers treated "thinking geometrically" as an ideal of rigorous reasoning. Modern historiography nevertheless warns against turning Euclid into an idol. His axiomatic system still depends on visual intuition, and some arguments appeal to an unformalized background of position and continuity. Yet these distinctly nonmodern features make the actual course of the history clearer: the deductive tradition was not completed at one stroke, but matured through repeated correction, clarification, and refinement.

The principal historical conclusion about the period from the ancient East to the Greek tradition can therefore be summarized as follows. Geometry was not a product of pure ideas created from nothing; it had a deep practical background. The turning point in Greek mathematics was the first systematic transformation of that practical knowledge into an axiomatic, proof-based subject. From surveying land to proving theorems, from empirical rules to a deductive system, and from "this procedure works" to "this must be so," geometry acquired, in this slow and profound transition, its oldest and most enduring identity. In this sense, geometry here truly became a science.

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## Exercises

**Exercise 11.1.** Complete the following statements from this section.

(1) The famous Plimpton 322 tablet is generally thought to be related to several ________ triples.
(2) The sixth-century-BCE figure traditionally regarded as a symbol of Greek geometry's movement from empirical measurement to deductive proof is ________.
(3) The theorem now called the ________ theorem is commonly associated with the Pythagorean school; historically, the crucial achievement was not simply knowing the conclusion but placing it within ________.
(4) In Greek philosophy and education, ________, definition, and conceptual distinction gradually became important rational activities, helping geometry become an exemplary form of knowledge.

**Exercise 11.2.** Answer briefly.

(1) How did geometry first arise?
(2) Briefly describe the significance of Euclid's *Elements*.

**Exercise 11.3.** Read the following passage and answer the questions.

> *A later tradition says that the entrance to Plato's Academy bore the words Ἀγεωμέτρητος μηδεὶς εἰσίτω, "Let no one ignorant of geometry enter." The story may not be reliable, and there is no certainty that these words really stood at the early Academy. Yet its long circulation shows that later generations often regarded geometry as an essential preparation for philosophical speculation and rational training. Even if the inscription is not strict historical fact, it still reflects one understanding of the mathematical spirit of Plato's Academy.*

(1) Do you think the story is true? Why can a story of this kind retain some value for historical interpretation even if it is not reliable fact?

(2) Using the passage and this section, explain why Greek geometry came to be regarded as an important form of rational training. Analyze the relations among geometry, philosophy, and the educational tradition.

## 11.2 Greek and Medieval Geometry: Zenith and Transmission

If Euclid established the deductive framework of plane geometry, the Hellenistic period of the following centuries carried geometric research to more intricate and refined objects. Its two most prominent figures were Archimedes of Syracuse (Ἀρχιμήδης ὁ Συρακόσιος) and Apollonius of Perga (Ἀπολλώνιος ὁ Περγαῖος). [11] Archimedes studied levers, floating bodies, and mechanical questions, while also developing geometry of extraordinary sophistication for areas, volumes, and the method of exhaustion. His treatments of the volume of a sphere, the area of a spherical cap, and the area of a parabolic segment are among many particularly subtle examples. From a modern perspective, much of his reasoning comes close to ideas of integration, although he had no unified language of analysis.

Archimedes' distinctive position derives from his embodiment both of the rigor of classical geometry and of the creativity of applied mathematics. In mechanics he did not merely make empirical estimates, but sought to place equilibrium, centers of gravity, buoyancy, and related questions within geometric reasoning. When treating areas and volumes, he did not simply give formulas, but used the method of exhaustion to approximate curvilinear figures. The underlying idea is crucial: an object that cannot be divided directly into finitely many simple figures can be bounded by a controlled sequence of approximations. Ancient Greece possessed neither limit notation nor modern integration, but this way of thinking prepared a deep intuition for later analysis.

Apollonius, by contrast, influenced later plane geometry most directly through his work on conic sections. His *Conics* [12] systematically treated diameters, tangents, conjugate diameters, and other properties of conics, elevating the ellipse, parabola, and hyperbola from a collection of scattered phenomena to a structurally rich family of objects. An important historical point follows: conics did not become genuinely important only with the later appearance of analytic geometry. They had already become central objects of advanced geometric research in late antiquity. Pascal's and Brianchon's theorems, projective definitions, and the orbital theory of modern celestial mechanics may all be viewed as renewals of this ancient inheritance under different historical conditions.

Hellenistic geometry also increasingly moved beyond elementary figures toward more complicated curves, surfaces, mechanical curves, and astronomical models. This development did not reject Euclid's tradition, but extended it. The *Elements* supplied a basic language and standard of proof, while Hellenistic mathematicians continually expanded the range of objects that could be studied in that language. Historically, classical geometry already contained a strong movement toward higher mathematics. The common impression that it studied only triangles and circles often reflects the selections of elementary textbooks rather than the actual character of ancient mathematics.

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[11] Archimedes of Syracuse lived c. 287–212 BCE; Apollonius of Perga, c. 262–c. 190 BCE.

[12] The original Greek title is Κωνικά; its Latin title is *Conica*, and the conventional English title is *Conics*. The work is traditionally described as consisting of eight books. Book VIII is lost, and part of the work was preserved through the Arabic tradition.

Around the second century CE, Claudius Ptolemy (Κλαύδιος Πτολεμαῖος) [13] made extensive use of geometric models in astronomy. In the later Greek mathematical tradition, Heron of Alexandria (Ἥρων ὁ Ἀλεξανδρεύς), Pappus of Alexandria (Πάππος ὁ Ἀλεξανδρεύς), and Proclus the Successor (Πρόκλος ὁ Διάδοχος) [14] played key roles, respectively, in applied geometry, the commentary tradition, and the preservation of materials for the history of mathematics. Pappus's *Mathematical Collection* [15] is especially important. It preserves many propositions and ideas of earlier geometers, and supplies sources or clues for numerous subjects that reappeared in the seventeenth and nineteenth centuries. Without compilers such as Pappus, much of advanced Greek geometry would probably have been lost.

The commentary tradition of this period deserves particular notice. To a modern reader, commentary can appear to be mere explanation rather than creation. In late antiquity and the Middle Ages, however, commentary was often the principal means of preserving, reorganizing, and reinterpreting knowledge. Much material that once circulated independently survived only through commentaries, compilations, and teaching texts. The history of geometry therefore belongs not only to those who proposed new theorems, but also to those who preserved terminology, organized propositions, explained methods, and transmitted a tradition to later civilizations. The importance of Pappus and Proclus should not be underestimated in this respect.

As the Western Roman Empire declined, the direct transmission of Greek geometry in Western Europe weakened for a time. The tradition did not end, but acquired new life in the Arabic-speaking world. From the eighth to the tenth centuries, the translation movement centered in Baghdad and elsewhere rendered many Greek mathematical works into Arabic. In the process geometry was not only preserved, but investigated further. Al-Ḥasan ibn al-Haytham (الحسن بن الهيثم), [16] for example, brought the analysis of visual rays, perspective, and projection to a more refined theoretical level through his work on optics. Discussion of the parallel postulate also continued in Arabic mathematics.

The importance of the translation movement should not be understood as simple conveyance. Every passage from Greek into Arabic and later from Arabic into Latin involved choices of terminology, conceptual reorganization, and changes of emphasis. Arabic-writing mathematicians did not merely study Euclid and Apollonius: they connected geometry with algebra, astronomy, optics, and the making of instruments. Ancient geometry thus remained a living resource in a new scholarly setting rather than becoming a museum relic in the Middle Ages.

Among medieval Islamic mathematicians particularly important to the history of geometry were ʿUmar ibn Ibrāhīm al-Khayyām (عمر بن إبراهيم الخيام) [17] and Naṣīr al-Dīn Muḥammad al-Ṭūsī (نصير الدين محمد الطوسي). [18] ʿUmar attempted to study the parallel postulate by means of a special quadrilateral, and Naṣīr further systematized the parallel problem and the foundations of geometry. Strictly speaking, they did not establish non-Euclidean geometry, but their work shows that the difficulty of the parallel postulate did not suddenly appear in the nineteenth century. It had already prompted sustained reflection in the Middle Ages. The later revolution concerning the independence of the postulate grew out of a long sequence of unsuccessful attempts, ancient and medieval, to prove it.

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[13] Ptolemy, c. 100–c. 170 CE, made extensive use of geometric models in astronomy. His principal work was originally entitled Μαθηματικὴ Σύνταξις; after its transmission through Arabic into Latin it became known as the *Almagest*.

[14] Heron of Alexandria is generally placed around the first century CE; Pappus of Alexandria, in the early fourth century; and Proclus the Successor lived from 410 to 485.

[15] Its Greek title is usually given as Μαθηματικὴ Συναγωγή; in English it is commonly called the *Collection* or the *Mathematical Collection*. It preserves a great deal of advanced Greek geometry, especially material on analysis, construction problems, conics, and several results associated with what is now called Pappus's theorem.

[16] His full name is Abū ʿAlī al-Ḥasan ibn al-Ḥasan ibn al-Haytham (أبو علي الحسن بن الحسن بن الهيثم). He is commonly called Ibn al-Haytham in English and Alhazen in the Latin tradition; he lived c. 965–c. 1040 and was an Arab scholar. For a discussion of Arabic names, see footnote 14 on page 160; the same naming principles apply here. As in that discussion, the following text uses the person's given name.

[17] His full name may be given as Ghiyāth al-Dīn Abū al-Fatḥ ʿUmar ibn Ibrāhīm al-Khayyām (غياث الدين أبو الفتح عمر بن إبراهيم الخيام). Commonly known as Omar Khayyam, he lived c. 1048–1131 and was a Persian scholar.

[18] His full name may be given as Naṣīr al-Dīn Muḥammad ibn Muḥammad ibn al-Ḥasan al-Ṭūsī (نصير الدين محمد بن محمد بن الحسن الطوسي). He lived from 1201 to 1274 and was a Persian scholar.

European Renaissance painting then gave geometry new impetus from another direction: the problem of perspective. In the fifteenth century Filippo Brunelleschi and Leon Battista Alberti [19] had mastered, in practice and theory, vanishing points, the horizon line, and foreshortening. Their object was artistic, but the questions beneath their work were profoundly geometric: how is a three-dimensional scene projected onto a two-dimensional plane, and which relations survive the projection? Mature projective geometry belongs to the seventeenth through nineteenth centuries, but without the practical questions posed by Renaissance perspective it might not have developed in the form it did.

Renaissance perspective reconnected ancient optics, pictorial practice, and geometric projection. Painters wanted to produce spatial depth on a plane; mathematicians later recognized that a projection destroys lengths, angles, and parallelism in general, but preserves collinearity, incidence, and certain deeper relations. Here lay the germ of projective geometry. Perspective was therefore not an application that followed projective geometry, but the practical soil from which projective thinking later grew. Such episodes recur throughout the history of geometry: the true source of a theory need not lie within pure theory, but may be a concrete difficulty in art, astronomy, instruments, mechanics, or measurement.

The period from the Hellenistic zenith to medieval transmission thus had two central characteristics. Classical geometry continually enlarged its range of objects, bringing conics, the method of exhaustion, and geometric models in astronomy to a high level of maturity. At the same time, geometry was not confined within one civilization; it was transmitted and transformed among the Greek, Arabic, and Latin worlds. Historically, the period did more than preserve ancient achievements. It prepared a long accumulation of material for several decisive problems of modern geometry: conics, projection, the parallel postulate, and mathematical optics.

Geometry after Euclid did not, then, advance along a single straight path. Late Hellenistic geometry had already brought sophisticated objects to a high degree of development, while medieval transmission and Renaissance practice preserved and recast them under very different conditions. Modern geometry could advance so rapidly not because an entirely new subject suddenly appeared, but because after a long transmission many old problems finally met new languages. Conics, the parallel postulate, projection, and the measurement of curvilinear figures each became an entrance to a modern geometric revolution.

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## Exercises

**Exercise 11.4.** Complete the following statements from this section.

(1) The Greek title of Euclid's *Elements* is ________, its Latin title is *Elementa*, and its conventional English title is *Elements*.
(2) The ancient Greek mathematician ________ systematically studied conic sections in his ________.
(3) Archimedes developed sophisticated techniques for areas, volumes, and the ________; from a modern perspective, many of his arguments come close to ideas of integration.
(4) Pappus's ________ preserves many propositions and ideas of earlier geometers and is an important source for advanced Greek geometry.
(5) From the eighth to the tenth centuries, the ________ movement in Baghdad and elsewhere rendered many Greek mathematical works into Arabic, giving Greek geometry new life in the Arabic-speaking world.
(6) The medieval Islamic mathematicians ʿUmar and Naṣīr both studied the ________, showing that the problem had a long prehistory before the formal establishment of non-Euclidean geometry.

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[19] Filippo Brunelleschi (1377–1446) was an Italian architect and engineer, generally regarded as one of the founders of early Renaissance architecture and associated with the rediscovery of linear perspective. Leon Battista Alberti (1404–1472) was an Italian humanist and architectural theorist. His *On Painting*, originally entitled *De pictura* in Latin and *Della pittura* in Italian, systematically discusses perspective, composition, and related problems in painting.

**Exercise 11.5.** Answer briefly.

(1) Name representative ancient Greek geometers. Compare the places of Archimedes and Apollonius in the history of geometry.
(2) The work of al-Ḥasan, ʿUmar, and Naṣīr shows that medieval transmission of geometry was not merely a passive preservation of Greek texts. Using optics, perspective, and the parallel postulate, explain how medieval mathematicians continued to advance geometric research.

**Exercise 11.6.** This section has followed geometry from its Greek zenith through medieval preservation, translation, and renewed investigation. Explain why "transmission" in the history of geometry does not mean simply copying old knowledge, but often involves reorganizing and reinterpreting it and posing its questions anew.

**Exercise 11.7.** Read the following passage and answer the questions.

> *Ancient Chinese mathematics also contained a rich body of geometry. In the Mathematical Treatise in Nine Sections (1247), Qin Jiushao gave the "procedure for finding an area from three oblique sides," a method of finding the area of a triangle from its three side lengths: "Add the squares of the shorter and longer sides and subtract the square of the middle side; halve the remainder and square it. From the product of the squares of the shorter and longer sides subtract that square; quarter the remainder to obtain the dividend. Take one as the coefficient of the squared unknown and extract the square root to obtain the area." In modern notation, if the side lengths are $a, b, c$, the area is*
>
> $$S = \sqrt{\frac{1}{4}\left[a^2c^2 - \left(\frac{a^2+c^2-b^2}{2}\right)^2\right]}.$$
>
> *This is equivalent to the formula usually attributed to Heron, but its form of expression and theoretical setting are different.*

(1) Prove that the "procedure for finding an area from three oblique sides" is equivalent to Heron's formula.
(2) Compare ancient Chinese and ancient Greek geometry. You may consider their objects of study, modes of expression, attitudes toward proof, practical settings, and ways of organizing knowledge.
(3) In light of the relevant historical settings, analyze possible reasons for the differences between ancient Chinese and ancient Greek geometry.

## 11.3 Geometry in the Early Modern Period: Analytic and Projective

If the fundamental spirit of Greek geometry was deduction, the most profound change after the seventeenth century was that geometry began to advance along two mutually influential paths. Along one, algebraic language increasingly entered geometry, so that points, curves, and surfaces could be described uniformly by coordinates and equations. The other grew from the study of projection, perspective, and conics, and gradually taught geometers that some relations do not depend on particular lengths and angles but remain stable under transformations. In retrospect these were not unrelated branches but two forces in the early modern transformation of geometry. The first continually enlarged the range of geometric objects; the second continually asked which properties truly belong to an object when its appearance and representation change.

The best-known turning point is usually associated with René Descartes and his publication of *La Géométrie* in 1637, [20] while Pierre de Fermat [21] independently developed similar ideas. Points acquired coordinates $(x,y)$, curves became sets of points satisfying equations, and geometric problems could be transformed systematically into algebraic ones. In a modern textbook this may look like nothing more than placing a coordinate system on the plane. Historically, however, it meant that geometry had for the first time acquired a symbolic language capable of treating curves in general. Circles, ellipses, hyperbolas, parabolas, and curves of higher degree could all be compared within one algebraic framework, so that their intersections, tangents, singularities, parameters, and degrees gradually became subjects of a common discussion.

The deeper setting for this change was the general transformation of European mathematical language in the sixteenth and seventeenth centuries. Coordinate methods could not have become general without a mature symbolic algebra, nor would there have been the same pressure to pass beyond the objects of traditional straightedge-and-compass construction without the constant appearance of curve problems in mechanics, astronomy, and optics. Analytic geometry did more than place number axes in a plane: it brought algebraic symbols, geometric objects, and problems of natural science into a common language. Its historical importance therefore extends far beyond a new set of computational techniques. It changed basic conceptions of what a curve is and how a geometric problem is organized. In classical geometry an unknown object was commonly a point, line, circle, or curve to be constructed; in analytic geometry it increasingly became a quantity satisfying specified equations. Intersections of curves could be treated as common solutions of a system, tangents through multiple roots, limits, or local variation, and locus problems by eliminating auxiliary variables. Such procedures developed further in calculus, analytic mechanics, and the theory of algebraic curves, becoming part of the shared language of modern mathematics.

It would nevertheless be inadequate to understand the period only as the triumph of analysis. While Descartes and Fermat were translating geometric objects into equations, another equally important viewpoint was taking shape. In 1639 Girard Desargues [22] published the *Brouillon project d'une atteinte aux evenemens des rencontres du Cone avec un Plan*, known in modern French as the *Brouillon project d'une atteinte aux événements des rencontres du cône avec un plan* and commonly shortened to *Brouillon project*, which already contained perspective triangles, elements at infinity, and the core idea of what was later called Desargues' theorem. In 1640, at the age of sixteen, Blaise Pascal [23] wrote his *Essai pour les coniques* (*Essay on Conics*) and proposed the celebrated theorem that bears his name. These works may look synthetic, but they too were unsettling older conceptions of figure. Conics were no longer isolated special figures; by projection, perspective, and degeneration they could be understood within a broader network of relations. If analytic methods unified curves as equations, these early projective ideas began to suggest that apparently different figures might be manifestations of one structure viewed in different ways.

What is most striking about seventeenth-century geometry is therefore not only the rise of coordinates, but the simultaneous expansion of the idea of a curve in two directions. Equations unified different curves as algebraic objects, while projection unified different figures as objects related under transformation. The former enabled researchers to treat a curve as a point set; the latter drew their attention to the possibility that different appearances can express one structure. Analytic and projective geometry did not begin as two completely separate historical episodes. Their interaction was already implicit in the early modern period: one increased the general power to handle curves, while the other weakened the absolute status of their classical metric features.

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[20] *La Géométrie* was an appendix to René Descartes's *Discourse on the Method*. Descartes (1596–1650) was a French philosopher and mathematician.

[21] Pierre de Fermat (1607–1665) was a French mathematician.

[22] Girard Desargues (1591–1661) was a French mathematician and engineer and one of the principal forerunners of modern projective geometry.

[23] Blaise Pascal (1623–1662) was a French mathematician, physicist, and philosopher.

Problems about curves in seventeenth-century science gave this interaction a concrete setting. In 1609 Johannes Kepler proposed that planetary orbits are ellipses. In 1687 Isaac Newton [24] used geometry and limiting ideas in his *Mathematical Principles of Natural Philosophy* [25] to show that an inverse-square central force gives rise to conic orbits. Thus conics were both mathematical objects and a natural language for celestial mechanics. Newton himself nevertheless used geometry extensively in the *Principia*, rather than the fully analytic presentation familiar from later textbooks. The seventeenth century was not one in which analytic methods immediately overwhelmed synthetic ones, but one in which several methods coexisted, competed, and were translated into one another. Equations offered generality, figures preserved intuition and structure, and the projective viewpoint quietly suggested that the nature of a curve need not be identical with its appearance in a particular coordinate system.

On the question of priority between Descartes and Fermat, historians now generally hold that both made independent and important contributions to analytic geometry, though in different styles. Descartes placed greater emphasis on a general method and the manipulation of equations; Fermat was more flexible in local problems and methods for tangents. The work of Desargues and Pascal likewise did not immediately produce a stable school, but prepared the ground for later projective thought. In the broader historical view, seventeenth-century geometry is not a single narrative. It was neither a simple replacement of a declining synthetic method by a rising analytic one, nor an isolated appearance of projective ideas. It was a transitional period in which curves, equations, perspective, and structural conceptions were simultaneously active and mutually influential.

In the nineteenth century these older tensions finally assumed the form of distinct disciplines. During the Napoleonic Wars, the descriptive geometry created by Gaspard Monge [26] trained a large body of French mathematicians who valued methods of projection. One of the most important was Jean-Victor Poncelet. [27] In 1822 Poncelet published his *Traité des propriétés projectives des figures* (*Treatise on the Projective Properties of Figures*), [28] arguing explicitly that geometry should study properties invariant under projection. The historical significance of this step was not merely the organization of a body of projective propositions, but the clear statement of a central nineteenth-century question. If analytic geometry tells us that one object may have different equations in different coordinate systems, then projective geometry asks which relations remain essential under a broader class of transformations.

The development of nineteenth-century projective geometry was not, however, a smooth one-way course. It inherited the wider vision opened by analytic geometry and algebraic methods, yet for a long period retained a markedly synthetic tendency. Geometers such as Poncelet and Jakob Steiner [29] often held that, powerful as coordinate calculation was, it could conceal the relations within a figure. For them, the distinctive value of projective geometry was its ability to reveal structure directly from points, lines, conics, projections, incidences, and dualities, without first reducing everything to equations and elimination. Poncelet deliberately wrote synthetically in his treatise, and Steiner became famous for his adherence to purely synthetic methods. This was more than an aesthetic preference. It renewed a fundamental question: what should a geometric proof be? A long algebraic calculation may be correct without making clear which relations in the figure actually do the work.

The resulting debate between synthetic and analytic methods should not be reduced to an opposition between drawing and calculation. The synthetic method organizes proof from relations within

[24] Isaac Newton (1643–1727) was an English mathematician, physicist, and astronomer.

[25] Its Latin title is *Philosophiæ Naturalis Principia Mathematica*; the work is usually called the *Principia* and first appeared in 1687.

[26] Gaspard Monge (1746–1818) was a French mathematician and one of the founders of descriptive geometry. His principal work in the subject is *Géométrie descriptive*.

[27] Jean-Victor Poncelet (1788–1867) was a French mathematician and engineer and a central figure in the nineteenth-century revival of projective geometry.

[28] The book explicitly emphasizes properties of figures that remain invariant under projection and is generally regarded as an important landmark in the systematization of modern projective geometry.

[29] Jakob Steiner (1796–1863) was a Swiss mathematician and a leading representative of nineteenth-century synthetic geometry.

a figure, emphasizing construction, correspondence, duality, intuitive continuity, and the internal connections among invariants. The analytic method uses coordinates, equations, determinants, and algebraic elimination to place geometric objects within a unified symbolic system. The first can keep geometric structure transparent and make the significance of a theorem readily visible; the second offers greater generality and extensibility, especially for curves of higher degree, surfaces in space, complicated intersections, and parameter families. The central question in the controversy was not which technique was more convenient, but how geometric knowledge should be organized: by preserving the intuitive relations of the figure as far as possible, or by using algebraic symbols to obtain a more general and stable expression?

Within the synthetic tendency, the work of Karl Georg Christian von Staudt [30] is especially noteworthy. He sought to detach projective geometry still further from metric notions, coordinates, and even some intuitive appeals to continuity, characterizing points, lines, harmonic relations, cross-ratios, and other concepts as far as possible through purely geometric incidence. This path is not always technically easy for a beginner, but it was conceptually important. It showed that projective geometry could not only be made analytic but could also be pushed toward a more thoroughly coordinate-free form. Nineteenth-century projective geometry contained a strong desire to separate the relations within a figure from particular measurements and coordinate representations, making incidence, correspondence, and invariants central to geometric exposition.

It would be equally misleading, however, to construe projective geometry as a simple rejection of analytic geometry. From the beginning, nineteenth-century projective geometry was deeply entangled with coordinates, algebra, and analysis. Poncelet himself did not discover and organize all his results without analytic thought. The work of August Ferdinand Möbius, Julius Plücker, Arthur Cayley, and Michel Chasles [31] made still clearer the deep connection between projective geometry and algebraic methods. Plücker coordinates and the concept of a dual curve gave points and lines symmetric places in algebraic expression. Cayley attempted to embed metric geometry once more within a projective framework, while Chasles systematically organized projective correspondences and the theory of conics. Projective geometry was therefore not merely a return to synthesis. It moved continually between synthetic intuition and analytic expression: synthesis helped geometers see which relations should be studied, while analysis enabled those relations to be generalized, calculated, and systematized.

Historically, the controversy did not end in the complete victory of either side. Geometry in the second half of the nineteenth century instead showed that each method had an irreplaceable role. Synthetic geometry retained the intuitive force of relations among figures, principles of duality, and invariants, permitting projective geometry to develop a conceptual system independent of measurement. Analytic geometry supplied a uniform language for general cases, enabling the systematic development of projective coordinates, homogeneous equations, curves of higher degree, and algebraic invariants. More importantly, the conflict made geometry self-conscious about its methods. Coordinates and algebra were no longer mere external calculational devices, nor was a purely synthetic proof merely an old style of construction. Mutual criticism and assistance produced not a division of geometry, but an expansion of its languages.

The principal contribution of nineteenth-century projective geometry was thus not just an additional body of theorems about projection and conics. It redefined the question of which quantities are worth studying in geometry. If lengths and angles are not preserved by projection, what should count as fundamental? Attention shifted from traditional metric notions to more stable relations such as collinearity, concurrence, cross ratio, involution, and pole–polar correspondence. In this framework

[30] Karl Georg Christian von Staudt (1798–1867) was a German mathematician whose purely synthetic treatment had an important influence on projective geometry. His works sought to establish projective geometry without metric or coordinate notions and to introduce concepts such as cross ratio from geometric relations.

[31] August Ferdinand Möbius (1790–1868) was a German mathematician; Julius Plücker (1801–1868), a German mathematician and physicist; Arthur Cayley (1821–1895), an English mathematician; and Michel Chasles (1793–1880), a French mathematician.

the cross ratio naturally became a central invariant. For four points $A,B,C,D$ on a line,

$$(A,B;C,D) = \frac{\overline{AC}}{\overline{BC}} \div \frac{\overline{AD}}{\overline{BD}}$$

is invariant under projective transformations. Duality, meanwhile, exhibits mirror-like correspondences between propositions about points and lines. For conics, Pascal's and Brianchon's theorems [32] reveal profound structures through inscribed and circumscribed hexagons, respectively. Pole and polar, quadratic point ranges, pencils of curves, and polarity carried the ancient study of conics far beyond its Greek state. The powerful attraction of these subjects in the nineteenth century was itself made possible by analytic methods, which had already placed conics, coordinate transformations, and families of curves in a broader setting. Projective geometry did not return to a world before analysis; standing after analysis, it reorganized the world of curves structurally.

The idea of elements at infinity deserves special emphasis. Two parallel lines do not meet in a Euclidean plane, but in a projective plane they may be regarded as meeting at a point at infinity. At first this can look like an artificial convention, yet it greatly unifies the forms of propositions: parallel and intersecting lines no longer require separate treatments, and the various cases of conics can be expressed in a more uniform framework. This unification was not a purely conceptual maneuver independent of analytic methods. Once coordinates and equations had accustomed mathematicians to placing objects within larger algebraic settings, elements at infinity could more readily be understood as a natural completion of a theory rather than a mysterious invention detached from representation and calculation. Analytic geometry removed many psychological obstacles to projective geometry; projective geometry in turn showed that, for the objects represented by coordinate equations to attain a higher unity, their structure had to be reorganized at infinity.

In 1872 Felix Klein [33] stated in the Erlangen Program [34] that a branch of geometry can be characterized by its allowed group of transformations and the corresponding invariants. Within this scheme, Euclidean, affine, and projective geometry are not isolated subjects but form a hierarchy from more restrictive to less restrictive. Euclidean transformations preserve lengths and angles; affine transformations abandon angles but preserve parallel structure; projective transformations also abandon parallelism while preserving such higher-level invariants as cross ratio. This formulation was so powerful because it summarized the long interaction of analytic methods and projective thought as a more general principle of classification. Geometries need not first be distinguished by having different objects; they may be distinguished by the transformations allowed and the invariants preserved. The core of geometry becomes not "what figure is drawn?" but "under which transformations do which relations remain stable?"

The Erlangen Program was thus not only a summary of projective geometry, but a methodological distillation of early modern geometry as a whole. It inherited from analytic methods the generality of objects and freedom of representation, but constrained that freedom once more through invariants and structure. Topology, differential geometry, and ideas of symmetry in modern physics can all be connected with this conception in broader senses. Klein's program does not encompass every direction in modern geometry, but it marks a highly self-conscious nineteenth-century summary: geometry can be developed by coordinates and equations, yet must identify its true core through transformations and invariants.

Many subjects of classical plane geometry were recast during the same period. Projective definitions of conics, the polar structure between lines and conics, the organization of dual propositions, and Poncelet's closure theorem [35] all show that apparently scattered results in plane geometry belong to a larger world of projective structures. Once these structures could also be translated into coordinates, algebraic curves, and metric concepts, classical theorems ceased to be mere fragments

[32] Brianchon's theorem is named after Charles Julien Brianchon (1783–1864), a French mathematician.

[33] Felix Klein (1849–1925) was a German mathematician.

[34] Its German title is *Vergleichende Betrachtungen über neuere geometrische Forschungen*; it is usually called the *Erlanger Programm*, or Erlangen Program.

[35] Poncelet's closure theorem is named after Jean-Victor Poncelet (1788–1867), the French mathematician and engineer introduced above.

from an earlier age. They became nodes through which different geometric languages illuminated one another. From the seventeenth to the nineteenth century, analytic and projective geometry were not parallel, unrelated paths. The former continually enlarged the algebraic expression of curves and spaces, enriching the range of geometric objects; the latter continually asked after the structural invariants behind those expressions, clarifying what should count as essential amid that richness. Through this long entanglement, early modern geometry gradually became a study of structures in the modern sense.

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## Exercises

**Exercise 11.8.** Complete the following statements from this section.

(1) René Descartes's publication of ________ in 1637 is generally regarded as one of the landmarks in the formation of analytic geometry.
(2) Analytic geometry assigns coordinates to points and treats a curve as the set of points satisfying an ________, thereby systematically converting geometric problems into algebraic ones.
(3) In 1639 Girard Desargues published the *Brouillon project*, which already contained perspective triangles, ________ elements, and the central idea of what later became Desargues' theorem.
(4) In 1640 Blaise Pascal wrote the ________, proposing the celebrated theorem that bears his name.
(5) Descriptive geometry is commonly associated with the French mathematician ________.
(6) The mathematician in the second half of the nineteenth century who advocated a purely geometric approach to projective geometry and importantly influenced the coordinate-free treatment of concepts such as cross ratio was ________.
(7) The mathematician who proposed the Erlangen Program and sought to view geometries uniformly through ________ was Felix Klein.

**Exercise 11.9.** Answer briefly.

(1) Besides providing a computational technique, what role did the analytic geometry represented by Descartes and Fermat play? Explain using this section.
(2) Briefly describe the controversy between the synthetic and analytic schools. Who represented each side, and how did their controversy ultimately advance geometry?

**Exercise 11.10.** Read the following passage and answer the questions.

> *Early in his career, Gaspard Monge joined the École Royale du Génie at Mézières, a school closely connected with military engineering. Initially employed as a draftsman, he developed methods for representing three-dimensional objects exactly on two-dimensional drawings while working on fortifications, spatial components, and engineering plans. These methods later became known as descriptive geometry and long served technical drawing, military construction, and engineering education. Within the French system of engineering education, descriptive geometry trained students to treat spatial objects through projections, sections, and multiple views, making projective relations an increasingly important geometric language.*

(1) Summarize the historical setting in which descriptive geometry arose. Why did it draw attention to projective relations?
(2) A discipline often emerges in close relation to its historical setting. Using this section and other relevant historical knowledge, analyze why projective geometry gradually arose in the modern period, and explain the relation between descriptive and projective geometry.

## 11.4 Classical Geometry in the Modern Era: Synthesis and Development

Although nineteenth-century geometry underwent analytic and projective transformations, classical plane geometry did not decline. On the contrary, it entered a new period of flourishing. The preceding section described deep changes in geometric method and structure; here older objects acquire new life under new points of view. Mathematicians found extraordinarily rich structures in triangle geometry, inversion, theories of conjugacy, and the special conics associated with a triangle. This body of work forms one of the most technically ingenious and aesthetically attractive parts of what is now called classical geometry. It retained the synthetic tradition's sensitivity to relations within figures, while continually absorbing ideas from projective geometry, transformations, and algebra. Far from merely repeating ancient material, it was a genuine revival born in a modern setting.

The revival was closely connected with nineteenth-century mathematical culture. Journals, competitions, schools, and mathematical communities allowed concrete problems to circulate rapidly, while projective geometry, algebraic methods, and transformations supplied new means of explaining traditional plane figures. A triangle was no longer only three sides and three angles: it generated numerous centers, conjugate points, associated conics, transformational correspondences, and invariant relations. The period was not the survival of an old geometry after the appearance of a new one, but an internal expansion of the traditional world of figures under the stimulus of modern methods. Objects that looked simple suddenly disclosed layer upon layer of interconnected structure.

The systematic study of triangle centers, for example, deepened markedly in the nineteenth century. The incenter, centroid, circumcenter, and orthocenter had long been known, but more specialized objects such as the Gergonne, Nagel, Brocard, Lemoine, and Feuerbach points [36] were progressively defined and studied in the modern era. After the 1840s, propositions about triangle centers multiplied; in the twentieth century this tradition eventually developed into the system organized by Clark Kimberling in the *Encyclopedia of Triangle Centers* (ETC). [37] Nineteenth-century geometers did not think of the objects through a systematic numbering like the one used today, but they already understood that a triangle was not merely an elementary figure. It was an experimental field capable of carrying deep structure. Auxiliary lines and circles, conjugacies, and special points continually formed new connections; the richness came not from complexity in the object itself, but from the growth of its network of relations.

The historical significance of the study of triangle centers is that it exhibits a mechanism by which objects proliferate within classical plane geometry. From one reference triangle, one can define angle bisectors, medians, altitudes, excircles, contact triangles, anticomplemental triangles, orthic triangles, and many other auxiliary objects. Each new class may in turn produce special points and lines. The multiplication can appear cumbersome, as if it consisted only in attaching more names to points. At a deeper level, however, it reveals a fundamental capacity of classical geometry: simple figures can generate complex structures through networks of relations, and local definitions can illuminate one another within a whole. Many objects later called classical centers did not arise from nothing; they emerged naturally from the long accumulation of relations internal to the figure. The subject was therefore more than one expression of nineteenth- century taste. In a certain sense, it exposed the deeper nature of classical plane geometry as a study of relations.

Research on triangles also became increasingly intertwined with theories of conjugacy. Isogonal and isotomic conjugacy, along with more general point correspondences, organized many special points into pairs and placed many formerly separate propositions within common frameworks. A point was no longer merely a location in a diagram: it could carry polar lines, loci, conjugate images,

[36] These names refer, respectively, to Joseph Diez Gergonne (1771–1859), a French mathematician; Christian Heinrich von Nagel (1803–1882), a German mathematician; Henri Brocard (1845–1922), a French mathematician; Émile Lemoine (1840–1912), a French mathematician; and Karl Wilhelm Feuerbach (1800–1834), a German mathematician.

[37] Clark Kimberling (born 1942) is an American mathematician and the compiler of the *Encyclopedia of Triangle Centers*.

and relations to curves. The formalism had not yet attained the abstraction of later modern geometry, but its methodological direction was already clear. Classical geometry was no longer satisfied with proving one proposition at a time; it sought the latent mechanisms by which propositions were generated. From the later viewpoint, this was a forerunner of a more structural understanding.

Inversion also acquired special importance during this period. Under an inversion with center $O$ and squared radius $k$, a point $P$ and its image $P'$ satisfy

$$OP \cdot OP' = k.$$

This simple relation produces subtle interchanges between lines and circles, allowing many apparently complicated configurations to be converted into more tractable ones. Jakob Steiner, [38] August Ferdinand Möbius, and many later plane geometers valued this method because it did more than provide a technique: it exhibited another system of invariants, one preserving angles but not lengths. Much later work on isogonal conjugacy, homothety, affine correspondences, and projective correspondences may likewise be seen as extending this transformational point of view. Compared with the classical tradition of construction, inversion made one fact especially clear: a problem may be difficult not because its objects are intrinsically complicated, but because they are being viewed in an unsuitable setting.

Inversion's historical significance also lies in making transformations concrete within elementary plane geometry. Compared with projective transformations, it retains a strong intuition from circle geometry; compared with Euclidean transformations such as translations and rotations, which preserve lengths and angles, it clearly exceeds the restrictions of rigid motion. A line can become a circle, and a circle a line or another circle. Geometers thereby learned that the difficulty of a problem often depends on the world of figures in which they choose to view it. Transforming a complicated object into a simple one became a characteristic nineteenth- century method. More than a technical shortcut, it trained sensitivity to invariants: although an object's appearance changes, certain relations survive, and those preserved relations are often precisely what deserves understanding. In this way inversion provides a natural bridge between elementary circle geometry and modern transformational thought.

The study of special conics associated with a triangle flourished at the same time. Rectangular hyperbolas, the Kiepert, Jerabek, and Feuerbach hyperbolas, [39] and many inscribed, circumscribed, and confocal conics made a conic no longer an isolated curve but a natural carrier of a triangle's internal structure. Historically, this research shows that classical plane geometry did not remain at the level of elementary techniques about triangles and circles. In the nineteenth century it developed a highly refined network of objects in which transformations, conjugacies, duality, cross ratio, and curves interwove to form a mature intermediate-level theory. Relative to the periods before the eighteenth century, the difference was not only a greater number of objects, but a denser organization of their relations. Points, lines, circles, and conics associated with a triangle began to occupy stable places in a common universe rather than serving as temporary pieces of isolated constructions.

These special curves deserve a place in history not because every one is equally fundamental, but because they express a way of conducting research: the triangle became a source of curves, point ranges, pencils of lines, and correspondences. Plane geometry and the theory of conics ceased to belong to unrelated compartments; they permeated one another through polars, conjugacies, isogonal relations, and projective transformations. This is why so many topics from the preceding chapters recur. Classical plane geometry is connected in depth with higher projective and algebraic structures. Nineteenth-century geometry flourished not only because mathematicians could devise new propositions, but because they possessed intermediate concepts that connected those propositions. The theory had not yet reached the unified abstraction of later modern geometry, but it had moved well beyond a mere collection of tricks.

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[38] Jakob Steiner (1796–1863) was a Swiss mathematician and a leading representative of nineteenth-century synthetic geometry.

[39] The Kiepert hyperbola is named after Ludwig Kiepert (1846–1934), a German mathematician; the Jeřábek hyperbola after Václav Jeřábek (1845–1931), a Czech mathematician whose name often appears without diacritics as Jerabek in English-language sources.

The period's prosperity was also tied to changing modes of mathematical communication. Concrete problems could be published quickly in journals, answered throughout a community, and circulated repeatedly through schools and competitions. Plane geometry acquired a strongly problem-driven atmosphere. This stimulated the discovery of many beautiful theorems and also fostered shared aesthetic standards: brevity, symmetry, unexpected concurrences and collinearities, and elegant connections with conics were increasingly prized as geometric qualities. The nineteenth-century revival was therefore not only a growth in the quantity of knowledge, but the maturation of a style. Many classical propositions circulated so widely because they possessed both technical substance and geometric brilliance, suggesting that plane figures contained an order deeper than the eye could immediately see.

Its very success nevertheless exposed limitations in nineteenth-century classical plane geometry. Once a great number of beautiful propositions had been discovered, research could easily slide into the accumulation of results without a more general framework. This helps explain why geometers of the late nineteenth and twentieth centuries increasingly interpreted the results through projective geometry, group theory, differential geometry, and algebraic geometry. An isolated trick is often short-lived; inclusion in a larger structure magnifies its meaning. Historically, nineteenth-century classical geometry was both a brilliant culmination and a turning point toward modern unifying theories. It amassed a wealth of material, and in doing so made the need for higher principles of organization especially apparent.

The new development of classical plane geometry should therefore be regarded both as a culmination and as the eve of a transition. At its height it accumulated a large body of material susceptible to reinterpretation; at the turning point it made increasingly clear that more ingenious local constructions alone would not suffice. Higher points of view were needed to organize the objects. Non-Euclidean geometry, axiomatization, and modern algebraic geometry became important in part because they supplied new forms of organization for this highly refined classical material. The revival of classical plane geometry was not an epilogue unrelated to modern mathematics, but more like an ornate vestibule filled with material. Beyond it, geometry entered the twentieth-century world of structures, spaces, and generality.

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## Exercises

**Exercise 11.11.** Complete the following statements from this section.

(1) From the nineteenth century onward, systematic study of triangle centers deepened markedly, producing such objects as the Gergonne, Nagel, Brocard, Lemoine, and ________ points.
(2) In the twentieth century, Clark Kimberling organized the ________ system of triangle centers.
(3) One important feature of inversion is that it generally does not preserve length but is closely related to ________. Theories associated with a triangle also include ________ conjugacy and ________ conjugacy.
(4) Modern classical geometry was not merely a repetition of ancient plane geometry because it placed greater emphasis on special structures, transformational methods, and their relation to ________.

**Exercise 11.12.** Answer briefly.

(1) Summarize the principal achievements of classical geometry in the modern era. Why was it not merely a repetition of ancient plane geometry?
(2) Why did research related to triangles and conics flourish in modern classical geometry?

**Exercise 11.13.** Using this section and other relevant historical knowledge, discuss how the research style of classical geometry in the modern period differed from the geometry of Euclid's *Elements*. What caused the difference?

## 11.5 Geometry on the Threshold of Modernity: Non-Euclidean Geometry and Axioms

For more than two thousand years, classical Euclidean geometry was regarded as the only correct mathematical description of space. That position met a fundamental challenge in the nineteenth century, centered on Euclid's fifth postulate, the parallel postulate. Mathematicians had long suspected that it might be derivable from the other postulates. From Proclus through Giovanni Girolamo Saccheri and Johann Heinrich Lambert, [40] attempts to prove it continued for centuries without success. On the surface this looks like one obstinate technical problem. At a deeper level, it forced mathematicians to inspect the logical dependencies within Euclid's system: which propositions genuinely use the parallel postulate, which conclusions follow from weaker assumptions, and whether denying the postulate immediately produces a contradiction. Through repeated questions of this kind, the problem gradually became one about the structure of an entire geometric system rather than about a single postulate.

These attempted proofs matter historically not merely because they failed. They repeatedly brought mathematicians to the same boundary. The authority of the Euclidean tradition was so great that many initially hoped to vindicate the parallel postulate as a necessary consequence of the others. Yet the long effort to defend the tradition steadily exposed the special status of the postulate. Non-Euclidean geometry did not arise from a sudden rejection of the past; it was a conceptual change produced by the inability of sustained attempts to preserve that past to succeed. The nineteenth-century breakthrough looked so radical precisely because, after defending Euclid with the greatest seriousness, mathematicians finally had to admit that geometry might have more than one possible set of beginnings.

The decisive breakthrough came in the first half of the nineteenth century. Carl Friedrich Gauss (Gauß) [41] recognized relatively early that the parallel postulate might be independent, but cautiously refrained from publishing a complete system. János Bolyai [42] independently published his new geometry in 1832 in an appendix to a work by his father, Farkas Bolyai. Nikolai Ivanovich Lobachevsky (Николай Иванович Лобачевский) [43] published papers in 1829–1830 and continued the theory in 1835 and 1840. Together their work showed that replacing the parallel postulate by the assertion that at least two lines through a point outside a given line do not meet that line yields a consistent geometric theory: non-Euclidean geometry. The shock was not simply that one conclusion had changed, but that the tone of an entire geometric world changed with it. Once the uniqueness of a parallel disappeared, familiar relations among triangles, angles, areas, and distances all had to be understood anew.

In hyperbolic non-Euclidean geometry the sum of the angles of a triangle is no longer identically $180^\circ$, but is strictly less than $180^\circ$. Its area is directly related to the angular defect:

$$S = R^2\big(\pi - (A+B+C)\big).$$

This formula holds under an appropriate choice of units, with $R$ related to the curvature scale. Such phenomena show clearly that geometry had ceased to be only the description of one real space and had become a name for multiple possible axiomatic systems. What had once appeared to be an obvious law of geometry was now a property within one particular system; relations among angle sums, areas, parallels, and distances could no longer be understood without an axiomatic background. The

[40] Giovanni Girolamo Saccheri (1667–1733) was an Italian Jesuit and mathematician. His *Euclides ab omni naevo vindicatus* sought to prove the parallel postulate by contradiction. Johann Heinrich Lambert (1728–1777) was a Swiss mathematician and physicist who also made important contributions to the problem.

[41] Carl Friedrich Gauss (1777–1855) was a German mathematician. His surname is written *Gauß* in German orthography and commonly *Gauss* in international mathematical literature.

[42] János Bolyai (1802–1860) was a Hungarian mathematician. His ideas on non-Euclidean geometry appeared in an appendix to his father Farkas Bolyai's *Tentamen*; this publication of 1832 is commonly called the *Appendix*.

[43] Nikolai Ivanovich Lobachevsky (1792–1856) was a Russian mathematician. His early papers on the new geometry appeared in 1829–1830, and he continued to develop the theory in 1835 and 1840.

revolution's deeper significance was not the addition of a new figure but a change in the understanding of axioms. An axiom need not be a self-evident physical truth; it may be a logical starting point that specifies a class of structures.

This was profoundly unsettling to nineteenth-century mathematicians. If equally rigorous reasoning could start from different parallel postulates and produce different but internally consistent geometries, the status of geometric truth had to be reinterpreted. Geometry was no longer uniquely a description of visually perceived space, but the study of necessary relations within a specified structure. This did not make the question of physical space disappear. Rather, it distinguished mathematically possible spaces from empirically actual space, making the relation of geometry to physics more subtle. From then on, one had to distinguish between asking in which system a statement holds and whether that system describes reality. The distinction had lasting consequences for the foundations of mathematics and the philosophy of science.

The change was intertwined with the development of curvature. In 1827 Gauss published his *General Investigations of Curved Surfaces*, introducing Gaussian curvature. If the principal curvatures of a surface at a point are $k_1, k_2$, then

$$K = k_1 k_2.$$

Spherical, plane, and hyperbolic surfaces can be distinguished by $K > 0$, $K = 0$, and $K < 0$. One of Gauss's most important insights was that curvature is not merely an account of how a surface looks when bent in an ambient space, but an intrinsic quantity determined by measurements within the surface. A geometry therefore need not depend on an external three-dimensional space, and the subject began to move toward the internal structures of its objects. In 1854 Bernhard Riemann [44] advanced the idea of a manifold in his celebrated habilitation lecture, explaining that a space may be described locally by coordinates while having a metric structure unlike that of ordinary Euclidean space. Historically, non-Euclidean and differential geometry were not isolated developments. They met deeply in the idea that space can have different intrinsic structures.

Riemann's ideas further enlarged the class of geometric objects. Geometry need not be confined to surfaces embedded in three-dimensional space; it could investigate the intrinsic metrics of abstract manifolds themselves. Space could be defined mathematically without dependence on everyday intuition. Curvature was no longer only a description of how a surface bent in an external space, but an indicator of the space's own internal structure. This transformation later acquired physical meaning in general relativity, but within the history of mathematics it first marked a movement from the study of figures toward the study of spatial structures. From points, lines, and circles in the classical plane to abstract spaces with local coordinates and global curvature, the objects of geometry expanded decisively in the nineteenth century. The expansion did not abandon old questions, but proceeded by reinterpreting their assumptions and their range.

The change had an enormous influence on the foundations of mathematics. Eugenio Beltrami [45] constructed a model of non-Euclidean geometry in 1868, establishing its consistency relative to Euclidean geometry. Klein used projective methods to explain relations among geometries, while Henri Poincaré [46] gave the disk and upper-half-plane models, providing clear visual representations of non-Euclidean geometry. The idea of a model thereby entered geometry at a fundamental level. A geometric theory's credibility no longer depended on its immediate intuitive plausibility, but on its logical coherence, the existence of definite models, and its relative consistency with other systems. In the classical period geometry was more readily understood as a direct description of space itself. By the late nineteenth century, geometers increasingly interpreted one system's axioms and theorems using objects in another, more familiar system. This transformed geometry and anticipated modern mathematics' general attitude toward formal systems and their interpretations.

[44] Bernhard Riemann (1826–1866) was a German mathematician. The work referred to here is his habilitation lecture, *Über die Hypothesen, welche der Geometrie zu Grunde liegen*, commonly translated as *On the Hypotheses Which Lie at the Foundations of Geometry*.

[45] Eugenio Beltrami (1835–1900) was an Italian mathematician.

[46] Henri Poincaré (1854–1912) was a French mathematician, physicist, and philosopher.

The appearance of models was a critical step in the foundations of modern mathematics. In the classical tradition, a geometry was commonly regarded as describing space directly. From the model point of view, the same axioms may be realized in different objects, and the consistency of one theory can be established relative to another accepted theory by interpreting the first within the second. This idea of relative consistency changed geometry and later influenced logic, set theory, and model theory. The non-Euclidean revolution therefore went far beyond geometry. It forced mathematicians to reconsider the relations among formal systems, interpretation, and truth. Geometry became not only a science of space, but a test case for how mathematics can be possible and rigorous.

In 1899 David Hilbert [47] published his *Foundations of Geometry*, [48] reorganizing the axioms of Euclidean geometry and deliberately distinguishing the intuitive meanings of such words as point, line, and plane from their formal roles. The study of geometric foundations thereby entered its modern stage. The non-Euclidean revolution, the theory of curvature, and the subsequent work of formal axiomatization together ended the ancient belief that Euclidean geometry was the only geometry and made the subject capable of examining its own premises. Hilbert did more than make Euclid's system more rigorous. He made explicit that point, line, and plane need carry no predetermined intuition: any objects satisfying the stated relations can form a geometric system. This position is at once continuous with and different from the spirit of Greek geometry. Both seek a deductive structure beginning from axioms, but modern axiomatization consciously separates what an object is from which relations it satisfies. Geometry now studied not only space but also the language and assumptions by which space is described.

Non-Euclidean geometry at first seemed radical not only because some of its propositions contradicted ordinary intuition, but because it unsettled a long-unspoken belief that mathematics naturally describes one unique space. Once that belief was loosened, geometry ceased to be only a theory of actual space and increasingly became a study of possible spatial structures. The consequences were far-reaching. Axioms changed from evident truths into theoretical starting points, and models, relative consistency, and formal interpretation entered the foundations of geometry. The true modernity of nineteenth-century geometry thus lay not merely in its new results, but in mathematicians' deliberate treatment of geometric systems themselves as objects of analysis.

The transformation left a revealing tension in the history of education. Euclidean geometry long remained in textbooks as an exemplary training in logical reasoning, while modern geometry continually reminded students that any geometric language describes only one among many structures. Geometry education since the twentieth century has accordingly borne two tasks: to preserve the intuitive training offered by classical proofs about figures, and to show that geometry is not confined to lines and circles in a plane. The present history stands at the end of the book for the same reason. After encountering a large body of plane geometry, the reader can look back and see how its objects were gradually placed within broader historical and structural settings. The beauty of classical propositions and the self-consciousness of modern axiomatics can then be understood not as separate experiences, but as different levels in one long history.

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## Exercises

**Exercise 11.14.** Complete the following statements from this section.

(1) The fifth postulate, which provoked centuries of controversy in classical Euclidean geometry, is the ________ postulate.
(2) In the nineteenth century, ________ and ________ independently established important theories of non-Euclidean geometry.
(3) The mathematician whose study of intrinsic surface geometry helped lay the foundations for later Riemannian geometry was ________.

[47] David Hilbert (1862–1943) was a German mathematician.

[48] Its German title is *Grundlagen der Geometrie*; the first edition appeared in 1899.

(4) In his celebrated habilitation lecture of 1854, Bernhard Riemann introduced the idea of a ________, explaining that a space may be described locally by coordinates although its metric structure need not be that of ordinary Euclidean space.
(5) In 1868 Eugenio Beltrami constructed a ________ of non-Euclidean geometry, establishing its consistency relative to Euclidean geometry.
(6) In 1899 David Hilbert published his ________, reorganizing the axioms of Euclidean geometry and bringing the foundations of geometry into their modern stage.

**Exercise 11.15.** Answer briefly.
(1) Explain the importance of axiomatization in the development of mathematics. Euclid constructed an early axiomatic system for geometry, and Hilbert later gave a new one. Compare them and explain their significance.
(2) Why was the parallel postulate long regarded as different from the other postulates? Describe the historical transition from attempts to prove it to the establishment of non-Euclidean geometry, and explain how that transition changed conceptions of space.
(3) Non-Euclidean geometry was not immediately accepted when it appeared in the nineteenth century. Why were the later models of Beltrami, Klein, and Poincaré important? How did Riemannian geometry further expand the conception of space opened by non-Euclidean geometry?

**Exercise 11.16.** Read the following passage and answer the questions.

> *Lobachevsky was a professor at Kazan University and later served for many years as its rector. He began thinking about the parallel postulate early and published papers on his new geometry in 1829–1830, among the earliest public printed accounts of non-Euclidean geometry. His work did not receive immediate recognition. When he submitted related work to the Saint Petersburg Academy of Sciences, Ostrogradsky (*Остроградский*)* [49] *gave a negative opinion. He held that some calculations were erroneous, the writing insufficiently rigorous, and much of the text difficult to understand, so that it did not deserve the Academy's attention. A pamphlet mocking Lobachevsky's paper later appeared in a Russian periodical, and for a long time his new geometry looked more like an eccentric heresy than a mathematical theory to be taken seriously. Lobachevsky nevertheless continued to publish and organize his theory, including his 1837 paper on "imaginary geometry" and the 1840 German summary Geometrical Investigations on the Theory of Parallels, published in Berlin. The latter made a deep impression on Gauss, but Lobachevsky did not live to see his geometry widely accepted.*

(1) Why might a new theory later shown to be highly valuable initially be regarded as insufficiently rigorous or heretical?
(2) Compare Gauss's caution, Lobachevsky's publication, and János Bolyai's independent work. Explain why discovery, publication, and acceptance are not the same event, and analyze the relation between mathematical innovation and scholarly acceptance.
(3) In light of the reception of non-Euclidean geometry, discuss the relations among intuition, axiomatic systems, models, and the mathematical community.

## 11.6 Toward Modern Geometry: Topology and Algebra

After non-Euclidean geometry had changed the understanding of space and axioms, the domain of geometry continued to expand rapidly. Among the most important modern developments were, on

[49] That is, Mikhail Vasilyevich Ostrogradsky (Михаил Васильевич Остроградский), 1801–1862, a mathematician of the Russian Empire.

the one hand, the deepening of algebraic and differential geometry and, on the other, topology's profound effect on geometric language. Strictly speaking, topology is not always regarded as a branch of geometry; it is more nearly a mathematical language of continuity, connectedness, and global structure. Yet precisely because it sets lengths, angles, and particular shapes temporarily aside and studies properties invariant under continuous deformation, topology radically changed modern geometry's understanding of spaces and objects. Before the nineteenth century, geometry still dealt largely with figures. By the turn of the twentieth century, it increasingly studied structures themselves: which properties survive continuous deformation, which can be expressed uniformly by systems of equations, and which objects should count as geometric even when they cannot be pictured intuitively. Geometry did not lose its classical origin in this change; it extended its questions about shape, relation, and invariance to a much wider world. Most noteworthy was the broadening of the very notion of a geometric object—from visible figures to abstract spaces, and from concrete curves to objects described by equations, homotopy types, group actions, or sheaf structures.

The beginnings of topology are often traced to Leonhard Euler's 1736 paper on the seven bridges of Königsberg. [50] The essential feature of that problem is not the length of a bridge or the width of a river, but the pattern of connection. This simple observation contains a far-reaching idea. If the essence of a problem is independent of lengths, angles, and areas and depends only on how things are connected, the study of spatial objects need not always be centered on measurement. In the late nineteenth century, through the work of Johann Benedict Listing, [51] Riemann, Poincaré, and others, topology gradually developed from problems at the boundaries of analysis, geometry, and mechanics into an independent discipline. Topologically, a circle and an ellipse do not differ essentially, while a sphere and a torus belong to different types because they have different numbers of holes. The structure of space need not always be a matter of measurement; it can concern relations invariant under continuous deformation.

Topology profoundly influenced modern geometry first by providing invariants that distinguish global types of spaces. The classification of closed surfaces, for example, says roughly that a connected compact surface without boundary can be distinguished topologically by its number of holes. The fundamental group translates the ways in which closed paths wind in a space into an algebraic object, making it possible to distinguish an annulus, sphere, projective plane, and other spaces more systematically. Once these ideas entered geometry, geometers no longer asked only about a space's local metric or curvature. They also asked whether it is connected or orientable, whether it has holes, and whether it admits nontrivial closed paths or covering spaces. Questions of this kind are not prominent in classical plane geometry, but are inescapable in modern differential geometry and geometric topology.

Topology also transformed the relation between local and global geometry. Curvature is local in differential geometry, but the global topology of a space constrains the ways in which curvature can occur. The Gauss–Bonnet theorem, [52] for example, relates the integral of a surface's curvature to its Euler characteristic, exhibiting a profound connection between a local geometric quantity and a global topological invariant. Later, de Rham cohomology, [53] characteristic classes, index theorems, and related theories showed that geometric tools such as differential forms, connections, and curvature can produce topological invariants, while topological invariants can in turn constrain the existence of geometric structures. Topology did not create a separate domain outside geometry, but entered geometry as a new language for understanding space globally. The reader will find an introduction to topology in section A.7 of the Appendix.

Historically, topology did not simply replace traditional geometry. It pushed an old question to an extreme: once every metric detail is removed, what remains that makes two spaces the same? Through this question geometry revealed more clearly its character as a structural science. It also

[50] Leonhard Euler (1707–1783) was a Swiss mathematician. His paper on the Königsberg bridge problem is commonly regarded as an early source of graph theory and topological thought.

[51] Johann Benedict Listing (1808–1882) was a German mathematician who systematically used the word *Topologie*.

[52] Bonnet here is Pierre Ossian Bonnet (1819–1892), a French mathematician; Gauss was introduced above.

[53] De Rham cohomology is named after Georges de Rham (1903–1990), a Swiss mathematician.

began to abandon the old requirement that every object must be drawable. The worth of a geometric object increasingly depended not on whether it could be represented fully by plane intuition, but on its behavior under a class of deformations or a collection of invariants.

In the twentieth century, Élie Cartan [54] integrated Lie theory, differential forms, moving frames, and related methods more deeply into geometry, giving differential geometry a more powerful structural language. For Cartan, a geometric space was often no longer just a set of points with coordinates, but an object endowed with local structure, a connection, and symmetries. Curvature, torsion, frames, and structural equations formed a common set of tools for analyzing the intrinsic properties of a space. This development is deeply connected with Riemannian geometry, the geometry of spacetime in general relativity, and twentieth-century gauge theory. If Klein emphasized geometry as the study of invariants under a transformation group, Lie and Cartan further showed that transformation groups can themselves become central geometric objects and can be studied through differential methods at the deepest levels of local structure.

At the same time, algebraic geometry carried the analytic idea of a figure defined by equations to a higher level. In its most elementary sense, algebraic geometry studies geometric objects defined by systems of polynomial equations; plane conics and cubics are among its earliest and most concrete examples. Historical exposition must nevertheless distinguish classical from modern algebraic geometry. The former was centered on curves, surfaces, intersections, singularities, tangents, genus, linear systems, and projective transformations, and remained closely connected with concrete figures and construction problems. The latter increasingly employed commutative algebra, topology, sheaves, schemes, cohomology, and functors, understanding a geometric object as a space organized by algebraic structures. The relation between them was not a simple rupture, but a change in language and level of objects. Classical algebraic geometry supplied concrete problems and geometric intuition; modern algebraic geometry placed them in more uniform and abstract frameworks.

Nineteenth-century algebraic geometry was still primarily classical. Through the work of Julius Plücker, Arthur Cayley, Alexander von Brill, Max Noether, and others, [55] the degree, singularities, duals, genus, and related notions of algebraic curves gradually took shape. Algebraic geometry developed from the study of particular curves into a systematic theory. The importance of this step lay not only in its wider range of objects, but in its large-scale treatment of a figure as the outward expression of an algebraic structure. A curve was no longer just a shape drawn on paper; it also carried functions, equations, intersection multiplicities, and singular behavior. Seeing and calculating acquired a new division of labor. Figures remained important, but became intuitive projections of algebraic structure rather than exhaustive accounts of the objects themselves.

The projective viewpoint is particularly natural in classical algebraic geometry. Curves that appear different in an affine plane can often be studied uniformly after homogenization in the projective plane $\mathbb{P}^2$. All nondegenerate plane conics over the complex numbers are projectively equivalent, while stable counts of intersections of higher-degree curves require projective completion. Bézout's theorem, [56] for example, says that under suitable conditions two plane algebraic curves of degrees $m$ and $n$ have $mn$ intersections in the projective plane when multiplicities and imaginary points are counted. Similarly, the Cayley–Bacharach theorem [57] shows that point configurations in the plane and curves of higher degree obey global constraints much stronger than the requirement that a curve pass through each point separately. These results are simultaneously algebraic and geometric. Classically they concern points, curves, and incidences; modernly they already touch such higher concepts as linear systems, codimension conditions, and global constraints.

[54] Élie Cartan (1869–1951) was a French mathematician whose work on Lie groups, differential forms, moving frames, and modern differential geometry had a deep influence.

[55] Alexander von Brill (1842–1935) was a German mathematician. The Noether meant here is Max Noether (1844–1921), a German mathematician, who must be distinguished from Emmy Noether. The historical family name sometimes had the form *Nöther*, but Max Noether himself, his mathematical publications, and modern histories of mathematics conventionally use *Noether*; this book therefore also uses *Noether*, not *Nöther*.

[56] The theorem is conventionally named after Étienne Bézout (1730–1783), a French mathematician.

[57] The theorem is conventionally named after Arthur Cayley (1821–1895), an English mathematician, and Isaak Bacharach (1854–1919), a German mathematician.

Algebraic geometry gradually entered its modern stage in the twentieth century. From the late nineteenth century onward, the geometric objects defined by polynomial equations were increasingly brought under the common name *algebraic variety*. Oscar Zariski [58] and others reorganized the theory of varieties on rigorous algebraic foundations, giving more precise treatments of singularities, local rings, valuations, and dimension. André Weil [59] further advanced abstract varieties and arithmetic geometry, enabling algebraic geometry to treat not only curves and surfaces over the complex numbers but systems of equations defined over general fields. This work had moved far from geometry as drawing, yet retained clear classical origins: curves, surfaces, intersections, singularities, and mappings remained basic objects, though their language grew more algebraic and more general.

Alexander Grothendieck [60] carried this modernization much further. His systematic use of schemes, sheaves, cohomology, functors, and related language made algebraic geometry a study not merely of solution sets of equations, but of whole spaces organized by algebraic data. Very roughly, the idea of a scheme is that a geometric object is determined not only by its points, but by the functions allowed on it, its local algebraic structures, and the ways in which those local pieces are glued. This language is undoubtedly abstract for a beginner, but its historical motivation was not a gratuitous pursuit of formal complexity. It was devised to treat equations over different fields, singular objects, problems in number theory, and many local-to-global phenomena within one framework. Modern algebraic geometry did not sever itself from classical algebraic geometry. Many modern notions still lead back to curves, tangents, duality, intersections, and projections in classical plane geometry. Modern theory places those objects in a broader structural setting and explains them more uniformly. Abstract language did not erase classical material; it made that material reappear in a more general form.

Developments in modern physics in turn reinforced the importance of these geometric ideas. Albert Einstein [61] used Riemannian geometry in 1915 to describe gravity and the curvature of spacetime in general relativity. From the second half of the twentieth century, this connection developed at still higher levels. Gauge theory describes physical fields through fiber bundles, connections, and curvature; string theory and related mathematics bring Calabi–Yau manifolds, [62] mirror symmetry, symplectic geometry, and complex algebraic geometry into the meeting ground of geometry and physics. These subjects lie far beyond the main scope of this book and cannot be developed here, but they show that modern geometry has not withdrawn from physical intuition. At a higher level of abstraction it has again become a language for describing natural structure.

From plane figures to the structure of spacetime, from conics to higher-dimensional manifolds, and from straightedge-and-compass constructions to symmetry and the geometrization of fields, the long course of geometric history is not simply a movement ever farther from intuition. It is an increasing ability to reconstruct intuition at more fundamental levels. Space is no longer merely a visible container, but a whole structure susceptible to axiomatic, curvature, topological, and algebraic treatment.

In the author's view, the relation between classical plane geometry and modern geometry should not be understood simply as the replacement of an old discipline by a new one. There is a connection, not always obvious but persistent, between them. Many objects later renamed, reorganized, and reinterpreted had already appeared in simpler forms within classical materials. The recurrent place in geometric history of conics, cross ratio, polarity, pencils of curves, homogeneous coordinates, and higher-degree curves may indicate that these concepts have always occupied transitional regions

[58] Oscar Zariski (1899–1986) was a Jewish mathematician born in the Russian Empire who later worked for many years in the United States. His birth name is usually recorded as Ascher or Oscher Zaritsky; he adopted the name Zariski after going to Italy, and it is the form standard in mathematical literature.

[59] André Weil (1906–1998) was a French mathematician.

[60] Alexander Grothendieck (1928–2014) was born in Germany and is best known for work within the French mathematical tradition.

[61] Albert Einstein (1879–1955) was a German-born theoretical physicist.

[62] These manifolds are named after Eugenio Calabi (1923–2023), an Italian-American mathematician, and Shing-Tung Yau (丘成桐; born 1949), a Chinese mathematician. They play important roles in complex geometry, algebraic geometry, and string theory.

between classical and modern geometry. Modern geometry greatly extended the range of study and changed its language, but many earlier outlines remain recognizable when one looks back along the path by which it arose.

The development of geometry need not be viewed as a continual negation of the past. It moved slowly among several levels: sometimes emphasizing intuitive figures, sometimes abstract structures; sometimes concrete objects, sometimes transformations and invariants; sometimes beginning with experience and construction, sometimes returning to axioms, models, and general spaces. From surveying land to proving theorems, from plane figures to abstract spaces, from straightedge-and-compass construction to axiomatic systems, and from particular curves to general structures, geometry passed through many turns while repeatedly circling similar questions. Behind changing shapes, are there stable relations? Beneath expressions in different languages, is there a deeper order worth recognizing?

For a student of geometry, its history is therefore more than supplementary background. It reminds us that many concepts now taken as natural were not always so, and that some questions which appear long dormant never entirely departed from later theory. Looking back over this book in that light, one can more readily understand why triangles, circles, conics, projective transformations, polars, cross-ratios, curves of higher degree, and related objects repeatedly return in different chapters. They are not necessarily isolated items of knowledge, but recurring nodes in geometry's long history. Here may lie part of the meaning of classical plane geometry. It can be admired as a comparatively complete world of its own, while remaining an old door into later geometry. The door's design is ancient, but the places to which it leads do not seem to have been closed.

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## Exercises

**Exercise 11.17.** Complete the following statements from this section.

(1) The beginnings of topology are often traced to Leonhard Euler's 1736 paper on the seven bridges of ________.
(2) Topology studies properties invariant under continuous deformation; a sphere and a torus, for example, belong to different types because they have different numbers of ________.
(3) The Gauss–Bonnet theorem relates the integral of a surface's curvature to its ________ characteristic, exhibiting a profound connection between local geometric quantities and global topological invariants.
(4) The French mathematician who closely connected Lie groups and differential forms with differential geometry and deeply influenced modern geometry was ________.
(5) Historical accounts distinguish classical and modern algebraic geometry. The former centers chiefly on ________, while the latter makes greater use of the language of ________.
(6) The twentieth-century mathematician who profoundly influenced modern algebraic geometry and advanced the foundation of scheme theory was ________.

**Exercise 11.18.** Answer briefly.

(1) Summarize the development and principal characteristics of modern geometry.
(2) Describe the relations between modern geometry and other disciplines.
(3) Divide the development of geometry into periods according to criteria you consider reasonable, and summarize the principal contents and features of each period.
(4) Give a rough classification of the branches of geometry, and describe their connections and differences.

**Exercise 11.19.** The same theorem often has proofs by different geometric methods, each with a distinctive character. Taking Miquel's theorem as an example and using the material of this book, give proofs from the viewpoints of classical plane geometry, analytic geometry, projective geometry, and algebraic geometry. Compare the methods. Can you give other examples of the same phenomenon?

**Exercise 11.20.** Consider the following three claims about projective geometry.

(1) Projective geometry merely adjoins points at infinity to the Euclidean plane for the convenience of studying plane geometry. Its objects are essentially those of the old Euclidean geometry, so it remains a branch of classical geometry.

(2) Projective geometry declares that any two lines meet at a point, thereby breaking free of the restriction of the parallel axiom, and should therefore be classified as a non-Euclidean geometry.

(3) Projective geometry introduced new algebraic structures, and modern algebraic geometry is built on projective space. Projective geometry is therefore essentially a kind of algebraic geometry.

Analyze and evaluate each claim, identifying what is reasonable and what is misleading in it.

**Exercise 11.21.** Problems in classical plane geometry can in principle be converted into problems about algebraic equations. It has therefore been claimed that traditional plane geometry has become a "dead subject." Do you agree? Discuss whether learning plane geometry remains meaningful and whether research in plane geometry remains meaningful, in light of classical geometry's place in mathematical education, mathematical thought, and modern mathematics.

**Exercise 11.22**.** In *The Structure of Scientific Revolutions*, Thomas Kuhn proposed his theory of paradigms, but did not use the history of mathematics to support it with examples. After reading relevant literature, discuss whether the history of mathematics can be described in terms of paradigms. In particular, can the development of geometry be understood as a succession of paradigm shifts? Write a short essay presenting your conclusions.

## Appendix A

# Preliminaries: An Outline of Modern Mathematics

This appendix introduces several subjects that the reader will need but may not have encountered at school: basic ideas from set theory, abstract algebra, topology, and analysis. It is not necessary to study the entire appendix before beginning the main text. The reader may instead consult the relevant parts as the need arises, while those already familiar with the material may use it for review. Since this is necessarily a rapid survey, we state only the definitions and results most useful in this book and omit some proofs.

The following guide indicates how much of each part is needed.

1. The whole of section A.1.1, on sets and mappings; the portions of section A.1.2 on equivalence relations and congruence classes modulo $a$; and the whole of section A.2.1, on vectors and their operations, supply notation, terminology, and basic tools used throughout the book. Readers are advised to make sure that they understand these parts. The treatment of vectors is compatible with both algebraic coordinates and geometric language. In particular, we define the cross product from geometric intuition and prove its properties to show that vectors and their operations can themselves be defined as geometric concepts. Thus the purely geometric chapters may use those properties without thereby invoking an algebraic method.

2. Groups also occur from time to time in the geometric chapters, chiefly to describe geometric transformations and their compositions. For this purpose, the notion of a group in section A.3.1 is essentially all that is needed; the remainder of the group-theoretic material need not be studied in depth.

3. Two other notions used along the geometric thread are fields, which occur in section 6.6.2, and elementary complex functions such as $\ln z$, which occur in the statement of Laguerre's theorem in section 6.4.2. For the former, it is enough to read the definition of a field in section A.4.1; for the latter, it is enough to read section A.9.1. Readers who do not wish to pursue these subjects may provisionally interpret the fields in question as $\mathbb{R}$ or $\mathbb{C}$, and may regard the complex functions as natural extensions of their real counterparts.

4. Only a little more is needed in the remaining geometric material. The topological model of the projective plane in section 3.1.1 uses the notion of homeomorphism from section A.7; that passage is optional small print and may also be understood intuitively in terms of continuous deformation. One proof of Laguerre's theorem in section 6.4.2 uses the Cauchy functional equation and mentions analyticity, as discussed in section A.9.2, but the text gives another proof that does not depend on that equation. The description of the one-dimensional projective transformation group in section 6.6.2 mentions group homomorphisms from section A.3.2; this too is optional and does not affect the main development. Some starred exercises use material from the appendix, but those exercises are themselves optional.

5. Chapter 10 combines geometric intuition with a substantial amount of algebra. On the set-theoretic side, it invokes order relations and the axiom of choice from section A.1.2 once each, and these may be consulted when needed. Nearly every part of the group theory in section A.3 is used, including the group actions of section A.3.3; these may be studied in advance. From ring theory, the principal requirements are division rings and fields as defined in section A.4.1; the quaternion division ring is used repeatedly as an example. The constructions in section A.4.3 provide concrete examples of division rings and fields but are not essential to the main ideas, while the integral-domain material in section A.4.2 is not needed there. The discussion of linear spaces over a general division ring in section A.5 explains the algebraic meaning of matrix descriptions over such a ring, including matrix rank. A reader may consult only the results of section A.5.3, provided that the matrix material of section A.2.2 is already familiar. Since standard university courses usually treat vector spaces over fields rather than general division rings, we develop the properties of linear spaces and linear maps

rather fully from sections A.5.1 and A.5.2 onward; readers may nevertheless use the stated results without following every proof.

6. To define complex functions and analyticity completely and rigorously, we use the topology of section A.7.1 and the continuity definitions of section A.7.2 to give a unified definition of limits, and we use the real derivatives and partial derivatives of section A.8 as preparation for complex differentiation. Readers who have not studied these subjects may temporarily pass over the foundations and read the conclusions about complex functions directly, interpreting a partial derivative intuitively as differentiation in one variable while the others are held fixed (or first reading its brief definition in section A.8). This is not fully rigorous, but is normally sufficient for the applications in the main text.

7. The remainder of section A.4.2 concerns integral domains, while section A.6 concerns polynomials. Partial orders, group theory, ring theory, polynomials, and point-set topology are included here as supplementary material; properties of real and complex derivatives may serve as auxiliary tools. The basic topology of surfaces in sections A.7.3 and A.7.4 chiefly supplies geometric intuition while defining the relevant properties rigorously.

Finally, the two sections on analysis require only differential properties such as differentiation and analyticity, so integration is not treated here.

# A.1 Sets and Mappings

## A.1.1 Basic Concepts of Sets and Mappings

### Basic properties of sets

At school one learns to call any object under consideration—a number, a vector, or even a book on a desk—an *element*. A specified collection, finite or infinite, of elements is a *set*. We shall not repeat the most elementary properties of sets, such as the notions of the empty set, intersection, union, and complement, but record several facts that may be new to the reader.

We use $\mathbb{N}$ for the natural numbers, including zero; $\mathbb{N}_+$ or $\mathbb{Z}_+$ for the positive integers; $\mathbb{Z}$ for the integers; $\mathbb{Q}$ for the rational numbers; $\mathbb{Q}_+$ for the positive rational numbers; $\mathbb{R}$ for the real numbers; and $\mathbb{C}$ for the complex numbers.[1]

**Definition A.1.1.** For sets $A, B$ with $A \cap B = \emptyset$, one may write $A \sqcup B$ or $A \dot{\cup} B$ in place of $A \cup B$ to emphasize that the two sets are disjoint. This is their *disjoint union*.

As with the summation and product signs $\sum$ and $\prod$, the symbols $\bigcup$, $\bigcap$, and $\bigsqcup$ denote unions, intersections, and disjoint unions over an indexed family. Thus $\bigsqcup_{\alpha \in I} X_\alpha$ is the disjoint union of all $X_\alpha$ with $\alpha \in I$.

**Definition A.1.2.** For sets $A, B$, the set $\{x \mid x \in A, x \notin B\}$ is their *set difference*, denoted by $A - B$ or $A \setminus B$.

**Example A.1.3.** If $A = \{0, 1, 2, 3\}$ and $B = \{2, 4\}$, then $A - B = \{0, 1, 3\}$.

**Example A.1.4.** The notation $\mathbb{R} \setminus \{0\}$ denotes the set of nonzero real numbers.

[1] The complex numbers are developed fully only in section A.9.1, but the reader is expected to have some elementary familiarity with them. In this book, "natural number" is used as a synonym for "nonnegative integer."

**Theorem A.1.5** (De Morgan's laws)**.** *Let $\{A_i \mid i \in I\}$ be a family of subsets of a universal set $X$, where $I$ is the set of indices of the $A_i$. Then*

$$X \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I}(X \setminus A_i), \quad X \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I}(X \setminus A_i).$$

**Definition A.1.6.** The *Cartesian product* of sets $X_1, X_2, \dots, X_n$ is

$$X_1 \times X_2 \times \cdots \times X_n := \{(x_1, x_2, \dots, x_n) \mid x_i \in X_i \text{ for every } i\}.$$

The Cartesian product of $n$ copies of a set $X$, $\underbrace{X \times X \times \cdots \times X}_{n \text{ copies of } X}$, is abbreviated to $X^n$.

**Example A.1.7.** For the real numbers, $\mathbb{R}^2 = \{(x,y) \mid x, y \in \mathbb{R}\}$ may be identified with the point set of the real Euclidean plane. We therefore use $\mathbb{R}^2$ to denote the real Euclidean plane. Similarly, $\mathbb{C}^2$ denotes the complex Euclidean plane. More generally, $\mathbb{R}^n$ and $\mathbb{C}^n$ denote real and complex $n$-dimensional Euclidean spaces.

Conventional symbols of the form $X^n$ do not invariably denote Cartesian products: for example, $\mathbb{S}^n$ in section A.7.3 denotes a sphere. For a general set $X$, however, $X^n$ will mean the Cartesian power unless stated otherwise.

**Definition A.1.8.** The number of elements of a set $X$, finite or infinite, is its *cardinality*, denoted by $|X|$.

### Basic concepts of mappings

**Definition A.1.9.** Let $X, Y$ be sets. A rule $f$ that assigns to every $x \in X$ a unique element $y \in Y$ is a *mapping* or *map* from $X$ to $Y$. The assigned element is denoted by $f(x)$, and the map is written $f: X \to Y, x \mapsto f(x)$.

(1) The element $f(x)$ is the *image* of $x$ under $f$, and $x$ is a *preimage* (or *inverse image*) of $f(x)$. The action of the map may be written $f: x \mapsto f(x)$.
(2) The set $X$ is the *domain* and $Y$ the *codomain* of $f$.
(3) For $A \subseteq X$, the set $\{f(a) \mid a \in A\}$ is the *image of* $A$, denoted by $f(A)$. In particular, $f(X) \subseteq Y$ is the *range* of $f$.
(4) For $A \subseteq Y$, the set $\{a \in X \mid f(a) \in A\}$ is the *preimage of* $A$, denoted by $f^{-1}(A)$.

**Remark.** The notation $f: X \to Y$, $x \mapsto f(x)$ records the domain $X$, the codomain $Y$, and the specific rule $f(x)$ for the action of $f$ on $x$. When only some of this information matters, one may write simply $f: X \to Y$ or $f: x \mapsto f(x)$.

The phrase "the preimage of $A$" does not require every member of $A$ to have a preimage. If no $x \in X$ satisfies $f(x) \in A$, then $f^{-1}(A) = \emptyset$.

**Theorem A.1.10.** *Let $f: X \to Y$ be a map.*
*(1) For $A, B \subseteq X$,*

$$f(A \cap B) \subseteq f(A) \cap f(B), \quad f(A \cup B) = f(A) \cup f(B).$$

*(2) For $C, D \subseteq Y$,*

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D), \quad f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D).$$

**Example A.1.11.** For $f: \mathbb{R} \times \mathbb{C} \to \mathbb{R}$, $(x,z) \mapsto x^2|z|$, the domain is the Cartesian product $\mathbb{R} \times \mathbb{C}$ of the real and complex number sets, the codomain is the real number set $\mathbb{R}$, and the rule is $f(x,z) = x^2|z|$. For example, $f(-2, \mathrm{i}+1) = (-2)^2|\mathrm{i}+1| = 4\sqrt{2}$. Its range is the set of nonnegative real numbers, $[0, +\infty)$.

**Definition A.1.12.** Let $f: X \to Y$.

(1) The map $f$ is a *surjection*, or is *surjective*, if for every $y \in Y$ there is an $x \in X$ with $f(x) = y$.
(2) The map $f$ is an *injection*, or is *injective*, if $x \neq x'$ implies $f(x) \neq f(x')$.
(3) The map $f$ is a *bijection*, or a *one-to-one correspondence*, if it is both injective and surjective.
(4) The map $f$ is a *constant map* if $f(x) = f(y)$ for all $x, y \in X$.

A map $f: X \to X$ is a *transformation* of $X$. A bijective transformation is a *one-to-one transformation*. If $f(x) = x$ for every $x \in X$, then $f$ is the *identity map*, denoted by $\mathrm{id}_X$, or simply id when no ambiguity can arise.

**Example A.1.13.** Let $M = \{1,2,3,\dots,n\}$, where $n \in \mathbb{Z}_+$. A bijection $\varphi: M \to M$ is commonly written

$$\varphi = \begin{pmatrix} 1 & 2 & \cdots & n \\ \varphi(1) & \varphi(2) & \cdots & \varphi(n) \end{pmatrix}$$

and is called a *permutation* of $n$ elements. The set of all such permutations is denoted by $\mathrm{S}_n$. The same permutation may have different displays; for instance, when $n = 3$,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \end{pmatrix}.$$

**Definition A.1.14.** For maps $f: X \to Y$ and $g: Y \to Z$, their *composition* is the map $g \circ f: X \to Z$ defined by $(g \circ f)(x) = g(f(x))$. If $f: X \to X$ is a transformation, the $n$-fold composite $\underbrace{f \circ f \circ \cdots \circ f}_{n \text{ copies of } f}$ is abbreviated to $f^n$.

**Definition A.1.15.** Let $f: X \to Y$. A map $g: Y \to X$ satisfying $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$ is the *inverse map* of $f$, denoted by $f^{-1}$.

**Theorem A.1.16.** *A map is bijective if and only if it has an inverse.*

**Theorem A.1.17.** *Let $f: X \to Y$ be a bijection and $g: Y \to X$ a map. If $g \circ f = \mathrm{id}_X$, then $f \circ g = \mathrm{id}_Y$; hence $g = f^{-1}$.*

**Proof.** For any $y \in Y$, the definition gives an $x \in X$ with $f(x) = y$. Then

$$(f \circ g)(y) = f(g(y)) = f(g(f(x))) = f((g \circ f)(x)) = f(x) = y.$$

Thus $f \circ g = \mathrm{id}_Y$. □

**Theorem A.1.18.** *For every map $f: X \to Y$, $f \circ \mathrm{id}_X = \mathrm{id}_Y \circ f = f$.*

**Theorem A.1.19.** *Composition of maps is associative. If $f: X \to Y$, $g: Y \to Z$, and $h: Z \to W$, then $h \circ (g \circ f) = (h \circ g) \circ f$.*

**Theorem A.1.20.** *If $f: X \to Y$ and $g: Y \to Z$ are bijective, then $g \circ f$ is bijective and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.*

**Definition A.1.21.** Let $f: X \to Y$ and $A \subseteq X$. The map $g: A \to Y$, $a \mapsto f(a)$, is the *restriction* of $f$ to $A$; this map $g$ is denoted by $f|_A$.

**Algebraic operations**

**Definition A.1.22.** An *algebraic operation* on a set $M$ is a rule assigning to each ordered pair $a,b\in M$ a unique element $d\in M$.[2] A set equipped with an algebraic operation is an *algebraic system*.

We may denote the algebraic operation of definition A.1.22 by $\circ$ and write the assigned element as $d=a\circ b$.

**Example A.1.23.** Ordinary addition is an algebraic operation on the positive integers, whereas ordinary subtraction is not, since $1-2=-1\notin\mathbb{Z}_+$. Ordinary multiplication is an algebraic operation on the rational numbers, whereas ordinary division is not, since division by zero is undefined.

**Example A.1.24.** Let $T(X)$ be the set of all transformations of a set $X$. Since $\sigma\circ\tau\in T(X)$ whenever $\sigma,\tau\in T(X)$, composition defines an algebraic operation on $T(X)$, given by $\sigma\circ\tau=\sigma\circ\tau$. We call it *multiplication of transformations*; when the meaning is clear, we write $\sigma\tau$ for $\sigma\circ\tau$.

**Theorem A.1.25.** *Multiplication of transformations is associative: for transformations $\sigma,\tau,\upsilon$ of $X$, $\sigma(\tau\upsilon)=(\sigma\tau)\upsilon$.*

**Definition A.1.26.** Suppose that $M$ and $M'$ carry operations $\circ$ and $\circ'$. A map $f: M\to M'$ satisfying

$$f(a)\circ' f(b)=f(a\circ b)\qquad\text{for all } a,b\in M$$

is a *homomorphism* of algebraic systems. If it is surjective, it is a *surjective homomorphism*; we then say that $M'$ is a *homomorphic image* of $M$ and write $M\sim M'$. If it is bijective, it is an *isomorphism*, and $M$ and $M'$ are *isomorphic*, written $M\cong M'$. The sets of all homomorphisms and all isomorphisms from $M$ to $M'$ are denoted by $\mathrm{Hom}(M,M')$ and $\mathrm{Iso}(M,M')$, respectively.

A homomorphism or isomorphism from an algebraic system to itself is an *endomorphism* or *automorphism*. Their sets are denoted by $\mathrm{End}(M)$ and $\mathrm{Aut}(M)$.

**Example A.1.27.** Let $M$ be the set of integers and let $M'$ be the even integers. The map $f: n\mapsto 2n$ from $M$ to $M'$ is an isomorphism for ordinary addition, but not for ordinary multiplication.

**Example A.1.28.** Let $M$ be the positive rational numbers, with ordinary multiplication as the algebraic operation. Then $f: a\mapsto\frac{1}{a}$ is an automorphism of $M$. It is not an automorphism for ordinary addition: for example, $f(2+3)=f(5)=\frac{1}{5}$, whereas $f(2)+f(3)=\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$.

**Theorem A.1.29.** *Let $M_1,M_2,M_3$ be sets endowed with algebraic operations. Isomorphism has the following properties.*

*(1) If $M_1\cong M_2$, then $M_2\cong M_1$.*
*(2) If $M_1\cong M_2$ and $M_2\cong M_3$, then $M_1\cong M_3$.*

When no confusion can arise, the symbol $\circ$ for an operation is often omitted: one writes $a\circ b=ab$ and calls the operation a *multiplication*.

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## Exercises

**Exercise A.1.** Prove theorems A.1.5, A.1.10, A.1.16, A.1.18 to A.1.20 and A.1.25 and example A.1.28.

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[2] In the language of maps, this is a map $f: M\times M\to M$.

**Exercise A.2.** For sets $A,B$, define $A \triangledown B = (A-B) \cup (B-A)$. Prove that
(1) $(A \triangledown B) \triangledown C = (A \triangledown C) \triangledown B$;
(2) $A \triangledown B = \emptyset$ if and only if $A = B$.

**Exercise A.3.** Find all maps $f:\ \mathbb{Q} \to \mathbb{R}$ satisfying $f(x+y) = f(x) + f(y)$ for all $x,y \in \mathbb{Q}$.

**Exercise A.4.** Let $S(X)$ and $T(X)$ denote the sets of all bijective transformations and all transformations of $X$, respectively. Prove that $S(X) \subseteq T(X)$ and that multiplication of transformations is an algebraic operation on $S(X)$.

**Exercise A.5.** Regard $\mathbb{Q}$ as an algebraic system under ordinary addition. Determine all of its automorphisms.

## A.1.2 Binary Relations

This section develops the language of relations on a set, including equivalence and order relations.

### Relations and equivalence relations

**Definition A.1.30.** Let $A,B$ be sets.
(1) A *binary relation* $R$ from $A$ to $B$ is a subset of $A \times B$. For $a \in A$ and $b \in B$, if $(a,b) \in R$, we say that $a,b$ are related by $R$ and write $aRb$; if $(a,b) \notin R$, they are not related by $R$, and we write $a \not{R} b$. A binary relation from a set $X$ to itself is called simply a *relation on* $X$.
(2) A relation $R$ on $X$ is an *equivalence relation* if it is *reflexive*, *symmetric*, and *transitive*; that is,
  (i) $xRx$ for every $x \in X$;
  (ii) for every $x,y \in X$, $xRy$ implies $yRx$;
  (iii) for every $x,y,z \in X$, $xRy$ and $yRz$ imply $xRz$.
The symbol $\sim$ is commonly used for an equivalence relation.
(3) Given an equivalence relation $\sim$ on $X$, for $x \in X$ the *equivalence class* of $x$ is $[x] := \{y \in X \mid x \sim y\}$. Thus $[x] \neq [y]$ exactly when $x \nsim y$. The set of all distinct equivalence classes is the *quotient set*, denoted by $X/\sim$, so that $X/\sim\, := \{[x] \mid x \in X\}$. The map $f:\ X \to X/\sim$, $x \mapsto [x]$, is readily seen to be surjective; it is the *natural projection* for $\sim$.

**Example A.1.31.**
(1) Equality of numbers is an equivalence relation.
(2) The relation $\leq$ on numbers is not an equivalence relation, since it is not symmetric: $1 \leq 2$ but $2 \nleq 1$.
(3) Similarity of plane triangles is an equivalence relation.
(4) Isomorphism of algebraic systems is an equivalence relation.
(5) On $\mathbb{Z}$, the relation $aRb$ if and only if $a-b=1$ is not an equivalence relation.
(6) Fix a number $a$ and a set of numbers $X$. Define a relation $R$ on $X$ by $xRy$ if and only if $x-y=na$ for some integer $n$. In this case, $x,y$ are said to be *congruent modulo* $a$. Congruence modulo $a$ is a *congruence* relation, commonly written $x \equiv y \pmod{a}$. It is easy to verify that congruence modulo $a$ is an equivalence relation. For example, $17 \equiv 5 \pmod{6}$. Equality of directed angles between directed lines is likewise defined modulo $2\pi$; see section 0.1.1.
(7) Congruence modulo 3 partitions $\mathbb{Z}$ into
$$[0] = \{\ldots,-6,-3,0,3,6,\ldots\},$$
$$[1] = \{\ldots,-5,-2,1,4,\ldots\},$$
$$[2] = \{\ldots,-4,-1,2,5,\ldots\}.$$
Thus the quotient set is $\{[0],[1],[2]\}$.

### Order relations

**Definition A.1.32.** Let $R$ be a relation on $X$.

(1) It is a *partial order* if it is reflexive, antisymmetric, and transitive:
   (i) $xRx$ for every $x \in X$;
   (ii) for every $x, y \in X$, $xRy$ and $yRx$ imply $x = y$;
   (iii) for every $x, y, z \in X$, $xRy$ and $yRz$ imply $xRz$.

   We commonly write an abstract partial order as $\preccurlyeq$, and write $y \succcurlyeq x$ equivalently to $x \preccurlyeq y$. A set equipped with a partial order is a *partially ordered set*, or *poset*.

(2) It is a *strict partial order* if it is irreflexive and transitive: $x \not R x$ for every $x \in X$, and, for all $x, y, z \in X$, $xRy$ and $yRz$ imply $xRz$. We commonly write such an order as $\prec$, with $y \succ x$ equivalent to $x \prec y$. A set $X$ equipped with $\prec$ is a *strictly partially ordered set*.

(3) A partial order is a *total order* if any two elements are *comparable*: for every $x, y \in X$, either $x \preccurlyeq y$ or $y \preccurlyeq x$. A set with a total order is a *totally ordered set*.

(4) A strict partial order is a *strict total order* if it satisfies comparability, more precisely *trichotomy*: for every $x, y \in X$, exactly one of $x \prec y$, $y \prec x$, and $x = y$ holds. A set $X$ equipped with such an order $\prec$ is a *strictly totally ordered set*.

**Remark.** A partial order $\preccurlyeq$ determines a strict partial order by $x \prec y$ if and only if $x \preccurlyeq y$ and $x \neq y$. Conversely, a strict partial order determines a partial order by $x \preccurlyeq y$ if and only if $x \prec y$ or $x = y$. If the partial order $\preccurlyeq$ is total, the corresponding strict partial order $\prec$ is a strict total order. Conversely, if $\prec$ is a strict total order, its corresponding partial order $\preccurlyeq$ is total. We shall ordinarily use $\preccurlyeq$ and $\prec$ for the corresponding nonstrict and strict relations, and use *order relation* collectively for all four notions.

**Definition A.1.33.** Let $P$ be a poset with order $\preccurlyeq$, and let $C \subseteq P$.

(1) An element $p \in P$ is an *upper bound* of $C$ if $c \preccurlyeq p$ for every $c \in C$, and a *lower bound* if $c \succcurlyeq p$ for every $c \in C$. An upper bound $p$ is the *supremum* of $C$, written $p = \sup C$, if $p \preccurlyeq s$ for every upper bound $s$ of $C$. Dually, a lower bound $p$ is the *infimum*, written $p = \inf C$, if $p \succcurlyeq i$ for every lower bound $i$. The set $C$ is *bounded* if it has both an upper and a lower bound, and *unbounded* otherwise.

(2) An element $p \in P$ is *maximal* if, for every $x \in P$, $p \preccurlyeq x$ implies $x = p$, and *minimal* if, for every $x \in P$, $p \succcurlyeq x$ implies $x = p$. It is the *greatest element* if $x \preccurlyeq p$ for every $x \in P$, and the *least element* if $x \succcurlyeq p$ for every $x \in P$.

(3) A subset $C \subseteq P$ on which $\preccurlyeq$ is a total order is a *chain*.

**Remark.** Upper and lower bounds of $C$ need not belong to $C$. A maximal element has no strictly larger element, whereas a greatest element is at least every element; the dual distinction holds for minimal and least elements. A greatest or least element, if it exists, is unique, but there may be several maximal or minimal elements. A supremum or infimum, if it exists, is also unique.

**Example A.1.34.** Let $X$ have at least two elements and let $P$ be the set of all its subsets. Inclusion $\subseteq$ is a partial order on $P$, and proper inclusion $\subsetneqq$ is the corresponding strict partial order. Neither is total: if $a \neq b$ are in $X$, the sets $\{a\}$ and $\{b\}$ are incomparable.

**Example A.1.35.** The usual relation $\leq$ on $\mathbb{R}$ is a total order, and $<$ is the corresponding strict total order. Bounds, suprema, and infima of subsets of $\mathbb{R}$ are always understood with respect to this order. Formally, we set $\sup C = +\infty$ when $C$ is unbounded above and $\inf C = -\infty$ when $C$ is unbounded below, with $-\infty < x < +\infty$ for every $x \in \mathbb{R}$.

**Theorem A.1.36** (Least-upper-bound property of the real numbers)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$, and every nonempty subset that is bounded below has an infimum in $\mathbb{R}$.*

**Remark.** This is the completeness property of the real numbers. Although intuitively natural, a full proof requires a rigorous construction of $\mathbb{R}$ and is omitted; readers need only know the result. The rational numbers do not have this property: the set $A=\{x\in\mathbb{Q}\,|\,x^2<2\}$ has neither a supremum nor an infimum in $\mathbb{Q}$; only as a subset of $\mathbb{R}$ does it have supremum $\sqrt{2}$ and infimum $-\sqrt{2}$.

**Theorem A.1.37** (Zorn's lemma)**.** *If every chain in a nonempty poset has an upper bound, then the poset has a maximal element.*

The proof of Zorn's lemma uses the *axiom of choice*; more precisely, over the usual basic axioms of set theory, Zorn's lemma is equivalent to that axiom.

An *axiom* is a basic proposition accepted without proof as a starting point for a theory. The standard foundational axioms of set theory are the *Zermelo–Fraenkel axioms*, abbreviated *ZF*. Adding the axiom of choice gives *ZFC*, or "ZF with Choice," which is the customary foundation for ordinary mathematics. We shall not list the other axioms here.

*Axiom of Choice.* For every family $\{A_i\,|\,i\in I\}$ of nonempty sets, with $I$ of arbitrary cardinality, there exists a choice function $f: I\to\bigcup_{i\in I}A_i$ such that $f(i)\in A_i$ for every $i\in I$.

For a finite family the choice is immediate. For an arbitrary infinite family, the statement can neither be proved nor refuted from the remaining ZF axioms, and is therefore adopted as an axiom in the usual ZFC framework, even though the infinite case may also seem intuitively plausible.

One may instead work with a different axiom. For example, adjoining the axiom of determinacy to ZF rules out the general axiom of choice, and hence Zorn's lemma.

*Axiom of Determinacy.* Given $A\subseteq\mathbb{N}^{\mathbb{N}}$, consider the infinite game in which two players alternately choose natural numbers $x_0,x_1,x_2,\ldots$, producing a sequence $x=(x_0,x_1,x_2,\ldots)$. The first player wins if $x\in A$, and the second otherwise. For every such $A$, one of the two players has a winning strategy. Here $\mathbb{N}^{\mathbb{N}}$ is the countable Cartesian power of $\mathbb{N}$, whose elements are infinite sequences of natural numbers $(a_0,a_1,a_2,a_3,\ldots)$ with $a_i\in\mathbb{N}$.

Several axiom systems for set theory are possible. In this book we follow the usual mathematical convention of working in ZFC, and hence assume Zorn's lemma.

The set-theoretic derivation of Zorn's lemma from the axiom of choice is too long for this survey and is omitted. When applying the lemma, remember that it produces a maximal element, not necessarily a greatest element.

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## Exercises

**Exercise A.6.** Answer the following conceptual questions.

(1) Verify directly that congruence modulo $a$ is an equivalence relation.

(2) Can the relation "one person loves another" be regarded as a binary relation? If so, is it an equivalence relation?

**Exercise A.7.** On $\mathbb{Z}$, let $\sim$ be congruence modulo $n$, where $n\in\mathbb{Z}$.

(1) For $n=4$, write out the quotient set $\mathbb{Z}/\sim$ explicitly.

(2) In $A=\mathbb{Z}/\sim$, for $a,b\in\mathbb{Z}$, define addition by $[a]+[b]=[a+b]$ and multiplication by $[a][b]=[ab]$. Determine whether these rules define algebraic operations on $A$.

**Exercise A.8.** On $\mathbb{N}_+$ define $m\preccurlyeq n$ if and only if $n=km$ for some $k\in\mathbb{N}_+$.

(1) Prove that $\preccurlyeq$ is a partial order but not a total order.

(2) For $C=\{2,3,4,6,12\}$, determine the maximal, minimal, greatest, and least elements, and the supremum and infimum, with respect to this order.

**Exercise A.9.** For a set $S$, let $P$ be the set of all its subsets.

(1) Prove that inclusion is a partial order on $P$, and determine when it is a total order.
(2) For a nonempty subset $A \subseteq P$, determine its supremum and infimum under inclusion.

**Exercise A.10.** Prove that every finite nonempty poset has a maximal and a minimal element. Prove further that if a finite poset has only one maximal element, then that element is greatest.

**Exercise A.11* (Well-ordering theorem).** A total order $\preccurlyeq$ on a set $X$ is a *well-order* if every nonempty subset of $X$ has a least element; then $X$ is *well-ordered*. Use Zorn's lemma to prove that every set admits a well-order.

## A.2 Vectors and Matrices over the Real Numbers

### A.2.1 Vectors and Their Operations

#### Review of vectors

We first recall the vectors in two- and three-dimensional space and the operations of addition, subtraction, scalar multiplication, and inner product encountered in secondary-school mathematics.

A vector is a quantity with both magnitude and direction. Consider two vectors $\boldsymbol{a},\boldsymbol{b}$ in two- or three-dimensional Euclidean space, represented geometrically by $\overrightarrow{A_1A_2}$ and $\overrightarrow{B_1B_2}$. Thus $\boldsymbol{a}$ has magnitude $|A_1A_2|$ and points from $A_1$ to $A_2$; $\boldsymbol{b}$ has magnitude $|B_1B_2|$ and points from $B_1$ to $B_2$. Their basic operations are as follows.

- The *magnitude* $|\boldsymbol{a}|$ is the size of the vector: $|\boldsymbol{a}| = |A_1A_2|$.
- For addition, translate the vectors so that the head of the first is the tail of the second. Their sum points from the tail of the first to the head of the second. Equivalently, choose $A_2C$ so that $\overrightarrow{A_2C} = \overrightarrow{B_1B_2}$; then $\boldsymbol{a}+\boldsymbol{b} = \overrightarrow{A_1C}$.
- Define $-\boldsymbol{b}$ to have the same magnitude as $\boldsymbol{b}$ and the opposite direction, so that $-\boldsymbol{b} = \overrightarrow{B_2B_1}$. Subtraction is defined by $\boldsymbol{a}-\boldsymbol{b} = \boldsymbol{a}+(-\boldsymbol{b})$. Equivalently, choose $A_1C$ so that $\overrightarrow{A_1C} = \overrightarrow{B_1B_2}$; then $\boldsymbol{a}-\boldsymbol{b} = \overrightarrow{CA_2}$.
- For a scalar $\lambda$, scalar multiplication produces a vector $\lambda\boldsymbol{a}$ parallel to the original vector, with its length scaled by the magnitude of $\lambda$. More precisely, if $\lambda > 0$, the direction is unchanged and the length is multiplied by $\lambda$; if $\lambda < 0$, the direction is reversed and the length is multiplied by $|\lambda|$. If $\lambda = 0$, then $\lambda\boldsymbol{a} = \boldsymbol{0}$ is the zero vector.
- The *dot product*, or *inner product*, is a scalar: $\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta$, where $\theta$ is the angle between the directions of $\boldsymbol{a}$ and $\boldsymbol{b}$, with $0 \le \theta \le 180°$.

In rectangular coordinates, vectors can be represented by their components. In the plane, let $\boldsymbol{i},\boldsymbol{j}$ be the unit vectors in the $x$- and $y$-directions. If $\boldsymbol{a} = a_1\boldsymbol{i}+a_2\boldsymbol{j}$, we write $\boldsymbol{a} = (a_1,a_2)$. In three-dimensional space, let $\boldsymbol{i},\boldsymbol{j},\boldsymbol{k}$ be the unit vectors in the $x$-, $y$-, and $z$-directions of the rectangular coordinate system. If $\boldsymbol{a} = a_1\boldsymbol{i}+a_2\boldsymbol{j}+a_3\boldsymbol{k}$, we write $\boldsymbol{a} = (a_1,a_2,a_3)$. Thus, in the plane, if $\boldsymbol{a} = (a_1,a_2)$ and $\boldsymbol{b} = (b_1,b_2)$, then

$$|\boldsymbol{a}| = \sqrt{a_1^2+a_2^2}, \quad \boldsymbol{a}\pm\boldsymbol{b} = (a_1\pm b_1, a_2\pm b_2),$$
$$\lambda\boldsymbol{a} = (\lambda a_1, \lambda a_2), \quad \boldsymbol{a}\cdot\boldsymbol{b} = a_1b_1+a_2b_2,$$

and, in space, if $\boldsymbol{a} = (a_1,a_2,a_3)$ and $\boldsymbol{b} = (b_1,b_2,b_3)$, then

$$|\boldsymbol{a}| = \sqrt{a_1^2+a_2^2+a_3^2}, \quad \boldsymbol{a}\pm\boldsymbol{b} = (a_1\pm b_1, a_2\pm b_2, a_3\pm b_3),$$
$$\lambda\boldsymbol{a} = (\lambda a_1, \lambda a_2, \lambda a_3), \quad \boldsymbol{a}\cdot\boldsymbol{b} = a_1b_1+a_2b_2+a_3b_3.$$

The basic laws include:

- $\boldsymbol{a}+\boldsymbol{b} = \boldsymbol{b}+\boldsymbol{a}$;
- $\lambda(\boldsymbol{a}+\boldsymbol{b}) = \lambda\boldsymbol{a}+\lambda\boldsymbol{b}$, and $(\lambda+\mu)\boldsymbol{a} = \lambda\boldsymbol{a}+\mu\boldsymbol{a}$;

- $\boldsymbol{a}\cdot\boldsymbol{b}=\boldsymbol{b}\cdot\boldsymbol{a}$;
- $\boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c})=\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\cdot\boldsymbol{c}$, and $(\boldsymbol{a}+\boldsymbol{b})\cdot\boldsymbol{c}=\boldsymbol{a}\cdot\boldsymbol{c}+\boldsymbol{b}\cdot\boldsymbol{c}$;
- $(\lambda\boldsymbol{a})\cdot\boldsymbol{b}=\boldsymbol{a}\cdot(\lambda\boldsymbol{b})=\lambda(\boldsymbol{a}\cdot\boldsymbol{b})$.

We also recall the *fundamental theorem of vectors* in the plane and in space. In the plane, choose two nonparallel vectors $\boldsymbol{e}_1,\boldsymbol{e}_2$. Every vector $\boldsymbol{v}$ is a linear combination of them: there exist scalars $a_1,a_2$ such that $\boldsymbol{v}=a_1\boldsymbol{e}_1+a_2\boldsymbol{e}_2$. In three-dimensional space, choose three noncoplanar vectors $\boldsymbol{e}_1,\boldsymbol{e}_2,\boldsymbol{e}_3$. Every vector $\boldsymbol{v}$ is a linear combination of them: there exist scalars $a_1,a_2,a_3$ such that $\boldsymbol{v}=a_1\boldsymbol{e}_1+a_2\boldsymbol{e}_2+a_3\boldsymbol{e}_3$.

### The cross product

The preceding material is familiar from secondary-school mathematics. We now introduce a new operation.

**Definition A.2.1.** For vectors $\boldsymbol{a},\boldsymbol{b}$ in three-dimensional space, their *vector product*, or *cross product*, is the vector $\boldsymbol{a}\times\boldsymbol{b}$ of magnitude $|\boldsymbol{a}||\boldsymbol{b}|\sin\theta$, where $\theta$ is the angle between their directions and $0\le\theta\le 180^\circ$. Its direction is perpendicular to both $\boldsymbol{a}$ and $\boldsymbol{b}$. The choice between the two possible directions depends on the *orientation* of the space.

In a right-handed space, use the right-hand rule: extend the right thumb and curl the other four fingers from $\boldsymbol{a}$ towards $\boldsymbol{b}$. The thumb then points in the direction of $\boldsymbol{a}\times\boldsymbol{b}$; see figure A.1a.[3] In a left-handed space, use the left-hand rule: extend the left thumb and curl the other four fingers from $\boldsymbol{a}$ towards $\boldsymbol{b}$. The thumb then points in the direction of $\boldsymbol{a}\times\boldsymbol{b}$; see figure A.1b.


(a) Cross product in a right-handed space (b) Cross product in a left-handed space

FIGURE A.1. Orientation conventions for the cross product

**Remark.** A rectangular coordinate system in space likewise has an orientation: it can be right-handed or left-handed. Let $\boldsymbol{i},\boldsymbol{j},\boldsymbol{k}$ be the unit vectors in the $x$-, $y$-, and $z$-directions, respectively. If rotation from $\boldsymbol{i}$ to $\boldsymbol{j}$ according to the right-hand rule makes the thumb point along $\boldsymbol{k}$, the coordinate system is right-handed. If rotation from $\boldsymbol{i}$ to $\boldsymbol{j}$ according to the left-hand rule makes the thumb point along $\boldsymbol{k}$, it is left-handed. The orientation of the coordinate system and that of the ambient space are distinct. The orientation of the space is independent of the orientation of a coordinate system in it. For example, even when right-handed coordinates are used in a left-handed space, the cross product is still defined by the left-hand rule.

The orientation of space is a convention chosen in advance, rather than an intrinsic choice already made for us. The usual convention is to use a right-handed space and right-handed coordinates, and we follow it here. Since the angle between $\boldsymbol{i}$ and $\boldsymbol{j}$ is $90^\circ$, the definition then gives $\boldsymbol{i}\times\boldsymbol{j}=\boldsymbol{k}$ and $\boldsymbol{j}\times\boldsymbol{i}=-\boldsymbol{k}$.

For planar vectors one also sometimes speaks of a cross product, but its meaning differs from that of the usual three-dimensional cross product. The cross product of two planar vectors is defined to be the scalar $\boldsymbol{a}\times\boldsymbol{b}=|\boldsymbol{a}||\boldsymbol{b}|\sin\angle(\boldsymbol{a},\boldsymbol{b})$. Unlike the angle in the preceding definition, this angle is directed;

[3] If the angle is 0 or $180^\circ$, the magnitude definition makes the cross product the zero vector, so its direction need not be considered. Otherwise, the two directions form an angle smaller than $180^\circ$ and a reflex angle greater than $180^\circ$. Curl the fingers through the smaller angle. The same convention applies to the left-hand rule below.

see section 0.1.1. Unless explicitly described as planar, a cross product below is the three-dimensional vector product.

The following geometric interpretation is immediate from the definition.

**Theorem A.2.2.** *For points $O,A,B$, $|\overrightarrow{OA}\times\overrightarrow{OB}|=2S_{\triangle AOB}$. For the planar scalar cross product, $\overrightarrow{OA}\times\overrightarrow{OB}=2[\triangle AOB]$.*

In coordinates the cross product can be calculated as follows.

**Theorem A.2.3.** *In right-handed rectangular coordinates in a right-handed space, if $\boldsymbol{a}=(a_1,a_2,a_3)$ and $\boldsymbol{b}=(b_1,b_2,b_3)$, then*

$$\boldsymbol{a}\times\boldsymbol{b}=(a_2b_3-a_3b_2,\ a_3b_1-a_1b_3,\ a_1b_2-a_2b_1).$$

**Proof.** Write $\boldsymbol{a}\times\boldsymbol{b}=\boldsymbol{c}=(c_1,c_2,c_3)$. Orthogonality gives

$$0=\boldsymbol{a}\cdot\boldsymbol{c}=a_1c_1+a_2c_2+a_3c_3,$$
$$0=\boldsymbol{b}\cdot\boldsymbol{c}=b_1c_1+b_2c_2+b_3c_3.$$

This system has three unknowns. For nonparallel $\boldsymbol{a},\boldsymbol{b}$, the solutions are scalar multiples of the displayed vector; explicitly,

$$(c_1,c_2,c_3)=\lambda(a_2b_3-a_3b_2,\ a_3b_1-a_1b_3,\ a_1b_2-a_2b_1),$$

where $\lambda$ remains to be determined from the magnitude condition. By the geometric definitions of the dot and cross products, if $\theta$ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$, then

$$\begin{aligned}|\boldsymbol{a}\times\boldsymbol{b}|&=|\boldsymbol{a}||\boldsymbol{b}|\sin\theta=\sqrt{|\boldsymbol{a}|^2|\boldsymbol{b}|^2-|\boldsymbol{a}|^2|\boldsymbol{b}|^2\cos^2\theta}=\sqrt{|\boldsymbol{a}|^2|\boldsymbol{b}|^2-(\boldsymbol{a}\cdot\boldsymbol{b})^2}\\&=\sqrt{(a_2b_3-a_3b_2)^2+(a_3b_1-a_1b_3)^2+(a_1b_2-a_2b_1)^2}.\end{aligned}$$

Thus $\lambda=\pm1$. To determine the sign, the right-handed basis $\boldsymbol{i},\boldsymbol{j},\boldsymbol{k}$ satisfies $(1,0,0)\times(0,1,0)=(0,0,1)$, and hence requires $\lambda=1$. Any triple $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ can be rotated so that $\boldsymbol{a},\boldsymbol{b}$ lie in the $xy$-plane, and this rotation varies continuously; consequently $\lambda$ cannot change sign discontinuously, so $\lambda=1$ throughout. If $\boldsymbol{a}\parallel\boldsymbol{b}$, then $\boldsymbol{a}\times\boldsymbol{b}=\boldsymbol{0}$, and the displayed coordinate formula holds directly. □

**Remark.** In rectangular coordinates, for planar vectors $\boldsymbol{a}=(a_1,a_2)$ and $\boldsymbol{b}=(b_1,b_2)$, the scalar cross product is $\boldsymbol{a}\times\boldsymbol{b}=a_1b_2-a_2b_1$, which, by definition, is the $z$-component of $(a_1,a_2,0)\times(b_1,b_2,0)$. As before, subsequent cross products are three-dimensional unless otherwise stated.

**Theorem A.2.4.** *For vectors $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ and a scalar $\lambda$:*

*(1) Anticommutativity: $\boldsymbol{a}\times\boldsymbol{b}=-\boldsymbol{b}\times\boldsymbol{a}$;*
*(2) $(\lambda\boldsymbol{a})\times\boldsymbol{b}=\boldsymbol{a}\times(\lambda\boldsymbol{b})=\lambda(\boldsymbol{a}\times\boldsymbol{b})$;*
*(3) Left and right distributivity: $\boldsymbol{a}\times(\boldsymbol{b}+\boldsymbol{c})=\boldsymbol{a}\times\boldsymbol{b}+\boldsymbol{a}\times\boldsymbol{c}$, and $(\boldsymbol{a}+\boldsymbol{b})\times\boldsymbol{c}=\boldsymbol{a}\times\boldsymbol{c}+\boldsymbol{b}\times\boldsymbol{c}$;*
*(4) Cyclic invariance of the scalar triple product: $\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})=\boldsymbol{b}\cdot(\boldsymbol{c}\times\boldsymbol{a})=\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$.* An expression $\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})$, involving both a dot product and a cross product, is the *scalar triple product*, or *mixed product*, of the vectors $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$, and may be denoted by $[\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}]$.
*(5) The expansion formula for the vector triple product, or* Lagrange formula *is*

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})=(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}-(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}.$$

This double cross product is the *vector triple product* of $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$.

**Proof.** The first two statements are immediate from the definition, and all five may also be checked from the coordinate formula in theorem A.2.3. We give geometric proofs of the last three. As in the synthetic parts of the main text, routine degenerate cases are omitted here; the coordinate calculation covers them all, and they may also be verified directly.

For the first distributive law, use the configuration of figure A.2a. Represent $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$, and $\boldsymbol{b}+\boldsymbol{c}$ by $\overrightarrow{OA},\overrightarrow{OB},\overrightarrow{OC}$, and $\overrightarrow{OD}$, so that $OCDB$ is a parallelogram. Project $\overrightarrow{OB},\overrightarrow{OC},\overrightarrow{OD}$ onto the plane $\pi$

perpendicular to $\boldsymbol{a}$, obtaining $\overrightarrow{OB'},\overrightarrow{OC'},\overrightarrow{OD'}$. Projection preserves their vector sum, $\overrightarrow{OB'}+\overrightarrow{OC'}=\overrightarrow{OD'}$, so $OB'D'C'$ is again a parallelogram. By the definition of the cross product,

$$|\boldsymbol{a}\times\boldsymbol{b}|=OA\cdot OB\sin\angle AOB=OA\cdot OB'.$$

The points $A,O,B,B'$ are coplanar, so $\boldsymbol{a}\times\boldsymbol{b}$ and $\boldsymbol{a}\times\overrightarrow{OB'}$ also have the same direction. Thus $\boldsymbol{a}\times\boldsymbol{b}=\boldsymbol{a}\times\overrightarrow{OB'}$. Similarly, $\boldsymbol{a}\times\boldsymbol{c}=\boldsymbol{a}\times\overrightarrow{OC'}$ and $\boldsymbol{a}\times\overrightarrow{OD}=\boldsymbol{a}\times\overrightarrow{OD'}$. Choose $\overrightarrow{OB}_1,\overrightarrow{OC}_1,\overrightarrow{OD}_1$ equal to $\boldsymbol{a}\times\overrightarrow{OB'},\boldsymbol{a}\times\overrightarrow{OC'},\boldsymbol{a}\times\overrightarrow{OD'}$, respectively. It remains to prove $\overrightarrow{OB}_1+\overrightarrow{OC}_1=\overrightarrow{OD}_1$.

By the definition of the cross product, $OB_1$ is perpendicular to the plane $AOB'$. Since $OA\perp\pi$, it follows that $OB_1$ lies in $\pi$; also $OB_1\perp OB'$ and $\frac{OB_1}{OB'}=OA$. Similarly, $OC_1\perp OC'$, $OD_1\perp OD'$, and $\frac{OC_1}{OC'}=\frac{OD_1}{OD'}=OA$. Consequently,

$$(B_1,C_1,D_1)=(\mathfrak{h}_{O,|OA|}\circ\mathfrak{r}_{O,90^\circ})(B',C',D');$$

that is, they are obtained from $B',C',D'$ by rotation through $90^\circ$ about $O$ followed by dilation by $|OA|$; see section 0.4. Thus $OB_1D_1C_1$ is a parallelogram and $\overrightarrow{OB}_1+\overrightarrow{OC}_1=\overrightarrow{OD}_1$. The other distributive law is proved in the same way.

For the scalar triple product, use figure A.2b. Let $\overrightarrow{OA},\overrightarrow{OB},\overrightarrow{OC}$ represent $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ and span the parallelepiped $AFEG\text{–}OBCD$; a *parallelepiped* is a prism with parallelogram bases, equivalently a hexahedron whose opposite faces are congruent parallelograms. Let $A'$ be the projection of $A$ onto the base plane $BOC$. Write $S_{OBDC}$ for the area of the parallelogram $OBDC$. By the definition of the cross product,

$$|\boldsymbol{b}\times\boldsymbol{c}|=OB\cdot OC\cdot\sin\angle BOC=S_{OBDC}.$$

Its direction is perpendicular to the plane $BOC$, that is, $\boldsymbol{b}\times\boldsymbol{c}\parallel\overrightarrow{A'A}$. Hence the projection of $\overrightarrow{OA}$ in the direction $\overrightarrow{A'A}$ is $\overrightarrow{A'A}$, and

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})=A'A\cdot S_{OBDC}=V_{AFEG\text{–}OBCD}$$

is the volume of the parallelepiped. Cyclically permuting $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ gives the same volume for $\boldsymbol{b}\cdot(\boldsymbol{c}\times\boldsymbol{a})$ and $\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$, so the three expressions are equal.

Here $V_{AFEG\text{–}OBCD}$ can be positive or negative: its sign depends on whether $\boldsymbol{b}\times\boldsymbol{c}$ points in the same direction as $\overrightarrow{A'A}$ or in the opposite direction. It is therefore called the *oriented volume* of the parallelepiped. A cyclic permutation preserves this sign as well, so the stated equality holds. Interchanging two factors, as in $\boldsymbol{a}\cdot(\boldsymbol{c}\times\boldsymbol{b})$, instead reverses the sign.

For the vector triple product, use figure A.2c. Represent $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ by $\overrightarrow{OA},\overrightarrow{OB},\overrightarrow{OC}$, respectively. Let $\pi$ be the plane containing $OB,OC$, and let $A'$ be the projection of $A$ onto $\pi$. By definition, $(\boldsymbol{b}\times\boldsymbol{c})\perp\pi$, so $(\boldsymbol{b}\times\boldsymbol{c})\parallel\overrightarrow{A'A}$. As in the proof of (3), the component of $\overrightarrow{OA}$ perpendicular to $\overrightarrow{AA'}$ is $\overrightarrow{OA'}$; therefore $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})=\overrightarrow{OA'}\times(\boldsymbol{b}\times\boldsymbol{c})$. Moreover, $\overrightarrow{OA'}\times(\boldsymbol{b}\times\boldsymbol{c})$ is parallel to $\overrightarrow{OA'}\times\overrightarrow{A'A}$ and hence perpendicular to the plane $OAA'$. Write the resulting vector as $\overrightarrow{OX}$. Then $X$ lies in $\pi$, so $\overrightarrow{OX},\overrightarrow{OB},\overrightarrow{OC}$ are coplanar. By the fundamental theorem for planar vectors, there are $\mu,\nu$ with $\overrightarrow{OX}=\mu\boldsymbol{b}+\nu\boldsymbol{c}$. Since $\overrightarrow{OX}\perp\boldsymbol{a}$, we have $0=\overrightarrow{OX}\cdot\boldsymbol{a}=(\mu\boldsymbol{b}+\nu\boldsymbol{c})\cdot\boldsymbol{a}=\mu(\boldsymbol{a}\cdot\boldsymbol{b})+\nu(\boldsymbol{a}\cdot\boldsymbol{c})$. Thus $\frac{\mu}{\nu}=-\frac{\boldsymbol{a}\cdot\boldsymbol{c}}{\boldsymbol{a}\cdot\boldsymbol{b}}$, and consequently there is a scalar $\lambda$ such that

$$\overrightarrow{OX}=\lambda\left[(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}-(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}\right].$$

Write $b,c$ for the magnitudes of $\boldsymbol{b},\boldsymbol{c}$ and $\hat{\boldsymbol{b}},\hat{\boldsymbol{c}}$ for their unit direction vectors. Let $D,E$ be the projections of $A$ onto $OB,OC$. The three-perpendiculars theorem gives $A'E\perp OC$ and $A'D\perp OB$, so $O,D,A',E$ lie on the circle $\odot(OA')$.[4] The geometric interpretation of the inner product now gives

$$\begin{aligned}|\overrightarrow{OX}|&=|\lambda[(c\cdot OE)\cdot b\hat{\boldsymbol{b}}-(b\cdot OD)\cdot c\hat{\boldsymbol{c}}]|=|\lambda bc(OE\cdot\hat{\boldsymbol{b}}-OD\cdot\hat{\boldsymbol{c}})|\\&=|\lambda bc(OD\cdot\hat{\boldsymbol{b}}-OE\cdot\hat{\boldsymbol{c}})|=|\lambda bc||\overrightarrow{DE}|=|\lambda bc||OA'|\sin\angle BOC.\end{aligned}$$

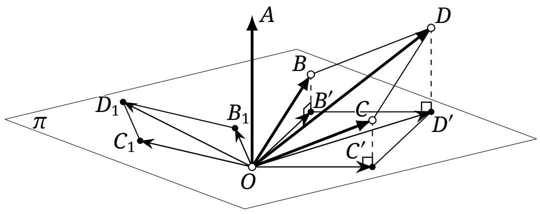
The final equality uses the sine rule. In the first equality on the second displayed line, only the magnitude is being considered: interchanging the magnitudes of the two vectors being subtracted,

[4]The three-perpendiculars theorem states that if $l$ lies in a plane $\pi$, while $m$ neither lies in $\pi$ nor is perpendicular to $\pi$, and $n$ is the projection of $m$ onto $\pi$, then $l\perp m\Longleftrightarrow l\perp n$. This is a basic result of secondary-school solid geometry.

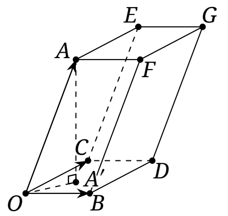
while leaving their directions fixed, does not change the resulting magnitude. On the other hand, since $\overrightarrow{OA'} \perp \boldsymbol{b}\times\boldsymbol{c}$,

$$|\overrightarrow{OX}| = |\overrightarrow{OA'} \times (\boldsymbol{b}\times\boldsymbol{c})| = |\overrightarrow{OA'}|\cdot|\boldsymbol{b}\times\boldsymbol{c}| = |OA'||bc\sin\angle BOC|.$$

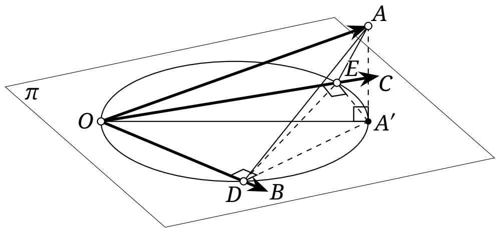
Thus $\lambda = \pm 1$. In the special case $\boldsymbol{b}\perp\boldsymbol{c}$ and $\boldsymbol{a}=\boldsymbol{b}$, the result is $-b^2\boldsymbol{c}$, which gives $\lambda = 1$; continuity prevents this sign from changing. Hence $\lambda = 1$, as required. □




(a) Distributivity




(b) Scalar triple product




(c) Vector triple product

FIGURE A.2. Geometric interpretations of theorem A.2.4

**Remark.** The vector triple product is generally not associative: the equality $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = (\boldsymbol{a}\times\boldsymbol{b})\times\boldsymbol{c}$ need not hold.

Determinants provide a convenient way to describe the coordinate calculation of a cross product. We therefore introduce determinants of orders two and three.

**Definition A.2.5.** The determinants of orders two and three are defined by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} := ad - bc, \qquad \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Combining theorem A.2.3 with the determinant notation of definition A.2.5 gives the following formulas, whose direct verification is left to the reader.

**Theorem A.2.6.** *For $\boldsymbol{a} = (a_1,a_2,a_3)$, $\boldsymbol{b} = (b_1,b_2,b_3)$, and $\boldsymbol{c} = (c_1,c_2,c_3)$ in rectangular coordinates, where $\boldsymbol{i},\boldsymbol{j},\boldsymbol{k}$ are the unit vectors in the x-, y-, and z-directions, respectively, we have:*

*(1)*

$$\boldsymbol{a}\times\boldsymbol{b} = \left( \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}, \begin{vmatrix} a_3 & a_1 \\ b_3 & b_1 \end{vmatrix}, \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \right) = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix};$$

*(2)*

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

*In rectangular coordinates, for planar vectors $\boldsymbol{a} = (a_1,a_2)$ and $\boldsymbol{b} = (b_1,b_2)$, one may put $\boldsymbol{a}\times\boldsymbol{b} = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$. In what follows, however, cross products are understood to be three-dimensional unless otherwise stated.*

Determinants make cross products and scalar triple products easy to remember and calculate. Their general definition is given in the next subsection.

### $n$-dimensional vectors

Planar and three-dimensional vectors extend directly to *vectors* in $n$-dimensional space. The geometric definitions of magnitude, addition, subtraction, scalar multiplication, and inner product given above are unchanged. Let $\boldsymbol{i}_1, \boldsymbol{i}_2, \ldots, \boldsymbol{i}_n$ be the unit vectors along the $n$ axes of a rectangular coordinate system. If $\boldsymbol{a} = a_1\boldsymbol{i}_1 + a_2\boldsymbol{i}_2 + \cdots + a_n\boldsymbol{i}_n$, we write $\boldsymbol{a} = (a_1, a_2, \ldots, a_n)$. The coordinate operations are analogous:

- Addition and subtraction: $(a_1, a_2, \ldots, a_n) \pm (b_1, b_2, \ldots, b_n) = (a_1 \pm b_1, a_2 \pm b_2, \ldots, a_n \pm b_n)$;
- scalar multiplication: $c(a_1, a_2, \ldots, a_n) = (ca_1, ca_2, \ldots, ca_n)$;
- inner product: $(a_1, a_2, \ldots, a_n) \cdot (b_1, b_2, \ldots, b_n) = a_1b_1 + a_2b_2 + \cdots + a_nb_n$;
- magnitude: $|(a_1, a_2, \ldots, a_n)| = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$.

The laws in the opening review, section A.2.1, remain valid. The cross product is special to dimension three; this operation does not extend readily to general $n$-dimensional space. The planar scalar cross product mentioned earlier is a distinct special convention introduced for convenience in some applications. A cross product also exists in seven dimensions, but we shall not need it and do not discuss it here.

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## Exercises

**Exercise A.12.** Verify theorems A.2.4 and A.2.6 directly from the definitions.

**Exercise A.13.** Let $O$ be the origin in three-dimensional rectangular coordinates and let $A(a_1, a_2, a_3)$, $B(b_1, b_2, b_3)$, and $C(c_1, c_2, c_3)$. Find the volume of the tetrahedron $O$–$ABC$.

**Exercise A.14** (Jacobi identity)**.** For vectors in three-dimensional space, prove

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) + \boldsymbol{c} \times (\boldsymbol{a} \times \boldsymbol{b}) + \boldsymbol{b} \times (\boldsymbol{c} \times \boldsymbol{a}) = \boldsymbol{0}.$$

**Exercise A.15.** Prove the following laws for cross products of three-dimensional vectors:

$$(\boldsymbol{a} \times \boldsymbol{b}) \cdot (\boldsymbol{c} \times \boldsymbol{d}) = (\boldsymbol{a} \cdot \boldsymbol{c})(\boldsymbol{b} \cdot \boldsymbol{d}) - (\boldsymbol{b} \cdot \boldsymbol{c})(\boldsymbol{a} \cdot \boldsymbol{d}), \tag{1}$$

$$(\boldsymbol{a} \times \boldsymbol{b}) \times (\boldsymbol{c} \times \boldsymbol{d}) = [\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{d}]\boldsymbol{c} - [\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}]\boldsymbol{d} = [\boldsymbol{a}, \boldsymbol{c}, \boldsymbol{d}]\boldsymbol{b} - [\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}]\boldsymbol{a}. \tag{2}$$

**Exercise A.16**.** Consult suitable references and study the definition and operational laws of the cross product in seven dimensions.

## A.2.2 Basic Concepts of Matrices

We now introduce the basic concepts of matrices. Matrices are closely connected with the linear spaces introduced later, but the treatment here is mainly concrete and computational, so we present it first.

### Matrices

**Definition A.2.7** (Row and column vectors)**.**

(1) An element of $\mathbb{R}^m$ of the form
$$\boldsymbol{v} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \in \mathbb{R}^m$$
where $a_i \in \mathbb{R}$ for $i = 1, 2, \ldots, m$, is a *column vector*. The same element is sometimes written as $\begin{bmatrix} a_1 & a_2 & \cdots & a_m \end{bmatrix}$, in which case it is called a *row vector* and denoted by $\boldsymbol{v}^\top$. We may also write $\boldsymbol{v} = (a_1, a_2, \ldots, a_m)$.

(2) Define two operations on $\mathbb{R}^m$: addition,
$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{bmatrix},$$
and scalar multiplication,
$$c \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} ca_1 \\ ca_2 \\ \vdots \\ ca_m \end{bmatrix} \qquad (c \in \mathbb{R}).$$
With these operations, $\mathbb{R}^m$ is said to form a linear space.

(3) Subtraction is obtained from these two operations: for a vector $\boldsymbol{v}$, define $-\boldsymbol{v} = (-1)\boldsymbol{v}$, and put $\boldsymbol{w} - \boldsymbol{v} = \boldsymbol{w} + (-\boldsymbol{v})$.

(4) Given a family $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ of vectors in $\mathbb{R}^m$, where $n$ is a positive integer, one can form a new vector
$$\boldsymbol{v} = x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n \qquad (x_1, x_2, \ldots, x_n \in \mathbb{R}).$$
This vector $\boldsymbol{v}$ is called a *linear combination* of $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$. A family of vectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is *linearly independent* if the equation in $x_1, \ldots, x_n$
$$x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n = \mathbf{0}$$
has only the zero solution, meaning that all the $x_i$ are zero. If the family is not linearly independent, it is *linearly dependent*. In that case, at least one of its vectors can be written as a linear combination of the others: any $\boldsymbol{v}_i$ with $x_i \neq 0$ can be so expressed.

(5) For two vectors in $\mathbb{R}^m$ with $\boldsymbol{v}^\top = \begin{bmatrix} v_1 & \cdots & v_m \end{bmatrix}$ and $\boldsymbol{w}^\top = \begin{bmatrix} w_1 & \cdots & w_m \end{bmatrix}$, define the inner product $\boldsymbol{v} \cdot \boldsymbol{w} = \sum_{i=1}^m v_i w_i$. The magnitude of $\boldsymbol{v}$ is defined by $|\boldsymbol{v}| := \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}}$. The vectors are *perpendicular*, or *orthogonal*, if $\boldsymbol{v} \cdot \boldsymbol{w} = 0$, written $\boldsymbol{v} \perp \boldsymbol{w}$.

**Remark.** Row and column vectors are two representations of the components of an $m$-dimensional vector as discussed in the preceding subsection. For vectors in general, it suffices to write $\boldsymbol{v} = (a_1, a_2, \ldots, a_m)$. In the context of matrices, the two forms in the definition make it convenient to perform the same operations with vectors and matrices; both forms are called vectors. In this context, "vector $\boldsymbol{v}$" normally means the column form, while $\boldsymbol{v}^\top$ denotes the row form. The latter often saves space in inline expressions. The symbol $\top$ is the transpose symbol, with the same meaning as for matrices below. Row and column vectors can also be written with parentheses, for example $\boldsymbol{v}^\top = \begin{pmatrix} a_1 & \cdots & a_m \end{pmatrix}$; this book consistently uses square brackets.

We next define matrices.

**Definition A.2.8** (Matrix)**.** For positive integers $m,n$, an $m\times n$ *matrix* over $\mathbb{R}$ is an array of $mn$ numbers in $m$ rows and $n$ columns, of the form

$$\boldsymbol{A}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \qquad a_{ij}\in\mathbb{R}.$$

It may be abbreviated to $\boldsymbol{A}=(a_{ij})_{m\times n}$, where $a_{ij}$ is the entry in row $i$ and column $j$, or to $\boldsymbol{A}=(a_{ij})$ when the numbers of rows and columns are clear from context. The same entry may be written $(\boldsymbol{A})_{ij}$. The set of all $m\times n$ real matrices is $\mathrm{M}_{m\times n}(\mathbb{R})$.

**Remark.** Matrices too may be written with parentheses or square brackets; this book consistently uses square brackets. When parentheses are used, they should not be confused with the permutation notation in section A.1.1.

A column vector in $\mathbb{R}^m$ can therefore be regarded as an $m\times 1$ matrix, and a row vector as a $1\times m$ matrix. In terms of the linear spaces just discussed, an $m\times n$ matrix $\boldsymbol{A}$ consists of $n$ column vectors in $\mathbb{R}^m$ placed together:

$$\boldsymbol{A}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}=\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n\end{bmatrix}, \quad \boldsymbol{v}_i=\begin{bmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{mi}\end{bmatrix}. \tag{$\clubsuit$}$$

It can also be regarded as $m$ row vectors in $\mathbb{R}^n$ placed together:

$$\boldsymbol{A}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}=\begin{bmatrix}\boldsymbol{w}_1^\top \\ \boldsymbol{w}_2^\top \\ \vdots \\ \boldsymbol{w}_m^\top\end{bmatrix}, \quad \boldsymbol{w}_i^\top=\begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in}\end{bmatrix}. \tag{$\diamond$}$$

We now define several special types of matrices.

**Definition A.2.9.**

(1) The matrix whose entries are all zero is the *zero matrix* $\mathbf{0}_{m\times n}$; when its size is clear, write simply $\mathbf{0}$ (some sources instead use $\mathbf{O}$).

(2) An $n\times n$ matrix is an $n$th-order *square matrix*.

(3) The entries in positions $(i,i)$, $i=1,2,\ldots,n$, constitute its *main diagonal*. A square matrix whose only possibly nonzero entries lie on the main diagonal is an $n$th-order *diagonal matrix*, or diagonal square matrix, and is denoted

$$\operatorname{diag}(a_1,a_2,a_3,\ldots,a_n):=\begin{bmatrix} a_1 & 0 & 0 & \cdots & 0 \\ 0 & a_2 & 0 & \cdots & 0 \\ 0 & 0 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_n \end{bmatrix}.$$

(4) An $n$th-order diagonal matrix all of whose diagonal entries are 1 is the $n$th-order *identity matrix*, or identity square matrix, denoted by $\mathbf{I}_n$ and abbreviated to $\mathbf{I}$ when $n$ is clear; thus $\mathbf{I}_n=\operatorname{diag}(\underbrace{1,1,\ldots,1}_{n\text{ entries}})$.

### Basic matrix operations

We define the following basic operations on matrices.

**Definition A.2.10.**

(1) Fix positive integers $m,n$. On the matrix space $\mathrm{M}_{m\times n}(\mathbb{R})$ there are two natural operations. For $\boldsymbol{A}=(a_{ij})$ and $\boldsymbol{B}=(b_{ij})$, define *addition* by $\boldsymbol{A}+\boldsymbol{B}=(a_{ij}+b_{ij})$. For a real scalar $c$ and $\boldsymbol{A}=(a_{ij})$, define *scalar multiplication* by $c\boldsymbol{A}=(ca_{ij})$. Define $-\boldsymbol{A}=(-1)\boldsymbol{A}$ and *subtraction* by $\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}+(-\boldsymbol{B})$; thus $(\boldsymbol{A}-\boldsymbol{B})_{ij}=(\boldsymbol{A})_{ij}-(\boldsymbol{B})_{ij}$.

(2) For $\boldsymbol{A}=(a_{ij})_{m\times n}$ and $\boldsymbol{B}=(b_{ij})_{n\times l}$, define the *matrix product* $\boldsymbol{C}=\boldsymbol{A}\boldsymbol{B}=(c_{ij})_{m\times l}$ by

$$c_{ij}=\sum_{k=1}^{n}a_{ik}b_{kj}=a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{in}b_{nj}.$$

(3) The *transpose* $\boldsymbol{A}^{\mathsf{T}}$ is given by $(\boldsymbol{A}^{\mathsf{T}})_{ij}=(\boldsymbol{A})_{ji}$: the rows of $\boldsymbol{A}$ become columns, and its columns become rows.

Addition is defined only when the two matrices have the same numbers of rows and columns, while $\boldsymbol{A}\boldsymbol{B}$ is defined only when the number of columns of $\boldsymbol{A}$ equals the number of rows of $\boldsymbol{B}$. The notation $\boldsymbol{v}^{\mathsf{T}}$ for a row vector above is precisely the transpose notation just defined. Regarding the vectors as matrices, their inner product is $\boldsymbol{v}\cdot\boldsymbol{w}=\boldsymbol{v}^{\mathsf{T}}\boldsymbol{w}=\boldsymbol{w}^{\mathsf{T}}\boldsymbol{v}$.

Linear independence of vectors was defined above. Using the matrix operations just defined, a family $\{\boldsymbol{v}_1,\boldsymbol{v}_2,\dots,\boldsymbol{v}_n\}$ is linearly independent if and only if the equation $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{0}$ in the column vector $\boldsymbol{x}=(x_i)_{n\times 1}$ has only the zero solution, where $\boldsymbol{A}=\begin{bmatrix}\boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n\end{bmatrix}$.

**Example A.2.11.**

$$\begin{bmatrix}2&3\\1&2\end{bmatrix}\begin{bmatrix}3&5&7\\4&6&8\end{bmatrix}=\begin{bmatrix}18&28&38\\11&17&23\end{bmatrix},\quad \begin{bmatrix}1&2&3\\2&3&4\end{bmatrix}^{\mathsf{T}}=\begin{bmatrix}1&2\\2&3\\3&4\end{bmatrix}.$$

**Theorem A.2.12.** *Matrix multiplication has the following properties.*

*(a) Associativity: for matrices $\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}$, $(\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}=\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C})$ and $(c\boldsymbol{A})\boldsymbol{B}=c(\boldsymbol{A}\boldsymbol{B})$ for $c\in\mathbb{R}$.*

*(b) Distributivity: for matrices $\boldsymbol{A},\boldsymbol{B},\boldsymbol{C}$, $\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C})=\boldsymbol{A}\boldsymbol{B}+\boldsymbol{A}\boldsymbol{C}$ and $(\boldsymbol{A}+\boldsymbol{B})\boldsymbol{C}=\boldsymbol{A}\boldsymbol{C}+\boldsymbol{B}\boldsymbol{C}$.*

*(c) For an nth-order square matrix $\boldsymbol{A}$, $\boldsymbol{A}\mathbf{I}_n=\mathbf{I}_n\boldsymbol{A}=\boldsymbol{A}$.*

**Proof.** The distributive laws, the identity laws, and the associativity of scalar multiplication follow directly from the definitions. It remains to prove $(\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}=\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C})$. Let $\boldsymbol{A}=(a_{ij})_{m\times n}$, $\boldsymbol{B}=(b_{ij})_{n\times k}$, and $\boldsymbol{C}=(c_{ij})_{k\times l}$. Then

$$((\boldsymbol{A}\boldsymbol{B})\boldsymbol{C})_{ij}=\sum_{u=1}^{k}(\boldsymbol{A}\boldsymbol{B})_{iu}(\boldsymbol{C})_{uj}=\sum_{u=1}^{k}\left(\sum_{v=1}^{n}a_{iv}b_{vu}\right)c_{uj}=\sum_{u=1}^{k}\sum_{v=1}^{n}a_{iv}b_{vu}c_{uj}.$$

The same calculation gives $(\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C}))_{ij}=\sum_{u=1}^{k}\sum_{v=1}^{n}a_{iv}b_{vu}c_{uj}$, proving the assertion. $\square$

**Remark.** Matrix multiplication is generally not commutative: $\boldsymbol{A}\boldsymbol{B}=\boldsymbol{B}\boldsymbol{A}$ need not hold.

**Theorem A.2.13.** *Matrix transposition has the following properties.*

*(a) For every matrix $\boldsymbol{A}$, $(\boldsymbol{A}^{\mathsf{T}})^{\mathsf{T}}=\boldsymbol{A}$.*

*(b) For matrices $\boldsymbol{A},\boldsymbol{B}$, $(\boldsymbol{A}+\boldsymbol{B})^{\mathsf{T}}=\boldsymbol{A}^{\mathsf{T}}+\boldsymbol{B}^{\mathsf{T}}$.*

*(c) For matrices $\boldsymbol{A},\boldsymbol{B}$, $(\boldsymbol{A}\boldsymbol{B})^{\mathsf{T}}=\boldsymbol{B}^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}}$.*

**Proof.** The first two identities are immediate from the definition. For the third, let $\boldsymbol{A}=(a_{ij})_{m\times n}$ and $\boldsymbol{B}=(b_{ij})_{n\times k}$. Then

$$((\boldsymbol{A}\boldsymbol{B})^{\mathsf{T}})_{ij}=(\boldsymbol{A}\boldsymbol{B})_{ji}=\sum_{l=1}^{n}a_{jl}b_{li},$$

$$(\boldsymbol{B}^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}})_{ij}=\sum_{l=1}^{n}(\boldsymbol{B}^{\mathsf{T}})_{il}(\boldsymbol{A}^{\mathsf{T}})_{lj}=\sum_{l=1}^{n}b_{li}a_{jl}=((\boldsymbol{A}\boldsymbol{B})^{\mathsf{T}})_{ij}.$$

□

Transposition gives rise to two special types of matrices.

**Definition A.2.14.** A matrix $\boldsymbol{A}$ is *symmetric* if $\boldsymbol{A} = \boldsymbol{A}^{\mathsf{T}}$, and *skew-symmetric* if $\boldsymbol{A} = -\boldsymbol{A}^{\mathsf{T}}$.

**Remark.** A symmetric or skew-symmetric matrix is square, and every diagonal entry of a skew-symmetric matrix is zero.

**Theorem A.2.15.** *An $n \times n$ matrix $\boldsymbol{A}$ is symmetric if and only if $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y} = \boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}$ for all column vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$; it is skew-symmetric if and only if $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y} = -\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}$ for all column vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.*

**Proof.** Let $\boldsymbol{e}_i$ be the column vector whose $i$th entry is 1 and whose other entries are zero, that is,

$$\boldsymbol{e}_i = \begin{bmatrix} 0 & \cdots & 0 & \underset{i\text{th entry}}{1} & 0 & \cdots & 0 \end{bmatrix}^{\mathsf{T}}.$$

Then $\boldsymbol{e}_i^{\mathsf{T}}\boldsymbol{A}\boldsymbol{e}_j = (\boldsymbol{A})_{ij}$, which immediately proves the sufficiency of each stated condition. Conversely, suppose first that $\boldsymbol{A}$ is symmetric. Since $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}$ is a scalar, equivalently a $1 \times 1$ matrix,

$$\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y} = (\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y})^{\mathsf{T}} = \boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}.$$

Similarly, if $\boldsymbol{A}$ is skew-symmetric, then

$$\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y} = (\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y})^{\mathsf{T}} = -\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}.$$

□

Matrix multiplication leads to the notion of an inverse.

**Definition A.2.16.** An $n$th-order square matrix $\boldsymbol{A}$ is *invertible* if there is an $n$th-order square matrix $\boldsymbol{B}$ such that $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{B}\boldsymbol{A} = \mathbf{I}_n$. The matrix $\boldsymbol{B}$ is called the *inverse matrix*, or *inverse*, of $\boldsymbol{A}$, and is denoted by $\boldsymbol{B} = \boldsymbol{A}^{-1}$.

**Theorem A.2.17.** *The inverse of a square matrix has the following properties.*

*(a) If it exists, it is unique.*
*(b) For invertible matrices $\boldsymbol{A}, \boldsymbol{B}$, $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.*
*(c) If $\boldsymbol{A}$ is invertible, then $(\boldsymbol{A}^{-1})^{\mathsf{T}} = (\boldsymbol{A}^{\mathsf{T}})^{-1}$.*
*(d) If a square matrix $\boldsymbol{A}$ has a left inverse $\boldsymbol{L}$ satisfying $\boldsymbol{L}\boldsymbol{A} = \mathbf{I}$, then it is invertible and $\boldsymbol{A}^{-1} = \boldsymbol{L}$.*

**Proof.** If $\boldsymbol{B}$ and $\boldsymbol{C}$ are both inverses of $\boldsymbol{A}$, then $\boldsymbol{B}(\boldsymbol{A}\boldsymbol{C}) = \boldsymbol{B}\mathbf{I} = \boldsymbol{B}$, while on the other hand $\boldsymbol{B}(\boldsymbol{A}\boldsymbol{C}) = (\boldsymbol{B}\boldsymbol{A})\boldsymbol{C} = \mathbf{I}\boldsymbol{C} = \boldsymbol{C}$. Therefore $\boldsymbol{B} = \boldsymbol{C}$.

Next,

$$(\boldsymbol{A}\boldsymbol{B})(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}) = \boldsymbol{A}(\boldsymbol{B}\boldsymbol{B}^{-1})\boldsymbol{A}^{-1} = \boldsymbol{A}\mathbf{I}\boldsymbol{A}^{-1} = \boldsymbol{A}\boldsymbol{A}^{-1} = \mathbf{I},$$

and

$$(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{A}\boldsymbol{B}) = \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} = \boldsymbol{B}^{-1}\mathbf{I}\boldsymbol{B} = \boldsymbol{B}^{-1}\boldsymbol{B} = \mathbf{I}.$$

Hence $\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ is the inverse of $\boldsymbol{A}\boldsymbol{B}$.

Moreover,

$$(\boldsymbol{A}^{-1})^{\mathsf{T}}\boldsymbol{A}^{\mathsf{T}} = (\boldsymbol{A}\boldsymbol{A}^{-1})^{\mathsf{T}} = \mathbf{I}.$$

The final assertion cannot yet be proved here; it will be proved in the next subsection. □

It is sometimes convenient to represent a matrix in blocks. Draw horizontal or vertical lines all the way across a matrix $\boldsymbol{A}$, thereby dividing its entries into rectangular blocks. Write each block as a matrix $\boldsymbol{A}_{ij}$. The resulting representation of $\boldsymbol{A}$ is a *block matrix*, and each $\boldsymbol{A}_{ij}$ is a *block*, or *submatrix*, of $\boldsymbol{A}$. As with ordinary matrices, we write $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$, where each column has $r$ blocks and each row has $s$ blocks:

$$\boldsymbol{A} = \left[\begin{array}{c|cc|c|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ \hline a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array}\right] \quad \longrightarrow \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} & \cdots & \boldsymbol{A}_{1s} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} & \cdots & \boldsymbol{A}_{2s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}_{r1} & \boldsymbol{A}_{r2} & \cdots & \boldsymbol{A}_{rs} \end{bmatrix}.$$

For example, partition the following $4\times 5$ matrix after rows 1,3 and columns 1,3:

$$\boldsymbol{A} = \left[\begin{array}{c|cc|cc} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ \hline a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ \hline a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \end{array}\right] \quad \longrightarrow \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} & \boldsymbol{A}_{13} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} & \boldsymbol{A}_{23} \\ \boldsymbol{A}_{31} & \boldsymbol{A}_{32} & \boldsymbol{A}_{33} \end{bmatrix},$$

where

$$\boldsymbol{A}_{11} = \begin{bmatrix} a_{11} \end{bmatrix}, \quad \boldsymbol{A}_{12} = \begin{bmatrix} a_{12} & a_{13} \end{bmatrix}, \quad \boldsymbol{A}_{13} = \begin{bmatrix} a_{14} & a_{15} \end{bmatrix},$$
$$\boldsymbol{A}_{21} = \begin{bmatrix} a_{21} \\ a_{31} \end{bmatrix}, \quad \boldsymbol{A}_{22} = \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}, \quad \boldsymbol{A}_{23} = \begin{bmatrix} a_{24} & a_{25} \\ a_{34} & a_{35} \end{bmatrix},$$
$$\boldsymbol{A}_{31} = \begin{bmatrix} a_{41} \end{bmatrix}, \quad \boldsymbol{A}_{32} = \begin{bmatrix} a_{42} & a_{43} \end{bmatrix}, \quad \boldsymbol{A}_{33} = \begin{bmatrix} a_{44} & a_{45} \end{bmatrix}.$$

A $1\times 1$ matrix and a scalar are distinct objects, but in a block representation a $1\times 1$ block may be written as its sole entry. For example, the same matrix can be displayed as

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & \boldsymbol{A}_{12} & \boldsymbol{A}_{13} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} & \boldsymbol{A}_{23} \\ a_{41} & \boldsymbol{A}_{32} & \boldsymbol{A}_{33} \end{bmatrix}.$$

The earlier displays (♣) and ($\diamond$) are also examples of block matrices. Their operational laws are entirely analogous to those of ordinary matrices, but the partitions must match appropriately.

**Theorem A.2.18.** *Block matrices satisfy the following rules.*

*(a) If $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and $\boldsymbol{B} = (\boldsymbol{B}_{ij})_{r\times s}$ have the same block partition, meaning that $\boldsymbol{A}_{ij}$ and $\boldsymbol{B}_{ij}$ have the same numbers of rows and columns for every $i,j$, then $\boldsymbol{A}+\boldsymbol{B} = (\boldsymbol{A}_{ij}+\boldsymbol{B}_{ij})_{r\times s}$.*

*(b) For a block matrix $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and a scalar $c$, one has $c\boldsymbol{A} = (c\boldsymbol{A}_{ij})_{r\times s}$.*

*(c) For $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$, one has $\boldsymbol{A}^\top = (\boldsymbol{A}_{ji}^\top)_{s\times r}$.*

*(d) If $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and $\boldsymbol{B} = (\boldsymbol{B}_{ij})_{s\times t}$ have compatible block partitions, so that $\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}$ is defined for every $1\le i\le r$, $1\le j\le t$, $1\le k\le s$, then*

$$\boldsymbol{A}\boldsymbol{B} = (\boldsymbol{C}_{ij})_{r\times t}, \qquad \boldsymbol{C}_{ij} = \sum_{k=1}^{s} \boldsymbol{A}_{ik}\boldsymbol{B}_{kj}.$$

These assertions follow directly from the definitions and are left for the reader to verify. Elementary row and column operations are also an important and basic part of linear algebra, but are omitted because they are not needed in this course. Matrix rank is important as well; we shall study it by means of linear maps in section A.5.3.

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## Exercises

**Exercise A.17.** Prove

(1) $(\boldsymbol{A}_1\boldsymbol{A}_2\cdots\boldsymbol{A}_n)^{-1} = \boldsymbol{A}_n^{-1}\boldsymbol{A}_{n-1}^{-1}\cdots\boldsymbol{A}_1^{-1}$;

(2) $(\boldsymbol{A}_1\boldsymbol{A}_2\cdots\boldsymbol{A}_n)^\top = \boldsymbol{A}_n^\top\boldsymbol{A}_{n-1}^\top\cdots\boldsymbol{A}_1^\top$.

**Exercise A.18.**

(1) Use the definition of matrix multiplication to prove parts (b) and (c) of theorem A.2.12.
(2) Use the definition of transposition to prove parts (a) and (b) of theorem A.2.13.
(3) Use the definition of a block matrix to prove theorem A.2.18.

**Exercise A.19.** For

$$\boldsymbol{A}=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix},\quad \boldsymbol{B}=\begin{bmatrix}0&0&-1\\0&2&0\\1&0&0\end{bmatrix},\quad \boldsymbol{C}=\begin{bmatrix}1&2&0&-1\\2&5&-2&0\\4&3&-2&1\end{bmatrix},$$

compute (1) $\boldsymbol{A}^{\mathsf{T}}+\boldsymbol{B}$; (2) $\boldsymbol{A}\boldsymbol{B}^{\mathsf{T}}\boldsymbol{C}$; (3) $(\boldsymbol{A}-\boldsymbol{B}+2\mathbf{I}_3)^{\mathsf{T}}\boldsymbol{C}$; (4) $\boldsymbol{B}^{-1}$; and (5) $\begin{bmatrix}\boldsymbol{A}&\mathbf{I}_3\\\mathbf{0}_{3\times3}&\boldsymbol{B}\end{bmatrix}\begin{bmatrix}\boldsymbol{C}\\\boldsymbol{C}\end{bmatrix}$.

**Exercise A.20.** Prove the following assertions.

(1) The square matrix $\boldsymbol{A}=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ of order two is invertible if and only if $ad\neq bc$, and when the inverse exists,
$$\boldsymbol{A}^{-1}=\frac{1}{ad-bc}\begin{bmatrix}d&-b\\-c&a\end{bmatrix}.$$
(2) A diagonal matrix is invertible if and only if all its diagonal entries are nonzero; when its inverse exists, $\operatorname{diag}(d_1,d_2,\dots,d_n)^{-1}=\operatorname{diag}(d_1^{-1},d_2^{-1},\dots,d_n^{-1})$.

**Exercise A.21.** Let $\boldsymbol{A}$ be a square matrix. Suppose that, for all column vectors $\boldsymbol{x},\boldsymbol{y}$, $\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}=0$ always implies $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}=0$. Prove that $\boldsymbol{A}$ is symmetric or skew-symmetric.

**Exercise A.22.** If $\boldsymbol{A}$ is invertible but $\boldsymbol{B}$ is not, can $\boldsymbol{A}\boldsymbol{B}$ be invertible?

**Exercise A.23** (Gram–Schmidt orthogonalization)**.**
Given a linearly independent family $\{\boldsymbol{a}_1,\boldsymbol{a}_2,\boldsymbol{a}_3,\boldsymbol{a}_4,\dots\}$, prove that the following construction produces an orthonormal, linearly independent family.[5]

(1) Choose a vector $\boldsymbol{a}_1$ and put $\boldsymbol{a}_1'=\boldsymbol{a}_1$.
(2) Put $\boldsymbol{a}_2'=\boldsymbol{a}_2-\frac{\boldsymbol{a}_1\cdot\boldsymbol{a}_2}{\boldsymbol{a}_1'\cdot\boldsymbol{a}_1'}\boldsymbol{a}_1'$, and prove that $\boldsymbol{a}_2'\perp\boldsymbol{a}_1'$.
(3) Put $\boldsymbol{a}_3'=\boldsymbol{a}_3-\frac{\boldsymbol{a}_1'\cdot\boldsymbol{a}_3}{\boldsymbol{a}_1'\cdot\boldsymbol{a}_1'}\boldsymbol{a}_1'-\frac{\boldsymbol{a}_2'\cdot\boldsymbol{a}_3}{\boldsymbol{a}_2'\cdot\boldsymbol{a}_2'}\boldsymbol{a}_2'$, and prove that $\boldsymbol{a}_3'\perp\boldsymbol{a}_1',\boldsymbol{a}_2'$.
(4) Put $\boldsymbol{a}_4'=\boldsymbol{a}_4-\frac{\boldsymbol{a}_1'\cdot\boldsymbol{a}_4}{\boldsymbol{a}_1'\cdot\boldsymbol{a}_1'}\boldsymbol{a}_1'-\frac{\boldsymbol{a}_2'\cdot\boldsymbol{a}_4}{\boldsymbol{a}_2'\cdot\boldsymbol{a}_2'}\boldsymbol{a}_2'-\frac{\boldsymbol{a}_3'\cdot\boldsymbol{a}_4}{\boldsymbol{a}_3'\cdot\boldsymbol{a}_3'}\boldsymbol{a}_3'$, and prove that $\boldsymbol{a}_4'\perp\boldsymbol{a}_1',\boldsymbol{a}_2',\boldsymbol{a}_3'$.
(5) Continue in the same way to obtain $\{\boldsymbol{a}_1',\boldsymbol{a}_2',\boldsymbol{a}_3',\boldsymbol{a}_4',\dots\}$. Prove that the normalized vectors $\left\{\frac{\boldsymbol{a}_1'}{|\boldsymbol{a}_1'|},\frac{\boldsymbol{a}_2'}{|\boldsymbol{a}_2'|},\frac{\boldsymbol{a}_3'}{|\boldsymbol{a}_3'|},\frac{\boldsymbol{a}_4'}{|\boldsymbol{a}_4'|},\dots\right\}$ are the required vectors.

## A.2.3 Determinants and Their Applications

### Definition of the determinant

**Definition A.2.19.** Let $\sigma$ be a permutation of $M=\{1,2,\dots,n\}$, where $n$ is a positive integer; that is, $\sigma\in\mathrm{S}_n$, in the sense of section A.1.1. It is *even* or *odd* according as it is a product of an even or odd number of interchanges of two elements (called transpositions), and its *sign* is $\operatorname{sgn}\sigma=1$ or $-1$,

[5]A family is orthonormal if every vector has magnitude 1 and every two distinct vectors are orthogonal.

respectively. Define the *determinant* operation by

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} := \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\, a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\Big(\prod_{i=1}^{n} a_{i,\sigma(i)}\Big).$$

For the corresponding square matrix $\boldsymbol{A}=(a_{ij})_{n\times n}$, this determinant may also be denoted by $\det\boldsymbol{A}$ or $|\boldsymbol{A}|$.

**Remark.** There are several equivalent definitions of a determinant. The one above is the *Leibniz definition of the determinant*. It is generally not the best introduction for a beginner, but is convenient for learning the subject rapidly in this course. The other definitions are not needed here and will not be presented.

**Example A.2.20.** There are two permutations in $S_2$: $\sigma_1=\begin{pmatrix}1&2\\1&2\end{pmatrix}$ and $\sigma_2=\begin{pmatrix}1&2\\2&1\end{pmatrix}$. The first is the identity map, a composition of zero transpositions, so $\operatorname{sgn}\sigma_1=1$. The second interchanges 1 and 2 and is odd. Thus the determinant of order two is

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \operatorname{sgn}(\sigma_1)\, a_{11}a_{22} + \operatorname{sgn}(\sigma_2)\, a_{12}a_{21} = a_{11}a_{22} - a_{12}a_{21}.$$

This agrees with the order-two definition in definition A.2.5.

**Example A.2.21.** There are six permutations in $S_3$:

$$\sigma_1=\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix},\quad \sigma_2=\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix},\quad \sigma_3=\begin{pmatrix}1&2&3\\3&2&1\end{pmatrix},$$
$$\sigma_4=\begin{pmatrix}1&2&3\\2&1&3\end{pmatrix},\quad \sigma_5=\begin{pmatrix}1&2&3\\2&3&1\end{pmatrix},\quad \sigma_6=\begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}.$$

Here $\sigma_1$ is the identity, obtained with no transpositions, so $\operatorname{sgn}\sigma_1=1$. The permutation $\sigma_2$ interchanges 2 and 3, so it is odd and $\operatorname{sgn}\sigma_2=-1$; similarly, $\operatorname{sgn}\sigma_3=\operatorname{sgn}\sigma_4=-1$. The permutation $\sigma_5$ first interchanges 1 and 3 and then interchanges 2 and 3, so it is even and $\operatorname{sgn}\sigma_5=1$; similarly, $\operatorname{sgn}\sigma_6=1$. Consequently,

$$\begin{aligned} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} &= \sum_{i=1}^{6} \operatorname{sgn}(\sigma_i) a_{1,\sigma_i(1)} a_{2,\sigma_i(2)} a_{3,\sigma_i(3)} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31} \\ &\quad - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}, \end{aligned}$$

in agreement with definition A.2.5.

**Theorem A.2.22.** *For every square matrix* $\boldsymbol{A}$, $\det(\boldsymbol{A})=\det(\boldsymbol{A}^{\mathsf{T}})$.

**Proof.** For $\boldsymbol{A}=(a_{ij})_{n\times n}$,

$$\begin{aligned} \det(\boldsymbol{A}^{\mathsf{T}}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} (\boldsymbol{A}^{\mathsf{T}})_{i,\sigma(i)} = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{\sigma(i),i} \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma^{-1}(i)} = \sum_{\tau\in S_n} \operatorname{sgn}(\tau) \prod_{i=1}^{n} a_{i,\tau(i)} = \det\boldsymbol{A}, \end{aligned}$$

where the first equality on the second line relabels the product index by $i\mapsto\sigma^{-1}(i)$; the next puts $\tau=\sigma^{-1}$ and uses $\operatorname{sgn}(\sigma^{-1})=\operatorname{sgn}(\sigma)$. □

**Theorem A.2.23.** *For square matrices $\boldsymbol{A}$,$\boldsymbol{B}$ of the same order,* $\det(\boldsymbol{A})\det(\boldsymbol{B}) = \det(\boldsymbol{AB})$.

**Proof.** Let the matrices have order $n$. Write $\boldsymbol{A} = (a_{ij})$ and $\boldsymbol{B} = (b_{ij})$, and put $\boldsymbol{C} = \boldsymbol{AB} = (c_{ij})$. We use $\sum_{\{k_i\}}$ to abbreviate the sum

$$\sum_{k_1=1}^{n}\sum_{k_2=1}^{n}\cdots\sum_{k_n=1}^{n} = \sum_{\substack{k_1,k_2,\ldots,k_n\\ \in\{1,2,\ldots,n\}}}.$$

If the set $I_i$ of possible values of each $k_i$ is finite, then sums and products may be interchanged according to the identity

$$\prod_{i=1}^{n}\Big[\sum_{k_i\in I_i} g_i(k_i)\Big] = \sum_{k_1\in I_1}\sum_{k_2\in I_2}\cdots\sum_{k_n\in I_n}\Big[\prod_{i=1}^{n} g_i(k_i)\Big].$$

This generalization of the distributive law is readily proved by induction. By the definition of the determinant,

$$\begin{aligned}
\det(\boldsymbol{AB}) &= \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n} c_{i,\sigma(i)} = \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n}\Big[\sum_{k_i=1}^{n} a_{ik_i}b_{k_i,\sigma(i)}\Big]\\
&= \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\sum_{\{k_i\}}\prod_{i=1}^{n} a_{ik_i}b_{k_i,\sigma(i)} = \sum_{\{k_i\}}\Big[\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n} b_{k_i,\sigma(i)}a_{ik_i}\Big]\\
&= \sum_{\{k_i\}}\Big[\Big(\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n} b_{k_i,\sigma(i)}\Big)\prod_{i=1}^{n} a_{ik_i}\Big].
\end{aligned}$$

The second line expands by the distributive identity, and the third interchanges the order of summation. If two values among the $k_i$ are equal, say $k_p = k_q$, the terms in the inner sum cancel in pairs. For each $\sigma\in S_n$, let $\sigma'\in S_n$ be its composition with the transposition of $p,q$. This adds one transposition, so $\operatorname{sgn}(\sigma') = -\operatorname{sgn}(\sigma)$, while $k_p = k_q$ gives

$$\prod_{i=1}^{n} b_{k_i,\sigma(i)} = \prod_{i=1}^{n} b_{k_i,\sigma'(i)}.$$

Thus the paired terms sum to zero. Only tuples with pairwise distinct $k_i$ contribute, and these are precisely $k_i = \tau(i)$ for $\tau\in S_n$. Hence

$$\begin{aligned}
\det(\boldsymbol{AB}) &= \sum_{\tau\in S_n}\Big[\Big(\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n} b_{\tau(i),\sigma(i)}\Big)\prod_{i=1}^{n} a_{i,\tau(i)}\Big]\\
&= \sum_{\tau\in S_n}\Big[\Big(\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n} b_{i,\sigma(\tau^{-1}(i))}\Big)\prod_{i=1}^{n} a_{i,\tau(i)}\Big]\\
&= \sum_{\tau\in S_n}\Big[\Big(\sum_{\rho\in S_n}\operatorname{sgn}(\rho)\operatorname{sgn}(\tau)\prod_{i=1}^{n} b_{i,\rho(i)}\Big)\prod_{i=1}^{n} a_{i,\tau(i)}\Big]\\
&= \sum_{\tau\in S_n}\Big[\Big(\sum_{\rho\in S_n}\operatorname{sgn}(\rho)\prod_{i=1}^{n} b_{i,\rho(i)}\Big)\operatorname{sgn}(\tau)\prod_{i=1}^{n} a_{i,\tau(i)}\Big]\\
&= \Big(\sum_{\rho\in S_n}\operatorname{sgn}(\rho)\prod_{i=1}^{n} b_{i,\rho(i)}\Big)\Big(\sum_{\tau\in S_n}\operatorname{sgn}(\tau)\prod_{i=1}^{n} a_{i,\tau(i)}\Big) = \det(\boldsymbol{A})\det(\boldsymbol{B}).
\end{aligned}$$

In the third line, put $\rho = \sigma\circ\tau^{-1}$, so $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\rho\circ\tau) = \operatorname{sgn}(\rho)\operatorname{sgn}(\tau)$. □

### The adjugate and the inverse

For an $n$th-order square matrix $\boldsymbol{A} = (a_{ij})_{n\times n}$, delete all entries in row $i$ and column $j$. The resulting $(n-1)$th-order square matrix $\boldsymbol{A}_{ij}$ is the *minor matrix* of $a_{ij}$. In the first array below, the row and column to be deleted are shown in red and blue, respectively:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & \cdots & a_{1,j-1} & a_{1j} & a_{1,j+1} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,j-1} & a_{i-1,j} & a_{i-1,j+1} & \cdots & a_{i-1,n} \\ a_{i1} & \cdots & a_{i,j-1} & a_{ij} & a_{i,j+1} & \cdots & a_{in} \\ a_{i+1,1} & \cdots & a_{i+1,j-1} & a_{i+1,j} & a_{i+1,j+1} & \cdots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & a_{nj} & a_{n,j+1} & \cdots & a_{nn} \end{bmatrix}.$$

Deleting this row and column gives

$$\boldsymbol{A}_{ij} = \begin{bmatrix} a_{11} & \cdots & a_{1,j-1} & a_{1,j+1} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,j-1} & a_{i-1,j+1} & \cdots & a_{i-1,n} \\ a_{i+1,1} & \cdots & a_{i+1,j-1} & a_{i+1,j+1} & \cdots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & a_{n,j+1} & \cdots & a_{nn} \end{bmatrix}.$$

The scalar $M_{ij} = (-1)^{i+j}\det(\boldsymbol{A}_{ij})$ is the *cofactor* of $a_{ij}$. The matrix $\boldsymbol{M} = (M_{ij})_{n\times n}$ is the *cofactor matrix* of $\boldsymbol{A}$; the *adjugate matrix* of $\boldsymbol{A}$ is $\boldsymbol{A}^* = \boldsymbol{M}^\top$.

Deleting the only entry of a matrix of order one leaves an empty matrix. We define the determinant of the empty matrix to be 1. Thus the cofactor of the only entry of a matrix of order one is 1, and its adjugate is the identity matrix $\mathbf{I}_1$ of order one.

The adjugate allows us to expand a determinant along a row or a column.

**Theorem A.2.24** (Laplace expansion). *Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ be a square matrix, and let $M_{ij}$ be the cofactor of $a_{ij}$. For every row $i$ and column $j$,*

$$\det(\boldsymbol{A}) = \sum_{k=1}^{n} a_{ik}M_{ik} = \sum_{k=1}^{n} a_{kj}M_{kj}.$$

*The first sum is the expansion along a row, and the second is the expansion along a column.*

**Proof.** We prove only the row expansion; the column expansion is analogous. Direct calculation gives

$$\det\boldsymbol{A} = \sum_{\sigma\in\mathrm{S}_n} \operatorname{sgn}(\sigma)\prod_{k=1}^{n} a_{k,\sigma(k)} = \sum_{j=1}^{n}\sum_{\substack{\sigma\in\mathrm{S}_n\\ \sigma(i)=j}} \operatorname{sgn}(\sigma)\prod_{k=1}^{n} a_{k,\sigma(k)} = \sum_{j=1}^{n} a_{ij}\sum_{\substack{\sigma\in\mathrm{S}_n\\ \sigma(i)=j}} \operatorname{sgn}(\sigma)\prod_{k\neq i} a_{k,\sigma(k)}.$$

To continue, we use the following fact about permutations. Each $\sigma\in\mathrm{S}_n$ satisfying $\sigma(i)=j$ restricts to a bijection

$$\{1,2,\ldots,n\}\setminus\{i\}\longrightarrow\{1,2,\ldots,n\}\setminus\{j\}.$$

Deleting $i$ and $j$ from the domain and range, respectively, and relabeling in order identifies it with a permutation $\tau\in\mathrm{S}_{n-1}$, for which $\operatorname{sgn}(\sigma)=(-1)^{i+j}\operatorname{sgn}(\tau)$. To verify the sign relation, start with $\tau\in\mathrm{S}_{n-1}$, adjoin an $n$th element, and move it into positions $i,j$ in the ordered domain and range, respectively. Specifically, set

$$\tau'(i) = \begin{cases} \tau(i), & i<n, \\ n, & i=n, \end{cases} \qquad \tau'\in\mathrm{S}_n,$$

and let

$$p_i = \begin{pmatrix} 1 & \cdots & i-1 & i & i+1 & i+2 & \cdots & n \\ 1 & \cdots & i-1 & n & i & i+1 & \cdots & n-1 \end{pmatrix}.$$

Then $\sigma = p_j^{-1}\circ\tau'\circ p_i$. Moving $n$ from the end into position $i$ by successively interchanging it with the preceding element uses $n-i$ transpositions, so $\operatorname{sgn}(p_i)=(-1)^{n-i}$, while plainly $\operatorname{sgn}(\tau')=\operatorname{sgn}(\tau)$.

Therefore

$$\operatorname{sgn}(\sigma) = (-1)^{n-j}\operatorname{sgn}(\tau)(-1)^{n-i} = (-1)^{i+j}\operatorname{sgn}(\tau).$$

Using this correspondence in the determinant calculation, we obtain

$$\det \boldsymbol{A} = \sum_{j=1}^{n} a_{ij} \sum_{\tau\in \mathrm{S}_{n-1}} (-1)^{i+j}\operatorname{sgn}(\tau)\prod_{k=1}^{n-1}(\boldsymbol{A}_{ij})_{k,\tau(k)} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(\boldsymbol{A}_{ij}) = \sum_{j=1}^{n} a_{ij}M_{ij}. \qquad \square$$

**Theorem A.2.25.** *For every square matrix $\boldsymbol{A}$, $\boldsymbol{AA}^* = \boldsymbol{A}^*\boldsymbol{A} = \det(\boldsymbol{A})\mathbf{I}$.*

**Proof.** Using the notation in the definition of the adjugate matrix,

$$(\boldsymbol{AA}^*)_{ij} = \sum_{k=1}^{n} a_{ik}(\boldsymbol{A}^*)_{kj}.$$

When $i = j$, this is the row-wise Laplace expansion of the determinant, so $(\boldsymbol{AA}^*)_{ii} = \det(\boldsymbol{A})$. When $i \neq j$, let $\boldsymbol{T}_{ij}$ be the matrix obtained from $\boldsymbol{A}$ by replacing row $j$ by row $i$. The row-wise Laplace expansion gives $(\boldsymbol{AA}^*)_{ij} = \det(\boldsymbol{T}_{ij})$. Rows $i$ and $j$ of $\boldsymbol{T}_{ij}$ are equal. For each $\sigma \in \mathrm{S}_n$, interchange $i$ and $j$ to obtain $\sigma'$. Then $\operatorname{sgn}(\sigma') = -\operatorname{sgn}(\sigma)$, while the corresponding products of entries are equal, so the paired contributions in the Leibniz sum cancel. Thus $\det(\boldsymbol{T}_{ij}) = 0$ and $(\boldsymbol{AA}^*)_{ij} = 0$. The column-wise Laplace expansion proves in the same way that $\boldsymbol{A}^*\boldsymbol{A} = \det(\boldsymbol{A})\mathbf{I}$. $\square$

We can now calculate the inverse matrix as follows.

**Corollary A.2.26.** *A square matrix $\boldsymbol{A}$ is invertible if and only if $\det(\boldsymbol{A}) \neq 0$; when its inverse exists, $\boldsymbol{A}^{-1} = \frac{\boldsymbol{A}^*}{\det(\boldsymbol{A})}$.*

**Proof.** Suppose that $\boldsymbol{A}$ has an inverse $\boldsymbol{B}$. Then, by theorem A.2.23,

$$1 = \det(\mathbf{I}) = \det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B}).$$

If $\det(\boldsymbol{A}) = 0$, this equality is impossible. Thus invertibility of $\boldsymbol{A}$ implies $\det(\boldsymbol{A}) \neq 0$. Conversely, if $\det(\boldsymbol{A}) \neq 0$, then by theorem A.2.25, $\boldsymbol{A}^{-1} = \frac{\boldsymbol{A}^*}{\det(\boldsymbol{A})}$ satisfies the definition of the inverse matrix. $\square$

Part (d) of theorem A.2.17, whose proof was deferred, can now be proved.

**Revisit** (theorem A.2.17(d))**.** *If a square matrix $\boldsymbol{A}$ has a left inverse $\boldsymbol{L}$ satisfying $\boldsymbol{LA} = \mathbf{I}$, then $\boldsymbol{A}$ is invertible and $\boldsymbol{L} = \boldsymbol{A}^{-1}$.*

**Proof.** Taking determinants in $\boldsymbol{LA} = \mathbf{I}$ gives $\det(\boldsymbol{L})\det(\boldsymbol{A}) = 1$, so $\det(\boldsymbol{A}) \neq 0$ and $\boldsymbol{A}$ is invertible: there exists $\boldsymbol{A}^{-1}$ with $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{AA}^{-1} = \mathbf{I}$. Associativity then gives

$$\boldsymbol{AL} = \boldsymbol{ALAA}^{-1} = \boldsymbol{A}\mathbf{I}\boldsymbol{A}^{-1} = \boldsymbol{AA}^{-1} = \mathbf{I}.$$

Thus $\boldsymbol{L}$ is also a right inverse, and hence $\boldsymbol{L} = \boldsymbol{A}^{-1}$. $\square$

**Theorem A.2.27.** *For a square matrix $\boldsymbol{A}$, the homogeneous matrix equation $\boldsymbol{Ax} = \mathbf{0}$ in the column vector $\boldsymbol{x}$ has a nonzero solution if and only if $\det(\boldsymbol{A}) = 0$.*

**Proof.** If $\det(\boldsymbol{A}) \neq 0$, then $\boldsymbol{A}^{-1}$ exists, and left multiplication of $\boldsymbol{Ax} = \mathbf{0}$ by $\boldsymbol{A}^{-1}$ gives $\mathbf{0} = \boldsymbol{A}^{-1}\boldsymbol{Ax} = \boldsymbol{x}$.

Conversely, suppose $\det(\boldsymbol{A}) = 0$. Write $\boldsymbol{A} = (a_{ij})$, $\boldsymbol{x}^\top = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$, and let $\boldsymbol{A}_{ij}$ denote the minor matrix of $a_{ij}$. We prove by induction on the order of $\boldsymbol{A}$ that $\boldsymbol{Ax} = \mathbf{0}$ has a nonzero solution.

For order one, $\det \boldsymbol{A} = 0$ means $\boldsymbol{A} = \mathbf{0}$, so any column of $\mathbb{R}^1$ is a solution. For order two, if $\boldsymbol{A}^* \neq \mathbf{0}$, any nonzero column of $\boldsymbol{A}^*$ is a solution because $\boldsymbol{AA}^* = \det(\boldsymbol{A})\mathbf{I} = \mathbf{0}$ by theorem A.2.25; if $\boldsymbol{A}^* = \mathbf{0}$, the definition of the adjugate gives $\boldsymbol{A} = \mathbf{0}$, and any column of $\mathbb{R}^2$ is a solution.

Now assume the result for matrices of order $n-1$ and let $\boldsymbol{A}$ have order $n$. As above, a nonzero column of $\boldsymbol{A}^*$ is a solution when $\boldsymbol{A}^* \neq \boldsymbol{0}$. Suppose therefore that $\boldsymbol{A}^* = \boldsymbol{0}$. Put

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{x}' \\ x_n \end{bmatrix}, \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{v}_1^\top & a_{1n} \\ \vdots & \vdots \\ \boldsymbol{v}_n^\top & a_{nn} \end{bmatrix},$$

where $\boldsymbol{x}'$ consists of the first $n-1$ entries of $\boldsymbol{x}$, and each $\boldsymbol{v}_i^\top$ is a row in $\mathbb{R}^{n-1}$. On setting $x_n = 0$, the required equations are $\boldsymbol{v}_i^\top \boldsymbol{x}' = 0$ for all $i$.

Since $\boldsymbol{A}^* = \boldsymbol{0}$, we have $\det(\boldsymbol{A}_{1n}) = 0$. The induction hypothesis applied to $\boldsymbol{A}_{1n}^\top$ gives a nonzero column $\boldsymbol{y} = \begin{bmatrix} y_2 & \cdots & y_n \end{bmatrix}^\top$ satisfying

$$\boldsymbol{0} = \boldsymbol{A}_{1n}^\top \boldsymbol{y} = \begin{bmatrix} \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \end{bmatrix} \boldsymbol{y} = y_2 \boldsymbol{v}_2 + \cdots + y_n \boldsymbol{v}_n. \tag{$\triangle$}$$

At least one $y_i$ is nonzero. After relabeling the last $n-1$ rows if necessary, take $y_n \neq 0$. Then

$$\boldsymbol{v}_n^\top \boldsymbol{x}' = -\frac{1}{y_n}(y_2\boldsymbol{v}_2 + \cdots + y_{n-1}\boldsymbol{v}_{n-1})^\top \boldsymbol{x}' \quad = -\frac{1}{y_n}(y_2\boldsymbol{v}_2^\top \boldsymbol{x}' + \cdots + y_{n-1}\boldsymbol{v}_{n-1}^\top \boldsymbol{x}').$$

Thus $\boldsymbol{v}_n^\top \boldsymbol{x}' = 0$ follows whenever $\boldsymbol{v}_i^\top \boldsymbol{x}' = 0$ for $2 \le i \le n-1$. It is therefore enough to solve the first $n-1$ equations, namely $\boldsymbol{A}_{nn}\boldsymbol{x}' = \boldsymbol{0}$. Again $\det(\boldsymbol{A}_{nn}) = 0$ because $\boldsymbol{A}^* = \boldsymbol{0}$, so the induction hypothesis supplies a nonzero $\boldsymbol{x}'$. Then $\boldsymbol{x} = \begin{bmatrix} \boldsymbol{x}' \\ 0 \end{bmatrix}$ is the required nonzero solution. □

There is a more direct proof using the general theory of linear systems, but that would require further notions such as rank. The induction above avoids introducing this additional material.

By theorem A.2.27 we obtain the following criterion immediately.

**Corollary A.2.28.** *The column vectors* $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ *in* $\mathbb{R}^n$ *are linearly independent if and only if* $\det[\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_n] \neq 0$.

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## Exercises

**Exercise A.24.** In theorem A.2.25 only $\boldsymbol{A}\boldsymbol{A}^* = \det(\boldsymbol{A})\mathbf{I}$ was verified in detail. Complete the proof by verifying $\boldsymbol{A}^*\boldsymbol{A} = \det(\boldsymbol{A})\mathbf{I}$ directly.

**Exercise A.25.** Prove the following assertions.

(1) If two rows or two columns of a square matrix $\boldsymbol{A}$ are identical, then $\det(\boldsymbol{A}) = 0$.
(2) A determinant is linear in each row separately; explicitly, for every $j$,

$$\begin{vmatrix} \boldsymbol{v}_1^\top \\ \vdots \\ \boldsymbol{v}_{j-1}^\top \\ a\boldsymbol{v}_j^\top + b(\boldsymbol{v}_j')^\top \\ \boldsymbol{v}_{j+1}^\top \\ \vdots \\ \boldsymbol{v}_n^\top \end{vmatrix} = a \begin{vmatrix} \boldsymbol{v}_1^\top \\ \vdots \\ \boldsymbol{v}_{j-1}^\top \\ \boldsymbol{v}_j^\top \\ \boldsymbol{v}_{j+1}^\top \\ \vdots \\ \boldsymbol{v}_n^\top \end{vmatrix} + b \begin{vmatrix} \boldsymbol{v}_1^\top \\ \vdots \\ \boldsymbol{v}_{j-1}^\top \\ (\boldsymbol{v}_j')^\top \\ \boldsymbol{v}_{j+1}^\top \\ \vdots \\ \boldsymbol{v}_n^\top \end{vmatrix}.$$

Here the changed row in each array is row $j$.

**Exercise A.26.** Find the determinants of

$$(1)\ \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}; \qquad (2)\ \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -2 & 2 \end{bmatrix}; \qquad (3)\ \begin{bmatrix} 6 & 0 & 3 & 1 \\ 0 & -2 & 1 & 3 \\ 3 & 1 & 4 & 0 \\ 2 & 3 & 0 & -4 \end{bmatrix}.$$

**Exercise A.27** (Cramer's rule)**.** Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ be a square matrix, and let $\boldsymbol{x} = (x_i)_{n\times 1}$ and $\boldsymbol{b} = (b_i)_{n\times 1}$ be column vectors. If the matrix equation $\boldsymbol{Ax} = \boldsymbol{b}$ in $\boldsymbol{x}$ has a unique solution, prove that it is given by $x_i = \frac{\det(\boldsymbol{B}_i)}{\det \boldsymbol{A}}$, where $\boldsymbol{B}_i$ is obtained from $\boldsymbol{A}$ by replacing column $i$ with $\boldsymbol{b}$.

**Exercise A.28.** Prove that a skew-symmetric matrix of odd order is not invertible.

## A.3 Basic Group Theory

### A.3.1 Basic Concepts of Groups

#### Definition of a Group

**Definition A.3.1.** A set $G$ with an algebraic operation $\circ$ is a *group*, denoted $(G,\circ)$ or simply $G$ when the operation is clear, if $\circ$ satisfies:
(1) associativity: $(a\circ b)\circ c = a\circ(b\circ c)$ for all $a,b,c\in G$;
(2) there is an *identity element* $e$ such that $a\circ e = e\circ a = a$;
(3) every $a\in G$ has an *inverse* $b$ such that $a\circ b = b\circ a = e$; the inverse of $a$ is denoted by $a^{-1}$.

**Remark.** It is enough to assume a left identity and left inverses, namely $e\circ a = a$ and $a^{-1}\circ a = e$; these imply $a\circ e = a$ and $a\circ a^{-1} = e$. Let $e$ be a left identity, let $a\in G$, and choose a left inverse $a^{-1}$ of $a$ and a left inverse $a'$ of $a^{-1}$. Thus $e\circ a = a$, $a^{-1}\circ a = e$ and $a'\circ a^{-1} = e$. By associativity,

$$\begin{aligned} a\circ e &= e\circ(a\circ e) = a'\circ a^{-1}\circ(a\circ e) = a'\circ(a^{-1}\circ a)\circ e \\ &= a'\circ e\circ e = a'\circ e = a'\circ(a^{-1}\circ a) = (a'\circ a^{-1})\circ a = e\circ a = a. \end{aligned}$$

Hence $e$ is also a right identity, with $a\circ e = a$. Moreover,

$$a\circ a^{-1} = e\circ(a\circ a^{-1}) = a'\circ a^{-1}\circ(a\circ a^{-1}) = a'\circ(a^{-1}\circ a)\circ a^{-1} = a'\circ e\circ a^{-1} = a'\circ a^{-1} = e.$$

Thus $a^{-1}$ is also a right inverse of $a$, with $a\circ a^{-1} = e$. □

The identity is unique: if $e,e'$ are identities, then $e\circ e' = e'$ by the left-identity law, while $e\circ e' = e$ by the right-identity law, so $e = e'$. Inverses are unique as well. If both $b$ and $a^{-1}$ invert $a$, then

$$b = b\circ e = b\circ(a\circ a^{-1}) = (b\circ a)\circ a^{-1} = e\circ a^{-1} = a^{-1}.$$

Thus $b = a^{-1}$, so the inverse of $a$ is unique. □

When no ambiguity arises, omit the symbol $\circ$, write $a\circ b = ab$, and call the operation *multiplication*. This is a name for the operation; it does not mean that the operation is ordinary multiplication of numbers. As with numbers, we use powers for repeated products of the same element:

$$a^n := \underbrace{aa\cdots a}_{n \text{ copies of } a}, \quad a^0 := e, \quad a^{-n} := (a^n)^{-1} = (a^{-1})^n,$$

where $n$ is a positive integer. The last equality, $(a^n)^{-1} = (a^{-1})^n$, holds because multiplication of either expression by $a^n$ is readily verified to give the identity $e$. The usual laws of exponents still hold, for example $a^n\circ a^m = a^{n+m}$ and $(a^n)^m = a^{nm}$, where $m,n$ are integers and $a,b$ are group elements. However, $(ab)^n = a^n b^n$ need not hold: $(ab)^n = \underbrace{abab\cdots ab}_{n \text{ copies of } ab}$, and multiplication in a group need not be commutative.

A group whose operation is written as multiplication is called a *multiplicative group*. Its identity may also be denoted by 1; this symbol denotes the group identity, not the number 1. For subsets $M_1,M_2$ of a group $G$, define their product by $M_1M_2 := \{xy \mid x\in M_1,\ y\in M_2\}$. *Subset multiplication is associative*: for $M_1,M_2,M_3\subseteq G$, $(M_1M_2)M_3 = M_1(M_2M_3)$.

**Theorem A.3.2.** *For elements $a,b,c$ of any group, $ab=ac$ implies $b=c$, and $ba=ca$ implies $b=c$.* These properties are called the *cancellation laws*.

**Proof.** If $ab=ac$, multiply both sides on the left by $a^{-1}$ to obtain $b=a^{-1}ab=a^{-1}ac=c$. If $ba=ca$, multiply both sides on the right by $a^{-1}$ to obtain $b=baa^{-1}=caa^{-1}=c$. □

**Theorem A.3.3.** *Let $a,b$ be two elements of a group $G$. Then $(ab)^{-1}=b^{-1}a^{-1}$. More generally, if $a_1,a_2,\ldots,a_n$ are $n$ elements of $G$, then $(a_1a_2\cdots a_n)^{-1}=a_n^{-1}\cdots a_2^{-1}a_1^{-1}$.*

**Proof.** Indeed,
$$(b^{-1}a^{-1})\cdot ab=b^{-1}(a^{-1}a)b=b^{-1}eb=b^{-1}b=e.$$
Uniqueness of inverses gives $(ab)^{-1}=b^{-1}a^{-1}$; the latter formula generalizes this identity. □

**Definition A.3.4.** A group $G$ is *abelian*, or *commutative*, if its operation is commutative. Otherwise it is *nonabelian*, or *noncommutative*. A group is *finite* or *infinite* according as its underlying set is finite or infinite; the number of elements in a finite group is its *order*, denoted $|G|$.

**Example A.3.5.** The integers $\mathbb{Z}$ form a group under addition $+$, with identity 0 and inverse $(-n)$ for each element $n$. The positive integers $\mathbb{N}_+$ do not form a group under addition, since they have neither an identity nor inverses. The additive group of integers is plainly abelian and infinite.

**Example A.3.6.** The nonzero real numbers $\mathbb{R}\setminus\{0\}$ form a group under multiplication $\times$, with identity 1 and inverse $\frac{1}{a}$ for each element $a$. It is plainly abelian and infinite. The real numbers $\mathbb{R}$ do not form a group under multiplication because 0 has no inverse.

**Example A.3.7.** The permutations of $n$ objects form the finite *symmetric group* $\mathrm{S}_n$ under composition.

**Example A.3.8.** The automorphisms of an algebraic structure $M$ form its *automorphism group* $\mathrm{Aut}(M)$ under composition. This is readily verified from the definition of an automorphism.

**Definition A.3.9.** For an element $a$ of a group $G$, its *order* is the least positive integer $n$ such that $a^n=e$, the identity element. If no such $n$ exists, the order of $a$ is infinite.

**Example A.3.10.** Let i be the imaginary unit. The set $G=\{1,\mathrm{i},-1,-\mathrm{i}\}$ forms a group under ordinary multiplication. Here $1^1=1$, $(-1)^2=1$, and $(\pm\mathrm{i})^4=1$, and the orders of $1,\mathrm{i},-1,-\mathrm{i}$ are $1,4,2,4$, respectively.

### Subgroups

**Definition A.3.11.** Let $(G,\circ)$ be a group. A set $K$ is a *subgroup* if $K\subseteq G$ and $K$ itself forms a group under $\circ$; we write $K\leq G$ or $G\geq K$. If also $K\neq G$, it is a *proper subgroup*, written $K<G$ or $G>K$.

**Theorem A.3.12.** *Let $H\leq G$, where $G$ is a group. The identity of the subgroup $H$ is the identity of $G$, and for an element $a$ of $H$, its inverse in $H$ is its inverse in $G$.*

**Proof.** Let $e,e'$ be the identities of $G,H$. Then $e'e'=e'=e'e$, so cancellation gives $e'=e$. If $a'$ is the inverse of $a$ in $H$ and $a^{-1}$ is its inverse in $G$, then $a'a=e=a^{-1}a$, and cancellation gives $a'=a^{-1}$. □

**Theorem A.3.13.** *A nonempty subset $H$ of a group $G$ is a subgroup if and only if both of the following hold:*
*(i) $ab \in H$ for every $a,b \in H$;*
*(ii) $a^{-1} \in H$ for every $a \in H$, where $a^{-1}$ means the inverse in $G$.*

**Proof.** If $H \leq G$, then the definition of a subgroup gives $ab \in H$ whenever $a,b \in H$, and theorem A.3.12 gives $a^{-1} \in H$ whenever $a \in H$. Conversely, suppose that the two conditions hold. Closure under multiplication shows that the operation of $G$ restricts to an operation on $H$, and its associativity on $G$ therefore implies its associativity on $H$. If $a \in H$, then $a^{-1} \in H$ by the second condition, and the first condition gives $aa^{-1} = e \in H$. Thus $H$ contains the identity and contains the inverse of each of its elements, so $H$ is a subgroup of $G$. □

**Theorem A.3.14.** *A nonempty subset $H$ of a group $G$ is a subgroup if and only if $ab^{-1} \in H$ whenever $a,b \in H$.*

**Proof.** If $H \leq G$, then $b^{-1} \in H$ by theorem A.3.13, and hence $ab^{-1} \in H$. Conversely, suppose $ab^{-1} \in H$ for all $a,b \in H$. Taking $b = a$ gives $aa^{-1} = e \in H$, and then $ea^{-1} = a^{-1} \in H$; this is condition (ii) of theorem A.3.13. It now follows that $b^{-1} \in H$ and $a(b^{-1})^{-1} = ab \in H$, which is condition (i) of theorem A.3.13. Thus $H \leq G$. □

**Theorem A.3.15.** *Let $H_1, H_2$ be subgroups of a group $G$. If $H_1 \cap H_2$ is nonempty, then $H_1 \cap H_2 \leq G$.*

**Proof.** It suffices to verify closure under multiplication and the existence of inverses. Let $a,b \in H_1 \cap H_2$. Since $H_1, H_2$ are subgroups, $ab$ belongs to both $H_1$ and $H_2$, giving closure under multiplication. Since $H_1, H_2$ are subgroups, $a^{-1}$ also belongs to both $H_1$ and $H_2$, giving the required inverses. □

**Example A.3.16.** It is readily verified that $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +)$.

**Example A.3.17.** The *quaternion group* $G = \{1, \mathbf{i}, \mathbf{j}, \mathbf{k}, -1, -\mathbf{i}, -\mathbf{j}, -\mathbf{k}\}$ is defined by the rules $(-x)y = x(-y) = -1(xy)$ and $(-1)(-x) = x$ for $x, y \in \{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, together with the multiplication table

| $\cdot$ | $1$ | $\mathbf{i}$ | $\mathbf{j}$ | $\mathbf{k}$ |
|---|---|---|---|---|
| $1$ | $1$ | $\mathbf{i}$ | $\mathbf{j}$ | $\mathbf{k}$ |
| $\mathbf{i}$ | $\mathbf{i}$ | $-1$ | $\boxed{\mathbf{k}}$ | $-\mathbf{j}$ |
| $\mathbf{j}$ | $\mathbf{j}$ | $-\mathbf{k}$ | $-1$ | $\mathbf{i}$ |
| $\mathbf{k}$ | $\mathbf{k}$ | $\mathbf{j}$ | $-\mathbf{i}$ | $-1$ |

To read this table, multiply an element of the first column by an element of the first row; their product is the entry at the corresponding intersection. The element from the first column comes first: the order must not be reversed, since multiplication in a group need not be commutative. For example, the boxed entry corresponds to $\mathbf{i}$ in the first column and $\mathbf{j}$ in the first row, so it reads $\mathbf{ij} = \mathbf{k}$.

One can verify that this is a finite nonabelian group. It has proper subgroups $\{1\}$, $\{1, -1\}$, $\{1, -1, \mathbf{i}, -\mathbf{i}\}$, $\{1, -1, \mathbf{j}, -\mathbf{j}\}$, and $\{1, -1, \mathbf{k}, -\mathbf{k}\}$. For example, consider $H = \{1, -1, \mathbf{i}, -\mathbf{i}\}$. The operation rules show that the product of any two elements of $H$ still belongs to $H$. The inverses of $1, -1, \mathbf{i}, -\mathbf{i}$ are $1, -1, -\mathbf{i}, \mathbf{i}$, respectively, and these also belong to $H$. Thus $H$ is a proper subgroup.

**Proposition A.3.18.** *Let $H \leq G$, where $G$ is a group. Then $a \in H$ if and only if $aH = H$.*

**Proof.** Suppose first that $a \in H$. Since $H$ is a subgroup, $ab \in H$ for every $b \in H$, and hence $aH \subseteq H$. Conversely, for every $b \in H$ we have $b = a(a^{-1}b)$, and $a^{-1}b \in H$ because $H$ is a subgroup. Thus $b \in aH$, so $H \subseteq aH$. Therefore $aH = H$.

Conversely, suppose that $aH = H$. Since $H$ is a subgroup, its identity $e$ belongs to $H$, and therefore $a = ae \in aH = H$. □

**Definition A.3.19.** Let $G$ be a group. An element $a \in G$ is *central* if $ag = ga$ for every $g \in G$. The set of all central elements is the *center* $\mathrm{Z}(G)$.

**Theorem A.3.20.** *For every group $G$, $\mathrm{Z}(G) \leq G$.*

**Proof.** The identity $e$ of $G$ plainly belongs to $\mathrm{Z}(G)$, so $\mathrm{Z}(G) \neq \emptyset$. If $a, b \in \mathrm{Z}(G)$, then for every $x \in G$ we have $ax = xa$ and $bx = xb$. Hence

$$(ab)x = a(bx) = axb = (xa)b = x(ab),$$
$$xa^{-1} = (a^{-1}a)(xa^{-1}) = a^{-1}(ax)a^{-1} = a^{-1}xaa^{-1} = a^{-1}x.$$

Thus $ab, a^{-1} \in \mathrm{Z}(G)$, and theorem A.3.13 gives $\mathrm{Z}(G) \leq G$. □

### Additive Groups

So far we have generally called a group's algebraic operation "multiplication." For an abelian group we sometimes use the sign + and call the operation "addition"; such a group is an *additive group*. Again, this name does not mean that the operation is ordinary addition of numbers. To conform with the notation for numbers, its identity is written 0 and called the *zero element*; the inverse of $a$ is written $-a$ and called its *additive inverse*. Thus

$$0 + a = a + 0 = a, \qquad a + (-a) = (-a) + a = 0.$$

As with numbers, we write $a + (-b)$ as $a - b$, obtaining a *subtraction* operation denoted by $-$. It is the inverse operation of addition. The following rules always hold in an additive group:

$$-a + a = a - a = 0, \qquad -(-a) = a,$$
$$-(a + b) = -a - b, \qquad -(a - b) = b - a,$$
$$a + b = c \quad \Longleftrightarrow \quad b = c - a.$$

The last rule is the additive form of the cancellation law in multiplicative notation. Similarly, powers in a multiplicative group become multiples:

$$0a := 0, \qquad na = \underbrace{a + a + \cdots + a}_{n \text{ copies of } a}, \qquad (-n)a := -(na) = n(-a),$$

where $n$ is a positive integer. The equality $-(na) = n(-a)$ holds because adding either expression to $na$ gives the zero element. In the definition $0a = 0$, the 0 on the left is the number zero, while the 0 on the right is the zero element of the group. In this notation the following laws hold:

$$ma + na = (m + n)a, \qquad m(na) = (mn)a, \qquad n(a + b) = na + nb,$$

where $m, n$ are integers and $a, b$ are group elements.

For a group $G$ written additively, the criteria for a nonempty subset $H$ to be a subgroup become the following.

**Theorem A.3.21.** *A nonempty subset $H$ of an additive group $G$ is a subgroup if and only if both of the following hold:*

*(i) $a + b \in H$ for every $a, b \in H$;*
*(ii) $-a \in H$ for every $a \in H$, where $-a$ is the inverse in $G$.*

**Theorem A.3.22.** *A nonempty subset $H$ of an additive group $G$ is a subgroup if and only if $a - b \in H$ whenever $a, b \in H$.*

## Exercises

**Exercise A.29.** Determine whether $(G,\circ)$ is a group in each case: (1) $G=\mathbb{R}_+$ and $\circ$ is ordinary multiplication; (2) $G$ is the set of all vectors in the Euclidean plane and $\circ$ is vector addition.

**Exercise A.30.** Prove example A.3.8.

**Exercise A.31.** Find the order of every element and all subgroups of $S_4$. Is $S_4$ abelian?

**Exercise A.32.** Prove that the intersection of any family of subgroups of a group is again a subgroup.

**Exercise A.33.** Prove that the product of two subgroups of an abelian group is a subgroup.

**Exercise A.34.** Prove that a finite nonempty algebraic system closed under an associative operation and satisfying cancellation has an identity and an inverse for every element, and hence forms a group. Deduce that a finite nonempty subset $H$ of a group $G$ satisfies $H\leq G$ if and only if $ab\in H$ for every $a,b\in H$.

**Exercise A.35.** Let the identity of a group $G$ be $e$, and let $a\in G$ have order $n$. Prove that:

(1) $a^m=e$ if and only if $n$ divides $m$;

(2) for an integer $k$, if $(k,n)$ denotes the greatest common divisor, then $a^k$ has order $\frac{n}{(k,n)}$.

## A.3.2 Homomorphisms, Isomorphisms, and Normal Subgroups

This subsection introduces basic facts about group homomorphisms, isomorphisms, and normal subgroups.

### Homomorphisms and isomorphisms

**Definition A.3.23.** Let $(G,\circ)$ and $(G',\circ')$ be groups. A map $f:G\to G'$ is a *group homomorphism* if $f(a)\circ' f(b)=f(a\circ b)$ for all $a,b\in G$. If $f$ is surjective, we say that $G$ is homomorphic to $G'$ and write $G\sim G'$. If $f$ is bijective, we say that $G$ and $G'$ are isomorphic, write $G\cong G'$, and call $f$ an *isomorphism* from $G$ to $G'$. We write $\mathrm{Hom}(G,G')$ and $\mathrm{Iso}(G,G')$ for the sets of homomorphisms and isomorphisms from $G$ to $G'$. A homomorphism or isomorphism from $G$ to itself is called an endomorphism or automorphism, respectively. The sets of these maps are $\mathrm{End}(G)$ and $\mathrm{Aut}(G)$, and $\mathrm{Aut}(G)$ is readily seen to be a group, called the automorphism group of $G$. For a homomorphism $f$ from an additive group $G$ to an additive group $G'$, if $f(a)$ is the zero element of $G'$ for every $a\in G$, then $f$ is the *zero homomorphism*; otherwise it is a *nonzero homomorphism*.

**Example A.3.24.** All integers form a group $G_1$ under addition, and all even integers form a group $G_2$ under addition. The map $\sigma:\ G_1\to G_2,\ x\mapsto 2x$, is an isomorphism, so $G_1\cong G_2$.

By definition, isomorphic groups have the same operational structure: knowing the group structure of either determines that of the other. A homomorphism is a weaker condition, but it still gives considerable information.

**Theorem A.3.25.** *Let $G,G'$ be algebraic systems. If $G\sim G'$ and $G$ is a group, then $G'$ is also a group.*

**Proof.** Since $G\sim G'$ and multiplication in $G$ is associative, multiplication in $G'$ is associative as well. Let $e$ be the identity of $G$, let $a'$ be any element of $G'$, and let $f:G\to G'$ be the surjective homomorphism, with $f(e)=e'$ and $f(a)=a'$. Then

$$a'=f(a)=f(ea)=e'a'. \qquad (\dagger)$$

Thus $e'$ is the identity of $G'$. Here and below, when the operation symbol is omitted, a product such as $e'a'$ is taken in $G'$, whereas $ea$ is taken in $G$; the ambient algebraic system is determined by context. If $f(a^{-1})=(a^{-1})'$, then

$$e'=f(e)=f(a^{-1}a)=(a^{-1})'a'. \qquad (\ddagger)$$

Thus $a'$ has inverse $(a^{-1})'$. Therefore $G'$ is a group. □

**Remark.** Surjectivity is essential in theorem A.3.25. For example, let $G$ be the multiplicative group of positive rationals and let $G'$ be the positive even integers with the operation $a'\circ b'=2$ for all $a',b'\in G'$. The constant map $f:\ G\to G'$, $x\mapsto 2$ for every $x\in G$, is a homomorphism but is not surjective, and $G'$ is not a group.

The preceding proof immediately gives the following consequence.

**Corollary A.3.26.** *Let $f$ be a homomorphism from a group $G$ to a group $G'$. The image of the identity of $G$ is the identity of $G'$, and the image of the inverse of an element $a$ of $G$ is the inverse of its image; that is, $f(a^{-1})=f(a)^{-1}$.*[6]

**Proof.** This follows from the proof of the preceding theorem. □

**Remark.** The proof of theorem A.3.25 requires $f$ to be surjective, but this corollary does not, because $G'$ is already assumed to be a group. Under this assumption, the arguments at (†) and (‡) still apply to $f$. The key distinction is that surjectivity ensures that an element $a'$ has a preimage $a$, whereas the corollary proceeds in the forward direction, asserting that $f(a^{-1})$ is the inverse of $f(a)$ in $G'$.

**Example A.3.27.** Let $n$ be a positive integer, and give $G'=\{0,1,2,\dots,n-1\}$ the operation $a\circ b=r$, where $r$ is the remainder when $a+b$ is divided by $n$. Let $G=(\mathbb{Z},+)$ and consider the map $f:G\to G'$, $x\mapsto x'$, where $x'$ is the remainder when $x$ is divided by $n$. The remainder of a sum is obtained by adding the remainders of the two summands and then taking the remainder modulo $n$ again. This congruence property shows that $f$ is a surjective homomorphism. Thus $G'$ is a group by theorem A.3.25.

**Theorem A.3.28.** *Let $f:G\to G'$ be a homomorphism between groups $G,G'$; it need not be surjective. Then:*

*(1) if $H\le G$, then $f(H)\le G'$ and $H\sim f(H)$;*
*(2) if $H'\le G'$, then $f^{-1}(H')\le G$, and $f$ induces a homomorphism from $f^{-1}(H')$ to $H'$.*

**Proof.** For $a',b'\in f(H)$, choose $a,b\in H$ with $f(a)=a'$ and $f(b)=b'$. Since $a,b\in H$ and $H\le G$, we have $ab\in H$ and $f(ab)=a'b'\in f(H)$. Thus $f(H)$ is closed under multiplication and $H\sim f(H)$. Closure under multiplication makes $f(H)$ an algebraic system; by theorem A.3.25, it is therefore a group. Since plainly $f(H)\subseteq G'$, it is a subgroup of $G'$.

For the inverse image, corollary A.3.26 shows that the identity of $G$ belongs to $f^{-1}(H')$, so this set is nonempty. Take $a,b\in f^{-1}(H')$ and put $f(a)=a'$, $f(b)=b'$. Then $f(ab^{-1})=f(a)f(b^{-1})=a'(b')^{-1}$ by corollary A.3.26. Since $H'$ is a subgroup, $a'(b')^{-1}\in H'$. Hence $ab^{-1}\in f^{-1}(H')$, so theorem A.3.14 gives $f^{-1}(H')\le G$, and $f$ plainly induces a homomorphism $f^{-1}(H')\to H'$. □

[6] When two groups occur, the inverse notation may refer to the inverse in $G$ or in $G'$, as the context makes clear. For example, $a^{-1}$ is the inverse of $a$ in $G$, whereas $f(a)^{-1}$ is the inverse of $f(a)$ in $G'$. This convention is used below as well.

**Definition A.3.29.** Write the multiplication in a group $G$ as $\circ$. On the same set define a new multiplication $\circ'$ by $a\circ' b = b\circ a$. The set $G$ equipped with $\circ'$ is readily seen to be a group, called the *opposite group* and denoted by $G^{\mathrm{op}}$. For groups $G,H$, a map $f: G\to H$ satisfying $f(ab) = f(b)f(a)$ for every $a,b\in G$ is an *antihomomorphism*; if bijective, it is an *anti-isomorphism*. If an anti-isomorphism $f: G\to H$ exists, $G$ and $H$ are said to be anti-isomorphic.

The following results follow readily from the definitions.

**Theorem A.3.30.** *A group $G$ and its opposite group $G^{\mathrm{op}}$ have the same elements, the same identity, and the same inverses, and they are anti-isomorphic.*

**Theorem A.3.31.** *If the groups $G$ and $H$ are anti-isomorphic, then $G\cong H^{\mathrm{op}}$.*

**Remark.** There is no separate universally used symbol for anti-isomorphism; in view of the preceding theorem, it is normally expressed as $G\cong H^{\mathrm{op}}$.

### Normal subgroups

**Definition A.3.32.** Let $H,G$ be groups with $H\le G$. For $a\in G$, define $aH := \{ax \mid x\in H\}$ as a *left coset* of $H$ in $G$, and $Ha := \{xa \mid x\in H\}$ as a *right coset* of $H$ in $G$.

Let $N,G$ be groups with $N\le G$. If $aN = Na$ for every $a\in G$, then $N$ is a *normal subgroup*, or invariant subgroup, of $G$, written $N\trianglelefteq G$ or $G\trianglerighteq N$; otherwise write $N\ntrianglelefteq G$ or $G\ntrianglerighteq N$. If in addition $N\ne G$, write $N\lhd G$ or $G\rhd N$ and call $N$ a *proper normal subgroup* of $G$. For a normal subgroup $N$, every left coset equals the corresponding right coset, that is, $gN = Ng$ for every $g\in G$; either is therefore simply called a *coset*.

**Remark.** The group $G$ itself and the subset $\{e\}$ consisting only of its identity are plainly normal subgroups of $G$, called the *trivial normal subgroups*. A normal subgroup distinct from both is *nontrivial*.

**Theorem A.3.33.** *For every group $G$, $\mathrm{Z}(G)\trianglelefteq G$.*

**Proof.** This follows immediately from the definitions of the center and a normal subgroup. $\square$

**Theorem A.3.34.** *Let $N\trianglelefteq G$, where $G$ is a group. For all $a,b\in G$, $(aN)(bN) = (ab)N$.* Here $(aN)(bN) := \{xy \mid x\in aN,\ y\in bN\}$ is called *coset multiplication*.

**Proof.** Since $N$ is normal, $(aN)(bN) = a(Nb)N = a(bN)N = (ab)NN = (ab)N$. $\square$

**Example A.3.35.** Consider the quaternion group $G = \{1,\mathbf{i},\mathbf{j},\mathbf{k},-1,-\mathbf{i},-\mathbf{j},-\mathbf{k}\}$ of example A.3.17. Every proper subgroup of $G$ is normal. For instance, for $N := \{1,-1,\mathbf{i},-\mathbf{i}\}$ and every $x\in G$, the multiplication rules readily give

$$xN = \{x,-x,x\mathbf{i},-x\mathbf{i}\} = \{x,-x,\mathbf{i}x,-\mathbf{i}x\} = Nx.$$

Thus $N\lhd G$.

**Theorem A.3.36.** *Let $G$ be a group and let $N\le G$. Then $N\trianglelefteq G$ if and only if $aNa^{-1}\subseteq N$ for every $a\in G$.*

**Proof.** If $N\trianglelefteq G$, then $aN = Na$, and right multiplication by $a^{-1}$ gives $aNa^{-1} = N$, and therefore $aNa^{-1}\subseteq N$. Conversely, if $aNa^{-1}\subseteq N$ for every $a\in G$, then $aN = aNa^{-1}a\subseteq Na$; applying the same hypothesis with $a^{-1}$ gives $Na = aa^{-1}Na\subseteq aN$. Hence $aN = Na$ for every $a$, so $N\trianglelefteq G$ by definition. $\square$

**Theorem A.3.37.** *Let $f: G \to G'$ be a surjective homomorphism between groups $G, G'$.*
*(1) If $N \trianglelefteq G$, then $f(N) \trianglelefteq G'$.*
*(2) If $N' \trianglelefteq G'$, then $f^{-1}(N') \trianglelefteq G$.*

**Proof.** If $N \trianglelefteq G$, then $f(N) \leq G'$ by theorem A.3.28. For $n' \in f(N)$ and $a' \in G'$, surjectivity provides $a \in G$ and $n \in N$ with $f(a) = a'$ and $f(n) = n'$. Since $N \trianglelefteq G$, we have $ana^{-1} \in N$, and hence $f(ana^{-1}) = a'n'(a^{-1})' = a'n'(a')^{-1} \in f(N)$, where $(a^{-1})' = f(a^{-1})$ as above. Thus $a'f(N)(a')^{-1} \subseteq f(N)$, and theorem A.3.36 shows that $f(N) \trianglelefteq G'$. The inverse-image assertion is analogous and is left to the reader. □

Finally, we introduce quotient groups.

**Theorem A.3.38.** *Let $N \trianglelefteq G$, where $G$ is a group. The set of all cosets of $N$ in $G$ forms a group under coset multiplication.* It is called the *quotient group* of $G$ by $N$ and is denoted by $G/N$.

**Proof.** Since multiplication of subsets of a group is associative, coset multiplication is associative as well. The coset $N$ is plainly the identity, because $(aN)N = a(NN) = aN$. Moreover, $(a^{-1}N)(aN) = a^{-1}aN = N$, so $a^{-1}N$ is the inverse of $aN$. Therefore $G/N$ is a group. □

The meaning of "quotienting out" a normal subgroup $N$ is to regard two elements $a, b \in G$ as equivalent when their quotient belongs to $N$, that is, when $at = b$ for some $t \in N$. Thus $N$ determines an equivalence relation $R$ on $G$ by $aRb \iff a^{-1}b \in N$. The quotient group is the quotient set for this relation. Its elements are the equivalence classes, each of which is a coset.

### The fundamental homomorphism theorem

**Definition A.3.39.** Let $f$ be a homomorphism from a group $G$ to a group $G'$. Then the *kernel* $\ker f$ is the inverse image of the identity of $G'$, that is, the set of all its preimages. The set of images under $f$ of all elements of $G$ is the *image* of $f$, denoted by $\operatorname{im} f$.

**Theorem A.3.40** (Fundamental homomorphism theorem)**.** *If $f: G \to G'$ is a surjective homomorphism between groups $G, G'$ and $N = \ker f$, then $\ker f \trianglelefteq G$ and $G/N \cong G'$.*

**Proof.** The singleton consisting of the identity of $G'$ is a normal subgroup of $G'$. Hence theorem A.3.37 and the definition of the kernel give $N = \ker f \trianglelefteq G$. Write $f: G \to G'$, $a \mapsto a'$, and define the correspondence $g: aN \mapsto f(a)$.

First, let $a, b \in G$, with $f(a) = a'$ and $f(b) = b'$. If $aN = bN$, left multiplication by $a^{-1}$ gives $a^{-1}bN = a^{-1}aN = N$. By proposition A.3.18, $a^{-1}b \in N$, and the definition of the kernel gives $(a')^{-1}b' = f(a^{-1}b) = e'$. Left multiplication by $a'$ gives $a' = b'$. Thus every left coset of $N$ has a unique image under the correspondence, so $g$ is indeed a map $G/N \to G'$.

Second, take any $a' \in G'$. Since $f$ is surjective, there is $a \in G$ with $f(a) = a'$, and then $aN$ is a preimage of $a'$ under $g$. Hence $g$ is surjective.

Third, let $a, b \in G$. If $a^{-1}b \in N$, then proposition A.3.18 gives $a^{-1}bN = N$, and left multiplication by $a$ gives $bN = aN$. Taking the contrapositive, if $aN \neq bN$, then $a^{-1}b \notin N$, so $(a')^{-1}b' = f(a^{-1}b) \notin f(N) = \{e'\}$. Thus $(a')^{-1}b' \neq e'$, and left multiplication by $a'$ gives $a' \neq b'$. Therefore $g$ is injective.

It follows that $g: G/N \to G'$ is a bijection. Finally, coset multiplication gives $g((aN)(bN)) = g(abN) = f(ab) = a'b'$, so $g$ is an isomorphism and $G/N \cong G'$. □

## Exercises

**Exercise A.36.**

(1) Prove part (2) of theorem A.3.37.
(2) Verify that the opposite group $G^{\mathrm{op}}$ in definition A.3.29 is a group.
(3) Prove theorems A.3.30 and A.3.31.

**Exercise A.37** (Lagrange's theorem)**.** Let $G$ be a finite group and let $H \leq G$. The number of distinct cosets of $H$ in $G$ is called the index of $H$ in $G$, denoted by $[G:H]$. Prove that $[G:H] = \frac{|G|}{|H|}$.

**Exercise A.38.** Let $f$ be a homomorphism from a group $G$ to a group $G'$. Prove $\ker f \leq G$ and $\operatorname{im} f \leq G'$.

**Exercise A.39.** Let $f: G \to G'$ be a group homomorphism. Prove that it is injective if and only if the inverse image of the identity $e'$ of $G'$ consists only of the identity $e$ of $G$.

**Exercise A.40.** Let $G = (\mathbb{Q} \setminus \{0\}, \times)$ and $G' = (\mathbb{Z}, +)$, and define $f:\ \mathbb{Q} \setminus \{0\} \to \mathbb{Z},\ 2^n \varepsilon \frac{b}{a} \mapsto n$, where $a,b$ are coprime positive odd integers, $\varepsilon \in \{\pm 1\}$, and $n$ is an integer. Prove that $f$ is a surjective homomorphism from $G$ to $G'$.

**Exercise A.41.** Prove $\operatorname{Aut}(\mathbb{Q}, +) \cong (\mathbb{Q} \setminus \{0\}, \times)$.

**Exercise A.42.** Let $G$ be a group. For each $g \in G$, define $\varphi_g:\ G \to G,\ x \mapsto gxg^{-1}$. Prove that $\varphi_g$ is an automorphism of $G$ and that all such maps form a normal subgroup of $\operatorname{Aut}(G)$. These are the *inner automorphisms*; their transformation group is the *inner automorphism group* of $G$, denoted by $\operatorname{Inn}(G)$.

**Exercise A.43.** Let $G$ be a group and $N \trianglelefteq G$. Prove that $G \sim G/N$.

*Hint.* Consider $h:\ a \mapsto aN$ and prove that it gives a homomorphism. This is the *natural homomorphism $G \to G/N$.*

**Exercise A.44.** Let $G, G'$ be groups with $G \sim G'$ and let the homomorphism have kernel $K$. Prove that two elements of $G$ have the same image in $G'$ if and only if they belong to the same coset of $K$.

### A.3.3 Group Actions, Direct Products, and Semidirect Products

This subsection supplies further material on group actions, direct products, and semidirect products. We begin with group actions.

**Definition A.3.41.** An *action* of a group $G$ on a set $X$ is a map $f: G \times X \to X$, $(g,x) \mapsto g \circ x$, so that $f(g,x) = g \circ x$, satisfying $e \circ x = x$ and $(gh) \circ x = g \circ (h \circ x)$ for every $x \in X$ and $g,h \in G$, where $e$ is the identity of $G$. We also say that $G$ *acts on* $X$. The symbol $\circ$ here denotes the action, not the multiplication in $G$. For this action and $x \in X$, the *orbit* of $x$, written $\operatorname{Orb}_G(x)$ or $Gx$, and its *stabilizer*, written $\operatorname{Stab}_G(x)$ or $G_x$, are defined by

$$\operatorname{Orb}_G(x) := \{g \circ x \mid g \in G\}, \quad \operatorname{Stab}_G(x) := \{g \in G \mid g \circ x = x\}.$$

For an ordered tuple $(x_1, \ldots, x_n)$ of elements of $X$, define its *pointwise stabilizer*, also written $G_{x_1,\ldots,x_n}$, by

$$\operatorname{Stab}_G(x_1, \ldots, x_n) := \{g \in G \mid g \circ x_i = x_i,\ i = 1, \ldots, n\}.$$

**Example A.3.42.** A group $G$ acts on itself by *conjugation*: for $g,x \in G$, define $g \circ x = gxg^{-1}$. The stabilizer of $x$ is

$$G_x = \{g \in G \mid gxg^{-1} = x\} = \{g \in G \mid gx = xg\},$$

and is also called the *centralizer* of $x$ in $G$.

**Example A.3.43.** As discussed in section 6.6.2, let $X = \mathbb{R} \cup \{\infty\}$, and write $G$ for the group of one-dimensional projectivities on $X$. Define the action of a transformation $g$ in $G$ on a point $x$ in $X$ to be the ordinary transformation, so that $g \circ x = g(x)$. Let $x_1, x_2, x_3$ be three distinct points of $X$. Since three points uniquely determine a projectivity, $\mathrm{Stab}_G(x_1, x_2, x_3) = \{\mathrm{id}_X\}$.

We next define direct products.

**Definition A.3.44.** For groups $G,H$, define multiplication on the set $G \times H$ by $(g_1,h_1)(g_2,h_2) = (g_1g_2, h_1h_2)$. Then $G \times H$ is readily seen to form a group, called the *direct product* of $G$ and $H$. Its identity is $(e_G, e_H)$, where $e_G, e_H$ are the identities of $G,H$, respectively, and the inverse of $(g,h)$ is $(g,h)^{-1} = (g^{-1}, h^{-1})$.

**Example A.3.45.** Consider the additive group of real numbers $\mathbb{R}^+ = (\mathbb{R}, +)$. The elements of the direct product $G = \mathbb{R}^+ \times \mathbb{R}^+$ are ordered pairs of real numbers $(a,b)$. The preceding definition used multiplicative notation; in additive notation the operation on $G$ is $(a,b) + (c,d) = (a+c, b+d)$.

**Remark.** The two factors embed naturally into $G \times H$, because $G \cong G \times \{e_H\}$ and $H \cong \{e_G\} \times H$, under the isomorphisms $g \mapsto (g, e_H)$ and $h \mapsto (e_G, h)$, respectively. Moreover, elements of $G \times \{e_H\}$ commute with those of $\{e_G\} \times H$: $(g,e_H)(e_G,h) = (e_G,h)(g,e_H)$. Thus a direct product can be understood as placing two groups together without interaction.

Direct products generalize to semidirect products.

**Definition A.3.46.** Let $N,H$ be groups. Suppose that to each $h \in H$ there corresponds an automorphism $\varphi_h : N \to N$ and that $\varphi : H \to \mathrm{Aut}(N)$, $h \mapsto \varphi_h$, is a group homomorphism. This gives an action of $H$ on the group $N$, with $h \circ n = \varphi_h(n)$. On $N \times H$, define

$$(n_1,h_1)(n_2,h_2) = (n_1\varphi_{h_1}(n_2), h_1h_2).$$

This operation makes $N \times H$ a group, called the *semidirect product* of $N$ and $H$ with respect to the action $\varphi$, denoted by $N \rtimes_\varphi H$; when the action is clear, abbreviate it to $N \rtimes H$.

**Remark.** In the semidirect product $N \rtimes_\varphi H$, the identity is $(e_N, e_H)$, and the inverse of an element $(n,h)$ is $(n,h)^{-1} = (\varphi_{h^{-1}}(n^{-1}), h^{-1})$. In particular, $N$ and $H$ may again be regarded as subgroups of $N \rtimes_\varphi H$, because $N \cong N \times \{e_H\}$ and $H \cong \{e_N\} \times H$ under $n \mapsto (n, e_H)$ and $h \mapsto (e_N, h)$, respectively. Unlike the direct product, elements of these subgroups need not commute; indeed the definition gives

$$(e_N,h)(n,e_H) = (\varphi_h(n), e_H)(e_N,h). \tag{$\spadesuit$}$$

Here $\varphi_h(e_N) = e_N$ by corollary A.3.26; thus $H$ acts on $N$ through automorphisms.

**Example A.3.47.** If the action $\varphi$ is trivial, meaning that $\varphi_h = \mathrm{id}_N$ for every $h \in H$, then multiplication in the semidirect product reduces to

$$(n_1,h_1)(n_2,h_2) = (n_1n_2, h_1h_2).$$

Hence $N \rtimes_\varphi H \cong N \times H$, so the direct product is a special case of the semidirect product.

**Example A.3.48.** Let $N$ be an abelian group, and let $H=\{e,\varepsilon\}$ be the two-element group, where $e$ is the identity and $\varepsilon^2=e$. Define $\varphi_e(n)=n$ and $\varphi_\varepsilon(n)=n^{-1}$. Since $N$ is abelian, the map $N\to N$, $n\mapsto n^{-1}$, is readily verified to be an automorphism. Hence one may form

$$G=N\rtimes_\varphi H=\{(n,e)\,|\,n\in N\}\cup\{(n,\varepsilon)\,|\,n\in N\}.$$

If $N$ contains some $n\neq n^{-1}$, this semidirect product is not a direct product.

The notation $N\rtimes H$ depends on the action of $H$ on $N$ as well as on the groups $N,H$ themselves. Different actions can give different semidirect-product groups. Another example is given in the exercises. We now interpret the semidirect product within an already given group.

**Definition A.3.49.** Let $G$ be a group and let $N,H\leq G$, with $N\trianglelefteq G$. If $G=NH$ and $N\cap H=\{e\}$, where $e$ is the identity of $G$, then every $g\in G$ has a unique expression $g=nh$ with $n\in N$, $h\in H$. In this case $G=N\rtimes H$ is an *internal semidirect product* of $N$ and $H$; *$G$ splits over $N$*, and $H$ is a *complement* of $N$ in $G$. The construction in definition A.3.46 is the corresponding external semidirect product.

**Remark.** The asserted uniqueness follows directly. If $n_1h_1=n_2h_2$, with $n_1,n_2\in N$ and $h_1,h_2\in H$, then $n_2^{-1}n_1=h_2h_1^{-1}$. The left side belongs to $N$ and the right side to $H$, so both belong to $N\cap H=\{e\}$. Thus $n_1=n_2$ and $h_1=h_2$.

**Example A.3.50.** Let $G$ be the group of isometries of the Euclidean plane that fix the origin, where an isometry preserves the distance between any two points. Its elements include the identity $e$, reflections $\sigma_l$ in lines $l$ through the origin, rotations $\rho_\theta$ through angles $\theta$ about the origin, and compositions of reflections and rotations. All rotations about the origin form the subgroup $N=\{\rho_\theta\,|\,\theta\in[0,2\pi)\}$. Take $H=\{e,\sigma_x\}$, where $\sigma_x$ is reflection in the $x$-axis.

For every rotation $\rho_\theta\in N$ and every $g\in G$, the conjugate $g\rho_\theta g^{-1}$ is again a rotation about the origin and hence lies in $N$. Thus theorem A.3.36 gives $N\trianglelefteq G$. Every element of $G$ is a rotation, a reflection, or a composition of the two, and every reflection can be written as the composition of a rotation and $\sigma_x$; hence $G=NH$. Plainly $N\cap H=\{e\}$. We therefore have the internal semidirect product $G=N\rtimes H$.

**Example A.3.51.** Not every normal subgroup gives an internal semidirect product. In the quaternion group $G=\{1,\mathbf{i},\mathbf{j},\mathbf{k},-1,-\mathbf{i},-\mathbf{j},-\mathbf{k}\}$, take $N=\{1,-1\}$. It is readily verified that $N\trianglelefteq G$. If $G=N\rtimes H$, then $H$ would have four elements and $N\cap H=\{1\}$. However, every subgroup of the quaternion group with four elements contains $-1$, so its intersection with $N$ is nontrivial. Thus $G$ is not the internal semidirect product of $N$ and any complement.

Finally, the internal and external constructions are essentially the same, viewed from different perspectives.

First, for an internal semidirect product, $N\trianglelefteq G$ implies that for every $h\in H$, conjugation $\varphi_h\colon N\to N$, $n\mapsto hnh^{-1}$, is an automorphism of $N$. Moreover, $\varphi\colon H\to\mathrm{Aut}(N)$, $h\mapsto\varphi_h$, is a group homomorphism. The external semidirect product $N\rtimes_\varphi H$ constructed from this action is isomorphic to $G$, by the specific isomorphism $N\rtimes_\varphi H\to G$, $(n,h)\mapsto nh$.

Conversely, let $N,H$ be groups and let $\varphi\colon H\to\mathrm{Aut}(N)$, $h\mapsto\varphi_h$, be a homomorphism, giving the external semidirect product $G=N\rtimes_\varphi H$. In $G$, put $N_0=N\times\{e_H\}$ and $H_0=\{e_N\}\times H$. It is readily verified that $N_0,H_0\leq G$, with $N_0\cong N$ and $H_0\cong H$. For every $(n,h)\in G$ and $(m,e_H)\in N_0$, (♠) gives $(n,h)(m,e_H)(n,h)^{-1}\in N_0$; thus theorem A.3.36 gives $N_0\trianglelefteq G$. Plainly $N_0\cap H_0=\{(e_N,e_H)\}$, and every $(n,h)\in N\rtimes_\varphi H$ has the unique expression $(n,h)=(n,e_H)(e_N,h)$. Therefore the external semidirect product $G$ can also be regarded as the internal semidirect product of $N_0,H_0$. □

An external semidirect product constructs a new group from two groups and an action; an internal semidirect product identifies this structure within an existing group. The underlying structure is the same.

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## Exercises

**Exercise A.45.** Verify carefully from the definitions that direct products, internal semidirect products, and external semidirect products all form groups.

**Exercise A.46.** Let $N,H$ both be normal subgroups of $G$, so $N \trianglelefteq G$ and $H \trianglelefteq G$, and suppose $G = NH$ and $N \cap H = \{e\}$, where $e$ is the identity. Prove that, in addition to $G \cong N \rtimes H$, we have $G \cong N \times H$. Then $G$ is the *internal direct product* of $N$ and $H$.

**Exercise A.47.** Show that the internal semidirect product in example A.3.50 is not an internal direct product.

**Exercise A.48.** Let $G$ be a group with identity $e$, and let $N,H \leq G$ with $N \trianglelefteq G$. Prove that the following conditions are equivalent:

(1) $G = NH$ and $N \cap H = \{e\}$;
(2) $G = HN$ and $N \cap H = \{e\}$;
(3) every element of $G$ has a unique expression $nh$, with $n \in N$, $h \in H$;
(4) every element of $G$ has a unique expression $hn$, with $h \in H$, $n \in N$;
(5) there is a group isomorphism $H \cong G/N$ given by $f: H \to G/N$, $h \mapsto hN$;
(6) there is a group homomorphism $f: G \to H$ such that $\ker f = N$ and $f|_H = \mathrm{id}_H$.

When these conditions hold, $G = N \rtimes H$.

**Exercise A.49** (Orbit–stabilizer formula)**.** If a finite group $G$ acts on a set $X$, prove that for every $x \in X$, $|G| = |Gx||G_x|$.

**Exercise A.50.** Let $\mathbb{R}^+$ be the additive group of real numbers and $\mathbb{R}^\times$ the multiplicative group of nonzero real numbers, and let

$$G = \{(f: \mathbb{R} \to \mathbb{R},\ x \mapsto ax + b) \mid a \in \mathbb{R}^\times,\ b \in \mathbb{R}\}.$$

Prove that $G$ forms a group and that $G \cong \mathbb{R}^+ \rtimes \mathbb{R}^\times$, where the action of $\mathbb{R}^\times$ on $\mathbb{R}^+$ is ordinary multiplication of numbers.

# A.4 Basic Ring Theory

## A.4.1 Basic Concepts of Rings and Fields

### Rings

**Definition A.4.1.** A nonempty set $R$ with two algebraic operations, called addition (usually written "+") and multiplication, is a *ring* if:

(1) addition makes $R$ an additive group;
(2) multiplication is associative, so $(ab)c = a(bc)$ for all $a,b,c \in R$;
(3) multiplication obeys the left and right distributive laws: for all $a,b,c \in R$, $a(b+c) = ab + ac$ and $(b+c)a = ba + ca$.

It is *commutative* if $ab = ba$ for all $a,b \in R$, and *noncommutative* otherwise. It is finite or infinite according as its underlying set is finite or infinite; the number of elements of a finite ring is its *order*, denoted $|R|$. A ring whose multiplication has an identity $e \in R$, so that $ea = ae = a$ for every $a \in R$, is a *unital ring*; the multiplicative identity is called the *identity element* of $R$, and is usually denoted by 1. A nonempty subset $S$ that is itself a ring under the inherited operations is a *subring*, written $S \leq R$ or

$R \geq S$. $R$ itself is also an additive group. To emphasize its group structure, we may write $R^+$ and call it the *additive group* of $R$.

**Remark.** In the definition of a unital ring, suppose a ring $R$ has a left identity $e$ satisfying $ea = a$ and a right identity $e'$ satisfying $ae' = a$. Using $e$ as a left identity gives $e \cdot e' = e'$, while using $e'$ as a right identity gives $e \cdot e' = e$, so $e' = e$. Moreover, if $e, e'$ are both identities, the same argument gives $e = e'$. Thus, if a ring has both a left identity and a right identity, they coincide and are unique.

In a unital ring, if an element $a$ has an inverse $b$, so that $ab = ba = e$ is the identity, we may still write $b = a^{-1}$.

**Example A.4.2.** The even integers form a commutative infinite ring under ordinary addition and multiplication. Since 1 is not even, this ring has neither a left nor a right identity. All integers $\mathbb{Z}$, however, form a unital ring under ordinary addition and multiplication, called the *ring of integers*.

**Example A.4.3.** Let $\mathbb{Z}$ be the ring of integers. On $R = \mathbb{Z} \times \mathbb{Z} = \{(a,b) \mid a,b \in \mathbb{Z}\}$, define addition and multiplication by

$$(a,b) + (c,d) = (a+c, b+d), \qquad (a,b)(c,d) = (ac, bd).$$

These operations readily make $R$ a ring with identity $(1,1)$. The subset $S = \mathbb{Z} \times \{0\}$ is closed under addition, multiplication, and additive inverses in $R$, so it is a subring. Its multiplicative identity is $(1,0)$. Thus a subring and its ambient ring can have different identities.

**Example A.4.4.** On $R = \mathbb{Z} \times \mathbb{Z} = \{(a,b) \mid a,b \in \mathbb{Z}\}$, define

$$(a,b) + (c,d) = (a+c, b+d), \qquad (a,b)(c,d) = (ac, ad).$$

Plainly $(R,+)$ is an additive group, and multiplication can be verified to satisfy associativity and the distributive laws. Thus $R$ is a ring under this addition and multiplication, and it is plainly noncommutative and infinite. Put $u = (1,0)$. For every $(a,b) \in R$, one can verify that $u(a,b) = (a,b)$, so $u$ is a left identity. If $(p,q)$ were a right identity, however, then $(a,b)(p,q) = (ap, aq) = (a,b)$ would have to hold for all $a,b$, which would require both $ap = a$ and $aq = b$; the latter cannot hold for arbitrary $a,b$. This shows that a ring may have a left identity without having a right identity. Replacing multiplication by $(a,b)(c,d) = (ac, bc)$ gives another ring, with right identity $(1,0)$ but no left identity.

Here are some rules for multiplication.

**Theorem A.4.5.** *Let* $0$ *be the zero element of a ring* $R$. *For all* $a,b,c \in R$:

*(1)* $0a = a0 = 0$, *where every* $0$ *denotes the zero element, not a number;*
*(2)* $(-a)b = a(-b) = -ab$;
*(3)* $(-a)(-b) = ab$;
*(4)* $c(a-b) = ca - cb$ *and* $(a-b)c = ac - bc$;
*(5)* *for positive integers* $m,n$,

$$\left(\sum_{i=1}^{m} a_i\right)\left(\sum_{j=1}^{n} b_j\right) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_j;$$

*(6)* $(ma)(nb) = (na)(mb) = (mn)(ab)$ *for integers* $m,n$.

**Proof.** For the first identity, $0a + 0a = (0+0)a = 0a$, so $0a = 0$; similarly, $a0 + a0 = a(0+0) = a0$, so $a0 = 0$. Next, $(-a)b + ab = (-a+a)b = 0b = 0$, and hence $(-a)b = -ab$; similarly $a(-b) = -ab$. It follows that $(-a)(-b) = a[-(-b)] = ab$. Also $c(a-b) = c[a + (-b)] = ca + c(-b) = ca - cb$, and the other distributive identity is analogous. Expanding completely by the distributive laws proves the double-sum identity. When $m,n$ are positive, the final identity is the special case $a_i = a$, $b_j = b$ of that formula. If either integer is zero, the result is plainly zero, so the identity holds, and if either is negative one extracts the minus sign and reduces to the positive case. □

In a ring, positive powers are likewise defined by repeated multiplication: for a positive integer $n$, $a^n := \underbrace{aa\cdots a}_{n \text{ copies of } a}$. In a unital ring we additionally define $a^0 := e$, the identity; here the exponent 0 is a number, not the zero element of the ring. A ring need not give every element a multiplicative inverse. If, however, an element $a$ of a unital ring has an inverse $b$, so $ab = ba = 1$, we call *a invertible*, write $b = a^{-1}$, and may introduce negative powers by $a^{-n} = (a^{-1})^n$.

The preceding discussion shows that most ordinary arithmetic rules hold in a ring, though not always. Since multiplication need not commute, identities such as $(ab)^n = a^n b^n$ and $(a+b)^2 = a^2 + 2ab + b^2$ need not hold: in general $(ab)^n = abab\cdots ab$ and $(a+b)^2 = a^2 + ab + ba + b^2$.

### Elementary properties of rings

To test whether a subset is a subring, it suffices to check closure, because its operations are inherited from the ring. The criteria for an additive subgroup are already known, so only closure under multiplication must be added. Thus the following subring criteria are readily proved.

**Theorem A.4.6.** *A nonempty subset $S$ of a ring $R$ is a subring if and only if all three conditions hold:*

(i) $a + b \in S$ *for all* $a, b \in S$*;*
(ii) $-a \in S$ *for all* $a \in S$*;*
(iii) $ab \in S$ *for all* $a, b \in S$*.*

**Theorem A.4.7.** *A nonempty subset $S$ of a ring $R$ is a subring if and only if $a - b, ab \in S$ whenever $a, b \in S$.*

**Theorem A.4.8.** *Let $S_1, S_2$ be subrings of a ring $R$. If $S_1 \cap S_2$ is nonempty, then $S_1 \cap S_2 \leq R$.*

Here is another elementary result.

**Theorem A.4.9.** *Let $R$ be a unital ring whose identity $e$ differs from its zero element* 0*, and define $R^\times := \{a \in R \mid \exists b \in R,\ ab = ba = e\}$. Then $R^\times$ forms a group under multiplication in $R$.* It is called the *group of invertible elements*, or *group of units*, of $R$.

**Proof.** Multiplication in $R$ already satisfies associativity, and the definition ensures the existence of an identity and inverses. If $a, b \in R^\times$, then $b^{-1}a^{-1}$ is the inverse of $ab$, so $R^\times$ is closed under multiplication and is therefore a group. □

**Definition A.4.10.** Let $R$ be a ring. An element $a \in R$ is a *central element* if $ar = ra$ for every $r \in R$. The set of all central elements is the *center* of $R$, denoted by $\mathrm{Z}(R)$.

**Theorem A.4.11.** *For any ring $R$, $\mathrm{Z}(R) \leq R$, and $\mathrm{Z}(R)$ is commutative.*

**Proof.** For $a, b \in \mathrm{Z}(R)$ and every $r \in R$,

$$(a+b)r = ar + br = ra + rb = r(a+b),$$
$$(ab)r = a(br) = arb = (ra)b = r(ab),$$
$$(-a)r = -ar = -ra = r(-a).$$

Thus $a+b, ab, -a \in \mathrm{Z}(R)$, so $\mathrm{Z}(R) \leq R$. By its definition, every central element commutes with every other element, and hence this subring is commutative. □

**Definition A.4.12.** If the largest order of an element of the additive group $R^+$ is $n$, then $n$ is the *characteristic* of $R$, denoted by $\operatorname{char} R$. If no such $n$ exists, the characteristic is said to be infinite, or zero.

**Remark.** In the definition of order, the identity of the group $R^+$ is the zero element of the ring: the order of $a$ is the least positive integer $n$ satisfying $na = 0$. The identity of the ring, by contrast, means its multiplicative identity.

**Theorem A.4.13.** *The characteristic of a unital ring is the additive order of its identity.*

**Proof.** If the identity $e$ has infinite order in $R^+$, then plainly $\operatorname{char} R$ is infinite. If the order of $e$ is the positive integer $n$, then for every nonzero element $a$, $na = (n \cdot e)a = 0a = 0$. Hence the order of $a$ in the additive group $R^+$ is at most $n$, as required. □

### Division rings and fields

**Definition A.4.14.** If $R$ is a unital ring with more than one element and every nonzero element of $R$ has an inverse, then $R$ is a *division ring*, also called a *skew field*. A commutative division ring is a *field*. Division subrings and subfields are defined in the same way as subgroups and subrings. Division rings and fields inherit the definition of characteristic from rings; by theorem A.4.13, it is the additive order of their identity.

**Theorem A.4.15.** *If $K$ is a division ring, then $K \setminus \{0\} = K^\times$ is a group under multiplication.*

**Proof.** This is immediate from the definition. □

**Example A.4.16.** The integers $\mathbb{Z}$ plainly form a ring under numerical addition and multiplication, but multiplicative inverses are generally fractions and need not be integers, so $\mathbb{Z}$ is not a field. The rational numbers $\mathbb{Q}$ do form a field under these operations, called the *rational number field*. Likewise, the real numbers $\mathbb{R}$ and complex numbers $\mathbb{C}$ form respectively the *real number field* and the *complex number field*.

**Example A.4.17.** The quaternion group was defined in example A.3.17. We now define the set

$$\mathbb{H} = \{a \cdot 1 + b \cdot \mathbf{i} + c \cdot \mathbf{j} + d \cdot \mathbf{k} \mid a,b,c,d \in \mathbb{R}\}$$

with the same multiplication rules for $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$ as in example A.3.17. Elements of $\mathbb{H}$ are called *quaternions*. Terms with zero coefficient may be omitted, for example $a1 + b\mathbf{i} = a1 + b\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$; the coefficient 1 may also be omitted, as in $a1 = a$, $1\mathbf{i} = \mathbf{i}$, $1\mathbf{j} = \mathbf{j}$, and $1\mathbf{k} = \mathbf{k}$.

We stipulate that $a_1 + b_1\mathbf{i} + c_1\mathbf{j} + d_1\mathbf{k} = a_2 + b_2\mathbf{i} + c_2\mathbf{j} + d_2\mathbf{k}$ if and only if $a_1 = a_2$, $b_1 = b_2$, $c_1 = c_2$, and $d_1 = d_2$ all hold, and define addition by

$$(a_1 + b_1\mathbf{i} + c_1\mathbf{j} + d_1\mathbf{k}) + (a_2 + b_2\mathbf{i} + c_2\mathbf{j} + d_2\mathbf{k}) = (a_1 + a_2) + (b_1 + b_2)\mathbf{i} + (c_1 + c_2)\mathbf{j} + (d_1 + d_2)\mathbf{k}.$$

Multiplication is defined by expanding with the ordinary distributive law and collecting terms using the quaternion-group multiplication rules. This resembles multiplication of complex numbers, but multiplication here is noncommutative. Explicitly,

$$\begin{aligned}
&(a_1 + b_1\mathbf{i} + c_1\mathbf{j} + d_1\mathbf{k})(a_2 + b_2\mathbf{i} + c_2\mathbf{j} + d_2\mathbf{k}) \\
&= (a_1a_2 - b_1b_2 - c_1c_2 - d_1d_2) + (a_1b_2 + b_1a_2 + c_1d_2 - c_2d_1)\mathbf{i} \\
&\quad + (a_1c_2 + a_2c_1 + d_1b_2 - d_2b_1)\mathbf{j} + (a_1d_2 + a_2d_1 + b_1c_2 - b_2c_1)\mathbf{k}.
\end{aligned}$$

These operations can be verified to satisfy the ring axioms, with identity 1. Moreover, the inverse is

$$(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k})^{-1} = \frac{a - b\mathbf{i} - c\mathbf{j} - d\mathbf{k}}{a^2 + b^2 + c^2 + d^2}.$$

Thus $\mathbb{H}$ is a division ring, called the *quaternion division ring*. Its multiplication is plainly noncommutative, so it does not form a field.

For any two elements $a,b$ of a field with $a \neq 0$, commutativity gives $a^{-1}b = ba^{-1}$; their common value is denoted $\frac{b}{a}$, thereby defining *division* in a field. The usual rules of arithmetic hold in a field; for example, $\frac{b}{a} + \frac{d}{c} = \frac{bc+ad}{ac}$. We shall not elaborate on them here.

**Theorem A.4.18.** *For a division ring $K$, its center $\mathrm{Z}(K)$ is a division subring of $K$ and is a field.*

**Proof.** By theorem A.4.11, the center $\mathrm{Z}(K)$ is a ring, and $1a = a1 = a$ shows that $1 \in \mathrm{Z}(K)$. For nonzero $a \in \mathrm{Z}(K)$ and every nonzero $k \in K$,

$$a^{-1}k = (a)^{-1}(k^{-1})^{-1} = (k^{-1}a)^{-1} = (ak^{-1})^{-1} = (k^{-1})^{-1}(a)^{-1} = ka^{-1}.$$

Also $a^{-1}0 = 0a^{-1} = 0$, so $a^{-1} \in \mathrm{Z}(K)$. Thus $\mathrm{Z}(K)$ is a division subring, and it is a field because its multiplication is commutative. □

Finally, we give an important result whose proof lies beyond the scope of the course. Interested readers may consult further references.

**Theorem A.4.19** (Wedderburn's little theorem)**.** *Every finite division ring is a field.*

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## Exercises

**Exercise A.51.** Prove theorems A.4.6 to A.4.8 and A.4.15 in detail.

**Exercise A.52.**

(1) On $\mathbb{R}$, define a new multiplication by $a \circ b = |a|b$ for $a,b \in \mathbb{R}$. Does ordinary addition together with this multiplication make $\mathbb{R}$ a ring?
(2) For the set $R = \{0\}$, do ordinary numerical addition and multiplication make $R$ a ring?
(3) Using the definitions of a ring and a field, prove that $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields under ordinary numerical addition and multiplication, while $\mathbb{Z}$ is a ring but not a field under these operations.

**Exercise A.53.** Let $K, K'$ be division rings and let $f : K \to K'$ be a bijection.

(1) If $f$ preserves addition, so that $f(a+b) = f(a) + f(b)$ for all $a,b \in K$, prove that the zero of $K$ maps to the zero of $K'$.
(2) If $f$ preserves multiplication, so that $f(ab) = f(a)f(b)$ for all $a,b \in K$, prove that the zero and identity of $K$ map respectively to the zero and identity of $K'$.

**Exercise A.54.**

(1) Let $\mathbb{Z}[\mathrm{i}] := \{a + b\mathrm{i} \mid a,b \in \mathbb{Z}\}$, where i is the imaginary unit. Does it form a ring under numerical addition and multiplication? Does it form a field? Is it a subring or a subfield of $\mathbb{C}$?
(2) Let $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} \mid a,b \in \mathbb{Q}\}$. Does it form a ring under numerical addition and multiplication? Does it form a field? Is it a subring or a subfield of $\mathbb{C}$?

**Exercise A.55.**

(1) If a ring $R$ has nonzero elements $a,b$ satisfying $ab = 0$, prove that $R$ cannot be a division ring.
(2) For $a,b$ in a division ring $K$, prove that $ab \neq 0$ if and only if both $a \neq 0$ and $b \neq 0$.

**Exercise A.56.** For quaternions $\alpha,\beta,\gamma$, prove $(\alpha\beta-\beta\alpha)^2\gamma=\gamma(\alpha\beta-\beta\alpha)^2$.

**Exercise A.57.** Let $G$ be an additive group. For any $f,g\in\mathrm{End}(G)$, define addition by $(f+g)a=f(a)+g(a)$ and multiplication by $(fg)a=f(g(a))$ for every $a\in G$. Prove that under these operations $\mathrm{End}(G)$ is a unital ring, the *endomorphism ring* of $G$.

## A.4.2 Commutative Rings and Integral Domains

This section records several notions concerning commutative rings and integral domains.

### Zero divisors

**Definition A.4.20.** Let $R$ be a ring. A nonzero $a\in R$ is a *left zero divisor* if $ab=0$ for some nonzero $b\in R$, and a *right zero divisor* if $ba=0$ for some nonzero $b\in R$. Both are called *zero divisors*. A ring with neither kind is said to have *no zero divisors* , or a ring without zero divisors. The *left cancellation law* holds if, for all $a,b,c\in R$, $a\neq 0$ and $ab=ac$ imply $b=c$; the *right cancellation law* holds if, for all $a,b,c\in R$, $a\neq 0$ and $ba=ca$ imply $b=c$. If both hold, $R$ satisfies the *cancellation laws*. A commutative unital ring with $1\neq 0$ and no zero divisors is an *integral domain*.

**Proposition A.4.21.**

*(1) A ring R has no zero divisors if and only if, for all $a,b\in R$, $ab=0$ implies $a=0$ or $b=0$.*
*(2) A ring without zero divisors satisfies both the left and right cancellation laws.*
*(3) Every division ring has no zero divisors.*

**Proof.** The first assertion is immediate. For the second, if $a\neq 0$ and $ab=ac$, then $a(b-c)=0$. Since there are no zero divisors, $b-c=0$, so $b=c$. Thus the left cancellation law holds, and the right cancellation law follows similarly.

For the third, let $K$ be a division ring and let $a,b\in K$ both be nonzero. If $ab=0$, multiplying the equation on the left by $a^{-1}$ gives $b=0$, a contradiction. Hence a product of nonzero elements is nonzero, and $K$ has neither left nor right zero divisors. □

### Integral domains

**Definition A.4.22.** Let $R$ be an integral domain.

(1) For $a,b\in R$, *a divides b*, written $a\,|\,b$, if $b=ac$ for some $c\in R$. We also call $a$ a *divisor* of $b$; if $a$ does not divide $b$, write $a\nmid b$.
(2) For $a,b\in R$, if $a\,|\,b$ and $b\,|\,a$, then $a,b$ are *associates*, written $a\sim b$.
(3) Let $p\in R$ be nonzero and not invertible.
 (a) The element $p$ is an *irreducible element* if it cannot be written as a product of two noninvertible elements; that is, if $p=ab$, then $a$ or $b$ must be invertible. If $p$ is not irreducible, it is a *reducible element*. We also say simply that an irreducible or reducible element is *irreducible* or *reducible*, respectively. A *nontrivial factorization* of an element is a factorization into a product of noninvertible factors. Plainly an element is irreducible if and only if it is nonzero, is not invertible, and has no nontrivial factorization.
 (b) The element $p$ is a *prime element* if, for every $a,b$, $p\,|\,ab$ implies $p\,|\,a$ or $p\,|\,b$.
(4) Let $a_1,\dots,a_n\in R$ be not all zero. If $d\,|\,a_i$ for every $i$, then $d$ is a *common divisor* of $a_1,\dots,a_n$. A common divisor $d'$ divisible by every common divisor of the $a_i$ is a *greatest common divisor*, denoted by $\gcd(a_1,\dots,a_n)$.

**Proposition A.4.23.** *Divisibility in an integral domain R has the following properties:*

*(1)* $1\,|\,a$, $a\,|\,0$, *and* $a\,|\,a$ *for every* $a\in R$*;*
*(2) for* $a,b,c\in R$*, if* $a\,|\,b$ *and* $b\,|\,c$*, then* $a\,|\,c$*;*
*(3) for* $a,b,c\in R$*, if* $a\,|\,b$ *and* $a\,|\,c$*, then* $a\,|\,(mb+nc)$ *for all* $m,n\in R$*;*
*(4) for* $a,b,c,d\in R$*, if* $a\,|\,b$ *and* $c\,|\,d$*, then* $ac\,|\,bd$*;*
*(5) for* $a,b\in R$*, if* $c\neq 0$*, then* $ac\,|\,bc$ *if and only if* $a\,|\,b$*;*
*(6) given any invertible element (or unit) u, an element* $a\in R$ *is a unit if and only if* $a\,|\,u$*; in particular, this holds if and only if* $a\,|\,1$*.*

**Proof.** For (1), $a=1a$ and $0=a0$ make the assertions immediate.

For (2), if $a\,|\,b$ and $b\,|\,c$, there are $u,v\in R$ such that $c=ub$ and $b=va$. Then $c=(uv)a$, so $a\,|\,c$.

For (3), write $b=ua$ and $c=va$. Then $mb+nc=(mu+nv)a$, and hence $a\,|\,(mb+nc)$.

For (4), write $b=ua$ and $d=vc$. Then $bd=uv(ac)$, so $ac\,|\,bd$.

For (5), cancellation gives $bc=uac \Longleftrightarrow b=ua$, as required.

For (6), if $a$ is invertible, then $u=a(a^{-1}u)$, so $a\,|\,u$. Conversely, if $a\,|\,u$, there is a $v$ such that $u=av$. Thus $1=a(vu^{-1})$, and $vu^{-1}$ is the inverse of $a$. □

**Proposition A.4.24.** *In an integral domain:*

*(1) nonzero* $a,b$ *are associates exactly when* $a=ub$ *for some unit* $u\in R^\times$*;*
*(2) association is an equivalence relation; in particular,* $a\,|\,b$, $b\,|\,c$ *imply* $a\,|\,c$*, and* $a\sim b$, $b\sim c$ *imply* $a\sim c$*;*
*(3) of two associates, one is a unit if and only if the other is a unit; one is irreducible if and only if the other is irreducible; and one is prime if and only if the other is prime;*
*(4) all units are associates, in particular with* $1$*;*
*(5) every prime element is irreducible;*
*(6) a greatest common divisor is unique up to association: if* $a_1,\ldots,a_n$ *have two greatest common divisors* $d,d'$*, then* $d\sim d'$*.*

**Proof.** If $a=ub$ with $u$ a unit, then $b|a$, while $b=u^{-1}a$ gives $a|b$. Conversely, if $a=cb$ and $b=da$, then $a=cda$; since $a\neq 0$, cancellation gives $cd=1$, so $c$ and $d$ are units. Reflexivity and symmetry are immediate, and transitivity follows from that of divisibility together with the definition.

Now let $a=ub$ with $u$ a unit. Plainly $a$ is a unit exactly when $b$ is. Each pair of elements $c,d$ satisfying $a=cd$ corresponds one-to-one to the equation $b=(u^{-1}c)d$, and multiplication by a unit does not change whether an element is a unit; hence $a$ is irreducible exactly when $b$ is. Since $a=ub$, we have $a\,|\,cd \Longleftrightarrow b\,|\,cd$, $a\,|\,c \Longleftrightarrow b\,|\,c$, and $a\,|\,d \Longleftrightarrow b\,|\,d$. Thus the prime-element property holds for both elements or for neither.

If $a,b$ are units, then $b=(ba^{-1})a$, and $ba^{-1}$ has inverse $ab^{-1}$, so $a\sim b$. If a prime $p$ satisfies $p=ab=1ab$, then $p\,|\,ab$. Hence $p\,|\,a$ or $p\,|\,b$. Without loss of generality, suppose $p\,|\,a$ and write $a=pc$. Hence $p=pcb$; cancellation of $p$ gives $cb=1$, and $b$ is a unit. Thus $p$ is irreducible. Finally, if $d,d'$ are both greatest common divisors of $a_1,\ldots,a_n$, their defining property gives $d\,|\,d'$ and $d'\,|\,d$, so $d\sim d'$. □

**Definition A.4.25.** An integral domain $R$ is a *unique factorization domain* (UFD) if both conditions hold:

(a) Every nonzero nonunit is a product of finitely many irreducible elements. An expression $a=p_1p_2\cdots p_m$ of this form is an *irreducible factorization* of $a$.
(b) If $a=p_1p_2\cdots p_m=q_1q_2\cdots q_n$ gives two irreducible factorizations of the same element $a$, then $m=n$, and there is a permutation $\sigma\in \mathrm{S}_n$ such that $p_i\sim q_{\sigma(i)}$ for every $i$.

**Proposition A.4.26.** *Every field is an integral domain and a UFD.*

**Proof.** A field is a commutative ring whose nonzero elements are all invertible, and therefore satisfies cancellation and is an integral domain. Since it has no nonzero nonunits, the unique-factorization conditions hold vacuously. □

**Proposition A.4.27.** *In a UFD R:*

(1) *for every $a \neq 0$, there are a unit $u$ and irreducibles $p_1,\ldots,p_m$ such that $a = up_1p_2\cdots p_m$; $m=0$ is allowed, in which case $a=u$. If the same element has two such factorizations $a = up_1p_2\cdots p_m = vq_1q_2\cdots q_n$, with $u,v$ units and $p_i,q_i$ irreducible, then $m=n$; if $m \neq 0$, there is a permutation $\sigma \in \mathrm{S}_n$ that satisfies $p_i \sim q_{\sigma(i)}$ for every $i$;*

(2) *every irreducible element is prime;*

(3) *any finite family of nonzero elements $a_1,\ldots,a_s \in R$, where $s \geq 1$, has a greatest common divisor.*

**Proof.** For existence in the first assertion, if $a$ is a unit take $m=0$ and $u=a$; if $a$ is not a unit, the definition gives an irreducible factorization $a = p_1\cdots p_m$, and we take $u=1$.

Suppose the unit $a$ has two factorizations of the stated form. If $m \geq 1$, then $p_1\cdots p_m = u^{-1}a$ is invertible, so $p_1\big((p_2\cdots p_m)a^{-1}u\big) = 1$. Thus $p_1$ is invertible, contrary to its irreducibility. Hence an invertible $a$ can only be written as $a=a$.

Now suppose a noninvertible $a$ has two factorizations of the stated form. Necessarily $m,n \geq 1$. Put $p_1' = v^{-1}up_1$, so that $p_1'p_2\cdots p_m = q_1q_2\cdots q_n$. Since $v^{-1}u$ is invertible, $p_1' \sim p_1$, and proposition A.4.24 shows that $p_1'$ is irreducible. Thus $p_1'p_2\cdots p_m = q_1q_2\cdots q_n$ gives two irreducible factorizations of the same element. The UFD definition gives $m=n$ and a permutation $\sigma$ such that $p_i \sim q_{\sigma(i)}$ for $i \geq 2$, and $p_1' \sim q_{\sigma(1)}$. Since $p_1 \sim p_1'$, we also have $p_1 \sim q_{\sigma(1)}$, proving the required uniqueness.

For the second assertion, let the irreducible $p$ divide $ab$. If $a=0$ or $b=0$, then $p$ divides the corresponding zero factor. Otherwise write $ab = pc$, where $c \neq 0$, and factor the nonzero elements $a,b,c$ as a unit times irreducibles. By the first assertion, $p$ is associated with one of the irreducible factors of $a$ or $b$. Without loss of generality, let $a = uq_1\cdots q_s$ and $q_i \sim p$. Then $q_i \mid a$ and $p \mid q_i$, so $p \mid a$. Thus $p$ is prime.

For the third assertion, use (1) to collect all irreducible factors occurring in the factorizations of $a_1,\ldots,a_s$ into association classes, and choose representatives $p_1,\ldots,p_t$. We may write

$$a_i = u_i \prod_{j=1}^{t} p_j^{e_{ij}}, \qquad u_i \in R^\times, \quad e_{ij} \in \mathbb{N},$$

where we set $e_{ij}=0$ if $p_j$ did not occur in the original factorization. Set $e_j = \min_i e_{ij}$ and $d = \prod_j p_j^{e_j}$. Then $d \mid a_i$ for every $i$. If $c$ is any common divisor, then $c \neq 0$ because $a_1 \neq 0$. Factor $c$ into a unit and irreducibles. Since irreducibles in a UFD are prime, each irreducible factor $q_j$ of $c$, with multiplicity $e_j'$ (that is, the factor $q_j^{e_j'}$), must occur in every $a_i$, with $e_j' \leq e_j$; hence $c \mid d$.[7] Therefore $d = \gcd(a_1,\ldots,a_s)$. □

**Example A.4.28.** For $a,b \in \mathbb{Z}$, the number-theoretic relation $a$ divides $b$, written $a \mid b$ and meaning $b = na$ for some $n \in \mathbb{Z}$, is precisely divisibility in the ring of integers, so all the corresponding ring-theoretic properties of divisibility carry over. Accordingly:

(1) the only units are 1 and $-1$, so $a,b$ are associates exactly when $a = \pm b$;

(2) an element $p$ is irreducible or prime exactly when $|p|$ is a prime number; equivalently, the positive prime elements of $\mathbb{Z}$ are the prime numbers (equivalently, its positive irreducible elements), while its positive reducible elements are the composite numbers;

(3) ring-theoretic gcds are the usual number-theoretic gcds, up to sign;

[7] Although the factorization of $c$ may contain a unit, an element and its product with a unit are still associates.

(4) the fundamental theorem of arithmetic states that *every positive integer greater than* 1 *is a product of finitely many primes, and this factorization is unique apart from the order of its factors*. This is precisely the assertion that $\mathbb{Z}$ is a UFD. Extending from positive to negative integers only introduces a factor of $-1$, hence an associate.

Bézout's theorem states that *if integers $a,b$, not both zero, have greatest common divisor $d$, then there are integers $s,t$ such that $as+bt=d$*. To prove it, we may assume $a \geq b \geq 0$: replacing a negative integer by its opposite leaves the gcd unchanged, and changing the sign of the corresponding coefficient returns to the original statement. If $b=0$, take $s=1, t=0$; if $b|a$, take $s=0, t=1$. Now suppose $b>0$ and $b$ does not divide $a$. The Euclidean algorithm, familiar from school, computes the greatest common divisor.[8] Perform the successive divisions with remainder as follows. Denote the quotients in order by $q_1, q_2, \ldots$ and the remainders by $r_1, r_2, \ldots$; also put $r_{-1}=a$ and $r_0=b$ to make the notation uniform. We obtain

$$a=q_1b+r_1, \quad b=q_2r_1+r_2, \quad r_1=q_3r_2+r_3, \quad \ldots,$$
$$\text{until} \quad r_{n-2}=q_nr_{n-1}+r_n, \quad r_{n-1}=q_{n+1}r_n+0.$$

Stop when the remainder is zero, and write $r_{n+1}=0$ to make the notation uniform. Here $n, q_1, \ldots, q_{n+1}$ and the preceding remainders $r_1, \ldots, r_n$ are positive integers, with $0 \leq r_{i+1} < r_i$ for $1 \leq i \leq n$ and $0 < r_1 < b$. The last nonzero remainder $r_n$ is the greatest common divisor of $a,b$.

Working backwards gives $r_n = r_{n-2} - q_n r_{n-1}$. If $n \geq 2$, substitute $r_{n-1} = r_{n-3} - q_{n-1} r_{n-2}$ from the preceding step and continue back-substitution in order. If $n=1$, the first step already gives $r_1 = a - q_1 b$. Thus in every case we eventually obtain $r_n = sa + tb$. □

In ring theory this Bézout property belongs more generally to principal ideal domains. We shall not introduce that general setting, since it is not needed; later we prove the property only in the particular cases where it is used.

The field of fractions generalizes the construction of the rational numbers from the integers to an arbitrary integral domain.

**Theorem A.4.29.** *Let $R$ be an integral domain and $R^\circ = \{b \in R \mid b \neq 0\}$. On the Cartesian product $R \times R^\circ$, define $(a,b) \sim (c,d) \iff ad = bc$. This is an equivalence relation. Write the equivalence class $[(a,b)]$ as $\frac{a}{b}$. Define addition and multiplication, respectively, by*

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

*Then the quotient set $(R \times R^\circ)/\sim$ forms a field under these two operations.* It is called the *field of fractions* of $R$ and denoted by $\mathrm{Frac}(R)$. *Abbreviate $\frac{a}{1} \in \mathrm{Frac}(R)$ to $a$. The map $f: R \to \mathrm{Frac}(R)$, $a \mapsto \frac{a}{1}$, preserves addition and multiplication, and identifies each $a \in R$ with a unique element $a \in \mathrm{Frac}(R)$. Thus $R$ may be regarded directly as a subring of* $\mathrm{Frac}(R)$.

**Proof.** Reflexivity and symmetry of $\sim$ are immediate. For transitivity, if $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$, then $ad = bc$ and $cf = de$. Hence $d(af - be) = adf - bde = bcf - bde = 0$. Since $d \neq 0$ and $R$ is an integral domain, cancelling $d$ gives $af = be$, so $(a,b) \sim (e,f)$. Thus transitivity holds.

Since $R$ is an integral domain, $b,d \neq 0$ implies $bd \neq 0$, so the results of the defined addition and multiplication still have valid nonzero denominators. If $\frac{a}{b} = \frac{a'}{b'}$ and $\frac{c}{d} = \frac{c'}{d'}$, that is, $ab' = a'b$ and $cd' = c'd$, direct expansion gives $(ad+bc)b'd' = (a'd'+b'c')bd$ and $(ac)b'd' = (a'c')bd$. Hence both operations are independent of the representatives.

Associativity, commutativity, and distributivity can be checked term by term using the corresponding laws in $R$. The zero and identity are $\frac{0}{1}$ and $\frac{1}{1}$, respectively, and $-\frac{a}{b} = \frac{-a}{b}$ gives the additive

[8] The algorithm uses the following fact: if division of $a$ by $b$ leaves remainder $r$, then $\gcd(a,b) = \gcd(b,r)$. Indeed, write $a = nb + r$, where $a,b,n,r \in \mathbb{Z}$. If $d$ is a common divisor of $a,b$, that is, $d|a$ and $d|b$, then $d|(a-nb)$, hence $d|r$; thus $d$ is a common divisor of $b,r$. Conversely, if $d|b$ and $d|r$, then $a = nb + r$ gives $d|a$, so $d$ is a common divisor of $a,b$. Therefore $(a,b)$ and $(b,r)$ have the same common divisors, and consequently the same greatest common divisor up to associates.

inverse. By definition, $\frac{a}{b} = \frac{0}{1}$ if and only if $a = 0$. Thus precisely the nonzero fractions admit representatives with $a \neq 0$; in that case $\frac{b}{a}$ has a valid denominator and is the multiplicative inverse of $\frac{a}{b}$. Therefore $\mathrm{Frac}(R)$ is a field.

Finally, $\frac{a}{1} = \frac{b}{1}$ if and only if $a = b$, so we may also regard $a \in R$ as the element $a$ of $\mathrm{Frac}(R)$. The equations $a + b = c$ and $ab = c$ in $R$ are precisely $\frac{a}{1} + \frac{b}{1} = \frac{c}{1}$ and $\frac{a}{1}\frac{b}{1} = \frac{c}{1}$ in $\mathrm{Frac}(R)$. Thus $R$ can indeed be regarded as a subring of $\mathrm{Frac}(R)$. $\square$

**Example A.4.30.** $\mathbb{Q} = \mathrm{Frac}(\mathbb{Z})$, with the usual addition and multiplication of numbers.

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## Exercises

**Exercise A.58.** Let $R$ be a ring with more than one element and no zero divisors. Prove:
(1) all nonzero elements of $R$ have the same order in its additive group;
(2) if $\mathrm{char} R$ is finite, it is prime.

**Exercise A.59.** Prove $R = \{a + b\sqrt{5}\mathrm{i} \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$, and determine whether $2 + \sqrt{5}\mathrm{i}$ divides 3 in it.

**Exercise A.60.** In an integral domain, let $a, b$ be not both zero and have a greatest common divisor. If $a \sim a'$ and $b \sim b'$, prove that $a', b'$ also have a greatest common divisor and that $\gcd(a, b) \sim \gcd(a', b')$.

**Exercise A.61.** Let $R$ be an integral domain in which every nonzero, noninvertible element can be written as a product of irreducible elements and every irreducible element is prime. Prove that $R$ is a unique factorization domain.

### A.4.3 Basic Constructions of Rings and Fields

This section supplies several constructions involving rings and fields, in order to provide concrete examples.

#### Special rings and fields

**Definition A.4.31.**
(1) A subset $R$ of $\mathbb{C}$ is a *number ring* if it contains the numbers 0 and 1 and is closed under numerical addition, subtraction, and multiplication.
(2) A subset $R$ of $\mathbb{C}$ is a *number field* if it contains the numbers 0 and 1 and is closed under numerical addition, subtraction, multiplication, and division when the divisor is nonzero.

These terms have the literal book-wide meaning explained at their first use in definition 6.6.25.

Equivalently, a number ring is a unital subring of $\mathbb{C}$ with the same identity, and a number field is a subfield of $\mathbb{C}$. The sets $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are number fields, whereas $\mathbb{Z}$ is not, since a quotient of two integers need not be an integer.

**Definition A.4.32.** For a positive integer $n$, let $\mathbb{Z}_n = \{\overline{0}, \overline{1}, \overline{2}, \ldots, \overline{n-1}\}$, where a bar denotes a congruence class modulo $n$: all integers having the same remainder as $a$ are treated as the element $\overline{a}$. For example, in $\mathbb{Z}_5$ one has $\overline{-3} = \overline{2} = \overline{7} = \overline{2}$. For $\overline{i}, \overline{j} \in \mathbb{Z}_n$, define addition by $\overline{i} + \overline{j} = \overline{i+j}$ and multiplication by $\overline{i} \cdot \overline{j} = \overline{ij}$. The properties of congruences show that these operations make $\mathbb{Z}_n$ a ring, with zero $\overline{0}$; the additive inverse of $\overline{i}$ is $\overline{-i}$. This finite commutative ring is the *ring of integers modulo n*.

**Theorem A.4.33.** *If $n$ is prime, the ring $\mathbb{Z}_n$ is a field; if $n$ is composite, $\mathbb{Z}_n$ is not a field.*

**Proof.** If $n=n_1n_2$ is composite with $1<n_1,n_2<n$, then $\overline{n_1}\neq\overline{0}$ and $\overline{n_2}\neq\overline{0}$ but $\overline{n_1}\cdot\overline{n_2}=\overline{n}=\overline{0}$. Were $\mathbb{Z}_n$ a field, and were $\overline{n_1}^{-1}$ the inverse of $\overline{n_1}$, left multiplication of the preceding equality by $\overline{n_1}^{-1}$ would give $\overline{n_1}^{-1}\cdot\overline{n_1}\cdot\overline{n_2}=\overline{0}$, and hence $\overline{n_2}=\overline{0}$, a contradiction. Therefore $\mathbb{Z}_n$ is not a field.

If $n$ is prime, then every integer $i$ with $1\leq i<n$ is coprime to $n$. Bézout's identity gives integers $s,t$ with $is+nt=1$, and hence $\overline{i}\cdot\overline{s}=\overline{is+nt}=\overline{1}$. Thus $\overline{i}$ has inverse $\overline{s}$, so $\mathbb{Z}_n$ is a field. □

The following more general result has a considerably more involved proof, which we omit.

**Theorem A.4.34.** A finite field with $q$ elements, where $q$ is a positive integer, is denoted $\mathbb{F}_q$ and called a *finite field of order $q$. For a finite field $\mathbb{F}_q$, there are a prime $p$ and a positive integer $n$ such that $q=p^n$. For fixed $q$, all fields $\mathbb{F}_q$ are the same up to field isomorphism.*

In particular, $\mathbb{F}_p\cong\mathbb{Z}_p$ for prime $p$. For a general order $q=p^n$, the structure of the field is more complicated and will not be developed here. We next introduce polynomial rings.

**Definition A.4.35.** Let $R$ be a ring and let $x$ be a formal symbol, called an *indeterminate*, that is stipulated to commute with every element of $R$, so $ax=xa$ for $a\in R$.

(1) A *polynomial* over $R$ in the indeterminate $x$ is an expression

$$f(x)=\sum_{i=0}^{n}a_ix^i=a_0x^0+a_1x^1+\cdots+a_nx^n,\qquad a_i\in R.$$

Here $n$ is a finite natural number.

(2) The expression $a_ix^i$ is a *term* of $f(x)$, specifically its term of degree $i$, and $a_i$ is that term's *coefficient*, or the coefficient of degree $i$ in $f(x)$; $a_0x^0$ is the *constant term*. Terms with zero coefficient may be omitted or inserted at will, and we use the conventions $0x^i=0$, $(-a)x^i=-ax^i$, and $a_0x^0=a_0$. When $R$ is unital with identity 1, we also write $1x^i=x^i$.

(3) The polynomial all of whose coefficients vanish, namely $f(x)=0$, is the *zero polynomial*; one with $f(x)\neq 0$ is a nonzero polynomial. A polynomial with only its constant coefficient possibly nonzero, namely $f(x)=a_0x^0=a_0$, is a *constant polynomial*, so every $a\in R$ may be regarded both as an element of $R$ and as a polynomial. A polynomial with precisely one nonzero coefficient, of the form $f(x)=ax^i$, is a *monomial*.

(4) For $f(x)\neq 0$ in (1), its *degree* is the degree of its highest term with nonzero coefficient, denoted by $\deg f(x)$ or $\deg f$: $\deg f(x)=\max_{a_i\neq 0}(i)$. If $\deg f=d$, then $f(x)$ is a polynomial of degree $d$, and $a_dx^d$ and $a_d$ are its *leading term* and *leading coefficient*. In particular, a nonzero constant polynomial has degree zero. The degree of the zero polynomial is undefined, though by convention we also set $\deg 0=-\infty$. A polynomial over a unital ring is *monic* if its leading coefficient is the identity of $R$.

(5) Let $f(x)=\sum_{i=0}^{m}a_ix^i$ and $g(x)=\sum_{i=0}^{n}b_ix^i$, with missing coefficients above the written highest degrees understood to be zero. They are equal if their coefficients of degree $i$ agree for every $i$. Their sum and product are defined by

$$f(x)+g(x)=\sum_{k=0}^{\max\{m,n\}}(a_k+b_k)x^k,\quad f(x)g(x)=\sum_{k=0}^{m+n}\Big[\sum_{i=0}^{k}a_ib_{k-i}\Big]x^k.$$

Even when $R$ is noncommutative, $x$ has been stipulated to commute with its elements; nevertheless, the order of the coefficient product $a_ib_{k-i}$ in the last formula must not be reversed. All polynomials over $R$ in the indeterminate $x$ form a ring under these addition and multiplication operations, called the *polynomial ring* over $R$ in $x$ and denoted by $R[x]$. Its zero is the zero polynomial 0.

(6) For $f(x)=\sum_{i=0}^{n}a_ix^i\in R[x]$, if a ring $S$ satisfies $R\leq S$, then for $s\in S$ define its *value on substitution at $s$* by $f(s):=\sum_{i=0}^{n}a_is^i$. When $R=S$, this is simply called the value of $f$ at $s$.

**Remark.** The coefficients $a_i$ in a polynomial over a ring need not be numbers in the ordinary sense, though the definition is exactly analogous to that of the polynomials studied at school. The symbol $a_0x^0$ is only a formal device for writing the constant term uniformly and is always understood to equal $a_0$. Thus, even when evaluating $f(x)$ at 0, polynomial notation uses the convention $a_0 0^0 = a_0$.

We shall study polynomials in detail later. Polynomial rings also lead to some frequently used notation.

(1) Let $R$ be a subring of a ring $S$, with the same identity. Let $x$ be an indeterminate and $\alpha \in S$ satisfy $\alpha r = r\alpha$ for every $r \in R$. Define

$$R[\alpha] := \{f(\alpha) \mid f(x) \in R[x]\}.$$

It is readily proved to be a subring of $S$, with addition and multiplication inherited from $S$. For example, in $\mathbb{Z}[\mathrm{i}]$, where i is the imaginary unit, $\mathrm{i}^2 = -1$ reduces the terms of $f(\mathrm{i})$ to a constant term and a term of "degree one" in i, so

$$\mathbb{Z}[\mathrm{i}] = \{a + b\mathrm{i} \mid a,b \in \mathbb{Z}\},$$

called the *ring of Gaussian integers*, with addition and multiplication inherited from the complex field.

(2) For a field $F$ and an indeterminate $x$, define

$$F(x) := \left\{ \frac{f(x)}{g(x)} \middle| f(x), g(x) \in F[x],\ g(x) \neq 0 \right\} = \mathrm{Frac}(F[x])$$

to be the *rational-function field* over $F$.

(3) Let $F$ be a subfield of a field $D$, with the same identity, let $x$ be an indeterminate, and let $\alpha \in D$. Define

$$F(\alpha) := \left\{ \frac{f(\alpha)}{g(\alpha)} \middle| f(x), g(x) \in F[x],\ g(\alpha) \neq 0 \right\}.$$

It is readily proved to be a subfield of $D$, with addition and multiplication inherited from $D$. A representative example is $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a,b \in \mathbb{Q}\}$.

### Ring homomorphisms, isomorphisms, and ideals

Homomorphisms and isomorphisms of rings are defined just as for groups, except that the map $f$ must preserve both addition and multiplication.

**Definition A.4.36.** Let $R$ and $R'$ be rings, with operations $+,\cdot$ and $+',\cdot'$, respectively. A map $f: R \to R'$ is a *ring homomorphism* if, for all $a,b \in R$, $f(a) +' f(b) = f(a+b)$ and $f(a) \cdot' f(b) = f(a \cdot b)$. If such a map is surjective, we say that $R$ is homomorphic to $R'$ and write $R \sim R'$. If $f$ is bijective, $R$ and $R'$ are *isomorphic*, written $R \cong R'$, and $f$ is called a *ring isomorphism* from $R$ to $R'$. The sets of all homomorphisms and isomorphisms from $R$ to $R'$ are denoted $\mathrm{Hom}(R,R')$ and $\mathrm{Iso}(R,R')$. A homomorphism or isomorphism from $R$ to itself is an *endomorphism* or *automorphism*; their sets are $\mathrm{End}(R)$ and $\mathrm{Aut}(R)$. Plainly $\mathrm{Aut}(R)$ forms a group, called the *automorphism group*.

For a homomorphism $f$ from $R$ to $R'$, if $f(a)$ is the zero of $R'$ for every $a \in R$, then $f$ is the *zero homomorphism*; otherwise it is a *nonzero homomorphism*.

**Remark.** Because the relevant ring is clear from context, we do not distinguish the symbols for the two additions and multiplications. Thus in $f(a)f(b) = f(ab)$, the multiplication on the left is in $R'$ and that inside $f$ is in $R$.

Analogously to the opposite group, the *opposite ring* $R^{\mathrm{op}}$ has the same underlying set and addition as $R$, but its multiplication is defined by $a \circ' b = ba$. A ring *antihomomorphism* $f: R \to R'$ satisfies

$$f(a+b) = f(a) +' f(b), \qquad f(a \cdot b) = f(b) \cdot' f(a),$$

where the primed operations belong to $R'$. If $f$ is bijective, it is an *anti-isomorphism*. If $R$ and $R'$ are anti-isomorphic, it is readily seen that $R \cong (R')^{\mathrm{op}}$.

The following results are entirely analogous to the group case and are readily proved by following the corresponding group arguments.

**Theorem A.4.37.** *Let $R$,$R'$ be sets each equipped with two algebraic operations, addition and multiplication. If there is a surjective homomorphism for both operations, so $R \sim R'$, and $R$ is a ring, then $R'$ is a ring.*

**Corollary A.4.38.**
*(1) If $f: R \to R'$ is a ring homomorphism, it sends the zero of $R$ to that of $R'$, and sends the additive inverse of $a$ to the additive inverse of its image: $f(-a) = -f(a)$.*
*(2) Let $R \sim R'$ and let $f$ be a corresponding surjective homomorphism. If $R$ is commutative, so is $R'$; if $R$ is unital, so is $R'$, and $f$ sends the identity of $R$ to the identity of $R'$.*

**Remark.** Assertion (1) agrees with the group case. For the multiplicative properties in (2), however, note the additional requirement that $f$ be surjective: we have not required multiplication in a ring to form a group. For example, $f(a) = 0$ for every $a \in R$ plainly defines a homomorphism but does not preserve the identity. Although (2) differs from the group case, it is also readily proved.

As for groups, an isomorphism of rings says that the two rings have exactly the same structure. Homomorphisms and isomorphisms of division rings or fields are defined in precisely the same way, since these are rings. The following properties are left as exercises.

**Theorem A.4.39.** *Let $K$,$K'$ be sets each equipped with two algebraic operations, addition and multiplication. Suppose that there is a surjective homomorphism for both operations, so $K \sim K'$, which is nonzero (that is, $K'$ contains more than its zero element). Then:*
*(1) if $K$ is a division ring, so is $K'$;*
*(2) if $K$ is a field, so is $K'$.*

**Theorem A.4.40.** *Suppose that division rings (or fields) satisfy $K \sim K'$ and that the corresponding surjective homomorphism $f: K \to K'$ is nonzero. Then $f$ preserves zero, identity, additive inverses, and multiplicative inverses: the zero and identity of $K$ map to the zero and identity of $K'$, and, for $a \in K$, $f(-a) = -f(a)$; for $a \in K$ with $a \neq 0$, $f(a^{-1}) = [f(a)]^{-1}$.*

As with normal subgroups and quotient groups, we can define ideals and quotient rings in parallel. First we define ideals.

**Definition A.4.41.** Let $R$ be a ring and let $N$ be an additive subgroup: a nonempty subset of $R$ that forms a group under its addition. It is a *left ideal* if $ra \in N$ for all $r \in R, a \in N$, in which case it satisfies the *left absorption law*; it is a *right ideal* if $ar \in N$ for all $r \in R, a \in N$, in which case it satisfies the *right absorption law*; and it is a *two-sided ideal*, or simply *ideal*, of $R$ if it is both a left and a right ideal of $R$; write $N \trianglelefteq R$ (and $N \ntrianglelefteq R$ otherwise). By definition, every ideal $N$ is plainly a subring of $R$.

For any ring $R$, the set $\{0\}$ consisting only of its zero is readily seen to be an ideal of $R$, since $0+0=0$ and $0 \cdot 0 = 0$; it is called the *zero ideal*. The ring $R$ itself is also an ideal of $R$, called its *unit ideal*. Together the zero and unit ideals are the *trivial ideals*; any other ideals are *nontrivial ideals*. An ideal $N$ unequal to $R$ itself is a *proper ideal* of $R$, written $N \triangleleft R$.

For any $a \in R$, an ideal containing $a$ always exists, since $R \trianglelefteq R$. The intersection of two ideals containing $a$ is readily seen to be an ideal of $R$, so the intersection of all ideals containing $a$ is the least ideal containing $a$. It is the *principal ideal* $\langle a \rangle$ generated by $a$. Similarly, $\langle a_1, \ldots, a_n \rangle$ is the least ideal containing all the displayed elements.

Let $N$ be an ideal of a ring $R$. Under addition, $N$ is a subgroup of $R$; since ring addition is commutative, $N$ is a normal subgroup of $R$. A coset of $N$ in $R$ is written $a+N$, where $a \in R$. What was called coset multiplication is called *coset addition* in the additive setting: for $a,b \in R$, $(a+N)+(b+N) = (a+b)+N$. The set $R/N$ of these cosets forms an additive group under coset addition, as established in section A.3.2. We now add a second operation so that, in parallel with the group construction, the set of cosets becomes a ring.

**Theorem A.4.42.** *Let $N$ be an ideal of a ring $R$, and let $a+N, b+N$ be arbitrary cosets, with $a,b \in R$.* Define *coset multiplication* by $(a+N)(b+N) = ab+N$. *Then the set of cosets $R/N$ forms a ring under coset addition and multiplication.* It is called the *quotient ring*, or *residue class ring*. *Moreover, $R \sim R/N$.*

**Proof.** Define $h: R \to R/N$, $a \mapsto a+N$. As in exercise A.43, this is a surjective homomorphism for both addition and multiplication, so $R \sim R/N$. Since $R$ is a ring, theorem A.4.37 shows that $R/N$ is a ring. □

**Remark.** The map $h$ above is called the *natural homomorphism* from $R$ to the quotient ring $R/N$.

In parallel with the fundamental homomorphism theorem for groups, we obtain the corresponding theorem for rings.

**Theorem A.4.43** (Fundamental homomorphism theorem for rings)**.** *Let $R, R'$ be rings with $R \sim R'$, and let $f$ be a corresponding surjective homomorphism. Then:*

*(1) Write $N = \ker f$.[9] Then $N \trianglelefteq R$, so $N$ is an ideal of $R$;*

*(2) $R/N \cong R'$.*

**Proof.** For (1), as in the multiplicative-group case in section A.3.2, $N$ is plainly an additive subgroup of $R$. For $a \in N$ and $r \in R$, write $f(a) = 0'$ and $f(r) = r'$. Then $f(ra) = r'0' = 0'$ and $f(ar) = 0'r' = 0'$, so $ra, ar \in N$. Hence $N \trianglelefteq R$.

For (2), define $g: R/N \to R'$, $r+N \mapsto f(r)$. The proof of the fundamental homomorphism theorem for groups shows that $g$ is an isomorphism of the additive groups $R/N$ and $R'$. By the definition of coset multiplication, $(r+N)(s+N) = rs+N$; also $f(rs) = f(r)f(s)$. Thus $g$ is an isomorphism from $R/N$ to $R'$, as required. □

The meaning of a quotient ring is direct: quotienting by an ideal $N$ means treating two elements of $R$ as the same object whenever their difference lies in $N$; each coset consists of all elements so identified.

**Example A.4.44.** For a positive integer $n$, define $n\mathbb{Z} := \{nm \mid m \in \mathbb{Z}\}$. Under ordinary numerical addition and multiplication it is a commutative ring and an ideal of the integer ring, $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$. Indeed, $n\mathbb{Z}$ consists of all integral multiples of $n$, and the quotient packages together integers that differ by a multiple of $n$, just as a residue class in $\mathbb{Z}_n$ treats them as one element. The former writes the elements so identified as a set, while the latter represents them by a single element. Alternatively, the map $f: \mathbb{Z} \to \mathbb{Z}_n$, $m \mapsto \overline{m}$, is plainly a surjective homomorphism with $\ker f = \{\ldots, -2n, -n, 0, n, 2n, \ldots\} = n\mathbb{Z}$; the fundamental homomorphism theorem for rings therefore gives $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

### Modules and algebras

Finally, we give a definition that will be needed later.

**Definition A.4.45.** Let $A$ be a commutative unital ring. An *$A$-module* is a set $M$ equipped with addition $+: M \times M \to M$, $(m,n) \mapsto m+n$, and scalar multiplication $\cdot: A \times M \to M$, $(a,m) \mapsto am$, such that:

(1) $M$ is an abelian group under addition;

(2) scalar multiplication is associative: $(ab)m = a(bm)$ for all $a,b \in A$ and $m \in M$;

(3) writing 1 for the multiplicative identity of $A$, we have $1 \cdot m = m$ for every $m \in M$;

[9] The identity in a multiplicative group corresponds to the zero in an additive group. Thus the *kernel* of a ring homomorphism is defined here as the inverse image of the zero.

(4) both distributive laws hold for all $a,b\in A$ and $m,n\in M$: $a(m+n)=am+an$ and $(a+b)m=am+bm$.

If there are $m_1,\dots,m_r\in M$ such that every $m\in M$ can be written as $m=a_1m_1+\dots+a_rm_r$, with $a_i\in A$, then $M$ is a *finitely generated A-module*, or simply *finitely generated*. Then $\{m_1,\dots,m_r\}$ is called a set of *generators* of $M$.

**Definition A.4.46.** Let $A$ be a commutative unital ring. For a commutative unital ring $B$ and a unit-preserving ring homomorphism $\imath: A\to B$, define scalar multiplication by $a\cdot b:=\imath(a)b$. The addition on $B$ and this scalar multiplication can be verified to make $B$ an $A$-module. Then $B$, together with $\imath$, is an *A-algebra*.

For $b_1,\dots,b_r\in B$, write $A[b_1,\dots,b_r]$ for the least subring of $B$ containing the image $\imath(A)$ and $b_1,\dots,b_r$; equivalently, take the intersection of all subrings containing them. If $B=A[b_1,\dots,b_r]$, call $B$ a *finitely generated A-algebra*, or simply finitely generated, and call $b_1,\dots,b_r$ its *generators* as an $A$-algebra. It is a *finite A-algebra* if it is finitely generated as an $A$-module.

**Remark.** Modules can also be defined over noncommutative rings, but that generality is not needed here, so we have used a simplified definition. In the noncommutative case scalar multiplication requires left and right versions; interested readers may consult further references. In fact, replacing “division ring” by “unital ring” in the definition of vector spaces over general division rings in section A.5 gives general unital left and right modules.

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## Exercises

**Exercise A.62.** Verify the concepts and theorems.

(1) Prove theorem A.4.37 and corollary A.4.38.
(2) Prove theorems A.4.39 and A.4.40.
(3) Prove that $R[\alpha]$ and $F(\alpha)$, defined on page 571, are respectively a subring of $S$ and a subfield of $D$.
(4) For a ring $R$ and $a\in R$, prove that intersections of ideals containing $a$ are ideals of $R$, justifying the definition of a principal ideal on page 572.

**Exercise A.63.** For a field $F$, prove that the set $N$ of all polynomials in $F[x]$ with zero constant term is an ideal of $F[x]$.

**Exercise A.64.** For positive integers $m,n$, prove that the rings $\mathbb{Z}_m,\mathbb{Z}_n$ satisfy $\mathbb{Z}_m\sim\mathbb{Z}_n$ if and only if $n$ is a divisor of $m$.

**Exercise A.65.** Find all subrings of $\mathbb{Z}_{10}$.

**Exercise A.66.** For a field $F$, prove that $f:\mathrm{Frac}(F)\to F$, $\frac{x}{y}\mapsto xy^{-1}$, is a natural field isomorphism. Thus every element $\frac{x}{y}$ of $\mathrm{Frac}(F)$ may be regarded as the element $xy^{-1}$ of $F$.

**Exercise A.67.** Determine $\mathrm{Aut}(\mathbb{Q}(\mathrm{i}))$.

**Exercise A.68.** Let $R$ be a unital ring. For every $r\in R^\times$, define $\varphi_r: R\to R$, $x\mapsto rxr^{-1}$. Prove that $\varphi_r$ is an automorphism of $R$ and that all the maps $\varphi_r$ form a normal subgroup of $\mathrm{Aut}(R)$. *The map $\varphi_r$ is called an* inner automorphism *of R, and all inner automorphisms of R form a transformation group, denoted by* $\mathrm{Inn}(R)$.

**Exercise A.69.** If $N$ is the kernel of a surjective ring homomorphism $f: R \to R'$, prove that $f$ is an isomorphism if and only if $N = \{0\}$.

**Exercise A.70.** Prove that if $B$ is a finite $A$-algebra, then $B$ is a finitely generated $A$-algebra.

# A.5 Linear Spaces over a Division Ring

## A.5.1 Basic Concepts of Linear Spaces

### Linear spaces

The linear spaces over $\mathbb{R}$ introduced earlier extend to linear spaces over an arbitrary division ring.

**Definition A.5.1.**

(1) Let $K$ be a division ring and $V$ a set, equipped with addition

$$+ : V \times V \longrightarrow V, \qquad (a,b) \longmapsto a+b,$$

and left scalar multiplication

$$\cdot : K \times V \longrightarrow V, \qquad (\lambda, v) \longmapsto \lambda v,$$

satisfying the following conditions.

(i) $V$ is an abelian group under addition, with identity denoted by $\mathbf{0}_V$, or simply $\mathbf{0}$.
(ii) Scalar multiplication is associative: $(\lambda\mu)v = \lambda(\mu v)$ for all $\lambda, \mu \in K$ and $v \in V$.
(iii) Writing 1 for the multiplicative identity of $K^\times$, we have $1 \cdot v = v$ for every $v \in V$.
(iv) The distributive laws hold: for all $\lambda, \mu \in K$ and $v, w \in V$, $(\lambda + \mu)v = \lambda v + \mu v$ and $\lambda(v+w) = \lambda v + \lambda w$.

Then $V$ is called a *left $K$-vector space*, or a left linear space over $K$, also called a *left $K$-linear space*.

(2) Multiplication in a division ring need not be commutative, so scalar multiplication can instead be defined on the right. Let $K$ be a division ring and $V$ a set, equipped with addition

$$+ : V \times V \longrightarrow V, \qquad (a,b) \longmapsto a+b,$$

and right scalar multiplication

$$\cdot : V \times K \longrightarrow V, \qquad (v, \lambda) \longmapsto v\lambda,$$

satisfying the following conditions.

(i) $V$ is an abelian group under addition, with identity denoted by $\mathbf{0}_V$, or simply $\mathbf{0}$.
(ii) Scalar multiplication is associative: $v(\lambda\mu) = (v\lambda)\mu$ for all $\lambda, \mu \in K$ and $v \in V$.
(iii) Writing 1 for the multiplicative identity of $K^\times$, we have $v \cdot 1 = v$ for every $v \in V$.
(iv) The distributive laws hold: for all $\lambda, \mu \in K$ and $v, w \in V$, $v(\lambda + \mu) = v\lambda + v\mu$ and $(v+w)\lambda = v\lambda + w\lambda$.

Then $V$ is called a *right $K$-vector space*, or a right linear space over $K$, also called a *right $K$-linear space*.

(3) When the side of scalar multiplication is clear, a left or right space over $K$ is called simply a linear space over $K$, or a $K$-linear space. Its elements are called vectors, and the additive identity $\mathbf{0}$ is the zero vector. The inverse operation to addition is subtraction: for a vector $v$, in a left space define $-v = (-1)v$, and in a right space define $-v = v(-1)$; in either case subtraction is defined by $w - v = w + (-v)$.

(4) To emphasize that $V$ is a left space over $K$, write ${}_K V$; to emphasize that it is a right space, write $V_K$. When the division ring $K$ and the side of scalar multiplication are clear from the context, simply say "the linear space $V$."

**Remark.** If the division ring $K$ is a field and $V$ is a left $K$-linear space, its left scalar multiplication $\lambda v$ naturally defines a right scalar multiplication by $v\lambda = \lambda v$. To verify that this makes $V$ a right space, the essential point is associativity:

$$(v\lambda)\mu = \mu(v\lambda) = \mu(\lambda v) = (\mu\lambda)v = v(\mu\lambda),$$

When $K$ is commutative, $\mu\lambda = \lambda\mu$, so $(v\lambda)\mu = v(\lambda\mu)$ holds. If $K$ is noncommutative, however, this associativity law need not hold. Conversely, for a right linear space over a field $F$, right scalar multiplication $v\lambda$ naturally defines left scalar multiplication by $\lambda v = v\lambda$, making $V$ a left space. Thus left and right spaces are naturally the same notion over a field; one usually makes no distinction and simply says "a linear space over $K$."

A linear space over a field is precisely the special case of the earlier notion of a module in which the underlying ring is a field. In fact, modules can be defined over general rings, in which case left and right modules must be distinguished. We needed only modules over commutative unital rings, so the earlier definition did not give the full generality.

**Example A.5.2.** Over a division ring $\mathbb{K}$, a vector in $V = \mathbb{K}^m$ has the form[10]

$$\boldsymbol{v} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix},$$

where $a_i \in \mathbb{K}$ for $i = 1,2,\ldots,m$. Sometimes the same element is written as $[a_1 \ \ a_2 \ \ \cdots \ \ a_m]$. To distinguish the two forms, the former is called a *column vector* and the latter a *row vector*; the latter form equals $\boldsymbol{v}^{\mathsf{T}}$. We also write simply $\boldsymbol{v} = (a_1, a_2, \ldots, a_m)$. If this is made a right vector space, its addition is defined by

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_m + b_m \end{bmatrix},$$

and scalar multiplication by

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} c = \begin{bmatrix} a_1 c \\ a_2 c \\ \vdots \\ a_m c \end{bmatrix} \qquad (c \in \mathbb{K}).$$

It is readily verified that $\mathbb{K}^m$ forms the linear space $\mathbb{K}^m_{\mathbb{K}}$ over $\mathbb{K}$. One may instead define left scalar multiplication, obtaining the left space ${}_{\mathbb{K}}\mathbb{K}^m$.

**Example A.5.3.** As for matrices over the real field, we can define matrices over a general division ring. Given a division ring $\mathbb{K}$ and positive integers $m,n$, an $m \times n$ matrix over $\mathbb{K}$ is an array of $m$ rows and $n$ columns, $\boldsymbol{A} = (a_{ij})_{m\times n}$ with $a_{ij} \in \mathbb{K}$, of the form

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

When its size is clear we write simply $\boldsymbol{A} = (a_{ij})$, and $a_{ij}$ or $(\boldsymbol{A})_{ij}$ denotes its entry in row $i$, column $j$. The set of all matrices over $\mathbb{K}$ with $m$ rows and $n$ columns is denoted by $\mathrm{M}_{m\times n}(\mathbb{K})$. Put $V = \mathrm{M}_{m\times n}(\mathbb{K})$. For fixed $m,n$, $V$ also forms a linear space over $\mathbb{K}$: define addition by $(a_{ij})_{m\times n} + (b_{ij})_{m\times n} = (a_{ij} + b_{ij})_{m\times n}$, and scalar multiplication by $(a_{ij})_{m\times n} c = (a_{ij} c)_{m\times n}$, where $c \in \mathbb{K}$. It is readily verified that $V$

[10] In what follows, matrices and row or column vectors consisting of concrete entries are written in bold; abstract vector elements of a linear space are not written in bold.

is a right linear space over $\mathbb{K}$. If instead left scalar multiplication is defined by $c(a_{ij})_{m\times n} = (ca_{ij})_{m\times n}$, then $V$ is a left $\mathbb{K}$-space. A column in $\mathbb{K}^n$ is an $n\times 1$ matrix and a row is a $1\times n$ matrix.

Vectors and matrices are not the only objects that can form linear spaces; any objects satisfying the definition do so.

**Example A.5.4.** Let $K$ be a division ring and let $V = K[x]$ be its set of polynomials. Define addition in $V$ to be polynomial addition and scalar multiplication to be the ordinary product of a polynomial and a scalar. Then $V$ is readily seen to be a linear space over $K$.

### Subspaces and dimension

The theories of left and right linear spaces are entirely parallel. Unless otherwise stated, we shall use right linear spaces in what follows; the theory of left linear spaces is entirely analogous.

**Definition A.5.5.**

(1) A subset $M$ of a right $\mathbb{K}$-space $V$ is a *subspace*, or linear subspace or vector subspace, if it is closed under scalar multiplication and vector addition in $V$ and is itself a right $\mathbb{K}$-space under these operations. Plainly $V$ is a subspace of itself. A subspace unequal to $V$ is a *proper subspace*, or proper linear subspace.

(2) A family $\{x_1,x_2,\dots,x_n\}$ of vectors in a right $\mathbb{K}$-space $V$ is *linearly independent* if the equation in $a_1,\dots,a_n\in\mathbb{K}$,
$$x_1a_1 + x_2a_2 + \cdots + x_na_n = \mathbf{0},$$
has only the solution $a_1 = a_2 = \cdots = a_n = 0$; otherwise the family is *linearly dependent*.

(3) For $n$ vectors $\{x_1,x_2,\dots,x_n\}$ in $V$, define their *span* by
$$\operatorname{span}\{x_1,x_2,\dots,x_n\} := \{x_1a_1 + x_2a_2 + \cdots + x_na_n \mid a_i\in\mathbb{K}\ (i = 1,2,\dots,n)\}.$$
It is a subspace of $V$, called the linear subspace spanned by the vectors $\{x_1,x_2,\dots,x_n\}$.

**Remark.** By definition, the zero vector together with any other vector is linearly dependent.

**Theorem A.5.6.** *If $M,N$ are subspaces of a vector space $V$, then $M\cap N$ is also a subspace of $V$.*

**Proof.** It suffices to verify closure under addition and scalar multiplication. If $u,v\in M\cap N$, then $u+v\in M$ because $M$ is a subspace; similarly $u+v\in N$, so $u+v\in M\cap N$. If $u\in M\cap N$ and $c\in\mathbb{K}$, then $uc\in M$ because $M$ is a subspace; similarly $uc\in N$, so $uc\in M\cap N$. □

**Theorem A.5.7.** *Let $\mathbb{K}$ be an infinite division ring, that is, one with infinitely many elements, and let $V$ be a left or right linear space over $\mathbb{K}$. Then $V$ cannot be written as the union of finitely many proper linear subspaces.*

**Proof.** It suffices to prove the assertion for right spaces; the proof for left spaces is analogous. Suppose, to the contrary, that $V = W_1\cup\cdots\cup W_s$, where every $W_i$ is proper. Delete redundant members one at a time until the cover is irredundant, meaning that $V \neq \bigcup_{j\neq i} W_j$ for every $i$. A single proper subspace cannot cover $V$, so $s\geq 2$. Since the cover is irredundant, each $W_i$ has an element not covered by any of the other $W_j$. We may therefore choose $a\in W_1 - \bigcup_{i\neq 1} W_i$ and $b\in W_2 - \bigcup_{i\neq 2} W_i$, and put $L := \{a+b\lambda \mid \lambda\in\mathbb{K}\}$. No $W_i$ contains two distinct points of $L$. Indeed, if $a+b\lambda, a+b\mu\in W_i$ with $\lambda\neq\mu$, then their difference $b(\lambda-\mu)$ lies in $W_i$. Since $\lambda-\mu$ is invertible, $b\in W_i$; then $a+b\lambda, b\in W_i$ gives $a\in W_i$. For $i=1$ this contradicts the choice of $b$, and for $i\neq 1$ it contradicts the choice of $a$. Thus each $W_i$ contains at most one point of $L$.

If $a+b\lambda_1 = a+b\lambda_2$, subtraction gives $b(\lambda_1-\lambda_2) = \mathbf{0}$, so $\lambda_1 = \lambda_2$. Thus $\lambda\mapsto a+b\lambda$ is injective, and since $\mathbb{K}$ is infinite, $L$ is infinite and cannot be covered by the finitely many sets $W_i\cap L$, each of which contains at most one point. This is a contradiction. □

**Definition A.5.8.** For subspaces $M,N$ of a linear space $V$, their *sum* $M+N$ is the least subspace containing both. If $M\cap N=0$, it is their *direct sum*, written $M\oplus N$. Here 0 denotes the subspace consisting only of the zero vector. More generally, for subspaces $M_1,M_2,\dots,M_n$ of $V$, their sum $M_1+M_2+\cdots+M_n$ is the least subspace containing all the $M_i$. It is a direct sum, written $M_1\oplus M_2\oplus\cdots\oplus M_n$, if

$$M_i\cap(M_1+\cdots+M_{i-1}+M_{i+1}+\cdots+M_n)=0$$

for every $i=1,\dots,n$.

**Remark.** The least subspace containing specified subspaces always exists: $V$ is one such subspace, and by theorem A.5.6 the intersection of all of them is again a subspace.

**Theorem A.5.9.** *Let $M_1,\dots,M_n$ be subspaces of a vector space $V$. If their sum $N=\bigoplus_{i=1}^n M_i$ is direct, every $x\in N$ has a unique expression $x=\sum_{i=1}^n x_i$ with $x_i\in M_i$.*

**Proof.** Let $S=\left\{\sum_{i=1}^n x_i \middle| x_i\in M_i,\ i=1,2,\dots,n\right\}$. This is readily seen to be a linear subspace of $V$. Plainly $M_i\subseteq S$, and $N$ is the least linear subspace containing all the $M_i$, so $N\subseteq S$. Conversely, since $N$ is a linear subspace and $x_i\in M_i\subseteq N$, the sum $x$ belongs to $N$. Hence $S\subseteq N$, so $S=N$ and the decomposition exists.

Suppose there is another decomposition $x=\sum_{i=1}^n y_i$, with $y_i\in M_i$, and put $z_i=x_i-y_i$. Then

$$\mathbf{0}=\sum_{i=1}^n(x_i-y_i)=z_1+z_2+\cdots+z_n,\Longrightarrow \qquad z_i=-\sum_{j\neq i}z_j.$$

In the last equality, the right side is a vector in $\bigoplus_{j\neq i}M_j$ and the left side is a vector in $M_i$. By definition, their only common element is the zero vector. Thus $z_i=\mathbf{0}$, so $x_i=y_i$, and the decomposition is unique. ☐

**Lemma A.5.10.** *Let $V=\operatorname{span}\{x_1,x_2,\dots,x_n\}$ be a linear space. If $b\in V$ and some $b_j\neq 0$ in $b=x_1b_1+\cdots+x_nb_n$, then $V=\operatorname{span}\{x_1,\dots,x_{j-1},b,x_{j+1},\dots,x_n\}$; that is, $x_j$ may be replaced by $b$. To emphasize that the linear space is over $K$, we may use the subscript in* $\operatorname{span}_K$.

**Proof.** Since $x_j=bb_j^{-1}-\sum_{i\neq j}x_ib_ib_j^{-1}$, the vector $x_j$ is spanned by the family obtained after the replacement. Conversely, $b$ is spanned by the original family. The two families therefore span the same vector space. ☐

**Definition A.5.11.** A linearly independent family $\{x_1,x_2,\dots,x_n\}$ in $V$ such that $V=\operatorname{span}\{x_1,x_2,\dots,x_n\}$ is a *basis* of $V$. *A linear space can have many different bases, but the number of vectors in a basis remains the same.* This number is its *dimension*, denoted by $\dim V$; write $\dim_K V$ to emphasize the division ring $K$. By definition, the set $\{\mathbf{0}\}$ containing only the zero vector is also a linear space. Its dimension is defined to be zero.

We prove that if another linearly independent family $\{y_1,\dots,y_m\}$ is a basis of $V$, then again $m=n$. Since $y_1\in V$, the first basis gives $y_1=\sum_{i=1}^n x_ia_i$. As $y_1\neq\mathbf{0}$, some $a_i\neq 0$; without loss of generality, let $a_1\neq 0$. By lemma A.5.10, $V=\operatorname{span}\{y_1,x_2,\dots,x_n\}$.

Now express $y_2$ using the family after replacement: $y_2=y_1b_1+x_2a_2'+x_3a_3'+\cdots+x_na_n'$. If all the $a_i'$ were zero, then $y_2=y_1b_1$, contradicting linear independence. Thus, without loss of generality, let $a_2'\neq 0$. Another application of lemma A.5.10 gives $V=\operatorname{span}\{y_1,y_2,x_3,\dots,x_n\}$.

Continue replacing vectors in this way. If $m>n$, after $n$ replacements we have $V=\operatorname{span}\{y_1,\dots,y_n\}$, so $y_{n+1}$ is dependent on the first $n$ vectors, a contradiction. Hence $m\leq n$. Similarly $n\leq m$, so $m=n$.☐

The replacement process gives the following theorem. Its proof repeats the successive replacements above and is therefore omitted.

**Theorem A.5.12** (Steinitz exchange lemma)**.** *Let $V$ be a vector space. If a family $\{y_1,\dots,y_m\}$ spans all of $V$ and another family $\{x_1,\dots,x_n\}$ is linearly independent, then $n \le m$, and $m-n$ vectors $z_{n+1},\dots,z_m$ can be chosen from $\{y_1,\dots,y_m\}$ so that $V = \operatorname{span}\{x_1,\dots,x_n,z_{n+1},\dots,z_m\}$.*

**Theorem A.5.13.** *Let $\{e_1,\dots,e_n\}$ be a basis of a right linear space $V$. For every vector $x \in V$, there are unique $x_1,\dots,x_n \in \mathbb{K}$ such that $x = \sum_{i=1}^{n} e_i x_i$.* The scalars $(x_1,\dots,x_n)$ are the *coordinates* of $x$ in that basis, often represented by the column vector $[x_1 \ \cdots \ x_n]^\mathsf{T} \in \mathbb{K}^n$.

**Proof.** Suppose $x$ has another set of coordinates $(y_1,\dots,y_n)$. Then

$$\mathbf{0} = \sum_{i=1}^{n} e_i x_i - \sum_{i=1}^{n} e_i y_i = \sum_{i=1}^{n} e_i (x_i - y_i).$$

By the definition of linear independence of a basis, $x_i - y_i = 0$, so $y_i = x_i$. □

**Remark.** For a left vector space, coordinates are usually written as the row vector $[x_1 \ \cdots \ x_n]$, with $x = \sum_{i=1}^{n} x_i e_i$.

**Example A.5.14.** In the right $\mathbb{K}$-linear space $V = \mathbb{K}^n$, take $n$ column vectors $\boldsymbol{e}_1,\dots,\boldsymbol{e}_n$, where

$$\boldsymbol{e}_j = \begin{array}{c} \begin{array}{ccccccc} & & & j\text{th} & & & \\ & & & \downarrow & & & \end{array} \\ \begin{bmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{bmatrix}^\mathsf{T} \end{array}.$$

Thus the $j$th component is 1, and all others are zero. It is readily seen that $\{\boldsymbol{e}_1,\dots,\boldsymbol{e}_n\}$ is a basis of $\mathbb{K}^n$, so $\dim \mathbb{K}^n = n$. If

$$M_i = \operatorname{span}\{\boldsymbol{e}_i\} = \{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^\mathsf{T} \mid x_i \in \mathbb{K},\ x_j = 0\ (j \ne i)\},$$

then $M_i$ is a subspace of $V$, $V = \bigoplus_{i=1}^{n} M_i$, and every $\boldsymbol{x} = [x_1 \ \cdots \ x_n]^\mathsf{T}$ has the unique decomposition $\boldsymbol{x} = \sum_{i=1}^{n} \boldsymbol{e}_i x_i$.

**Lemma A.5.15.** *Let $V$ be an $n$-dimensional vector space, and let $x_1,\dots,x_m$ be linearly independent vectors in $V$, where $m < n$. Then we can add $n-m$ vectors $x_{m+1},\dots,x_n$ in $V$ such that $\{x_1,\dots,x_n\}$ is a basis of $V$.*

**Proof.** Let $M = \operatorname{span}\{x_1,\dots,x_m\}$. Since $\dim M = m < n = \dim V$, choose $x_{m+1} \in V \setminus M$. It is independent of $x_1,\dots,x_m$, by definition (otherwise $x_{m+1} \in M$). If $\operatorname{span}\{x_1,\dots,x_{m+1}\}$ still does not span $V$, continue adding vectors in the same way.

Alternatively, choose a basis of $V$ and apply the Steinitz exchange lemma to it and $\{x_1,\dots,x_m\}$. □

**Lemma A.5.16.** *Let $V$ be a finite-dimensional vector space, and put $n = \dim V$. Every linearly independent family of $n$ vectors in $V$ is a basis of $V$.*

**Proof.** Let $M$ be the subspace spanned by these $n$ linearly independent vectors. If $M \subsetneqq V$, any $x \in V \setminus M$ would be independent of the original $n$ vectors by the argument of the preceding lemma, forcing $\dim V \ge n+1$, a contradiction. □

**Theorem A.5.17** (Dimension formula)**.** *For subspaces $M, N$ of a finite-dimensional vector space $V$,*

$$\dim(M+N) = \dim M + \dim N - \dim(M \cap N).$$

**Proof.** Since $M\cap N$ is a linear subspace, it has a basis $u_1,\dots,u_r$. By lemma A.5.15, add $v_1,\dots,v_p$ to obtain the basis $\{u_1,\dots,u_r,v_1,\dots,v_p\}$ of $M$. Adding $w_1,\dots,w_q$ instead gives the basis $\{u_1,\dots,u_r,w_1,\dots,w_q\}$ of $N$. Thus $\dim(M\cap N)=r$, $\dim M=r+p$, and $\dim N=r+q$.

We prove that $\{u_1,\dots,u_r,v_1,\dots,v_p,w_1,\dots,w_q\}$ is a basis of $M+N$. This family spans $M+N$: every $x\in M+N$ has the form $x=m+n$, with $m\in M$ and $n\in N$, and $m,n$ can be expanded in the displayed bases. If the union were dependent, there would be scalars $a_i,b_j,c_k\in\mathbb{K}$, not all zero, such that

$$\sum_{k=1}^{q} w_k c_k = -\sum_{i=1}^{r} u_i a_i - \sum_{j=1}^{p} v_j b_j.$$

The left side lies in $N$ and the right side in $M$, so both lie in $M\cap N$. Thus there are scalars $d_i\in\mathbb{K}$ such that

$$\sum_{k=1}^{q} w_k c_k = \sum_{i=1}^{r} u_i d_i.$$

The vectors $u_i,w_k$ are linearly independent, so all $c_k=d_i=0$. Similarly, all $b_j=0$, leaving $\sum_{i=1}^{r} u_i a_i = \mathbf{0}$, which forces all $a_i=0$. This contradicts the assumption that the scalars are not all zero. Therefore $\{u_1,\dots,u_r,v_1,\dots,v_p,w_1,\dots,w_q\}$ is a basis of $M+N$, so $\dim(M+N)=r+p+q$, as required. □

This subsection has used right linear spaces, so "span" uses right scalar multiplication. In the noncommutative case a left vector space instead uses

$$\operatorname{span}\{x_1,\dots,x_n\} := \{a_1x_1+\cdots+a_nx_n \mid a_i\in\mathbb{K}\},$$

with scalars on the left. We generally continue to use right spaces as examples. When the distinction needs emphasis, write $\operatorname{span}^{\mathrm{R}}$ and $\operatorname{span}^{\mathrm{L}}$ for right and left spans, respectively; likewise use $\dim^{\mathrm{R}}$ and $\dim^{\mathrm{L}}$.

The dimension of a linear space may be infinite. For example, the set of polynomials $K[x]$ over a division ring $K$, regarded as a linear space, has basis $\{1,x,x^2,x^3,\dots,x^n,\dots\}$. Also, a set $\mathbb{K}^n$ need not be regarded only as a linear space over $\mathbb{K}$, as the following example shows.

**Example A.5.18.** $V=\mathbb{C}^n$ is a vector space over $\mathbb{C}$. If $\boldsymbol{e}_j$ denotes the vector whose $j$th component is 1 and whose other components are 0, then $\{\boldsymbol{e}_1,\dots,\boldsymbol{e}_n\}$ is a basis and $\dim_{\mathbb{C}} V=n$.

The same set $V$ is also a linear space over $\mathbb{R}$. Write $\boldsymbol{e}_j$ for the vector whose $j$th component is 1 and whose other components are 0, and $\boldsymbol{e}_j'$ for the vector whose $j$th component is i and whose other components are 0. No two nonzero real numbers $a_1,a_2$ satisfy $\boldsymbol{e}_j a_1+\boldsymbol{e}_j' a_2=\mathbf{0}$, since $a_1+a_2\mathrm{i}=0$ has no nonzero real solution. Hence $\{\boldsymbol{e}_1,\dots,\boldsymbol{e}_n,\boldsymbol{e}_1',\dots,\boldsymbol{e}_n'\}$ is a basis of $V$ as a linear space over $\mathbb{R}$, and $\dim_{\mathbb{R}} V=2n$.

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## Exercises

**Exercise A.71.** Write the left-space versions of every result in this subsection stated only for right linear spaces, giving the forms explicitly.

**Exercise A.72.** Decide whether each is a vector space and, if it is, find its dimension when that dimension is finite.

(1) $\mathbb{R}^n$ over $\mathbb{C}$;
(2) $\mathbb{R}^n$ over $\mathbb{Q}[\sqrt{2}]$;
(3) all real-valued functions with the same open domain $X$, over $\mathbb{R}$;
(4) the set $\mathbb{C}$ of complex numbers over $\mathbb{R}$.

**Exercise A.73.** Consider the quaternion division ring $\mathbb{H}$. For $\mathbb{K}=\mathbb{R},\mathbb{C},\mathbb{H}$ in turn, determine whether $\mathbb{H}^n$ can be a left $\mathbb{K}$-space and whether it can be a right $\mathbb{K}$-space. Whenever it can, find its dimension.

**Exercise A.74.** If a linear space satisfies $V=\bigoplus_{i=1}^n V_i$, prove $\dim V=\sum_{i=1}^n \dim V_i$.

**Exercise A.75.** For right linear spaces $V_1,V_2$ over $\mathbb{K}$, define their *direct product* $V_1\times V_2 := \{(v_1,v_2) \mid v_1\in V_1, v_2\in V_2\}$, with addition given by $(v_1,v_2)+(v_1',v_2')=(v_1+v_1',v_2+v_2')$, right scalar multiplication by $c\in\mathbb{K}$ given by $(v_1,v_2)c=(v_1c,v_2c)$, and zero vector $(\mathbf{0}_{V_1},\mathbf{0}_{V_2})$. Prove:

(1) $V_1\times V_2$ is also a right $\mathbb{K}$-linear space;

(2) $\dim(V_1\times V_2)=\dim V_1+\dim V_2$.

## A.5.2 Linear Maps and Linear Spaces

### Quotient spaces

We first define quotient spaces, another way to construct a linear space and one that will also be used below in the study of linear maps.

**Definition A.5.19.** For a right linear space $V$ over $\mathbb{K}$ and a subspace $U$, define the *quotient space* $V/U$ as follows. Each element of $V/U$ is an equivalence class $[v]$ of a vector in $V$, where the equivalence relation on vectors of $V$ is given by $v_1\sim v_2$ if and only if $v_1-v_2\in U$.

For $v,w\in V$, define the operations on $V/U$ naturally from those on $V$: addition is $[v]+[w]=[v+w]$, and right scalar multiplication is $[v]c=[vc]$. Its zero is $[\mathbf{0}_V]$, which is also the class $[u]$ of any $u\in U$. It is readily verified that $V/U$ forms a right linear space over $\mathbb{K}$.

**Remark.** The linear space itself is an additive group, and $V/U$ is its quotient group. Since addition is commutative, $(U,+)\trianglelefteq(V,+)$. For $v\in V$, the coset $v+U=\{v+u \mid u\in U\}$ is exactly the equivalence class $[v]$. The quotient group $(V/U,+)$ is the set of all these distinct cosets. Moreover, if $v_1-v_2\in U$, then $(v_1-v_2)c\in U$ because $U$ is closed under scalar multiplication. Hence $[v]c=[vc]$ is well defined and gives the quotient group its natural linear-space structure.

**Theorem A.5.20.** *Let $V$ be a finite-dimensional vector space, and let $U\subseteq V$ be a subspace. Then*
$$\dim(V/U)=\dim V-\dim U.$$

**Proof.** Let $\{u_1,\dots,u_r\}$ be a basis of $U$. By lemma A.5.15, choose suitable $v_i\in V-U$ to extend it to a basis $\{u_1,\dots,u_r,v_1,\dots,v_s\}$ of $V$. Thus $\dim V=r+s$ and $\dim U=r$. Every $x\in V$ has an expression $x=\sum_{i=1}^r u_ia_i+\sum_{i=1}^s v_ib_i$, and therefore
$$[x]=\left[\sum_{i=1}^r u_ia_i+\sum_{i=1}^s v_ib_i\right]=\sum_{i=1}^r [u_i]a_i+\sum_{i=1}^s [v_i]b_i=[\mathbf{0}]+\sum_{i=1}^s [v_i]b_i=\sum_{i=1}^s [v_i]b_i.$$
Thus $\operatorname{span}\{[v_1],\dots,[v_s]\}=V/U$.

Now consider the case $[x]=[\mathbf{0}]$. If $[\mathbf{0}]=[\sum_{i=1}^s v_ib_i]$, then $\sum_{i=1}^s v_ib_i\in U$, so $\sum_{i=1}^s v_ib_i=\sum_{i=1}^r u_ic_i$ for some scalars $c_i\in\mathbb{K}$. Since $\{u_1,\dots,u_r,v_1,\dots,v_s\}$ is a basis of $V$, its independence forces every $b_i=0$. Hence $\{[v_1],\dots,[v_s]\}$ is a basis of $V/U$, so $\dim(V/U)=s$, as required. □

**Example A.5.21.** In the linear space $V=\mathbb{R}^2$ over $\mathbb{R}$, take the subspace $U=\left\{\begin{bmatrix} t\\ 0\end{bmatrix} \middle| t\in\mathbb{R}\right\}$. For any given $\boldsymbol{v}=\begin{bmatrix} a\\ b\end{bmatrix}\in V$, its equivalence class is readily seen to be $[\boldsymbol{v}]=\left\{\begin{bmatrix} t\\ b\end{bmatrix} \middle| t\in\mathbb{R}\right\}$. Take $[\boldsymbol{e}_2]=\left\{\begin{bmatrix} t\\ 1\end{bmatrix} \middle| t\in\mathbb{R}\right\}$. It can serve as a basis of $V/U$, since the preceding class $[\boldsymbol{v}]\in V/U$ satisfies $[\boldsymbol{v}]=[\boldsymbol{e}_2]b$. Thus $\dim(V/U)=1$, verifying $\dim(V/U)=\dim V-\dim U=2-1$.

**Linear maps over a division ring**

We now study maps of linear spaces, beginning with maps between right vector spaces.

**Definition A.5.22.** Let $V, W$ be right linear spaces over a division ring $\mathbb{K}$. A map $f: V \to W$ is *right linear* if $f(xa+yb) = f(x)a + f(y)b$ for all $x, y \in V$ and $a, b \in \mathbb{K}$. A right linear map $V \to V$ is a *right linear transformation*. If such an $f$ is bijective, $V, W$ are *linearly isomorphic*, or simply isomorphic when there is no ambiguity, written $V \cong W$; $f$ is a *linear isomorphism* between them.

**Remark.** For left linear spaces, the condition for a *left linear map* and a *left linear transformation* is $f(ax+by) = af(x) + bf(y)$. Over a field, or when the side is clear, we simply say linear map and linear transformation.

**Proposition A.5.23.** *A linear map $f: V_{\mathbb{K}} \to W_{\mathbb{K}}$ has the following basic properties:*

*(1)* $f(\mathbf{0}_V) = \mathbf{0}_W$*;*
*(2)* $\ker f := \{v \mid f(v) = \mathbf{0}_W\}$ *is a subspace of* $V$*,* called the *kernel* of $f$;
*(3)* $\operatorname{im} f := \{f(v) \mid v \in V\}$ *is a subspace of* $W$*,* called the *image* of $f$;
*(4) its values on a basis determine it uniquely: if* $\{e_1, \ldots, e_n\}$ *is a basis of* $V$ *and* $v = \sum_{i=1}^{n} e_i a_i$*, with* $a_i \in \mathbb{K}$*, then* $f(v) = \sum_{i=1}^{n} f(e_i) a_i$*;*
*(5) it is injective exactly when* $\ker f = 0$*, and surjective exactly when* $\operatorname{im} f = W$*;*
*(6) if* $f$ *is bijective and* $V, W$ *have finite dimensions, then* $\dim V = \dim W$.

**Proof.** For (1), the definition gives $f(\mathbf{0}) = f(\mathbf{0}+\mathbf{0}) = f(\mathbf{0})2$, so $f(\mathbf{0}) = \mathbf{0}$.

For (2), if $u, v \in \ker f$, then $f(ua+vb) = f(u)a + f(v)b = \mathbf{0}$, so $ua + vb \in \ker f$. Thus the kernel is closed under addition and scalar multiplication.

For (3), if $u', v' \in \operatorname{im} f$, there are $u, v \in V$ with $f(u) = u'$ and $f(v) = v'$. Then $u'a + v'b = f(ua+vb) \in \operatorname{im} f$, so the image is closed under addition and scalar multiplication.

Assertion (4) is immediate.

For (5), if $f$ is injective, (1) shows that necessarily $\ker f = 0$. Conversely, if $\ker f = 0$, take any $u', v' \in f(V) \subseteq W$ and write $f(u) = u'$, $f(v) = v'$. We have $u' - v' = f(u-v)$, so $u' = v' \iff f(u-v) = \mathbf{0}_W$. Since $\ker f = 0$, this gives $u' = v' \iff u = v$. Hence $f$ is injective. The assertion that $f$ is surjective if and only if $\operatorname{im} f = W$ is immediate.

For (6), it is readily proved that a basis $\{e_1, \ldots, e_n\}$ of $V$ is sent by $f$ to $\{f(e_1), \ldots, f(e_n)\}$, which is a basis of $W$. The detailed verification is left to the reader. $\square$

**Remark.** A linear space is an abelian group under addition, and a linear map $f: V \to W$ is also a group homomorphism $(V, +) \to (W, +)$. Its kernel and image are precisely its kernel and image as a group homomorphism, and a linear isomorphism is also an isomorphism of the additive groups. Thus we borrow $\operatorname{Hom}(V, W)$ for the set of all linear maps $V \to W$, and $\operatorname{Iso}(V, W)$ for the set of all linear isomorphisms. Every linear map is a group homomorphism, but a homomorphism $(V, +) \to (W, +)$ need not be linear. In the context of linear spaces, these notations require preservation of scalar multiplication as well.

Ordinary linear maps require the two spaces to be over the same division ring and to use the same side. When these data need emphasis, we attach them to the notation; for example, $\operatorname{Iso}^{\mathrm{R}}_{\mathbb{K}}(V, W)$ denotes the set of isomorphisms between right linear spaces over $\mathbb{K}$. For left linear spaces, property (4) becomes $f[\sum_{i=1}^{n} a_i e_i] = \sum_{i=1}^{n} a_i f(e_i)$.

**Theorem A.5.24** (Rank–nullity theorem)**.** *For a linear map $f: V \to W$ between finite-dimensional vector spaces on the same side over the same division ring,*

$$\dim V = \dim \ker f + \dim \operatorname{im} f.$$

Here $\dim\ker f$ is the *nullity* of $f$ and $\dim\operatorname{im} f$ is its *rank*, denoted by $\operatorname{rank} f$.[11]

**Proof.** Regard $V, W$ as additive groups and $f$ as a group homomorphism. The fundamental homomorphism theorem gives $(V,+)/\ker f \cong (\operatorname{im} f,+)$, with the natural isomorphism $\sigma : V/\ker f \to \operatorname{im} f$, $(v+\ker f) \mapsto f(v)$. One can check that $\sigma$ also preserves scalar multiplication, so it is a linear isomorphism $V/\ker f \cong \operatorname{im} f$. By theorem A.5.20,

$$\dim\operatorname{im} f = \dim(V/\ker f) = \dim V - \dim\ker f,$$

which is the required identity. □

**Corollary A.5.25.** *Let $V, W$ be vector spaces on the same side over a division ring $\mathbb{K}$. If $\dim V = \dim W < \infty$, every injective linear map $f\colon V \to W$ is bijective.*

**Proof.** Injectivity gives $\dim\ker f = 0$, so rank–nullity yields $\dim W = \dim\operatorname{im} f$. Since $\operatorname{im} f \subseteq W$, a basis of $\operatorname{im} f$ consists of $\dim W$ independent vectors and, by lemma A.5.16, is also a basis of $W$. Hence $\operatorname{im} f = W$, so $f$ is surjective and therefore bijective. □

### Dual spaces

We continue to use right vector spaces as the principal case.

**Definition A.5.26.** For a right $\mathbb{K}$-space $V$, its *dual space* is $V^* = \operatorname{Hom}^{\mathrm{R}}_{\mathbb{K}}(V,\mathbb{K})$, the set of all right linear maps $f : V_{\mathbb{K}} \to \mathbb{K}$, where $\mathbb{K}$ itself is regarded as the right space $\mathbb{K}^1_{\mathbb{K}}$. The star is a usual notation for a dual space. For $f, g \in V^*$ define pointwise addition by $(f+g)(x) = f(x) + g(x)$ for every $x \in V$; plainly $f+g \in V^*$. For $c \in \mathbb{K}$ define left scalar multiplication by $(cf)(x) = cf(x)$ for every $x \in V$. This gives $cf \in V^*$, because $(cf)(xa) = cf(xa) = c(f(x)a) = (cf(x))a$ for every $a \in \mathbb{K}$. Thus $V^*$ is naturally a left $\mathbb{K}$-space. If $\dim^{\mathrm{R}}_{\mathbb{K}} V = n$ and $\{e_1,\dots,e_n\}$ is a basis of $V$, every $x \in V$ has a unique expression $x = \sum_{i=1}^n e_i x^i$. The *dual basis* $\{e^1, e^2, \dots, e^n\}$ in $V^*$ is defined by $e^i(e_j) = \delta^i{}_j$, where $\delta^i{}_j$ is the *Kronecker delta*, equal to 1 if $i=j$ and to 0 otherwise. These functionals form a basis of $V^*$. Every $f \in V^*$ has a unique expression $f = \sum_{i=1}^n f_i e^i$, with $f_i \in \mathbb{K}$; the tuple $(f_1, f_2, \dots, f_n)$ gives the coordinates of $f$ in this dual basis. Since $V^*$ is a left linear space, these coordinates are often written as the row vector $[f_1 \ \cdots \ f_n]$.

**Remark.** Superscripts on $e^i$ are labels, not powers. The dual basis is often labeled with upper indices and the basis of $V$ with lower indices. The coordinates $x^i$ of $x \in V$ then use upper indices, while the coordinates $f_i$ of a vector in $V^*$ use lower indices. Every summed index consequently appears once up and once down, making the notation clearer. One may instead use lower indices throughout and denote dual basis vectors by $e_i^*$. Likewise, $\delta^i{}_j, \delta_i{}^j, \delta_{ij}, \delta^{ij}$ all mean 1 for $i=j$ and 0 otherwise; only their index placement differs. The dual of a left space is analogously a right space.

We verify that $\{e^1, e^2, \dots, e^n\}$ really is a basis of $V^*$. For any $f \in V^*$, put $f_i = f(e_i)$. Then for $x \in V$,

$$\begin{aligned}\left[\sum_{i=1}^n f_i e^i\right]\left[\sum_{j=1}^n e_j x^j\right] &= \sum_{i=1}^n f_i e^i\left[\sum_{j=1}^n e_j x^j\right] = \sum_{i=1}^n f_i \sum_{j=1}^n e^i(e_j x^j) = \sum_{i=1}^n f_i \sum_{j=1}^n e^i(e_j) x^j \\ &= \sum_{i=1}^n f_i \sum_{j=1}^n \delta_{ij} x^j = \sum_{i=1}^n f_i x^i = \sum_{i=1}^n f(e_i) x^i = f\left[\sum_{i=1}^n e_i x^i\right] = f(x).\end{aligned}$$

Hence the $e^i$ span $V^*$. If $\sum_{i=1}^n c_i e^i = \mathbf{0}$, evaluation at $e_j$ on both sides gives $0 = \sum_{i=1}^n c_i e^i(e_j) = \sum_{i=1}^n c_i \delta^i{}_j = c_j$. Thus every coefficient $c_i = 0$, so they are also linearly independent. □

Once $\{e^1, \dots, e^n\}$ is known to be a basis, the following properties are immediate.

[11] To emphasize that $f$ is a left or right linear map between linear spaces over a particular division ring $\mathbb{K}$, one may again use subscripts and superscripts. For example, $\operatorname{rank}^{\mathrm{R}}_{\mathbb{K}} f$ emphasizes that $V, W$ are both regarded as right $\mathbb{K}$-linear spaces.

**Proposition A.5.27.** *Let $V$ be an $n$-dimensional right vector space over a division ring $\mathbb{K}$, with basis $\{e_1,\dots,e_n\}$ and dual basis $\{e^1,\dots,e^n\}$. For $f\in V^*$ and $x\in V$, write $f_i=f(e_i)$ and $x=\sum_{i=1}^n e_ix^i$. Then*

*(1) the space and its dual have the same dimension:* $\dim^{\mathrm{L}}_{\mathbb{K}} V^*=\dim^{\mathrm{R}}_{\mathbb{K}} V$*;*
*(2) in the basis $\{e^1,\dots,e^n\}$ of $(V_{\mathbb{K}})^*$, $f=\sum_{i=1}^n f_ie^i$, where $f_i=f(e_i)$;*
*(3) if $x\in V_{\mathbb{K}}$ has coordinates $(x^1,\dots,x^n)$ and $f$ has coordinates $(f_1,\dots,f_n)$ in the dual space, then $f(x)=\sum_{i=1}^n f_ix^i$.*

**Example A.5.28.** For $V=\mathbb{R}^3$ with standard basis

$$\boldsymbol{e}_1=\begin{bmatrix}1\\0\\0\end{bmatrix},\qquad \boldsymbol{e}_2=\begin{bmatrix}0\\1\\0\end{bmatrix},\qquad \boldsymbol{e}_3=\begin{bmatrix}0\\0\\1\end{bmatrix},$$

the dual basis, regarded as linear maps $V\to\mathbb{R}$, is represented by

$$\boldsymbol{e}^1=\begin{bmatrix}1&0&0\end{bmatrix},\qquad \boldsymbol{e}^2=\begin{bmatrix}0&1&0\end{bmatrix},\qquad \boldsymbol{e}^3=\begin{bmatrix}0&0&1\end{bmatrix}.$$

Here $\boldsymbol{e}^i$, as a linear map $V\to\mathbb{R}$, acts on a column vector $\boldsymbol{X}\in\mathbb{R}^3$ by $\boldsymbol{X}\mapsto\boldsymbol{e}^i\boldsymbol{X}$, and $\boldsymbol{e}^i\boldsymbol{e}_j=\delta^i{}_j$, so these do form the dual basis.

For general right linear spaces $V,W$ over a division ring $\mathbb{K}$, $\mathrm{Hom}^{\mathrm{R}}_{\mathbb{K}}(V,W)$ naturally has addition $(f+g)(x)=f(x)+g(x)$ but does not in general carry a natural $\mathbb{K}$-space structure. For example, defining $(cf)(x)=cf(x)$ would require a natural left scalar multiplication on the right space $W$. The dual is special because $W=\mathbb{K}$ has its own left multiplication, so $cf(x)\in\mathbb{K}$. When $\mathbb{K}$ is a field, left and right scalar multiplication agree and $\mathrm{Hom}_{\mathbb{K}}(V,W)$ does naturally form a $\mathbb{K}$-space. After bases have been selected, maps can also be represented by matrices, and the matrix set is a vector space; that basis-dependent fact does not furnish an intrinsic $\mathbb{K}$-space structure on $\mathrm{Hom}^{\mathrm{R}}_{\mathbb{K}}(V,W)$ over a general noncommutative division ring.

**Definition A.5.29.** For a right linear map $T:V\to W$, its *dual map* is

$$T^*:W^*\to V^*,\qquad \varphi\mapsto T^*(\varphi)=\varphi\circ T.$$

The star is also a usual notation for a dual map.

**Proposition A.5.30.** *The dual of a right linear map is left linear.*

**Proof.** Using the notation of the definition, preservation of addition is easy to verify. To check preservation of scalar multiplication, let $c\in\mathbb{K}$. Then $T^*(c\varphi)=(c\varphi)\circ T=c(\varphi\circ T)=cT^*(\varphi)$. □

**Theorem A.5.31.** *Let $\mathbb{K}$ be a division ring, let $V,W$ be finite-dimensional right $\mathbb{K}$-vector spaces, and let $f:V\to W$ be a right linear map. Then* $\dim^{\mathrm{L}}_{\mathbb{K}}\operatorname{im}f^*=\dim^{\mathrm{R}}_{\mathbb{K}}\operatorname{im}f$*, so* $\operatorname{rank}f=\operatorname{rank}f^*$.

**Proof.** Put $r=\dim^{\mathrm{R}}_{\mathbb{K}}\operatorname{im}f$ and $m=\dim^{\mathrm{R}}_{\mathbb{K}}W$, and define the *annihilator* of $\operatorname{im}f$ in $W^*$ by

$$(\operatorname{im}f)^\circ=\{\varphi\in W^*\mid\varphi(w)=0\text{ for every }w\in\operatorname{im}f\}.$$

By definition, $\ker f^*=(\operatorname{im}f)^\circ$, since $\varphi\in\ker f^*$ exactly when $\varphi\circ f=0$.

We now compute the dimension of $(\operatorname{im}f)^\circ$. Choose a right basis $\{w_1,\dots,w_r\}$ of $\operatorname{im}f$. Extend it to obtain $\{w_1,\dots,w_r,w_{r+1},\dots,w_m\}$ as a right basis of $W$. Let the dual basis be $\{w^1,\dots,w^m\}$. Every $\varphi\in W^*$ can be written uniquely as $\varphi=\sum_{i=1}^m a_iw^i$, and then $\varphi(w_j)=a_j$.

Consequently, $\varphi\in(\operatorname{im}f)^\circ$ exactly when $a_1=\cdots=a_r=0$, so $(\operatorname{im}f)^\circ=\operatorname{span}^{\mathrm{L}}\{w^{r+1},\dots,w^m\}$. Thus $\dim^{\mathrm{L}}_{\mathbb{K}}\ker f^*=\dim^{\mathrm{L}}_{\mathbb{K}}(\operatorname{im}f)^\circ=m-r$. Rank–nullity now gives

$$\dim^{\mathrm{L}}_{\mathbb{K}}\operatorname{im}f^*=\dim^{\mathrm{L}}_{\mathbb{K}}W^*-\dim^{\mathrm{L}}_{\mathbb{K}}\ker f^*=m-(m-r)=r.$$

□

**Theorem A.5.32.** *If $V$ is a finite-dimensional right $\mathbb{K}$-vector space, then $V^{**} = (V^*)^*$ is naturally a right $\mathbb{K}$-vector space, and there is a natural isomorphism $V \cong V^{**}$.*

**Proof.** Define $\eta : V \to V^{**}$, $x \mapsto v_x$, by requiring $v_x(f) = f(x)$ for $f \in V^*$. For $a, b \in \mathbb{K}$ and $x, y \in V$, we have[12]

$$(\eta(xa+yb))(f) = f(xa+yb) = f(x)a + f(y)b = (\eta(x))(f)a + (\eta(y))(f)b,$$

so $\eta$ is right linear.

For a given $x \in V \setminus \{\mathbf{0}_V\}$, extend it to a basis $\{x, e_2, \dots, e_n\}$ of $V$. By proposition A.5.23(4), there is an $f \in V^*$ with $f(x) = 1$ and $f(e_i) = 0$. Thus for $x \neq y$, there is an $f$ such that $(v_x - v_y)f = f(x-y) = f(x) - f(y) \neq 0$, so $\eta$ is injective. Since $V$ and $V^{**}$ have the same finite dimension, corollary A.5.25 makes $\eta$ bijective; it is therefore an isomorphism $V \to V^{**}$. □

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## Exercises

**Exercise A.76.**

(1) Prove proposition A.5.23(6) and proposition A.5.27 in detail.

(2) Verify directly that every dual basis vector $e^i$ specified in definition A.5.26 is indeed a linear map $V \to \mathbb{K}$.

**Exercise A.77.** Write explicitly the left-linear-space versions of all results in this subsection stated only for right linear spaces.

**Exercise A.78.** Let $U$ be a subspace of a linear space $V$ and define the *quotient map* $\pi : V \to V/U$, $v \mapsto [v]$. Prove that $\pi$ is linear and surjective, and find $\ker \pi$.

**Exercise A.79.** Let $f : V \to W$ be a linear map between finite-dimensional linear spaces. Prove that $f$ is an isomorphism if and only if it carries a basis of $V$ to a basis of $W$.

**Exercise A.80.** Let $f$ be a linear transformation of a finite-dimensional linear space $V$, with $f^2 = f$. Prove $V = \ker f \oplus \operatorname{im} f$.

**Exercise A.81.** Let $V$ be an $n$-dimensional right $\mathbb{K}$-linear space and let $f \in V^*$ be a nonzero map. Prove that $\ker f$ is an $(n-1)$-dimensional subspace of $V$. Conversely, if $U$ is an $(n-1)$-dimensional subspace of $V$, prove that there is a nonzero $f \in V^*$ with $U = \ker f$.

**Exercise A.82.** Let $V$ be an $n$-dimensional right $\mathbb{K}$-linear space, and let $f, g \in V^*$. If $\ker f = \ker g$, prove that $f = ag$ for some $a \in \mathbb{K}$.

**Exercise A.83.** For linear maps $f : U_{\mathbb{K}} \to V_{\mathbb{K}}$ and $g : V_{\mathbb{K}} \to W_{\mathbb{K}}$, prove that their dual maps satisfy $(g \circ f)^* = f^* g^*$.

**Exercise A.84.** In the right linear space $V = \mathbb{H}^3$ over the quaternion division ring $\mathbb{H}$, take three column vectors

$$\boldsymbol{e}_1 = \begin{bmatrix} 1 & \mathbf{i} & 0 \end{bmatrix}^{\mathsf{T}}, \quad \boldsymbol{e}_2 = \begin{bmatrix} 0 & 1 & \mathbf{j} \end{bmatrix}^{\mathsf{T}}, \quad \boldsymbol{e}_3 = \begin{bmatrix} \mathbf{k} & 0 & 1 \end{bmatrix}^{\mathsf{T}}.$$

Write $V^*$ for its dual.

[12] Recall that $V^{**}$ is a right linear space, with right scalar multiplication defined by $(\eta(x)a)f = (\eta(x))(f)a$.

(1) Verify that $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ is a basis of $V$, and find the corresponding dual basis explicitly as maps in $\mathrm{Hom}(V, \mathbb{H})$.
(2) Find the coordinates, in the dual basis from (1), of the map in $V^*$

$$f : (\boldsymbol{e}_1 x + \boldsymbol{e}_2 y + \boldsymbol{e}_3 z) \longmapsto (x - 2\mathbf{k}y + (\mathbf{i} - \mathbf{j})z),$$

where $(x, y, z)$ is the coordinate triple of the vector in $V$.

**Exercise A.85.** Let $V$ be a finite-dimensional linear space, and let $M, N$ be subspaces of $V$ with $M \subseteq N$. Prove that $M$ is a subspace of $N$, and that $\dim(V/M) - \dim(V/N) = \dim(N/M)$.

### A.5.3 Matrices and Linear Maps

**Matrices over a division ring**

As preparation, we extend the earlier definitions in full to a general division ring. Matrices over $\mathbb{K}$ have already been defined; the associated concepts are the same as for real matrices. A *zero matrix* has only zero elements of $\mathbb{K}$ as its entries and is denoted $\mathbf{0}_{m\times n}$, or simply $\mathbf{0}$ when its size is clear. A matrix with $n$ rows and $n$ columns is a *square matrix* of order $n$. A square matrix whose off-diagonal entries vanish is *diagonal*, with the same diag notation as before. The diagonal matrix of order $n$ all of whose diagonal entries are the identity 1 of $\mathbb{K}$ is the *identity matrix* $\mathbf{I}_n$, or simply $\mathbf{I}$. *Matrix multiplication* is

$$(a_{ij})_{m\times n}(b_{ij})_{n\times l} = \left[\sum_{k=1}^{n} a_{ik}b_{kj}\right];$$

over a noncommutative division ring $K$, the order of $a_{ik}b_{kj}$ must not be reversed. The *transpose* is still defined by $(\boldsymbol{A}^\top)_{ij} = (\boldsymbol{A})_{ji}$. A square matrix $\boldsymbol{A}$ of order $n$ is invertible if a square matrix $\boldsymbol{B}$ of order $n$ satisfies $\boldsymbol{AB} = \boldsymbol{BA} = \mathbf{I}_n$; its *inverse* is then $\boldsymbol{B} = \boldsymbol{A}^{-1}$. A matrix equal to its transpose is *symmetric*, while one whose sum with its transpose is zero is *skew-symmetric*. *Block matrices* are defined as before.

The corresponding properties are almost the same, with proofs entirely analogous to the earlier ones.

**Theorem A.5.33.**
*($a^1$) Matrix multiplication is associative: $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$ for matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$.*
*($a^2$) For matrices $\boldsymbol{A}, \boldsymbol{B}$ over $\mathbb{K}$ and a scalar $c \in \mathbb{K}$, left and right scalar multiplication satisfy $(c\boldsymbol{A})\boldsymbol{B} = c(\boldsymbol{AB})$ and $\boldsymbol{A}(\boldsymbol{B}c) = (\boldsymbol{AB})c$.*
*(b) For matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$, the distributive laws hold: $\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC}$ and $(\boldsymbol{A} + \boldsymbol{B})\boldsymbol{C} = \boldsymbol{AC} + \boldsymbol{BC}$;*
*(c) For a square matrix $\boldsymbol{A}$ of order $n$, $\boldsymbol{A}\mathbf{I}_n = \mathbf{I}_n\boldsymbol{A} = \boldsymbol{A}$.*

**Theorem A.5.34.**
*(a) For every matrix $\boldsymbol{A}$, $(\boldsymbol{A}^\top)^\top = \boldsymbol{A}$.*
*(b) For matrices $\boldsymbol{A}, \boldsymbol{B}$, $(\boldsymbol{A} + \boldsymbol{B})^\top = \boldsymbol{A}^\top + \boldsymbol{B}^\top$.*
*($c^1$) For matrices $\boldsymbol{A}, \boldsymbol{B}$* over a field, *$(\boldsymbol{AB})^\top = \boldsymbol{B}^\top\boldsymbol{A}^\top$.*
*($c^2$) For matrices $\boldsymbol{A}, \boldsymbol{B}$ over a general division ring $\mathbb{K}$,*

$$(\boldsymbol{AB})^\top_{\mathbb{K}} = ((\boldsymbol{B}^\top)(\boldsymbol{A}^\top))_{\mathbb{K}^{\mathrm{op}}},$$

*where the entries on the right are unchanged but are regarded as elements of $\mathbb{K}^{\mathrm{op}}$ for multiplication; the resulting arrays are formally identical.*[13]

[13]For $\boldsymbol{A} = (a_{ij}) \in \mathrm{M}_{m\times n}(\mathbb{K})$ and $\boldsymbol{B} = (b_{ij}) \in \mathrm{M}_{n\times p}(\mathbb{K})$, this means $[(\boldsymbol{AB})_{\mathbb{K}^{\mathrm{op}}}]_{ij} := \sum_{k=1}^{n}(b_{kj}a_{ik})_{\mathbb{K}}$: the scalar products on the right are taken in $\mathbb{K}$.

**Remark.** Property (c) of theorem A.2.13 used commutativity and is therefore valid as stated only over a field. Over a general division ring the opposite division ring is needed to reverse the order of the entries as well.

**Theorem A.5.35.** *Let $\mathbb{K}$ be a division ring. An $n \times n$ matrix $\boldsymbol{A}$ over $\mathbb{K}$ is symmetric exactly when*

$$(\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y})_{\mathbb{K}} = (\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x})_{\mathbb{K}^{\mathrm{op}}}$$

*for all column vectors $\boldsymbol{x},\boldsymbol{y} \in \mathbb{K}^n$. It is skew-symmetric exactly when*

$$(\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y})_{\mathbb{K}} = -(\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x})_{\mathbb{K}^{\mathrm{op}}}$$

*for all column vectors $\boldsymbol{x},\boldsymbol{y} \in \mathbb{K}^n$. The subscript $\mathbb{K}^{\mathrm{op}}$ means that the same entries are regarded as elements of the opposite division ring when they are multiplied; the resulting matrix has the same formal array of entries.*

**Theorem A.5.36.**

- (a) *If a matrix has an inverse, that inverse is unique.*
- (b) *If $\boldsymbol{A},\boldsymbol{B}$ are both invertible, then $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.*
- ($c^1$) *For an invertible square matrix $\boldsymbol{A}$ over* a field, $(\boldsymbol{A}^{-1})^{\mathsf{T}} = (\boldsymbol{A}^{\mathsf{T}})^{-1}$.
- ($c^2$) *For an invertible square matrix $\boldsymbol{A}$ over a general division ring $\mathbb{K}$,*

$$(\boldsymbol{A}_{\mathbb{K}}^{-1})^{\mathsf{T}} = (\boldsymbol{A}^{\mathsf{T}})_{\mathbb{K}^{\mathrm{op}}}^{-1}.$$

  *In the latter formula, the entries of $\boldsymbol{A}^{\mathsf{T}}$ are regarded as elements of $\mathbb{K}^{\mathrm{op}}$ when the inverse is taken; the resulting matrix has the same formal array of entries.*
- (d) *If a square matrix $\boldsymbol{A}$ has a left inverse $\boldsymbol{L}$, so that $\boldsymbol{L}\boldsymbol{A} = \mathbf{I}$, then the inverse of $\boldsymbol{A}$ exists and equals $\boldsymbol{L}$.*

**Remark.** Property (d) cannot be proved using determinants as before; its proof is given below. Also, ($c^1$) requires commutativity, since its derivation uses property (c) of theorem A.5.34. For a general division ring, the opposite division ring is needed to reverse the multiplication order.

**Theorem A.5.37.**

- (a) *Let $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and $\boldsymbol{B} = (\boldsymbol{B}_{ij})_{r\times s}$. If, for every $i,j$, the blocks $\boldsymbol{A}_{ij},\boldsymbol{B}_{ij}$ have the same numbers of rows and columns, that is, $\boldsymbol{A},\boldsymbol{B}$ have the same block structure, then $\boldsymbol{A}+\boldsymbol{B} = (\boldsymbol{A}_{ij}+\boldsymbol{B}_{ij})_{r\times s}$.*
- (b) *If $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and $c$ is a scalar, then $c\boldsymbol{A} = (c\boldsymbol{A}_{ij})_{r\times s}$ and $\boldsymbol{A}c = (\boldsymbol{A}_{ij}c)_{r\times s}$.*
- (c) *If $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$, then $\boldsymbol{A}^{\mathsf{T}} = (\boldsymbol{A}_{ji}^{\mathsf{T}})_{s\times r}$.*
- (d) *Let $\boldsymbol{A} = (\boldsymbol{A}_{ij})_{r\times s}$ and $\boldsymbol{B} = (\boldsymbol{B}_{ij})_{s\times t}$. If the block products $\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}$ are defined for every $1 \le i \le r$, $1 \le j \le t$, and $1 \le k \le s$, then $\boldsymbol{A}\boldsymbol{B} = (\boldsymbol{C}_{ij})_{r\times t}$, where the blocks are $\boldsymbol{C}_{ij} = \sum_{k=1}^{s}\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}$.*

**Example A.5.38.** Over a division ring, the square matrix $\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of order two is invertible if and only if either $c = 0$ and $a,d \neq 0$, or $c \neq 0$ and $b \neq ac^{-1}d$.

**Proof.** If $\boldsymbol{A}' = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$ is the inverse of $\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\boldsymbol{A}\boldsymbol{A}' = \mathbf{I}$ gives

$$aa' + bc' = 1, \quad ab' + bd' = 0, \quad ca' + dc' = 0, \quad cb' + dd' = 1,$$

while $\boldsymbol{A}'\boldsymbol{A} = \mathbf{I}$ gives

$$a'a + b'c = 1, \quad a'b + b'd = 0, \quad c'a + d'c = 0, \quad c'b + d'd = 1.$$

If $c = 0$, then $cb' + dd' = 1$ gives $dd' = 1$, so invertibility requires $d \neq 0$. Also, $a'a + b'c = 1$ gives $a'a = 1$, so invertibility requires $a \neq 0$. Conversely, in that case direct multiplication verifies

$$\boldsymbol{A}' = \begin{bmatrix} a^{-1} & -a^{-1}bd^{-1} \\ 0 & d^{-1} \end{bmatrix}.$$

If $c \neq 0$, put $s = b - ac^{-1}d$. Were $s = 0$, left multiplication of $ca' + dc' = 0$ by $ac^{-1}$ would give

$$0 = ac^{-1}(ca' + dc') = aa' + ac^{-1}dc' = aa' + bc' = 0,$$

contradicting $aa' + bc' = 1$. Thus $s \neq 0$, and direct multiplication verifies

$$\boldsymbol{A}' = \begin{bmatrix} -c^{-1}ds^{-1} & c^{-1} + c^{-1}ds^{-1}ac^{-1} \\ s^{-1} & -s^{-1}ac^{-1} \end{bmatrix}.$$

□

The *determinant* requires commutative multiplication and is therefore used only for matrices over fields:

$$\det((a_{ij})_{n\times n}) := \sum_{\sigma \in \mathrm{S}_n} \operatorname{sgn}(\sigma)\, a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} = \sum_{\sigma \in \mathrm{S}_n} \operatorname{sgn}(\sigma) \left[\prod_{i=1}^{n} a_{i,\sigma(i)}\right].$$

For matrices over a field, all the determinant properties in section A.2.3 remain exactly the same, so we shall not repeat them.

We also give an important result here.

**Theorem A.5.39.** *The set of all invertible matrices of order n over a division ring $\mathbb{K}$ forms a group under matrix multiplication.* It is called the *general linear group* of dimension $n$, denoted by $\mathrm{GL}(n, \mathbb{K})$.

**Proof.** The group axioms follow immediately from theorems A.5.33 and A.5.36. □

**Theorem A.5.40.** *For a division ring $\mathbb{K}$ and a positive integer n,*

$$\mathrm{Z}(\mathrm{GL}(n, \mathbb{K})) = \{\lambda \mathbf{I}_n \mid \lambda \in \mathrm{Z}(\mathbb{K})^\times\}.$$

**Proof.** For $n = 1$ this is just multiplication in $\mathbb{K}$ and is immediate. Let $n \geq 2$, and let $\boldsymbol{E}_{ij}$ be a square matrix of order $n$, with $i \neq j$, having entry 1 in position $(i,j)$ and zero elsewhere. For every $r \in \mathbb{K}$,

$$(\mathbf{I} + r\boldsymbol{E}_{ij})(\mathbf{I} - r\boldsymbol{E}_{ij}) = \mathbf{I} - r^2\boldsymbol{E}_{ij}^2 = \mathbf{I},$$

so $\mathbf{I} + r\boldsymbol{E}_{ij} \in \mathrm{GL}(n, \mathbb{K})$. If $\boldsymbol{A} = (a_{ij}) \in \mathrm{Z}(\mathrm{GL}(n, \mathbb{K}))$, then

$$(\mathbf{I} + r\boldsymbol{E}_{ij})\boldsymbol{A} = \boldsymbol{A}(\mathbf{I} + r\boldsymbol{E}_{ij}) \Longrightarrow \quad r\boldsymbol{E}_{ij}\boldsymbol{A} = \boldsymbol{A}r\boldsymbol{E}_{ij} \Longrightarrow \quad \delta_{pi} r a_{jq} = a_{pi} r \delta_{jq}.$$

Here the last equality compares the entries in row $p$, column $q$ of the preceding matrices, and the *Kronecker delta* is $\delta_{ij} := \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$ Taking $p = i$, $q \neq j$, and $r = 1$ gives $a_{jq} = 0$ for every $j \neq q$. Taking $p = i$, $q = j$, and $r = 1$ gives $a_{ii} = a_{jj}$. Hence $a_{ii} = \lambda$ for every $i$, for some $\lambda \in \mathbb{K}^\times$; that is, $\boldsymbol{A} = \lambda\mathbf{I}$. Finally, taking $p = i$, $q = j$ with arbitrary $r$ gives $r\lambda = \lambda r$, so $\lambda \in \mathrm{Z}(\mathbb{K})^\times$. Conversely, for $\lambda \in \mathrm{Z}(\mathbb{K})^\times$, it is easy to verify that $\lambda\mathbf{I}$ is invertible and commutes with all other matrices. □

### Matrix representations of linear maps

Linear maps can be represented by matrices. First fix the following notation.

Consider right linear spaces $V, W$ over a division ring $\mathbb{K}$ and a linear map $f : V \to W$. Choose a basis $\{v_1, \ldots, v_n\}$ of $V$ and a basis $\{w_1, \ldots, w_m\}$ of $W$. Express each $f(v_j) \in W$ uniquely in the basis of $W$ as

$$f(v_j) = w_1 a_{1j} + w_2 a_{2j} + \cdots + w_m a_{mj} = \sum_{i=1}^{m} w_i a_{ij}.$$

The *representing matrix* of $f$ in the bases $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ is $\boldsymbol{A} = (a_{ij})_{m\times n} \in \mathrm{M}_{m\times n}(\mathbb{K})$. Thus the $j$th column of $\boldsymbol{A}$ is the column vector of coefficients expressing $f(v_j)$ in the basis of $W$.

The basis vectors of an $n$-dimensional linear space are often formally arranged in a row. For example, the basis of $V$ above can be arranged as $\boldsymbol{V} = [v_1 \ \ v_2 \ \ \cdots \ \ v_n]$, and that of $W$ as $\boldsymbol{W} = [w_1 \ \ w_2 \ \ \cdots \ \ w_m]$.

If the basis of $V$ is changed to $\boldsymbol{V}' = [v_1' \ \ v_2' \ \ \cdots \ \ v_n']$ with $v_j' = \sum_{i=1}^{n} v_i P_{ij}$, the coefficients form the *transition matrix* $\boldsymbol{P} = (P_{ij})_{n\times n} \in \mathrm{M}_{n\times n}(\mathbb{K})$ of the change of basis. Formally, $\boldsymbol{V}' = \boldsymbol{VP}$.

**Lemma A.5.41.** *Every transition matrix is invertible, and its inverse is the transition matrix for the reverse change of basis.*

**Proof.** Continue with the notation above. Every vector of the old basis can be expressed in the new basis, so there is a matrix $\boldsymbol{R}$ such that $\boldsymbol{V} = \boldsymbol{V}'\boldsymbol{R}$. Then $\boldsymbol{V} = \boldsymbol{V}'\boldsymbol{R} = \boldsymbol{V}(\boldsymbol{PR})$. Uniqueness of coordinates in the basis $\boldsymbol{V}$ gives $\boldsymbol{PR} = \mathbf{I}_n$. Similarly, $\boldsymbol{V}' = \boldsymbol{VP} = \boldsymbol{V}'(\boldsymbol{RP})$ gives $\boldsymbol{RP} = \mathbf{I}_n$. Thus $\boldsymbol{P}^{-1} = \boldsymbol{R}$. □

**Theorem A.5.42.** *Let $f : V \to W$ be a linear map between right linear spaces, with a chosen basis in each of $V, W$. If the coordinates of $x \in V$ in its basis are arranged as a column vector $\boldsymbol{X}$, and $\boldsymbol{A}$ is the representing matrix of $f$, then the coordinates of $f(x)$ in the basis of $W$ form the column vector $\boldsymbol{AX}$.*

**Proof.** Writing $\boldsymbol{X} = [x_1 \ \ \cdots \ \ x_n]^{\mathsf{T}}$,

$$f(x) = \sum_{i=1}^{n} f(v_i)x_i = \sum_{i=1}^{n}\sum_{j=1}^{m} w_j a_{ji} x_i = \sum_{j=1}^{m} w_j \left[\sum_{i=1}^{n} a_{ji} x_i\right].$$

The coefficient of $w_j$ is therefore $\sum_{i=1}^{n} a_{ji} x_i$; these coordinates, arranged as a column, form $\boldsymbol{AX}$. □

**Corollary A.5.43.** *Let $U, V, W$ be right linear spaces with chosen bases. If the linear maps $f : U \to V$ and $g : V \to W$ have representing matrices $\boldsymbol{A}, \boldsymbol{B}$, respectively, then the composition $g \circ f : U \to W$ has representing matrix $\boldsymbol{BA}$.*

**Proof.** Let $\boldsymbol{X}$ be the coordinate column of $x$ in the basis of $U$. Then the coordinate column of $f(x)$ in the basis of $V$ is $\boldsymbol{AX}$, so the coordinate column of $g(f(x))$ in the basis of $W$ is $\boldsymbol{B}(\boldsymbol{AX}) = (\boldsymbol{BA})\boldsymbol{X}$. Therefore the representing matrix of $g \circ f$ is $\boldsymbol{BA}$. □

**Theorem A.5.44** (Change-of-basis formula)**.** *Let $f : V \to W$ be a linear map between right linear spaces. Fix a basis in each of $V, W$, and let $\boldsymbol{A}$ represent $f$ in these bases. Change the basis of $V$ with transition matrix $\boldsymbol{P} \in \mathrm{GL}(n, \mathbb{K})$, and change the basis of $W$ with transition matrix $\boldsymbol{Q} \in \mathrm{GL}(m, \mathbb{K})$. Then, in the new bases:*

*(1) if the coordinates of a vector $x \in V$ in the old basis form a column vector $\boldsymbol{X}$, its coordinates in the new basis form the column vector $\boldsymbol{X}' = \boldsymbol{P}^{-1}\boldsymbol{X}$;*
*(2) the representing matrix of $f$ is $\boldsymbol{A}' = \boldsymbol{Q}^{-1}\boldsymbol{AP}$.*

**Proof.** Arrange the old bases of $V, W$ as rows $\boldsymbol{V}, \boldsymbol{W}$ and the new bases as rows $\boldsymbol{V}', \boldsymbol{W}'$. In the old bases, let the coordinates of $x \in V$ and $f(x) \in W$ form column vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. Then $x = \sum_{i=1}^{n} v_i x_i = \boldsymbol{VX}$, and similarly $f(x) = \boldsymbol{WY}$. Let the coordinates of $x, f(x)$ in the new bases form column vectors $\boldsymbol{X}', \boldsymbol{Y}'$, respectively.

For (1),

$$\boldsymbol{V}'\boldsymbol{X}' = x = \boldsymbol{VX} = \boldsymbol{V}'(\boldsymbol{P}^{-1}\boldsymbol{X}).$$

Since $\boldsymbol{V}'$ is a basis, the coordinates of $x$ in $\boldsymbol{V}'$ are unique; hence $\boldsymbol{X}' = \boldsymbol{P}^{-1}\boldsymbol{X}$.

For (2), similarly $\boldsymbol{Y}' = \boldsymbol{Q}^{-1}\boldsymbol{Y}$. By the definition of the representing matrix, $\boldsymbol{Y} = \boldsymbol{AX}$, so

$$\boldsymbol{Y}' = \boldsymbol{Q}^{-1}\boldsymbol{Y} = \boldsymbol{Q}^{-1}\boldsymbol{AX} = \boldsymbol{Q}^{-1}\boldsymbol{APX}'.$$

Thus $\boldsymbol{A}' = \boldsymbol{Q}^{-1}\boldsymbol{AP}$ makes $\boldsymbol{Y}' = \boldsymbol{A}'\boldsymbol{X}'$.

Alternatively, use $\boldsymbol{W}=\boldsymbol{W}'\boldsymbol{Q}^{-1}$, that is, $w_k=\sum_{j=1}^m w_j'(\boldsymbol{Q}^{-1})_{jk}$, and calculate directly from the definition:

$$\begin{aligned} f(v_i') &= f\Big(\sum_{j=1}^n v_j P_{ji}\Big)=\sum_{j=1}^n f(v_j)P_{ji}=\sum_{j=1}^n\Big(\sum_{k=1}^m w_k a_{kj}\Big)P_{ji}=\sum_{j=1}^n\sum_{k=1}^m\sum_{l=1}^m w_l'(\boldsymbol{Q}^{-1})_{lk}a_{kj}P_{ji}\\ &=\sum_{l=1}^m w_l'\Big(\sum_{k=1}^m\sum_{j=1}^n(\boldsymbol{Q}^{-1})_{lk}a_{kj}P_{ji}\Big)=\sum_{l=1}^m w_l'(\boldsymbol{Q}^{-1}\boldsymbol{AP})_{li}. \end{aligned}$$ □

**Remark.** For linear maps between left spaces the theory is parallel, with bases arranged as columns, for example $\boldsymbol{V}=\begin{bmatrix}v_1\\ \vdots\\ v_n\end{bmatrix}$, and coordinates as rows, for example $\boldsymbol{X}=[x_1\ \cdots\ x_n]$.

(1) If $f(v_i)=\sum_{j=1}^m a_{ij}w_j$, then its representing matrix is $\boldsymbol{A}=(a_{ij})_{n\times m}\in\mathrm{M}_{n\times m}(\mathbb{K})$.
(2) For the coordinate rows $\boldsymbol{X}$ of $x\in V$ and $\boldsymbol{Y}$ of $f(x)\in W$, formally $x=\boldsymbol{XV}$ and $f(x)=\boldsymbol{YW}$, with $\boldsymbol{Y}=\boldsymbol{XA}$.
(3) Transition matrices multiply on the left: $\boldsymbol{V}'=\boldsymbol{PV}$ and $\boldsymbol{W}'=\boldsymbol{QW}$, that is, $v_i'=\sum_{j=1}^n P_{ij}v_j$ and $w_i'=\sum_{j=1}^m Q_{ij}w_j$. The change-of-basis formulas are $\boldsymbol{X}'=\boldsymbol{XP}^{-1}$ for the coordinates of a vector in $V$, and $\boldsymbol{A}'=\boldsymbol{PAQ}^{-1}$ for the representing matrix.
(4) Over a field the two final formulas differ but give the same result: matrices are transposed and the order of matrix multiplication is reversed. For example, the $\boldsymbol{P},\boldsymbol{Q},\boldsymbol{A}$ multiplying on the left here are the transposes of the $\boldsymbol{P},\boldsymbol{Q},\boldsymbol{A}$ that multiplied on the right earlier. Over a field there is no essential distinction between the two forms; the earlier convention with coordinate columns is more common.

For a linear space $V$, elements of its dual space $V^*$ can also be represented by matrices. Take $V$ to be a right space and arrange its basis as the row $\boldsymbol{V}=[e_1\ \cdots\ e_n]$. Since $V^*$ is a left space, arrange its dual basis as the column $\boldsymbol{E}=\begin{bmatrix}e^1\\ \vdots\\ e^n\end{bmatrix}$, and the coordinates of a linear map $f\in V^*$ in that basis as the row $\boldsymbol{F}=[f_1\ \cdots\ f_n]$.

**Theorem A.5.45.** *Let $V$ be a right vector space. If $\boldsymbol{X}$ is the coordinate column of $x\in V$ and $\boldsymbol{F}$ is the coordinate row of $f\in V^*$, then $f(x)=\boldsymbol{FX}$.*

**Proof.** This is precisely proposition A.5.27(3). □

Now change the basis of $V$ to $\boldsymbol{V}'=[e_1'\ \cdots\ e_n']$, with transition matrix $\boldsymbol{P}=(P^i{}_j)\in\mathrm{GL}(n,\mathbb{K})$,[14] so that $e_j'=\sum_{i=1}^n e_iP^i{}_j$, or $\boldsymbol{V}'=\boldsymbol{VP}$. The dual basis changes correspondingly to $\boldsymbol{E}'=\begin{bmatrix}e'^1\\ \vdots\\ e'^n\end{bmatrix}$. The transition matrix $\boldsymbol{Q}=(Q^i{}_j)\in\mathrm{GL}(n,\mathbb{K})$ of the left vector space multiplies on the left: $e'^i=\sum_{j=1}^n Q^i{}_je^j$, or $\boldsymbol{E}'=\boldsymbol{QE}$. We need to determine the relation between $\boldsymbol{Q}$ and $\boldsymbol{P}$, and the transformation formula for $\boldsymbol{F}$.

**Theorem A.5.46** (Dual change-of-basis formula)**.** *If a right vector space $V$ changes basis with transition matrix $\boldsymbol{P}\in\mathrm{GL}(n,\mathbb{K})$, then:*
*(1) the transition matrix of $V^*$ as a left space is $\boldsymbol{Q}=\boldsymbol{P}^{-1}$;*
*(2) if the old coordinates of $f\in V^*$ are arranged as the row $\boldsymbol{F}$, its new coordinates are the row $\boldsymbol{F}'=\boldsymbol{FP}$.*

**Proof.** Using the notation above,

$$\delta^i{}_j=e'^i(e_j')=\sum_{k=1}^n Q^i{}_ke^k\left(\sum_{l=1}^n e_lP^l{}_j\right)=\sum_{k=1}^n\sum_{l=1}^n Q^i{}_k\delta^k{}_lP^l{}_j=\sum_{k=1}^n Q^i{}_kP^k{}_j=(\boldsymbol{QP})_{ij}.$$

[14] Upper and lower indices are distinguished here in discussing the dual space; this is why $P^i{}_j$ is written with an upper and a lower index.

Thus $\boldsymbol{Q}=\boldsymbol{P}^{-1}$. The coordinate transformation formula for $f$ is the left-space version of theorem A.5.44(1). □

**Remark.** Consequently $\boldsymbol{F}'\boldsymbol{X}'=(\boldsymbol{FP})(\boldsymbol{P}^{-1}\boldsymbol{X})=\boldsymbol{FX}$, as required because $f(x)$ is unchanged by a change of basis: it is a fixed element of $\mathbb{K}$, independent of its coordinate representation.

**Theorem A.5.47.** *If a right linear map $f:V\to W$ is represented by $\boldsymbol{A}$, then its dual $f^*:W^*\to V^*$, viewed as a left linear map in the dual bases of $W^*,V^*$, is represented by the same matrix $\boldsymbol{A}$.*

**Proof.** With the preceding notation, write $\boldsymbol{A}=(a^i{}_j)_{m\times n}$. Then the basis of $V$ satisfies $f(v_j)=\sum_{i=1}^m w_i a^i{}_j$. Since $W^*$ is a left linear space, for $\varphi\in W^*$ its coordinates in the dual basis of $W^*$ form the row vector $\Phi=[\varphi_1\ \cdots\ \varphi_m]$, and proposition A.5.27 gives $\varphi_i=\varphi(w_i)$. The coordinates of $f^*(\varphi)=\varphi\circ f\in V^*$ in the dual basis of $V^*$ form the row vector $\Psi=[\psi_1\ \cdots\ \psi_n]$. Again by proposition A.5.27, $\psi_j=[f^*(\varphi)](v_j)=\varphi(f(v_j))$. Thus

$$\psi_j=\varphi\Big(\sum_{i=1}^m w_i a^i{}_j\Big)=\sum_{i=1}^m \varphi(w_i)a^i{}_j=\sum_{i=1}^m \varphi_i a^i{}_j.$$

This corresponds precisely to the matrix form $\Psi=\Phi\boldsymbol{A}$, which is the action on coordinates of a representing matrix between left vector spaces. □

### Matrices as linear maps

Once bases are chosen, a linear map $f$ of right vector spaces is associated with a matrix $\boldsymbol{A}$, and a matrix can also be regarded as a linear map. This viewpoint conveniently proves the matrix property that was left unproved earlier.

**Revisit** (theorem A.5.36(d))**.** *If an $n\times n$ matrix $\boldsymbol{A}$ over a division ring has a left inverse $\boldsymbol{L}$, so that $\boldsymbol{LA}=\mathbf{I}_n$, then it also has a right inverse, equal to its left inverse: $\boldsymbol{AL}=\mathbf{I}_n$.*

**Proof.** Let $\boldsymbol{A},\boldsymbol{L}$ represent transformations $f,g$ of the same $n$-dimensional right $\mathbb{K}$-space $V$. Plainly $\mathbf{I}_n$ represents the identity map $\mathrm{id}_V$. If $\boldsymbol{LA}=\mathbf{I}_n$, then corollary A.5.43 gives $g\circ f=\mathrm{id}$. For $x,y\in V$, if $f(x)=f(y)$, then $(g\circ f)(x)=(g\circ f)(y)$, so $x=y$; hence $f$ is injective. Corollary A.5.25 makes it surjective and therefore bijective. The general property of inverse maps in theorem A.1.17 then gives $f\circ g=\mathrm{id}_V$, so $\boldsymbol{AL}=\mathbf{I}_n$. □

Matrix equations $\boldsymbol{Ax}=\mathbf{0}$ in a column vector $\boldsymbol{x}$ can also be related to linear maps. For example, regard $\boldsymbol{x}$ as a vector in the linear space $\mathbb{K}^n_{\mathbb{K}}$. The matrix $\boldsymbol{A}$ gives the linear map $f:\boldsymbol{x}\mapsto\boldsymbol{Ax}$, with $\ker f=\{\boldsymbol{x}|\boldsymbol{Ax}=\mathbf{0}\}$ and $\operatorname{im} f=\{\boldsymbol{Ax}\,|\,\boldsymbol{x}\in\mathrm{M}_{n\times1}(\mathbb{K})\}$. We now study matrices further by using linear maps.

For the moment, we first regard matrices over $\mathbb{K}$ simply as arrays of elements, without regard to left- or right-linear-space structures.

**Definition A.5.48.** Let $\boldsymbol{A}=(a_{ij})\in\mathrm{M}_{m\times n}(\mathbb{K})$. Here we first regard $\boldsymbol{A}$ only as an $m\times n$ array of elements of $\mathbb{K}$. Write its $i$th row as $\boldsymbol{R}_i=[a_{i1}\ a_{i2}\ \cdots\ a_{in}]$ and its $j$th column as $\boldsymbol{C}_j=[a_{1j}\ a_{2j}\ \cdots\ a_{mj}]^{\mathsf{T}}$. Define the following spaces.

(1) The *left row space* is $\mathrm{Row}_{\mathrm{L}}(\boldsymbol{A}):=\mathrm{span}^{\mathrm{L}}_{\mathbb{K}}\{\boldsymbol{R}_1,\boldsymbol{R}_2,\ldots,\boldsymbol{R}_m\}$.
(2) The *right row space* is $\mathrm{Row}_{\mathrm{R}}(\boldsymbol{A}):=\mathrm{span}^{\mathrm{R}}_{\mathbb{K}}\{\boldsymbol{R}_1,\boldsymbol{R}_2,\ldots,\boldsymbol{R}_m\}$.
(3) The *left column space* is $\mathrm{Col}_{\mathrm{L}}(\boldsymbol{A}):=\mathrm{span}^{\mathrm{L}}_{\mathbb{K}}\{\boldsymbol{C}_1,\boldsymbol{C}_2,\ldots,\boldsymbol{C}_n\}$.
(4) The *right column space* is $\mathrm{Col}_{\mathrm{R}}(\boldsymbol{A}):=\mathrm{span}^{\mathrm{R}}_{\mathbb{K}}\{\boldsymbol{C}_1,\boldsymbol{C}_2,\ldots,\boldsymbol{C}_n\}$.
(5) The *right null space* is $\mathrm{Nul}_{\mathrm{R}}(\boldsymbol{A}):=\{\boldsymbol{x}\in\mathbb{K}^n\,|\boldsymbol{Ax}=\mathbf{0}\}$, where $\boldsymbol{x}$ is regarded as a column vector.
(6) The *left null space* is $\mathrm{Nul}_{\mathrm{L}}(\boldsymbol{A}):=\{\boldsymbol{y}\in\mathbb{K}^m\,|\boldsymbol{yA}=\mathbf{0}\}$, where $\boldsymbol{y}$ is regarded as a row vector.

**Remark.** Over a commutative field $\mathbb{K}$, the left and right row spaces coincide by commutativity, and their common space is called the *row space* of $\boldsymbol{A}$, written $\operatorname{Row}(\boldsymbol{A})$. The left and right column spaces likewise coincide, and are called the *column space* of $\boldsymbol{A}$, written $\operatorname{Col}(\boldsymbol{A})$. Even over a field the left and right null spaces remain different. Unless specified otherwise, "null space" means the right null space, written $\operatorname{Nul}(\boldsymbol{A})$.

The following assertions are easily verified.

**Proposition A.5.49.** *For a matrix $\boldsymbol{A} \in \mathrm{M}_{m\times n}(\mathbb{K})$:*

*(1)* $\operatorname{Row}_{\mathrm{L}}(\boldsymbol{A})$ *is a linear subspace of $\mathbb{K}^n$ viewed as a left $\mathbb{K}$-linear space, and* $\operatorname{Row}_{\mathrm{R}}(\boldsymbol{A})$ *is a subspace of $\mathbb{K}^n$ viewed as a right $\mathbb{K}$-linear space;*
*(2)* $\operatorname{Col}_{\mathrm{L}}(\boldsymbol{A})$ *is a linear subspace of $\mathbb{K}^m$ viewed as a left $\mathbb{K}$-linear space, and* $\operatorname{Col}_{\mathrm{R}}(\boldsymbol{A})$ *is a linear subspace of $\mathbb{K}^m$ viewed as a right $\mathbb{K}$-linear space;*
*(3)* $\operatorname{Nul}_{\mathrm{L}}(\boldsymbol{A})$ *is a subspace of $\mathbb{K}^m$ viewed as a left $\mathbb{K}$-linear space, and* $\operatorname{Nul}_{\mathrm{R}}(\boldsymbol{A})$ *is a subspace of $\mathbb{K}^n$ viewed as a right $\mathbb{K}$-linear space.*

These spaces are therefore linear spaces, and we can discuss their dimensions.

**Definition A.5.50.** For a matrix $\boldsymbol{A} \in \mathrm{M}_{m\times n}(\mathbb{K})$, the dimensions of its left row space, right row space, left column space, and right column space are called, respectively, its *left row rank*, *right row rank*, *left column rank*, and *right column rank*. When $\mathbb{K}$ is commutative, the left and right row ranks agree and are called the *row rank*; the left and right column ranks agree and are called the *column rank*.

The theorem below will show that the left row rank equals the right column rank. Their common value is called the *right rank* of $\boldsymbol{A}$, denoted by $\operatorname{rank}_{\mathrm{R}}\boldsymbol{A}$. It will also show that the right row rank equals the left column rank; their common value is the *left rank* of $\boldsymbol{A}$, denoted by $\operatorname{rank}_{\mathrm{L}}\boldsymbol{A}$. When $\mathbb{K}$ is commutative, the left and right ranks agree and are called the *rank*, denoted by $\operatorname{rank}\boldsymbol{A}$.

**Theorem A.5.51** (Fundamental theorem of linear algebra)**.** *Let $\mathbb{K}$ be a division ring. For $\boldsymbol{A} \in \mathrm{M}_{m\times n}(\mathbb{K})$:*

*(1)* $\dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Col}_{\mathrm{R}}(\boldsymbol{A}) = \dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Row}_{\mathrm{L}}(\boldsymbol{A}) := \operatorname{rank}_{\mathrm{R}}(\boldsymbol{A})$;
*(2)* $\dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Nul}_{\mathrm{R}}(\boldsymbol{A}) + \operatorname{rank}_{\mathrm{R}}(\boldsymbol{A}) = n$;
*(3)* $\dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Nul}_{\mathrm{L}}(\boldsymbol{A}) + \operatorname{rank}_{\mathrm{R}}(\boldsymbol{A}) = m$;
*(1′)* $\dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Col}_{\mathrm{L}}(\boldsymbol{A}) = \dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Row}_{\mathrm{R}}(\boldsymbol{A}) := \operatorname{rank}_{\mathrm{L}}(\boldsymbol{A})$;
*(2′)* $\dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Nul}_{\mathrm{R}}(\boldsymbol{A}^{\mathsf{T}}) + \operatorname{rank}_{\mathrm{L}}(\boldsymbol{A}) = m$;
*(3′)* $\dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Nul}_{\mathrm{L}}(\boldsymbol{A}^{\mathsf{T}}) + \operatorname{rank}_{\mathrm{L}}(\boldsymbol{A}) = n$.

**Proof.** We first prove the first three identities. By theorem A.5.47, when $\boldsymbol{A}$ represents the right linear map $T_{\boldsymbol{A}} : \mathbb{K}^n_{\mathbb{K}} \to \mathbb{K}^m_{\mathbb{K}}$, we have $\operatorname{im} T_{\boldsymbol{A}} = \operatorname{Col}_{\mathrm{R}}(\boldsymbol{A})$. The same matrix also represents the dual map $T^*_{\boldsymbol{A}} : (\mathbb{K}^m_{\mathbb{K}})^* \to (\mathbb{K}^n_{\mathbb{K}})^*$, with $\operatorname{im} T^*_{\boldsymbol{A}} = \operatorname{Row}_{\mathrm{L}}(\boldsymbol{A})$. Thus theorem A.5.31 proves the first identity.

Regard $\boldsymbol{A}$ as the right linear map $T_{\boldsymbol{A}} : \mathbb{K}^n_{\mathbb{K}} \to \mathbb{K}^m_{\mathbb{K}}$ given by $\boldsymbol{X} \mapsto \boldsymbol{A}\boldsymbol{X}$. Then $\ker T_{\boldsymbol{A}} = \operatorname{Nul}_{\mathrm{R}}(\boldsymbol{A})$ and $\operatorname{im} T_{\boldsymbol{A}} = \operatorname{Col}_{\mathrm{R}}(\boldsymbol{A})$, so the rank–nullity theorem gives

$$\dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Nul}_{\mathrm{R}}(\boldsymbol{A}) + \dim_{\mathbb{K}}^{\mathrm{R}} \operatorname{Col}_{\mathrm{R}}(\boldsymbol{A}) = n,$$

proving the second identity.

Next regard $\boldsymbol{A}$ as the left linear map $S_{\boldsymbol{A}} : {}_{\mathbb{K}}\mathbb{K}^m \to {}_{\mathbb{K}}\mathbb{K}^n$ given by $\boldsymbol{Y} \mapsto \boldsymbol{Y}\boldsymbol{A}$. Then $\ker S_{\boldsymbol{A}} = \operatorname{Nul}_{\mathrm{L}}(\boldsymbol{A})$ and $\operatorname{im} S_{\boldsymbol{A}} = \operatorname{Row}_{\mathrm{L}}(\boldsymbol{A})$, so rank–nullity gives

$$\dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Nul}_{\mathrm{L}}(\boldsymbol{A}) + \dim_{\mathbb{K}}^{\mathrm{L}} \operatorname{Row}_{\mathrm{L}}(\boldsymbol{A}) = m,$$

proving the third identity.

Finally, apply the first three identities just proved to the transpose matrix $\boldsymbol{A}^{\mathsf{T}} \in \mathrm{M}_{n\times m}(\mathbb{K})$. Under transposition, the right column space of $\boldsymbol{A}^{\mathsf{T}}$ corresponds to the right row space of $\boldsymbol{A}$, and the left row space of $\boldsymbol{A}^{\mathsf{T}}$ corresponds to the left column space of $\boldsymbol{A}$. The last three identities follow. □

For linear spaces over a field, this relation is much simpler and can be written directly as follows.

**Theorem A.5.52** (Fundamental theorem of linear algebra)**.** *Let* $\mathbb{F}$ *be a field. For every matrix* $\boldsymbol{A} \in \mathrm{M}_{m\times n}(\mathbb{F})$, $\mathrm{Nul}_{\mathrm{L}}(\boldsymbol{A}) = \mathrm{Nul}_{\mathrm{R}}(\boldsymbol{A}^{\top})$,[15] *and*

$$\dim_{\mathbb{F}} \mathrm{Col}(\boldsymbol{A}) = \dim_{\mathbb{F}} \mathrm{Row}(\boldsymbol{A}) = n - \dim_{\mathbb{F}} \mathrm{Nul}_{\mathrm{R}}(\boldsymbol{A}) = m - \dim_{\mathbb{F}} \mathrm{Nul}_{\mathrm{L}}(\boldsymbol{A}) := \mathrm{rank}(\boldsymbol{A}).$$

The preceding results immediately give the following corollary.

**Corollary A.5.53.** *For a square matrix* $\boldsymbol{A}$ *of order* $n$ *over a division ring, the following are equivalent:*
*(1)* $\boldsymbol{A}$ *is invertible;*
*(2)* $\mathrm{rank}_{\mathrm{R}} \boldsymbol{A} = n$*;*
*(3) the equation* $\boldsymbol{Ax} = \boldsymbol{0}$ *in an unknown column vector* $\boldsymbol{x}$ *has only the solution* $\boldsymbol{x} = \boldsymbol{0}$.
*When the division ring is a commutative field, these are also equivalent to (4)* $\det \boldsymbol{A} \neq 0$.

**Theorem A.5.54.** *For a square matrix* $\boldsymbol{A}$ *over a field, the matrix equation* $\boldsymbol{Ax} = \boldsymbol{0}$ *in a column vector* $\boldsymbol{x}$ *has a nonzero solution if and only if* $\det(\boldsymbol{A}) = 0$.

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## Exercises

**Exercise A.86.** Prove theorem A.5.39, proposition A.5.49, and corollary A.5.53.

**Exercise A.87.** Write the left-space versions of every assertion in this subsection stated only for right spaces.

**Exercise A.88.** For $\boldsymbol{A} = (a_{ij})_{m\times n}$ and $\boldsymbol{B} = (b_{ij})_{n\times l}$, define a new operation $\circ$ giving the $m \times l$ matrix $\boldsymbol{A}\circ\boldsymbol{B} = (\sum_{k=1}^{n} b_{kj} a_{ik})_{m\times l}$. Prove $\boldsymbol{A}\circ\boldsymbol{B} = (\boldsymbol{B}^{\top}\boldsymbol{A}^{\top})^{\top}$.

**Exercise A.89.** Over the finite field $\mathbb{F}_5$, let $\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}$ and the column vector $\boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$.
(1) Compute, in order, $\det\boldsymbol{A}$, $\boldsymbol{b}^{\top}\boldsymbol{b}$, $\boldsymbol{A}^2\boldsymbol{b}$, and $\boldsymbol{b}^{\top}(\boldsymbol{A}^{\top} + \mathbf{I}_3)$.
(2) Is $\boldsymbol{A}$ invertible? If so, find its inverse.
(3) Solve $\boldsymbol{Ax} = \boldsymbol{b}$ for the unknown column vector $\boldsymbol{x}$.

**Exercise A.90.** For the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & \mathbf{i} \\ \mathbf{j} & \mathbf{k} \end{bmatrix}$ over the quaternion division ring $\mathbb{H}$, find its left and right row spaces, its left and right column spaces, and its left and right null spaces, together with the dimensions of all six spaces. What are its left rank and its right rank?

**Exercise A.91.** For a block matrix $\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}$ over a division ring, suppose that the block $\boldsymbol{A}$ is invertible and that $\boldsymbol{S} = \boldsymbol{D} - \boldsymbol{CA}^{-1}\boldsymbol{B}$ is invertible. Prove

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{A}^{-1} + \boldsymbol{A}^{-1}\boldsymbol{BS}^{-1}\boldsymbol{CA}^{-1} & -\boldsymbol{A}^{-1}\boldsymbol{BS}^{-1} \\ -\boldsymbol{S}^{-1}\boldsymbol{CA}^{-1} & \boldsymbol{S}^{-1} \end{bmatrix}.$$

**Exercise A.92.** In the linear space $V = \mathbb{F}_5^2$ over $\mathbb{F}_5$, take the standard basis $\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and let the transition matrix be $\boldsymbol{P} = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$.
(1) Find the original dual basis, the new basis of $V$, and the new dual basis of $V^*$.

[15] Strictly speaking, these are not identical: one is a linear space of row vectors and the other a linear space of column vectors. If a row vector and a column vector are regarded as two representations of the same vector, then the two linear spaces agree.

(2) If $f \in V^*$ has coordinates $(4,2)$ in the original dual basis, find its coordinates in the new basis.

**Exercise A.93.** Prove:
(1) over a field of characteristic not 2, a skew-symmetric matrix of odd order is not invertible;
(2) over a field of characteristic 2, a symmetric matrix of odd order with all diagonal entries zero is not invertible.

Since $1=-1$ in characteristic 2, skew-symmetric matrices are then symmetric; in summary, every odd-order skew-symmetric matrix with zero diagonal is singular.

## A.6 Basic Properties of Polynomials

Recall the general definition of a polynomial from definition A.4.35. We now develop some of its basic properties. By definition, the degree of the zero polynomial is undefined. Throughout this section, however, we set $\deg 0=-\infty$, with the conventions $-\infty<n$, $\max\{-\infty,n\}=n$, $-\infty+n=-\infty$, and $(-\infty)+(-\infty)=-\infty$ for every integer $n$. These conventions let the usual degree formulas include the zero polynomial without separate cases.

### A.6.1 Basic Properties of One-Variable Polynomials

We begin with some general properties of polynomials.

**Theorem A.6.1.** *Let $R$ be a ring without zero divisors and let $f,g\in R[x]$. Then $\deg(fg)=\deg f+\deg g$; in particular, the product of two nonzero polynomials is nonzero.*

**Proof.** If either $f$ or $g$ is the zero polynomial, then $fg=0$, and the equality holds by convention. Suppose now that both are nonzero, with leading terms $a_mx^m$ and $b_nx^n$, respectively. The highest possible term of $fg$ is $a_mb_nx^{m+n}$. Since $R$ has no zero divisors and $a_m,b_n\neq 0$, we have $a_mb_n\neq 0$. Thus this term does not vanish, and no term of degree greater than $m+n$ exists. Hence $\deg(fg)=m+n$. No commutativity assumption is needed. ☐

**Proposition A.6.2.** *For a ring $R$ and $f,g\in R[x]$, $\deg(f+g)\le\max\{\deg f,\deg g\}$. The condition $\deg f\neq\deg g$ is sufficient, though not necessary, for equality.*

**Proof.** If $f=0$ or $g=0$, the conclusion follows directly from the convention. If $f+g=0$, the degree on the left is $-\infty$, so the inequality again holds. In all remaining cases, every coefficient of degree greater than $\max\{\deg f,\deg g\}$ vanishes in both $f$ and $g$, and hence also in $f+g$.

For the second assertion, suppose without loss of generality that $\deg f<\deg g=n$. The coefficient of $x^n$ in $f$ is zero, so the coefficient of $x^n$ in $f+g$ equals the nonzero leading coefficient of $g$. Therefore $\deg(f+g)=n$. ☐

**Theorem A.6.3.** *For a ring $R$ and its polynomial ring $R[x]$:*
*(1) $R[x]$ has an identity exactly when $R$ does, and when both have identities, the identity of $R$, viewed as a constant polynomial, is the identity of $R[x]$;*
*(2) $R[x]$ is commutative exactly when $R$ is commutative;*
*(3) $R[x]$ has no zero divisors exactly when $R$ has no zero divisors.*

**Proof.** For (1), if $R$ has an identity, its constant polynomial is plainly the identity of $R[x]$. Conversely, let $e(x)=\sum_{i=0}^{n}a_ix^i$ be an identity of $R[x]$, and denote its constant term by $e_0$. For every constant polynomial $a\in R$, we have $e(x)a=ae(x)=a$. Comparing constant terms gives $e_0a=ae_0=a$, so $e_0$

is an identity of $R$. Its constant polynomial is consequently an identity of $R[x]$. Uniqueness of the identity in a ring with identity gives $e(x) = e_0$.

For (2), if $R$ is commutative, the products of coefficients commute, so the definition of polynomial multiplication makes $R[x]$ commutative. Conversely, if $R[x]$ is commutative, the elements of $R$, viewed as constant polynomials, commute as well.

For (3), if $R$ has no zero divisors, theorem A.6.1 says that the product of two nonzero polynomials is nonzero, so $R[x]$ has no zero divisors. Conversely, if $R[x]$ has no zero divisors, two nonzero elements of $R$, viewed as constant polynomials, have nonzero product. Thus $R$ has no zero divisors either. $\square$

**Theorem A.6.4.** *A ring $R$ is an integral domain exactly when $R[x]$ is an integral domain. If $R$ is an integral domain, the units of $R[x]$ are exactly the units of $R$, that is, $R[x]^\times = R^\times$. In particular, a polynomial ring over a field is an integral domain.*

**Proof.** The equivalence between $R$ and $R[x]$ being integral domains follows from theorem A.6.3; it remains to prove the assertion about units. When $R$ is an integral domain, if $f(x)g(x) = 1$, then the degree formula gives $\deg f + \deg g = 0$, so $f$ and $g$ are constant polynomials and are inverses of each other in $R$. Conversely, every unit of $R$ is plainly a unit of $R[x]$. Finally, every field is an integral domain. $\square$

### Division, the factor theorem, and multiplicity

Division with remainder is not always possible for polynomials over a ring, but it is always possible for a monic polynomial over an integral domain. One can also define left and right division with remainder over a general ring, but we shall not discuss them, since they are not needed.

**Theorem A.6.5.** *Let $R$ be an integral domain, and let $f, g \in R[x]$, with $g$ monic. There are unique $q, r \in R[x]$ such that $f = qg + r$ and $\deg r < \deg g$.*

**Proof.** We first prove existence. If $f = 0$, take $q = r = 0$. If $f \neq 0$ and $\deg f < \deg g$, take $q = 0$ and $r = f$. Henceforth write $f = a_m x^m +$ lower-degree terms and $g = x^n +$ lower-degree terms, with $m \geq n$.

Put $f_1 = f - a_m x^{m-n} g$. The second term on the right has the same leading term as $f$, so $f_1 = 0$ or $\deg f_1 < m$. By induction on $\deg f$, we may write $f_1 = q_1 g + r$, with $\deg r < n$; when $r = 0$, this also holds because $\deg r = -\infty$. Thus $f = (q_1 + a_m x^{m-n})g + r$, proving existence.

For uniqueness, suppose $f = qg + r = q'g + r'$, where the degrees of $r, r'$ are both less than $\deg g$. Subtraction gives $(q - q')g = r' - r$. If $q - q' \neq 0$, then theorem A.6.1 makes the degree of the left side at least $\deg g$. If the right side is nonzero, proposition A.6.2 makes its degree less than $\deg g$, a contradiction. If the right side is zero, cancellation in the integral domain $R[x]$ also gives $q = q'$. Thus $q = q'$ and $r = r'$ in all cases. $\square$

**Theorem A.6.6** (Polynomial division)**.** *Let $F$ be a field, and let $f, g \in F[x]$, with $g \neq 0$. There are unique $q, r \in F[x]$ such that $f = qg + r$ and $\deg r < \deg g$.*

**Proof.** Write $g = b_n x^n +$ lower-degree terms. Since $F$ is a field, $b_n^{-1}$ exists. Put $\tilde{g} := b_n^{-1} g = x^n +$ lower-degree terms. Then $\tilde{g}$ is monic and $\deg \tilde{g} = \deg g$. The preceding theorem gives $Q, r \in F[x]$ such that $f = Q\tilde{g} + r$ and $\deg r < \deg \tilde{g} = \deg g$. Thus $f = (b_n^{-1} Q g) + r$; taking $q = b_n^{-1} Q$ establishes the existence of $q, r$.

If two expressions satisfy the requirements, so that $f = qg + r = q'g + r'$, then $f = (b_n q)\tilde{g} + r = (b_n q')\tilde{g} + r'$. Uniqueness in the preceding theorem gives $r' = r$ and $b_n q' = b_n q$; cancellation in the latter equality gives $q' = q$. $\square$

The expression $f = qg + r$, with $\deg r < \deg g$, is the *division with remainder* of $f$ by $g$; $q$, $r$, $g$, and $f$ are respectively the *quotient*, *remainder*, *divisor*, and *dividend*. If $r = 0$, then $g$ *divides* $f$, written

$g\,|\,f$; because $R[x]$ is an integral domain, this is the divisibility relation defined earlier for an integral domain. Note that a field guarantees that $b_n^{-1}$ always exists; over a general integral domain, division by a nonmonic polynomial need not be possible.

**Corollary A.6.7** (Remainder theorem). *If $R$ is an integral domain, $f\in R[x]$, and $a\in R$, then the remainder on division of $f$ by $(x-a)$ is $f(a)$.*

**Proof.** By division with remainder, there are $q\in R[x]$ and a constant $r\in R$ such that $f=(x-a)q+r$. Substituting $x=a$ gives $r=f(a)$. ☐

**Corollary A.6.8** (Factor theorem). *If $R$ is an integral domain, $f\in R[x]$, and $a\in R$, then $f(a)=0 \Longleftrightarrow (x-a)\,|\,f$.*

**Proof.** By the remainder theorem, $r=0 \Longleftrightarrow f(a)=0$. ☐

**Proposition A.6.9.** *If $R$ is an integral domain, $0\neq f,g\in R[x]$, and $f\,|\,g$, then* $\deg f\leq \deg g$.

**Proof.** By hypothesis, there is a nonzero polynomial $h$ with $g=fh$, and hence $\deg g=\deg f+\deg h\geq \deg f$. ☐

In view of the preceding results, let $R$ be an integral domain, let $f\in R[x]$ be nonzero, and let $a\in R$. By the degree formula, any nonnegative integer $m$ satisfying $(x-a)^m\,|\,f$ is at most $\deg f$. Write $\nu_a(f)$ for the greatest natural number $m$ such that $(x-a)^m\,|\,f$. By the factor theorem, $\nu_a(f)=0$ if and only if $f(a)\neq 0$. If $\nu_a(f)=m>0$, then $a$ is called a *root of multiplicity $m$* or a *zero of multiplicity $m$* of $f$ (or simply a *root* or *zero* when $m$ is not emphasized). The number $m$ is called the *multiplicity* of $a$ as a root of $f$. When $m\geq 2$, $a$ is called a *multiple root* of $f$.

**Example A.6.10.** In $\mathbb{Q}[x]$:
(1) $x^4+x+1=(x^2-1)(x^2+1)+(x+2)$, so division by $x^2+1$ leaves remainder $x+2$;
(2) $x^3-3x+2=(x-1)^2(x+2)$, so $x=1$ is a double root;
(3) no rational number $a\in\mathbb{Q}$ satisfies $a^2=2$, so $x^2-2$ has no root in $\mathbb{Q}$. Viewed as a polynomial in $\mathbb{R}[x]$, however, it has roots $\pm\sqrt{2}$.

**Proposition A.6.11.** *Let $R$ be an integral domain and let $f,g\in R[x]$ be nonzero. For every $a\in R$, $\nu_a(fg)=\nu_a(f)+\nu_a(g)$.*

**Proof.** Write $f=(x-a)^m f_0$ and $g=(x-a)^n g_0$, where $f_0(a)g_0(a)\neq 0$. Then $fg=(x-a)^{m+n}f_0g_0$ and $(f_0g_0)(a)=f_0(a)g_0(a)\neq 0$. Thus the greatest power of $x-a$ that divides $fg$ is precisely $m+n$. ☐

**Proposition A.6.12.** *Let $R$ be an integral domain, let $0\neq f\in R[x]$, and let $a_1,\ldots,a_s\in R$ be distinct. Then $\nu_{a_1}(f)+\cdots+\nu_{a_s}(f)\leq \deg f$. In particular, a nonzero polynomial of degree $n$ has at most $n$ distinct roots in $R$.*

**Proof.** Put $m_i=\nu_{a_i}(f)$. By definition, $(x-a_1)^{m_1}\,|\,f$. Remove this factor from $f$ and substitute $a_i$ in the quotient for $i>1$. $a_i-a_1\neq 0$, so proposition A.6.11 shows that $a_i$ retains multiplicity $m_i$ in the quotient. Repeating this argument gives $\prod_{i=1}^{s}(x-a_i)^{m_i}\,|\,f$. By divisibility, the degree of the polynomial on the left is no greater than that of the polynomial on the right, giving the required inequality. In particular, if $a_1,\ldots,a_s$ are all roots, then every $m_i\geq 1$, so $s\leq \deg f$; that is, a polynomial of degree $n$ has no more than $\deg f$ distinct roots. ☐

**Theorem A.6.13** (Vieta's formulas)**.** *Let $R$ be a commutative ring with identity, and let $f(x) = \sum_{i=0}^{n} a_i x^i \in R[x]$, where $a_n \neq 0$. If $r_1, \dots, r_n \in R$, not necessarily distinct, satisfy $f(x) = a_n \prod_{i=1}^{n}(x - r_i)$, then, for $k = 1, \dots, n$,*

$$a_{n-k} = (-1)^k a_n \sum_{1 \le i_1 < \cdots < i_k \le n} r_{i_1} \cdots r_{i_k}.$$

*In particular,*

$$a_{n-1} = -a_n(r_1 + \cdots + r_n), \quad a_0 = (-1)^n a_n r_1 \cdots r_n.$$

**Proof.** In expanding $\prod_{i=1}^{n}(x - r_i)$, a term in $x^{n-k}$ is obtained by choosing the constant term $-r_i$ from exactly $k$ of the $n$ factors and choosing $x$ from every remaining factor. Hence the coefficient of $x^{n-k}$ is $(-1)^k \sum_{1 \le i_1 < \cdots < i_k \le n} r_{i_1} \cdots r_{i_k}$. Multiplying this coefficient by $a_n$ and comparing coefficients of like powers on the two sides gives the result. □

### Bézout's theorem

**Lemma A.6.14.** *Let $R$ be an integral domain. Two associated monic polynomials in $R[x]$ are equal.*

**Proof.** By theorem A.6.4, $R[x]$ is an integral domain and $R[x]^\times = R^\times$. Thus polynomials satisfy $f \sim g$ if and only if there is a unit $u \in R^\times$ such that $f = ug$. The degree formula gives $\deg f = \deg u + \deg g$. Since $u$ is a constant polynomial, $\deg u = 0$, so $\deg f = \deg g$. Since $f, g$ are monic, comparison of leading coefficients gives $1 = u \cdot 1$, and hence $u = 1$ and $f = g$. □

**Theorem A.6.15** (Bézout's theorem)**.** *Let $F$ be a field and let $f, g \in F[x]$ be not both zero. Then $f, g$ have a unique monic polynomial as their greatest common divisor.* Hereafter we denote it by $\gcd(f, g)$. *Moreover, there are $u, v \in F[x]$ such that* $\gcd(f, g) = uf + vg$.

**Proof.** Without loss of generality, suppose $g \neq 0$. Put $r_{-1} = f$ and $r_0 = g$, and repeatedly apply division with remainder:

$$r_{-1} = q_1 r_0 + r_1, \quad r_0 = q_2 r_1 + r_2, \quad \dots, \quad r_{s-1} = q_{s+1} r_s + r_{s+1},$$

where $r_s \neq 0$, $r_{s+1} = 0$, and the degree of each nonzero remainder is less than that of the preceding divisor. Nonnegative integer degrees cannot decrease strictly forever, so this process terminates after finitely many steps.

If $r_1 = 0$, then $r_0 = g$ is already a greatest common divisor of $f, g$. Multiplying by the inverse $c$ of its leading coefficient gives the monic greatest common divisor $cg$, with $cg = 0f + cg$. Any two greatest common divisors are associated, and two associated monic polynomials must be equal. We now consider $r_1 \neq 0$; then $s \geq 1$.

By proposition A.4.23(3), each division identity shows that a polynomial dividing both the divisor and the dividend also divides the remainder. Conversely, one dividing both the divisor and the remainder also divides the dividend. Thus $h$ is a common divisor of $r_{i-1}, r_i$ if and only if it is a common divisor of $r_i, r_{i+1}$. Successive applications show that the common divisors of $f, g$ are precisely the common divisors of $r_s, r_{s+1} = 0$, namely the divisors of $r_s$. Hence $r_s$ is a greatest common divisor of $f, g$.

Finally, start with $r_s = r_{s-2} - q_s r_{s-1}$ and successively substitute $r_i = r_{i-2} - q_i r_{i-1}$ to obtain

$$r_s = u_0 r_{-1} + v_0 r_0 = u_0 f + v_0 g,$$

where $u_0, v_0 \in F[x]$. Multiplying both sides of $r_s = u_0 f + v_0 g$ by the inverse of the leading coefficient of $r_s$ gives the monic greatest common divisor and the identity $\gcd(f, g) = uf + vg$. Any two greatest common divisors are associated, and two associated monic polynomials must be equal. Thus this monic polynomial is unique. □

We have met Bézout's theorem for the integer ring in number theory, and now for divisibility of polynomials over a field. The polynomial theorem does not imply the integer version: $\mathbb{Z}$ is not a field, so the theorem does not apply there, while any two nonzero integers become associates over $\mathbb{Q}$.

Thus treating integers as constant polynomials in $\mathbb{Z}[x]$ or $\mathbb{Q}[x]$ does not derive the integer theorem. Bézout's theorem is in fact a common property of the so-called principal ideal domains (PIDs). Their general theory is not needed here, so we shall not discuss it.

### Primitive polynomials

**Definition A.6.16.** Let $R$ be an integral domain. A nonzero polynomial $f \in R[x]$ is called a *primitive polynomial* (or $f$ is called *primitive*) if the coefficients of all its terms have no common nonunit divisor.

Let $R$ be a unique factorization domain and let $f = \sum_{i=0}^{n} a_i x^i \in R[x]$ be nonzero. By proposition A.4.27, the nonzero coefficients $a_i$ of $f$ have a greatest common divisor, called the *content* of $f$ and hereafter denoted by $\mathrm{c}(f)$; it is uniquely determined as an equivalence class of associates. If $\mathrm{c}(f)$ is a unit, then $f$ is called a *primitive polynomial*, or simply *primitive*.

We shall study primitive polynomials only over a unique factorization domain $R$, since greatest common divisors of elements of such an $R$ always exist. The theory can in fact be extended to so-called greatest common divisor domains (GCD domains), in which any two elements are required to have a greatest common divisor, but which need not be unique factorization domains. This general theory is not needed later, and the extension would require introducing further theory, so we focus only on unique factorization domains.

**Proposition A.6.17.** *Let $R$ be a unique factorization domain and $0 \neq f \in R[x]$.*
*(1) $f$ is primitive exactly when $\mathrm{c}(f) \sim 1$;*
*(2) if $f$ is primitive and $0 \neq a \in R$, then $\mathrm{c}(af) \sim a$.*

**Proof.** The first assertion is the definition together with proposition A.4.24(4).

For the second, let the coefficients of $f$ be $b_i$, and put $d = \mathrm{c}(af)$. Since $a \mid ab_i$ for every $i$, we have $a \mid d$, so write $d = ae$. By the definition of content, $d \mid ab_i$ for every $i$; thus $ae \mid ab_i$. Canceling $a$ in the integral domain gives $e \mid b_i$ for every $i$. Since $f$ is primitive, every common divisor of all the $b_i$ is a unit. Therefore $e$ is a unit, and $d = ae \sim a$. □

**Proposition A.6.18.** *For a unique factorization domain $R$ and $0 \neq f \in R[x]$, the content $\mathrm{c}(f)$ always exists, and $f = \mathrm{c}(f) f_0$ for some primitive $f_0 \in R[x]$.*

**Proof.** By proposition A.4.27(3), the content always exists. Once a representative of $\mathrm{c}(f)$ is chosen, it divides every coefficient of $f$. Thus every nonzero $f \in R[x]$ can be written as $f = \mathrm{c}(f) f_0$. We prove that $f_0$ is primitive.

Otherwise $\mathrm{c}(f_0)$ would be nonzero and a nonunit. Since $R$ is a UFD, factor $\mathrm{c}(f_0)$ and choose an irreducible factor $p$. Then $p$ divides every coefficient of $f_0$, so $p\,\mathrm{c}(f)$ is a common divisor of all the coefficients of $f$. The definition of a greatest common divisor gives $p\,\mathrm{c}(f) \mid \mathrm{c}(f)$. Cancel the nonzero $\mathrm{c}(f)$ using proposition A.4.23(5) to obtain $p \mid 1$. By proposition A.4.23(6), this contradicts the fact that $p$ is a nonunit. □

**Corollary A.6.19.** *Let $R$ be a unique factorization domain and let $f \in R[x]$ be a nonconstant polynomial. If $f$ is irreducible, then $f$ is primitive.*

**Proof.** Write $f = \mathrm{c}(f) f_0$. Since $f$ is nonconstant, $\deg f > 0$, so $\deg f_0 = \deg f > 0$. Thus $f_0$ has positive degree and is therefore a nonunit by theorem A.6.4. If $\mathrm{c}(f)$ were also a nonunit, $f$ would be a product of nonunits, contradicting its irreducibility. □

**Theorem A.6.20** (Gauss's lemma)**.** *Let $R$ be a unique factorization domain. The product of two primitive polynomials in $R[x]$ is primitive.*

**Proof.** Write $f=\sum_{i=0}^{m} a_i x^i$ and $g=\sum_{j=0}^{n} b_j x^j$, both primitive. If $fg$ is not primitive, then $\mathrm{c}(fg)$ is a nonunit. Choose a factor $p$ in an irreducible factorization of $\mathrm{c}(fg)$; it divides all the coefficients of $fg$. By proposition A.4.27(2), $p$ is prime.

Since $f$ is primitive, if $p$ divided every $a_i$, then $p \mid \mathrm{c}(f)$ and proposition A.4.23(6) would make $p$ a unit, contrary to its irreducibility. Thus there is a least $i$ with $p \nmid a_i$. Similarly, choose the least $j$ with $p \nmid b_j$. Consider the coefficient $c_{i+j}=\sum_{r+s=i+j} a_r b_s$ of $x^{i+j}$ in $fg$.[16] If $r<i$, then $p \mid a_r$; if $r>i$, then $s=i+j-r<j$, so $p \mid b_s$.

Thus every term other than $a_i b_j$ is divisible by $p$. On the other hand, $p \nmid a_i$ and $p \nmid b_j$, so the primality of $p$ gives $p \nmid a_i b_j$. Hence $p \nmid c_{i+j}$, contradicting the fact that $p$ divides all the coefficients of $fg$. Therefore $fg$ is primitive. □

**Corollary A.6.21.** *For nonzero $f,g \in R[x]$ over a unique factorization domain $R$,* $\mathrm{c}(fg) \sim \mathrm{c}(f)\,\mathrm{c}(g)$.

**Proof.** Write $f=\mathrm{c}(f) f_0$ and $g=\mathrm{c}(g) g_0$ with $f_0, g_0$ primitive. Gauss's lemma makes $f_0 g_0$ primitive. Since $fg=\mathrm{c}(f)\,\mathrm{c}(g) f_0 g_0$, proposition A.6.17(2) shows that its content is associated with $\mathrm{c}(f)\,\mathrm{c}(g)$. □

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## Exercises

**Exercise A.94.** In $\mathbb{Z}_6[x]$, let $f=2x+1$ and $g=3x+1$. Compute $\deg f$, $\deg g$, $\deg(fg)$, and $\deg(f+g)$. Does the degree formula above hold?

**Exercise A.95.** Perform the following divisions with remainder in $\mathbb{Q}[x]$, and decide whether they can also be performed with remainder in $\mathbb{Z}[x]$:
(1) divide $x^5+2x^3-x+3$ by $x^2-x+1$;
(2) divide $x^4+x^2+1$ by $2x-1$.

**Exercise A.96.** If $R$ is an integral domain, $0 \neq f, g \in R[x]$, and $f \mid g$, prove $\nu_a(f) \leq \nu_a(g)$.

**Exercise A.97.** Let $R$ be an integral domain, let $0 \neq f, g \in R[x]$ with $f+g \neq 0$, and let $a \in R$. Prove $\nu_a(f+g) \geq \min\{\nu_a(f), \nu_a(g)\}$, and show that $\nu_a(f) \neq \nu_a(g)$ is sufficient, but not necessary, for equality.

**Exercise A.98** (Lagrange interpolation formula)**.** Let $F$ be a field, let $a_1, \ldots, a_n \in F$ be distinct, and let $b_1, \ldots, b_n \in F$. Prove that there exists a unique polynomial $f \in F[x]$ of degree less than $n$ satisfying $f(a_i)=b_i$ for every $i$, and that it is

$$f(x)=\sum_{i=1}^{n} b_i \prod_{j \neq i} \frac{x-a_j}{a_i-a_j}.$$

**Exercise A.99.** Let $F$ be a field and let $a_1, \ldots, a_n \in F$ be distinct. Put $g(x)=\prod_{i=1}^{n}(x-a_i)$. Prove that the remainder when $f \in F[x]$ is divided by $g$ is

$$r(x)=\sum_{i=1}^{n} f(a_i) \prod_{j \neq i} \frac{x-a_j}{a_i-a_j}.$$

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[16] In the sum, a coefficient whose degree does not occur in the original polynomial is regarded as zero.

**Exercise A.100.** Let $f = 12x^4 - 18x^2 + 30x$ and $g = 10x^3 + 15x^2 - 5$.

(1) Regard $f, g$ as polynomials in $\mathbb{Q}[x]$. Find $\gcd(f,g)$ and $u, v \in \mathbb{Q}[x]$ such that $uf + vg = \gcd(f,g)$.
(2) Regard $f, g$ as polynomials in $\mathbb{Z}[x]$. Compute directly $\mathrm{c}(f)$, $\mathrm{c}(g)$, $fg$, and $\mathrm{c}(fg)$, and verify Gauss's lemma.

## A.6.2 Factorization of One-Variable Polynomials

### Factorization over a field

Fields have the richest properties, so divisibility, greatest common divisors, associates, and irreducibility are conveniently studied over them. We therefore consider polynomial factorization over a field first.

**Theorem A.6.22.** *Every irreducible element of a polynomial ring over a field is prime.*

**Proof.** Consider the polynomial ring $F[x]$ over a field $F$. Suppose that $p$ is irreducible, $p \mid fg$, and $p \nmid f$. Since $\gcd(p,f) \mid p$, write $p = u\gcd(p,f)$. If $\gcd(p,f)$ were not a unit, irreducibility of $p$ would make $u$ a unit, and hence $p \mid \gcd(p,f)$. Since $\gcd(p,f) \mid f$, this would imply $p \mid f$, a contradiction. Thus $\gcd(p,f)$ is a unit. By theorem A.6.4, $\gcd(p,f) \in F^\times$, so it is a constant polynomial. Multiplying by a suitable nonzero invertible constant, we may take $\gcd(p,f) = 1$; the replacement is associated with the original greatest common divisor.

Bézout's theorem supplies $u, v \in F[x]$ such that $up + vf = 1$. Multiplying by $g$ gives $upg + vfg = g$. Both terms on the left are divisible by $p$, because $p \mid p$ and $p \mid fg$. Therefore $p \mid g$. $\square$

**Theorem A.6.23.** *For every field $F$, the one-variable polynomial ring $F[x]$ is a unique factorization domain.*

**Proof.** We first prove existence. Let $f \in F[x]$ be nonzero and a nonunit. By theorem A.6.4, its degree is positive. We argue by induction on $\deg f$. If $f$ is irreducible, the assertion already holds. If it is reducible, write $f = gh$ with both $g$ and $h$ nonunits. Then $0 < \deg g, \deg h < \deg f$. By the induction hypothesis, both $g$ and $h$ factor into irreducibles, and combining their products gives a factorization of $f$.

For uniqueness, suppose $p_1 \cdots p_m = q_1 \cdots q_n$, with all $p_i, q_j$ irreducible, and argue by induction on $m+n$. By theorem A.6.22, $p_1$ is prime, so it divides some $q_j$. Without loss of generality, suppose $p_1 \mid q_1$ (reorder the $q_j$ if necessary). Since $q_1$ is irreducible and $p_1$ is a nonunit, $p_1 \sim q_1$. Write $p_1 = uq_1$ with $u$ a unit, and cancel $q_1$ to obtain

$$up_2 \cdots p_m = q_2 \cdots q_n.$$

If $m = 1$, the left side is the unit $u$, so the product on the right is a unit. Every factor of a unit product is a unit: multiplying the inverse of the whole product by all the other factors gives an inverse of that factor. Thus $n = 1$, since otherwise the irreducibility of $q_2, \ldots, q_n$ would be contradicted, and the right side is the empty product 1. Similarly, $n = 1$ implies $m = 1$. If $m, n > 1$, then $up_2$ remains irreducible because $up_2 \sim p_2$, by proposition A.4.24. The induction hypothesis applied to the displayed equality gives $m - 1 = n - 1$ and matches the remaining factors pairwise up to associates after reordering. Together with $p_1 \sim q_1$, this proves uniqueness. $\square$

**Remark.** Uniqueness in a unique factorization domain is understood up to the order of the factors and multiplication by units. In $F[x]$ the units are precisely the nonzero constants. Thus, in $\mathbb{Q}[x]$, the associated polynomials $x - 1$ and $2x - 2$ should not be regarded as essentially different irreducible factors. In $\mathbb{Z}[x]$, however, $2x - 2 = 2(x - 1)$: the constant 2 is irreducible in both $\mathbb{Z}$ and $\mathbb{Z}[x]$, so the polynomial factors as the product of the constant irreducible 2 and the linear polynomial $x - 1$.

**Formal derivatives and multiple roots**

For real and complex functions, derivatives will later be defined by local limits in sections A.8 and A.9. For general polynomials, the formal derivative can be defined directly from the coefficients over any coefficient ring. When $F=\mathbb{R}$ or $F=\mathbb{C}$, this gives the function derivative defined later. Formal derivatives help detect multiple roots of polynomials over a field and help factor polynomials.

We first define the formal derivative for polynomials over a general ring with identity, then focus on polynomials over a field.

**Definition A.6.24.** Let $R$ be a ring. For $f(x)=\sum_{i=0}^{n} a_i x^i \in R[x]$, its *formal derivative* is $f'(x)=\sum_{i=1}^{n} i a_i x^{i-1}$. We abbreviate $f'(x)$ to $f'$. Here $i$ in $ia_i$ is an integer and does not denote multiplication by an element of $R$: in additive-group notation, $ia_i$ is the sum of $i$ copies of $a_i$.

The formal derivative $f'$ is itself a polynomial, so it may be formally differentiated again. Write $(f')'=f''$ and call it the *second formal derivative* of $f$. Inductively, after $f^{(n-1)}$ has been defined, define the *nth formal derivative* by $f^{(n)}=(f^{(n-1)})'$. For $a\in R$, write its value at $a$ as $f^{(n)}(a)$, and adopt the convention $f^{(0)}=f$.

**Proposition A.6.25.** *Let $R$ be a ring. For $f,g\in R[x]$ and $a,b\in R$,*

*(1)* $(af+bg)'=af'+bg'$;

*(2)* $(fg)'=f'g+fg'$;

*(3) if $R$ is commutative and $n$ is a positive integer, $(f^n)'=nf'f^{n-1}$. Here $f^n$ is the product of $n$ copies of $f$,[17] while $nf$ means the sum of $n$ copies in the polynomial ring.*

**Proof.** Part (1) is easy. For (2), let $f=\sum_i a_i x^i$ and $g=\sum_j b_j x^j$. The coefficient of $x^r$ in $fg$ is $\sum_{i+j=r} a_i b_j$. Thus the coefficient of $x^{r-1}$ in $(fg)'$ is

$$r\sum_{i+j=r} a_i b_j=\sum_{i+j=r} i a_i b_j+\sum_{i+j=r} j a_i b_j,$$

which is precisely the coefficient of $x^{r-1}$ in $f'g+fg'$.

For (3), use induction and (2). For $n=2$, $(f^2)'=ff'+f'f=2ff'$. Suppose it is known for $n-1$. Then

$$(f^n)'=(ff^{n-1})'=f'f^{n-1}+f(f^{n-1})'=f'f^{n-1}+f\cdot(n-1)f'f^{n-2}=nf'f^{n-1}.$$

□

**Theorem A.6.26.** *Let $R$ be an integral domain, $0\neq f\in R[x]$, and $a\in R$. Then $a$ is a multiple root of $f$ exactly when $f(a)=f'(a)=0$.*

**Proof.** If $(x-a)^2 \mid f$, write $f=(x-a)^2h$. The product rule in proposition A.6.25 gives $f'=2(x-a)h+(x-a)^2h'$, so $f(a)=f'(a)=0$. Conversely, suppose that $f(a)=f'(a)=0$. By the factor theorem, write $f=(x-a)g$. Differentiating gives $f'=g+(x-a)g'$, and substitution of $a$ gives $g(a)=f'(a)=0$. A second application of the factor theorem yields $(x-a)\mid g$, and hence $(x-a)^2\mid f$. Thus $a$ is a multiple root. □

**Theorem A.6.27.** *Let $R$ be an integral domain with* $\operatorname{char} R=0$, *and let $f\in R[x]$ be nonzero and $a\in R$. Then $a$ is a root of multiplicity $m$ exactly when*

$$f(a)=f'(a)=f''(a)=\cdots=f^{(m-1)}(a)=0,\qquad f^{(m)}(a)\neq 0. \tag{♠}$$

[17] The power $f^0$ is understood as an empty product: when $n=1$, $(f^1)'=1f'\cdot f^0=f'$.

**Proof.** Suppose $a$ is a root of multiplicity $m$, so $f(x) = (x-a)^m g(x)$. By proposition A.6.25,

$$f' = m(x-a)^{m-1}g + (x-a)^m g' = (x-a)^{m-1}(mg + (x-a)g').$$

The expression in parentheses takes the value $mg(a)$ at $x=a$. Since $g(a) \neq 0$ and $\operatorname{char} R = 0$, we have $mg(a) \neq 0$. Hence $\nu_a(f') = \nu_a(f) - 1$ whenever $\nu_a(f) \geq 1$.

Repeating this argument for the root $a$ of multiplicity $m$ gives $\nu_a(f^{(k)}) = m-k$ for $0 \leq k \leq m$. Therefore $f^{(k)}(a) = 0$ for $k < m$, but $f^{(m)}(a) \neq 0$. Conversely, assume equation (♠) and put $\nu_a(f) = k$. If $k < m$, the preceding calculation gives $f^{(k)}(a) \neq 0$; if $k > m$, it gives $f^{(m)}(a) = 0$. Both contradict the displayed conditions, so $k = m$. □

**Proposition A.6.28.** *Let $F$ be a field with* $\operatorname{char} F = 0$*, and let $f \in F[x]$ be nonzero. Then $f$ has no repeated irreducible factor*[18] *if and only if* $\gcd(f,f') = 1$.

**Proof.** First, every polynomial of positive degree has an irreducible factor. This follows by induction on degree: if the polynomial itself is irreducible, the assertion holds; if it is reducible, write it as the product of two factors of smaller positive degree, and apply the induction hypothesis to one of them.

If $f$ has a repeated irreducible factor $p$, write $f = p^2 h$. Then $f' = 2pp'h + p^2 h'$, so $p | f'$. Thus $p$ is a common divisor of $f, f'$. Since $p$ is irreducible, $p \nsim 1$ (otherwise $p$ would be a unit), and the monic greatest common divisor satisfies $\gcd(f,f') \neq 1$.

Conversely, if $\gcd(f,f') \neq 1$, this monic polynomial is a nonunit. Since every polynomial of positive degree has an irreducible factor, the preceding argument lets us choose an irreducible $p$ dividing it. Then $p$ divides both $f$ and $f'$. Among the positive integers $m$ for which $p^m | f$, choose the largest; such a maximum exists because $m \deg p \leq \deg f$.

Write $f = p^m h$. Then $m \geq 1$ and $p \nmid h$. We must show that $m > 1$. Suppose to the contrary that $m = 1$. Then $f' = p'h + ph'$, and $p | f'$ gives $p | p'h$. By theorem A.6.22, $p$ is prime; since $p \nmid h$, we obtain $p | p'$. Let the leading term of $p$ be $a_n x^n$, with $n > 0$. In $p'$ this term becomes $na_n x^{n-1}$, and $\operatorname{char} F = 0$ guarantees $na_n \neq 0$, so $p' \neq 0$. Since $\deg p' < \deg p$, it is impossible that $p | p'$. Thus $m \geq 2$, and $f$ has a repeated irreducible factor. □

**Remark.** In the preceding two results, the hypotheses $\operatorname{char} R = 0$ and $\operatorname{char} F = 0$ ensure that $b \neq 0$ implies $mb \neq 0$ for $m \in \mathbb{Z}_+$. The two results need not hold for an integral domain or field of nonzero characteristic.

### Algebraically closed fields

**Definition A.6.29.** A field $F$ is *algebraically closed* if every nonconstant polynomial in the one-variable polynomial ring $F[x]$ has at least one root in $F$.

**Theorem A.6.30.** *For a field $F$, the following are equivalent:*

*(a) $F$ is algebraically closed;*
*(b) every nonzero $f \in F[x]$ is a product of finitely many linear polynomials and a nonzero constant; the number of linear factors may be zero;*
*(c) every irreducible polynomial in $F[x]$ is linear.*

**Proof.** Suppose $F$ is algebraically closed, and take a nonconstant polynomial $f$. Choose a root $a$. By the factor theorem, $(x-a) | f$. The quotient on division by $x-a$ has degree $\deg f - 1$, so induction factors $f$ completely into linear factors. Thus (a) implies (b).

Suppose (b) holds. If an irreducible polynomial $f$ had degree greater than one, it would have a linear factor; this factor and the product of the remaining factors would give a nontrivial factorization, a contradiction. Thus (b) implies (c).

[18] A repeated irreducible factor means a factor of the form $q^r$, where $q$ is irreducible and $r$ is an integer greater than 1.

Suppose (c) holds. By theorem A.6.23, every nonconstant polynomial $f$ can be factored into irreducible factors. Each factor is linear, so $f$ has a root. Hence (c) implies (a). ☐

**Proposition A.6.31.** *Let $F$ be an algebraically closed field. A nonzero polynomial $f \in F[x]$ has no repeated irreducible factor if and only if it has no multiple root.*

**Proof.** When $F$ is algebraically closed, every irreducible factor is linear. Thus $f$ has no repeated irreducible factor if and only if $f$ has no multiple root. ☐

**Corollary A.6.32.** *If $F$ is an algebraically closed field, every polynomial $f \in F[x]$ of positive degree $n$ has a unique expression*

$$f(x) = c\prod_{i=1}^{s}(x-a_i)^{m_i},$$

*where $c \in F^\times$, the $a_1,\dots,a_s \in F$ are pairwise distinct, the $m_i$ are positive integers, and $m_1+\cdots+m_s = n$. Uniqueness disregards the order of the linear factors.*

**Proof.** By theorem A.6.30, $f$ is a product of $n$ linear factors. Combining equal roots gives the displayed form, and the degree formula gives $m_1+\cdots+m_s = n$. Uniqueness follows because $F[x]$ is a UFD and, by lemma A.6.14, associated monic linear polynomials are equal. ☐

**Corollary A.6.33** (Vieta's formulas)**.** *Let $F$ be an algebraically closed field. Suppose that $f(x) = \sum_{i=0}^{n} a_i x^i \in F[x]$ has degree $n$, with $a_n \neq 0$. Its roots $r_1,\dots,r_n \in F$, repeated according to multiplicity, satisfy, for $k = 1,\dots,n$,*

$$\sum_{1\le i_1<\cdots<i_k\le n} r_{i_1}\cdots r_{i_k} = (-1)^k \frac{a_{n-k}}{a_n}.$$

*In particular,*

$$a_{n-1} = -a_n(r_1+\cdots+r_n), \quad a_0 = (-1)^n a_n r_1\cdots r_n.$$

**Proof.** By corollary A.6.32, knowing the roots gives $f(x) = a_n\prod_{i=1}^{n}(x-r_i)$. Apply the preceding Vieta theorem for a commutative ring with identity. The earlier formula was written in the form $a_{n-k} = \cdots$ to avoid the issue of zero divisors. Over a field there are no zero divisors, so we may divide by $a_n$ and move it to the other side. ☐

**Theorem A.6.34** (Fundamental theorem of algebra)**.** *The field $\mathbb{C}$ is algebraically closed.*

**Proof.** Although called a "fundamental theorem," its standard proofs require further results of real or complex analysis. Those tools lie beyond the scope of these notes, so the proof is omitted. ☐

**Corollary A.6.35.** *Every complex polynomial $f$ of positive degree $n$ has a unique expression*

$$f(x) = c\prod_{i=1}^{s}(x-a_i)^{m_i},$$

*where $c \in \mathbb{C}^\times$, the $a_1,\dots,a_s \in \mathbb{C}$ are pairwise distinct, the $m_i$ are positive integers, and $m_1+\cdots+m_s = n$. Here uniqueness disregards the order of the linear factors.*

**Example A.6.36.** The real field $\mathbb{R}$ is not algebraically closed: for example, $x^2+1$ has no root in $\mathbb{R}$. In $\mathbb{C}[x]$, however, $x^2+1 = (x-\mathrm{i})(x+\mathrm{i})$.

### Factorization over a UFD

Factorization is conveniently studied over a field, but the general case is more complicated. Here we consider polynomials over a UFD $R$. Primitive polynomials connect them with polynomials over the field of fractions $\mathrm{Frac}(R)$, and also help the later passage from one variable to several variables.

**Lemma A.6.37.** *Let $R$ be a unique factorization domain and $F = \mathrm{Frac}(R)$. Every nonzero $f \in F[x]$ can be written $f = \lambda g$, where $\lambda \in F^\times$ and $g \in R[x]$ is primitive. If there are two representations $f = \lambda g = \mu h$ with $g,h$ both primitive, then $\lambda/\mu \in R^\times$ and $g,h$ are associated.*[19]

**Proof.** Write the coefficients of $f$ as $\frac{a_i}{b_i}$. Since each $b_i \neq 0$ and $R$ has no zero divisors, take $d = \prod_i b_i \neq 0$. Then $b_i | d$ for every $i$, so $d\frac{a_i}{b_i} \in R$ and hence $df \in R[x]$. By proposition A.6.18, dividing $df$ by its content gives a primitive polynomial $g \in R[x]$. Therefore $f = \lambda g$, where $\lambda = \mathrm{c}(df)d^{-1} \in F^\times$, since both $\mathrm{c}(df)$ and $d$ are nonzero in $R$.

For the second assertion, suppose $f = \lambda g = \mu h$. Then $g = \alpha h$ with $\alpha = \lambda^{-1}\mu \in F^\times$. Write $\alpha = \frac{a}{b}$, where $a,b \in R$ have no common irreducible factor, canceling such factors first if necessary. Thus $bg = ah$. If an irreducible $p \in R$ divides $a$, then $p \nmid b$, and by proposition A.4.27 it is prime. Since $p$ divides the product of $a$ with every coefficient of $h$, the equality $bg = ah$ shows that it divides the product of $b$ with every coefficient of $g$. Primality and $p \nmid b$ force $p$ to divide every coefficient of $g$, contradicting primitivity. Thus $a$ is a unit; similarly, $b$ is a unit. Consequently $\alpha \in R^\times$, and $g,h$ are associated. $\square$

**Theorem A.6.38.** *Let $R$ be a unique factorization domain, $F = \mathrm{Frac}(R)$, and let $f \in R[x]$ be primitive of positive degree. Then $f$ is reducible in $R[x]$ exactly when it is reducible in $F[x]$.*

**Proof.** A nontrivial factorization $f = gh$ in $R[x]$ cannot have a nonunit constant factor: such a constant would divide every coefficient of $f$, contradicting primitivity. Thus $g,h$ both have positive degree, and the factorization remains nontrivial in $F[x]$.

Conversely, suppose $f = gh$ in $F[x]$, with $g,h$ of positive degree. By lemma A.6.37, write $g = \alpha g_0$ and $h = \beta h_0$ with primitive $g_0, h_0 \in R[x]$. Then $f = \alpha\beta g_0 h_0$. Gauss's lemma makes $g_0 h_0$ primitive, and lemma A.6.37 gives $\alpha\beta \in R^\times$. Hence $f = (\alpha\beta g_0)h_0$ is a nontrivial factorization in $R[x]$. $\square$

The preceding two results extend to greatest common divisor domains. We have stated them only for UFDs to avoid introducing additional theory. The next property is special to UFDs.

**Theorem A.6.39.** *For a ring $R$, the polynomial ring $R[x]$ is a unique factorization domain exactly when $R$ is a unique factorization domain.*

**Proof.** First suppose that $R$ is a UFD and put $F = \mathrm{Frac}(R)$. We first prove that a factorization exists. For a nonzero nonunit $f \in R[x]$, write $f = \mathrm{c}(f)f_0$, with $f_0$ primitive. If $\mathrm{c}(f)$ is a nonunit, factor it into irreducible elements of $R$; its irreducible factors remain irreducible as constant polynomials in $R[x]$, because the degree formula permits only constant factors. If $f_0$ is a unit, the factorization of $\mathrm{c}(f)$ already gives one of $f$: the remaining unit can be absorbed into an irreducible factor, which remains irreducible.

Otherwise $f_0$ has positive degree: a constant polynomial has itself as content, so a primitive constant is a unit. By theorem A.6.23, factor it in the UFD $F[x]$ as $f_0 = p_1 \cdots p_s$, where $s \geq 1$ and each $p_i$ is irreducible in $F[x]$. For each $i$, lemma A.6.37 supplies $\lambda_i \in F^\times$ such that $P_i = \lambda_i p_i \in R[x]$ is primitive. Thus $f_0 = \delta P_1 \cdots P_s$, where $\delta = (\lambda_1 \cdots \lambda_s)^{-1} \in F^\times$. Gauss's lemma makes $P_1 \cdots P_s$ primitive, and lemma A.6.37 then gives $\delta \in R^\times$. Absorb $\delta$ into $P_1$. Each $P_i$ is irreducible in $R[x]$ by theorem A.6.38, so together with the factorization of $\mathrm{c}(f)$ this proves existence.

[19]Strictly from the original definition, $R$ is not a subset of $\mathrm{Frac}(R)$. In theorem A.4.29, however, $\frac{a}{1} \in \mathrm{Frac}(R)$ was abbreviated to $a$, identifying $a \in R$ with $a \in \mathrm{Frac}(R)$. Equivalently, $f : R \to \mathrm{Frac}(R)$, $a \mapsto \frac{a}{1}$, is an injective ring homomorphism. Thus $R$ is regarded directly as a subring of $\mathrm{Frac}(R)$. We use this convention from now on.

For uniqueness, corollary A.6.19 shows that the irreducibles of $R[x]$ fall into two classes: constant irreducibles from $R$, and primitive irreducible polynomials of positive degree. Suppose that the same nonzero element has two irreducible factorizations

$$f = a_1 \cdots a_r p_1 \cdots p_m = b_1 \cdots b_s q_1 \cdots q_n$$

with constant irreducibles $a_i, b_j \in R$ and positive-degree primitive irreducibles $p_i, q_j \in R[x]$. Gauss's lemma makes $P = p_1 \cdots p_m$ and $Q = q_1 \cdots q_n$ primitive. By proposition A.6.17(2), $\mathrm{c}(f) \sim a_1 \cdots a_r$ and $\mathrm{c}(f) \sim b_1 \cdots b_s$. Thus $a_1 \cdots a_r \sim b_1 \cdots b_s$. Uniqueness in $R$ gives $r = s$ and a permutation $\sigma \in \mathrm{S}_r$ with $a_i \sim b_{\sigma(i)}$. Associated elements differ by multiplication by a unit. Replace associated pairs $a_i, b_{\sigma(i)}$ by equal elements, absorb their unit factors into a single unit, and use theorem A.6.4, which shows that $R[x]$ is an integral domain, to cancel the equal constant factors. We obtain $p_1 \cdots p_m = e q_1 \cdots q_n$ with $e \in R^\times$.

Viewed in the UFD $F[x]$, this equality gives $m = n$ and a permutation $\tau \in \mathrm{S}_m$ with $p_i \sim_{F[x]} q_{\tau(i)}$. Thus $p_i = \lambda_i q_{\tau(i)}$ for some $\lambda_i \in F^\times$, for every $i$. Since $p_i, q_j$ are primitive, lemma A.6.37 gives $\lambda_i \in R^\times$, so $p_i \sim_{R[x]} q_{\tau(i)}$. This proves uniqueness in $R[x]$.

Conversely, suppose $R[x]$ is a UFD. By theorem A.6.4, $R$ is first of all an integral domain. Take any nonzero nonunit $a \in R$ and regard it as a constant polynomial in $R[x]$. It can be factored into irreducible elements in $R[x]$. Since $\deg a = 0$, theorem A.6.1 shows that every factor must be a constant polynomial, hence an element of $R$. Thus this very factorization is an irreducible factorization of $a$ in $R$. Its uniqueness also follows from uniqueness in $R[x]$. Therefore $R$ is a unique factorization domain. □

## Exercises

**Exercise A.101.** For $f(x) = 12x^4 - 24x^3 + 24x^2 - 24x + 12 \in \mathbb{Z}[x]$:

(1) prove that $\mathbb{Z}[x]$ is a unique factorization domain;
(2) use the formal-derivative criterion to determine whether $x = 0$ and $x = 1$ are zeros of $f(x)$, respectively, and find their multiplicities;
(3) factor $f(x)$ into irreducibles;
(4) regard $f(x)$ in turn as an element of $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$, and factor it into irreducibles again.

**Exercise A.102.** Prove that a polynomial of degree 2 or 3 over a field is reducible exactly when it has a root.

**Exercise A.103.** Let $n$ be a positive integer and $a \in \mathbb{C} \setminus \{0\}$. Without explicitly finding its roots, prove that $x^n - a \in \mathbb{C}[x]$ has no multiple root.

**Exercise A.104.** Let $F$ be a field of characteristic zero, and suppose $0 \neq f \in F[x]$ has no repeated irreducible factor. Prove that:

(1) $uf + vf' = 1$ for some $u, v \in F[x]$;
(2) for $g \in F[x]$, if $f \mid g^m$ for some positive integer $m$, then $f \mid g$.

**Exercise A.105.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Q}[x]$, where $a_0, a_1, \ldots, a_n \in \mathbb{Z}$ and $a_n a_0 \neq 0$. If $f$ has a root $\frac{p}{q}$, where $p, q$ are coprime, $\gcd_{\mathbb{Z}}(p, q) = 1$, prove in $\mathbb{Z}$ that $p \mid a_0$ and $q \mid a_n$.

**Exercise A.106** (Eisenstein's criterion)**.** Let $n \geq 1$, let $R$ be a UFD, let $p$ be prime in $R$, and let $f(x) = \sum_{i=0}^{n} a_i x^i \in R[x]$, with $a_n \neq 0$. If $p \nmid a_n$, $p \mid a_i$ for $0 \leq i < n$, and $p^2 \nmid a_0$, prove that $f$ is irreducible in $\mathrm{Frac}(R)[x]$.

**Exercise A.107*.** For a commutative ring $R$ with identity and $f(x) = \sum_{n\geq 0} a_n x^n \in R[x]$, with only finitely many nonzero $a_n$, define the $k$th *Hasse derivative* by

$$\mathrm{D}^{(k)} f(x) = \sum_{n\geq k} \binom{n}{k} a_n x^{n-k}.$$

Here $\binom{n}{k}$ is the binomial coefficient, also sometimes written as the combination number $\mathrm{C}_n^k$. It is an integer in $\mathbb{Z}$; before $a_n$, it means the sum of that many copies of $a_n$, not multiplication in the ring $R$. We also set $\mathrm{D}^{(0)} f = f$.

(1) For $f,g \in R[x]$, prove $\mathrm{D}^{(k)}(f+g) = \mathrm{D}^{(k)} f + \mathrm{D}^{(k)} g$ and $\mathrm{D}^{(k)}(fg) = \sum_{i=0}^{k} \mathrm{D}^{(i)} f\, \mathrm{D}^{(k-i)} g$.
(2) For every $a \in R$, prove $f(x+a) = \sum_{k\geq 0} (\mathrm{D}^{(k)} f(a)) x^k$.
(3) Let $R = F$ be any field, let $a \in F$, and let $0 \neq f \in F[x]$. Prove that $\nu_a(f) = m$ exactly when

$$\mathrm{D}^{(0)} f(a) = \mathrm{D}^{(1)} f(a) = \cdots = \mathrm{D}^{(m-1)} f(a) = 0, \qquad \mathrm{D}^{(m)} f(a) \neq 0.$$

### A.6.3 Multivariable Polynomials

Having studied polynomials in one variable, we now use them to study polynomials in several variables.

**Definition A.6.40.** Let $R$ be a ring. The one-variable polynomial ring $R[x_1]$ has already been defined. For integers $r \geq 2$, define recursively

$$R[x_1,\dots,x_r] := (R[x_1,\dots,x_{r-1}])[x_r].$$

This is the *polynomial ring in r indeterminates* over $R$. The *indeterminates* are $x_1,\dots,x_r$. In the one-variable definition, $x_1$ commutes with elements of $R$; recursively, all the indeterminates commute with one another and with $R$. Thus $x_i x_j = x_j x_i$ and $x_i r = r x_i$ for $r \in R$ and every $i,j$. For an $r$-tuple of nonnegative integers $\boldsymbol{\alpha} = (\alpha_1,\dots,\alpha_r) \in \mathbb{N}^r$, write $x^{\boldsymbol{\alpha}} := x_1^{\alpha_1} \cdots x_r^{\alpha_r}$ and $|\boldsymbol{\alpha}| := \alpha_1 + \cdots + \alpha_r$. We call $\boldsymbol{\alpha}$ a *multi-index*, and a polynomial of the form $a_{\boldsymbol{\alpha}} x^{\boldsymbol{\alpha}}$, where $a_{\boldsymbol{\alpha}} \in R$, a *monomial*; $|\boldsymbol{\alpha}|$ is its *total degree*.

**Theorem A.6.41.** *Let R be a ring and $S = R[x_1,\dots,x_r]$. Then:*
*(1) R has an identity if and only if S has an identity. In this case the identity of R, regarded as a constant polynomial, is the identity of S.*
*(2) R is commutative if and only if S is commutative.*
*(3) R has no zero divisors if and only if S has no zero divisors.*
*(4) R is an integral domain if and only if S is an integral domain; in this case $R^\times = S^\times$.*
*(5) R is a unique factorization domain if and only if S is a unique factorization domain.*
*In particular, $F[x_1,\dots,x_r]$ is a UFD and has units $F^\times$ for every field F.*

**Proof.** Use the recursive definition of multivariable polynomials. Recursive application of theorem A.6.3 proves (1), (2), and (3). Recursive application of theorem A.6.4 and theorem A.6.39 proves (4) and (5), respectively. □

**Proposition A.6.42.** *Let R be a ring. Every $f \in R[x_1,\dots,x_r]$ can be written uniquely as a finite sum $f = \sum_{\boldsymbol{\alpha}} a_{\boldsymbol{\alpha}} x^{\boldsymbol{\alpha}}$, where $a_{\boldsymbol{\alpha}} \in R$ and only finitely many coefficients are nonzero.* Each $a_{\boldsymbol{\alpha}} x^{\boldsymbol{\alpha}}$ is a *term* of the polynomial, and $|\boldsymbol{\alpha}|$ is the *total degree* of that term. *Two polynomials are equal if and only if their corresponding coefficients are equal.*

**Proof.** We argue by induction on the number $r$ of indeterminates. The case $r = 1$ is the definition of a polynomial in one indeterminate. Suppose the assertion holds for $r-1$ indeterminates. By the recursive definition, $f$ has a unique expression

$$f = f_0 + f_1 x_r + \cdots + f_n x_r^n, \qquad f_i \in R[x_1,\dots,x_{r-1}].$$

By the induction hypothesis, each $f_i$ in turn has a unique expansion of the required form. This gives both the required expression for $f$ and its uniqueness. □

**Remark.** The recursive definition also lets us speak of terms in a selected $x_i$. Here the polynomial is viewed in the ring $(R[x_1,\dots,x_{i-1},x_{i+1},\dots,x_r])[x_i]$. Without qualification, however, a term means $a_{\boldsymbol\alpha} x^{\boldsymbol\alpha}$.

**Proposition A.6.43.** *Let $R$ be a ring, and let $f,g \in R[x_1,\dots,x_r]$. If $f=\sum_{\boldsymbol\alpha} a_{\boldsymbol\alpha} x^{\boldsymbol\alpha}$ and $g=\sum_{\boldsymbol\beta} b_{\boldsymbol\beta} x^{\boldsymbol\beta}$, with absent coefficients understood to be zero, then the following formulas hold.*[20]

$$f+g=\sum_{\boldsymbol\gamma}(a_{\boldsymbol\gamma}+b_{\boldsymbol\gamma})x^{\boldsymbol\gamma}, \quad fg=\sum_{\boldsymbol\alpha,\boldsymbol\beta} a_{\boldsymbol\alpha} b_{\boldsymbol\beta} x^{\boldsymbol\alpha+\boldsymbol\beta}=\sum_{\boldsymbol\gamma}\left(\sum_{\boldsymbol\alpha+\boldsymbol\beta=\boldsymbol\gamma} a_{\boldsymbol\alpha} b_{\boldsymbol\beta}\right)x^{\boldsymbol\gamma}.$$

*All multi-indices range over $\mathbb{N}^r$. When $R$ is noncommutative, the coefficient product $a_{\boldsymbol\alpha} b_{\boldsymbol\beta}$ cannot be reversed to $b_{\boldsymbol\beta} a_{\boldsymbol\alpha}$.*

**Proof.** Induct on the number of indeterminates. For $r=1$ these are the definitions of addition and multiplication of one-variable polynomials. Suppose the formulas hold for $r-1$ indeterminates. By the recursive definition, $f,g$ have unique expressions $f=\sum_i f_i x_r^i$ and $g=\sum_j g_j x_r^j$, where $f_i, g_j \in R[x_1,\dots,x_{r-1}]$. Regarding these as polynomials in $x_r$, the one-variable definitions of addition and multiplication give

$$f+g=\sum_k (f_k+g_k)x_r^k, \quad fg=\sum_k\left(\sum_{i+j=k} f_i g_j\right)x_r^k.$$

Expanding $f_k, g_k$ by the induction hypothesis and combining the multi-indices gives the stated formulas. □

**Definition A.6.44.** For a multivariable polynomial $f=\sum_{\boldsymbol\alpha} a_{\boldsymbol\alpha} x^{\boldsymbol\alpha}$:

(a) its *support* is $\{\boldsymbol\alpha \mid a_{\boldsymbol\alpha} \neq 0\}$;

(b) if $f \neq 0$, its *total degree* $\deg f$ is the greatest $|\boldsymbol\alpha|$ for which $a_{\boldsymbol\alpha} \neq 0$. The degree of the zero polynomial is undefined, but we retain the formal convention $\deg 0 = -\infty$;

(c) $f$ is a *homogeneous polynomial of degree $d$*, or a *homogeneous form of degree $d$*, if every nonzero term has total degree $d$. The zero polynomial is regarded as homogeneous of every degree. The sum of all degree-$d$ terms of $f$ is its *homogeneous component of degree $d$*.

**Proposition A.6.45.** *Every multivariable polynomial $f$ has a unique family of homogeneous components $\{f_d \mid d \geq 0\}$, where $f_d$ is its degree-$d$ component, only finitely many $f_d$ are nonzero, and $f=\sum_{d\geq 0} f_d$. If $f \neq 0$, let $n$ be the greatest $d$ for which $f_d \neq 0$; then $f=f_0+\cdots+f_n$ and $\deg f = n$. If $f=0$, then every $f_d=0$.*

**Proof.** Group the finite sum in proposition A.6.42 according to the total degree of each monomial. Uniqueness follows from uniqueness of all coefficients. □

**Theorem A.6.46.** *If $R$ is a ring without zero divisors and $f,g \in R[x_1,\dots,x_r]$ are nonzero, then* $\deg(fg)=\deg f+\deg g$.

**Proof.** Put $\deg f = m$ and $\deg g = n$, and let their highest nonzero homogeneous components be $f_m$ and $g_n$, respectively. Since $R$ has no zero divisors, theorem A.6.41 says that $R[x_1,\dots,x_r]$ has no zero divisors, so $f_m g_n \neq 0$. This is the degree-$(m+n)$ component of $fg$, and $fg$ has no term of degree greater than $m+n$. The conclusion follows. □

[20] Here $\boldsymbol\alpha+\boldsymbol\beta$ means ordinary vector addition in $\mathbb{N}^r$.

**Definition A.6.47.** Let $R \leq S$ be commutative rings and $s_1,\ldots,s_r \in S$. For $f = \sum_{\boldsymbol{\alpha}} a_{\boldsymbol{\alpha}} x^{\boldsymbol{\alpha}} \in R[x_1,\ldots,x_r]$, define the *evaluation* obtained by substituting each $s_i$ for $x_i$:

$$f(s_1,\ldots,s_r) = \sum_{\boldsymbol{\alpha}} a_{\boldsymbol{\alpha}} s_1^{\alpha_1} \cdots s_r^{\alpha_r}.$$

When $S = R$, this is the *value* of $f$ at the point $(s_1,\ldots,s_r)$. We also write $s^{\boldsymbol{\alpha}} := s_1^{\alpha_1} \cdots s_r^{\alpha_r}$, where $\boldsymbol{\alpha} = (\alpha_1,\ldots,\alpha_r) \in \mathbb{N}^r$.

If $\alpha_i = 0$, the factor $x_i^{\alpha_i}$ is omitted. Formally we write $x_i^0 = 1$, even when the value substituted for $x_i$ is 0. Here 1 denotes an empty product and does not require either $R$ or $S$ to have an identity. In particular, when $\boldsymbol{\alpha} = \mathbf{0}$, $a_{\mathbf{0}} x^{\mathbf{0}} = a_{\mathbf{0}}$.

**Proposition A.6.48.** *For commutative rings $R \leq S$ and elements $s_1,\ldots,s_r \in S$, the map*

$$\mathrm{ev}_{s_1,\ldots,s_r} : R[x_1,\ldots,x_r] \longrightarrow S, \qquad f \longmapsto f(s_1,\ldots,s_r),$$

*is a ring homomorphism preserving addition and multiplication.* The symbol ev denotes the *evaluation map. If $R,S$ have the same identity, it is the unique ring homomorphism $R[x_1,\ldots,x_r] \to S$ that preserves the identity, is the identity on R, and sends $x_i$ to $s_i$.*

**Proof.** Termwise addition proves additivity. Multiplicativity follows from the distributive law and $s^{\boldsymbol{\alpha}} s^{\boldsymbol{\beta}} = s^{\boldsymbol{\alpha}+\boldsymbol{\beta}}$. When $R$ and $S$ have the same identity, the images of constants and of each $x_i$ are, by definition, the constants themselves and $s_i$, respectively. Every polynomial is obtained from constants and the $x_i$ by finitely many additions and multiplications; consequently any homomorphism with the stated properties must be the displayed evaluation map. $\square$

The identity hypothesis in the last assertion is needed because otherwise an indeterminate $x_i$ need not itself belong to the polynomial ring under the conventions used here.

Apply the preceding proposition, in the commutative unital case, to the subring $\imath(A)$ of the $A$-algebra $B$. The notation in definition A.4.46 then takes the concrete form

$$A[b_1,\ldots,b_r] = \{f(b_1,\ldots,b_r) \mid f \in \imath(A)[x_1,\ldots,x_r]\}.$$

The right side is a subring containing $\imath(A)$ and all the $b_i$; every subring containing those elements is closed under finite sums and products and therefore contains the right side. It is thus exactly the previously defined smallest subring.

**Theorem A.6.49** (Polynomial identity principle)**.** *Let $R$ be an infinite integral domain. If $f \in R[x_1,\ldots,x_r]$ vanishes at every point of $R^r$,*[21] *then $f = 0$. Thus two polynomials over $R$ agree as polynomials exactly when they have the same value at every point of $R^r$.*

**Proof.** Induct on $r$. When $r = 1$, a nonzero polynomial $f$ vanishing at every point of the infinite set $R$ would have more than $\deg f$ distinct roots, contradicting proposition A.6.12.

Suppose the result holds for $r-1$ variables. Expand $f$ in $x_r$ as $f = f_0 + f_1 x_r + \cdots + f_n x_r^n$. Each coefficient satisfies $f_i \in R[x_1,\ldots,x_{r-1}]$. Take an arbitrary $a = (a_1,\ldots,a_{r-1}) \in R^{r-1}$. The one-variable polynomial

$$f(a_1,\ldots,a_{r-1},x_r) = \sum_{i=0}^{n} f_i(a) x_r^i$$

vanishes at every point of the infinite set $R$. By the case $r = 1$, it is the zero polynomial, so $f_i(a) = 0$ for every $i$. Since $a$ was arbitrary, the induction hypothesis gives $f_i = 0$ for every $i$, and therefore $f = 0$.

Applying this result to the difference of two polynomials proves the final assertion. $\square$

Finally, recall that the fundamental theorem of algebra guarantees factorization into a product of linear polynomials only for polynomials in one variable. Multivariable polynomials with complex coefficients need not factor into linear factors.

[21] $R^r$ denotes the $r$-fold Cartesian product of $R$. Each point has the form $(x_1,\ldots,x_r)$ with $x_i \in R$.

## Exercises

**Exercise A.108.** For $f(x,y,z) = 3x^2y - 2xyz^2 + y^4 + 5z - 7 \in \mathbb{C}[x,y,z]$, find:

(1) its support;
(2) the multi-index and total degree of every nonzero term;
(3) its total degree;
(4) its homogeneous components;
(5) its degree in each of the three indeterminates, regarding it in turn as a one-variable polynomial in $x$, in $y$, and in $z$.

**Exercise A.109*.**

(1) Prove that $xy - 1 \in \mathbb{C}[x,y]$ is irreducible.
(2) Prove that the polynomial $f(x,y,z)$ in exercise A.108 is irreducible.

**Exercise A.110.** Let $F$ be a field, $a,b \in F$, and $f \in F[x,y]$.

(1) Prove that $f(a,b) = 0$ exactly when $f(x,y) = (x-a)q(x,y) + (y-b)r(x,y)$ for some $q,r \in F[x,y]$.
(2) In homomorphism and ideal notation, prove $\ker \mathrm{ev}_{a,b} = \langle x-a, y-b\rangle$.

**Exercise A.111.** Let $R$ be an infinite integral domain and $d \in \mathbb{N}$. Prove that $f \in R[x_1,\ldots,x_r]$ is homogeneous of degree $d$ exactly when

$$f(ta_1,\ldots,ta_r) = t^d f(a_1,\ldots,a_r)$$

for every $(t,a_1,\ldots,a_r) \in R^{r+1}$.

**Exercise A.112.** Let $F$ be a field and $\boldsymbol{A} = (a_{ij}) \in \mathrm{GL}(r,F)$. Prove that the change of variables $x_i \mapsto \sum_{j=1}^{r} a_{ij}x_j$ induces an automorphism of $F[x_1,\ldots,x_r]$, whose inverse is induced by the change of variables of the same form using $\boldsymbol{A}^{-1}$, and that it preserves the degree of homogeneous forms.

**Exercise A.113.** Let $F$ be algebraically closed and $n \geq 1$. Prove that every nonzero binary homogeneous form of degree $n$ is a product of $n$ linear homogeneous forms, counted with multiplicity.

**Exercise A.114.** Replace “division ring” by “ring” in the matrices defined in example A.5.3. Matrix addition and multiplication are unchanged, although the resulting set no longer carries the corresponding vector-space structure. The set $\mathrm{M}_n(R)$ of square matrices of order $n$ over $R$, with matrix addition and multiplication, is a ring. Prove

$$\mathrm{M}_n(R)[x_1,\ldots,x_r] \cong \mathrm{M}_n(R[x_1,\ldots,x_r]),$$

and give the isomorphism and its inverse explicitly.

# A.7 An Introduction to Topology

This section briefly introduces some topology. We shall not give a full account of the definitions and properties of the various kinds of topological spaces. Their geometric ideas, such as embedded surfaces, and precise language, such as limits and connectedness, will nevertheless help in our study.

## A.7.1 Basic Definitions in Point-Set Topology

We give precise definitions of some basic concepts of topology.

**Topological spaces**

**Definition A.7.1.**

(1) Let $X$ be a set. A *topological space* is a pair $(X,\mathcal{T})$, where $\mathcal{T}$ is a collection of subsets of $X$ such that
 (i) $\emptyset, X \in \mathcal{T}$;
 (ii) a union of any finite or infinite family of members of $\mathcal{T}$ belongs to $\mathcal{T}$;
 (iii) a finite intersection of members of $\mathcal{T}$ belongs to $\mathcal{T}$.

 Members of $\mathcal{T}$ are *open sets* in $X$; a subset of $X$ is *closed* if and only if its complement is open. When the context is clear, we usually abbreviate the topological space $(X,\mathcal{T})$ to the topological space $X$.

(2) Let $(X,\mathcal{T})$ be a topological space and $A \subseteq X$. Declare $U \subseteq A$ to be open in $A$ if and only if there is an open set $V$ in $X$ such that $U = A \cap V$. The resulting topology is the *subspace topology* that $A$ inherits from $X$.

**Proposition A.7.2.** *Let $X$ be a topological space and $A \subseteq Y \subseteq X$, with $Y$ carrying the subspace topology. Then the topology that $A$ inherits through $Y$ is the same as the topology that it inherits directly from $X$.*

**Proof.** Let $\mathcal{T}_X$ be the topology on $X$. The topology induced on $Y$ is $\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}_X\}$. Inducing a topology on $A$ in turn gives

$$\mathcal{T}_A = \{A \cap V \mid V \in \mathcal{T}_Y\} = \{A \cap (Y \cap U) \mid U \in \mathcal{T}_X\}.$$

Since $A \subseteq Y$, we have $A \cap (Y \cap U) = A \cap U$. Thus $\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}_X\}$, the topology induced directly from $X$. ☐

**Proposition A.7.3.**

*(1) In a topological space, arbitrary intersections and finite unions of closed sets are closed.*

*(2) For a topological space $X$ and a subset $A \subseteq X$ with its subspace topology, $C \subseteq A$ is closed in $A$ exactly when $C = A \cap F$ for some closed $F \subseteq X$.*

**Proof.** The first assertion follows by taking complements in the open-set axioms and applying De Morgan's laws. For the second, if $C$ is closed in $A$, then $A \setminus C = A \cap U$ for some open set $U$ in $X$. Hence

$$C = A \setminus (A \setminus C) = A \setminus (A \cap U) = A \setminus U = A \cap (X \setminus U).$$

The last equality holds because $x \in A$ and $x \notin U$ exactly when $x \in A$ and $x \in X \setminus U$. Take $F = X \setminus U$. ☐

**Definition A.7.4.**

(1) Let $X$ be a topological space. An *open cover* of $E \subseteq X$ is a family $\{U_i \mid i \in I\}$ of open sets in $X$, indexed by a set $I$, such that $E \subseteq \bigcup_{i\in I} U_i$. A subfamily that still covers $E$ is a *subcover*; it is a *finite subcover* if it has finitely many members.

(2) In (1), another open cover $\{V_j \mid j \in J\}$ of $E$ *refines* $\{U_i \mid i \in I\}$ if, for every $j \in J$, some $i \in I$ satisfies $V_j \subseteq U_i$. When covering an open set $E = U$, we tacitly require all members of the cover to be contained in $U$.

(3) A topological space $X$ is *compact* if every open cover of $X$ has a finite subcover: whenever a family of open sets $\{U_\alpha \mid \alpha \in A\}$ satisfies $X = \bigcup_{\alpha \in A} U_\alpha$, there are finitely many $\alpha_1, \alpha_2, \ldots, \alpha_r \in A$ such that $X = \bigcup_{i=1}^{r} U_{\alpha_i}$. We then call $X$ a *compact space*. A subset of $X$ is a *compact subset* if it is compact when endowed with the subspace topology.

(4) For a topological space $X$, a family $\mathcal{B}$ of open sets is a *basis for the topology* if, for every open $U \subseteq X$ and every $x \in U$, some $B \in \mathcal{B}$ satisfies $x \in B \subseteq U$.

**Proposition A.7.5.**

(1) *A family $\mathcal{B}$ of open sets in a topological space $X$ is a basis exactly when every open set is a union of finitely or infinitely many members of $\mathcal{B}$.*

(2) *If $\mathcal{B}$ is a basis and $\{U_i \mid i \in I\}$ is an open cover of a subset $U \subseteq X$, then*
$$\mathcal{C} = \{B \in \mathcal{B} \mid \exists i \in I,\ B \subseteq U_i\}$$
*is a refinement. A finite subcover from this refinement yields a finite subcover from the original cover.*

**Proof.** If for each $x \in U$ there is a $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U$, then $U = \bigcup_{x \in U} B_x$. Conversely, if $U$ is a union of members of $\mathcal{B}$, every $x \in U$ belongs to one of those members, which is contained in $U$.

For the second assertion, given $x \in U$, first choose $U_i$ containing $x$ and then a basis element $B$ with $x \in B \subseteq U_i$. Thus $\mathcal{C}$ covers $U$; every member of $\mathcal{C}$ is contained in some $U_i$, so $\mathcal{C}$ refines the original cover. From a finite subcover by basis elements, choose one corresponding $U_i$ for each member; these finitely many $U_i$ cover $U$. □

**Metric spaces**

We shall not study general topological spaces below, but shall instead equip metric spaces with a topology.

**Definition A.7.6.**

(1) A *metric* on a set $X$ is a map $d : X \times X \to \mathbb{R}$ such that, for all $x, y, z \in X$,
   (i) $d(x,y) \geq 0$;
   (ii) $d(x,y) = 0 \Longleftrightarrow x = y$;
   (iii) $d(x,y) = d(y,x)$;
   (iv) $d(x,z) \leq d(x,y) + d(y,z)$.

   The pair $(X,d)$ is a *metric space*. For $x \in X$ and $r > 0$, the *open ball* with centre $x$ and radius $r$ is $\mathrm{B}_d(x,r) := \{y \in X \mid d(x,y) < r\}$. Let $A \subseteq X$. If there are $P_0 \in X$ and $R > 0$ such that $d(P,P_0) \leq R$ for every $P \in A$, then $A$ is a *bounded set* in $X$; otherwise it is an *unbounded set*.

(2) For a metric space $(X,d)$, a subset $U \subseteq X$ is *open* if, for every $x \in U$, there is an $r > 0$ such that $\mathrm{B}_d(x,r) \subseteq U$. That is, the open balls form a basis. One can verify that these open sets form a topology on $X$; this is the *metric topology* induced by $d$.

(3) On $\mathbb{R}^n$ (or $\mathbb{C}^n$), for arbitrary points $\boldsymbol{x} = (x_1, \ldots, x_n)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$, the *Euclidean metric* is
$$d(\boldsymbol{x},\boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\| := \sqrt{\sum_{j=1}^{n} |x_j - y_j|^2}.$$
   It is readily verified that this is a metric. The resulting topology is called the *usual topology*, or *Euclidean topology*, on $\mathbb{R}^n$ (or $\mathbb{C}^n$). Unless otherwise stated, the topological spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ are understood to carry the usual topology.

Let us verify that a metric does indeed define a topology. For $\emptyset$, there are no points to check, so the condition holds vacuously. For $X$, clearly $\mathrm{B}(x,r) \subseteq X$ for every $x \in X$ and every $r > 0$.

If $x$ belongs to a union $U = \bigcup_\alpha U_\alpha$ of open sets, then $x \in U_{\alpha_0}$ for some $\alpha_0$. Since $U_{\alpha_0}$ is open, some $r > 0$ satisfies $\mathrm{B}(x,r) \subseteq U_{\alpha_0} \subseteq U$. Thus $U$ is open.

Finally, consider a finite intersection $U = \bigcap_{i=1}^{r} U_i$ of open sets. If $x \in U$, then $x \in U_i$ for every $i$, so there are radii $r_i > 0$ with $\mathrm{B}(x,r_i) \subseteq U_i$. Taking $r = \min_i r_i$, we have $\mathrm{B}(x,r) \subseteq \mathrm{B}(x,r_i) \subseteq U_i$ for every $i$, and therefore $\mathrm{B}(x,r) \subseteq U$. Thus $U$ is open. □

**Remark.** A set may be both open and closed, as are $\emptyset$ and $X$, or it may be neither, as is $[a,b)$ in $\mathbb{R}$. Every open ball about $a$ contains points less than $a$, and every open ball about $b$ contains points of $[a,b)$, so neither this interval nor its complement $(-\infty,a) \cup [b,+\infty)$ is open.

Openness may change on passing to a subspace. The set $[0,0.5)$ is not open in $\mathbb{R}$, but it is open in $I = [0,1]$, since $[0,0.5) = [0,1] \cap (-1,0.5)$ and $(-1,0.5)$ is open in $\mathbb{R}$.

Only finite intersections of open sets are required to be open. For example, the open sets $U_n = (-1/n, 1/n)$ in $\mathbb{R}$ have intersection $\bigcap_{n=1}^{\infty} U_n = \{0\}$, which is not open: every open ball about 0 contains other points. Likewise, only finite unions of closed sets are guaranteed to be closed; an infinite union need not be closed.

We state the following result without proof.

**Theorem A.7.7** (Heine–Borel theorem). *A subset $A$ of finite-dimensional real Euclidean space $\mathbb{R}^n$ is compact exactly when it is closed in $\mathbb{R}^n$ and bounded.*

### Interior, boundary, and closure

We next define some further basic concepts in a topological space.

**Definition A.7.8.** Let $A$ be a subset of a topological space $X$.

(1) A point $P \in A$ is an *interior point* of $A$ relative to $X$ if there is an open set $U$ in $X$ with $P \in U \subseteq A$. The set of all such points is the *interior* of $A$ relative to $X$, denoted by $\operatorname{int}_X A$, abbreviated to $\operatorname{int} A$ when $X$ is clear, or written $A^\circ$. Interior points of $X \setminus A$ are *exterior points* of $A$ relative to $X$; they form its *exterior* $\operatorname{ext}_X A$, abbreviated to $\operatorname{ext} A$ when $X$ is clear.

(2) A point $P \in X$ is a *boundary point* of $A$ relative to $X$ if every open set $U$ in $X$ containing $P$ satisfies both $U \cap A \neq \emptyset$ and $U \cap (X \setminus A) \neq \emptyset$. Their set is the *boundary* of $A$ relative to $X$, $\partial_X A$, abbreviated to $\partial A$ when $X$ is clear.

(3) A point $P \in X$ is an *adherent point* of $A$ if every open set containing $P$ has nonempty intersection with $A$; all such points form the *closure* of $A$. A point $P \in X$ is an *accumulation point*, or *limit point*, of $A$ if every open set containing $P$ has nonempty intersection with $A \setminus \{P\}$; all such points form the *derived set* of $A$.

**Remark.** Interior points, accumulation points, and boundary points depend on the ambient space. For example, $\partial_{\mathbb{R}}[0,1] = \{0,1\}$, whereas $\partial_{[0,1]}[0,1] = \emptyset$.

**Proposition A.7.9.** *A subset $A$ of a topological space $X$ is open exactly when $A = \operatorname{int}_X A$.*

**Proof.** This follows immediately from the definition of an interior point. □

**Proposition A.7.10.** *For a topological space $X$ and every subset $A \subseteq X$, we have $X = \operatorname{int}_X A \sqcup \partial_X A \sqcup \operatorname{ext}_X A$.*

**Proof.** Fix $P \in X$. If there is an open set $U$ containing $P$ with $U \subseteq A$, then $P$ is interior; if there is such a $U$ with $U \subseteq X \setminus A$, then $P$ is exterior. If both cases occurred, there would be open sets $U_1, U_2$ containing $P$ with $U_1 \subseteq A$ and $U_2 \subseteq X \setminus A$, and hence $P \in U_1 \cap U_2 \subseteq A \cap (X \setminus A) = \emptyset$, a contradiction.

If neither case occurs, every open set $U$ containing $P$ is contained neither in $A$ nor in $X \setminus A$. It therefore contains both a point of $A$ and a point of $X \setminus A$, so $P$ is a boundary point. The definitions of interior, exterior, and boundary points are mutually exclusive, proving the disjoint decomposition. □

**Proposition A.7.11.** *Let $A, B$ be subsets of a topological space $X$. Writing $A^\circ$, $\overline{A}$, $A'$, and $\partial A$ for the interior, closure, derived set, and boundary:*

*(1) $\overline{A} = \overline{\overline{A}} = A \cup A'$ and $(A^\circ)^\circ = A^\circ$;*

*(2) $A^\circ = X \setminus \overline{(X \setminus A)}$ and $\overline{A} = X \setminus (X \setminus A)^\circ$;*

*(3) $\partial A = \overline{A} \cap \overline{(X \setminus A)} = \overline{A} \setminus A^\circ = \partial(X \setminus A)$;*

*(4) $A$ is open exactly when $A = A^\circ$, and closed exactly when $A = \overline{A}$;*

*(5) every closed set containing $A$ contains $\overline{A}$, and every open set contained in $A$ is contained in $A^\circ$. Thus $\overline{A}$ is the least closed set containing $A$, and $A^\circ$ is the greatest open set contained in $A$;*

(6) *$A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$ and $A^\circ \subseteq B^\circ$;*
(7) *$\overline{A \cup B} = \overline{A} \cup \overline{B}$ and $(A \cap B)^\circ = A^\circ \cap B^\circ$. More generally, the closure of a finite union is the union of the closures, and the interior of a finite intersection is the intersection of the interiors.*

**Proof.** We prove only the closure portions of (5), (6), and (7), which may be used explicitly later; the remaining assertions are left as exercises.

For (5), $x \notin \overline{A}$ exactly when some open set containing $x$ is disjoint from $A$. Hence $X \setminus \overline{A}$ is a union of these open sets and $\overline{A}$ is closed. If a closed set $F$ contains $A$ and $x \notin F$, then $X \setminus F$ is an open set containing $x$ and disjoint from $A$; thus $x \notin \overline{A}$. Consequently every closed set containing $A$ also contains $\overline{A}$.

For (6), if $A \subseteq B$, then $A \subseteq B \subseteq \overline{B}$, and since $\overline{B}$ is closed, (5) gives $\overline{A} \subseteq \overline{B}$.

For (7), (5) shows that $\overline{A}$ and $\overline{B}$ are closed, so $\overline{A} \cup \overline{B}$ is a closed set, plainly containing $A \cup B$. Applying (5) gives $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$. Conversely, since $A \subseteq A \cup B$, (6) gives $\overline{A} \subseteq \overline{A \cup B}$, and similarly $\overline{B} \subseteq \overline{A \cup B}$. Therefore $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$. Thus $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Induction gives the finite-union version. The restriction to finitely many sets is essential because an infinite union of closed sets need not be closed. □

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## Exercises

**Exercise A.115.** Let $A$ be a subset of a topological space $X$. Prove:
(1) $A$ is open in $X$ if and only if every open set in the subspace topology on $A$ is open in $X$;
(2) $A$ is closed in $X$ if and only if every closed set in the subspace topology on $A$ is closed in $X$.

**Exercise A.116.** Let $\mathcal{B}$ be a basis for a topological space $X$. For $A \subseteq X$, prove that $\{B \cap A \mid B \in \mathcal{B}\}$ is a basis for the subspace topology on $A$.

**Exercise A.117.** Prove the following statements about compact spaces.
(1) A compact subset of a metric space is closed.
(2) A closed subset of a compact space is compact.
(3) A finite union of compact subsets of a topological space is compact.
(4) A finite subset of a topological space is compact.

**Exercise A.118.** For a set $X$, define the map

$$d : X \times X \longrightarrow \{0,1\}, \qquad (x,y) \mapsto \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

(1) Prove that $d$ is a metric on $X$.
(2) Prove that every subset of $X$ is open in the topology induced by $d$.
(3) Determine the interior, exterior, boundary, closure, and derived set of an arbitrary $A \subseteq X$ in the topology of (2).

**Exercise A.119.** Let $X$ be a topological space and $B \subseteq A \subseteq X$, with $A, B$ carrying the subspace topologies.
(1) Prove that the closure of $B$ in $A$ is the intersection of $A$ with the closure of $B$ in $X$.
(2) Is $\operatorname{int}_A B = A \cap \operatorname{int}_X B$ always true? Explain your answer.

**Exercise A.120.** Prove all assertions in proposition A.7.11.

### A.7.2 Basic Properties of Point-Set Topology

Continuing the preceding subsection, we give some further common concepts in topology.

#### Maps between topological spaces

Continuity and limits can be defined precisely in topological spaces.

**Definition A.7.12.**

(1) For topological spaces $X,Y$, a map $f:X\to Y$ is a *continuous map*, or is *continuous*, if $f^{-1}(V)$ is open in $X$ for every open $V\subseteq Y$. For a map $f:I\to Y$ with $I\subseteq X$, continuity is understood using the subspace topology on $I$.

(2) Let $(x_n)=x_1,x_2,\dots,x_n,\dots$ be a sequence of points of $X$. For $x\in X$, if every open set $U\subseteq X$ containing $x$ admits an $N\in\mathbb{N}$ such that $x_n\in U$ for every $n\geq N$, then $(x_n)$ *converges* to $x$. The point $x$ is its *limit*, written $x_n\to x$ $(n\to\infty)$ or $\lim_{n\to\infty}x_n=x$.

(3) Let $X,Y$ be topological spaces, $D\subseteq X$, and $f:D\to Y$, and let $a\in X$ be an accumulation point of $D$. For $y\in Y$, if every open set $V\subseteq Y$ containing $y$ admits an open set $U\subseteq X$ containing $a$ such that $f(U\cap(D\setminus\{a\}))\subseteq V$, we say that $f(x)$ tends to $y$ as $x\to a$, or that $y$ is the *limit* of $f(x)$ as $x\to a$. We write $\lim_{x\to a}f(x)=y$ or $f(x)\to y$ $(x\to a)$. The point $a$ need not belong to $D$.

(4) Two topological spaces $X,X'$ are *topologically homeomorphic*, or simply *homeomorphic*, if there is a bijection $f:X\to X'$ such that both $f$ and $f^{-1}$ are continuous. We write $X\cong X'$ and call $f$ the corresponding *homeomorphism*. Homeomorphism is an equivalence relation.

**Remark.** On Euclidean spaces the usual elementary functions—exponential, logarithmic, trigonometric, and power functions, together with their arithmetic combinations and compositions—are generally continuous on their domains; we shall use this without proof.

**Proposition A.7.13.** *A map $f:X\to Y$ between topological spaces $X,Y$ is continuous exactly when, for every $x\in X$ and every open set $V\subseteq Y$ containing $f(x)$, some open set $U\subseteq X$ containing $x$ satisfies $f(U)\subseteq V$.*

**Proof.** If $f$ is continuous, then $f^{-1}(V)$ is open in $X$. Since $f(x)\in V$, we may take $U=f^{-1}(V)$.

Conversely, let $V$ be any open subset of $Y$. For every $x\in f^{-1}(V)$ there is an open set $U_x\subseteq X$ such that $f(U_x)\subseteq V$, or equivalently $U_x\subseteq f^{-1}(V)$. Therefore

$$f^{-1}(V)=\bigcup_{x\in f^{-1}(V)}U_x.$$

The right-hand side is a union of open sets and is therefore open. □

**Proposition A.7.14.** *A map $f:X\to Y$ between topological spaces is continuous if and only if the preimage of every closed set in $Y$ is closed in $X$.*

**Proof.** Use $f^{-1}(Y\setminus F)=X\setminus f^{-1}(F)$ together with the definition of a closed set to see that the open-set criterion is equivalent to this closed-set criterion. □

**Proposition A.7.15.** *A composition of continuous maps between topological spaces is continuous.*

**Proof.** Let $f:X\to Y$ and $g:Y\to Z$ be continuous maps. Since $(g\circ f)^{-1}(W)=f^{-1}(g^{-1}(W))$, for every open set $W\subseteq Z$, the set $g^{-1}(W)$ is open in $Y$, and hence $(g\circ f)^{-1}(W)$ is open in $X$. Thus $(g\circ f)$ is continuous. □

**Theorem A.7.16.** *A convergent sequence in a metric space has a unique limit.*

**Proof.** Let $(X,d)$ be a metric space, and suppose that $x_n \to a$ and $x_n \to b$ for distinct $a,b \in X$. By the definition of a metric, $d(a,b) > 0$. Put $\epsilon = \frac{1}{3} d(a,b) > 0$. Since $x_n \to a$, there is $N_1 \in \mathbb{N}$ such that $n \geq N_1 \Longrightarrow x_n \in \mathrm{B}(a,\epsilon)$. Since $x_n \to b$, there is $N_2 \in \mathbb{N}$ such that $n \geq N_2 \Longrightarrow x_n \in \mathrm{B}(b,\epsilon)$. Choose $n \geq \max\{N_1,N_2\}$. Then

$$d(a,b) \leq d(a,x_n) + d(x_n,b) < 2\epsilon = \frac{2}{3} d(a,b),$$

a contradiction. Hence $a = b$. □

**Theorem A.7.17.** *Let $(X,d_X),(Y,d_Y)$ be metric spaces, let $D \subseteq X$, let a be an accumulation point of D, and let $f : D \to Y$ be a map. If $\lim_{x\to a} f(x)$ exists, it is unique.*

**Proof.** Suppose that $\lim_{x\to a} f(x)$ has two different values $L_1, L_2$, with $L_1 \neq L_2$. Then $d_Y(L_1,L_2) > 0$. Put $\epsilon = \frac{1}{3} d_Y(L_1,L_2) > 0$. Since $f(x) \to L_1$, there is $\delta_1 > 0$ such that, for every $x \in D$, $0 < d_X(x,a) < \delta_1 \Longrightarrow f(x) \in \mathrm{B}_Y(L_1,\epsilon)$. Similarly, there is $\delta_2 > 0$ such that, for every $x \in D$, $0 < d_X(x,a) < \delta_2 \Longrightarrow f(x) \in \mathrm{B}_Y(L_2,\epsilon)$. Because $a$ is an accumulation point of $D$, there is $x \in D \setminus \{a\}$ such that $d_X(x,a) < \min\{\delta_1,\delta_2\}$. Therefore

$$d_Y(L_1,L_2) \leq d_Y(L_1,f(x)) + d_Y(f(x),L_2) < 2\epsilon = \frac{2}{3} d_Y(L_1,L_2),$$

a contradiction. Hence $L_1 = L_2$. □

**Theorem A.7.18.** *Let $X \cong Y$ be homeomorphic topological spaces and $f : X \to Y$ a corresponding homeomorphism. Then X is compact if and only if Y is compact. The map f sends open subsets of X to open subsets of Y, closed subsets of X to closed subsets of Y, and compact subsets of X to compact subsets of Y.*

**Proposition A.7.19.** *For a finite positive integer n, $\mathbb{C}^n \cong \mathbb{R}^{2n}$. Consequently, a subset of $\mathbb{C}^n$ is compact exactly when it is closed and bounded.*

**Proof.** Define

$$f : (z_1,z_2,\ldots,z_n) \longmapsto (\mathrm{Re}\, z_1, \mathrm{Im}\, z_1, \mathrm{Re}\, z_2, \mathrm{Im}\, z_2, \ldots, \mathrm{Re}\, z_n, \mathrm{Im}\, z_n).$$

This map is a homeomorphism; this is verified directly and will not be expanded here. By theorem A.7.18 it preserves compact and closed sets. Moreover, one verifies $d_{\mathbb{C}}(x,y) = d_{\mathbb{R}}(f(x),f(y))$, so $f$ preserves the metric and hence boundedness. The Heine–Borel theorem now shows that a subset of $\mathbb{C}^n$ is compact if and only if it is closed and bounded. □

**Remark.** Boundedness is not a topological property, and homeomorphisms need not preserve it. The preceding proof uses the special properties of this particular homeomorphism to preserve boundedness.

### Product and quotient topologies

We define the product and quotient topologies here; concrete examples will follow in the next subsection.

**Definition A.7.20.** Let $X, Y$ be topological spaces and $Z = X \times Y$ their Cartesian product. Declare every set $U \times V$, with $U$ open in $X$ and $V$ open in $Y$, to be open in $Z$, and take all unions of such sets to be the open sets of $Z$. Equivalently, the sets $U \times V$ form a basis. This is the *product topology* on $X \times Y$.

The preceding definition of openness is equivalently characterized by the following condition.

**Proposition A.7.21.** *Let $X, Y$ be topological spaces and equip $X \times Y$ with the product topology. A set $W \subseteq X \times Y$ is open exactly when, for every $(x,y) \in W$, there are open $U \subseteq X$ and $V \subseteq Y$ such that $(x,y) \in U \times V \subseteq W$.*

**Proof.** Suppose $W$ is open in $X \times Y$. By definition, there are families of open sets $\{U_\alpha \subseteq X \mid \alpha\}$ and $\{V_\alpha \subseteq Y \mid \alpha\}$, with index $\alpha$ ranging over $\alpha \in A$, such that $W = \bigcup_{\alpha \in A}(U_\alpha \times V_\alpha)$. Every $(x,y) \in W$ belongs to this union, so there is $\alpha_0 \in A$ with $(x,y) \in U_{\alpha_0} \times V_{\alpha_0} \subseteq W$. Conversely, suppose that for every $(x,y) \in W$ there are open sets $U_{x,y} \subseteq X$ and $V_{x,y} \subseteq Y$ satisfying $(x,y) \in U_{x,y} \times V_{x,y} \subseteq W$. Then

$$W = \bigcup_{(x,y) \in W} (U_{x,y} \times V_{x,y}),$$

so $W$ is open. □

These sets do form a topology on $X \times Y$. The empty set is $\emptyset \times \emptyset$, and $X \times Y$ itself is also of the basic form in the definition. Thus the empty set and the whole space belong to the product topology. Each open set is a union of sets of the form $U \times V$; a union of open sets is still a union of such sets, and hence is open by definition.

Finally, let $W_1, W_2$ be open in the product topology and $(x,y) \in W_1 \cap W_2$. Choose basic open sets $U_1 \times V_1 \subseteq W_1$ and $U_2 \times V_2 \subseteq W_2$, both containing $(x,y)$. Then

$$(x,y) \in (U_1 \cap U_2) \times (V_1 \cap V_2) \subseteq W_1 \cap W_2,$$

so $W_1 \cap W_2$ is open in the product topology as well. □

**Definition A.7.22.** Let $X$ be a topological space, $Y$ a set, and $f : X \to Y$ a surjection. The *quotient topology* induced by $f$ declares $U \subseteq Y$ open exactly when $f^{-1}(U)$ is open in $X$; then $Y$ is the *quotient space* of $X$ with respect to $f$. For an equivalence relation $\sim$ on $X$, the quotient set $X/\sim$ receives the *quotient topology* from the natural projection $f : X \to X/\sim$, $x \mapsto [x]$.

The quotient construction also gives a topology. Since $f$ is surjective, $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(Y) = X$, so $\emptyset$ and $Y$ are open in the quotient topology.

For any family of sets $\{U_i \mid i \in I\}$, with $I$ the index set,

$$f^{-1}\left(\bigcup_{i \in I} U_i\right) = \bigcup_{i \in I} f^{-1}(U_i).$$

If every $U_i$ is open in the quotient topology, the right side is a union of open sets in $X$, so $\bigcup_{i \in I} U_i$ is open in the quotient topology.

For any finite family $U_1, \ldots, U_m$,

$$f^{-1}(U_1 \cap \cdots \cap U_m) = f^{-1}(U_1) \cap \cdots \cap f^{-1}(U_m).$$

If every $U_i$ is open in the quotient topology, the right side is a finite intersection of open sets; therefore $U_1 \cap \cdots \cap U_m$ is open in the quotient topology. □

The quotient topology can be used to construct adjunction spaces.

**Definition A.7.23.** Let $\mathcal{T}_X, \mathcal{T}_Y$ be topologies on sets $X, Y$, respectively. On the disjoint union $X \sqcup Y$, define the topology $\mathcal{T} = \{U \sqcup V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$, called the *topological sum* of $X, Y$.

Let $X, Y$ be topological spaces, $A \subseteq Y$ a subspace, and $f : A \to X$ a continuous map. Define an equivalence relation $\sim$ on $X \sqcup Y$ as follows: for every $x \in f(A)$, $\{x\} \sqcup f^{-1}(x)$ is one equivalence class; for every $x \notin (A \sqcup f(A))$, the singleton $\{x\}$ is an equivalence class.[22] The quotient $(X \sqcup Y)/\sim$, with the quotient topology, is denoted by $X \sqcup_f Y$ and is called the *adjunction space* obtained by gluing $X$ and $Y$ along $f$.

## Connectedness

We can picture a space as being "connected," but this also has a precise definition.

[22] $f$ need not be injective, so one cannot declare $\{x\} \sqcup \{f(x)\}$ to be an equivalence class for each $x \in A$.

**Definition A.7.24.**

(1) A topological space $X$ is *connected*, or is a *connected space*, if there are no two nonempty disjoint open sets $U,V \subseteq X$ with $X = U \sqcup V$. Otherwise it is *disconnected*, or is a *disconnected space*.

(2) A subset $A$ of a topological space $X$ is connected if it is connected in the subspace topology inherited from $X$. A connected subset $C$ of $X$ is a *connected component* if it is not properly contained in any larger connected subset.

(3) A topological space $X$ is *path-connected*, or is a *path-connected space*, if, for every $P,Q \in X$, there is a continuous map $\gamma : [0,1] \to X$ with $\gamma(0) = P$ and $\gamma(1) = Q$. Such a $\gamma$ is a *path* from $P$ to $Q$. A subset $A$ of $X$ is path-connected if it is path-connected in the subspace topology inherited from $X$.

Intuitively, connectedness says that a space cannot be split into two completely separated nonempty open pieces. Its connected components are the maximal pieces obtained from connectedness; $X$ is connected exactly when it has one connected component, provided $X$ is nonempty.

We give some simple properties below. Proofs of several illustrate how to reason about connectedness; other proofs are omitted.

**Lemma A.7.25.** *For real numbers $a < b$, each of $[a,b]$, $(a,b)$, $[a,b)$, and $(a,b]$ is connected.*

**Proof.** We first prove that $[a,b]$ is connected. Suppose to the contrary that $[a,b] = U \sqcup V$ for nonempty open sets $U,V$. Choose $x \in U$ and $y \in V$. Without loss of generality, assume $x < y$ (otherwise interchange $U,V$). Put $S = U \cap [x,y]$. The set $S$ is nonempty because $x \in S$, and it is bounded because $S \subseteq [x,y]$. By the least-upper-bound property of the real numbers, $c = \sup S$ exists. Since $x$ and $y$ are respectively a lower and an upper bound of $S$, we have $x \le c \le y$. Thus $c \in [x,y] \subseteq [a,b] = U \sqcup V$, so either $c \in U$ or $c \in V$.

If $c \in U$, then $c \ne y$ because $y \in V$, and hence $c < y$. By the definitions of an open set in $\mathbb{R}$ and of the subspace topology, there is $\epsilon > 0$ such that $(c-\epsilon, c+\epsilon) \cap [a,b] \subseteq U$. Set $z = c + \frac{1}{2}\min\{\epsilon, y-c\}$. Then $z \in U$ and $c < z < y$, so $z \in U \cap [x,y] = S$; but $z > c$, contradicting the fact that $c$ is an upper bound of $S$.

If $c \in V$, there is likewise $\epsilon > 0$ such that $(c-\epsilon, c+\epsilon) \cap [a,b] \subseteq V$. Since $c = \sup S$, the number $c - \epsilon$ is not an upper bound of $S$. Hence some $s \in S$ satisfies $c - \epsilon < s \le c$. Thus $s \in (c-\epsilon, c] \subseteq (c-\epsilon, c+\epsilon)$; also $s \in S \subseteq [x,y] \subseteq [a,b]$, and consequently $s \in (c-\epsilon, c+\epsilon) \cap [a,b] \subseteq V$. This contradicts $s \in S \subseteq U$. Therefore $[a,b]$ is connected.

Now consider $(a,b)$. Suppose to the contrary that there are open sets $U,V$ such that $(a,b) = U \sqcup V$. Choose $x \in U$ and $y \in V$; without loss of generality, assume $x < y$. Then $[x,y] \subseteq (a,b)$ and

$$[x,y] = (U \cap [x,y]) \sqcup (V \cap [x,y]).$$

The two sets on the right are nonempty because they contain $x$ and $y$, respectively, and by proposition A.7.2 both are open in $[x,y]$. This expresses $[x,y]$ as the disjoint union of two nonempty open sets, contradicting the connectedness of $[x,y]$. The arguments for $[a,b)$ and $(a,b]$ are the same. □

**Theorem A.7.26.** *Every path-connected topological space is connected.*

**Proof.** Let $X$ be path-connected. If $X$ were disconnected, there would be nonempty open sets $U,V \subseteq X$ such that $X = U \sqcup V$. Choose $P \in U$ and $Q \in V$. Path-connectedness gives a continuous path $\gamma : [0,1] \to X$ with $\gamma(0) = P$ and $\gamma(1) = Q$. Since $\gamma$ is continuous, the definition of continuity shows that the inverse images $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are open in $[0,1]$, and

$$[0,1] = \gamma^{-1}(X) = \gamma^{-1}(U) \cup \gamma^{-1}(V),$$
$$\gamma^{-1}(U) \cap \gamma^{-1}(V) = \gamma^{-1}(U \cap V) = \gamma^{-1}(\emptyset) = \emptyset.$$

Moreover, $0 \in \gamma^{-1}(U)$ and $1 \in \gamma^{-1}(V)$, so $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are both nonempty. This splits $[0,1]$ into two disjoint nonempty open sets, contradicting the connectedness of $[0,1]$ given by lemma A.7.25. Hence $X$ is connected. ☐

**Theorem A.7.27.** *Every connected open subset of $X = \mathbb{R}^n$ or $\mathbb{C}^n$ is path-connected.*

**Proof.** The proof is omitted. ☐

**Proposition A.7.28.** *A topological space is connected exactly when it has no nonempty proper subset that is both open and closed.*

**Proof.** If $A \subseteq X$ is both open and closed, then $A$ and $X \setminus A$ are disjoint open sets, so $X = A \sqcup (X \setminus A)$. This expresses $X$ as the disjoint union of two open sets. Thus the absence of a nonempty proper subset $A$ that is both open and closed is equivalent to connectedness. ☐

**Proposition A.7.29.** For a topological space $X$ and a map $f : X \to Y$, call $f$ *locally constant* if, for every $x \in X$, there is an open set $U_x$ in $X$ such that $x \in U_x \subseteq X$ and $f(U_x)$ is a singleton; that is, $f$ takes one value throughout $U_x$. *If $X$ is nonempty and connected, every locally constant map $f : X \to Y$ is constant on all of $X$, so $f(X)$ is a singleton.*

**Proof.** Choose $y \in f(X) \subseteq Y$. For every $x \in f^{-1}(y)$, local constancy provides an open set $U_x$ containing $x$ such that $f(U_x) = \{y\}$. Consequently,

$$f^{-1}(y) = \bigcup_{x \in f^{-1}(y)} U_x$$

is open. Now take any $x \in X \setminus f^{-1}(y)$. Again there is an open set $U_x$ containing $x$ on which $f$ is constantly equal to $f(x)$. Since $f(x) \neq y$, one has $U_x \subseteq X \setminus f^{-1}(y)$; by the same reasoning, $X \setminus f^{-1}(y)$ is open. Hence $f^{-1}(y)$ is closed as well as open. It is nonempty because $y \in f(X)$, so proposition A.7.28 gives $f^{-1}(y) = X$. Thus $f$ is constant. ☐

**Theorem A.7.30.** *Let $X \cong Y$ be homeomorphic topological spaces and let $f : X \to Y$ be a corresponding homeomorphism. Then $X$ is connected if and only if $Y$ is connected; $X$ is path-connected if and only if $Y$ is path-connected; and $f$ sends the connected components of $X$ to the connected components of $Y$. The two spaces have the same number of connected components.*

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## Exercises

**Exercise A.121.** Prove theorems A.7.18 and A.7.30.

**Exercise A.122.** Let $X, Y, Z$ be topological spaces. Prove that both projection maps $\pi_X : X \times Y \to X$, $(x,y) \mapsto x$, and $\pi_Y : X \times Y \to Y$, $(x,y) \mapsto y$, are continuous. Prove also that a map $f : Z \to X \times Y$ is continuous if and only if both $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

**Exercise A.123.** Let $X$ be a topological space and $\mathscr{T}$ any topology on $Y$. If $f : X \to (Y, \mathscr{T})$ is a continuous surjection, prove that every member of $\mathscr{T}$ is open in the quotient topology on $Y$ induced by $f$.

**Exercise A.124.** Let $X$ be a topological space and $f: X \to Y$ a surjection, and give $Y$ the quotient topology induced by $f$. Prove that, for every topological space $Z$ and map $g: Y \to Z$, $g$ is continuous if and only if $g \circ f$ is continuous.

**Exercise A.125.** Let $f: X \to Y$ be continuous. Prove that if $X$ is compact, then $f(X)$ is compact. Deduce that homeomorphisms preserve compactness.

**Exercise A.126.** Give an example showing that a homeomorphism need not preserve boundedness.

**Exercise A.127.** Give $X = [0,1)$ the subspace topology inherited from $\mathbb{R}$. Define an equivalence relation $\sim$ on $\mathbb{R}$ by $x \sim y$ if and only if $x - y$ is an integer, and form the quotient space $Y = \mathbb{R}/\sim$. Consider $f: X \to Y$, $t \mapsto [t]$. Show that $f$ is a continuous bijection but $f^{-1}$ is not continuous.

**Exercise A.128.** Prove that a continuous map sends connected spaces to connected spaces and path-connected spaces to path-connected spaces.

**Exercise A.129.** Consider the following subset of $\mathbb{R}^2$:

$$X = \left\{ \left( x, \sin\frac{1}{x} \right) \middle| 0 < x \leq 1 \right\} \cup \{(0,x) \mid -1 \leq x \leq 1\}.$$

Is $X$ connected? Is it path-connected?

### A.7.3 Basic Properties of Embedded Surfaces

**Geometric examples**

We use concrete surfaces in $\mathbb{R}^n$ to illustrate homeomorphisms, product topologies, and quotient topologies. The concepts in the preceding subsection were given for completeness. A reader seeking a quick introduction can instead begin with geometric intuition, treating the objects below as geometric objects without first considering the definition of a topology. Spaces such as $\mathbb{R}^n$ and $\mathbb{C}^n$ are topological spaces, and surfaces embedded in them inherit their subspace topologies. We first describe some objects intuitively and then give definitions.

For points $A, B \in \mathbb{R}^n$, the segment usually said to join them is the *closed line segment* they determine, denoted by $[A, B]$.

In real Euclidean space, write $\mathbb{S}^n$ for an $n$-dimensional *sphere*. Thus $\mathbb{S}^1$ is a *circle*, and $\mathbb{S}^2$ is the usual sphere in three-dimensional space. Here $n$ is the dimension of the sphere itself, not of its ambient space. Roughly, a surface in three-dimensional space is two-dimensional, and a curve is one-dimensional: a small piece near a point of a surface resembles part of a two-dimensional plane, while a short piece of a curve resembles part of a one-dimensional line. A precise definition of dimension is given below.

In $n$-dimensional real Euclidean space, the region inside the $(n-1)$-sphere $\mathbb{S}^{n-1}$, excluding the sphere itself, is an $n$-dimensional topological space called an *open n-disk*, denoted by $\mathring{\mathbb{D}}^n$. Adding its boundary gives the *closed n-disk*, or simply *n-disk*, denoted by $\mathbb{D}^n$. Thus $\mathbb{D}^1$ is a line segment, $\mathbb{D}^2$ is a circular disk including its boundary and interior, and $\mathbb{D}^3$ is a solid ball including its boundary and interior.[23]

A more precise definition follows.

[23] Some authors write $\mathbb{D}^n$ for the open disk and $\overline{\mathbb{D}}^n$ for the closed disk; this book uses the convention in the text.

**Definition A.7.31.**

(1) For points $\boldsymbol{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, define the open disk, disk, and sphere, respectively, by
$$\mathring{\mathbb{D}}^n = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, |\boldsymbol{x}| < 1\}, \quad \mathbb{D}^n = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, |\boldsymbol{x}| \leq 1\}, \quad \mathbb{S}^{n-1} = \{\boldsymbol{x} \in \mathbb{R}^n \,|\, |\boldsymbol{x}| = 1\}.$$
As subsets of $\mathbb{R}^n$, they naturally inherit their topologies from $\mathbb{R}^n$, and $\mathring{\mathbb{D}}^n = \mathrm{int}_{\mathbb{R}^n} \mathbb{D}^n$.

(2) For points $\boldsymbol{x} = (x_1, \ldots, x_n)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, define the closed line segment joining them by
$$[\boldsymbol{x}, \boldsymbol{y}] = \{(1-t)\boldsymbol{x} + t\boldsymbol{y} \,|\, 0 \leq t \leq 1\}.$$

A homeomorphism is a bijection for which both the map and its inverse are continuous. For $f: S \to S'$, requiring $f$ to be bijective already rules out collapsing a whole piece of a surface to a point or gluing originally distinct parts together. Requiring $f$ to be continuous means that points of $S$ which are close to one another remain close after mapping to $S'$; requiring $f^{-1}$ to be continuous means that points of $S'$ which are close to one another have preimages which remain close in $S$.

Thus a homeomorphism preserves only the intrinsic pattern of connection, not lengths, angles, areas, curvature, or the concrete shape of the surface in three-dimensional space. Flattening, stretching, or bending a sphere does not change its topological type. The familiar picture of deforming modelling clay allows stretching and bending, but no tearing or gluing. This is only an aid to understanding: the precise condition is a continuous bijection with continuous inverse.[24]

Collapsing an annulus to a point would require gluing up its hole, so the two are not homeomorphic; turning an annulus into a line segment would require tearing it, so these are not homeomorphic either. The saying that a topologist cannot distinguish a coffee cup from a doughnut means that the coffee cup and the doughnut are homeomorphic: both are tori (defined below).

In topology, descriptions of types of surface are often understood up to homeomorphism, without specifying a concrete shape. For example, spaces homeomorphic to the $\mathring{\mathbb{D}}^n$, $\mathbb{D}^n$, and $\mathbb{S}^n$ defined above are also often called open disks, disks, and spheres, respectively. The previously defined $\mathrm{B}(x, r) = \{y \in \mathbb{R}^n \,|\, \|y - x\| < r\}$ is likewise homeomorphic to an open $n$-disk.

**Example A.7.32.** As an example of a homeomorphism, define an equivalence relation on the real line $\mathbb{R}$ by $x \sim y$ if and only if $x - y \in \mathbb{Z}$. The corresponding quotient topology $\mathbb{R}/\sim$ is also denoted by $\mathbb{R}/\mathbb{Z}$. Then $\mathbb{R}/\mathbb{Z} \cong \mathbb{S}^1$, because all points differing by an integer translation are identified; this may be visualized as turning the line into a circle. Explicitly, a homeomorphism is
$$f: \mathbb{R}/\mathbb{Z} \longrightarrow \mathbb{S}^1, \qquad [x] \longmapsto (\cos 2\pi x, \sin 2\pi x).$$

The product topology can be pictured directly as a Cartesian product.

**Example A.7.33.** Cylinders.

(1) The topology of a *cylinder* in $\mathbb{R}^3$ is given by
$$\mathbb{S}^1 \times [0,1] = \{(x_1, x_2, x_3) \,|\, (x_1, x_2) \in \mathbb{S}^1,\ x_3 \in [0,1]\} \cong \{(x_1, x_2, x_3) \,|\, x_1^2 + x_2^2 = 1, x_3 \in [0,1]\}.$$
One can prove that the product topology on $\mathbb{S}^1 \times [0,1]$ is also its subspace topology as a subset of $\mathbb{R}^3$.

(2) For $b > a > 0$, a planar *annulus* is the part of the plane between two concentric circles, including the circles themselves:
$$A = \{(x, y) \in \mathbb{R}^2 \,|\, a \leq \sqrt{x^2 + y^2} \leq b\}.$$

[24]This popular description is closer to the topological notion of *isotopy*, more precisely ambient isotopy. Roughly, it adds a time dimension to homeomorphism by requiring a continuous deformation between the two objects. Knots, for example, are all homeomorphic to a circle, but can be wound in complicated ways: an arbitrary knot cannot be untied without cutting it, and different knots are generally inequivalent under isotopy. As an intuitive, not entirely precise picture, knowing only that two objects are homeomorphic places no condition on intermediate states when one tries to imagine the correspondence as a continuous motion; one may therefore imagine intermediate states that pass through themselves. We need not give the precise definition of isotopy here.

Then $A \cong \mathbb{S}^1 \times [0,1]$; intuitively, the height of the cylinder is flattened into the plane. In fact, one can verify the homeomorphism

$$f : A \longrightarrow \mathbb{S}^1 \times [0,1],$$
$$(r\cos\theta, r\sin\theta) \mapsto \Big((\cos\theta,\sin\theta), \frac{r-a}{b-a}\Big).$$

Its inverse is

$$f^{-1}((\cos\theta,\sin\theta),t) = ((a+(b-a)t)\cos\theta, (a+(b-a)t)\sin\theta).$$

**Example A.7.34.** The $n$-dimensional *torus* is $\mathbb{T}^n = \underbrace{\mathbb{S}^1 \times \mathbb{S}^1 \times \cdots \times \mathbb{S}^1}_{n \text{ factors}}$ with the product topology. The surface $\mathbb{T}^2$ is the familiar tire-shaped or swim-ring-shaped torus that can be embedded in $\mathbb{R}^3$: a point on it may be regarded as determined by a point on a circular cross-section together with a point on the circle traced by the center of that cross-section.

It can also be described by the equation of a surface of revolution. For $R > r > 0$,

$$\mathbb{T}^2 \cong \big\{(x,y,z) \in \mathbb{R}^3 \;\big|\; x = (R + r\cos v)\cos u, \quad y = (R + r\cos v)\sin u, \quad z = r\sin v; \quad u,v \in [0,2\pi)\big\}.$$

A point on $\mathbb{S}^1$ has the angular parametrization $(\cos\theta, \sin\theta)$. Sending the angular parameters $(\theta_1, \theta_2)$ of $\mathbb{S}^1 \times \mathbb{S}^1$ to the parameters $(u,v)$ in the displayed surface gives the stated homeomorphism.

Another earlier construction was the quotient topology. Ignoring its topological structure for the moment, one may view $X/\sim$ as the quotient set of $X$ under the equivalence relation $\sim$. Geometrically, this glues equivalent points into a single point.

**Example A.7.35.** Collapsing the boundary of $\mathbb{D}^2$ to a single point gives a sphere. More precisely, define $\sim$ by declaring all points on the boundary of the disk equivalent. This quotient-topology description gives $\mathbb{S}^2 \cong \mathbb{D}^2/\sim$. An explicit homeomorphism can be given as follows. Let $(r,\theta)$ be the polar coordinates of a point of the disk, whose Cartesian coordinates are $(r\cos\theta, r\sin\theta)$. Writing its equivalence class under $\sim$ as $[(r,\theta)]$, we obtain the homeomorphism to $\mathbb{S}^2$

$$f : \mathbb{D}^2/\sim \longrightarrow \mathbb{S}^2,$$
$$[(r,\theta)] \mapsto (\sin(\pi r)\cos\theta, \sin(\pi r)\sin\theta, \cos(\pi r)).$$

**Example A.7.36.** For the square region $I^2 = [0,1] \times [0,1]$, the equivalence relation $(0,y) \sim (1,y)$ glues corresponding points of one pair of opposite sides and gives a cylinder, so $I^2/\sim\; \cong \mathbb{S}^1 \times [0,1]$. The equivalence relation $(0,y) \sim (1,y)$, $(x,0) \sim (x,1)$ also glues the corresponding points of the other pair, on the cylinder already formed, and gives a torus. Thus, for this relation, $I^2/\sim\; \cong \mathbb{T}^2$. We do not write the homeomorphisms explicitly here.

### Boundary and interior of a surface

We often speak of a surface, its boundary, and the interior it encloses. We now give precise definitions of these intuitive notions.

**Definition A.7.37.**

(1) Let $M \subseteq \mathbb{R}^n$ have the subspace topology inherited from $\mathbb{R}^n$. If every $P \in M$ admits an open set $U$ in $M$ containing $P$, an open set $V$ in $\mathbb{D}^m$, and a homeomorphism $f : U \to V$, then $M$ is an *embedded m-manifold* in $\mathbb{R}^n$.[25] If $m = n-1$, it is an embedded *hypersurface* in $\mathbb{R}^n$.

(2) For an embedded manifold $M \subseteq \mathbb{R}^n$ and $P \in M$, choose a local homeomorphism $f$ as above. The point $P$ is an *interior point of the manifold* if and only if $f(P) \in \mathring{\mathbb{D}}^m$, and a *boundary point of the manifold* if and only if $f(P) \in \partial_{\mathbb{R}^m}\mathbb{D}^m$. All its interior points form the *interior* $M^\circ$, and all its boundary points form the *boundary* $\partial M$.

[25] $V$ is open in the subspace topology that $\mathbb{D}^m$ inherits from $\mathbb{R}^m$; this does not mean that $V$ is open in $\mathbb{R}^m$ and contained in $\mathbb{D}^m$. Likewise, $U$ is open in the topological space $M$.

(3) An embedded manifold $M \subseteq \mathbb{R}^n$ is *without boundary* if and only if $\partial M = \emptyset$; otherwise it is *with boundary*.
(4) A manifold $M$ is *compact* if it is compact as a topological space.

**Remark.**
(1) This definition also explains dimension: near each point, an embedded $m$-manifold resembles part of $m$-dimensional Euclidean space. Thus $\mathbb{S}^{n-1} \subseteq \mathbb{R}^n$ is an $(n-1)$-dimensional hypersurface.
(2) "Without boundary" is an intrinsic property of the manifold; it does not say that $M$, as a subset of $\mathbb{R}^n$, has no boundary. For the closed hypersurfaces discussed below, one usually has $\operatorname{int}_{\mathbb{R}^n} M = \emptyset$ and $\partial_{\mathbb{R}^n} M = M$. Intuitively, an $(n-1)$-dimensional hypersurface has no $n$-dimensional volume, and every ambient open ball about one of its points also contains points outside it.
(3) Having a boundary and being bounded are different notions. The former means, intuitively, that the surface has an edge; the latter means that it is contained in a finite region. For example, deleting an open disk from an infinite plane in $\mathbb{R}^3$, homeomorphic to $\mathbb{R}^2$, makes the boundary of the disk the boundary of the resulting surface. The surface thus has a boundary but is still unbounded in $\mathbb{R}^3$.
(4) A manifold need not first be defined inside $\mathbb{R}^n$; it can be defined intrinsically. We use the embedded definition here for simplicity. For an embedded manifold $M$, another topological space homeomorphic to $M$ may itself be regarded as a manifold, independently of the space in which $M$ is embedded. In fact, every usual finite-dimensional *topological manifold* $M$ admits an integer $n$ and an $m$-manifold $M'$ embedded in $\mathbb{R}^n$ with $M \cong M'$. Different embeddings give embedded manifolds of the same dimension.

**Proposition A.7.38.** *In the preceding definition of a manifold $M$, the distinction between interior and boundary points is independent of the chosen local homeomorphism $f$, and $M = M^\circ \sqcup \partial M$.*

**Proof.** The first assertion requires the invariance of boundary theorem from topology; we do not prove it here. The decomposition $M = M^\circ \sqcup \partial M$ is immediate from the definition. □

Intuitively, the boundary of a surface means the edge up to which the surface extends, rather than the surface itself. For example, the surface of the two-dimensional sphere $\mathbb{S}^2$ is intuitively closed: it has no edge at which it ends, so it is a surface without boundary. For the two-dimensional disk $\mathbb{D}^2$, however, what we ordinarily call its "surface" is the entire circular disk, and the edge to which it extends is the circle $\mathbb{S}^1$.

**Example A.7.39.** The boundary of a ball is a sphere, $\partial \mathbb{D}^m = \mathbb{S}^{m-1}$. A sphere has no boundary, $\partial \mathbb{S}^m = \emptyset$. The boundary of a cylinder consists of its two end circles:

$$\partial(\mathbb{S}^1 \times [0,1]) = (\mathbb{S}^1 \times \{0\}) \cup (\mathbb{S}^1 \times \{1\}).$$

**Theorem A.7.40.** *For an embedded manifold $M \subseteq \mathbb{R}^n$, the following are equivalent: $M$ is a compact manifold; $M$ is compact as a topological space; $M$ is a compact subset of $\mathbb{R}^n$; and $M$ is closed and bounded in $\mathbb{R}^n$.*

Accordingly, for an embedded manifold in Euclidean space, compactness may be pictured as being closed and bounded.

**Theorem A.7.41** (Jordan–Brouwer separation theorem)**.** *Let $n \geq 2$, and let $M \subseteq \mathbb{R}^n$ be a connected, compact, embedded $(n-1)$-manifold without boundary. Then $\mathbb{R}^n \setminus M$ has exactly two connected components, a bounded component $\Omega_{\text{in}}$ and an unbounded component $\Omega_{\text{out}}$, and*

$$\partial_{\mathbb{R}^n} \Omega_{\text{in}} = \partial_{\mathbb{R}^n} \Omega_{\text{out}} = M.$$

The bounded component $\Omega_{\text{in}}$ is the *interior region* enclosed by $M$, and the unbounded component $\Omega_{\text{out}}$ is its *exterior region*. The interior region together with the hypersurface itself, $\Omega_{\text{in}} \cup M$, is the *closed region bounded by* $M$.

**Remark.** There are now three different notions of "interior":

(1) $M^\circ$ consists of the intrinsic interior points of the manifold;
(2) $\operatorname{int}_{\mathbb{R}^n} M$ is the interior of $M$ as a subset of $\mathbb{R}^n$, and is usually empty for a hypersurface;
(3) $\Omega_{\text{in}} \subseteq \mathbb{R}^n \setminus M$ is the $n$-dimensional interior region bounded by a boundaryless hypersurface, and only this "interior" lies outside the surface itself.

The proof of Jordan–Brouwer is far beyond our scope, but its content is the familiar fact that a closed hypersurface separates space into an inside and an outside.

**Example A.7.42.** For the embedded unit sphere $\mathbb{S}^{n-1} = \{(x_1,\dots,x_n) \in \mathbb{R}^n \,|\, x_1^2 + \cdots + x_n^2 = 1\}$, the interior and exterior regions are

$$\Omega_{\text{in}} = \{(x_1,\dots,x_n) \in \mathbb{R}^n \,|\, x_1^2 + \cdots + x_n^2 < 1\} = \mathring{\mathbb{D}}^n,$$
$$\Omega_{\text{out}} = \{(x_1,\dots,x_n) \in \mathbb{R}^n \,|\, x_1^2 + \cdots + x_n^2 > 1\}.$$

They have the common boundary $\partial_{\mathbb{R}^n} \Omega_{\text{in}} = \partial_{\mathbb{R}^n} \Omega_{\text{out}} = \mathbb{S}^{n-1}$.

### Triangulations and orientability of surfaces

**Definition A.7.43.**

(1) Let $\Delta = \{(x,y) \in \mathbb{R}^2 \,|\, x \ge 0, y \ge 0, x+y \le 1\}$ be the standard *closed triangle*, with the subspace topology inherited from $\mathbb{R}^2$ and vertices $\boldsymbol{e}_0 = (0,0)$, $\boldsymbol{e}_1 = (1,0)$, and $\boldsymbol{e}_2 = (0,1)$; the three closed segments joining pairs of vertices are its *edges*. A triangle has three edges.
(2) Let $M$ be a two-dimensional surface. If $T \subseteq M$ and $f : \Delta \to T$ is a homeomorphism, then $(T,f)$ is a *triangle* in $M$. The points $f(\boldsymbol{e}_0), f(\boldsymbol{e}_1), f(\boldsymbol{e}_2)$ are its *vertices*, the images of the three edges of $\Delta$ are its *edges*, and $f(\operatorname{int}\Delta)$ is its *relative interior*.
(3) For a two-dimensional surface $M$, a finite or infinite family of triangles $\{(T_i,f_i) \,|\, i\}$, with index $i \in A$, is a *triangulation* of $M$ if all of the following hold:
   (i) $M = \bigcup_{i\in A} T_i$;
   (ii) the intersection of two distinct triangles is empty, a common vertex, or a common edge;
   (iii) each edge belongs to exactly one or two triangles;
   (iv) the triangulation is locally finite: for every $P \in M$, there is an open set $U$ in $M$ containing $P$ such that only finitely many triangles $T_i$ satisfy $U \cap T_i \neq \emptyset$.

   An edge in one triangle is a *boundary edge*; one in two is an *interior edge*. A triangulation with finitely many triangles is a *finite triangulation*; otherwise it is an *infinite triangulation*.

We give the following properties without proof.

**Proposition A.7.44.** *For a triangulation of a surface $M$, the relative interiors of distinct triangles are disjoint; the union of all boundary edges of a triangulation is precisely the boundary of the surface; and if $f : M \to N$ is a homeomorphism of surfaces, it carries a triangulation of $M$ to a triangulation of $N$, preserving the numbers of vertices, edges, and triangles.*

**Theorem A.7.45** (Radó's triangulation theorem)**.** *Every two-dimensional surface $M$ embedded in some $\mathbb{R}^n$ admits a triangulation. The triangulation is finite if and only if $M$ is compact.*

We can now define orientations of two-dimensional surfaces precisely.

**Definition A.7.46.**

(1) Let $T$ be a triangle with vertices $A_1,A_2,A_3$. An *orientation of a triangle* is a cyclic order of its vertices. Thus $(A_1,A_2,A_3)$, $(A_2,A_3,A_1)$, and $(A_3,A_1,A_2)$ give one orientation, whereas $(A_1,A_3,A_2)$, $(A_3,A_2,A_1)$, and $(A_2,A_1,A_3)$ give the opposite orientation. A triangle therefore has exactly two opposite orientations. The oriented triangle $(A_1,A_2,A_3)$ induces the edge directions $A_1 \to A_2$, $A_2 \to A_3$, and $A_3 \to A_1$.

(2) Orientations on two triangles sharing an edge are compatible when they induce opposite directions on that edge.

A given triangulation of a two-dimensional surface $M$ is *orientable* if each triangle can be oriented so that the two triangles at every interior edge have compatible orientations there. The surface $M$ is *orientable* if it has an orientable triangulation; otherwise it is *nonorientable*. For an orientable surface $M$, a compatible choice of directions in one of its orientable triangulations is an *orientation* of $M$.

**Theorem A.7.47.** *A two-dimensional surface M is orientable if and only if every triangulation of it is orientable. If M is connected and orientable, its triangulation has exactly two orientations, obtained from one another by reversing the directions of all triangles simultaneously.*

Intuitively, orientability means that the two sides of a surface can be chosen continuously and kept separate: a path that does not cross the boundary cannot pass from one side to the other. For a triangulated surface, an orientation of each triangle selects a normal direction by the right- or left-hand rule. Applying the right-hand rule throughout a compatible choice of orientations gives a continuous normal direction to the surface. Reversing all triangle orientations simultaneously and again using the right-hand rule gives the opposite continuous normal direction. Orientations can also be defined in general dimensions, but not all higher-dimensional manifolds are triangulable, making an elementary definition difficult; we shall not need it here.

The Jordan–Brouwer separation theorem gives an intuitive picture for a two-dimensional surface embedded in three-dimensional space: the two regions into which it separates the space meet its two sides, suggesting that the surface is orientable. In fact, we have the following result.

**Theorem A.7.48.** *Let M be a compact, connected, two-dimensional surface without boundary. Then M is orientable if and only if it is homeomorphic to a two-dimensional surface embedded in* $\mathbb{R}^3$.

The sphere is orientable. A frequently mentioned example is the *Klein bottle*, obtained intuitively by gluing one pair of opposite sides of a square in matching directions and the other pair in opposite directions. Precisely, it is

$$([0,1]\times[0,1])/((0,y)\sim(1,y),(x,0)\sim(1-x,1)).$$

The Klein bottle is compact, but it cannot be embedded in $\mathbb{R}^3$; an embedding requires at least $\mathbb{R}^4$ (we omit the proof). It is nonorientable.

**Example A.7.49.** The sphere $\mathbb{S}^2$ is orientable.

**Proof.** Choose four points $A_0,A_1,A_2,A_3$ on $\mathbb{S}^2$. Give the triangles the orientations

$$(A_1,A_2,A_3), \quad (A_0,A_3,A_2), \quad (A_0,A_1,A_3), \quad (A_0,A_2,A_1).$$

One can verify that they form an orientable triangulation of $\mathbb{S}^2$. Their compatibility can be checked directly. For example, the first two induce the opposite directions $A_2 \to A_3$ and $A_3 \to A_2$ on their common edge $[A_2,A_3]$. □

**Example A.7.50.** Intuitively, a *Möbius band* is formed by reversing one of a pair of sides of a rectangle and joining the pair. Precisely, let $Q=[0,1]\times[-1,1]$ and define the equivalence relation $(0,t)\sim(1,-t)$. The quotient space $M=Q/\sim$ is the Möbius band. Its indicated identification glues a pair of opposite sides after reversing one of them. The resulting surface is not orientable.

**Proof.** Orientability is independent of the triangulation, so use the following one. For $i=0,1,2,3$, put $A_i=(i/3,1)$ and $B_i=(i/3,-1)$. For $i=0,1,2$, take the unoriented triangles

$$\Delta_i=[B_i,B_{i+1},A_{i+1}], \qquad \Gamma_i=[B_i,A_{i+1},A_i].$$

These six triangles triangulate $Q$, and under the quotient $A_0\sim B_3$, $B_0\sim A_3$; the boundary edges $[A_0,B_0]$ and $[B_3,A_3]$ become one interior edge. Hence they induce a triangulation of the Möbius band $M$.

Suppose this triangulation is orientable, and give $\Delta_0$ the orientation $(B_0,B_1,A_1)$. Adjacent triangles must induce opposite directions on their common edge, so compatibility successively forces

$$\Gamma_0=(B_0,A_1,A_0), \qquad \Gamma_1=(B_1,A_2,A_1),$$
$$\Delta_1=(B_1,B_2,A_2), \qquad \Gamma_2=(B_2,A_3,A_2),$$
$$\Delta_2=(B_2,B_3,A_3).$$

For example, $\Delta_0,\Gamma_0$ share $[B_0,A_1]$, while $\Delta_0,\Gamma_1$ share $[B_1,A_1]$; the subsequent steps are similar. Now $\Gamma_0$ induces $A_0\to B_0$ on $[A_0,B_0]$, and $\Delta_2$ induces $B_3\to A_3$ on $[B_3,A_3]$. Under the gluing, $A_0\sim B_3$ and $B_0\sim A_3$, these become the same direction on the common edge, namely $(A_0=B_3)\to(B_0=A_3)$, contradicting compatibility. If $\Delta_0$ is initially given the opposite orientation, all directions in the argument reverse simultaneously and the result is unchanged. $\square$

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## Exercises

**Exercise A.130.** Prove from the definition that relative interiors of distinct triangles in a triangulation are disjoint.

**Exercise A.131.** If a finite triangulation has $F$ triangles, $E_1$ interior edges, and $E_2$ boundary edges, prove $3F=2E_1+E_2$.

**Exercise A.132.** Put $I=[0,1]$. Give explicit homeomorphisms realizing the identifications mentioned in example A.7.36:

$$I^2/((0,y)\sim(1,y))\cong\mathbb{S}^1\times I, \quad I^2/((0,y)\sim(1,y),(x,0)\sim(x,1))\cong\mathbb{T}^2.$$

**Exercise A.133.** For $\mathbb{S}^1\subseteq\mathbb{R}^2$, $\mathbb{D}^2\subseteq\mathbb{R}^2$, $\mathbb{S}^2\subseteq\mathbb{R}^3$, and $\mathbb{S}^1\times[0,1]\subseteq\mathbb{R}^3$, determine both the intrinsic manifold boundary and the boundary as a subset of the ambient Euclidean space. If a surface encloses an interior region in the corresponding Euclidean space, determine that region as well.

**Exercise A.134.** Prove that the intrinsic boundary of the Möbius band is homeomorphic to $\mathbb{S}^1$.

### A.7.4 Classification of Surfaces

A quantity preserved by every homeomorphism is a *topological invariant*. Intuitively, it is a property unchanged by continuous deformation without gluing or tearing. Compactness, boundary, and connectedness are examples for surfaces. The intuitive statement that a continuously changing surface keeps its topological type is not unconditional: if a loop is pinched down to a point, for example, the limiting surface need not be homeomorphic to the original one. Away from such limiting degenerations, the intuition is appropriate for the situations considered here. Building on theorems A.7.18 and A.7.30, we use the following additional invariants.

**Theorem A.7.51.** *Let $X \cong Y$ be homeomorphic embedded manifolds or topological manifolds, and let $f: X \to Y$ be a corresponding homeomorphism. Then:*
*(1) $X, Y$ have the same dimension.*
*(2) $f$ sends intrinsic interior and boundary points of $X$ to intrinsic interior and boundary points of $Y$, respectively. In particular, $X$ has boundary if and only if $Y$ has boundary.*
*(3) $X$ is orientable if and only if $Y$ is orientable.*

We next study further constructions. A classic geometric construction glues two surfaces together: adding a handle to a cup, for example, creates a "hole." We begin by defining the gluing operation precisely.

**Definition A.7.52.** Let $M,N$ be connected two-dimensional embedded manifolds or topological manifolds. Choose disks $D_M \subseteq M^\circ$ and $D_N \subseteq N^\circ$, with $D_M, D_N \cong \mathbb{D}^2$, and remove their interiors. Both resulting boundaries $\partial D_M, \partial D_N$ are homeomorphic to $\mathbb{S}^1$. Choose a homeomorphism $f: \partial D_N \to \partial D_M$. Their *connected sum* is the adjunction space

$$M\#N := (M \setminus \operatorname{int} D_M) \sqcup_f (N \setminus \operatorname{int} D_N) = \big((M \setminus \operatorname{int} D_M) \sqcup (N \setminus \operatorname{int} D_N)\big) / \big(x \sim f(x),\ x \in \partial D_N\big).$$

**Theorem A.7.53.** *For connected surfaces $M,N$, up to homeomorphism, $M\#N$ is independent of the disks $D_M, D_N$ and gluing map $f$ chosen in the definition.*

**Proposition A.7.54.** *For connected surfaces $M,N$, their connected sum $M\#N$ is orientable exactly when both $M$ and $N$ are orientable.*

**Proposition A.7.55.** *For every connected two-dimensional surface $M$, $M\#\mathbb{S}^2 \cong M$.*

**Proof.** Deleting the interior of a disk from a sphere leaves a space homeomorphic to a disk $D'$; we omit that elementary verification. Thus the connected sum removes the interior of a disk $D$ from $M$ and glues a disk back along its boundary by a homeomorphism $f: \partial D' \to \partial D$. By theorem A.7.53, choose $D'$ so that $D, D'$ are congruent disks, and let $g$ match their corresponding points. Taking $f$ to be the restriction of $g$ to $\partial D'$, the glued surface is plainly $M$ again. $\square$

Intuitively, a connected sum removes a disk from each of two surfaces and glues corresponding positions on the disk boundaries. Connected sums can be extended to higher dimensions, but one cannot simply remove an arbitrary topological $n$-disk: unexpected behaviour can occur in higher dimensions. The preceding theorem A.7.53 always holds in two dimensions; we do not introduce connected sums in general dimensions.

### Boundary components, handles, and crosscaps

We now give further precise descriptions of two-dimensional surfaces.

**Theorem A.7.56.** *The boundary of a compact two-dimensional surface $M$ is a finite disjoint union of circles. Explicitly, $\partial M$ has finitely many connected components $C_1, \ldots, C_b$, with $\partial M = C_1 \sqcup \cdots \sqcup C_b$ and $C_i \cong \mathbb{S}^1$.* The number of connected components of $\partial M$ is the *number of boundary components* of $M$; when $\partial M = \emptyset$, this number is defined to be zero.

**Example A.7.57.** The spaces $\mathbb{S}^2$, $\mathbb{D}^2$, and $\mathbb{S}^1 \times [0,1]$ have respectively 0, 1, and 2 boundary components: the sphere has no boundary, the disk has one boundary circle, and the cylinder has its upper and lower boundary circles.

The surface of a pair of trousers has one boundary component at the waist and one at each leg, hence three in all. Topologically, if $D_1, D_2, D_3$ are three pairwise disjoint closed disks in $\mathbb{S}^2$, each homeomorphic to $\mathbb{D}^2$, then $\mathbb{S}^2 \setminus (\operatorname{int} D_1 \sqcup \operatorname{int} D_2 \sqcup \operatorname{int} D_3)$ is called a *pair of pants*. It is homeomorphic to the surface of a pair of trousers when features such as buttons and pockets are ignored.

**Definition A.7.58.**

(1) Removing the interior of a disk $\mathbb{D}^2$ from a torus $\mathbb{T}^2$ gives the surface $\mathbb{T}^2 \setminus \operatorname{int}\mathbb{D}^2$, called a *handle*; it has exactly one boundary component.[26] A *crosscap* here means a surface homeomorphic to the Möbius band in example A.7.50; it too has exactly one boundary component.[27]

(2) Let $M$ be a compact connected two-dimensional surface. Take a disk $\mathbb{D}^2$ contained in $M^\circ$, remove its interior, and attach a handle $\mathcal{H}$ along the resulting boundary circle. This is called *adding a handle* to $M$. More precisely, choose a homeomorphism $f : \partial\mathcal{H} \to \partial\mathbb{D}^2$; the resulting surface is $(M \setminus \operatorname{int}\mathbb{D}^2) \sqcup_f \mathcal{H}$.

(3) Let $M$ be a compact connected two-dimensional surface. Take a disk $\mathbb{D}^2$ contained in $M^\circ$, remove its interior, and attach a crosscap $\mathcal{C}$ along the resulting boundary circle. This is called *adding a crosscap* to $M$. More precisely, choose a homeomorphism $f : \partial\mathcal{C} \to \partial\mathbb{D}^2$; the resulting surface is $(M \setminus \operatorname{int}\mathbb{D}^2) \sqcup_f \mathcal{C}$.

**Theorem A.7.59.** *For a compact connected two-dimensional surface $M$, adding a handle gives a surface homeomorphic to $M \# \mathbb{T}^2$; adding a crosscap gives one homeomorphic to $M \# \mathbb{RP}^2$.*

**Remark.** Here $\mathbb{RP}^2$ is the *real projective plane* discussed in Chapter 3, equipped with its quotient topology. It may be defined topologically as in theorem 3.1.4, or, using the coordinates of $\mathbb{R}^3$, more directly by

$$\mathbb{RP}^2 := \{(x_1,x_2,x_3) \in \mathbb{R}^3 \mid (x_1,x_2,x_3) \neq (0,0,0)\} / ((x_1,x_2,x_3) \sim (\lambda x_1, \lambda x_2, \lambda x_3),\ \lambda \in \mathbb{R}^\times),$$

where $\mathbb{R}^3 \setminus \{(0,0,0)\}$ first inherits the subspace topology of $\mathbb{R}^3$, after which the quotient topology is imposed. The geometric result theorem 3.1.3 gives $\mathbb{RP}^2 \cong \mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$, where $\boldsymbol{x} \sim -\boldsymbol{x}$ identifies points symmetric about the center of the sphere. The homeomorphism is explicitly $f : \mathbb{RP}^2 \to \mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$, given by

$$[(x_1,x_2,x_3)] \mapsto \left[\left(\frac{x_1}{\sqrt{x_1^2+x_2^2+x_3^2}}, \frac{x_2}{\sqrt{x_1^2+x_2^2+x_3^2}}, \frac{x_3}{\sqrt{x_1^2+x_2^2+x_3^2}}\right)\right].$$

*The space $\mathbb{RP}^2$ is also a connected, compact, two-dimensional surface*, so the following discussion applies to it. Although it appears boundaryless in the projective treatment, compactness can also be seen from its quotient of the closed and bounded sphere $\mathbb{S}^2$ in $\mathbb{R}^3$. *Alternatively, it can in fact be embedded in $\mathbb{R}^5$ as a closed and bounded subset (we do not prove this).*

**Proof of theorem A.7.59.** By definition, $\mathbb{T}^2 \setminus \operatorname{int}\mathbb{D}^2$ is a handle. The following lemma identifies $\mathbb{RP}^2 \setminus \operatorname{int}\mathbb{D}^2$ with a crosscap. The claim therefore follows directly from the definition of connected sum. □

**Lemma A.7.60.** *Removing the interior of a disk from the real projective plane leaves a crosscap:*

$$\mathbb{RP}^2 \setminus \operatorname{int}\mathbb{D}^2 \cong ([0,1] \times [-1,1]) / ((0,t) \sim (1,-t)).$$

**Proof.** We first give the geometric picture. Since $\mathbb{RP}^2 \cong \mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$, it suffices to consider the sphere with antipodal points identified. Take two disks $D_1, D_2$ on $\mathbb{S}^2$ that are centrally symmetric about the center of the sphere. Under $\boldsymbol{x} \sim -\boldsymbol{x}$, we have $D_1 \sim D_2$, so removing the interior of one disk in $\mathbb{S}^2/(\boldsymbol{x} \sim -\boldsymbol{x})$ amounts to removing the interiors of these two disks from the sphere and then

[26] A disk here is not restricted to the standard set given by the equations in definition A.7.31: it means a subset $D \subseteq \mathbb{T}^2$ with $D \cong \mathbb{D}^2$. The same convention applies below.

[27] The single boundary of a handle is intuitive. In the model of example A.7.50, identifying the rectangle's sides by $(0,t) \sim (1,t)$ would give a cylinder with two boundary components. The reversed gluing of the Möbius band instead joins the original upper and lower boundaries, making one boundary component. "Crosscap" can have different meanings in different contexts; here it denotes a piece homeomorphic to the Möbius band.

identifying antipodal points. Removing the interiors of disks around the north and south poles leaves an equatorial annulus, homeomorphic to a cylinder. Thus

$$\mathbb{R}\mathbb{P}^2 \setminus \operatorname{int}\mathbb{D}^2 \cong (\mathbb{S}^2 \setminus (\operatorname{int}D_1 \cup \operatorname{int}D_2))/(\boldsymbol{x} \sim -\boldsymbol{x}) \cong (\mathbb{S}^1 \times [-1,1])/(\boldsymbol{x} \sim -\boldsymbol{x}).$$

Here points centrally symmetric about the center of the cylinder are identified. Cut the cylinder along two centrally symmetric generators. Each of the two resulting parts is homeomorphic to a rectangle when the cut edges are included. Since centrally symmetric points in the two rectangle interiors are identified, it suffices to retain one rectangle, homeomorphic to $[0,1]\times[-1,1]$. The two new cut edges were centrally symmetric, so the quotient adds the boundary identification $(0,t)\sim(1,-t)$. This is precisely the Möbius band. □

**Alternative proof.** Alternatively, one can write the homeomorphism explicitly, although we would prefer the reader to understand the geometric idea. Algebraically, put $X = \mathbb{R}\mathbb{P}^2 = (\mathbb{R}^3 \setminus \{O\})/\sim$ and let $M$ be the Möbius band. Take

$$D = \{[(x,y,z)] \in X \mid x^2+y^2 \le z^2\};$$

this is a disk because $g:\mathbb{D}^2 \to D$, $(u,v)\mapsto[(u,v,1)]$, is a homeomorphism. Define

$$f: M \longrightarrow X \setminus \operatorname{int}D, \qquad [(s,t)] \longmapsto [(\cos\pi s, \sin\pi s, t)].$$

One can verify that $f$ respects the equivalence classes and is a bijection. Its expression uses elementary functions, so it is continuous (we omit the proof); one can also verify that its inverse is continuous. □

**Remark.** Another common picture of adding a handle attaches the two ends of a cylinder to the two boundary circles left after deleting the interiors of two disjoint disks from the original surface $M$. One must require an untwisted, orientation-compatible attachment. The definition using a handle or the connected sum $M\#\mathbb{T}^2$ is more precise.

One can explain the intuition informally as follows. Let $M$ be a connected surface and $D_1, D_2$ two disjoint disks contained in $M^\circ$. Write $A = \mathbb{S}^1 \times [0,1]$ for the cylinder, and choose a disk $D$ in $M$ containing both $D_1, D_2$. Then $P = D \setminus (\operatorname{int}D_1 \cup \operatorname{int}D_2)$ is the pair of pants defined above. Attaching the two ends of the cylinder to $\partial D_1, \partial D_2$ in $M$ can be done in two stages: first attach $A$ to $\partial D_1 \sqcup \partial D_2$ of $P$ to obtain $A'$, then attach $A'$ to the boundary $\partial D$ of $M \setminus \operatorname{int}D$. Intuitively, attaching the cylinder without a twist to $\partial D_1, \partial D_2$ of the pair of pants gives $\mathbb{T}^2 \setminus \operatorname{int}D$.

We now state the classification theorem without proof.

**Theorem A.7.61** (Classification theorem for compact surfaces)**.** *Let $M$ be a connected compact two-dimensional surface with $b$ boundary components.*

*(1) If $M$ is orientable, there is a unique $g \in \mathbb{N}$ such that $M$ is homeomorphic to the surface obtained by starting with a sphere, adding $g$ handles, and removing the interiors of $b$ pairwise disjoint disks lying in its interior. Explicitly,*

$$M \cong (\mathbb{S}^2\#\underbrace{\mathbb{T}^2\#\mathbb{T}^2\#\cdots\#\mathbb{T}^2}_{g\text{ copies}}) \setminus \left(\bigsqcup_{i=1}^{b} \operatorname{int}D_i\right).$$

*Here $D_i \cong \mathbb{D}^2$.* The number $g$ is the *genus*, or *number of handles*, of $M$.

*(2) If $M$ is nonorientable, there is a unique positive integer $k$ such that $M$ is homeomorphic to the surface obtained by starting with a sphere, adding $k$ crosscaps, and removing the interiors of $b$ pairwise disjoint disks lying in its interior. Explicitly,*

$$M \cong (\underbrace{\mathbb{R}\mathbb{P}^2\#\mathbb{R}\mathbb{P}^2\#\cdots\#\mathbb{R}\mathbb{P}^2}_{k\text{ copies}}) \setminus \left(\bigsqcup_{i=1}^{b} \operatorname{int}D_i\right).$$

*Here $D_i \cong \mathbb{D}^2$.* The number $k$ is the *nonorientable genus*, or *number of crosscaps*, of $M$.

**Remark.** Removing the interior of a disk from a sphere leaves a space homeomorphic to a disk. Thus taking a connected sum with $\mathbb{S}^2$ does not change a surface: one removes a disk interior and then attaches a disk. Although (2) says "starting with a sphere," its formula therefore need not begin with $\mathbb{S}^2\#\mathbb{R}\mathbb{P}^2\#\cdots$. In (1), $\mathbb{S}^2\#\cdots$ is written to include the case $g=0$.

Thus, intuitively, the topological type is determined by the number $b$ of boundary "openings" and by the number of handles or crosscaps. For a compact two-dimensional surface without boundary, the everyday number of "holes" is its genus.

### Euler characteristic

The Euler characteristic is a topological invariant that describes a geometric property of an object. We develop this in more detail.

**Definition A.7.62.** Let $M$ be a compact two-dimensional surface and choose a finite triangulation of $M$. Let $V,E,F$ be its numbers of distinct vertices, edges, and triangles, respectively. Define the *Euler characteristic* of $M$ by $\chi(M) := V-E+F$.

**Proposition A.7.63.** *The Euler characteristic is independent of the finite triangulation and is a topological invariant of compact two-dimensional surfaces.*

**Proof.** First consider how $V-E+F$ changes under a local subdivision. Adding a vertex in the relative interior of a triangle and joining it to the three old vertices gives $\Delta V=1$, $\Delta E=3$, and $\Delta F=2$. Thus the total change is $\Delta\chi(M)=\Delta(V-E+F)=1-3+2=0$. If a vertex is added in the interior of an edge, the old edge is divided into two. For a boundary edge its one incident triangle is divided into two, so $\Delta V=1$, $\Delta E=2$, and $\Delta F=1$, giving $\Delta\chi(M)=0$. For an interior edge its two incident triangles are each divided into two, so $\Delta V=1$, $\Delta E=3$, and $\Delta F=2$, giving $\Delta\chi(M)=0$. Thus subdivision does not change $V-E+F$.

In fact, any two finite triangulations of a compact two-dimensional surface have a finite common subdivision after suitable adjustments that preserve their respective combinatorial types. We omit the full proof, whose idea is as follows. Draw both families of edges on $M$. After a small adjustment that does not change either combinatorial type, no vertex lies in the interior of an edge of the other triangulation, and noncoincident edges from the two systems meet transversely in only finitely many points. Retain any common edge segment and split it at its endpoints. Taking all old vertices and all intersections as new vertices, split every old edge there, then subdivide each resulting polygonal region into triangles. This yields a common subdivision. Hence all finite triangulations give the same value of $V-E+F$.

Finally, if $f:M\to N$ is a homeomorphism, it carries a finite triangulation of $M$ to one of $N$ and preserves the numbers of vertices, edges, and triangles. Therefore $\chi(M)=\chi(N)$. □

**Theorem A.7.64.** *The Euler characteristics of some basic surfaces are* $\chi(\mathbb{D}^2)=1$*,* $\chi(\mathbb{S}^2)=2$*,* $\chi(\mathbb{T}^2)=0$*, and* $\chi(\mathbb{R}\mathbb{P}^2)=1$.

**Proof.**

(1) For $\mathbb{D}^2$, choose three points on the boundary and join them along the boundary. The resulting single triangle triangulates the disk, so $\chi(\mathbb{D}^2)=3-3+1=1$.

(2) For $\mathbb{S}^2$, choose four points. Every pair determines an edge, giving six edges, and every three determine a triangular face. The four triangles form a triangulation, so $\chi(\mathbb{S}^2)=4-6+4=2$.

(3) For the torus, example A.7.36 gives $\mathbb{T}^2=[0,3]^2/\sim$, with $(0,y)\sim(3,y)$ and $(x,0)\sim(x,3)$. This is the same construction as the one using $[0,1]^2$, with the square scaled by a factor of three. Divide $[0,3]^2$ into a $3\times3$ grid and draw a diagonal in every small square. More precisely, for $i,j=0,1,2$, take

$$T_{ij}^{(1)}=[(i,j),(i+1,j),(i+1,j+1)],\qquad T_{ij}^{(2)}=[(i,j),(i,j+1),(i+1,j+1)],$$

where $[A,B,C]$ denotes the triangle with vertices $A,B,C$.

Now take the quotient. The original 18 triangles remain distinct, so $F=18$. The square has 21 interior edges and 12 boundary edges; the boundary edges pair up under the quotient map, so $E=21+12/2=27$. There are four interior vertices; the four corner vertices become one, and the eight other boundary vertices pair up to give four. Thus $V=4+1+4=9$, and $\chi(\mathbb{T}^2)=9-27+18=0$.

(4) The explicit triangulation of the Möbius band in example A.7.50 has six vertices and six triangles after taking the quotient identifying the two sides of the rectangle. Its six boundary edges account for one edge on each triangle; if $E_{\text{in}}$ is the number of interior edges, each triangle has three edges and each interior edge belongs to two triangles. Thus $3\times 6=2E_{\text{in}}+6$, so $E_{\text{in}}=6$ and $E=12$. Hence its Euler characteristic is $6-12+6=0$. By lemma A.7.60, capping its sole boundary circle with a disk gives $\mathbb{RP}^2$; the first part of the next theorem adds 1, and therefore $\chi(\mathbb{RP}^2)=0+1=1$.

□

**Theorem A.7.65.** *Let $M$ be a compact two-dimensional surface.*

*(1) If $D\subseteq M^\circ$ is a disk, then $\chi(M\setminus \operatorname{int}D)=\chi(M)-1$. Conversely, if $C$ is a connected component of $\partial M$, choose a disk $\mathbb{D}^2$ and a homeomorphism $f:\partial\mathbb{D}^2\to C$. After gluing along $C$, $\chi(M\sqcup_f \mathbb{D}^2)=\chi(M)+1$.*

*(2) If $N$ is also a compact two-dimensional surface, then the connected sum satisfies $\chi(M\#N)=\chi(M)+\chi(N)-2$.*

*(3) Adding a handle to $M$ decreases its Euler characteristic by 2; adding a crosscap to $M$ decreases it by 1.*

**Proof.** First consider the count when gluing along boundary circles. Subdivide the two boundary circles to be glued into the same number of edges, so that the gluing map matches vertices and edges. If each circle has $n$ vertices and $n$ edges, gluing identifies exactly $n$ pairs of vertices and $n$ pairs of edges and does not change the number of triangles. Thus $V$ and $E$ both decrease by $n$, and $V-E+F$ is unchanged.

(1) Since $M$ is obtained by gluing $D$ to $M\setminus\operatorname{int}D$ along their boundary,

$$\chi(M)=\chi(M\setminus\operatorname{int}D)+\chi(D)=\chi(M\setminus\operatorname{int}D)+1.$$

The converse follows similarly.

(2) A connected sum deletes the interior of a disk from each of $M,N$. By (1), deleting a disk interior decreases the Euler characteristic of a surface by 1, so

$$\chi(M\#N)=(\chi(M)-1)+(\chi(N)-1)=\chi(M)+\chi(N)-2.$$

(3) We have proved that adding a handle is equivalent to taking the connected sum with $\mathbb{T}^2$, so $\chi(M\#\mathbb{T}^2)=\chi(M)+0-2=\chi(M)-2$. Adding a crosscap is equivalent to taking the connected sum with $\mathbb{RP}^2$, so $\chi(M\#\mathbb{RP}^2)=\chi(M)+1-2=\chi(M)-1$.

□

**Theorem A.7.66** (Euler's formula)**.** *Let $M$ be a connected compact two-dimensional surface with $b$ boundary components.*

*(1) If $M$ is orientable of genus $g$, then $\chi(M)=2-2g-b$.*

*(2) If $M$ is nonorientable of nonorientable genus $k$, then $\chi(M)=2-k-b$.*

**Proof.** Apply the classification theorem for compact surfaces, adding handles or crosscaps successively and deleting the interiors of disks. □

## Exercises

**Exercise A.135.** Prove proposition A.7.54.

**Exercise A.136.** Find the Euler characteristic and genus of each of the following: a cylinder, a pair of pants, a cylinder # a pair of pants, and a pair of pants # a torus.

**Exercise A.137.** Define the Klein bottle by

$$K := ([0,1]\times[0,1])/((0,y)\sim(1,y),\ (x,0)\sim(1-x,1)).$$

Prove that it is nonorientable and has Euler characteristic 0. Find its nonorientable genus.

**Exercise A.138.** Prove $\mathbb{RP}^2\#\mathbb{T}^2 \cong \mathbb{RP}^2\#\mathbb{RP}^2\#\mathbb{RP}^2$.

**Exercise A.139.** A nondegenerate conic in $\mathbb{CP}^2$ can be regarded as a two-dimensional real surface that is orientable, compact, and connected. Find its Euler characteristic and genus. *Hint.* $\mathbb{CP}^1 \cong \mathbb{S}^2$.

# A.8 Derivatives of Real-Variable Functions

This section briefly introduces derivatives of functions. Readers wishing to learn more may consult books on the subject.

### Derivatives of one-variable functions

One-variable derivatives are introduced in high school, so our treatment here is brief. We do not carefully prove the analytic details underlying continuity and related concepts. The definitions of sequence limits, continuity, and function limits are inherited directly from the topology of $\mathbb{R}$ with its usual topology. An infinite series $\sum_{n=1}^{\infty} a_n$ is defined as the limit, as $n\to\infty$, of the sequence $(\sum_{m=1}^{n} a_m)_n$ of partial sums. This supplies the definition for completeness; readers need not pursue it further here.

**Definition A.8.1.** Suppose $y=f(x)$ is defined near $x_0$. If the limit

$$\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} = \lim_{\Delta x\to 0}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$$

exists, then $f$ is *differentiable at* $x_0$, and the limit is its *derivative* $f'(x_0)$, also written $y'\big|_{x=x_0}$, $\frac{\mathrm{d}y}{\mathrm{d}x}\big|_{x=x_0}$, or $\frac{\mathrm{d}f}{\mathrm{d}x}(x_0)$.

If $y=f(x)$ is defined and differentiable at every point of an interval $I$, pairing each value $x_0$ with its corresponding derivative gives a new function $x_0\mapsto f'(x_0)$, the *derivative function* of $f(x)$, also denoted by $y'$, $f'(x)$, $\frac{\mathrm{d}y}{\mathrm{d}x}$, or $\frac{\mathrm{d}f}{\mathrm{d}x}$. The function $f(x)$ is an *antiderivative* of $f'(x)$.

**Theorem A.8.2.** *The rules for differentiating functions, with x as the independent variable, are*

*(1)* $(f(x)\pm g(x))' = f'(x)\pm g'(x)$;
*(2)* $[f(x)\cdot g(x)]' = f'(x)\cdot g(x)+f(x)\cdot g'(x)$;
*(3)* $\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)\cdot g(x)-f(x)\cdot g'(x)}{g^2(x)}$;
*(4)* $[f(g(x))]' = f'(g(x))\cdot g'(x)$.

**Theorem A.8.3.** *The following are standard derivatives, with $x$ as the independent variable and $a$ a constant:*

*(1)* $(c)'=0$*, where $c$ is constant;*
*(2)* $(x^a)'=ax^{a-1}$*. Thus the derivative defined by limits over the real field agrees, on polynomials, with the formal derivative defined earlier;*
*(3)* $(\mathrm{e}^x)'=\mathrm{e}^x$*,* $(a^x)'=a^x\ln a$*;*
*(4)* $(\ln x)'=\frac{1}{x}$*,* $(\log_a x)'=\frac{1}{x\ln a}$*;*
*(5)* $(\sin x)'=\cos x$*,* $(\cos x)'=-\sin x$*, and* $(\tan x)'=\left(\frac{\sin x}{\cos x}\right)'=\frac{1}{\cos^2 x}=\sec^2 x$*.*

**Theorem A.8.4.** *In rectangular coordinates, the tangent to the curve $y=f(x)$ at $(x_0,f(x_0))$ has slope $f'(x_0)$.*

These definitions and elementary facts are standard high-school material, so their proofs are omitted.

The derivative function leads in turn to higher derivatives.

**Definition A.8.5.** If $y=f(x)$ is differentiable on an interval and its derivative $f'(x)$ is also differentiable there, the *second derivative* is $(f'(x))'$, denoted by $y''$, $f''(x)$, $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$, or $\frac{\mathrm{d}^2f}{\mathrm{d}x^2}$. Continuing inductively, if $f^{(n-1)}(x)$ is differentiable on the interval, its derivative $[f^{(n-1)}(x)]'$ is the *nth derivative function*, denoted by $y^{(n)}$, $f^{(n)}(x)$,[28] $\frac{\mathrm{d}^ny}{\mathrm{d}x^n}$, or $\frac{\mathrm{d}^nf}{\mathrm{d}x^n}$. Its value at $x=x_0$ is the *nth derivative* of $f(x)$ at that point, denoted by $y^{(n)}\big|_{x=x_0}$, $f^{(n)}(x_0)$, $\frac{\mathrm{d}^ny}{\mathrm{d}x^n}\big|_{x=x_0}$, or $\frac{\mathrm{d}^nf}{\mathrm{d}x^n}(x_0)$. The derivative and derivative function in definition A.8.1 are also called the first derivative and the first derivative function, respectively. When no ambiguity arises, "derivative function" may be shortened to "derivative."

Rules for higher derivatives follow recursively from the first-derivative rules.

**Example A.8.6.** Twice applying the product rule gives

$$\begin{aligned}[f(x)\cdot g(x)]'' &= [f'(x)\cdot g(x)+f(x)\cdot g'(x)]' \\ &= [f''(x)\cdot g(x)+f'(x)\cdot g'(x)]+[f'(x)\cdot g'(x)+f(x)\cdot g''(x)] \\ &= f''(x)\cdot g(x)+2f'(x)\cdot g'(x)+f(x)\cdot g''(x).\end{aligned}$$

This resembles the binomial expansion of a square and leads, by induction, to the following formula; we omit the formal induction.

**Theorem A.8.7** (Leibniz's formula)**.** *The rule for the higher derivatives of a product is*

$$[f(x)g(x)]^{(n)}=\sum_{k=0}^{n}\mathrm{C}_n^k f^{(k)}(x)g^{(n-k)}(x).$$

Here $f^{(0)}(x)$ is the "zeroth derivative," meaning that no differentiation is performed, so it is the original function. The number $\mathrm{C}_n^k=\frac{n!}{k!(n-k)!}$ is the binomial coefficient.

Another useful formula is the Taylor series.

**Theorem A.8.8.** *Suppose that $f$ has derivatives of every order at $x=x_0$ and satisfies certain additional conditions. Then, on some open set containing $x_0$ (see the following remark for the conditions),*

$$\begin{aligned}f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n &= f(x_0)+f'(x_0)(x-x_0) \\ &+\frac{1}{2!}f''(x_0)(x-x_0)^2+\frac{1}{3!}f'''(x_0)(x-x_0)^3+\cdots.\end{aligned}$$

[28]One can use $n$ primes for an $n$th derivative, but when the order is large or denoted by a letter, a parenthesized superscript is used instead. For example, $f'''(x)$ and $f^{(3)}(x)$ both denote the third derivative function of $f(x)$. The notation $f^{(0)}(x)$ means no differentiation, namely $f(x)$ itself. Do not mistake this footnote's number, 28, for part of the derivative notation.

This expansion is the *Taylor series of $f$ at $x_0$*.

**Remark.** Equality between a function and its Taylor series requires additional conditions in analysis, which space does not permit us to develop here. One rigorous criterion extends the function to the complex domain and applies theorem A.9.13 in the next section. For this course, the functions used later will generally have Taylor expansions whenever they have derivatives of every order, although mathematically this requires a strong hypothesis.

Space also prevents us from proving this theorem. The function $f(x)$ and its Taylor series have the same derivatives of every order at $x_0$, so the Taylor series may be understood intuitively as a polynomial approximation to the function.

**Example A.8.9.** The Taylor series of several elementary functions are

$$\sin x = \frac{1}{1!}x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots,$$
$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots,$$
$$\mathrm{e}^x = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots.$$

### Derivatives of multivariable functions

One-variable derivatives extend to functions of several variables.

**Definition A.8.10.** For $I_i \subseteq \mathbb{R}$, $i = 1,2,\dots,n$, the Cartesian product gives a map

$$f : I_1 \times I_2 \times \cdots \times I_n \longrightarrow \mathbb{R},$$
$$(x_1,x_2,\dots,x_n) \longmapsto f(x_1,x_2,\dots,x_n).$$

This maps an ordered tuple of real numbers $(x_1,x_2,\dots,x_n)$ to a real number and is called an *$n$-variable real-valued function*. Limits of sequences, continuity, and limits of functions are inherited from the topology on $\mathbb{R}^n$. Suppose $f(x_1,x_2,\dots,x_n)$ is defined near $\boldsymbol{x}_0 = (x_{10},x_{20},\dots,x_{n0})$. Vary one variable $x_i$ while fixing $x_1,\dots,x_{i-1},x_{i+1},\dots,x_n$. Differentiating with respect to $x_i$ as for a one-variable function gives the *partial derivative* of $f$ with respect to $x_i$:

$$\left.\frac{\partial f}{\partial x_i}\right|_{\boldsymbol{x}=\boldsymbol{x}_0} := \lim_{\Delta x_i \to 0} \frac{f(x_{10},\dots,x_{i-1,0},x_{i0}+\Delta x_i,x_{i+1,0},\dots,x_{n0}) - f(x_{10},x_{20},\dots,x_{n0})}{\Delta x_i}.$$

The correspondence between each $\boldsymbol{x}_0$ in a region and the partial derivative there gives the *partial derivative function*, denoted by $\frac{\partial f}{\partial x_i}$, also called simply the partial derivative when no ambiguity arises.

As for one variable, higher partial derivatives can be defined. In $\frac{\partial}{\partial x_i}(\#)$, the part denoted by # in parentheses is to be differentiated with respect to $x_i$. This notation can emphasize that part, or accommodate an expression too long to place conveniently after $\partial$. The same convention applies to one-variable derivatives: for example, $\frac{\mathrm{d}^2}{\mathrm{d}x^2}(\ln^2 x)$ is the second derivative of $\ln^2 x$ with respect to $x$. We define

$$\frac{\partial^2 f}{\partial x_i^2} := \frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_i}\right), \quad \frac{\partial^n f}{\partial x_i^n} := \frac{\partial}{\partial x_i}\left(\frac{\partial^{n-1} f}{\partial x_i^{n-1}}\right).$$

Having several variables also allows mixed partial derivatives:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} := \frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_j}\right).$$

Combining higher partial derivatives and mixed partial derivatives gives, for example,

$$\frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k} = \frac{\partial}{\partial x_i}\left(\frac{\partial^2 f}{\partial x_j \partial x_k}\right), \quad \frac{\partial^5 f}{\partial x_i^2 \partial x_j \partial x_k^2} = \frac{\partial^2}{\partial x_i^2}\left[\frac{\partial}{\partial x_j}\left(\frac{\partial^2 f}{\partial x_k^2}\right)\right], \quad \frac{\partial^{(n+m)} f}{\partial x_i^n \partial x_j^m} = \frac{\partial^n}{\partial x_i^n}\left(\frac{\partial^m f}{\partial x_j^m}\right).$$

We write $\frac{\partial f}{\partial x_i}(\boldsymbol{x})$ for the value of the partial derivative at $\boldsymbol{x} = (x_1,\dots,x_n)$, and likewise for the other derivatives.

Roughly speaking, to find a partial derivative with respect to one variable, treat the other variables as constant parameters and differentiate as for a one-variable function.

**Example A.8.11.** For the three-variable function $f(x,y,z)=x^3z+\cos(xy)\sin z+2$,

(1) $\frac{\partial f}{\partial x}=3x^2z-y\sin(xy)\sin z$, $\frac{\partial f}{\partial y}=-x\sin(xy)\sin z$, and $\frac{\partial f}{\partial z}=x^3+\cos(xy)\cos z$;

(2)
$$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}[3x^2z-y\sin(xy)\sin z]=6xz-y^2\cos(xy)\sin z;$$

(3)
$$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial}{\partial x}[-x\sin(xy)\sin z]=-\sin(xy)\sin z-xy\cos(xy)\sin z;$$

(4)
$$\frac{\partial^2 f}{\partial y\partial x}=\frac{\partial}{\partial y}[3x^2z-y\sin(xy)\sin z]=-\sin(xy)\sin z-xy\cos(xy)\sin z;$$

(5)
$$\frac{\partial^3 f}{\partial z\partial y\partial x}=\frac{\partial}{\partial z}\left(\frac{\partial^2 f}{\partial y\partial x}\right)=\frac{\partial}{\partial z}[-\sin(xy)\sin z-xy\cos(xy)\sin z]=\cdots.$$

Parts (3) and (4) of the example suggest that the order of mixed partial derivatives can be interchanged, but this is not always true: $\frac{\partial^2 f}{\partial x_i\partial x_j}=\frac{\partial^2 f}{\partial x_j\partial x_i}$ need not hold. Using the topological terminology of section A.7.1, one useful sufficient condition is the following.

**Theorem A.8.12** (Schwarz's theorem)**.** *Let $f:U\to\mathbb{R}$, where $U\subseteq\mathbb{R}^n$ is open. If both mixed partials $\frac{\partial^2 f}{\partial x_i\partial x_j}$ and $\frac{\partial^2 f}{\partial x_j\partial x_i}$ exist and are continuous, then $\frac{\partial^2 f}{\partial x_i\partial x_j}=\frac{\partial^2 f}{\partial x_j\partial x_i}$.*

For every use in the main text, interchanging the order of partial derivatives is legitimate, so we do not pursue the subtleties further.

**Theorem A.8.13** (Chain rule)**.** *Let $z=f(u_1,u_2,\dots,u_m)$. Each $u_j$ is a function of $\boldsymbol{x}=(x_1,x_2,\dots,x_n)$, with $u_j=u_j(\boldsymbol{x})=u_j(x_1,\dots,x_n)$. Under suitable conditions,*
$$\frac{\partial z}{\partial x_i}=\sum_{j=1}^{m}\left[\frac{\partial f}{\partial u_j}(u_1(\boldsymbol{x}),\dots,u_m(\boldsymbol{x}))\right]\cdot\frac{\partial u_j}{\partial x_i}. \tag{A.1}$$

**Remark.** The conditions here are usually stated in terms of differentiability. Within the scope of this appendix, however, we can give the following stronger sufficient condition. Let $U\subseteq\mathbb{R}^n$ and $V\subseteq\mathbb{R}^m$ be open, with
$$\boldsymbol{u}:U\to V,\qquad \boldsymbol{x}\mapsto u(\boldsymbol{x})=(u_1(\boldsymbol{x}),\dots,u_m(\boldsymbol{x})),\quad f:V\to\mathbb{R}.$$
If every $u_j$ has continuous first partial derivatives on $U$, and $f$ has continuous first partial derivatives on $V$, then $z=f\circ\boldsymbol{u}$ has continuous first partial derivatives on $U$ and satisfies equation (A.1). These conditions are always met in the applications in the main text, although mathematically they are strong.

The proof is beyond the course requirements, but the formula can be understood through rates of change. In one variable, if $y=f(u)$ and $u=g(x)$, then $y=f(g(x))$ is a function of $x$. A small change $\Delta x$ produces $\Delta u=g(x+\Delta x)-g(x)$ and then $\Delta y=f(u+\Delta u)-f(u)$. Intuitively, $g$ scales the change in $x$ up or down to give $\Delta u$, and $f$ then converts $\Delta u$ to $\Delta y$. Thus the rate of change of $y$ with respect to $x$ follows the inner function and then the outer function: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}u}\Big|_{u=u(x)}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}$.

The multivariable chain rule follows the same idea, except that changes can pass through several intermediate variables simultaneously. The rate of change along one independent variable is the weighted sum of the rates along all these paths. For each original variable $u_j$, multiply the partial derivative with respect to $u_j$ by the partial derivative of $u_j$ with respect to $x_i$ to obtain the contribution along one path. Summing the contributions from all $u_j$ gives the final partial derivative.

**Example A.8.14.** Let $t = f(u,v,w) = u^2 + \mathrm{e}^v \sin w$, where $u = x^2 + y^2$, $v = \ln(xy)$, $w = xz$, and $x,y,z$ depend on $s$, so $x = x(s)$, $y = y(s)$, and $z = z(s)$. Then

$$\begin{aligned}\frac{\mathrm{d}t}{\mathrm{d}s} &= \frac{\partial f}{\partial u}\frac{\mathrm{d}u}{\mathrm{d}s} + \frac{\partial f}{\partial v}\frac{\mathrm{d}v}{\mathrm{d}s} + \frac{\partial f}{\partial w}\frac{\mathrm{d}w}{\mathrm{d}s} = \frac{\partial f}{\partial u}\left(\frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial u}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s} + \frac{\partial u}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}s}\right)\\ &\quad + \frac{\partial f}{\partial v}\left(\frac{\partial v}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial v}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s} + \frac{\partial v}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}s}\right) + \frac{\partial f}{\partial w}\left(\frac{\partial w}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial w}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s} + \frac{\partial w}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}s}\right)\\ &= 2u\left(2x\frac{\mathrm{d}x}{\mathrm{d}s} + 2y\frac{\mathrm{d}y}{\mathrm{d}s}\right) + \mathrm{e}^v \sin w\left(\frac{1}{x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}s}\right) + \mathrm{e}^v \cos w\left(z\frac{\mathrm{d}x}{\mathrm{d}s} + x\frac{\mathrm{d}z}{\mathrm{d}s}\right).\end{aligned}$$

**Applications of partial derivatives**

1. A function $F(x_1,\dots,x_n)$ is *homogeneous of degree k* if, for every $\lambda > 0$,

$$F(\lambda x_1,\dots,\lambda x_n) = \lambda^k F(x_1,\dots,x_n).$$

Homogeneous functions have the following important property.

**Theorem A.8.15** (Euler's theorem on homogeneous functions)**.** *Let $F(x_1,\dots,x_n)$ be homogeneous of degree k, and suppose all its first partial derivatives exist and are continuous. Then*

$$\sum_{i=1}^{n} x_i \frac{\partial F}{\partial x_i} = kF(x_1,\dots,x_n).$$

**Proof.** Since $F(x_1,\dots,x_n)$ is homogeneous of degree $k$, for every $\lambda > 0$ we have

$$F(\lambda x_1,\dots,\lambda x_n) = \lambda^k F(x_1,\dots,x_n).$$

Differentiate both sides with respect to $\lambda$, treating $x_1,\dots,x_n$ as constants, to obtain

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}F(\lambda x_1,\dots,\lambda x_n) = k\lambda^{k-1}F(x_1,\dots,x_n).$$

By the chain rule, the left-hand side expands as

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}F(\lambda x_1,\dots,\lambda x_n) = \sum_{i=1}^{n} x_i \frac{\partial F}{\partial x_i}(\lambda x_1,\dots,\lambda x_n).$$

Substituting $\lambda = 1$ gives

$$\sum_{i=1}^{n} x_i \frac{\partial F}{\partial x_i}(x_1,\dots,x_n) = kF(x_1,\dots,x_n).$$

□

2. An *implicit function* is a one-variable function $y = f(x)$ specified by an equation $F(x,y) = 0$ rather than explicitly. Under suitable conditions, differentiating both sides with respect to $x$ gives

$$\frac{\mathrm{d}F}{\mathrm{d}x} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y}.$$

Here $F_x = \frac{\partial F}{\partial x}$ and $F_y = \frac{\partial F}{\partial y}$.

**Example A.8.16.** The inverse-sine function $y = \arcsin x$ is the inverse of the sine function and is determined by $\sin y - x = 0$. Differentiating gives

$$\cos y \frac{\mathrm{d}y}{\mathrm{d}x} - 1 = 0 \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$

Similarly, $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$ and $(\arctan x)' = \frac{1}{1+x^2}$; their proofs are left as exercises.

This extends to several variables. For a map $\boldsymbol{f}: \mathbb{R}^n \to \mathbb{R}^m$ and $\boldsymbol{x} = (x_1,\dots,x_n)$, write $\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}),\dots,f_m(\boldsymbol{x}))$. The matrix $\left(\frac{\partial f_i}{\partial x_j}\Big|_{\boldsymbol{x}}\right)_{m\times n}$ of partial derivatives at the point $\boldsymbol{x}$, whose entry in row $i$ and column $j$ is the value of $\frac{\partial f_i}{\partial x_j}$ at $\boldsymbol{x}$, is the *Jacobian matrix* (named after Jacobi) of $\boldsymbol{f}$ at $\boldsymbol{x}$. It is denoted by $\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}$ or $\frac{\partial(f_1,\dots,f_m)}{\partial(x_1,\dots,x_n)}$. When $n = m$, its determinant is the *Jacobian determinant*. With this notation we can state the following result.

**Theorem A.8.17** (Implicit function theorem)**.** *Let $k \geq 1$ be an integer and let $\Omega \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open. Suppose each function $F_i : \Omega \to \mathbb{R}$, $(\boldsymbol{x},\boldsymbol{y}) \mapsto F_i(\boldsymbol{x},\boldsymbol{y})$, has continuous partial derivatives of all orders at most $k$ on $\Omega$. Write $\boldsymbol{F} = (F_1,\dots,F_m)$, a map $\Omega \to \mathbb{R}^m$. Let $(\boldsymbol{a},\boldsymbol{b}) \in \Omega$, where $\boldsymbol{a} \in \mathbb{R}^n$ and $\boldsymbol{b} \in \mathbb{R}^m$, and suppose*

$$\boldsymbol{F}(\boldsymbol{a},\boldsymbol{b}) = \boldsymbol{0}, \qquad \det\left(\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{y}}(\boldsymbol{a},\boldsymbol{b})\right) \neq 0.$$

*Then there are open sets $U \subseteq \mathbb{R}^n$ containing $\boldsymbol{a}$ and $V \subseteq \mathbb{R}^m$ containing $\boldsymbol{b}$, with $U \times V \subseteq \Omega$, such that $\boldsymbol{F}(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{0}$ uniquely determines a function $\boldsymbol{f} = (f_1,\dots,f_m) : U \to V$, with $f_i : U \to \mathbb{R}$. It satisfies $\boldsymbol{f}(\boldsymbol{a}) = \boldsymbol{b}$, and, for every $(\boldsymbol{x}_0,\boldsymbol{y}_0) \in U \times V$:*

*(1) $\boldsymbol{F}(\boldsymbol{x}_0,\boldsymbol{y}_0) = \boldsymbol{0} \Longleftrightarrow \boldsymbol{y}_0 = \boldsymbol{f}(\boldsymbol{x}_0) = (f_1(\boldsymbol{x}_0),\dots,f_m(\boldsymbol{x}_0))$;*
*(2) every $f_i$ has continuous derivatives of all orders at most $k$;*
*(3) the Jacobian matrix of first derivatives is*

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}(\boldsymbol{x}_0) = -\left[\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{y}}(\boldsymbol{x}_0,\boldsymbol{f}(\boldsymbol{x}_0))\right]^{-1}\left[\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{x}}(\boldsymbol{x}_0,\boldsymbol{f}(\boldsymbol{x}_0))\right].$$

The hypotheses in both applications above are almost always satisfied in the main text, although they are mathematically substantial conditions.

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## Exercises

**Exercise A.140.** Find $(x^x)'$.

**Exercise A.141.** For $y = e^{2x}\sin x$, find the first three derivatives with respect to $x$.

**Exercise A.142.** Find the derivatives of $\arcsin x$ and $\arctan x$.

**Exercise A.143.** Find the Taylor series at $x = 1$ of $\frac{1}{x}$, $\ln x$, and $x^\alpha$, where $\alpha \in \mathbb{R}$.

**Exercise A.144.** Let $f(x,y,z) = \frac{x^2}{z}\cos y + e^{xy}\ln(yz)$.
(1) Find the partial derivatives of $f$ with respect to $x,y,z$.
(2) Find $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial x \partial y}$, $\frac{\partial^2 f}{\partial y \partial x}$, and $\frac{\partial^3 f}{\partial z \partial x \partial y}$.
(3) Find $\frac{\partial^4 f}{\partial x \partial y^2 \partial z}$, $\frac{\partial^4 f}{\partial y^2 \partial z \partial x}$, and $\frac{\partial^4 f}{\partial z \partial x \partial y^2}$, and check whether they agree.

**Exercise A.145.** If $z = f(u,v)$, $u = x+y$, and $v = x-y$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

**Exercise A.146.** Let $t = f(u,v,w) = u^2 v + e^v \ln(uv)\sin w$, where $u = x^2+2y$, $v = \sqrt{z}$, $w = xyz$, and $x,y,z$ depend on $s$, so $x = x(s)$, $y = y(s)$, and $z = z(s)$. Find $\frac{dt}{ds}$.

**Exercise A.147.** Let $w = f(x,y,z)$ be a function on $\mathbb{R}^3$, homogeneous of degree $k$, whose partial derivatives all exist. Prove that its partial derivative functions $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$ are homogeneous of degree $k-1$.

**Exercise A.148.** For $\boldsymbol{x} = (x_1,x_2,x_3)$, define

$$F_1(x_1,x_2,x_3) = e^{x_1} + x_2 + x_3\cos x_1 - 2,$$
$$F_2(x_1,x_2,x_3) = 3x_1\ln x_3 + e^{x_2} + x_3^2 - 2,$$

and write $\boldsymbol{F}(\boldsymbol{x}) = (F_1(\boldsymbol{x}),F_2(\boldsymbol{x})) \in \mathbb{R}^2$. Use the natural domains of these functions.

(1) Compute the Jacobian matrix $\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{x}}$.
(2) Is there an open set $U$ containing $\boldsymbol{x} = (0,0,1)$ and a unique pair of functions $f, g$ such that, for every $\boldsymbol{x} \in U$, $\boldsymbol{F}(\boldsymbol{x}) = \boldsymbol{0} \Longleftrightarrow x_2 = f(x_1), x_3 = g(x_1)$? If so, find $f'(0), g'(0), f''(0), g''(0)$.
(3) Find a specific point $\boldsymbol{x}_0$ satisfying $\boldsymbol{F}(\boldsymbol{x}_0) = \boldsymbol{0}$ such that no such $U$ exists when the point $\boldsymbol{x}$ in (2) is replaced by $\boldsymbol{x}_0$.

# A.9 Complex Numbers and Analytic Functions

This section outlines the basics of complex numbers and some elementary concepts concerning complex-variable functions that will be used later. Here a complex-variable function means a function whose independent and dependent variables are complex (see definition A.9.4). The expression refers to these functions themselves, rather than to the university course in complex analysis. That course generally studies their properties systematically, including complex integration and residue theory. This section does not attempt to cover all the basic content of that course; it introduces only some elementary concepts concerning complex-variable functions.

## A.9.1 Complex Numbers and Elementary Complex Functions

### Complex numbers

We begin by briefly recalling the definition familiar from high school.

**Definition A.9.1.** Let the *imaginary unit* i satisfy $\mathrm{i}^2 = -1$. The set of *complex numbers* is

$$\mathbb{C} := \{z = x + \mathrm{i}y \mid x, y \in \mathbb{R}\}.$$

For $z = x + \mathrm{i}y$, its *real part* and *imaginary part* are $\mathrm{Re}(z) := x$ and $\mathrm{Im}(z) := y$. If $y = 0$, then $z$ is real; if $y \neq 0$, it is *imaginary*; and if $x = 0$ and $y \neq 0$, it is *purely imaginary*. Thus every complex number is either real or imaginary in this sense.

Treating i as a number subject to $\mathrm{i}^2 = -1$ gives the four arithmetic operations. For $a, b, c, d \in \mathbb{R}$,

$$(a + \mathrm{i}b) \pm (c + \mathrm{i}d) = (a \pm c) + \mathrm{i}(b \pm d),$$
$$(a + \mathrm{i}b) \cdot (c + \mathrm{i}d) = (ac + \mathrm{i}bc + a \cdot \mathrm{i}d + \mathrm{i}b \cdot \mathrm{i}d) = (ac - bd) + \mathrm{i}(bc + ad),$$
$$\frac{1}{c + \mathrm{i}d} = \frac{c - \mathrm{i}d}{(c + \mathrm{i}d)(c - \mathrm{i}d)} = \frac{c - \mathrm{i}d}{c^2 + d^2} = \frac{c}{c^2 + d^2} + \mathrm{i}\frac{-d}{c^2 + d^2},$$
$$\frac{a + \mathrm{i}b}{c + \mathrm{i}d} = (a + \mathrm{i}b) \cdot \frac{1}{c + \mathrm{i}d} \qquad (c + \mathrm{i}d \neq 0).$$

We shall also use the following constructions.

**Definition A.9.2.** Let $z = x + \mathrm{i}y$, with $x, y \in \mathbb{R}$.
(1) Its *complex conjugate* is $x - \mathrm{i}y$, denoted by $\bar{z}$ or $z^*$.
(2) In a Cartesian plane, let the horizontal coordinate represent the real part and the vertical coordinate the imaginary part. Thus the point $(x, y)$ represents $z = x + \mathrm{i}y$. The plane is then called the *complex plane*; the $x$-axis and $y$-axis are respectively the *real axis* and the *imaginary axis*. They are perpendicular and intersect at the *origin*.
  (a) The distance from a point of the complex plane to the origin is the *modulus* of the complex number $z$ represented by that point, denoted by $|z|$. Clearly $|z| = \sqrt{x^2 + y^2}$.
  (b) An angle $\theta$ through which the positive real axis may be rotated counterclockwise to the ray through $z$ is an *argument* of $z$, denoted by $\theta = \arg z$. Thus $\theta + 2\pi n$ is also an argument whenever $n \in \mathbb{Z}$, so $\arg z$ has infinitely many values. The argument of 0 is undetermined, or undefined. For $z \neq 0$, the unique argument in $(-\pi, \pi]$ is the *principal argument* $\mathrm{Arg} z$. The interval $[0, 2\pi)$ is more commonly used in physics. When $x \neq 0$, one has $\tan(\arg z) = \frac{y}{x}$.

(c) If $|z| = r$ and $\arg z = \theta$, then $z$ may also be written as $r\mathrm{e}^{\mathrm{i}\theta}$, where

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta. \tag{A.2}$$

This is *Euler's formula*. Thus $r\mathrm{e}^{\mathrm{i}\theta} = r\cos\theta + \mathrm{i}r\sin\theta$; this representation is the *exponential form* of $z$.

(d) Adjoining a point at infinity, denoted by $\infty$, to the complex plane gives the *extended complex plane* $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. The point at infinity is approached by ordinary complex numbers moving away from the origin in any direction. Its modulus is infinite, and its argument is undetermined, or undefined. Thus, for example, $-\infty = \infty$ because both have infinite modulus, and $a + \infty = a - \infty = \infty$.

Give $\mathbb{C}$ its usual topology. A set $U \subseteq \hat{\mathbb{C}}$ is declared open exactly when $U \cap \mathbb{C}$ is open in $\mathbb{C}$ and, if $\infty \in U$, there is an $R > 0$ such that $\{z \mid |z| > R\} \subseteq U$.

**Remark.** The real Taylor series

$$\sin x = \frac{1}{1!}x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots,$$
$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots,$$
$$\mathrm{e}^x = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots$$

give, on extending their domains to complex numbers,

$$\cos\theta + \mathrm{i}\sin\theta = 1 + \frac{1}{1!}(\mathrm{i}\theta) + \frac{1}{2!}(\mathrm{i}\theta)^2 + \frac{1}{3!}(\mathrm{i}\theta)^3 + \cdots = \mathrm{e}^{\mathrm{i}\theta}.$$

This was Euler's reason for defining the exponential form of a complex number. The formulas originally hold only over the real numbers, however, so the notation $\mathrm{e}^{\mathrm{i}\theta}$ should be regarded as a definition rather than a deduction, albeit one highly consistent with the existing real number system.

Finally, we also give a topological model of the extended complex plane.

**Theorem A.9.3.** *The extended complex plane is homeomorphic to the sphere:* $\hat{\mathbb{C}} \cong \mathbb{S}^2$.

**Proof.** Let $N$ and $S$ be the north and south poles of the unit sphere, and let $\pi$ be the tangent plane at $S$. For every $P \in \mathbb{S}^2 \setminus \{N\}$, the line $NP$ meets $\pi$ in a unique point $P'$. Identifying $\pi$ with the complex plane gives a bijection $\mathbb{S}^2 \setminus \{N\} \to \mathbb{C}$. As $P \to N$, the point $P'$ tends to the single point at infinity. Geometrically, one can already see intuitively that this projection and its inverse are continuous, so the extended correspondence is a homeomorphism.

Notice that $\hat{\mathbb{C}}$ has only one point at infinity, whereas the projective plane used in the main text adds a point at infinity to every line. □

**Remark.** Here is a more direct, rigorous verification of continuity, although the reader should first understand the geometric intuition above. Write $\mathbb{S}^2 = \{(x,y,u) \in \mathbb{R}^3 \mid x^2 + y^2 + u^2 = 1\}$, with north pole $N = (0,0,1)$ and south pole $S = (0,0,-1)$. Identify the tangent plane at $S$, $\pi = \{(X,Y,-1) \mid X, Y \in \mathbb{R}\}$, with $\mathbb{C}$ by associating $(X,Y,-1)$ with $X + \mathrm{i}Y$. The construction gives the map $f : \mathbb{S}^2 \longrightarrow \hat{\mathbb{C}}$ defined by

$$f(x,y,u) = \frac{2(x+\mathrm{i}y)}{1-u} \quad ((x,y,u) \neq N), \quad f(N) = \infty.$$

It is easy to verify that its inverse $g : \hat{\mathbb{C}} \longrightarrow \mathbb{S}^2$ is

$$g(w) = \left( \frac{4\operatorname{Re} w}{|w|^2+4}, \frac{4\operatorname{Im} w}{|w|^2+4}, \frac{|w|^2-4}{|w|^2+4} \right) \quad (w \in \mathbb{C}), \quad g(\infty) = N.$$

The existence of this inverse shows that $f$ is bijective.

We first verify that $f$ is continuous. At $P = (x,y,u) \neq N$, $1 - u \neq 0$, so on $\mathbb{S}^2 \setminus \{N\}$ the coordinates of $f$ are obtained by the four arithmetic operations and are continuous. We omit the proof that these operations are continuous. It remains to check $N$. For $P = (x,y,u) \neq N$, the identity $x^2 + y^2 + u^2 = 1$

gives, in the metric on $\mathbb{R}^3$,

$$\|P-N\|^2=2(1-u), \quad |f(P)|^2=\frac{4(1+u)}{1-u}=\frac{16}{\|P-N\|^2}-4.$$

Every open set $V$ containing $\infty$ contains, by definition, a basic open set $U_R=\{\infty\}\cup\{z\,|\,|z|>R\}$ for some $R>0$. Put $\delta=\frac{4}{\sqrt{R^2+4}}$. If $0<\|P-N\|<\delta$, then $|f(P)|^2>\frac{16}{\delta^2}-4=R^2$, so $|f(P)|>R$. Also $f(N)=\infty$, and hence $f(\mathrm{B}(N,\delta)\cap\mathbb{S}^2)\subseteq U_R\subseteq V$. This proves continuity at $N$.[29] Thus $f$ is continuous on $\mathbb{S}^2$.

We next verify that $g$ is continuous. At a finite point $w\in\mathbb{C}$, the denominators of its three coordinate functions are positive. These functions are combinations of the four arithmetic operations, so they are continuous. We need only check $\infty$. Let $V$ be an open set in $\mathbb{S}^2$ containing $N$. Since $\mathbb{S}^2$ has the subspace topology from $\mathbb{R}^3$, there is a $\delta>0$ such that $\mathrm{B}(N,\delta)\cap\mathbb{S}^2\subseteq V$. For $w\in\mathbb{C}$, write $u=\frac{|w|^2-4}{|w|^2+4}$ for the third coordinate of $g(w)$. Since $g(w)\in\mathbb{S}^2$, calculation gives

$$\|g(w)-N\|=\sqrt{2(1-u)}=\frac{4}{\sqrt{|w|^2+4}}.$$

Choose $R>0$ sufficiently large that $\frac{4}{\sqrt{R^2+4}}<\delta$. If $|w|>R$, then $\|g(w)-N\|<\frac{4}{\sqrt{R^2+4}}<\delta$, and hence $g(w)\in V$. Also $g(\infty)=N\in V$, so $g(U_R)\subseteq V$. This proves continuity at $\infty$, and therefore on all of $\hat{\mathbb{C}}$.

**Elementary complex functions**

Limits and continuity in $\mathbb{C}$ are those inherited from its usual Euclidean topology. We now define the derivative of a complex-variable function.

**Definition A.9.4.** Let $D\subseteq\mathbb{C}$ be open and let $f:D\to\mathbb{C}$, $w=f(z)$. Its independent and dependent variables are complex; such a function is called a *complex-variable function*. Correspondingly, a function whose independent variable is real is a *real-variable function*.

At a point $z_0\in D$, consider the limit

$$\lim_{\Delta z\to 0}\frac{\Delta w}{\Delta z}=\lim_{\Delta z\to 0}\frac{f(z_0+\Delta z)-f(z_0)}{\Delta z}. \tag{A.3}$$

If this limit exists, the function is *differentiable at* $z_0$. The limit is its *derivative* at $z_0$, with the same notation as for real functions, for example $f'(z_0)$ or $\left.\frac{\mathrm{d}f}{\mathrm{d}z}\right|_{z=z_0}$. As in the real case, one can define the *derivative function* $f'(z)$.

A function differentiable at every point of an open set is *holomorphic*, or *analytic*, on that set. It is analytic at $z_0$ if there is an open set $U$ in its domain containing $z_0$ on which it is analytic.[30] As with real functions, higher derivatives such as $f''(z)$ can also be defined.

On the extended complex plane, if no value at infinity is specified, one may define $f(\infty)=\lim_{z\to\infty}f(z)$ whenever this limit exists. If $f$ is defined on an open set containing $\infty$ and $f(\infty)\in\mathbb{C}$, then $f$ is *analytic at infinity* when

$$g(z)=\begin{cases} f(1/z), & z\neq 0,\\ f(\infty), & z=0\end{cases}$$

is analytic at 0.

The definition resembles that for real functions, but the complex difference quotient must have the same limit along every route to 0. For example, the limit with the real part held at zero along the imaginary axis must agree with the limit with the imaginary part held at zero along the real axis. With

[29] The open balls $\mathrm{B}(x,\delta)$ form a basis for the topology of $\mathbb{R}^3$. The subspace topology on $\mathbb{S}^2$ therefore has a basis consisting of sets $\mathrm{B}(x,\delta)\cap\mathbb{S}^2$. At infinity in $\hat{\mathbb{C}}$, the basic open sets are the $U_R$. Thus we are applying the criterion in proposition A.7.13. The verification for $g$ below is similar: for each open set $V\subseteq\mathbb{S}^2$ containing $N$, it suffices to find $R>0$ such that $g(U_R)\subseteq V$.

[30] Complex analysis textbooks generally define analytic and holomorphic functions in different ways, but the two notions are equivalent, so we do not distinguish them in this book.

$f(z)=|z|$ at $z=1$, letting $\Delta z$ tend to zero along the real axis gives

$$\lim_{\Delta z\to 0}\frac{|1+\Delta z|-1}{\Delta z}=\lim_{x\to 0}\frac{(1+x)-1}{x}=1,$$

whereas letting $\Delta z$ tend to zero along the imaginary axis gives

$$\lim_{\Delta z\to 0}\frac{|1+\Delta z|-1}{\Delta z}=\lim_{y\to 0}\frac{\sqrt{1+y^2}-1}{\mathrm{i}y}=0.$$

Thus complex differentiability is much more restrictive than real differentiability.

We shall discuss criteria for complex differentiability and analyticity later. First we define some elementary complex functions; their definitions closely resemble those over the real numbers.

**Definition A.9.5.** For $z=x+\mathrm{i}y$, with $x,y\in\mathbb{R}$, define:

(a) the *power function*: for a positive integer $n$, $z^n=\underbrace{z\cdot z\cdots z}_{n\text{ copies of }z}$; for a negative integer $n$, $z^n=\frac{1}{z^{-n}}$ when $z\neq 0$; and $z^0=1$ when $z\neq 0$;

(b) the *exponential function*

$$\mathrm{e}^z=\mathrm{e}^{x+\mathrm{i}y}:=\mathrm{e}^x\cdot\mathrm{e}^{\mathrm{i}y}=\mathrm{e}^x(\cos y+\mathrm{i}\sin y),$$

also denoted by $\exp z$;

(c) the *trigonometric functions*: the *sine function* $\sin z:=\frac{\mathrm{e}^{\mathrm{i}z}-\mathrm{e}^{-\mathrm{i}z}}{2\mathrm{i}}$, the *cosine function* $\cos z:=\frac{\mathrm{e}^{\mathrm{i}z}+\mathrm{e}^{-\mathrm{i}z}}{2}$, and the *tangent function* $\tan z:=\frac{\sin z}{\cos z}$ when $\cos z\neq 0$.

These three kinds of elementary functions extend their real counterparts coherently: the exponential function extends by Euler's formula, the trigonometric functions are defined through the complex exponential, and the tangent remains the ratio of the sine to the cosine. Familiar identities and differentiation rules remain valid, including $(z^n)'=nz^{n-1}$, $\sin^2 z+\cos^2 z=1$, and $(\cos z)'=-\sin z$.

**Definition A.9.6.** For $z\in\mathbb{C}$, the *complex logarithm* $\ln z$ is the inverse relation of the exponential function: every $w$ satisfying $\mathrm{e}^w=z$ is a value of $\ln z$.

**Theorem A.9.7.** *The complex logarithm* $\ln z$ *is defined when* $z\neq 0$. *If* $z=r\mathrm{e}^{\mathrm{i}\theta}$, *where* $r>0$ *and* $\theta\in\mathbb{R}$, *then* $\ln z=\ln|z|+\mathrm{i}\arg z$.

**Proof.** Let $u,v$ be the real and imaginary parts of $w$, respectively. From $\mathrm{e}^w=\mathrm{e}^u\cdot\mathrm{e}^{\mathrm{i}v}=r\mathrm{e}^{\mathrm{i}\theta}$ we obtain $u=\ln r$ and $v=\theta+2n\pi$. □

The function $\ln z$ has infinitely many values because $\arg z$ is multivalued; such a function is called a *multivalued function*. In general, discussing a multivalued function requires specifying a *single-valued branch*; we do not discuss this further here. In later applications the logarithm gives angle values, and angles differing by $2\pi$ represent the same angle, in agreement with the preceding result. Its multivaluedness therefore does not affect statements within the scope of those applications.

**Definition A.9.8.** The *square-root function* $\sqrt{z}$ is also multivalued. It is the inverse relation of the power function $z^2:=z\cdot z$: every $w$ satisfying $w^2=z$ is a value of $\sqrt{z}$.

**Theorem A.9.9.** *Let* $z=r\mathrm{e}^{\mathrm{i}\theta}$, *where* $r>0$ *and* $\theta\in\mathbb{R}$. *Then* $\sqrt{z}=\sqrt{r}\mathrm{e}^{\mathrm{i}\theta/2},\sqrt{r}\mathrm{e}^{\mathrm{i}(\theta/2+\pi)}$ *has two values.*

**Proof.** Let $w=\rho\mathrm{e}^{\mathrm{i}\phi}$, where $\rho>0$ and $\phi\in\mathbb{R}$. Directly from the definition of the complex power function, $w^2=\rho^2\mathrm{e}^{2\mathrm{i}\phi}$. Equality of moduli requires $\rho=\sqrt{r}$. Since arguments differing by $2\pi$ are equivalent, equality of arguments requires $2\phi=\theta+2n\pi$, where $n\in\mathbb{Z}$. This gives the two arguments $\phi=\frac{\theta}{2}$ when $n=0$ and $\phi=\frac{\theta}{2}+\pi$ when $n=1$; all other integer values of $n$ are equivalent to one of these two. □

**Definition A.9.10.** The complex *inverse trigonometric functions* are also multivalued, and are defined as the inverse relations of the corresponding trigonometric functions. For a given $z$, every $w$ satisfying $\sin w = z$ is a value of the *arcsine function* $\arcsin z$; every $w$ satisfying $\cos w = z$ is a value of the *arccosine function* $\arccos z$; and every $w$ satisfying $\tan w = z$ is a value of the *arctangent function* $\arctan z$.

**Theorem A.9.11.** *The inverse trigonometric functions satisfy*

$$\arcsin z = \frac{1}{\mathrm{i}} \ln\left(\mathrm{i}z + \sqrt{1-z^2}\right), \quad \arccos z = \frac{1}{\mathrm{i}} \ln\left(z + \sqrt{z^2-1}\right), \quad \arctan z = \frac{1}{2\mathrm{i}} \ln \frac{1+\mathrm{i}z}{1-\mathrm{i}z}.$$

**Proof.** Solve the defining formulas of the trigonometric functions for $w$. The calculations are left to the reader. □

The inverse trigonometric functions are also multivalued. In the relevant geometric applications, values of arcsin and arccos differing by $2\pi$, and values of arctan differing by $\pi$, represent the same angle in the corresponding sense. However, different values of arcsin or arccos need not differ by a whole number of periods. We must therefore choose a single value consistent with the particular geometric configuration; this is what is meant by choosing a *single-valued branch*. This book does not develop the general theory of single-valued branches.

On domains where they are defined, the preceding elementary functions are analytic. For a multivalued function one must choose a branch and exclude an appropriate branch cut, or work on a Riemann surface. We do not develop these matters further, because in the applications in this course the formulas for real functions can be used directly. The differentiation rules likewise carry over directly; for example, $(\ln z)' = \frac{1}{z}$ and $(\cos z)' = -\sin z$.

Multivariable complex functions can likewise be defined as in the real case. Within the scope of this book, combining the definitions for one complex variable with the rules for partial derivatives gives rules entirely analogous to those above, so we do not elaborate. Continuity and limits in $\mathbb{C}^n$ are inherited from the corresponding topological space. For example, if $f(z_1, z_2) = z_2 z_1^2 + \mathrm{i}\sin z_2$, then $\frac{\partial f}{\partial z_1} = 2z_2 z_1$ and $\frac{\partial f}{\partial z_2} = z_1^2 + \mathrm{i}\cos z_2$.

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## Exercises

**Exercise A.149.** Using the definition of the exponential form of a complex number, prove:
(1) $z_1 \cdot z_2 = |z_1||z_2| \mathrm{e}^{\mathrm{i}(\arg z_1 + \arg z_2)}$;
(2) $\dfrac{z_1}{z_2} = \dfrac{|z_1|}{|z_2|} \mathrm{e}^{\mathrm{i}(\arg z_1 - \arg z_2)}$;
(3) $(r\mathrm{e}^{\mathrm{i}\theta})^n = r^n \mathrm{e}^{\mathrm{i}n\theta}$ for every integer $n$.

**Exercise A.150.** From the definitions, prove over $\mathbb{C}$ that:
(1) $\cos^2 z + \sin^2 z = 1$;
(2*) $(z^n)' = nz^{n-1}$ for every nonzero integer $n$;
(3*) $(\mathrm{e}^z)' = \mathrm{e}^z$;
(4*) $(\sin z)' = \cos z$, $(\cos z)' = -\sin z$, and $(\tan z)' = \frac{1}{\cos^2 z}$.
*Hint.* For the last three parts, one may use $\lim_{z \to 0} \frac{\mathrm{e}^z - 1}{z} = 1$.

**Exercise A.151.** Prove theorem A.9.11.

**Exercise A.152.** For each of $w = |z|$, $w = z^*$, and $w = |z|^2$, determine where it is differentiable and where it is analytic.

**Exercise A.153*.** Adjoin one point $\infty$ to $\mathbb{C}^n$, forming $\widehat{\mathbb{C}^n}$. Declare a subset $U$ open exactly when both of the following hold:

(i) $\mathbb{C}^n \cap U$ is open;
(ii) if $\infty \in U$, then $\widehat{\mathbb{C}^n} \setminus U$ is compact in $\mathbb{C}^n$.

Prove $\widehat{\mathbb{C}^n} \cong \mathbb{S}^{2n}$.

## A.9.2 Analytic Functions and Their Derivatives

### Analytic functions of one complex variable

We consider the differentiability condition for a general function of one complex variable. For $z = x + \mathrm{i}y$, where $x, y \in \mathbb{R}$, write $w = f(x,y) = u(x,y) + \mathrm{i}v(x,y)$, where $u$ and $v$ are real-variable functions. Put $\Delta z = \Delta x + \mathrm{i}\Delta y$, with $\Delta x, \Delta y \in \mathbb{R}$, and consider two ways of tending to zero. First, hold $\Delta y = 0$ and let $\Delta x \to 0$. Then

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta x \to 0} \frac{\Delta u + \mathrm{i}\Delta v}{\Delta x} = \frac{\partial u}{\partial x} + \mathrm{i}\frac{\partial v}{\partial x}.$$

Next, hold $\Delta x = 0$ and let $\Delta y \to 0$. Then

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta y \to 0} \frac{\Delta u + \mathrm{i}\Delta v}{\mathrm{i}\Delta y} = \frac{\partial v}{\partial y} - \mathrm{i}\frac{\partial u}{\partial y}.$$

Equating the real and imaginary parts yields the following condition.

**Theorem A.9.12** (Cauchy–Riemann conditions)**.** *For $z = x + \mathrm{i}y$, where $x, y \in \mathbb{R}$, let the complex-variable function be $w = f(x,y) = u(x,y) + \mathrm{i}v(x,y)$. A necessary condition for differentiability at $z$ is*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{A.4}$$

**Remark.** The equations in equation (A.4) are the *Cauchy–Riemann equations*. These conditions are necessary for differentiability. Conversely, under suitable continuity conditions on the partial derivatives they are also sufficient. For example, if $f$ satisfies the Cauchy–Riemann equations and its real and imaginary parts $u, v$ have continuous first partial derivatives, then $f(z)$ is differentiable.

The rules for derivatives of differentiable complex functions are entirely analogous to those for real functions. For example, $(f(z) + g(z))' = f'(z) + g'(z)$ and $(f(z)g(z))' = f'(z)g(z) + f(z)g'(z)$; we do not give detailed proofs.

As an illustration, consider solving the Cauchy functional equation: *What are the functions $f(x)$ over $\mathbb{C}$ that satisfy $f(x) + f(y) = f(x+y)$ for every $x, y$?*

(1) Substituting $x = y = 0$ gives $2f(0) = f(0)$, hence $f(0) = 0$.
(2) Substituting $y = -x$ gives $f(x) + f(-x) = f(0) = 0$, so $f(-x) = -f(x)$.
(3) Substituting $y = mx$, with $m$ a positive integer, and iterating addition gives

$$f(mx) = f(\underbrace{x + \cdots + x}_{m \text{ copies of } x}) = \underbrace{f(x) + \cdots + f(x)}_{m \text{ copies of } f(x)} = mf(x).$$

(4) Replacing $x$ in $f(mx) = mf(x)$ by $y/m$ gives $f(y) = mf(\frac{y}{m})$. Thus $f(\frac{x}{n}) = \frac{1}{n}f(x)$ for every nonzero integer $n$.
(5) Combining steps (3) and (4) gives $f(\frac{m}{n}x) = \frac{m}{n}f(x)$ for all positive integers $m, n$. Using step (2) as well gives $f(\frac{m}{n}x) = \frac{m}{n}f(x)$ for integers $m, n$ with $n \neq 0$. Hence $f(\alpha x) = \alpha f(x)$ for every $\alpha \in \mathbb{Q}$. Taking $x = 1$ shows that on $\mathbb{Q}$ one has $f(\alpha) = C\alpha$, where $C = f(1)$.
(6) We now pass to the real numbers. Step (5) gives the solution on the rational numbers, but the preceding arguments do not restrict $f(x)$ for a general irrational $x$. Assume, however, that $f$ is continuous. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, rational sequences approaching an arbitrary real number extend the relation $f(x) = Cx$ from the rational numbers to all the real numbers. Thus $f(x) = Cx$ holds throughout the real numbers. A fully rigorous deduction from continuity is outside the scope of this book; interested readers may consult further references.

(7) Taking $x = \mathrm{i}$ in $f(\alpha x) = \alpha f(x)$ gives $f(\mathrm{i}\alpha) = C'\alpha$ on $\mathbb{Q}$, where $C' = f(\mathrm{i})$ is another constant. As in step (6), continuity gives $f(\mathrm{i}x) = C'x$ for all $x \in \mathbb{R}$.

(8) Therefore $f(x+\mathrm{i}y) = f(x) + f(\mathrm{i}y) = Cx + C'y$ for real $x, y$. Continuity alone does not imply $C' = \mathrm{i}C$, but differentiability does. Write $C = a + \mathrm{i}b$ and $C' = c + \mathrm{i}d$, where $a, b, c, d \in \mathbb{R}$. Then $f(x + \mathrm{i}y) = u(x,y) + \mathrm{i}v(x,y)$, where $u(x,y) = ax + cy$ and $v(x,y) = bx + dy$. The Cauchy–Riemann conditions require
$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$
Solving these gives $a = d$ and $c = -b$, and hence
$$f(x+\mathrm{i}y) = (a+\mathrm{i}b)x + (-b+\mathrm{i}a)y = (a+\mathrm{i}b)(x+\mathrm{i}y).$$
Thus $f(z) = C_0 z$ throughout $\mathbb{C}$, where $C_0$ is a constant. □

As for real-variable functions, Taylor series can also be defined for complex-variable functions.

**Theorem A.9.13.** *If $f$ is analytic on an open set $D \subseteq \mathbb{C}$, then it has continuous derivatives of every order throughout $D$. For $z_0 \in D$,*
$$\begin{aligned} f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n &= f(z_0) + f'(z_0)(z-z_0) \\ &+ \frac{1}{2!}f''(z_0)(z-z_0)^2 + \frac{1}{3!}f'''(z_0)(z-z_0)^3 + \cdots. \end{aligned} \tag{A.5}$$
*whenever $|z - z_0| < R$, where $R$ is the least distance from $z_0$ to the boundary of $D$. Here the boundary is understood in the topology inherited from $\mathbb{C}$ or $\widehat{\mathbb{C}}$. If $D = \mathbb{C}$, take $R = \infty$; the expansion then holds for every $z \in \mathbb{C}$.*

The displayed expansion is the *Taylor series of $f$ at $z_0$*.

Infinite sums in the form of power series are also analytic.

**Theorem A.9.14.** *Let $z_0$ and $a_n$ for $n \geq 0$ be complex numbers. Suppose the power series $I = a_0 + \sum_{n=1}^{\infty} a_n(z-z_0)^n$ converges in the open disk $D : |z-z_0| < R$, where $R > 0$.*[31] *Then:*

*(1) $I(z)$ is complex analytic on $D$;*

*(2) it may be differentiated term by term on $D$:*
$$I'(z) = \sum_{n=0}^{\infty} \left(a_n(z-z_0)^n\right)' = a_1 + \sum_{n=2}^{\infty} a_n n (z-z_0)^{n-1};$$

*(3) if there is a natural number $r$ such that $a_n = 0$ for all $n < r$, then the part remaining after factoring out $(z-z_0)^r$ is also analytic; that is, $a_r + \sum_{n=r+1}^{\infty} a_n(z-z_0)^{n-r}$ is also analytic on $D$.*

The proofs of the preceding two theorems lie beyond the scope of this course, but they imply the following local factorization.

**Theorem A.9.15.** *Let $D \subseteq \mathbb{C}$ be open, $z_0 \in D$, and let $f : D \to \mathbb{C}$ be analytic. Suppose there is no open neighborhood $U \subseteq D$ of $z_0$ on which $f$ is identically zero. Then there are a unique natural number $r$, an open neighborhood $U \subseteq D$ of $z_0$, and an analytic function $u$ on $U$ such that $f(z) = (z-z_0)^r u(z)$ and $u(z_0) \neq 0$. For different admissible open sets $U$ containing $z_0$, the exponent $r$ is the same, and this relation uniquely determines the function $u$. If $r \geq 1$, $U$ may be chosen so that $z_0$ is the only zero of $f$ in $U$.*

The integer $r$ is the *order of vanishing of $f$ at $z_0$*; when $r \geq 1$, the point $z_0$ is a *zero of order $r$*.

**Proof.** By theorem A.9.13, on a sufficiently small disk $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$, and the hypothesis ensures that not all $a_n$ vanish. Let $r$ be the least index for which $a_r \neq 0$. Factoring out $(z-z_0)^r$, theorem A.9.14 shows that $u(z) = \sum_{n=r}^{\infty} a_n(z-z_0)^{n-r}$ is also analytic, and $u(z_0) = a_r \neq 0$. Suppose also that $f = (z-z_0)^s v(z)$ with $v(z_0) \neq 0$. The constant term of the Taylor expansion of $v(z)$ is nonzero, so the degree of the first nonzero Taylor term of $f$ is precisely $s$. This forces $s = r$. Shrink the open sets

[31] Convergence means that the sequence $\{\sum_{m=1}^{n} a_m(z-z_0)^m\}_n$ has a limit.

under consideration to an open disk $D'$ on which both representations are defined. On $D' \setminus \{z_0\}$ we have $u = f/(z-z_0)^r = v$; continuity then gives equality at $z_0$ as well. Thus $u$ is unique. Since $u(z_0) \neq 0$, continuity allows the disk to be shrunk to an open set on which $u$ is nowhere zero. When $r \geq 1$, $z_0$ is then the only zero of $f$ in that disk. □

**Theorem A.9.16.** *Let $f$ be analytic on a connected open set $D \subseteq \mathbb{C}$. If the zero set of $f$ has an accumulation point in D, then $f$ vanishes identically on D. If $f'(z) = 0$ holds identically on D, then $f$ is constant on D.*

**Proof.** If the zeros of $f$ accumulate at $z_0 \in D$, continuity first gives $f(z_0) = 0$. If $f$ were not identically zero on any open set containing $z_0$, theorem A.9.15 would give an open set containing $z_0$ with no other zero. This contradicts the definition of an accumulation point. Thus $f$ is identically zero on some open set containing $z_0$.

Put
$$E = \{z \in D \mid \text{there is an open set } U \subseteq D \text{ containing } z \text{ with } f|_U = 0\}.$$
By definition, $E$ is nonempty and open. If $z \notin E$ and $f(z) \neq 0$, continuity gives an open set $W \subseteq D$ containing $z$ on which $f$ is nowhere zero, so $W \cap E = \emptyset$. If $z \notin E$ and $f(z) = 0$, theorem A.9.15 gives an open set $W \subseteq D$ containing $z$ in which $z$ is the only zero. Since $z \notin E$, again $W \cap E = \emptyset$. This proves that $D \setminus E$ is open as well. Thus $E$ is both open and closed, and connectedness, by proposition A.7.28, forces $E = D$. Consequently $f(D) = \{0\}$.

If $f'(z) = 0$, then for each $z_0 \in D$, theorem A.9.13 gives an open set $U \subseteq D$ containing $z_0$ on which $f$ has the Taylor expansion $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$. By theorem A.9.14, $f'(z) = 0$ implies $a_n n = 0$ for every $n \geq 1$, and hence $a_n = 0$ for every $n \geq 1$. Thus $f(z) = a_0$ throughout $U$. The function is therefore locally constant, and proposition A.7.29 shows that it is constant. □

**Example A.9.17.** Solve the Cauchy functional equation again: an analytic function $f : \mathbb{C} \to \mathbb{C}$ satisfies $f(z+w) = f(z) + f(w)$ $(\forall z, w)$.

**Solution.** Putting $z = w = 0$ gives $f(0) = 2f(0)$, hence $f(0) = 0$. Now fix $w$ and differentiate both sides with respect to $z$, obtaining $f'(z+w) = f'(z)$. Since $w$ is arbitrary, $f'(z) = C$ is constant. Thus $(f(z) - Cz)' = f'(z) - C = 0$, and theorem A.9.16 gives a constant $C'$ such that $f(z) - Cz = C'$. Since $f(0) = 0$, we have $C' = 0$, so $f(z) = Cz$. This replaces the earlier lengthy continuity argument. □

### Analytic functions of several complex variables

We next discuss properties of analytic functions of several complex variables. Let $\Omega \subseteq \mathbb{C}^n$ be open and let $\boldsymbol{F} : \Omega \to \mathbb{C}^m$. Separating real and imaginary parts identifies $\mathbb{C}^n$ with $\mathbb{R}^{2n}$. The independent variable of $\boldsymbol{F}$ is $z = (z_1, \ldots, z_n) \in \Omega$; write $\boldsymbol{F}(z) = (F_1(z), \ldots, F_m(z))$, where $z_j = x_j + \mathrm{i}y_j$ and $F_\alpha = u_\alpha + \mathrm{i}v_\alpha$. If every $F_i$ is analytic in every $z_j$, then $\boldsymbol{F}$ is a *complex-analytic map*, or simply an *analytic map*. The one-variable result theorem A.9.13, together with the Cauchy–Riemann equations, extends to the following statements; their full proofs are omitted.

**Theorem A.9.18.** *Let $\Omega \subseteq \mathbb{C}^n$ be open. If $\boldsymbol{F} : \Omega \to \mathbb{C}^m$ is complex analytic, then every component $F_\alpha$ has continuous partial derivatives and mixed partial derivatives of all orders with respect to each $z_j$ throughout $\Omega$; the order of mixed partial derivatives may be interchanged.*

**Theorem A.9.19** (Cauchy–Riemann conditions)**.** *Let $\Omega \subseteq \mathbb{C}^n$ be open. If $\boldsymbol{F} : \Omega \to \mathbb{C}^m$ is a complex-analytic map, then for each $\alpha, j$ the one-variable Cauchy–Riemann conditions hold:*
$$\frac{\partial u_\alpha}{\partial x_j} = \frac{\partial v_\alpha}{\partial y_j}, \qquad \frac{\partial u_\alpha}{\partial y_j} = -\frac{\partial v_\alpha}{\partial x_j}. \tag{A.6}$$
*Conversely, if all the real coordinate functions of $\boldsymbol{F}$, namely $u_\alpha, v_\alpha$, have continuous first partial derivatives and satisfy these equations, then $\boldsymbol{F}$ is complex analytic.*

For the complex-analytic map $\boldsymbol{F}$, arrange its partial derivatives at a point $\boldsymbol{z}$ in the matrix $\left(\frac{\partial F_i}{\partial z_j}(\boldsymbol{z})\right)_{m\times n}$. The entry in row $i$, column $j$ is the value of $\frac{\partial F_i}{\partial z_j}$ at $\boldsymbol{z}$. This is the *Jacobian matrix* of $\boldsymbol{F}$ at $\boldsymbol{z}$, denoted by $\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{z}}$ or $\frac{\partial(F_1,\ldots,F_m)}{\partial(z_1,\ldots,z_n)}$. When $n=m$, its determinant is the *Jacobian determinant*. We may also consider only some of the partial derivatives, for example $\frac{\partial(F_1,\ldots,F_m)}{\partial(z_1,\ldots,z_\alpha)}$ with $\alpha<n$. We can now state the implicit function theorem for complex-analytic maps.

**Theorem A.9.20** (Implicit function theorem)**.** *Let $\Omega\subseteq\mathbb{C}^n\times\mathbb{C}^m$ be open and let $\boldsymbol{F}:\Omega\to\mathbb{C}^m$, $(\boldsymbol{z},\boldsymbol{w})\mapsto\boldsymbol{F}(\boldsymbol{z},\boldsymbol{w})$, be complex analytic, where $\boldsymbol{z}\in\mathbb{C}^n$ and $\boldsymbol{w}\in\mathbb{C}^m$. Let $(\boldsymbol{a},\boldsymbol{b})\in\Omega$, where $\boldsymbol{a}\in\mathbb{C}^n$ and $\boldsymbol{b}\in\mathbb{C}^m$, and suppose*

$$\boldsymbol{F}(\boldsymbol{a},\boldsymbol{b})=\boldsymbol{0},\qquad \det\left(\frac{\partial\boldsymbol{F}}{\partial\boldsymbol{w}}(\boldsymbol{a},\boldsymbol{b})\right)\neq 0.$$

*Then there are open sets $U\subseteq\mathbb{C}^n$ and $V\subseteq\mathbb{C}^m$ containing $\boldsymbol{a}$ and $\boldsymbol{b}$, respectively, with $U\times V\subseteq\Omega$. On $U\times V$, the equation $\boldsymbol{F}(\boldsymbol{z},\boldsymbol{w})=\boldsymbol{0}$ uniquely determines a complex-analytic map $\boldsymbol{w}=\boldsymbol{\varphi}(\boldsymbol{z})$. Moreover,*

$$\frac{\partial\boldsymbol{\varphi}}{\partial\boldsymbol{z}}(\boldsymbol{z})=-\left[\frac{\partial\boldsymbol{F}}{\partial\boldsymbol{w}}(\boldsymbol{z},\boldsymbol{\varphi}(\boldsymbol{z}))\right]^{-1}\left[\frac{\partial\boldsymbol{F}}{\partial\boldsymbol{z}}(\boldsymbol{z},\boldsymbol{\varphi}(\boldsymbol{z}))\right].$$

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## Exercises

**Exercise A.154.** Prove that an analytic function on a connected open set whose imaginary part vanishes identically must be constant.

**Exercise A.155.** Let the real-variable functions $u(x,y)$ and $v(x,y)$ be the real and imaginary parts of an analytic function $w_1$, and let $w_2$ be analytic. Can $u^2$ be the real part of $w_2$? Can $uv$? Justify your answers.

**Exercise A.156.** Consider $f:\mathbb{C}\to\mathbb{C}$. For $z=x+\mathrm{i}y$ with $x,y\in\mathbb{R}$, let $f(z)=\frac{x^3+\mathrm{i}y^3}{x^2+y^2}$ when $z\neq 0$, and $f(z)=0$ when $z=0$. Prove that $f(z)$ satisfies the Cauchy–Riemann equations at $z=0$, and determine whether it is analytic at $z=0$.

**Exercise A.157.** Suppose two power series converge and are equal throughout a closed disk containing $z_0$:

$$a_0+\sum_{n=1}^{\infty}a_n(z-z_0)^n=b_0+\sum_{n=1}^{\infty}b_n(z-z_0)^n.$$

Prove that their corresponding coefficients agree: $a_n=b_n$ for $n=0,1,\ldots$.

**Exercise A.158.** For each function below, find its Taylor expansion at $z=0$, specify its range of convergence, and find its order of vanishing at $z=0$: (1) $z^3\sin z$; (2) $(1-\cos z)^2$; (3) $\sin z-z$.

**Exercise A.159.** Let analytic functions $f,g$ be not identically zero on an open disk containing $z_0$. Prove that the order of vanishing of $f(z)g(z)$ at $z_0$ is the sum of the orders of vanishing of $f$ and $g$ there.

**Exercise A.160.** Let $f$ be analytic on an open disk $D$, and suppose its zero set in $D$ is not all of $D$. Prove that $f$ has only finitely many zeros in every compact subset of $D$, although it may have infinitely many zeros in a general subset of $D$.

**Exercise A.161.** For
$$\boldsymbol{F}:\mathbb{C}^2 \longrightarrow \mathbb{C}, \qquad (z_1, z_2) \longmapsto z_1^2 + z_1 z_2 + z_2,$$
use the implicit function theorem to determine the points in whose neighborhood one can take an open disk on which $\boldsymbol{F}(z_1, z_2) = 0$ uniquely defines an analytic function $z_1 = \varphi(z_2)$. Wherever $\varphi$ is uniquely determined, find $\varphi'(z)$.

**Exercise A.162.** Use the real implicit function theorem together with the multivariable Cauchy–Riemann equations to prove the complex-analytic implicit function theorem.

## Appendix B

# Hints and Solutions to Selected Exercises

The solutions are arranged in the order in which the corresponding exercises occur in the book. A few entries give only a hint or a route to a proof.

## Chapter 0

**Solution to Exercise 0.43.** Let the circle $\odot(CFT)$ meet $EF$ and $CB$ again at $X$ and $Y$, respectively. Then

$$\measuredangle REF = \measuredangle DEF = \measuredangle DCF = \measuredangle TCF = \measuredangle TXF,$$

so $RE \parallel TX$; similarly, $RB \parallel TY$. Moreover,

$$\measuredangle BEF = \measuredangle BCF = \measuredangle YCF = \measuredangle YXF,$$

and hence $BE \parallel YX$. Thus the corresponding sides of $\triangle XYT$ and $\triangle EBR$ are parallel, so the two triangles are homothetic. Their corresponding vertices are joined by the concurrent lines $XE = EF$, $YB = BC$, and $RT$. Since $S = BC \cap EF$, their common point is $S$, and consequently $R, S, T$ are collinear.

**Solution to Exercise 0.50.** Let $X', Y', Z'$ be the midpoints of $DI, EJ, FK$, respectively. Echols's theorem shows that $\triangle X'Y'Z'$ is equilateral. By the centroid property, each vertex of $\triangle XYZ$ is the point dividing, in the ratio $2:1$, the segment joining the corresponding vertices of $\triangle ABC$ and $\triangle X'Y'Z'$. The assertion now follows from theorem 0.2.12.

**Solution to Exercise 0.73.** For $\triangle ABC$, let $Ni, H, O$ be its nine-point center, orthocenter, and circumcenter, respectively. With respect to the orthic triangle, $Ni$ is the circumcenter and $H$ is an isocenter. The Euler–Chapple formula gives $NiH^2 = R'^2 - 2R'\rho$, where $R'$ is the circumradius of the orthic triangle. Since $OH = 2NiH$ and $R = 2R'$, we obtain $OH^2 = R^2 - 4R\rho$.

## Chapter 1

**Solution to Exercise 1.9.** Let the common focus be $F$, denote the two ellipses by $\alpha_1, \alpha_2$, their auxiliary circles by $\odot O_1, \odot O_2$, and their other foci by $U, V$, respectively.

Suppose first that $\alpha_1$ and $\alpha_2$ are tangent at $K$, and let $P$ be the projection of $F$ on their common tangent at $K$. By theorem 1.3.5, $P \in \odot O_1$ and $P \in \odot O_2$; the proof of that theorem also gives $PO_1 \parallel UK$ and $PO_2 \parallel VK$. Since $O_1O_2 \parallel UV$, the triples $U, V, K$ and $P, O_1, O_2$ are each collinear. Hence $\odot O_1$ and $\odot O_2$ are tangent. The converse is proved in the same way.

**Solution to Exercise 1.24.**

(1) Use proposition 1.4.1 to construct two diameters of the ellipse. Their intersection is the center $O$.
(2) Draw a circle centered at $O$ that meets the ellipse in four points. These points form a rectangle. By symmetry, the two angle bisectors of its diagonals are the major and minor axes of the ellipse.
(3) With either minor-axis vertex as center and the semimajor axis as radius, draw a circle. Its intersections with the major axis are the two foci of the ellipse.

# Chapter 2

**Solution to Exercise 2.9.** Put $S = m \cap l$, and let $T$ be the intersection of $m$ with the other asymptote. By proposition 2.1.15, $QS = PT$; proposition 2.1.16 then gives

$$\operatorname{dist}(P,l) \cdot \operatorname{dist}(Q,l) = PS \cdot QS \cdot \cos^2 \angle(n,l) = PS \cdot PT \cdot \cos^2 \angle(n,l) = \text{const}.$$

**Solution to Exercise 2.22.** We give the argument for an ellipse; the hyperbolic case is analogous. In figure B.1, let $F'$ be the other focus of $\alpha$, let $A, A'$ be its major-axis vertices, and let $X$ be the projection of $F'$ on the tangent at $P$. By theorem 1.3.5, $OX = OA$. The definition of conjugate diameters and the diameter property give $OQ \parallel PX$.

Let $Z$ be the reflection of $F'$ in $X$. By the optical property, $Z, P, F$ are collinear. Hence $OX$ is a midline of $\triangle FF'Z$, and so $OX \parallel PF$. Since $Y \in OQ \cap PF$, together with $OQ \parallel PX$ this shows that $XOYP$ is a parallelogram. Consequently $PY = OX = OA$.

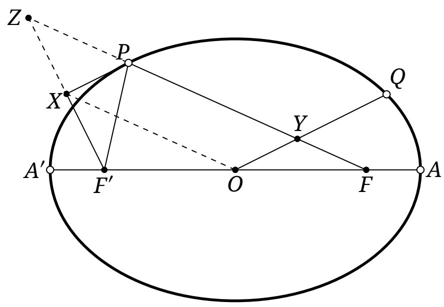


FIGURE B.1

**Solution to Exercise 2.23.** As in figure B.2, let $\odot C$ be the circumcircle of $\triangle AOB$. Since $\angle AOB =$ const, the sine rule gives

$$AC = OC = BC = \frac{AB}{2\sin\angle AOB} = \text{const}.$$

Draw $CH \perp AB$ at $H$. Then $\angle ACH = \frac{1}{2}\angle ACB = \angle AOB = \text{const}$, and therefore $CH = AC\cos\angle ACH =$ const and $CP = \sqrt{CH^2 + PH^2} = \text{const}$. Since $AC, CP, AP$ are all fixed, $\angle ACP = \text{const}$.

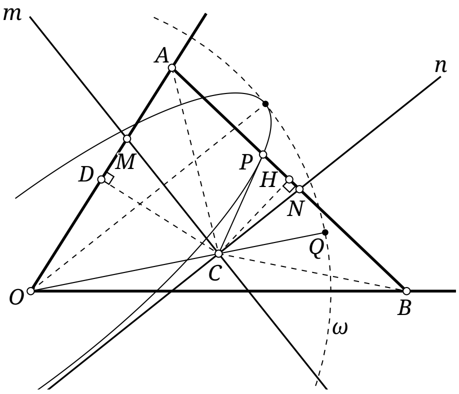


FIGURE B.2

Draw $CD \perp AO$ at $D$; then $D$ is the midpoint of $AO$. Draw a line $m$ through $M \in AO$ so that $\measuredangle DCM = \measuredangle ACP/2 = \text{const}$. It follows that

$$\measuredangle MCO = \measuredangle DCO + \measuredangle MCD = \frac{1}{2}(\measuredangle ACO + \measuredangle PCA) = \frac{1}{2}\measuredangle PCO.$$

The line $AO$ is fixed, so the direction of its perpendicular $DC$ is fixed. Since $\angle MCD$ is constant, the direction of $m$ is fixed as well. Through $C$ draw $n \perp m$, meeting $AB$ at $N$. The lines $m,n$ are respectively the internal and external angle bisectors of $\angle OCP$, and the direction of $n$ is also fixed.

Let $Q = \mathsf{f}_n(P)$. Since $n$ is the external angle bisector of $\angle OCP$, the points $O,C,Q$ are collinear, and $OQ = OC + CQ = \text{const}$. Thus $Q$ moves on a fixed circle $\omega$ centered at $O$. Let $\hat{\boldsymbol{n}},\hat{\boldsymbol{m}}$ be unit vectors along $n,m$, respectively. Then the ratios of the projections of $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ in the directions of $n$ and $m$ are, respectively,

$$\frac{\overrightarrow{OP}\cdot\hat{\boldsymbol{n}}}{\overrightarrow{OQ}\cdot\hat{\boldsymbol{n}}} = \frac{OC+CP}{OC+CQ} = \frac{OC+CP}{OC+CP} = 1,$$
$$\frac{\overrightarrow{OP}\cdot\hat{\boldsymbol{m}}}{\overrightarrow{OQ}\cdot\hat{\boldsymbol{m}}} = \frac{OC-CP}{OC+CQ} = \frac{OC-CP}{OC+CP} = \text{const.}$$

By proposition 2.2.11, the locus of $P$ is therefore an ellipse, $\omega$ is its auxiliary circle, and the ratio of its semimajor axis to its semiminor axis is $\left|\frac{\overrightarrow{OQ}\cdot\hat{\boldsymbol{m}}}{\overrightarrow{OP}\cdot\hat{\boldsymbol{m}}}\right|$.

# Chapter 3

**Solution to Exercise 3.24.** Let $l$ be the tangent at $P$, and let $\theta_1,\theta_2,\theta_3$ be the directed angles made with the principal axis by $OP,n,l$, respectively. By example 3.3.21, $\tan\theta_1\tan\theta_3 = e^2-1$. Since $n$ is the normal, $\theta_2 = 90^\circ+\theta_3$, and hence $\tan\theta_2\tan\theta_3 = -\cot\theta_3\tan\theta_3 = -1$. Therefore $\frac{\tan\theta_1}{\tan\theta_2} = 1-e^2$.

**Solution to Exercise 3.40.** Let $\infty$ and $\infty'$ be the points at infinity on $l$ and $l^*$, respectively. Then

$$(AA',BB') = \frac{\overline{BA}}{\overline{BA'}} \div \frac{\overline{B'A}}{\overline{B'A'}} = \left(\frac{\overline{BA}}{\overline{BA'}}\right)^2 = (AA'B)^2 = (AA',B\infty)^2.$$

Similarly,

$$(A^*A'^*,B^*B'^*) = (A^*A'^*B^*)^2 = (A^*A'^*,B^*\infty')^2.$$

Cross-ratios are preserved by a projectivity, so

$$(AA',B\infty) = (A^*A'^*,B^*\infty').$$

The negative sign is excluded by the order of the corresponding points.[1] It follows that $\infty' = \infty^*$.

Because $M$ is the midpoint of $AA'$, the points $A,A',M,\infty$ form a harmonic range. Their images $A^*,A'^*,M^*,\infty^*$ therefore also form a harmonic range. Since $\infty^* = \infty'$, the point $M^*$ is the midpoint of $A^*A'^*$; hence $M^* = M'$.

**Solution to Exercise 3.50.** Put $X = B'C' \cap BC$, and let $PA,PB,PC$ meet $B'C'$ at $D,E,F$, respectively. Apply Desargues's involution theorem to the complete quadrangle $ABCP$ and the line $B'C'$. The opposite sides $BC,AP$ meet $B'C'$ at $X,D$, respectively; $CA,BP$ meet it at $B',E$; and $AB,CP$ meet it at $C',F$. Thus $(X,D)$, $(B',E)$, and $(C',F)$ are three conjugate pairs of one involution on $B'C'$.

Projecting the point row from $P$ gives the conjugate pairs $(PX,PD=PA)$, $(PB',PE=PB)$, and $(PC',PF=PC)$ of an involution on the pencil at $P$. Reflection in $l$ is also an involution, and it sends $PB'$ to $PB$ and $PC'$ to $PC$. Two conjugate pairs determine an involution on a pencil, so the former involution is precisely reflection in $l$. Thus $PX$ and $PA$ are symmetric about $l$, whence $PX = PA'$. It follows that $X = A'$, and therefore $A',B',C'$ are collinear.

# Chapter 4

[1] More precisely, that order rules out $(AA',B\infty) = -(A^*A'^*,B^*\infty')$.

**Solution to Exercise 4.17.** By theorem 4.2.11, for a given triangle $ABC$ and a fixed point $P$ on its circumcircle, there is a unique parabola with focus $P$ tangent to all three sidelines of $\triangle ABC$. Indeed, reflect $P$ in the three sidelines to obtain $P_A, P_B, P_C$. Simson's theorem shows that these three points are collinear, and their common line is the directrix of the parabola.

**Solution to Exercise 4.22.** Let $M$ be the midpoint of $PQ$, and join $AM$. Then $AM$ is a diameter of the parabola. If it meets the parabola at $N$, proposition 2.3.6 shows that the tangent $l$ at $N$ is parallel to $PQ$, while proposition 2.3.7 shows that $N$ is the midpoint of $AM$.

Let $l$ meet $AP, AQ$ at $X, Y$, respectively. Then $X, Y$ are the midpoints of $AP, AQ$. Since $O$ is the circumcenter of $\triangle APQ$, we have $AX \perp OX$ and $AY \perp OY$. Thus $A, X, Y, O$ are concyclic on $\odot(AO)$. By theorem 4.2.11, $F \in \odot(AXY) = \odot(AO)$, and hence $AF \perp OF$.

**Solution to Exercise 4.23.** Construct the parabola $\alpha$ tangent to the three sidelines of $\triangle ABC$ and to $l_1$. By theorem 1.5.7, $H$ lies on the directrix of $\alpha$; together with theorem 4.2.12, this shows that $l_2$ is also tangent to $\alpha$.

By theorem 4.2.11, the circumcircles of $\triangle A_1A_2H$, $\triangle B_1B_2H$, and $\triangle C_1C_2H$ all pass through the focus $F$ of $\alpha$. Since all three circles pass through $F, H$, their centers lie on the perpendicular bisector of $FH$. Those centers are precisely $A_0, B_0, C_0$, so the three points are collinear.

**Solution to Exercise 4.34.** Put $T = PQ \cap m$ and $A_0 = m \cap BC$. The points $P, Q, X, A_0$ are concyclic: indeed, $A_0$ is the vertex opposite $X$ in the circumcircle mid-arc triangle of $\triangle PQX$. Denote their circle by $\omega$. Since $m$ is perpendicular to the chord $PQ$ of $\omega$, it is a diameter line of $\omega$.

Let $\omega \cap m = A_0, A_1$ and $\mathscr{C} \cap m = U, V$. The intersecting chords theorem, applied to $\mathscr{C}$ and $\omega$, gives

$$\overline{TU} \cdot \overline{TV} = \overline{TP} \cdot \overline{TQ} = \overline{TA_0} \cdot \overline{TA_1}.$$

Since $U, V, A_0$ are fixed, the correspondence $[T] \to A_1$ is a projectivity on $m$. Moreover, $m = A_0A_1$ is a diameter line of $\omega$, so $A_1X \perp XA_0 = BC$. Thus $m[A_1] \to BC[X]$ is a perspectivity, and consequently $m[T] \to BC[X]$ is a projectivity.

In the same way, $m[T] \to CA[Y]$ is a projectivity. Hence $BC[X] \barwedge CA[Y]$. In the limiting case $X \to C$, one also has $Y \to C$; therefore the intersection of the two carrier lines is fixed under this projectivity. The correspondence is consequently a perspectivity, so $XY$ passes through a fixed point $K$.

We have proved that $l(P)$ passes through $K$. For $P = U$ and $P = V$, the line $l(P)$ is respectively $\mathscr{S}(U)$ and $\mathscr{S}(V)$. Hence $K = \mathscr{S}(U) \cap \mathscr{S}(V)$. By proposition 4.2.3, $\mathscr{S}(U) \perp \mathscr{S}(V)$. Let $H$ be the orthocenter and put $E = \mathscr{S}(U) \cap UH$ and $F = \mathscr{S}(V) \cap VH$. Steiner's theorem says that $E, F$ are the images of $U, V$, respectively, under $\mathfrak{h}_{H,1/2}$. Hence $E, F$ are antipodal points of $\mathscr{N}$. Since $EK \perp FK$, it follows that $K \in \mathscr{N}$.

**Solution to Exercise 4.47.** Put $Q = \mathbf{g}P$. Let $\triangle Q_aQ_bQ_c$ be the pedal triangle of $Q$, let $\triangle A'B'C'$ be the circumcevian triangle of $P$, and let $\triangle P_aP_bP_c$ be the antipedal triangle of $P$ with respect to $\triangle A'B'C'$. Clearly $P_bP_c \perp PA' = PA$. On the other hand, proposition 4.4.7 gives $PA \perp Q_bQ_c$, and hence $P_bP_c \parallel Q_bQ_c$. The other two pairs of corresponding sides are parallel in the same way. The two triangles are therefore homothetic.

# Chapter 5

**Solution to Exercise 5.11.** The point $N$ is the homothety center of $\triangle A_1A_2A_3$ and $\triangle ABC$, and $Y_3, G_c$ are corresponding points under this homothety. Hence $NG_c$ passes through $Y_3$, and $Y_3$ is the midpoint of $NG_c$.

Recall the proof of proposition 5.2.7: $IM_c \parallel CN$, while $I$ is the midpoint of $G_cZ_3$. Here $Z_3$ is the antipode of $G_c$ on the incircle and lies on $CN$; see lemma 5.2.6. It follows that $IM_c$ passes through the midpoint of $NG_c$, namely $Y_3$. Thus $Y_3 \in IM_c$.

**Solution to Exercise 5.27.** Let $H$ be the orthocenter of $\triangle ABC$. The points $G,H,O,Et,Ht,To$ are collinear. Denote by $R,R_1,R_2$ the circumradii of the original, orthic, and tangential triangles, respectively. The circumcircle of the orthic triangle is the nine-point circle, so $R_1 = \frac{R}{2}$.

The orthic and tangential triangles are homothetic. Their circumcenters are $Ni,To$, and their incenters are $H,O$, respectively. Therefore

$$\frac{\overline{HtNi}}{\overline{HtTo}} = \frac{\overline{HtH}}{\overline{HtO}} = \frac{R_1}{R_2} = \frac{R}{2R_2}. \tag{$\diamondsuit$}$$

The point $Et$ is the internal similicenter of the tangential triangle, and the circumcircle of $\triangle ABC$ is the incircle of that triangle. Hence $\frac{\overline{EtO}}{\overline{EtTo}} = -\frac{R}{R_2}$. Together with ($\diamondsuit$), this yields

$$-\frac{\overline{EtO}}{\overline{EtTo}} = 2\frac{\overline{HtNi}}{\overline{HtTo}} = 2\frac{\overline{HtH}}{\overline{HtO}}. \tag{$\spadesuit$}$$

We also have

$$\overline{HNi} = \overline{NiO}, \qquad \overline{HG} = 2\overline{GO}. \tag{$\clubsuit$}$$

Once $H,O,Ht$ are fixed, ($\spadesuit$) and ($\clubsuit$) determine the positions of $G,Ni,Et,To$. A direct calculation now gives $(G,Et;O,Ht) = -1$; we leave it to the reader.

**Solution to Exercise 5.37.** The proof of proposition 5.5.7 shows that the extangents, intangents, and tangential triangles have the common homothety center $In$. The points $Cl,Ic,Ht$ are corresponding points under these homotheties: each is the homothety center of the corresponding triangle and the orthic triangle.

**Solution to Exercise 5.65.** For $\triangle ABC$, the complements of $H,Al,Br_3$ are respectively the circumcenter $O$, the Lemoine point $L$, and the Brocard midpoint $Br_{\mathrm{m}}$. The points $O,L,Br_{\mathrm{m}}$ are collinear.

**Solution to Exercise 5.66.** Suppose first that the original triangle is acute. It is the excentral triangle of its orthic triangle. The circumcenter and the Lemoine point of the original triangle are, respectively, the Bevan point and the Mittenpunkt of the orthic triangle. Together with the orthocenter of the orthic triangle, these are precisely the three points in proposition 5.4.8, so they are collinear.

When the original triangle is obtuse, these relationships do not hold. The reader can nevertheless readily prove the result again by following the argument of proposition 5.4.8, taking the orthic triangle as the reference triangle.

# Chapter 6

**Solution to Exercise 6.1.** Let the two conjugate lines meet at $P$, let the tangents from $P$ touch the conic at $A,B$, and let the conjugate lines meet $AB$ at $M,N$, respectively. By the definition of conjugate lines, $N \in \mathfrak{p}(M)$, and hence $(MN,AB) = -1$. Therefore $(PM,PN;PA,PB) = -1$, as required.

**Solution to Exercise 6.3.** Choose $X \in \alpha$, and let $Y$ be the second intersection of $BX$ with $\alpha$. Let $Z,W$ be the second intersections of $CX,CY$ with $\alpha$, respectively. Finally, let $Z'$ be the second intersection of $BW$ with $\alpha$, and put $T = CA \cap BW$.

Since $\mathfrak{p}(B) = AC$, both $(CB,CA,CX,CY)$ and $(CB,CA,CZ',CW)$ are harmonic pencils. As $CY = CW$, three corresponding lines in these two pencils coincide; therefore the fourth lines coincide as well. Thus $CX = CZ'$, so $C,X,Z'$ are collinear and consequently $Z = Z'$. Hence $B,Z,W$ are collinear.

The same argument gives the collinearities $X,A,W$ and $Y,A,Z$. It follows that $\triangle ABC$ is the diagonal triangle of the complete quadrangle $XYWZ$.

**Solution to Exercise 6.4.** Let $A$ be the vertex of $\alpha$ and put $T = \mathfrak{p}^{-1}(QN)$. By symmetry, $T \in AN$. The harmonic property of pole and polar gives $(NT,A\infty_{AL}) = -1$, so $A$ is the midpoint of $NT$. Moreover, $Q \in QN = \mathfrak{p}(T)$ implies $T \in \mathfrak{p}(Q)$.

Put $S = QN \cap \mathfrak{p}(Q)$, and let the segment $QS$ meet $\alpha$ at $P$. The same harmonic property, together with proposition 3.2.18, gives $NP^2 = NQ \cdot NS$. Since $\triangle NQL \sim \triangle NTS$,

$$NL = NQ \cdot \frac{NS}{NT} = \frac{NP^2}{NT} = \frac{NP^2}{2NA}.$$

By proposition 2.3.2, the last expression is the focal parameter.

**Solution to Exercise 6.22.** We first prove that a conic through the orthocenter is a rectangular hyperbola. Let a conic pass through the vertices of $\triangle ABC$ and its orthocenter $H$. Let $X$ be the point at infinity on one of its asymptotes,[2] and let $Y$ be the point at infinity in the direction perpendicular to the direction of $X$. Choose the altitudes $AA', BB'$ of $\triangle ABC$. As in figure B.3, the lines $BX, AX, HY, CY$ form the quadrilateral $SVTU$.

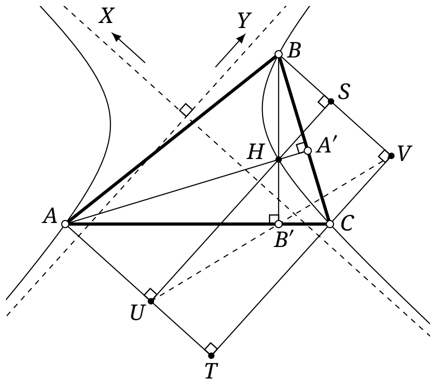


FIGURE B.3

The quadruples $A, U, B', H$, $H, A', S, B$, and $B', C, V, B$ are cyclic. Hence

$$\angle AB'U = \angle AHU = \angle A'HS = \angle A'BS = \angle CB'V,$$

and therefore $U, V, B'$ are collinear.

Apply the converse of Pascal's theorem to the hexagon $AXBYHC$. It follows that these six points lie on one conic, so $Y \in \mathfrak{C}(ABCHX)$. Thus the two asymptotic directions of the conic are perpendicular, and the conic is a rectangular hyperbola.

Conversely, if a rectangular hyperbola passes through $A, B, C$, it has points at infinity $X, Y$ in two perpendicular directions. A similar argument shows that it must also pass through $H$.

**Solution to Exercise 6.11.** First consider the general case of an ellipse. Let $O$ be the center of $\alpha$, let $a, b$ be its semimajor and semiminor axes, respectively, and let $\triangle ABC$ be self-polar with respect to $\alpha$. Put $\omega = \odot(ABC)$. We may assume that $A$ is outside the ellipse. Set $D = OA \cap BC$, and let $A'$ be the other intersection of the line $OA$ with $\omega$. Write $OA \cap \alpha = U, U'$ and $BC \cap \alpha = X, Y$. Through $O$, draw a parallel to $BC$ meeting $\alpha$ at $V$, and put $p = OU$, $q = OV$.

Since $\mathfrak{p}_\alpha(A) = BC$, the lines $AX, AY$ are tangent to $\alpha$. By proposition 2.2.14, $D$ is the midpoint of $XY$, and $OU, OV$ form a pair of conjugate radii. Also, theorem 6.1.7 gives $(UU', AD) = -1$. Together with the fact that $O$ is the midpoint of $UU'$, this yields $\overline{OA} \cdot \overline{OD} = p^2$. Similarly, since $B, C$ are conjugate and $D$ is the midpoint of $XY$, we obtain $\overline{DB} \cdot \overline{DC} = \overline{DX}^2 = -\overline{DX} \cdot \overline{DY}$.

Applying the power-of-a-point theorem to $\omega$ gives $\overline{DA} \cdot \overline{DA'} = \overline{DB} \cdot \overline{DC} = -\overline{DX} \cdot \overline{DY}$. On the other hand, $\overline{DU} \cdot \overline{DU'} = \overline{OD}^2 - p^2 = \overline{OD} \cdot \overline{AD}$. The power theorem for conics therefore gives

$$\frac{p^2}{q^2} = \frac{\overline{DU} \cdot \overline{DU'}}{\overline{DX} \cdot \overline{DY}} = \frac{\overline{OD} \cdot \overline{AD}}{-\overline{DA} \cdot \overline{DA'}} = \frac{\overline{OD}}{\overline{DA'}}.$$

[2] The four points $A, B, C, H$ form a concave quadrilateral, so the conic through them in the configuration under consideration is a hyperbola.

Combining this with $\overline{OA}\cdot\overline{OD}=p^2$, we obtain $\overline{OA}\cdot\overline{DA'}=q^2$. Applying the power-of-a-point theorem once more and using proposition 2.2.18, we find

$$\operatorname{pow}_\omega(O)=\overline{OA}\cdot\overline{OA'}=\overline{OA}(\overline{OD}+\overline{DA'})=p^2+q^2=a^2+b^2,$$

where $a,b$ are the semimajor and semiminor axes, respectively. By the Monge circle theorem, the director circle of $\alpha$ is centered at $O$ and has squared radius $a^2+b^2$. Hence proposition 4.5.4 shows that it is orthogonal to $\omega$.

For a hyperbola, the same calculation gives $\operatorname{pow}_\omega(O)=p^2+q^2$, but the radii must now be taken on the original hyperbola, and imaginary radii are allowed. Here an imaginary radius means the following: if the real radius has length $p$, and the real radius in the other conjugate direction on the conjugate hyperbola has length $q_0$, then $q^2=-q_0^2$. By proposition 2.1.24, $p^2+q^2=p^2-q_0^2=a^2-b^2$, where $a,b$ are the semi-transverse and semi-conjugate axes, respectively. This is again the squared radius of the director circle, so the conclusion still holds.

When the director circle is a point circle or an imaginary circle, the statement is understood by the corresponding limit or in the complex Euclidean plane. The case of a parabola follows by taking a limit of central conics. The director circle then becomes the directrix, and the conclusion is that the center of $\omega$ lies on the directrix.

**Solution to Exercise 6.29.** Let $\triangle M_aM_bM_c$ be the medial triangle of $\triangle ABC$, and let $\triangle A_1B_1C_1$ be the anticevian triangle of $\mathcal{S}p$ with respect to $\triangle M_aM_bM_c$. The anticevian triangle of $G$ with respect to $\triangle M_aM_bM_c$ is $\triangle ABC$.

By theorem 6.3.8, the eight points $\mathcal{S}p,G,A,B,C,A_1,B_1,C_1$ lie on one conic $\alpha$. Since $\mathcal{S}p$ is the incenter of $\triangle M_aM_bM_c$, the triangle $\triangle A_1B_1C_1$ is its excentral triangle; hence $\mathcal{S}p$ is the orthocenter of $\triangle A_1B_1C_1$. By exercise 6.22, $\alpha$ is a rectangular hyperbola. Applying exercise 6.22 once more, now to $\triangle ABC$, shows that $H\in\alpha$.

**Solution to Exercise 6.44.** Draw the two tangents from $F$ to $\alpha$. They are the lines $t_3,t_4$ through the circular points $I,J$, and they meet $m$ at $M_3,M_4$, respectively. Let $M_1',M_2'$ be the intersections of $M_1F,M_2F$ with the line at infinity.

We use the following projective fact: if $l_1,l_2,l_3,l_4$ are four fixed tangents to a conic and another tangent meets them at $L_1,L_2,L_3,L_4$, then $(L_1,L_2;L_3,L_4)$ is independent of that variable tangent. It follows here that $(M_1,M_2;M_3,M_4)=\text{const}$. Projection from $F$ gives $(M_1,M_2,M_3,M_4)\overset{(F)}{\doublebarwedge}(M_1',M_2',I,J)$, so $(M_1',M_2';I,J)=\text{const}$. By theorem 6.4.16, $\angle M_1FM_2=\text{const}$.

**Solution to Exercise 6.45.** Let $\triangle ABC$ be circumscribed about the parabola, let $F$ be its focus, and let $I,J$ be the circular points. The triangle $\triangle IJF$ is also circumscribed about the parabola. Thus $\triangle ABC$ and $\triangle IJF$ are two triangles circumscribed about the same conic. By exercise 6.24, the six points $A,B,C,I,J,F$ lie on one circle, namely the circumcircle of $\triangle ABC$.

**Solution to Exercise 6.51.** The point $P$ lies on the Euler circle of $\triangle ABC$, which is the circumcircle of its medial triangle $\triangle M_aM_bM_c$. Hence the reflections of $P$ in the three sidelines of the medial triangle are collinear: their line is the Steiner line of $P$ with respect to the nine-point circle. Hence there is a parabola with focus $P$ and this line as directrix, tangent to all three sidelines of $\triangle M_aM_bM_c$ (why?).

Take an arbitrary line tangent to this parabola. Let it meet $AB,AC$ at $C_1,B_1$, respectively, put $Q=B_1B\cap C_1C$, and set $A_1=AQ\cap BC$. Consider the complete quadrilateral $\mathscr{a}$ formed by $\triangle M_aM_bM_c$ and the line at infinity, and the complete quadrilateral $\mathscr{b}$ formed by $\triangle A_1B_1C_1$ and the trilinear polar of $Q$. It is easy to verify that they satisfy the hypotheses of example 6.5.12, since their diagonal trilaterals are both $\triangle ABC$. It follows that $\triangle A_1B_1C_1$ is circumscribed about the parabola and that the Miquel point of $\mathscr{b}$ is $P$.[3]

[3]After the properties of rectangular hyperbolas have been developed, one obtains a wider statement. For every point on the circumrectangular hyperbola of $\triangle ABC$ centered at $P$, its trilinear polar has the same property: the complete quadrilateral formed by that trilinear polar and the three sidelines of the cevian triangle of its trilinear pole has Miquel point $P$.

**Solution to Exercise 6.57.** Apply the standard polarity centered at $P$, and let $\alpha'$ be the image of $\alpha$. The lines $\flat(A), \flat(B)$ become two perpendicular tangents to $\alpha'$. Their intersection is $\flat(AB)$, which therefore lies on the Monge circle $\omega'$ of $\alpha'$. Notice that $P \notin \alpha$, so $\alpha'$ is necessarily a central conic.

The line $\flat(Q)$ passes through $\flat(AB)$ and is perpendicular to $PQ$. By theorem 1.3.6, it is always tangent to a conic $\beta$ having $P$ as a focus. By theorem 6.5.5, the inverse polar image $\flat(\beta)$ is a circle, and this circle is precisely the locus of $Q$.

**Solution to Exercise 6.69.** Consider the map $f : PA \mapsto PB$. It is an involution. Indeed, if the slope of $PA$ is $k$, then the slope of $f(PA)$ is $-\frac{c+ak}{a+bk}$. It is readily verified that this map preserves cross-ratios: assign slopes to four lines and expand the definition directly using theorem 3.2.7. Moreover, $k_1, k_2$ occur symmetrically in $a(k_1+k_2)+b(k_1k_2)+c=0$, so $f \circ f = \mathrm{id}$. The rest is the same as in the preceding proof.

**Solution to Exercise 6.71.** Let $\triangle LMN$ be the medial triangle of $\triangle ABC$. Suppose $l$ meets $BC, CA, AB$ at $A_1, B_1, C_1$, respectively, and let $A', B', C'$ be the midpoints of $AA_1, BB_1, CC_1$. Then $n = \overline{A'B'C'}$, and $A', B', C'$ lie on $NL, LM, MN$, respectively, as in figure B.4.

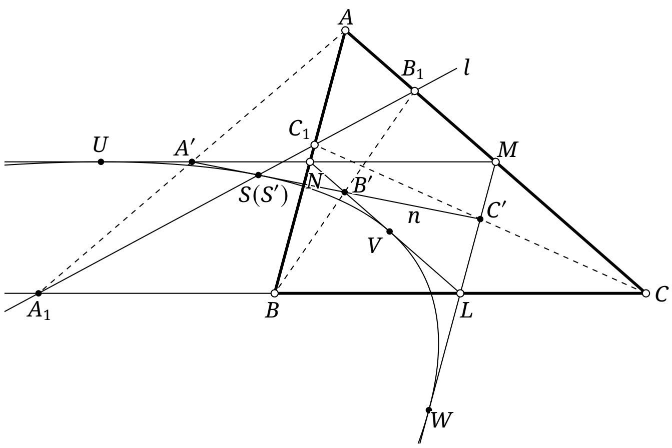


FIGURE B.4

Clearly,

$$[A'] \overset{(A)}{\barwedge} [A_1] \barwedge [l] \barwedge [B_1] \overset{(B)}{\barwedge} [B'].$$

By the projective definition of a conic, the variable line $n = A'B'$ envelops a conic $\alpha$ tangent to $NL$ and $LM$; cyclically, it is also tangent to $MN$. As $l$ tends to the line at infinity, so does $n$. Therefore $\alpha$ is tangent to $l_\infty$ and is a parabola. More precisely, in this limit $n$ becomes parallel to $l$, so the axis of $\alpha$ is parallel to $l$. Thus $X$ is the point of $\alpha$ at infinity.

Let $n$ touch $\alpha$ at $S$, and let $l$ meet $\alpha$ at $X, S'$. It remains to prove that $S = S'$. Since $n$ and $ML$ are tangents to $\alpha$, we have the projective chain

$$\alpha[S] \barwedge \alpha[n] \barwedge [C'] \overset{(C)}{\barwedge} [C_1] \barwedge [l].$$

On the other hand, because $l$ passes through $X \in \alpha$, projection from $X$ gives $[l] \barwedge [S']$. We have therefore obtained a projectivity $\alpha[S] \barwedge \alpha[S']$; it suffices to show that this projectivity is the identity.

Let $l$ tend to a line through $B$. Then $n \to NL$. Since $n$ and $NL$ are tangents to $\alpha$, their intersection $B'$ tends to the point of contact of $NL$ with $\alpha$. But when $l$ passes through $B$, one has $B' = l \cap NL$. Thus the intersection $l \cap n$ is then the point where $n$ touches $\alpha$, and $S = S'$. The same conclusion holds when $l$ passes through $A$ or $C$. We have found three fixed points of the projectivity $[S] \barwedge [S']$. Since three corresponding point pairs determine a projectivity, it must be the identity. Hence $S = S'$ for every $l$.

**Solution to Exercise 6.78.** We first establish a lemma.

*Let $\alpha$ be a central conic other than a circle, with center $O$, and let $l$ be the tangent at a variable point $P \in \alpha$. As $P$ ranges over $\alpha$, the distance from $O$ to $l$ can assume any prescribed value at no more than four points.*

*Proof.* For an ellipse, symmetry reduces the proof to showing that this distance is monotonically increasing as $P$ moves from the upper vertex $M$ to the right-hand vertex $N$. Drop perpendiculars from the foci $F_1, F_2$ and the center $O$ to the tangent, with feet $A, B, D$, respectively. By proposition 1.3.7, the product $AF_1 \cdot BF_2$ is constant, whereas $OD = \frac{AF_1+BF_2}{2}$. An interior extremum of $OD$ could therefore occur only when $AF_1 = BF_2$, or when $AF_1$ itself is extremal (equivalently, when $BF_2$ is extremal). Clearly, none of these conditions occurs while $P$ lies between $M$ and $N$ (excluding the endpoints). Thus $OD$ is strictly monotone as $P$ moves from $M$ to $N$, and symmetry gives at most four points altogether. The proof for a hyperbola is identical.

Return to the problem. If $T$ lies on the Gauss line of $ABCD$, there is a central conic centered at $T$ and tangent to the four sidelines of $ABCD$. If the four sidelines are tangent to one circle, the center of that circle plainly satisfies the required equal-distance conditions.

Suppose now that the quadrilateral is not circumscribed about a circle. By the lemma, equality of the distances from the center to one pair of opposite sidelines means that those sidelines are either parallel or mirror images in an axis of the conic. If one pair is parallel, $ABCD$ is a proper trapezoid or a parallelogram. Otherwise the two pairs of opposite sidelines are symmetric about the two axes of the conic, as in figure B.6; by proposition 0.3.8, the quadrilateral is cyclic. The converse is not difficult to prove.

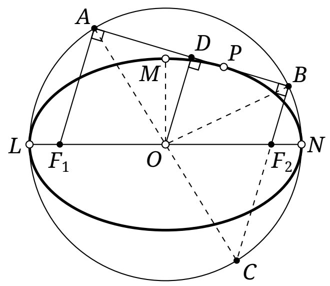


FIGURE B.5

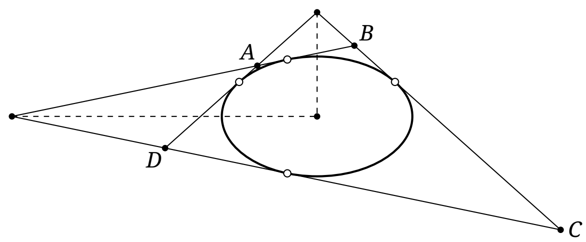


FIGURE B.6

**Solution to Exercise 6.79.** By hypothesis, $O$ is equidistant from $AX, BX$, and also from $AY, BY$. Thus $O$ is equidistant from each pair of opposite sidelines of the quadrilateral whose sides, in order, lie on $AX, BY, XB, AY$. This is not the quadrilateral $AXBY$. More explicitly, if $M = AX \cap BY$ and $N = AY \cap BX$, it is the quadrilateral $AMBN$. Figure B.7 illustrates one possible configuration. Its Gauss line is $CZ$. Such a quadrilateral cannot be a trapezoid or a parallelogram.

By exercise 6.78, there are therefore two possibilities. First, its four vertices may lie on one circle. The bisectors of the angles formed by its pairs of opposite sides are then $PO$ and $QO$, and proposition 0.3.8 shows that they are perpendicular. Second, its four sidelines may be tangent to one circle, as in figure B.7. This circle is an excircle: not all four points of contact lie on the corresponding side segments. Equality of tangent lengths then gives $AX + XB = AY + YB$. The converse follows in the same way.

**Solution to Exercise 6.77.** See theorems 8.3.8 and 8.3.9.

**Solution to Exercise 6.88.** Put $F = AC \cap BD$. We have $F_1P \perp F_2P$, while the harmonic property of the complete quadrilateral gives $P(AB, EF) = -1$. By proposition 3.2.17, the lines $PE, PF$ are the internal and external bisectors of $\angle F_1PF_2$. Since example 6.7.18 also gives $PE \perp PF$, we obtain $\angle F_2PE = 45^\circ$.

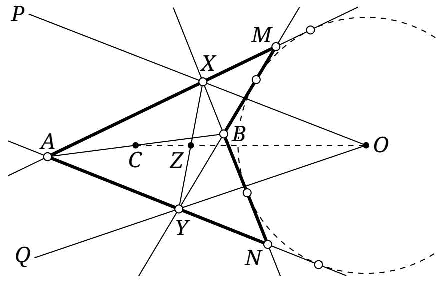


FIGURE B.7

Let $G$ be the second intersection of $PE$ with $\odot O$. Then $\measuredangle APG = 45°$. The inscribed-angle theorem shows that $G$ is one of two fixed points satisfying $GO \perp F_1F_2$. By Brocard's theorem, $\mathfrak{p}(E) = PG$, so $E \in \mathfrak{p}(G)$. As $G$ is one of two fixed points, $\mathfrak{p}(G)$ is one of two fixed lines. These are the required loci.

**Solution to Exercise 6.94.** As in figure B.8, put $L = AX \cap BC$. It is enough to prove that $X,U,V,T$ are concyclic. Since $T$ is the Miquel point of the degenerate quadrilateral $ABDC$, $\angle LDT = \angle ACT = 180° - \angle TXA$. Hence $L,X,D,T$ are concyclic.

Apply the dual Desargues involution theorem to the degenerate quadrilateral $ABDC$ and the point $X$. The line pairs $(XA,XD)$, $(XB,XC)$, and $(XU,XV)$ belong to one involution. Therefore $(L,D)$, $(B,C)$, and $(U,V)$ are conjugate pairs of one involution $f$ on $BC$.

Let $I,J$ be the circular points, and apply Desargues's involution theorem to the quadrangle $IJXT$ and the line $BC$. The pairs $(B,C)$ and $(L,D)$ belong to the resulting involution $g$ on $BC$: indeed, each conjugate pair of $g$ is the pair of intersections with $BC$ of a circle through $X,T$.

The involutions $f,g$ have two conjugate pairs in common, so $f = g$. Consequently $(U,V)$ is a conjugate pair of $g$, and the construction of $g$ shows that $U,V,T,X$ are concyclic.

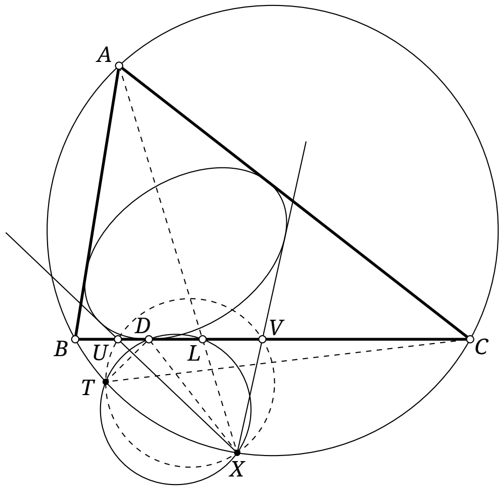


FIGURE B.8

# Chapter 7

**Solution to Exercise 7.4.** As in figure B.9, let $l \cap \mathscr{C} = T,T'$, let $P'$ be the projection of $A$ on $l$, and let $X'$ be the projection of $A$ on $PX$. It is easy to prove that $B'C' \parallel BC \parallel AX'$. Indeed, let $C''$ be the projection of $O$ on $AC$. Since the points $A,B',W,C'$ are concyclic, we have $\measuredangle C'B'A = \measuredangle C'WA = \measuredangle C''OA = \measuredangle CBA$, and hence $B'C' \parallel BC$.

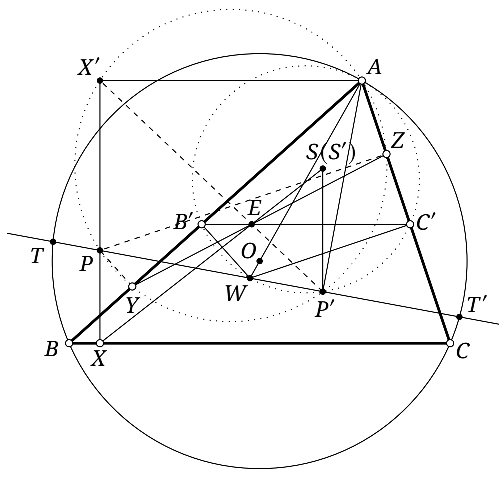


FIGURE B.9

The five points $A,B',C',W,P'$ are concyclic on $\odot(AW)$, while $A,Z,Y,P,P',X'$ are concyclic on $\odot(AP)$. It follows that $\measuredangle P'YZ = \measuredangle P'AZ = \measuredangle P'C'B'$. Thus $P',Y,C',E$ are concyclic, and consequently $\measuredangle P'EC' = \measuredangle P'YC = \measuredangle P'X'A$. Together with $B'C' \parallel AX'$, this shows that $P',E,X'$ are collinear.

Through $P'$ draw the perpendicular to $AB$, meeting $XE$ at $S'$. We have $\triangle S'P'E \sim \triangle XX'E$, so

$$S'P' = \frac{XX' \cdot EP'}{EX'} = \frac{XX' \cdot \mathrm{dist}(B'C',P')}{\mathrm{dist}(B'C',AX')} = \text{const.}$$

Hence $S'$ is fixed. Now take $P=T$ and $P=T'$. In these two cases $X,Y,Z,E$ lie respectively on the Simson lines $\mathscr{S}(T)$ and $\mathscr{S}(T')$. By corollary 7.1.9, $S' = \mathfrak{o}(TT') = S$.

**Solution to Exercise 7.16.** Let the quadrilateral be $ABCD$ and put $S = BC \cap AD$. By proposition 7.2.21, the corresponding isoconjugations $\mathbf{j}_1,\mathbf{j}_2$ in $\triangle ABS, \triangle CDS$ determine the isoconjugation $\mathbf{j}$ of the quadrilateral: if $P \in \gamma$, then $\mathbf{j}_1(P) = \mathbf{j}_2(P)$. Thus, when $P \in l$, the condition $P \in \gamma$ requires $\mathbf{j}P \in \mathbf{j}_1(l) \cap \mathbf{j}_2(l)$. For $l$ in general position, $\mathbf{j}_1(l)$ and $\mathbf{j}_2(l)$ are circumconics of $\triangle ABS$ and $\triangle CDS$, respectively. After their common point $S$ is excluded, their other three intersections are the candidate positions for $\mathbf{j}(P)$.

On the other hand, by the definition of isoconjugation, the involutions induced by $\mathbf{j}_1,\mathbf{j}_2$ on the pencil at $S$ both have $(SX,SX')$ as a corresponding pair and also have $(SA=SD,SB=SC)$ as a corresponding pair. Hence they are the same involution. If $Q$ is an intersection of $\mathbf{j}_1(l)$ and $\mathbf{j}_2(l)$ other than $S$, it follows that $\mathbf{j}_1(Q)$ and $\mathbf{j}_2(Q)$ are the same point of $l$. Therefore each intersection of $\mathbf{j}_1(l)$ and $\mathbf{j}_2(l)$ other than $S$ corresponds to one of the original intersections in $l \cap \gamma$.

**Solution to Exercise 7.22.**

(1) Put $\mathscr{W} = \mathrm{pen}(AB,BC,CD,DA)$. For any point $Z$, Plücker's theorem says that $\mathscr{W}$ induces an involution $g_Z$ on the pencil at $Z$; two of its degenerate members give the corresponding line pairs $(ZA,ZC)$ and $(ZB,ZD)$. Suppose that $Z \in \Gamma$, and let $E,F,G,H$ be the orthogonal projections of $Z$ onto $AB,BC,CD,DA$, respectively. By theorem 4.4.19, the points $E,F,G,H$ are concyclic. Perpendicularity also shows that the four quadruples $(Z,A,E,H)$, $(Z,B,E,F)$, $(Z,C,F,G)$, and $(Z,D,G,H)$ are cyclic. Hence

$$\measuredangle HEF = \measuredangle HEA + \measuredangle BEF = \measuredangle HZA + \measuredangle BZF,$$
$$\measuredangle HGF = \measuredangle HGD + \measuredangle CGF = \measuredangle HZD + \measuredangle CZF.$$

Since $\measuredangle HEF = \measuredangle HGF$, rearranging gives $\measuredangle AZB + \measuredangle CZD = 0$. Thus the two line pairs $(ZA,ZC)$ and $(ZB,ZD)$ have the same angle bisectors.

An involution is determined by two corresponding pairs, so $g_Z$ is the reflection in an angle bisector of $\angle(ZA,ZC)$. By theorem 4.4.20, some conic in $\mathscr{W}$ has foci $U,U'$. The little Poncelet theorem says that the two tangents from $Z$ to this conic form one corresponding pair under $g_Z$, while $(ZU,ZU')$ forms another. Likewise, $ZV$ and $ZV'$ correspond under $g_Z$. It follows that $Z$ possesses an isogonal conjugate with respect to the quadrilateral $UVU'V'$. The converse is similar.

(2) Since $f$ is an opposite similarity, for $X \neq U,V$ we have $\measuredangle U'XV' = -\measuredangle Uf^{-1}(X)V$. Part (1) and the cyclicity criterion therefore give, for $X \in \omega \setminus \{U,V\}$,
$$X \in \Gamma \iff \measuredangle UXV = \measuredangle Uf^{-1}(X)V \iff f^{-1}(X) \in \omega \iff X \in f(\omega).$$
Thus the finite points of $\omega \cap \Gamma$ other than $U,V$ are precisely $W,Z$, and $WZ$ is the radical axis of $\omega$ and $f(\omega)$.

Choose two general circles $\omega_1,\omega_2$ through $U,V$, with centers $O_1,O_2$, and let the two radical axes of $\omega_i$ and $f(\omega_i)$ meet at $T$. Let $\omega$ be a circle through $U,V$, with center $O$, and put $\lambda = \dfrac{\overline{O_1O}}{\overline{O_1O_2}}$. The three circles pass through $U$. By Stewart's theorem, for any point $Y$ we have
$$YO^2 = (1-\lambda)YO_1^2 + \lambda YO_2^2 - \lambda(1-\lambda)O_1O_2^2,$$
$$UO^2 = (1-\lambda)UO_1^2 + \lambda UO_2^2 - \lambda(1-\lambda)O_1O_2^2.$$
Consequently,
$$\begin{aligned}\operatorname{pow}_\omega(Y) = YO^2 - UO^2 &= (1-\lambda)(YO_1^2 - UO_1^2) + \lambda(YO_2^2 - UO_2^2)\\ &= (1-\lambda)\operatorname{pow}_{\omega_1}(Y) + \lambda\operatorname{pow}_{\omega_2}(Y).\end{aligned}$$
Let $k$ be the similarity ratio of $f$. An opposite similarity sends the center of a circle to the center of its image and multiplies both radii and distances from centers by $k$. Hence
$$\begin{aligned}\operatorname{pow}_{f(\omega)}(Y) = k^2\operatorname{pow}_\omega(f^{-1}(Y)) &= k^2\big[(1-\lambda)\operatorname{pow}_{\omega_1}(f^{-1}(Y)) + \lambda\operatorname{pow}_{\omega_2}(f^{-1}(Y))\big]\\ &= (1-\lambda)\operatorname{pow}_{f(\omega_1)}(Y) + \lambda\operatorname{pow}_{f(\omega_2)}(Y).\end{aligned}$$
The same conclusion can also be obtained from exercise 4.79. Since $\operatorname{pow}_{\omega_i}(T) = \operatorname{pow}_{f(\omega_i)}(T)$ for $i=1,2$, we have $\operatorname{pow}_\omega(T) = \operatorname{pow}_{f(\omega)}(T)$ for every $\omega$. Thus all the lines $WZ$ pass through $T$. Now take the circle $\eta$ through $U,V,T$. Then $\operatorname{pow}_\eta(T) = \operatorname{pow}_{f(\eta)}(T) = 0$, and the first half of this part gives $T \in \Gamma$. Therefore $T = T_{UV}$.

(3) In the parabolic limiting case of theorem 4.4.20, the focus of a parabola inscribed in $ABCD$ and the point at infinity $N$ on its axis are isogonally conjugate. Hence $N \in \Gamma$. Put $K = UV \cap U'V'$. By theorem 7.2.22, $K \in \Gamma$, and consequently $K = U * V$.

Let $W$ tend to $N$ along $\Gamma$, take $\omega = \odot(UVW)$, and denote by $Z$ the other finite intersection of $\omega$ and $f(\omega)$. Every circle passes through the circular points $I,J$, so the limit of $\omega$ contains $l_\infty$, while its other component is $UV$. Similarly, the limit of $f(\omega)$ is $l_\infty \cup U'V'$. It follows that $Z \to K$, and hence $WZ \to NK$. Since $T_{UV} \in WZ$ throughout, $T_{UV} \in NK$. Therefore $T_{UV} = N*K = N*(U*V)$.

(4) By definition, $A*(A*B) = B$; in particular, $N*(N*A) = A$. By parts (2) and (3), the point of $\Gamma$ on the circle through $A,B,C$ other than those three points is $h(A,B,C) := (N*(A*B))*C$. Thus $h(A,B,C)$ is symmetric in $A,B,C$. Put $Z = h(P,Q,R)$. Then $Z*R = N*(P*Q)$ and $Z*Q = N*(P*R)$, so
$$\begin{aligned}(P*Q)*(R*S) &= (N*(Z*R))*(R*S) = h(Z,R,R*S) = h(R,R*S,Z)\\ &= (N*(R*(R*S)))*Z = (N*S)*Z = (N*(Q*(Q*S)))*Z\\ &= h(Q,Q*S,Z) = h(Z,Q,Q*S) = (N*(Z*Q))*(Q*S) = (P*R)*(Q*S).\end{aligned}$$
This proves $(P*Q)*(R*S) = (P*R)*(Q*S)$.

(5) We may suppose that the point not yet known to lie on $\Gamma$ is $P_{33}$. From the intersections of $l_1,l_2,m_1,m_2$ with $\Gamma$, we have $P_{11}*P_{12} = P_{13}$, $P_{21}*P_{22} = P_{23}$, $P_{11}*P_{21} = P_{31}$, and $P_{12}*P_{22} = P_{32}$. Applying part (4) to $P_{11},P_{12},P_{21},P_{22}$ gives $P_{13}*P_{23} = P_{31}*P_{32}$. The point on the left lies on $m_3$, while the point on the right lies on $l_3$. It is therefore $P_{33} = l_3 \cap m_3$, and $P_{33} \in \Gamma$.

*Note.* In fact, every nonsingular cubic—that is, every cubic without a singular point—can be realized as the locus of points possessing isoconjugates under some isoconjugation of some quadrilateral. A projective transformation can then carry it to an isogonal-conjugacy locus. This proves the three-line form of the *Cayley–Bacharach theorem*: if the nine points $P_{ij} = l_i \cap m_j$ determined by six lines $l_1, l_2, l_3, m_1, m_2, m_3$ are distinct and eight of them lie on a cubic $\Gamma$, then the ninth also lies on $\Gamma$.

**Solution to Exercise 7.28.** By corollary 7.4.20, and regarding the center of a parabola as a point at infinity, the isotomic conjugate of the perspector of every parabolic inconic lies at infinity. Apply an affine transformation taking the reference triangle to an equilateral triangle. For an equilateral triangle the isotomic and isogonal conjugations coincide, and the isogonal conjugates of points at infinity lie on the circumcircle. Thus, in the equilateral case, the required perspectors trace the circumcircle. Applying the inverse affine transformation shows that in the original triangle they trace the Steiner circumellipse.

**Solution to Exercise 7.31.** Let $Q$ be the fourth point of $\mathscr{C} \cap \alpha$, apart from $A, B, C$. By corollary 2.4.2, the axes of $\alpha$ are parallel to the bisectors of the angle between $AB$ and $CQ$. Thus the hypothesis that the axis directions are fixed for a given $l$ forces $Q$ to be fixed.

By theorem 7.3.19, $P \in \mathscr{T}(Q)$. Since $P$ ranges over $l$, it follows that $l = \mathscr{T}(Q)$. But $Q \in \mathscr{C}$, so another application of theorem 7.3.19 shows that $l$ passes through the perspector of $\mathscr{C}$. This perspector is the Lemoine point of $\triangle ABC$.

**Solution to Exercise 7.40.** By proposition 5.7.12, the Lemoine point $L$ is the perspector of the Brocard ellipse. Moreover, proposition 5.7.16 gives $\mathbf{t}L = Br_3$, while proposition 5.7.14 says that $Br_{\mathrm{m}}$ is the center of the Brocard ellipse. The claim now follows from corollary 7.4.20.

**Solution to Exercise 7.52.** Let $G$ be the centroid. Then

$$I \star Mi = I \times (I \div Mi)^{\mathbf{c}} = I \times (Ge \div G)^{\mathbf{c}} = I \times (Mi \div G) = I \times (I \div Ge) = \mathbf{g}Ge = In.$$

The second and fourth equalities use the identity $I \times G = Mi \times Ge$ proved in example 7.5.21; the third uses proposition 5.4.3.

# Chapter 8

**Solution to Exercise 8.9.** As in figure 8.12, let $H$ be the orthocenter of $\triangle APB$. Then $H \in \alpha$, and $P, D, H$ are collinear. Put $E = AP \cap BH$ and $F = BP \cap AH$. Brocard's theorem gives $EF = \mathfrak{p}(D) = l$. Since $E, F, A, B$ are concyclic,

$$\measuredangle(EF, AB) = \measuredangle(EF, EA) + \measuredangle(EA, AB) = \measuredangle FEA + \measuredangle EAB = \measuredangle PBA + \measuredangle PAB.$$

**Solution to Exercise 8.19.** Continue with the notation in the proof of proposition 8.2.7. Let the tangent to $\alpha$ at $A$ meet the directrix $LR_A$ at $N$, and let $l$ be the normal at $A$. By theorem 1.3.3, $AP \perp PN$, so $A, P, N, R_A$ are concyclic. Moreover, $\triangle PAR_A \sim \triangle QII_D$, and consequently $\measuredangle(PA, l) = \measuredangle PNA = \measuredangle PR_AA = \measuredangle II_DQ$. Since $PA \parallel II_D$, it follows that $l \parallel I_DQ$.

**Solution to Exercise 8.20.** Let $R$ be the circumradius of $\triangle ABC$, and let $O, Ni$ be its circumcenter and nine-point center. Let the foci and center of $\Gamma$ be $P, Q$ and $O'$, let its major and minor axes have lengths $2a, 2b$, and write $PQ = 2c = 2\sqrt{a^2 - b^2}$. Denote its inner auxiliary circle by $\omega$. The condition that $\omega$ and $\mathscr{N}$ are tangent can be written in the equivalent forms

$$NiO' \pm b = r_{\mathscr{N}} = \frac{R}{2} \quad \Longleftrightarrow \quad b^2 = \left(NiO' - \frac{R}{2}\right)^2.$$

Equivalently, using Aiyar's theorem,

$$b^2 = \left(\frac{OP \cdot OQ}{2R} - \frac{R}{2}\right)^2 \quad \Longleftrightarrow \quad 4(bR)^2 = (R^2 - OP \cdot OQ)^2.$$

On the other hand, theorem 8.2.5 gives $4(bR)^2 = (R^2 - OP^2)(R^2 - OQ^2)$. Thus tangency is equivalent to

$$(R^2 - OP^2)(R^2 - OQ^2) = (R^2 - OP \cdot OQ)^2.$$

After simplification, this is just $OP=OQ$. In other words, $O$ lies on the perpendicular bisector of $PQ$, as required.

**Solution to Exercise 8.21.** Apply polarity with respect to the incircle $\odot I$. The triangle $\triangle ABC$ becomes its intouch triangle $\triangle A'B'C'$, while $\Gamma$ becomes a fixed inconic $\gamma$ of $\triangle A'B'C'$. Let $P,Q$ be its foci, and let $M$ be its center.

Suppose the circumcircle $\Omega$ of the original triangle $\triangle ABC$ is tangent to a circle $\omega$. Under polarity they become conics $\Omega'$ and $\omega'$ that are still tangent. By proposition 6.5.4, $\Omega'$ and $\omega'$ have a common focus; hence exercise 1.9 shows that their auxiliary circles are tangent. Combining exercise 4.1 and proposition 6.5.4, the auxiliary circle of $\Omega'$ is precisely the nine-point circle $\mathcal{N}'$ of $\triangle A'B'C'$. Thus, to prove that $\Omega$ is tangent to two fixed circles, it is enough to prove that $\mathcal{N}'$ is always tangent to two fixed circles.

The center of $\mathcal{N}'$ is the nine-point center $Ni'$ of $\triangle A'B'C'$, while the circumcenter of $\triangle A'B'C'$ is the original incenter $I$. By theorem 8.2.6, $Ni'M \cdot 2r_{\mathcal{N}'} = IP \cdot IQ$. Furthermore, $2r_{\mathcal{N}'} = r_{\mathscr{C}_{\triangle A'B'C'}} = r_{\odot I}$ is fixed, and $M,P,Q$ are fixed. Hence $Ni'M$ and $r_{\mathcal{N}'}$ are both constant. The circle $\mathcal{N}'$ is therefore tangent to two fixed circles. This proves the first assertion.

For the second assertion, suppose that $I$ lies on the major axis of $\Gamma$, and assume without loss of generality that $\Gamma$ is an ellipse. Its polar image $\gamma$ is then an inscribed ellipse with $I$ on its minor axis. The auxiliary circle $\alpha$ of $\Gamma$ becomes an ellipse $\alpha'$ whose major axis is the minor axis of $\gamma$. Thus the auxiliary circle $\alpha_1$ of $\alpha'$ is the inner auxiliary circle of $\gamma$. Since $I$ is the circumcenter of $\triangle A'B'C'$, exercise 8.20 shows that $\alpha_1$ is tangent to $\mathcal{N}'$. Reversing the polarity, the original circumcircle is tangent to $\alpha$. The hyperbolic and parabolic cases require only the corresponding minor modifications of exercise 8.20; the essential argument is the same, so we do not repeat it.

**Solution to Exercise 8.23.** As in figure B.10, let $M_1,M_2$ be the midpoints of $AC,BD$, respectively. Through $U$ draw $UL \parallel VW$, meeting $\alpha$ again at $L$, and put $T = \mathrm{t}_\alpha(U) \cap VW$. The proof of proposition 8.3.7 shows that $M_1M_2$ bisects $VW$ and is a diameter of $\alpha$. Since $UL \parallel VW$, the diameter property shows that $M_1M_2$ also bisects $UL$. Moreover, proposition 8.3.7 gives $UL \parallel \mathrm{t}_\alpha(M_2) = M_2M_2$. Hence $M_2(M_2M_1,UL) = -1$. Notice that

$$M_2(M_2,M_1,U,L) \overline{\wedge} U(M_2,M_1,U,L) \overline{\wedge} (F,E,T,\infty_{VW}),$$

so $(FE,T\infty_{VW}) = -1$. Therefore $T$ is the midpoint of $EF$.

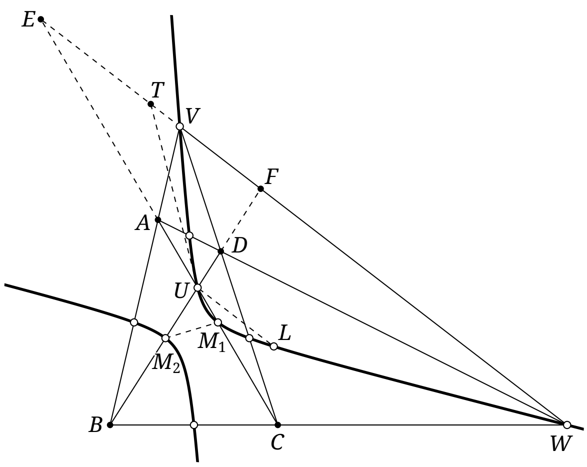


FIGURE B.10

**Solution to Exercise 8.26.** Let $\alpha$ be the nine-point conic of the complete quadrangle $DBCE$. Then $A,F,M,N \in \alpha$, and theorem 8.3.8 shows that $\alpha$ is a rectangular hyperbola. By theorem 8.3.5, $MN$ is a diameter. It also bisects the chord $AF$ of $\alpha$ at $L$, so proposition 8.1.24 yields $LM \cdot LN = LA^2$.

**Solution to Exercise 8.27.** By theorem 8.3.5, the center of the nine-point conic $\alpha$ of $ABCD$ is also $M$. By corollary 2.4.2, an axis of a conic $\beta$ through $ABCD$ is parallel to an asymptote $l$ of $\alpha$. Let the center of $\beta$ tend to infinity along $\alpha$. When its center $O$ becomes a point at infinity, $O$ also lies on the asymptote $l$, and $\beta$ becomes a parabola. Its axis is parallel to $l$ and therefore can only coincide with $l$. Hence the axis of the parabola passes through $M$.

**Solution to Exercise 8.37.** Let $F_1,F_2$ be the common foci of $\alpha,\beta$, and let $O$ be their common center. Let the tangents to $\beta$ at $X,Y$ meet at $B$; this is precisely the pole specified in the exercise. Put $P = \mathfrak{f}_{XB}(XY)\cap\mathfrak{f}_{YB}(XY)$. Then $B$ is an isocenter of $\triangle XYP$. By proposition 8.4.5, the lines $XP,YP$ are tangent to $\alpha$. The optical property shows that $BX$ bisects the angle between $XF_1,XF_2$, while $YB$ bisects the angle between $YF_1,YF_2$.

For a proof using example 6.7.17, observe also that $F_1B$ bisects the angle between $F_1X,F_1Y$. Hence $B$ is equidistant from the four lines $XF_1,XF_2,YF_1,YF_2$, which are therefore tangent to one circle centered at $B$. Let $H$ be the projection of $B$ on $XY$. By example 6.7.17, $BH$ bisects $\angle F_1HF_2$, so $XY$ bisects its exterior angle. The optical property now identifies $H$ as the point $A$ at which $XY$ is tangent to $\alpha$. Thus $BA\perp XY$.

For a proof using exercise 6.79, the fact that $\beta$ is an ellipse gives $F_1X+F_2X=F_1Y+F_2Y$. By exercise 6.79, the line $BO$ passes through the midpoint $M$ of $XY$. Consider the dual pencil $\mathscr{P}$ determined by the degenerate complete quadrilateral of lines $XP,YP,XA,YA$; the coincident lines $XA,YA$ give the fixed point of contact $A$. We have $\alpha\in\mathscr{P}$, and it has center $O$. Since $M$ is the midpoint of the diagonal $XY$ of the degenerate quadrilateral $XAYP$, theorem 6.7.4 identifies $OM$ as its Gauss line. Consequently, there is a conic $\omega\in\mathscr{P}$ centered at $B$. By corollary 7.4.21, the conic centered at $B$ and tangent to $PX,XY,YP$ is unique: it is the in- or excircle of $\triangle PXY$ centered at $B$. Its point of contact with $XY$ is therefore the fixed point $A$, so $BA\perp XY$.

**Solution to Exercise 8.43.** Let $a,b$ be the semimajor and semiminor axes of the outer ellipse $\alpha$, and let $a_0,b_0$ be those of the inner ellipse $\beta$. Denote their common foci by $F_1,F_2$ and their common center by $O$, and put $\rho=\frac{ab}{\sqrt{b^2-b_0^2}}$.

We first prove a lemma. *Suppose a chord $AB$ of $\alpha$ is tangent to $\beta$. Let $t$ be the tangent to $\alpha$ at $A$, put $h=\operatorname{dist}(O,t)$, and let $\eta$ be the angle between the ray $AB$ and the inward normal ray to $\alpha$ at $A$. Then $h=\rho\cos\eta$.*

*Proof of the lemma.* Put $u=AF_1$, $v=AF_2$, and $\varphi=\frac{1}{2}\angle F_1AF_2$. By the optical property, the distances from the two foci to $t$ are $u\cos\varphi$ and $v\cos\varphi$. Since $O$ is the midpoint of $F_1F_2$ and $u+v=2a$, it follows that $h=a\cos\varphi$. Applying proposition 1.3.7 first to the tangent $t$ of $\alpha$ and then to the tangent $AB$ of $\beta$ gives

$$uv\cos^2\varphi=b^2,\qquad uv\sin(\eta+\varphi)\sin(\eta-\varphi)=b_0^2.$$

Subtracting and simplifying the trigonometric expressions yields $uv\cos^2\eta=b^2-b_0^2$. Eliminating $uv$ and $\varphi$ from this identity, the first displayed identity, and $h=a\cos\varphi$ gives $h=\rho\cos\eta$, as claimed.

Return now to the problem. Let the polygon be $P_1\cdots P_n$, let its interior angles be $\theta_1,\dots,\theta_n$, and let its perimeter be $L$. Write $h_i$ for the distance from $O$ to the tangent to $\alpha$ at $P_i$. Here $P_{i+n}=P_i$, and the other subscripts are read cyclically in the same way. By proposition 8.4.5, the inward normal ray at $P_i$ bisects the interior angle of the polygon. The lemma therefore gives $h_i=\rho\cos\frac{\theta_i}{2}$.

Let $O_i$ be the projection of $O$ on $P_iP_{i-1}$. Then

$$\overline{P_iO_i}+\overline{P_iO_{i+1}}=2h_i\cos\frac{\theta_i}{2},$$

where $\overline{P_iO_i}$ is positive when it has the same direction as $\overline{P_iP_{i-1}}$, and $\overline{P_iO_{i+1}}$ is positive when it has the same direction as $\overline{P_iP_{i+1}}$. Hence

$$\begin{aligned}L&=\sum_{i=1}^{n}P_iP_{i+1}=\sum_{i=1}^{n}(\overline{P_iO_{i+1}}+\overline{O_{i+1}P_{i+1}})=\sum_{i=1}^{n}(\overline{P_iO_i}+\overline{P_iO_{i+1}})\\&=2\sum_{i=1}^{n}h_i\cos\frac{\theta_i}{2}=2\rho\sum_{i=1}^{n}\cos^2\frac{\theta_i}{2}=\rho\Big(n+\sum_{i=1}^{n}\cos\theta_i\Big).\end{aligned}$$

By theorem 8.4.9, the perimeter $L$ is constant, while $\rho$ and $n$ are fixed. Therefore $\sum_{i=1}^{n}\cos\theta_i$ is constant.

**Solution to Exercise 8.44.** By definition, *Na* is the perspector. By corollary 7.4.20 and propositions 5.2.3 and 5.4.3, the center is *Mi*. It remains to prove that the ellipse passes through *Fe*.

For a triangle $\triangle ABC$, let $I$ be its incenter and let $I_a,I_b,I_c$ be its excenters. Since $\triangle ABC$ and $\triangle I_aI_bI_c$ are perspective from the incenter, there is a circumellipse $\alpha$ of $\triangle ABC$ tangent to the three sides of $\triangle I_aI_bI_c$. By proposition 8.4.5, there is an inellipse $\beta$ of $\triangle ABC$ confocal with $\alpha$. Moreover, proposition 8.4.6 and the optical property show that $\beta$ touches the three sidelines of $\triangle ABC$ at the respective orthogonal projections of $I_a,I_b,I_c$. Thus $\beta$ is the Mandart ellipse of $\triangle ABC$.

Let $O$ be the common center of $\alpha$ and $\beta$. By the optical property, the normals to $\alpha$ at $A,B,C$ concur at the incenter $I$. Consider the Apollonius hyperbola $\gamma$ of $I$ with respect to $\alpha$. It is a rectangular hyperbola through $A,B,C,I,O$, and its asymptotes are parallel to the common axes of the two ellipses. By propositions 8.1.14 and 8.1.15, its center is the Feuerbach point *Fe*.

Apply a stretch $f$, along the common axes and fixing $O$, that sends the inner ellipse $\beta$ to a circle. Then $f(\beta)$ is the incircle of $f(\triangle ABC)$, so $O$ is the incenter of $f(\triangle ABC)$. The curve $f(\gamma)$ still passes through $O$, and its asymptotes remain perpendicular. Thus $f(\gamma)$ is a rectangular circumhyperbola of $f(\triangle ABC)$ through its incenter. Its center is consequently the Feuerbach point $Fe'$ of $f(\triangle ABC)$, so $Fe' \in f(\beta)$. But the center of $f(\gamma)$ is $f(Fe)$. Hence $f(Fe)\in f(\beta)$, and therefore $Fe \in \beta$.

**Solution to Exercise 8.45.** Let $\triangle ABC$ be the original triangle, let $I$ be its incenter, and let $O$ be the common center of the two ellipses.

(1) By exercise 8.43, the sum $\cos A+\cos B+\cos C$ is constant. The identity

$$\sum_{\text{cyc}}\cos A = 4\prod_{\text{cyc}}\sin\frac{A}{2}+1,$$

together with proposition 0.6.16, proves the assertion.

(2) By proposition 8.4.6 and the optical property—or, equivalently, by the proof of theorem 8.4.10—the excentral triangle of $\triangle ABC$ is the triangle formed by the tangents to $\alpha$ at $A,B,C$. Hence the excenters of $\triangle ABC$ lie on the polar curve of $\beta$ with respect to $\alpha$.

(3) By exercise 8.44, $Fe\in\beta$, and the proof of that exercise shows that *Fe* is the center of the Apollonius hyperbola $\gamma$ of $I$ with respect to $\alpha$. By theorem 8.3.13, the positions of this center and of $I$ are related by a fixed stretch along the two axes. Therefore $I$ moves on a fixed ellipse.

# Chapter 9

**Solution to Exercise 9.31.** First prove sufficiency. Since $\mathscr{A}$ does not belong to the codomain, $\mathscr{A}$ and $f(\mathscr{H})$ are distinct members of $\mathfrak{B}(\mathscr{A}\cap\mathscr{H};\mathscr{A}\vee\mathscr{H})$. Consequently,

$$\mathscr{A}\cap f(\mathscr{H})=\mathscr{A}\cap\mathscr{H},\qquad \mathscr{A}\vee f(\mathscr{H})=\mathscr{A}\vee\mathscr{H}.$$

The inverse map therefore satisfies analogous conditions: after interchanging $\mathscr{H}$ and $f(\mathscr{H})$, we still have

$$\dim(\mathscr{A}\cap f(\mathscr{H}))=\mu-2,\quad \mathscr{H}\in\mathfrak{B}(\mathscr{A}\cap f(\mathscr{H});\mathscr{A}\vee f(\mathscr{H})).$$

If $\mathscr{A}\nsubseteq\mathscr{N}$, then $\dim(\mathscr{A}\cap\mathscr{N})\geq\dim(\mathscr{A}\cap\mathscr{H})=\mu-2$. Also $\mathscr{A}\nsubseteq\mathscr{N}$ implies $\mathscr{A}\vee\mathscr{N}\supsetneq\mathscr{N}$, so $\dim(\mathscr{A}\vee\mathscr{N})>\mu$. Hence

$$\dim(\mathscr{A}\cap\mathscr{N})=(\mu-1)+\mu-\dim(\mathscr{A}\vee\mathscr{N})\leq\mu-2.$$

Since $\mathscr{A}\cap\mathscr{H}\subseteq\mathscr{A}\cap\mathscr{N}$, we must have $\mathscr{A}\cap\mathscr{N}=\mathscr{A}\cap\mathscr{H}$. For every $\mathscr{H}$, the subnet $\mathscr{A}\cap\mathscr{H}$ is contained in every member of the domain pencil, so $\mathscr{A}\cap\mathscr{H}\subseteq\mathscr{K}$. Both sides have dimension $\mu-2$, and therefore $\mathscr{A}\cap\mathscr{N}=\mathscr{A}\cap\mathscr{H}=\mathscr{K}$, whence $\mathscr{K}\subsetneq\mathscr{A}$. The same argument applies to the codomain using the inverse map. Thus, for the domain, either $\mathscr{K}\subsetneq\mathscr{A}$ or $\mathscr{A}\subseteq\mathscr{N}$; for the codomain, either $\mathscr{K}'\subsetneq\mathscr{A}$ or $\mathscr{A}\subseteq\mathscr{N}'$.

*Case 1:* $\mathscr{K},\mathscr{K}' \subsetneq \mathscr{A}$. For every domain member, $\mathscr{A}\cap\mathscr{H}=\mathscr{K}$, and for every codomain member, $\mathscr{A}\cap f(\mathscr{H})=\mathscr{K}'$. Hence $\mathscr{K}=\mathscr{K}'$. The hypothesis shows that every codomain member is contained in $\mathscr{A}\vee\mathscr{H}\subseteq\mathscr{A}\vee\mathscr{N}$. The images span all of $\mathscr{N}'$, so $\mathscr{N}'\subseteq\mathscr{A}\vee\mathscr{N}$. Joining both sides with $\mathscr{A}$ gives $\mathscr{A}\vee\mathscr{N}'\subseteq\mathscr{A}\vee\mathscr{N}$. Since the inverse map satisfies analogous conditions, we also have $\mathscr{A}\vee\mathscr{N}\subseteq\mathscr{A}\vee\mathscr{N}'$. Thus $\mathscr{A}\vee\mathscr{N}=\mathscr{A}\vee\mathscr{N}'$. Furthermore, we already have $\mathscr{A}\cap\mathscr{N}=\mathscr{K}$, and similarly $\mathscr{A}\cap\mathscr{N}'=\mathscr{K}'=\mathscr{K}$. Therefore

$$\mathscr{A}\cap\mathscr{N}=\mathscr{A}\cap\mathscr{N}'=\mathscr{A}\cap\mathscr{K}=\mathscr{K}.$$

These are the hypotheses of case (1) of theorem 9.4.10.

*Case 2:* $\mathscr{A}\subseteq\mathscr{N},\mathscr{N}'$. The equality $\mathscr{A}\vee f(\mathscr{H})=\mathscr{A}\vee\mathscr{H}$ shows that the codomain is contained in $\mathscr{N}$. The images span all of $\mathscr{N}'$, so $\mathscr{N}'\subseteq\mathscr{N}$. Conversely, the same argument gives $\mathscr{N}\subseteq\mathscr{N}'$, and hence $\mathscr{N}=\mathscr{N}'$. The hypothesis implies that $\mathscr{A}\cap\mathscr{K}$ is contained in every codomain member, so $\mathscr{A}\cap\mathscr{K}\subseteq\mathscr{K}'$, and hence $\mathscr{A}\cap\mathscr{K}\subseteq\mathscr{A}\cap\mathscr{K}'$. The analogous conditions for the inverse map also give $\mathscr{A}\cap\mathscr{K}'\subseteq\mathscr{A}\cap\mathscr{K}$. Therefore $\mathscr{A}\cap\mathscr{K}=\mathscr{A}\cap\mathscr{K}'$. Here $\mathscr{A}$ is a hypernet of $\mathscr{N}$ and does not belong to the domain, so $\mathscr{K}\nsubseteq\mathscr{A}$. Thus $\mathscr{A}\subsetneq\mathscr{A}\vee\mathscr{K}\subseteq\mathscr{N}$. Since $\mathscr{A}$ is a hypernet of $\mathscr{N}$, we have $\mathscr{A}\vee\mathscr{K}=\mathscr{N}$, and similarly $\mathscr{A}\vee\mathscr{K}'=\mathscr{N}$. Also $\mathscr{A}\vee\mathscr{N}=\mathscr{N}$ and $\mathscr{N}\nsubseteq\mathscr{A}$, so

$$\mathscr{A}\vee\mathscr{K}=\mathscr{A}\vee\mathscr{K}'=\mathscr{A}\vee\mathscr{N}=\mathscr{N}.$$

This is case (2) of theorem 9.4.10.

*Case 3:* $\mathscr{A}\cap\mathscr{N}=\mathscr{K}$ *but* $\mathscr{A}\subseteq\mathscr{N}'$. Then $\mathscr{A}\cap\mathscr{H}=\mathscr{K}$ for every domain member, and $\mathscr{A}\cap f(\mathscr{H})=\mathscr{K}$ for every codomain member. Thus $\mathscr{K}$ is contained in every codomain member, so $\mathscr{K}\subseteq\mathscr{K}'$. Their dimensions are equal, and hence $\mathscr{K}=\mathscr{K}'$. Therefore $\mathscr{A}$ contains the center of the domain pencil and is also a hypernet of $\mathscr{N}'$, so $\mathscr{A}\in\mathfrak{B}(\mathscr{K}';\mathscr{N}')$, a contradiction. Similarly, we cannot have $\mathscr{A}\subseteq\mathscr{N}$ but $\mathscr{A}\cap\mathscr{N}'=\mathscr{K}'$.

For necessity, suppose case (1) of theorem 9.4.10 holds and $\mathscr{K}\subsetneq\mathscr{A}$. Then

$$\mathscr{A}\cap\mathscr{N}=\mathscr{A}\cap\mathscr{N}'=\mathscr{A}\cap\mathscr{K}=\mathscr{K}.$$

For any $\mathscr{H}\in\mathfrak{B}(\mathscr{K};\mathscr{N})$, we thus have $\mathscr{A}\cap\mathscr{H}=\mathscr{K}$, so $\dim(\mathscr{A}\cap\mathscr{H})=\mu-2$. Also $\dim(\mathscr{A}\vee\mathscr{H})=\mu$, and case (1) gives $\mathscr{K}\subseteq f(\mathscr{H})\subseteq\mathscr{A}\vee\mathscr{H}$. Therefore $f(\mathscr{H})\in\mathfrak{B}(\mathscr{A}\cap\mathscr{H};\mathscr{A}\vee\mathscr{H})$.

If case (2) of theorem 9.4.10 holds and $\mathscr{A}\subseteq\mathscr{N}$, then $\mathscr{A}$ is a hypernet of $\mathscr{N}$. Since $\mathscr{A}$ does not belong to the domain, for every $\mathscr{H}\in\mathfrak{B}(\mathscr{K};\mathscr{N})$ we have $\mathscr{H}\neq\mathscr{A}$. Hence $\dim(\mathscr{A}\cap\mathscr{H})=\mu-2$ and $\mathscr{A}\vee\mathscr{H}=\mathscr{N}$. By case (2), $\mathscr{A}\cap\mathscr{H}=\mathscr{A}\cap f(\mathscr{H})$, so

$$\mathscr{A}\cap\mathscr{H}\subseteq f(\mathscr{H})\subseteq\mathscr{N}=\mathscr{A}\vee\mathscr{H}.$$

It follows that $f(\mathscr{H})\in\mathfrak{B}(\mathscr{A}\cap\mathscr{H};\mathscr{A}\vee\mathscr{H})$.

# Chapter 10

**Solution to Exercise 10.1.** The necessity uses only the basic axioms. The sufficiency under condition (a) uses the basic axioms and Desargues's theorem, whereas the sufficiency under condition (b) uses the basic axioms and the stronger theorem of Pappus.

For necessity, suppose that $f$ is a perspectivity. The line joining any point to $X$ intersects $n$ at $X$ itself, so $f(X)=X$. Under $l \overset{(P)}{\overline{\overline{\wedge}}} m$, the point $X$ is sent to $PX\cap m$, and under $m \overset{(Q)}{\overline{\overline{\wedge}}} n$ this point is sent to $Q(PX\cap m)\cap n$. Thus we must have $X=Q(PX\cap m)\cap n$, or equivalently $Q,X,PX\cap m$ are collinear. If $l,m,n$ are concurrent, then $PX\cap m=X$, and $Q,X,X$ are plainly collinear. Otherwise it implies that $P,Q,X$ are collinear.

Now prove sufficiency. Suppose first that (a) holds, as in figure B.11a. Fix $A\in l$ and let $B\in l$ vary. Put $A'=PA\cap m$, $B'=PB\cap m$, $A''=QA'\cap n$, and $B''=QB'\cap n$. Thus $l(A,B) \overset{(P)}{\overline{\overline{\wedge}}} m(A',B') \overset{(Q)}{\overline{\overline{\wedge}}} n(A'',B'')$. Let $R=AA''\cap PQ$. We claim that $R\in BB''$, which will give $l(A,B) \overset{(R)}{\overline{\overline{\wedge}}} n(A'',B'')$. Apply Desargues's theorem to $\triangle AA'A''$ and $\triangle BB'B''$. The lines $AB,A'B',A''B''$ concur at $X$. Hence the three points $AA''\cap BB''$, $P$, and $Q$ are collinear. In other words, $AA'',PQ,BB''$ are concurrent, proving the claim.

Suppose next that (b) holds, as in figure B.11b, so $P,Q,X$ are collinear. Put $Y = m\cap n$, $Z = m\cap l$, and take the fixed point $R = PY \cap QZ$. For a variable point $A \in l$, let $A' = PA \cap m$ and $A'' = QA' \cap n$. Then $l[A] \overset{(P)}{\barwedge} m[A'] \overset{(Q)}{\barwedge} n[A'']$. To prove $l[A] \overset{(R)}{\barwedge} n[A'']$, it remains to show that $A, A'', R$ are collinear. Apply Pappus's theorem to the two collinear triples $P,Q,X$ and $Z,Y,A'$. The three Pappus points are $PY \cap QZ = R$, $PA' \cap XZ = A$, and $QA' \cap XY = A''$, so they are collinear, as required.


(a) (b)

FIGURE B.11

**Solution to Exercise 10.2.** This proof requires the basic axioms and Desargues's theorem.

As in figure B.12, choose $A, A' \in l$, keeping $A$ fixed and allowing $A'$ to vary. Put $B = PA\cap m$, $B' = PA' \cap m$, $C = QB \cap n$, and $C' = QB' \cap n$. Then $l(A,A') \overset{(P)}{\barwedge} m(B,B') \overset{(Q)}{\barwedge} n(C,C')$. Let $P' = PQ \cap n$. Since $P,Q,X$ are not collinear, $P' \neq X$, and hence $P' \notin l$. Define $D = P'A \cap QC$, $D' = P'A' \cap QC'$, and $Z = m \cap l$.

Apply the converse of Desargues's theorem to $\triangle ABD$ and $\triangle A'B'D'$. The intersections $P,Q,P'$ of their corresponding sides are collinear, so $DD', m, l$ are concurrent. Their common point is $Z$, and therefore $D, D', Z$ are collinear. Once $A$ is fixed, the line $DZ$ is fixed. Writing $m_1 = DD'$, we obtain $l(A,A') \overset{(P')}{\barwedge} m_1(D,D') \overset{(Q)}{\barwedge} n(C,C')$, and this composition induces the original projectivity. Repeating the symmetric construction with $Q' = PQ \cap l$ gives a line $m'$ for which $l \overset{(P')}{\barwedge} m' \overset{(Q')}{\barwedge} n$ induces the same projectivity.

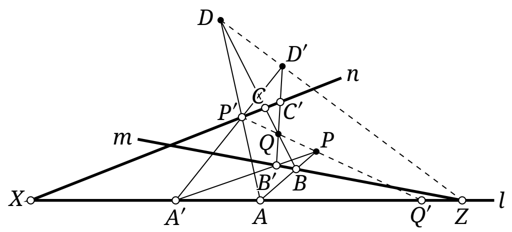


FIGURE B.12

**Solution to Exercise 10.28.** Assume A. We prove C* from C; the reverse implication follows by duality. We already know that C implies B, namely Desargues's theorem, and therefore B* also holds.

Let $a,b,c$ be three lines concurrent at $O$, and let $a',b',c'$ be three lines concurrent at $O'$. Put $P = a\cap b'$, $Q = a' \cap b$, $R = a \cap c'$, $S = a' \cap c$, $T = b \cap c'$, and $W = b' \cap c$. We must prove that $PQ, RS, TW$ are concurrent.

Apply Pappus's theorem to the collinear triples $Q,O,T$ and $P,O',W$. It shows that $a\cap a'$, $QW \cap PT$, and $c \cap c'$ are collinear. These are the intersections of corresponding sides of $\triangle PRT$ and $\triangle QSW$. The converse of Desargues's theorem therefore gives the concurrence of $PQ, RS, TW$. Dualizing this argument proves C from C*.

**Solution to Exercise 10.32.** Although $\mathbb{H}$ is a noncommutative division ring by theorem 10.5.11, the dual statement naturally belongs to $\mathbb{P}^2(\mathbb{H}^{\mathrm{op}})$. One can verify that quaternionic conjugation $f:\ \mathbb{H} \longrightarrow \mathbb{H}^{\mathrm{op}}$, given by

$$a+b\mathbf{i}+c\mathbf{j}+d\mathbf{k} \longmapsto \overline{a+b\mathbf{i}+c\mathbf{j}+d\mathbf{k}} = a-b\mathbf{i}-c\mathbf{j}-d\mathbf{k},$$

reverses multiplication on $\mathbb{H}$ and therefore gives the isomorphism $\mathbb{H} \cong \mathbb{H}^{\mathrm{op}}$. Hence the dual statement is equivalent to the original one.

**Solution to Exercise 10.44.** For $P \in \pi$, define $P_x = \underbracket{\infty_y\infty_z P} \cap x$, and define $P_y, P_z$ cyclically.

First suppose that exactly two of $\infty_x, \infty_y, \infty_z$ lie on $\pi$; all three cannot lie there. Without loss of generality, $\infty_x \notin \pi$. If $\pi \cap x = A = \boldsymbol{x}^{-1}(m,0,0)$, the basic incidence axiom A gives $\mathrm{x}_x(P_x) = m$. We may therefore use the parametrization

$$\mathrm{x}_x(P) = 0 \cdot s + 0 \cdot t + m, \quad \mathrm{x}_y(P) = a_y s + b_y t + c_y, \quad \mathrm{x}_z(P) = a_z s + b_z t + c_z.$$

The other two possibilities are identical after permuting the coordinates.

Now suppose that $\pi$ omits at least two of the three points at infinity; without loss of generality, $\infty_y, \infty_z \notin \pi$. Define $P_{xz} = \infty_y P \cap \underbracket{xz}$. As in the first case,

$$\mathrm{x}_z(P) = \mathrm{x}_z(P_z) = \mathrm{x}_z(P_{xz}), \quad \mathrm{x}_x(P) = \mathrm{x}_x(P_x) = \mathrm{x}_x(P_{xz}).$$

There is a point $Q \in \Pi_\infty$ such that $Q \notin \underbracket{xz} \cup \underbracket{xy} \cup \pi$, with one exception: $\mathbb{K}$ has only two elements (that is, $\mathbb{K} = \mathbb{F}_2$) and $\infty_x \in \pi$. In that exceptional case the points of $\pi$ can be listed directly. For example, if $x \subset \pi$, its finite points are $(0,0,0)$, $(1,0,0)$, $(0,1,1)$, and $(1,1,1)$, and the required parametrization is plainly

$$\mathrm{x}_x(P) = s + 0 \cdot t + 0, \qquad \mathrm{x}_y(P) = \mathrm{x}_z(P) = 0 \cdot s + t + 0.$$

We may now treat all remaining cases uniformly. Let $\varphi:\ \pi \longrightarrow \underbracket{xz}$ be the perspective affine map with center $\infty_y$, and let $\psi:\ \pi \longrightarrow \underbracket{xz}$ be the perspective affine map with center $Q$. Write $\psi(P) = R = \boldsymbol{x}^{-1}(s,0,t)$. The map $\varphi \circ \psi^{-1}:\ R \longmapsto P_{xz}$ is a two-dimensional perspective affine transformation of $\underbracket{xz}$. By theorem 10.6.3,

$$\mathrm{x}_x(P) = \mathrm{x}_x(P_{xz}) = a_x s + b_x t + c_x, \quad \mathrm{x}_z(P) = \mathrm{x}_z(P_{xz}) = a_z s + b_z t + c_z.$$

The planes $\underbracket{xz}$ and $\underbracket{xy}$ acquire natural two-dimensional projective coordinates by retaining, respectively, the $x,z$ and $x,y$ coordinates. Their unit points have three-dimensional coordinates $\boldsymbol{x}(E_{xz}) = (1,0,1)$ and $\boldsymbol{x}(E_{xy}) = (1,1,0)$. Their points at infinity on the $x$-axis coincide, as do their origins. Moreover, $E_{xz}E_{xy}$ and $\infty_y\infty_z$ have a common point $S$ at infinity. Let $\theta:\ \underbracket{xz} \longrightarrow \underbracket{xy}$ be the perspective affine map with center $S$, and put $R' = \theta(R)$. By theorem 10.3.4,

$$\mathrm{x}_x(R') = \mathrm{x}_x(R) = s, \qquad \mathrm{x}_y(R') = \mathrm{x}_z(R) = t, \tag{†}$$

while of course $\mathrm{x}_z(R') = 0$.[4]

Let $\phi':\ \pi \longrightarrow \underbracket{xy}$ be the perspective affine map with center $\infty_z$. Then $\phi' \circ \psi^{-1} \circ \theta^{-1}:\ R' \longmapsto P_{xy}$, where $P_{xy}$ is defined analogously to $P_{xz}$ and has the analogous coordinate properties, is a perspective affine transformation. Combining theorem 10.6.3 with (†) gives

$$\mathrm{x}_x(P) = \mathrm{x}_x(P_{xz}) = a'_x \mathrm{x}_x(R') + b'_x \mathrm{x}_y(R') + c'_x,$$
$$\mathrm{x}_y(P) = \mathrm{x}_y(P_{xy}) = a_y \mathrm{x}_x(R') + b_y \mathrm{x}_y(R') + c_y = a_y s + b_y t + c_y.$$

Although the first expression for $\mathrm{x}_x(P)$ appears different from the one obtained above, the derivation forces $a'_x = a_x$, $b'_x = b_x$, and $c'_x = c_x$. The second expression supplies the required affine form of $\mathrm{x}_y(P)$.

Conversely, it is straightforward to verify that every parametrization of the stated form represents a finite plane; we leave this verification to the reader.

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[4]In theorem 10.3.4, the two coordinate axes are identified isomorphically in this situation. The four base points determine the entire projective coordinate system, so all remaining corresponding points have the same coordinates when regarded in the two-dimensional systems.

# Appendix A

**Solution to Exercise A.3.** The answer is $f(x)=Cx$, where $C$ is a constant. See the discussion on page 642 of section A.9. Although that discussion is stated over the complex numbers, its opening argument gives exactly the result needed here.

**Solution to Exercise A.5.** For arbitrary $a,b\in\mathbb{Q}$, an automorphism $f$ must satisfy $f(a+b)=f(a)+f(b)$. This is the functional equation in exercise A.3, so $f(x)=Cx$. Here the constant $C$ cannot be zero and must, as required, be a nonzero rational number. Thus

$$\operatorname{Aut}(\mathbb{Q})=\{f: x\mapsto Cx \mid C\in\mathbb{Q}, C\neq 0\}.$$

**Solution to Exercise A.21.** First suppose that $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=0$ for every column vector $\boldsymbol{u}$. Then, for arbitrary column vectors $\boldsymbol{x},\boldsymbol{y}$,

$$0=(\boldsymbol{x}+\boldsymbol{y})^{\mathsf{T}}\boldsymbol{A}(\boldsymbol{x}+\boldsymbol{y})=\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}+\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}+\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}=\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}+\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}.$$

Thus $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}=-\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}$. This holds for all $\boldsymbol{x},\boldsymbol{y}$, and theorem A.2.15 shows that $\boldsymbol{A}$ is skew-symmetric.

Now suppose there is a column vector $\boldsymbol{u}$ for which $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}\neq 0$. For an arbitrary column vector $\boldsymbol{v}$, put $\boldsymbol{y}=\boldsymbol{v}-\frac{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}}\boldsymbol{u}$. Here both $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}$ and $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}$ are scalars. We then have

$$\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{y}=\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}-\frac{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=0.$$

This implies

$$0=\boldsymbol{y}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}-\frac{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}}\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}-\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}.$$

Therefore $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}$ for every $\boldsymbol{v}$.

Fix any $\boldsymbol{x}$. The expression

$$(\boldsymbol{x}+t\boldsymbol{u})^{\mathsf{T}}\boldsymbol{A}(\boldsymbol{x}+t\boldsymbol{u})=(\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u})t^2+(\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}+\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x})t+\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}$$

is a quadratic polynomial in $t$ whose leading coefficient $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}$ is nonzero. Hence some $t$ makes it nonzero. By the preceding argument, for every $\boldsymbol{v}$,

$$0=\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}(\boldsymbol{x}+t\boldsymbol{u})-(\boldsymbol{x}+t\boldsymbol{u})^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}=(\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}+t\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u})-(\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}+t\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}).$$

Because $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{u}=\boldsymbol{u}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}$, the terms containing $t$ cancel, leaving $\boldsymbol{v}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{x}^{\mathsf{T}}\boldsymbol{A}\boldsymbol{v}$. This holds for all $\boldsymbol{x},\boldsymbol{v}$, so theorem A.2.15 shows that $\boldsymbol{A}$ is symmetric.

**Solution to Exercise A.23.** First, an orthonormal family is linearly independent. Indeed, if

$$x_1\boldsymbol{v}_1+\cdots+x_n\boldsymbol{v}_n=\boldsymbol{0}$$

for orthonormal vectors $\boldsymbol{v}_1,\dots,\boldsymbol{v}_n$, taking the dot product with $\boldsymbol{v}_i$ gives $x_i=0$. Thus this equation has only the zero solution.

Second, a direct substitution in the stated recursion shows that the vectors $\boldsymbol{a}'_1,\boldsymbol{a}'_2,\boldsymbol{a}'_3,\dots$ are pairwise orthogonal.

Third, assume that $\boldsymbol{a}_1,\boldsymbol{a}_2,\boldsymbol{a}_3,\dots$ are linearly independent. Then all the vectors $\boldsymbol{a}'_1,\boldsymbol{a}'_2,\boldsymbol{a}'_3,\dots$ constructed by this method are nonzero. For if $\boldsymbol{a}'_i=\boldsymbol{0}$, then $\boldsymbol{a}_i$ is a linear combination of $\boldsymbol{a}'_1,\dots,\boldsymbol{a}'_{i-1}$; by construction, each of those vectors is a linear combination of $\boldsymbol{a}_1,\dots,\boldsymbol{a}_{i-1}$. This would make $\boldsymbol{a}_i$ a linear combination of $\boldsymbol{a}_1,\dots,\boldsymbol{a}_{i-1}$, contrary to the assumed linear independence.

Thus the $\boldsymbol{a}'_i$ form a nonzero pairwise orthogonal family and are therefore linearly independent. Dividing each vector by its magnitude gives the required orthonormal family.

**Solution to Exercise A.27.** Define

$$\boldsymbol{C}_i = \begin{bmatrix} 1 & \cdots & 0 & x_1 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & x_{i-1} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & x_i & 0 & \cdots & 0 \\ 0 & \cdots & 0 & x_{i+1} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & x_n & 0 & \cdots & 1 \end{bmatrix},$$

where the displayed replacement column is column $i$, and the row containing $x_i$ is row $i$. Also define

$$\boldsymbol{B}_i = \begin{bmatrix} a_{11} & \cdots & a_{1,i-1} & b_1 & a_{1,i+1} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,i-1} & b_{i-1} & a_{i-1,i+1} & \cdots & a_{i-1,n} \\ a_{i1} & \cdots & a_{i,i-1} & b_i & a_{i,i+1} & \cdots & a_{in} \\ a_{i+1,1} & \cdots & a_{i+1,i-1} & b_{i+1} & a_{i+1,i+1} & \cdots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{n,i-1} & b_n & a_{n,i+1} & \cdots & a_{nn} \end{bmatrix},$$

where again the displayed replacement column is column $i$, and the row containing $b_i$ is row $i$. Since $\boldsymbol{Ax} = \boldsymbol{b}$, the definition of matrix multiplication gives $\boldsymbol{AC}_i = \boldsymbol{B}_i$. From the definition of the determinant, $\det \boldsymbol{C}_i = x_i$: among $\sigma \in \mathrm{S}_n$, only the identity permutation $\sigma = \mathrm{id}$ can have all the entries $c_{i,\sigma(i)}$ nonzero. That is, only the product of the diagonal entries contributes to the determinant. Taking determinants and using theorem A.2.23, we obtain

$$\det \boldsymbol{B}_i = \det(\boldsymbol{AC}_i) = \det(\boldsymbol{A})\det(\boldsymbol{C}_i) = x_i \det \boldsymbol{A}.$$

It follows that $x_i = \frac{\det \boldsymbol{B}_i}{\det \boldsymbol{A}}$.

**Solution to Exercise A.38.** Let $e, e'$ be the identities of $G, G'$, respectively. Since $f$ is a group homomorphism, corollary A.3.26 gives $f(e) = e'$. Thus $e \in \ker f$, so the kernel is nonempty. If $a, b \in \ker f$, then $f(a) = f(b) = e'$, and corollary A.3.26 gives

$$f(ab^{-1}) = f(a)f(b^{-1}) = f(a)f(b)^{-1} = e'{e'}^{-1} = e'.$$

The subgroup criterion theorem A.3.14 therefore gives $\ker f \leq G$.

Next consider the image $\operatorname{im} f$. Since $f(e) = e' \in \operatorname{im} f$, the image contains the identity. For arbitrary $f(a), f(b) \in \operatorname{im} f$, corollary A.3.26 gives $f(a)f(b)^{-1} = f(a)f(b^{-1}) = f(ab^{-1}) \in \operatorname{im} f$. Another application of theorem A.3.14 gives $\operatorname{im} f \leq G'$.

**Solution to Exercise A.53.** If $f$ preserves addition, it is an isomorphism between the additive groups of the two division rings. By corollary A.3.26, it preserves their identity element, which in additive notation is zero.

If $f$ preserves multiplication, then $f(0) = f(0a) = f(0)f(a)$ for every $a \in K$. Because $f$ is bijective, $f(a)$ can be any element of $K'$. Taking $f(a)$ to be the zero of $K'$ gives $f(0) = 0$. The restriction of $f$ to $K \setminus \{0\}$ therefore gives an isomorphism $K^\times \to K'^\times$. By corollary A.3.26, this restriction preserves the multiplicative identity, so $f(1) = 1$.

**Solution to Exercise A.55.** For (1), if $R$ were a division ring and nonzero elements $a, b$ satisfied $ab = 0$, then $a^{-1}$ would exist. Multiplying both sides on the left by $a^{-1}$ gives $a^{-1}ab = 0$, and hence $b = 0$, a contradiction.

For (2), to prove sufficiency, suppose $a, b \neq 0$ and, to the contrary, $ab = 0$. Left multiplication by $a^{-1}$ gives $b = 0$, a contradiction. To prove necessity, suppose $ab \neq 0$ and, to the contrary, at least one of $a, b$ is zero. Then immediately $ab = 0$, again a contradiction.

**Solution to Exercise A.64.** Write

$$\mathbb{Z}_m = \{\overline{0}, \overline{1}, \ldots, \overline{m-1}\}, \qquad \mathbb{Z}_n = \{\tilde{0}, \tilde{1}, \ldots, \widetilde{n-1}\}.$$

Here the overline and the tilde have analogous meanings but distinguish congruence classes in the two different rings. Suppose first that $\mathbb{Z}_m \sim \mathbb{Z}_n$, and let $f : \mathbb{Z}_m \to \mathbb{Z}_n$ be a surjective homomorphism.

The image of the identity is again the identity, that is, $f(\overline{1}) = \tilde{1}$. Hence, for every integer $a$,

$$f(\overline{a}) = f(a\overline{1}) = af(\overline{1}) = a\tilde{1} = \tilde{a}.$$

Thus necessarily $f : \overline{x} \mapsto \tilde{x}$. In particular, $f(\overline{m}) = f(\overline{0}) = \tilde{0}$, and hence $\widetilde{m} = \tilde{0}$. Since $\widetilde{m} = \tilde{0}$, the definition of congruence implies that $n$ divides $m$, since only $\widetilde{kn} = \tilde{0}$ for $k \in \mathbb{Z}$.

Conversely, if $n$ divides $m$, it is also easy to prove that the map above is a surjective homomorphism.

**Solution to Exercise A.93.** For a skew-symmetric $n \times n$ matrix $\boldsymbol{A}$, $\boldsymbol{A}^\top = -\boldsymbol{A}$, and therefore

$$\det\boldsymbol{A} = \det\boldsymbol{A}^\top = \det(-\boldsymbol{A}) = \det((-\mathbf{I}_n)(\boldsymbol{A})) = \det(-\mathbf{I}_n)\det\boldsymbol{A} = (-1)^n \det\boldsymbol{A}.$$

If $n$ is odd and the field has characteristic different from 2, then $(-1)^n = -1$; over a field of characteristic 2, one instead has $-1 = 1$. In the former case the equality above gives $\det\boldsymbol{A} = -\det\boldsymbol{A}$, whence $\det\boldsymbol{A} = 0$. Thus $\boldsymbol{A}$ is singular.

If instead the field has characteristic 2, then $-1 = 1$. Writing $\boldsymbol{A} = (a_{ij})_{n\times n}$, we have

$$\det\boldsymbol{A} = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)} = \sum_{\sigma\in S_n} \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

If there is an $i$ such that $i = \sigma(i)$, then $a_{i,\sigma(i)} = 0$, so this permutation $\sigma$ contributes zero to the determinant. If $\sigma^{-1}$ is the inverse permutation, then

$$\prod_{i=1}^{n} a_{i,\sigma^{-1}(i)} = \prod_{j=1}^{n} a_{\sigma(j),j} = \prod_{j=1}^{n} a_{j,\sigma(j)},$$

where the first equality is the substitution $i = \sigma(j)$ and the last uses $\boldsymbol{A}^\top = \boldsymbol{A}$. Whenever $\sigma \neq \sigma^{-1}$, the contributions of $\sigma$ and $\sigma^{-1}$ cancel because $a + a = 0$ for every $a$ in a field of characteristic 2.

Finally, if $\sigma = \sigma^{-1}$ and there is no $i = \sigma(i)$, then the permutation must interchange the elements of $\{1, 2, \ldots, n\}$ in pairs. No such permutation exists when $n$ is odd. Thus every term in the determinant either vanishes or cancels with another term, so $\det\boldsymbol{A} = 0$. Hence $\boldsymbol{A}$ is not invertible.

**Solution to Exercise A.140.**

$$(x^x)' = (e^{x\ln x})' = e^{x\ln x}(x\ln x)' = x^x\left[(x)'\ln x + x(\ln x)'\right] = x^x(\ln x + 1).$$

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